Polyfold Theory, proof by proof

The formal proof of Hofer–Wysocki–Zehnder's Polyfold and Fredholm Theory, Part I and Part II, Chapters 7–10, in words

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Part I. The sc-calculus (HWZ Chapter 1, §§1.1–1.4)

Chapter 1 of Part I fixes the vocabulary of the whole theory: sc-Banach spaces, sc-operators, sc-subspaces and quotients, sc-Fredholm operators and their stability under sc^+-perturbations, partial quadrants, sc^0 and sc^1 maps, the tangent construction and the inductive definitions of sc^k and sc-smooth, the relation between sc-differentiability and classical differentiability on the levels (Propositions 1.2.1–1.2.4 (preprint 1.16–1.19) and Corollary 1.2.5 (preprint 1.20), quoted from the 2010 paper), the chain rule, the degeneracy index of a point of a partial quadrant and boundary recognition. The formal proof establishes all of it, Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the text's own hypothesis (C^{l+1} from U_{m+l} to F_m for $l \le k$), by the 2010 paper's induction (S_k) on elementary maps (§I.17). Example 1.1.3 (preprint 1.2), the weighted Sobolev spaces, is not formalized.

Two decisions shape everything. First, an sc-Banach space is an *indexed family* of Banach spaces E_m with injective compact inclusions $E_{m+1} \to E_m$, not one space with a chain of subspaces (§I.1). "The point x of E_0 lies in E_m" then means that x is in the image of E_m, and every map has level maps $f_m : E_m \to F_m$ determined by injectivity. Second, the text's standing hypothesis "U relatively open in a partial quadrant" enters the proofs of §§1.2–1.3 only through two facts about each level set U_m: it is the intersection of an open and a convex set, and derivatives within it are unique. These two facts make up the notion of an *sc-domain* (§I.10), which the tangent construction and the level shift preserve by a two-line argument, so the inductions of §1.2 run on sc-domains, and the text's tuples (U, C, E) are a special case.

Four proofs take a different road from the text's. The chain rule replaces the two integrals of §1.4.2 by the one-dimensional mean value inequality along a segment, and does not need the first integral at all (§I.12). The compactness argument behind Propositions 1.2.1–1.2.3 (preprint 1.16–1.18), the subsequence $h_n \to h$ in E_m for unit vectors $h_n \in E_{m+1}$, is isolated once as the *compact-open lemma*, and its C^k version replaces the multilinear maps $\Gamma(x)$ and the differentiation under the integral sign of the 2010 paper (§I.13). Theorem 1.3.5 (preprint 1.25), which the text quotes from another paper, is proved by the argument of §2.4 (preprint §2.3), for sc^1 germs (§I.19). And the induction (S_k) behind Proposition 1.2.4 (preprint 1.19) runs on the order s over all elementary maps $d^j f(\pi_0 z)(\pi_1 z, \dots, \pi_j z)$ at once, rather than on a description of the 2^{k+1} components of $T^{k+1} f$ (§I.17). Smaller departures are flagged where they occur: the complement in Proposition 1.1.11 (preprint 1.10) is built from a right inverse rather than a lifted basis; the regularizing step of §1.4.1 is a bootstrap; $k = k'$ in Lemma 1.3.4 (preprint 1.24) is derived from the lemma, not proved first. What Mathlib lacked, the Riesz–Schauder theorem, the stability of Fredholm operators under compact perturbations and the invariance of the Fredholm index, is proved in §I.7; the index statements of the exposition's Lemma 2.D and Proposition 1.1.11 (preprint 1.10) follow in §I.6 and §I.9. The lemmas that the exposition adds to make the text's remarks precise are proved where they belong: uniqueness of the sc-structure in finite dimensions (§I.1), sc-splittings as sc-projections (§I.3), the smooth points of a quotient (§I.5), the shift and tangent of a tuple (§I.10), compact-open continuity and $\mathrm{sc}^1 = C^1$ on quadrants in finite dimensions (§I.11).

Map.

- `Chapter1/ScBanach`: scales and sc-Banach spaces, the constant structure, sums and shifts; sc-operators, sc-isomorphisms, sc^+-operators; sc-subspaces, sc-splittings, Proposition 1.1.7 (preprint 1.6); quotients, Proposition 1.1.8 (preprint 1.7).
- `Chapter1/FredholmPerturbation`: Fredholm operators between Banach spaces; Riesz–Schauder; compact perturbations.
- `Chapter1/FredholmIndex`: the index of a linear map, its additivity, its invariance under small and compact perturbations; the index on the levels of an sc-Fredholm operator; the index in Proposition 1.1.11 (preprint 1.10).
- `Chapter1/ExpositionLemmas`: the exposition's Lemmas 1.A(2)–(3), 2.B, 2.C, 3.F, and $\mathrm{sc}^1 = C^1$ in finite dimensions on quadrants.
- `Chapter1/ScFredholm`: Definition 1.1.9 (preprint 1.8), the index, the level-wise Fredholm property; Propositions 1.1.10 and 1.1.11 (preprint 1.9 and 1.10).
- `Chapter1/ScSmooth`: level sets, the tangent TE and TU, level maps, sc^0; sc^1 with its derivative family, uniqueness of Df, sc^k and sc-smooth; partial quadrants, tuples, sc-domains.
- `Chapter1/ChainRule`: Theorem 1.3.1 (preprint 1.21) and its iterates.
- `Chapter1/Properties`: the compact-open continuity of Df; the base cases of Propositions 1.2.1, 1.2.4 (preprint 1.16, 1.19) and Corollary 1.2.5 (preprint 1.20).
- `Chapter1/Levels`: the compact-open lemma; the level operators $Df(x) : E_m \to F_m$; Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) on all levels; Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) under the stronger hypothesis "C^{k+1} on every level".
- `Chapter1/LevelsWeak`: Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20) as stated: the derivative tower, elementary maps, their sc^1 step and tangent maps, the sum rule for sc^k maps, the induction (S_k).
- `Chapter1/Degeneracy`: Definition 1.3.3 (preprint 1.23) and Lemma 1.3.4 (preprint 1.24).
- `Chapter2/DegeneracyIndex`: Theorem 1.3.5 (preprint 1.25), by the argument of §2.4 (preprint §2.3).

I.1 Scales and sc-Banach spaces (Definition 1.1.1 (preprint 1.1))

The scale. A **scale of Banach spaces** consists of real Banach spaces E_m, one for each $m \in \mathbb{N}$, together with continuous linear maps $\mathrm{incl}_m : E_{m+1} \to E_m$ that are injective and compact operators `Chapter1/ScBanach · ScScale`. Compactness is Mathlib's: the preimage of some neighbourhood of 0 is mapped into a compact set `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator`. The composite inclusion $\iota_m : E_m \to E_0$ is defined by recursion, $\iota_0 = \mathrm{id}$ and $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$ `Chapter1/ScBanach · ScScale.toBase`; it is injective `Chapter1/ScBanach · ScScale.toBase_injective`, and the composite $E_{m+k} \to E_m$ is defined likewise `Chapter1/ScBanach · ScScale.inclK` and is compatible with ι `Chapter1/ScBanach · ScScale.toBase_inclK`. The text's E_m is here the image $\iota_m(E_m) \subseteq E_0$; the text's $E_m \supset E_{m+1}$ is the inclusion of images, and "$x \in E_m$" for $x \in E_0$ is "$x \in \mathrm{range}\,\iota_m$". The **smooth points** E_∞ form the subspace of E_0 of points that lift to every level, $E_\infty = \bigcap_m \mathrm{range}\,\iota_m$ `Chapter1/ScBanach · ScScale.smooth` `Chapter1/ScBanach · ScScale.mem_smooth_iff`, and $E_\infty \cap E_m$ is its preimage in E_m `Chapter1/ScBanach · ScScale.smoothLvl`; a smooth point of E_m lifts to every E_{m+k} `Chapter1/ScBanach · ScScale.exists_inclK_of_mem_smoothLvl`.

The density axiom. An **sc-Banach space** is a scale in which $E_\infty \cap E_m$ is dense in E_m for every m `Chapter1/ScBanach · ScBanach`. This is condition (2) of Definition 1.1.1 (preprint 1.1) exactly as the text states it. The weaker statement that E_{m+k} has dense image in E_m follows `Chapter1/ScBanach · ScBanach.denseRange_inclK` `Chapter1/ScBanach · ScBanach.denseRange_incl`; it does not obviously imply the axiom, which is why the strong form is the definition.

Why a family. Nothing in Chapter 1 uses more than the family: the text's identification of E_{m+1} with a subset of E_m is the injectivity of incl_m, a field of the structure, and its "the inclusion is continuous" is part of compactness. The gain is that each level is a Banach space in its own right, with its own norm, and Mathlib's analysis applies on each level without carrying a filtration along.

The constant structure. For a finite-dimensional normed space V the family $E_m = V$ with identity inclu-

sions is a scale: the identity of a finite-dimensional space is a compact operator Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · isCompactOperator_id_iff_finiteDimensional Chapter1/ScBanach · ScBanach.constScale. Here ι_m is the identity Chapter1/ScBanach · ScBanach.constScale_toBase, every point is smooth Chapter1/ScBanach · ScBanach.constScale_smoothLvl, and density is trivial, so V is an sc-Banach space Chapter1/ScBanach · ScBanach.const Chapter1/ScBanach · ScBanach.const_toBase.

Only the constant structure (Lemma 1.A(2)–(3) of the exposition). The text remarks that a finite-dimensional space has precisely one sc-structure. For families this reads: if E is an sc-Banach space with E_0 finite-dimensional, the inclusions identify E with the constant structure on E_0. First, every point is smooth: E_∞ is a subspace of the finite-dimensional E_0, hence closed Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional, and dense by the axiom at level 0, hence everything Chapter1/ExpositionLemmas · ScBanach.smooth_eq_top_of_finiteDimensional. So every $x \in E_0$ lifts to every level, and $\iota_m : E_m \to E_0$, injective by construction, is bijective Chapter1/ExpositionLemmas · ScBanach.toBase_bijective_of_finiteDimensional. A continuous linear bijection between Banach spaces is an isomorphism (open mapping theorem) Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective; the isomorphisms ι_m commute with the inclusions by the very definition $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$, so they form an sc-isomorphism of E onto the constant structure on E_0 Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional_apply. In particular all the norms $|\cdot|_m$ are equivalent to $|\cdot|_0$. The second remark, that on an infinite-dimensional space the constant structure fails the compactness axiom, becomes: if an inclusion $\mathrm{incl}_m : E_{m+1} \to E_m$ of a scale is onto, then E_m is finite-dimensional. Indeed incl_m is then a continuous linear bijection, hence an isomorphism e, and $\mathrm{id}_{E_m} = \mathrm{incl}_m \circ e^{-1}$ is compact as a compact operator composed with a bounded one Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm; a Banach space with compact identity is finite-dimensional Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id Chapter1/ExpositionLemmas · ScScale.finiteDimensional_of_surjective_incl.

Direct sums. For scales E, F, the family $E_m \times F_m$ with the product inclusions is a scale: the product of two compact operators is compact, because the preimage of a product of neighbourhoods is mapped into the product of two compact sets Chapter1/ScBanach · ScBanach.prodScale. Its ι_m is the pair of the two Chapter1/ScBanach · ScBanach.prodScale_toBase, and its smooth points are the pairs of smooth points Chapter1/ScBanach · ScBanach.prodScale_smoothLvl; since a product of dense sets is dense, $E \oplus F$ is an sc-Banach space Chapter1/ScBanach · ScBanach.prod Chapter1/ScBanach · ScBanach.prod_smoothLvl.

The level shift. The family E_{m+1} with the inclusions incl_{m+1} is a scale E^1 Chapter1/ScBanach · ScBanach.shiftScale. Its composite inclusion to its base E_1, followed by ι_1, is ι_{m+1} Chapter1/ScBanach · ScBanach.shiftScale_toBase, and its smooth points at level m are the smooth points of E at level $m+1$ Chapter1/ScBanach · ScBanach.shiftScale_smoothLvl, so E^1 is an sc-Banach space Chapter1/ScBanach · ScBanach.shift Chapter1/ScBanach · ScBanach.shift_toBase Chapter1/ScBanach · ScBanach.shift_smoothLvl; E^k is the k-fold iterate Chapter1/ScBanach · ScBanach.shiftK.

Level sets. For $U \subseteq E_0$ the level set $U_m \subseteq E_m$ is the preimage $\iota_m^{-1}(U)$ Chapter1/ScSmooth · ScBanach.lvlSet; $U_0 = U$ Chapter1/ScSmooth · ScBanach.lvlSet_zero, $U_{m+1} = \mathrm{incl}_m^{-1}(U_m)$ Chapter1/ScSmooth · ScBanach.lvlSet_succ, and level sets preserve inclusions, intersections, openness, closedness and convexity Chapter1/ScSmooth · ScBanach.lvlSet_mono Chapter1/ScSmooth · ScBanach.lvlSet_inter Chapter1/ScSmooth · ScBanach.isOpen_lvlSet Chapter1/ScSmooth · ScBanach.isClosed_lvlSet Chapter1/ScSmooth · ScBanach.convex_lvlSet. This is the text's filtration $A_m = A \cap E_m$ of a subset. The level sets of $U_1 \subseteq E_1$, taken in the shifted space E^1, are the U_{m+1} Chapter1/Levels · ScBanach.shift_lvlSet, which is the text's $(A^1)_m = A_{1+m}$.

I.2 Sc-operators, sc-isomorphisms, sc⁺-operators (Definitions 1.1.4–1.1.5 (preprint 1.3–1.4))

An **sc-operator** $T : E \to F$ is a family of continuous linear maps $T_m : E_m \to F_m$ commuting with the inclusions, $\mathrm{incl}_m^F \circ T_{m+1} = T_m \circ \mathrm{incl}_m^E$ Chapter1/ScBanach · ScOperator. Then $\iota_m T_m = T_0 \iota_m$

by induction Chapter1/ScBanach · ScOperator.toBase_comm, so the whole family is determined by the base map T_0 Chapter1/ScBanach · ScOperator.ext_base: this is the sense in which the text's single linear map T "with $T(E_m) \subset F_m$ and $T : E_m \to F_m$ continuous" is the same thing. The text's formulation is also available as a predicate on a continuous linear $T_0 : E_0 \to F_0$, "for every m there is a continuous linear T_m with $\iota_m T_m = T_0 \iota_m$" Chapter1/ScBanach · IsScOperator, and the two notions pass into each other Chapter1/ScBanach · ScOperator.isScOperator Chapter1/ScBanach · IsScOperator.toScOperator Chapter1/ScBanach · IsScOperator.toScOperator_zero. The identity is an sc-operator, sc-operators compose and add Chapter1/ScBanach · ScOperator.id Chapter1/ScBanach · ScOperator.comp Chapter1/ScBanach · ScOperator.add, and they map smooth points to smooth points, since a lift of x to level j is carried by T_j to a lift of $T_0 x$ Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl.

An **sc-isomorphism** is a family of continuous linear equivalences $E_m \simeq F_m$ commuting with the inclusions Chapter1/ScBanach · ScIso; it has an underlying sc-operator Chapter1/ScBanach · ScIso.toScOperator, an inverse Chapter1/ScBanach · ScIso.symm, composites Chapter1/ScBanach · ScIso.trans and the identity Chapter1/ScBanach · ScIso.refl, and ι_m commutes with it and with its inverse Chapter1/ScBanach · ScIso.toBase_comm Chapter1/ScBanach · ScIso.symm_toBase_comm. The text defines an sc-isomorphism as a bijective sc-operator whose inverse is an sc-operator; the continuity of the inverse on each level, which there follows from the open mapping theorem, is here part of the data.

An **sc$^+$-operator** $E \to F$ is an sc-operator $E \to F^1$ into the shifted space, i.e. continuous linear maps $S_m : E_m \to F_{m+1}$ commuting with the inclusions Chapter1/ScBanach · ScPlus. The text's first condition, $S(E_m) \subset F_{m+1}$, is contained in the second and is dropped. The inclusion $F^1 \to F$, level by level $\mathrm{incl}_m : F_{m+1} \to F_m$, is itself an sc-operator Chapter1/ScBanach · ScBanach.shiftIncl, and composing gives the sc-operator $E \to F$ underlying S Chapter1/ScBanach · ScPlus.toScOperator Chapter1/ScBanach · ScPlus.toScOperator_apply. Its level-m map is the compact incl_m after the bounded S_m, hence compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm Chapter1/ScBanach · ScPlus.isCompactOperator: the text's remark that sc$^+$-operators are sc-compact. In terms of images, S raises the level: if $x \in \operatorname{range} \iota_m$ then $S_0 x \in \operatorname{range} \iota_{m+1}$ Chapter1/ScFredholm · ScPlus.toScOperator_mem_range_toBase_succ.

I.3 Sc-subspaces and sc-splittings (Definition 1.1.6 (preprint 1.5))

For a subspace $A \subseteq E_0$ its **level-m part** $A_m \subseteq E_m$ is the preimage of A under ι_m Chapter1/ScBanach · ScBanach.levelSub Chapter1/ScBanach · ScBanach.mem_levelSub; $A_0 = A$ Chapter1/ScBanach · ScBanach.levelSub_zero, the inclusions map A_{m+1} into A_m Chapter1/ScBanach · ScBanach.incl_mem_levelSub, and A_m is closed when A is, as the preimage of a closed set under a continuous map Chapter1/ScBanach · ScBanach.isClosed_levelSub.

A is an **sc-subspace** if A is closed and, for every m, A_m lies in the closure of $A_m \cap E_\infty$ Chapter1/ScBanach · ScBanach.IsScSubspace Chapter1/ScBanach · ScBanach.IsScSubspace.isClosed. The text asks that (A_m) be an sc-structure on A; of the conditions in that phrase, the closedness of A_m in E_m was just proved, and the compactness of the restricted inclusions $A_{m+1} \to A_m$ is automatic, since the restriction of a compact operator to a subspace, with values in a closed subspace, is compact. This is how the scale structure on a closed A is built: levels A_m, inclusions the restrictions of incl_m, injective because incl_m is, compact by the remark Chapter1/ScBanach · ScBanach.ofSubspaceScale. Its composite inclusion is that of E on the underlying points Chapter1/ScBanach · ScBanach.ofSubspaceScale_coe_toBase, and its smooth points are the points of A_m that are smooth in E Chapter1/ScBanach · ScBanach.mem_ofSubspaceScale_smoothLvl Chapter1/ScBanach · ScBanach.ofSubspaceScale_smoothLvl. The density condition of `IsScSubspace` is then exactly the density axiom for this scale, so an sc-subspace is an sc-Banach space with the induced structure Chapter1/ScBanach · ScBanach.ofSubspace.

$E = F \oplus G$ is an **sc-splitting** if F and G are sc-subspaces and, for every m, F_m and G_m are complementary subspaces of E_m Chapter1/ScBanach · ScBanach.IsScSplitting. The text asks for a *topological* direct sum $E_m = F_m \oplus G_m$. Since F_m and G_m are closed and complementary, the projection onto F_m along

G_m is continuous by the closed graph theorem, which is Mathlib's "closed complemented from complementary closed subspaces" `Chapter1/ScBanach · ScBanach.IsScSplitting.closedComplemented`; so nothing is lost by asking only for complementarity. The splitting is symmetric in F and G `Chapter1/ScBanach · ScBanach.IsScSplitting.symm`.

Sc-splittings are sc-projections (Lemma 2.B of the exposition). The proof of Proposition 1.1.7 (preprint 1.6) in the text passes from "$P \circ P = P$, continuous on every level" to "a sc-projection" and an sc-complement $(1-P)(E)$ without comment. Both directions are proved. (i) Given an sc-splitting $E = A \oplus B$, on each level A_m and B_m are complementary and closed, hence topological complements `Mathlib/Analysis/Normed/Module/Complemented · Submodule.IsCompl.isTopCompl_of_isClosed Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.isTopCompl`, so the projection P_m of E_m onto A_m along B_m is continuous `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL`. These projections commute with the inclusions: write $x = P_{m+1}x + (x - P_{m+1}x)$ with the first term in A_{m+1} and the second in B_{m+1} `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_sub_projectionL`; incl_m carries them into A_m and B_m (§I.3), where P_m is the identity, resp. zero `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_iff Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_apply_eq_zero_iff`, so $P_m(\mathrm{incl}_m x) = \mathrm{incl}_m(P_{m+1}x)$. Hence the P_m form an sc-operator P `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_apply` with $P_m^2 = P_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_idem`, $P_m(E_m) = A_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.range_scProj` and $\ker P_m = B_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.ker_scProj`. (ii) Conversely let P be an sc-operator with $P_0^2 = P_0$. Idempotence passes to every level by injectivity of ι_m, since $\iota_m P_m P_m x = P_0 P_0 \iota_m x = P_0 \iota_m x = \iota_m P_m x$ `Chapter1/ExpositionLemmas · ScOperator.idem_lvl`. The level-m part of $P_0(E_0)$ is $P_m(E_m)$: if $\iota_m x = P_0 y$ then $P_0 \iota_m x = \iota_m x$, i.e. $\iota_m P_m x = \iota_m x$, so $x = P_m x$ `Chapter1/ExpositionLemmas · ScOperator.levelSub_range_eq`; the level-m part of $\ker P_0$ is $\ker P_m$ (§I.6, `Chapter1/ScFredholm · ScOperator.ker_lvl`). $P_0(E_0) = \{x : P_0 x = x\}$ is closed, and $\ker P_0$ is closed. Density of smooth points: $x \in P_m(E_m)$ satisfies $x = P_m x$ and is the limit of $P_m z_n$ for smooth $z_n \to x$, and the $P_m z_n$ are smooth points of $P_m(E_m)$ (§I.2); $x \in \ker P_m$ satisfies $x = x - P_m x$ and is the limit of $z_n - P_m z_n$, smooth points of $\ker P_m$. Finally $P_m(E_m)$ and $\ker P_m$ are complementary because P_m is idempotent `Mathlib/LinearAlgebra/Projection · LinearMap.IsIdempotentElem.isCompl`. So $E = P_0(E_0) \oplus \ker P_0$ is an sc-splitting `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`, and together: $E = A \oplus B$ is an sc-splitting iff an idempotent sc-operator has range A and kernel B `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`. The text's $(1-P)(E)$ is $\ker P$. The formal Proposition 1.1.7 (preprint 1.6) (§I.4) builds its complement directly and does not use this.

I.4 Proposition 1.1.7 (preprint 1.6): finite-dimensional sc-subspaces

Statement. Let $F \subseteq E_0$ be finite-dimensional. Then F is an sc-subspace if and only if $F \le E_\infty$ `Chapter1/ScBanach · ScBanach.proposition_1_6_iff`; and if $F \le E_\infty$ there is a G with $E = F \oplus G$ an sc-splitting `Chapter1/ScBanach · ScBanach.proposition_1_6`.

The equivalence. If F is an sc-subspace, the density condition at level 0 says $F \subseteq \overline{F \cap E_\infty}$. But $F \cap E_\infty$ is a subspace of the finite-dimensional F, hence finite-dimensional, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional`; so $F \subseteq F \cap E_\infty$, i.e. $F \le E_\infty$. Conversely, if $F \le E_\infty$ then F is closed (finite-dimensional) and every point of F_m is already a smooth point of F_m, so the density condition holds trivially.

The complement. Mathlib provides a continuous linear projection $p : E_0 \to F$ with $p|_F = \mathrm{id}$, from the closed complementation of finite-dimensional subspaces (Hahn–Banach) `Mathlib/Analysis/LocallyConvex/HahnBanach · Submodule.ClosedComplemented.of_finiteDimensional`. This is the text's $P(x) = \sum \lambda_i(x) e_i$. Put $G := \ker p$. Since $F \le E_\infty$, the restriction of ι_m to F_m is a bijection onto F (injective by injectivity of ι_m, surjective because every point of F lifts to level m) `Chapter1/ScBanach · ScBanach.levelRestrict Chapter1/ScBanach · ScBanach.levelRestrict_bijective Chapter1/ScBanach · ScBanach.levelEquiv`; through this linear isomorphism $F_m \cong F$ the projection lifts to a continuous linear $P_m : E_m \to F_m$, $P_m = (\iota_m|_{F_m})^{-1} \circ p \circ \iota_m$

(continuous because F_m is finite-dimensional) Chapter1/ScBanach · ScBanach.finProj. It fixes F_m Chapter1/ScBanach · ScBanach.finProj_apply_of_mem, and $P_m x = 0$ iff $p(\iota_m x) = 0$ Chapter1/ScBanach · ScBanach.finProj_eq_zero_iff, so $G_m = \ker P_m$. Now F is an sc-subspace by the first part, and G is closed as a kernel. For the density of $G_m \cap E_\infty$ in G_m: given $g \in G_m$, let $Q := 1 - P_m$ on E_m; then $Qg = g$, Q maps E_m into G_m, and Q maps smooth points to smooth points, since $P_m x$ is smooth ($F \le E_\infty$) and smooth points form a subspace Chapter1/ScBanach · ScScale.smoothLvl_submodule_closed_sub. Hence $g = Qg$ lies in the closure of $Q(E_\infty \cap E_m) \subseteq G_m \cap E_\infty$, because $E_\infty \cap E_m$ is dense and Q is continuous. This is the text's $g_n = (1 - P) f_n \to g$. Finally F_m and $G_m = \ker P_m$ are complementary because P_m is a projection onto F_m Mathlib/LinearAlgebra/Projection · isCompl_of_proj.

I.5 Proposition 1.1.8 (preprint 1.7): quotients

Statement. Let $A \subseteq E_0$ be closed. The quotients E_m/A_m form an sc-Banach space E/A Chapter1/ScBanach · ScBanach.quotient, the quotient maps form an sc-operator $E \to E/A$ Chapter1/ScBanach · ScBanach.quotientMap, and, as a statement, there are an sc-Banach space Q and an sc-operator $q : E \to Q$ with every q_m surjective and $\ker q_m = A_m$ Chapter1/ScBanach · ScBanach.proposition_1_7. The text assumes A is an sc-subspace; only closedness is used.

The inclusions. A_m is closed in E_m (§I.3), so E_m/A_m is a Banach space. The map $E_{m+1}/A_{m+1} \to E_m/A_m$ is induced by incl_m: it is well defined because incl_m maps A_{m+1} into A_m Chapter1/ScBanach · ScBanach.quotIncl Chapter1/ScBanach · ScBanach.quotIncl_mk. It is injective: if $\mathrm{incl}_m x \in A_m$ then $\iota_{m+1} x = \iota_m(\mathrm{incl}_m x) \in A$, so $x \in A_{m+1}$ Chapter1/ScBanach · ScBanach.quotIncl_injective. This injectivity is the text's identification of E_m/A_m with the subspace $\{x + A : x \in E_m\}$ of E/A, and it holds precisely because A_m is the *full* preimage of A. It is compact: the quotient map $E_m \to E_m/A_m$ after the compact incl_m is compact, and the induced map on E_{m+1}/A_{m+1} sends the image of a neighbourhood of 0 (which is a neighbourhood, since quotient maps are open) into the same compact set Chapter1/ScBanach · ScBanach.quotIncl_isCompactOperator. So the quotients form a scale Chapter1/ScBanach · ScBanach.quotientScale, whose composite inclusion sends the class of x to the class of $\iota_m x$ Chapter1/ScBanach · ScBanach.quotientScale_toBase_mk. The text's argument with a sequence $x_k + A_{m+1}$ of norm ≤ 1 and representatives of norm ≤ 2 is the sequential form of the same fact.

Density. The quotient map $E_m \to E_m/A_m$ is continuous and surjective, so the image of the dense set $E_\infty \cap E_m$ is dense. A class $[x]$ with x smooth is a smooth point of the quotient scale: a lift y of x to level j gives the lift $[y]$ of $[x]$. So the smooth points of E_m/A_m are dense Chapter1/ScBanach · ScBanach.quotient. This is the inclusion "$\{x + A : x \in E_\infty\} \subseteq (E/A)_\infty$" of the text's proof, which is all its density claim needs.

The smooth points of the quotient (Lemma 2.C of the exposition). The text also asserts that the two sets are equal. Let A be an sc-subspace and $x \in E_0$ with, for every j, a point $x_j \in E_j$ such that $\iota_j x_j - x \in A$. We find a smooth y with $y - x \in A$ Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset. *The differences.* $d_j := \mathrm{incl}_j x_{j+1} - x_j \in E_j$ has $\iota_j d_j = \iota_{j+1} x_{j+1} - \iota_j x_j = (\iota_{j+1} x_{j+1} - x) - (\iota_j x_j - x) \in A$, so $d_j \in A_j$. *A uniform bound for the inclusions.* Put $P_j := \prod_{i<j} \max(1, \|\mathrm{incl}_i\|) \ge 1$ Chapter1/ExpositionLemmas · ScBanach.inclBound Chapter1/ExpositionLemmas · ScBanach.one_le_inclBound. By induction, $\|\iota_j v\| \le P_j \|v\|$ for $v \in E_j$ Chapter1/ExpositionLemmas · ScBanach.norm_toBase_le and $\|\mathrm{incl}_J^{J+k} v\| \le P_{J+k} \|v\|$ for the composite inclusion $E_{J+k} \to E_J$ Chapter1/ExpositionLemmas · ScBanach.norm_inclK_le, each step multiplying by $\|\mathrm{incl}_i\| \le \max(1, \|\mathrm{incl}_i\|)$ Chapter1/ExpositionLemmas · ScBanach.norm_incl_le. This replaces the exposition's $M_j = \max_{i \le j} \|E_j \to E_i\|$, and avoids comparing levels $J + k$ with j. *Smooth corrections.* Since A is an sc-subspace, $A_j \cap E_\infty$ is dense in A_j, so there are $c_j \in A_j \cap E_\infty$ with $\|d_j - c_j\| \le 2^{-j}/P_j$ Mathlib/Topology/MetricSpace/Pseudo/Defs · Metric.mem_closure_iff. Then $b_j := d_j - c_j \in A_j$ has $P_j \|b_j\| \le 2^{-j}$. *The series.* $\sum_j \iota_j b_j$ converges absolutely in E_0, its terms being bounded by 2^{-j} Mathlib/Analysis/Normed/Group/InfiniteSum · Summable.of_norm_bounded; let S be its sum and $y := x_0 + S$. Its partial sums lie in A, which is closed, so $S \in A$ (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · HasProd.tendsto_prod_

nat), and $y - x = (x_0 - x) + S \in A$. *$y$ is smooth.* Fix J. The tail $t := \sum_k \mathrm{incl}_J^{J+k} b_{J+k}$ converges in E_J (terms bounded by $2^{-J}2^{-k}$), and $\iota_J t = \sum_k \iota_{J+k} b_{J+k}$ because ι_J is continuous and linear Mathlib/Topology/Algebra/InfiniteSum/Module · ContinuousLinearMap.map_tsum and $\iota_J \circ \mathrm{incl}_J^{J+k} = \iota_{J+k}$ (§I.1). The head telescopes: $\sum_{j<J} \iota_j d_j = \iota_J x_J - x_0$ (the additive form of Mathlib/Algebra/BigOperators/Group/Finset/Basic · Finset.prod_range_div), so $\sum_{j<J} \iota_j b_j = (\iota_J x_J - x_0) - \sum_{j<J} \iota_j c_j$, and the last sum is smooth, hence $\iota_J w$ for some $w \in E_J$. Splitting S into head and tail (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · Multipliable.prod_mul_tprod_nat_add) gives $y = \iota_J(x_J - w + t)$. As J was arbitrary, $y \in E_\infty$. *As a statement about E/A.* The class of x is a smooth point of E/A iff for every j it is the image of a class $[z]$, $z \in E_j$, i.e. iff $\iota_j z - x \in A$ for some z (§I.5 and Mathlib/LinearAlgebra/Quotient/Defs · Submodule.Quotient.eq); by the above this happens iff $x + A$ contains a smooth point Chapter1/ExpositionLemmas · ScBanach.mk_mem_quotient_smooth_iff. This is the text's equality.

I.6 Definition 1.1.9 (preprint 1.8): sc-Fredholm operators

Level kernels and ranges. For an sc-operator T, the kernel of T_m is the level-m part of the kernel of T_0 Chapter1/ScFredholm · ScOperator.ker_lvl, and the range of T_m lies in the level-m part of the range of T_0 Chapter1/ScFredholm · ScOperator.range_lvl_le; both from $\iota_m T_m = T_0 \iota_m$ and injectivity.

The definition. An **sc-Fredholm splitting** of $T : E \to F$ consists of subspaces $K, X \subseteq E_0$ and $C, Y \subseteq F_0$ such that $E = K \oplus X$ and $F = C \oplus Y$ are sc-splittings, $K = \ker T_0$, $Y = \operatorname{range} T_0$, K and C are finite-dimensional, and T_m restricts to a bijection $X_m \to Y_m$ for every m Chapter1/ScFredholm · ScFredholmSplitting; T is **sc-Fredholm** if such a splitting exists Chapter1/ScFredholm · IsScFredholm. The text's condition (3), "$T : X \to Y$ is an sc-isomorphism", is here the bijectivity on each level; the continuity of the inverses $Y_m \to X_m$ is automatic by the open mapping theorem between the closed subspaces X_m and Y_m, so nothing is lost.

The index. The index of a splitting is $\dim K - \dim C$ Chapter1/ScFredholm · ScFredholmSplitting.index, and the index of T is $\dim \ker T_0 - \dim(F_0/\operatorname{range} T_0)$ Chapter1/ScFredholm · scIndex. They agree: $K = \ker T_0$, and C is a complement of $Y = \operatorname{range} T_0$ in F_0, so $C \cong F_0/Y$ Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl Chapter1/ScFredholm · ScFredholmSplitting.index_eq. So the index does not depend on the splitting, which the text does not state.

The remarks after the definition. K is a finite-dimensional sc-subspace, hence $K \le E_\infty$ by Proposition 1.1.7 (preprint 1.6) Chapter1/ScFredholm · ScFredholmSplitting.K_le_smooth. $\ker T_m = K_m$ Chapter1/ScFredholm · ScFredholmSplitting.ker_lvl and $\operatorname{range} T_m = Y_m$: one inclusion from above, the other because T_m maps X_m onto Y_m Chapter1/ScFredholm · ScFredholmSplitting.range_lvl. The text's "$F = C \oplus T(E_m)$" (read F_m) is the level-m part of the splitting $F = C \oplus Y$ together with $\operatorname{range} T_m = Y_m$. Finally, every T_m is a Fredholm operator in Mathlib's sense: its kernel K_m injects into the finite-dimensional K, its range Y_m is closed, and Y_m has the finite-dimensional complement C_m (which injects into C), so its range has finite codimension Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_isCompl Chapter1/ScFredholm · ScFredholmSplitting.isFredholm, using the criterion of §I.7. This is the fact the text uses in the first line of §1.4.1.

The index on every level (Lemma 2.D of the exposition). C is a finite-dimensional sc-subspace, so $C \le F_\infty$ by Proposition 1.1.7 (preprint 1.6) Chapter1/FredholmIndex · ScFredholmSplitting.C_le_smooth. For any subspace $A \le E_\infty$, ι_m restricts to a bijection $A_m \to A$ (injective as ι_m is, surjective because every point of A lifts), so $\dim A_m = \dim A$ Chapter1/FredholmIndex · ScBanach.finrank_levelSub_of_le_smooth. Hence $\dim \ker T_m = \dim K_m = \dim K$, and $F_m/\operatorname{range} T_m = F_m/Y_m \cong C_m$ (complements, Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl) has dimension $\dim C$. So T_m is Fredholm with index $\dim K - \dim C$ in the sense of §I.7 Chapter1/FredholmIndex · ScFredholmSplitting.fredholmIndex_lvl, which by the previous paragraph is the index of T for every splitting Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl; on level 0 this is the definition, $\mathrm{ind}(T)$ being the Fredholm index of T_0 Chapter1/FredholmIndex · scIndex_eq_fredholmIndex. The text states the splittings on each level but not the index.

I.7 Fredholm operators between Banach spaces: Riesz–Schauder and compact perturbations

Mathlib has a notion of Fredholm operator `Mathlib/Analysis/Normed/Operator/Fredholm/Basic · IsFredholm`: a continuous linear $u : E \to F$ that is a strict map, with closed range, finite-dimensional kernel, range of finite codimension `Mathlib/RingTheory/Finiteness/Cofinite · CoFG` and closed complemented kernel. It does not have the two theorems that §1.4.1 quotes as classical, that $1 +$ compact is Fredholm and that compact perturbations preserve the property. Both are proved here, for real Banach spaces.

The criterion. A continuous linear map with closed range between Banach spaces is a strict map `Mathlib/Topology/Maps/Strict/Basic · IsStrictMap`: by the open mapping theorem its corestriction to the range is an open quotient map `Mathlib/Topology/Maps/Strict/Module · isStrictMap_iff_isOpenQuotientMap_rangeRestrict` `Chapter1/FredholmPerturbation · isStrictMap_of_isClosed_range`. Hence a continuous linear map with finite-dimensional kernel, closed range and range of finite codimension is Fredholm, the kernel being closed complemented because it is finite-dimensional `Chapter1/FredholmPerturbation · isFredholm_of_banach`. A finite-rank operator is compact: the image of the unit ball is bounded in the finite-dimensional range, so its closure is compact `Chapter1/FredholmPerturbation · isCompactOperator_of_finiteDimensional_range`.

Riesz–Schauder: the kernel. Let $c : E \to E$ be compact and $N := \ker(1 + c)$. On N the operator c is $-\mathrm{id}$, and c restricts to a compact operator of the closed subspace N `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.restrict`; so id_N is compact, and N is finite-dimensional `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id` `Chapter1/FredholmPerturbation · finiteDimensional_ker_one_add`.

The range is closed. Take a closed complement G of N (which exists since N is finite-dimensional). $1 + c$ is bounded below on G: otherwise there are unit vectors $g_n \in G$ with $(1 + c)g_n \to 0$; the cg_n lie in a compact set `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_compact`, so along a subsequence $cg_n \to y$ `Mathlib/Topology/Sequences · IsCompact.tendsto_subseq`, hence $g_n = (1+c)g_n - cg_n \to -y$; then $-y \in G$ (closed), $\|-y\| = 1$, and $(1+c)(-y) = 0$, so $-y \in G \cap N = 0$, a contradiction `Chapter1/FredholmPerturbation · exists_bound_below_on_compl`. A bounded-below operator from a Banach space has closed range (it is antilipschitz and uniformly continuous), and $(1+c)(G) = (1+c)(E)$ since $1 + c$ kills N `Chapter1/FredholmPerturbation · isClosed_range_one_add`.

The range has finite codimension. Suppose $R := \mathrm{range}(1+c)$ has infinite codimension. Pick, for every proper subspace, a vector outside it `Chapter1/FredholmPerturbation · pick` `Chapter1/FredholmPerturbation · pick_notMem`, and build the chain $M_0 = R$, $M_{j+1} = M_j + \mathbb{R}\,\mathrm{pick}(M_j)$ `Chapter1/FredholmPerturbation · chain`. Each M_j is R plus a finite-dimensional subspace `Chapter1/FredholmPerturbation · chain_eq_sup`, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_sup_finiteDimensional` `Chapter1/FredholmPerturbation · chain_isClosed`, and proper, since R has infinite codimension `Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_codisjoint` `Chapter1/FredholmPerturbation · chain_ne_top`; the chain is increasing `Chapter1/FredholmPerturbation · chain_mono`. Riesz's lemma inside a subspace: for closed $M \subsetneq M'$ there is a unit vector of M' at distance $\geq 1/2$ from M (take $x \in M' \setminus M$, a point $y \in M$ with $\|x-y\| < 2\,\mathrm{dist}(x, M)$ `Mathlib/Topology/MetricSpace/HausdorffDistance · infDist_lt_iff`, and normalize $x-y$) `Chapter1/FredholmPerturbation · exists_unit_far`. This gives unit vectors $u_j \in M_{j+1}$ at distance $\geq 1/2$ from M_j. For $j < l$, $u_j + (1+c)u_l - (1+c)u_j \in M_l$, and $u_l - [u_j + (1+c)u_l - (1+c)u_j] = -(cu_l - cu_j)$, so $\|cu_l - cu_j\| \geq 1/2$. But the cu_j lie in a compact set and so have a convergent subsequence, two of whose terms are within $1/2$ of each other: a contradiction `Chapter1/FredholmPerturbation · coFG_range_one_add`. Hence $1 + c$ is Fredholm `Chapter1/FredholmPerturbation · isFredholm_one_add`.

Compact perturbations. Let $u : E \to F$ be Fredholm and k compact. Mathlib gives a quasi-inverse $v : F \to E$ of u, with $vu - 1$ and $uv - 1$ of finite rank `Mathlib/Analysis/Normed/Operator/Fredholm/Basic · isFredholm_iff_exists_isQuasiInverse`; a finite-rank difference from the identity is compact `Chapter1/FredholmPerturbation · isCompactOperator_sub_one_of_equiv`. Then $v(u + k) = 1 + c_1$ and $(u + k)v = 1 + c_2$ with $c_1 = (vu - 1) + vk$ and $c_2 = (uv - 1) + kv$ compact. The kernel of $u + k$ lies in the finite-

dimensional kernel of $1 + c_1$; the range of $u + k$ contains the range of $1 + c_2$, which is closed and of finite codimension, so range$(u + k)$ has finite codimension `Mathlib/RingTheory/Finiteness/Cofinite · CoFG.of_le` and is closed, as a subspace containing a closed subspace of finite codimension `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_mono_of_finiteDimensional_quotient`. By the criterion, $u + k$ is Fredholm `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`.

The index. Mathlib has no index; it is built here, first for linear maps over a field. For $u : E \to F$ put $\operatorname{ind}(u) := \dim \ker u - \dim(F/\operatorname{range} u) \in \mathbb{Z}$ `Chapter1/FredholmIndex · linIndex`, meaningful when u has *finite index*, i.e. finite-dimensional kernel and cokernel `Chapter1/FredholmIndex · HasFiniteIndex`. *Additivity.* If $u : E \to F$ and $v : F \to G$ have finite index, so has vu, and $\operatorname{ind}(vu) = \operatorname{ind}(u) + \operatorname{ind}(v)$ `Chapter1/FredholmIndex · HasFiniteIndex.comp_and_linIndex Chapter1/FredholmIndex · HasFiniteIndex.comp Chapter1/FredholmIndex · linIndex_comp`. The proof is the dimension count along the exact sequence $0 \to \ker u \to \ker vu \to \operatorname{range} u \cap \ker v \to F/\operatorname{range} u \to G/\operatorname{range} vu \to G/\operatorname{range} v \to 0$, done as four applications of rank–nullity `Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · LinearMap.finrank_range_add_finrank_ker` and two of the third isomorphism theorem `Mathlib/LinearAlgebra/Isomorphisms · Submodule.quotientQuotientEquivQuotient`: to u restricted to $\ker vu$ (kernel $\ker u$, range $\operatorname{range} u \cap \ker v$), to the quotient map $F \to F/\operatorname{range} u$ restricted to $\ker v$, and to v followed by $G \to G/\operatorname{range} vu$ (kernel $\operatorname{range} u + \ker v$); finiteness of the new kernel and cokernel comes out of the same maps `Chapter1/FredholmIndex · finite_of_finite_ker_range Chapter1/FredholmIndex · finite_of_finite_quotient`. Conversely, if vu and uv have finite index then so has u, since $\ker u \subseteq \ker vu$ and $\operatorname{range} uv \subseteq \operatorname{range} u$ `Chapter1/FredholmIndex · finiteDimensional_ker_of_comp Chapter1/FredholmIndex · finiteDimensional_coker_of_comp Chapter1/FredholmIndex · HasFiniteIndex.of_comp_of_comp`. *Index zero.* A linear equivalence has index 0 `Chapter1/FredholmIndex · HasFiniteIndex.equiv Chapter1/FredholmIndex · linIndex_equiv`. If $w - 1$ has finite-dimensional range R, then $\ker w \subseteq R$, $\operatorname{range} w + R = E$ and $R \cap \operatorname{range} w = w(R)$, so $\dim \ker w$ and $\dim(E/\operatorname{range} w)$ both equal $\dim R - \dim w(R)$ and $\operatorname{ind}(w) = 0$ `Chapter1/FredholmIndex · HasFiniteIndex.of_finiteRange_sub_id_and_linIndex Chapter1/FredholmIndex · linIndex_eq_zero_of_finiteRange_sub_id`. If a is bijective and r has finite rank, $a + r = a(1 + a^{-1}r)$ has index $0 + 0$ by additivity `Chapter1/FredholmIndex · HasFiniteIndex.of_bijective_add_and_linIndex`.

Banach spaces. For $u : E \to F$ continuous linear, the **Fredholm index** is $\operatorname{ind}(u)$ `Chapter1/FredholmIndex · fredholmIndex`; a Fredholm operator has finite index `Chapter1/FredholmIndex · IsFredholm.hasFiniteIndex`. *Small perturbations.* Let v be a quasi-inverse of u and $\|v\|\,\|s\| < 1$. Then $\|vs\|, \|sv\| < 1$ `Mathlib/Analysis/Normed/Operator/Basic · ContinuousLinearMap.opNorm_comp_le`, so $1 + vs$ and $1 + sv$ are invertible by the Neumann series `Mathlib/Analysis/SpecificLimits/Normed · isUnit_one_sub_of_norm_lt_one Chapter1/FredholmIndex · bijective_one_add_of_norm_lt`. Since $vu - 1$ and $uv - 1$ have finite rank `Chapter1/FredholmIndex · finiteDimensional_range_sub_id_of_equiv`, $v(u + s) = (1 + vs) + (vu - 1)$ and $(u + s)v = (1 + sv) + (uv - 1)$ are bijections plus finite-rank maps, hence of index 0 `Chapter1/FredholmIndex · hasFiniteIndex_comp_add_of_norm_lt`; the same with $s = 0$ for vu and uv. By the converse above, $u + s$, u and v have finite index, and additivity gives $\operatorname{ind}(u + s) + \operatorname{ind}(v) = 0 = \operatorname{ind}(u) + \operatorname{ind}(v)$, so $\operatorname{ind}(u + s) = \operatorname{ind}(u)$ `Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt`. *Compact perturbations.* Let u be Fredholm and k compact, and $\varphi(t) := \operatorname{ind}(u + tk)$ for $t \in \mathbb{R}$. At each t_0, $u + t_0 k$ is Fredholm (the previous paragraph, $t_0 k$ being compact `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.smul`), with a quasi-inverse v_0; for t so close to t_0 that $\|v_0\|\,\|(t - t_0)k\| < 1$, $\varphi(t) = \operatorname{ind}((u + t_0 k) + (t - t_0)k) = \varphi(t_0)$. So φ is locally constant `Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.iff_eventually_eq` on the connected $\mathbb{R}$, hence constant `Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.apply_eq_of_preconnectedSpace`, and $\operatorname{ind}(u + k) = \varphi(1) = \varphi(0) = \operatorname{ind}(u)$ `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`. This replaces the citation of Kato IV.5.26 in the exposition's (F2).

I.8 Proposition 1.1.10 (preprint 1.9): sc-Fredholm operators are regularizing

Statement. Let T have an sc-Fredholm splitting, $e \in E_0$, and $T_0 e \in \operatorname{range} \iota_m$. Then $e \in \operatorname{range} \iota_m$ `Chapter1/ScFredholm · proposition_1_9`.

Proof (the text's). Write $T_0 e = \iota_m y$. Since $F_m = C_m \oplus Y_m$, $y = c + y'$ with $c \in C_m$ and $y' \in Y_m$, and $y' = T_m x'$ for some $x' \in X_m$ by the bijectivity of $T_m : X_m \to Y_m$. Then $T_0(e - \iota_m x') = \iota_m c$, so $\iota_m c$ lies in C and in $Y = \operatorname{range} T_0$; these are complementary on level 0, so $\iota_m c = 0$. Hence $e - \iota_m x' \in \ker T_0 = K \leq E_\infty$ (§I.6), so it lifts to some $z \in E_m$, and $e = \iota_m(z + x')$. The step "$c = 0$" is where $C \cap Y = 0$ enters, as the exposition notes.

I.9 Proposition 1.1.11 (preprint 1.10): compact perturbations of sc-Fredholm operators

Statement. If $T : E \to F$ is sc-Fredholm and $S : E \to F$ is an sc$^+$-operator, then $T + S$ is sc-Fredholm `Chapter1/ScFredholm · proposition_1_10`. Here S is read as the sc-operator $E \to F$ underlying it (§I.2), and $T + S$ is the sum of sc-operators.

The bootstrap. The text says: if $y = (T + S)e \in Y_m$ then, since T is regularizing, $e \in E_m$. But $Te = y - Se$ with Se only one level up. The formal proof iterates: if $(T + S)_0 e \in \operatorname{range} \iota_{m+1}$, then a fortiori $(T + S)_0 e \in \operatorname{range} \iota_m$ `Chapter1/ScFredholm · ScBanach.range_toBase_succ_le`, so $e \in \operatorname{range} \iota_m$ by induction, so $S_0 e \in \operatorname{range} \iota_{m+1}$ (§I.2), so $T_0 e = (T + S)_0 e - S_0 e \in \operatorname{range} \iota_{m+1}$, and Proposition 1.1.10 (preprint 1.9) gives $e \in \operatorname{range} \iota_{m+1}$ `Chapter1/ScFredholm · ScPlus.regularizing_add`. So $T + S$ is regularizing.

The kernel. Each $(T + S)_m = T_m + S_m$ is Fredholm, as a compact perturbation (§I.7) of the Fredholm operator T_m (§I.6) `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`. Let $K' := \ker(T + S)_0$, finite-dimensional. Every $x \in K'$ is smooth: $(T + S)_0 x = 0 \in \operatorname{range} \iota_m$ for every m, so $x \in \operatorname{range} \iota_m$ by the bootstrap. By Proposition 1.1.7 (preprint 1.6) there is an sc-splitting $E = K' \oplus X'$ `Chapter1/ScBanach · ScBanach.proposition_1_6`. (The text's $K_m = K_{m+1}$ is the same bootstrap.)

The range. Let $Y' := \operatorname{range}(T + S)_0$, closed since $(T + S)_0$ is Fredholm. The range of $(T + S)_m$ is exactly Y'_m: one inclusion from §I.6; conversely if $\iota_m y = (T + S)_0 e$ then e lifts to level m by the bootstrap, and $y = (T + S)_m e'$ by injectivity of ι_m. Y' is an sc-subspace: for $y = (T + S)_m e \in Y'_m$, approximate e by smooth points $e_n \in E_m$; then $(T+S)_m e_n \to y$ and the $(T+S)_m e_n$ are smooth points of Y'_m (sc-operators preserve smooth points, §I.2). This is the text's argument "$y_n = (T + S)e_n \in Y_\infty$, $y_n \to y$".

The complement. F_0/Y' is finite-dimensional. Let $p : F_0 \to F_0/Y'$ be the quotient map. The image $p(F_\infty)$ is a subspace of the finite-dimensional F_0/Y', dense because F_∞ is dense in F_0 and p is a continuous surjection, hence closed, hence everything. Instead of lifting a basis, take a linear right inverse g of $p|_{F_\infty}$ and let $\sigma := F_\infty \hookrightarrow F_0$ composed with g, so $p \circ \sigma = \mathrm{id}$; put $C' := \operatorname{range} \sigma$, a finite-dimensional subspace of F_∞. On every level Y'_m and C'_m are complementary: disjoint because an element of $C'_m \cap Y'_m$ is $\sigma(q)$ with $p\sigma(q) = q = 0$; spanning because for $z \in F_m$ the point $c := \sigma(p(\iota_m z))$ is smooth, so it lifts to level m, and $z - c \in Y'_m$. So $F = C' \oplus Y'$ is an sc-splitting (C' is an sc-subspace by Proposition 1.1.7 (preprint 1.6) `Chapter1/ScBanach · ScBanach.proposition_1_6_iff`).

The bijections. $(T + S)_m$ maps X'_m into Y'_m; it is injective on X'_m because its kernel on level m is K'_m, which meets X'_m in 0; and it is onto Y'_m because every $e \in E_m$ splits as $k + x$ with $k \in K'_m$, $x \in X'_m$ and $(T + S)_m e = (T + S)_m x$. This is condition (3) of Definition 1.1.9 (preprint 1.8), which the text does not verify; the exposition supplies it with the open mapping theorem, the formalization needs only the bijections (§I.6). Hence $T + S$ is sc-Fredholm `Chapter1/ScFredholm · proposition_1_10`.

The index. The text does not say that the index is preserved; the exposition does. The index of $T+S$ is the

Fredholm index of $(T+S)_0 = T_0+S_0$ (§I.6); T_0 is Fredholm and S_0 compact, so by the invariance under compact perturbations (§I.7) $\mathrm{ind}(T+S) = \mathrm{ind}(T)$ `Chapter1/FredholmIndex · proposition_1_10_index`. On every level, $T+S$ being sc-Fredholm, its level map $(T+S)_m$ is Fredholm of index $\mathrm{ind}(T+S) = \mathrm{ind}(T)$ by the index on every level (§I.6) `Chapter1/FredholmIndex · proposition_1_10_index_lvl`.

I.10 Partial quadrants, tuples and sc-domains (Definition 1.1.12 (preprint 1.11))

The quadrant. In $\mathbb{R}^n \oplus W$, with $\mathbb{R}^n$ carrying its constant structure, the quadrant on level m is the set of (a, w) with $a_i \geq 0$ for all i `Chapter1/ScSmooth · quadrant`; it is convex `Chapter1/ScSmooth · quadrant_convex`, closed `Chapter1/ScSmooth · quadrant_isClosed`, has non-empty interior (the point $(1, \dots, 1, 0)$) `Chapter1/ScSmooth · quadrant_interior_nonempty`, and the level sets of the level-0 quadrant are the level quadrants `Chapter1/ScSmooth · lvlSet_quadrant`.

Partial quadrants. $C \subseteq E_0$ is a **partial quadrant** if $C = T_0^{-1}(\text{quadrant})$ for some n, some sc-Banach space W and some sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ `Chapter1/ScSmooth · IsPartialQuadrant`. The text asks in addition that C be closed and convex; both follow. Indeed the level set C_m is T_m^{-1} of the level-m quadrant `Chapter1/ScSmooth · IsPartialQuadrant.lvlSet_eq`, hence convex `Chapter1/ScSmooth · IsPartialQuadrant.convex_lvlSet`, closed `Chapter1/ScSmooth · IsPartialQuadrant.isClosed_lvlSet` and with non-empty interior `Chapter1/ScSmooth · IsPartialQuadrant.interior_lvlSet_nonempty`; for $m = 0$ this is the text's "closed convex" `Chapter1/ScSmooth · IsPartialQuadrant.convex` `Chapter1/ScSmooth · IsPartialQuadrant.isClosed`. A convex set with non-empty interior has unique differentiability at every point of its closure `Mathlib/Analysis/Calculus/TangentCone/Real · uniqueDiffWithinAt_convex`, so C_m has unique differentiability at each of its points `Chapter1/ScSmooth · IsPartialQuadrant.uniqueDiffWithinAt_lvlSet`; this is the exposition's Lemma 3.D, that derivatives on a quadrant are unique, in Mathlib's language. The whole space is a partial quadrant with $n = 0$ `Chapter1/ScSmooth · isPartialQuadrant_univ`.

Tuples. (U, C, E) is a tuple, U a **relatively open subset of the partial quadrant** C, if C is a partial quadrant and $U = O \cap C$ for an open $O \subseteq E_0$ `Chapter1/ScSmooth · IsRelOpenInQuadrant`.

Sc-domains. $U \subseteq E_0$ is an **sc-domain** if for every m the level set U_m is the intersection of an open and a convex subset of E_m, and U_m has unique differentiability at each of its points `Chapter1/ScSmooth · IsScDomain`. A relatively open subset of a partial quadrant is an sc-domain: $U_m = O_m \cap C_m$ with O_m open and C_m convex, and unique differentiability of C_m at a point survives intersection with a neighbourhood `Chapter1/ScSmooth · IsRelOpenInQuadrant.isScDomain`. The property passes to the tangent: $(TU)_m = U_{m+1} \times E_m$ (§I.11) is $(O \times E_m) \cap (C \times E_m)$, and unique differentiability of a product is the product of the two `Chapter1/ScSmooth · IsScDomain.tangentSet`; and to the level shift, since the level sets of U_1 in E^1 are the U_{m+1} `Chapter1/Levels · IsScDomain.shift`. This is what the proofs of §§1.2–1.3 use in place of the text's claim that $T(U, C, E)$ is again a tuple (Definition 1.1.14 (preprint 1.13)). Every theorem of §§1.2–1.3 below is stated for sc-domains; the tuples are a special case.

The shift and the tangent of a tuple (Lemmas 3.A(2) and 3.B of the exposition). The text's claim is nevertheless true, and is proved with Chapter 2, where retracts need it. If $T : E \to \mathbb{R}^n \oplus W$ presents C, its levels T_{m+1} form an sc-isomorphism $T^1 : E^1 \to \mathbb{R}^n \oplus W^1$ `Chapter2/LevelPolyfold · ScIso.shift`, and $C_1 = T_1^{-1}(\text{quadrant})$ by the level description of a partial quadrant above, so C_1 is a partial quadrant in E^1 `Chapter2/LevelPolyfold · IsPartialQuadrant.shift`; with $U = O \cap C$, $U_1 = O_1 \cap C_1$ is relatively open in it `Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift`. For the tangent, $T_{m+1} \times T_m$ maps $(TE)_m = E_{m+1} \times E_m$ onto $(\mathbb{R}^n \times W_{m+1}) \times (\mathbb{R}^n \times W_m)$, and the reassociation $((a, w), q) \mapsto (a, (w, q))$ is an sc-isomorphism onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ `Chapter2/TangentBundle · tangentQuadrantIso`; the composite presents $TC = \{(x, h) : x \in C_1\}$, since the quadrant condition only looks at the $\mathbb{R}^n$-coordinates of $T_1 x$ `Chapter2/TangentBundle · IsPartialQuadrant.tangentSet`. And $TU = (O_1 \times E_0) \cap TC$ is relatively open in TC `Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet`. The identity $T(U^1) = (TU)^1$ is §I.11's `Chapter1/Levels · ScBanach.shift_tangentSet`.

I.11 Sc⁰ maps, the tangent, sc¹ maps, sc^k (Definitions 1.1.13–1.1.15 and the unnumbered definition after them (preprint Definitions 1.12–1.15))

Level maps. For $f : E_0 \to F_0$ and $x \in E_m$, the level map $f_m(x)$ is the inverse image under ι_m of $f(\iota_m x)$, when $f(\iota_m x) \in \text{range}\, \iota_m$, and a junk value otherwise `Chapter1/ScSmooth · levelMap`; so $\iota_m f_m(x) = f(\iota_m x)$ whenever the right side lifts `Chapter1/ScSmooth · toBase_levelMap`, $f_0 = f$ `Chapter1/ScSmooth · levelMap_zero`, and $f_m(x)$ is characterized by $\iota_m y = f(\iota_m x)$ `Chapter1/ScSmooth · levelMap_eq_of_toBase_eq`. Maps are total functions on E_0; only the values on the domain U will matter.

Sc⁰. f is sc^0 **on** U if for every m and $x \in U_m$ the point $f(\iota_m x)$ lifts to F_m, and f_m is continuous on U_m `Chapter1/ScSmooth · IsSc0` `Chapter1/ScSmooth · IsSc0.toBase_levelMap`. Then f is continuous on U `Chapter1/ScSmooth · IsSc0.continuousOn_base`, f_m maps U_m into V_m when f maps U into V `Chapter1/ScSmooth · IsSc0.levelMap_mapsTo`, the property passes to subsets `Chapter1/ScSmooth · IsSc0.mono` and depends only on the values on U `Chapter1/ScSmooth · IsSc0.congr`. Level maps compose: $(g \circ f)_m = g_m \circ f_m$ on U_m `Chapter1/ScSmooth · IsSc0.levelMap_comp`, so compositions of sc^0 maps are sc^0 `Chapter1/ScSmooth · IsSc0.comp`. The identity is sc^0 `Chapter1/ScSmooth · isSc0_id`, and an sc-operator is sc^0 on all of E_0 with level maps T_m `Chapter1/ScSmooth · ScOperator.isSc0` `Chapter1/ScSmooth · ScOperator.levelMap_eq`. The text's definition says "$f(U_m) \subset V_m$" with V undefined; V is U', and the target tuple enters only as $f(U) \subseteq U'$.

The tangent. $TE := E^1 \oplus E$, with levels $E_{m+1} \times E_m$ `Chapter1/ScSmooth · ScBanach.tangent` `Chapter1/ScSmooth · ScBanach.tangent_toBase`; $TU \subseteq (TE)_0 = E_1 \times E_0$ is the set of (x, h) with $x \in U_1$ `Chapter1/ScSmooth · ScBanach.tangentSet`, with level-m part $U_{m+1} \times E_m$ `Chapter1/ScSmooth · ScBanach.mem_lvlSet_tangentSet` `Chapter1/ScSmooth · ScBanach.lvlSet_tangentSet`, monotone in U `Chapter1/ScSmooth · ScBanach.tangentSet_mono`. The level-1 part of TU is $T(U_1)$ in $T(E^1) = (TE)^1$, the identity $T(U^1) = (TU)^1$ that the inductions of §1.2 use `Chapter1/Levels · ScBanach.shift_tangentSet`.

Sc¹. Given a family $D : E_1 \to \mathcal{L}(E_0, F_0)$, the tangent map is $T_D f(x, h) = (f_1(x), D(x)h)$ `Chapter1/ScSmooth · tangentMapWith`. f is sc^1 **on** U **with derivatives** D if (1) f is sc^0 on U; (2) for every $x \in U_1$ the map $f \circ \iota_1 : E_1 \to F_0$ has derivative $D(x) \circ \iota_1$ at x within U_1 `Mathlib/Analysis/Calculus/FDeriv/Defs · HasFDerivWithinAt`; (3) $T_D f$ is sc^0 on TU `Chapter1/ScSmooth · IsSc1With`. Condition (2) is the text's limit $|f(x+h) - f(x) - Df(x)h|_0/|h|_1 \to 0$ over $h \in E_1$ with $x + h \in U_1$, in Mathlib's words. f is sc^1 **on** U if some D works `Chapter1/ScSmooth · IsSc1`; `scDeriv` is a chosen such D `Chapter1/ScSmooth · scDeriv` `Chapter1/ScSmooth · IsSc1.isSc1With_scDeriv` and `tangentMap` the tangent map built from it `Chapter1/ScSmooth · tangentMap` `Chapter1/ScSmooth · IsSc1.tangentMap_isSc0`. If f maps U into V then $T_D f$ maps TU into TV `Chapter1/ScSmooth · IsSc1With.tangentMapWith_mapsTo` `Chapter1/ScSmooth · IsSc1.tangentMap_mapsTo`; level 0 of (3) says $(x, h) \mapsto D(x)h$ is continuous on $U_1 \times E_0$ `Chapter1/ScSmooth · IsSc1With.continuousOn_deriv`; and both notions depend only on the values of f on U `Chapter1/ScSmooth · IsSc1With.congr` `Chapter1/ScSmooth · IsSc1.congr`.

Uniqueness of $Df(x)$. The text writes "the" operator $Df(x)$ and "the" tangent map. If D and D' both satisfy (2) at $x \in U_1$ and U_1 has unique differentiability at x, then $D(x) \circ \iota_1 = D'(x) \circ \iota_1$ `Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq`; since ι_1 has dense range (§I.1) and both operators are continuous, $D(x) = D'(x)$ `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on` `Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt`. Hence on a set U whose U_1 has unique differentiability everywhere (an sc-domain), the chosen derivative agrees on U_1 with any admissible family `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn` and the chosen tangent map agrees on TU with any $T_D f$ `Chapter1/ScSmooth · IsSc1.tangentMap_eqOn`. This is the exposition's Lemma 3.E, and the formal proofs invoke it wherever the text says "the tangent map". The uniform version of (3) at level 0: for $x \in U_1$, a compact $K \subseteq E_0$ and $c > 0$ there is $\delta > 0$ such that $\|D(x')w - D(x)w\| \le c$ for all $x' \in U_1$ within δ of x and all $w \in K$; this is the continuity of $(x', w) \mapsto D(x')w - D(x)w$ on $U_1 \times E_0$ turned uniform over the compact K `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually` `Chapter1/Properties · IsSc1With.exists_delta_deriv`, and it is the text's remark that $x \mapsto Df(x)$ is continuous for the compact-open topology.

Compact-open continuity as an equivalence (Lemma 3.F of the exposition). For a metric space X,

$s \subseteq X$ and $A : X \to \mathcal{L}(Y, Z)$: $(x, h) \mapsto A(x)h$ is continuous on $s \times Y$ if and only if for every $x \in s$, compact $K \subseteq Y$ and $\varepsilon > 0$, eventually for x' near x in s, $\|A(x')w - A(x)w\| < \varepsilon$ for all $w \in K$ `Chapter1/ExpositionLemmas · continuousOn_apply_iff_uniformOnCompact`. (□) As above, on the subtype s: for each $w \in K$, $(x', w') \mapsto A(x')w' - A(x)w'$ is continuous with value 0 at (x, w), so it is small near (x, w), and compactness of K makes the neighbourhood of x uniform in w `Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually`; neighbourhoods in the subtype are neighbourhoods within s `Mathlib/Topology/NhdsWithin · eventually_nhds_subtype_iff`. (□) Continuity within $s \times Y$ at (x, w) is tested on sequences `Mathlib/Order/Filter/AtTopBot/CountablyGenerated · tendsto_iff_seq_tendsto`: let $(x_n, w_n) \to (x, w)$ with $x_n \in s$. The set $K = \{w\} \cup \{w_n\}$ is compact `Mathlib/Topology/Compactness/Compact · Filter.Tendsto.isCompact_insert_range`, so $\|A(x_n)w_n - A(x)w_n\| \to 0$ by the hypothesis, and $A(x)w_n \to A(x)w$ by continuity of $A(x)$; adding, $A(x_n)w_n \to A(x)w$. For an sc^1 map, level 0 of condition (3) is the left side for Df on U_1, so $x \mapsto Df(x)$ is compact-open continuous on U_1 `Chapter1/ExpositionLemmas · IsSc1With.deriv_uniformOnCompact`. The text's other half, that norm continuity can fail, needs the shift map of the exposition's §6 and is not formalized.

Sc^k and sc-smooth. By recursion on k: sc^0 is sc^0, and f is sc^{k+1} on U if it is sc^1 on U and the chosen tangent map is sc^k on TU `Chapter1/ScSmooth · IsSck`; f is **sc-smooth** on U if it is sc^k for every k `Chapter1/ScSmooth · IsScSmooth`. An sc^k map is sc^0 `Chapter1/ScSmooth · IsSck.isSc0`, sc^{k+1} implies sc^k `Chapter1/ScSmooth · IsSck.of_succ Chapter1/ScSmooth · IsSck.of_le`, and f is sc-smooth iff it is sc^1 with sc-smooth tangent map `Chapter1/ScSmooth · isScSmooth_iff`. On an sc-domain the notion depends only on the values on U, by induction with the uniqueness of the tangent map `Chapter1/ChainRule · IsSc1.tangentMap_congr Chapter1/ChainRule · IsSck.congr`. **Finite dimensions.** The text's remark that sc^1 maps between finite-dimensional spaces are the C^1 maps holds on every sc-domain, in particular on relatively open subsets of quadrants, with boundary. Let V, W carry their constant structures and $U \subseteq V$ be an sc-domain. All level sets of U are U itself `Chapter1/ExpositionLemmas · ScBanach.lvlSet_const` and all $f \circ \iota_m$ are f. If f is sc^1, then $f \circ \iota_1 = f$ is C^1 within $U_1 = U$ by Proposition 1.2.1 (preprint 1.16) (§I.14, `Chapter1/Properties · IsSc1With.contDiffOn_base`); if f is C^1 within U, it is C^1 on every level, so sc^1 by Corollary 1.2.5 (preprint 1.20) at $k = 0$ (§I.17, `Chapter1/Properties · corollary_1_20_zero`) `Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_const`. Mathlib's "C^1 within U" means, on a set with unique derivatives, differentiable within U at every point with $x \mapsto$ fderivWithin $f\,U\,x$ continuous on U `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_fderivWithin Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const`: at a boundary point of a quadrant this is the one-sided derivative along U, unique because the quadrant has interior (§I.10). Likewise sc-smooth is C^∞ within U, by Proposition 1.2.3 (preprint 1.18) at $m = 0$ (§I.16) and the sc-smooth case of Corollary 1.2.5 (preprint 1.20) `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`. For the text's tuples: `Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_of_isRelOpenInQuadrant Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant`. The open case is also checked in the test file `Test · isSc1_iff_contDiffOn_of_finiteDimensional Test · isScSmooth_iff_contDiffOn_of_finiteDimensional`. By §I.1 every sc-structure on a finite-dimensional space is the constant one up to sc-isomorphism.

I.12 Theorem 1.3.1 (preprint 1.21): the chain rule

Statement. Let $U \subseteq C \subseteq E$ and $V \subseteq D \subseteq F$ be relatively open subsets of partial quadrants, f sc^1 on U with $f(U) \subseteq V$, and g sc^1 on V. Then $g \circ f$ is sc^1 on U and $T(g \circ f) = Tg \circ Tf$ on TU `Chapter1/ChainRule · theorem_1_21`. The same holds for sc-domains `Chapter1/ChainRule · theorem_1_21'`, and the existence part needs still less: if g is sc^1 on V with derivatives Dg, f is sc^1 on U with derivatives Df, $f(U) \subseteq V$, and V_1 is open-in-convex, then $g \circ f$ is sc^1 on U with derivatives $x \mapsto Dg(f_1(x)) \circ Df(x)$, with no hypothesis on U `Chapter1/ChainRule · IsSc1With.comp`. The text's third tuple $W \subset Q \subset G$ plays no role.

The derivative. Fix $x \in U_1$, let $y_0 := f_1(x) \in V_1$ and K a compact set containing $Df(x)(\iota_1 \bar{B}_1)$, which exists because ι_1 is compact `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_compact`. Write $V_1 = O \cap C$ with O open and C convex, and choose $\rho > 0$ with the ρ-ball around y_0 inside O. Given $c > 0$, two tolerances are fixed. First, by the uniform smallness of $[Dg(y) - Dg(y_0)](w + b)$ for $y \in V_1$ near y_0, $w \in K$ and b small, which is the continuity of

$(y,b,w) \mapsto [Dg(y)-Dg(y_0)](w+b)$ on $V_1 \times F_0 \times K$ made uniform over K Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually Chapter1/ChainRule · IsSc1With.exists_delta, there is δ with this expression of norm $\le c/2$. Second, $\varepsilon := \min(\delta/2, (c/2)/(\|Dg(y_0)\|+1))$. For $z \in U_1$ close to x three things hold eventually: $\|f_1(z) - y_0\|_1 < \min(\rho,\delta)$ (continuity of f_1 on U_1), and $r := f(\iota z) - f(\iota x) - Df(x)\iota(z-x)$ has $\|r\| \le \varepsilon\|z-x\|_1$ (condition (2) for f Mathlib/Analysis/Calculus/FDeriv/Defs · hasFDerivWithinAt_iff_isLittleO). Let $k := f_1(z) - y_0 \in F_1$. The segment $y_0 + tk$, $t \in [0,1]$, lies in V_1: inside the ρ-ball, hence in O, and in C by convexity, since both endpoints are in C. The text states this inclusion without comment; it is the one place where the structure of V enters.

The mean value inequality. The text's second integral is replaced by: for $\varphi(t) := g(\iota(y_0 + tk)) - t\,Dg(y_0)(\iota k)$ on $[0,1]$, the derivative within $[0,1]$ is $[Dg(y_0+tk) - Dg(y_0)](\iota k)$ (condition (2) for g along the segment, composed with the affine curve), and if its norm is $\le C$ on $[0,1]$ then $\|\varphi(1) - \varphi(0)\| \le C$ Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le Chapter1/ChainRule · IsSc1With.norm_sub_le_of_segment. Now write $\iota k = f(\iota z) - f(\iota x) = s(w_1 + b)$ with $s := \|z-x\|_1$, $w_1 := Df(x)\iota(u)$ for the unit vector $u := s^{-1}(z-x)$, so $w_1 \in K$, and $b := s^{-1}r$, so $\|b\| \le \varepsilon \le \delta/2 < \delta$. For every t, $y_0 + tk$ is within δ of y_0, so $\|[Dg(y_0+tk) - Dg(y_0)](\iota k)\| \le (c/2)\,s$, and the mean value inequality gives

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0)(\iota k)\| \le \tfrac{c}{2}\|z-x\|_1.$$

(When $z = x$ both sides vanish.) The text's first integral is not needed: $\|Dg(y_0)r\| \le \|Dg(y_0)\|\,\varepsilon\|z - x\|_1 \le \frac{c}{2}\|z-x\|_1$ directly, because $Dg(y_0)$ is bounded on F_0. Adding the two estimates, since $\iota k = Df(x)\iota(z-x) + r$,

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0)Df(x)\iota(z-x)\| \le c\|z-x\|_1$$

eventually, which is condition (2) for $g \circ f$ at x with the operator $Dg(y_0) \circ Df(x)$ Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp. The text's subsequence argument for the uniform convergence of the second integrand is the compactness step above; the text's appeal to Proposition 1.2.3 (preprint 1.18) to make $g : V_1 \to G_0$ of class C^1 is not used.

The tangent map. $g \circ f$ is sc^0 as a composite. The tangent map of $g \circ f$ with the composed derivatives equals $T_{Dg} \circ T_D f$ on TU, since $(g \circ f)_1 = g_1 \circ f_1$ on U_1 Chapter1/ChainRule · IsSc1With.tangentMapWith_comp; the right side is a composite of sc^0 maps ($T_D f$ maps TU into TV), hence sc^0: condition (3) Chapter1/ChainRule · IsSc1With.comp. For the chosen derivatives, $T(g \circ f) = Tg \circ Tf$ on TU follows from the uniqueness of the derivative on U_1 (§I.11), which is where the quadrant structure of U enters Chapter1/ChainRule · theorem_1_21' Chapter1/ChainRule · theorem_1_21.

Iterating. If U, V are sc-domains, f is sc^k on U with $f(U) \subseteq V$ and g is sc^k on V, then $g \circ f$ is sc^k on U: induction on k, the step being the theorem for the sc^1 parts, the induction hypothesis for $Tg \circ Tf$ on the sc-domains TU, TV, and the identity $T(g \circ f) = Tg \circ Tf$ with the independence of sc^k from the values off TU Chapter1/ChainRule · IsSck.comp Chapter1/ChainRule · IsSck.comp_of_isRelOpenInQuadrant. Sc-smooth maps compose to sc-smooth maps Chapter1/ChainRule · IsScSmooth.comp. This is the exposition's Corollary 4.A.

I.13 The compact-open lemma and the level operators $Df(x) : E_m \to F_m$

The proofs of Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) in the 2010 paper all rest on one mechanism, which the formal proof isolates.

The lemma, $k = 0$. Let $A : X \to \mathcal{L}(Y,Z)$ with $(x,y) \mapsto A(x)y$ continuous on $s \times Y$, and $\iota : Y' \to Y$ a compact operator. Then $x \mapsto A(x) \circ \iota$ is continuous on s into $\mathcal{L}(Y',Z)$ with the operator norm Chapter1/Levels · continuousOn_clm_comp_of_isCompactOperator. Proof: the image of the unit ball of Y' under ι lies in a compact K; by uniform continuity on compact sets Chapter1/Levels · exists_delta_of_continuousOn, for $x' \in s$ near x and all $w \in K$, $\|A(x')w - A(x)w\| \le \varepsilon/2$, so $\|(A(x') - A(x)) \circ \iota\| \le \varepsilon/2$ on unit vectors. This

is the 2010 paper's subsequence argument (unit vectors $h_n \in E_{m+1}$ have a subsequence converging in E_m) in uniform form.

The lemma, C^k. Let s be open-in-convex with unique differentiability at every point, and $(x, y) \mapsto A(x)y$ of class C^k on $s \times Y$. Then $x \mapsto A(x) \circ \iota$ is C^k on s `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. Induction on k over all Z and A. For $k+1$: the partial derivative in x, $B(x)y := D_x(A(x)y)$, is C^k in (x, y) `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn`, linear and continuous in y for $x \in s$ (differentiate the linearity identities and use uniqueness of derivatives), so $B : s \to \mathcal{L}(Y, \mathcal{L}(X, Z))$; by the induction hypothesis $x \mapsto B(x) \circ \iota$ is C^k, and so is its flip $G'(x) \in \mathcal{L}(X, \mathcal{L}(Y', Z))$. G' is the derivative of $x \mapsto A(x) \circ \iota$: for $z \in s$ near x the segment from x to z lies in s (open-in-convex), and for a unit vector $v \in Y'$ the mean value inequality applied to $\tau \mapsto A(x + \tau(z-x))(\iota v) - \tau B(x)(\iota v)(z-x)$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le`, with the uniform continuity of B on $\{x\} \times \iota(\bar{B}_1)$, bounds $\|A(z)\iota v - A(x)\iota v - B(x)(\iota v)(z-x)\| \le c\|z-x\|$ uniformly in v. So $x \mapsto A(x) \circ \iota$ is differentiable within s with a C^k derivative, hence C^{k+1} `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`. The 2010 paper's Proposition 2.3 carries the same information in the multilinear maps $\Gamma(x)(h_1, \dots, h_k, h)$ and a differentiation under the integral sign; the lemma replaces both.

The level operators. Let f be sc^1 on U with derivatives D, and $x \in E_{m+1}$. The **level-m operator** $Df(x) : E_m \to F_m$ is the continuous linear T with $\iota_m \circ T = D(\iota x) \circ \iota_m$, when such T exists (it is unique by injectivity), and 0 otherwise `Chapter1/Levels · levelDeriv` `Chapter1/Levels · toBase_levelDeriv`; on level 0 it is $D(x)$ itself `Chapter1/Levels · levelDeriv_zero`. For $x \in U_{m+1}$ it exists: the level-m tangent map sends $(x, v) \in U_{m+1} \times E_m$ to a pair whose second component $T_m(v)$ satisfies $\iota_m T_m(v) = D(\iota x)\iota_m v$; T_m is linear (by injectivity of ι_m) and continuous (condition (3) on level m) `Chapter1/Levels · IsSc1With.exists_levelDeriv` `Chapter1/Levels · IsSc1With.toBase_levelDeriv`. So the level-m tangent map is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.levelMap_tangentMapWith`, and condition (3) says exactly that $(x, h) \mapsto Df(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv`: this is condition (2) of Proposition 1.2.1 (preprint 1.16). By the compact-open lemma with $\iota = \mathrm{incl}_m : E_{m+1} \to E_m$, the map $x \mapsto Df(x) \circ \mathrm{incl}_m$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$ in the operator norm `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv_clm`: the continuity of df in condition (1) of Proposition 1.2.1 (preprint 1.16). The base case, $x \mapsto D(x) \circ \iota_1$ continuous $U_1 \to \mathcal{L}(E_1, F_0)$, is also proved directly `Chapter1/Properties · IsSc1With.continuousOn_deriv_clm`, and with it $f \circ \iota_1 : U_1 \to F_0$ is C^1 `Chapter1/Properties · IsSc1With.contDiffOn_base`.

Level-raising: differentiability of $f : U_{m+1} \to F_m$. Let U be an sc-domain. For $x \in U_{m+1}$, the map $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ has derivative $Df(x) \circ \mathrm{incl}_m$ at x within U_{m+1} `Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level`. This is the 2010 paper's integral representation $f(x+k) - f(x) = \int_0^1 Df(x + \tau k)k \, d\tau$. For $z \in U_{m+1}$ near x and $k := z - x$, the segment $x + \tau k$ lies in U_{m+1} (open-in-convex). The integrand $g(\tau) := Df(x + \tau k)(\mathrm{incl}_m k)$ is continuous into F_m by condition (3) on level m, so its Bochner integral over $[0, 1]$ exists in F_m. Its image under ι_m is the integral of $D(\iota(x + \tau k))\iota_{m+1}k$ in F_0, which by the fundamental theorem of calculus for the C^1 curve $\tau \mapsto f(\iota(x + \tau k))$ (condition (2) for f, composed with the affine curve) `Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus · integral_eq_sub_of_hasDerivAt_of_le` equals $f(\iota z) - f(\iota x) = \iota_m(f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x))$. By injectivity, $f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) = \int_0^1 g$ in F_m. Then

$$\left\| f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) - Df(x)\mathrm{incl}_m k \right\|_m \le \sup_\tau \|(Df(x + \tau k) - Df(x)) \circ \mathrm{incl}_m\| \, \|k\|_{m+1},$$

and the supremum tends to 0 by the norm continuity just proved. The 2010 paper forms the integral in F_0 and remarks that the integrand is continuous into F_m; here it is formed in F_m and pushed down. Consequently $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ is C^1 within U_{m+1} `Chapter1/Levels · IsSc1With.contDiffOn_level`, which is condition (1) of Proposition 1.2.1 (preprint 1.16) in full.

The level operator is the derivative of f_m. If $f_m : U_m \to F_m$ happens to be differentiable at $\mathrm{incl}_m x$ within U_m with derivative L, then $L = Df(x)$: both $\iota_m \circ L \circ \mathrm{incl}_m$ and $D(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at x within U_{m+1}, hence equal by uniqueness; and since incl_m has dense range, $\iota_m \circ L = D(\iota x) \circ \iota_m$ on all of E_m `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on Chapter1/Levels · IsSc1With.levelDeriv_eq`. This is the exposition's Lemma 3.H and the key to Proposition 1.2.4 (preprint 1.19).

I.14 Proposition 1.2.1 (preprint 1.16): sc¹ in terms of the levels

The level conditions. For a family $D_m(x) \in \mathcal{L}(E_m, F_m)$, $x \in E_{m+1}$, the **level conditions** are: (a) for every m and $x \in U_{m+1}$, $f_m \circ \mathrm{incl}_m$ has derivative $D_m(x) \circ \mathrm{incl}_m$ at x within U_{m+1}; (b) for every m, $(x, h) \mapsto D_m(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1Levels`. These are the text's (1) and (2) with $m - 1$ renamed m and the extensions $Df(x)$ made an explicit family. From them, $x \mapsto D_m(x) \circ \mathrm{incl}_m$ is norm continuous (compact-open lemma) `Chapter1/Levels · IsSc1Levels.continuousOn_clm`, so $f : U_{m+1} \to F_m$ is C^1 on an sc-domain `Chapter1/Levels · IsSc1Levels.contDiffOn`: the rest of the text's (1).

Statement. For an sc-domain U and f sc^0 on U: f is sc^1 on U iff some family D_m satisfies the level conditions `Chapter1/Levels · proposition_1_16`.

(□). The level operators of §I.13 satisfy (a) (level-raising) and (b) (condition (3)) `Chapter1/Levels · IsSc1With.isSc1Levels`.

(□), and what the text hides. The 2010 paper calls this direction clear. It is, once one knows that the operators $D_m(x)$ for different m are restrictions of one operator, which the text's notation $Df(x)$ assumes. Proof: for $x \in U_{m+1}$, both $\iota_m \circ D_m(x) \circ \mathrm{incl}_m$ and $D_0(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at x within U_{m+1} (compose (a) at level m with ι_m, and (a) at level 0 with the inclusion $E_{m+1} \to E_1$), so they agree by uniqueness, and since incl_m has dense range, $\iota_m \circ D_m(x) = D_0(\iota x) \circ \iota_m$ `Chapter1/Levels · IsSc1Levels.toBase_apply`. Then f is sc^1 with derivatives D_0: condition (2) is (a) at level 0; condition (3) at level m reads, after the compatibility, $(x, h) \mapsto (f_{m+1}(x), D_m(x)h)$ on $U_{m+1} \times E_m$, which is continuous by (b) and the continuity of f_{m+1} `Chapter1/Levels · IsSc1Levels.isSc1With`.

The "in particular". If f is sc^1 with derivatives D and $x \in U_1$ is a smooth point, then $D(x) : E_0 \to F_0$ is an sc-operator: lift x to $y \in U_{m+1}$; the level-m tangent map at (y, v) has second component $T_m(v)$ with $\iota_m T_m(v) = D(x)\iota_m v$, linear and continuous in v, which is the level-m map required by `IsScOperator` `Chapter1/Properties · IsSc1With.scOperator_deriv`.

I.15 Proposition 1.2.2 (preprint 1.17): lifting the index

Statement. If U is an sc-domain in E and f is sc^k on U, then $f_1 : E_1 \to F_1$ is sc^k on U_1, as a map between the shifted spaces E^1 and F^1 `Chapter1/Levels · IsSck.shift`; sc-smooth maps lift to sc-smooth maps `Chapter1/Levels · IsScSmooth.shift`. The target V of the text plays no role.

Proof. The level maps of f_1, computed in the shifted spaces, are the level maps of f shifted by one `Chapter1/Levels · IsSc0.levelMap_shift`, so an sc^0 map lifts to an sc^0 map, with no hypothesis on U `Chapter1/Levels · IsSc0.shift`. For sc^1 with derivatives D: the derivatives of f_1 are the level-1 operators $Df(x) : E_1 \to F_1$, $x \in U_2$; condition (2) for f_1 is the level-raising statement of §I.13 at $m = 1$, and condition (3) is the shift of condition (3) for f, since $T(U_1) = (TU)^1$ (§I.11) and the level tangent maps are $(x, h) \mapsto (f_{m+2}(x), Df(x)h)$ `Chapter1/Levels · IsSc1With.shift`. For sc^{k+1}: the tangent map of f_1 agrees on $T(U_1)$ with the level-1 map of Tf (uniqueness of derivatives on the sc-domain U_1), which is sc^k by the induction hypothesis applied to Tf on TU `Chapter1/Levels · IsSck.shift`. This is the 2010 paper's argument, with the compatibility $T(U^1) = (TU)^1$ and the restriction of Df proved rather than assumed.

I.16 Proposition 1.2.3 (preprint 1.18): sc^k maps are C^k from U_{m+k} to F_m

Statement. Let U be an sc-domain and f sc^k on U. Then $f_m \circ (E_{m+k} \to E_m) : U_{m+k} \to F_m$ is C^k within U_{m+k} for every m `Chapter1/Levels · IsSck.contDiffOn_levels`; $f : U_{m+l} \to F_m$ is C^l for $l \le k$ `Chapter1/Levels · IsSck.contDiffOn_levels_of_le`; in particular $f \circ \iota_k : U_k \to F_0$ is C^k `Chapter1/Levels · IsSck.contDiffOn_base`; and an sc-smooth map is C^k from U_{m+k} to F_m for all m, k `Chapter1/Levels · IsScSmooth.contDiffOn_levels`.

Proof. The 2010 paper proves the case $m = 0$ and reduces to it with Proposition 1.2.2 (preprint 1.17) applied m times. The formal proof handles all levels at once, by induction on k over all sc-domains, with a compatible level-lowering operator $\rho : E_n \to E_m$, $n = m + k$, $\iota_m \rho = \iota_n$, as a parameter `Chapter1/Levels · IsSck.contDiffOn_levels'`; this avoids identifying $(E^m)_k$ with E_{m+k}. For $k = 0$, ρ is the identity and the claim is the continuity of f_m. For $k + 1$: write $\rho = \mathrm{incl}_m \circ \rho'$ with $\rho' : E_{m+k+1} \to E_{m+1}$ `Chapter1/Levels · ScBanach.incl_shift_inclK` `Chapter1/Levels · ScBanach.shift_inclK_mem_lvlSet`. By the level-raising statement of §I.13, $f_m \circ \rho$ has derivative $x \mapsto Df(\rho' x) \circ \rho$ on U_{m+k+1}, where $Df(\cdot)$ is the level-m operator. It remains to see that this derivative is C^k into $\mathcal{L}(E_{m+k+1}, F_m)$. The induction hypothesis for the sc^k map Tf on the sc-domain TU, with the level-lowering operator of TE (which is the pair of those of E^1 and E `Chapter1/Levels · ScBanach.tangent_inclK`), says that the level-m tangent map composed with it, $(x, h) \mapsto (f_{m+1}(\rho' x), Df(\rho' x)(E_{m+k} \to E_m)h)$, is C^k on $U_{m+k+1} \times E_{m+k}$; its second component is $(x, h) \mapsto A(x)h$ with $A(x) := Df(\rho' x) \circ (E_{m+k} \to E_m)$. The compact-open lemma with $\iota = \mathrm{incl}_{m+k} : E_{m+k+1} \to E_{m+k}$ makes $x \mapsto A(x) \circ \mathrm{incl}_{m+k} = Df(\rho' x) \circ \rho$ of class C^k `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. A map that is differentiable within an sc-domain with a C^k derivative is C^{k+1} `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`.

I.17 Proposition 1.2.4 (preprint 1.19) and Corollary 1.2.5 (preprint 1.20): sc^{k+1} from classical smoothness on the levels

Statement. Let U be an sc-domain and f sc^0 on U. If for every $m \ge 0$ and $0 \le l \le k$ the map $f_m \circ (E_{m+l} \to E_m) : U_{m+l} \to F_m$ is C^{l+1} within U_{m+l}, then f is sc^{k+1} on U `Chapter1/LevelsWeak · proposition_1_19_text`. This is the text's statement; its "f induces a map $f : U_{m+l} \to F_m$" is the sc^0 hypothesis (values of level m on U_m; the continuity is then the case $l = 0$). The proof is the 2010 paper's (its Proposition 2.4) and lives in `Chapter1/LevelsWeak`; an earlier, shorter proof under a stronger hypothesis is kept and described at the end of this section.

Why the hypothesis does not induct. The obvious induction would show that f is sc^1 and that Tf satisfies the hypothesis with $k - 1$. It does not. The map $Tf : (TU)_{m+l} \to (TF)_m$ has the second component $(x, h) \mapsto Df(x)h$ on $U_{m+l+1} \times E_{m+l} \to F_m$, and $Df(x)$ on E_{m+l} is the derivative of $f : U_{m+l} \to F_m$ (§I.13), a C^{l+1} map; so this component is only C^l, one order short of the C^{l+1} that the hypothesis for Tf would ask. What survives the induction is a description of the components of the iterated tangent maps: they are built from the derivatives $d^j f(x)$ of f itself, evaluated at points and vectors of various levels, and the regularity of f on the coarser levels supplies the missing order. The 2010 paper's inductive statement (S_k) carries exactly this description. The formal proof organizes it in six steps: (1) the derivatives $d^j f(x)$ on all level pairs, with their compatibility, form a *derivative tower*; (2) an *elementary map* is $d^j f$ evaluated at sc-operators; (3) it is sc^0, and sc^1 with an explicit derivative, by the level criterion of Proposition 1.2.1 (preprint 1.16); (4) its tangent map is the pair of an elementary map and a sum of elementary maps; (5) by induction on s, every elementary map with $j + s \le k + 1$ is sc^s; (6) f is the elementary map with $j = 0$.

The derivative tower. A derivative tower of order $k + 1$ for f on U `Chapter1/LevelsWeak · IsDerivTower`

consists of base-level maps $\Phi_i(x)(h_1,\dots,h_i) \in F_0$ ($x, h_a \in E_0$) and, for $x \in U_{m+l}$, continuous i-linear maps $\Psi_{i,m,l}(x)$ on E_{m+l} with values in F_m, given for $i \le l+1$ and $i \le k+1$, such that: $\Phi_0 = f$; $\iota_m \Psi_{i,m,l}(x)(h_1,\dots,h_i) = \Phi_i(\iota_{m+l}x)(\iota_{m+l}h_1,\dots,\iota_{m+l}h_i)$, i.e. all the Ψ are one map seen on different levels; $\Psi_{i,m,l}$ is continuous on U_{m+l}; and for $i \le l$, $i \le k$ it is differentiable within U_{m+l} at every point, with derivative $\xi \mapsto \Psi_{i+1,m,l}(x)(\xi,\cdot)$. The ranges of i are exactly what "$f : U_{m+l} \to F_m$ is C^{l+1}" provides; the compatibility is the text's use of one symbol $d^j f(x)$ on all levels. Two values with the same images in E_0 agree, by injectivity of ι_m `Chapter1/LevelsWeak · IsDerivTower.eq_of_toBase`.

The tower of a map satisfying the hypothesis. $\Psi_{i,m,l}$ is Mathlib's i-th derivative within U_{m+l} of $f_m \circ (E_{m+l} \to E_m)$ `Chapter1/LevelsWeak · levelDerivTower` `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin`, and Φ_i is the i-th derivative of $f : U_{i-1} \to F_0$ at the lifts of its arguments to E_{i-1} (arbitrary off the image of E_{i-1}) `Chapter1/LevelsWeak · baseDerivTower`. For $l > k$ the hypothesis is used with order $k+1$: $f : U_{m+l} \to F_m$ is the C^{k+1} map $f : U_{m+k} \to F_m$ restricted along the inclusion $E_{m+l} \to E_{m+k}$, and U_{m+l} is the preimage of U_{m+k} `Chapter1/LevelsWeak · ScBanach.inclLE` `Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet`; so $f : U_{m+l} \to F_m$ is $C^{\min(l,k)+1}$ for all m, l `Chapter1/LevelsWeak · contDiffOn_levels_min`. Continuity of $\Psi_{i,m,l}$ for $i \le \min(l,k)+1$ is the continuity of the iterated derivatives of a $C^{\min(l,k)+1}$ map `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn.continuousOn_iteratedFDerivWithin`; for $i \le \min(l,k)$ the i-th derivative is C^1 `Mathlib/Analysis/Calculus/ContDiff/Comp · ContDiffWithinAt.iteratedFDerivWithin_right`, and its derivative is the $(i+1)$-st derivative with the new vector in front `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_succ_apply_left`. Both use the uniqueness of derivatives on the level sets of the sc-domain. The compatibility `Chapter1/LevelsWeak · levelDerivTower_toBase` is proved in three steps, for $i \le l+1$ (so that $i-1 \le m+l$): (i) composing on the left with the continuous linear ι_m commutes with iterated derivatives within a set `Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_left`; (ii) on U_{m+l}, $\iota_m \circ f_m \circ (E_{m+l} \to E_m)$ equals $(f : U_{i-1} \to F_0) \circ (E_{m+l} \to E_{i-1})$, since f is sc^0 `Chapter1/ScSmooth · IsSc0.toBase_levelMap`, and a derivative within a set depends only on the values on the set `Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_congr`; (iii) composing on the right with the continuous linear inclusion $E_{m+l} \to E_{i-1}$ commutes as well, the preimage of U_{i-1} being U_{m+l} `Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_right`. With the lifts in the definition of Φ_i `Chapter1/LevelsWeak · baseDerivTower_toBase` this gives the tower `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`.

Shifted spaces. E^p is the sc-Banach space with $(E^p)_l = E_{p+l}$ by definition `Chapter1/LevelsWeak · ScBanach.shiftAdd` (the p-fold iterate of the shift of §I.1 has these levels only up to rewriting $p+l$). The identifications $E \cong E^0$ and $(E^p)^1 \cong E^{p+1}$ are sc-isomorphisms whose level maps are identities transported along $l = 0+l$, resp. $p+(l+1) = p+1+l$ `Chapter1/LevelsWeak · ScBanach.castCLE` `Chapter1/LevelsWeak · ScBanach.shiftAddZeroIso` `Chapter1/LevelsWeak · ScBanach.shiftAddShiftIso`; they are used only through their compatibility with the inclusions into the base level `Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm`. For $b \le a$ the inclusion $E_a \to E_b$ is compatible with the ι's `Chapter1/LevelsWeak · ScBanach.toBase_inclLE`.

Elementary maps. Elementary data on an sc-Banach space X `Chapter1/LevelsWeak · ElemData` are an sc-operator $\pi_0 : X \to E^p$ (the base point), i sc-operators $\pi_a : X \to E^{n_a}$ (the vectors) and a target shift m. The elementary map is

$$g(z) = d^i f(\pi_0 z)(\pi_1 z,\dots,\pi_i z) := \Phi_i(\pi_0 z)(\pi_1 z,\dots,\pi_i z),$$

lifted to $F_m = (F^m)_0$ `Chapter1/LevelsWeak · ElemData.map`, on the domain $W = \pi_0^{-1}(U_p)$ `Chapter1/LevelsWeak · ElemData.dom`, whose level-l set is $\pi_0^{-1}(U_{p+l})$ `Chapter1/LevelsWeak · ElemData.mem_lvlSet_dom`. The exposition's map of type $(p,l;n_1,\dots,n_j)$ on $U^p \oplus E^{n_1} \oplus \cdots \oplus E^{n_j}$ is the case $X = E^p \oplus E^{n_1} \oplus \cdots$ with the projections as slots. The conditions are `Chapter1/LevelsWeak · ElemData.Good`

$$m+i \le p, \qquad m+i \le n_a+1 \ (1 \le a \le i), \qquad i \le k.$$

The second is the text's $j+l-1 \le n_i$. The first is $j+l \le k+1$ for the base point $x \in U_{k+1}$, which the

text derives from $j+l-1 \le n_i \le k$ (and states separately, as $l \le k+1$, only for $j=0$ in the exposition). The third bounds the order of derivative that will be used. The text's upper bound $n_i \le k$ is not needed.

On level l, the base point and the vectors pushed down to a common level q ($q \le p+l$, $q \le n_a+l$) are $\rho_0 = (E_{p+l} \to E_q) \circ \pi_0$ and $\rho_a = (E_{n_a+l} \to E_q) \circ \pi_a$ `Chapter1/LevelsWeak · ElemData.ρ₀` `Chapter1/LevelsWeak · ElemData.ρ`. Seen in F_0, g at a point of level l is Φ_i at the base-level images of $\pi_0 z$ and $\pi_a z$ `Chapter1/LevelsWeak · ElemData.map_toBase`, and its level-l map is $z \mapsto \Psi_{i,m+l,\lambda}(\rho_0 z)(\rho_1 z, \dots, \rho_i z)$ for any λ with $i \le \lambda+1$, $m+l+\lambda \le p+l$ and $m+l+\lambda \le n_a+l$ `Chapter1/LevelsWeak · ElemData.levelMap_eq`. The first two conditions make $\lambda = i-1$ admissible, and $i \le k$ gives $i \le k+1$, so the tower value exists. Hence g is sc^0: its values at level-l points lie in F_{m+l}, and its level-l map is continuous on W_l as the evaluation of the continuous $\Psi_{i,m+l,i-1} \circ \rho_0$ at the continuous $(\rho_1, \dots, \rho_i)$ `Chapter1/LevelsWeak · ElemData.isSc0`.

The sc^1 step. For $z \in W_{l+1}$ the candidate derivative $Dg(z) : X_l \to F_{m+l}$ is the derivative (3.5) of the exposition `Chapter1/LevelsWeak · ElemData.deriv` `Chapter1/LevelsWeak · ElemData.deriv_apply`:

$$Dg(z)\zeta = \Psi_{i+1,m+l,i}(\rho_0 z)(\rho_0\zeta, \rho_1 z, \dots, \rho_i z) + \sum_{a=1}^{i} \Psi_{i,m+l,i-1}(\rho_0 z)(\rho_1 z, \dots, \rho_a\zeta, \dots, \rho_i z),$$

the first term on the uniform level $m+l+i$, the sum on $m+l+i-1$. The conditions put every argument where it belongs: $\rho_0 z$ and $\rho_a z$ (with z of level $l+1$) in E_{m+l+i}, because $m+i \le p+1$ and $m+i \le n_a+1$; $\rho_0\zeta$ (with ζ of level l) in E_{m+l+i}, because $m+i \le p$; $\rho_a\zeta$ in $E_{m+l+i-1}$, because $m+i-1 \le n_a$; and $\Psi_{i+1,m+l,i}$ exists because $i+1 \le k+1$, which is where $i \le k$ enters. On the base level, $Dg(z)\zeta$ is $\Phi_{i+1}(\pi_0 z)(\pi_0\zeta, \pi z) + \sum_a \Phi_i(\pi_0 z)(\pi z$ with $\pi_a z := \pi_a\zeta)$ `Chapter1/LevelsWeak · ElemData.deriv_toBase`. The level criterion of Proposition 1.2.1 (preprint 1.16) (§I.14) asks two things.

Differentiability of $g : W_{l+1} \to F_{m+l}$ with derivative $Dg(z)$ restricted to X_{l+1}. On W_{l+1}, the level-l map of g is $z' \mapsto \Psi_{i,m+l,i}(\rho_0 z')(\rho_1 z', \dots, \rho_i z')$ with the slots taken at level $l+1$ (the formula above with $\lambda = i$) `Chapter1/LevelsWeak · ElemData.levelMap_incl_eq`. In the tower, $\Psi_{i,m+l,i}$ is differentiable with derivative $\Psi_{i+1,m+l,i}$ ($i \le i$ and $i \le k$); the slots are continuous linear; and the evaluation $(A, h) \mapsto A(h_1, \dots, h_i)$ of continuous multilinear maps is differentiable with derivative $(B, \eta) \mapsto B(h) + \sum_a A(h$ with $h_a := \eta_a)$ `Chapter1/LevelsWeak · hasFDerivAt_cmm_eval`, proved from Mathlib's derivative of a multilinear map in its arguments `Mathlib/Analysis/Calculus/FDeriv/Analytic · ContinuousMultilinearMap.hasFDerivAt` and the linearity in A (the evaluation written as a flipped multilinear map `Chapter1/LevelsWeak · cmm_eval_eq`). The chain rule gives a derivative whose first term is that of Dg (the base point restricted from level $l+1$ to level l is the same vector `Chapter1/LevelsWeak · ElemData.ρ₀_incl`) and whose sum terms are values of $\Psi_{i,m+l,i}$; these equal the $\Psi_{i,m+l,i-1}$ terms of Dg because they have the same images in E_0 (the tower's compatibility) `Chapter1/LevelsWeak · ElemData.hasFDerivWithinAt_level`.

Continuity of $(z, \zeta) \mapsto Dg(z)\zeta$ on $W_{l+1} \times X_l$: $\Psi_{i+1,m+l,i}$ and $\Psi_{i,m+l,i-1}$ are continuous on their level sets, and the slots, the insertion of $\rho_a\zeta$ and the evaluation are continuous `Chapter1/LevelsWeak · ElemData.continuousOn_deriv`.

So g satisfies the level conditions `Chapter1/LevelsWeak · ElemData.isSc1Levels` `Chapter1/Levels · IsSc1Levels`, and when W is an sc-domain it is sc^1 with derivative Dg at level 0 `Chapter1/Levels · IsSc1Levels.isSc1With` `Chapter1/LevelsWeak · ElemData.isSc1With`. This is the text's "Γ is sc^1", with the level criterion in place of the direct check of Definition 1.1.15 (preprint 1.14) that the exposition carries out.

The tangent map of an elementary map. On $TX = X^1 \oplus X$ a slot $\pi : X \to E^n$ gives two slots: π on the first factor, $TX \to E^{n+1}$ through $(E^n)^1 \cong E^{n+1}$ `Chapter1/LevelsWeak · ScOperator.tangentX`, and π on the second factor, $TX \to E^n$ `Chapter1/LevelsWeak · ScOperator.tangentH`. Three kinds of elementary data on TX arise from g. The first, g', has all slots on the first factor, with levels $p+1$, n_a+1 and target shift $m+1$ `Chapter1/LevelsWeak · ElemData.shiftData`: it is g one level up. The second, g_0, has the base point π_0 on the first factor (level $p+1$), $i+1$ vectors, namely π_0 on the second factor (level p) followed by the π_a on the first (levels n_a+1), and target shift m `Chapter1/LevelsWeak · ElemData.derivData`: it is the term $d^{i+1}f(x)(\delta x, h)$. The third, g_a for $1 \le a \le i$, is like g' but with π_a moved to the second factor (level n_a) and target

shift m Chapter1/LevelsWeak · ElemData.slotData: it is the term $d^i f(x)(h \text{ with } h_a := \delta h_a)$. All satisfy the conditions, which is the text's "one checks that the new indices satisfy the required inequalities": for g', $m+1+i \le p+1$ and $m+1+i \le n_a+2$ Chapter1/LevelsWeak · ElemData.Good.shiftData; for g_0, $m+i+1 \le p+1$ (for the base point and for the new first vector, of level p) and $m+i+1 \le n_a+2$, and $i+1 \le k$, which has to be assumed Chapter1/LevelsWeak · ElemData.Good.derivData; for g_a, $m+i \le p+1$, $m+i \le n_a+1$ for the moved slot and $m+i \le n_b+2$ for the others Chapter1/LevelsWeak · ElemData.Good.slotData. The domain of all three (they share the base point π_0 on the first factor) is TW Chapter1/LevelsWeak · ElemData.tangentSet_dom.

If W is an sc-domain and $i+1 \le k$, the tangent map of g agrees on TW with the pair (g' carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$) Chapter1/LevelsWeak · ElemData.tangentMap_eqOn. Proof: by the uniqueness of the derivative on the sc-domain W, the tangent map is (g on level 1, Dg) Chapter1/ScSmooth · IsSc1.tangentMap_eqOn; the first components agree in F_0, both being Φ_i at the base-level images of the level-1 slots Chapter1/LevelsWeak · ElemData.map_toBase Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm; the second components agree in F_0 term by term, by the base-level formula for Dg Chapter1/LevelsWeak · ElemData.deriv_toBase.

Sums. The constant map 0 is sc^k, being the zero sc-operator Chapter1/LevelsWeak · ScOperator.zeroOp Chapter1/LevelsWeak · IsSck.zero. If g_1, g_2 are sc^k on an sc-domain, g_1+g_2 is the addition $Y \oplus Y \to Y$ Chapter1/LevelsWeak · ScOperator.addOp, an sc-operator and hence sc-smooth Chapter2/TangentBundle · ScOperator.isScSmooth, composed with the pairing (g_1, g_2), which is sc^k Chapter2/StrongBundle · IsSck.scPair; so it is sc^k by the chain rule (§I.12, Chapter1/ChainRule · IsSck.comp) Chapter1/LevelsWeak · IsSck.add. Finite sums follow by induction over the index set Chapter1/LevelsWeak · IsSck.sum. Scalar multiples are not needed: the text's "linear combinations" are sums.

The induction (S_k)**.** *Claim:* for every s, every sc-Banach space X and all elementary data on X satisfying the conditions, with sc-domain W and $i+s \le k+1$, the elementary map is sc^s on W Chapter1/LevelsWeak · ElemData.isSck. The induction is on s, for all X, i and data at once. For $s=0$ this is the sc^0 statement. For $s+1$: g is sc^1 by the sc^1 step, and it remains to show that its tangent map is sc^s. If $s=0$, the tangent map of an sc^1 map is sc^0 Chapter1/ScSmooth · IsSc1.tangentMap_isSc0. If $s \ge 1$, then $i+1 \le k$ (from $i+s+1 \le k+1$), so the tangent map agrees on TW with (g' carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$); TW is an sc-domain Chapter1/ScSmooth · IsScDomain.tangentSet, equal to the domain of the three new data Chapter1/LevelsWeak · IsScDomain.of_eq Chapter1/LevelsWeak · IsSck.of_eq_set, and sc^s passes to maps that agree on the domain Chapter1/ChainRule · IsSck.congr. The pair is sc^s if its two components are Chapter2/StrongBundle · IsSck.scPair. Now g' and the g_a have i vectors and $i+s \le k+1$, and g_0 has $i+1$ vectors and $(i+1)+s \le k+1$, so all are sc^s by the induction hypothesis (applied on $X' = TX$); g' composed with the sc-isomorphism $(F^m)^1 \cong F^{m+1}$, an sc-operator Chapter1/ScBanach · ScIso.toScOperator, is sc^s by the chain rule; and $g_0 + \sum_a g_a$ by the sum rule.

This is the text's induction, run on s over all elementary maps rather than on the description of $T^{k+1}f$. The 2010 paper proves that every term of $\pi \circ T^{k+1}f$ is sc^1, hence $T^{k+1}f$ is, and that the terms of $T^{k+2}f$ are again elementary with admissible indices; the formal proof shows that an elementary map is sc^{s+1} because its tangent map is made of elementary maps that are sc^s. The computations are the same — the derivative (3.5), the three kinds of new terms and their indices — and the formal version never names $T^{k+1}f$ or its 2^{k+1} components.

Proposition 1.2.4 (preprint 1.19). f is the elementary map on $X = E$ with no vectors, $p = m = 0$ and $\pi_0 : E \cong E^0$ Chapter1/LevelsWeak · ElemData.base; its domain is U Chapter1/LevelsWeak · ElemData.base_dom and the conditions hold trivially Chapter1/LevelsWeak · ElemData.base_good. With the tower of f (step 1) and $s = k+1$, $0+(k+1) \le k+1$, this elementary map is sc^{k+1}; composed with the sc-isomorphism $F^0 \cong F$ it agrees with f on U, since $\Phi_0 = f$; so f is sc^{k+1} Chapter1/LevelsWeak · proposition_1_19_text. If the hypothesis holds for every l, then f is sc^{k+1} for every k, i.e. sc-smooth Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_text.

Corollary 1.2.5 (preprint 1.20). For a finite-dimensional target V with its constant structure, the level maps of $f : E_0 \to V$ are $f \circ \iota_m$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq Chapter1/ScBanach · ScBanach.const_

toBase. Suppose $f \circ \iota_l$ is C^{l+1} within U_l for $0 \le l \le k$. Then f is sc^0 with no further hypothesis: f is continuous on $U = U_0$ (the case $l = 0$), so each $f \circ \iota_m$ is continuous on U_m. And the hypothesis of Proposition 1.2.4 (preprint 1.19) holds: $f_m \circ (E_{m+l} \to E_m) = (f \circ \iota_l) \circ (E_{m+l} \to E_l)$, a C^{l+1} map composed with a continuous linear one, on U_{m+l}, the preimage of U_l Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet. So f is sc^{k+1} Chapter1/LevelsWeak · corollary_1_20_text. If $f \circ \iota_l$ is C^{l+1} within U_l for every l, then f is sc-smooth (for $k = 0$ the corollary gives sc^1, hence sc^0) Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_finiteDimensional_text; and if f is C^∞ on U, it is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional (proved with the stronger version below, since C^∞ on U restricts to C^{k+1} on every U_m). The two sc-smooth statements are the exposition's; the second is what the theory uses (bump functions, Chapter 5).

The earlier version, under a stronger hypothesis. Chapter1/Levels proves Proposition 1.2.4 (preprint 1.19) under the hypothesis that every level map $f_m : U_m \to F_m$ is C^{k+1} within U_m Chapter1/Levels · proposition_1_19. For $k \ge 1$ this asks more than the text (C^{k+1} instead of C^1 on $U_m \to F_m$ itself), and in exchange the induction needs no (S_k); for $k = 0$ the hypotheses coincide.

$k = 0$. Suppose every f_m is C^1 within U_m. Put $Df(x) :=$ the derivative of $f = f_0$ within U at $\iota_1 x$, for $x \in E_1$ Mathlib/Analysis/Calculus/FDeriv/Defs · fderivWithin. Condition (2) holds because $f \circ \iota_1$ is the restriction of the differentiable f along the continuous linear ι_1. For condition (3) at level m: the derivative $df_m(x)$ of f_m within U_m at $x \in U_m$ is compatible with Df, $\iota_m \circ df_m(x) = Df(\iota_m x) \circ \iota_m$, because $\iota_m \circ f_m = f \circ \iota_m$ on U_m and derivatives within U_m are unique Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq; so the level-m tangent map is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$, continuous on $U_{m+1} \times E_m$ because f_m is C^1 (its derivative is continuous into $\mathcal{L}(E_m, F_m)$, and evaluation is continuous) Chapter1/Properties · proposition_1_19_zero. This is the text's proof of (S_0).

$k + 1$ from k. Suppose every f_m is C^{k+2}. Then f is sc^1 by the case $k = 0$, with the derivative Df above, and we must show Tf is sc^{k+1} on TU. By the induction hypothesis (for the sc-domain TU and the sc^0 map Tf) it suffices that the level maps of Tf are C^{k+1}. The level-m map of Tf is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ with the level operator (§I.13), and the level operator at $x \in U_{m+1}$ is the derivative of f_m at $\mathrm{incl}_m x$ (Lemma 3.H, Chapter1/Levels · IsSc1With.levelDeriv_eq). So the level-m map of Tf is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$ on $U_{m+1} \times E_m$, whose components are C^{k+1}: the first as f_{m+1} is, the second as the derivative of a C^{k+2} map is C^{k+1} and evaluation is bilinear Chapter1/Levels · proposition_1_19. This is where the stronger hypothesis is used, and where the text's hypothesis would fail (see "Why the hypothesis does not induct").

Sc-smooth, and the corollary. If f maps each U_m into F_m and every f_m is C^∞, then f is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_levels. For a finite-dimensional target: C^1 on every U_m gives sc^1 Chapter1/Properties · corollary_1_20_zero, C^{k+1} on every U_m gives sc^{k+1} Chapter1/Levels · corollary_1_20, in particular C^{k+1} on $U = U_0$ does, since restricting along ι_m preserves C^{k+1} Chapter1/Levels · corollary_1_20_base.

I.18 Definition 1.3.3 (preprint 1.23) and Lemma 1.3.4 (preprint 1.24): the degeneracy index

Definitions. For $p = (a, w) \in \mathbb{R}^n \oplus W$ let $\#(p)$ be the number of indices i with $a_i = 0$ Chapter1/Degeneracy · quadrantDeg. For an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ presenting a partial quadrant C and $x \in E_0$, let $d_T(x) := \#(T_0 x)$ Chapter1/Degeneracy · degIndexWith. The **degeneracy index** $d_C(x)$ is $d_T(x)$ for a presentation T of C chosen once and for all, and 0 if C is not a partial quadrant Chapter1/Degeneracy · degIndex; by Lemma 1.3.4 (preprint 1.24) below it equals $d_T(x)$ for every presentation and every $x \in C$ Chapter1/Degeneracy · degIndex_eq, and on the quadrant itself it is $\#$ Chapter1/Degeneracy · degIndex_quadrant. The text's interior points, boundary points and corners are the points with $d_C = 0$, 1, ≥ 2; Chapter 2 makes the boundary a definition.

Lemma 1.3.4 (preprint 1.24), core form. Let $S : \mathbb{R}^n \times W_0 \to \mathbb{R}^{n'} \times W_0'$ be a continuous linear equiv-

alence with $q \in Q \iff Sq \in Q'$ for the two quadrants. Then $\#(p) = \#(Sp)$ for every $p \in Q$ `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`. Only level 0 and linear algebra are used; the sc-structure plays no role. Proof. For $p \in Q$ let $E_p := \{(b, v) : b_i = 0 \text{ whenever } p_i = 0\}$, the kernel of the surjection $(b, v) \mapsto (b_i)_{i \in I(p)}$ `Chapter1/Degeneracy · coordMap` `Chapter1/Degeneracy · coordMap_surjective` `Chapter1/Degeneracy · mem_ker_coordMap`, so that $\dim(V/E_p) = \#(p)$ `Chapter1/Degeneracy · finrank_quotient_ker_coordMap`: this makes the text's "codimension of E_x" precise. For $u \in E_p$, $p + tu \in Q$ for all t near 0: coordinate by coordinate, if $p_i = 0$ then $u_i = 0$ and the coordinate stays 0; if $p_i > 0$ it stays positive for small t by continuity `Chapter1/Degeneracy · eventually_add_smul_mem_quadrant`. Hence $Sp + tSu \in Q'$ for t near 0, so for each i with $(Sp)_i = 0$ we have $0 \le t(Su)_i$ for all t near 0, of both signs, which forces $(Su)_i = 0$ `Chapter1/Degeneracy · eq_zero_of_eventually_nonneg_mul`; so $S(E_p) \subseteq E'_{Sp}$ `Chapter1/Degeneracy · map_ker_coordMap_le`. The same argument for S^{-1} gives the reverse inclusion (the text concludes equality from one inclusion "since S is an isomorphism", which for infinite-dimensional W needs this second step), so $S(E_p) = E'_{Sp}$ and S induces a linear isomorphism $V/E_p \cong V'/E'_{Sp}$; comparing dimensions, $\#(p) = \#(Sp)$.

Lemma 1.3.4 (preprint 1.24). If T and T' both present C, then $S := T'_0 \circ T_0^{-1}$ carries the quadrant onto the quadrant (both are the image of C), and the core form at $p = T_0 x$ gives $d_T(x) = d_{T'}(x)$ for $x \in C$ `Chapter1/Degeneracy · lemma_1_24`. The text proves first that $k = k'$; here this is a consequence: $T_0^{-1}(0) \in C$ and $\#(0) = n$ in $\mathbb{R}^n \oplus W$ `Chapter1/Degeneracy · quadrantDeg_zero`, so the lemma at this point gives $n = n'$ `Chapter1/Degeneracy · lemma_1_24_dim`.

I.19 Theorem 1.3.5 (preprint 1.25): boundary recognition

Statement. Let $U \subseteq C \subseteq E$ and $U' \subseteq C' \subseteq E'$ be relatively open subsets of partial quadrants, f sc^1 on U with $f(U) \subseteq U'$, g sc^1 on U' with $g(U') \subseteq U$, $g \circ f = \mathrm{id}$ on U and $f \circ g = \mathrm{id}$ on U'. Then $d_C(x) = d_{C'}(f(x))$ for every $x \in U$ `Chapter2/DegeneracyIndex · theorem_1_25`. The text's "germ of a local sc-diffeomorphism" is undefined in Chapter 1 and is read this way, with U, U' as small as one likes; the text has sc-smooth where sc^1 suffices, so the formal statement is stronger. The text gives no proof, referring to "[20], Theorem 1.19"; the formal proof is the argument of §2.4 (preprint §2.3) (Lemma 2.4.9 (preprint 2.29), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)) specialized to the identity retraction, and lives in the Chapter 2 file because it uses the retraction vocabulary. Smooth points are not needed; level-1 points suffice.

The subspaces E_x. For a presentation T of C and $x \in E_0$, let $E_x := \{v \in E_0 : (T_0 v)_i = 0 \text{ whenever } (T_0 x)_i = 0\}$, the kernel of a continuous linear map to a finite-dimensional space `Chapter2/DegeneracyIndex · minimalSubspaceWith`; its codimension is $d_T(x)$ `Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith`. Two facts about the index: it is upper semicontinuous, $d_T(y) \le d_T(x)$ for y near x (a nonzero coordinate stays nonzero) `Chapter2/DegeneracyIndex · eventually_degIndexWith_le`; and every $x \in C$ is the limit of points of C on level 1 with the same index, namely $T_0^{-1}(a, w_n)$ with $w_n \in W_1$ converging to w in W_0, which exist by density `Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith`. These are the exposition's Lemma 5.A, with level 1 in place of smooth.

The sign argument. Let f be sc^1 on $U \subseteq C$ with values in C', $x \in U_1$, and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x + tv \in U_1$ for t near 0 (the quadrant coordinates as in §I.18, and the openness of U_1) `Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet`, so $f(\iota(x + tv)) \in C'$ and, for each j with $(T'_0 f(x))_j = 0$, the function $t \mapsto (T'_0 f(\iota(x + tv)))_j$ is ≥ 0 with a zero at $t = 0$: a local minimum, so its derivative at 0, which is $(T'_0 Df(x)\iota_1 v)_j$ by condition (2), vanishes `Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero` `Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1`. So $Df(x)$ maps $E_x \cap \iota_1(E_1)$ into $E'_{f(x)}$. Since f is differentiable only along E_1, the density of $E_x \cap \iota_1(E_1)$ in E_x is needed to conclude $Df(x)(E_x) \subseteq E'_{f(x)}$; it holds because E_x is the kernel of a continuous linear map to a finite-dimensional space and $\iota_1(E_1)$ is dense `Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase` `Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace`. The text asserts the corresponding density ("the closure of $(T_x^R O)_\infty$ is $T_x^R O$") without proof.

One inequality at level-1 points. With g the inverse, the chain rule gives $Dg(f_1(x)) \circ Df(x) = \mathrm{id}$ on E_0 at $x \in U_1$ `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`, so $Df(x)$ is injective and $E_x \cong Df(x)(E_x) \leq E'_{f(x)}$ sits inside a subspace of codimension $d_{C'}(f(x))$, while $Dg(f_1(x))$ maps $E'_{f(x)}$ into E_x by the sign argument for g; comparing codimensions, $d_C(x) \leq d_{C'}(f(x))$ `Chapter2/DegeneracyIndex · degIndexWith_le_of_germ`. Applying this to g at $f_1(x)$ gives the reverse inequality, so $d_C(x) = d_{C'}(f(x))$ for $x \in U_1$ `Chapter2/DegeneracyIndex · theorem_1_25_lvl1`.

General points. For $x \in U$, approximate x by level-1 points $x_n \in U$ of C with $d_C(x_n) = d_C(x)$; then $f(x_n) \to f(x)$ (continuity of f on U), so $d_{C'}(f(x_n)) \leq d_{C'}(f(x))$ eventually by semicontinuity, and $d_{C'}(f(x_n)) = d_C(x_n) = d_C(x)$ by the level-1 case; hence $d_C(x) \leq d_{C'}(f(x))$. The symmetric argument for g at $f(x)$ gives equality `Chapter2/DegeneracyIndex · theorem_1_25`.

I.20 What the formalization taught us

- **The chain rule needs one integral less than the text writes, and one hypothesis more than it states.** The term $Dg(f(x))[f(x+h) - f(x) - Df(x)h]$ is $o(|h|_1)$ because $Dg(f(x))$ is bounded, with no integral `Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp`; the segment from $f(x)$ to $f(x+h)$ lies in V_1 only because V_1 is open-in-convex and $f : U_1 \to F_1$ is continuous, which the text does not say. The whole proof uses, of the quadrant structure of U, only the uniqueness of derivatives, and that only for $T(g \circ f) = Tg \circ Tf$ `Chapter1/ChainRule · theorem_1_21'`.
- **"The" derivative $Df(x)$ is a theorem, not a notation.** It is unique because E_1 is dense in E_0 and U_1 has unique differentiability `Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt`; without it the tangent map, hence sc^2, is undefined. The same density argument shows that the extensions $Df(x) : E_{m-1} \to F_{m-1}$ of Proposition 1.2.1 (preprint 1.16), written with one symbol in the text, are restrictions of one operator `Chapter1/Levels · IsSc1Levels.toBase_apply`, which is what makes the "clear" direction of that proposition work.
- **One compactness argument carries §1.2.** The compact-open lemma `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator` turns joint continuity (or C^k-ness) of $(x,h) \mapsto Df(x)h$ into norm continuity (or C^k-ness) of $x \mapsto Df(x)|_{E_{m+1}}$; Propositions 1.2.1, 1.2.2 and 1.2.3 (preprint 1.16, 1.17 and 1.18) are this lemma plus bookkeeping of levels.
- **Proposition 1.2.4 (preprint 1.19) does not induct on its own hypothesis.** If f satisfies it with k, then Tf in general does not satisfy it with $k-1$: the second component $(x,h) \mapsto Df(x)h : U_{m+l+1} \times E_{m+l} \to F_m$ is one order short. The stronger hypothesis "C^{k+1} on every level" makes the induction two lines `Chapter1/Levels · proposition_1_19`; the text's needs the 2010 paper's (S_k), which remembers f through the elementary maps $d^j f$ on mixed levels `Chapter1/LevelsWeak · ElemData.isSck` `Chapter1/LevelsWeak · proposition_1_19_text`. Written out, (S_k) needs three things the text leaves implicit: one $d^j f(x)$ on all levels `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`, the base-point condition $j + l \leq k+1$ in place of the bound $n_i \leq k$ (which is not needed) `Chapter1/LevelsWeak · ElemData.Good`, and the derivative of the multilinear evaluation `Chapter1/LevelsWeak · hasFDerivAt_cmm_eval`.
- **Regularizing is a bootstrap.** In §1.4.1 the step "$e \in E_m$ by Proposition 1.9" needs m applications, since Se is only one level above e `Chapter1/ScFredholm · ScPlus.regularizing_add`.
- **The index is an addition, and an easy one once additivity is there.** Neither Proposition 1.1.11 (preprint 1.10) nor the remark after Definition 1.1.9 (preprint 1.8) mentions the index of $T + S$ or of the level operators; with the additivity of the index `Chapter1/FredholmIndex · linIndex_comp` and a quasi-inverse, the index is locally constant along $u + tk$, hence invariant `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`, and the sc-statements follow on every level `Chapter1/FredholmIndex · proposition_1_10_index_lvl`.
- **The text's quick remarks are all true, and some need real arguments.** "Precisely one sc-structure" in finite dimensions `Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional`; "$(E/A)_\infty$ consists of the classes of E_∞", a Mittag-Leffler series `Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset`; "it is a sc-projection" `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`; "sc^1 is C^1 in finite dimensions", which at the boundary of a quadrant means derivatives within U `Chap-`

ter1/ExpositionLemmas · isSc1_iff_fderivWithin_const.

- **Mathlib had Fredholm operators but not Riesz–Schauder.** The theorem that $1 +$ compact is Fredholm, and the stability under compact perturbations, had to be proved Chapter1/FredholmPerturbation · isFredholm_one_add Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator; the finite codimension of the range comes from a chain of closed subspaces and Riesz's lemma Chapter1/FredholmPerturbation · coFG_range_one_add.
- **The degeneracy index knows nothing of the sc-structure.** Lemma 1.3.4 (preprint 1.24) holds for any linear equivalence of the base levels carrying one quadrant onto the other Chapter1/Degeneracy · quadrantDeg_eq_of_equiv, and $k = k'$ is its value at the origin Chapter1/Degeneracy · lemma_1_24_dim. Boundary recognition needs only sc^1 and only level-1 points, not smooth ones Chapter2/DegeneracyIndex · theorem_1_25.

I.21 The 2021 edition: Remark 1.1.2 and Proposition 1.1.11 with the index

The Springer edition (2021) changes Chapter 1 in two places. Remark 1.1.2 (p. 5) is new, and the statement of Proposition 1.1.11 (p. 9; preprint 1.10) now includes the index.

Remark 1.1.2. The remark makes "Banach space" mean a *Banachable* space, a vector space whose topology is induced by some complete norm, and says that all norms inducing the topology are equivalent "in view of the open mapping theorem". The formalization models one vector space with two norms as two normed spaces E_1, E_2 and a linear equivalence $e : E_1 \to E_2$, the identity of the underlying space. It records the remark's content in two readings. *(1) Same topology, equivalent norms* Chapter1/Book2021 · remark_1_1_2_2021. If e and e^{-1} are continuous, they are bounded, $\|ex\| \le \|f\|\,\|x\|$ and $\|x\| = \|g(ex)\| \le \|g\|\,\|ex\|$ for the continuous linear maps f, g they define, and $c = \max(\|f\|, \|g\|) + 1$ serves for both inequalities. No completeness enters. *(2) The open mapping theorem* Chapter1/Book2021 · remark_1_1_2_2021_openMapping. If both norms are complete and e is continuous, the bijective continuous linear map e is a linear homeomorphism by the open mapping theorem Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective. So e^{-1} is continuous, and (1) applies. As we read it, the sentence of the remark is true without the open mapping theorem. The theorem is what makes two complete norms equivalent when one of them dominates the other.

Proposition 1.1.11 (preprint 1.10), 2021 statement. Let $T : E \to F$ be sc-Fredholm and $S : E \to F$ an sc^+-operator. Then $T + S$ is sc-Fredholm, $\mathrm{ind}(T + S) = \mathrm{ind}(T)$, and every level map $(T + S)_m : E_m \to F_m$ is Fredholm of index $\mathrm{ind}(T)$ Chapter1/Book2021 · proposition_1_1_11_2021. Nothing new has to be proved, because the formal statement is the conjunction of three earlier theorems: - the sc-Fredholm property of §I.9 Chapter1/ScFredholm · proposition_1_10; - the index on level 0, from the invariance of the Fredholm index under compact perturbations Chapter1/FredholmIndex · proposition_1_10_index; - the index on every level Chapter1/FredholmIndex · proposition_1_10_index_lvl.

The 2021 proof in Appendix 1.4.1 is the earlier one and does not argue the index. The index statement, the exposition's addition in the earlier editions, is thus now the book's own. The level statement remains the formalization's.

Part II. Differential geometry based on retracts (HWZ Chapter 2, §§2.1, 2.3–2.7 (preprint §§2.1–2.6))

Chapter 2 builds a differential geometry whose local models are not open sets of Banach spaces but images of sc-smooth retractions, sets whose "dimension" may jump. The formal proof establishes the chapter in its order: retractions and retracts and the independence of their tangent from the retraction (§§II.2–II.4), M-polyfolds with their filtration, metrizability and tangent bundle (§§II.5–II.9), sub-M-polyfolds (§II.10), the degeneracy index with the boundary recognition argument (§§II.11–II.13), tame retracts and the faces of tame M-polyfolds (§§II.14–II.16), products and fibred products (§II.17), and strong bundles with their pull-backs (§§II.18–II.20). What is not formalized is the classical Proposition 2.1.2 (preprint 2.2) (motivation only), Lemma 2.4.9 (preprint 2.29) in its formulation by sc-smooth paths (its content, the sign arguments, is formalized and used instead), the second statement of Theorem 2.4.12 (preprint 2.31) in general (for tame retracts its conclusion, Proposition 2.5.6's (preprint 2.36) partial quadrant, is proved directly), the openness of $p : TX \to X_1$, and the categories $\mathcal{R}$, $\mathcal{BR}$, $\mathcal{SBR}$.

Three choices shape the formal route. First, a retraction is a total map $r : E_0 \to E_0$ with an explicit domain U, and the tangent TO of a retract is described as the *fixed-point set* of Tr in TU (§II.2); this description, which the text's proofs use without stating it, makes Propositions 2.1.11 and 2.1.13 (preprint 2.4 and 2.6) one-line consequences of the chain rule. Second, an M-polyfold is a space with *one* atlas, and every notion is defined through the maximal atlas of all charts compatible with it (§II.5); this needs the fact, not in the text, that compatibility is a local property, proved from the chain rule and the locality of sc-smoothness (§II.1). Third, where the text computes in the model $C = [0,\infty)^k \oplus W$, the formal statements carry a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant and its coordinate functionals $\lambda_i = (T\,\cdot)_i$; the subspaces E_x and cones C_x are defined through T, and are shown not to depend on it (§II.14).

Six proofs take a different road from the text's (or the exposition's). Theorem 2.3.10 (preprint 2.16) (metrizability of the levels) is proved without Lemma 2.7.3 (preprint 2.56) and Proposition 2.7.4 (preprint 2.57), which need a regularity of X_m the text does not check: a partition of unity of X pulled back to X_m gives an explicit metric, and the same argument proves Smirnov's theorem, which Mathlib lacks (§II.7). Proposition 2.4.10 (preprint 2.30) is proved from the sign arguments at points on level 1, not from Lemma 2.4.9 (preprint 2.29) at smooth points (§II.13). Theorem 2.4.12 (preprint 2.31) is proved for every chart of every M-polyfold by a direct isomorphism of quotients, without reducing to a chart that realizes the index (§II.13). Proposition 2.5.3 (preprint 2.35) uses only the second condition of tameness (§II.14). Proposition 2.5.6's (preprint 2.36) partial quadrant is proved for tame retracts without the equality case of Theorem 2.4.12 (preprint 2.31), the smooth vectors of its proof being written down explicitly (§II.14). The second sentence of Proposition 2.5.16 (preprint 2.44) is proved by removing all coordinates of σ at once rather than by the exposition's induction on $\#\sigma$ (§II.16).

Map.

- `Chapter2/Retracts`: restriction and locality of sc^k maps; retractions, retracts, Proposition 2.1.11 (preprint 2.4), Definitions 2.1.12, 2.1.14 (preprint 2.5, 2.7), Propositions 2.1.13, 2.3.1 (preprint 2.6, 2.10), Theorem 2.1.15 (preprint 2.8); local models.
- `Chapter2/TangentBundle`: the tangent of a retract as a retract, the tangent functor, locality of Tf; the tangent bundle TX, Proposition 2.3.15 (1)–(2) (preprint 2.19), TX as an M-polyfold.
- `Chapter2/MPolyfold`: charts, atlases, M-polyfolds, Definition 2.3.8 (preprint 2.14), the levels X_m, Lemma 2.3.9 (preprint 2.15), sub-M-polyfolds (Definition 2.3.18 (preprint 2.20)), a retract as an M-polyfold.
- `Chapter2/LevelPolyfold`: X^1 as an M-polyfold; the locality of Definition 2.3.8 (preprint 2.14).
- `Chapter2/Metrizable`: Smirnov's theorem, Theorem 2.3.10 (preprint 2.16), Proposition 2.3.15 (3) (preprint 2.19), Proposition 2.3.19 (3)–(4) (preprint 2.21).
- `Chapter2/TangentMap`: the tangent map $Tf : TX \to TY$ and the chain rule.
- `Chapter2/SubMPolyfold`: Proposition 2.3.19 (1)–(2) (preprint 2.21), Lemma 2.4.6 (preprint 2.26).
- `Chapter2/DegeneracyIndex`: E_x, the two-sided sign argument, the core of Theorem 2.4.12 (preprint 2.31), Theorem 1.3.5 (preprint 1.25), Definition 2.4.1 (preprint 2.22), Lemma 2.4.3 (preprint 2.23), Proposition 2.4.4 (preprint 2.24), Corollary 2.4.13 (preprint 2.32).
- `Chapter2/Boundary`: the boundary, the one-sided sign argument, Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31).
- `Chapter2/Tame`: the definitions of §2.5 (preprint §2.4) and of §2.6's (preprint §2.5) local objects.
- `Chapter2/TameRetracts`: Propositions 2.5.3, 2.5.6 (preprint 2.35, 2.36), the local geometry of faces, Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Propositions 2.5.14, 2.5.15 (preprint 2.42, 2.43) (the retraction onto a face).
- `Chapter2/TameFaces`: independence of the presentation; Proposition 2.5.15 (preprint 2.43) (faces are tame), Proposition 2.5.16 (preprint 2.44) (first sentence), Theorem 2.5.17 (preprint 2.45).
- `Chapter2/PartialCone`: Proposition 2.5.6 (preprint 2.36), $T^R_x O$ an sc-subspace and $C_x O$ a partial quadrant.
- `Chapter2/CornerGerms`: Proposition 2.5.16 (preprint 2.44) (second sentence), the corners f_σ.
- `Chapter2/FibredProduct`: products of M-polyfolds (the exposition's Lemma 4.D), Proposition 2.5.20 (preprint 2.46).
- `Chapter2/StrongBundle`, `Chapter2/StrongBundlePullback`: §2.6 (preprint §2.5).
- `Chapter2/StrongBundleOne`: the M-polyfold $Y[1]$, $P[1]$, the chart independence of $Y_{m,k}$, sc$^+$-sections into $Y[1]$.

II.1 Restriction and locality of sc^k maps

The text restricts sc-smooth maps to smaller sets ("$r : U \to U \cap U'$ is sc-smooth") and glues local sc-smoothness without comment. Both need an argument, because the tangent map of an sc^1 map is built from a *chosen* derivative family (Part I, §I.11), and a chosen derivative on U and one on $U' \subseteq U$ need not be the same objects.

Restriction. If f is sc^1 on U with derivative family D and $U' \subseteq U$, the same D works on U' `Chapter2/Retracts · IsSc1With.of_subset` `Chapter2/Retracts · IsSc1.of_subset`. When U' is an sc-domain (unique differentiability on every level, Part I, §I.10), derivatives at points of U'_1 are unique, so the tangent maps built on U' and on U agree on TU' `Chapter2/Retracts · IsSc1.tangentMap_eqOn_of_subset`. By induction on k, using that TU' is again an sc-domain and that the tangent maps agree there, an sc^k map restricts to sc^k on every smaller sc-domain `Chapter2/Retracts · IsSck.of_subset`, and so does an sc-smooth map `Chapter2/Retracts · IsScSmooth.of_subset`; sc-smoothness depends only on the values on U `Chapter2/Retracts · IsScSmooth.congr`. Relatively open subsets of quadrants, and their intersections with open sets, are sc-domains `Chapter2/Retracts · IsRelOpenInQuadrant.inter_isOpen` `Chapter2/Retracts · IsScDomain.inter_isOpen`, and $T(U \cap W) = TU \cap (W_1 \times E_0)$ `Chapter2/Retracts · ScBanach.tangentSet_inter_isOpen`.

Locality. sc^0 is local: continuity on each level is a local property `Chapter2/Retracts · IsSc0.of_locally`. For sc^1, suppose every $x \in U$ has an open $W_x \ni x$ with f sc^1 on $U \cap W_x$. Two such local structures give

the same derivative at a common point, by uniqueness on the sc-domain `Chapter2/Retracts · IsSc1With.eq_of_eq_near`. Glue: at a point $x \in U_1$ take the derivative of the structure on $U \cap W_{\iota_1 x}$. This family D agrees near every point with a local derivative family; the three conditions of Definition 1.1.15 (preprint 1.14) (the sc^0 property, differentiability on U_1, and the sc^0 property of the tangent map on TU) are local and hold `Chapter2/Retracts · IsSc1.of_locally`. The induction on k applies the same argument to the tangent maps on TU `Chapter2/Retracts · IsSck.of_locally`, and gives the sc-smooth case `Chapter2/Retracts · IsScSmooth.of_locally`. The identity is sc^1 with derivative the identity `Chapter2/Retracts · isSc1With_id`, its tangent map is the identity on TU `Chapter2/Retracts · tangentMap_id_eqOn`, and it is sc-smooth on every sc-domain `Chapter2/Retracts · isSck_id Chapter2/Retracts · isScSmooth_id`.

II.2 Retractions, retracts and their tangent (Definitions 2.1.1, 2.1.3, 2.1.12 (preprint 2.1, 2.3, 2.5); Proposition 2.1.11 (preprint 2.4))

Definition 2.1.1 (preprint 2.1). An *sc-smooth retraction* on a partial quadrant $C \subseteq E_0$ is a set U relatively open in C and a map $r : E_0 \to E_0$ with $r(U) \subseteq U$, r sc-smooth on U and $r(r(x)) = r(x)$ for $x \in U$ `Chapter2/Retracts · ScRetraction`. Since U is relatively open, it is an sc-domain and r is sc^1 `Chapter2/Retracts · ScRetraction.isScDomain Chapter2/Retracts · ScRetraction.isSc1`. The *retract* is $O = r(U)$ `Chapter2/Retracts · ScRetraction.retract`. It is contained in U and is exactly the fixed-point set of r in U: if $x = r(y)$ then $r(x) = r(r(y)) = x$ `Chapter2/Retracts · ScRetraction.retract_subset Chapter2/Retracts · ScRetraction.apply_eq_self_of_mem_retract Chapter2/Retracts · ScRetraction.mem_retract_iff`. On level 1, r_1 fixes the points of O_1 `Chapter2/Retracts · ScRetraction.levelMap_one_eq_self`, and $O_m \subseteq U_m$ `Chapter2/Retracts · ScRetraction.lvlSet_retract_subset`.

Definition 2.1.3 (preprint 2.3). (O, C, E) is an *sc-smooth retract* if O is the retract of some sc-smooth retraction on C `Chapter2/Retracts · IsScRetract Chapter2/Retracts · ScRetraction.isScRetract`. Every relatively open $U \subseteq C$ is one, with $r = \mathrm{id}$ `Chapter2/Retracts · ScRetraction.id Chapter2/Retracts · IsRelOpenInQuadrant.isScRetract`. A *local model* is a triple (E, C, O) with C a partial quadrant and O a retract in C `Chapter2/Retracts · LocalModel`; these are the models of M-polyfold charts.

The tangent of a retraction. Put $Tr(TU) := \{Tr(p) : p \in TU\} \subseteq E_1 \times E_0$ `Chapter2/Retracts · ScRetraction.tangent`. The chain rule applied to $r \circ r = r$ (Part I, Theorem 1.3.1 (preprint 1.21)) gives $Tr \circ Tr = Tr$ on TU `Chapter2/Retracts · ScRetraction.tangentMap_tangentMap Chapter1/ChainRule · theorem_1_21`, since Tr maps TU into TU `Chapter2/Retracts · ScRetraction.tangentMap_mapsTo`. Hence the **fixed-point description**: $p \in Tr(TU)$ iff $p \in TU$ and $Tr(p) = p$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff`. Base points of $Tr(TU)$ lie in O_1 `Chapter2/Retracts · ScRetraction.fst_mem_of_mem_tangent`. For $x \in O_1$ the fixed-point condition reads $Dr(x)h = h$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff_of_mem`, and the second component of $Tr \circ Tr = Tr$ at (x, h) gives $Dr(x) \circ Dr(x) = Dr(x)$ `Chapter2/Retracts · ScRetraction.scDeriv_comp_self` — the remark after Definition 2.1.3 (preprint 2.3) that $Dr(x)$ is a projection. So for $x \in O_1$, $(x, h) \in Tr(TU)$ iff h is in the range of $Dr(x)$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff_mem_range`, and $Tr(TU) = \{(r(x), Dr(x)h) : x \in U_1, h \in E_0\}$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff_exists`.

Proposition 2.1.11 (preprint 2.4). *If $r(U) \subseteq r'(U')$ then $r' \circ r = r$ on U*: for $y \in U$, $r(y) \in r'(U')$ is a fixed point of r' `Chapter2/Retracts · ScRetraction.comp_eq_self_of_retract_subset`. *And then $Tr(TU) \subseteq Tr'(TU')$*: let $p \in Tr(TU)$, so $p \in TU$ and $Tr(p) = p$. Its base point lies in $O_1 \subseteq U'_1$, so $p \in TU'$. The map r sends U into U' (into $O \subseteq U'$), so the chain rule applies to $r' \circ r$ on U: $T(r' \circ r)(p) = Tr'(Tr(p)) = Tr'(p)$; and $T(r' \circ r) = Tr$ on TU because $r' \circ r = r$ on U (tangent maps depend only on the values on U). So $Tr'(p) = Tr(p) = p$, which is the fixed-point condition for r' `Chapter2/Retracts · ScRetraction.tangent_subset_of_retract_subset`. Applied in both directions, two retractions with the same retract have the same tangent image `Chapter2/Retracts · proposition_2_4`. This is the text's proof, with its step "$Tr'(x, h) = (x, h)$" made the definition of membership.

Definition 2.1.12 (preprint 2.5). TO is $Tr(TU)$ for a retraction chosen (by choice) among those onto

O, and the empty set if O is not a retract `Chapter2/Retracts · retractTangent`; by Proposition 2.1.11 (preprint 2.4) it equals $Tr(TU)$ for *every* retraction onto O `Chapter2/Retracts · retractTangent_eq`. The tangent space at $x \in O_1$ is the fibre $T_xO = \{h : (x,h) \in TO\}$ `Chapter2/Retracts · tangentSpace`, the range of $Dr(x)$ `Chapter2/Retracts · tangentSpace_eq_range`, a linear subspace `Chapter2/Retracts · tangentSpace_eq_coe_submodule` (closed, as the range of a continuous projection).

II.3 Sc-smooth maps between retracts (Proposition 2.1.13 (preprint 2.6), Definition 2.1.14 (preprint 2.7), Theorem 2.1.15 (preprint 2.8), Proposition 2.3.1 (preprint 2.10))

Proposition 2.1.13 (preprint 2.6). Let r on U and s on V be retractions onto the same O. Since s takes values in O, which r fixes, $(f \circ r) \circ s = f \circ s$ on V `Chapter2/Retracts · ScRetraction.comp_comp_eqOn`. (1) If $f \circ r$ is sc-smooth on U, then $f \circ r \circ s$ is sc-smooth on V by the chain rule (s maps V into $O \subseteq U$), hence so is $f \circ s$ `Chapter2/Retracts · proposition_2_6_1` `Chapter1/ChainRule · IsScSmooth.comp`. (2) For $p \in TO$: $p \in TV$ and $Ts(p) = p$ (fixed-point description for s, after Proposition 2.1.11 (preprint 2.4)); so $T(f \circ r)(p) = T(f \circ r)(Ts(p)) = T((f \circ r) \circ s)(p) = T(f \circ s)(p)$ `Chapter2/Retracts · proposition_2_6_2`. (3) If f maps O into $O' = r'(U')$, then $r' \circ f \circ r = f \circ r$ on U; the chain rule gives $Tr'(T(f \circ r)(p)) = T(f \circ r)(p)$ for $p \in TU$, and $T(f \circ r)(p) \in TU'$, so $T(f \circ r)(p) \in TO'$ by the fixed-point description `Chapter2/Retracts · proposition_2_6_3`.

Definition 2.1.14 (preprint 2.7). f is *sc-smooth from* (O,C,E) *to* (O',C',E') if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth on U for every retraction r onto O `Chapter2/Retracts · IsScSmoothRetractMap`. For a retract O this is equivalent to "$f(O) \subseteq O'$ and $f \circ r$ sc-smooth for one r", by Proposition 2.1.13 (1) (preprint 2.6) `Chapter2/Retracts · isScSmoothRetractMap_iff_exists`. Quantifying over all retractions makes the definition independent of choices without further argument; the target triple enters only through $f(O) \subseteq O'$. The *tangent map* Tf is $T(f \circ r)$ for the chosen retraction `Chapter2/Retracts · retractTangentMap`; on TO it equals $T(f \circ r)$ for every r onto O, by Proposition 2.1.13 (2) (preprint 2.6) `Chapter2/Retracts · retractTangentMap_eqOn`, it maps TO into TO' by (3) `Chapter2/Retracts · IsScSmoothRetractMap.tangentMap_mapsTo`, and $Tf(x,h) = (f(x), D(f \circ r)(x)h)$ `Chapter2/Retracts · retractTangentMap_apply`. An sc-smooth map between retracts is continuous on O (level 0 of the sc^0 property of $f \circ r$, and $f = f \circ r$ on O) `Chapter2/Retracts · IsScSmoothRetractMap.continuousOn` and maps points of O on level m to points on level m `Chapter2/Retracts · IsScSmoothRetractMap.mem_range_toBase`.

Theorem 2.1.15 (preprint 2.8). Let $O = r(U)$, $O' = r'(U')$ be retracts, f sc-smooth from O to O' and g from O' to O''. Since $f \circ r$ takes values in O', which r' fixes, $(g \circ r') \circ (f \circ r) = (g \circ f) \circ r$ on U `Chapter2/Retracts · ScRetraction.comp_comp_comp_eqOn`. The left side is sc-smooth by the chain rule ($f \circ r$ maps U into $O' \subseteq U'$, and $g \circ r'$ is sc-smooth on U'), so $g \circ f$ is sc-smooth from O to O'' `Chapter2/Retracts · IsScSmoothRetractMap.comp`. For the tangent maps, at $p \in TO$: $T(g \circ f)(p) = T((g \circ f) \circ r)(p) = T(g \circ r')(T(f \circ r)(p)) = Tg(Tf(p))$, the last step because $Tf(p) \in TO'$ `Chapter2/Retracts · retractTangentMap_comp`. Together `Chapter2/Retracts · theorem_2_8`. The text gives no proof ("follows from Theorem 1.21"); this is the one it has in mind, with $r' \circ f \circ r = f \circ r$ made explicit. Further: the identity is sc-smooth `Chapter2/Retracts · isScSmoothRetractMap_id`, a retraction is sc-smooth as a map $(U,C,E) \to (O,C,E)$ and the inclusion $O \to U$ as a map $(O,C,E) \to (U,C,E)$ `Chapter2/Retracts · ScRetraction.isScSmoothRetractMap` `Chapter2/Retracts · ScRetraction.isScSmoothRetractMap_inclusion`, and sc-smoothness depends only on $f|_O$ `Chapter2/Retracts · IsScSmoothRetractMap.congr`.

Proposition 2.3.1 (preprint 2.10). (1) For $W \subseteq E_0$ open, $U \cap r^{-1}(W)$ is relatively open in C because r is continuous on U `Chapter2/Retracts · ScRetraction.relOpen_inter_preimage`; the restriction of r to it is a retraction `Chapter2/Retracts · ScRetraction.restrict` (sc-smooth by §II.1) with retract $O \cap W$ `Chapter2/Retracts · ScRetraction.restrict_retract`. So $(O \cap W, C, E)$ is a retract `Chapter2/Retracts · proposition_2_10_1` `Chapter2/Retracts · LocalModel.restrict`; "an open subset O' of O" is $O \cap W$ in the subspace topology. (2) If $V = r(U)$ and s is sc-smooth from (V,C,E) to itself with $s \circ s = s$ on V, then $\rho = s \circ r$ is sc-smooth

on U (that is what sc-smoothness of s on the retract V means), maps U into $s(V) \subseteq V \subseteq U$, and is idempotent: $\rho(\rho(x)) = s(r(s(r(x)))) = s(s(r(x))) = s(r(x))$ because r fixes $s(r(x)) \in V$. Its retract is $s(V)$ `Chapter2/Retracts · proposition_2_10_2`. As the text's proof.

Restriction and locality between retracts. An sc-smooth map between retracts restricts to $O \cap W$: a retraction onto $O \cap W$ composed with r is handled by the chain rule `Chapter2/Retracts · IsScSmoothRetractMap.restrict`; the target may be shrunk to any set containing the image `Chapter2/Retracts · IsScSmoothRetractMap.mono_target`. Conversely, if every point of O has an open W with f sc-smooth on $(O \cap W, C, E)$, then f is sc-smooth on (O, C, E): for a retraction r onto O and $y \in U$, the set W around $r(y)$ pulls back under the continuous r to an open $u \ni y$; on $U \cap u$ the map $f \circ r$ equals $(f \circ r') \circ r$ with r' the restriction of r onto $O \cap W$, which is sc-smooth, and the locality of §II.1 concludes `Chapter2/Retracts · IsScSmoothRetractMap.of_locally`. The text uses this silently every time it says "the definition does not depend on the choice of charts".

II.4 The tangent functor on retracts

The tangent of a partial quadrant. If $C = T_0^{-1}(\text{quadrant})$ for $T : E \cong \mathbb{R}^n \oplus W$, then $TC = C_1 \times E_0$ is a partial quadrant in $TE = E^1 \oplus E$: compose the tangent of T, an sc-isomorphism $TE \cong T(\mathbb{R}^n \oplus W)$, with the re-association $((a, w), (b, v)) \mapsto (a, (w, (b, v)))$ onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ `Chapter2/TangentBundle · tangentQuadrantIso` `Chapter2/TangentBundle · IsPartialQuadrant.tangentSet`. TU is relatively open in TC `Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet` (this is Lemma 3.B of the exposition of Chapter 1; Part I, V1.36). Sc-operators are sc-smooth on every sc-domain `Chapter2/TangentBundle · ScOperator.isScSmooth`.

Tr is a retraction. Tr is sc-smooth on TU (Part I: the tangent map of an sc-smooth map is sc-smooth) `Chapter2/Retracts · LocalModel.tangent_isScSmooth`, maps TU into itself and is idempotent; so it is an sc-smooth retraction on TC `Chapter2/TangentBundle · ScRetraction.tangentRetraction` with retract TO `Chapter2/TangentBundle · ScRetraction.tangentRetraction_retract`. Hence (TO, TC, TE) is a retract `Chapter2/TangentBundle · IsScRetract.tangentSet` `Chapter2/TangentBundle · LocalModel.tangent`, base points of TO lie in O_1 and $(x, 0) \in TO$ for $x \in O_1$ `Chapter2/TangentBundle · retractTangent_fst_mem` `Chapter2/TangentBundle · mk_zero_mem_retractTangent`.

Tf is sc-smooth. If f is sc-smooth from O to O' then Tf is sc-smooth from (TO, TC, TE) to (TO', TC', TE'): with the retraction Tr onto TO, $Tf \circ Tr = T(f \circ r) \circ Tr = T(f \circ r \circ r) = T(f \circ r)$ on TU, which is sc-smooth `Chapter2/TangentBundle · IsScSmoothRetractMap.tangent`. Tf depends only on $f|_O$ `Chapter2/TangentBundle · retractTangentMap_congr`, and a map equal to the identity on O has tangent map the identity on TO `Chapter2/TangentBundle · retractTangentMap_id`. With Theorem 2.1.15 (preprint 2.8) this is the tangent functor $T : \mathcal{R} \to \mathcal{R}$ of the text; the bundle structure of Definition 2.1.16 (preprint 2.9) is §II.32 (Tr a bundle retraction, Tf a bundle map), the category $\mathcal{BR}$ is not formalized.

Locality of the tangent. For W open, $T(O \cap W) = TO \cap (W_1 \times E_0)$: with the restricted retraction, a fixed point of Tr over a base point in W_1 is a fixed point of the restricted Tr, the two tangent maps agreeing on the smaller domain (§II.1) `Chapter2/TangentBundle · retractTangent_inter_isOpen` `Chapter2/TangentBundle · retractTangent_mono_inter`. And Tf computed on $(O \cap W, C, E)$ agrees with Tf computed on (O, C, E) on $T(O \cap W)$ `Chapter2/TangentBundle · retractTangentMap_inter_isOpen`. These two facts are what Definitions 2.3.12–2.3.14 (preprint 2.17–2.18) use silently when the chain rule is applied on a triple overlap.

II.5 Charts, atlases and M-polyfolds (Definitions 2.3.2–2.3.8 (preprint 2.11–2.14))

Definition 2.3.2 (preprint 2.11). A *chart* on X is a local model (O, C, E) and a partial equivalence φ between X and E_0 with open source V and target O, continuous on V, with inverse continuous on O `Chapter2/MPolyfold · MChart`. Images of open subsets of V are open subsets $O \cap W$ of

O Chapter2/MPolyfold · MChart.exists_isOpen_image_eq, hence retracts (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/MPolyfold · MChart.isScRetract_image, and charts restrict to open subsets of their sources Chapter2/MPolyfold · MChart.restrict. The overlap $\varphi(V \cap V')$ Chapter2/MPolyfold · MChart.overlap is a retract Chapter2/MPolyfold · MChart.isScRetract_overlap, the transition map is $\psi \circ \varphi^{-1}$ Chapter2/MPolyfold · MChart.transition, mapping $\varphi(V \cap V')$ into $\psi(V \cap V')$ Chapter2/MPolyfold · MChart.transition_mapsTo, and two charts are *compatible* if both transition maps are sc-smooth between the overlap retracts Chapter2/MPolyfold · MChart.Compatible. Compatibility is symmetric, reflexive, and passes to restrictions Chapter2/MPolyfold · MChart.Compatible.symm Chapter2/MPolyfold · MChart.compatible_self Chapter2/MPolyfold · MChart.Compatible.restrict.

Compatibility is local. Let φ, ψ be charts and S a family of charts each compatible with φ and with ψ, whose sources cover $V \cap V'$. Then φ and ψ are compatible Chapter2/MPolyfold · MChart.compatible_of_locally. By the locality between retracts (§II.3) it suffices to show that $\psi \circ \varphi^{-1}$ is sc-smooth near every point $\varphi(x)$ of the overlap. Take $\chi \in S$ around x and the triple overlap $s = V \cap V' \cap V''$, with $\varphi(s) = \varphi(V \cap V') \cap W$ for an open W. On $\varphi(s)$, $\psi \circ \varphi^{-1} = (\psi \circ \chi^{-1}) \circ (\chi \circ \varphi^{-1})$ Chapter2/MPolyfold · MChart.transition_eq_comp_on; the two factors are sc-smooth on the restricted retracts $\varphi(s)$ and $\chi(s)$, and Theorem 2.1.15 (preprint 2.8) composes them. The reverse transition is the same argument with the roles exchanged. This is the step that makes "the equivalence class of atlases" a usable notion; the text does not discuss it.

Definition 2.3.3 (preprint 2.12). An *atlas* is a set of pairwise compatible charts covering X Chapter2/MPolyfold · MAtlas; two atlases are *equivalent* if their union is an atlas Chapter2/MPolyfold · MAtlas.Equiv, i.e. if every chart of one is compatible with every chart of the other Chapter2/MPolyfold · MAtlas.equiv_iff. The relation is reflexive and symmetric Chapter2/MPolyfold · MAtlas.Equiv.refl Chapter2/MPolyfold · MAtlas.Equiv.symm.

Definition 2.3.4 (preprint 2.13). An *M-polyfold* structure on a Hausdorff paracompact space is an atlas Chapter2/MPolyfold · MPolyfold. Its *charts* are all charts compatible with every chart of the atlas Chapter2/MPolyfold · MPolyfold.IsChart Chapter2/MPolyfold · MPolyfold.isChart_of_mem. By locality of compatibility (with S the atlas), any two of them are compatible Chapter2/MPolyfold · MPolyfold.IsChart.compatible; restrictions of charts are charts Chapter2/MPolyfold · MPolyfold.IsChart.restrict; every point lies in one Chapter2/MPolyfold · MPolyfold.exists_isChart_mem_source; and equivalent atlases have the same charts, again by locality Chapter2/MPolyfold · MPolyfold.isChart_congr. So the maximal atlas represents the equivalence class, and every later definition (levels, index, tangent bundle) quantifies over it.

Definition 2.3.8 (preprint 2.14). $f : X \to Y$ is *sc-smooth on the open set* U if it is continuous on U and for all charts φ of X, ψ of Y and every open $V \subseteq U \cap \operatorname{dom}\varphi$ with $f(V) \subseteq \operatorname{dom}\psi$, the representation $\psi \circ f \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V), C, E)$ into (O', C', E') Chapter2/MPolyfold · MPolyfold.IsScSmoothOn; *sc-smooth* means sc-smooth on X Chapter2/MPolyfold · MPolyfold.IsScSmoothMap. Continuity is a field: the text's pairs of charts with $f(V) \subseteq V'$ exist around every point only for continuous f. The notion passes to open subsets and to maps agreeing on U Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.mono Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.congr. *One pair of charts suffices*: if every $x \in U$ has charts φ, ψ and an open $V \ni x$ with $\psi \circ f \circ \varphi^{-1}$ sc-smooth on $\varphi(V)$, then f is sc-smooth on U Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts. For arbitrary charts φ', ψ' and V', locality reduces to a neighbourhood $\varphi'(V \cap V')$ of each point, where $\psi' \circ f \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$, three sc-smooth maps between retracts composed by Theorem 2.1.15 (preprint 2.8). This is the text's "the definition does not depend on the choice of charts". The identity is sc-smooth Chapter2/MPolyfold · MPolyfold.isScSmoothMap_id, and compositions are sc-smooth by the same three-factor argument with a chart of the middle space Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.comp Chapter2/MPolyfold · MPolyfold.IsScSmoothMap.comp.

II.6 The filtration X_m and the M-polyfold X^1 (Lemma 2.3.9 (preprint 2.15))

Levels. x is on level m if $\varphi(x) \in \iota_m(E_m)$ for some chart φ around x Chapter2/MPolyfold · MPolyfold.levelSet. *Independence of the chart*: if ψ is another chart around x, the transition map $\varphi \circ \psi^{-1}$ is sc-smooth between the overlap retracts, hence maps points on level m to points on level m (§II.3), and

$\varphi(x) = (\varphi \circ \psi^{-1})(\psi(x))$ Chapter2/MPolyfold · MPolyfold.mem_levelSet_iff. $X_0 = X$, $X_{m+1} \subseteq X_m$, and $X_\infty = \bigcap_m X_m$ Chapter2/MPolyfold · MPolyfold.levelSet_zero Chapter2/MPolyfold · MPolyfold.levelSet_succ_subset Chapter2/MPolyfold · MPolyfold.smoothSet. For $x \in V_m$ the lift of $\varphi(x)$ to E_m Chapter2/MPolyfold · MPolyfold.levelLift satisfies $\iota_m(\text{lift}) = \varphi(x)$ and is compatible with the inclusions Chapter2/MPolyfold · MPolyfold.toBase_levelLift Chapter2/MPolyfold · MPolyfold.levelLift_eq_incl. X_m carries the topology generated by the sets $\{x \in V : \text{lift}(x) \in W\}$, W open in E_m Chapter2/MPolyfold · MPolyfold.levelBasis Chapter2/MPolyfold · MPolyfold.levelTopology; that this family is a basis is not needed.

Lemma 2.3.9 (preprint 2.15). The preimage under $X_{m+1} \to X_m$ Chapter2/MPolyfold · MPolyfold.levelIncl of a generating set $\{\text{lift}_m \in W\}$ is the generating set $\{\text{lift}_{m+1} \in \text{incl}_m^{-1}(W)\}$, by the compatibility of lifts with inclusions; so the inclusion is continuous Chapter2/MPolyfold · lemma_2_15. This is the text's proof.

The space X_m. As a type, X_m is the subtype of points on level m with this topology Chapter2/LevelPolyfold · MPolyfold.Level. The inclusion $X_m \to X$ is continuous: for V open in X and $x \in X_m$, a chart φ around x gives $\varphi(V \cap \operatorname{dom}\varphi) = O \cap W$, and the generating set $\{\text{lift}_m \in W_m\}$ is a neighbourhood of x inside the preimage of V Chapter2/LevelPolyfold · MPolyfold.continuous_levelVal Chapter2/LevelPolyfold · MPolyfold.isOpen_level_basic. Since this inclusion is injective and continuous into the Hausdorff X, X_m is Hausdorff (an instance in `LevelPolyfold`, the text's argument with disjoint neighbourhoods).

The level shift of the local data. An sc-isomorphism shifts to $E^1 \to F^1$ Chapter2/LevelPolyfold · ScIso.shift, so C_1 is a partial quadrant in E^1 and U_1 relatively open in it Chapter2/LevelPolyfold · IsPartialQuadrant.shift Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift. A retraction r induces the retraction r_1 on U_1 — sc-smooth by Proposition 1.2.2 (preprint 1.17) (Part I, §I.15), idempotent by the injectivity of ι_1 Chapter2/LevelPolyfold · ScRetraction.shift — with retract O_1 Chapter2/LevelPolyfold · ScRetraction.shift_retract Chapter2/LevelPolyfold · IsScRetract.shift Chapter2/LevelPolyfold · LocalModel.shift. An sc-smooth map between retracts is sc^0 on O, maps O_m into O'_m and is continuous there Chapter2/LevelPolyfold · IsScSmoothRetractMap.isSc0 Chapter2/LevelPolyfold · IsScSmoothRetractMap.mapsTo_levelMap Chapter2/LevelPolyfold · IsScSmoothRetractMap.continuousOn_levelMap, and its level-1 map is sc-smooth from O_1 to O'_1: with the retraction r_1, $f_1 \circ r_1 = (f \circ r)_1$, sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift.

The M-polyfold X^1. For a chart φ of X, $x \mapsto \text{lift}_1(x)$ on V_1 with model (O_1, C_1, E^1) is a chart of X_1 Chapter2/LevelPolyfold · MPolyfold.levelChart (with a junk value for the inverse off O_1, since X_1 might be empty Chapter2/LevelPolyfold · MPolyfold.levelChartInv); its overlap with another level chart is the level-1 part of the overlap Chapter2/LevelPolyfold · MPolyfold.levelChart_overlap, its transition map the level-1 map of the transition map Chapter2/LevelPolyfold · MPolyfold.levelChart_transition, and so any two level charts are compatible by the level shift of sc-smooth maps Chapter2/LevelPolyfold · MPolyfold.levelChart_compatible. They form an atlas Chapter2/LevelPolyfold · MPolyfold.levelAtlas, and X_1 with it is the M-polyfold X^1 Chapter2/LevelPolyfold · MPolyfold.shift Chapter2/LevelPolyfold · MPolyfold.shift_isChart. Its paracompactness, an instance argument, is supplied by Theorem 2.3.10 (preprint 2.16) (§II.7). Only $m = 1$ is constructed; the levels of X^1 are those of X with their topology and charts, so the iterates are the X_m (§II.32; gap G.6, closed).

II.7 Metrizability: Smirnov's theorem and Theorem 2.3.10 (preprint 2.16) (Theorem 2.7.1 (preprint 2.55))

The text proves Theorem 2.3.10 (preprint 2.16) with Smirnov's theorem (2.55), Michael's characterization of paracompactness (Lemma 2.7.3 (preprint 2.56)) and Proposition 2.7.4 (preprint 2.57) (a regular space with a locally finite closed cover by paracompact subspaces is paracompact). Proposition 2.7.4 (preprint 2.57) needs X_m to be regular, which the text does not check. The formal proof needs neither 2.56 nor 2.57.

The metrization lemma. Let Y have open subspaces V_i, each metrizable with a metric d_i, and continuous $\rho_i : Y \to [0,\infty)$ with $\operatorname{supp}\rho_i \subseteq V_i$, the supports locally finite and $\rho_i(y) > 0$ for some i at every y `Chapter2/Metrizable · metrizableSpace_of_partition`. Put $\delta_i(x,y) := \min(d_i(x,y),1)$ for $x,y \in V_i$ and 1 otherwise `Chapter2/Metrizable · truncDist` — nonnegative, at most 1, symmetric, zero only on the diagonal, and satisfying the triangle inequality inside V_i `Chapter2/Metrizable · truncDist_nonneg` `Chapter2/Metrizable · truncDist_le_one` `Chapter2/Metrizable · truncDist_triangle` `Chapter2/Metrizable · truncDist_eq_zero` — and the *cone metric*

$$e_i(x,y) := |\rho_i(x) - \rho_i(y)| + \min(\rho_i(x),\rho_i(y))\,\delta_i(x,y), \qquad D(x,y) := \sum_i e_i(x,y).$$

The sum is finite (only the i with x or y in $\operatorname{supp}\rho_i$ contribute). $D(x,x) = 0$ and D is symmetric. *Triangle inequality:* termwise, by an elementary inequality for $a,b,c \geq 0$ and δ's bounded by 1, which needs the triangle inequality for δ_i only when $\min(\rho_i(x),\rho_i(z)) > 0$ and $\rho_i(y) > 0$, i.e. when all three points lie in V_i `Chapter2/Metrizable · cone_triangle`. *Separation:* if $D(x,y) = 0$, take i with $\rho_i(x) > 0$; then $\rho_i(y) = \rho_i(x)$ and $\delta_i(x,y) = 0$, so $x = y$. *Topology:* $y \mapsto D(x,y)$ is continuous (locally a finite sum of continuous functions; the term $\min(\rho_i(x),\rho_i(y))\delta_i(x,y)$ is continuous where $y \in V_i$ and vanishes near points outside $\operatorname{supp}\rho_i$), so D-balls are open; conversely, if s is open and $x \in s$, take i with $\rho_i(x) > 0$ and a d_i-ball of radius $r \leq 1$ around x inside s; then $D(x,y) < \frac{1}{2}\rho_i(x)r$ forces $\rho_i(y) > \frac{1}{2}\rho_i(x)$ (so $y \in V_i$) and $\frac{1}{2}\rho_i(x)\delta_i(x,y) < \frac{1}{2}\rho_i(x)r$, so $d_i(x,y) < r$ and $y \in s$. Mathlib then builds a metric space structure with this distance and the given topology `Mathlib/Topology/MetricSpace/Defs · MetricSpace.ofDistTopology`.

Pulled back, and Smirnov. If $\pi : Y \to X$ is continuous into a paracompact Hausdorff X and every point of X has an open U with $\pi^{-1}(U)$ metrizable, then Y is metrizable: a partition of unity of X subordinate to these U `Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate`, composed with π, satisfies the hypotheses of the lemma with $V_i = \pi^{-1}(U_i)$ `Chapter2/Metrizable · metrizableSpace_of_pullback`. With $\pi = \mathrm{id}$ this is **Smirnov's theorem** (Theorem 2.7.1 (preprint 2.55)) `Chapter2/Metrizable · metrizableSpace_of_locally`. Metrizable spaces are paracompact (Stone; in Mathlib through the instance for pseudo-emetric spaces) `Chapter2/Metrizable · paracompactSpace_of_metrizableSpace`.

Theorem 2.3.10 (preprint 2.16). *X is metrizable*: chart domains are homeomorphic to subsets O of the Banach space E_0 `Chapter2/Metrizable · MChart.sourceHomeomorph` `Chapter2/Metrizable · MChart.metrizableSpace_source`, and Smirnov applies `Chapter2/Metrizable · MPolyfold.metrizableSpace`. $V_m \cong O_m$: the lift $x \mapsto \mathrm{lift}_m(x)$ is a topological embedding of $V \cap X_m$ into E_m `Chapter2/Metrizable · MPolyfold.isEmbedding_levelLift`. Its injectivity is that of φ; the induced topology is finer than that of X_m because generating sets of X_m inside V are preimages of open sets; it is coarser because a generating set $\{\mathrm{lift}_m^{\psi} \in W\}$ of another chart, intersected with V, is the preimage under $\mathrm{lift}_m^{\varphi}$ of the open set $\{z \in \varphi(V \cap V')_m : (\psi \circ \varphi^{-1})_m(z) \in W\}$, open since the level-$m$ map of the transition is continuous on the level-m part of the overlap retract. *X_m is metrizable*: apply the pulled-back Smirnov to the continuous inclusion $X_m \to X$ and the chart domains V, whose preimages $V \cap X_m$ embed in E_m `Chapter2/Metrizable · MPolyfold.metrizableSpace_level`; hence paracompact `Chapter2/Metrizable · MPolyfold.paracompactSpace_level`. X_∞: the text does not define its topology; it is taken to be the initial topology of the inclusions $X_\infty \to X_m$ `Chapter2/Metrizable · MPolyfold.SmoothLevel` `Chapter2/Metrizable · MPolyfold.smoothIncl` `Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq`. It embeds in the countable product $\prod_m X_m$, which is metrizable `Chapter2/Metrizable · metrizableSpace_pi_nat`, so X_∞ is metrizable `Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel`. Together `Chapter2/Metrizable · theorem_2_16`. *No hypothesis is added.* As a by-product every subset of an M-polyfold is metrizable, hence paracompact `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`, which discharges the paracompactness of sub-M-polyfolds (§II.10).

II.8 The tangent bundle TX (Definitions 2.3.12–2.3.14 (preprint 2.17–2.18), Proposition 2.3.15 (preprint 2.19))

Tuples and transport. A tangent tuple is (x, φ, h) with $x \in X_1$, φ a chart of the structure around x and $(\mathrm{lift}_1 x, h) \in TO$ `Chapter2/TangentBundle · MPolyfold.TangentPre`; its vector is $(\mathrm{lift}_1 x, h)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.vec` `Chapter2/TangentBundle · MPolyfold.TangentPre.vec_mem`. For a vector of TO over a point of an open $\varphi(s)$, membership in TO and in $T(\varphi(s))$ agree (locality of the tangent, §II.4) `Chapter2/TangentBundle · MPolyfold.TangentPre.mem_retractTangent_image_iff`. The transport of the tuple to a chart ψ around x is $T(\psi \circ \varphi^{-1})(\mathrm{vec})$, the tangent map computed on the overlap retract `Chapter2/TangentBundle · MPolyfold.TangentPre.transport`. Transport to φ itself does nothing, by $T(\mathrm{id}) = \mathrm{id}$ on the overlap $\varphi(V)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_self` `Chapter2/TangentBundle · MChart.transition_self_eqOn`; the transported vector lies in TO' `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_mem` and has base point $\mathrm{lift}_1^{\psi}(x)$ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_fst`.

The chain rule for transports. Transporting to ψ and then by $T(\chi \circ \psi^{-1})$ is transporting to χ `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport`. On the triple overlap $s = V \cap V' \cap V''$ the transition maps are sc-smooth from $\varphi(s)$ to $\psi(s)$ and from $\psi(s)$ to $\chi(s)$ `Chapter2/TangentBundle · MChart.Compatible.transition_image` `Chapter2/TangentBundle · MChart.image_eq_overlap_inter`, $\chi \circ \varphi^{-1} = (\chi \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1})$ there, and Theorem 2.1.15 (preprint 2.8) gives $T(\chi \circ \varphi^{-1}) = T(\chi \circ \psi^{-1}) \circ T(\psi \circ \varphi^{-1})$ on $T(\varphi(s))$. The three tangent maps here are computed on $\varphi(s)$ and $\psi(s)$, the transports on the full overlaps; the locality of Tf (§II.4) identifies them. The text never checks this.

Definition 2.3.12 (preprint 2.17). Two tuples are *equivalent* if their points agree and the transport of the first to the chart of the second is the second `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel`. Equivalent tuples transport to the same vector in every chart (chain rule for transports) `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.transport`; hence the relation is reflexive, symmetric and transitive `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.refl` `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.symm` `Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.trans`, an equivalence relation `Chapter2/TangentBundle · MPolyfold.TangentPre.setoid`.

Definition 2.3.14 (preprint 2.18). TX is the quotient `Chapter2/TangentBundle · MPolyfold.TangentSpace` `Chapter2/TangentBundle · MPolyfold.TangentSpace.mk`, with $p : TX \to X^1$, $[(x, \varphi, h)] \mapsto x$ `Chapter2/TangentBundle · MPolyfold.tangentProj`, and $T_x X = p^{-1}(x)$ `Chapter2/TangentBundle · MPolyfold.tangentFiber`. For a chart φ: $TV = p^{-1}(V_1)$ `Chapter2/TangentBundle · MPolyfold.tangentChartSet` and $T\varphi : TV \to TO$ sends a class to the transport of a representative to φ, well defined by the previous paragraph `Chapter2/TangentBundle · MPolyfold.tangentChartMap` `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk`; $T\varphi[(x, \varphi, h)] = (\varphi(x), h)$ `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_self`, $T\varphi$ lands in TO over the lift of the base point `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mem` `Chapter2/TangentBundle · MPolyfold.tangentChartMap_fst`, and it is injective on TV: equal images have equal base points (injectivity of φ and ι_1) and then the representatives are equivalent `Chapter2/TangentBundle · MPolyfold.tangentChartMap_injOn`. Its inverse on TO is $a \mapsto [(\varphi^{-1}(\iota_1 a_1), \varphi, a_2)]$ `Chapter2/TangentBundle · MPolyfold.TangentPre.ofVec` `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec` `Chapter2/TangentBundle · MPolyfold.mk_ofVec_tangentChartMap`, and in another chart $T\psi \circ (T\varphi)^{-1} = T(\psi \circ \varphi^{-1})$ on $T(\varphi(V \cap V'))$ `Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec_of_mem` `Chapter2/TangentBundle · MPolyfold.mk_ofVec_mem_tangentChartSet_iff`. The topology is generated by the sets $\{\alpha \in TV : T\varphi(\alpha) \in W\}$, W open `Chapter2/TangentBundle · MPolyfold.tangentBasis` `Chapter2/TangentBundle · MPolyfold.isOpen_tangent_basic`; TV itself is open `Chapter2/TangentBundle · MPolyfold.isOpen_tangentChartSet`. The vector space structure of $T_x X$ is not formalized; $T_x X$ is identified with $T_{\varphi(x)} O$ by $T\varphi$.

Proposition 2.3.15 (preprint 2.19). (2) *p is continuous*: the preimage of a generating set $\{\mathrm{lift}_1^{\psi} \in W\}$ of X_1 is the generating set of TX given by ψ and $W \times E_0$ `Chapter2/TangentBundle · MPolyfold.continuous_tangentProj`; so is $TX \to X$ `Chapter2/TangentBundle · MPolyfold.continuous_levelVal_tangentProj`. Openness of p is not proved. (1) *TX is Hausdorff* (the instance `T2Space` in `TangentBundle`): two classes over the same point lie in one TV and have distinct images under the injective $T\varphi$, separated by open

sets of $E_1 \times E_0$ whose preimages are basic; classes over distinct points are separated by preimages of disjoint open sets of X under the continuous $TX \to X$. (3) TX *is metrizable*: apply the pulled-back Smirnov theorem to $TX \to X$ and the chart domains V, whose preimages TV are homeomorphic to subsets of $E_1 \times E_0$ through the charts of §II.8's next paragraph `Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace` `Chapter2/Metrizable · MChart.metrizableSpace_source`; hence paracompact `Chapter2/Metrizable · MPolyfold.paracompactSpace_tangentSpace` `Chapter2/Metrizable · proposition_2_19_3`.

TX as an M-polyfold. For a chart φ of X, $(TV, T\varphi, (TO, TC, TE))$ is a chart of TX `Chapter2/TangentBundle · MPolyfold.tangentChart` `Chapter2/TangentBundle · MPolyfold.tangentChartInv`; its overlap with another is $T(\varphi(V \cap V'))$ `Chapter2/TangentBundle · MPolyfold.tangentChart_overlap` and its transition map is $T(\psi \circ \varphi^{-1})$ — formula (5) `Chapter2/TangentBundle · MPolyfold.tangentChart_transition` — which is sc-smooth between the tangent retracts by the tangent functor (§II.4); so the charts are compatible `Chapter2/TangentBundle · MPolyfold.tangentChart_compatible`, form an atlas `Chapter2/TangentBundle · MPolyfold.tangentAtlas` and make TX an M-polyfold `Chapter2/TangentBundle · MPolyfold.tangentPolyfold` `Chapter2/TangentBundle · MPolyfold.tangentPolyfold_isChart`. $p : TX \to X^1$ *is sc-smooth*: in the charts $T\varphi$ and the level chart of φ it is the projection $TO \to O_1$, $(a, h) \mapsto a$, which is sc-smooth between the retracts (TO, TC, TE) and (O_1, C_1, E^1) because, with the retraction Tr onto TO, $\mathrm{pr}_1 \circ Tr = r_1 \circ \mathrm{pr}_1$ on TU, a composite of the sc-operator pr_1 `Chapter2/TangentBundle · ScBanach.tangentFst` and the sc-smooth r_1 `Chapter2/TangentBundle · MPolyfold.isScSmoothRetractMap_tangent_fst`; one pair of charts per point suffices (§II.5) `Chapter2/TangentBundle · MPolyfold.tangentProj_isScSmooth`.

II.9 The tangent map of an sc-smooth map between M-polyfolds

The text uses Tf for maps between M-polyfolds from §2.4 (preprint §2.3) on without defining it. *The chain rule on chart images, with locality*: for F sc-smooth on $\varphi(s_1)$ into a retract O_1 mapping $\varphi(t)$ into O_1', G sc-smooth on O_1' and H sc-smooth on $\varphi(s_2)$ with $H = G \circ F$ on $\varphi(t)$ ($t \subseteq s_1 \cap s_2$ open), $TH = TG \circ TF$ on $T(\varphi(t))$, the three tangent maps computed on their own domains `Chapter2/TangentMap · MChart.retractTangentMap_comp_image` — Theorem 2.1.15 (preprint 2.8) on $\varphi(t)$ plus the locality of Tf. *Levels*: an sc-smooth map maps X_m into Y_m, because its coordinate representations do `Chapter2/TangentMap · MPolyfold.IsScSmoothOn.mapsTo_levelSet` `Chapter2/TangentMap · MPolyfold.IsScSmoothMap.mapsTo_levelSet`. *The image vector*: for a tuple (x, φ, h) and a chart ψ around $f(x)$, $T(\psi \circ f \circ \varphi^{-1})(\mathrm{vec})$, computed on $\varphi(V \cap f^{-1}(V'))$ `Chapter2/TangentMap · MPolyfold.TangentPre.fdom` `Chapter2/TangentMap · MPolyfold.TangentPre.ftransport`, lies in TO' over $\psi(f(x))$ `Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_mem` `Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_fst`; changing ψ changes it by T of the transition map `Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_transport`, changing φ by transport `Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_eq_of_transport`, and equivalent tuples give the same image `Chapter2/TangentMap · MPolyfold.TangentPre.Rel.ftransport` — all by the chain rule with locality. The image tuple $(f(x), \psi, \cdot)$ `Chapter2/TangentMap · MPolyfold.TangentPre.fimageWith` therefore has a class independent of ψ and of the representative `Chapter2/TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq` `Chapter2/TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq_of_rel`.

Tf. With a chosen ψ, $Tf[(x, \varphi, h)]$ is the class of the image tuple `Chapter2/TangentMap · MPolyfold.tangentMap`, computable in any chart around $f(x)$ `Chapter2/TangentMap · MPolyfold.tangentMap_mk`. It covers f `Chapter2/TangentMap · MPolyfold.levelVal_tangentProj_tangentMap`, maps T_xX into $T_{f(x)}Y$ `Chapter2/TangentMap · MPolyfold.tangentMap_mapsTo_fiber`, and in charts $T\psi \circ Tf = T(\psi \circ f \circ \varphi^{-1}) \circ T\varphi$ `Chapter2/TangentMap · MPolyfold.tangentChartMap_tangentMap`. $T(\mathrm{id}) = \mathrm{id}$ `Chapter2/TangentMap · MPolyfold.tangentMap_id`, and **the chain rule** $T(g \circ f) = Tg \circ Tf$ holds: with charts ψ around $f(x)$ and χ around $g(f(x))$ and the open set $t = \mathrm{fdom}(\varphi, f, \psi) \cap f^{-1}(\mathrm{fdom}(\psi, g, \chi))$, the chain rule with locality gives $T(\chi \circ g \circ f \circ \varphi^{-1}) = T(\chi \circ g \circ \psi^{-1}) \circ T(\psi \circ f \circ \varphi^{-1})$ on $T(\varphi(t))$ `Chapter2/TangentMap · MPolyfold.tangentMap_comp`. Tf is continuous: near a class in TV, the preimage of a generating set of TY contains the generating set of TX given by φ and the preimage under the continuous $T(\psi \circ f \circ \varphi^{-1})$ `Chapter2/TangentMap · MPolyfold.continuous_tangentMap`; and Tf is sc-smooth $TX \to TY$, its representation in the charts $T\varphi$, $T\psi$ being $T(\psi \circ f \circ \varphi^{-1})$, sc-smooth by the tangent functor `Chapter2/TangentMap · MPolyfold.tangentMap_isScSmooth`.

II.10 Sub-M-polyfolds (Definition 2.3.18 (preprint 2.20), Proposition 2.3.19 (preprint 2.21))

Definition 2.3.18 (preprint 2.20). $A \subseteq X$ is a *sub-M-polyfold* if every $a \in A$ has an open $V \ni a$ and a map r with $r(V) \subseteq V$, sc-smooth on V, idempotent on V, with $r(V) = A \cap V$ `Chapter2/MPolyfold · MPolyfold.IsSubMPolyfold`; X is one `Chapter2/MPolyfold · MPolyfold.isSubMPolyfold_univ`. The data at a point are packaged `Chapter2/SubMPolyfold · MPolyfold.SubData` `Chapter2/SubMPolyfold · MPolyfold.IsSubMPolyfold.data`: r maps V into A, fixes $A \cap V$, and is continuous on V `Chapter2/SubMPolyfold · MPolyfold.SubData.apply_mem` `Chapter2/SubMPolyfold · MPolyfold.SubData.apply_eq_self` `Chapter2/SubMPolyfold · MPolyfold.SubData.continuousOn`.

Proposition 2.3.19 (preprint 2.21), the structure. *Shrinking.* For a chart φ, put $W := (V \cap \operatorname{dom}\varphi) \cap r^{-1}(V \cap \operatorname{dom}\varphi)$ `Chapter2/SubMPolyfold · MPolyfold.SubData.shrink`: open (continuity of r) `Chapter2/SubMPolyfold · MPolyfold.SubData.isOpen_shrink`, containing every $a \in A \cap V \cap \operatorname{dom}\varphi$ `Chapter2/SubMPolyfold · MPolyfold.SubData.mem_shrink`, with $r(W) \subseteq W$ and $r(W) = W \cap A$ `Chapter2/SubMPolyfold · MPolyfold.SubData.mapsTo_shrink` `Chapter2/SubMPolyfold · MPolyfold.SubData.image_shrink`. *The retract.* $\rho := \varphi \circ r \circ \varphi^{-1}$ is sc-smooth from $\varphi(W)$ to itself — Definition 2.3.8 (preprint 2.14) for r in the charts φ, φ — idempotent, and $\rho(\varphi(W)) = \varphi(W \cap A)$ `Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth` `Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem` `Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image`; so $\varphi(W \cap A)$ is a retract by Proposition 2.3.1 (2) (preprint 2.10) `Chapter2/SubMPolyfold · MPolyfold.SubData.isScRetract_image`. *The chart.* With the subspace topology on A, φ restricted to $W \cap A$ is a chart of A with model $(\varphi(W \cap A), C, E)$ `Chapter2/SubMPolyfold · MPolyfold.subChart` `Chapter2/SubMPolyfold · MPolyfold.subChartInv`. *Compatibility.* The overlap of two such charts is $\varphi((W \cap W') \cap A)$, a subset of $\varphi(V \cap V')$ `Chapter2/SubMPolyfold · MPolyfold.subChart_overlap`. An sc-smooth map between retracts is sc-smooth on every sub-retract $O_1 \subseteq O$: for a retraction s onto O_1, $f \circ s = (f \circ r) \circ s$, sc-smooth by the chain rule `Chapter2/SubMPolyfold · IsScSmoothRetractMap.mono_source` — this is the text's "$(\psi' \circ \psi^{-1}) \circ s'' = (\varphi' \circ \varphi^{-1}) \circ s''$". The transition maps of the sub-charts are restrictions of those of X, hence sc-smooth `Chapter2/SubMPolyfold · MPolyfold.subChart_compatible`. They form an atlas `Chapter2/SubMPolyfold · MPolyfold.subAtlas`, and A is an M-polyfold `Chapter2/SubMPolyfold · MPolyfold.subPolyfold` `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart`. The structure requires A to be paracompact (an instance argument); it is, as a subset of the metrizable X (§II.7), and for closed A also elementarily `Chapter2/SubMPolyfold · MPolyfold.paracompactSpace_of_isClosed`.

(1) and (2). The inclusion $A \to X$ is sc-smooth: in the sub-chart and φ it is the identity of $\varphi(W \cap A)$ into O; one pair of charts per point suffices `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_inclusion_isScSmooth`; it is a homeomorphism onto its image by the choice of topology. For any data (V, r), the map $i^{-1} \circ r$ (a total map with a junk value off V) `Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto` `Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto_val` is sc-smooth on V: near $x \in V$ take a chart φ' around x, a chart φ around $r(x)$ and $V_0 = V \cap \operatorname{dom}\varphi' \cap r^{-1}(\operatorname{dom}\varphi)$; on $\varphi'(V_0)$ the representation in φ' and the sub-chart on φ is $\varphi \circ r \circ \varphi'^{-1}$, sc-smooth by Definition 2.3.8 (preprint 2.14) for r `Chapter2/SubMPolyfold · MPolyfold.subPolyfold_retraction_isScSmooth`.

Proposition 2.3.19 (3), (4) (preprint 2.21), in a chart. Let r be a retraction onto O, s sc-smooth from (O, C, E) to itself with $s \circ s = s$, and $t := s \circ r$, a retraction onto $s(O)$ (Proposition 2.3.1 (2) (preprint 2.10)) `Chapter2/Metrizable · ScRetraction.compRetraction` `Chapter2/Metrizable · ScRetraction.compRetraction_retract`. For $a \in s(O)_1$ put $P := Dr(a)$, $Q := Dt(a) = D(s \circ r)(a)$ — the text's "$Ds(a)$". Both are idempotent (§II.2). From $r \circ t = t$ and $t \circ r = t$ on U (s takes values in O, which r fixes; and $r \circ r = r$), the chain rule at the point a `Chapter2/DegeneracyIndex · scDeriv_comp_apply` gives $PQ = Q$ and $QP = Q$. If a is smooth, P and Q are sc-operators (Part I, Proposition 1.2.1 (preprint 1.16)), and so is $P - Q$; identities between base maps of sc-operators hold on every level `Chapter2/Metrizable · ScOperator.lvl_comp_of_base` `Chapter2/Metrizable · ScOperator.lvl_comp_eq_zero_of_base` `Chapter2/Metrizable · ScOperator.lvl_eq_add_of_base`. Then Q and $P - Q$ are commuting idempotent sc-operators with $Q(P - Q) = 0$, their ranges are sc-subspaces (Part I, §I.3: sc-splittings and sc-projections `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`), and on every level the level parts of QE and $(P-Q)E$ are disjoint with sum the level part of PE. Since $T_aO = PE$ and $T_a s(O) = QE$ (§II.2), this is the sc-splitting $T_aO = T_aA \oplus (Dr(a) - Dt(a))E$

Chapter2/Metrizable · proposition_2_21_3. For (4), s is sc-smooth from O onto $s(O)$ (shrinking the target), and $Ts(a,h) = (a, Qh)$ for $h \in T_aO$ because $Qh = QPh$; as Q maps PE onto $QE = Q(PE)$, $Ts(a)$ maps T_aO onto $T_as(O)$ Chapter2/Metrizable · proposition_2_21_4. The transport to T_aX is not formalized.

II.11 The degeneracy index of an M-polyfold (Definitions 2.4.1, 2.4.5 (preprint 2.22, 2.25); Lemmas 2.4.3, 2.4.6 (preprint 2.23, 2.26); Proposition 2.4.4 (preprint 2.24))

A retract is contained in its quadrant, and chart values lie in C Chapter2/DegeneracyIndex · LocalModel.O_subset_C Chapter2/DegeneracyIndex · MChart.apply_mem_C Chapter2/DegeneracyIndex · LocalModel.isPartialQuadrant.

Definition 2.4.1 (preprint 2.22). $d(x,\varphi) := d_C(\varphi(x))$ Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex, and $d_X(x)$ is the least $d(x,\varphi)$ over the charts of the structure around x Chapter2/DegeneracyIndex · MPolyfold.degIndex; $d_X(x) \le d(x,\varphi)$ for every such chart and the minimum is attained Chapter2/DegeneracyIndex · MPolyfold.degIndex_le Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq.

Lemma 2.4.3 (preprint 2.23). d_C is upper semicontinuous: coordinates positive at x stay positive nearby, so the vanishing set can only shrink Chapter2/DegeneracyIndex · eventually_degIndexWith_le; composed with the continuous chart, $d(\cdot,\varphi)$ is upper semicontinuous on V Chapter2/DegeneracyIndex · MPolyfold.eventually_chartDegIndex_le, and with a chart realizing $d_X(x)$, $d_X(y) \le d(y,\varphi) \le d(x,\varphi) = d_X(x)$ near x Chapter2/DegeneracyIndex · lemma_2_23. Hence $\{d_X < k\}$ is open and $\{d_X \ge k\}$ closed Chapter2/Boundary · MPolyfold.isOpen_setOf_degIndex_lt Chapter2/Boundary · MPolyfold.isClosed_setOf_le_degIndex.

Proposition 2.4.4 (preprint 2.24). An sc-diffeomorphism of open sets is f sc-smooth on U, g sc-smooth on V, mapping into each other and mutually inverse Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn.symm. For a chart ψ of Y, $\psi \circ f$ on $U \cap f^{-1}(\mathrm{dom}\,\psi)$, with inverse $g \circ \psi^{-1}$, is a chart of X Chapter2/DegeneracyIndex · MPolyfold.transportChart; it is compatible with every chart φ of the atlas of X, because its transition maps with φ are $\psi \circ f \circ \varphi^{-1}$ and $\varphi \circ g \circ \psi^{-1}$, sc-smooth by Definition 2.3.8 (preprint 2.14) for f and for g Chapter2/DegeneracyIndex · MPolyfold.transportChart_isChart. Its index at x is $d(f(x),\psi)$ Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex_transportChart. With ψ realizing $d_Y(f(x))$: $d_X(x) \le d_Y(f(x))$ Chapter2/DegeneracyIndex · MPolyfold.degIndex_le_of_isScDiffeoOn, and with g the reverse Chapter2/DegeneracyIndex · proposition_2_24. The text's "from the definitions one deduces immediately" is this construction.

Definition 2.4.5 (preprint 2.25). $\partial X = \{d_X \ge 1\}$ Chapter2/DegeneracyIndex · MPolyfold.boundary; without boundary means $d_X \equiv 0$, equivalently $\partial X = \emptyset$ Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary Chapter2/DegeneracyIndex · MPolyfold.isWithoutBoundary_iff. ∂X and the corners $\{d_X \ge 2\}$ are closed, the interior $\{d_X = 0\}$ is open Chapter2/Boundary · MPolyfold.isClosed_boundary Chapter2/Boundary · MPolyfold.corners Chapter2/Boundary · MPolyfold.isClosed_corners Chapter2/Boundary · MPolyfold.isOpen_interior, and sc-diffeomorphisms preserve ∂X by Proposition 2.4.4 (preprint 2.24) Chapter2/Boundary · MPolyfold.IsScDiffeoOn.mem_boundary_iff.

Lemma 2.4.6 (preprint 2.26). For a sub-M-polyfold A and $a \in A$: take a chart φ of X realizing $d_X(a)$ and the data at a; the sub-chart built on φ has the same quadrant C and the same value $\varphi(a)$, so $d_A(a) \le d_C(\varphi(a)) = d_X(a)$ Chapter2/SubMPolyfold · lemma_2_26.

A retract as an M-polyfold. A nonempty retract O, as a subtype, has the single chart id Chapter2/MPolyfold · LocalModel.selfChart, compatible with itself, and is an M-polyfold Chapter2/MPolyfold · LocalModel.toMPolyfold (Hausdorff and paracompact as a subspace of E_0); so is a nonempty relatively open $U \subseteq C$ Chapter2/MPolyfold · IsRelOpenInQuadrant.toMPolyfold Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.selfChart_isChart — the text's X_C.

II.12 The subspaces E_x, the cones C_x and the sign arguments

Coordinates. For a presentation T of C and $x \in E_0$, the coordinates vanishing at x form a continuous linear surjection $E_0 \to \mathbb{R}^{I(x)}$, $v \mapsto ((Tv)_i)_{i\in I(x)}$ `Chapter2/DegeneracyIndex · coordProj` `Chapter2/DegeneracyIndex · coordMapL` `Chapter2/DegeneracyIndex · coordMapL_surjective`. Its kernel is E_x `Chapter2/DegeneracyIndex · minimalSubspaceWith` `Chapter2/DegeneracyIndex · mem_minimalSubspaceWith`, closed `Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith`, and by the first isomorphism theorem $E_0/E_x \cong \mathbb{R}^{I(x)}$, so $\dim(E_0/E_x) = d_C(x)$ `Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith`. The cone $C_x = \{v : (Tv)_i \geq 0 \text{ for } i \in I(x)\}$ `Chapter2/Tame · partialQuadrantAtWith` is convex and closed and contains E_x `Chapter2/Tame · convex_partialQuadrantAtWith` `Chapter2/Boundary · isClosed_partialQuadrantAtWith` `Chapter2/Tame · minimalSubspaceWith_subset_partialQuadrantAtWith`. *Approximation*: if $Tx = (a, w)$, the points $T^{-1}(a, w')$ with $w' \in W_1$ (resp. W_∞) converge to x as $w' \to w$ and have the same vanishing coordinates, so x is a limit of points on level 1 (resp. smooth points) of the same index, in C iff x is `Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith` `Chapter2/TameRetracts · exists_tendsto_smooth_degIndexWith`.

Density on level 1. For a continuous linear $L : E_0 \to V$, V finite-dimensional, the level-1 points of $\ker L$ are dense in $\ker L$ `Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase`. The image S of E_1 is dense in E_0; L maps S onto the finite-dimensional $L(S)$, which has a linear, hence continuous, right inverse g with values in S. For $v \in \ker L$ take $u_k \in S$ with $u_k \to v$; then $L(u_k) \to 0$, $g(L(u_k)) \to 0$, and $u_k - g(L(u_k)) \in S \cap \ker L$ converges to v. This is the text's "the closure of $(T^R_x O)_\infty$ is $T^R_x O$" in the form the proofs need; the level-1 points of C_x are dense in C_x likewise `Chapter2/Boundary · partialQuadrantAtWith_subset_closure`.

The two-sided sign argument. Let f be sc¹ on U relatively open in C, with values in C', $x \in U_1$ and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x + tv \in U_1$ for $|t|$ small: the coordinates in $I(x)$ stay 0, the others stay positive, and U_1 is relatively open `Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet`. For j with $(T' f(x))_j = 0$ the real function $g(t) = (T' f(\iota_1(x + tv)))_j$ is ≥ 0 near 0 (values in C'), vanishes at 0, and has derivative $(T' Df(x)v)_j$ at 0 (Definition 1.1.15's (preprint 1.14) derivative along E_1, composed with the path); a local minimum has zero derivative `Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero`. So $Df(x)v \in E'_{f(x)}$ `Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1`, and by density and continuity of $Df(x)$ on E_0, $Df(x)$ maps E_x into $E'_{f(x)}$ `Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace`.

The one-sided sign argument. For $v \in E_1$ with $\iota_1 v \in C_x$, $x + tv \in U_1$ for small $t \geq 0$ `Chapter2/Boundary · eventually_add_smul_mem_lvlSet_of_cone`, the same g is ≥ 0 on $[0, \varepsilon)$ with $g(0) = 0$, so its one-sided derivative is ≥ 0 `Chapter2/Boundary · nonneg_of_hasDerivWithinAt_Ici`; hence $Df(x)v \in C'_{f(x)}$ `Chapter2/Boundary · IsSc1With.deriv_mem_partialQuadrant_of_lvl1`, and $Df(x)$ maps C_x into $C'_{f(x)}$ `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant`. These are the derivative-level content of Lemma 2.4.9 (1) and (2) (preprint 2.29); the lemma itself, about sc-smooth paths, is not formalized.

The chain rule at a point. For sc¹ maps f, g on relatively open sets with $f(U) \subseteq U'$, $Dg(f(x))Df(x) = D(g \circ f)(x)$ at $x \in U_1$ `Chapter2/DegeneracyIndex · scDeriv_comp_apply`; if $g \circ f = \mathrm{id}$ on U then $g_1(f_1(x)) = x$ and $Dg(f(x))Df(x) = \mathrm{id}$ `Chapter2/DegeneracyIndex · levelMap_comp_eq_self` `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`.

II.13 Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31), Corollary 2.4.13 (preprint 2.32)

Definition 2.4.7 (preprint 2.27), Remark 2.4.8 (preprint 2.28). For a retraction r, a presentation T and $x \in O_1$: $T^R_x O := Dr(x)E_0 \cap E_x$ and $C_x O := Dr(x)E_0 \cap C_x$ `Chapter2/Tame · reducedTangentSpaceWith` `Chapter2/Tame · partialConeWith`, defined at every point on level 1; $T^R_x O \subseteq C_x O$ `Chapter2/Tame · reducedTangentSpaceWith_subset_partialConeWith`. The number $\dim(T_x O / T^R_x O)$ is $\dim(A/(A \cap B))$ for $A = T_x O$, $B = E_x$ `Chap-`

ter2/Boundary · quotDim; since $A \to E_0/E_x$ has kernel $A \cap E_x$, it is at most $d_C(x)$ and finite Chapter2/Boundary · quotDim_inf_minimalSubspace_le Chapter2/Boundary · quotDim_reducedTangentSpaceWith_le.

Germs between retracts. A germ of sc-diffeomorphisms between $O = r(U)$ and $O' = r'(U')$ is a pair F, G with $F \circ r$, $G \circ r'$ sc¹ (sc¹ suffices here; the text has sc-smooth), $F(O) \subseteq O'$, $G(O') \subseteq O$, inverse to each other on O, O' Chapter2/Boundary · ScRetraction.IsDiffeoGerm; an sc-smooth map between retracts with an sc-smooth inverse is one Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm, and two compatible charts give one between the overlap retracts, with the restricted retractions Chapter2/Boundary · MChart.Compatible.isDiffeoGerm. Since $(G \circ r') \circ (F \circ r) = r$ on U Chapter2/Boundary · ScRetraction.IsDiffeoGerm.comp_eq, the chain rule at $x \in O_1$, $y = (F \circ r)_1(x) \in O_1'$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.image_mem Chapter2/Boundary · ScRetraction.IsDiffeoGerm.image_image, gives

$$D(G \circ r')(y)\, D(F \circ r)(x) = Dr(x), \qquad D(F \circ r)(x)\, D(G \circ r')(y) = Dr'(y)$$

Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_deriv_inv. So $D := D(F \circ r)(x)$ maps T_xO into T_yO' Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range, and by the sign arguments for $F \circ r$ (with values in $O' \subseteq C'$) it maps E_x into E_y' and C_x into C_y' Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_minimalSubspace Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_partialQuadrant.

Proposition 2.4.10 (preprint 2.30). $D(T_x^R O) = T_y^R O'$ and $D(C_x O) = C_y O'$, with $Tf(x) = D$ Chapter2/Boundary · proposition_2_30_reduced Chapter2/Boundary · proposition_2_30_cone. "$\subseteq$": D maps T_xO into T_yO' and E_x (resp. C_x) into E_y' (resp. C_y'). "$\supseteq$": for $w \in T_y^R O'$, the vector $D'w$, $D' = D(G \circ r')(y)$, lies in T_xO and in E_x by the same facts for the inverse germ, and $D(D'w) = Dr'(y)w = w$ because w is in the range of the projection $Dr'(y)$. This proof does not pass through Lemma 2.4.9 (preprint 2.29).

The core of Theorem 2.4.12 (preprint 2.31). Along a germ F, $\dim(T_xO/T_x^R O) \le d_{C'}(F(x))$ for $x \in O_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.quotDim_le. With $P = Dr(x)$, $P' = Dr'(y)$: D maps $T_x = PE$ into $T_y = P'E'$ and D' maps back, $D'D = P$ is the identity on T_x and $DD' = P'$ the identity on T_y. The map $\Phi : T_x \to T_y/(T_y \cap E_y')$, $v \mapsto [Dv]$, is onto ($[w] = \Phi(D'w)$) with kernel $T_x \cap E_x$ ($Dv \in E_y'$ implies $v = D'Dv \in E_x$). Hence $T_x/T_x^R \cong T_y/T_y^R$, which embeds in E_0'/E_y', of dimension $d_{C'}(y)$. The same argument with U_0 relatively open in C in place of the retract O is the form used for Corollary 2.4.13 (preprint 2.32) and Theorem 1.3.5 (preprint 1.25) Chapter2/DegeneracyIndex · degIndexWith_le_of_germ: there $D'D = \mathrm{id}$, so $E_0/E_x \cong T_y/T_y^R \hookrightarrow E_0'/E_y'$ and $d_C(x) \le d_{C'}(F(x))$.

Theorem 2.4.12 (preprint 2.31). Let φ be a chart of an M-polyfold around x with $\varphi(x) = \iota_1 z$, r a retraction onto its model and T a presentation. For every other chart ψ around x, the overlaps are $O \cap W$ and $O' \cap W'$, the transition maps form a germ between the restricted retractions, and the restricted retraction has the same derivative at z Chapter2/Boundary · ScRetraction.scDeriv_restrict Chapter2/Boundary · reducedTangentSpaceWith_restrict; so the core gives $\dim(T_zO/T_z^R O) \le d(x,\psi)$ Chapter2/Boundary · MPolyfold.quotDim_le_chartDegIndex. Taking ψ realizing $d_X(x)$: $\dim(T_zO/T_z^R O) \le d_X(x)$ Chapter2/Boundary · theorem_2_31. For a retract O as an M-polyfold with the chart id this is the text's statement $\dim(T_xO/T_x^R O) \le d_O(x)$, at points on level 1 Chapter2/Boundary · theorem_2_31_retract. The text instead reduces to a chart realizing $d_O(x)$ and counts linearly independent $\mathbb{R}^d$-components; the quotient isomorphism makes the reduction unnecessary. The second statement (equality makes C_xO a partial quadrant) is not formalized: it needs the sc-structure of T_xO and smooth lifts of a basis of $T_xO/T_x^R O$.

Corollary 2.4.13 (preprint 2.32). Let U be nonempty and relatively open in C, and X_U its M-polyfold Chapter2/DegeneracyIndex · corollary_2_32. The chart id gives $d_{X_U} \le d_C$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le. At a point x on level 1, a chart ψ realizing $d_{X_U}(x)$ is compatible with id, so $F = \psi$ (on $U_0 = \mathrm{dom}\,\psi$, relatively open in C) and $G = \psi^{-1}$ form a germ from U_0 to the retract $\psi(V)$, and the core gives $d_C(x) \le d_{C'}(\psi(x)) = d_{X_U}(x)$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le_degIndex_of_lvl1. For a general x, approximate by level-1 points $\sigma_i \in U$ with $d_C(\sigma_i) = d_C(x)$ (§II.12); by Lemma 2.4.3 (preprint 2.23), $d_{X_U}(\sigma_i) \le d_{X_U}(x)$

eventually, so $d_C(x) = d_C(\sigma_i) \le d_{X_U}(\sigma_i) \le d_{X_U}(x)$. The text approximates by smooth points; level 1 suffices. **Theorem 1.3.5 (preprint 1.25)** (Part I, §I.19) is proved in the same file by the same argument for a germ between relatively open subsets of two quadrants `Chapter2/DegeneracyIndex · theorem_1_25_lvl1` `Chapter2/DegeneracyIndex · theorem_1_25`.

II.14 Tame retractions; Propositions 2.5.3 and 2.5.6 (preprint 2.35 and 2.36)

Definition 2.5.1 (preprint 2.33). C_e and E_e are the cone and subspace of §II.12 for the presentation T. They do not depend on it: for two presentations T, T' of C and $e \in C$, $v \in E_e$ (for T) gives $e+tv \in C$ for $|t|$ small `Chapter2/TameFaces · eventually_add_smul_mem_of_minimalSubspace`; for j with $(T'e)_j = 0$, $(T'(e+tv))_j = t(T'v)_j \ge 0$ for $t = \pm\delta/2$, so $(T'v)_j = 0$ `Chapter2/TameFaces · minimalSubspaceWith_le`; by symmetry the two subspaces agree `Chapter2/TameFaces · minimalSubspaceWith_eq`. This is the argument of Lemma 1.3.4 (preprint 1.24).

Definition 2.5.2 (preprint 2.34). r is *tame for* T if (1) $d_C(r(x)) = d_C(x)$ on U and (2) at every smooth $x \in O_1$ some $A \le E_x$ makes $E = Dr(x)E_0 \oplus A$ an sc-splitting `Chapter2/Tame · ScRetraction.IsTameWith`; *tame* if tame for every presentation `Chapter2/Tame · ScRetraction.IsTame`, and by the independence of E_x tameness for one presentation suffices `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`. A *tame retract* is the retract of a tame retraction `Chapter2/Tame · IsTameRetract` `Chapter2/Tame · IsTameRetract.isScRetract`. Identities of relatively open sets are tame ($T_xU = E$, $A = 0$) `Chapter2/Tame · ScRetraction.id_isTame`, and restrictions of tame retractions to $U \cap r^{-1}(V)$ are tame (same derivatives) `Chapter2/TameFaces · ScRetraction.IsTameWith.restrict`.

Proposition 2.5.3 (preprint 2.35). *$Dr(x)E_x \subseteq E_x$ for every retraction at every $x \in O_1$*: the two-sided sign argument for r, with $r(x) = x$ `Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace`. At a smooth x, $Dr(x)$ is an sc-operator `Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv`, so $E = Dr(x)E \oplus (1 - Dr(x))E$ is an sc-splitting `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. *The proposition* `Chapter2/TameRetracts · proposition_2_35`: with $A \le E_x$ from tameness (2), $E_0 = Dr(x)E_0 \oplus A$ algebraically (the level-0 part of the sc-splitting); write $v = t+a$; then $(1-Dr(x))v = (1-Dr(x))a = a - Dr(x)a \in E_x$, because $a \in E_x$ and $Dr(x)a \in E_x$. Only condition (2) is used; the text uses condition (1) for $Dr(x)E_x \subseteq E_x$. In coordinates: $\lambda_j(Dr(x)v) = \lambda_j(v)$ for $j \in I(x)$ `Chapter2/TameRetracts · coord_scDeriv_eq`.

Proposition 2.5.6 (preprint 2.36). *(13) at smooth points.* If $V = A' \oplus B$ and $B \le K$, then $A'/(A' \cap K) \cong V/K$ `Chapter2/TameRetracts · quotDim_eq_finrank_quotient_of_isCompl`; with $A' = T_xO$, $B = A$ and $K = E_x$ this gives $\dim(T_xO/T_x^RO) = \dim(E_0/E_x) = d_C(x)$ `Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq` — the text's equations (15)–(16). *A tame chart realizes the index* `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`: let φ be a chart of any M-polyfold whose model has a tame retraction r. At y with $\varphi(y)$ smooth, $d(y,\varphi) = \dim(T/T^R) \le d_X(y)$ by Theorem 2.4.12 (preprint 2.31), and $d_X(y) \le d(y,\varphi)$ by definition. At a general y, with $\varphi(y) = o$, approximate o by smooth points σ_i of C with the same vanishing coordinates; then $r(\sigma_i)$ are smooth points of O (an sc^0 map preserves smoothness `Chapter2/TameRetracts · IsSc0.mem_smooth` `Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth`) converging to $r(o) = o$, with $d_C(r(\sigma_i)) = d_C(\sigma_i) = d_C(o)$ by tameness (1); so the points $\varphi^{-1}(r(\sigma_i))$ converge to y and have $d_X = d(\cdot,\varphi) = d_C(o)$, and Lemma 2.4.3 (preprint 2.23) gives $d_C(o) \le d_X(y)$. For charts of the structure modelled on a tame retract `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`. Applied to the retract itself as an M-polyfold `Chapter2/TameRetracts · ScRetraction.toLocalModel` `Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith`: (14) $d_O = d_C$ on O `Chapter2/TameRetracts · proposition_2_36_degIndex` and (13) $\dim(T_xO/T_x^RO) = d_O(x) = d_C(x)$ at smooth points `Chapter2/TameRetracts · proposition_2_36`. This is the text's proof, generalized from the retract to any chart; Proposition 2.5.14 (preprint 2.42) needs the general form.

Splitting off finitely many coordinates. Let $B \le A$ be sc-subspaces of E, $\ell_1, \dots, \ell_d$ continuous function-

als on E_0 with $B = A \cap \bigcap_j \ker \ell_j$, and $e_1, \dots, e_d \in A$ smooth with $\ell_i(e_j) = \delta_{ij}$. Each e_j has a lift to every level m, compatible with the inclusions `Chapter2/PartialCone · smoothLift` `Chapter2/PartialCone · incl_smoothLift`. On level m the maps $v \mapsto ((\ell_j v)_j,\ v - \sum_j \ell_j(v) e_j)$ and $(a,u) \mapsto u + \sum_j a_j e_j$ are continuous linear $A_m \to \mathbb{R}^d \times B_m$ and back `Chapter2/PartialCone · splitFwd` `Chapter2/PartialCone · splitBwd` — the bracket lies in B because every ℓ_i kills it `Chapter2/PartialCone · sub_sum_mem` — and mutually inverse `Chapter2/PartialCone · splitBwd_splitFwd` `Chapter2/PartialCone · splitFwd_splitBwd`; they commute with the inclusions, so they form an sc-isomorphism $A \cong \mathbb{R}^d \oplus B$ `Chapter2/PartialCone · splitIso`, A and B with the sc-structures induced from E `Chapter1/ScBanach · ScBanach.ofSubspace`. It carries $\{v \in A : \ell_j(v) \geq 0\ \forall j\}$ onto $[0,\infty)^d \oplus B$, so that set is a partial quadrant in A `Chapter2/PartialCone · isPartialQuadrant_ofSubspace`. This is the map S of the proof of Theorem 2.4.12 (preprint 2.31), with the e_j smooth as the exposition requires.

Proposition 2.5.6 (preprint 2.36), the remaining statements `Chapter2/PartialCone · proposition_2_36_partialCone`. Let r be tame for T, $x \in O_1$ smooth, $P = Dr(x)$ and $I(x) = \{i_1, \dots, i_d\}$. $T_xO = PE$ is an sc-subspace, the range of the sc-projection P (Proposition 2.5.3 (preprint 2.35) above) `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. The vectors $\hat{e}_j := P(T^{-1}(e_{i_j}, 0))$ `Chapter2/TameRetracts · coordVec` lie in T_xO; they are smooth, because $T^{-1}(e_{i_j}, 0)$ is smooth `Chapter2/PartialCone · coordVec_mem_smooth` and P is an sc-operator at the smooth point x `Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv` `Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl`; and $\lambda_{i_k}(\hat{e}_j) = \lambda_{i_k}(T^{-1}(e_{i_j}, 0)) = \delta_{jk}$ by Proposition 2.5.3 (preprint 2.35) in coordinates `Chapter2/TameRetracts · coord_scDeriv_eq`. By definition $T^R_xO = T_xO \cap \bigcap_k \ker \lambda_{i_k}$ `Chapter2/TameRetracts · coordL`. *T^R_xO is an sc-subspace.* It is closed, as an intersection of closed subspaces `Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith`. For the density of smooth points, let $u \in T^R_xO \cap E_m$ with $Tu = (a,w)$ on level m, and put $\phi(w') := P(T^{-1}(a,w'))$, a continuous map $W_m \to E_m$. For smooth w' the point $\phi(w')$ is smooth, lies in T_xO, and lies in E_x, because $T^{-1}(a,w')$ has the same vanishing coordinates in $I(x)$ as u and $PE_x \subseteq E_x$ `Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace`. And $\phi(w) = Pu = u$, since $u \in T_xO$ and P is a projection `Chapter2/Retracts · ScRetraction.scDeriv_comp_self`. As W_∞ is dense in W_m, u lies in the closure of $\phi(W_\infty) \subseteq T^R_xO \cap E_\infty$ (this is the exposition's Lemma 3.A (1) in the case at hand). *The quadrant.* The splitting with $A = T_xO$, $B = T^R_xO$, $\ell_j = \lambda_{i_j}$ and $e_j = \hat{e}_j$ makes $\{v \in T_xO : \lambda_{i_j}(v) \geq 0\ \forall j\}$ a partial quadrant in T_xO, and this set is $C_xO = T_xO \cap C_x$. The text deduces the claim from the equality case of Theorem 2.4.12 (preprint 2.31), which needs smooth lifts $\hat{v}^j$ of a basis of T_xO/T^R_xO (the exposition chooses them by a density argument); for a tame retraction they are the explicit $\hat{e}_j$, and the general equality case, still not formalized, is not needed.

II.15 Faces: local geometry, Lemma 2.5.12 (preprint 2.40), Corollary 2.5.13 (preprint 2.41), Proposition 2.5.14 (preprint 2.42)

Definitions 2.5.7, 2.5.10 (preprint 2.37, 2.39). X is *tame* if an atlas equivalent to its atlas has only tame models `Chapter2/Tame · MPolyfold.IsTame`; charts of an equivalent atlas are charts of the structure `Chapter2/TameRetracts · MPolyfold.isChart_of_mem_equiv`, so every point lies in a chart of the structure with a tame model `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`. A *face* is the closure of the connected component in $\{d_X = 1\}$ of a point with $d_X = 1$ `Chapter2/Tame · MPolyfold.IsFace`, closed `Chapter2/Tame · MPolyfold.IsFace.isClosed` and inside ∂X `Chapter2/Boundary · MPolyfold.IsFace.subset_boundary`; X is *face-structured* if every x lies in exactly $d_X(x)$ faces `Chapter2/Tame · MPolyfold.IsFaceStructured`.

Coordinates of faces. $I(z) = \{i : \lambda_i(z) = 0\}$ `Chapter2/TameRetracts · zeroSet` with $d_C = \#I$ `Chapter2/TameRetracts · degIndexWith_eq_card_zeroSet`; the extended face f_i is the hyperplane $\{\lambda_i = 0\}$, closed `Chapter2/TameRetracts · coordHyper` `Chapter2/TameRetracts · isClosed_coordHyper`; $I(z) \subseteq I(e)$ for z near e `Chapter2/TameRetracts · eventually_zeroSet_subset`. *A connected piece of $\{d_C = 1\}$ lies in one face* (exposition Lemma 4.A): if $S \subseteq C$ is preconnected with $d_C = 1$ on S and $\lambda_i(p) = 0$ at some $p \in S$, then $S \subseteq f_i$ `Chapter2/TameRetracts · subset_coordHyper_of_isPreconnected`. Points of S have $I = \{i\}$ or $\lambda_i > 0$ `Chap-`

ter2/TameRetracts · zeroSet_eq_singleton, so S is covered by the open sets $u = \{\lambda_l > 0 \text{ for } l \neq i\}$ and $v = \{\lambda_i > 0\}$, which are disjoint on S (a point in both would have $d_C = 0$); preconnectedness puts S in one of them Mathlib/Topology/Connected/Clopen · isPreconnected_iff_subset_of_disjoint, and $p \in u$.

Rays stay in their face (exposition Lemma 4.B). If f is sc¹, $x + tv \in U_1$ for small $t \geq 0$ and a functional L vanishes on $f(x + tv)$ for $t \in [0, \varepsilon)$, then $L(Df(x)v) = 0$ Chapter2/TameRetracts · IsSc1With.apply_deriv_eq_zero_of_ray. Now let r be tame, $x \in O_1$ smooth, $i \in I(x)$, $\lambda_i(y) = 0$ and $\lambda_l(y) > 0$ for the other $l \in I(x)$ Chapter2/TameRetracts · ray_zeroSet. For small $\tau > 0$, $x + \tau y \in U$ and $I(x + \tau y) = \{i\}$, so $d_C(r(x + \tau y)) = 1$ by tameness (1), and $I(r(x + \tau y)) \subseteq I(x)$ by continuity of r at $x = r(x)$. The image of $(0, \tau_0]$ under $\tau \mapsto r(x + \tau y)$ is preconnected with $d_C = 1$, so it lies in one f_j with $j \in I(x)$. Differentiating $\lambda_j(r(x + \tau y)) = 0$ at $\tau = 0$ gives $\lambda_j(Dr(x)y) = 0$, and $\lambda_j(Dr(x)y) = \lambda_j(y)$ by Proposition 2.5.3 (preprint 2.35) Chapter2/TameRetracts · coordL; as $\lambda_l(y) > 0$ for $l \in I(x) \setminus \{i\}$, $j = i$. This is the step of Lemma 2.5.12 (preprint 2.40) that the text omits, and it needs tameness (2).

The local structure of a face (exposition Lemma 4.C) Chapter2/TameRetracts · local_face_structure. Let $S = \{z \in O : d_C(z) = 1\}$ Chapter2/TameRetracts · ScRetraction.faceStratum and $e \in O$ with $\lambda_i(e) = 0$. (a) Choose ε so that $B = B_\varepsilon(e) \cap C \subseteq U$ and $I(z), I(r(z)) \subseteq I(e)$ on B. (b) $A_i \cap B = \{z \in B : I(z) = \{i\}\}$ is convex Chapter2/TameRetracts · convex_ball_inter_zeroSet_eq, hence connected, and r maps it into S; a smooth $u = r(T^{-1}(a, w')) \in O \cap B$ with $I(u) = I(e)$ and the ray of Lemma 4.B from u in the direction $T^{-1}(b, 0)$, $b = 1$ on $I(e) \setminus \{i\}$ and 0 elsewhere Chapter2/TameRetracts · coordVec Chapter2/TameRetracts · rayDir Chapter2/TameRetracts · exists_lvl1_coordVec, produces a point $p \in S$ with $\lambda_i(p) = 0$ in $r(A_i \cap B)$; so $r(A_i \cap B)$ lies in the component of S through p. (c) A point $e' \in f_i \cap B$ is the limit of points of $A_i \cap B$ (perturb the coordinates of $I(e') \setminus \{i\}$ upward), so $r(e')$ lies in the closure of that component.

Lemma 2.5.12 (preprint 2.40). A *face of the retract* is $\overline{\hat{F}} \cap O$ for a component $\hat{F}$ of S Chapter2/TameRetracts · ScRetraction.IsRetractFace. A component of S through a point with $\lambda_i = 0$ lies in f_i Chapter2/TameRetracts · faceStratum_degIndexWith Chapter2/TameRetracts · connectedComponentIn_faceStratum_subset, and a component of S in f_i whose closure contains e is the component of Lemma 4.C at e Chapter2/TameRetracts · connectedComponentIn_eq_of_mem_closure. *The face through $x \in S$, $\lambda_i(x) = 0$, is the component of $f_i \cap O$ through x* Chapter2/TameRetracts · closure_connectedComponentIn_faceStratum: it is connected, lies in $f_i \cap O$, is closed in it, and is relatively open in it by Lemma 4.C(c); a preconnected subset meeting a relatively clopen set lies in it Chapter2/TameRetracts · subset_of_relClopen_of_isPreconnected. Conversely, the component of $f_i \cap O$ through any $e \in O$ with $\lambda_i(e) = 0$ is the face through the point p of Lemma 4.C at e Chapter2/TameRetracts · connectedComponentIn_coordHyper_eq. Together: the faces of the retract are the components of the sets $f_i \cap O$ Chapter2/TameRetracts · lemma_2_40. If the components of $f_i \cap O$ and $f_j \cap O$ through x coincide, a point z of the face in S has $I(z) = \{i\}$ and is a limit of points with $I = \{j\}$, which near z forces $j = i$ Chapter2/TameRetracts · lemma_2_40_injective. The faces of the M-polyfold O (Definition 2.5.10 (preprint 2.39), with $d_O = d_C$ by Proposition 2.5.6 (preprint 2.36)) are exactly the faces of the retract, transferred along the subtype inclusion Chapter2/TameRetracts · image_val_connectedComponentIn Chapter2/TameRetracts · image_val_closure Chapter2/TameRetracts · LocalModel.isFace_toMPolyfold_iff.

Corollary 2.5.13 (preprint 2.41). The map $i \mapsto$ (component of $f_i \cap O$ through x) sends $I(x)$ onto the faces through x (Lemma 2.5.12 (preprint 2.40)) and is injective (its second statement), so x lies in exactly $\#I(x) = d_C(x)$ faces of the retract Chapter2/TameRetracts · corollary_2_41_retract, and the M-polyfold O is face-structured Chapter2/TameRetracts · LocalModel.isFaceStructured_toMPolyfold Chapter2/TameRetracts · corollary_2_41.

Proposition 2.5.14 (preprint 2.42). Let X be tame. *The global bound* Chapter2/TameRetracts · MPolyfold.faces_encard_le: in a tame chart φ around x, $o = \varphi(x)$, Lemma 4.C gives for each $i \in I(o)$ a component $\hat{F}_i$ of S containing all points of O near o with $I = \{i\}$. A face $F = \overline{G}$ through x has points $y \in G$ near x; $d_C(\varphi(y)) = d_X(y) = 1$ (the tame chart realizes d_X) and $I(\varphi y) \subseteq I(o)$, so $I(\varphi y) = \{i\}$ for some $i \in I(o)$, and the preconnected set $\varphi^{-1}(\hat{F}_i) \subseteq \{d_X = 1\}$ meets G, hence lies in G. Distinct faces give distinct i; so at most $d_X(x)$ faces pass through x. *The local statement*: the tame chart realizes d_X on its domain, the M-polyfold O has $d_O = d_C$ (Proposition 2.5.6 (preprint 2.36)), and O is face-structured

(Corollary 2.5.13 (preprint 2.41)) Chapter2/TameRetracts · proposition_2_42. The neighbourhood V of the text is represented by O.

II.16 Faces are tame sub-M-polyfolds: Propositions 2.5.15, 2.5.16 (preprint 2.43, 2.44) and Theorem 2.5.17 (preprint 2.45)

The projection onto a face. $P_i := T^{-1} \circ (a \mapsto a \text{ with } a_i = 0) \circ T$ is an sc-operator Chapter2/TameRetracts · killCoordL Chapter2/TameRetracts · killCoordModel Chapter2/TameRetracts · killCoord which kills the i-th coordinate and keeps the others Chapter2/TameRetracts · killCoord_coord Chapter2/TameRetracts · killCoord_coord_self, fixes f_i Chapter2/TameRetracts · killCoord_eq_self and maps C into C Chapter2/TameRetracts · killCoord_mem.

Proposition 2.5.15 (preprint 2.43), the retraction. Let F be a face of the retract, through $x \in S$ with $I(x) = \{i\}$, so $F \subseteq f_i$ Chapter2/TameRetracts · proposition_2_43. For each $e \in F$, Lemma 4.C gives ε_e with $B_e = B_{\varepsilon_e}(e) \cap C \subseteq U$ and $r(z) \in F$ for $z \in B_e \cap f_i$ (the component of Lemma 4.C at e is the one of F). Put $U^* := \bigcup_{e\in F}(B_{\varepsilon_e}(e) \cap P_i^{-1}(B_{\varepsilon_e}(e))) \cap C$: relatively open in C, contains F (as $P_i e = e$), and for $z \in U^*$ the point $P_i z$ lies in some $B_e \cap f_i$, so $r(P_i z) \in F$. Then $s := r \circ P_i$ is sc-smooth on U^* (chain rule), maps U^* into $F \subseteq U^*$, and fixes F; so it is a retraction onto F, and $U^* \subseteq U$. The preimages under P_i are needed because P_i need not be norm-decreasing for the norm of E_0; the exposition's "$|P_1 z - e|_0 \le |z - e|_0$" holds only for special norms. *F is a sub-M-polyfold of O*: in the chart id of O, s restricted to $O \cap U^*$ is an sc-smooth retraction onto F Chapter2/TameRetracts · LocalModel.selfChart_symm_apply Chapter2/TameRetracts · LocalModel.isSubMPolyfold_of_isFace.

The face as a model space. Insert and delete a coordinate: $\mathbb{R}^k \to \mathbb{R}^{k+1}$, $b \mapsto (b_1, \dots, 0_i, \dots, b_k)$, and back Chapter2/TameFaces · insertCoordL Chapter2/TameFaces · removeCoordL, as sc-operators on $\mathbb{R}^k \oplus W$ Chapter2/TameFaces · insertCoordModel Chapter2/TameFaces · removeCoordModel; composed with T they give $\iota : \mathbb{R}^k \oplus W \to E$ onto f_i and $\pi : E \to \mathbb{R}^k \oplus W$ Chapter2/TameFaces · faceIncl Chapter2/TameFaces · faceProj, with $\pi\iota = 1$, $\iota\pi = 1$ on f_i Chapter2/TameFaces · faceProj_faceIncl Chapter2/TameFaces · faceIncl_faceProj, $\iota(C') \subseteq C$ and $\pi(C) \subseteq C'$ for $C' = [0,\infty)^k \oplus W$ Chapter2/TameFaces · faceIncl_mem Chapter2/TameFaces · faceProj_mem, and $d_C(y) = d_{C'}(\pi y) + 1$ on f_i Chapter2/TameFaces · degIndexWith_eq_quadrantDeg_faceProj_add_one. The derivative of an sc-operator is itself Chapter2/TameFaces · ScOperator.scDeriv_eq, and the chain rule at a point holds between arbitrary sc-domains Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain.

Proposition 2.5.15 (preprint 2.43), the tame model Chapter2/TameFaces · proposition_2_43_tameModel. With U^* and s as above, put $U' := \iota^{-1}(U^*)$ (relatively open in C') and $r' := \pi \circ r \circ \iota$ on U'. Since $r \circ \iota$ maps U' into $F \subseteq f_i$, where $\iota\pi = 1$, r' is an sc-smooth retraction with retract $\pi(F)$, and ι maps it back into F. *Tameness (1)*: for $z \in U'$, $d_{C'}(r'z) + 1 = d_C(r(\iota z)) = d_C(\iota z) = d_{C'}(z) + 1$. *Tameness (2)*: at a smooth $x \in \pi(F)$, ιx is a smooth point of O and $Dr'(x)v = \pi(Dr(\iota x)(\iota v))$ by the chain rule; for j with $x_j = 0$ the coordinate $j' = i.\text{succAbove}(j)$ vanishes at ιx, and Proposition 2.5.3 (preprint 2.35) for r gives $\lambda_{j'}(Dr(\iota x)\iota v) = \lambda_{j'}(\iota v) = v_j$; so $\lambda'_j \circ Dr'(x) = \lambda'_j$, and $(1 - Dr'(x))E'$ is an sc-complement of $T_x\pi(F)$ inside E'_x. The exposition instead intersects the complement A of T_xO with f_1; that is not needed. And $d_{C'}(\pi y) + 1 = d_C(y)$ on F.

Proposition 2.5.15 (preprint 2.43), faces are tame Chapter2/TameFaces · LocalModel.proposition_2_43_isTame. For a face F of a tame retract O (paracompact, being closed), the chart $\pi : F \to \pi(F)$ Chapter2/TameFaces · MPolyfold.faceChart Chapter2/TameFaces · MPolyfold.faceChartInv is compatible with the sub-charts of F built on the chart id of O, its transition maps being the sc-operators ι and π Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart; a chart compatible with all sub-charts built on one chart of X is a chart of the induced structure, by locality of compatibility Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible. Its model is the tame retract $(\pi(F), C', E')$, so F is tame (a space all of whose points lie in tame charts is tame Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart), and the tame chart realizes the index: $d_F(x) = d_{C'}(\pi x) = d_C(x) - 1 = d_O(x) - 1$ (equation (14) twice). The text calls this last

statement obvious.

Proposition 2.5.16 (preprint 2.44), first sentence `Chapter2/TameFaces · MPolyfold.proposition_2_44`. In a tame chart φ around x: $\varphi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O (Proposition 2.5.14 (preprint 2.42) with Corollary 2.5.13 (preprint 2.41)), and each face of O is a tame sub-M-polyfold with $d_F + 1 = d_O$, where $d_O(\varphi x) = d_X(x)$.

The corner as a model space. For $\sigma \subseteq \{1, \dots, n\}$ list the coordinates outside σ as $j \mapsto l_j$ `Chapter2/CornerGerms · cornerIdx` `Chapter2/CornerGerms · cornerIdx_injective` `Chapter2/CornerGerms · exists_cornerIdx`. Deleting the coordinates in σ and inserting zeros there `Chapter2/CornerGerms · cornerRemoveL` `Chapter2/CornerGerms · cornerInsertL` are mutually inverse on $\{a : a_l = 0,\ l \in \sigma\}$ `Chapter2/CornerGerms · cornerRemoveL_cornerInsertL` `Chapter2/CornerGerms · cornerInsertL_cornerRemoveL`; as sc-operators on $\mathbb{R}^n \oplus W$ and $\mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerInsertModel` `Chapter2/CornerGerms · cornerRemoveModel`, composed with T, they give $\iota_\sigma : \mathbb{R}^{n-\#\sigma} \oplus W \to E$ onto the corner $f_\sigma = \bigcap_{i\in\sigma} f_i$ and $\pi_\sigma : E \to \mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl` `Chapter2/CornerGerms · cornerProj`, with $\pi_\sigma\iota_\sigma = 1$ and $\iota_\sigma\pi_\sigma = 1$ on f_σ `Chapter2/CornerGerms · cornerProj_cornerIncl` `Chapter2/CornerGerms · cornerIncl_cornerProj`, $\iota_\sigma(C') \subseteq C$ and $\pi_\sigma(C) \subseteq C'$ for $C' = [0,\infty)^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl_mem` `Chapter2/CornerGerms · cornerProj_mem`, and $d_C(y) = d_{C'}(\pi_\sigma y) + \#\sigma$ on f_σ: the vanishing coordinates of y are those of σ together with the images l_j of the vanishing coordinates of $\pi_\sigma y$, disjointly `Chapter2/CornerGerms · degIndexWith_eq_quadrantDeg_cornerProj_add_card`. The sc-projection $P_\sigma = \iota_\sigma \circ \pi_\sigma$ `Chapter2/CornerGerms · cornerKill` sets the coordinates in σ to zero `Chapter2/CornerGerms · cornerKill_coord_mem`, fixes f_σ `Chapter2/CornerGerms · cornerKill_eq_self` and maps C into C `Chapter2/CornerGerms · cornerKill_mem`. For $\sigma = \{i\}$ these are the ι, π and P_i of Proposition 2.5.15 (preprint 2.43).

The retraction onto a corner `Chapter2/CornerGerms · corner_retraction`. Let r be tame for T, $o \in O$ and $\sigma \subseteq I(o)$. For each $i \in \sigma$, Lemma 4.C at o `Chapter2/TameRetracts · local_face_structure` gives a face F_i of the retract with $o \in F_i \subseteq f_i$, containing a point with $d_C = 1$, and a ball around o in which r maps the points of $C \cap f_i$ into F_i. Distinct i give distinct faces: a point $p \in F_i$ with $d_C(p) = 1$ has $I(p) = \{i\}$, so $F_i = F_j$ forces $i = j$ `Chapter2/TameRetracts · zeroSet_eq_singleton`. Take a ball B around o inside all these balls with $B \cap C \subseteq U$, an open W_s with $W_s \cap U = r^{-1}(B) \cap U$ (continuity of r), and put $V := B \cap P_\sigma^{-1}(B \cap W_s)$, an open set containing o (since $P_\sigma o = o = r(o)$). On $V \cap C$ the map $s := r \circ P_\sigma$ is sc-smooth by the chain rule. For $z \in V \cap C$, $P_\sigma z \in C \cap f_\sigma \cap B$, so $r(P_\sigma z) \in F_i \subseteq f_i$ for every $i \in \sigma$: $s(z) \in O \cap f_\sigma$. Moreover $s(z) \in B$ (as $P_\sigma z \in W_s$), $P_\sigma s(z) = s(z)$, and $s(z) \in W_s$ (as $r(s(z)) = s(z) \in B$); so s maps $V \cap C$ into itself. On $f_\sigma \cap O$, P_σ and r are the identity, so $s \circ s = s$: s is an sc-smooth retraction with retract $O \cap V \cap f_\sigma$, and this set equals $O \cap V \cap \bigcap_{i\in\sigma} F_i$ (a point of $O \cap V \cap f_\sigma$ is fixed by r, which maps it into each F_i; conversely $F_i \subseteq f_i$). The exposition proceeds by induction on $\#\sigma$, identifying the germ of $F_1 \cap F_2$ with a face of F_1; the formal proof removes all coordinates of σ at once.

The tame model of a corner `Chapter2/CornerGerms · corner_tameModel`. Let s be an sc-smooth retraction on $U_s \subseteq U$, equal to $r \circ P_\sigma$ there, with retract $F \subseteq O \cap f_\sigma$; write $U_s = W_0 \cap C$ with W_0 open, and put $U' := \iota_\sigma^{-1}(W_0) \cap C'$ and $r' := \pi_\sigma \circ r \circ \iota_\sigma$ on U'. For $z \in U'$, $r(\iota_\sigma z) = s(\iota_\sigma z) \in F$ (as $P_\sigma\iota_\sigma = \iota_\sigma$), so r' maps U' into $\pi_\sigma(F) \subseteq U'$; it is sc-smooth by the chain rule and idempotent because $\iota_\sigma\pi_\sigma = 1$ on F and $r = 1$ on $F \subseteq O$; its retract is $\pi_\sigma(F)$. *Tameness (1):* $d_{C'}(r'z) + \#\sigma = d_C(r(\iota_\sigma z)) = d_C(\iota_\sigma z) = d_{C'}(z) + \#\sigma$ (the index formula on f_σ twice, and tameness (1) of r) `Chapter1/Degeneracy · degIndex_eq`. *Tameness (2):* at a smooth x of the retract, $\iota_\sigma x$ is a smooth point of O, and $Dr'(x)v = \pi_\sigma(Dr(\iota_\sigma x)\,\iota_\sigma v)$ by the chain rule between sc-domains `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain` `Chapter2/TameFaces · ScOperator.scDeriv_eq`; for j with $x_j = 0$ the coordinate l_j vanishes at $\iota_\sigma x$ `Chapter2/CornerGerms · coord_cornerIncl_idx`, and Proposition 2.5.3 (preprint 2.35) gives $\lambda_{l_j}(Dr(\iota_\sigma x)\,\iota_\sigma v) = \lambda_{l_j}(\iota_\sigma v) = v_j$ `Chapter2/TameRetracts · coord_scDeriv_eq`; so $(1 - Dr'(x))E'$ lies in E'_x and is an sc-complement of $T_x\pi_\sigma(F)$ `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. And $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ for $y \in F$. This is the proof of the tame model of Proposition 2.5.15 (preprint 2.43) with σ in place of $\{i\}$.

Proposition 2.5.16 (preprint 2.44), second sentence `Chapter2/CornerGerms · MPolyfold.proposition_2_44_corner`. Let X be tame and $x \in X$, and take a chart $\varphi : V \to O$ of the structure around x with a tame retraction r

onto O, tame for a presentation T Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart. The face germs at x are the faces F_i, $i \in I(\varphi x)$, of the retract (Corollary 2.5.13 (preprint 2.41): there are $d_X(x)$ of them), and σ is a set of such indices. The retraction onto the corner gives the pairwise distinct faces $F_i \ni \varphi(x)$, $i \in \sigma$, an open $B \ni \varphi(x)$ and s with retract $O \cap B \cap \bigcap_{i\in\sigma} F_i$. Put $A := \{y \in V : \varphi(y) \in B,\ \varphi(y) \in F_i\ \forall i \in \sigma\}$; its germ at x is the germ F^σ_x. Since $\varphi(A)$ is the retract of s, A is a sub-M-polyfold by the local criterion of Theorem 2.5.17 (preprint 2.45) Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local (paracompact as every subset Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype). The chart $\pi_\sigma \circ \varphi$ of A onto the retract $\pi_\sigma(\varphi(A))$ of r' Chapter2/TameFaces · MPolyfold.faceChart is compatible with the sub-charts of A built on φ, its transitions being ι_σ and π_σ Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart, hence a chart of the induced structure Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible; its model is tame Chapter2/TameFaces · ScRetraction.IsTameWith.isTame, so A is tame Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart. Both tame charts realize the index Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith, so for $a \in A$: $d_A(a) + \#\sigma = d_{C'}(\pi_\sigma\varphi(a)) + \#\sigma = d_C(\varphi(a)) = d_X(a)$; in particular $d_A(x) = d_X(x) - \#\sigma$, and the codimension, the number of quadrant coordinates removed in the tame model, is $\#\sigma$. The text states the proposition without proof.

Theorem 2.5.17 (preprint 2.45). *The interior of a face.* $F \cap \{d_X = 1\} = G$ for $F = \overline{G}$ Chapter2/TameRetracts · MPolyfold.theorem_2_45_interior: at $y \in F$ with $d_X(y) = 1$, in a tame chart with $I(\varphi y) = \{i\}$, all points of $\{d_X = 1\}$ near y lie in $\varphi^{-1}(\hat{F})$ for the single component $\hat{F}$ of Lemma 4.C at $\varphi(y)$, a connected subset of $\{d_X = 1\}$ containing y; it meets G (as $y \in \overline{G}$), hence lies in G. *A local criterion for sub-M-polyfolds* Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local: if near each $a \in A$, in a chart φ, $\varphi(A \cap V)$ agrees on an open $B \ni \varphi(a)$ with the retract of an sc-smooth retraction s contained in O, then A is a sub-M-polyfold, with the retraction $\varphi^{-1} \circ s \circ \varphi$ on $V' = \varphi^{-1}(B \cap W_0 \cap s^{-1}(B))$ — shrunk so that s maps it into itself, a step the exposition's "transport by φ^{-1}" omits. *G is a sub-M-polyfold* Chapter2/TameFaces · MPolyfold.theorem_2_45_isSubMPolyfold_interior: at $y \in G$, the local face $\overline{\hat{F}} \cap O$ agrees with $\varphi(G)$ on the ball of Lemma 4.C (both consist of the points of S there), and Proposition 2.5.15 (preprint 2.43) gives its retraction. *The face-structured case.* Near $x \in F$, in a tame chart, $\varphi(F)$ agrees on a ball with a face of the retract Chapter2/TameFaces · MPolyfold.exists_localFace_of_isFaceStructured: the map $i \mapsto F_i$ (the closure of the component of $\varphi^{-1}(\hat{F}_i)$) from $I(\varphi x)$ to the faces through x is onto (as in Proposition 2.5.14 (preprint 2.42)), and since there are exactly $d_X(x) = \#I(\varphi x)$ faces through x, it is injective; so the points of G near x lie in $\varphi^{-1}(\hat{F}_j)$ for the one j with $F = F_j$. Hence F is a sub-M-polyfold Chapter2/TameFaces · MPolyfold.theorem_2_45_isSubMPolyfold_face, and with the charts $\pi \circ \varphi$ onto the restrictions of the tame retractions r' it is tame with $d_X(x) = d_F(x) + 1$ Chapter2/TameFaces · MPolyfold.theorem_2_45_isTame_face. The local and global face counts of the theorem are Proposition 2.5.14 (preprint 2.42) and the definition of face-structured.

II.17 Products and fibred products: Lemma 4.D and Proposition 2.5.20 (preprint 2.46)

Products of partial quadrants. For presentations $T : E \cong \mathbb{R}^n \oplus W$ and $T' : E' \cong \mathbb{R}^{n'} \oplus W'$ of C and C', concatenating the coordinates ($\mathbb{R}^n \times \mathbb{R}^{n'} \cong \mathbb{R}^{n+n'}$ Chapter2/FibredProduct · finAppendL) gives a presentation $E \oplus E' \cong \mathbb{R}^{n+n'} \oplus (W \oplus W')$ Chapter2/FibredProduct · prodPresentation of $C \times C'$ Chapter2/FibredProduct · prod_eq_prodPresentation, so $C \times C'$ is a partial quadrant Chapter2/FibredProduct · IsPartialQuadrant.prod and $U \times U'$ is relatively open in it Chapter2/FibredProduct · IsRelOpenInQuadrant.prod. The vanishing coordinates of (x, x') are those of x followed by those of x', so $d_{C\times C'}(x, x') = d_C(x) + d_{C'}(x')$ Chapter2/FibredProduct · degIndexWith_prodPresentation Chapter2/FibredProduct · degIndex_prod, and the minimal subspace is $E_x \oplus E'_{x'}$ Chapter2/FibredProduct · minimalSubspaceWith_prodPresentation.

Products of retractions. $f \times f'$ is sc-smooth when f and f' are Chapter2/FibredProduct · IsScSmooth.prodMap, with derivative $Df(x) \times Df'(x')$ Chapter2/FibredProduct · scDeriv_prodMap; so $r \times r'$ is an sc-smooth retraction Chapter2/FibredProduct · ScRetraction.prod with retract $O \times O'$ Chapter2/FibredProduct · ScRetraction.prod_

retract Chapter2/FibredProduct · IsScRetract.prod, and products of sc-smooth maps between retracts are sc-smooth Chapter2/FibredProduct · IsScSmoothRetractMap.prodMap. Products of sc-subspaces and of sc-splittings are again such Chapter2/FibredProduct · IsScSubspace.prod Chapter2/FibredProduct · IsScSplitting.prod; hence if r, r' are tame for T, T', then $r \times r'$ is tame for the product presentation Chapter2/FibredProduct · ScRetraction.IsTameWith.prod: (1) is the additivity of the index, and for (2) the complement of $T_{(x,x')}(O \times O') = T_xO \oplus T_{x'}O'$ is $A \oplus A'$ inside $E_x \oplus E'_{x'}$.

Lemma 4.D (the product M-polyfold). $X \times Y$ is paracompact, being metrizable by Theorem 2.3.10 (preprint 2.16) Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The product charts Chapter2/FibredProduct · MChart.prod have the product retracts as models Chapter2/FibredProduct · LocalModel.prod, their overlaps are products Chapter2/FibredProduct · MChart.overlap_prod, and products of compatible charts are compatible Chapter2/FibredProduct · MChart.Compatible.prod; so they form an atlas of an M-polyfold structure on $X \times Y$ Chapter2/FibredProduct · MPolyfold.prod, in which products of charts of the factors are charts Chapter2/FibredProduct · MPolyfold.IsChart.prod. In a product chart the index is additive Chapter2/FibredProduct · MPolyfold.chartDegIndex_prod; choosing charts that realize $d_X(x)$ and $d_Y(y)$ gives $d_{X\times Y}(x,y) \le d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le. If X and Y are tame, products of tame charts are tame charts, so $X \times Y$ is tame, and since a tame chart realizes the index (§II.14), $d_{X\times Y}(x,y) = d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.isTame_prod. The text's "$d_{X\times Y} = d_X + d_Y$" is thus a statement about tame spaces. The projections are sc-smooth Chapter2/FibredProduct · MPolyfold.isScSmoothMap_fst Chapter2/FibredProduct · MPolyfold.isScSmoothMap_snd and so is the pairing $z \mapsto (f(z), g(z))$ of sc-smooth maps Chapter2/FibredProduct · MPolyfold.IsScSmoothOn.prodMk.

Proposition 2.5.20 (preprint 2.46). A *local sc-diffeomorphism* $f : X \to Z$ is read as: every point has an open neighbourhood U that f maps sc-diffeomorphically onto an open subset Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo; such an f is continuous and sc-smooth Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.continuous Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.isScSmoothMap. The fibred product $X_f \times_g Y = \{(x,y) : f(x) = g(y)\}$ Chapter2/FibredProduct · MPolyfold.fibredProduct is closed in $X \times Y$ Chapter2/FibredProduct · MPolyfold.isClosed_fibredProduct, hence paracompact, which the text does not mention. *Sub-M-polyfold.* Near (x,y), with $f|_U : U \to W$ an sc-diffeomorphism and $V = g^{-1}(W)$, the map $R(x',y') = ((f|_U)^{-1}(g(y')), y')$ on $U \times V$ is sc-smooth, idempotent, with image the fibred product near (x,y); in a product chart it is an sc-smooth retraction Chapter2/FibredProduct · MPolyfold.fibredSubDataL, and so the fibred product is a sub-M-polyfold Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct (symmetrically when g is the local sc-diffeomorphism Chapter2/FibredProduct · MPolyfold.fibredSubDataR Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct'). *π_2 is a local sc-diffeomorphism.* On the fibred product near (x,y), π_2 is sc-smooth with inverse $y' \mapsto ((f|_U)^{-1}(g(y')), y')$, a pairing of sc-smooth maps with values in the fibred product, hence sc-smooth into the sub-M-polyfold by Proposition 2.3.19 (2) (preprint 2.21) Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct; symmetrically for π_1 when g is a local sc-diffeomorphism Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_fst_fibredProduct. The exposition factors this inverse through $y' \mapsto (x, y')$, which is not sc-smooth unless x is a smooth point (a constant map is sc^0 only when its value lies in X_∞); the formula above has the same values and avoids it. *The index.* Proposition 2.4.4 (preprint 2.24) along π_2 gives $d_{X_f\times_g Y}(x,y) = d_Y(y)$ for arbitrary Y Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_snd, and along π_1, when g is a local sc-diffeomorphism, $= d_X(x)$ Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_fst. *Tameness.* If Y is tame, the charts $\psi \circ \pi_2$, ψ a tame chart of Y restricted to $g^{-1}(W)$, are charts of the fibred product with tame models Chapter2/FibredProduct · MPolyfold.isTame_fibredProduct. Together: Proposition 2.5.20 (preprint 2.46) Chapter2/FibredProduct · MPolyfold.proposition_2_46, and in its last statement $d = d_X = d_Y$ for arbitrary X, Y and, for tame X, Y, $2d = d_{X\times Y}$ by Lemma 4.D Chapter2/FibredProduct · MPolyfold.proposition_2_46_of_isLocalScDiffeo.

II.18 Strong bundle retractions (Definitions 2.6.1–2.6.3 (preprint 2.47–2.49))

$U \lhd F$. The double filtration is $U_m \times F_k$ `Chapter2/Tame · doubleLevel`. $C \times F_0$ is a partial quadrant in $E \oplus F$, through $(\mathbb{R}^n \oplus W) \oplus F \cong \mathbb{R}^n \oplus (W \oplus F)$ `Chapter2/StrongBundle · prodQuadrantIso` `Chapter2/StrongBundle · IsPartialQuadrant.prod_univ`, and $U \times F_0$ is relatively open in it `Chapter2/StrongBundle · IsRelOpenInQuadrant.prod_univ` — so "sc-smooth on $(U \lhd F)[i] = U \oplus F^i$" makes sense (exposition Lemma 1.B of Chapter 1). Pairings of sc^k maps are sc^k, with the tangent map rearranged by an sc-operator `Chapter2/StrongBundle · scPair` `Chapter2/StrongBundle · tangentProdSwap` `Chapter2/StrongBundle · IsSck.scPair` `Chapter2/StrongBundle · IsScSmooth.scPair`, and the projections and inclusions of products are sc-operators `Chapter2/StrongBundle · ScBanach.prodFstOp` `Chapter2/StrongBundle · ScBanach.prodSndOp` `Chapter2/StrongBundle · ScBanach.prodInlOp` `Chapter2/StrongBundle · ScBanach.prodShiftInclOp`.

Definition 2.6.1 (preprint 2.47). A strong bundle map maps $U \times F_0$ into $U' \times F_0'$, has first component independent of h and second linear in h, maps $U_m \times F_k$ into $U_m' \times F_k'$ for $k \le m+1$, is sc-smooth on $U \times F_0$, and its fibre-lifted map on $U \times F_1$ `Chapter2/Tame · shiftFiber` is sc-smooth for $E \oplus F^1$ `Chapter2/Tame · IsStrongBundleMap`; a strong bundle retraction is an idempotent one `Chapter2/Tame · IsStrongBundleRetraction`. *Lemma 5.A.* The base $r(u) = R(u,0)_1$ `Chapter2/Tame · IsStrongBundleMap.base` `Chapter2/StrongBundle · IsStrongBundleRetraction.apply_fst` is an sc-smooth retraction on U: sc-smooth as R composed with $u \mapsto (u,0)$ and the first projection, idempotent by the first component of $R \circ R = R$ `Chapter2/StrongBundle · IsStrongBundleRetraction.baseRetraction`. For $u \in O$, $R(u,h) = (u, \Gamma(u,h))$, so the second component of $R \circ R = R$ reads $\Gamma(u,\Gamma(u,h)) = \Gamma(u,h)$: $\Gamma(u,\cdot)$ is a projection `Chapter2/StrongBundle · IsStrongBundleRetraction.fiber_idem`. R is an sc-smooth retraction on $U \times F_0$ in $C \times F_0$ `Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetraction` and, through the lift of the fibre component to F_1 (possible since R preserves the double filtration at $(0,1)$ `Chapter2/StrongBundle · IsStrongBundleRetraction.snd_mem_range` `Chapter2/StrongBundle · IsStrongBundleRetraction.toBase_shiftFiber_snd`), on $U \times F_1$ for $E \oplus F^1$ `Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne`.

Definition 2.6.2 (preprint 2.48). K is a local strong bundle retract if $K = R(U \times F_0)$ `Chapter2/Tame · IsStrongBundleRetract`, with $K_{m,k}$, $K[i]_m$ and $K[1] \subseteq E_0 \times F_1$ `Chapter2/Tame · strongLevel` `Chapter2/Tame · strongLevelShift` `Chapter2/Tame · strongShiftOne` and the projection p `Chapter2/Tame · strongProj`. $K = \{(u,h) : u \in O, \Gamma(u,h) = h\}$ `Chapter2/StrongBundle · IsStrongBundleRetraction.mem_image_iff`. *Lemma 5.B.* $K[1]$ is the image of R on $U \times F_1$ `Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne_retract`, so K and $K[1]$ are retracts `Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract` `Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract_one`; the base $p(K)$ is O `Chapter2/StrongBundle · IsStrongBundleRetract.base` `Chapter2/StrongBundle · IsStrongBundleRetract.base_eq`; and $p : K[i] \to O$ is sc-smooth, because $p \circ R = r \circ \mathrm{pr}_1$ is `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst` `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one`, as is the inclusion $K[1] \to K[0]$ `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl`.

Definition 2.6.3 (preprint 2.49). An sc-smooth section is sc-smooth from (O, C, E) into $(K, C \times F_0, E \oplus F)$ with $f(x)_1 = x$ `Chapter2/Tame · IsScSmoothSection`; an sc^+-section is sc-smooth into $K[1]$ `Chapter2/Tame · IsScPlusSection` `Chapter2/Tame · IsScPlusSection.mem`. Composed with the inclusion $K[1] \to K[0]$ (Theorem 2.1.15 (preprint 2.8)), an sc^+-section is an sc-smooth section `Chapter2/StrongBundle · IsScPlusSection.isScSmoothSection`.

II.19 Strong bundles over M-polyfolds (Definitions 2.6.4–2.6.7 (preprint 2.50–2.53))

Definition 2.6.4 (preprint 2.50). For $P : Y \to X$ with linear structures on the fibres, a strong bundle chart is a chart φ of X (of the structure — the text says only "a homeomorphism onto O"), a strong bundle retraction R on $U \subseteq C$ with base retract O, and a homeomorphism

$\Phi : P^{-1}(V) \to K$ with $\Phi(y)_1 = \varphi(P(y))$, linear on fibres Chapter2/StrongBundle · StrongBundleChart Chapter2/StrongBundle · StrongBundleChart.K. K is a local strong bundle retract and a retract on $[0]$ and $[1]$ Chapter2/StrongBundle · StrongBundleChart.isStrongBundleRetract Chapter2/StrongBundle · StrongBundleChart.isScRetract Chapter2/StrongBundle · StrongBundleChart.isScRetract_one Chapter2/StrongBundle · StrongBundleChart.model, the source $P^{-1}(V)$ is open Chapter2/StrongBundle · StrongBundleChart.isOpen_source, $\Phi(P^{-1}(V)) = K$ and $p(K) = O$ Chapter2/StrongBundle · StrongBundleChart.image_source Chapter2/StrongBundle · StrongBundleChart.base_K. "Bounded on the fibres" needs no hypothesis: the restriction of the continuous Φ to a fibre is continuous Chapter2/StrongBundle · StrongBundleChart.continuous_fiber. Two charts are compatible if the transitions $\Phi' \circ \Phi^{-1}$ Chapter2/StrongBundle · StrongBundleChart.overlap Chapter2/StrongBundle · StrongBundleChart.transition and back are sc-smooth between the overlaps on $[0]$ and, fibre-lifted, on $[1]$, and carry fibre components in F_1 to F_1' Chapter2/StrongBundle · StrongBundleChart.Compatible Chapter2/StrongBundle · StrongBundleChart.Compatible.symm; the last condition, implicit in the text, is what makes $(\Phi' \circ \Phi^{-1})[1]$ the restriction of $\Phi' \circ \Phi^{-1}$. Atlases and their equivalence Chapter2/StrongBundle · StrongBundleAtlas Chapter2/StrongBundle · StrongBundleAtlas.Equiv; Banachable fibres Chapter2/StrongBundle · IsBanachable.

Definition 2.6.5 (preprint 2.51). A strong bundle is a continuous surjection with Banachable fibres and a strong bundle atlas, Y Hausdorff paracompact Chapter2/StrongBundle · StrongBundle. Each Φ is a chart of Y with model $(K, C \times F_0, E \oplus F)$ Chapter2/StrongBundle · StrongBundleChart.toMChart, compatibility of strong bundle charts on $[0]$ is compatibility of these charts Chapter2/StrongBundle · StrongBundleChart.Compatible.toMChart, and they form the atlas of the M-polyfold $Y[0]$ Chapter2/StrongBundle · StrongBundle.atlas0 Chapter2/StrongBundle · StrongBundle.toMPolyfold0 Chapter2/StrongBundle · StrongBundle.isChart_toMChart. $P : Y[0] \to X$ is sc-smooth: in the charts Φ and φ it is $p : K \to O$ (Lemma 5.B), and one pair of charts per point suffices Chapter2/StrongBundle · StrongBundle.isScSmoothMap_proj. The double filtration $Y_{m,k}$ is defined through all the charts of the atlas around a point Chapter2/StrongBundle · StrongBundle.doubleLevel.

$Y_{m,k}$ does not depend on the chart Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff. Let y lie in the domain of a chart Φ of the atlas with $\Phi(y) \in E_m \times F_k$, $k \le m+1$, and let Φ' be another chart around y; then $\Phi'(y) = (\Phi' \circ \Phi^{-1})(\Phi(y))$ Chapter2/StrongBundlePullback · StrongBundleChart.transition_apply. *The base component:* $\Phi'(y)_1 = \varphi'(\varphi^{-1}(\Phi(y)_1))$, and the base transition, sc-smooth and so sc^0, maps points of level m to points of level m Chapter2/MPolyfold · MChart.transition_apply. *The fibre component, $k \le m$:* $\Phi(y)$ lies on level k of $E \oplus F$ (levels decrease Chapter2/StrongBundleOne · StrongBundle.range_toBase_mono), and the $[0]$-transition, sc^0 on the overlap retract Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap, keeps it on level k. *The case $k = m+1$:* the fibre component of $\Phi(y)$ lies in F_1, so its lift $\Phi(y)[1]$ to $E_0 \times F_1$ Chapter2/StrongBundleOne · StrongBundleChart.liftOne Chapter2/StrongBundleOne · StrongBundleChart.toBase_liftOne is a point of level m of $E \oplus F^1$ (level m of F^1 is F_{m+1} Chapter1/ScBanach · ScBanach.shiftScale_toBase); the $[1]$-transition, sc^0 for $E \oplus F^1$ on the overlap Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one, keeps it on level m, and its value is the lift $\Phi'(y)[1]$, because the F_1-condition of compatibility puts the fibre component of $\Phi'(y)$ in F_1'. So $\Phi'(y) \in E_m' \times F_{m+1}'$. The case $k \le m$ uses the $[0]$-transition only; the case $k = m+1$ uses exactly the F_1-condition made explicit in Definition 2.6.4 (preprint 2.50).

The set $Y[1]$ and its topology. $Y_{0,1}$ is the set of points whose fibre component lies in F_1 in every chart of the atlas around them Chapter2/StrongBundleOne · StrongBundle.oneSet; by the F_1-condition one chart suffices Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff. $Y[1]$ is $Y_{0,1}$ as a type Chapter2/StrongBundleOne · StrongBundle.One with the topology generated by the sets $\{y \in \mathrm{dom}\,\Phi : \Phi(y)[1] \in W\}$, W open in $E_0 \times F_1$, as for X_m in §2.3 (preprint §2.2) Chapter2/StrongBundleOne · StrongBundle.isOpen_one_basic. The inclusion $Y[1] \to Y$ is continuous Chapter2/StrongBundleOne · StrongBundle.continuous_oneVal: inside the domain of a chart Φ, an open subset of Y is the preimage under Φ of an open $W \subseteq E_0 \times F_0$ (Φ is a homeomorphism onto K Chapter2/MPolyfold · MChart.exists_isOpen_image_eq), and its points in $Y[1]$ form the basic set of $(1 \times \iota)^{-1}(W)$. A continuous injection into the Hausdorff space Y makes $Y[1]$ Hausdorff Mathlib/Topology/Separation/Hausdorff · T2Space.of_injective_continuous.

The charts $\Phi[1]$ Chapter2/StrongBundleOne · StrongBundle.oneChart. $y \mapsto \Phi(y)[1]$ maps $P^{-1}(V) \cap Y[1]$ into $K[1]$ Chapter2/StrongBundleOne · StrongBundleChart.liftOne_mem, with inverse $z \mapsto \Phi^{-1}(z_1, \iota z_2)$, a point of $Y_{0,1}$

Chapter2/StrongBundleOne · StrongBundle.symm_mem_oneSet Chapter2/StrongBundleOne · StrongBundle.oneChartInv Chapter2/StrongBundleOne · StrongBundle.liftOne_symm. It is continuous by the definition of the topology. Its inverse is continuous: a basic open set $\{\Phi'(y)[1] \in W\}$ pulls back to the set of $z \in K[1]$ in the $[1]$-part of the overlap with $(\Phi' \circ \Phi^{-1})[1](z) \in W$; the overlap is $K \cap W_0$ for an open W_0 Chapter2/StrongBundlePullback · StrongBundleChart.overlap_eq_inter and the $[1]$-transition is continuous on it, so this set is open in $K[1]$. The overlap of $\Phi[1]$ and $\Phi'[1]$ is the $[1]$-part of the overlap of Φ and Φ' Chapter2/StrongBundleOne · StrongBundle.oneChart_overlap, and their transition map is the $[1]$-transition $(\Phi' \circ \Phi^{-1})[1]$ Chapter2/StrongBundleOne · StrongBundle.oneChart_transition, sc-smooth in both directions by Definition 2.6.4 (preprint 2.50); so the charts $\Phi[1]$ are compatible Chapter2/StrongBundleOne · StrongBundle.oneChart_compatible and form an atlas of $Y[1]$ Chapter2/StrongBundleOne · StrongBundle.atlas1. $Y[1]$ is metrizable: P restricted to $Y[1]$ is continuous into the paracompact Hausdorff X, and over the domain V of a chart of X its preimage is the source of a chart $\Phi[1]$, homeomorphic to a retract and so metrizable Chapter2/Metrizable · MChart.metrizableSpace_source; the pull-back form of Smirnov's theorem applies Chapter2/Metrizable · metrizableSpace_of_pullback Chapter2/StrongBundleOne · StrongBundle.metrizableSpace_one. Hence $Y[1]$ is paracompact Chapter2/StrongBundleOne · StrongBundle.paracompactSpace_one, which an M-polyfold must be and the text does not mention, and $Y[1]$ **is an M-polyfold** Chapter2/StrongBundleOne · StrongBundle.toMPolyfold1 whose charts include the $\Phi[1]$ Chapter2/StrongBundleOne · StrongBundle.isChart_oneChart.

$P[1]$ and the inclusion $Y[1] \to Y[0]$. In the charts $\Phi[1]$ and φ, $P[1]$ is the projection $K[1] \to O$ of Lemma 5.B Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one, and one pair of charts per point suffices Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts; so $P[1] : Y[1] \to X$ **is sc-smooth** Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_proj1. In the charts $\Phi[1]$ and Φ the inclusion $Y[1] \to Y[0]$ is $(u, h) \mapsto (u, \iota h)$ Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl, so it is sc-smooth Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_oneVal (not stated in the text; it gives that an sc⁺-section is an sc-smooth section).

Definition 2.6.6 (preprint 2.52). A section has $P \circ f = \mathrm{id}$ Chapter2/StrongBundle · IsSection; it is sc-smooth if sc-smooth into $Y[0]$ Chapter2/StrongBundle · StrongBundle.IsScSmoothSection, and an sc⁺-section if in every chart its fibre values lie in F_1 and its fibre-lifted representative Chapter2/StrongBundle · StrongBundle.localRepOne is sc-smooth into $K[1]$ Chapter2/StrongBundle · StrongBundle.IsScPlusSection. *This is the text's definition* Chapter2/StrongBundleOne · StrongBundle.isScPlusSection_iff: f is an sc⁺-section iff it is a section and $f = \iota \circ g$ for an sc-smooth $g : X \to Y[1]$, ι the inclusion. Forward: the values of f lie in $Y_{0,1}$ (one chart suffices), which defines g; on the domain of a chart φ of X under a strong bundle chart Φ, $g = \Phi[1]^{-1} \circ (\text{the local representative}) \circ \varphi$, which gives the continuity of g, and the representative of g in the charts φ, $\Phi[1]$ is the chart-wise representative, sc-smooth by hypothesis. Backward: the fibre values lie in F_1 because g takes values in $Y_{0,1}$, and the chart-wise representative is the representative of g in φ, $\Phi[1]$, sc-smooth by Definition 2.3.8 (preprint 2.14) (a chart with empty domain is handled separately).

Definition 2.6.7 (preprint 2.53). $f^*Y = \{(z, y) : P(y) = f(z)\}$ Chapter2/StrongBundle · pullbackSpace with the projections Chapter2/StrongBundle · pullbackProj Chapter2/StrongBundle · pullbackProj', $P \circ P' = f \circ P_f$ Chapter2/StrongBundle · pullback_comm; P_f is continuous and f^*Y closed in $Z \times Y$ Chapter2/StrongBundle · continuous_pullbackProj Chapter2/StrongBundle · isClosed_pullbackSpace.

II.20 Proposition 2.6.8 (preprint 2.54): the pull-back strong bundle

The pulled-back retraction. For a retraction r' on U' and $g : O' \to O$ sc-smooth between retracts, $R_g(w, h) := (r'(w), \Gamma(g(r'(w)), h))$ Chapter2/StrongBundle · pullbackRetraction, where $g(r'(w)) \in O \subseteq U$ Chapter2/StrongBundle · pullback_mem_O Chapter2/StrongBundle · pullback_mem_U. It is a strong bundle retraction Chapter2/StrongBundle · isStrongBundleRetraction_pullback: it maps $U' \times F_0$ into itself; its first component does not depend on h and its second is linear in h; it preserves the double filtration (r' and $g \circ r'$ preserve levels, and R preserves the double filtration); on $[0]$ it is the pairing of $r' \circ \mathrm{pr}_1$ with the second component of $R \circ ((g \circ r') \circ \mathrm{pr}_1, \mathrm{pr}_2)$, sc-smooth by the chain rule and the pairing lemma, and likewise on $[1]$; and $R_g \circ R_g = R_g$ because $r' \circ r' = r'$ and $\Gamma(o, \cdot)$ is a projection for $o = g(r'(w)) \in O$ (Lemma

5.A). Its image is $K_g = \{(w,h) : w \in O', (g(w),h) \in K\}$ Chapter2/StrongBundle · pullback_image Chapter2/StrongBundle · isStrongBundleRetract_pullback.

The fibres. $P_f^{-1}(z) \to P^{-1}(f(z))$, $(z,y) \mapsto y$, is a bijection Chapter2/StrongBundlePullback · pullbackFiberEquiv; transporting the linear structure along it Chapter2/StrongBundlePullback · pullbackFiber_addCommGroup Chapter2/StrongBundlePullback · pullbackFiber_module makes it a linear homeomorphism Chapter2/StrongBundlePullback · pullbackFiberCLE, so the fibres of f^*Y are Banachable.

The charts. The data: a strong bundle chart c of Y, a chart ψ of Z with $f(\operatorname{dom}\psi) \subseteq \operatorname{dom}\varphi$, a retraction r_Z onto the model of ψ, and the sc-smoothness of $g = \varphi \circ f \circ \psi^{-1}$ Chapter2/StrongBundlePullback · PullbackChartData Chapter2/StrongBundlePullback · PullbackChartData.g Chapter2/StrongBundlePullback · PullbackChartData.g_smooth. The chart $\Phi_f(z,y) = (\psi(z), \mathrm{pr}_F\Phi(y))$ maps $P_f^{-1}(\operatorname{dom}\psi)$ into K_g — because $\Phi(y) = (g(\psi(z)),\cdot) \in K$ Chapter2/StrongBundlePullback · PullbackChartData.g_apply Chapter2/StrongBundlePullback · PullbackChartData.mem_Kg — with inverse $(w,h) \mapsto (\psi^{-1}(w), \Phi^{-1}(g(w),h))$ Chapter2/StrongBundlePullback · PullbackChartData.symm_mem_pullback Chapter2/StrongBundlePullback · PullbackChartData.invFun; both are continuous, it covers ψ and is linear on fibres, so it is a strong bundle chart of f^*Y Chapter2/StrongBundlePullback · PullbackChartData.toChart Chapter2/StrongBundlePullback · PullbackChartData.image_eq_Kg.

Compatibility. Sc-operators and pairings are sc-smooth between retracts Chapter2/StrongBundlePullback · ScOperator.isScSmoothRetractMap Chapter2/StrongBundlePullback · IsScSmoothRetractMap.scPair Chapter2/StrongBundlePullback · isScRetract_univ, and overlaps of strong bundle charts are retracts on $[0]$ and $[1]$ Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one. On the overlap of two pulled-back charts the transition map is

$$(w,h) \mapsto (\psi_2\psi_1^{-1}(w),\ \mathrm{pr}_F(\Phi_2\Phi_1^{-1})(g_1(w),h))$$

Chapter2/StrongBundlePullback · PullbackChartData.transitionFormula Chapter2/StrongBundlePullback · PullbackChartData.transition_eqOn: its first component is the transition of Z composed with the projection, its second the fibre component of the transition of Y composed with $(w,h) \mapsto (g_1(w),h)$ Chapter2/StrongBundlePullback · PullbackChartData.gPair, which maps the overlap into the overlap of Y's charts Chapter2/StrongBundlePullback · PullbackChartData.gPair_mem. Each factor is sc-smooth between retracts, and Theorem 2.1.15 (preprint 2.8) and the pairing lemma give the sc-smoothness on $[0]$ Chapter2/StrongBundlePullback · PullbackChartData.transition_isScSmooth; the same with F^1 on $[1]$ Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_eqOn Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_isScSmooth; fibre components in F_1 go to F_1 because the transitions of Y have this property Chapter2/StrongBundlePullback · PullbackChartData.transition_snd_mem_range. So any two pulled-back charts are compatible Chapter2/StrongBundlePullback · PullbackChartData.compatible_toChart.

Proposition 2.6.8 (preprint 2.54). f^*Y is paracompact: Z and $Y[0]$ are metrizable (Theorem 2.3.10 (preprint 2.16)), so $Z \times Y$ and its subspace f^*Y are metrizable, hence paracompact Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace — a requirement of Definition 2.6.5 (preprint 2.51) that the text does not mention. The pulled-back charts, built around every point from a chart of Y's atlas and a chart of Z restricted to $f^{-1}(\operatorname{dom}\varphi)$, cover f^*Y and form an atlas Chapter2/StrongBundlePullback · pullbackAtlas. With the continuous surjection P_f and its Banachable fibres this is a strong bundle structure on $f^*Y \to Z$ Chapter2/StrongBundlePullback · proposition_2_54 ("the easy proof is left to the reader").

II.21 What the formalization taught us

- **Theorem 2.3.10 (preprint 2.16) needs no added hypothesis, and not the text's route.** The text's proof applies Proposition 2.7.4 (preprint 2.57) to X_m without checking that X_m is regular. A partition of unity of X, pulled back along the continuous inclusion, metrizes X_m directly Chapter2/Metrizable · metrizableSpace_of_pullback, and the same lemma proves

Smirnov's theorem `Chapter2/Metrizable · metrizableSpace_of_locally`, Proposition 2.3.15 (3) (preprint 2.19) `Chapter2/Metrizable · proposition_2_19_3` and the paracompactness of f^*Y `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace`. With it every paracompactness hypothesis carried by the constructions is discharged `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`.

- **Compatibility is local, and the text needs it.** That the charts compatible with an atlas are compatible with each other is not in the text; it is the chain rule plus the locality of sc-smoothness `Chapter2/MPolyfold · MChart.compatible_of_locally`, and every chart-independence claim of §2.3 (preprint §2.2) rests on it.
- TO **is a fixed-point set.** Proposition 2.1.11 (preprint 2.4) and Proposition 2.1.13 (preprint 2.6) are one-line consequences of $p \in Tr(TU) \iff Tr(p) = p$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff`, which the text's proofs use without stating.
- **The tangent bundle needs the locality of** Tf**.** That Definition 2.3.12's (preprint 2.17) relation is an equivalence relation needs the chain rule on the triple overlap and the agreement of Tf computed on an open subset with Tf computed on the whole retract `Chapter2/TangentBundle · retractTangentMap_inter_isOpen` `Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport`.
- **Level** 1 **suffices for §2.4 (preprint §2.3).** Proposition 2.4.10 (preprint 2.30), Theorem 2.4.12 (preprint 2.31) and Corollary 2.4.13 (preprint 2.32) hold at points on level 1 and for sc^1 germs `Chapter2/Boundary · proposition_2_30_reduced` `Chapter2/Boundary · theorem_2_31` `Chapter2/DegeneracyIndex · corollary_2_32`; the text's "closure of $(T^R_xO)_\infty$" is the density of level-1 points in the kernel of a finite-rank map `Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase`.
- **Proposition 2.5.3 (preprint 2.35) needs only tameness (2), and Lemma 2.5.12 (preprint 2.40) needs (2) through it.** $Dr(x)E_x \subseteq E_x$ holds for every retraction `Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace`; that the retraction keeps points of a face in that face is the ray lemma `Chapter2/TameRetracts · ray_zeroSet`, which uses Proposition 2.5.3 (preprint 2.35).
- **Three slips in the exposition.** "$|P_1 z - e|_0 \le |z - e|_0$" in the exposition's proof of Proposition 2.5.15 (preprint 2.43) is false for a general norm on E_0; the formal neighbourhood intersects balls with their P_i-preimages `Chapter2/TameRetracts · proposition_2_43`. The "transport by φ^{-1}" in Theorem 2.5.17 (preprint 2.45) needs a shrinking so that the retraction maps its domain into itself `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local`. In Proposition 2.5.20 (preprint 2.46) the inverse of π_2 cannot be factored through the constant-in-x map $y' \mapsto (x, y')$, which is not sc-smooth at a non-smooth x `Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct`.
- **Tameness is needed only for the additivity of the index under products.** $d_{X\times Y} \le d_X + d_Y$ in general `Chapter2/FibredProduct · MPolyfold.degIndex_prod_le`, with equality for tame factors `Chapter2/FibredProduct · MPolyfold.isTame_prod`; the identities $d = d_X = d_Y$ of Proposition 2.5.20 (preprint 2.46) hold without tameness `Chapter2/FibredProduct · MPolyfold.proposition_2_46_of_isLocalScDiffeo`.
- **Definition 2.6.4 (preprint 2.50) needs two explicit conditions.** The base homeomorphism of a strong bundle chart must be a chart of X, and the $[1]$-transition must preserve fibre components in F_1 `Chapter2/StrongBundle · StrongBundleChart.Compatible`. The second condition is exactly what makes $Y_{0,1}$, and so the M-polyfold $Y[1]$, independent of the chart `Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff` `Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff`.
- **Proposition 2.5.6 (preprint 2.36) does not need the equality case of Theorem 2.4.12 (preprint 2.31).** For a tame retraction the smooth vectors of the splitting are $Dr(x)T^{-1}(e_{i_j}, 0)$, dual to the coordinates by Proposition 2.5.3 (preprint 2.35), and T^R_xO is an sc-subspace because $Dr(x)$ preserves E_x `Chapter2/PartialCone · proposition_2_36_partialCone`.
- **Corners at once, not by induction.** The intersections F^σ_x of Proposition 2.5.16 (preprint 2.44) are retracts of $r \circ P_\sigma$, by Lemma 4.C for each $i \in \sigma$; no identification of the germ of $F_1 \cap F_2$ with a face of F_1 is needed `Chapter2/CornerGerms · corner_retraction`.

II.22 The 2021 edition: Proposition 2.1.13 and local completeness (§2.1)

The Springer edition (2021) adds to §2.1 a theory of *local completeness*: Proposition 2.1.4, Definition 2.1.5, Remark 2.1.6, Corollary 2.1.7, Proposition 2.1.8, Lemma 2.1.9 and Proposition 2.1.10, pp. 22–25. It also

changes one statement, Proposition 2.1.13.

Proposition 2.1.13 (preprint 2.6), the 2021 statement. For sc-smooth retractions r, s onto the same retract O, a retraction ρ onto O', and f with $f(O) \subseteq O'$, three things hold `Chapter2/Book2021 · proposition_2_1_13_2021`: - (1) $f \circ r$ is sc-smooth iff $f \circ s$ is; - (2) if $f \circ r$ is sc-smooth, $T(f \circ r) = T(f \circ s)$ on TO; - (3) if $f \circ r$ is sc-smooth, $T(f \circ r)$ maps TO into TO'.

The 2021 statement of (3) carries the hypothesis "$f \circ r$ sc-smooth", without which $T(f \circ r)$ is not defined. The formal statement of §II.3 had read (3) this way from the start `Chapter2/Retracts · proposition_2_6_3`, so the 2021 theorem is the conjunction of the three earlier ones. The first is used in both directions `Chapter2/Retracts · proposition_2_6_1` `Chapter2/Retracts · proposition_2_6_2`.

Complete metrics on a subspace. "W admits a metric d defining its topology for which K is complete" is a metric structure on W whose topology is the given one, with K complete for it `Chapter2/LocalTopology · AdmitsCompleteMetricOn`. A metric space admits its own metric `Chapter2/LocalTopology · admitsCompleteMetricOn_of_isComplete`, and such a W is metrizable `Chapter2/LocalTopology · AdmitsCompleteMetricOn.metrizableSpace`. The property passes to smaller sets `Chapter2/LocalTopology · AdmitsCompleteMetricOn.mono` and along embeddings: an embedding $j : T' \to T$ pulls the metric back, and a closed subset of a complete set is complete `Chapter2/LocalTopology · isComplete_inter_of_isClosed` `Chapter2/LocalTopology · AdmitsCompleteMetricOn.of_isEmbedding`.

Definition 2.1.5. X is *locally complete* if every x has open $V \ni x$ and W with $\mathrm{cl}_X(V) \subseteq W$ such that W admits a metric defining its topology for which $\mathrm{cl}_X(V)$ is complete `Chapter2/LocalTopology · IsLocallyComplete`. The text's $V(x) \subseteq W(x)$ follows from $\mathrm{cl}_X(V) \subseteq W$. The text also assumes X Hausdorff; the formal definition does not, because it follows. - *W arbitrarily small* `Chapter2/LocalTopology · IsLocallyComplete.exists_subset`. Given a neighbourhood N of x, take a metric ball V' around x in W whose closure in W lies in $W \cap N$. Replace W by the open $W \cap \mathrm{int}\, N$ and V by $V \cap V'$; the metric restricts. - *Regular* `Chapter2/LocalTopology · IsLocallyComplete.regularSpace`. A neighbourhood N of x contains the closed neighbourhood $\mathrm{cl}_X(V)$ for a $W \subseteq N$; this is where the global condition $\mathrm{cl}_X(V) \subseteq W$ enters. - T_1 `Chapter2/LocalTopology · IsLocallyComplete.t1Space`. Points are closed in the metrizable W. For $y \neq x$, a neighbourhood of x missing y comes from W when $y \in W$; otherwise W itself serves. - *Hausdorff* `Chapter2/LocalTopology · IsLocallyComplete.t2Space`: regular and T_1.

Witnesses: a complete metric space ($W = V = X$) `Chapter2/LocalTopology · isLocallyComplete_of_completeSpace`, $\mathbb{R}$ `Chapter2/LocalTopology · isLocallyComplete_real`, and a metric space whose points have open $V \subseteq W$ with $\mathrm{cl}(V) \subseteq W$ and $\mathrm{cl}(V)$ complete for the given metric `Chapter2/LocalTopology · isLocallyComplete_of_metric`.

Remark 2.1.6. If $\mathrm{cl}_X(V) \subseteq W$, the closure of V in the subspace W is $\mathrm{cl}_X(V)$ `Chapter2/LocalTopology · remark_2_1_6_2021`. The modified notion asks only $V \subseteq W$, with completeness of $\mathrm{cl}_W(V)$ `Chapter2/LocalTopology · IsLocallyCompleteRel`. Local completeness implies it, by the identity just proved `Chapter2/LocalTopology · IsLocallyComplete.isLocallyCompleteRel`. The remark's "weaker" is formalized as this implication; that the modified notion is *strictly* weaker is §II.31 (gap G.76, closed).

Proposition 2.1.4. Let $O = r(U)$ be a retract with the metric of E_0. *Closed pieces of O:* for ε small, $O \cap \bar{B}_\varepsilon(x)$ is closed in E_0 `Chapter2/LocalTopology · ScRetraction.exists_isClosed_retract_inter_closedBall`. Choose δ with $\bar{B}_\delta(x) \cap C \subseteq U$ (U is relatively open in the closed C). A limit v of points $v_k \in O \cap \bar{B}_\varepsilon(x)$ lies in $C \cap \bar{B}_\varepsilon(x) \subseteq U$, and $v_k = r(v_k) \to r(v)$ by continuity of r on U, so $v = r(v) \in O$. *The neighbourhoods* `Chapter2/LocalTopology · IsScRetract.exists_isComplete_closure`: V the open $\varepsilon/2$-ball and W the open ε-ball in O, inside any given neighbourhood. The closure of V in O lies in $O \cap \bar{B}_{\varepsilon/2}(x)$, closed in the Banach space E_0 and therefore complete. This proves the proposition `Chapter2/LocalTopology · proposition_2_1_4_2021`. The text's set V' with $r(V') \subseteq V'$ is not needed, because the closed balls already give closed pieces of O.

Corollary 2.1.7. By Proposition 2.1.4, with the metric d_O restricted to W `Chapter2/LocalTopology · isLocallyComplete_of_metric`, a retract is locally complete `Chapter2/LocalTopology · corollary_2_1_`

7_2021. So is E_0, the retract of the identity Chapter2/LocalTopology · isLocallyComplete_univ_retract.

Proposition 2.1.8. *Embeddings with locally closed image* Chapter2/LocalTopology · IsLocallyComplete.of_isEmbedding. Let $j : Y \to X$ be an embedding with $j(Y) = U \cap Z$, U open and Z closed. For $y \in Y$, take W, V around $j(y)$ with $W \subseteq U$. The preimages $j^{-1}(W)$ and $j^{-1}(V)$ serve for y, for two reasons. First, the closure of $j^{-1}(V)$ corresponds to $\mathrm{cl}_X(V) \cap j(Y)$, and $j(Y) \cap W = Z \cap W$ is closed in W. Second, the embedding $j^{-1}(W) \to W$ has closed image, so the metric of W pulls back with the completeness of the closure. Parts (1)–(3) follow: - (1) a locally closed subspace Chapter2/LocalTopology · proposition_2_1_8_1_2021; - (2) a homeomorphic space, the image being everything Chapter2/LocalTopology · proposition_2_1_8_2_2021; - (3) an open subspace Chapter2/LocalTopology · proposition_2_1_8_3_2021.

The open-cover statement Chapter2/LocalTopology · proposition_2_1_8_4_2021. Let $y \in Y_i$, with $U \ni y$ open, $U \subseteq Y_i$ and $\mathrm{cl}_Y(U) \subseteq Y_i$ (condition (b)), and W', V' for y in the locally complete Y_i. Put $V = V' \cap U$ and $W = W'$, both open in Y because Y_i is. Then $\mathrm{cl}_Y(V) \subseteq \mathrm{cl}_Y(U) \subseteq Y_i$ by (b), so $\mathrm{cl}_Y(V)$ is the closure of V in Y_i and lies in W. The metric of $W' \subseteq Y_i$ is transported to $W \subseteq Y$, the identity being a homeomorphism. No Hausdorff hypothesis is used anywhere.

Lemma 2.1.9 and Proposition 2.1.10. *First countable* Chapter2/LocalTopology · lemma_2_1_9_2021: the neighbourhood filter of x in X is the image of that of x in the open, metrizable W, and the latter is countably generated. *Completely regular* Chapter2/LocalTopology · IsLocallyComplete.completelyRegularSpace. For x outside a closed A, take $W \subseteq X \setminus A$ (arbitrarily small W). In the metrizable W, a continuous $g : W \to [0, 1]$ has $g(x) = 0$ and $g = 1$ off V. Extended by 1 on $X \setminus \mathrm{cl}_X(V)$, it is continuous on the two open sets W and $X \setminus \mathrm{cl}_X(V)$, which cover X because $\mathrm{cl}_X(V) \subseteq W$. Mathlib's complete regularity uses this normalization ($f(x) = 0$, $f = 1$ on A); the text's β is $1 - f$. With Hausdorff, the space is $T_{3\frac{1}{2}}$ Chapter2/LocalTopology · proposition_2_1_10_2021. The text argues continuity with sequences; the formal proof works with filters and open covers, so first countability is not needed for it.

II.23 The 2021 edition: local path-connectedness and local contractibility (§2.2)

§2.2 of the 2021 edition, pp. 30–33, is new.

Lemma 2.2.1. Let U be relatively open in a partial quadrant C and connected. For $k \in \mathbb{N}$, $U_k = U \cap E_k$ carries the topology of E_k Chapter1/ScSmooth · ScBanach.lvlSet. For $k = \infty$ the text does not say which topology U_∞ carries. The formalization takes the topology of all the norms: U_∞ is the set of compatible sequences $(v_m) \in \prod_m E_m$ with $v_0 \in U$ Chapter2/LocalTopology · ScBanach.lvlSetInfty, with the product topology. A smooth point x gives its sequence of lifts Chapter2/LocalTopology · ScBanach.smoothSeq Chapter2/LocalTopology · ScBanach.toBase_smoothSeq Chapter2/LocalTopology · ScBanach.smoothSeq_mem Chapter2/LocalTopology · ScBanach.smoothSeq_zero. The image of U_∞ in E_0 is $U \cap E_\infty$ Chapter2/LocalTopology · ScBanach.image_lvlSetInfty.

The proof uses three facts: - two points of U_∞ whose base points span a segment inside U are joined in U_∞ by the segment, which is continuous into every level Chapter2/LocalTopology · ScBanach.joinedIn_lvlSetInfty_of_segment; - smooth points are dense in every level C_m Chapter2/LocalTopology · IsPartialQuadrant.exists_smooth_near, the text's "U_∞ is dense in every U_k"; - every $z \in U$ has a ball with $B_\varepsilon(z) \cap C \subseteq U$, a convex set Chapter2/LocalTopology · IsRelOpenInQuadrant.exists_ball_inter_subset.

The case $k = \infty$ Chapter2/LocalTopology · lemma_2_2_1_infty_2021. Fix a point a of U_∞, and say that $z \in U$ is good if near z every point of U_∞ is joined to a. Joining propagates inside a convex $B_\varepsilon(z) \cap C \subseteq U$ by segments, and smooth points are dense, so the set of good points is open and closed in U. It is nonempty, so it is all of U because U is connected. *The case $k \in \mathbb{N}$* Chapter2/LocalTopology · lemma_2_2_1_2021: project the path-connected U_∞ to U_k and join points of U_k to nearby smooth points by segments. The text's proof

replaces a path in U_0 by a polygon through smooth points, using compactness of the path; the clopen argument avoids both the path and the polygon.

Lemma 2.2.2. For $x \in O$ and $V = V' \cap U$ with $V' \ni x$ open in E_0, the set $W = V \cap r^{-1}(V)$ has these properties `Chapter2/LocalTopology · lemma_2_2_2_2021`: - it is relatively open in C, by continuity of r on U; - it contains x, since $r(x) = x$; - $W \cap O = V \cap O$, because points of O are fixed; - $r(W) \subseteq W$, since $r(r(y)) = r(y) \in V$; - $r(W) = O \cap V$.

Definition 2.2.3 and Theorem 2.2.4. Mathlib's local path-connectedness is the text's: every open neighbourhood contains a path-connected open neighbourhood `Chapter2/LocalTopology · definition_2_2_3_2021`. A retract is locally path-connected `Chapter2/LocalTopology · theorem_2_2_4_2021`. This was proved earlier for Chapter 7 (2017, p. 183, recalls it without proof) `Chapter7/LocConnected · IsScRetract.locallyPathConnectedSpace`: points o, p of O near each other are joined in O by $\tau \mapsto r(o + \tau(p - o))$, the segment lying in a convex piece of U. The text goes through Lemma 2.2.2 and the component of $W(x)$ containing x.

Definition 2.2.5 and Proposition 2.2.6. *Almost locally contractible* `Chapter2/LocalTopology · IsAlmostLocallyContractible`: for z and open $V \ni z$, an open W with $z \in W \subseteq V$ and a continuous $\Phi : [0,1] \times W \to Z$ with values in V, $\Phi(0, \cdot) = \mathrm{id}$, $\Phi(t, z) = z$ and $\Phi(1, \cdot) \equiv z$. For a retract `Chapter2/LocalTopology · proposition_2_2_6_2021`, choose ε with $B_\varepsilon(z) \cap C \subseteq U$ and $r(B_\varepsilon(z) \cap C) \subseteq V$, by continuity of r at $z = r(z)$. Then $\Phi(t, y) = r((1-t)y + tz)$ on $W = B_\varepsilon(z) \cap O$ is the text's homotopy: the segment stays in the convex $B_\varepsilon(z) \cap C$, and r fixes y and z. The model E_0 itself is a witness `Chapter2/LocalTopology · isAlmostLocallyContractible_univ`.

II.24 The 2021 edition: paracompactness and M-polyfolds (§2.3, Appendix 2.7.2)

The footnote to Definition 2.3.4 (preprint 2.13). The 2021 footnote says that paracompactness does not include the Hausdorff property; it asks only that every open cover have a locally finite open refinement. In the formalization, this paracompactness* is defined with covers as sets of open sets `Chapter2/LocalTopology · IsLocallyFiniteSetFamily` `Chapter2/LocalTopology · IsParacompactStar` and coincides with Mathlib's paracompactness, which does not include the Hausdorff property either `Chapter2/LocalTopology · isParacompactStar_iff`. So "Hausdorff paracompact" *is exactly the hypothesis of the M-polyfold structure (§II.5)* `Chapter2/LocalTopology · definition_2_3_4_2021`, *and* $\mathbb{R}$ *is paracompact* `Chapter2/LocalTopology · isParacompactStar_real`. The footnote changes nothing for M-polyfolds, whose Hausdorff property is required separately.

The convention of Appendix 2.7.2. The appendix (p. 72; 2017 Appendix 2.6.2, p. 51) states the convention with a locally finite open *subcovering* `Chapter2/LocalTopology · IsParacompactBySubcover`. Read literally, it excludes $\mathbb{R}$ `Chapter2/LocalTopology · not_isParacompactBySubcover_real`. The cover by the intervals $(-n-1, n+1)$ has no locally finite subcover: a subcover is infinite, since $\mathbb{R}$ is unbounded, and the infinitely many intervals of it all contain 0. So the wording does not hold as we read it, and we read it as "refinement", as in the footnote (possible misprint).

Lemmas 2.7.2 and 2.3.5. In a regular space every open $U \ni x$ contains an open $V \ni x$ with $\mathrm{cl}(V) \subseteq U$, because closed neighbourhoods form a basis `Chapter2/LocalTopology · lemma_2_7_2_2021`. The text states this for regular Hausdorff spaces. A Hausdorff paracompact space is normal, hence regular, and the shrinking can be done inside any given neighbourhood `Chapter2/LocalTopology · exists_isOpen_closure_subset_of_paracompact`. This is Lemma 2.3.5 for M-polyfolds, $V(x)$ "arbitrarily small" `Chapter2/LocalTopology · lemma_2_3_5_2021`.

Proposition 2.3.6. A chart domain is homeomorphic to a retract, hence locally complete by Corollary 2.1.7 and Proposition 2.1.8 (2) `Chapter2/LocalTopology · MChart.isLocallyComplete_source`. The chart domains form an open cover of the M-polyfold X, and condition (b) of the open-cover part of Proposition 2.1.8 holds by Lemma 2.3.5. So X is locally complete `Chapter2/LocalTopology · MPolyfold.isLocallyComplete`, with

$W(x)$ arbitrarily small `Chapter2/LocalTopology · proposition_2_3_6_2021`. The text argues directly in a chart, with Proposition 2.1.4 and Lemma 2.3.5. "The topology on level 0" in the text is the topology of X.

Proposition 2.3.7. Every M-polyfold is locally path-connected `Chapter2/LocalTopology · proposition_2_3_7_2021`. As for Theorem 2.2.4, this was already proved for Chapter 7 `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`, by transporting the local path-connectedness of retracts through charts.

II.25 The 2021 edition: tame sub-M-polyfolds (Definitions 2.5.18, 2.5.19)

Definition 2.5.18. The structure induced on a sub-M-polyfold Y (§II.10) is *tame* if the M-polyfold Y, paracompact as a subspace, is tame `Chapter2/TameBoundary · MPolyfold.HasTameInducedStructure`.

Definition 2.5.19. Condition (3) asks for an sc-complement of T_yY in T_yX inside the reduced tangent space T^R_yX. The formalization has no linear structure on T_yX (gap G.7), so (3) is stated in a chart. For retracts $Q \subseteq O$ in C, a presentation T of C, and a level-1 point z of Q, there must be an sc-subspace A of E with $A \subseteq T_zO \cap E_z$ and $(T_zO)_m = (T_zQ)_m \oplus A_m$ on every level `Chapter2/TameBoundary · MPolyfold.HasReducedComplementAt`. (E_z is the subspace of §II.12 and $T_zO \cap E_z$ the reduced tangent space of Definition 2.4.7 (preprint 2.27).) This is literally Chapter 4's general-position condition, restated so that Chapter 2 does not import Chapter 4. $A = 0$ serves for $Q = O$ `Chapter2/TameBoundary · MPolyfold.hasReducedComplementAt_self`.

The data at a point consist of sub-M-polyfold data (V, r), so that $r(V) = Y \cap V$, together with two conditions `Chapter2/TameBoundary · MPolyfold.TameSubData`: - $d_X(r(x)) = d_X(x)$ on V; - at every smooth $y \in Y \cap V$, the chart form of (3) for some chart around y, with Q the chart image of Y near y.

Y is a tame sub-M-polyfold of the tame X if every point of Y has such data `Chapter2/TameBoundary · MPolyfold.IsTameSubMPolyfold`; it is then a sub-M-polyfold `Chapter2/TameBoundary · MPolyfold.IsTameSubMPolyfold.isSubMPolyfold`. Satisfiability: in a tame X, $Y = X$ with $r = \mathrm{id}$ has tame induced structure, the sub-charts built on tame charts having the same tame models `Chapter2/TameBoundary · MPolyfold.hasTameInducedStructure_univ`. It is a tame sub-M-polyfold, with $Q = O$ and $A = 0$ in every chart `Chapter2/TameBoundary · MPolyfold.isTameSubMPolyfold_univ`. The independence of (3) of the chart, the remark that a tame sub-M-polyfold is tame, and the two examples in $[0,\infty)^2 \times \mathbb{R}$ are §II.35 (gap G.77, closed).

II.26 The 2021 edition: faces of open sets and the corner axioms (§2.7.4, Propositions 2.7.5, 2.7.7)

Appendix §2.7.4 is new in the 2021 edition. It builds from a tame M-polyfold X a space bX of "points with a chosen face" and a map $B : bX \to X$. The formalization proves all of it for an abstract $d : X \to \mathbb{N}$ satisfying *corner axioms*, then specializes to the degeneracy index of a tame M-polyfold. The same proofs serve for the object spaces of tame ep-groupoids in 2021 §7.6.5.

Faces of open sets. For an open V and $p \in V$ with $d(p) = 1$, the face of V through p is the closure in V of the connected component of $V \cap \{d = 1\}$ through p `Chapter2/TameBoundary · TameBoundary.faceAt` `Chapter2/TameBoundary · TameBoundary.IsFaceIn`. $F_X(y, V)$ is the set of faces of V through y `Chapter2/TameBoundary · TameBoundary.facesThrough`. For an M-polyfold the faces of X itself are those of Definition 2.5.10 (preprint 2.39) `Chapter2/TameBoundary · MPolyfold.isFaceIn_univ_iff`. The text's faces of V are those of V with its induced structure, whose degeneracy index is $d_X|V$ (2021 Proposition 2.4.4), so defining them from d_X is the same thing.

The corner axioms `Chapter2/TameBoundary · TameBoundary.IsCornerIndex`. First, d is upper semicontinuous (Lemma 2.4.3 (preprint 2.23)). Second, every x has an open D such that every $y \in D$ has *local pieces* in arbitrarily small neighbourhoods N: an open $B \ni y$ in N and $d(y)$ preconnected sets $\ell_i \subseteq N \cap \{d = 1\}$

accumulating at y and covering $B \cap \{d = 1\}$ Chapter2/TameBoundary · TameBoundary.LocalPieces. Third, the pieces are separated in D: no preconnected subset of $D \cap \{d = 1\}$ meets two of them. In a tame chart the pieces are the images under r of the sets $\{z \in C : |z - \varphi(y)| < \varepsilon,\ I(z) = \{i\}\}$, $i \in I(\varphi(y))$, from the local face structure of §II.15 Chapter2/TameBoundary · MPolyfold.exists_localPieces_chart. So the degeneracy index of a tame M-polyfold satisfies the axioms Chapter2/TameBoundary · MPolyfold.isCornerIndex_degIndex, and $d \equiv 0$ does trivially Chapter2/TameBoundary · TameBoundary.isCornerIndex_zero. Faces of nested opens are compatible: a face of V meets an open $U \subseteq V$ in a union of faces of U Chapter2/TameBoundary · TameBoundary.faceAt_inter_eq. A face θ of U lies in a unique face of V, the union θ' of the faces of V containing it Chapter2/TameBoundary · TameBoundary.faceExt Chapter2/TameBoundary · TameBoundary.faceExt_unique. Every face of V through a point of U comes from a face of U Chapter2/TameBoundary · TameBoundary.exists_faceExt_eq.

Proposition 2.7.5. *Face-structured neighbourhoods exist* Chapter2/TameBoundary · TameBoundary.exists_nbhd_forall_isFaceStructuredOpen. Take D from the axioms. For an open $V \subseteq D$ and $y \in V$, the local pieces of y inside V lie on faces of V, and every face through y contains a piece. Distinct pieces lie on distinct faces: a face through two pieces would give a preconnected subset of $D \cap \{d = 1\}$ meeting both, which separation forbids. So y lies on exactly $d(y)$ faces of V. Then $V = D \cap \{d \le d(x)\}$, open by semicontinuity, gives (1) and (2). For (3), on any open $V \ni x$, take the set B of the local pieces of x inside V: a face of V through a point of B contains one of the pieces of x, which accumulate at x, and faces are closed in V Chapter2/TameBoundary · TameBoundary.exists_nbhd_faces_mem. Together these prove the proposition Chapter2/TameBoundary · TameBoundary.proposition_2_7_5 Chapter2/TameBoundary · MPolyfold.proposition_2_7_5_2021.

Definition 2.7.6 and Proposition 2.7.7. The definitions: - an open V is *face-structured* if every $y \in V$ lies on exactly $d(y)$ faces of V Chapter2/TameBoundary · TameBoundary.IsFaceStructuredOpen Chapter2/TameBoundary · TameBoundary.faceStructuredOpens; - N_X is the set of the nonempty face-structured V with $d|V$ bounded Chapter2/TameBoundary · TameBoundary.faceStructuredNbhds; - $N_{X,x}$ is the set of those $V \in N_X$ with $x \in V$ and $d \le d(x)$ on V Chapter2/TameBoundary · TameBoundary.faceStructuredNbhdsAt.

$N_X = \bigcup_x N_{X,x}$, because a $V \in N_X$ contains a point where the bounded $d|V$ is maximal Chapter2/TameBoundary · TameBoundary.faceStructuredNbhds_eq_iUnion. An open subset of a face-structured set is face-structured: its faces through a point are the traces of those of the larger set, by the local pieces Chapter2/TameBoundary · TameBoundary.IsFaceStructuredOpen.mono. Hence Proposition 2.7.7: - $N_{X,x}$ is nonempty (Proposition 2.7.5) Chapter2/TameBoundary · TameBoundary.faceStructuredNbhdsAt_nonempty, so N_X covers X Chapter2/TameBoundary · TameBoundary.sUnion_faceStructuredNbhds; - nonempty open subsets of members of N_X are in N_X Chapter2/TameBoundary · TameBoundary.mem_faceStructuredNbhds_of_subset; - nonempty intersections of members of N_X are in N_X Chapter2/TameBoundary · TameBoundary.inter_mem_faceStructuredNbhds; - $N_{X,x}$ is closed under intersections Chapter2/TameBoundary · TameBoundary.inter_mem_faceStructuredNbhdsAt.

For tame M-polyfolds this is Chapter2/TameBoundary · MPolyfold.proposition_2_7_7_2021.

Quadrants. In $[0,\infty)^n \oplus W$, d counts the vanishing half-line coordinates Chapter2/TameBoundaryExample · TameBoundaryExample.dq_eq. $\{d = 1\}$ is the union of the strata $\{p_k = 0,\ p_l > 0\ (l \ne k)\}$ Chapter2/TameBoundaryExample · TameBoundaryExample.part Chapter2/TameBoundaryExample · TameBoundaryExample.dq_eq_one_iff. A preconnected subset of $\{d = 1\}$ lies in one stratum Chapter2/TameBoundaryExample · TameBoundaryExample.subset_part, so a face through a point of the k-th stratum lies in $\{p_k = 0\}$ Chapter2/TameBoundaryExample · TameBoundaryExample.faceAt_subset_coord_eq_zero. A point of an open V with $y_k = 0$ lies on the face of V through a nearby point of the k-th stratum Chapter2/TameBoundaryExample · TameBoundaryExample.exists_part_mem_faceAt. Hence every open subset of a quadrant is face-structured Chapter2/TameBoundaryExample · TameBoundaryExample.isFaceStructuredOpen_quadrant.

II.27 The 2021 edition: L_X, F_X, bX and $U(x, V, \theta)$ (Definitions 2.7.8–2.7.11)

Definition 2.7.8. There is a morphism $(x, V) \to (y, W)$ iff $V \subseteq W$, x lies on $d(x)$ faces of W, and each of them contains y Chapter2/TameBoundary · TameBoundary.IsMor. The objects are the pairs (x, V) with

$d(x) \geq 1$ and $V \in N_{X,x}$ Chapter2/TameBoundary · TameBoundary.LObj. There is at most one morphism between two objects, so L_X is a preorder, a thin category. *Composition* Chapter2/TameBoundary · TameBoundary.IsMor.trans follows the text's argument. Given $(x, V) \to (y, W) \to (z, Q)$, every face γ of Q through x extends a face θ of W through x Chapter2/TameBoundary · TameBoundary.exists_faceExt_eq. That face contains y, so $y \in \gamma$, and then $z \in \gamma$.

Definition 2.7.9. The functor sends (x, V) to $F_X(x, V)$ and a morphism to $\theta \mapsto \theta'$, the face of W containing θ Chapter2/TameBoundary · TameBoundary.boundaryFunctor Chapter2/TameBoundary · MPolyfold.LocalBoundaryCategory Chapter2/TameBoundary · MPolyfold.boundaryStructureFunctor. It is the text's "face θ' of W through x and y with $\theta \cap \theta'$ open in θ" Chapter2/TameBoundary · TameBoundary.faceExt_unique. Identities and composition hold because extending twice is extending once. $F_X(\varphi)$ is injective, the text's injection (2.43) Chapter2/TameBoundary · TameBoundary.boundaryFunctor_map_injective. It is bijective when the point does not change, since both sets have $d(x)$ elements Chapter2/TameBoundary · TameBoundary.boundaryFunctor_map_bijective.

Definition 2.7.10. An admissible triple is an object (x, V) with a face $\theta \in F_X(x, V)$ Chapter2/TameBoundary · TameBoundary.BTriple. $(x, V, \theta) \sim (x', V', \theta')$ iff $x = x'$ and some $V'' \in N_{X,x}$, $V'' \subseteq V \cap V'$, has a face through x sent to θ and to θ' Chapter2/TameBoundary · TameBoundary.BRel. Reflexivity, symmetry and transitivity hold Chapter2/TameBoundary · TameBoundary.BRel.refl Chapter2/TameBoundary · TameBoundary.BRel.symm Chapter2/TameBoundary · TameBoundary.BRel.trans; for transitivity, intersect the two V'' and use the bijectivity at a fixed point. bX is the quotient Chapter2/TameBoundary · TameBoundary.bSetoid Chapter2/TameBoundary · TameBoundary.BSpace Chapter2/TameBoundary · MPolyfold.TameBoundarySpace, and $B[x, V, \theta] = x$ Chapter2/TameBoundary · TameBoundary.bMap Chapter2/TameBoundary · MPolyfold.tameBoundaryMap. The image of B is $\{d \geq 1\} = \partial X$ Chapter2/TameBoundary · TameBoundary.range_bMap Chapter2/TameBoundary · MPolyfold.range_tameBoundaryMap. *Fibres.* $B^{-1}(x)$ has exactly $d(x)$ points Chapter2/TameBoundary · TameBoundary.encard_bMap_preimage Chapter2/TameBoundary · MPolyfold.encard_tameBoundaryMap_preimage: for a fixed $V \in N_{X,x}$, the classes over x correspond to the $d(x)$ faces of V through x, because every triple over x is equivalent to one with this V and distinct faces give distinct classes. This is the set-level content of Theorem 2.7.13 (4).

Definition 2.7.11. $U(x, V, \theta)$ is the set of classes with a representative (y, V', θ') such that $y \in \theta$, there is a morphism $(y, V') \to (x, V)$, and it sends θ' to θ Chapter2/TameBoundary · TameBoundary.uSet. The formalization reads membership as "for some representative". $\Phi_{(x,V,\theta)}$ is B restricted to it, with values in θ Chapter2/TameBoundary · TameBoundary.phi. It is injective: two classes over the same y in $U(x, V, \theta)$ are sent to the same face θ of V, and F_X of the morphism to (x, V) is injective Chapter2/TameBoundary · TameBoundary.phi_injective.

The image. Put $V^x = \{y \in V : \text{every face of } V \text{ through } y \text{ contains } x\}$ Chapter2/TameBoundary · TameBoundary.reach, an open set Chapter2/TameBoundary · TameBoundary.isOpen_reach. For $y \in V$ there is a morphism $(y, V') \to (x, V)$ exactly when $y \in V^x$. So the image of $\Phi_{(x,V,\theta)}$ is $\theta \cap V^x$ Chapter2/TameBoundary · TameBoundary.bMap_image_uSet. If V is centred at x, i.e. every face of V contains x Chapter2/TameBoundary · TameBoundary.IsCenteredAt, then $V^x = V$ and Φ is a bijection onto θ Chapter2/TameBoundary · TameBoundary.phi_bijective_of_isCenteredAt. V^x is itself centred at x Chapter2/TameBoundary · TameBoundary.isCenteredAt_reach and lies in $N_{X,x}$ Chapter2/TameBoundary · TameBoundary.reach_mem_faceStructuredNbhdsAt. So the centred neighbourhoods are cofinal, which is what the text's "take V sufficiently small" (p. 85) provides. The text calls $\Phi_{(x,V,\theta)}$ a canonical bijection for every triple (2.44); §II.28 shows that some such restriction is needed.

II.28 The 2021 edition: (2.44) and an example in $[0, \infty)^3$

Let $X = [0, \infty)^3 \oplus W$, W any sc-Banach space; the trivial W gives the setting $[0, \infty)^3$. Its degeneracy index satisfies the corner axioms Chapter2/TameBoundaryExample · TameBoundaryExample.dq Chapter2/TameBoundaryExample · TameBoundaryExample.isCornerIndex_dq. Take: - $x = (0, 0, 1)$, a corner point with $d(x) = 2$; - $V = \{p_0 < 1/2,\ 1/2 < p_1 + p_2 < 3/2\}$, a convex open slab around the segment from x to $(0, 1, 0)$

Chapter2/TameBoundaryExample · TameBoundaryExample.slab.

On V, $d \leq 2$. V is face-structured, like every open subset of a quadrant (§II.26), so $V \in N_{X,x}$ Chapter2/TameBoundaryExample · TameBoundaryExample.slab_mem_faceStructuredNbhdsAt. The open segment $\{(0,t,1-t)\}$ lies in $\{d = 1\}$, and the face θ of V through it contains its endpoints x and $(0,1,0)$ Chapter2/TameBoundaryExample · TameBoundaryExample.slabFace. (x,V,θ) is an admissible triple Chapter2/TameBoundaryExample · TameBoundaryExample.counterTriple. The point $(0,1,0)$ also lies on the face of V through the nearby points of the stratum $\{p_2 = 0\}$. That face lies in $\{p_2 = 0\}$, so it misses x Chapter2/TameBoundaryExample · TameBoundaryExample.exists_face_not_mem. Hence $(0,1,0) \notin V^x$, and $(0,1,0) \in \theta$ is not in the image $\theta \cap V^x$ of $\Phi_{(x,V,\theta)}$ Chapter2/TameBoundaryExample · TameBoundaryExample.pt_010_not_mem_image. So $\Phi_{(x,V,\theta)}$ is not onto θ Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective_2021, and not every Φ is a bijection Chapter2/TameBoundaryExample · TameBoundaryExample.not_forall_bijective_phi_2021. As we read it, (2.44) does not hold for every $V \in N_{X,x}$. It holds for centred V (§II.27). The example does not depend on reading membership in $U(x,V,\theta)$ as "for some representative", because Φ misses $(0,1,0)$ already on the level of points.

II.29 The 2021 edition: the topology of bX (Lemma 2.7.15, Propositions 2.7.12, 2.7.16, Theorem 2.7.13 (1))

The topology. $\mathcal{U} \subseteq bX$ is in T_{bX} if every class $p \in \mathcal{U}$ has a representative (x,V,θ) and a set $O \ni p$, $O \subseteq \mathcal{U} \cap U(x,V,\theta)$, with $B(O)$ open in θ Chapter2/TameBoundaryTopology · TameBoundary.IsBOpen. We read O as containing p: otherwise $O = \emptyset$ would make every set open. Two tools: - a triple can be shrunk into any open $N \ni x$, keeping its class and shrinking its U-set Chapter2/TameBoundaryTopology · TameBoundary.BTriple.exists_shrink; - equivalent triples have a common refinement whose U-set lies in both Chapter2/TameBoundaryTopology · TameBoundary.BRel.exists_common.

If two triples with the same V have meeting U-sets, their faces agree Chapter2/TameBoundaryTopology · TameBoundary.eq_of_mem_uSet. These give the alternative characterization: $\mathcal{U} \in T_{bX}$ iff every $p \in \mathcal{U}$ has a representative with $U(x,V,\theta) \subseteq \mathcal{U}$ Chapter2/TameBoundaryTopology · TameBoundary.isBOpen_iff. In that form the axioms of a topology are immediate. For intersections, two representatives of p have a common refinement with U-set inside both Chapter2/TameBoundaryTopology · TameBoundary.bTopology Chapter2/TameBoundaryTopology · TameBoundary.isOpen_iff_isBOpen Chapter2/TameBoundaryTopology · TameBoundary.isOpen_iff_forall_uSet. This proves Lemma 2.7.15 Chapter2/TameBoundaryTopology · TameBoundary.lemma_2_7_15 Chapter2/TameBoundaryTopology · MPolyfold.lemma_2_7_15_2021. The text's proof of the intersection axiom uses the sc-smoothness of transition maps; this one uses only faces.

Proposition 2.7.12, topological part. The sets $U(x,V,\theta)$ are open Chapter2/TameBoundaryTopology · TameBoundary.isOpen_uSet and form a basis Chapter2/TameBoundaryTopology · TameBoundary.isTopologicalBasis_uSet, and B is continuous Chapter2/TameBoundaryTopology · TameBoundary.continuous_bMap. *The germ of a face:* if θ is a face of $U \subseteq V$ through y, then near y the extended face θ' of V coincides with θ Chapter2/TameBoundaryTopology · TameBoundary.exists_nbhd_faceExt_inter_subset. Hence Φ is open: $B(\mathcal{U} \cap U(x,V,\theta))$ is open in θ for open $\mathcal{U}$ Chapter2/TameBoundaryTopology · TameBoundary.exists_isOpen_bMap_image. With injectivity and continuity, every $\Phi_{(x,V,\theta)}$ is an open embedding into θ Chapter2/TameBoundaryTopology · TameBoundary.isOpenEmbedding_phi, and a homeomorphism onto θ for V centred at x Chapter2/TameBoundaryTopology · TameBoundary.phiHomeomorph. For two triples, the image of $U(x,V,\theta) \cap U(x',V',\theta')$ is open in θ and in θ', and the transition map is the identity on points of X, both maps being restrictions of B Chapter2/TameBoundaryTopology · TameBoundary.proposition_2_7_12 Chapter2/TameBoundaryTopology · MPolyfold.proposition_2_7_12_2021. Not part of this section: the sc-diffeomorphism half of Proposition 2.7.12. For centred representatives it is proved with the M-polyfold structure of bX (§II.36), for every representative in §II.37 (gap G.78, closed).

Proposition 2.7.16. *Hausdorff.* Two classes over different points of the Hausdorff X are separated by preimages under the continuous B. Two classes over the same x have representatives with a

common centred V and different faces, and their U-sets are disjoint (Hausdorff instance in `Chapter2/TameBoundaryTopology.lean`). *Metrizable* `Chapter2/TameBoundaryTopology · TameBoundary.proposition_2_7_16`. For $x \in \partial X$ and a centred $W \in N_{X,x}$, write (x, W, θ) for the triple of a face θ of W through x `Chapter2/TameBoundaryTopology · TameBoundary.tripleOf`. The sets $U(x, W, \theta)$ cover $B^{-1}(W)$ `Chapter2/TameBoundaryTopology · TameBoundary.exists_mem_uSet_tripleOf`; they are open, pairwise disjoint and metrizable through Φ. So $B^{-1}(W)$ embeds into $X \times \mathbb{R}^{F_X(x,W)}$ by B and the indicators of these clopen pieces, and is metrizable `Chapter2/TameBoundaryTopology · TameBoundary.metrizableSpace_preimage`. The pull-back form of Smirnov's theorem along B (§II.7) `Chapter2/Metrizable · metrizableSpace_of_pullback` then metrizes bX. For tame M-polyfolds, bX is Hausdorff and metrizable `Chapter2/TameBoundaryTopology · MPolyfold.proposition_2_7_16_2021`. The text's route instead builds a locally finite refinement from one of the cover (V_x) of ∂X, proves paracompactness, and applies Nagata–Smirnov.

Theorem 2.7.13 (1) and the topological characterization. A topology in which every $U(x, V, \theta)$ is open, and every Φ maps open sets to relatively open subsets of θ, is T_{bX} `Chapter2/TameBoundaryTopology · TameBoundary.eq_of_isOpen_uSet`. With the results above this gives the topological part of the theorem `Chapter2/TameBoundaryTopology · TameBoundary.theorem_2_7_13_topology` `Chapter2/TameBoundaryTopology · MPolyfold.theorem_2_7_13_2021`: - bX is metrizable; - the U-sets are open and the Φ are open embeddings; - T_{bX} is unique with these properties; - B is continuous, with image ∂X and fibres of $d_X(x)$ points.

The tame boundary of Definition 2.7.14, as a continuous map, is B `Chapter2/TameBoundaryTopology · MPolyfold.tameBoundary`. Parts (2)–(4) (the tame M-polyfold structure, B sc-smooth, d_{bX}) and the iterated boundaries are §II.36. *Witness.* The quadrant $[0, \infty)^n \oplus W$ is a tame M-polyfold `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold` `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_isTame`. Its corner has exactly n preimages in bX `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_encard_preimage`, so $bX \neq \emptyset$ for $n \geq 1$ `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_tameBoundarySpace_nonempty`. It is a tame sub-M-polyfold of itself `Chapter2/TameBoundary · MPolyfold.quadrantPolyfold_univ`.

II.30 What the 2021 additions taught us

- **Local completeness needs no Hausdorff hypothesis.** The global condition $\mathrm{cl}_X(V) \subseteq W$ gives regularity, and the metrizable W gives T_1 `Chapter2/LocalTopology · IsLocallyComplete.t2Space`. Proposition 2.1.8, Lemma 2.1.9 and complete regularity hold without it.
- **One permanence lemma.** Parts (1)–(3) of Proposition 2.1.8 are one statement about embeddings with locally closed image `Chapter2/LocalTopology · IsLocallyComplete.of_isEmbedding`. Proposition 2.3.6 is then the open-cover part applied to chart domains.
- **"Subcovering" in Appendix 2.7.2 does not hold as we read it.** With it, $\mathbb{R}$ would not be paracompact `Chapter2/LocalTopology · not_isParacompactBySubcover_real`. The footnote to Definition 2.3.4 has "refinement", which is Mathlib's notion `Chapter2/LocalTopology · isParacompactStar_iff` (possible misprint).
- **U_∞ needs a topology.** Lemma 2.2.1 for $k = \infty$ is proved for the topology of all the norms `Chapter2/LocalTopology · lemma_2_2_1_infty_2021`, by a clopen argument rather than polygons.
- **§2.7.4 is combinatorics of d.** Every statement of the appendix about faces, L_X, bX and its topology follows from the upper semicontinuity of d and separated local pieces `Chapter2/TameBoundary · TameBoundary.IsCornerIndex`, which is why it applies to ep-groupoids as well.
- **(2.44) needs V centred.** $\Phi_{(x,V,\theta)}$ has image $\theta \cap V^x$ `Chapter2/TameBoundary · TameBoundary.bMap_image_uSet`, and in $[0, \infty)^3$ it is not onto for a thin slab `Chapter2/TameBoundaryExample · TameBoundaryExample.phi_not_surjective_2021`. Centred neighbourhoods are cofinal `Chapter2/TameBoundary · TameBoundary.reach_mem_faceStructuredNbhdsAt`.
- **The topology of bX needs no sc-calculus.** Lemma 2.7.15 and Proposition 2.7.16 are proved from faces and the pull-back form of Smirnov's theorem `Chapter2/TameBoundaryTopology · TameBoundary.metrizableSpace_preimage`. The text uses the sc-smoothness of transition maps and a paracompactness argument.

II.31 The 2021 edition, continued: Remark 2.1.6 is strict (the K-topology)

§II.22 proved the implication of Remark 2.1.6. That the modified notion is *strictly* weaker needs an example, which the text does not give.

A criterion. If every point has an open neighbourhood W that is completely metrizable, the modified definition holds with $V = W$, the closure of W in itself being W `Chapter2/LocalTopologyExample · isLocallyCompleteRel_of_forall_isCompletelyMetrizable`. A space embedded in $\mathbb{R}$ onto the complement of a sequence (a_n) is completely metrizable: $y \mapsto (y, ((y - a_n)^{-1})_n)$ is a closed embedding into $\mathbb{R} \times \mathbb{R}^{\mathbb{N}}$, with image $\{(u, z) : z_n(u - a_n) = 1 \text{ for all } n\}$ `Chapter2/LocalTopologyExample · isCompletelyMetrizableSpace_of_isEmbedding_compl_range`. This replaces the theorem that G_δ subsets of complete metric spaces are completely metrizable, which Mathlib lacks.

The K-topology. Let $K = \{1/(n+1)\}$ `Chapter2/LocalTopologyExample · kSeq` and give $\mathbb{R}$ the topology generated by the usual open sets and $\mathbb{R} \setminus K$ `Chapter2/LocalTopologyExample · kTop` `Chapter2/LocalTopologyExample · RealKTop`. It is finer than the usual topology, so Hausdorff. The open sets $\mathbb{R} \setminus K$ and $\mathbb{R} \setminus \{0\}$ cover it, carry the usual subspace topology, and are complements of sequences; by the criterion the modified definition holds `Chapter2/LocalTopologyExample · RealKTop.isLocallyCompleteRel`. But 0 and the closed set K cannot be separated: every closed neighbourhood of 0 contains irrational points arbitrarily close to some $1/(n+1)$, hence meets K `Chapter2/LocalTopologyExample · RealKTop.not_regularSpace`. A locally complete space is regular (§II.22), so this one is not locally complete `Chapter2/LocalTopologyExample · RealKTop.not_isLocallyComplete`.

Remark 2.1.6, strictness. Some Hausdorff space satisfies the modified definition and not Definition 2.1.5 `Chapter2/LocalTopologyExample · remark_2_1_6_strict_2021`. So the remark holds as we read it, and the example shows that the global condition $\mathrm{cl}_X(V) \subseteq W$ is what makes a locally complete space regular.

II.32 The 2021 edition: bundle retractions and the levels of X^1 (Definition 2.1.16, after Definition 2.3.8)

Definition 2.1.16 (preprint 2.9). A retraction R of $E \oplus F$ is a bundle retraction covering the retraction r of E if its domain is $U \oplus F$, U the domain of r, and $R(u, v) = (r(u), \rho(u)v)$ there, with $\rho(u)$ a continuous linear map `Chapter2/TangentLift · ScRetraction.IsBundleRetractionOver`. A map Φ is a bundle map on B covering a if $\Phi(u, v) = (a(u), \phi(u)v)$ on B `Chapter2/TangentLift · IsBundleMapOn`; then the first component of Φ is a of the first component `Chapter2/TangentLift · IsBundleMapOn.fst_eq`. The text writes "$p' \circ \Phi = p$", which we read as $p' \circ \Phi = a \circ p$ (possible misprint). Sc-smoothness and $R \circ R = R$ are part of the data, as for every retraction (§II.2). The tangent retraction is a bundle retraction covering the level-1 retraction r^1: $Tr(u, h) = (r(u), Dr(u)h)$ `Chapter2/TangentLift · ScRetraction.tangentRetraction_isBundleRetractionOver`. The tangent map of an sc-smooth map between retracts is a bundle map covering f^1 `Chapter2/TangentLift · IsScSmoothRetractMap.retractTangentMap_isBundleMapOn`. The categories $\mathcal{R}$ and $\mathcal{BR}$ and the forgetful functors only name these objects and maps; they are not formalized.

The levels of X^1 (gap G.6). §II.6 built the M-polyfold X^1 on X_1; its levels are those of X shifted by one as sets. Their topologies agree too. In a chart, the level-m lift of a point is moved to another chart by the level-m map of the transition map `Chapter2/TangentLift · MPolyfold.levelLift_transition`, and the level-m lift of the level chart φ^1 of X^1 is the level-$(m+1)$ lift of φ, since $(E^1)_m = E_{m+1}$ `Chapter2/TangentLift · MPolyfold.levelLift_levelChart`. The identity of points is therefore continuous from $(X^1)_m$ to X_{m+1} and back, chart by chart, for an arbitrary chart of X^1 through level maps of transition maps `Chapter2/TangentLift · MPolyfold.shiftLevelMap` `Chapter2/TangentLift · MPolyfold.continuous_shiftLevelMap` `Chapter2/TangentLift · MPolyfold.shiftLevelInv` `Chapter2/TangentLift · MPolyfold.continuous_shiftLevelInv`. So $(X^1)_m \cong X_{m+1}$ `Chapter2/TangentLift · MPolyfold.shiftLevelHomeomorph`, with the charts of X_{m+1} `Chapter2/`

TangentLift · MPolyfold.shift_level_eq. Composing these homeomorphisms, each iterate $(X^1)^1 \cdots$ is X_m with its topology and charts, which is the text's statement after Definition 2.3.8.

II.33 The 2021 edition: Theorem 2.3.16 and Proposition 2.3.17, $T(X^i) \equiv (TX)^i$

In a local model. For an sc¹ map f, the tangent map of the level-1 map f^1 is the level-1 map of Tf on $T(U^1) = (TU)^1$ Chapter2/TangentLift · tangentMap_levelMap_one Chapter2/TangentLift · toBase_tangentMap_levelMap_one. For a retract, $T(O^1) = (TO)^1$ Chapter2/TangentLift · retractTangent_shift_of_isScRetract, and for sc-smooth maps between retracts $T(f^1)$ is again the level-1 map of Tf Chapter2/TangentLift · toBase_retractTangentMap_shift.

Theorem 2.3.16, bundle form. For an M-polyfold X, the tangent charts belong to the structure of TX (§II.8), $p : TX \to X^1$ is sc-smooth, and in a tangent chart p is the first projection onto the level chart φ^1. Every model (TO, TC, TE) is a bundle retract, Tr covering r^1, and the transition map of two tangent charts is a bundle map covering the level-1 map of the transition map of the charts Chapter2/TangentLift · theorem_2_3_16_2021. The fibres T_xX carry no linear structure of their own (gap G.7), so "M-polyfold bundle" is read through the models and transition maps.

Proposition 2.3.17 for $i = 1$. A tangent vector q of X^1 at $x \in (X^1)_1 = X_2$, read in the level chart φ^1 of a chosen chart φ of X around x Chapter2/TangentLift · MPolyfold.someChart, is $(\varphi^1(x), h')$ with $h' \in E_1$; it gives the tuple (x, φ, h') of TX Chapter2/TangentLift · MPolyfold.TangentPre.lower. Read in any chart ψ of X, this tuple is the inclusion of q read in ψ^1, because $T(\psi \circ \varphi^{-1})$ on $(TO)^1$ is the tangent map of $(\psi \circ \varphi^{-1})^1$ Chapter2/TangentLift · MPolyfold.TangentPre.transport_lower. So it defines an inclusion $\iota : T(X^1) \to TX$ Chapter2/TangentLift · MPolyfold.tangentLower. Its values lie on level 1 of TX, giving $\Phi : T(X^1) \to (TX)^1$ Chapter2/TangentLift · MPolyfold.tangentShiftMap, a bijection Chapter2/TangentLift · MPolyfold.tangentShiftEquiv and a homeomorphism Chapter2/TangentLift · MPolyfold.tangentShiftHomeomorph. In every chart ψ around the base point, $T\psi(\Phi(\alpha))$ is the inclusion of $T(\psi^1)(\alpha)$ Chapter2/TangentLift · MPolyfold.tangentChartMap_tangentLower, so Φ does not depend on the chosen chart; in the charts $T(\varphi^1)$ and $(T\varphi)^1$ it is the identity of $T(O^1) = (TO)^1$ Chapter2/TangentLift · MPolyfold.levelChart_tangentShiftMap. Hence Φ is an sc-diffeomorphism Chapter2/TangentLift · MPolyfold.tangentShiftEquiv_isScDiffeoOn, it covers the identity of $(X^1)^1$, and the level-1 map $p^1 : (TX)^1 \to (X^1)^1$ is sc-smooth Chapter2/TangentLift · MPolyfold.tangentProjShift Chapter2/TangentLift · proposition_2_3_17_2021.

$i = 2$. A bijection of M-polyfolds sc-smooth in both directions raises to an sc-diffeomorphism of the lifts Chapter2/TangentLift · MPolyfold.levelOneEquiv Chapter2/TangentLift · MPolyfold.levelOneEquiv_isScDiffeoOn. Composing Φ for X^1 with the raise of Φ^{-1} gives $T(X^2) \cong (TX)^2$ Chapter2/TangentLift · MPolyfold.tangentShiftEquivTwo, covering the identity of $(X^1)^2 = (X^2)^1$ Chapter2/TangentLift · MPolyfold.tangentProjShiftTwo Chapter2/TangentLift · proposition_2_3_17_two_2021.

Every i. To speak of X^i as a function of i, an M-polyfold is bundled with its space Chapter2/TangentLiftAll · BundledMPolyfold; then $X^0 = X$, $X^{i+1} = (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.lift, and TX is again bundled Chapter2/TangentLiftAll · BundledMPolyfold.tangent. Sc-diffeomorphisms and sc-smooth maps are packaged with their index raises Chapter2/TangentLiftAll · MPolyfold.ScEquiv Chapter2/TangentLiftAll · MPolyfold.ScEquiv.shift Chapter2/TangentLiftAll · MPolyfold.ScMap, with Φ and p as instances Chapter2/TangentLiftAll · MPolyfold.tangentShiftScEquiv Chapter2/TangentLiftAll · MPolyfold.tangentProjScMap. By recursion, $\Phi_0 = \mathrm{id}$ and $\Phi_{i+1} = (\Phi_i)^1 \circ \Phi$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv; likewise $J_i : (X^1)^i \to (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.shiftLiftEquiv and p^i Chapter2/TangentLiftAll · BundledMPolyfold.tangentProjLift. The inclusions $X^i \to X$ Chapter2/TangentLiftAll · BundledMPolyfold.incl are injective, and $T(X^i) \to TX$ is the composite of the inclusions ι Chapter2/TangentLiftAll · BundledMPolyfold.tangentIncl. With these, Φ_i and J_i are sc-diffeomorphisms, the projections are sc-smooth, $J_i \circ p^i \circ \Phi_i = p$, Φ_i commutes with the inclusions into TX and with those of the two towers $T(X^{i+1}) \to T(X^i)$ and $(TX)^{i+1} \to (TX)^i$, J_i is the identity on points, and

these properties determine Φ_i Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_unique Chapter2/TangentLiftAll · proposition_2_3_17_all_2021. Φ_1 is the Φ above Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_one. Because $\iota \circ T(f^1) = Tf \circ \iota$ for every sc-smooth f Chapter2/TangentLiftAll · MPolyfold.tangentLower_tangentMap, Φ_i is natural in X Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_natural. A bundled M-polyfold exists, the half-line Chapter2/TangentLiftAll · BundledMPolyfold.nonempty. Linearity of Φ_i on the fibres is not stated, since T_xX has no linear structure here (gap G.7); in charts Φ_i is a composite of index raises of identities. Lemma 4.4.4 for every i uses this identification (Chapter 4).

II.34 The 2021 edition: Remark 2.4.2, the diagonal half-line

The text's example is $\Delta = \{(s,s,s) : s \geq 0\}$ in $C = [0,\infty)^3$; here $C = [0,\infty)^3 \oplus W$ for any sc-Banach space W, with $W = 0$ the text's case. The constant linear maps between $\mathbb{R}$ and $\mathbb{R}^3$ are sc-operators Chapter2/TangentLift · ScOperator.ofConst, and replacing the three coordinates by their average Chapter2/TangentLift · diagonalAverage is an sc-smooth retraction of C onto $\Delta \oplus W$ Chapter2/TangentLift · diagonalRetraction Chapter2/TangentLift · mem_diagonalRetraction_retract_iff, the orthogonal projection onto the diagonal; the text gives no retraction. Its retract is a local model Chapter2/TangentLift · diagonalModel and an M-polyfold Chapter2/TangentLift · diagonalPolyfold.

$(s,w) \mapsto ((s,s,s),w)$ Chapter2/TangentLift · diagonalEmbed, with inverse the first coordinate Chapter2/TangentLift · diagonalProject, is an sc-diffeomorphism from the half-line M-polyfold $[0,\infty) \oplus W$ onto $\Delta \oplus W$ Chapter2/TangentLift · diagonalEmbed_isScDiffeoOn. In C, $d_C((s,s,s),w)$ is 3 at $s = 0$ and 0 otherwise Chapter2/TangentLift · degIndex_diagonal_quadrant; on the M-polyfold, d_Δ is 1 at $s = 0$ and 0 otherwise, by the invariance of the index under sc-diffeomorphisms (Proposition 2.4.4 (preprint 2.24), §II.11) applied to the half-line Chapter2/TangentLift · diagonalPolyfold_degIndex. One chart, defined on all of $\Delta \oplus W$, has the model $[0,\infty) \oplus W$, and in it the chart index is d_Δ Chapter2/TangentLift · diagonalHalfLineChart Chapter2/TangentLift · diagonalHalfLineChart_spec Chapter2/TangentLift · remark_2_4_2_2021. So the defining chart realizes d_C, the more economical one d_Δ, as the remark says.

II.35 The 2021 edition: tame sub-M-polyfolds, continued (Definition 2.5.19 and the text after it)

§II.25 stated condition (3) of Definition 2.5.19 in one chart around each smooth point, with one presentation of its partial quadrant.

Chart independence. A point is smooth iff its image in one chart is a smooth point of the model Chapter2/TameSub · MPolyfold.mem_smoothSet_iff_of_isChart. Condition (3) in a chart is the general-position condition of Chapter 4 for the retraction of the model and the chart image of Y, word for word. Chapter 4's chart change for that condition (MPolyfold.isGeneralPositionRetractAt_chart_change, Chapter 4) moves it to every chart $\varphi \ni y$, every presentation of the partial quadrant of φ, every level-1 representative, and the chart image of any other sub-M-polyfold retraction onto Y near y Chapter2/TameSub · MPolyfold.TameSubData.hasReducedComplementAt_of_isChart_2021, in particular of the same one Chapter2/TameSub · MPolyfold.TameSubData.complement_forall_charts_2021. (3) stays a statement in charts, since T_yX has no linear structure here (gap G.7).

A tame sub-M-polyfold is tame. Sc-splittings compose: if $E = M \oplus B$ is an sc-splitting and $M_m = N_m \oplus A_m$ on every level for sc-subspaces $N, A \subseteq M$, then $E = N \oplus (A \oplus B)$ is one Chapter2/TameSub · ScBanach.levelSub_sup_of_le_of_isScSplitting Chapter2/TameSub · ScBanach.isScSplitting_sup_of_complementIn. In a chart φ with a tame model, whose tame retraction is t, the chart image $\varphi(W \cap Y)$ is the retract of $(\varphi r \varphi^{-1}) \circ t$. That retraction is tame: the index condition comes from (2) of Definition 2.5.19 and $d_X = d_C \circ \varphi$ in a tame chart; the sc-complement of the tangent space of $\varphi(W \cap Y)$ inside E_x is $A \oplus B$, with A from (3) in the chart φ (chart independence) and B the complement of T_xO from the tameness of t Chapter2/TameSub · MPolyfold.TameSubData.isTameRetract_image. So the sub-charts of Y built on tame charts of

X have tame models, and Y has a tame induced structure (Definition 2.5.18) Chapter2/TameSub · MPolyfold.IsTameSubMPolyfold.hasTameInducedStructure_2021. The text calls this not difficult to see and gives no proof.

The examples. In $X = [0,\infty)^n \oplus W$, the fixed set of an idempotent sc-operator L mapping the quadrant into itself Chapter2/TameSub · TameSubExample.fixSet is a sub-M-polyfold, L restricted to the quadrant being a retraction Chapter2/TameSub · TameSubExample.isSubMPolyfold_fixSet; if L keeps the $\mathbb{R}^n$-component, L preserves d_X (condition (2)) and (3) holds in the identity chart, so the fixed set is a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.isTameSubMPolyfold_fixSet. For $n = 2$, $A = \{(x,y,0)\}$ Chapter2/TameSub · TameSubExample.setA is the case $L(a,w) = (a,0)$: a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.isTameSubMPolyfold_setA_2021 with tame induced structure Chapter2/TameSub · TameSubExample.hasTameInducedStructure_setA_2021. $B = \{(x,x,0)\}$ Chapter2/TameSub · TameSubExample.setB is a sub-M-polyfold Chapter2/TameSub · TameSubExample.isSubMPolyfold_setB. No retraction onto B with (1) and (2) is defined near the corner: for small $t > 0$ the point $(t,0,0)$ has $d_X = 1$, while d_X takes only the values 0 and 2 on B Chapter2/TameSub · TameSubExample.not_mem_tameSubData_setB_2021. So B is not a tame sub-M-polyfold Chapter2/TameSub · TameSubExample.not_isTameSubMPolyfold_setB_2021, and (3) plays no role. Yet $t \mapsto (t,t,0)$ is an sc-diffeomorphism from the half-line onto B with its induced structure Chapter2/TameSub · TameSubExample.halfLineToB_isScDiffeoOn_2021, which is therefore tame Chapter2/TameSub · TameSubExample.hasTameInducedStructure_setB_2021. For $W = \mathbb{R}$ these are the text's statements Chapter2/TameSub · TameSubExample.examples_A_B_2021.

II.36 The 2021 edition: the tame M-polyfold bX (Theorem 2.7.13 (2)–(4), Definition 2.7.14, (2.45)–(2.47))

Local faces in a tame chart. Every point of a tame M-polyfold has tame charts with domain inside a given neighbourhood Chapter2/TameBoundaryStructure · MPolyfold.exists_tame_chart_subset, and near a point of a face θ of a face-structured open V the face is, in such a chart, the zero set of one coordinate, as in Theorem 2.5.17 (preprint 2.45) (§II.16) Chapter2/TameBoundaryStructure · MPolyfold.exists_localFace_of_faceIn.

Chart data. A chart datum consists of a triple (x,V,θ) with V centred at x (§II.27), a chart φ of X, an open V_1 inside its domain, an sc-isomorphism T of its model space with $\mathbb{R}^{k+1} \oplus W$ and a coordinate i vanishing on $T\varphi(\theta \cap V_1)$, such that the face projection π of Proposition 2.5.15 (preprint 2.43) maps $\varphi(\theta \cap V_1)$ onto a tame retract of $[0,\infty)^k \oplus W$, and $d_X = d \circ \pi \circ \varphi + 1$ on $\theta \cap V_1$ Chapter2/TameBoundaryStructure · MPolyfold.BChartData. Its chart is $p \mapsto \pi(\varphi(B(p)))$ on $U(x,V,\theta) \cap B^{-1}(V_1)$ Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart. Two such charts have the transition map $\pi' \circ (\varphi' \circ \varphi^{-1}) \circ \iota$, ι the face inclusion, which is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart_transition_isScSmooth, so they are compatible Chapter2/TameBoundaryStructure · MPolyfold.BChartData.chart_compatible. By the local faces, every point of θ lies in the V_1 of a datum with the triple (x,V,θ), and every class has a centred representative Chapter2/TameBoundaryStructure · TameBoundary.BSpace.exists_centred_rep; so the charts cover bX Chapter2/TameBoundaryStructure · MPolyfold.exists_bChartData and form an atlas Chapter2/TameBoundaryStructure · MPolyfold.bAtlas.

Theorem 2.7.13 (2)–(4). T_{bX} is metrizable (§II.29), hence paracompact Chapter2/TameBoundaryStructure · MPolyfold.paracompactSpace_tameBoundarySpace, and with the atlas bX is an M-polyfold Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold, tame because the models are tame retracts Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_isTame (2). In these charts B is the face inclusion ι followed by φ^{-1}, so B is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryMap_isScSmooth (3). The index of the face model is one less than that of the quadrant: $d_X(B(p)) = d_{bX}(p) + 1$ Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap, which is (4) as printed Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap_sub_one.

The characterization, centred representatives. A face θ of a face-structured open V is a sub-M-polyfold of X Chapter2/TameBoundaryStructure · MPolyfold.isSubMPolyfold_of_isFaceIn, which we take as "the structure induced from V" Chapter2/TameBoundaryStructure · MPolyfold.facePolyfold. For V centred at x, $\Phi_{(x,V,\theta)}$ Chapter2/

TameBoundaryStructure · MPolyfold.phiTot is an sc-diffeomorphism of $U(x,V,\theta)$ onto θ, because in the charts above it is the face inclusion Chapter2/TameBoundaryStructure · MPolyfold.phi_isScDiffeoOn. Every M-polyfold structure on bX with the topology T_{bX} for which all these Φ are sc-diffeomorphisms is equivalent to this one Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_unique, which has the property Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_satisfies. The sc-half of Proposition 2.7.12 for centred representatives follows Chapter2/TameBoundaryStructure · MPolyfold.proposition_2_7_12_scDiffeo_2021, and one statement collects (2)–(4), the characterization and its uniqueness Chapter2/TameBoundaryStructure · MPolyfold.theorem_2_7_13_structure_2021.

Definition 2.7.14 and the iterated boundaries. The tame boundary is bX with this structure and the sc-smooth B Chapter2/TameBoundaryStructure · MPolyfold.definition_2_7_14_2021. To iterate b, a tame M-polyfold is bundled with its space Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle, so that bX is again such a bundle Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.b. Then $b^0X = X$, $b^{k+1}X = b^k(bX)$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iter, and $B^0 = \mathrm{id}$, $B^{k+1} = B_X \circ B^k_{bX}$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iterMap, which unwinds to the text's $B^k = B_X \circ B_{bX} \circ \cdots \circ B_{b^{k-1}X}$. By (3) and (4) and induction, B^k is sc-smooth Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.iterMap_isScSmooth, $d_X(B^k(p)) = d_{b^kX}(p) + k$ Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.degIndex_iterMap, and the image of B^k is $\{d_X \geq k\}$, by induction from the image ∂X of B (§II.29) and the index formula Chapter2/TameBoundaryStructure · MPolyfold.TameMPolyfoldBundle.range_iterMap.

Witness. For the tame quadrant $[0,\infty)^n \oplus W$, $n \geq 1$, chart data exist Chapter2/TameBoundaryStructure · MPolyfold.quadrantPolyfold_bChartData_nonempty, and the n points of bX over the corner have $d_{bX} = n-1$ Chapter2/TameBoundaryStructure · MPolyfold.quadrantPolyfold_degIndex_tameBoundary.

II.37 The 2021 edition: Proposition 2.7.12 and the characterization for every representative

Centred refinements. Every triple s has an equivalent triple s' centred at its point with $U(s') \subseteq U(s)$ Chapter2/TameBoundaryNonCentred · TameBoundary.BTriple.exists_centred_subset: replace V by V^x (§II.27). On the image $\theta \cap V^x$ of $\Phi_{(x,V,\theta)}$ Chapter2/TameBoundaryNonCentred · MPolyfold.reachSet, an open subset of θ Chapter2/TameBoundaryNonCentred · MPolyfold.isOpen_reachSet equal to θ for centred V Chapter2/TameBoundaryNonCentred · MPolyfold.reachSet_eq_univ_of_isCenteredAt, the inverse is a total map Chapter2/TameBoundaryNonCentred · TameBoundary.phiInv Chapter2/TameBoundaryNonCentred · TameBoundary.phiInv_spec.

Sc-smoothness. Φ is sc-smooth on $U(x,V,\theta)$, as B followed by the inclusion into the sub-M-polyfold θ Chapter2/TameBoundaryNonCentred · MPolyfold.phiTot_isScSmoothOn. Near each point y of $\theta \cap V^x$, Φ^{-1} is $\Phi^{-1}_{s'}$ for a centred refinement s' at y, composed with the inclusion of an open piece of θ into the face of s'; that inclusion is sc-smooth because a map into a sub-M-polyfold is sc-smooth as soon as it is sc-smooth into X. This replaces the open sc-embeddings of faces in the text's argument Chapter2/TameBoundaryNonCentred · MPolyfold.phiInv_isScSmoothOn. So Φ is an sc-diffeomorphism of $U(x,V,\theta)$ onto the open set $\theta \cap V^x$ Chapter2/TameBoundaryNonCentred · MPolyfold.phi_isScDiffeoOn_nonCentred_2021.

Proposition 2.7.12 as printed. For every pair of representatives, the images of $U \cap U'$ lie in $\theta \cap V^x$ and $\theta' \cap V'^{x'}$ and are open there, and the transition map is the composite of an inverse and a forward map, hence an sc-diffeomorphism, with inverse the reverse transition map and the identity on points of X Chapter2/TameBoundaryNonCentred · MPolyfold.proposition_2_7_12_scDiffeo_nonCentred_2021.

The characterization. An M-polyfold structure on bX in which every $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto $\theta \cap V^x$ is equivalent to that of §II.36 Chapter2/TameBoundaryNonCentred · MPolyfold.tameBoundaryPolyfold_unique_allReps_2021, which has the property Chapter2/TameBoundaryNonCentred · MPolyfold.tameBoundaryPolyfold_satisfies_allReps_2021. With "onto θ" it does not hold as we read it for every representative: for the slab triple of §II.28, Φ does not map $U(x,V,\theta)$ onto θ Chapter2/TameBoundaryNonCentred · MPolyfold.not_surjOn_

`phiTot_2021`. The strongest version that holds is the one onto $\theta \cap V^x$, which for centred V is the printed statement (§II.36).

What these additions taught us. - **"Weaker" is strict**, through a classical non-regular space `Chapter2/LocalTopologyExample · remark_2_1_6_strict_2021`; the global condition of Definition 2.1.5 is what gives regularity. - **The levels of X^1 are those of X**, chart by chart `Chapter2/TangentLift · MPolyfold.shiftLevelHomeomorph`, so the iterates realize the X_m of the text. - $T(X^i) \equiv (TX)^i$ **is one statement for all** i once M-polyfolds are bundled with their spaces `Chapter2/TangentLiftAll · proposition_2_3_17_all_2021`; fibre linearity waits for a linear structure on T_xX (gap G.7). - **Condition (3) of Definition 2.5.19 is Chapter 4's general-position condition**, and its chart change is Chapter 4's `Chapter2/TameSub · MPolyfold.TameSubData.hasReducedComplementAt_of_isChart_2021`. - bX **is built from face charts of centred triples** `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold`; non-centred representatives need the image $\theta \cap V^x$ `Chapter2/TameBoundaryNonCentred · MPolyfold.phi_isScDiffeoOn_nonCentred_2021`.

Part III. Sc-Fredholm theory and the implicit function theorem (HWZ Chapter 3, §§3.1–3.8)

Chapter 3 is the reason for the first two chapters: it defines the sections of strong bundles to which an implicit function theorem applies, the *sc-Fredholm sections*, and proves that theorem. The engine is a fixed point problem with finitely many parameters, $w = B(a, w)$, $a \in \widetilde{C} \subseteq \mathbb{R}^n$, $w \in W$, where $B(a, \cdot)$ contracts with an arbitrarily small constant near 0 *on every level* W_m of the scale. Its solution $w = \delta(a)$ is the *solution germ*; the germ implicit function theorem (Theorem 3.3.6 (preprint 3.33)) says that δ is as regular as B. Around it the chapter builds: germs and their calculus (§III.1), sections of strong bundles in charts (§III.2), fillers (§§III.3–III.4), basic germs and their linearization (§III.5), sc-Fredholm germs and sections with the stability theorem 3.15 (§III.6); the germ implicit function theorem itself (§§III.7–III.11); local regularity and compactness, Theorems 3.5.1 and 3.1.13 (preprint 3.40 and 3.12) (§§III.12–III.13); the finite-dimensional geometry of good position and the inverse and implicit function theorems in partial quadrants, which the formalization proved first and the rest uses (§§III.14–III.17); the basic class and local solvability (§§III.18–III.20); the linearization of an sc-Fredholm section at a zero, good position, and the graph description of the solution set of Theorem 3.6.8 (preprint 3.54) (§§III.21–III.23); the tangent germ property, its invariance, and the manifold structure of solution sets (§III.24); and Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61) of the basic germ criterion (§III.25). Sections III.27–III.39 were added after the part was first assembled, for the Lean written since: tameness under sc-diffeomorphisms (§III.27), Corollary 3.4.3 (preprint 3.39) (§III.28), good complements whose projection maps the quadrant into itself and local solvability with such a complement (§III.29), the index in Proposition 3.6.5 (preprint 3.52) (§III.30), the linearization at a zero in charts and its chart independence (§III.31), Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros (§§III.32–III.33), Theorem 3.6.8 (2) (preprint 3.54) (§§III.34–III.36), the implicit function theorems 3.54, 3.23, 3.19, 3.24, 3.56, 3.57 assembled (§III.37), the basic germ criterion, Theorem 3.7.1 (preprint 3.58) (§III.38), and what this addendum taught us (§III.39). Sections III.40–III.46 were added in a second addendum: Theorem 3.1.21 (preprint 3.18) (§III.40); Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, by an implicit function theorem at the nearby zero (§§III.41–III.43); the dimension of the solution set as the Fredholm index and its local constancy (§III.44); the kernel as tangent space, good position at interior zeros, and the filler condition of §III.33 as a theorem (§III.45); and what the second addendum taught us (§III.46). Section III.47 repairs a tameness hypothesis, and §III.48 records the switch of Definition 3.1.11 (preprint 3.11) to the text's germ reading; the passages above that use the filling or the sc$^+$-section have been rewritten for it.

What is proved, and what is not. The germ implicit function theorem is proved *with the text's hypotheses*: no hypothesis had to be added to Theorems 3.3.2–3.3.7 (preprint 3.29–3.34) and Theorem 3.35 (preprint; not in the 2021 edition). What the formalization adds is precision about *where* the solution germ is regular. A germ is formalized as a fixed sequence of sets, and the proof shows that δ is sc^k on the germ shifted by $s_k = 2^k - 1$ levels; "if f is sc-smooth so is its solution germ" therefore holds for every k on a germ depending on k (§III.11). Elsewhere hypotheses had to be added: the good complement must be *closed*

in Lemma 3.6.9 (preprint 3.55) and in Definition 3.2.1 (preprint 3.25); and Theorem 3.2.3 (preprint 3.27) needs two regularity conditions on its data, each of which the text's proof asserts and each of which fails in an example (§III.24) — for solution sets both are theorems (§§III.29, III.35–III.36). These are the only ones. Definition 3.1.11 (preprint 3.11) is formalized as the text writes it, with the filling and the sc$^+$-section sc-smooth *germs* whose levels may shrink, and every theorem built on it holds in that reading with the text's conclusions (§§III.6, III.48); the first formalization's whole-chart reading was narrower. Definition 6.2.1 (preprint 6.1) (the index of the germ at a smooth point) and Proposition 6.2.2 (preprint 6.2) (its constancy along sc-smooth paths), from Chapter 6 of the text, are proved with the text's hypotheses — tame germ data, no bump functions, the locally defined sc$^+$-sections of Definition 6.2.1 (preprint 6.1) — for paths on $\mathbb{R}$, by the text's device run on the restriction of the bundle to a chart domain; the first proof, with global sections, assumed bump functions (§III.49). The implicit function theorems of §§III.23–III.45 were first stated with a tameness hypothesis that no strong bundle over a nonempty base satisfies, and were vacuous; they now assume germ data and good position in tame charts, as the text does (§III.47). Theorem 3.6.8 (1) (preprint 3.54) was first proved only under condition (3) of Definition 3.1.4 (preprint 3.4) at the nearby zeros (§III.33); it is now proved with the text's hypotheses, by an argument that is not the text's (§§III.41–III.43), and the condition is a theorem (§III.45). The local solvability theorems 3.43 and 3.47 are proved in full, their clause (1) ("the solution map is an sc-smooth germ") in the level-wise sense with k-dependent neighbourhoods (§III.20). Of Theorem 3.6.8 (preprint 3.54) the graph description of the solution set near a zero in good position is §III.23. Its conclusion (2), the manifold structure, and the global Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57) and Corollary 3.1.27 (preprint 3.24) are proved with the text's hypotheses (§§III.34–III.37); its conclusion (1), good position at the nearby zeros, first under the added filler condition (§III.33), then with the text's hypotheses (§III.43), and with it the local Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) (§§III.37, III.43), with the dimension of the solution set equal to the Fredholm index (§III.44). Theorem 3.1.21 (preprint 3.18) is §III.40. The basic germ criterion (Theorem 3.7.1 (preprint 3.58)) is proved in the coordinates of its proof (§III.38), with Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61) (§III.25).

Seven structural choices. (1) Germs are fixed sequences of sets, and the calculus the text uses on them silently is proved (§III.1). (2) Contraction germs carry explicit contraction regions instead of the text's balls, because the regions of the iterated germ $f^{(1)}$ must be unbounded in the tangent directions (§III.7). (3) "A strong bundle chart" in the definitions of §3.1 is a chart *compatible* with the atlas (§III.2). (4) Tameness, which the text assumes throughout, is used only in Theorem 3.6.8 (preprint 3.54), and only for the chart of the germ data at the zero: through Proposition 2.5.3 (preprint 2.35), $(1 - Dr(0))E \subseteq E_0$, and through the tameness of the model of the chart $\psi \circ \varphi$ (§III.27). The theorems assume germ data and good position in tame charts, as Definitions 3.6.3, 3.6.4 and 3.6.6 (preprint 3.50, 3.51 and 3.53) do; the first formalization assumed every compatible chart tame, which cannot be met (§III.47). (5) The coordinate change of Theorem 3.4.2 (preprint 3.38) is linear, which makes the transport of solution sets in §3.5 linear algebra. (6) Classical regularity in the appendix is Mathlib's C^k within a set, and the iterations of the text's Lemmas 3.8.20 and 3.8.26 (preprint 3.76 and 3.82) are replaced by bootstrapping a derivative formula. (7) In Definition 3.1.11 (preprint 3.11) the filling g and the sc$^+$-section s are germs on germs of neighbourhoods of their own, and the basic germ is shrunk to the germ G'' of the points that ψ^{-1} sends into both, so that the nearby zeros produced by the implicit function theorem are points where g and s are sc^1 (§§III.6, III.48).

Map.

- `Chapter3/Germ`, `Chapter3/GermLevels`: Definitions 3.1.1, 3.1.2 (preprint 3.1, 3.2); the calculus of germs; Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs.
- `Chapter3/StrongBundleSections`: the section algebra, compatible charts, Definitions 3.1.3, 3.1.6, 3.1.11, 3.1.16 (preprint 3.3, 3.6, 3.11, 3.14), Theorem 3.1.18 (preprint 3.15).
- `Chapter3/Filler`, `Chapter3/FillerLinearAlgebra`, `Chapter3/FillerLinearization`: Definition 3.1.4 (preprint 3.4), Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17), Proposition 3.1.5 (preprint 3.5), Remark 3.6.2 (preprint 3.49).
- `Chapter3/BasicGermLinearization`: Lemma 3.1.9 (preprint 3.9) at 0, Proposition 3.1.10 (preprint

3.10).

- `Chapter3/ContractionGerm`, `Chapter3/GermIFT`: Definition 3.3.1 (preprint 3.28), Theorems 3.3.2, 3.3.3 (preprint 3.29, 3.30), Lemma 3.3.4 (preprint 3.31), the induction (Lemma 3.3.5 (preprint 3.32), Theorem 3.3.6 (preprint 3.33)), Corollary 3.3.7 (preprint 3.34), Theorem 3.35 (preprint; not in the 2021 edition).
- `Chapter3/LocalCompactness`, `Chapter3/SectionCompactness`: the parametrized fixed point theorem, Definition 3.1.7 (preprint 3.7), Theorem 3.5.1 (preprint 3.40), Theorem 3.1.13 (preprint 3.12), Corollary 3.1.15 (preprint 3.13).
- `Chapter3/GoodPosition`, `Chapter3/GoodPositionGraph`: Definition 3.1.24 (preprint 3.21), Proposition 3.1.25 (preprint 3.22), Lemmas 3.5.9, 3.8.8 (preprint 3.45, 3.65), Proposition 3.8.12 (preprint 3.69), Lemma 3.8.13 (preprint 3.70).
- `Chapter3/QuadrantInverse`, `Chapter3/QuadrantIFT`: Theorems 3.8.15, 3.8.22 (preprint 3.72, 3.78) with their lemmas.
- `Chapter3/BasicClass`, `Chapter3/SolutionLinearization`, `Chapter3/LocalSolvability`: Theorem 3.4.2 (preprint 3.38), Corollary 3.5.3 (preprint 3.41), Lemmas 3.5.5, 3.5.8, 3.5.10 (preprint 3.42, 3.44, 3.46), Theorem 3.5.6 (preprint 3.43), Theorem 3.5.11 (preprint Corollary 3.47).
- `Chapter3/FredholmGermLinearization`, `Chapter3/GoodPositionGerm`, `Chapter3/GoodPositionOpen`, `Chapter3/GlobalIFT`: Proposition 3.6.5 (preprint 3.52), Definition 3.6.6 (preprint 3.53), Lemma 3.6.9 (preprint 3.55), Theorem 3.6.8 (preprint 3.54) (the linear part and the graph description), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57) from the tangent germ property.
- `Chapter3/TangentGerm`: Definition 3.2.1 (preprint 3.25), Proposition 3.2.2 (preprint 3.26), Theorem 3.2.3 (preprint 3.27).
- `Chapter3/BasicGermCriterion`, `Chapter3/InverseFamily`: Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) (abstract form), Lemma 3.7.4 (preprint 3.61).
- Added later (§§III.27–III.39): `Chapter2/TameInvariance` (tameness under sc-diffeomorphisms); `Chapter3/BasicClassComposition` (Corollary 3.4.3 (preprint 3.39), the calculus of strong bundle isomorphisms); `Chapter3/GoodComplementProjection`, `Chapter3/LocalSolvabilityClosed` (good complements with a projection, Theorems 3.5.6, 3.5.11 (preprint 3.43, 3.47) with a closed projecting complement); `Chapter3/FredholmGermIndex` (Proposition 3.6.5 (preprint 3.52) with its index); `Chapter3/LinearizationCharts` (the linearization at a zero in charts, chart independence, Definition 3.6.6 (preprint 3.53) from a non-centred chart); `Chapter3/GlobalIFTFull`, `Chapter3/GlobalIFTNearbyGraph`, `Chapter3/GlobalIFTNearbyKernel`, `Chapter3/GlobalIFTNearbySurj`, `Chapter3/GlobalIFTNearbyPerturb` (Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros); `Chapter3/GlobalIFTCenter`, `Chapter3/SolutionGermRegularity` (Theorem 3.6.8 (2) (preprint 3.54) at the centre); `Chapter3/SolutionGermRecentre` (the re-centring, regular data, Proposition 3.8.17 (preprint 3.73) / Lemma 3.8.21 (preprint 3.77) with weak hypotheses); `Chapter3/GlobalIFTTheorems` (Theorem 3.2.3 (preprint 3.27) with locally constant dimension, Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19), Corollary 3.1.27 (preprint 3.24), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)); `Chapter3/BasicGermCriterionProof`, `Chapter3/BasicGermCriterionIndex` (Theorem 3.7.1 (preprint 3.58)).
- Added in the second addendum (§§III.40–III.46): `Chapter3/ParamPerturbation` (Theorem 3.1.21 (preprint 3.18), the pull-back bundle, inert parameters, the linearization of the perturbed section); `Chapter3/GlobalIFTNearbyOntoLemmas`, `Chapter3/GlobalIFTNearbyOntoGerm`, `Chapter3/GlobalIFTNearbyOntoXi`, `Chapter3/GlobalIFTNearbyOntoCurve`, `Chapter3/GlobalIFTNearbyOnto` (Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses, Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19) with them); `Chapter3/GlobalIFTIndex` (the index of the linearization in every chart, dimension = index, the index locally constant, the kernel as tangent space, good position at interior zeros, the filler condition as a theorem).
- Added with the repair of the tameness hypothesis (§III.47): `Chapter3/TameGerms` (tame charts, Definitions 3.6.3, 3.6.4, 3.6.6 (preprint 3.50, 3.51, 3.53) in a tame chart, good position read in a tame chart, Theorem 3.1.18 (preprint 3.15) for tame germ data); `Chapter3/TameVacuity` (a non-tame retraction

onto a product with a point).

- Added with the germ reading of Definition 3.1.11 (preprint 3.11) (§III.48): `Chapter3/GermVsChart` (a germ sc-smooth on no neighbourhood); `Chapter3/GermFilling`, `Chapter3/GermFillingChart` (Definitions 3.1.6 and 3.1.11 (preprint 3.6 and 3.11) on germs, the bridges to the whole-chart notions); `Chapter3/GermCalculus`, `Chapter3/GermCalculusParam`, `Chapter3/GermPushforward`, `Chapter3/GermFillerLinearization` (sums, chain rules with whole-chart maps, the fill defect, push-forwards and products of germs, Proposition 3.1.5 (preprint 3.5) for filling germs); `Chapter3/BasicGermShrink`, `Chapter3/BasicGermShrinkIFT` (the shrunk basic germ G'' and Corollary 3.5.3 (preprint 3.41) on it); and the files `Chapter3/…G` (`FredholmSectionsG`, `TameGermsG`, `GoodPositionGermG`, `FredholmGermLinearizationG`, `GlobalIFTNearbyGraphG`, `GlobalIFTTheoremsG`, …), where the canonical names of Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4, 3.6.6 (preprint 3.11, 3.14, 3.50, 3.51, 3.53) and of the theorems built on them now live. Their original files (`StrongBundleSections`, `GlobalIFT`, …) keep the whole-chart versions, named `…WholeChart` / `…_wholeChart`.
- Added in the fourth addendum (§§III.47, III.49): `Chapter3/TameGermData` (condition (3) of Definition 3.1.4 (preprint 3.4) as a condition on the chart; a tame atlas whose chart carries no germ data); `Chapter3/IndexAlongPaths`, `Chapter3/IndexAlongPathsDevice` (Definition 6.2.1 (preprint 6.1), bump functions on $\mathbb{R}^n \times X$, the section along a path, the slice of a section of the pull-back, Proposition 6.2.2 (preprint 6.2)).
- Added in the fifth addendum (§III.49): `Chapter3/RestrictOpen` (the restriction of M-polyfolds and strong bundles to open sets); `Chapter3/RestrictOpenGerms` (germ data and the index on the restriction); `Chapter3/IndexLocallyConstantLocal` (sc$^+$-sections through the fibre points from a chart over the whole base, the index along the zero set without bump functions); `Chapter3/IndexAlongPathsLocal` (Definition 6.2.1 (preprint 6.1) with local sections, Proposition 6.2.2 (preprint 6.2) with the text's hypotheses); `Chapter3/IndexAlongPathsLocalBump` (the two readings of Definition 6.2.1 (preprint 6.1) agree under bump functions).

Throughout, P and Q are finite-dimensional normed spaces with the constant sc-structure (the text's $\mathbb{R}^n$ and $\mathbb{R}^N$), $\widetilde{C} \subseteq P$ a set containing 0, convex, with nonempty interior (the text's $[0,\infty)^k \oplus \mathbb{R}^{n-k}$), W an sc-Banach space, $E = P \oplus W$ with levels $P \times W_m$ `Chapter3/ContractionGerm · ScBanach.paramProd`. Because all levels of P coincide, the condition "$\widetilde{C} \times W$ is quadrant-like" is just "$0 \in \widetilde{C}$, convex, nonempty interior" `Chapter3/ContractionGerm · isQuadrantLike_const_iff` `Chapter3/ContractionGerm · IsQuadrantLike.paramProd`.

III.1 Germs and their calculus (Definitions 3.1.1, 3.1.2 (preprint 3.1, 3.2))

Definition 3.1.1 (preprint 3.1). An *sc-germ of neighbourhoods* around $0 \in C$ is a family of sets $U_m \subseteq E_m$, each of the form $O \cap C_m$ with O open in E_m, containing 0, and nested along the inclusions: $\iota U_{m+1} \subseteq U_m$ `Chapter3/Germ · ScNbhdGerm`. Its *tangent germ* has levels $U_{m+1} \times E_m$ `Chapter3/Germ · ScNbhdGerm.tangent`. The text's example, a relatively open $U \ni 0$ with $U_m = U \cap E_m$, is `ofSet` `Chapter3/Germ · ScNbhdGerm.ofSet`. The germ calculus needs one property of C: every level C_m is convex with nonempty interior and $0 \in C$ ("quadrant-like") `Chapter3/Germ · IsQuadrantLike`. Partial quadrants have it `Chapter3/Germ · IsPartialQuadrant.isQuadrantLike`, and it passes to TC `Chapter3/Germ · IsQuadrantLike.tangentSet`. It makes every U_m a set of unique differentiability `Chapter3/Germ · ScNbhdGerm.uniqueDiffWithinAt`, the intersection of an open and a convex set `Chapter3/Germ · ScNbhdGerm.relOpenConvex`.

A germ is a *fixed* sequence of sets. The text says "the size of the sets U_m does not matter", i.e. it reads germs up to shrinking; the formalization cannot, and must say on which sequence each statement holds. One trivial remark becomes important in §III.9: the re-indexed sequence $(U_{m+1})_m$ is in general not a germ, since U_{m+1} is relatively open in C_{m+1} and not in C_m; it is one when all levels coincide, i.e. for germs in the finite-dimensional parameter space P.

Definition 3.1.2 (preprint 3.2). f is an *sc^0-germ* on $G = (U_m)$ if $f(U_m) \subseteq F_m$ and the level map $f_m : U_m \to F_m$ is continuous for every m `Chapter3/Germ · IsSc0Germ`. It is an *sc^1-germ with derivative family* D if it is sc^0, differentiable at every $x \in U_1$ within U_1 in the sense of Definition 1.1.15

(preprint 1.14) (increments in E_1, remainder measured in F_0) with derivative $D(x)$, and its tangent map $(x,h) \mapsto (f(x), D(x)h)$ is an sc⁰-germ on the tangent germ Chapter3/Germ · IsSc1GermWith; an sc¹-germ if some such D exists Chapter3/Germ · IsSc1Germ, in which case a derivative family is chosen Chapter3/Germ · scDerivGerm and gives the tangent germ Chapter3/Germ · tangentMapGerm. On quadrant-like C the derivative is unique on U_1 Chapter3/Germ · IsSc1GermWith.eq_of_uniqueDiffWithinAt, so the choice does not matter there Chapter3/Germ · IsSc1Germ.tangentMapGerm_eqOn. Then sc^k is defined by recursion — sc⁰ for $k = 0$, and sc¹ with sc^k tangent germ for $k+1$ Chapter3/Germ · IsSckGerm — and sc-smooth means sc^k for all k Chapter3/Germ · IsScSmoothGerm. These are the notions of Part I, §I.11, with $U \cap E_m$ replaced by U_m, and the bridge is exact: an sc^k map on a relatively open $U \ni 0$ is an sc^k-germ on `ofSet U` Chapter3/Germ · IsSck.isSckGerm Chapter3/Germ · IsScSmooth.isScSmoothGerm.

The calculus of germs. The text restricts, composes and pairs germs without comment. All of it is proved, with the proofs of Part I, §§I.11–I.12. (1) sc^k-germs depend only on the values on U_0 Chapter3/Germ · IsSckGerm.congr and restrict to smaller germs Chapter3/Germ · IsSckGerm.mono — by induction on k, using that the tangent germs of the smaller germ and of the larger one agree on the smaller tangent germ Chapter3/Germ · IsSc1Germ.tangentMapGerm_mono. (2) *The chain rule.* A composition $g \circ f$ makes sense level by level only if f maps U_m into the domain V_m of g for every m Chapter3/Germ · ScNbhdGerm.MapsToGerm; then sc⁰-germs compose Chapter3/Germ · IsSc0Germ.comp, the derivative of $g \circ f$ at $x \in U_1$ is $Dg(f(x)) \circ Df(x)$ — proved as in §I.12 with the mean value inequality along segments in V_1 and the uniform smallness of $Dg(y) - Dg(y_0)$ on compact sets Chapter3/Germ · IsSc1GermWith.hasFDerivWithinAt_comp — so $g \circ f$ is sc¹ with $T(g \circ f) = Tg \circ Tf$ on the tangent germ Chapter3/Germ · IsSc1GermWith.comp Chapter3/Germ · IsSc1Germ.comp, and by induction on k it is sc^k Chapter3/Germ · IsSckGerm.comp Chapter3/Germ · IsScSmoothGerm.comp. (3) An sc-operator T is an sc¹-germ with derivative T itself Chapter3/Germ · ScOperator.isSc1GermWith, its tangent map is the sc-operator (T, T) Chapter3/Germ · ScOperator.tangentOp, and so it is an sc-smooth germ on every germ Chapter3/Germ · ScOperator.isSckGerm Chapter3/Germ · ScOperator.isScSmoothGerm; composing with one on the outside preserves sc^k Chapter3/Germ · IsSckGerm.operator_comp. (4) The pairing (f, g) of two sc^k-germs is an sc^k-germ Chapter3/Germ · pairMap Chapter3/Germ · IsSckGerm.pair, through the reshuffle $TF \oplus TG \cong T(F \oplus G)$ Chapter3/Germ · tangentProdComm. (5) Where the text says "near 0" it shrinks germs: for an sc⁰-germ f with $f(0) = 0$ and a germ G' the sets $\{x \in U_m : f_m(x) \in U'_m\}$ form the largest sub-germ on which f maps into G' Chapter3/GermFilling · ScNbhdGerm.preimage Chapter3/GermFilling · ScNbhdGerm.preimage_mapsToGerm.

Propositions 1.2.1–1.2.3 (preprint 1.16–1.18) for germs. The text applies these to germs ("recalling Proposition 1.17" in Lemma 3.1.9 (preprint 3.9); Lemma 3.3.4 (preprint 3.31); Corollary 3.3.7 (preprint 3.34)) but states them for relatively open sets. For an sc¹-germ with derivative family D and $x \in U_{m+1}$, $D(x)$ restricts to a bounded level operator $E_m \to F_m$ Chapter3/GermLevels · IsSc1GermWith.exists_levelDeriv Chapter3/GermLevels · IsSc1GermWith.toBase_levelDeriv, the level-m tangent map is $(x,h) \mapsto (f_{m+1}(x), D_m(x)h)$ Chapter3/GermLevels · IsSc1GermWith.levelMap_tangentMapWith, jointly continuous Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv and continuous in x in the operator norm (the compact-open lemma of §I.13) Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv_clm; and $f : U_{m+1} \to F_m$ is differentiable within U_{m+1} with derivative $D_m(x)$, by the integral representation $f(x+k) - f(x) = \int_0^1 D(x+\tau k)k\,d\tau$ in F_m (Proposition 1.2.1 (preprint 1.16)) Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level. The shifted sequence $(U_{m+1})_m$ is a germ in E^1 around $0 \in C^1$ Chapter3/GermLevels · ScNbhdGerm.shift, and an sc^k-germ on G is an sc^k-germ $G^1 \to F^1$ (Proposition 1.2.2 (preprint 1.17)) Chapter3/GermLevels · IsSc0Germ.shift Chapter3/GermLevels · IsSc1GermWith.shift Chapter3/GermLevels · IsSckGerm.shift. An sc^k-germ is C^k from U_{m+k} to F_m (Proposition 1.2.3 (preprint 1.18)), by the proof of §I.16 with U_m for $U \cap E_m$ Chapter3/GermLevels · IsSckGerm.contDiffOn_levels' Chapter3/GermLevels · IsSckGerm.contDiffOn_levels. sc^k depends on a germ only through its sets Chapter3/GermLevels · IsSckGerm.of_U_eq.

III.2 Sections of strong bundles in charts (Definition 3.1.3 (preprint 3.3))

Let $P : Y \to X$ be a strong bundle over an M-polyfold (Part II, §II.19). A strong bundle chart $(\Phi, P^{-1}(V), K, U \lhd F)$ has a model $K = R(U \lhd F)$ for a strong bundle retraction R. Its base and fibre parts are read off: $r(u) = (R(u,0))_1$ `Chapter3/Filler · StrongRetr.r`, $\rho(u)h = (R(u,h))_2$ `Chapter3/Filler · StrongRetr.ρ`, and $R(u,h) = (r(u), \rho(u)h)$ because the base component of a strong bundle map does not depend on the fibre variable `Chapter3/Filler · StrongRetr.apply_eq`. Comparing components of $R \circ R = R$ gives $r \circ r = r$ `Chapter3/Filler · StrongRetr.r_r` and $\rho(r(u))\rho(u) = \rho(u)$ `Chapter3/Filler · StrongRetr.ρ_r_ρ`; for $x \in O = r(U)$, $r(x) = x$ `Chapter3/Filler · StrongRetr.r_eq_self` and $\rho(x)$ is a projection `Chapter3/Filler · StrongRetr.ρ_ρ`; R fixes the points of K `Chapter3/Filler · StrongRetr.apply_eq_self_of_mem_K`, so $\rho(x)h = h$ for $(x,h) \in K$ `Chapter3/Filler · StrongRetr.ρ_eq_self_of_mem_K`. $\rho(u)$ is linear by Definition 2.6.1 (preprint 2.47) and continuous because R is sc^0 on level 0 `Chapter3/Filler · StrongRetr.ρ_continuous`; it is a bounded operator `Chapter3/Filler · StrongRetr.ρCLM` — the text's "$\rho(u) : F \to F$ is a bounded linear operator", without comment.

Principal parts and the section algebra. For a section f (a map with $P \circ f = \mathrm{id}$), $\Phi \circ f \circ \varphi^{-1}(z) = (z, \mathbf{f}(z))$; $\mathbf{f}$ is its *principal part* `Chapter3/StrongBundleSections · StrongBundleChart.principal` `Chapter3/StrongBundleSections · StrongBundleChart.apply_section`. The fibrewise sum $f + s$ is defined with the vector space structure of the fibres $P^{-1}(x)$ `Chapter3/StrongBundleSections · sectionAdd` and is a section `Chapter3/StrongBundleSections · IsSection.sectionAdd`. Since Φ is linear on the fibres `Chapter3/StrongBundleSections · StrongBundleChart.snd_add` and maps the zero of $P^{-1}(x)$ to $(\varphi(x), 0)$ `Chapter3/StrongBundleSections · StrongBundleChart.snd_fiberZero`, the principal part of $f + s$ is $\mathbf{f} + \mathbf{s}$ `Chapter3/StrongBundleSections · StrongBundleChart.principal_sectionAdd` and $f(x) = 0$ iff $\mathbf{f}(\varphi(x)) = 0$ `Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff` `Chapter3/StrongBundleSections · IsZeroAt`. The fibres of K over O are linear: $(u,a), (u,b) \in K$ give $(u, a+b) \in K$, because membership is $\rho(u)a = a$ `Chapter3/StrongBundleSections · StrongBundleChart.add_mem_K`. *The sum rule*, used silently in "$f + s$ is an sc-smooth section", is pairing (Part I) followed by the addition sc-operator $F \oplus F \to F$ `Chapter3/StrongBundleSections · ScBanach.addOp` `Chapter3/StrongBundleSections · IsScSmooth.add` `Chapter3/StrongBundleSections · IsScSmooth.sub` `Chapter3/StrongBundleSections · IsScSmooth.neg`; a map sc-smooth into F^1 is sc-smooth into F (compose with the inclusion) `Chapter3/StrongBundleSections · IsScSmooth.toBase_shift`.

Which charts. "A strong bundle chart" in Definitions 3.1.3, 3.1.6 and 3.1.11 (preprint 3.3, 3.6 and 3.11) must mean a chart *compatible* with the atlas of P `Chapter3/StrongBundleSections · StrongBundle.IsChart` `Chapter3/StrongBundleSections · StrongBundle.isChart_of_mem`: a chart of the atlas need not be centred at the point in question, and the notions must not depend on the atlas within its class. A compatible chart is a chart of the M-polyfold $Y[0]$ `Chapter3/StrongBundleSections · StrongBundle.IsChart.toMChart`. The price is that representatives of sections must be sc-smooth in every compatible chart. For an sc-smooth section, $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$ by Definition 2.3.8 (preprint 2.14) as it stands `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothRetractMap`. For an sc$^+$-section, sc-smooth into $Y[1]$, the principal part takes values in F_1 in every compatible chart `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.principal_mem_range`, and $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$: in a chart Φ' of the atlas this is the definition, and in the given chart it is the composition with the change of chart $(\Phi \circ \Phi'^{-1})[1]$ on $K[1]$, glued over the atlas by locality `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.isScSmoothRetractMap_one`. This is precisely the compatibility on $[1]$ of Definition 2.6.4 (preprint 2.50), used here for the first time.

Definition 3.1.3 (preprint 3.3). (f, x_0) is an *sc-smooth section germ* if x_0 is smooth and in some compatible strong bundle chart with $\varphi(x_0) = 0$ the map $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods G of 0 with $G_0 \subseteq U$, whose levels may shrink with m `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSectionGerm` — the text's germ reading, as for Definition 3.1.11 (preprint 3.11) (§III.48); until the fourth addendum the germ was the chart's $U \cap E_m$, which is stricter (§III.48). The text's "continuous section on a neighbourhood V" is the section on the chart domain; near x_0 continuity is a consequence. An sc-smooth section is an sc-smooth section germ at every smooth point with a centred chart: its repre-

sentative composed with r is sc-smooth on the relatively open U, and §III.1's bridge applies on the germ $U \cap E_m$ `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothSectionGerm`. No theorem assumes the definition.

III.3 Fillers (Definitions 3.1.4, 3.1.6 (preprint 3.4, 3.6); Lemmas 3.1.19, 3.1.20 (preprint 3.16, 3.17); Remark 3.6.2 (preprint 3.49))

Definition 3.1.4 (preprint 3.4). For the principal parts $f, g : E_0 \to F_0$ (with $O = r(U)$, $K = R(U \lhd F)$, $0 \in U$), *g fills f* if (1) $f = g$ on O near 0; (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; (3) the map $y \mapsto [1 - \rho(r(y))]g(y)$ is sc¹ on U, and its derivative at 0 restricts to a topological linear isomorphism $\ker Dr(0) \to \ker \rho(0)$ `Chapter3/Filler · IsFillerOf`. The text's condition (3) presupposes the linearization; requiring the map to be sc¹ loses nothing, since it is sc-smooth when g and R are. The sc-smoothness of *g* itself is added in Definition 3.1.6 (preprint 3.6). **Definition 3.1.6 (preprint 3.6)**: a *filled version* of **f** is a germ $(g, 0)$, and it is formalized as one, a *filling germ* `Chapter3/GermFilling · IsFillingGerm`: g is an sc-smooth germ (§III.1) on a germ G with $G_0 \subseteq U$, whose levels may shrink with m; (1) and (2) hold near 0; and in (3) the fill defect $k(y) = [1 - \rho(r(y))]g(y)$ `Chapter3/GermFilling · fillDefect` is an sc¹-germ on G with a derivative family Dk whose value $Dk(0)$ — unique (§III.1) — restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$. The first formalization took a filler sc-smooth on the whole U of the chart `Chapter3/StrongBundleSections · IsFilledVersion`, on the ground that a chart can always be restricted to $U \cap W$. That is the special case \$G = \$ `ofSet U` `Chapter3/GermFillingChart · IsFilledVersion.isFillingGerm` `Chapter3/GermFillingChart · IsFillingGerm.isFilledVersion`, and it is narrower: restricting a chart on level 0 never makes the levels shrink, and there are sc-smooth germs with shrinking levels that are sc-smooth on no neighbourhood of 0 `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth` (§III.48). The lemmas below about fillers on U are the whole-chart ones; their germ forms, which the canonical notions use, are cited next to them.

Lemma 3.1.19 (preprint 3.16). For a section s of $K \to O$ (that is, $(x, s(x)) \in K$ for $x \in O$), the extension $\tilde{s} = s \circ r$ satisfies $\tilde{s} = s$ on O (as $r(x) = x$ there), and $R(r(u), s(r(u))) = (r(u), s(r(u)))$, i.e. $\rho(r(u))\tilde{s}(u) = \tilde{s}(u)$, because $s(r(u))$ is a point of K and R fixes K `Chapter3/Filler · lemma_3_16`. The text's computation uses $s(x) \in K$ without saying so; it is the hypothesis that s is a section. **Lemma 3.1.20 (preprint 3.17).** If g fills f and s is a section of $K \to O$, then $g + s \circ r$ fills $f + s$ `Chapter3/Filler · lemma_3_17`: (1) on O, $g + s \circ r = f + s$; (2) by Lemma 3.1.19 (preprint 3.16), $(g + \tilde{s})(y) = \rho(r(y))(g + \tilde{s})(y)$ is equivalent to $g(y) = \rho(r(y))g(y)$; (3) by the same identity $[1 - \rho(r(y))](g + \tilde{s})(y) = [1 - \rho(r(y))]g(y)$ on U, so the two maps of (3) coincide on U and have the same derivative at 0 (derivatives on U_1 are unique). *For a filling germ* g on G and s sc-smooth on U the proof is the same: the identity of (3) holds on U, hence on G_0, so the derivative family Dk of the fill defect of g is one of $g + s \circ r$, a filling germ on the same G `Chapter3/GermCalculus · IsFillingGerm.add_section`; for s itself a germ on G_s the filling lives on $G \cap r^{-1}(G_s)$ `Chapter3/GermCalculus · IsFillingGerm.add_section_germ`. The regularity statements of both lemmas — $\tilde{s}$ sc$^+$, $g + \tilde{s}$ sc-smooth — are proved where they are used, in Theorem 3.1.18 (preprint 3.15) (§III.6). Condition (1) only involves f on O `Chapter3/StrongBundleSections · IsFillerOf.congr_left`.

Remark 3.6.2 (preprint 3.49). On a smaller neighbourhood W of 0 condition (1) holds on all of $O \cap W$ and (2) for all $y \in U \cap W$: intersect the two neighbourhoods given by (1) and (2) `Chapter3/FillerLinearization · IsFillerOf.remark_3_49`.

III.4 The linearization of a filler (Proposition 3.1.5 (preprint 3.5))

The text derives, "by linearization at $y = 0$", the matrix of $Dg(0)$ in the splittings $E = Dr(0)E \oplus \ker Dr(0)$, $F = \rho(0)F \oplus \ker \rho(0)$, and from it Proposition 3.1.5 (preprint 3.5). The formal proof has two halves: the derivative identities, and the linear algebra.

The derivative identities. r is sc-smooth on U `Chapter3/FillerLinearization · StrongRetr.r_isScSmooth` with

$r(0) = 0$ since $0 \in O$ Chapter3/FillerLinearization · StrongRetr.r_zero, and the chain rule applied to $r \circ r = r$ makes $Dr(0)$ idempotent Chapter3/FillerLinearization · StrongRetr.scDeriv_r_idem; $\rho(0)$ is idempotent Chapter3/FillerLinearization · StrongRetr.ρCLM_idem. The fibre part $(u,k) \mapsto \rho(u)k$ is sc¹ on $U \oplus F$ Chapter3/FillerLinearization · StrongRetr.fiber_isSc1, and its derivative at $(0,0)$ is $(h,k) \mapsto \rho(0)k$: in the fibre direction because $k \mapsto \rho(0)k$ is linear, in the base direction because $\rho(u)0 = 0$; both directional derivatives are computed on the dense level 1 and extended by continuity Chapter3/FillerLinearization · StrongRetr.scDeriv_fiber_zero. By the chain rule for $(u,k) \mapsto \rho(u)k$ composed with $y \mapsto (r(y), g(y))$ at the point $(r(0), g(0)) = (0,0)$ — this is where $g(0) = 0$ enters — the derivative of $y \mapsto [1 - \rho(r(y))]g(y)$ at 0 is $(1 - \rho(0))Dg(0)$ Chapter3/FillerLinearization · IsFillerOf.scDeriv_sub_eq Chapter3/FillerLinearization · scDeriv_comp_apply'. Next, $[1 - \rho(r(y))]g(r(y)) = 0$ near 0, from $g = f$ on O and $\rho(x)f(x) = f(x)$ on O (f a section of K) and the continuity of r; linearizing gives $(1 - \rho(0))Dg(0)Dr(0) = 0$ Chapter3/FillerLinearization · IsFillerOf.deriv_comp_r_eq_zero, i.e. $\rho(0)Dg(0)Dr(0) = Dg(0)Dr(0)$ Chapter3/FillerLinearization · IsFillerOf.ρ_deriv_comp_r. Condition (3) becomes: $(1 - \rho(0))Dg(0)$ restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$ Chapter3/FillerLinearization · IsFillerOf.exists_kerEquiv. Finally $f \circ r = g \circ r$ near 0, so $D(f \circ r)(0) = Dg(0)Dr(0)$ — the text's "$Tg(0) = Tf(0)$ on T_0O" Chapter3/FillerLinearization · IsFillerOf.scDeriv_section_comp_r.

The linear algebra. Let p, q be bounded idempotents on Banach spaces E, F and $L : E \to F$ bounded with $(1-q)Lp = 0$ and $(1-q)L : \ker p \to \ker q$ a topological isomorphism C. Then $A := L|_{\operatorname{range} p} : \operatorname{range} p \to \operatorname{range} q$ is well defined Chapter3/FillerLinearAlgebra · fillerRestrict. Two facts carry everything: if $(1-q)Lw = 0$ then $w \in \operatorname{range} p$ (injectivity of C, applied to the $\ker p$-component of w) Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_of; and $(1-q)Lz = y$ is solvable in $\ker p$ for $y \in \ker q$ (surjectivity of C) Chapter3/FillerLinearAlgebra · fillerRestrict_exists_of. From them: A is onto iff L is Chapter3/FillerLinearAlgebra · fillerRestrict_surjective_iff; the inclusion $\ker A \to \ker L$ is a bijection — the text's $\ker Dg(0) = \ker Tf(0) \oplus \{0\}$ Chapter3/FillerLinearAlgebra · fillerRestrict_kerEquiv; $\operatorname{range} A = \operatorname{range} L \cap \operatorname{range} q$ Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_iff and $\operatorname{range} q/\operatorname{range} A \cong F/\operatorname{range} L$ Chapter3/FillerLinearAlgebra · fillerRestrict_cokerEquiv. So the indices are equal as $\dim\ker - \dim\operatorname{coker}$ with no finiteness assumption Chapter3/FillerLinearAlgebra · fredholmIndex_fillerRestrict. Closedness of the ranges transfers both ways: $\operatorname{range} A$ is closed in $\operatorname{range} q$ when $\operatorname{range} L$ is Chapter3/FillerLinearAlgebra · fillerRestrict_isClosed_range_of, and conversely $\operatorname{range} L$ is the preimage of $\operatorname{range} A$ under the continuous $M(y) = y - LC^{-1}(1-q)y$ Chapter3/FillerLinearAlgebra · isClosed_range_of_fillerRestrict — this is where "topological" in (3) is used. Hence A is Fredholm iff L is Chapter3/FillerLinearAlgebra · fillerRestrict_isFredholm_iff, and the three statements together Chapter3/FillerLinearAlgebra · proposition_3_5. The text's proof factorizes $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix} = \begin{pmatrix} 1 & BC^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$ and uses the additivity of the index; the isomorphisms of kernels and cokernels avoid both.

Proposition 3.1.5 (preprint 3.5). For a filler g of the section f of $K \to O$, g sc-smooth with $g(0) = 0$: with $p = Dr(0)$, $q = \rho(0)$, $L = Dg(0)$ the hypotheses of the linear algebra hold by the identities above, and $Tf(0) := Dg(0)|_{Dr(0)E} : Dr(0)E \to \rho(0)F$ is onto iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, with equal index Chapter3/FillerLinearization · IsFillerOf.proposition_3_5. (The text writes the target of the lower right entry as "$(1 - D\rho(0))F$"; it is $(1 - \rho(0))F$.) *For a filling germ* (§III.3) the proof is the same with any derivative family Dg of g on any germ inside U: the fill defect is an sc-smooth germ on the germ of g, by the chain rule with the whole-chart outer map R (§III.48) Chapter3/GermCalculus · fillDefect_isScSmoothGerm, and the value at 0 of its derivative family is $(1 - \rho(0))Dg(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, which replaces `scDeriv_sub_eq`; so $Tf(0) = Dg(0)|_{Dr(0)E}$ is onto, Fredholm, of the index of $Dg(0)$ iff $Dg(0)$ is Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_5. $Dg(0)$ does not depend on the family or the germ Chapter3/GoodPositionGermG · IsSc1GermWith.deriv_zero_eq.

III.5 Basic germs, Lemma 3.1.9 (preprint 3.9) and Proposition 3.1.10 (preprint 3.10)

Definition 3.1.7 (preprint 3.7) (and 3.37). $f : P \times W_0 \to Q \times W_0$ is a *basic germ* on a germ G around 0 in $\widetilde{C} \oplus W$, with contraction map B, if f is an sc-smooth germ with $f(0) = 0$ whose W-component is $w - B(a, w)$ on U_0, and B is an sc^0-contraction germ in the sense of Definition 3.3.1 (preprint 3.28) (§III.7) `Chapter3/LocalCompactness · IsBasicGerm`. ($B = w - f_W$ is then itself an sc-smooth germ `Chapter3/LocalSolvability · IsBasicGerm.B_isScSmoothGerm`.)

Lemma 3.1.9 (preprint 3.9). The text: for every $\varepsilon > 0$ and m, $|D_2B(a,w)\zeta|_m \le \varepsilon|\zeta|_m$ when $(a, w) \in E_{m+1}$ is close to 0 in E_m; in particular $D_2B(0,0) = 0$. Its proof: for $\zeta \in W_{m+1}$ the difference quotient of $t \mapsto B(a, w + t\zeta)$ is bounded by the contraction estimate, and the derivative is the limit (this needs $B : U_{m+1} \to W_m$ differentiable, Proposition 1.2.1 (preprint 1.16) for germs — "recalling Proposition 1.17" in the text); then density of W_{m+1} in W_m. Only sc^1 is used. The formalization proves the three forms the chapter needs. (a) On level m in the working form of §III.7, with the constant $\frac{1}{2}$: at $(a, w) \in U_{m+1}$ with $a \in N(m+1) \cap \widetilde{C}$, $w \in O(m+1)$ and $\iota w \in O(m)$, $\|D_2B(a,w)\|_{L(W_m)} \le \frac{1}{2}$ `Chapter3/ContractionGerm · ContractionGerm.norm_levelDeriv_snd_le`; a linear map of norm $\le \frac{1}{2}$ then has no nonzero fixed point `Chapter3/ContractionGerm · ContractionGerm.eq_zero_of_norm_le_half`. (b) $D_2B(0,0) = 0$, from the difference quotients along $t \mapsto (0, t\zeta)$, bounded by $\varepsilon|\zeta|$ for every ε, then density `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`. (c) On level 0 with any ε: $|D_2B(x)\zeta|_0 \le \varepsilon|\zeta|_0$ for $x \in U_1$ small *on level* 1 `Chapter3/SolutionLinearization · IsSc0ContractionGerm.exists_norm_deriv_snd_le` — the curve $t \mapsto x + t(0, \zeta)$, $\zeta \in W_1$, must stay in U_1, where B is differentiable.

Proposition 3.1.10 (preprint 3.10). Let $\widetilde{C}$ be a partial quadrant, h a basic germ, s an sc^1-germ into $(Q \oplus W)^1$ (an sc$^+$-germ) with derivative family Ds, and $f = h + \iota s$ sc^1 with derivative family Df. Then $Df(0)$ is an sc-operator, sc-Fredholm of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · proposition_3_10`. *Proof.* The derivative of an sc^1-germ at the smooth point 0 is an sc-operator, its level-m map being the fibre component of the level-m tangent map at $(0, \cdot)$ (the remark after Proposition 1.2.1 (preprint 1.16)) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`. The operator (32) of the text, $T_0(\delta a, \delta w) = (0, \delta w)$ `Chapter3/BasicGermLinearization · ScBanach.projW0p`, is sc-Fredholm with kernel $P \oplus 0$ and cokernel $Q \oplus 0$ `Chapter3/BasicGermLinearization · ScBanach.projWSplitting Chapter3/BasicGermLinearization · ScBanach.isScFredholm_projW0p`, of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · ScBanach.scIndex_projW0p`. The text says "$Df(0)$ is an sc$^+$-perturbation of (32)"; the formal proof builds the sc$^+$-operator $S = Df(0) - T_0$ level by level `Chapter3/BasicGermLinearization · proposition_3_10_scPlus`: its Q-component is sc$^+$ because Q has all levels equal; $\mathrm{pr}_W Dh(0)(\delta a, 0) = -D_1B(0)\delta a = -DB(0)(\delta a, 0)$ has values in W_∞ ($(\delta a, 0)$ is smooth and $DB(0)$ an sc-operator) and its level maps $E_m \to W_{m+1}$ factor through $DB(0)_{m+1}$; $\iota Ds(0)$ is sc$^+$ because $Ds(0)$ is an sc-operator into $(Q \oplus W)^1$; and $\mathrm{pr}_W Dh(0)(0, \delta w) = \delta w$ because $D_2B(0) = 0$ (Lemma 3.1.9 (preprint 3.9) (b)). The identities $Df(0) = Dh(0) + \iota Ds(0)$ and $\mathrm{pr}_W Dh(0) = \mathrm{pr}_W - DB(0)$ are derivative identities within U_1, extended to E_0 because two bounded operators agreeing on the dense E_1 are equal `Chapter3/BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one`. Proposition 1.1.11 (preprint 1.10) with its index statement concludes `Chapter1/ScFredholm · proposition_1_10 Chapter1/FredholmIndex · proposition_1_10_index`. For a basic germ alone ($s = 0$) `Chapter3/FredholmGermLinearization · ScOperator.zero Chapter3/FredholmGermLinearization · proposition_3_10_basic`, with its level-0 consequence: $Dh(0)$ is Fredholm of index $\dim P - \dim Q$ `Chapter3/FredholmGermLinearization · proposition_3_10_basic_level_zero`.

The second statement of Proposition 3.1.10 (preprint 3.10) ($Df(a, w)$ Fredholm on every level m for (a, w) small) is proved on level 0 for the germs of Theorem 3.4.2 (preprint 3.38), at points of U_1 small on level 1 (§III.19); on levels $m \ge 1$ it is not formalized.

III.6 Sc-Fredholm germs and sections; Theorem 3.1.18 (preprint 3.15) (Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4 (preprint 3.11, 3.14, 3.50, 3.51))

Definition 3.1.11 (preprint 3.11). A *strong bundle isomorphism* $\Psi : U \lhd F \to U' \lhd F'$ is a pair of strong bundle maps (Definition 2.6.1 (preprint 2.47)) inverse to each other on $U \oplus F$ and $U' \oplus F'$ `Chapter3/StrongBundleSections · IsStrongBundleIso`; its base map is ψ. A principal part $k : U \to F$ is *conjugated to a basic germ* if there are a strong bundle isomorphism with $\psi(0) = 0$ onto U' relatively open in $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant of a finite-dimensional P), a basic germ h on a germ G with $G_0 \subseteq U'$, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0 `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`; this depends only on k on U `Chapter3/StrongBundleSections · IsConjugateToBasicGerm.congr`. (f, x_0) is an *sc-Fredholm germ* if in a compatible chart centred at x_0 the principal part has a filling germ g (§III.3), and there is a local sc$^+$-section s — an sc-smooth germ into F^1 on a germ G_s with $(G_s)_0 \subseteq U$ `Chapter3/GermFilling · IsScPlusGerm` — with $\iota s(0) = g(0)$ such that $g - \iota s$ is conjugated to a basic germ `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm`. This is the text's reading: g and s are germs, on germs of neighbourhoods of their own whose levels may shrink with the level, while Ψ is an isomorphism of the whole local bundles and the conjugation identity is asked near 0 only, so it depends only on the germ of $g - \iota s$ `Chapter3/GermFillingChart · IsConjugateToBasicGerm.congr_nhds`. The first formalization asked g and s to be sc-smooth on $U \cap E_m$ for every m `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart`. That notion implies the germ reading `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGermWholeChart.isScFredholmGerm` and is exactly the germ reading with g and s sc-smooth on U `Chapter3/GermFillingChart · StrongBundle.isScFredholmGerm_iff_germG_wholeChart`; whether every germ datum can be traded for a whole-chart one is open (§III.48). Definitions 3.1.16, 3.6.3, 3.6.4, 3.6.6 (preprint 3.14, 3.50, 3.51, 3.53) and every theorem of this part built on them are stated for the germ reading; the whole-chart notions and theorems are kept under the names `…WholeChart`, `…_wholeChart`. The text's $C' = T\psi(0)C$, used in the proof of Theorem 3.6.8 (preprint 3.54), is not part of the data; it is a theorem (§III.23).

Definition 3.1.16 (preprint 3.14). f is an *sc-Fredholm section* if it is sc-smooth, regularizing ($x \in X_m$ and $f(x) \in Y_{m,m+1}$ imply $x \in X_{m+1}$) and an sc-Fredholm germ at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection`. The text assumes a tame bundle; nothing in Definitions 3.1.3–3.1.16 (preprint 3.3–3.14) or in Theorem 3.1.18 (preprint 3.15) uses tameness, so it is not assumed. (Definitions 3.6.3, 3.6.4 (preprint 3.50, 3.51) restate 3.11, 3.14.) A zero of a regularizing section is a smooth point: the zero section lies in $Y_{m,m+1}$ over X_m `Chapter3/SectionCompactness · StrongBundle.fiberZero_mem_doubleLevel`, so a zero on level m is on level $m + 1$, and by induction on every level `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`.

Theorem 3.1.18 (preprint 3.15). If f is an sc-Fredholm section and s an sc$^+$-section, then $f + s$ is an sc-Fredholm section `Chapter3/FredholmSectionsG · theorem_3_15`. *Proof.* The text says "$f + s$ is an sc-smooth section which is also regularizing" and checks the germ property; each clause is a proof.

Sc-smooth. In a chart the representative of $f+s$ is $z \mapsto (z, \mathbf{f}(z)+\mathbf{s}(z))$. The map $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$, $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$ (§III.2), and composing the latter with $K[1] \subseteq K$ and adding fibre components (sum rule) gives an sc-smooth map with values in K, since its fibres are linear `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.add_isScSmoothRetractMap`; continuity of $f + s$ on each chart domain is read off the same representative `Chapter3/StrongBundleSections · StrongBundle.continuousOn_sectionAdd` `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.continuousOn_source`. So $f + s$ is sc-smooth `Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.sectionAdd`.

Regularizing. For $x \in X_m$, the representative of s maps O_m into $K[1]_m$, so $\mathbf{s}(\varphi(x)) \in F_{m+1}$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.snd_mem_range`. If $(f + s)(x) \in Y_{m,m+1}$ then $\mathbf{f}(\varphi(x)) = (\mathbf{f} + \mathbf{s})(\varphi(x)) - \mathbf{s}(\varphi(x)) \in F_{m+1}$, i.e. $f(x) \in Y_{m,m+1}$ (Φ is linear on fibres `Chapter3/StrongBundleSections · StrongBundleChart.snd_sub`), and $x \in X_{m+1}$ `Chapter3/FredholmSectionsG · StrongBundle.regularizing_sectionAdd`.

The germ property at a smooth point x `Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd`.

Take the chart, filling germ g on G_g, sc⁺-germ t_0 on G_{t_0} with $\iota t_0(0) = g(0)$ and conjugation Ψ for (f, x). Let $t = \mathbf{s}$ be the principal part of s in this chart; it is a section of $K \to O$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.mem_K` with lift t_1 to F_1, $\iota t_1 = t$ on O `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.toBase_principalLift`. Then: $\tilde{t} = t \circ r$ is sc-smooth on the whole U, and $t_1 \circ r$ is sc-smooth into F^1 on U (sc-smoothness on the retract O, Definition 2.1.14 (preprint 2.7)) — s is a section of the bundle, not a germ; $g + \tilde{t}$ is a filling germ of $\mathbf{f} + \mathbf{s}$ on the same G_g (Lemma 3.1.20 (preprint 3.17) for germs, §III.3 `Chapter3/GermCalculus · IsFillingGerm.add_section`, with the filler condition (1) transported along $\mathbf{f} + \mathbf{s}$ on O `Chapter3/FredholmSectionsG · IsFillingGerm.congr`); $t_1 \circ r + t_0$ is an sc⁺-germ on the same G_{t_0}, a germ plus a whole-chart map (§III.48), with value $(g + \tilde{t})(0)$ at 0 `Chapter3/FredholmSectionsG · StrongBundle.sectionAdd_chartData_germ`; and $(g + \tilde{t}) - \iota(t_1 \circ r + t_0) = g - \iota t_0$ *on* U (where $\iota t_1(r(u)) = t(r(u))$ because $r(u) \in O$), which is conjugated to a basic germ by the same Ψ. This last identity holds on U only, which is why the conjugation condition is stated for $u \in U$. The germs of the data do not change, so nothing of g or t_0 away from G_g, G_{t_0} is used; the whole-chart proof `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart.sectionAdd` is the same with the sum rule on U in place of the sum of a germ and a whole-chart map. □

III.7 Contraction germs: Definition 3.3.1 (preprint 3.28) and its working form

Definition 3.3.1 (preprint 3.28). For a germ G around 0 in $\widetilde{C} \oplus W$, $B : P \times W_0 \to W_0$ is an *sc⁰-contraction germ* (i.e. $f(a, w) = w - B(a, w)$ is one `Chapter3/ContractionGerm · contractionMap`) if B is an sc⁰-germ on G, $B(0, 0) = 0$, and for every $\varepsilon > 0$ and m there is $r > 0$ such that every (a, w) with $a \in \widetilde{C}$, $|a| < r$, $|w|_m < r$ lies in U_m and $|B_m(a, w) - B_m(a, w')|_m \le \varepsilon |w - w'|_m$ for such points `Chapter3/ContractionGerm · IsSc0ContractionGerm`. (The text: "for all (a, w), (a, w') on level m sufficiently close to $(0, 0)$"; that these points are in the domain U_m is presupposed.) B_m is the level-m map `Chapter3/ContractionGerm · levelB`, with $B_m(0) = 0$ `Chapter3/ContractionGerm · levelB_zero`.

The working form. The induction of §III.10 builds from f a contraction germ $f^{(1)}$ whose fibre is $TW = W^1 \oplus W$ and whose map is affine in the second factor; its natural contraction region is unbounded there, which balls cannot express. So the proofs work with explicit regions: a *contraction germ in working form* consists of B and G as above with open neighbourhoods $N(0) \supseteq N(1) \supseteq \cdots$ of 0 in P, open $O(m) \ni 0$ and closed $K(m) \subseteq O(m)$ in W_m with $\iota K(m+1) \subseteq K(m)$, such that for $a \in N(m) \cap \widetilde{C}$: $\{a\} \times O(m) \subseteq U_m$, $B_m(a, \cdot)$ maps $K(m)$ into $K(m)$ and is a $\frac{1}{2}$-contraction on $O(m)$ `Chapter3/ContractionGerm · ContractionGerm`.

Lemma: every sc⁰-contraction germ has a working form `Chapter3/GermIFT · IsSc0ContractionGerm.toContractionGerm`. Apply Definition 3.3.1 (preprint 3.28) with $\varepsilon = \frac{1}{2}$ to get radii r_m `Chapter3/GermIFT · IsSc0ContractionGerm.rad Chapter3/GermIFT · IsSc0ContractionGerm.rad_spec`. Make them decrease fast enough for the inclusions: $r'_0 = r_0$, $r'_{m+1} = \min(r_{m+1}, r'_m/(\|\iota_m\| + 1))$ `Chapter3/GermIFT · IsSc0ContractionGerm.rad' Chapter3/GermIFT · IsSc0ContractionGerm.norm_incl_le_of_le`. Put $O(m) = \{|w|_m < r'_m\}$, $K(m) = \{|w|_m \le r'_m/2\}$; then $|\iota w|_m \le \|\iota_m\| r'_{m+1}/2 \le r'_m/2$ for $w \in K(m+1)$. Since B_m is continuous at 0 with $B_m(0) = 0$ `Chapter3/GermIFT · IsSc0ContractionGerm.levelB_zero_eq`, there is $\rho^0_m > 0$ with $|B_m(x)|_m < r'_m/4$ for $x \in U_m$, $|x|_m < \rho^0_m$ `Chapter3/GermIFT · IsSc0ContractionGerm.exists_cont_rad Chapter3/GermIFT · IsSc0ContractionGerm.contRad`. Let ρ_m decrease with $\rho_m \le \min(\rho^0_m, r'_m)$ `Chapter3/GermIFT · IsSc0ContractionGerm.prad Chapter3/GermIFT · IsSc0ContractionGerm.prad_anti` and $N(m) = \{|a| < \rho_m\}$. For $a \in N(m) \cap \widetilde{C}$ and $w \in K(m)$, $|B_m(a, w)|_m \le |B_m(a, w) - B_m(a, 0)|_m + |B_m(a, 0)|_m \le \frac{1}{2} \cdot \frac{r'_m}{2} + \frac{r'_m}{4} = \frac{r'_m}{2}$.

III.8 Existence of the solution germ: the parametrized fixed point theorem and Theorem 3.3.2 (preprint 3.29)

The parametrized Banach fixed point theorem. The text invokes "the parametrized version of Banach's fixed point theorem" in Theorems 3.3.2 and 3.5.1 (preprint 3.29 and 3.40) without stating it; Mathlib has the theorem without the parameter. Let X be a complete metric space, $K \subseteq X$ closed and nonempty, and $g_l : K \to K$ ($l \in \Lambda'$, a subset of a topological space) $\frac{1}{2}$-contractions such that $l \mapsto g_l(x)$ is continuous for every $x \in K$. Then the fixed point $\phi(l) \in K$ is unique in K and continuous in l: from $d(\phi(l), \phi(l_0)) \le \frac{1}{2} d(\phi(l), \phi(l_0)) + d(g_l\phi(l_0), g_{l_0}\phi(l_0))$ one gets $d(\phi(l), \phi(l_0)) \le 2d(g_l(\phi(l_0)), g_{l_0}(\phi(l_0)))$, which tends to 0 by continuity at the single point $\phi(l_0)$ `Chapter3/LocalCompactness · exists_fixedPoint_continuousOn`.

Theorem 3.3.2 (preprint 3.29). In the working form, for $a \in N(m) \cap \widetilde{C}$ the map $B_m(a,\cdot) : K(m) \to K(m)$ is a $\frac{1}{2}$-contraction of a closed subset of W_m, with a fixed point $\delta_m(a) \in K(m)$ `Chapter3/ContractionGerm · ContractionGerm.exists_fixed` `Chapter3/ContractionGerm · ContractionGerm.fixLvl`, which is the only fixed point in the larger set $O(m)$ (two fixed points in $O(m)$ satisfy $|w - w'|_m \le \frac{1}{2}|w - w'|_m$) `Chapter3/ContractionGerm · ContractionGerm.fixed_unique` `Chapter3/ContractionGerm · ContractionGerm.eq_fixLvl`. For $a \in N(m+1) \cap \widetilde{C}$, $\iota\delta_{m+1}(a) \in K(m) \subseteq O(m)$ is a fixed point of $B_m(a,\cdot)$ because B commutes with the inclusions; hence $\iota\delta_{m+1}(a) = \delta_m(a)$ `Chapter3/ContractionGerm · ContractionGerm.incl_fixLvl`. So $\delta := \delta_0$ `Chapter3/ContractionGerm · ContractionGerm.sol` maps $V_m := N(m) \cap \widetilde{C}$ into W_m with level-m map δ_m `Chapter3/ContractionGerm · ContractionGerm.levelMap_sol`. Each δ_m is continuous on V_m by the parametrized fixed point theorem ($g_a = B_m(a,\cdot)$ on $K(m)$, continuous in a because B_m is) `Chapter3/ContractionGerm · ContractionGerm.continuousOn_fixLvl`. Write $\mathrm{germ}(s) = (N(m+s) \cap \widetilde{C})_m$ `Chapter3/ContractionGerm · ContractionGerm.germ` — legitimately a germ because P is finite-dimensional — and call δ a *solution germ* on a germ V if it is an sc^0-germ there with $\delta(0) = 0$, $(a, \delta_m(a)) \in U_m$ on V_m and $B(a, \delta(a)) = \delta(a)$ on V_0 `Chapter3/ContractionGerm · ContractionGerm.IsSolutionGerm`. Then: δ is a solution germ on $\mathrm{germ}(0)$, and every $w \in O(0)$ with $B(a,w) = w$, $a \in N(0) \cap \widetilde{C}$, equals $\delta(a)$ `Chapter3/ContractionGerm · theorem_3_29` `Chapter3/ContractionGerm · ContractionGerm.eq_sol`. The property passes to every $\mathrm{germ}(s)$ `Chapter3/ContractionGerm · ContractionGerm.isSolutionGerm_germ`. In the text's form, from Definition 3.3.1 (preprint 3.28) with balls: the working form of §III.7 gives the solution germ, unique among solutions w with $|w|_0 < r_0'$ `Chapter3/GermIFT · theorem_3_29_of_isSc0ContractionGerm`. The text's "uniquely determined" leaves this region implicit (Remark 3.3.8 (preprint 3.36) names it, $|w|_0 \le \tau_0$).

III.9 The solution germ is sc^1: Theorem 3.3.3 (preprint 3.30)

The text cites [21, Theorem 2.3], "the hard part of the regularity theory", and gives no proof. The formal proof shows that δ is sc^1 on the germ shifted by one level, $V_m = N(m+1) \cap \widetilde{C}$, and the shift is forced by the argument.

The operators. Let D be the derivative family of B. For $a \in N(m+1) \cap \widetilde{C}$ the point $(a, \delta_{m+1}(a))$ lies in U_{m+1}, and the level-m operator of §III.1 there is $L_m(a) := DB(a, \delta_{m+1}(a)) : P \times W_m \to W_m$ `Chapter3/ContractionGerm · ContractionGerm.Lop`, with fibre part $T_m(a) = D_2B(a, \delta_{m+1}(a))$ `Chapter3/ContractionGerm · ContractionGerm.Top` and $L_m(a)(h,\zeta) = L_m(a)(h,0) + T_m(a)\zeta$ `Chapter3/ContractionGerm · ContractionGerm.Lop_apply`. By Lemma 3.1.9 (preprint 3.9) (a), $\|T_m(a)\| \le \frac{1}{2}$ `Chapter3/ContractionGerm · ContractionGerm.norm_Top_le`, so $1 - T_m(a)$ is invertible (Neumann series) `Chapter3/ContractionGerm · ContractionGerm.isUnit_one_sub_Top`, and the candidate derivative is $A_m(a) := (1 - T_m(a))^{-1} L_m(a)(\cdot, 0) \in L(P, W_m)$ `Chapter3/ContractionGerm · ContractionGerm.Aop`. It satisfies the *key identity* $L_m(a)(h, A_m(a)h) = A_m(a)h$ `Chapter3/ContractionGerm · ContractionGerm.Lop_Aop`, and is the unique solution ζ of $L_m(a)(h,\zeta) = \zeta$ `Chapter3/ContractionGerm · ContractionGerm.eq_Aop`. Because the level operators commute with the inclusions `Chapter3/ContractionGerm · ContractionGerm.toBase_Lop`, so do the A_m: $\iota A_m(a)h = A_0(a)h$ `Chapter3/ContractionGerm · ContractionGerm.toBase_Aop`.

Differentiability at $a_0 \in N(2) \cap \widetilde{C}$ `Chapter3/ContractionGerm · ContractionGerm.hasFDerivWithinAt_sol`. Let $x_1 = (a_0, \delta_1(a_0)) \in U_1$, $A_1 = A_1(a_0)$, $A_0 = A_0(a_0)$. For a near a_0 put $h = a - a_0$ and $k = (h, A_1 h) \in E_1$, so $|k|_1 \le (1 + \|A_1\|)|h|$. *This is where $a_0 \in N(2)$ is needed*: A_1 exists only because $(a_0, \delta_2(a_0)) \in U_2$, and without it the increment $A_0 h$ need not lie in W_1, so the sc¹-differentiability of B at x_1, which controls only increments in E_1, could not be used. For h small, $x_1 + k \in U_1$ and $y := \delta(a_0) + A_0 h \in O(0)$. By the sc¹ property, $B(a, y) = B(a_0, \delta(a_0)) + D(x_1)\iota k + \text{rem} = \delta(a_0) + L_0(a_0)(h, A_0 h) + \text{rem} = y + \text{rem}$ with $|\text{rem}|_0 = o(|h|)$, using the key identity on level 0. So y is an approximate fixed point of $B(a, \cdot)$, and the $\frac{1}{2}$-contraction on $O(0)$ gives $|\delta(a) - y|_0 \le \frac{1}{2}|\delta(a) - y|_0 + |\text{rem}|_0$, i.e. $|\delta(a) - \delta(a_0) - A_0 h|_0 \le 2|\text{rem}|_0 = o(|h|)$.

The tangent map is sc⁰. On level m of the tangent of germ(1), i.e. on $(N(m+2) \cap \widetilde{C}) \times P$, the tangent map is $(a, h) \mapsto (\delta_{m+1}(a), A_m(a)h)$ `Chapter3/ContractionGerm · ContractionGerm.tangent_toBase_sol`. Its first component is continuous (§III.8); the second because $(x, v) \mapsto D_m(x)v$ is jointly continuous and $\zeta(a) = A_m(a)h$ satisfies $\zeta(a) = L_m(a)(h, 0) + T_m(a)\zeta(a)$, whence $|\zeta(a) - \zeta(a')|_m \le 2|L_m(a)(h,0) - L_m(a')(h,0)|_m + 2|(T_m(a) - T_m(a'))\zeta(a')|_m$ `Chapter3/ContractionGerm · ContractionGerm.continuousOn_Aop`. With the derivative family $a \mapsto A_0(a)$ on germ(1) `Chapter3/ContractionGerm · ContractionGerm.solDeriv`, δ is an sc¹-germ there `Chapter3/ContractionGerm · ContractionGerm.isSc1GermWith_sol` — **Theorem 3.3.3 (preprint 3.30)**, on the shifted germ `Chapter3/ContractionGerm · theorem_3_30`. At parameters $a_0 \in N(1) \setminus N(2)$ the argument does not apply, and whether δ is differentiable there is neither needed nor addressed.

III.10 The contraction germ $f^{(1)}$: Lemma 3.3.4 (preprint 3.31)

Lemma 3.3.4 (preprint 3.31). The text: if f is of class sc^k, $k \ge 1$, and δ of class sc^j, then $f^{(1)}(v, b, u, w) = (u - B(v, u), w - DB(v, \delta(v))(b, w))$ is an sc⁰-contraction germ of class sc^{min(k–1, j)} whose solution germ is $T\delta$ — "as one verifies by comparing Tf with $f^{(1)}$", and "from the uniqueness". The formal version: if B is sc^{k+1} with derivative family D and δ is sc^j on germ(s), then $f^{(1)}$ is a contraction germ in working form of class sc^{min(k, j)} with solution germ $T\delta$ on $(N(2+s) \cap \widetilde{C}) \times P$ `Chapter3/GermIFT · ContractionGerm.lemma_3_31`. The parameters are $(v, b) \in \widetilde{C} \times P \subseteq P \times P$ (quadrant-like `Chapter3/GermIFT · ContractionGerm.isQuadrantLike_next`), the unknowns $(u, w) \in TW$, with levels $W_{m+1} \times W_m$.

Step 1: the germ and the regions. The germ of $f^{(1)}$ is $U_m^{(1)} = \{((v, b), (u, w)) : (v, u) \in U_{m+1},\ v \in N(m+1+s)\}$ `Chapter3/GermIFT · ContractionGerm.nextG`, and the regions are $N^{(1)}(m) = N(m+2+s) \times P$, $O^{(1)}(m) = O(m+1) \times W_m$, $K^{(1)}(m) = K(m+1) \times W_m$ `Chapter3/GermIFT · ContractionGerm.next`. They are unbounded in b and w, necessarily: the solution must be $(\delta(v), D\delta(v)b)$ for all $b \in P$.

Step 2: the map. With the reshuffling sc-operator $\sigma((v, b), (u, w)) = ((v, u), (b, w))$ `Chapter3/GermIFT · tangentShuffle` and $\Gamma((v, b), (u, w)) = ((v, \delta(v)), (b, w))$, where $v \mapsto \delta(v)$ is regarded as the *shifted* germ into W^1 `Chapter3/GermIFT · ContractionGerm.nextΓ`, formula (38) is $B^{(1)} = (\mathrm{pr}_1 \circ TB \circ \sigma, \mathrm{pr}_2 \circ TB \circ \Gamma)$ `Chapter3/GermIFT · ContractionGerm.nextB` `Chapter3/GermIFT · ContractionGerm.nextB_apply`. On level m, for $v \in N(m+1) \cap \widetilde{C}$, $B_m^{(1)}((v, b), (u, w)) = (B_{m+1}(v, u), L_m(v)(b, w))$ `Chapter3/GermIFT · ContractionGerm.levelB_next`: the point $(v, \delta_{m+1}(v))$ lies in U_{m+1}, which is where the parameter region $N(m+1+\cdot)$ comes from.

Step 3: contraction and invariance. For $(v, b) \in N^{(1)}(m)$ and $(u, w), (u', w') \in O^{(1)}(m)$, the two components contract by $\frac{1}{2}$: $B_{m+1}(v, \cdot)$ on $O(m+1)$, and $T_m(v)$ with $\|T_m(v)\| \le \frac{1}{2}$ (Lemma 3.1.9 (preprint 3.9) (a)); $K(m+1) \times W_m$ is mapped into itself; $B^{(1)}(0) = 0$ `Chapter3/GermIFT · ContractionGerm.nextB_zero`.

Step 4: the class of $B^{(1)}$. TB is an sc^k-germ on the tangent germ; σ, the projections $(v, b, u, w) \mapsto v$ into P^1 and $\mapsto (b, w)$ `Chapter3/GermIFT · projV` `Chapter3/GermIFT · projBW` and the identification $P^1 \oplus W^1 = (P \oplus W)^1$ `Chapter3/GermIFT · shiftParamOp` are sc-operators; Γ is the pairing of (b, w) with $(v, \delta(v))$, which is sc^j because δ is and so is its shift into W^1 (Proposition 1.2.2 (preprint 1.17) for germs) `Chap-`

ter3/GermIFT · ContractionGerm.nextΓ_isSckGerm; and Γ maps $U_m^{(1)}$ into $U_{m+1} \times E_m$ level by level Chapter3/GermIFT · ContractionGerm.levelMap_nextΓ Chapter3/GermIFT · ContractionGerm.nextΓ_mapsToGerm, as does σ Chapter3/GermIFT · ContractionGerm.tangentShuffle_mapsToGerm. The chain rule and the pairing rule of §III.1 give $B^{(1)}$ sc^j Chapter3/GermIFT · ContractionGerm.nextB_isSckGerm.

Step 5: the solution germ is $T\delta$. For $v \in N(2+s) \cap \widetilde{C}$ and $b \in P$, the point $(\delta_1(v), A_0(v)b) \in O(1) \times W_0$ is a fixed point of $B^{(1)}(v, b, \cdot)$ — the first component by $B_1(v, \delta_1(v)) = \delta_1(v)$, the second by the key identity of §III.9 — so by the uniqueness of Theorem 3.3.2 (preprint 3.29) *in* $O^{(1)}(0)$ it is the solution germ of $f^{(1)}$ at (v, b) Chapter3/GermIFT · ContractionGerm.next_sol, and since $A_0(v) = D\delta(v)$ (Theorem 3.3.3 (preprint 3.30)) this is $T\delta(v, b)$ Chapter3/GermIFT · ContractionGerm.next_sol_eq_tangentMapWith. The text's "from the uniqueness" is true because the regions of Step 1 contain $T\delta(v, b)$ for every b.

III.11 The induction: Lemma 3.3.5 (preprint 3.32), Theorem 3.3.6 (preprint 3.33), Corollary 3.3.7 (preprint 3.34), Theorem 3.35 (preprint)

The text proves Lemma 3.3.5 (preprint 3.32) by iterating $f \mapsto f^{(1)}$ and Theorem 3.3.6 (preprint 3.33) by contradiction (if δ were sc^j but not sc^{j+1}, $f^{(j)}$ would be sc¹, so $T^j\delta$ would be sc¹ by Theorem 3.3.3 (preprint 3.30)). The formal proof is the same argument as one induction on j, with the statement quantified over *all* contraction germs, since $f^{(1)}$ has parameter space $P \times P$ and fibre TW and the induction hypothesis must apply to it. It also keeps track of the germ.

Claim (S_j): for every contraction germ in working form, over any finite-dimensional parameter space, of class sc^k with $k \geq j$, the solution germ is an sc^j-germ on germ(s) *for some s* Chapter3/GermIFT · ContractionGerm.sol_isSckGerm. For $j = 0$ take $s = 0$ (Theorem 3.3.2 (preprint 3.29)). *Step $j \to j+1$* Chapter3/GermIFT · ContractionGerm.isSckGerm_succ_of_next: let B be sc^k, $k = k' + 1 \geq j + 1$. (1) By (S_j) for f, δ is sc^j on germ(s). (2) By Lemma 3.3.4 (preprint 3.31) with this s, $f^{(1)}$ is a contraction germ of class sc^{min(k', j)} = sc^j. (3) By (S_j) for $f^{(1)}$, its solution germ is sc^j on its germ(s'), whose level m is $(N(m + s' + 2 + s) \cap \widetilde{C}) \times P$. (4) By Theorem 3.3.3 (preprint 3.30) δ is sc¹ on germ(1), hence on germ$(s + s' + 1)$ (restriction), whose tangent germ has level m equal to the set in (3). (5) On that set the solution germ of $f^{(1)}$ is $T\delta$ (Lemma 3.3.4 (preprint 3.31), Step 5); the identification of $T\widetilde{C} = \widetilde{C} \times P$ inside TP with $\widetilde{C} \times P$ inside $P \times P$ is the identity on every level, an sc-isomorphism Chapter3/GermIFT · constTangentOp' through which the chain rule transports the germ. (6) So δ is sc¹ with sc^j tangent germ: sc^{j+1} on germ$(s + s' + 1)$. Since the proof does the same for every contraction germ, s and s' are the same function of j, so $s_0 = 0$, $s_{j+1} = 2s_j + 1$: $s_j = 2^j - 1$.

Theorem 3.3.6 (preprint 3.33) follows from (S_k): there is s such that δ is a solution germ and an sc^k-germ on germ(s) Chapter3/GermIFT · theorem_3_33; if B is sc-smooth this holds for every k Chapter3/GermIFT · theorem_3_33_smooth; and from Definition 3.3.1 (preprint 3.28) with balls through the working form Chapter3/GermIFT · theorem_3_33_of_isSc0ContractionGerm. Lemma 3.3.5 (preprint 3.32) is contained in the induction: its step from j to $j+1$ is items (2)–(3), without the iterated germs $f^{(j)}$. **Corollary 3.3.7 (preprint 3.34)**: if B is sc-smooth, for m and k take s for k; by Proposition 1.2.3 (preprint 1.18) for germs, δ maps $V_{m,k} := N(m + k + s) \cap \widetilde{C}$ into W_m by a C^k map Chapter3/GermIFT · corollary_3_34 Chapter3/GermLevels · IsSckGerm.contDiffOn_levels, the inclusions of the constant structure being the identity Chapter3/GermIFT · ScBanach.const_inclK. **Theorem 3.35 (preprint)**: for $\widetilde{C}$ a partial quadrant (which is quadrant-like), the same Chapter3/GermIFT · theorem_3_35 Chapter3/GermIFT · theorem_3_35_smooth.

What the shrinking means. Definition 3.1.2 (preprint 3.2) calls δ sc-smooth on a germ if it is sc^k *on that germ* for every k. The proof gives, for each k, the germ germ$(2^k - 1)$; these shrink with k, their intersection is in general not a neighbourhood of 0, and no diagonal choice gives one germ, because being sc^k on a germ involves every level of it. So Theorem 3.3.6 (preprint 3.33) and the first sentence

of Theorem 3.35 (preprint) hold for germs read up to shrinking, and the last sentences of Corollary 3.3.7 (preprint 3.34) and Theorem 3.35 (preprint) hold in the sense of Corollary 3.3.7 (preprint 3.34) — for every k on a germ depending on k — and are proved neither by the text nor here as `IsScSmoothGerm` on one germ. In the text's reading of germs the two senses coincide. Whether the shrinking is a property of contraction germs or only of the method is not decided: in the toy model of the exposition (Chapter3.md §3) the solution germ is sc-smooth on the original germ.

III.12 Local regularity and compactness: Theorem 3.5.1 (preprint 3.40)

Theorem 3.5.1 (preprint 3.40) (text): for $f = h + s$ on a relatively open $U \ni 0$ in C, h a basic germ, s an sc$^+$-germ with $s(0) = 0$, and $S = \{f = 0\}$, there are relatively open neighbourhoods $U \supseteq O(0) \supseteq O(1) \supseteq \cdots$ of 0 with level 0 with $\mathrm{cl}(S \cap O(m)) \subseteq C_m$. Of f only the W-component matters: S lies in the set of solutions of $w = B(a, w) - \mathbf{s}(a, w)$, $\mathbf{s} = P_W \circ s$. The formal statement uses only that B is an sc^0-contraction germ (not sc-smooth), that $\mathbf{s}$ is an sc^0-germ into W^1 with $\mathbf{s}(0) = 0$ (the sc$^+$ property level by level), and that $\widetilde{C}$ is closed (true for partial quadrants; the theorem is proved for convex $\widetilde{C}$, and closedness makes the solution set closed in E_0, where the text takes closures in C_0). It is *stronger* than the text's in the way Theorem 3.1.13 (preprint 3.12) needs: for any set S of solutions in U_0 there are nested open $O(i) \ni 0$ in E_0 and compact $K_i \subseteq E_i$ with $\mathrm{cl}_{E_0}(S \cap O(i)) = \iota_i(K_i)$ `Chapter3/LocalCompactness · theorem_3_40_compactness`; in particular $\mathrm{cl}(S \cap O(i)) \subseteq E_i$ `Chapter3/LocalCompactness · theorem_3_40_regularity`.

Step 0: the level-m contractions with a parameter. Let r_m be the radii of Definition 3.3.1 (preprint 3.28) for $\varepsilon = \frac{1}{2}$ and $c_m = \min(r_m, r_0/(\|\iota_m\| + 1))$, $\iota_m : W_m \to W_0$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.crad`, so that $|\iota_m y|_0 < r_0$ whenever $|y|_m \le c_m/2$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.norm_toBase_lt_rad` (the text's $\tau_1' < \tau_0/c_1$). Points with $a \in \widetilde{C}$, $|a|, |w|_m < r_m$ lie in U_m and satisfy the contraction estimate `Chapter3/LocalCompactness · IsSc0ContractionGerm.mem_G_of_lt` `Chapter3/LocalCompactness · IsSc0ContractionGerm.contr_of_lt`. Choose $\rho_m \in (0, c_m]$ with $|B_m(a, 0)|_m \le c_m/8$ for $|a| < \rho_m$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_cprad` `Chapter3/LocalCompactness · IsSc0ContractionGerm.cprad`. On $\Sigma_m = \{(a, z) : a \in \widetilde{C}, |a| < \rho_m, |z|_m \le c_m/8\}$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.Sigma` the maps $w \mapsto B_m(a, w) - z$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap` send the closed ball $\bar{B}_m(c_m/2)$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.cBall` into itself ($\frac{1}{2} \cdot \frac{c_m}{2} + \frac{c_m}{8} + \frac{c_m}{8} = \frac{c_m}{2}$) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_mapsTo`, are $\frac{1}{2}$-contractions `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_contr` and depend continuously on (a, z) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_continuousOn`. The parametrized fixed point theorem gives a continuous $\delta_m : \Sigma_m \to \bar{B}_m(c_m/2)$ with $\delta_m(a, z) = B_m(a, \delta_m(a, z)) - z$, unique in the ball `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_fixed`.

Step 1: the lifts — the induction the text leaves to the reader. Let $\mathrm{Sol}(\tau)$ be the solutions $x = (a, w)$ with $a \in \widetilde{C}$, $|a| \le \tau$, $|w|_0 \le \tau$ `Chapter3/LocalCompactness · solSet`. *Claim:* for every m there are $\tau_m > 0$ and a map $L_m : E_0 \to E_m$ such that for every $x \in \mathrm{Sol}(\tau_m)$: $\iota_m L_m x = x$, $L_m x \in U_m$, $|(L_m x)_W|_m \le c_m/2$, and L_m is continuous on $\mathrm{Sol}(\tau_m)$ *for the level-0 topology of the source* `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_lift`. For $m = 0$, $L_0 = \mathrm{id}$. Given L_m: by continuity of $\mathbf{s}_m : U_m \to W_{m+1}$ at 0 (where $\mathbf{s}_m(0) = 0$ `Chapter3/LocalCompactness · levelMap_s_zero`) there is η with $|\mathbf{s}_m(y)|_{m+1} < c_{m+1}/8$ for $|y|_m < \eta$; by continuity of L_m at 0 there is θ with $|L_m x|_m < \eta$ for $x \in \mathrm{Sol}(\tau_m)$, $|x|_0 < \theta$. Put $\tau_{m+1} = \min(\tau_m, \rho_{m+1}/2, \theta/2)$ and, for $x = (a, w) \in \mathrm{Sol}(\tau_{m+1})$, $z = \mathbf{s}_m(L_m x)$, so $(a, z) \in \Sigma_{m+1}$, and $L_{m+1} x := (a, \delta_{m+1}(a, z))$. *It lifts x:* $w' = \iota_{m+1}\delta_{m+1}(a, z)$ solves $w' = B(a, w') - \mathbf{s}(x)$ (apply ι_{m+1}, which commutes with B, and use $\iota_m L_m x = x$); so does w; both are smaller than r_0 on level 0 by the choice of c_{m+1}, so the level-0 contraction gives $w' = w$ — the text's (43), on every level. *Continuity:* $x \mapsto (a, \mathbf{s}_m(L_m x))$ is continuous from the level-0 topology into Σ_{m+1}, and δ_{m+1} is continuous. The induction step needs the continuity of L_m: the bound $|\mathbf{s}_m(L_m x)|_{m+1} \le c_{m+1}/8$ requires $L_m x$ small on level m, whereas x is small only on level 0. In the text's level-1 step $L_0 = \mathrm{id}$ and this does not show; existence, bound and continuity of the

lifts have to be proved together.

Step 2: compactness. $\mathrm{Sol}(\tau)$ is closed in E_0 ($\widetilde{C}$ closed, B and $\mathbf{s}$ continuous on level 0) `Chapter3/LocalCompactness · IsSc0ContractionGerm.isClosed_solSet`. For $x \in \mathrm{Sol}(\tau_{m+1})$ the W-component is ι_{m+1} of a vector of norm $\le c_{m+1}/2$, and $\iota_{m+1} : W_{m+1} \to W_0$ is compact `Chapter3/LocalCompactness · ScBanach.toBase_succ_isCompactOperator`; with $|a| \le \tau_{m+1}$ in the finite-dimensional P, a sequence of lifts has a subsequence converging in E_{m+1} (parameters in a compact ball, W_0-parts in a compact image, and the continuity of the lift for the level-0 topology) `Chapter3/LocalCompactness · IsSc0ContractionGerm.isCompact_lift_image`.

Step 3: the neighbourhoods. $O(i)$ are the balls of radius ϱ_i, $\varrho_0 = \tau_1$, $\varrho_{i+1} = \min(\varrho_i, \tau_{i+2})$ — one level ahead of the text's $O(m)$, because compactness in E_i comes from boundedness in E_{i+1}. Then $\mathrm{cl}(S \cap O(i)) \subseteq \mathrm{Sol}(\tau_{i+1})$, every point of it is ι_i of the image under $E_{i+1} \to E_i$ of its lift $L_{i+1}x$, and K_i is the compact set of these images intersected with the closed preimage of $\mathrm{cl}(S \cap O(i))$.

For a basic germ plus an sc$^+$-germ (S the zero set of $h + \iota s$; its Q-component only shrinks S) this is the **local Theorem 3.1.13 (preprint 3.12)** `Chapter3/LocalCompactness · theorem_3_12_local`, and **Corollary 3.1.15 (preprint 3.13) in local form**: solutions $x_k \to 0$ in E_0 lie in $\mathrm{Sol}(\tau_m)$ for k large, and $L_m x_k \to L_m 0 = 0$ in E_m `Chapter3/LocalCompactness · corollary_3_13_local`.

III.13 Theorem 3.1.13 (preprint 3.12) and Corollary 3.1.15 (preprint 3.13) for sections

Theorem 3.1.13 (preprint 3.12). Let f be a section of a strong bundle with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. Then there are nested open $O(i) \ni x_0$ in X such that $\mathrm{cl}_X\{x \in O(i) \mid f(x) = 0\}$ is the image of a compact subset of X_i (with the level topology of Part II, §II.6) under the inclusion $X_i \to X$ `Chapter3/SectionCompactnessG · theorem_3_12`. Of "f an sc-smooth section of a tame strong bundle" only $P \circ f = \mathrm{id}$ is used; the regularity enters through the germ data, and of the filling germ g only its continuity on level 0 near 0. The text calls the theorem "an immediate consequence of Theorem 3.40"; the proof makes explicit six steps.

(i) *Normalizations.* $f(x_0) = 0$ gives $\mathbf{f}(0) = 0$, so $g(0) = 0$ (filler (1)) and $s(0) = 0$ ($\iota s(0) = g(0)$). (ii) *The zeros correspond.* $f(x) = 0 \iff \mathbf{f}(\varphi(x)) = 0$ (§III.2) $\iff g(\varphi(x)) = 0$ (filler (1)) near x_0; zeros of g in U near 0 lie in O (filler (2)); and $g(u) = 0 \iff (h + t)(\psi(u)) = 0$, $t(\psi(u)) := (\Psi(u, s(u)))_2$, because $\Psi(u,k) = (\psi(u), \Gamma(u,k))$ `Chapter3/SectionCompactness · IsStrongBundleMap.apply_eq` with $\Gamma(u,\cdot)$ linear `Chapter3/SectionCompactness · IsStrongBundleMap.snd_sub` and injective `Chapter3/SectionCompactness · IsStrongBundleIso.snd_eq_zero_iff`. (iii) *t is an sc$^+$-germ with $t(0) = 0$:* the base map of a strong bundle map is sc-smooth `Chapter3/SectionCompactness · IsStrongBundleMap.base_isScSmooth`, $\psi'(0) = 0$ `Chapter3/SectionCompactness · IsStrongBundleIso.base_zero'`, and the fibre component of $\Psi[1]$ lifts that of Ψ to F_1' `Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd`. Theorem 3.5.1 (preprint 3.40) needs t only as an sc^0-germ into F'^1 on the germ of the basic germ. With s a germ on G_s, t is one on the germ of the points of the germ of h that ψ^{-1} maps into G_s level by level `Chapter3/GermFilling · ScNbhdGerm.preimage` (germ composition with the whole-chart ψ^{-1} and $\Psi[1]$, §III.48), and the basic germ restricts to that germ `Chapter3/BasicGermShrink · IsBasicGerm.mono` (a form of the shrinking of §III.48). (iv) *Local compactness for a subset.* The local theorem for an arbitrary subset of the solution set of $h + \iota t$ `Chapter3/SectionCompactness · theorem_3_12_local_subset` is applied on the restricted basic germ to $\psi\varphi$ of the zeros near x_0; its $O'(i)$ are intersected with a small ball, so that E_0-closures stay inside $\psi(U)$ and the region of (ii) (the closure of a subset of the solution set is still compact). (v) *Back to X.* The E_0-closure consists of zeros of $h + \iota t$, a limit of zeros in the domain of continuity being a zero `Chapter3/SectionCompactness · eq_of_mem_closure_of_continuousOn`, hence of zeros of g in U, which lie in O. Here g must be continuous at the limits $\psi^{-1}(u')$: the level-0 neighbourhood of the filler and conjugation identities is chosen inside the open set O_g with $(G_g)_0 = O_g \cap C$, so the limits lie in $(G_g)_0$, where the sc^0-germ g is continuous. The whole-chart theorem `Chapter3/SectionCompactness · theorem_3_12_wholeChart` took t sc-smooth on all of U' and g con-

tinuous on U; nothing of either away from a neighbourhood of 0 was used. An X-closure of a subset of the chart domain could a priori leave it; compactness in E_0 shows it does not. (vi) *Levels.* A compact subset of E_i is carried to a compact subset of X_i because ψ^{-1} is sc⁰ and the inverse chart $O_i \to X_i$ `Chapter3/SectionCompactness · MPolyfold.levelChartSymm` is continuous for the level topology of X_i, the transition maps being sc⁰ `Chapter3/SectionCompactness · MPolyfold.continuousOn_levelChartSymm` — the paragraph after Definition 2.3.8 (preprint 2.14), for every level.

Corollary 3.1.15 (preprint 3.13). If $f(x_k) = 0$ and $x_k \to x_0$, fix m and take $O(m)$ and the compact $K_m \subseteq X_m$; for k large $x_k \in O(m)$, so $x_k = \iota y_k$ with $y_k \in K_m$, and $x_0 = \iota y_0$. X_m is not known to be first countable here (its metrizability, Theorem 2.3.10 (preprint 2.16), is not used), so subsequences are avoided: a sequence in a compact set whose images under a continuous injection converge, converges `Chapter3/SectionCompactness · tendsto_of_isCompact_of_tendsto_map`. So $y_k \to y_0$ in X_m `Chapter3/SectionCompactnessG · corollary_3_13`. The text states the corollary for an arbitrary limit x while the hypothesis concerns the germ at x_0; for an sc-Fredholm section and any limit x of zeros, x is a zero (the principal part is continuous in a chart `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`), hence a smooth point (regularizing), hence the germ property holds at x and the first part applies `Chapter3/Chapter5BridgeG · corollary_3_13_section`.

III.14 Good position (Definition 3.1.24 (preprint 3.21), Lemma 3.8.8 (preprint 3.65), Proposition 3.1.25 (preprint 3.22), Lemma 3.5.9 (preprint 3.45), Proposition 3.8.12 (preprint 3.69))

Definition 3.1.24 (preprint 3.21). P is a *good complement* of the finite-dimensional N with respect to C if $E_0 = N \oplus P$ algebraically and for some $\varepsilon > 0$, $|p|_0 \le \varepsilon |n|_0$ implies ($n \in C \iff n + p \in C$) `Chapter3/GoodPosition · IsGoodComplement`; N is *in good position* if $N \cap C$ has nonempty interior in N and N has a good complement `Chapter3/GoodPosition · IsInGoodPosition`. The text asks for an sc-complement (or, for non-smooth N, a topological one); none of Proposition 3.1.25 (preprint 3.22), Lemmas 3.5.9, 3.8.8 (preprint 3.45, 3.65) or Proposition 3.8.12 (preprint 3.69) uses more than the algebraic splitting, so only that is required. Where closedness is needed (Lemma 3.6.9 (preprint 3.55), Definition 3.2.1 (preprint 3.25)) it is added there.

Set-up in coordinates. After a presentation $E \cong \mathbb{R}^n \oplus W$, $C = \{\ell_i \ge 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$ `Chapter3/GoodPosition · coordFun` `Chapter3/GoodPosition · coordFun_surj`. Inside N, $Q := N \cap C$ is the cone $\{\ell_i \ge 0\}$ `Chapter3/GoodPosition · quadCone`, closed `Chapter3/GoodPosition · quadCone_isClosed`. Let Σ be the indices i with $\ell_i(a) = 0$ for some nonzero $a \in Q$ `Chapter3/GoodPosition · vanIdx`; by definition, for $i \notin \Sigma$ and $x \in Q \setminus 0$, $\ell_i(x) > 0$ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`. Let $N \cap W = \{x \in N : \ell_i(x) = 0 \ \forall i\}$ `Chapter3/GoodPosition · nullSpace` and $Q_\Sigma = (\ell_i)_{i \in \Sigma} : N \to \mathbb{R}^\Sigma$ `Chapter3/GoodPosition · sigmaCoord`.

Lemma 3.8.8 (preprint 3.65). A good complement satisfies $\ell_i(p) = 0$ for $p \in P$, $i \in \Sigma$: take $a \in Q \setminus 0$ with $\ell_i(a) = 0$, scale p so that $|p| \le \varepsilon |a|$; $a \in C$ gives $a + p \in C$, so $\ell_i(p) = \ell_i(a + p) \ge 0$; the same for $-p$ `Chapter3/GoodPosition · lemma_3_65`. *Consequences.* Q_Σ is onto: for $k \in \mathbb{R}^\Sigma$ pick v with $\ell_i(v) = k_i$ and split $v = n + p$; then $Q_\Sigma(n) = k$ (the text's (55), preprint p. 125) `Chapter3/GoodPosition · sigmaCoord_surjective`. If an interior point y of Q has $\ell_i(y) = 0$, then $\ell_i = 0$ on N ($y \pm tu \in Q$) `Chapter3/GoodPosition · restrFun_eq_zero_of_interior`.

Proposition 3.1.25 (preprint 3.22), the normal form. There are k, a subspace N_0 and a linear isomorphism $e : N \to \mathbb{R}^k \times N_0$ with $Q = e^{-1}([0,\infty)^k \times N_0)$, $k = \dim N - \dim(N \cap W)$, the $\mathbb{R}^k$-component of e vanishing exactly on $N \cap W$, and the vanishing coordinates of $e(x)$ as many as the vanishing $\ell_i(x)$ for $x \in Q \setminus N \cap W$ `Chapter3/GoodPosition · QuadrantNormalForm` `Chapter3/GoodPosition · exists_quadrantNormalForm`. Three cases. *Case A, $Q = N$*: every ℓ_i is ≥ 0 on a vector space, hence 0; $k = 0$ `Chapter3/GoodPosition · quadrantNormalForm_of_eq_univ` (a case the text's case distinction does not single out). *Case B, $Q \ne N$, $\Sigma = \emptyset$*: an interior point y is nonzero with all $\ell_i(y) > 0$; for $u \in N$ the set $\{t : u + ty \in Q\}$

is closed, contains all large t and is bounded below; at its minimum t^* some $\ell_i(u+t^*y) = 0$, so $u+t^*y = 0$: N is the line $\mathbb{R}y$ and Q a half-line `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx_empty` (the text's Lemma 3.8.11 (preprint 3.68), second case). *Case* $C, \Sigma \neq \emptyset$: first $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in \Sigma\}$ — move from such an x towards an interior point y along $x+sy$; at the first parameter s^* with $x+s^*y \in Q$, if $s^* > 0$ the point $z = x + s^*y$ is nonzero (else $\ell_i(x) < 0$ for $i \in \Sigma$), so it has $\ell_i(z) > 0$ for all i ($\ell_i(z) \geq s^*\ell_i(y) > 0$ in Σ, by definition of Σ outside), and finitely many strict inequalities survive a small step back, contradicting minimality `Chapter3/GoodPosition · mem_quadCone_of_vanIdx`. This replaces the text's extreme rays (Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64), cited from [21], and 3.66–3.68). Then a linear right inverse g of Q_Σ gives $e(x) = (Q_\Sigma x, x - gQ_\Sigma x) \in \mathbb{R}^\Sigma \times \ker Q_\Sigma$ `Chapter3/GoodPosition · splitEquiv`, and $\ker Q_\Sigma = N \cap W$ `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx`. For a partial quadrant in good position this applies in a presentation `Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm`, and choosing a basis of N_0 makes $(N, N \cap C)$ isomorphic to $(\mathbb{R}^{\dim N}, [0,\infty)^k \oplus \mathbb{R}^{\dim N - k})$ through an sc-isomorphism of constant structures `Chapter3/GoodPosition · scIsoOfLinearEquiv` `Chapter3/GoodPosition · preimage_eq_of_normalForm`: $N \cap C$ **is a partial quadrant in** N `Chapter3/GoodPosition · proposition_3_22`.

Lemma 3.5.9 (preprint 3.45) (linear form). A continuous linear isomorphism $S : E_0 \to E_0'$ with $x \in C \iff Sx \in C'$ carries a good complement P of N to one of $S(N)$, with $\varepsilon' = \varepsilon/(\|S\|\|S^{-1}\| + 1)$: for $|Sp| \leq \varepsilon'|Sn|$, $|p| \leq \|S^{-1}\|\varepsilon'\|S\||n| \leq \varepsilon|n|$ `Chapter3/GoodPosition · IsGoodComplement.map`; and the interior condition is transported by the homeomorphism $S|_N$ `Chapter3/GoodPosition · IsInGoodPosition.map`. The text states it for $T\varphi(0)$; since the coordinate change of Theorem 3.4.2 (preprint 3.38) is linear, this is the form §§III.18–III.23 need.

Proposition 3.8.12 (preprint 3.69). In the model $\mathbb{R}^n \oplus W$, for $x \in N \cap C$: $d_{N\cap C}(x) = d_C(x)$ if $x \notin N \cap W$, and $d_{N\cap C}(x) + \dim(N \cap W) = \dim N$ if $x \in N \cap W$ `Chapter3/GoodPosition · proposition_3_69` — read off the normal form ($d_{N\cap C}$ does not depend on the presentation, Lemma 1.3.4 (preprint 1.24)).

III.15 The degeneracy index along a graph: Lemma 3.8.13 (preprint 3.70)

Lemma 3.8.13 (preprint 3.70). In the model, for N in good position with good complement Y and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$, and $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $v \notin N \cap W$ `Chapter3/GoodPositionGraph · lemma_3_70`. The text distinguishes $\dim N = 1$ (a contradiction argument with $\tau(v) = o(|v|)$) and $\dim N \geq 2$ (the sign of a directional derivative). The formal proof unifies them.

Ray decomposition. From the normal form, $u_i = e^{-1}(\mathbf{e}_i, 0)$ are finitely many nonzero points of Q, and every $v \in Q \setminus N \cap W$ has some i and $s > 0$ with $v - su_i \in Q$ and $s|u_i| \leq c|v|$ (take $s = \lambda_i > 0$ for $e(v) = (\lambda, n_0)$) `Chapter3/GoodPositionGraph · exists_rayDecomp` — the text's basis e^j of $\widetilde{N}$. *Smallness.* $D\tau(0) = 0$ and continuity of $D\tau$ on the convex Q make τ η-Lipschitz on $Q \cap B(\rho)$ for every η `Chapter3/GoodPositionGraph · exists_lipschitz_small` `Chapter3/GoodPositionGraph · quadCone_convex`. *The margin.* Choose $\eta \leq \varepsilon$ with $\eta\|\ell_j\||u_i| < \ell_j(u_i)$ for all i and $j \notin \Sigma$ — a positive lower bound of finitely many positive numbers `Chapter3/GoodPositionGraph · exists_pos_le_of_forall_pos`. *Conclusion* `Chapter3/GoodPositionGraph · lemma_3_70_core`. For $|v| < \rho/(1+c)$: $|\tau(v)| \leq \varepsilon|v|$, so $v + \tau(v) \in C$ (good complement). For $j \in \Sigma$, $\ell_j(\tau(v)) = 0$ (Lemma 3.8.8 (preprint 3.65)). For $v \notin N \cap W$, $w = v - su_i \in Q$, the segment $[w, v]$ lies in $Q \cap B(\rho)$ — which is why the radius is $\rho/(1+c)$, the text's "close to 0" — and for $j \notin \Sigma$,

$$\ell_j(v + \tau(v)) = \ell_j(w + \tau(w)) + s\ell_j(u_i) + \ell_j(\tau(v) - \tau(w)) \geq 0 + s\ell_j(u_i) - \|\ell_j\|\eta s|u_i| > 0.$$

So v and $v + \tau(v)$ have the same vanishing coordinates, and Proposition 3.8.12 (1) (preprint 3.69) gives the lemma in the model, with the coordinate functionals `Chapter3/GoodPositionGraph · coordCLM`. (The level Y_1 of the text's τ plays no role.)

III.16 The quadrant inverse function theorem (Theorem 3.8.15 (preprint 3.72), Proposition 3.8.17 (preprint 3.73), Lemmas 3.8.18, 3.8.21 (preprint 3.74, 3.77), Corollary 3.8.19 (preprint 3.75))

Interior points. For the model quadrant, x is interior iff all its half-line coordinates are positive `Chapter3/QuadrantInverse · interior_quadrant`, so $d_C(x) = 0$ exactly at interior points `Chapter3/QuadrantInverse · degIndex_eq_zero_iff`. A linear isomorphism S with $S(C) = C$ preserves d_C (Lemma 1.3.4 (preprint 1.24) for the presentation $T \circ S$) `Chapter3/QuadrantInverse · ScIso.ofCLE Chapter3/QuadrantInverse · degIndex_map_eq`. A linear map carrying a set with nonempty interior onto itself is onto `Chapter3/QuadrantInverse · surjective_of_image_eq` — the text's "$\mathbb{R}^n = C - C$".

Proposition 3.8.17 (preprint 3.73). The hypotheses (`InvData`): U relatively open and convex in the partial quadrant C of a finite-dimensional V; $f : U \to C$ of class C^j, $j \geq 1$; $\|Df(x) - 1\| \leq \frac{1}{2}$; and x interior iff $f(x)$ interior — the only consequence of $d_C(x) = d_C(f(x))$ the proof uses `Chapter3/QuadrantInverse · InvData`. The text's $0 \in U$, $f(0) = 0$ are not needed. Conclusion: $f(U)$ is relatively open in C, f is injective on U, with an inverse of class C^j on $f(U)$ `Chapter3/QuadrantInverse · proposition_3_73`.

Lemma 3.8.18 (preprint 3.74). $x \mapsto x - f(x)$ is a $\frac{1}{2}$-contraction on the convex U (mean value inequality) `Chapter3/QuadrantInverse · InvData.norm_sub_sub_le`, which gives (70): $\frac{1}{2}|x - x'| \leq |f(x) - f(x')| \leq \frac{3}{2}|x - x'|$ `Chapter3/QuadrantInverse · InvData.half_norm_le Chapter3/QuadrantInverse · InvData.norm_sub_le`, and injectivity `Chapter3/QuadrantInverse · InvData.injOn`. *Openness*: if $B(x', 3r) \cap C \subseteq U$ then $B(f(x'), r) \cap C \subseteq f(U)$ `Chapter3/QuadrantInverse · InvData.ball_inter_subset_image`. The set $A = f(\bar{B}(x', 2r) \cap C)$ is compact ($\bar{B}(x', 2r) \cap C \subseteq U$ is compact in finite dimensions). Interior points first: for an interior $y_1 \in B(y', r)$ choose an interior $x_0 \in U$ with $f(x_0) \in B(y', r)$ — the text takes any interior $x_0 \in B(x', 3r) \cap C$, which does not suffice — and follow the segment y_t from $f(x_0)$ to y_1, which stays interior. $\{t : y_t \in A\}$ is closed and contains 0; at its maximum $t^* < 1$, $y_{t^*} = f(x^*)$ with x^* interior and $|x^* - x'| < 2r$, and at the interior point x^*, $Df(x^*)$ is a unit `Chapter3/QuadrantInverse · InvData.isUnit_deriv` and U a neighbourhood `Chapter3/QuadrantInverse · InvData.U_mem_nhds`, so the classical inverse function theorem maps neighbourhoods onto neighbourhoods `Chapter3/QuadrantInverse · InvData.map_nhds_eq`, pushing t^* further — a contradiction. The compactness of A replaces the text's Cauchy sequence and also gives the remaining points by density. So $f(U)$ is relatively open `Chapter3/QuadrantInverse · InvData.image_relOpen`. **Corollary 3.8.19 (preprint 3.75)**: f and its inverse `Chapter3/QuadrantInverse · InvData.inv` (2-Lipschitz by (70) `Chapter3/QuadrantInverse · InvData.norm_inv_sub_le`) are continuous and inverse to each other `Chapter3/QuadrantInverse · corollary_3_75`.

Lemma 3.8.21 (preprint 3.77). The inverse g is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$ `Chapter3/QuadrantInverse · InvData.hasFDerivWithinAt_inv`: with $x_0 = g(y_0)$, $x_0 + \delta = g(y_0 + h)$, the quotient $|g(y_0 + h) - g(y_0) - Df(x_0)^{-1}h|/|h|$ is $|\delta|/|h|$ (bounded by (70)) times the remainder quotient of f at x_0. The lemma's extra hypotheses (relatively open image, homeomorphism, invertible derivative) are consequences of those of Proposition 3.8.17 (preprint 3.73). *Regularity (the text: Lemma 3.8.20 (preprint 3.76) on the iterated tangent maps $T^k f$).* $x \mapsto [Df(x)]^{-1}$ is C^{j-1} (inversion is smooth on units) `Chapter3/QuadrantInverse · InvData.contDiffOn_inverse_deriv`, so $Dg = [Df]^{-1} \circ g$ and $g \in C^i$, $i < j$, give $g \in C^{i+1}$: g is C^j `Chapter3/QuadrantInverse · InvData.contDiffOn_inv`. Lemma 3.8.20 (preprint 3.76) is not needed.

Theorem 3.8.15 (preprint 3.72). U a relatively open neighbourhood of 0 in C, f of class C^j with $f(U) \subseteq C$, $f(0) = 0$, $Df(0)C = C$, $d_C(x) = d_C(f(x))$ on U: then f is a C^j-diffeomorphism between relatively open neighbourhoods U', V' of 0 `Chapter3/QuadrantInverse · theorem_3_72`. *Proof.* $Df(0)$ is onto, hence invertible, with $Df(0)^{-1}C = C$; $\tilde{f} = Df(0)^{-1}f$ maps into C, $D\tilde{f}(0) = 1$, and preserves d_C ($d_C \circ Df(0)^{-1} = d_C$), hence interior points. On $U' = B(0, \rho) \cap C$ with $\|D\tilde{f} - 1\| \leq \frac{1}{2}$, Proposition 3.8.17 (preprint 3.73) applies, and $V' = Df(0)\tilde{f}(U')$ is relatively open since $Df(0)$ is a linear homeomorphism preserving C. The text's hypothesis (2) is used only through interior points; some such hypothesis is needed ($f(x, y) = (x, y + x^2)$ on $[0, \infty)^2$ satisfies (1) but $f(C)$ is not a neighbourhood of 0 in C).

III.17 The implicit function theorem in partial quadrants (Theorem 3.8.22 (preprint 3.78), Lemmas 3.8.23–3.8.26 (preprint 3.79–3.82))

Theorem 3.8.22 (preprint 3.78). U a relatively open neighbourhood of 0 in the partial quadrant C of a finite-dimensional V, $f : U \to F$ of class C^j, $j \geq 1$, with $f(0) = 0$, $Df(0)$ onto, $K = \ker Df(0)$ in good position with good complement Y. Then there are a relatively open $V' \ni 0$ in $K \cap C$, $\tau : K \to Y$ of class C^j on V' and $\varepsilon, \sigma > 0$ with (1) $\tau(0) = 0$, $D\tau(0) = 0$, and $k + \tau(k) \in U$, $f(k + \tau(k)) = 0$ for $k \in V'$; (2) $|\tau(k)| \leq \sigma$ for $|k| \leq \varepsilon$; (3) if $k \in V'$, $|k| \leq \varepsilon$, $y \in Y$, $|y| \leq \sigma$, $k + y \in U$ and $f(k + y) = 0$, then $y = \tau(k)$ `Chapter3/QuadrantIFT · theorem_3_78`. The two bold clauses are not in the text; the first is the content of the theorem (its proof has $\tau(k) = B(k, \tau(k))$), the second is needed for $f(k + y)$ to make sense.

Normalization (71)–(72). With $L = (Df(0)|_Y)^{-1}$, in coordinates $x = k + y \in K \oplus Y$, $Lf(k + y) = y - B(k, y)$ with $B(k, y) := y - Lf(k+y)$, $B(0, 0) = 0$, $DB(0, 0) = 0$. The data after normalization: on $K \times Y$ a convex set C_p with nonempty interior, a relatively open $U_p \ni 0$, B of class C^j on U_p with $B(0) = 0$, $DB(0) = 0$, the good-position estimate and the interior condition `Chapter3/QuadrantIFT · ContractionData`. Only convexity and interior are used; the quadrant enters through good position.

Lemmas 3.8.23, 3.8.24 (preprint 3.79, 3.80): the constants `Chapter3/QuadrantIFT · Radii` `Chapter3/QuadrantIFT · ContractionData.exists_radii`. With ε the good-position constant and $c_0 = \varepsilon/(2(1 + \varepsilon)) \leq \frac{1}{2}$ `Chapter3/QuadrantIFT · Radii.c₀`, continuity of DB at 0 gives a, b with $\varepsilon a \leq b$, the region $R = \{|k| \leq a, |y| \leq b\} \cap C_p \subseteq U_p$ `Chapter3/QuadrantIFT · Radii.R` `Chapter3/QuadrantIFT · Radii.R_sub_U` and $\|DB\| \leq c_0$ on R. R is convex `Chapter3/QuadrantIFT · Radii.R_convex`, so the mean value inequality gives $|B(q) - B(p)| \leq c_0|q - p|$ on R `Chapter3/QuadrantIFT · Radii.norm_sub_le` `Chapter3/QuadrantIFT · Radii.norm_B_sub_le_snd` and $|B(p)| \leq c_0|p|$ `Chapter3/QuadrantIFT · Radii.norm_B_le`. For $k \in V_0 = \{k \in K \cap C : |k| < a\}$ `Chapter3/QuadrantIFT · Radii.V₀` and $|y| \leq \varepsilon|k|$: good position puts $(k, y) \in C_p$, $|y| \leq \varepsilon a \leq b$, so $(k, y) \in R$ and $|B(k, y)| \leq c_0(1 + \varepsilon)|k| = \frac{\varepsilon}{2}|k|$. So $B(k, \cdot)$ maps $X_k = \bar{B}_Y(\varepsilon|k|)$ into itself `Chapter3/QuadrantIFT · Radii.mapsTo_ball` and has a fixed point $\tau(k)$ there `Chapter3/QuadrantIFT · Radii.exists_fixed` `Chapter3/QuadrantIFT · Radii.τ`; any fixed point y with $(k, y) \in R$ equals $\tau(k)$ `Chapter3/QuadrantIFT · Radii.eq_τ`. (In Lemma 3.8.24 (preprint 3.80) the maximum is over $B((1 + \varepsilon)a) \cap C$, where DB is defined.)

Lemma 3.8.25 (preprint 3.81): continuity. $|\tau(k)| \leq \frac{\varepsilon}{2}|k|$ `Chapter3/QuadrantIFT · Radii.norm_τ_le`. If $|k_0| \leq 2|k|$, then $(k, \tau(k_0)) \in R$ and $|\tau(k) - \tau(k_0)| \leq \frac{1}{2}|\tau(k) - \tau(k_0)| + c_0|k - k_0|$, so $|\tau(k) - \tau(k_0)| \leq \varepsilon|k - k_0|$ — the text's estimate, whose argument needs $(k, \tau(k_0))$ in the domain. If $|k_0| > 2|k|$, then $|k - k_0| \geq \frac{1}{2}|k_0|$ and $|\tau(k) - \tau(k_0)| \leq \frac{\varepsilon}{2}(|k| + |k_0|) \leq \frac{3\varepsilon}{2}|k - k_0|$. In all cases the constant is $1 + 2\varepsilon$ `Chapter3/QuadrantIFT · Radii.norm_τ_sub_le`, and τ is continuous `Chapter3/QuadrantIFT · Radii.continuousOn_τ`. Since X_k depends on k, the parametrized fixed point theorem of §III.8 does not apply, and continuity is proved by hand, as in the text.

Lemma 3.8.26 (preprint 3.82): regularity. With $D_1B = DB \circ \mathrm{inl}$, $D_2B = DB \circ \mathrm{inr}$ `Chapter3/QuadrantIFT · Radii.D₁` `Chapter3/QuadrantIFT · Radii.D₂`, $\|D_2B\| \leq \frac{1}{2}$ on R `Chapter3/QuadrantIFT · Radii.norm_D₂_le`, $M = (1 - D_2B)^{-1}$ (Neumann) `Chapter3/QuadrantIFT · Radii.M` and $\Phi = MD_1B$ `Chapter3/QuadrantIFT · Radii.Φ`: from the fixed-point identities and the little-o composition of B with the Lipschitz $k' \mapsto (k', \tau(k'))$, τ is differentiable within V_0 with $D\tau(k) = \Phi(k, \tau(k))$ `Chapter3/QuadrantIFT · Radii.hasFDerivWithinAt_τ` `Chapter3/QuadrantIFT · Radii.fderivWithin_τ`. Φ is C^{j-1} on R `Chapter3/QuadrantIFT · Radii.contDiffOn_Φ`, so $\tau \in C^i$, $i < j$, gives $D\tau = \Phi \circ (\mathrm{id}, \tau) \in C^i$ and $\tau \in C^{i+1}$: τ is C^j on all of V_0 `Chapter3/QuadrantIFT · Radii.contDiffOn_τ`. The text iterates $f \mapsto f^{(1)}$ (the finite-dimensional model of Lemma 3.3.4 (preprint 3.31)) and concludes "on a possibly smaller neighbourhood"; in finite dimensions there are no levels to shift and no shrinking is needed. *Assembly* `Chapter3/QuadrantIFT · ContractionData.exists_solution`: $V' = V_0$, $\sigma = b$, ε below a; (1) $Lf(k + \tau(k)) = \tau(k) - B(k, \tau(k)) = 0$ and L is injective; (3) $f(k + y) = 0$ means $B(k, y) = y$ with $(k, y) \in R$.

III.18 Weak stability of basic germs: Theorem 3.4.2 (preprint 3.38)

Theorem 3.4.2 (preprint 3.38) (text): for a basic germ f and the principal part s of an sc$^+$-section with $s(0)=0$ there is a strong bundle isomorphism Φ covering an sc-diffeomorphism φ such that $(\Phi_*(f+s))^1$, with the levels raised by one, is a basic germ, and $n-N=n'-N'$. The text's proof: with $A=P_W\circ D_2s(0)$, $1+A$ is sc-Fredholm of index 0, with sc-splittings $W=C\oplus X=R\oplus Z$, $C=\ker(1+A)$, $R=\operatorname{range}(1+A)$; $P_W\circ s(a,w)=Aw+S(a,w)$ with S a small contraction in w on every level $m\geq 1$; and with $L=(1+A)|_X : X\to R$ the equation becomes $x-L^{-1}P_2\bar{B}(a,c+x)=0$ in the new parameters (a,c) and the new fibre variable x. The formal proof follows it with two observations: the coordinate change is *linear*, and the index raise is a *change of sc-structure*.

The linear part of s. For an sc^1-germ s into W^1, $A=D_2s(0)$ is an sc$^+$-operator whose level-m map is the level operator of Proposition 1.2.1 (preprint 1.16) at $0\in U_{m+1}$ `Chapter3/BasicClass · scPlusOfDeriv`. The remainder $S(a,w)=s(a,w)-Aw$ on level $m+1$ `Chapter3/BasicClass · scPlusRemainder` is, for a and w small depending on m and ε, an ε-contraction in w: $D_2S(0)=0$ on level $m+1$, the derivative of $s:U_{m+1}\to W_{m+1}$ is continuous in the operator norm (§III.1), and the mean value inequality on a ball for fixed a concludes `Chapter3/BasicClass · scPlusRemainder_contraction`. The partial derivative of S in a does not vanish and plays no role. The text gives this one sentence ("S is … an arbitrary small contraction on every level $m\geq 1$").

The data and the splitting. The data of the theorem are the germ G, the W-component s of the sc$^+$-germ with $s(0)=0$, and an sc-Fredholm splitting $W=C\oplus X=R\oplus Z$ of $T=1+A$ `Chapter3/BasicClass · StabilityData`. Such data exist: the identity is sc-Fredholm of index 0 (the trivial splitting) `Chapter3/BasicClass · isScFredholm_id` `Chapter3/BasicClass · scIndex_id`, so $1+A$ is sc-Fredholm by Proposition 1.1.11 (preprint 1.10) `Chapter3/BasicClass · StabilityData.exists_of_scPlus`. With the projections $P_1 : W_m\to X_m$ along C_m, $P_2 : W_m\to R_m$ along Z_m `Chapter3/BasicClass · StabilityData.projX` `Chapter3/BasicClass · StabilityData.projR`, $L=T|_X : X_m\to R_m$ is bijective `Chapter3/BasicClass · StabilityData.Lm_bijective`, a topological isomorphism by the open mapping theorem `Chapter3/BasicClass · StabilityData.Lequiv`, and $L^{-1}P_2$ is an sc-operator $W\to X$ `Chapter3/BasicClass · StabilityData.LinvPOp`. The finite-dimensional C, $Z\subseteq W_\infty$ lift to every level `Chapter3/BasicClass · ScBanach.finLift`.

The coordinate changes. On the index-raised spaces, the new source is $E'=(P\oplus C)\oplus X^1$ `Chapter3/BasicClass · StabilityData.E'` and the new target $F'=(Q\oplus Z)\oplus X^1$ `Chapter3/BasicClass · StabilityData.F'` — the text's $\tau : Z\to C$, making the new finite-dimensional target $\mathbb{R}^N\oplus C$, is not applied. The base change $\psi(a,w)=((a,(1-P_1)w),P_1w)$ `Chapter3/BasicClass · StabilityData.split` and its inverse $\psi^{-1}((a,c),x)=(a,c+x)$ `Chapter3/BasicClass · StabilityData.glue` are sc-operators inverse to each other `Chapter3/BasicClass · StabilityData.glue_split` `Chapter3/BasicClass · StabilityData.split_glue`; so are the fibre change $\Psi(q,w)=((q,(1-P_2)w),L^{-1}P_2w)$ `Chapter3/BasicClass · StabilityData.fiber` and $\Psi^{-1}((q,z),x)=(q,z+(1+A)x)$ `Chapter3/BasicClass · StabilityData.fiberInv` `Chapter3/BasicClass · StabilityData.fiber_fiberInv` `Chapter3/BasicClass · StabilityData.fiberInv_fiber`, and Ψ vanishes only at 0 `Chapter3/BasicClass · StabilityData.fiber_eq_zero_iff`. No general strong bundle isomorphism is needed.

The push-forward. The perturbed germ is $f+\iota s$ with the inclusion $W^1\to Q\oplus W$ `Chapter3/BasicClass · ScBanach.inrOp` `Chapter3/BasicClass · StabilityData.fsum`, sc-smooth by the pairing and operator rules `Chapter3/BasicClass · StabilityData.fsum_isScSmoothGerm`. Its germ of neighbourhoods on E' is $\psi(U_{m+1})$, the preimage of U_{m+1} under ψ^{-1} `Chapter3/BasicClass · StabilityData.pushGerm` `Chapter3/BasicClass · StabilityData.glue_mapsToGerm`, and the push-forward is $(\Phi_*(f+s))^1=\Psi\circ(f+\iota s)^1\circ\psi^{-1}$ with the level-shifted operators `Chapter3/BasicClass · ScOperator.shiftOp` `Chapter3/BasicClass · StabilityData.pushMap`. It is sc-smooth: $(f+\iota s)^1$ is sc-smooth on the shifted germ (Proposition 1.2.2 (preprint 1.17) for germs) and Ψ, ψ^{-1} are sc-operators `Chapter3/BasicClass · StabilityData.pushMap_isScSmoothGerm`. With $\bar{B}=B-S$ `Chapter3/BasicClass · StabilityData.Bbar`, the W-component of $(f+\iota s)^1$ at a level-1 point is $(1+A)w-\bar{B}(a,w)$ `Chapter3/BasicClass · StabilityData.snd_levelMap_fsum`, so the X-component of the push-forward is $x-\hat{B}((a,c),x)$ with $\hat{B}((a,c),x)=L^{-1}P_2\bar{B}(a,c+x)$ `Chapter3/BasicClass · StabilityData.pushB`

Chapter3/BasicClass · StabilityData.snd_pushMap — the text's "desired formula". $\hat{B}$ is an sc⁰-contraction germ on the index-raised structure: on level m of E', i.e. level $m+1$ of W, B contracts by Definition 3.1.7 (preprint 3.7) and S by the remainder estimate, and $L^{-1}P_2$ is bounded Chapter3/BasicClass · StabilityData.levelMap_pushB Chapter3/BasicClass · StabilityData.pushB_isSc0ContractionGerm. So **the push-forward is a basic germ with contraction map** $\hat{B}$ Chapter3/BasicClass · theorem_3_38. Its zeros correspond to those of $f+\iota s$ under ψ, since Ψ is a linear isomorphism Chapter3/BasicClass · StabilityData.pushMap_eq_zero_iff. *The index*: $\operatorname{ind}(1+A) = \operatorname{ind}(1) = 0$ (Proposition 1.1.11 (preprint 1.10) with its index), i.e. $\dim C = \dim Z$, so $n'-N' = (n+\dim C)-(N+\dim Z) = n-N$ Chapter3/BasicClass · StabilityData.finrank_Cs_eq_finrank_Zs.

The same on level 0. For §§III.19–III.20 the coordinate changes are also needed before the index raise: ψ, ψ^{-1}, Ψ, Ψ^{-1} on every level Chapter3/BasicClass · StabilityData.split₀ Chapter3/BasicClass · StabilityData.glue₀ Chapter3/BasicClass · StabilityData.fiber₀, the germ $\psi(U_m)$ Chapter3/BasicClass · StabilityData.pushGerm₀, and $g = \Phi_*(f+s) = \Psi\circ(f+\iota s)\circ\psi^{-1}$ on level 0 Chapter3/BasicClass · StabilityData.pushMap₀, sc-smooth Chapter3/BasicClass · StabilityData.pushMap₀_isScSmoothGerm, whose level-1 map is the index-raised push-forward Chapter3/BasicClass · StabilityData.pushMap_eq_levelMap_pushMap₀.

III.19 The linearization at nearby solutions: Corollary 3.5.3 (preprint 3.41)

Corollary 3.5.3 (preprint 3.41). For the data of Theorem 3.4.2 (preprint 3.38) with $f := h$ basic and $D(h+\iota s)(0)$ surjective, there is $r>0$ such that every zero x of $h+\iota s$ in U_0 with $|x|_0 < r$ is $\iota(x_1)$ with $x_1 \in U_1$, and $D(h+\iota s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$ Chapter3/SolutionLinearization · StabilityData.corollary_3_41. The text's sc⁺-map s has an $\mathbb{R}^N$-component; it can be absorbed into the basic germ (as in §III.23), so the data carry only the W-component.

The derivative on U_1. For $x \in U_1$, $D(h+\iota s)(x) = Dh(x)+\iota Ds(x)$ — uniqueness of the derivative within U_1, then density of E_1 Chapter3/SolutionLinearization · StabilityData.Dfs'_eq — and $\operatorname{pr}_W Dh(x) = \operatorname{pr}_W - DB(x)$, from $\operatorname{pr}_W h = \operatorname{pr}_W - B$ on U_0 Chapter3/SolutionLinearization · StabilityData.snd_Df_eq; $D_2B(0)=0$ Chapter3/SolutionLinearization · StabilityData.DB_zero_snd. *The decomposition.* $D(h+\iota s)(x) = D(h+\iota s)(0) + S(x) + K(x)$ with $S(x)(\alpha,\zeta) = (0, -D_2B(x)\zeta)$ Chapter3/SolutionLinearization · StabilityData.Sop Chapter3/SolutionLinearization · StabilityData.Dfs'_eq_add of norm $\le \varepsilon$ for $x \in U_1$ small on level 1, by Lemma 3.1.9 (preprint 3.9) (c) Chapter3/SolutionLinearization · StabilityData.exists_norm_Sop_le, and $K(x)$ Chapter3/SolutionLinearization · StabilityData.Kop compact: the sum of finite-rank operators through Q and through P and of $\iota(Ds(x)-Ds(0))$ through the compact inclusion $W_1 \to W_0$ Chapter3/SolutionLinearization · StabilityData.Kop_isCompactOperator.

Fredholm with index $n-N$ *at every small* $x \in U_1$. $D(h+\iota s)(0)$ is Fredholm of index $n-N$: $Dh(0)$ is (Proposition 3.1.10 (preprint 3.10) for the basic germ, §III.5) and $\iota Ds(0)$ is compact Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs. A small perturbation of a Fredholm operator is Fredholm: if v is a quasi-inverse of u and $\|v\|\|s\| < 1$, then $(1+vs)^{-1}v$ is a quasi-inverse of $u+s$ (Neumann series) Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt, with the same index Chapter3/SolutionLinearization · isFredholm_fredholmIndex_add_of_norm_lt; a compact perturbation does not change the index Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator. So $D(h+\iota s)(x)$ is Fredholm of index $n-N$ for *every* $x \in U_1$ small on level 1, not only at solutions Chapter3/SolutionLinearization · StabilityData.exists_isFredholm_Dfs'.

Surjectivity: the compactness argument. Let N be a closed complement of $\ker D(h+\iota s)(0)$. Then $\ker D(h+\iota s)(x) \cap N = 0$ for $x \in U_1$ small on level 1 Chapter3/SolutionLinearization · StabilityData.exists_ker_Dfs'_inf_eq_bot. Otherwise there are $x_k \to 0$ in E_1 and unit vectors $(\alpha_k,\zeta_k) \in N$ in the kernels; the W-component of the equation reads $\zeta_k = D_1B(x_k)\alpha_k + D_2B(x_k)\zeta_k - \iota Ds(x_k)(\alpha_k,\zeta_k)$. α_k has a convergent subsequence (P finite-dimensional); $Ds(x_k)(\alpha_k,\zeta_k)$ is bounded in W_1 (the derivatives near 0 are uniformly bounded Chapter3/SolutionLinearization · IsSc1GermWith.exists_norm_deriv_le), so

its image in W_0 has a convergent subsequence; $D_1B(x_k)\alpha_k \to D_1B(0)\alpha$ by the joint continuity of the tangent map `Chapter3/SolutionLinearization · IsSc1GermWith.tendsto_deriv`; and $\|D_2B(x_k)\| \to 0$. So ζ_k converges, $(\alpha_k,\zeta_k) \to (\alpha,z_0) \in N$ of norm 1 in $\ker D(h+\iota s)(0)$ — a contradiction. (The text uses the Neumann series $\zeta_k = \sum(D_2B(x_k))^l z_k$; here $|\zeta_k - z_k| \le \varepsilon_k \to 0$ suffices.) Then $\dim\ker D(h+\iota s)(x) \le \dim\ker D(h+\iota s)(0) = n-N = \mathrm{ind}$, and an operator of finite index with $\dim\ker \le \mathrm{ind}$ is onto `Chapter3/SolutionLinearization · HasFiniteIndex.surjective_of_finrank_ker_le` — instead of the text's "$Df(x)|_N$ has index 0 and is injective". *Zeros are on level* 1 *and small there*: by the lifts of Theorem 3.5.1 (preprint 3.40), continuous at 0 (§III.12); the text's detour through "bounded on level 2, hence convergent on level 1" is not needed.

III.20 Local solvability: Lemma 3.5.10 (preprint 3.46), Theorem 3.5.6 (preprint 3.43), Lemmas 3.5.5, 3.5.8 (preprint 3.42, 3.44), Theorem 3.5.11 (preprint Corollary 3.47)

Lemma 3.5.10 (preprint 3.46) (Appendix 3.8.3 (preprint 3.8.2)). If $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$, the image of the graph map `Chapter3/LocalSolvability · graphMap` (with $(\alpha,L\alpha) \in \widetilde{C}\oplus W$ iff $\alpha \in \widetilde{C}$ `Chapter3/LocalSolvability · graphMap_mem_prod_iff`), has a good complement Y' with respect to $\widetilde{C}\oplus W$, then $\widetilde{K}$ has a good complement Z with respect to $\widetilde{C}$ such that $Z\times W$ is a good complement of K' `Chapter3/LocalSolvability · lemma_3_46`. *Proof.* $Z' := \mathrm{pr}_P(Y')$ satisfies $P = \widetilde{K}+Z'$; choose $Z\subseteq Z'$ with $P = \widetilde{K}\oplus Z$ and a linear right inverse $\rho : Z' \to Y'$ of the projection (bounded, finite dimensions); for $|z| \le \varepsilon'|\alpha|$, $\varepsilon' = \varepsilon/(\|\rho\|+1)$, $y = \rho(z) = (z,w) \in Y'$ has $|y| \le \varepsilon|(\alpha,L\alpha)|$, so $(\alpha,L\alpha)\in C'$ iff $(\alpha,L\alpha)+y \in C'$, i.e. $\alpha\in\widetilde{C}$ iff $\alpha+z\in\widetilde{C}$. The text's intermediate complement Y_0 is not needed. Good position of K' gives good position of $\widetilde{K}$ `Chapter3/LocalSolvability · lemma_3_46_isInGoodPosition`.

Projecting solutions into the quadrant. A gap in the text's proof of Theorem 3.5.6 (3) (preprint 3.43): a solution $a = k+z$ ($k\in\widetilde{K}$, $z\in Z$) of $H(a)=0$ is matched with the solution of Theorem 3.8.22 (preprint 3.78) by its uniqueness clause, which needs $k\in\widetilde{C}$ — not verified in the text. It holds because z is small compared with k: from $H(a) = 0 = H(0)$ and $DH(0)k = 0$, $|DH(0)z| = o(|a|)$; $DH(0)$ is injective on Z, hence bounded below there `Chapter3/LocalSolvability · exists_norm_le_of_injOn`; so $|z| \le \varepsilon|k|$ near 0, and the good-complement property gives $k\in\widetilde{C}$ from $k+z = a\in\widetilde{C}$ `Chapter3/LocalSolvability · exists_proj_mem_of_eq_zero`.

The data of Theorem 3.5.6 (preprint 3.43). The text states it for $g = \Phi_*(f+s)$; the formal statement is for any sc-smooth germ g on G with $g(0) = 0$ whose index-raised germ g^1 on the shifted germ is a basic germ with contraction map B `Chapter3/LocalSolvability · ShiftedBasicData`. Its solution germ δ (of the working form of B) takes values in W_1 `Chapter3/LocalSolvability · ShiftedBasicData.δ`, with $\mathrm{pr}_W g(a,\delta(a)) = 0$ on level 0 `Chapter3/LocalSolvability · ShiftedBasicData.snd_g_sol` and uniqueness in the contraction region `Chapter3/LocalSolvability · ShiftedBasicData.eq_δ`. Put $H(a) = \mathrm{pr}_Q\, g(a,\delta(a))$ `Chapter3/LocalSolvability · ShiftedBasicData.H`. *Regularity of* H (the text: "H restricted to U_i' is of class C^{i-1}"): for every $j \ge 1$ there is a relatively open $V \ni 0$ in $\widetilde{C}$ on which H is C^j — by Corollary 3.3.7 (preprint 3.34) for g^1, δ lifts to a C^j map into W_{2j}, and $g : E_{2j} \to F_j$ is C^j by Proposition 1.2.3 (preprint 1.18) for germs `Chapter3/LocalSolvability · ShiftedBasicData.H_contDiffOn`. A single C^1 neighbourhood V_1 carries the derivative data `Chapter3/LocalSolvability · ShiftedBasicData.Deriv` `Chapter3/LocalSolvability · ShiftedBasicData.exists_deriv`: $D\delta(0) : P \to W_1$, $L = \iota D\delta(0)$, and $DH(0) = \mathrm{pr}_Q Dg(0)(\alpha,L\alpha)$ `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀` `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_H`, by the chain rule for $a\mapsto g(a,\delta(a))$ `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_comp`.

Lemma 3.5.8 (preprint 3.44). "Recalling from Lemma 3.9 that $D_2B(0) = 0$": Lemma 3.1.9 (preprint

3.9) gives this for g^1, on level 1; the kernel of $Dg(0)$ is computed on level 0. So first the W-component of $Dg(0)(0,\zeta)$ is ζ for $\zeta \in W_1$, by Lemma 3.1.9 (preprint 3.9) (b) for g^1, and then for all $\zeta \in W_0$ by density Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_inr. Differentiating $\mathrm{pr}_W g(a,\delta(a)) = 0$ at 0 gives $\mathrm{pr}_W Dg(0)(\alpha, L\alpha) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_graph. Hence $Dg(0)(\alpha,\zeta) = 0$ iff $\zeta = L\alpha$ and $DH(0)\alpha = 0$: $\ker Dg(0) = \{(\alpha, D\delta(0)\alpha) : \alpha \in \ker DH(0)\}$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq, and $DH(0)$ is onto when $Dg(0)$ is Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀_surjective.

Theorem 3.5.6 (preprint 3.43) Chapter3/LocalSolvability · theorem_3_43. Hypotheses: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position to $\widetilde{C} \oplus W$, and *the regularity of Lemma 3.5.5 (preprint 3.42) as an explicit hypothesis*: for every $\varepsilon > 0$ there is $r > 0$ such that every zero x of g in U_0 with $|x|_0 < r$ has W-component $\iota(w_1)$, $|w_1|_1 < \varepsilon$. *The choices* Chapter3/LocalSolvability · ShiftedBasicData.Sol Chapter3/LocalSolvability · ShiftedBasicData.exists_sol: $\widetilde{K} = \ker DH(0)$ with K' its graph (Lemma 3.5.8 (preprint 3.44)); the good complement Z of $\widetilde{K}$ with $Z \oplus W$ good for K' (Lemma 3.5.10 (preprint 3.46)); and the solution $\sigma : V' \to Z$ of $H(k+\sigma(k)) = 0$ with its uniqueness constants, from Theorem 3.8.22 (preprint 3.78) applied to H (§III.17). Then $\tau(y) = (\sigma(v), \delta(v+\sigma(v)) - Lv)$ for $y = (v, Lv)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ on $V = \pi^{-1}(V')$, relatively open in $K' \cap C'$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.V Chapter3/LocalSolvability · ShiftedBasicData.Sol.V_relOpen, with values in $Z \oplus W$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_mem_Y, so that $y + \tau(y) = (a, \delta(a))$ with $a = v + \sigma(v)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.coe_add_τ. (3), first half: $y + \tau(y)$ lies in U_0 and is a zero, since $\mathrm{pr}_W g(a,\delta(a)) = 0$ and $\mathrm{pr}_Q g(a,\delta(a)) = H(a) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.sol_mem_and_eq_zero. Regularity: τ is C^1 on V Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_contDiffOn with W-component on level 1 Chapter3/LocalSolvability · ShiftedBasicData.Sol.snd_τ_mem_range, and $D\tau(0)\eta = -D\delta(0)\pi(\eta) + D\delta(0)\pi(\eta) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.hasFDerivWithinAt_τ Chapter3/LocalSolvability · ShiftedBasicData.Sol.fderivWithin_τ — (2). (3), second half Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ: a zero $x = (a, w)$ near 0 has, by the regularity hypothesis, w on level 1 and small there, so $w = \delta(a)$ by the uniqueness of the solution germ on level 1, so $H(a) = 0$; write $a = k + z$ with the projection along Z Chapter3/LocalSolvability · ShiftedBasicData.Sol.projK; then $k \in \widetilde{K} \cap \widetilde{C}$ (the gap above) and $z = \sigma(k)$ by the uniqueness clause of Theorem 3.8.22 (preprint 3.78); so $x = y + \tau(y)$ for $y = (k, Lk)$. *Clause (1), "τ is an sc-smooth germ"*, is proved level-wise: for every $k \geq 1$ there are a relatively open $V_k \subseteq V$ containing 0 and a C^k map $\tau_k : V_k \to E_k = P \oplus W_k$ with $\iota_k \tau_k = \tau$ on V_k Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level — the reading of Corollary 3.3.7 (preprint 3.34) and Definition 3.2.1 (ii) (preprint 3.25). The proof applies Theorem 3.8.22 (preprint 3.78) to H at regularity k on the neighbourhood of its regularity lemma, and identifies the C^k solution σ_k with σ near 0 through the uniqueness clause (3) of Theorem 3.8.22 (preprint 3.78) — which needs σ continuous at 0 with $\sigma(0) = 0$, so that $|\sigma(y)|$ is below the uniqueness constant for y small; the text says nothing about why the solutions produced for the different k coincide. The W-component $\iota\delta(a) - L\pi(y)$, $a = \pi y + \sigma(\pi y)$, lifts to W_k through the C^k lift of δ into W_{2k}, with L the derivative of that lift at 0 composed with the inclusion (uniqueness of the derivative of $\iota\delta$ at 0). The full statement is Chapter3/LocalSolvability · theorem_3_43_sc_smooth; the statement without (1) is kept for Theorem 3.6.8 (preprint 3.54).

Lemma 3.5.5 (preprint 3.42). For the push-forward $g = \Phi_*(f+s)$ of Theorem 3.4.2 (preprint 3.38) — packaged as the data of Theorem 3.5.6 (preprint 3.43), g on level 0 with index-raised basic germ $(\Phi_*(f+s))^1$ Chapter3/LocalSolvability · StabilityData.toShiftedBasicData — the regularity hypothesis holds: a zero x' of g in $\psi(U_0)$ with $|x'|_0$ small has $\psi^{-1}(x')$ a zero of $f + \iota s$, which by the lifts of Theorem 3.5.1 (preprint 3.40), continuous at 0, lies on level 1 and is small there; ψ is linear on level 1 Chapter3/LocalSolvability · StabilityData.lemma_3_42.

Theorem 3.5.11 (preprint Corollary 3.47) Chapter3/LocalSolvability · theorem_3_47_sc_smooth Chapter3/LocalSolvability · theorem_3_47. For a basic germ f and the data of Theorem 3.4.2 (preprint 3.38) with

$D(f+\iota s)(0)$ surjective and kernel K in good position to $C = \widetilde{C} \oplus W$: a good complement Y, V relatively open in $K \cap C$, $\sigma : V \to Y$ of class C^1 with values on level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$, the graph consisting of zeros of $f + \iota s$, an open $U'' \ni 0$ in which every zero lies on the graph, and for every k a C^k lift of σ into E_k near 0. *Proof.* Theorem 3.5.6 (preprint 3.43) for $g = \Phi_*(f+s)$, with Lemma 3.5.5 (preprint 3.42) for its regularity hypothesis. Since the coordinate change is linear, $Dg(0) = \Psi \circ D(f+\iota s)(0) \circ \psi^{-1}$ (chain rule for sc-operators) `Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀`, so surjectivity and the kernels correspond, $\ker Dg(0) = \psi(K)$; good position of $\psi(K)$ follows by Lemma 3.5.9 (preprint 3.45) (finite dimensionality of K from Lemma 3.5.8 (preprint 3.44)); and the conclusion is transported back by ψ^{-1}: $Y = \psi^{-1}(Y')$, $V = \psi^{-1}(V')$, $\sigma = \psi^{-1}\tau\psi$. The text transports the solution set back with the nonlinear φ and justifies the good position with the degeneracy index (Lemma 3.8.13 (preprint 3.70), Theorem 3.8.15 (preprint 3.72)); with ψ linear neither is needed. The transport uses that $\widetilde{C} \oplus C$ is a partial quadrant for finite-dimensional C `Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ`.

III.21 The linearization at a zero (Proposition 3.6.5 (preprint 3.52)) and good position of a germ (Definition 3.6.6 (preprint 3.53))

Proposition 3.6.5 (preprint 3.52) (text): if f is an sc-Fredholm section of a tame strong bundle and $f(x) = 0$, then $Tf(x) : T_xX \to Y_x$ is an sc-Fredholm operator. Formalized in the chart of Definition 3.1.11 (preprint 3.11) centred at a zero x_0, with the level-0 Fredholm property `Chapter3/FredholmGermLinearizationG · proposition_3_52` `Chapter3/FredholmGermLinearizationG · proposition_3_52_local`: $Dg(0) : E_0 \to F_0$ and $Tf(x_0) = Dg(0)|_{T_0O} : T_0O \to K_0$ are Fredholm with $\operatorname{ind} Tf(x_0) = \operatorname{ind} Dg(0)$. Here $Dg(0)$ is the value at 0 of a derivative family Dg of the filling germ on a germ G with $G_0 \subseteq U$, the same for every family and every germ (§III.4); the statement exposes G and Dg, where the whole-chart statement `Chapter3/FredholmGermLinearization · proposition_3_52_wholeChart` has the derivative on U, a junk value for a germ. (The sc-Fredholm property of $Tf(x)$ would need $T_0O = Dr(0)E$ and $K_0 = \rho(0)F$ as sc-Banach spaces, not set up. The statements quantify the index as "$n - N$ for some natural numbers n, N", which every integer satisfies; the identification with $\dim P - \dim Q$ of the conjugating basic germ is made inside the proof, by `proposition_3_10_basic_level_zero`, and not exported.)

Step (a): the derivative of the conjugation `Chapter3/FredholmGermLinearizationG · IsConjugateToBasicGerm.scDeriv_isFredholm`. Let $k = g - \iota s$, an sc¹-germ on $G_g \cap G_s$ (§III.48) with $k(0) = 0$ ($s(0) = g(0)$ in Definition 3.1.11 (preprint 3.11), $g(0) = \mathbf{f}(0) = 0$ by the filler) and $(\Psi(u, k(u)))_2 = h(\psi(u))$ near 0. Linearizing needs: the chain rule for the germ $h \circ \psi$ on a germ small enough that ψ maps it into the germ of h (§III.1, `ScNbhdGerm.preimage`); the derivative of $u \mapsto \Gamma(u)k(u)$ at 0, where $\Gamma(u) = (\Psi(u,\cdot))_2$ is a bounded operator `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM` and $D(\mathrm{pr}_{F'} \circ \Psi)(0,0)(h,k) = \Gamma(0)k$ (as in §III.4) `Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_zero`, so the derivative is $\Gamma(0)Dk(0)$ because $k(0) = 0$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp` — for the germ k by the chain rule with the whole-chart outer map $\mathrm{pr}_{F'} \circ \Psi$, the two derivative families agreeing at 0 on $G_k \cap \psi^{-1}(G)$ `Chapter3/FredholmGermLinearizationG · IsStrongBundleIso.fiberCLM_comp_deriv_zero_eq_germ`, so no shrinking of the basic germ is needed at 0; and the invertibility of $D\psi(0)$ and $\Gamma(0)$ (inverses $D\psi'(0)$, $\Gamma'(0)$, chain rule for $\psi' \circ \psi = \mathrm{id}$). Then $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, so $Dk(0) = \Gamma(0)^{-1}Dh(0)D\psi(0)$ is Fredholm with the index of $Dh(0)$: Fredholm operators and their index are stable under composition with isomorphisms on both sides `Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv`, and $Dh(0)$ is Fredholm by Proposition 3.1.10 (preprint 3.10) for the basic germ `Chapter3/FredholmGermLinearization · proposition_3_10_basic_level_zero`. *Step (b)*: $Dg(0) = Dk(0) + \iota Ds(0)$ — for germs this is algebra at 0: $g - \iota s$ has the derivative family $Dg - \iota Ds$ on $G_g \cap G_s$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift` `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus` (on U it was the uniqueness of the derivative within U_1 and density) — a compact perturbation since $\iota Ds(0)$ is the level-0 map of an sc⁺-operator (s is an sc¹-germ into F^1 and 0 is smooth); the index is unchanged (Part I, §I.9). *Step (c)*: Proposition 3.1.5

(preprint 3.5) (§III.4) transfers both to $Tf(x_0)$; it needs $g(0) = 0$.

Definition 3.6.6 (preprint 3.53) (text): (f, x) with $f(x) = 0$ is in good position if (1) $f'(x) : T_xX \to Y_x$ is onto and (2) if $d_X(x) \geq 1$, $\ker f'(x)$ is in good position to the partial cone C_xX in T_xX. In the chart: Definition 3.1.24 (preprint 3.21) *inside a subspace* $T \subseteq E_0$ — $N \subseteq T$, interior of $N \cap C$ in N nonempty, and a good complement $P \subseteq T$ of N in T `Chapter3/GoodPositionGerm · IsInGoodPositionIn` (for $T = E_0$ it is Definition 3.1.24 (preprint 3.21) `Chapter3/GoodPositionGerm · isInGoodPositionIn_top_iff`); and the germ is in good position if $Tf(0) = Dg(0)|_{T_0O}$ is onto and $\ker Dg(0)$ is in good position to $T_0O \cap C$ inside $T_0O = Dr(0)E$ `Chapter3/GoodPositionGerm · IsGoodPositionAt` `Chapter3/GoodPositionGerm · IsGoodPositionAt.surjective_fillerRestrict`. Here $\ker Dg(0) \subseteq T_0O$, because $(1 - \rho(0))Dg(0)$ is injective on $\ker Dr(0)$ (filler (3)) and $(1-\rho(0))Dg(0)Dr(0) = 0$ `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and it is the kernel of $Tf(0)$ `Chapter3/GoodPositionGerm · IsFillerOf.map_ker_fillerRestrict` — the text's "$\ker Tf(0) \oplus \{0\} = \ker Dg(0)$ by the properties of the filler". The case distinction $d_X(x) \geq 1$ is dropped: at an interior point the partial cone is the whole tangent space and (2) holds for every subspace. For a filling germ the condition is one on the operator $Dg(0)$ (the text's $f'(x)$ is defined through the filler at the centre only), which the chart notion takes as a parameter `Chapter3/GoodPositionGermG · IsGoodPositionAtGerm` — the notion above is the case of the derivative on U `Chapter3/GoodPositionGermG · isGoodPositionAt_iff_germ` — and the two kernel facts hold with the same proofs `Chapter3/GoodPositionGermG · IsFillingGerm.ker_deriv_le_range` `Chapter3/GoodPositionGermG · IsFillingGerm.map_ker_fillerRestrict`. For a section, (f, x_0) is in good position if it has the data of Definition 3.1.11 (preprint 3.11) at x_0 and a derivative family Dg of the filling germ on a germ inside U, with the germ in good position at $Dg(0)$ in that chart `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm`; as $Dg(0)$ does not depend on the family, "for some family" and "for every family" agree `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.forall_isGoodPositionAtGerm`; it is then an sc-Fredholm germ `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.isScFredholmGerm`. The definition is in a chart, not in T_xX; the transport of C_xX to T_xX (Definition 2.5.8 (preprint 2.38)) is not formalized.

III.22 Good position along a graph: Lemma 3.6.9 (preprint 3.55)

Lemma 3.6.9 (preprint 3.55) (text, Appendix 3.8.4 (preprint 3.8.3)): for N in good position to C with good complement Y, $V \subseteq N \cap C$ relatively open and $\tau : V \to Y_1$ of class C^1 with $\tau(0) = 0$, $D\tau(0) = 0$, for v near 0: (1) $v + \tau(v) \in C_1$; (2) Y is a topological complement of $N_v = \{n + D\tau(v)n\}$; (3) for some $\gamma_v > 0$, $n \in C_x \iff n + y \in C_x$ for $n \in N_v$, $y \in Y$, $|y|_0 \leq \gamma_v|n|_0$, $x = v + \tau(v)$. *Formalized* `Chapter3/GoodPositionOpen · lemma_3_55` in the model $\mathbb{R}^n \oplus W$, with the core in coordinates for any C cut out by functionals `Chapter3/GoodPositionOpen · lemma_3_55_core`: for N in good position with good complement Y, Y **closed**, and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$ and Y is a good complement of N_v (the subspace $\{n + Ln\}$ for $L = D\tau(v)$ `Chapter3/GoodPositionOpen · graphSpace`) with respect to the partial quadrant $C_{v+\tau(v)}$.

The added hypothesis. The step "$D\tau(v)n \in Y$ for all $n \in N$" needs Y closed: the derivative within $N \cap C$ at a point of unique differentiability of a map with values in a closed subspace takes values there, because the tangent cone of a subset of a closed subspace lies in it `Chapter3/GoodPositionOpen · tangentConeAt_subset_of_subset_closed` `Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo`; for a non-closed algebraic complement it fails. The text's $Y = Y_0$ is a topological complement, hence closed. With Y closed, (2) is linear algebra: $E_0 = N_v \oplus Y$ whenever $E_0 = N \oplus Y$ and L takes values in Y `Chapter3/GoodPositionOpen · graphSpace_isCompl`, and N_v is closed (finite-dimensional) `Chapter3/GoodPositionOpen · graphSpace_isClosed`.

The estimate. (1) is Lemma 3.8.13's (preprint 3.70) smallness: $|\tau(v)| \leq \varepsilon|v|$ near 0 and good position. For (3), take $\varepsilon_0 = \min(\varepsilon, 1)$ and the neighbourhood where $\|D\tau(v)\| \leq \varepsilon_0/4$; then $\gamma = \varepsilon_0/4$ works *for every* v, since $\gamma(1 + \varepsilon_0/4) + \varepsilon_0/4 \leq \varepsilon_0$ (the text: $\gamma_v(1 + \varepsilon) + \varepsilon < \gamma$, with γ_v chosen after v). If $v \in N \cap W$, $C_x = C$ and the estimate transfers the good-complement property from N to N_v. If $v \notin N \cap W$, no normal form is needed: if $\ell_i(v) = 0$ then $i \in \Sigma$, so ℓ_i vanishes on Y (Lemma 3.8.8 (preprint 3.65)), and $\ell_i(v + \tau(v)) = 0$ iff $\ell_i(v) = 0$ (Lemma 3.8.13 (preprint 3.70)); so $\ell_i(h + y) = \ell_i(h)$ for every coordinate i that defines C_x, and (3) holds for every γ. The text's cases $\dim N = 1$, $\dim N > 1$ are not needed,

and the sc-level of the values of τ plays no role. The formal τ is given on all of $N \cap C$ rather than on a neighbourhood V; the conclusion is local.

III.23 Theorem 3.6.8 (preprint 3.54): the linear part and the graph description of the solution set

Theorem 3.6.8 (preprint 3.54) (text): for an sc-Fredholm section f of a tame strong bundle with $f(x) = 0$ and (f, x) in good position there is an open $V \ni x$ such that $S = \{y \in V : f(y) = 0\}$ satisfies (1) (f, y) is in good position at every $y \in S$, and (2) S is a sub-M-polyfold with the structure of a manifold with boundary with corners. The proof: the linear part (surjectivity and good position transported from T_xX to the chart and the push-forward); (3), the graph description of S from Theorem 3.5.11 (preprint Corollary 3.47) for $\Psi_*(g) = h + t$; then Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) for (1), and Theorem 3.2.3 (preprint 3.27) for (2). Formalized: the linear part and (3).

The linear part `Chapter3/GoodPositionGermG · theorem_3_54_linear`. In the chart, with the germ in good position and $(1 - Dr(0))E \subseteq E_0$ (the minimal subspace at the corner 0; Proposition 2.5.3 (preprint 2.35) for the tame retraction r `Chapter3/GoodPositionGermG · theorem_3_54_linear_of_isTameWith Chapter2/TameRetracts · proposition_2_35`): $Dg(0)$ is onto (Proposition 3.1.5 (preprint 3.5) for the filling germ, §III.4, from the surjectivity of $Tf(0)$), and $\ker Dg(0)$ is in good position to C in E; only the filler at 0 enters, and r is whole-chart, so $Dr(0)$ is a genuine derivative. The latter is the *transport lemma*: for a bounded idempotent π with $(1-\pi)E_0$ inside the minimal subspace at 0, good position inside πE_0 gives good position in E_0, with the good complement $P \oplus (1-\pi)E_0$ and the constant $\varepsilon/(\|\pi\| + 1)$ `Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition`. Two points the text passes over: for $p = z + k$ with $|p| \le \varepsilon'|n|$ the T_0O-component is $z = Dr(0)p$, so $|z| \le \|Dr(0)\|\varepsilon'|n|$ — the splitting must be topological; and $\ker Dr(0) \subseteq E_0$ is used only as "the $\mathbb{R}^n$-coordinates of k vanish, so $n+z+k \in C \iff n+z \in C$". *This inclusion is the only place where tameness enters.*

$C' = T\psi(0)C$. Not part of the data of a conjugation (§III.6); it is a theorem: for an sc-diffeomorphism ψ between relatively open subsets of partial quadrants with $\psi(0) = 0$, $x \in C \iff D\psi(0)x \in C'$, by the one-sided sign argument for ψ and ψ^{-1} (Part II, §II.12) `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant`, since $C_0 = C$ at the corner `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero Chapter3/GoodPositionGerm · scDeriv_mem_iff_of_inverse`.

The push-forward. $\Psi_*(g)(u') = (\Psi(\psi^{-1}(u'), g(\psi^{-1}(u'))))_2$ `Chapter3/GermPushforward · pushforward` is an sc-smooth germ on the germ of the points that ψ^{-1} maps into the germ of g `Chapter3/GermPushforward · IsStrongBundleIso.pullGerm Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isScSmoothGerm` (chain rules with the whole-chart Ψ and ψ^{-1}, §III.48), with $\Psi_*(g)(0) = 0$ and $D(\Psi_*g)(0) = \Gamma(0)Dg(0)D\psi(0)^{-1}$ for its explicit derivative family `Chapter3/GermPushforward · IsStrongBundleIso.pushforwardDeriv_zero` ($Dk(0) = \Gamma(0)Dg(0)$ for $k(u) = (\Psi(u, g(u)))_2$) `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization`; so it is onto `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_surjective`, its kernel $D\psi(0)(\ker Dg(0))$ is in good position to C' (Lemma 3.5.9 (preprint 3.45)) `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_ker_isInGoodPosition`, and it is Fredholm (transported from $Dg(0)$, Proposition 3.6.5 (preprint 3.52)) — the hypotheses of Theorem 3.5.11 (preprint Corollary 3.47), for *any* conjugating isomorphism `Chapter3/GoodPositionGermG · theorem_3_54_pushforward`, and in bundle form `Chapter3/GoodPositionGermG · theorem_3_54_pushforward_section`. The text instead obtains the Fredholm property of $D(h + t)(0)$ from Propositions 3.1.10 and 1.1.11 (preprint 3.10 and 1.10).

The graph description `Chapter3/GlobalIFTG · theorem_3_54_graph`. Let the charts of the bundle have tame base retractions (`htame`, the hypothesis of the first formalization; it cannot be met, and it is replaced by good position in a tame chart, §III.47), $f(x_0) = 0$, (f, x_0) in good position. The proof, in the order of the Lean.

1. *The chart and the identities.* Take the chart of Definition 3.6.6 (preprint 3.53) centred at x_0, filling germ g on $G_{g'}$, sc$^+$-germ t_0 on G_s and conjugation Ψ covering ψ, with inverse Ψ' covering ψ'. *Shrink the*

basic germ to the germ G'' of the points of G that ψ' sends into $G_g \cap G_s$ level by level: same Ψ, h, B and conjugation identity `Chapter3/BasicGermShrink · ConjugationData.shrink` (§III.48); below, G denotes G''. There is an open $N_1 \ni 0$ on which $\mathbf{f} = g$ on O, zeros of g in U lie in O, and $\Psi(u, (g - \iota t_0)(u))_2 = h(\psi(u))$ (filler (1), (2) and the conjugation).

2. *The sc⁺-germ t.* $t(u') = \Psi[1](\psi'(u'), t_0(\psi'(u')))_2$ is an sc-smooth germ into $(Q \oplus W)^1$ on G: $t_0 \circ \psi'$ is a germ on the preimage of G_s, which contains $G = G''$, and the whole-chart $\Psi[1]$ is composed on the outside (§III.48). (With t_0 sc-smooth on U, t was sc-smooth on all of U'.) On $U \cap N_1$, $\Psi_*(g)(\psi(u)) = h(\psi(u)) + \iota t(\psi(u))$, by the linearity of the fibre maps.
3. *Absorbing the $\mathbb{R}^N$-component.* The data of Theorem 3.4.2 (preprint 3.38) carry only the W-component of an sc⁺-germ. Let t_Q, t_W be the two components of t, through the sc-operators $(q, w) \mapsto (q, 0)$ and $(q, w) \mapsto w$ `Chapter3/GlobalIFT · ScBanach.qPartOp Chapter3/GlobalIFT · ScBanach.wPartOp`. Then $h' = h + (t_Q, 0)$ is again a basic germ with the same contraction B (Definition 3.1.7 (preprint 3.7) constrains only the W-component), and with the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) for $s = t_W$ (they exist, §III.18) one has $\mathcal{S}.\mathrm{fsum}\, h' = h' + \iota t_W = h + \iota t$.
4. *The hypotheses of Theorem 3.5.11 (preprint Corollary 3.47).* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.5.3 (preprint 2.35)), so the push-forward step applies: $D(\Psi_* g)(0)$ is surjective with kernel in good position. Since $\mathcal{S}.\mathrm{fsum}\, h' = \Psi_*(g)$ only on a neighbourhood W' of 0, the two derivatives at 0 are identified by the uniqueness of the derivative within G_1'' and density of E_1 `Chapter3/GlobalIFT · scDeriv_eq_of_eqOn_nhds`; $\Psi_*(g)$ is a germ on the preimage of G_g, which contains G''.
5. *Theorem 3.5.11 (preprint Corollary 3.47) for $h + t$* gives Y, V_0, σ (good complement, C^1, level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$), the graph consisting of zeros, and U'' in which every zero lies on the graph.
6. *The neighbourhoods* (the text does not say which V). Let $\Omega = U'' \cap W'$ and $V' = V_0 \cap (1 + \sigma)^{-1}(\Omega)$, relatively open in $K \cap C'$ because $v \mapsto v + \sigma(v)$ is continuous. Let $V_x = \{y \in \mathrm{dom}\,\varphi : \varphi(y) \in N_1,\ \psi\varphi(y) \in \Omega \cap G_0\}$, open and containing x_0 by continuity of $\psi \circ \varphi$. $D\sigma(0) = 0$ persists on the smaller V' (unique differentiability at 0 of V').
7. *The conclusions.* $\psi \circ \varphi$ is injective on V_x (φ is, and $\psi' \circ \psi = \mathrm{id}$ on U). For $y \in V_x$: $f(y) = 0 \iff \mathbf{f}(\varphi y) = 0 \iff g(\varphi y) = 0 \iff (h + \iota t)(\psi\varphi y) = 0$ (items 1–2). If $f(y) = 0$, then $\psi\varphi(y)$ is a zero in $U'' \cap G_0$, hence $v + \sigma(v)$ with $v \in V_0$, and $v \in V'$ since $v + \sigma(v) \in \Omega$. Conversely, for $v \in V'$, $u = \psi'(v + \sigma(v))$ lies in $U \cap N_1$, $g(u) = 0$, so $u \in O$ and $u = \varphi(y)$ for some y, which lies in V_x and is a zero.

So $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$. When this section was written, (1) and (2) were not formalized; they are now: the kernel and good position at the nearby zeros in §§III.32–III.33 (the surjectivity of $f'(y)$ under an added hypothesis), the data of Definition 3.2.1 (preprint 3.25) at every zero in §§III.34–III.36, and the theorem in §III.37. The graph description also records, since then, that Y is closed, that the kernel is a smooth subspace and the level-wise lifts of σ (Theorem 3.5.11 (1) (preprint Corollary 3.47)), and it is a corollary of the version with a good complement whose projection maps C' into itself (§III.29).

Theorems 3.6.10 and 3.6.11 (preprint 3.56 and 3.57), the step from the tangent germ property. With $S = \{f = 0\}$ `Chapter3/GlobalIFT · solutionSet`: if $S \neq \emptyset$ has at every point regular data of Definition 3.2.1 (preprint 3.25) (§III.24), then S is a sub-M-polyfold whose induced structure is tame and an n-dimensional manifold with boundary with corners (Theorem 3.2.3 (preprint 3.27)) `Chapter3/GlobalIFT · theorem_3_57_of_tangentGerm`; if X has no boundary, the induced structure is without boundary, since $d_S \leq d_X = 0$ (Lemma 2.4.6 (preprint 2.26)) `Chapter2/SubMPolyfold · lemma_2_26 Chapter3/GlobalIFT · theorem_3_56_of_tangentGerm`. The first step of the text's proof — good position at every zero gives the tangent germ property — was not formalized when this section was written; the theorems with the text's hypotheses are in §III.37.

III.24 The tangent germ property: Definition 3.2.1 (preprint 3.25) and Theorem 3.2.3 (preprint 3.27)

Definition 3.2.1 (preprint 3.25). The data at $x \in M$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart`: a chart $\varphi : (U,x) \to (O,0)$ onto a tame retract (O,C,E); an n-dimensional smooth $N \subseteq E$ in good position to C with a good complement Y, Y **closed**; a relatively open $V \ni 0$ in $N \cap C$; $\delta : V \to E$ continuous with values in Y; (i) $\varphi(M \cap U) = \{v + \delta(v) : v \in V\}$; (ii) $\delta(0) = 0$, $D\delta(0) = 0$ and δ an sc^∞-germ, read as: for every m, k a relatively open $V_{m,k} \ni 0$ in $N \cap C$ with $\delta(V_{m,k}) \subseteq E_m$ and $\delta : V_{m,k} \to E_m$ of class C^k `Chapter3/TangentGerm · IsScInftyGermAt`. This k-dependent reading is what the germ implicit function theorem delivers (§III.11); the reading "sc-smooth on one fixed germ" implies it by Proposition 1.2.3 (preprint 1.18) for germs `Chapter3/TangentGerm · IsScSmoothGerm.isScInftyGermAt`, as does sc-smoothness on V as a set `Chapter3/TangentGerm · IsScSmooth.isScInftyGermAt`, so the theorem covers all three. M has the property if $M \subseteq X_\infty$ and every point has such data `Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty`. With these data: $N \cap C$ is a partial quadrant in N (Proposition 3.1.25 (preprint 3.22)) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isPartialQuadrant`, and N, Y are topological complements — this is where Y closed is used — so the projection $\pi : E \to N$ along Y is continuous `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isTopCompl Chapter3/TangentGerm · MPolyfold.TangentGermChart.π`; with $\Gamma(v) = v + \delta(v)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ`, $\pi\Gamma = \mathrm{id}$ and $\varphi(y) = \Gamma(\pi\varphi(y))$ for $y \in M \cap U$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`.

Proposition 3.2.2 (preprint 3.26) `Chapter3/TangentGerm · proposition_3_26`. For an sc-diffeomorphism $\psi : X \to X'$ (sc-smooth on all of X with sc-smooth inverse ψ' on all of X') and $\psi(M) = M'$, M' has the n-dimensional tangent germ property iff M has it. One direction suffices, applied to ψ and ψ' `Chapter3/TangentGerm · proposition_3_26_image`: $\psi(M) \subseteq X'_\infty$ because sc-smooth maps preserve the levels, and the data at $\psi(x)$ are the transported chart $\varphi \circ \psi^{-1}$ `Chapter2/DegeneracyIndex · transportChart`, whose model is that of φ `Chapter3/TangentGerm · MPolyfold.transportChart_model_0`, with the same N, Y, V, δ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.transport`: condition (i) is transported because $\varphi\psi^{-1}(\psi(M) \cap \psi(U)) = \varphi(M \cap U)$. As in the text; tameness of X plays no role.

Two gaps in the text's proof of Theorem 3.2.3 (preprint 3.27), and the conditions that close them `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`. (a) The text asserts that δ is an sc^∞-germ near every $v_0 \in V$, by linearizing at v_0 and claiming that a certain map $N' \to N$ is an isomorphism. That needs δ differentiable at v_0, known only near 0, and fails: in $X = \mathbb{R}^2$ take M the graph of g with $g(s) = s^2$ near 0 and $g(s) = 4 + (s-2)^{1/3}$ near 2, a smooth embedded curve with a vertical tangent at $(2,4)$; with $\varphi = \mathrm{id}$, N the horizontal axis, Y the vertical one and $\delta = g$, the data at 0 satisfy Definition 3.2.1 (preprint 3.25), M has the tangent germ property (at $(2,4)$ use a rotated chart), but δ is not differentiable at 2 and the chart $\pi\varphi$ on all of $M \cap U$ is not smoothly compatible with the chart at $(2,4)$. So condition (a): δ is sc-smooth on V as a map on a set. (b) The text's $\Sigma = \{v + y \in C : v \in V, y \in Y\}$ is claimed relatively open in C, so that $t(v+w) = v + \delta(v)$ is defined near 0 in O. False: $E = \mathbb{R}^3$, $C = [0,\infty)^3$, $N = \mathbb{R}(1,1,1)$, $Y = \{x + y - z = 0\}$ is a good complement, and $\pi(x,y,z) = (x+y-z)(1,1,1)$, so $\pi z \in C$ iff $x + y \geq z$; Σ is not open at $(1,0,1)$ and points $(\varepsilon, 0, \varepsilon + \varepsilon^2) \notin \Sigma$ approach 0. So condition (b): $\pi(O) \subseteq C$; then $\Sigma \cap O = \{z \in O : \pi z \in V\}$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig` is relatively open in O `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig_eq`. (A good complement with $\pi'(C) \subseteq N \cap C$ always exists, via the normal form, but then $M \cap U$ must be rewritten as a graph along it with the quadrant inverse function theorem, and the regularity problem of (a) returns; this is not done.)

Theorem 3.2.3 (preprint 3.27) `Chapter3/TangentGerm · theorem_3_27`. If $M \neq \emptyset$ has at every point data satisfying (a), (b), then:

M is a sub-M-polyfold `Chapter3/TangentGerm · theorem_3_27_isSubMPolyfold`, with the text's retraction. $t(v+w) = v + \delta(v)$, i.e. $t = \Gamma \circ \pi$ on $\Sigma \cap O$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t`, is idempotent with image $\Gamma(V)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_t Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_eq_self_of_mem`. It is sc-smooth: π is a continuous linear map into a finite-dimensional space, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmooth`; $\Gamma =$

$\iota + \delta$ is sc-smooth on V by (a) and because the inclusion of a smooth finite-dimensional subspace is sc-smooth, its level maps being the linear lifts $N \to E_m$ `Chapter3/TangentGerm · finSmoothLift` `Chapter3/TangentGerm · isScSmooth_subtype_val` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmooth`; and for a retraction R onto $\Sigma \cap O$, $t \circ R = \Gamma \circ \pi \circ R$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_isScSmoothRetractMap`. The text says nothing about either. On $U_0 = \varphi^{-1}(\Sigma \cap O)$, open `Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀`, the map $s = \varphi^{-1} \circ t \circ \varphi$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s` is an sc-smooth retraction `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn` onto $M \cap U_0$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s` — Definition 2.3.18 (preprint 2.20) at x.

The charts $\pi \circ \varphi$. On $M \cap U_0$, $y \mapsto \pi\varphi(y)$ is a chart with model $(V, N \cap C, N)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.model`, a tame retract (the identity retraction) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract`, with inverse $v \mapsto \varphi^{-1}(v + \delta(v))$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chartInv` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart`, a finite-dimensional corner chart of dimension n `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner` containing x `Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source`. It is compatible with every chart $\varphi'|_{W \cap M}$ of the induced structure of Proposition 2.3.19 (preprint 2.21): the transition maps are $\varphi' \circ \varphi^{-1} \circ \Gamma$ and $\pi \circ \varphi \circ \varphi'^{-1}$, compositions of sc-smooth maps between retracts `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmoothRetractMap` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmoothRetractMap` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart`. So it is a chart of the induced structure — **Theorem 3.2.3 (2) (preprint 3.27)** `Chapter3/TangentGerm · MPolyfold.theorem_3_27_isChart` `Chapter2/SubMPolyfold · subPolyfold`; these charts are pairwise compatible by locality `Chapter3/TangentGerm · MPolyfold.tangentCharts_compatible` and form an atlas `Chapter3/TangentGerm · MPolyfold.germAtlas` equivalent to the induced one `Chapter3/TangentGerm · MPolyfold.theorem_3_27_atlas_equiv`, with tame models: **the induced structure is tame** `Chapter3/TangentGerm · MPolyfold.theorem_3_27_isTame`, and **sc-smoothly equivalent to an n-dimensional manifold with boundary with corners** `Chapter3/TangentGerm · MPolyfold.theorem_3_27_isManifoldWithCorners`. "Manifold with boundary with corners" belongs to Chapter 4; here it is an equivalent atlas of charts on relatively open subsets of n-dimensional partial quadrants `Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners`, whose transition maps are classically smooth because sc-smooth maps between finite-dimensional constant structures are C^∞ `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`. Tameness of X is not used: the text's "since O is a tame retract, there exists … a tame sc-smooth retraction r" is Proposition 2.3.1 (1) (preprint 2.10) for any retract. *The degeneracy index*: since $\pi\varphi$ has a tame model, $d_M(y) = d_{N \cap C}(\pi\varphi(y))$ (Proposition 2.5.6 (preprint 2.36)) `Chapter3/TangentGerm · MPolyfold.theorem_3_27_degIndex`; Lemma 3.8.13 (preprint 3.70) relates this to $d_C(v + \delta(v)) = d_X(y)$ in the model $\mathbb{R}^n \oplus W$ only.

III.25 Lemmas 3.7.2–3.7.4 (preprint 3.59–3.61): towards the basic germ criterion

Lemmas 3.7.2 and 3.7.3 (preprint 3.59 and 3.60), abstract form. In the proof of Theorem 3.7.1 (preprint 3.58) the family $L(b) = P_Y \circ Df(b)|_X : X \to Y$ is shown to be injective for b near 0 (Lemma 3.7.2 (preprint 3.59)) and to satisfy uniform lower bounds $|L(b)h|_m \geq c_m|h|_m$ (Lemma 3.7.3 (preprint 3.60)) — the hypotheses of Lemma 3.7.4 (preprint 3.61). Both are sequence arguments, and all they use of the compactness condition (2) of Theorem 3.7.1 (preprint 3.58) is a property of the family itself: if $b_j \to b$, (h_j) is bounded in X_m and $L(b_j)h_j \to 0$ in Y_m, then (h_j) has a convergent subsequence in X_m `Chapter3/BasicGermCriterion · FamilyCompactness`. (For $L = P_Y \circ Df|_X$ it follows from (2), since $Df(b_j)h_j - L(b_j)h_j = (1 - P_Y)Df(b_j)h_j$ lies in the finite-dimensional smooth complement A and is bounded on every level.) The two lemmas are formalized for an abstract family of sc-operators with jointly continuous level maps. *Lemma 3.7.2 (preprint 3.59)*: if $L(0)$ is injective, so is $L(b)$ for b near 0 — otherwise $b_j \to 0$ and unit vectors h_j with $L(b_j)h_j = 0$ give, by the compactness property, a limit h with $|h| = 1$ and $L(0)h = 0$ `Chapter3/BasicGermCriterion · lemma_3_59`. *Lemma 3.7.3 (preprint 3.60)*: on a compact parameter set K on which every

$L(b)_m$ is injective there is $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ — otherwise $b_j \in K$ and unit h_j with $|L(b_j)h_j|_m \to 0$ give, after subsequences, $b_j \to b \in K$, $h_j \to h$, $|h|_m = 1$, $L(b)h = 0$ `Chapter3/BasicGermCriterion · lemma_3_60`. The text's step "$P_Y \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism" is needed only to feed Lemma 3.7.4 (preprint 3.61). The setting of Theorem 3.7.1 (preprint 3.58) (the splitting as coordinates, the complement A, the index computation), the strong bundle map $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ and the contraction estimate from condition (3) were not formalized when this section was written; they are now, in the coordinates of the text's proof, and there the hypotheses of the two lemmas are derived from those of the theorem (§III.38).

Lemma 3.7.4 (preprint 3.61) (quoted by the text from [25, Proposition 4.8] without proof). Let V be a germ of relatively open neighbourhoods of 0 in a partial quadrant $\widetilde{C}$ of a finite-dimensional G, $V \oplus E$ the product germ with levels $V_m \times E_m$ `Chapter3/InverseFamily · ScNbhdGerm.prodFull`, and $L(v) : E \to F$ sc-isomorphisms such that $\Phi(v,h) = L(v)h$ `Chapter3/InverseFamily · famMap` is an sc-smooth germ on $V \oplus E$ and $c_m|h|_m \le |L(v)h|_m$ for $v \in V_0$. Then $\Lambda(v,k) = L(v)^{-1}k$ `Chapter3/InverseFamily · famInv` is an sc-smooth germ on $V \oplus F$ `Chapter3/InverseFamily · lemma_3_61`. Operator-norm continuity of $v \mapsto L(v)$ is not assumed.

sc⁰. On level m, with $h_0 = L(v_0)^{-1}k_0$, $|L(v)^{-1}k - L(v_0)^{-1}k_0|_m \le c_m^{-1}(|k-k_0|_m + |(L(v_0)-L(v))h_0|_m)$, and $v \mapsto L(v)h_0$ is continuous `Chapter3/InverseFamily · famInv_isSc0Germ`.

Differentiability `Chapter3/InverseFamily · famInv_hasFDerivWithinAt`. The candidate is $D\Lambda(v,k)(\delta v,\delta k) = L(v)^{-1}(\delta k - D_1\Phi(v,L(v)^{-1}k)\delta v)$ `Chapter3/InverseFamily · famInvDeriv`. With $h = \Lambda(v,k)$ and $w = D\Lambda(v,k)(\delta v,\delta k)$, $\Lambda(v',k') - \Lambda(v,k) - w = L(v')^{-1}[(L(v)-L(v'))w] - L(v')^{-1}r$, where $r = \Phi(v',h) - \Phi(v,h) - D_1\Phi(v,h)\delta v = o(|\delta v|)$ is the remainder of Φ along $(\delta v, 0)$ with h fixed, so that only the finite-dimensional variable moves. The term $(L(v)-L(v'))w$ is $o(|(\delta v,\delta k)|_1)$ *because the increment is measured on level* 1: $w/|(\delta v,\delta k)|_1$ lies in the compact set $D\Lambda(v,k)(\iota(B_1))$ ($\iota : F_1 \to F_0$ compact), and $(L(v)-L(v'))u \to 0$ uniformly for u in a compact set — a tube lemma for jointly continuous maps vanishing on $\{x_0\} \times K$ `Chapter3/InverseFamily · exists_forall_norm_le_of_isCompact`. Strong continuity plus the compactness of $F_1 \to F_0$ replace the operator-norm continuity; the differentiability of Λ as a map $(V \oplus F)_0 \to E_0$ would need the latter and is not claimed.

The tangent map and the induction. $T\Lambda((v,k),(\delta v,\delta k)) = (\Lambda(v,k), \Lambda(v,\delta k - (T\Phi((v,\Lambda(v,k)),(\delta v,0)))_2))$, a composition of Λ, $T\Phi$ and sc-operators (the projections `Chapter3/InverseFamily · tangentBaseOp` `Chapter3/InverseFamily · tangentInlOp` `Chapter3/InverseFamily · tangentFiberOp` and subtraction `Chapter3/InverseFamily · ScBanach.subOp`); it is sc⁰ `Chapter3/InverseFamily · famInv_tangent_isSc0`, so Λ is sc¹ `Chapter3/InverseFamily · famInv_isSc1GermWith`; and if Φ is sc^{k+1}, then Λ (sc^k by induction) and $T\Phi$ (sc^k) make $T\Lambda$ sc^k, so Λ is sc^{k+1} `Chapter3/InverseFamily · famInv_isSckGerm`. The sc¹ step uses no quadrant hypothesis; the induction uses the composition rules for germs.

III.26 What the formalization taught us

1. **The germ implicit function theorem holds with the text's hypotheses**, but on shrinking germs: the solution germ is sc^k on the germ shifted by $2^k - 1$ levels. In the text's reading of germs (up to shrinking) this is the theorem; read with fixed germs, "if f is sc-smooth so is its solution germ" holds for every k on a germ depending on k. Every later "sc-smooth germ" (Definition 3.2.1 (ii) (preprint 3.25), Theorem 3.5.6 (1) (preprint 3.43), Theorem 3.5.11 (1) (preprint Corollary 3.47)) must be read the same way. Whether the shrinking is intrinsic is open.
2. **Theorem 3.3.3 (preprint 3.30) has a short proof**, which the text cites from [21]: an approximate fixed point and the $\frac{1}{2}$-contraction on level 0, at parameters where the level-1 operators exist.
3. **The contraction regions must be explicit** for "from the uniqueness we conclude $\delta^{(1)} = T\delta$" (Lemma 3.3.4 (preprint 3.31)) to be true.
4. **Several "immediate consequences" are proofs of their own**: the compactness of Theorem 3.1.13 (preprint 3.12) (not in Theorem 3.5.1 (preprint 3.40)), the induction of Theorem 3.5.1 (preprint 3.40)

(lifts, bounds and continuity together), the sc-smoothness and regularizing property in Theorem 3.1.18 (preprint 3.15), the sc$^+$ property in Proposition 3.1.10 (preprint 3.10), the projection of solutions into the quadrant in Theorem 3.5.6 (preprint 3.43).

5. **Three added hypotheses**: Y closed in Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25); and the two regularity conditions of Theorem 3.2.3 (preprint 3.27), each with a counterexample to the step of the text's proof it replaces.
6. **Tameness is nearly unused**: not in §3.1 at all, and in §3.6 only through $(1-Dr(0))E \subseteq E_0$ and, for the tangent germ property of the solution set, the tameness of the model of the chart $\psi \circ \varphi$ (§III.27).
7. **Linear coordinate changes** in Theorem 3.4.2 (preprint 3.38) make the text's degeneracy-index argument for Theorem 3.5.11 (preprint Corollary 3.47) unnecessary.
8. **The extreme-ray lemmas and the iterated tangent maps are not needed**: Proposition 3.1.25 (preprint 3.22) by a segment argument, the regularity in Theorems 3.8.15 and 3.8.22 (preprint 3.72 and 3.78) by bootstrapping, without shrinking.
9. **What remained of Theorem 3.6.8 (preprint 3.54)** when this list was written — the re-centring of good position at the nearby zeros, and the level-wise regularity of the solution map in the data of Definition 3.2.1 (preprint 3.25) at every zero — is done in §§III.31–III.37, with one added hypothesis for Theorem 3.6.8 (1) (preprint 3.54); see §III.39.

III.27 Tameness is carried by sc-diffeomorphisms, and by open subsets

The text never states that tameness is invariant under sc-diffeomorphisms, or that an open piece of a tame retract is a tame retract, but it uses both. In the proof of Theorem 3.6.8 (preprint 3.54) the chart $\psi \circ \varphi$ onto $\psi(O)$ must be a chart of the tame M-polyfold X for Definition 3.2.1 (preprint 3.25) to apply; in the proof of Proposition 3.2.2 (preprint 3.26) it writes "$X' = \psi(X)$ is tame"; and wherever a chart is restricted to a smaller domain (the shrinking of the data of Definition 3.2.1 (preprint 3.25) in §§III.36–III.37) the model of the restricted chart must again be a tame retract. Both facts are proved here, with Definition 2.5.2 (preprint 2.34) as the formalization states it (Part II).

The objects. An sc-diffeomorphism between relatively open subsets $U \subseteq C$, $U' \subseteq C'$ of partial quadrants is a pair $\psi : U \to U'$, $\psi' : U' \to U$ of sc-smooth maps, each mapping its domain into the other's, inverse to each other on U and on U' `Chapter2/TameInvariance · IsScDiffeoQuadrant` (its inverse is again one `Chapter2/TameInvariance · IsScDiffeoQuadrant.symm`). A retraction $r : U \to U$ is *tame* for a presentation T of C if (1) $d_C(r(x)) = d_C(x)$ for $x \in U$ and (2) at every smooth point x of $O = r(U)$ on level 1 the tangent space $T_xO = Dr(x)E$ has an sc-complement A contained in the minimal subspace E_x `Chapter2/Tame · ScRetraction.IsTameWith`; it is tame if it is tame for every presentation `Chapter2/Tame · ScRetraction.IsTame`, and (O, C, E) is a tame retract if O is the image of a tame retraction `Chapter2/Tame · IsTameRetract`.

The transported retraction. For an sc-smooth retraction r with domain U, put $r' = \psi \circ r \circ \psi' : U' \to U'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport`. It maps U' into U' because $\psi'(U') \subseteq U$, $r(U) \subseteq U$ and $\psi(U) \subseteq U'$; it is sc-smooth by the chain rule applied twice; and it is idempotent, $r'(r'(u')) = \psi(r(\psi'\psi(r(\psi'u')))) = \psi(r(r(\psi'u'))) = r'(u')$, since $\psi'\psi = \mathrm{id}$ on U and r is idempotent. Its image is $\psi(O)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport_retract`: $r'(u') = \psi(r(\psi'u'))$ with $r(\psi'u') \in O$; conversely, for $z \in O$, $r'(\psi z) = \psi(r(z)) = \psi(z)$.

Condition (1). The degeneracy index is preserved by ψ, $d_{C'}(\psi(u)) = d_C(u)$ for $u \in U$: this is Theorem 1.3.5 (preprint 1.25) `Chapter2/TameInvariance · IsScDiffeoQuadrant.degIndex_eq` `Chapter2/DegeneracyIndex · theorem_1_25`. Then for $u' \in U'$: $d_{C'}(r'(u')) = d_C(r(\psi'u')) = d_C(\psi'u') = d_{C'}(\psi\psi'u') = d_{C'}(u')$, the middle step being condition (1) for r.

The derivative of r'. On level 0, $D\psi'(\psi(x)) \circ D\psi(x) = 1$ for $x \in U_1$ and $D\psi(\psi'(y)) \circ D\psi'(y) = 1$ for $y \in U'_1$, by the chain rule for $\psi' \circ \psi = \mathrm{id}$ and $\psi \circ \psi' = \mathrm{id}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_inv_deriv` `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_deriv_inv` `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`. If $y \in U'_1$ and $\psi'(y) \in O$, then with $x = \psi'_1(y)$,

$$Dr'(y) = D\psi(x) \circ Dr(x) \circ D\psi'(y)$$

Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport. The chain rule is applied to $\psi \circ (r \circ \psi')$ and then to $r \circ \psi'$ Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain; the outer derivative of ψ is taken at the level-1 point of $r \circ \psi'$ over y, and this point is x because r fixes $\psi'(y) \in O$. Without $\psi'(y) \in O$ the formula would involve Dr at $\psi'(y)$ and $D\psi$ at the different point $r(\psi'(y))$.

Sc-isomorphisms carry sc-splittings. Let $D : E \to E'$ be an sc-operator with an sc-operator inverse D'. Its level maps are topological linear isomorphisms $E_m \cong E'_m$ Chapter2/TameInvariance · scIsoLevel (inverse to each other by the injectivity of the inclusions into level 0); they carry $A_m = A \cap E_m$ onto $(DA)_m$ Chapter2/TameInvariance · levelSub_map_eq and smooth points to smooth points Chapter2/TameInvariance · mem_smoothLvl_of_isScOperator. Hence an sc-subspace A (closed, with the smooth points of A_m dense in A_m for every m) is carried to the sc-subspace DA: $DA = (D')^{-1}(A)$ is closed, and the image of the closure of the smooth points of A_m lies in the closure of the image Chapter2/TameInvariance · IsScSubspace.map. An sc-splitting $E = F \oplus A$ (sc-subspaces, complementary on every level) is carried to the sc-splitting $E' = DF \oplus DA$, since an isomorphism of each level carries complements to complements Chapter2/TameInvariance · IsScSplitting.map.

Condition (2) Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameWith_transport. Let T, T' be presentations of C, C', r tame for T, and y a smooth point of $\psi(O)$ on level 1. Then $x = \psi'_1(y)$ lies on level 1 of O (if $y = \psi(z)$ with $z \in O$, then $\psi'(y) = z$) and is smooth, because sc-smooth maps carry smooth points to smooth points. Tameness of r at x gives an sc-splitting $E = T_xO \oplus A$ with $A \subseteq E_x$. Because x and y are smooth points, $D\psi(x)$ and $D\psi'(y)$ are sc-operators (the remark after Proposition 1.2.1 (preprint 1.16)) Chapter1/Properties · IsSc1With.scOperator_deriv — at a point that is only on level 1 they need not be — and they are inverse to each other. From the formula for $Dr'(y)$ and $D\psi'(y)D\psi(x) = 1$: $T_y\psi(O) = \operatorname{range} Dr'(y) = D\psi(x)(\operatorname{range} Dr(x)) = D\psi(x)(T_xO)$. So $E' = T_y\psi(O) \oplus D\psi(x)A$ is an sc-splitting. Finally $D\psi(x)A \subseteq E'_y$: the derivative of an sc¹-map between relatively open subsets of partial quadrants maps the minimal subspace E_x into $E'_{\psi(x)}$ (the one-sided sign argument, and density) Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace, which needs $\psi(U) \subseteq C'$, and $\psi(x) = y$. Since T' was arbitrary and the presentation T of C is available, the transported retraction of a tame retraction is tame Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport, and **the image $(\psi(O), C', E')$ of a tame retract under an sc-diffeomorphism is a tame retract**, with retraction $\psi \circ r \circ \psi^{-1}$ Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image.

Open subsets. For an open $W \subseteq E_0$, the restriction of r to $U \cap r^{-1}(W)$ is an sc-smooth retraction onto $O \cap W$ (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/Retracts · ScRetraction.restrict. It is tame if r is Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame: condition (1) is inherited pointwise; for (2), at a smooth point x of $O \cap W$ on level 1 the derivative of the restricted retraction is $Dr(x)$, because the two maps agree on the smaller relatively open domain, which contains x since $r(x) = x \in W$, and the derivative at a level-1 point depends only on the map near the point Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen; so the complement A of r at x serves. Hence $O \cap W$ is a tame retract whenever O is Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen. For a chart φ and an open s inside its source, $\varphi(s) = O \cap W$ for an open W Chapter2/MPolyfold · MChart.exists_isOpen_image_eq, so the model of the restricted chart is a tame retract when that of φ is Chapter3/SolutionGermRecentre · MChart.isTameRetract_image.

These are used in §III.34 (the chart $\psi \circ \varphi$ of Theorem 3.6.8 (preprint 3.54) has the tame model $\psi(O)$) and in §§III.36–III.37 (the shrunk data of Definition 3.2.1 (preprint 3.25) keep tame models). The text's sentence "$X' = \psi(X)$ is tame" in Proposition 3.2.2 (preprint 3.26) is not needed there (§III.24): the tameness Definition 3.2.1 (preprint 3.25) asks for is that of the chart models, which the transport of charts leaves unchanged.

III.28 Corollary 3.4.3 (preprint 3.39): composition with a given conjugation

Corollary 3.4.3 (preprint 3.39) (text): if the sc-germ g is equivalent, by a strong bundle isomorphism Φ, to a basic germ and s is an sc$^+$-germ, there is a strong bundle map Ψ such that $((\Psi\circ\Phi)_*(g+s))^1$ is a basic germ. The text's proof: $\Phi_*(g)=h$ is basic, $t=\Phi_*(s)$ is an sc$^+$-germ with $t(0)=0$, $\Phi_*(g+s)=h+t$, and Theorem 3.4.2 (preprint 3.38) gives Ψ with $(\Psi_*(h+t))^1$ basic; take $\Psi\circ\Phi$. Four things are silent: that Theorem 3.4.2's (preprint 3.38) coordinate change is a strong bundle *isomorphism*; that the index raise of a strong bundle isomorphism is one between the shifted structures; that strong bundle isomorphisms and push-forwards compose; and that $\Phi_*(g+s)=h+t$ holds only where the conjugation identity holds. The formalization proves all four and states the corollary with the composite explicit.

Linear strong bundle isomorphisms. Two sc-operators $S : E \to E'$, $T : F \to F'$ give the sc-operator $S\oplus T$ `Chapter3/BasicClassComposition · ScOperator.prodMap`, and if $S(U)\subseteq U'$ the map $(u,k)\mapsto(Su,Tk)$ is a strong bundle map $U \lhd F \to U' \lhd F'$ (Definition 2.6.1 (preprint 2.47) as formalized `Chapter2/Tame · IsStrongBundleMap`): its base component does not depend on k, it is linear in k, it preserves the double filtration because sc-operators preserve the levels, it is the sc-operator $S\oplus T$ on $(U\lhd F)[0]$, and on $(U\lhd F)[1]$ it is $S\oplus T^1$ with T^1 the shifted operator `Chapter3/BasicClassComposition · ScOperator.isStrongBundleMap_prodMap`. With inverse sc-operators S' (inverse to S on U, U') and T' it is a strong bundle isomorphism `Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap`.

Theorem 3.4.2's (preprint 3.38) coordinate change as an isomorphism. In §III.18 the change of coordinates of Theorem 3.4.2 (preprint 3.38) is the pair of linear sc-isomorphisms `$□ = $ split`, `$□ = $ fiber` (inverses `glue`, `fiberInv`) on the index-raised spaces `Chapter3/BasicClass · StabilityData.split` `Chapter3/BasicClass · StabilityData.fiber`. Put $\Lambda((a,w),(q,w'))=(\psi(a,w),\Psi(q,w'))$ `Chapter3/BasicClassComposition · StabilityData.linIso`. For every U relatively open in the shifted quadrant $(\widetilde{C}\oplus W)^1$, $\psi(U)=\mathrm{glue}^{-1}(U)$ is relatively open in $(\widetilde{C}\oplus C)\oplus X^1$ (`glue` is continuous and preserves the quadrant coordinates) `Chapter3/BasicClassComposition · StabilityData.isRelOpen_glue_preimage`, and $\Lambda : U\lhd(\mathbb{R}^N\oplus W)^1 \to \psi(U)\lhd((\mathbb{R}^N\oplus Z)\oplus X^1)$ is a strong bundle isomorphism by the preceding paragraph `Chapter3/BasicClassComposition · StabilityData.linIso_isStrongBundleIso`. Before the index raise, the same construction with the level-0 operators `split₀`, `fiber₀` gives Λ_0 `Chapter3/BasicClassComposition · StabilityData.linIso₀`, a strong bundle isomorphism $U\lhd(\mathbb{R}^N\oplus W)\to\psi(U)\lhd((\mathbb{R}^N\oplus Z)\oplus X)$ — the text's "strong bundle map covering ψ" `Chapter3/BasicClassComposition · StabilityData.linIso₀_isStrongBundleIso` — and its index raise is Λ, because `split` and `fiber` are `split₀` and `fiber₀` on the levels $m+1$ `Chapter3/BasicClassComposition · StabilityData.strongShift_linIso₀`.

The index raise. For $\Psi : E_0\times F_0 \to E_0'\times F_0'$ let Ψ^1 be its level-1 map on $E_1\times F_1$ `Chapter3/BasicClassComposition · strongShift`. For a strong bundle map it exists on $U_1\oplus F_1$ and satisfies $\iota\Psi^1(p)=\Psi(\iota p)$, because Ψ maps $U_1\oplus F_1$ into $U_1'\oplus F_1'$ (the filtration with $k=1\le m+1$) `Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift`; its base map is the level-1 map of the base map of Ψ `Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_base_strongShift`. **Ψ^1 is a strong bundle map** $U^1\lhd F^1\to U'^1\lhd F'^1$ of the shifted structures `Chapter3/BasicClassComposition · IsStrongBundleMap.shift`: independence of the base component and fibre linearity are read through the injective inclusions; the double filtration $(U^1\lhd F^1)_{m,k}=U_{m+1}\oplus F_{k+1}$ is preserved because $k+1\le(m+1)+1$ whenever $k\le m+1$, so the filtration property of Ψ applies; $(U^1\lhd F^1)[0]$ is the shift of $(U\lhd F)[0]$, so Ψ^1 is sc-smooth there by Proposition 1.2.2 (preprint 1.17) `Chapter1/Levels · IsScSmooth.shift`; and $(U^1\lhd F^1)[1]$, with levels $E_{m+1}\oplus F_{m+2}$, is the shift of $(U\lhd F)[1]$, on which $(\Psi^1)[1]$ agrees with $(\Psi[1])^1$ (the lift of the fibre component to F_2' is the same element, by the injectivity of $F_2'\to F_1'\to F_0'$), so Proposition 1.2.2 (preprint 1.17) applies again. For a strong bundle isomorphism with inverse Ψ', Ψ'^1 is the inverse of Ψ^1, again checked after the inclusion into level 0 `Chapter3/BasicClassComposition · IsStrongBundleIso.shift`. The text uses this whenever it writes $((\Psi\circ\Phi)_*(\cdot))^1$.

Composition. The base map of $\Lambda\circ\Phi$ is the composite of the base maps on U `Chapter3/`

BasicClassComposition · IsStrongBundleMap.base_comp, and $(\Lambda \circ \Phi)[1] = \Lambda[1] \circ \Phi[1]$ on $U \oplus F_1$ Chapter3/BasicClassComposition · IsStrongBundleMap.shiftFiber_comp; so the composite is a strong bundle map — fibre linearity because over u it is the fibre map of Λ over $\varphi(u)$ after that of Φ over u, sc-smoothness on $[0]$ and $[1]$ by the chain rule Chapter3/BasicClassComposition · IsStrongBundleMap.comp — and the composite of strong bundle isomorphisms is one, with the inverses composed in the opposite order Chapter3/BasicClassComposition · IsStrongBundleIso.comp. Push-forwards compose: $(\Lambda \circ \Phi)_* g = \Lambda_*(\Phi_* g)$ on the target of Λ, since the base of $(\Lambda \circ \Phi)^{-1}$ is $\varphi^{-1} \circ \lambda^{-1}$ and $\Phi(\varphi^{-1}(w), \cdot)$ lies over w Chapter3/BasicClassComposition · IsStrongBundleIso.pushforward_comp Chapter3/GermPushforward · pushforward. The index raise commutes with composition, $(\Lambda \circ \Phi)^1 = \Lambda^1 \circ \Phi^1$ on $U_1 \oplus F_1$ Chapter3/BasicClassComposition · IsStrongBundleMap.strongShift_comp.

Corollary 3.4.3 (preprint 3.39) as formalized Chapter3/BasicClassComposition · corollary_3_39. Let U be relatively open in a partial quadrant C of E with $0 \in U$, $g : U \to F$ sc-smooth (the text's "sc-germ"; sc-smoothness is what makes $(g+s)^1$ take values on level 1), d conjugation data of g (§III.30: a strong bundle isomorphism $\Phi : U \lhd F \to U' \lhd (\mathbb{R}^N \oplus W)$ covering φ with $\varphi(0) = 0$ and a basic germ h on a germ G inside U', with $(\Phi(u, g(u)))_2 = h(\varphi(u))$ for $u \in U$ near 0; the text's "equivalent by a strong bundle isomorphism to the basic germ $\Phi_* g$"), and $s : U \to F^1$ sc-smooth into F^1 with $s(0) = 0$ (the text's sc⁺-germ $s : O(C, 0) \to (F, 0)$). The proof, in the order of the Lean:

1. *The sc⁺-germ $t = \Phi[1]_* s$.* Put $t(u') = (\Phi[1](\varphi^{-1}u', s(\varphi^{-1}u')))_2$ Chapter3/BasicClassComposition · ConjugationData.pushPlus. It is sc-smooth from U' into $(\mathbb{R}^N \oplus W)^1$: the pairing of φ^{-1} with $s \circ \varphi^{-1}$ maps U' into $U \oplus F^1$, and $\Phi[1]$ is sc-smooth there by Definition 2.6.1 (preprint 2.47); restricted to the germ $G \subseteq U'$ it is an sc-smooth germ. And $t(0) = 0$, from $\varphi^{-1}(0) = 0$, $s(0) = 0$ and $\Phi(0,0)_2 = 0$ (fibre linearity).
2. *Absorbing the $\mathbb{R}^N$-component.* As in Theorem 3.6.8 (preprint 3.54) (§III.23, step 3), $h' = h + (t_{\mathbb{R}^N}, 0)$ Chapter3/BasicClassComposition · ConjugationData.absorb Chapter3/GlobalIFT · ScBanach.qPartOp is a basic germ with the same contraction map: it is an sc-smooth germ (pairing and the sum operator), $h'(0) = h(0) + 0 = 0$, and its W-component is that of h. The W-component t_W is an sc-smooth germ into W^1 vanishing at 0, so the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) exist for $s = t_W$ on the germ G of h Chapter3/BasicClass · StabilityData.exists_of_scPlus, and $\mathcal{S}.\mathrm{fsum}\, h' = h + \iota t$ Chapter3/BasicClass · StabilityData.fsum.
3. *$\Phi_*(g+s) = h + t$ near 0.* For $u \in U$ at which the conjugation identity holds, $\Phi_*(g + \iota s)(\varphi(u)) = (\Phi(u, g(u) + \iota s(u)))_2$ because $\varphi^{-1}\varphi(u) = u$; by fibre linearity this is $(\Phi(u, g(u)))_2 + (\Phi(u, \iota s(u)))_2 = h(\varphi(u)) + \iota t(\varphi(u))$, where $\iota(\Phi[1](u, s(u)))_2 = (\Phi(u, \iota s(u)))_2$ Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd. So $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ at the points $\varphi(u)$, $u \in U$ near 0 — not everywhere, as the text's "clearly" suggests.
4. *The composite.* $\Gamma = \Lambda \circ \Phi^1$ Chapter3/BasicClassComposition · ConjugationData.compIso, with inverse $\Phi'^1 \circ \Lambda^{-1}$ Chapter3/BasicClassComposition · ConjugationData.compIsoInv, is a strong bundle isomorphism $U^1 \lhd F^1 \to \psi(U'^1) \lhd ((\mathbb{R}^N \oplus Z) \oplus X^1)$ Chapter3/BasicClassComposition · ConjugationData.compDomain: the index raise of Φ (relatively open U_1' in the shifted quadrant) followed by Λ, composed by the composition lemma. Its base map is $u \mapsto \psi(\varphi_1(u))$, which vanishes at 0; $\Gamma_* = \Lambda_* \circ \Phi^1_*$ (composition of push-forwards); and the germ of $\mathcal{S}.\mathrm{pushMap}\, h'$ lies in the target, because `glue` maps the level-1 part of G into U_1'.
5. *The basic germ.* $\mathcal{S}.\mathrm{pushMap}\, h' = \mathrm{fiber} \circ (\mathcal{S}.\mathrm{fsum}\, h')^1 \circ \mathrm{glue}$ Chapter3/BasicClass · StabilityData.pushMap is a basic germ by Theorem 3.4.2 (preprint 3.38) (§III.18) Chapter3/BasicClass · theorem_3_38.
6. *The conjugation identity on level 1.* For $u \in U^1$ near 0: $(\Gamma(u, (g+s)^1(u)))_2 = \mathcal{S}.\mathrm{pushMap}\, h'(\gamma(u))$. Since $\gamma(u) = \mathrm{split}(\varphi_1(u))$ and $\mathrm{glue} \circ \mathrm{split} = \mathrm{id}$, both sides are `fiber` applied to elements of $(\mathbb{R}^N \oplus W)_1$, which are compared after the injective inclusion into level 0: $\iota(\Phi^1(u, (g+s)^1(u)))_2 = (\Phi(\iota u, (g+s)(\iota u)))_2$ Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift_snd, which is $\mathcal{S}.\mathrm{fsum}\, h'(\varphi(\iota u))$ by step 3, and this is ι of $(\mathcal{S}.\mathrm{fsum}\, h')^1(\varphi_1(u))$. "Near 0" means: ιu lies in the neighbourhood where the conjugation identity holds (continuity of $\iota : E_1 \to E_0$), and $\varphi_1(u)$ lies in the level-1 part of the germ G, a relatively open condition at 0 by the continuity of φ on level 1.

The conclusion of `corollary_3_39` collects these: $\mathcal{S}$ with $\mathcal{S}.G = G$ and $\mathcal{S}.s = t_W$, $t(0) = 0$, h' basic,

$\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ near 0, Γ a strong bundle isomorphism onto a relatively open subset of $(\widetilde{C}\oplus C)\oplus X^1$ with $\gamma(0)=0$ and $\Gamma_* = \Lambda_* \circ \Phi_*^1$, the basic germ $\mathcal{S}.\mathrm{pushMap}\, h'$ inside the target, and the identity of step 6. In the form of Definition 3.1.11 (preprint 3.11): $(g+s)^1 : U^1 \to F^1$ **is conjugated to a basic germ** on the shifted structures E^1, F^1 `Chapter3/BasicClassComposition · corollary_3_39_isConjugate`, with parameter space $\mathbb{R}^n \oplus C$, target $\mathbb{R}^N \oplus Z$ and fibre X^1 (the text's $\tau : Z \to C$ is not applied, as in §III.18). The text's form "$\Gamma = \Psi \circ \Phi$, then raise the index" is the same map: $\Lambda_0 \circ \Phi$ is a strong bundle isomorphism on level 0 and $(\Lambda_0 \circ \Phi)^1 = \Lambda_0^1 \circ \Phi^1 = \Lambda \circ \Phi^1$ on $U_1 \oplus F_1$ `Chapter3/BasicClassComposition · ConjugationData.compIso_eq_strongShift`. No hypothesis is added beyond the two readings ("sc-germ" as sc-smooth on U; "equivalent by a strong bundle isomorphism" as conjugation data). The first reading is the whole-chart one of §III.6, narrower than the text's germs; no theorem of the germ reading uses Corollary 3.4.3 (preprint 3.39) (the facts at the nearby zeros come from the chart at x_0 through the linear coordinate change of Theorem 3.4.2 (preprint 3.38), §III.43).

III.29 Good complements with a projection into the quadrant; Theorems 3.5.6, 3.5.11 and 3.6.8 (3) (preprint 3.43, 3.47 and 3.54) with a closed projecting complement

Two steps of the text depend on the choice of a good complement without saying so. In the proof of Theorem 3.2.3 (preprint 3.27) the retraction $t(v+w) = v + \delta(v)$ is defined on $\Sigma = \{v + y \in C : v \in V,\ y \in Y\}$, which the text calls relatively open in C; §III.24 shows that this fails for the good complement $Y = \{x + y - z = 0\}$ of $N = \mathbb{R}(1,1,1)$ in $C = [0,\infty)^3$, and makes the remedy a hypothesis: condition (b) of the regularity conditions, $\pi(O) \subseteq C$ for the projection π onto N along Y `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`. In the proof of Theorem 3.5.6 (preprint 3.43) (Lemma 3.5.10 (preprint 3.46)) a good complement Z of $\widetilde{K} = \ker DH(0)$ is produced by projecting a good complement of K'. This section shows that for solution sets the complement can be chosen once and for all, at the bottom of the chain Theorem 3.8.22 (preprint 3.78) □ Theorem 3.5.6 (preprint 3.43) □ Theorem 3.5.11 (preprint Corollary 3.47) □ Theorem 3.6.8 (3) (preprint 3.54), so that its projection maps the partial quadrant into itself; condition (b) then holds for the chart of §III.34 without any hypothesis (Findings, Chapter 3, items 130–133). Along the way the closedness of the complement, which Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25) need (items 49, 81), is recorded in the statements (item 93).

The projection property. For subspaces N, Y of a vector space and a set C, *N projects C into itself along Y* if $z \in C$, $k \in N$ and $z - k \in Y$ imply $k \in C$ `Chapter3/GoodComplementProjection · ProjectsInto`. The definition avoids naming the projection; when $E = N \oplus Y$ it says exactly that the projection π onto N along Y maps C into C `Chapter3/GoodComplementProjection · ProjectsInto.projectionOnto_mem` `Chapter3/GoodComplementProjection · projectsInto_of_projectionOnto_mem` ($z - \pi z \in Y$, and conversely $z - k \in Y$, $k \in N$ force $\pi z = k$). It is carried by a linear isomorphism S with $x \in C \iff Sx \in C'$: given $z' \in C'$, $k \in N$ and $z' - Sk = Sy$ with $y \in Y$, the point $S^{-1}z'$ lies in C and $S^{-1}z' - k = y$, so $k \in C$ and $Sk \in C'$ `Chapter3/GoodComplementProjection · ProjectsInto.map`, in particular for continuous linear isomorphisms `Chapter3/GoodComplementProjection · ProjectsInto.mapL` — the analogue of Lemma 3.5.9 (preprint 3.45) (§III.14).

The finite-dimensional lemma. *If the finite-dimensional subspace N is in good position to the partial quadrant C of the sc-Banach space E, then N has a closed good complement Y such that the projection onto N along Y maps C into $N \cap C$* `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`. The text has no such statement; Definition 3.1.24 (preprint 3.21) and Proposition 3.1.25 (preprint 3.22) do not need it. The proof works in the coordinates of §III.14: after a presentation, $C = \{\ell_i \geq 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$, the cone $Q = N \cap C$ is $\{x \in N : \ell_i(x) \geq 0\}$, Σ is the set of indices vanishing at some nonzero point of Q, and for a set S of indices $Q_S = (\ell_i)_{i\in S} : N \to \mathbb{R}^S$ `Chapter3/GoodComplementProjection · coordOn` (for $S = \Sigma$ it is the map Q_Σ of Proposition 3.1.25 (preprint 3.22) `Chapter3/GoodComplementProjection · sigmaCoord_eq_coordOn`).

Step 1: the choice of coordinates `Chapter3/GoodComplementProjection · exists_coordSet`. Wanted is a set S of indices with (1) Q_S onto, (2) $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in S\}$, (3) every ℓ_i, $i \notin S$, positive on $Q \setminus 0$. Property (3) holds for $S = \Sigma$ by the definition of Σ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`, and (1) for $S = \Sigma$ is the consequence of Lemma 3.8.8 (preprint 3.65) proved in §III.14 `Chapter3/GoodPosition · sigmaCoord_surjective`. Three cases, those of the normal form. If $Q = N$, take $S = \Sigma$; (2) is trivial. If $\Sigma \neq \emptyset$, take $S = \Sigma$; (2) is the segment argument of Proposition 3.1.25 (preprint 3.22) `Chapter3/GoodPosition · mem_quadCone_of_vanIdx`. In the remaining case, $\Sigma = \emptyset$ and $Q \neq N$, the choice $S = \Sigma$ fails, and here the sketch of Finding 83, which takes $S = \Sigma$ throughout, misses a case (item 131). With $S = \emptyset$, (2) would say $Q = N$; and the formula of Step 4 below would reduce to an arbitrary projection onto N, which need not map C into C: for $C = [0, \infty)^2$ and $N = \mathbb{R}(1, 1)$ every nonzero point of Q has both coordinates positive, so $\Sigma = \emptyset$; the line $\mathbb{R}(1, 2)$ is a good complement of N (for $|s| \leq \varepsilon|t|$ the coordinates of $t(1, 1) + s(1, 2)$ have the sign of t), and the projection along it maps $(0, 1) = -(1, 1) + (1, 2)$ to $-(1, 1) \notin C$. The formal proof keeps one coordinate. Some $x_1 \in N$ lies outside C, so there is at least one coordinate, j_0 say, and $S = \{j_0\}$. An interior point y of Q is nonzero — if 0 were interior, Q would contain a ball of N around 0 and, being closed under positive scaling, all of N `Chapter3/GoodComplementProjection · mem_quadCone_of_zero_mem_interior Chapter3/GoodPosition · smul_mem_quadCone` — so $\ell_{j_0}(y) > 0$ by (3), and $Q_{\{j_0\}}$ is onto: $k \mapsto (k/\ell_{j_0}(y))\, y$ is a preimage. Property (2) for $S = \{j_0\}$ is the next step.

Step 2: the segment argument for any S `Chapter3/GoodComplementProjection · mem_quadCone_of_coordSet`. Suppose (3) holds for S, some $i_0 \in S$ is positive at a point $y \in Q$, and $x \in N$ has $\ell_i(x) \geq 0$ for $i \in S$. The set $T = \{s \geq 0 : x + sy \in Q\}$ is nonempty (for $i \in S$, $\ell_i(x) + s\ell_i(y) \geq 0$ for all $s \geq 0$; for $i \notin S$, $\ell_i(y) > 0$, so large s work), closed and bounded below; let $s^* = \min T$. If $s^* > 0$, the point $z = x + s^* y$ is nonzero, since $\ell_{i_0}(z) \geq s^* \ell_{i_0}(y) > 0$; so by (3) all ℓ_i, $i \notin S$, are positive at z and, by continuity, at $x + sy$ for s near s^*; the coordinates in S stay ≥ 0 for all $s \geq 0$; so a point $s_1 = \max(0, s^* - \delta/2) < s^*$ lies in T, contradicting minimality. Hence $s^* = 0$ and $x \in Q$. This is the argument of Proposition 3.1.25 (preprint 3.22), Case C (§III.14), with the positivity outside S as the only input; with $S = \{j_0\}$ and y interior it gives (2) in the third case of Step 1.

Step 3: a lower bound `Chapter3/GoodComplementProjection · exists_coord_ge`. A coordinate positive on $Q \setminus 0$ is bounded below by $c|x|$ on Q, $c > 0$: its minimum on the compact set $Q \cap \{|x| = 1\}$ is positive, and Q is a cone.

Step 4: the projection and the estimate `Chapter3/GoodComplementProjection · exists_proj_quadCone`. Take S from Step 1, a continuous projection $p_N : E_0 \to N$ (Hahn–Banach, N finite-dimensional), a linear right inverse g of Q_S, and

$$\pi(z) = p_N(z) + g((\ell_i(z - p_N z))_{i \in S}).$$

Then π is continuous, $\pi = \mathrm{id}$ on N (the bracket vanishes), and $\ell_i(\pi z) = \ell_i(p_N z) + \ell_i(z - p_N z) = \ell_i(z)$ for $i \in S$, so for $z \in C$ the coordinates of πz in S are ≥ 0 and $\pi z \in Q$ by (2): $\pi(C) \subseteq N \cap C$. The kernel of π is a complement of N, and it is a good complement. For $p \in \ker \pi$ the coordinates in S vanish ($\ell_i(p) = \ell_i(\pi p) = 0$). Let c_i be the constants of Step 3 for $i \notin S$ and $\varepsilon' = \min_i c_i/(\|\ell_i\| + 1)$ `Chapter3/GoodComplementProjection · exists_pos_le_of_finite`. Let $x \in N$, $p \in \ker \pi$, $|p| \leq \varepsilon'|x|$. If $x \in C$: for $i \in S$, $\ell_i(x + p) = \ell_i(x) \geq 0$; for $i \notin S$, $\ell_i(x + p) \geq c_i|x| - \|\ell_i\|\,|p| \geq c_i|x| - \|\ell_i\|\varepsilon'|x| \geq 0$. If $x + p \in C$: the coordinates of x in S are those of $x + p$, hence ≥ 0, and $x \in Q$ by (2). The reverse implication needs no estimate.

Step 5: partial quadrants. For a partial quadrant presented by an sc-isomorphism T the functionals are the coordinate functions of T, continuous `Chapter3/GoodComplementProjection · continuous_coordFun` and jointly onto `Chapter3/GoodPosition · coordFun_surj`; good position supplies a good complement P with its constant and the interior point. Step 4 then gives a continuous projection $\pi : E_0 \to N$ with $\pi(C) \subseteq C$ whose kernel is a good complement `Chapter3/GoodComplementProjection · exists_proj_goodComplement` ($\ker \pi$ is a topological complement of N, as the kernel of a continuous projection). With $Y = \ker \pi$, closed as the kernel of a continuous map, the projection property is immediate: $z \in C$, $k \in N$, $z - k \in Y$ give $k = \pi k = \pi z \in C$ `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`.

Theorem 3.5.6 (preprint 3.43) with a prescribed good complement. In the text's proof the complement Z of $\widetilde{K} = \ker DH(0)$ in $\mathbb{R}^{n'}$ comes from a good complement of $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$ through Lemma 3.5.10 (preprint 3.46) (§III.20). Nothing later uses that origin. *For every good complement Z of $\widetilde{K}$ (constant ε), $Z \oplus W$ is a good complement of K' with respect to $\widetilde{C} \oplus W$, with constant $\varepsilon/(\|L\| + 1)$* `Chapter3/GoodComplementProjection · IsGoodComplement.prod_top_graph`. The splitting: $K' \cap (Z \oplus W) = 0$ because $\widetilde{K} \cap Z = 0$, and $(a, w) = (\alpha, L\alpha) + (z, w - L\alpha)$ for $a = \alpha + z$. The estimate: with the maximum norm, $|(\alpha, L\alpha)| \le (\|L\| + 1)|\alpha|$, so $|(z, w)| \le \varepsilon/(\|L\| + 1) \cdot |(\alpha, L\alpha)|$ gives $|z| \le \varepsilon|\alpha|$, and $(\alpha, L\alpha) + (z, w) \in \widetilde{C} \oplus W$ iff $\alpha + z \in \widetilde{C}$ iff $\alpha \in \widetilde{C}$ iff $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ `Chapter3/LocalSolvability · graphMap_mem_prod_iff`. This is the second half of the proof of Lemma 3.5.10 (preprint 3.46) for an arbitrary Z; Lemma 3.5.10 (preprint 3.46) itself is needed only for the good position of $\widetilde{K}$. The projection property passes the same way: if $z \in \widetilde{C} \oplus W$ and $z - (\alpha, L\alpha) \in Z \oplus W$, then $z_1 - \alpha \in Z$, so $\alpha \in \widetilde{C}$ when $(\widetilde{K}, Z)$ has the property, and $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ `Chapter3/GoodComplementProjection · ProjectsInto.prod_top_graph`; the projection along $Z \oplus W$ is $(a, w) \mapsto (\pi_Z a, L\pi_Z a)$, and $\widetilde{C} \oplus W$ constrains only the first component.

So the data of the proof of Theorem 3.5.6 (preprint 3.43) (§III.20: the derivative data, the graph description $K' = \operatorname{graph}(L|_{\widetilde{K}})$ of Lemma 3.5.8 (preprint 3.44), a complement Z, and the solution σ of $H(k + \sigma(k)) = 0$ from Theorem 3.8.22 (preprint 3.78) with its uniqueness constants `Chapter3/LocalSolvability · ShiftedBasicData.Sol`) exist with any prescribed good complement Z of $\widetilde{K}$ `Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_of_goodComplement`: K' is the graph `Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq`, so $\widetilde{K}$ is in good position (Lemma 3.5.10 (preprint 3.46)) `Chapter3/LocalSolvability · lemma_3_46_isInGoodPosition`, $Z \oplus W$ is a good complement of K' by the lemma above, and Theorem 3.8.22 (preprint 3.78) is applied to H with Z `Chapter3/QuadrantIFT · theorem_3_78`. Choosing Z by the finite-dimensional lemma in $\mathbb{R}^{n'}$ (constant sc-structure, partial quadrant $\widetilde{C}$) gives data whose Z projects $\widetilde{C}$ into $\widetilde{K} \cap \widetilde{C}$ `Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_proj`.

Theorem 3.5.6 (preprint 3.43), closed and projecting. With the hypotheses of Theorem 3.5.6 (preprint 3.43) (§III.20: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position, the regularity of Lemma 3.5.5 (preprint 3.42) as a hypothesis), the conclusions of the level-wise form `Chapter3/LocalSolvability · theorem_3_43_sc_smooth` hold with a good complement $Y = Z \oplus W$ that is **closed** `Chapter3/LocalSolvabilityClosed · theorem_3_43_closed` — Z is finite-dimensional, hence closed, and $Z \times W_0$ is closed — and, with the data of the last paragraph, that has the **projection property** for $\widetilde{C} \oplus W$ `Chapter3/GoodComplementProjection · theorem_3_43_proj` (rewrite K' as the graph and apply the passage of the property to $Z \oplus W$). The other conclusions (V, τ with values in Y and W-components on level 1, C^1, $\tau(0) = 0$, $D\tau(0) = 0$, the graph of zeros, the C^k lifts τ_k, the neighbourhood U'' in which every zero lies on the graph) are those of §III.20, read off the same data `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level` `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ`.

Theorem 3.5.11 (preprint Corollary 3.47), closed and projecting `Chapter3/LocalSolvabilityClosed · theorem_3_47_proj`. For a basic germ f, the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) with $D(f + \iota s)(0)$ onto and kernel K in good position to $\widetilde{C} \oplus W$: the conclusions of the level-wise Theorem 3.5.11 (preprint Corollary 3.47) `Chapter3/LocalSolvability · theorem_3_47_sc_smooth` hold with Y closed and K projecting $\widetilde{C} \oplus W$ into itself along Y. The proof is that of §III.20 with the two new properties carried along. Let S be the linear coordinate change $\psi(a, w) = ((a, (1 - P_1)w), P_1 w)$ of Theorem 3.4.2 (preprint 3.38) on level 0, a continuous linear isomorphism with inverse the gluing map. The push-forward $g = \Phi_*(f + s)$, packaged as data of Theorem 3.5.6 (preprint 3.43) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`, has $Dg(0) = \Psi \circ D(f + \iota s)(0) \circ S^{-1}$ `Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀`, so $\ker Dg(0) = S(K)$, $Dg(0)$ is onto, and K is finite-dimensional (Lemma 3.5.8 (preprint 3.44) for g and the injectivity of S^{-1}). S carries $\widetilde{C} \oplus W$ onto $(\widetilde{C} \oplus C_s) \oplus X$, where C_s is the finite-dimensional space of the splitting of Theorem 3.4.2 (preprint 3.38) (§III.18) and $\widetilde{C} \oplus C_s$ is a partial quadrant of $\mathbb{R}^n \oplus C_s$ `Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ`; so $S(K)$ is in good position (Lemma 3.5.9 (preprint

3.45), §III.14). Theorem 3.5.6 (preprint 3.43), closed and projecting, for g, with Lemma 3.5.5 (preprint 3.42) for its regularity hypothesis `Chapter3/LocalSolvability · StabilityData.lemma_3_42`, gives Y', V', τ. Transported back: $Y = S^{-1}(Y')$ is a good complement of $K = S^{-1}(S(K))$ (Lemma 3.5.9 (preprint 3.45)) `Chapter3/GoodPosition · IsGoodComplement.map`, closed as the image of a closed set under a linear homeomorphism, and projecting, by the transport of the projection property `Chapter3/GoodComplementProjection · ProjectsInto.mapL`; $V = (S|_K)^{-1}(V')$ and $\sigma = S^{-1} \circ \tau \circ S|_K$. The W-component of $\sigma(y)$ is $c + w$ with $c \in C_s \subseteq W_\infty$ `Chapter3/BasicClass · StabilityData.Cs_le_smooth` and w the level-1 X-component of τ, so it lies on level 1; $D\sigma(0) = S^{-1} \circ D\tau(0) \circ S|_K = 0$, by the chain rule within V, which has unique differentiability at 0 as a relatively open subset of the convex cone $K \cap C$ with interior; the graph points are zeros because the linear fibre map of Theorem 3.4.2 (preprint 3.38) is injective; the lifts are ψ^{-1} on level k applied to τ_k; and $U'' := S^{-1}(U'')$. The form with only the closedness is a corollary `Chapter3/LocalSolvabilityClosed · theorem_3_47_closed`.

Theorem 3.6.8 (3) (preprint 3.54), closed and projecting `Chapter3/GlobalIFTG · theorem_3_54_graph_proj`. The graph description of §III.23 — for a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the chart φ centred at x_0, the conjugating sc-diffeomorphism ψ, the germ $h + t$ with surjective linearization and kernel K in good position, Y, V', σ, and the open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$ — is proved with Theorem 3.5.11 (preprint Corollary 3.47) closed and projecting in place of the plain Theorem 3.5.11 (preprint Corollary 3.47), and the statement now records five more facts. (i) The inverse ψ' of ψ, sc-smooth on U', with $\psi'\psi = \mathrm{id}$ on U and $\psi\psi' = \mathrm{id}$ on U' (the base maps of the conjugating strong bundle isomorphism and its inverse); §III.34 needs it to make $\psi \circ \varphi$ a chart. (ii) Y is closed, and (iii) K projects $C' = \widetilde{C} \oplus W$ into itself along Y, both from Theorem 3.5.11 (preprint Corollary 3.47) — the shrinking of V' in step 6 of §III.23 ($V' = V_0 \cap O_1$ with O_1 open, chosen so that $v + \sigma(v)$ stays in $\Omega = U'' \cap W'$) does not touch Y. (iv) K is a smooth subspace `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth` (proved in §III.34). (v) Every $v + \sigma(v)$, $v \in V'$, is a zero of $h + t$ in the germ, and for every $k \geq 1$ the C^k lift σ_k of Theorem 3.5.11 (1) (preprint Corollary 3.47) survives the shrinking: on $V_k \cap O_1$, relatively open in $K \cap C'$, containing 0 and contained in V', it is still C^k and still lifts σ. The theorem of §III.23, with the same statement except the projection property, is a corollary `Chapter3/GlobalIFTG · theorem_3_54_graph`.

Where the formal route departs from the text. The text never chooses the good complement; Definition 3.2.1 (preprint 3.25) lets it be any good complement, and for an arbitrary subset with the tangent germ property the repair of the claim "Σ is relatively open" would need a re-graphing over a better complement, which is not done ($\pi(O) \subseteq C$ remains a hypothesis of Theorem 3.2.3 (preprint 3.27), §III.24). For solution sets the complement is not given but produced, at Theorem 3.8.22 (preprint 3.78), and every link of the chain accepts a prescribed complement; so the repair is a choice made at the bottom of the chain, not a change in the proof of Theorem 3.2.3 (preprint 3.27) (item 130). The closedness of Y is part of the text's sc-complement and is recorded because Lemma 3.6.9 (preprint 3.55) and Definition 3.2.1 (preprint 3.25) use it at the nearby zeros (items 49, 81, 93).

III.30 Proposition 3.6.5 (preprint 3.52) with its index

Proposition 3.6.5 (preprint 3.52) (text): the linearization $Tf(x) : T_xX \to Y_x$ of an sc-Fredholm section at a zero is an sc-Fredholm operator; in the proof (and in Theorem 3.6.8 (preprint 3.54)) its index is the index $n - N$ of the basic germ $h : O(([0,\infty)^k \oplus \mathbb{R}^{n-k}) \oplus W, 0) \to (\mathbb{R}^N \oplus W, 0)$ to which $g - s$ is conjugated. The statements of §III.21 state the index as "$n - N$ for some natural numbers n, N", which every integer is. The reason is that the conjugation of Definition 3.1.11 (preprint 3.11) is an existential proposition `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`, so the dimensions of the basic germ cannot be named outside it. Here the witnesses become data.

Conjugation data `Chapter3/FredholmGermIndex · ConjugationData`: finite-dimensional normed spaces P_n (the text's $\mathbb{R}^n$) and Q (the text's $\mathbb{R}^N$), an sc-Banach space W, a partial quadrant $\widetilde{C} \subseteq P_n$, a relatively open $U' \subseteq \widetilde{C} \oplus W$, a strong bundle isomorphism $\Psi : U \lhd F \to U' \lhd (Q \oplus W)$ with inverse Ψ' and

$\psi(0) = 0$ for its base map, a germ G with $G_0 \subseteq U'$, a basic germ h on G with contraction map B, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0. A map is conjugated to a basic germ iff such data exist `Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty`.

The derivative of a conjugated map `Chapter3/FredholmGermIndexG · ConjugationData.scDeriv_isFredholm_index`. Let U be relatively open in the partial quadrant C, $0 \in U$, k an sc¹-germ with derivative family Dk on a germ G_k with $(G_k)_0 \subseteq U$, $k(0) = 0$, and d conjugation data of k. Then $Dk(0) : E_0 \to F_0$ is Fredholm of index $\dim P_n - \dim Q$; for k sc-smooth on U this is the whole-chart form `Chapter3/FredholmGermIndex · ConjugationData.scDeriv_isFredholm_index_wholeChart`. The proof is that of §III.21, step (a), with the dimensions kept:

(a) $Dh(0)$ is Fredholm of index $\dim P_n - \dim Q$ (Proposition 3.1.10 (preprint 3.10) for the basic germ on level 0) `Chapter3/FredholmGermLinearization · proposition_3_10_basic_level_zero`.

(b) $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, with $\Gamma(0) = (\Psi(0,\cdot))_2$ the fibre operator of Ψ at 0 `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM`. The left side is the derivative at 0 of $u \mapsto (\Psi(u, k(u)))_2$, because $k(0) = 0$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp`. For the right side, let N_1 be an open neighbourhood of 0 on which the conjugation identity holds; the germ $\{U \cap N_1\}$ has a subgerm G_1 mapped by ψ into G level by level `Chapter3/GermFilling · ScNbhdGerm.preimage`; on G_1 the chain rule for germs gives the derivative family $Dh(\psi_1(x))D\psi(x)$ of $h \circ \psi$ `Chapter3/Germ · IsSc1GermWith.comp`; and $u \mapsto (\Psi(u, k(u)))_2$, an sc¹-germ on G_k by the chain rule with the whole-chart outer map (§III.48), agrees with $h \circ \psi$ on $G_1 \cap G_{k'}$ so the two derivative families agree at 0 (uniqueness of the derivative of a germ at points of U_1) `Chapter3/Germ · IsSc1Germ.scDerivGerm_eq0n`, with $\psi_1(0) = 0$.

(c) $D\psi(0)$ and $D\psi'(0)$ are inverse to each other (chain rule for $\psi' \circ \psi = \mathrm{id}$ on U and $\psi \circ \psi' = \mathrm{id}$ on U') `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`, and so are $\Gamma(0)$ and the fibre operator $\Gamma'(0)$ of Ψ' at 0: $\Psi'(\Psi(0, y)) = (0, y)$ with $\Psi(0, y) = (0, \Gamma(0)y)$ because $\psi(0) = 0$, and symmetrically.

(d) Hence $Dk(0) = \Gamma(0)^{-1} \circ Dh(0) \circ D\psi(0)$, a Fredholm operator composed with isomorphisms on both sides, of the same index `Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv`.

Consequently **the index $n - N$ does not depend on the conjugation**: two conjugation data d, d' of the same k give $\dim d.P_n - \dim d.Q = \mathrm{ind}\, Dk(0) = \dim d'.P_n - \dim d'.Q$ `Chapter3/FredholmGermIndexG · ConjugationData.index_eq`. This is the text's remark after Theorem 3.4.2 (preprint 3.38) that "the Fredholm index is invariant under strong bundle isomorphisms", for the conjugations of one germ.

Proposition 3.6.5 (preprint 3.52) in the chart, with the index `Chapter3/FredholmGermIndexG · proposition_3_52_local_index`. Hypotheses: U relatively open in C, R a strong bundle retraction on $U \lhd F$ with base retraction r, $0 \in U$ and $0 \in O = r(U)$; g a filling germ of the principal part $\hat{f}$ (§III.3) and Dg a derivative family of g on a germ G with $G_0 \subseteq U$; $\hat{f}(0) = 0$ and $(x, \hat{f}(x)) \in K$ for $x \in O$; s an sc⁺-germ with derivative family Ds and $\iota s(0) = g(0)$; and conjugation data d of $k = g - \iota s$. Conclusion: $Dg(0) : E_0 \to F_0$ and $Tf(0) = Dg(0)|_{T_0O} : T_0O \to K_0$ `Chapter3/FillerLinearAlgebra · fillerRestrict` are Fredholm with index $\dim d.P_n - \dim d.Q$. Proof: $g(0) = \hat{f}(0) = 0$ (filler condition (1) at 0), so k is an sc¹-germ on $G \cap G_s$ with the family $Dg - \iota Ds$ and $k(0) = 0$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift`, and $Dk(0)$ is Fredholm of index $\dim d.P_n - \dim d.Q$ by the preceding theorem. Since 0 is a smooth point, $Ds(0) : E \to F^1$ is an sc-operator (the remark after Proposition 1.2.1 (preprint 1.16), for germs) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`, so $\iota Ds(0)$ is the level-0 map of an sc⁺-operator and compact. $Dg(0) = Dk(0) + \iota Ds(0)$ is then the value at 0 of the identity $Dg = (Dg - \iota Ds) + \iota Ds$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus`; on U it was the uniqueness of derivatives within U_1 and the density of E_1 `Chapter3/BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one`. A compact perturbation of a Fredholm operator is Fredholm of the same index `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator` `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`. Proposition 3.1.5 (preprint 3.5) (§III.4) transfers the Fredholm property and the index to $Tf(0)$ `Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_5`.

Proposition 3.6.5 (preprint 3.52), with the index Chapter3/FredholmGermIndexG · proposition_3_52_index. Let f be a section of the strong bundle $P : Y \to X$ with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. In the strong bundle chart of Definition 3.1.11 (preprint 3.11) centred at x_0, with the filling germ g of $\hat{f}$, its germ G and a derivative family Dg, and the local sc$^+$-germ s, conjugation data of $g - s$ exist, $Dg(0)$ and $Tf(x_0)$ are Fredholm, and **for every** conjugation data d of $g - s$ their index is $\dim d.P_n - \dim d.Q$. Proof: unpack Definition 3.1.11 (preprint 3.11); $0 \in O$ because the chart is centred at x_0; $\hat{f}(0) = 0$ from $f(x_0) = 0$; $(x, \hat{f}(x)) \in K$ because the principal part of a section lies in the retract K of the chart; data d_0 exist by the equivalence above and give the Fredholm property; and the local theorem applied to an arbitrary d gives the index. No hypothesis is added. As in §III.21, only the level-0 Fredholm property is proved (T_0O and K_0 are not set up as sc-Banach spaces). The statements of §III.21 stand next to these, with the same proofs and without the dimensions; the whole-chart forms are kept as well Chapter3/FredholmGermIndex · proposition_3_52_index_wholeChart.

III.31 The linearization at a zero in charts, its chart independence, and Definition 3.6.6 (preprint 3.53) from a non-centred chart

Definition 3.6.6 (preprint 3.53) is stated in T_xX, with the linearization $f'(x) : T_xX \to Y_x$ and the partial cone C_xX of Definition 2.5.8 (preprint 2.38). The text defines $f'(x)$ only at the *centre* of a chart, before Definition 3.1.4 (preprint 3.4), as the restriction of $D(\mathbf{f} \circ r)(0)$ to T_0O, identifies it with $Dg(0)|_{T_0O}$ for a filler g, and appeals to Proposition 2.4.10 (preprint 2.30) for the independence of T^R_xX and C_xX of the chart. That $f'(x)$ itself, and the two conditions of Definition 3.6.6 (preprint 3.53), do not depend on the chart is used — the proof of Theorem 3.6.8 (preprint 3.54) repeats "as above" the argument at x_0 in a chart centred at a nearby zero y — but not proved (Finding 110). The formalization (§III.21) states Definition 3.6.6 (preprint 3.53) in the chart of Definition 3.1.11 (preprint 3.11) centred at the zero; this section proves the local form of the missing independence at a zero, in charts, and the assembly step that Theorem 3.6.8 (1) (preprint 3.54) needs: Definition 3.6.6 (preprint 3.53) at a zero y from data in a chart which is *not* centred at y (the chart at x_0). Throughout, z is a point on level 1 (the text has smooth points), and tameness is not used.

Linear algebra of conjugate operators. Let $L : T \to K$ and $L' : T' \to K'$ be bounded operators between normed spaces and $A : T \cong T'$, $\Gamma : K \cong K'$ topological linear isomorphisms with $L'A = \Gamma L$. Then $L' = \Gamma L A^{-1}$ Chapter3/LinearizationCharts · eq_comp_of_conj, so L is onto iff L' is Chapter3/LinearizationCharts · surjective_iff_of_conj; $\ker L' = A(\ker L)$ Chapter3/LinearizationCharts · ker_eq_map_of_conj; for complete spaces L is Fredholm iff L' is, with the same index, because Fredholm operators and their index are stable under composition with isomorphisms on both sides Chapter3/LinearizationCharts · isFredholm_iff_of_conj Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv; and if $D \subseteq K$ is dense and contained in the range of L, then $\Gamma(D)$ is dense in K' and contained in the range of L' Chapter3/LinearizationCharts · dense_image_of_conj. The isomorphisms that occur are restrictions of bounded operators $a : E_0 \to E_0'$, $a' : E_0' \to E_0$ which map a subspace T into T' and T' into T and are inverse to each other there Chapter3/LinearizationCharts · restrictEquiv. A subspace $N \subseteq E_0$ is transported by $A : T \cong T'$ to $A(N) = \{Ax : x \in T \cap N\} \subseteq T'$ Chapter3/LinearizationCharts · transportSub, with $y \in A(N) \iff A^{-1}y \in N$ for $y \in T'$ Chapter3/LinearizationCharts · mem_transportSub_iff and $A^{-1}(A(N)) = N$ for $N \subseteq T$ Chapter3/LinearizationCharts · transportSub_symm.

Good position inside a subspace is transported Chapter3/LinearizationCharts · IsInGoodPositionIn.transport. Good position inside T (§III.21, Chapter3/GoodPositionGerm · IsInGoodPositionIn) asks: $N \subseteq T$, the interior of $N \cap C$ in N nonempty, and a complement $P \subseteq T$ of N *in* T with the ε-condition of Definition 3.1.24 (preprint 3.21). If $A : T \cong T'$ respects the cones ($x \in C \iff Ax \in C'$ for $x \in T$) and N is in good position to C inside T, then $A(N)$ is in good position to C' inside T'. *The interior*: A restricts to a homeomorphism $N \cong A(N)$ which carries $N \cap C$ onto $A(N) \cap C'$, so it carries interior points to interior points. *The complement*: $A(P)$; $A(N) \cap A(P) = 0$ and $A(N) + A(P) = T'$ by pulling back with A^{-1}. *The constant*: with $a = \|A\|$, $b = \|A^{-1}\|$, take $\varepsilon' = \varepsilon/(ab+1)$; for $y = Ax$, $q = Ap$ with $|q| \le \varepsilon'|y|$,

$|p| \le b|q| \le b\varepsilon' a|x| \le \varepsilon|x|$, so $x \in C \iff x+p \in C$, i.e. $y \in C' \iff y+q \in C'$ — the constant of Lemma 3.5.9 (preprint 3.45), no finite dimensionality needed (Finding 115). Applied to A and A^{-1} this gives an equivalence `Chapter3/LinearizationCharts · isInGoodPositionIn_transport_iff`.

Two derivative tools at a level-1 point. (i) *Uniqueness*: if $f_1 = f_2$ on a relatively open $V \subseteq U_1 \cap U_2$ containing z, then $Df_1(z) = Df_2(z)$: both are derivatives within V_1 (level 1), V is an sc-domain with unique differentiability, and two operators $E_0 \to F_0$ that agree after composition with the dense inclusion $E_1 \to E_0$ are equal `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`. (ii) *The fibre derivative at a point of the zero section* `Chapter3/LinearizationCharts · scDeriv_fiber_of_vanishing`: let Φ be sc¹ on a set $D \subseteq E \oplus F$ containing $\{(u,0) : u \in S\}$ (S relatively open, $z \in S_1$) and the whole fibre $\{z\} \times F$, with $\Phi(u,0) = 0$ on S and $\Phi(z,k) = \Gamma k$ for a bounded Γ. Then $D\Phi(z,0)(h,k) = \Gamma k$. Proof: along $k_1 \mapsto (z,k_1)$, $k_1 \in F_1$, the composite is $\Gamma \circ \iota$, so $D\Phi(z,0)$ restricted to the fibre direction is Γ after composition with the inclusion, hence equal to Γ by density; along $u_1 \mapsto (u_1,0)$, $u_1 \in S_1$, the composite vanishes, and unique differentiability of S_1 at z gives zero in the base direction. This is the generalization from 0 to z of the identities of §III.4 and §III.21; it is the reason why only the fibre part survives at a zero (Finding 112).

The linearization at a point of the retract, without a filler. For the principal part $\mathbf{f}$ of a section in a strong bundle chart with retraction $R(u,h) = (r(u), \rho(u)h)$ and a level-1 point z of $O = r(U)$, put

$$\mathrm{Lin}(z) = D(\mathbf{f} \circ r)(z)|_{T_zO}, \qquad T_zO = Dr(z)E_0$$

`Chapter3/LinearizationCharts · linAt`; $Dr(z)$ is idempotent `Chapter3/LinearizationCharts · scDeriv_r_idem_at`. Three facts make it the right object.

Its values lie in $K_z = \rho(z)F$ at a zero `Chapter3/LinearizationCharts · rho_scDeriv_comp_r` (the text's "$\rho(0)Tf(0) = Tf(0)$", now at any zero). Since $(r(u), \mathbf{f}(r(u))) \in K$, $\rho(r(u))\mathbf{f}(r(u)) = \mathbf{f}(r(u))$ on U `Chapter3/Filler · ρ_eq_self_of_mem_K`; so $\mathbf{f} \circ r$ equals $\Psi_2(u) = (R(r(u), \mathbf{f}(r(u))))_2$, and their derivatives at z agree by (i). The chain rule for $\Psi_2 = (\text{fibre part of } R) \circ (r, \mathbf{f} \circ r)$ at z, where $(r, \mathbf{f} \circ r)$ passes through $(z,0)$ (because $r(z) = z$ and $\mathbf{f}(z) = 0$), and (ii) for the fibre map $(u,k) \mapsto \rho(u)k$, which vanishes on the zero section and is $\rho(z)$ on the fibre over z, give $D(\mathbf{f} \circ r)(z)v = \rho(z)D(\mathbf{f} \circ r)(z)v$. The fibre K_z is the fibre of K over z `Chapter3/LinearizationCharts · mem_rangeS_ρCLM_iff`.

The bridge to fillers `Chapter3/LinearizationCharts · scDeriv_comp_r_eq_of_eventuallyEq` (the text's "$Tg(0) = Tf(0)$ on T_0O", now at any point): if g is sc¹ and agrees with $\mathbf{f}$ on O near z, say on $O \cap W$ with W open, then $\mathbf{f} \circ r = g \circ r$ on the relatively open $U \cap r^{-1}(W)$, so by (i) and the chain rule (r maps z to itself on level 1) $D(\mathbf{f} \circ r)(z) = Dg(z)Dr(z)$. Hence $\mathrm{Lin}(z) = Dg(z)$ on T_zO `Chapter3/LinearizationCharts · linAt_eq_scDeriv`, by idempotence of $Dr(z)$. At a centre, for a filled version, $\mathrm{Lin}(0)$ is the $Tf(0) = Dg(0)|_{T_0O}$ of Proposition 3.1.5 (preprint 3.5) `Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt` `Chapter3/FillerLinearAlgebra · fillerRestrict`. At $z \neq 0$ the full $Dg(z) : E \to F$ is not related to $\mathrm{Lin}(z)$ (condition (3) of the filler holds at 0 only), so the relevant kernel is $\ker Dg(z) \cap T_zO = \ker \mathrm{Lin}(z)$ (Finding 111). *For a filling germ g* on G the bridge needs two things the whole-chart one did not: $Dg(z)$ exists only for z on level 1 of G, and the chain rule for $g \circ r$ is the inner one on the germ $r^{-1}(G)$, which is defined when $r(0) = 0$, i.e. $0 \in O$ `Chapter3/LinearizationChartsG · scDeriv_comp_r_eq_germ`. Under these two conditions $\mathrm{Lin}(z) = Dg(z)$ on T_zO `Chapter3/LinearizationChartsG · linAt_eq_germDeriv` `Chapter3/LinearizationChartsG · StrongBundle.linAt_eq_germDeriv_chart`, and at the centre, which lies in every germ, $\mathrm{Lin}(0) = Tf(0)$ `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt`. Both hold wherever the bridge is used: the chart is centred at a zero, and the points $z \neq 0$ are the level-1 lifts of nearby zeros, which lie in the germ of g by the shrinking of the basic germ (§§III.32, III.48).

Fibres of strong bundle charts and of transitions. For a strong bundle chart $\Phi : P^{-1}(V) \to K$ covering φ: points of K lie over O `Chapter3/LinearizationCharts · StrongBundleChart.fst_mem_0`; $\Phi^{-1}(x,\cdot)$ is additive and homogeneous on the fibre of K over x, since Φ is a fibrewise linear bijection of $P^{-1}(\varphi^{-1}(x))$ onto that fibre `Chapter3/LinearizationCharts · StrongBundleChart.symm_add` `Chapter3/LinearizationCharts · StrongBundleChart.symm_smul`; the fibre of K is closed under scalars and contains $(u, \rho(u)k)$ `Chapter3/LinearizationCharts · StrongBundleChart.mem_K_ρ`. For a second chart Φ' the transition

$\Theta = \Phi' \circ \Phi^{-1}$ covers the base transition $\tau = \varphi' \circ \varphi^{-1}$ `Chapter3/LinearizationCharts · StrongBundleChart.transition_fst`, is linear on the fibres `Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_add` `Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_smul`, maps the zero section to the zero section `Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_zero` and K into K' `Chapter3/LinearizationCharts · StrongBundleChart.transition_mem_K`, and is inverted by the reverse transition `Chapter3/LinearizationCharts · StrongBundleChart.transition_transition`. Over a point x of the base overlap, $\Gamma(x)k = (\Theta(x, \rho(x)k))_2$ is a bounded operator $F \to F'$ — linear by the fibre linearity, continuous as a composition of $\rho(x)$, Φ^{-1} and Φ' `Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM` — which on K_x is the transition itself `Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM_apply_of_mem`. The text's "bounded linear on the fibres" (Definition 2.6.4 (preprint 2.50)) is about Φ only; the linearity of Φ^{-1} on the fibres of K is this lemma (Finding 113). Finally, if $\varphi(V \cap V') = O \cap W$ with W open, the overlap of K is $K \cap (W \times F)$ `Chapter3/LinearizationCharts · StrongBundleChart.overlap_eq_inter_prod`.

The chart change at a zero `Chapter3/LinearizationCharts · linAt_chart_change`. *Statement.* Let c, c' be strong bundle charts whose transition Θ is sc-smooth on $[0]$ (as a map between the overlap retracts of K and K'), f a section whose representatives $z \mapsto (z, \mathbf{f}(z))$ are sc-smooth $O \to K$ and $O' \to K'$, y a zero of f in both chart domains, and z, z' level-1 points over $\varphi(y)$, $\varphi'(y)$. Then there are topological isomorphisms $A : T_zO \cong T_{z'}O'$ and $\Gamma : K_z \cong K'_{z'}$, with $\Gamma \circ \mathrm{Lin}(z) = \mathrm{Lin}'(z') \circ A$, and $x \in C_z \iff Ax \in C'_{z'}$, for $x \in T_zO$, where C_z is the partial quadrant at $\varphi(y)$ of a presentation of C `Chapter2/Tame · partialQuadrantAtWith`. The proof, in the order of the Lean:

1. *The base transition as a germ of sc-diffeomorphisms.* The base charts are compatible (both are charts of X, `Chapter2/MPolyfold · IsChart.compatible`). Choose open W, W' with $\varphi(V \cap V') = O \cap W$ and $\varphi'(V \cap V') = O' \cap W'$ `Chapter2/MPolyfold · exists_isOpen_image_eq` and restrict the base retractions to $U \cap r^{-1}(W)$, $U' \cap r'^{-1}(W')$ (Proposition 2.3.1 (1) (preprint 2.10)); their retracts are the overlaps. Then τ, $\tau' = \varphi \circ \varphi'^{-1}$ form a germ of sc-diffeomorphisms between them `Chapter2/Boundary · ScRetraction.IsDiffeoGerm`, and $\tau \circ r_W$ maps z to z' on level 1. The derivatives of the restricted retraction and of $\mathbf{f} \circ r$ at z are those on the full domains, by (i) (Finding 114).
2. *The transition near the zero section.* Restrict the strong bundle retraction to $W \times F$; its retract is the overlap of K, so $\Theta \circ R_W$ is sc-smooth and its fibre part Φ is sc¹. $\Phi(z, k) = \Gamma(z)k$ because $R(z, k) = (z, \rho(z)k)$; $\Phi(u, 0) = 0$ because $R(u, 0) = (r(u), 0)$ and Θ preserves the zero section.
3. *The section in the two charts.* For u in the restricted domain, $r(u) = \varphi(y_1)$ for some $y_1 \in V \cap V'$, and both $\mathbf{f}' \circ r' \circ \tau \circ r_W$ and $\Phi \circ (r, \mathbf{f} \circ r)$ take at u the value $(\Phi'(f(y_1)))_2$ (the principal parts are the fibre parts of $\Phi \circ f \circ \varphi^{-1}$ and $\Phi' \circ f \circ \varphi'^{-1}$, `Chapter3/StrongBundleSections · StrongBundleChart.apply_section'`). By (i) their derivatives at z agree.
4. *The key identity.* The chain rule on both sides, with $(r, \mathbf{f} \circ r)$ passing through $(z, 0)$ since $\mathbf{f}(z) = 0$, and (ii) for Φ give
$$D(\mathbf{f}' \circ r')(z')\, D(\tau \circ r_W)(z)v = \Gamma(z)\, D(\mathbf{f} \circ r)(z)v.$$
The derivative of $x \mapsto \Gamma(x)\mathbf{f}(z)$ in the base direction has dropped out *because* $\mathbf{f}(z) = 0$; away from zeros the linearization is not chart independent (Finding 112).
5. *The isomorphism A.* $A_0 = D(\tau \circ r_W)(z)$ maps T_zO into $T_{z'}O'$ and $A'_0 = D(\tau' \circ r'_{W'})(z')$ maps back `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range`; on these subspaces they are inverse `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv` (with the idempotence of $Dr(z)$); A is the induced isomorphism (`restrictEquiv`).
6. *The isomorphism Γ.* $\Gamma(z)$ maps K_z into $K'_{z'}$ (the transition maps K into K' and covers τ), the fibre operator $\Gamma'(z')$ of the reverse transition maps back, and they are inverse on these fibres (the reverse transition inverts the transition).
7. *Conclusions.* For $x \in T_zO$, $\mathrm{Lin}(z)x = D(\mathbf{f} \circ r)(z)x$ and $\mathrm{Lin}'(z')(Ax) = D(\mathbf{f}' \circ r')(z')A_0x$, so step 4 is $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(z')A$. The cones: Proposition 2.4.10 (preprint 2.30) for the partial cone, applied to the germ of step 1 `Chapter2/Boundary · proposition_2_30_cone`, gives $A_0(T_zO \cap C_z) = T_{z'}O' \cap C'_{z'}$ for the formalized partial cone `Chapter2/Tame · partialConeWith`, which is the cone condition on T_zO.

Hypotheses. Of the compatibility of Definition 2.6.4 (preprint 2.50) only the sc-smoothness of Θ on $[0]$ is

used — not on $[1]$, not the reverse transition (Finding 116). For two strong bundle charts of P it holds because both are charts of the M-polyfold $Y[0]$, hence compatible `Chapter3/LinearizationCharts · StrongBundle.IsChart.transition_isScSmoothRetractMap`; and for an sc-smooth section the representatives are sc-smooth in every strong bundle chart of P, so no hypothesis is added.

Chart independence `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. For two strong bundle charts of P around a zero y of an sc-smooth section, with level-1 points z, z' over $\varphi(y)$, $\varphi'(y)$: the linearizations $L = \mathrm{Lin}(z) : T_zO \to K_z$ and $L' = \mathrm{Lin}'(z') : T_{z'}O' \to K'_{z'}$ (corestricted to the fibres by the first fact above) are simultaneously surjective, simultaneously Fredholm with the same index, and $\ker L$ is in good position to C_z inside T_zO iff $\ker L'$ is in good position to $C'_{z'}$ inside $T_{z'}O'$. The four spaces are complete, being ranges of the idempotents $Dr(z)$, $\rho(z)$ and their primed versions `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`; the rest is the conjugation algebra, with $\ker L' = A(\ker L)$ and the transport of good position.

Definition 3.6.6 (preprint 3.53) at a zero, from a non-centred chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`. *Statement.* Let f be an sc-smooth section, y a zero with (f, y) an sc-Fredholm germ (Definition 3.1.11 (preprint 3.11)), c a strong bundle chart of P around y (not necessarily centred at y), z on level 1 over $\varphi(y)$, and g sc¹ on U agreeing with $\mathbf{f}$ on O near $\varphi(y)$ — the case in view was the chart at x_0 with a whole-chart filled version. Assume (a) the range of $Dg(z)|_{T_zO}$ contains $\rho(z)\hat{k}$ for every smooth $\hat{k} \in F_\infty$, and (b) $\ker Dg(z) \cap T_zO$ is in good position to $C_{\varphi(y)}$ inside T_zO. Then (f, y) is in good position in the whole-chart sense `Chapter3/GoodPositionGerm · StrongBundle.IsGoodPositionGermWholeChart`, in the chart c' centred at y given by Definition 3.1.11 (preprint 3.11), with its filled version g'. *Proof.* The chart change between c at z and c' at 0 gives A, Γ with $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(0)A$; the bridges give $\mathrm{Lin}(z) = Dg(z)$ on T_zO and $\mathrm{Lin}'(0) = Dg'(0)$ on T_0O' (filler condition (1)). *Surjectivity of $Tf(y) = Dg'(0)|_{T_0O'}$ onto K'_0.* By Proposition 3.6.5 (preprint 3.52) in the centred chart c' `Chapter3/FredholmGermLinearizationG · proposition_3_52_local` it is Fredholm, so its range S is closed in the closed subspace K'_0, hence closed in F'. The map $k \mapsto \Gamma(\rho(z)k)$ is continuous $F \to F'$; by (a) and the conjugation it maps F_∞ into S; it maps F onto K'_0 (preimages by Γ^{-1}, which lands in K_z where $\rho(z)$ is the identity); since F_∞ is dense in F, every point of K'_0 is a limit of points of S, hence in S. So only a dense subset of K_z in the range is needed (Finding 117 (b)). *The kernel.* (b) is transported by A to good position inside T_0O' with respect to $C'_0 = C'$ (the partial quadrant at the corner is C' itself `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero`); and the transported subspace is $\ker Dg'(0)$: $\ker Dg'(0) \subseteq T_0O'$ by condition (3) of the filler g' *at its centre* `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and for $w \in T_0O'$, $Dg'(0)w = 0 \iff \Gamma(Dg(z)A^{-1}w) = 0 \iff A^{-1}w \in \ker Dg(z)$. This is where the hypothesis is on $\ker Dg(z) \cap T_zO$ and not on $\ker Dg(z)$: at $z \neq 0$ condition (3) is not available and $\ker Dg(z)$ need not lie in T_zO (Finding 117 (a)). Variants: with all of K_z in the range `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_chart`, and with the hypotheses on $\mathrm{Lin}(z)$ itself, taking $g = \mathbf{f} \circ r$, which agrees with $\mathbf{f}$ on O because r is the identity there `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_linAt` `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`.

In the germ reading the assembly is the same, with g a germ with derivative family Dg, z on level 1 of its germ, and the bridge for germs at z and at 0; the conclusion is good position at $Dg'(0)$ for the filling germ g' at y. The dense form needs the closed range of $Dg'(0)|_{T_0O'}$, which is now a hypothesis at chart level `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart_dense`, supplied at section level by Proposition 3.6.5 (preprint 3.52) for germs in a chart centred at y `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. The forms with all of K_z in the range need no Proposition 3.6.5 (preprint 3.52) at all: $\mathrm{Lin}'(0) = \Gamma\,\mathrm{Lin}(z)A^{-1}$ is then onto K'_0 directly `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart` (the whole-chart variant had gone through the dense one).

What remains is to supply (a) and (b) in the chart at x_0, at the zeros $z_0 = \varphi(y)$ near 0: (b) with the text's hypotheses (§III.32), (a) first under an added hypothesis (§III.33) and then with the text's hypotheses (§III.43).

III.32 Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros: Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) on the graph, and the kernel

The text proves property (4) of Theorem 3.6.8 (preprint 3.54) — at every zero $y \in V$, with $\psi\varphi(y) = v+\sigma(v)$, "ker $f'(y)$ is in good position to $C_y X$ and $f'(y)$ is surjective" — in one sentence: it follows from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55). Unpacked, three steps are needed beyond the graph description of §III.23. First, Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) have to be applied at the graph points, Lemma 3.6.9 (preprint 3.55) on the relatively open V' of Theorem 3.5.11 (preprint Corollary 3.47) and for a general presentation of C' (Finding 92). Second, Lemma 3.6.9 (preprint 3.55) is about the tangent space $N_v = \{n + D\sigma(v)n\}$ of the graph, not about a kernel; the identification $N_v = \ker D(h+t)(v+\sigma(v))$ is not commented on (Finding 91 (a), 101). Third, the statements live in the push-forward chart $\psi \circ \varphi$, about $D(h+t)$ on E', while Definition 3.6.6 (preprint 3.53) at y (via §III.31) asks for statements about $Dg(z_0)$ on $T_{z_0}O$ in the chart at x_0 (Findings 102–104). This section proves all of it for the kernel; the surjectivity of $f'(y)$ is §III.33. No hypothesis beyond those of the graph description is used.

Lemma 3.6.9 (preprint 3.55) for a relatively open domain and a general presentation `Chapter3/GlobalIFTFull · lemma_3_55_relOpen`. *Statement.* Let $C = \{x : \ell_i(x) \geq 0\}$ be a partial quadrant presented by an sc-isomorphism T, with the coordinate functionals $\ell_i(x) = (Tx)_i$ `Chapter3/GlobalIFTFull · coordCLMWith`, N finite-dimensional in good position to C with a good complement Y, Y **closed**, V relatively open in $N \cap C$ with $0 \in V$, and $\sigma : N \to E_0$ with values in Y, C^1 on V, $\sigma(0) = 0$, $D\sigma(0) = 0$ (within V). Then there is $\rho > 0$ such that for $v \in V$ with $|v| < \rho$ and $z = v + \sigma(v)$: $z \in C$; Y is a good complement of $N_v = \{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace` with respect to the partial quadrant C_z; and N_v is in good position to C_z. *Proof.* $D = N \cap C$ is convex with nonempty interior in N (good position), so D and $V = D \cap O$ have unique differentiability. *The extension.* The text's Lemma 3.6.9 (preprint 3.55) (§III.22, `Chapter3/GoodPositionOpen · lemma_3_55_core`) wants a map on the whole cone. Take a ball $B(0,r) \subseteq O$ and a smooth bump b equal to 1 on $\bar{B}(0,r/2)$ and 0 outside $B(0,r)$, and $\Phi(v) = b(v)v$. Φ maps D into V: D is closed under nonnegative scaling, and $|b(v)v| < r$. So $\tau = \sigma \circ \Phi$ is C^1 on the cone, agrees with σ on $B(0,r/2)$, and there its derivatives within the cone are those of σ within V (a neighbourhood of the point inside O does not change derivatives within a set); in particular $\tau(0) = 0$, $D\tau(0) = 0$. *Lemma 3.6.9 (preprint 3.55) in coordinates* for τ, with the good-complement estimate of (N, Y) rewritten in the ℓ_i and the joint surjectivity of the coordinates ($T^{-1}(k, 0)$ has coordinates k), gives ρ_1, the membership $z \in C$, and the good complement Y of N_v with respect to C_z. *The interior*, which Lemma 3.6.9 (preprint 3.55) does not state but good position requires: by the continuity of $D\sigma$ at 0 (a C^1 map on a set of unique differentiability) there is ρ_2 with $\|D\sigma(v)\| < \varepsilon$, the good-complement constant of (N, Y). $L = D\sigma(v)$ takes values in Y — this is where Y closed is used `Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo` — so $n \mapsto n + Ln$ is injective ($N \cap Y = 0$), a linear homeomorphism $N \cong N_v$ in finite dimensions. It carries the interior of D to an open subset of N_v contained in C_z: for w in the interior of D, $w \in C$ and $|Lw| \leq \varepsilon|w|$, so $w + Lw \in C$ by the good-complement property, and $C \subseteq C_z$. Take $\rho = \min(\rho_1, r/2, \rho_2)$.

Corollary 3.5.3 (preprint 3.41) at the graph points `Chapter3/GlobalIFTFull · StabilityData.exists_surjective_Dfs'_graph`. For the data of Theorem 3.4.2 (preprint 3.38) with $D(h+s)(0)$ surjective and a continuous σ on $V \ni 0$, $\sigma(0) = 0$, whose graph points $v + \sigma(v)$ are zeros in the germ: there is $\rho > 0$ such that for $v \in V$, $|v| < \rho$, the zero $v + \sigma(v)$ lies on level 1, at a point x_1 of the level-1 germ, and $D(h+s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$. Corollary 3.5.3 (preprint 3.41) `Chapter3/SolutionLinearization · StabilityData.corollary_3_41` gives a radius r for the zeros, and continuity of $v \mapsto v + \sigma(v)$ at 0 gives ρ.

Property (4) in the push-forward chart `Chapter3/GlobalIFTFullG · theorem_3_54_nearby`. *Statement.* For a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the data of the graph description (§III.23, `Chapter3/GlobalIFTG · theorem_3_54_graph`: chart, ψ, $h+t$, K, closed

Y, V', σ, V_x), a presentation T of $C' = \widetilde{C} \oplus W$, and an open $V_{x,1} \subseteq V_x$ containing x_0 such that for every zero $y \in V_{x,1}$, with $\psi\varphi(y) = v + \sigma(v)$, $v \in V'$: the point lies on level 1 and $D(h+t)$ there is surjective Fredholm of index $n-N$; $v+\sigma(v) \in C'$; Y is a good complement of N_v with respect to $C'_{v+\sigma(v)}$, and N_v is in good position to it. Only f a section is assumed, not an sc-Fredholm section. *Proof.* Take the radii ρ_1 (Corollary 3.5.3 (preprint 3.41) at the graph points) and ρ_2 (Lemma 3.6.9 (preprint 3.55), relatively open form), $\rho = \min(\rho_1, \rho_2)$, and the projection π onto K along Y, continuous because Y is closed and K finite-dimensional. Put $V_{x,1} = V_x \cap (\psi\varphi)^{-1} B(0, \rho/(\|\pi\| + 1))$, open since $\psi \circ \varphi$ is continuous on V_x. For a zero $y \in V_{x,1}$, $\psi\varphi(y) = v + \sigma(v)$ by the graph description, $v = \pi(\psi\varphi(y))$ because $\sigma(v) \in Y$, so $|v| \le \|\pi\|\,|\psi\varphi(y)| < \rho$, and both lemmas apply at v.

Derivative identities at a nearby zero. Four lemmas move these statements from E' and N_v to E and $\ker Dg(z_0)$.

Uniqueness at a point of C_1 `Chapter3/GlobalIFTNearbyGraph · scDeriv_eq_of_eqOn_nhds_at`: two maps with sc¹-derivatives D_1, D_2 at $x \in C_1$ that agree on $W_1 \cap C_1$ for an open $W_1 \ni x$ of E_1 have $D_1 = D_2$ — uniqueness of the derivative within the quadrant on level 1, then density of E_1. The text writes $\Psi_*(g) = h + t$ as an identity; formally they agree only on $U' \cap W'$, and this lemma identifies their derivatives at every nearby point (Finding 104).

The fibre derivative of a strong bundle map at $(u_0, 0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_at`: for a strong bundle map Ψ with fibre operators $\Gamma(u) = (\Psi(u,\cdot))_2$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM` and $u_0 \in U_1$, $D(\mathrm{pr}_{F'} \circ \Psi)(u_0,0)(h,k) = \Gamma(u_0)k$; hence the derivative of $u \mapsto \Gamma(u)k(u)$ at a zero u_0 of k is $\Gamma(u_0)Dk(u_0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_comp_at` — §III.21's identity at 0, moved to u_0.

The chain rule for the push-forward at a zero `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_pushforward_at`: for a strong bundle isomorphism Ψ covering ψ with inverse Ψ' covering ψ', and a zero $u_0 \in U_1$ of the sc-smooth g, $\Psi_*(g) = \kappa \circ \psi'$ with $\kappa(u) = (\Psi(u, g(u)))_2$, so

$$D(\Psi_* g)(\psi(u_0)) = \Gamma(u_0) \circ Dg(u_0) \circ D\psi'(\psi(u_0)).$$

$D\psi'(\psi(u_0))$ and $D\psi(u_0)$ are inverse on level 0 `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_base_inv_base`, and $\Gamma(u_0)$ is injective because Ψ' inverts Ψ on the fibres `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.fiberCLM_injective`; hence $\ker Dg(u_0) = D\psi'(\psi(u_0))(\ker D(\Psi_* g)(\psi(u_0)))$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.ker_scDeriv_eq_map`. *For a germ g* with derivative family Dg the formula is an identity of the explicit derivative family of $\Psi_*(g)$ (§III.23), valid at every level-1 zero u_0 and for every family `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.pushforwardDeriv_at`, and the kernel formula follows as before `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.ker_deriv_eq_map_germ`; that g is sc¹ at u_0 then means that u_0 lies on level 1 of the germ of g.

Corresponding partial quadrants `Chapter3/GlobalIFTNearbyGraph · scDeriv_mem_partialQuadrantAtWith_iff`: for an sc-diffeomorphism $\psi : U \to U'$ between relatively open subsets of partial quadrants with inverse ψ' and $z \in U'_1$, $x \in C'_z \iff D\psi'(z)x \in C_{\psi'(z)}$: the one-sided sign argument `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant` for ψ' at z gives one direction, for ψ at $\psi'(z)$ the other, and $D\psi(\psi'(z))D\psi'(z) = 1$. At the centre this was "$C' = T\psi(0)C$" (§III.23).

Tangent vectors of a family of points of a retract `Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed`: let ρ be sc¹ on U and $\Theta_1 : N \to E_1$ differentiable at v within V (unique differentiability at v), with values in U_1 and $\rho(\iota\Theta_1(w)) = \iota\Theta_1(w)$ for $w \in V$. Then $D\rho(\Theta_1(v))$ fixes $\iota(D\Theta_1(v)n)$ for every n: differentiate $\rho \circ \iota \circ \Theta_1 = \iota \circ \Theta_1$ at v. For ρ a retraction onto O: the tangent vectors of a C^1 family of points of O lie in $T_{z_0}O$.

$N_v = \ker D(h+s)(v+\sigma(v))$ (Finding 101). *The inclusion* `Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_le_ker_Dfs'`. For the solution graph σ of Theorem 3.5.11 (preprint Corollary 3.47) on V, a C^1-lift $\sigma_1 : V_1 \to E_1$ of σ on a relatively open $V_1 \subseteq V$ (Theorem 3.5.11 (1) (preprint Corollary

3.47), §III.20), $v \in V_1$ and the level-1 point x_1 over $v+\sigma(v)$ in the germ: $N_v \subseteq \ker D(h+s)(x_1)$. *Proof.* $h+s$ is differentiable on level 1 only, so the graph is lifted: the kernel K is a smooth subspace `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth`, hence lifts linearly to E_1 `Chapter3/BasicClass · finLift`, and $\Gamma_1 = \text{lift} + \sigma_1$ is a C^1 map $V_1 \to E_1$ with $\iota\Gamma_1(w) = w + \sigma(w)$ and $\Gamma_1(v) = x_1$. $V_1 = V \cap O_1$ `Chapter3/GlobalIFTNearbyGraph · relOpen_cone_inter` has unique differentiability at its points `Chapter3/GlobalIFTNearbyGraph · uniqueDiffWithinAt_relOpen_cone`, and $D\sigma(v) = \iota \circ D\sigma_1(v)$ `Chapter3/GlobalIFTNearbyGraph · fderivWithin_eq_toBase_comp`. The composite $(h+s)\circ\iota\circ\Gamma_1$ vanishes on V_1, and near v (on $V_1 \cap \Gamma_1^{-1}$ of the open set defining the level-1 germ) Γ_1 maps into the level-1 germ, where $h+s$ has the derivative $D(h+s)(x_1)$ composed with ι `Chapter3/SolutionLinearization · StabilityData.isSc1GermWith_Dfs'`. The chain rule and uniqueness give $D(h+s)(x_1)(n + D\sigma(v)n) = 0$ for $n \in K$. *Equality* `Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_eq_ker_Dfs'` under the conclusions of Corollary 3.5.3 (preprint 3.41) at x_1 (Fredholm, index $\dim P - \dim Q$, surjective) and with the closed good complement Y: $D\sigma(v)$ takes values in Y, so $\dim N_v = \dim K$ `Chapter3/GlobalIFTNearbyGraph · finrank_graphSpace_of_compl`; for a surjective Fredholm operator $\dim\ker = \operatorname{ind}$ `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`, so $\dim K = \operatorname{ind} D(h+s)(0) = \dim P - \dim Q = \operatorname{ind} D(h+s)(x_1) = \dim\ker D(h+s)(x_1)$, and an inclusion of finite-dimensional spaces of the same dimension is an equality.

Good position restricts to a subspace containing the kernel `Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le`: if N is in good position to C in E_0 and $N \subseteq T$, a good complement P restricts to the good complement $P \cap T$ of N in T, with the same constant ($n+p \in T$ and $n \in T$ give $p \in T$).

The graph description with the filler data exposed `Chapter3/GlobalIFTNearbyGraphG · theorem_3_54_chart_graph`. This is the body of the graph description (§III.23) stated in the chart, with the filling germ g of $\mathbf{f}$ on G_g and a derivative family Dg, the sc$^+$-germ t_0 on G_s, the conjugation Ψ (base ψ, inverse ψ'), the basic germ on a germ G that ψ' maps into $G_g \cap G_s$ level by level ($G \subseteq G''$, §III.48), and the tameness consequence $(1 - Dr(0))E \subseteq E_0$ as hypotheses, so that the later steps can use g at the points z_0. Its conclusions: the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) on G itself and the basic germ $h' = h + (t_{\mathbb{R}^N}, 0)$ with $\mathcal{S}.\text{fsum}\, h' = \Psi_*(g)$ on $U' \cap W'$; $D(h+t)(0)$ surjective with kernel K in good position; Y (closed), V', σ with its C^k-lifts; an open $N_1 \ni 0$ on which the filler identities (1), (2) and the conjugation identity hold, with $\psi'(U' \cap W') \subseteq N_1$; and an open $N_2 \subseteq N_1$ with $\psi(U \cap N_2)$ inside the germ and $U' \cap W'$, every zero of g in $U \cap N_2$ in O, $g(u) = 0 \iff (h+t)(\psi(u)) = 0$ on $O \cap N_2$, and $\psi(\{u \in O \cap N_2 : g(u) = 0\}) = \{v + \sigma(v) : v \in V'\}$. The proof is that of §III.23 (items 1–7), in the chart; $D(h+t)(0) = D(\Psi_* g)(0)$ for the explicit family is part of the conclusion.

The kernel at the nearby zeros, in the chart at x_0 `Chapter3/GlobalIFTNearbyKernelG · theorem_3_54_chart_kernel`. *Statement.* In the data of the preceding theorem there is an open $N_3 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_3$ of g there are $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point $z_{0,1}$ over z_0, **on level 1 of the germ G_g of the filling**, and $x_1 = \psi_1(z_{0,1})$ in the level-1 germ over $v+\sigma(v)$, with ($Dg(z_{0,1})$ the value of the derivative family): (i) $D(h+t)(x_1)$ surjective Fredholm of index $n-N$; (a) $N_v = \ker D(h+t)(x_1)$; (b) $D(\Psi_* g)(x_1) = D(h+t)(x_1)$ and $\ker Dg(z_{0,1}) = D\psi'(x_1)(\ker D(h+t)(x_1))$, and $Dg(z_{0,1})$ is surjective; (c) $\ker Dg(z_{0,1}) \subseteq T_{z_0}O = Dr(z_{0,1})E$; (d) $\ker Dg(z_{0,1})$ is in good position to C_{z_0} inside $T_{z_0}O$. *Proof.*

1. *The neighbourhood.* Take the C^1-lift σ_1 on $V_1 = V' \cap O_1$ (Theorem 3.5.11 (1) (preprint Corollary 3.47) with $k=1$), the radii ρ_1 (Corollary 3.5.3 (preprint 3.41) at the graph points), ρ_2 (Lemma 3.6.9 (preprint 3.55) relatively open, for the presentation of $C' = \widetilde{C} \oplus W$), ρ_3 with $B(0,\rho_3) \subseteq O_1$, and $\rho = \min(\rho_1, \rho_2, \rho_3)$. With the projection π onto K along Y, let $N_3 = N_2 \cap O_3$ where $O_3 \cap U = \psi^{-1}(B(0, \rho/(\|\pi\|+1))) \cap U$ (continuity of ψ on U). For a zero $z_0 \in O \cap N_3$ of g, $\psi(z_0) = v + \sigma(v)$ with $v \in V'$, and as in `theorem_3_54_nearby` $|v| < \rho$, so $v \in V_1$ and both lemmas apply.
2. *(i), and the level-1 points.* Corollary 3.5.3 (preprint 3.41) at v gives x_1 on level 1 of $\mathcal{S}.G = G \subseteq G''$ (§III.48, `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_41_shrink`); x_1 lies in U_1', so $z_{0,1} = \psi_1'(x_1)$ is a level-1 point over $\psi'(\psi(z_0)) = z_0$, and $\psi_1(z_{0,1}) = x_1$ `Chapter2/DegeneracyIndex · levelMap_comp_eq_self`. Since ψ_1' maps G_1'' into $(G_g)_1$, $z_{0,1}$ lies in $(G_g)_1$ `Chapter3/`

BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem: this is the step that needs the shrinking of the basic germ, since g is differentiable only there. (With g sc-smooth on U, any level-1 point would do.)

3. *(a)* is $N_v = \ker D(h+t)(x_1)$ above, fed with the lift and (i).
4. $D(\Psi_* g)(x_1) = D(h+t)(x_1)$: the two maps agree on $U' \cap W'$, so the uniqueness lemma at x_1 applies on the open set of E_1' cut out by $W' \cap U_o$ (with $U' = U_o \cap C'$) and the open set defining the level-1 germ G_1; $\Psi_*(g)$ is an sc¹-germ on the preimage of G_g, which contains G Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isSc1GermWith (whole-chart: sc-smooth on U' Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization).
5. *(b)*: the kernel formula is the chain rule for the push-forward at the zero $z_{0,1}$ of g, in the germ form above. *Surjectivity of* $Dg(z_{0,1})$: given k, solve $D(h+t)(x_1)w = \Gamma(z_0)k$ by (i); then $\Gamma(z_0)Dg(z_{0,1})D\psi'(x_1)w = \Gamma(z_0)k$, and $\Gamma(z_0)$ is injective. This is the surjectivity of the *filled* $Dg(z_0) : E \to F$, not of $Tf(z_0) = Dg(z_0)|_{T_{z_0}O} : T_{z_0}O \to K_{z_0}$ (§III.33).
6. *(c), the tangency to* O (Finding 102). Condition (3) of Definition 3.1.4 (preprint 3.4), which gave $\ker Dg(0) \subseteq T_0 O$ at the centre, is not available at z_0. Instead: the transported retraction $r' = \psi \circ r \circ \psi'$ on U' Chapter2/TameInvariance · IsScDiffeoQuadrant.transport fixes every graph point, because each $w + \sigma(w)$, $w \in V'$, is $\psi(u)$ for a zero $u \in O$ of g (the graph description), and $r(u) = u$. The lifted graph map $\Gamma_1 = \text{lift} + \sigma_1$ maps V_1 into U_1', so the tangent-vector lemma for r' gives $Dr'(x_1)$ fixing $n + D\sigma(v)n$ for every $n \in K$, i.e. $Dr'(x_1)$ is the identity on N_v. The derivative of the transported retraction is $Dr'(x_1) = D\psi(z_{0,1})Dr(z_{0,1})D\psi'(x_1)$ Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport (§III.27); applying $D\psi'(x_1)$ and using that it inverts $D\psi(z_{0,1})$, $Dr(z_{0,1})$ fixes $D\psi'(x_1)m$ for every $m \in N_v$. By (a) and (b), these are exactly the elements of $\ker Dg(z_{0,1})$, so $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$.
7. *(d)* (Finding 103). Lemma 3.6.9 (preprint 3.55) gives N_v in good position to $C'_{v+\sigma(v)}$ in E'. The isomorphism $S = D\psi'(x_1) : E_0' \cong E_0$ (inverse $D\psi(z_{0,1})$, Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.derivInvEquiv) carries $C'_{v+\sigma(v)}$ onto C_{z_0} by the corresponding-quadrants lemma, so Lemma 3.5.9 (preprint 3.45) Chapter3/GoodPosition · IsInGoodPosition.map gives $S(N_v) = \ker Dg(z_{0,1})$ (by (a), (b)) in good position to C_{z_0} in E; by (c) and the restriction lemma it is in good position inside $T_{z_0}O$.

Theorem 3.6.8 (1) (preprint 3.54) at the nearby zeros, the kernel, for the section Chapter3/GlobalIFTNearbySurjG · theorem_3_54_nearby_kernel. *Statement.* For a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the data of the graph description, with the chart c centred at x_0, its filling germ g with a derivative family and the sc⁺-germ t_0 exposed (and K smooth, Y closed, the lifts), and an open $V_{x,2} \subseteq V_x$ containing x_0, such that for every zero $y \in V_{x,2}$: $\mathbf{f} = g$ on O near $\varphi(y)$, $g(\varphi(y)) = 0$, and with $z_0 = \varphi(y)$ the conclusions (i), (a)–(d) of the chart statement hold, with $z_{0,1}$ on level 1 of the germ of g. *Proof.* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.5.3 (preprint 2.35), Chapter2/TameRetracts · proposition_2_35, at the smooth corner point 0 of the tame base retraction; §III.23). The conjugation data are first shrunk to G'' (§III.48), so that the chart statements apply; they give N_1, N_2, N_3, with $\psi(U \cap N_2) \subseteq G_0''$; put $V_x = \varphi^{-1}(N_2)$ and $V_{x,2} = \varphi^{-1}(N_3)$ within the chart domain, both open by continuity of φ. On V_x, $\mathbf{f}(\varphi(y)) = g(\varphi(y))$ (filler (1) on N_1), so $f(y) = 0 \iff g(\varphi(y)) = 0 \iff (h+t)(\psi\varphi(y)) = 0$ Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff; $\psi \circ \varphi$ is injective on V_x; and the image of the zero set is the graph. For a zero $y \in V_{x,2}$, $\varphi(y) \in O \cap N_3$ is a zero of g, and the kernel statement applies. The hypotheses are those of the graph description: **the kernel part of Theorem 3.6.8 (1) (preprint 3.54) needs no added hypothesis**, and of f only that it is a section (the regularity enters through the filling germ g, at 0 and at the lifts $z_{0,1}$ of the nearby zeros, which lie in its germ).

III.33 Theorem 3.6.8 (1) (preprint 3.54) under the filler condition at the nearby zeros, and the λ-extension

At a zero y near x_0, with $z_0 = \varphi(y)$ in the chart at x_0, §III.32 gives the kernel condition (b) of the assembly step of §III.31 and the surjectivity of the *filled* $Dg(z_0) : E \to F$. Condition (a) asks for more: that $Tf(z_0) = Dg(z_0)|_{T_{z_0}O}$ maps onto $K_{z_0} = \rho(z_0)F$ (or onto a dense part of it). The text obtains it from Corollary 3.5.3 (preprint 3.41). This section shows exactly what is missing, proves Theorem 3.6.8 (1) (preprint 3.54) under the **added hypothesis** that closes the gap, and records the formalized part of an attempt to close it without that hypothesis (Findings 100, 105–107).

The block form at a zero. With respect to $E = T_{z_0}O \oplus \ker Dr(z_0)$ and $F = K_{z_0} \oplus \ker \rho(z_0)$ ($Dr(z_0)$ and $\rho(z_0)$ are idempotent),

$$Dg(z_0) = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}, \qquad A = Tf(z_0), \quad C = (1-\rho(z_0))Dg(z_0)|_{\ker Dr(z_0)} : \ker Dr(z_0) \to \ker \rho(z_0).$$

The lower left block vanishes at a zero: $Dg(z_0)Dr(z_0) = D(\mathbf{f} \circ r)(z_0)$ (the bridge, §III.31) takes values in K_{z_0} (the values of the linearization at a zero, §III.31). If $Dg(z_0)$ is onto, C is onto, and (Finding 100) $b \mapsto [Bb]$ induces an isomorphism $\ker C \cong \operatorname{coker} A$: so $Tf(z_0)$ **is onto if and only if** C **is injective** — condition (3) of Definition 3.1.4 (preprint 3.4) at z_0. Definition 3.1.4 (preprint 3.4) imposes it at 0 only, and it does not propagate: $u \mapsto Dg(u)$ is continuous for the strong operator topology, not in norm, and a strongly continuous family of surjections may pass from an isomorphism to a non-injective operator. In index terms: Corollary 3.5.3 (preprint 3.41) gives $\operatorname{ind} Dg(z_0) = \dim K$ at all nearby zeros, $\operatorname{ind} Dg(z_0) = \operatorname{ind} A + \operatorname{ind} C = \dim \ker A$ when C is onto, and $\operatorname{ind}(f, y) = \operatorname{ind} A = \dim K - \dim \operatorname{coker} A$; so $\operatorname{coker} Tf(z_0) = 0$ is the local constancy $\operatorname{ind}(f, y) = \operatorname{ind}(f, x_0)$ along the solution set — the index stability of Proposition 6.2.2 (preprint 6.2) of the text, whose proof uses the same step ("F' surjective" for the intrinsic linearization, from the implicit function theorem for the filled one). The text's "(4) follows from Corollary 3.41 and Lemma 3.55" therefore has a gap for the surjectivity of $f'(y)$; the same sentence stands in the book's arXiv version (Theorem 3.13 there, with Definition 3.13, the filling, still imposing (3) at 0 only; Finding 157). Formalized of this paragraph is the direction used:

Proposition 3.1.5 (1) (preprint 3.5) at a zero of a filler `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`. *Statement.* Let U be relatively open in a partial quadrant, R a strong bundle retraction on $U \lhd F$, $\mathbf{f}$ with $\mathbf{f} \circ r$ sc¹ and $(x, \mathbf{f}(x)) \in K$ on O, g sc¹, z on level 1 with $\iota z \in O$, $\mathbf{f}(\iota z) = 0$, $\mathbf{f} = g$ on O near ιz. If $Dg(z)$ is surjective and $(1-\rho(z))Dg(z)$ is injective on $\ker Dr(z)$ (in the form: $Dr(z)b = 0$ and $\rho(z)Dg(z)b = Dg(z)b$ imply $b = 0$), then for every $k \in F$, $\rho(z)k = Dg(z)x$ for some $x \in T_zO$. For a germ g, with $Dg(z)$ the value of a derivative family at a level-1 point of its germ, the proof is the same with the bridge for germs (§III.31) `Chapter3/GlobalIFTNearbySurjG · exists_range_deriv_eq_ρ_of_normal_injective_germ`. *Proof.* Write $Dg(z)e = \rho(z)k$, $a = Dr(z)e$, $b = e - a \in \ker Dr(z)$ (idempotence). $Dg(z)a = D(\mathbf{f}\circ r)(z)e$ lies in K_z (bridge and values at a zero), so $\rho(z)Dg(z)a = Dg(z)a$; then $\rho(z)Dg(z)b = \rho(z)(\rho(z)k - Dg(z)a) = \rho(z)k - Dg(z)a = Dg(z)b$ ($\rho(z)$ is linear and idempotent on the fibre). So $b = 0$, $e = a \in T_zO$. The converse direction and the index identity above are not formal statements; they explain why the hypothesis is the right one.

The added hypothesis `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. (f, x_0) is in good position *with a filler satisfying condition (3) at the nearby zeros* if x_0 is smooth and there are the data of Definition 3.6.6 (preprint 3.53) in the chart form (a strong bundle chart centred at x_0, tame since §III.47, a filling germ g of $\mathbf{f}$ with a derivative family Dg on a germ G inside U, a local sc⁺-germ s with $s(0) = g(0)$ and $g - s$ conjugated to a basic germ, the germ in good position at $Dg(0)$) together with an open $N \ni 0$ such that at every level-1 point z_1 *of* G with $\iota z_1 \in O \cap N$ and $g(\iota z_1) = 0$, $(1-\rho(\iota z_1))Dg(z_1)$ is injective on $\ker Dr(z_1)$. The condition is imposed only where Dg is defined, which is where the lifts of the nearby zeros lie (§III.32); the whole-chart form imposed it at every level-1 point over $O \cap N$ and implies it `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilledWholeChart.isGoodPositionGermFilled`. It implies Definition 3.6.6 (preprint 3.53) `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.`

`isGoodPositionGerm` (forget N). According to the docstring, the fillers of the applications satisfy (3) at every point; Definition 3.1.4 (preprint 3.4) as stated does not ask for it. It is recorded as an added hypothesis in the docstrings and in the Findings (items 100, 153).

Theorem 3.6.8 (1) (preprint 3.54) `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one`. *Statement.* Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position with a filler satisfying condition (3) at the nearby zeros. Then there is an open $V_x \ni x_0$ such that (f, y) is in good position (Definition 3.6.6 (preprint 3.53), `IsGoodPositionGerm`) at every zero $y \in V_x$. *Proof.*

1. In the chart c at x_0 with the filling germ g: $(1 - Dr(0))E \subseteq E_0$ from tameness (Proposition 2.5.3 (preprint 2.35)); the germ G of the condition is intersected with the germ of g, and the basic germ is shrunk to the points that ψ' maps into this intersection and into the germ of s (§III.48); then the graph description in the chart and the kernel statement of §III.32 give N_1, N_2, N_3. Put $V_x = \varphi^{-1}(N_3 \cap N)$ in the chart domain, with N the neighbourhood of the added hypothesis.
2. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O \cap N_3$, $\mathbf{f}(z_0) = 0$, and $g(z_0) = \mathbf{f}(z_0) = 0$ by filler (1) on N_1; $\mathbf{f} = g$ on O near z_0. The kernel statement gives the level-1 point $z_{0,1}$, on level 1 of G, $Dg(z_{0,1})$ surjective, $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$, and $\ker Dg(z_{0,1})$ in good position to C_{z_0} inside $T_{z_0}O$.
3. y is a smooth point, because f is regularizing `Chapter3/TameGermsG · IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so (f, y) is an sc-Fredholm germ (Definition 3.1.16 (preprint 3.14)); and $\mathbf{f} \circ r$ is sc^1 for an sc-smooth section `Chapter3/GlobalIFTNearbySurj · StrongBundle.IsScSmoothSection.isSc1_principal_comp_r`.
4. Apply the assembly step of §III.31 in the chart c at $z_{0,1}$ with g, in its germ form, the closed range in the tame chart at y coming from Proposition 3.6.5 (preprint 3.52) for germs `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. *Condition (a)*: the added hypothesis at $z_{0,1}$ (with $\iota z_{0,1} = z_0 \in O \cap N$, $g(z_0) = 0$) is the injectivity of C, and Proposition 3.1.5 (1) (preprint 3.5) at z_0 turns the surjectivity of $Dg(z_{0,1})$ into $\rho(z_0)k \in Dg(z_{0,1})(T_{z_0}O)$ for every k, smooth or not. *Condition (b)*: $\ker Dg(z_{0,1}) \cap T_{z_0}O = \ker Dg(z_{0,1})$ by the tangency (c), and the good position is (d).

So Theorem 3.6.8 (1) (preprint 3.54) holds under condition (3) of the filler at the nearby zeros; the kernel half holds with the text's hypotheses (§III.32). When this section was written, **the surjectivity of $f'(y)$ at the nearby zeros with Definition 3.1.4 (preprint 3.4) as stated was open in the formalization**; it is proved in §§III.41–III.43 by another argument, and the condition (3) at the nearby zeros turns out to be a consequence of good position (§III.45). The global theorems are not affected: they assume good position at every zero, which supplies at each x_0 the nearby good position that the regularity argument needs (§III.36–§III.37, Finding 156).

The λ-extension: what the perturbation argument of Proposition 6.2.2 (preprint 6.2) gives. The text's device for index stability (Proposition 6.2.2 (preprint 6.2)) is to add finitely many parameters λ multiplying sc^+-sections, so that the extended problem is surjective, and to apply the implicit function theorem to it. In the chart at x_0 it can be set up without new bundle isomorphisms: for a finite-dimensional $\Lambda_0 \subseteq F_\infty$, the sc^+-germ $\tilde{e}(\lambda, u) = \rho(r(u))(\lambda_1(\pi_K\psi(u)) + \hat{\jmath}(\lambda_2))$ (Finding 105). Two points are not addressed by the text, and the first is formalized.

The obstruction at the corner `Chapter3/GlobalIFTNearbyPerturb · mem_lineality_of_goodComplement_prod`. Theorem 3.5.11 (preprint Corollary 3.47) for the extended problem needs its kernel in good position to the extended partial quadrant at the corner. Let $C \subseteq V$ contain 0 and be closed under nonnegative scaling, $N \subseteq V$ a subspace, and $P \subseteq \mathbb{R} \times V$ a subspace with the ε-condition of Definition 3.1.24 (preprint 3.21) for $\mathbb{R} \times N$ `Chapter3/GlobalIFTNearbyPerturb · prodSub` with respect to $\mathbb{R} \times C$ `Chapter3/GlobalIFTNearbyPerturb · prodCone`. Then every $p = (\mu, y) \in P$ has $y \in C$ and $-y \in C$. *Proof.* $n = (1, 0)$ lies in $\mathbb{R} \times N$ and in $\mathbb{R} \times C$, $|n| = 1$; for $t = \pm\varepsilon/|p|$, $|tp| \le \varepsilon|n|$, so $n + tp \in \mathbb{R} \times C$, i.e. $ty \in C$, and scaling by $1/|t|$ gives $\pm y \in C$. Hence a good complement projects into the lineality space $C \cap (-C)$, and $\mathbb{R} \times N$ is in good position only if every $v \in V$ is $n + y$ with $n \in N$, $y \in C \cap (-C)$ `Chapter3/GlobalIFTNearbyPerturb · sup_lineality_eq_top_of_goodComplement_prod`. For the naive extension (parameters multiplying smooth vectors with a factor vanishing at 0) the kernel is $\mathbb{R} \times K$, so good position fails unless K maps onto all boundary coordinates — the text's "has

a complement contained in $T^{R\prime\prime}$ is exactly the statement that the complement lies in the lineality space.

The fix: a tilted kernel `Chapter3/GlobalIFTNearbyPerturb · isInGoodPosition_extendedKernel`. In $(\Lambda \times P_n) \oplus W$ with the partial quadrant $(\Lambda \times \widetilde{C}) \oplus W$ (the parameters λ unconstrained), let K, Y' be subspaces of $P_n \oplus W$ and $j : \Lambda \to P_n \oplus W$ linear with $K \cap j(\Lambda) = 0$, $(K + j(\Lambda)) \oplus Y' = P_n \oplus W$, Y' in the lineality space of $\widetilde{C} \oplus W$ ($v + y \in \widetilde{C} \oplus W \iff v \in \widetilde{C} \oplus W$ for $y \in Y'$), and $K + j(\Lambda)$ containing an interior point of $\widetilde{C} \oplus W$. Then the kernel $\widetilde{N} = \{((\lambda, a), w) : (a, w) + j(\lambda) \in K\}$ of $((\lambda, a), w) \mapsto D(h + t)(0)((a, w) + j(\lambda))$ `Chapter3/GlobalIFTNearbyPerturb · extendedKernel Chapter3/GlobalIFTNearbyPerturb · extProj Chapter3/GlobalIFTNearbyPerturb · extParam` is in good position. *Proof.* Choose a complement Λ_c of $\ker j$; every element of $j(\Lambda)$ is $j(\mu)$ with $\mu \in \Lambda_c$. *Interior*: if $k_0 + j(\mu_0)$ is interior, $n_0 = ((-\mu_0, k_0 + j(\mu_0)))$ lies in $\widetilde{N}$ and its $(P_n \oplus W)$-part is interior, an open condition. *The complement* $\Lambda_c \times Y'$ `Chapter3/GlobalIFTNearbyPerturb · extendedCompl`: an element of both has $(a, w) \in (K + j(\Lambda)) \cap Y' = 0$, then $j(\lambda) \in K \cap j(\Lambda) = 0$, $\lambda \in \ker j \cap \Lambda_c = 0$; and every $x = ((\lambda, a), w)$ splits by decomposing $(a, w) + j(\lambda) = k + j(\mu) + y$. *The constant*: since Y' lies in the lineality space, adding an element of $\Lambda_c \times Y'$ never changes membership in the quadrant, so the ε-condition holds for every ε (the proof takes $\varepsilon = 1$). The directions $j(\Lambda) \subseteq T_0(\psi(O)) \cap E'_\infty$ with $K + j(\Lambda) + E'_0 = E'$ exist by tameness ($T_0O + E_0 = E$) and would be realized by the sc$^+$-perturbations $\rho(r(u))Dg(0)D\psi^{-1}(0)j(\lambda)$.

Why the argument does not close the gap (Findings 105–106). Carried through (not formalized: the extended basic germ and its sc$^+$-perturbation, its derivative at the zeros, Theorem 3.5.11 (preprint Corollary 3.47) for it, and the tangency argument), the argument gives for each finite-dimensional $\Lambda_0 \subseteq F_\infty$ a neighbourhood $V(\Lambda_0)$ of x_0 with $\rho(z_0)\Lambda_0 \subseteq \operatorname{range} Tf(z_0)$ at every zero in $V(\Lambda_0)$. The assembly step at a fixed zero y needs $\rho(z_0)\hat{k}$ in the range for *all* smooth $\hat{k}$, and $V(\Lambda_0)$ comes from Theorem 3.5.11 (preprint Corollary 3.47) for the Λ_0-extended problem with no control in Λ_0: the radii of Theorems 3.5.1, 3.5.6, 3.5.11 (preprint 3.40, 3.43, 3.47) would have to be uniform in the finite-dimensional parameter space, which the statements do not provide; $\operatorname{coker} Tf(z_0)$ is finite-dimensional and spanned by smooth vectors, but the spanning set depends on y. The formalization of this route was stopped there; §§III.41–III.43 take another one.

III.34 Theorem 3.6.8 (2) (preprint 3.54) at the centre: the data of Definition 3.2.1 (preprint 3.25) for the solution set at x_0

The text ends the proof of Theorem 3.6.8 (preprint 3.54) with $\psi \circ \varphi(\{y \in V : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$ and concludes that the solution set has the d-dimensional tangent germ property, $d = \dim \ker f'(x)$, so that Theorem 3.2.3 (preprint 3.27) applies. Definition 3.2.1 (preprint 3.25) asks for much more than the graph identity (§III.24): a chart of X centred at the point onto a *tame* retract, a *smooth* finite-dimensional subspace N in good position with a *closed* good complement Y, a relatively open $V \ni 0$ in $N \cap C$, and $\delta : V \to Y$ continuous with the graph identity, $\delta(0) = 0$, $D\delta(0) = 0$ within $N \cap C$, and the sc$^\infty$-germ property in the k-dependent reading `Chapter3/TangentGerm · MPolyfold.TangentGermChart Chapter3/TangentGerm · IsScInftyGermAt`. This section assembles all of it at the centre x_0 from the graph description of §III.29, adding the three facts the text leaves silent: the chart is $\psi \circ \varphi$ and it is a chart of X; its model $\psi(O)$ is a tame retract; and $K = \ker D(h + t)(0)$ consists of smooth points. With the projecting complement of §III.29 the data also satisfy condition (b) of the regularity conditions. The data at the other zeros and condition (a) come in §§III.36–III.37.

The sc$^\infty$-germ property from the level-wise lifts. Theorem 3.5.11 (1) (preprint Corollary 3.47) gives, for every $k \geq 1$, a relatively open $V_k \ni 0$ in the cone and a C^k map $\sigma_k : V_k \to E_k$ with $\iota_k \sigma_k = \sigma$; Definition 3.2.1 (ii) (preprint 3.25) asks, for every level m and every k, for a relatively open $V_{m,k} \ni 0$ on which σ takes values in E_m and is C^k into E_m. Given m and k, take $j = \max(m, k) + 1$, $V_{m,k} = V_j$, and compose σ_j with the inclusion $E_j \to E_m$, a continuous linear map compatible with the inclusions into E_0 `Chapter3/SolutionGermRegularity · exists_clm_toBase_eq` (an iterated composition of the level inclusions): it is a lift of σ

into E_m, of class C^j, hence C^k, and it coincides with the level map of σ on V_j by the injectivity of the inclusions `Chapter3/SolutionGermRegularity · isScInftyGermAt_of_lifts`. So the level-wise reading of Theorem 3.5.11 (1) (preprint Corollary 3.47) (§III.20) is exactly what Definition 3.2.1 (ii) (preprint 3.25) needs (item 94 (a)).

The kernel is smooth `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth`. For a basic germ f with contraction map B and the data $\mathcal{S}$ of Theorem 3.4.2 (preprint 3.38) (sc$^+$-germ s into W^1), every $x = (a, w) \in \ker D(f + \iota s)(0)$ is a smooth point of $\mathbb{R}^n \oplus W$. The text takes the kernel of an sc-Fredholm operator to be smooth without comment. *The kernel equation.* $D(f + \iota s)(0) = Df(0) + \iota Ds(0)$ `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq`, and the W-component of $Df(0)x$ is $w - DB(0)x$ `Chapter3/SolutionLinearization · StabilityData.snd_Df_eq`. Lemma 3.1.9 (preprint 3.9) at $(0, 0)$ gives $DB(0)(0, w) = 0$ `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`, so $DB(0)x = DB(0)(a, 0)$, and the W-component of the kernel equation reads

$$w = DB(0)(a, 0) - \iota Ds(0)x.$$

The bootstrap. $DB(0)$ is an sc-operator (the derivative of an sc^1-germ at the smooth point 0) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`, and so is $Ds(0) : E \to W^1$, which raises the level by one. By induction on m, $w \in W_m$: if $w = \iota y$ with $y \in W_m$, then $(a, 0)$ lies on level $m + 1$ (the parameter space has all levels equal), so $DB(0)(a, 0) \in W_{m+1}$, and $(a, y) \in E_m$ gives $Ds(0)x \in W_{m+1}$; hence $w \in W_{m+1}$. So $x \in E_\infty$.

The chart $\psi \circ \varphi$. An sc-smooth map g on a relatively open U is an sc-smooth map between retracts $(O, C, E) \to (O', C', E')$ for every retract $O \subseteq U$ mapped into O': for every sc-smooth retraction R onto O, $g \circ R$ is sc-smooth by the chain rule `Chapter3/GlobalIFTCenter · IsScSmooth.isScSmoothRetractMap_of_subset`. For a chart $\varphi : V \to O$ of X and an sc-diffeomorphism $\psi : U \to U'$ between relatively open subsets of partial quadrants with inverse ψ' (§III.27) `Chapter2/TameInvariance · IsScDiffeoQuadrant` and $O \subseteq U$, given that $\psi(O)$ is a retract of C', the map $\psi \circ \varphi$ is a chart onto the model $(\psi(O), C', E')$, with inverse $\varphi^{-1} \circ \psi'$ and the source of φ `Chapter3/GlobalIFTCenter · MChart.compDiffeo` (continuity of both maps: ψ, ψ' are sc^0, hence continuous on level 0). *It is a chart of the M-polyfold structure* `Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart`: for a chart φ' of the atlas, the overlap of $\psi\varphi$ with φ' is ψ of the overlap of φ with φ', and the two transition maps are $(\varphi' \circ \varphi^{-1}) \circ \psi'$ and $\psi \circ (\varphi \circ \varphi'^{-1})$; both are compositions of sc-smooth maps between retracts — the transitions of φ and φ', compatible because both are charts of X, and ψ' resp. ψ restricted to the overlap retracts by the lemma above.

The tame model. For the base retraction r of the centred strong bundle chart, tame by the hypothesis on the charts, and the sc-diffeomorphism ψ of the conjugation, $(\psi(O), C', E')$ is a tame retract with the retraction $\psi \circ r \circ \psi'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image` (§III.27); the text takes this for granted when it calls $\psi \circ \varphi$ a chart of the tame M-polyfold (item 95).

Condition (b) from the choice of complement. For data of Definition 3.2.1 (preprint 3.25) whose complement Y has the projection property for C, $\pi(O) \subseteq C$: the retract lies in C, and π maps C into C `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.π_mem_of_projectsInto`. So the regularity conditions reduce to condition (a), the sc-smoothness of δ on V in the set sense `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.isRegular_of_projectsInto`.

Theorem 3.6.8 (2) (preprint 3.54) at the centre `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart_proj`. *Let f be a section of a strong bundle whose charts compatible with the atlas have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position. Then there are an open $V_x \ni x_0$ and a dimension d such that $S_{V_x} = \{y \in V_x : f(y) = 0\}$ has at x_0 the data of Definition 3.2.1 (preprint 3.25) of dimension d, with $\pi(O) \subseteq C$ for their projection and retract.* The proof fills the fields of the data one by one from the graph description, closed and projecting (§III.29) `Chapter3/GlobalIFTG · theorem_3_54_graph_proj`:

1. *The chart.* ψ and ψ' form an sc-diffeomorphism $U \to U'$ (item (i) of §III.29); $O = r(U) \subseteq U$ since the retraction maps U into U `Chapter3/Filler · StrongRetr.r_mem`; $\psi(O)$ is a tame retract (above, with the retract of the base retraction identified with $O = r(U)$ `Chapter3/`

`StrongBundleSections · StrongBundleChart.r_image`); the chart is $\chi = \psi \circ \varphi$, a chart of X, containing x_0, with $\chi(x_0) = \psi(0) = 0$.
2. *The subspace.* $N = K = \ker D(h+t)(0)$, finite-dimensional because $D(h+t)(0)$ is Fredholm `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`, smooth (above), in good position to C' (the hypothesis of Theorem 3.5.11 (preprint Corollary 3.47) established in §III.23); $d := \dim K$.
3. *The complement and the domain.* Y closed, a good complement; $V = V'$, relatively open in $K \cap C'$, containing 0; $\delta = \sigma$, continuous (it is C^1), with values in Y.
4. *The graph identity.* $\chi(S_{V_x} \cap \text{source}) = \psi\varphi(S_{V_x})$, since V_x lies in the source of φ; this is the graph identity of Theorem 3.6.8 (3) (preprint 3.54).
5. *Regularity at* 0. The sc$^\infty$-germ property from the lifts (above); $\sigma(0) = 0$; and $D\sigma(0) = 0$ within the whole cone $K \cap C'$, from $D\sigma(0) = 0$ within $V' = O \cap (K \cap C')$ with O open, since derivatives within a set do not change when the set is intersected with a neighbourhood of the point.
6. *Condition (b).* From the projection property of Y, by the lemma above.

Without the last item this is the plain form `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart`. *What the statement gives about d.* The dimension is existentially quantified in the statement; the proof takes $d = \dim\ker D(h+t)(0)$, the dimension of the kernel of the linearization of the push-forward, and the statement does not identify d with $\dim\ker f'(x_0)$ (the docstring's "$n = \dim\ker f'(x_0)$" is not part of the statement). The local constancy of §III.37 makes d intrinsic in another way: it is the dimension of every datum of Definition 3.2.1 (preprint 3.25) for the solution set at points of the chart domain.

What is missing at this stage, and where it is supplied. Condition (a), the sc-smoothness of σ on V' in the set sense, is not provided by Theorem 3.5.11 (1) (preprint Corollary 3.47), which gives the sc$^\infty$-germ property at 0 only (item 82); it is obtained in §III.36 by re-centring the data at the nearby zeros, which needs the data of this section at every zero near x_0, hence good position there (Theorem 3.6.8 (1) (preprint 3.54), §§III.32–III.33). The solution set here is the one in V_x; §III.37 restricts the chart so that it describes the global solution set.

III.35 The re-centring lemma

The text's proof of Theorem 3.2.3 (preprint 3.27) begins (preprint p. 84) by showing that the map δ of Definition 3.2.1 (preprint 3.25) is an sc$^\infty$-germ near *every* point $v_0 \in V$, not only near 0: take the point $x_0 = \varphi^{-1}(v_0 + \delta(v_0))$ of M, the data (ψ, N', Y', V', τ) of Definition 3.2.1 (preprint 3.25) at x_0, and consider $w \mapsto \pi \circ \varphi \circ \psi^{-1}(w + \tau(w))$; its linearization at 0 is claimed to be an isomorphism $N' \to N$, so it is a local diffeomorphism near 0, and $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$ shows that δ is regular near v_0. For an arbitrary set with the tangent germ property the claim is false: δ need not even be differentiable at v_0 (the curve with a vertical tangent of §III.24), and that is why condition (a) of `IsRegular` had to be added to Theorem 3.2.3 (preprint 3.27). What the argument really needs is that δ be C^1 at v_0. The formal *re-centring lemma* is the text's argument under that hypothesis, for two graph descriptions of the same set: one in a chart χ_1 by a map σ_1 of class C^1 on all of its domain, one in a chart χ_2 centred at the point over v_0 by a map σ_2 with the level-wise C^k-lifts of Theorem 3.5.11 (1) (preprint Corollary 3.47) at 0. Its conclusion is that σ_1 is C^k into every level $E_{1,m}$ near v_0, and that the two subspaces have the same dimension. Section III.36 applies it to the shrunk data of Definition 3.2.1 (preprint 3.25); for the solution set of an sc-Fredholm section, §III.37 supplies the second description at every nearby zero. Four ingredients are proved first: the inverse function machinery of Proposition 3.8.17 (preprint 3.73) with the hypotheses its proofs use, a local inverse theorem for C^k homeomorphisms between relatively open subsets of convex sets, Sard's lemma in fixed dimension, and two facts about relatively open subsets of convex sets.

The inverse function machinery with the hypotheses its proofs use `Chapter3/SolutionGermRecentre · InvDataW`. On a finite-dimensional space V, let $U \subseteq V$ be convex with unique differentiability at each of its points, $f : V \to V$ of class C^j on U with $j \geq 1$, $\|Df(x) - 1\| \leq \frac{1}{2}$ for $x \in U$ (derivatives within U), and let the image $f(U)$ have unique differentiability at each of its points.

These are the data of Proposition 3.8.17 (preprint 3.73) as formalized in §III.16 (`InvData`, `Chapter3/QuadrantInverse · InvData`) with the partial quadrant removed: `InvData` asks for a partial quadrant C, U relatively open in C and convex, $f(U) \subseteq C$, and that x be an interior point of C iff $f(x)$ is; from the last hypothesis it derives that $f(U)$ is relatively open in C `Chapter3/QuadrantInverse · InvData.image_relOpen`, and the only use of that openness in the regularity proof is the unique differentiability of $f(U)$. The weak data keep convexity and the derivative bound and replace the quadrant hypotheses by unique differentiability of U and of $f(U)$. The proofs are those of §III.16, retold here because they now stand on their own.

The bi-Lipschitz estimate (Lemma 3.8.18 (preprint 3.74), (70)). $f - \mathrm{id}$ has derivative $Df - 1$ of norm $\leq \frac{1}{2}$ on the convex U, so by the mean value inequality it is a $\frac{1}{2}$-contraction there: $|(f(x') - x') - (f(x) - x)| \leq \frac{1}{2}|x' - x|$ `Chapter3/SolutionGermRecentre · InvDataW.norm_sub_sub_le`. Writing $x' - x = (f(x') - f(x)) - ((f(x') - x') - (f(x) - x))$ gives $\frac{1}{2}|x' - x| \leq |f(x') - f(x)|$ `Chapter3/SolutionGermRecentre · InvDataW.half_norm_le`, hence f is injective on U `Chapter3/SolutionGermRecentre · InvDataW.injOn`. *The derivative is invertible*: $Df(x) = 1 - (1 - Df(x))$ with $\|1 - Df(x)\| \leq \frac{1}{2} < 1$ is a unit by the Neumann series `Chapter3/SolutionGermRecentre · InvDataW.derivUnit Chapter3/SolutionGermRecentre · InvDataW.isUnit_deriv`. *The inverse* g on $f(U)$ is the inverse of f on U (a choice of preimage, unique by injectivity) `Chapter3/SolutionGermRecentre · InvDataW.inv Chapter3/SolutionGermRecentre · InvDataW.inv_apply`; the estimate gives $|g(y') - g(y)| \leq 2|y' - y|$ `Chapter3/SolutionGermRecentre · InvDataW.norm_inv_sub_le`, so g is Lipschitz, hence continuous on $f(U)$ `Chapter3/SolutionGermRecentre · InvDataW.continuousOn_inv`.

Lemma 3.8.21 (preprint 3.77) `Chapter3/SolutionGermRecentre · InvDataW.hasFDerivWithinAt_inv`: g is differentiable within $f(U)$ at every $y \in f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$. With $x = g(y)$: as $y' \to y$ in $f(U)$, $g(y') \to x$ in U (continuity), so $f(g(y')) - f(x) - Df(x)(g(y') - x) = o(|g(y') - x|)$, and $|g(y') - x| \leq 2|y' - y|$ turns this into $o(|y' - y|)$; applying the bounded operator $Df(x)^{-1}$ and using $f(g(y')) = y'$ gives $g(y') - g(y) - Df(x)^{-1}(y' - y) = o(|y' - y|)$. No hypothesis on $f(U)$ enters here.

Proposition 3.8.17 (preprint 3.73), the regularity of the inverse `Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv`: g is C^j on $f(U)$. First, $x \mapsto [Df(x)]^{-1}$ is C^{j-1} on U: the derivative within U of a C^j map is C^{j-1} there because U has unique differentiability, and inversion is smooth near the units `Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inverse_deriv`. Then by induction on $i \leq j$: g is C^0 (continuous); if g is C^i with $i < j$, then g has the derivative $y \mapsto [Df(g(y))]^{-1}$ at every point of $f(U)$, which is C^i as the composite of the C^{j-1} map $x \mapsto [Df(x)]^{-1}$ with g, and a map with a C^i derivative on a set of unique differentiability is C^{i+1} there `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`. This last step is the one place where the unique differentiability of $f(U)$ is used. It is the bootstrapping of §III.16 in place of the text's iterated tangent maps (Lemma 3.8.20 (preprint 3.76)).

Relatively open subsets of convex sets. Let $D \subseteq V_1$ be convex with nonempty interior and $A = O \cap D$ with O open. Then A has unique differentiability at each of its points (a convex set with interior has it, and intersecting with an open set keeps it) `Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex`; and if A is nonempty, it has nonempty interior: for $a \in A$ and an interior point w of D, the points $a + t(w - a)$ with $0 < t$ small lie in O by continuity and in the interior of D, because the open segment from a point of the closure of a convex set to an interior point lies in the interior `Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex`.

Sard's lemma in fixed dimension `Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty`. If $f : V_1 \to V_2$ between finite-dimensional spaces is differentiable on s and $f(s)$ has nonempty interior, then $\dim V_2 \leq \dim V_1$. Suppose $\dim V_1 < \dim V_2$. There is a linear surjection $p : V_2 \to V_1$ (send the first $\dim V_1$ vectors of a basis of V_2 to a basis of V_1 and the rest to 0) `Chapter3/SolutionGermRecentre · exists_linearMap_surjective`. The map $f \circ p$ on $p^{-1}(s) \subseteq V_2$ has the derivative $Df(p(x)) \circ p$, an endomorphism of V_2 whose range lies in the range of $Df(p(x))$, of dimension $\leq \dim V_1 < \dim V_2$; so it is not injective and its determinant vanishes. By Mathlib's change-of-variables estimate, a set on which a map is differentiable with vanishing Jacobian determinant has an image of Haar measure zero `Mathlib/MeasureTheory/Function/Jacobian · MeasureTheory.addHaar_image_eq_zero_of_det_fderivWithin_eq_zero`. But $f(s) \subseteq$

$(f \circ p)(p^{-1}(s))$ since p is onto, and the nonempty open interior of $f(s)$ has positive Haar measure: a contradiction. The lemma uses only differentiability (not continuity of the derivative) and only the finite dimension of both spaces.

A C^k homeomorphism with invertible derivative has a C^k inverse `Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse`. Let $D_1 \subseteq V_1$, $D_2 \subseteq V_2$ be convex with nonempty interior, A relatively open in D_1, β of class C^k on A with $k \geq 1$, $\beta(A)$ relatively open in D_2, α continuous on $\beta(A)$ with $\alpha(\beta(w)) = w$ for $w \in A$, and $D\beta(w_0) = L$ (derivative within A) a linear isomorphism $V_1 \to V_2$ at some $w_0 \in A$. Then there is a relatively open $B' \subseteq \beta(A)$ in D_2 containing $\beta(w_0)$ on which α is C^k. *Proof.* Normalize: $\hat{\beta} = L^{-1} \circ \beta$ is C^k on A with $D\hat{\beta}(w_0) = 1$. Its derivative is continuous within A (C^k, $k \geq 1$, on a set of unique differentiability), so $\|D\hat{\beta}(w) - 1\| \leq \frac{1}{2}$ for $w \in A$ in a ball $B(w_0, r_1)$; take $r \leq r_1$ so small that $B(w_0, r) \subseteq O$, where $A = O \cap D_1$. The set $U = D_1 \cap B(w_0, r)$ is convex, lies in A, is relatively open in D_1 (hence has unique differentiability), and the derivatives of $\hat{\beta}$ within U and within A agree there (locality of the derivative within a set), so $\|D\hat{\beta} - 1\| \leq \frac{1}{2}$ on U. Its image $\beta(U) = \beta(A) \cap \alpha^{-1}(B(w_0, r))$ — because α inverts β on A — is relatively open in D_2 by the continuity of α, so $\beta(U)$, and with it $\hat{\beta}(U) = L^{-1}(\beta(U))$, has unique differentiability. So $(U, \hat{\beta}, k)$ are weak inverse data, and $\hat{\beta}^{-1}$ is C^k on $\hat{\beta}(U)$ (Proposition 3.8.17 (preprint 3.73) above). On $B' = \beta(U)$, $\alpha = \hat{\beta}^{-1} \circ L^{-1}$, since $\alpha(\beta(w)) = w = \hat{\beta}^{-1}(\hat{\beta}(w))$; hence α is C^k on B'. The relative openness of $\beta(U)$ comes from the continuity of the given inverse α, not from a degeneracy index; that is the point of the weak data.

The re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. Let χ_1, χ_2 be charts of the M-polyfold X `Chapter2/MPolyfold · MPolyfold.IsChart.compatible`, with models (O_i, C_i, E_i), and $S \subseteq X$. Suppose:

- (in χ_1) $K_1 \subseteq E_1$ is a finite-dimensional smooth subspace in good position to C_1, Y_1 a closed algebraic complement of K_1, V_1 relatively open in $K_1 \cap C_1$, $\sigma_1 : K_1 \to E_{1,0}$ with values in Y_1 and of class C^1 on V_1; $V_{x,1}$ is open in X inside the domain of χ_1 and $\chi_1(S \cap V_{x,1}) = \{v + \sigma_1(v) \mid v \in V_1\}$;
- (in χ_2) the same kind of data K_2, Y_2, $V_2 \ni 0$, σ_2 with values in Y_2, continuous on V_2, $\sigma_2(0) = 0$, $D\sigma_2(0) = 0$ within V_2, and for every $k \geq 1$ a relatively open $V_{2,k} \subseteq V_2$ containing 0 and a map $\sigma_{2,k} : K_2 \to E_{2,k}$ of class C^k on $V_{2,k}$ with $\iota_k \sigma_{2,k} = \sigma_2$ there (the lifts of Theorem 3.5.11 (1) (preprint Corollary 3.47)); $V_{x,2}$ open inside the domain of χ_2 with $\chi_2(S \cap V_{x,2}) = \{w + \sigma_2(w) \mid w \in V_2\}$;
- $y \in V_{x,1} \cap V_{x,2}$ with $\chi_2(y) = 0$, and $v_0 \in V_1$ with $\chi_1(y) = v_0 + \sigma_1(v_0)$.

Then $\dim K_1 = \dim K_2$, and for every level m and every k there is a relatively open $B' \subseteq V_1$ in $K_1 \cap C_1$ containing v_0 on which σ_1 takes values in $E_{1,m}$ and its level map $\sigma_1 : B' \to E_{1,m}$ is of class C^k.

The hypotheses are exactly what each step uses: the closedness of Y_i makes the projection π_i onto K_i along Y_i continuous; good position gives $K_i \cap C_i$ a nonempty interior in K_i (for unique differentiability and for Sard); smoothness of K_i gives the linear lifts of K_i to every level `Chapter3/BasicClass · ScBanach.finLift`. Note that V_1 need not contain 0 and σ_1 need not vanish anywhere: χ_1 is an arbitrary chart in which S is a C^1 graph.

Step 1: notation and the points of S. Write $\Gamma_1(v) = v + \sigma_1(v)$, $\Gamma_2(w) = w + \sigma_2(w)$, and π_1, π_2 for the continuous projections along Y_1, Y_2, so that $\pi_i \Gamma_i = \mathrm{id}$. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` For $v \in V_1$ the point $\chi_1^{-1}(\Gamma_1(v))$ lies in $S \cap V_{x,1}$ and χ_1 maps it back to $\Gamma_1(v)$, which therefore lies in O_1; likewise for $w \in V_2$. Since $\sigma_2(0) = 0$, $\Gamma_2(0) = 0$, and both $\chi_1^{-1}(\Gamma_1(v_0))$ and $\chi_2^{-1}(\Gamma_2(0))$ are y.

Step 2: the transition and the retractions onto the overlaps. The charts are compatible, so the transition $T = \chi_1 \circ \chi_2^{-1}$ is an sc-smooth map from the overlap retract $\chi_2(V_{\chi_2} \cap V_{\chi_1})$ to $\chi_1(V_{\chi_1} \cap V_{\chi_2})$, and $T' = \chi_2 \circ \chi_1^{-1}$ in the other direction `Chapter2/MPolyfold · MChart.transition Chapter2/MPolyfold · MChart.overlap`. The overlaps are sc-retracts `Chapter2/MPolyfold · MChart.isScRetract_overlap`: choose sc-smooth retractions r_2, r_1 onto them,

with domains U_{r_2}, U_{r_1}. Then $T\circ r_2$ is sc-smooth on U_{r_2}, $T'\circ r_1$ on U_{r_1}, $T(\chi_2(q))=\chi_1(q)$ on the domain of χ_2, and $T'\circ T=\mathrm{id}$ on the overlap. Here the formal proof departs from the text, which treats $\varphi\circ\psi^{-1}$ as a map one can differentiate along the graph; a transition is only an sc-smooth map between retracts, and every derivative below is taken of $T\circ r_2$ or $T'\circ r_1$, on relatively open subsets of partial quadrants.

Step 3: the parameter sets and the maps β, α. Let $A=\{w\in V_2\mid \chi_2^{-1}(\Gamma_2(w))\in V_{x,1}\}$ and $B_s=\{v\in V_1\mid \chi_1^{-1}(\Gamma_1(v))\in V_{x,2}\}$: the parameters whose points of S lie in both chart neighbourhoods. By continuity of $w\mapsto\chi_2^{-1}(\Gamma_2(w))$ on V_2 (and of its counterpart), A is relatively open in $K_2\cap C_2$, B_s in $K_1\cap C_1$; $0\in A$ and $v_0\in B_s$. Define

$$\beta(w)=\pi_1(T(\Gamma_2(w))),\qquad \alpha(v)=\pi_2(T'(\Gamma_1(v))).$$

For $w\in A$, with $q=\chi_2^{-1}(\Gamma_2(w))\in S\cap V_{x,1}\cap V_{x,2}$: $T(\Gamma_2(w))=\chi_1(q)$ lies on the graph of σ_1, say $=\Gamma_1(v)$ with $v\in V_1$; applying π_1 gives $\beta(w)=v$, so $T(\Gamma_2(w))=\Gamma_1(\beta(w))$, $\beta(w)\in B_s$, $\alpha(\beta(w))=w$, and $\Gamma_2(w)$ lies in the overlap. Symmetrically, for $v\in B_s$: $T'(\Gamma_1(v))=\Gamma_2(\alpha(v))$, $\alpha(v)\in A$, $\beta(\alpha(v))=v$. Moreover $\beta(0)=v_0$, and α is continuous on B_s ($\pi_2\circ\chi_2\circ\chi_1^{-1}\circ\Gamma_1$). Consequently, for every $A'\subseteq A$, $\beta(A')=\{v\in B_s\mid\alpha(v)\in A'\}$, which is relatively open in $K_1\cap C_1$ whenever A' is relatively open in $K_2\cap C_2$. This is the text's "local diffeomorphism" in its topological half, obtained without any derivative.
`Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`

Step 4: the lifted graph maps. For $j\geq 1$ let $A_j=A\cap V_{2,j}$ and $\Gamma_{2,j}(w)=\lambda_j(w)+\sigma_{2,j}(w)\in E_{2,j}$, where $\lambda_j: K_2\to E_{2,j}$ is the linear lift `Chapter3/BasicClass · ScBanach.finLift`. Then A_j is relatively open and contains 0, $\Gamma_{2,j}$ is C^j on A_j with $\iota_j\Gamma_{2,j}=\Gamma_2$ there, $\Gamma_{2,j}$ maps A_j into the level-j part of the domain of r_2, and $r_2(\Gamma_2(w))=\Gamma_2(w)$ for $w\in A_j$ (the graph points lie in the overlap, where r_2 is the identity). Its derivative within A_j at 0 is λ_j: indeed $\iota_j\circ D\sigma_{2,j}(0)=D\sigma_2(0)$ within A_j, which equals the derivative within V_2 (the two sets agree near 0) and so vanishes; ι_j is injective, so $D\sigma_{2,j}(0)=0$. Here the unique differentiability of A_j at 0 is that of a relatively open subset of the convex cone $K_2\cap C_2$ with interior `Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex`.

Step 5: β is C^j on A_j. On A_j, $\beta=\pi_1\circ((T\circ r_2)\circ\iota_j)\circ\Gamma_{2,j}$, because r_2 fixes $\Gamma_2(w)=\iota_j\Gamma_{2,j}(w)$. The map $(T\circ r_2)\circ\iota_j$ is C^j on the level-j part of U_{r_2} — Proposition 1.2.3 (preprint 1.18): an sc^j map is C^j from level j to level 0 `Chapter1/Levels · IsSck.contDiffOn_base` — and π_1 is continuous linear; so β is C^j on A_j. This is where the lifts are indispensable: σ_2 itself is only continuous into level 0, and $T\circ r_2$ is not differentiable on level 0.

Step 6: the chain rule at 0. Take $j=2$; then β is C^2, in particular C^1, on A_2, with derivative $D\beta=D\beta(0)$ within A_2 at 0. Read the lift on level 1: $\Gamma_{2,1}=\iota_{1,2}\circ\Gamma_{2,2}$, with $\iota_1\Gamma_{2,1}=\Gamma_2$ on A_2, values in the level-1 part of U_{r_2}, derivative $\iota_{1,2}\lambda_2$ at 0, and $\iota_1\iota_{1,2}\lambda_2(n)=n$ for $n\in K_2$. Let $\mathcal{D}=D(T\circ r_2)(\Gamma_{2,1}(0))$, the sc-derivative at the level-1 point over 0. The map $w\mapsto T(r_2(\iota_1\Gamma_{2,1}(w)))$ has, by the chain rule for the sc^1 map $T\circ r_2$ (differentiable from level 1 to level 0) and the differentiable $\Gamma_{2,1}$, the derivative $n\mapsto\mathcal{D}n$ at 0 within A_2. On A_2 this map equals $\Gamma_1\circ\beta$ (Step 3 and $r_2\circ\Gamma_2=\Gamma_2$), and $\Gamma_1\circ\beta$ has, at 0, the derivative $n\mapsto D\beta\,n+D\sigma_1(v_0)D\beta\,n$ by the chain rule — here the C^1-property of σ_1 *at* v_0, within V_1, is used, together with $\beta(A_2)\subseteq V_1$ and $\beta(0)=v_0$. By uniqueness of the derivative within A_2 at 0,

$$\mathcal{D}n=D\beta\,n+D\sigma_1(v_0)\,D\beta\,n\qquad(n\in K_2).$$

Step 7: $D\beta(0)$ is injective. Let $D\beta\,n=0$; then $\mathcal{D}n=0$. The composite $(T'\circ r_1)\circ(T\circ r_2)$ is sc-smooth on U_{r_2} ($T\circ r_2$ maps U_{r_2} into the overlap retract of χ_1, hence into U_{r_1}), and it equals r_2 there: $r_2(u)$ lies in the overlap, $T(r_2(u))$ in the other overlap where r_1 is the identity, and $T'\circ T=\mathrm{id}$ on the overlap. By the chain rule for sc^1 maps between relatively open subsets of partial quadrants `Chapter2/DegeneracyIndex · scDeriv_comp_apply`, $D((T'\circ r_1)\circ(T\circ r_2))(\Gamma_{2,1}(0))n=D(T'\circ r_1)(\cdot)\,\mathcal{D}n=0$. Since the composite agrees with r_2 on the relatively open U_{r_2}, its derivative at the level-1 point is

$Dr_2(\Gamma_{2,1}(0))$ Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen. And $Dr_2(\Gamma_{2,1}(0))n = n$: the family $w \mapsto \iota_1\Gamma_{2,1}(w)$ consists of fixed points of r_2, so the derivative of r_2 fixes its tangent vectors at 0, which are the vectors $\iota_1\iota_{1,2}\lambda_2(n) = n$ Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed — the tangent space K_2 of the graph lies in the tangent space of the retract. So $n = 0$. Hence $D\beta(0)$ is injective and $\dim K_2 \le \dim K_1$.

Step 8: surjectivity by counting dimensions. β is differentiable on A_2, and $\beta(A_2)$ is a nonempty relatively open subset of the convex cone $K_1 \cap C_1$ (Step 3; it contains $\beta(0) = v_0$), which has nonempty interior in K_1 by good position; so $\beta(A_2)$ has nonempty interior Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex, and Sard's lemma in fixed dimension gives $\dim K_1 \le \dim K_2$ Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty. So $\dim K_1 = \dim K_2$, the injective $D\beta(0)$ is bijective, and $L = D\beta(0) : K_2 \to K_1$ is a linear isomorphism. This is the first conclusion of the lemma, and also the reason why the dimension of a set with two such descriptions cannot jump.

Step 9: the inverse, and the regularity of σ_1. Fix m and k and take $j = m + k + 2$. By Step 5, β is C^j, hence $C^{\max(k,1)}$, on A_j, $\beta(A_j) \subseteq B_s$, and its derivative within A_j at 0 is again L (the derivatives within A_j and within A_2 agree at 0, both sets being relatively open near 0 in the same cone). Now apply the local inverse theorem above with $D_1 = K_2 \cap C_2$, $D_2 = K_1 \cap C_1$ (convex with interior), $A = A_j$, the continuous inverse α on $\beta(A_j)$ and $w_0 = 0$ Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse: there is a relatively open $B' \subseteq \beta(A_j)$ in $K_1 \cap C_1$ containing v_0 on which α is $C^{\max(k,1)}$, with $\alpha(B') \subseteq A_j$. Next the level-m map

$$\Psi(w) = (T \circ r_2)_m(\iota_{m,k+2}\,\Gamma_{2,j}(w)) - \lambda_m(\beta(w)) \in E_{1,m}, \qquad w \in A_j,$$

where $\iota_{m,k+2} : E_{2,m+k+2} \to E_{2,m}$ and λ_m is the lift of K_1 to level m. It is C^k on A_j: the sc^{k+2} map $T \circ r_2$ is C^{k+2} from level $m+k+2$ to level m (Proposition 1.2.3 (preprint 1.18) Chapter1/Levels · IsSck.contDiffOn_levels), $\Gamma_{2,j}$ is C^j into level $j = m+k+2$, and $\lambda_m \circ \beta$ is C^j. Including into level 0, $\iota_m\Psi(w) = T(r_2(\Gamma_2(w))) - \beta(w) = \Gamma_1(\beta(w)) - \beta(w) = \sigma_1(\beta(w))$. Hence for $v \in B'$, $v = \beta(\alpha(v))$ and $\sigma_1(v) = \iota_m\Psi(\alpha(v))$: σ_1 takes values in $E_{1,m}$ on B', its level-m map is $\Psi \circ \alpha$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq, which is C^k on B' as a composite of C^k maps, and $B' \subseteq B_s \subseteq V_1$. This is the text's identity $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$, written on level m through the lifts.

Comparison with the text. Four things in the formal argument are not in the sketch of preprint p. 84. (a) The isomorphism claim needs σ_1 differentiable at v_0, which for an arbitrary set with the tangent germ property fails; here σ_1 is assumed C^1 on all of V_1, and §III.36 arranges this by shrinking V_1 to the neighbourhood of 0 on which the germ property makes δ of class C^1. (b) Only the *injectivity* of the linearization comes from the chain rule, and it uses that the tangent space K_2 of the second graph lies in the tangent space of the overlap retract (Step 7). The symmetric argument for surjectivity would differentiate $T' \circ \Gamma_1 = \Gamma_2 \circ \alpha$ at v_0; but $T' \circ r_1$ is differentiable only from level 1 to level 0, so this would need σ_1 differentiable *into level* 1 at v_0, which is known near the centre of χ_1 (Theorem 3.5.11 (1) (preprint Corollary 3.47)) and not at v_0. Surjectivity therefore comes from the dimension count by Sard's lemma (Step 8), which at the same time shows that the dimension is locally constant. (c) The regularity of $\beta^{-1} = \alpha$ is not the quadrant inverse function theorem 3.72 (§III.16): that theorem asks for $d_C(x) = d_C(\beta(x))$ (formally: interior points to interior points), which here would be a statement about the intrinsic degeneracy index of S — not available. It is Lemma 3.8.21 (preprint 3.77) and the bootstrapping of Proposition 3.8.17 (preprint 3.73) under the weak data, with the relative openness of $\beta(U)$ supplied by the continuity of the inverse α. (d) The neighbourhood B' depends on (m,k) through the lift $\Gamma_{2,m+k+2}$; this is harmless, because sc-smoothness in the set sense is a local property of the level maps (§III.36).

III.36 Regular data by re-centring, and Theorem 3.2.3 (preprint 3.27) with locally constant dimension

This section turns the re-centring lemma of §III.35 into condition (a) of `IsRegular` (§III.24) and adapts Theorem 3.2.3 (preprint 3.27) to data whose dimension varies from point to point. The tool is the shrinking of the data of Definition 3.2.1 (2) (preprint 3.25): the chart is restricted to an open subset of its domain and the parameter set V to a smaller relatively open set, with the same N, Y and δ. The data at a point x are shrunk into the neighbourhood of 0 where δ is C^1; then, at every point of the shrunk graph, the data at that point (which a set with the tangent germ property has everywhere) provide the second graph description that the re-centring lemma needs.

Open subsets of tame retracts (told in §III.27). The restriction of a tame retraction $r : U \to U$ to $U \cap r^{-1}(W)$, W open (Proposition 2.3.1 (1) (preprint 2.10)) `Chapter2/Retracts · ScRetraction.restrict`, is tame: the degeneracy-index condition of Definition 2.5.2 (preprint 2.34) is inherited pointwise, and at a smooth point x of $O \cap W$ the sc-complement of T_xO of the original retraction serves, because the derivative of the restricted retraction at x is $Dr(x)$ — the two maps agree on a relatively open neighbourhood `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen` `Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`. So $O \cap W$ is a tame retract `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`, and the model $\varphi(s)$ of a chart restricted to an open s of its domain is tame when the model of φ is, since $\varphi(s) = O \cap W$ for some open W `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq` `Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`. The text uses this silently whenever it shrinks a chart of a tame M-polyfold.

Shrinking the data `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink`. Let $D = (\varphi, N, Y, V, \delta)$ be data of Definition 3.2.1 (2) (preprint 3.25) for M at x of dimension n `Chapter3/TangentGerm · MPolyfold.TangentGermChart`, s an open subset of the domain of φ containing x, $V' \subseteq V$ relatively open in $N \cap C$ containing 0, and M' a set with $\varphi(M' \cap s) = \{v + \delta(v) \mid v \in V'\}$. Then the chart φ restricted to s `Chapter2/MPolyfold · MChart.restrict` — a chart of the structure `Chapter2/MPolyfold · MPolyfold.IsChart.restrict` with a tame model by the paragraph above — together with the same N, Y, δ and the parameter set V', are data of Definition 3.2.1 (2) (preprint 3.25) for M' at x of dimension n. Every field is inherited: good position, the closed good complement, $\delta(0) = 0$, $D\delta(0) = 0$ and the germ property of δ concern N, Y, δ and the cone $N \cap C$ only, which are unchanged, and continuity on V' follows from continuity on V. The set M' may differ from M; this is how §III.37 passes between the solution set in a neighbourhood and the global solution set. The shrunk data have the source s, the model $\varphi(s)$, the same chart values, the same N, δ and projection π, and the parameter set V' `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_source` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_model_0` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_apply` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_N` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_V` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_δ` `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π`. They are regular `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular` as soon as δ is sc-smooth on V' in the set sense and the original data satisfy condition (b), $\pi(O) \subseteq C$: condition (b) passes to the smaller retract $\varphi(s) \subseteq O$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`.

The shrinking data `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_shrinkData`. For an open $W \ni x$ in X and an open $O'' \ni 0$ in N, let

$$s = \{y \in \operatorname{dom}\varphi \cap W \mid \pi\varphi(y) \in O''\}, \qquad V' = \{v \in V \cap O'' \mid \varphi^{-1}(v + \delta(v)) \in W\}.$$

Then s is open (the preimage of O'' under the continuous $\pi \circ \varphi$ on the open $\operatorname{dom}\varphi \cap W$) and contains x ($\pi\varphi(x) = \pi(0) = 0$); V' is relatively open in $N \cap C$ because $v \mapsto \varphi^{-1}(v + \delta(v))$ is continuous on V `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_continuousOn`, and contains 0 ($\varphi^{-1}(0) = x \in W$); $s \subseteq W$, $V' \subseteq V \cap O''$; and $\varphi(M \cap s) = \{v + \delta(v) \mid v \in V'\}$. For the last identity: a point $y \in M \cap s$ has $\varphi(y) = \Gamma(v)$ with $v = \pi\varphi(y) \in V$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`, $v \in O''$ because $y \in s$, and $\varphi^{-1}(\Gamma(v)) = y \in W$; conversely for $v \in V'$ the point $\varphi^{-1}(\Gamma(v))$ lies in M and in

the domain Chapter3/TangentGerm · MPolyfold.TangentGermChart.symm_Γ_mem, in W by definition of V', and $\pi\varphi$ of it is $v \in O''$ since $\pi\Gamma = \mathrm{id}$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_Γ.

The lifts from the germ property. Definition 3.2.1 (ii) (preprint 3.25), in the reading of §III.24 Chapter3/TangentGerm · IsScInftyGermAt, gives for every m, k a relatively open $V_{m,k} \ni 0$ on which δ takes values in E_m and is C^k into E_m. With $m = k$: for every $k \geq 1$ the level map of δ on $V_{k,k} \cap V$ is a C^k-lift δ_k of δ into E_k — the hypothesis of the re-centring lemma on the centred side Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_lift. With $(m, k) = (0, 1)$: δ is C^1 on a relatively open neighbourhood of 0 in $N \cap C$ Chapter3/TangentGerm · IsScInftyGermAt.contDiffOn_one. And $D\delta(0) = 0$ within V, since the data give it within $N \cap C$ and V agrees with $N \cap C$ near 0 Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.fderivWithin_V_zero.

Condition (a) of `IsRegular` by re-centring Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally. Let X be an M-polyfold (Hausdorff, paracompact), $M \subseteq X$, D data of Definition 3.2.1 (2) (preprint 3.25) for M at x of dimension n satisfying condition (b), $\pi(O) \subseteq C$, and $U \ni x$ open such that every point of $M \cap U$ has data of Definition 3.2.1 (2) (preprint 3.25) for M of *some* dimension. Then there are data D' for M at x of dimension n which are regular (`IsRegular`: (a) and (b)), whose chart domain lies in U, and such that every datum of Definition 3.2.1 (2) (preprint 3.25) for M at a point of M in the chart domain of D' has dimension n. (The proof constructs D' as the shrinking of D, with the same N, Y, δ; the statement records only the regularity, the domain and the dimension.)

Step 1: shrinking into the C^1 neighbourhood. By the germ property, δ is C^1 on $V_1 = O_1 \cap (N \cap C)$ with O_1 open and $0 \in O_1$. The shrinking data for $W = U$ and $O'' = O_1$ give $s \subseteq U$ and $V' \subseteq V \cap O_1$, so $V' \subseteq V_1$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_subset, with $\varphi(M \cap s) = \Gamma(V')$; let D_1 be the shrunk data for M itself. Then δ is C^1 on all of D_1's parameter set V': this is hypothesis (a) of the re-centring lemma on the side of D_1, and the reason for the shrinking.

Step 2: the re-centring at every point. Let $y \in M \cap s$ and let D'' be any data of Definition 3.2.1 (2) (preprint 3.25) for M at y, of dimension n'. Put $v_0 = \pi\varphi(y)$; then $v_0 \in V'$ and $\varphi(y) = \Gamma(v_0)$. The re-centring lemma applies with \$□ _1 = \$ the chart of D_1 (graph of δ over V' on its whole domain s, δ of class C^1 on V') and \$□ _2 = \$ the chart of D'', centred at y (graph of δ'' over V'' on its whole domain, $\delta''(0) = 0$, $D\delta''(0) = 0$ within V'', the lifts from the germ property of D''); all the remaining hypotheses — smooth finite-dimensional N, closed good complement, good position, relatively open parameter sets — are fields of the two data Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels. It gives $n = \dim N = \dim N'' = n'$, and for every m, k a relatively open $B' \subseteq V'$ in $N \cap C$ containing v_0 on which δ takes values in E_m and is C^k into E_m.

Step 3: δ is sc-smooth on V' in the set sense. V' is a relatively open subset of the partial quadrant $N \cap C$ of N with the constant sc-structure Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_isRelOpenInQuadrant, all of whose levels are V'. By Proposition 1.2.4 (preprint 1.19) in the form "maps every level into the corresponding level, and the level maps are C^∞" Chapter1/Levels · IsScSmooth.of_contDiffOn_levels, it suffices that for every m: (i) $\delta(V') \subseteq E_m$, and (ii) the level map $\delta : V' \to E_m$ is C^k for every k. For $v \in V'$ the point $y = \varphi^{-1}(\Gamma(v))$ lies in $M \cap s \subseteq M \cap U$, so by hypothesis there are data at y of some dimension, and Step 2 applies with $v_0 = \pi\varphi(y) = v$. For (i), the set B' for $(m, 0)$ contains v, and δ maps it into E_m. For (ii), fix k; around every $v \in V'$ the set $B' = O_B \cap (N \cap C)$ for (m, k) contains v, and the level map is C^k on $O_B \cap V' \subseteq B'$; a map that is C^k near every point of a set is C^k on the set Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_of_locally_contDiffOn. Here the dependence of B' on (m, k) (§III.35, Step 9) does no harm: the property is checked one (m, k) at a time, and locally.

Step 4: conclusion. With (a) from Step 3 and (b) inherited from D, the shrunk data D_1 are regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular; their chart domain is $s \subseteq U$; and by Step 2 every datum at a point of $M \cap s$ has dimension n.

What the statement gives for an arbitrary subset. The hypotheses of the theorem are about M and D only; no solution set, section or Fredholm property enters. If M has the n-dimensional tangent germ property Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty, every point of M has data of dimension n, so the

hypothesis "every point of $M \cap U$ has data of some dimension" holds with $U = X$. Hence, read off the statement (this is not a separate Lean theorem): *for any M with the tangent germ property, any data at x that satisfy condition (b) can be shrunk to regular data at x, of the same dimension.* Condition (a) of `IsRegular` is therefore not an independent hypothesis of Theorem 3.2.3 (preprint 3.27): if at every point of M some n-dimensional data satisfy (b), the regular data that Theorem 3.2.3 (preprint 3.27) asks for exist at every point `Chapter3/TangentGerm · theorem_3_27`. (The inclusion $M \subseteq X_\infty$ of Definition 3.2.1 (1) (preprint 3.25) is not used.) The counterexample to the text's step "δ is an sc$^\infty$-germ near every v_0" (§III.24) is not contradicted: there the data are regular only after shrinking V away from the point with a vertical tangent, which is what Step 1 does. What remains a genuine restriction is condition (b), which fails for some good complements (§III.24) and is obtained for solution sets by the choice of the complement (§III.29, §III.34).

The dimension is locally constant. The last clause of the theorem says more than the text: around every point where regular data exist, all data of Definition 3.2.1 (2) (preprint 3.25) at nearby points of M have the same dimension. The dimension of a set with the tangent germ property at varying dimensions is therefore locally constant along M — but not necessarily globally constant, and Theorem 3.2.3 (preprint 3.27) as formalized fixes n (§III.24). For the solution set of an sc-Fredholm section, whose dimension $\dim \ker f'(x)$ can change between components, the fixed-dimension theorem would need S split first; instead Theorem 3.2.3 (preprint 3.27) is restated for locally constant dimension.

Manifolds with corners of locally constant dimension `Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc`. An M-polyfold structure is that of a manifold with boundary with corners of locally constant dimension if it has an equivalent atlas all of whose charts have models (O, C, E) with O relatively open in the partial quadrant C and E_0 finite-dimensional — the definition of §III.24 `Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners` without the requirement that every chart have dimension n. A manifold with corners of dimension n is one of locally constant dimension `Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCorners.isManifoldWithCornersLoc`.

Theorem 3.2.3 (preprint 3.27) with locally constant dimension `Chapter3/GlobalIFTTheorems · theorem_3_27_loc`. Let X be an M-polyfold, $M \subseteq X$ nonempty, and suppose every $x \in M$ has regular data of Definition 3.2.1 (2) (preprint 3.25) for M of some dimension $n(x)$. Then M is a sub-M-polyfold of X; its induced structure (Proposition 2.3.19 (preprint 2.21)) is tame and that of a manifold with boundary with corners of locally constant dimension; and for every $x \in M$, all regular data d at x of any dimension and every base point $a_0 \in M$ (the value of the chart inverse off its domain), the chart $y \mapsto \pi\varphi(y)$ of d is a chart of the induced structure whose domain contains x. The proof is that of Theorem 3.2.3 (preprint 3.27) (§III.24) with the dimension quantified pointwise; every step there is either pointwise or does not use n.

Sub-M-polyfold `Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isSubMPolyfold`. Definition 2.3.18 (preprint 2.20) is pointwise: at $a \in M$ take regular data d (of whatever dimension); the open set $U_0 = \varphi^{-1}(\Sigma \cap O)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_U₀` and the sc-smooth retraction $s = \varphi^{-1} \circ t \circ \varphi$ onto $M \cap U_0$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s` are those of §III.24, exactly as in the fixed-dimension proof `Chapter3/TangentGerm · theorem_3_27_isSubMPolyfold`.

The charts and the atlas. Let the charts be all maps $\pi\varphi$ of all regular data at all points of M, of all dimensions, with all base points `Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart`. Each is a chart of the induced structure `Chapter2/SubMPolyfold · subPolyfold`: it is compatible with every chart $\varphi'|_{W \cap M}$ of the induced atlas, by the lemma of §III.24, whose proof never compares dimensions `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart` `Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isChart`. Any two of them are compatible, by the locality of compatibility: both are compatible with every chart of the induced atlas, whose domains cover M `Chapter2/SubMPolyfold · MPolyfold.subAtlas` `Chapter2/MPolyfold · MChart.compatible_of_locally` `Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc_compatible`. The argument never mentions the dimension of

either chart. They form an atlas: every $y \in M$ has regular data, and the chart of these data, with the base point a fixed element of M (here $M \neq \emptyset$ is used), contains y Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source Chapter3/GlobalIFTTheorems · MPolyfold.germAtlasLoc. This atlas is equivalent to the induced atlas of Proposition 2.3.19 (preprint 2.21), since each of its charts is a chart of the induced structure Chapter2/MPolyfold · MAtlas.equiv_iff Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_atlas_equiv.

Tame, and a manifold with corners. The model of each chart $\pi\varphi$ is $(V, N \cap C, N)$ with the identity retraction, a tame retract Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract; so the induced structure is tame, witnessed by the equivalent atlas just built Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isTame. Each model is relatively open in the partial quadrant $N \cap C$ of the finite-dimensional N Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner; so the induced structure is that of a manifold with boundary with corners of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isManifoldWithCornersLoc. The assembled statement collects these, with the chart property of every $\pi\varphi$ Chapter3/GlobalIFTTheorems · theorem_3_27_loc.

As for Theorem 3.2.3 (preprint 3.27), the hypotheses are the regularity conditions at every point; by the observation above, condition (a) there can be replaced by condition (b) plus shrinking. Theorem 3.6.11 (preprint 3.57) (§III.37) applies this version to the solution set, where regular data exist at every zero by §§III.34–III.36 and the dimension $\dim \ker f'(x)$ is locally constant by Step 2 of the theorem above; the fixed-dimension statement for each part S_n of the solution set is obtained from Theorem 3.2.3 (preprint 3.27) itself.

III.37 The implicit function theorems assembled: Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19), Corollary 3.1.27 (preprint 3.24), Theorems 3.6.10, 3.6.11 (preprint 3.56, 3.57)

The text derives all its implicit function theorems from Theorem 3.6.8 (preprint 3.54): (1) good position at every zero near x, (2) the sub-M-polyfold structure of the solution set near x, through the tangent germ property and Theorem 3.2.3 (preprint 3.27). In the formalization the pieces are: the data of Definition 3.2.1 (preprint 3.25) at the centre, with condition (b) (§III.34); good position at the nearby zeros, under an added hypothesis (§III.33); the re-centring, which turns data at all nearby points into regular data and shows that the dimension is locally constant (§§III.35–III.36); and Theorem 3.2.3 (preprint 3.27), with fixed dimension (§III.24) or with locally constant dimension Chapter3/GlobalIFTTheorems · theorem_3_27_loc (§III.36). This section assembles them. Throughout, f is a section of a strong bundle $P : Y \to X$ over an M-polyfold, "the charts have tame base retractions" means that the base retraction of every strong bundle chart compatible with the atlas is tame (the formal reading of "tame strong bundle", §III.23), $S = \{x \in X : f(x) = 0\}$ is the global solution set Chapter3/GlobalIFT · solutionSet, and good position is Definition 3.6.6 (preprint 3.53) in the chart form Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm (§III.21).

The data for the global solution set Chapter3/GlobalIFTTheoremsG · theorem_3_54_tangentGermChart_solutionSet. *If the charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) is in good position, then S has at x_0 data of Definition 3.2.1 (preprint 3.25) of some dimension d with $\pi(O) \subseteq C$.* §III.34 gives data D for $S_{V_x} = \{y \in V_x : f(y) = 0\}$. Restrict its chart to the open set $s = \text{source} \cap V_x \ni x_0$, keeping N, Y, V, δ Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink Chapter2/MPolyfold · MChart.restrict: the graph identity holds for the global S because $S \cap s = S_{V_x} \cap \text{source}$. The restricted model $\varphi(s)$ is a tame retract (an open subset of a tame retract, §III.36), and condition (b) passes to it, since $\varphi(s) \subseteq O$ Chapter2/MPolyfold · MChart.image_subset_model and the projection is unchanged Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π.

Theorem 3.6.8 (2) (preprint 3.54), from the nearby good position Chapter3/GlobalIFTTheoremsG · theorem_3_54_two. *Let the charts have tame base retractions, $f(x_0) = 0$, (f, x_0) in good position, and let every zero of f in*

an open $V_{x_0} \ni x_0$ be in good position. Then there is a dimension d and regular data of Definition 3.2.1 (preprint 3.25) of dimension d for S at x_0 (condition (a): δ sc-smooth on V in the set sense; condition (b): $\pi(O) \subseteq C$) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`*, on a chart whose source lies in V_{x_0}; and every datum of Definition 3.2.1 (preprint 3.25) for S at a point of S in that source has dimension d.* Only that f is a section is assumed; the regularity of f enters through the good position at x_0 and at the nearby zeros, which carry the sc-Fredholm germ data. *Proof.* Take the data D at x_0 above. Every $y \in S \cap V_{x_0}$ is a zero in good position, so the same theorem at y gives data of Definition 3.2.1 (preprint 3.25) for S at y, of some dimension. These are the hypotheses of the re-centring theorem of §III.36 `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally` for D (which satisfies (b)) and the open set V_{x_0}: it shrinks D to regular data D' with source in V_{x_0}, of the same dimension d, along which every datum has dimension d. This is the text's "proceeding as above we can construct a map σ" at every zero, made into a proof: the regularity condition (a), an added hypothesis of Theorem 3.2.3 (preprint 3.27) for general sets (§III.24), is a theorem for solution sets (items 150–151).

Theorem 3.6.8 (1) and (2) (preprint 3.54) `Chapter3/GlobalIFTTheoremsG · theorem_3_54`*. Let f be an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and let (f, x_0) be in good position* ***with a filler that satisfies condition (3) of Definition 3.1.4 (preprint 3.4) at the zeros of the filler near*** 0 `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`*. Then there is an open $V_x \ni x_0$ such that (1) (f, y) is in good position at every zero $y \in V_x$, and (2) the conclusions of Theorem 3.6.8 (2) (preprint 3.54) above hold with $V_{x_0} = V_x$.* (1) is §III.33 `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one`; (2) is the previous theorem, fed with (1), with the strengthened good position weakened to good position `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm` and the section property of the sc-Fredholm section. The filler condition was an **added hypothesis** when this form was proved: Definition 3.1.4 (preprint 3.4) imposes condition (3) at 0 only, and the surjectivity of $f'(y)$ at the nearby zeros is equivalent to it there (§III.33; items 100, 105–106); the kernel half of (1) needs no added hypothesis (§III.32). Since the second addendum the condition is a consequence of good position (§III.45), and Theorem 3.6.8 (preprint 3.54) is proved with the text's hypotheses (§III.43). Part (3) of the text's proof, the graph description, is §III.29 `Chapter3/GlobalIFTG · theorem_3_54_graph_proj`, and part (4), the kernel at the nearby zeros in the chart at x_0, is §III.32 `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_nearby_kernel`. The text's neighbourhood V is the V_x of (3), shrunk in (2) to the chart source of the re-centred data.

Theorem 3.1.26 (preprint 3.23), boundary case, from the nearby good position `Chapter3/GlobalIFTTheoremsG · theorem_3_23_of_nearby`*. Let f be an sc-Fredholm section, the charts with tame base retractions, $f(x) = 0$, (f, x) in good position ($f'(x)$ onto with kernel in good position, Definition 3.6.6 (preprint 3.53)), and every zero in an open $V_{x_0} \ni x$ in good position. Then there are an open U with $x \in U \subseteq V_{x_0}$ and a dimension d such that every zero in U is a smooth point, $S(f, U) = \{y \in U : f(y) = 0\}$ is a sub-M-polyfold of X, and its induced M-polyfold structure (Proposition 2.3.19 (preprint 2.21))* `Chapter2/SubMPolyfold · subPolyfold` *is tame and equivalent to that of a d-dimensional manifold with boundary with corners* `Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners`*. Proof.* Theorem 3.6.8 (2) (preprint 3.54) at x gives regular data D of dimension d with source $U \subseteq V_{x_0}$ and the local constancy of the dimension; let $S' = S(f, U)$. At every $y \in S'$, Theorem 3.6.8 (2) (preprint 3.54) at y, with the open neighbourhood $U \cap V_{x_0}$ in place of V_{x_0}, gives regular data D_y for S with source in $U \cap V_{x_0}$, of a dimension d_y; since $y \in S \cap U$, the local constancy along D gives $d_y = d$. Because the source of D_y lies in U, $S' \cap \text{source} = S \cap \text{source}$, so D_y restricted to its own source is a datum for S' `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink`, still regular `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`. So every point of S' has regular data of dimension d, $S' \ni x$ is nonempty, and Theorem 3.2.3 (preprint 3.27) with fixed dimension (§III.24) `Chapter3/TangentGerm · theorem_3_27` gives the sub-M-polyfold with its tame induced structure equivalent to a d-dimensional manifold with boundary with corners. The zeros are smooth points because an sc-Fredholm section is regularizing `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`. The text's "a uniquely determined structure … the same as the one defined in Proposition 2.21" is the induced structure itself, with the equivalent atlas of the charts $\pi \circ \varphi$; uniqueness is not stated separately.

Theorem 3.1.26 (preprint 3.23) Chapter3/GlobalIFTTheoremsG · theorem_3_23. *The same under the added filler hypothesis at x instead of the nearby good position, with in addition: every zero in U is in good position.* Theorem 3.6.8 (1) (preprint 3.54) (§III.33) supplies the open V_x of nearby good position, and the previous theorem applies with $V_{x_0} = V_x$.

Theorem 3.1.22 (preprint 3.19), interior case. The first clause, "$f'(x_0)$ is sc-Fredholm", is Proposition 3.6.5 (preprint 3.52) in the chart at x_0 Chapter3/GlobalIFTTheoremsG · theorem_3_19_linearization Chapter3/FredholmGermLinearizationG · proposition_3_52, applied to the sc-Fredholm germ contained in the good position: $Dg(0)$, for a derivative family Dg of the filling germ (§III.21), and $Tf(x_0) = Dg(0)|_{T_0O}$ are Fredholm on level 0 with the same index. The statement inherits the index clause of that proposition, "$= n - N$ for some natural numbers n, N", which carries no information; the index as the $n - N$ of the conjugating basic germ is §III.30 Chapter3/FredholmGermIndexG · proposition_3_52_index. *The theorem* Chapter3/GlobalIFTTheoremsG · theorem_3_19: *let $d_X \equiv 0$ Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary, f an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position with the filler condition at the nearby zeros. Then there are an open $U \ni x_0$ and d such that every zero in U is in good position (in particular $f'(y)$ is onto) and a smooth point, and $S(f, U)$ is a sub-M-polyfold whose induced structure is equivalent to a d-dimensional manifold with boundary with corners and has degeneracy index 0 everywhere. Proof.* Theorem 3.1.26 (preprint 3.23) gives everything but the last clause; for that, Lemma 2.4.6 (preprint 2.26), $d_S(y) \le d_X(y)$ Chapter2/SubMPolyfold · lemma_2_26, and $d_X = 0$. So the interior case is the boundary case plus Lemma 2.4.6 (preprint 2.26); $d_X \equiv 0$ is used for nothing else. *Two readings and two omissions* (item 154), all resolved in the second addendum. (a) The text's hypothesis is "$f'(x_0)$ surjective"; good position adds condition (2) of Definition 3.6.6 (preprint 3.53), which the text calls an empty condition when $d_X(x) = 0$. Formally condition (2) is stated in the chart for the chart's partial quadrant, so this form takes good position as in the boundary case; that the condition is void at $d_X = 0$ in a tame chart, and Theorem 3.1.22 (preprint 3.19) with "$f'(x_0)$ surjective", are §III.45. The filler condition at the nearby zeros was the added hypothesis of Theorem 3.6.8 (1) (preprint 3.54); the form with the text's hypotheses is §III.43. (b) "The dimension agrees with the Fredholm index": this statement quantifies d existentially; the proof takes $d = \dim\ker D(h+t)(0)$ (§III.34), and $\operatorname{ind} D(h+t)(0) = \dim P - \dim Q = \operatorname{ind} f'(x_0)$ holds for the conjugation data of the proof (§§III.18, III.30); the graph description with the conjugation exposed, and $\dim\ker = \operatorname{ind}$ for the surjective $D(h+t)(0)$, give the dimension as the index in §III.44. (c) "$\ker f'(y) = T_yS(f,U)$": in the chart at x_0, $\ker D(h+t)(z) = N_v$, the tangent space of the graph (§III.32); the chart form of the statement is §III.45, and the identification with T_yS as a subspace of T_yX needs the linear structure of T_yX (gap G.13 of Chapter 2) and is not formalized.

Theorem 3.6.11 (preprint 3.57), with the text's hypotheses Chapter3/GlobalIFTTheoremsG · theorem_3_57. *Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with (f, x) in good position at every zero x, and let $S \neq \emptyset$. Then S is a sub-M-polyfold of X, its induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension* Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc *(an equivalent atlas of charts on relatively open subsets of partial quadrants of finite-dimensional spaces), and at every $x \in S$ there are regular data of Definition 3.2.1 (preprint 3.25) of a dimension $d(x)$ such that every datum at a point of S in their chart source has dimension $d(x)$. Proof.* At every $x \in S$, Theorem 3.6.8 (2) (preprint 3.54) with $V_{x_0} = X$ — the hypothesis "every zero near x in good position" is part of the global hypothesis — gives the regular data and the local constancy. Theorem 3.2.3 (preprint 3.27) with locally constant dimension (§III.36) Chapter3/GlobalIFTTheorems · theorem_3_27_loc then gives the sub-M-polyfold, the tameness and the corner atlas. **No hypothesis is added**: good position at every zero is the text's hypothesis, and it supplies at every point the nearby good position that the re-centring needs, so the gap of Theorem 3.6.8 (1) (preprint 3.54) does not enter (item 156). $S \neq \emptyset$ is needed for the formal charts, whose inverses need a point of S as a value off the chart (for $S = \emptyset$ the statement says nothing). "Tame strong bundle" is read as above. The text states no dimension; the theorem says that the dimension is locally constant, which the text's tangent germ property of a fixed dimension d near each point implies but does not state (item 152).

Corollary 3.1.27 (preprint 3.24) Chapter3/GlobalIFTTheoremsG · corollary_3_24 is Theorem 3.6.11 (preprint 3.57) without the last clause: if at every zero $f'(x)$ is onto with kernel in good position, the solution set is a

sub-M-polyfold with tame induced structure equivalent to a manifold with boundary with corners of locally constant dimension.

Theorem 3.6.10 (preprint 3.56), $\partial X = \emptyset$, with the text's hypotheses `Chapter3/GlobalIFTTheoremsG · theorem_3_56`. *Under the hypotheses of Theorem 3.6.11 (preprint 3.57) and $d_X \equiv 0$, S is a sub-M-polyfold whose induced structure is equivalent to a manifold with corners of locally constant dimension and has degeneracy index 0 everywhere*: Theorem 3.6.11 (preprint 3.57) and Lemma 2.4.6 (preprint 2.26), as for Theorem 3.1.22 (preprint 3.19). (The statement keeps only these two conclusions; the tameness of the induced structure and the regular data are those of Theorem 3.6.11 (preprint 3.57).) The text's hypothesis "$f'(x)$ surjective at every zero" is read as good position, which is the text's Definition 3.6.6 (preprint 3.53) for $d_X(x) = 0$; that condition (2) is void in a tame chart at $d_X = 0$, and Theorem 3.6.10 (preprint 3.56) with "$f'(x)$ surjective" and the tameness of the induced structure, are §III.45.

The parts of fixed dimension `Chapter3/GlobalIFTTheoremsG · theorem_3_57_dim`. Under the hypotheses of Theorem 3.6.11 (preprint 3.57) (without $S \neq \emptyset$), let S_n be the set of points of S at which S has data of Definition 3.2.1 (preprint 3.25) of dimension n `Chapter3/GlobalIFTTheorems · solutionSetDim`, and U_n the union of the sources of the regular data of dimension n along which the dimension is constantly n `Chapter3/GlobalIFTTheorems · solutionSetDimOpen`, an open set `Chapter3/GlobalIFTTheorems · isOpen_solutionSetDimOpen`. *Then S is the union of the S_n, the S_n are pairwise disjoint, $S_n = S \cap U_n$ and $S_n = S \cap F_n$ for a closed F_n (so S_n is open and closed in S), and each nonempty S_n is a sub-M-polyfold whose induced structure is tame and equivalent to an n-dimensional manifold with boundary with corners. Proof.* At every $x \in S$, Theorem 3.6.8 (2) (preprint 3.54) with $V_{x_0} = X$ gives regular data D_x of a dimension $d(x)$ with the local constancy. *The dimension at a point is well defined*: two data at x of dimensions n, m both have the dimension $d(x)$ by the local constancy along D_x, since x lies in its source; so the S_n cover S (by D_x) and are pairwise disjoint. $S_n = S \cap U_n$: for $x \in S_n$, $d(x) = n$ and D_x is one of the data whose sources make up U_n; conversely, if $x \in S$ lies in the source of regular data of dimension n along which the dimension is n, then $d(x) = n$ and D_x witnesses $x \in S_n$. *S_n is closed in S*: $S_n = S \setminus \bigcup_{m \neq n} U_m$, since a point of S lies in exactly one S_m; take F_n the complement of that open union. *Each nonempty S_n*: at $x \in S_n$, restrict D_x (of dimension n) to a datum for S_n — $S_n \cap$ source $= S \cap$ source because every point of S in the source has dimension n — still regular `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`, and apply Theorem 3.2.3 (preprint 3.27) with fixed dimension n `Chapter3/TangentGerm · theorem_3_27`.

Where the formal route departs from the text. Two steps of the text's proof are replaced (items 150, 157): the regularity of the solution germ away from the centre, which the proof of Theorem 3.2.3 (preprint 3.27) asserts for any set and which fails for general sets (§III.24), is proved for solution sets by re-centring with the C^1-property of σ on V' and the level-wise lifts at the nearby zero (§§III.35–III.36); and the relative openness of Σ, false for an arbitrary good complement, is obtained by choosing the complement (§III.29). With these, the global theorems 3.56, 3.57 and Corollary 3.1.27 (preprint 3.24) are proved with the text's hypotheses, and the local Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) under the hypothesis that the zeros near the point are in good position, which is Theorem 3.6.8 (1) (preprint 3.54); this section proves that one under the filler condition at the nearby zeros, which Definition 3.1.4 (preprint 3.4) does not impose, and §III.43 with the text's hypotheses. Not stated here: the uniqueness of the manifold structure separately from the equivalence of atlases; the dimension d of these statements is existentially quantified (only as the common dimension of all data of Definition 3.2.1 (preprint 3.25) for S near the point) — its identification with $\dim \ker f'(x) = \operatorname{ind} f'(x)$ is §III.44, and $\ker f'(y) = T_y S$ in the chart is §III.45.

III.38 The basic germ criterion: Theorem 3.7.1 (preprint 3.58)

Theorem 3.7.1 (preprint 3.58) (text): let U be a relatively open neighbourhood of 0 in a partial quadrant $C \subseteq E$ and $f : U \to F$ sc-smooth, with (1) $Df(x)$ sc-Fredholm at the smooth points, of constant index; (2) an sc-splitting $E = B \oplus X$, B finite-dimensional containing $\ker Df(0)$, with a compactness property: for $b_j \to b$ in $B \cap U$ small and (η_j) bounded in X_m with $Df(b_j)\eta_j = y_j + z_j$, $y_j \to 0$ in F_m, (z_j)

bounded in F_{m+1}, (η_j) has a convergent subsequence in X_m; (3) for every m and $\varepsilon > 0$, $|[D_2 f(b,0) - D_2 f(b,x)]h|_m \le \varepsilon |h|_m$ for b small and $x \in X_{m+1}$ small on level m. Then $f - f(0)$ is conjugated near 0 to a basic germ. The proof: $Y = Df(0)X$, $F = A \oplus Y$, P the projection onto Y; Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) and an index argument make $L(b) = P \circ Df(b)|_X$ sc-isomorphisms with uniform lower bounds; Lemma 3.7.4 (preprint 3.61) makes $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ a strong bundle isomorphism; and $H(b,x) = x - L(b)^{-1}Ph(b,x)$, $h = f - f(0)$, is a contraction by (3).

The coordinates. The formalization works in the coordinates of the text's proof: $E = \mathbb{R}^n \oplus X$ with $\mathbb{R}^n$ a finite-dimensional normed space P (constant sc-structure), $C = \widetilde{C} \oplus X$ with $\widetilde{C}$ a partial quadrant of P, $F = \mathbb{R}^N \oplus Y$ with $\mathbb{R}^N = Q$ finite-dimensional, and "$P\circ$" the Y-component. Condition (3) becomes `Chapter3/BasicGermCriterionProof · DerivCondition`: for every m and $\varepsilon > 0$ there is $r > 0$ with $|Df(b,0)(0,h) - Df(b,x)(0,h)|_m \le \varepsilon |h|_m$ for $h \in X_m$, $b \in \widetilde{C}$, $|b| < r$, $x \in X_{m+1}$, $|x|_m < r$, $(b,0)$ and (b,x) in U_{m+1} (the level operators of Proposition 1.2.1 (preprint 1.16)). Condition (2) becomes `Chapter3/BasicGermCriterionIndex · CompactnessCondition`: there is $r > 0$ such that for every m, every $b_j \to b_0$ in $\widetilde{C}$ with $(b_j, 0) \in U$ and $|b_0| < r$, every bounded $(\eta_j) \subseteq X_m$ with $Df(b_j,0)(0,\eta_j) = y_j + \iota z_j$, $y_j \to 0$ in F_m and (z_j) bounded in F_{m+1}, (η_j) has a subsequence converging in X_m. (The smallness of b is read as uniform in m, as the order of the text's quantifiers suggests; Lemma 3.7.3 (preprint 3.60) needs the property on every level on one neighbourhood.)

Theorem 3.7.1 (preprint 3.58) as formalized `Chapter3/BasicGermCriterionIndex · theorem_3_58`. Let U be relatively open in $\widetilde{C} \oplus X$ with $0 \in U$ and $f : U \to Q \oplus Y$ sc-smooth such that (1) for every $b \in \widetilde{C}$ with $(b,0) \in U$ the linearization $Df(b,0)$, an sc-operator `Chapter3/BasicGermCriterionIndex · derivParamOp`, is sc-Fredholm with the sc-index of $Df(0)$; (2) $h \mapsto (Df(0)(0,h))_2$ is bijective $X_0 \to Y_0$, and the compactness condition holds; (3) the derivative condition holds. Then there are an open $V_o \ni 0$ in P and sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ for $b \in V_o \cap \widetilde{C}$, such that the push-forward $k(b,x) = ((f - f(0))_1(b,x), L(b)^{-1}(f - f(0))_2(b,x))$ is a basic germ on $U \cap (V_o \oplus X)$ with contraction map $H(b,x) = x - L(b)^{-1}(f(b,x) - f(0))_2$, Φ is a strong bundle isomorphism covering the identity, and $f - f(0)$ is conjugated to a basic germ (Definition 3.1.11 (preprint 3.11)). The hypotheses differ from the text's in three places: (1) is asked only at the points $(b,0)$, $b \in \widetilde{C}$, which is all the proof uses; "constant index" is equality with the index at 0; and the text's "$B \supseteq \ker Df(0)$, $Y = Df(0)X$" is replaced by its consequence, the bijectivity of $P \circ Df(0)|_X$ on level 0. The reduction of a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$ to these coordinates is not formalized.

The proof has two halves, as in the text.

First half: Theorem 3.7.1 (preprint 3.58) from the family of sc-isomorphisms.

Theorem 3.7.1 (preprint 3.58), given the family `Chapter3/BasicGermCriterionProof · theorem_3_58_of_family`. In addition to the coordinates, f sc-smooth on U and (3), suppose: V_o open with $0 \in V_o$ and $(b,0) \in U$ for $b \in V_o \cap \widetilde{C}$; sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ on level 0 for $b \in V_o \cap \widetilde{C}$; and for every m a constant $c_m > 0$ with $c_m |h|_m \le |L(b)h|_m$ for $b \in V_o \cap \widetilde{C}$, $h \in X_m$. Then, with $U' = U \cap (V_o \oplus X)$, the conclusions above hold.

Step 1: $(b,h) \mapsto L(b)h$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus X$ `Chapter3/BasicGermCriterionProof · famMap_isScSmooth_of_deriv` (the text: "since B is a finite-dimensional sc-smooth space, the map $(b,x) \mapsto P \circ Df(b)x$ is sc-smooth"). It is $\mathrm{pr}_Y \circ Tf \circ J$ with the sc-operator $J(b,h) = ((b,0),(0,h))$ into the tangent space `Chapter3/BasicGermCriterionProof · tangentInjOp`; J maps into the tangent set because $(b,0) \in U$ lies on every level; and the tangent map of the sc-smooth f is sc-smooth.

Step 2: $(b,y) \mapsto L(b)^{-1}y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ `Chapter3/BasicGermCriterionProof · famInv_isScSmooth_of`. This is Lemma 3.7.4 (preprint 3.61) (§III.25) `Chapter3/InverseFamily · lemma_3_61` for the germ $\{V_o \cap \widetilde{C}\}$ with the uniform lower bounds, translated between maps on sets and germs on the special germ of a set: an sc^k-germ on $\{U\}$ is an sc^k-map on U `Chapter3/BasicGermCriterionProof · IsSckGerm.isSck_ofSet Chap-`

ter3/BasicGermCriterionProof · IsScSmoothGerm.isScSmooth_ofSet, by induction on k: the sc^0 case is a matter of definitions Chapter3/BasicGermCriterionProof · IsSc0Germ.isSc0_ofSet; for the step, the tangent germ of $\{U\}$ is the germ of the tangent set, and the derivative family of the germ agrees with the derivative of the map on the tangent set; the converse is `IsSck.isSckGerm` (§III.1).

Step 3: the strong bundle isomorphism Φ. For a family $M(b) : Y \to X$ of sc-isomorphisms put $\Phi_M((b,z),(c,y)) = ((b,z),(c,M(b)y))$ Chapter3/BasicGermCriterionProof · fiberIsoMap. If $(b,y) \mapsto M(b)y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ and the parameters of U' lie in $V_o \cap \widetilde{C}$, Φ_M is a strong bundle map $U' \lhd (Q \oplus Y) \to U' \lhd (Q \oplus X)$ covering the identity Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleMap: it is linear on the fibres; it preserves the filtration because $M(b)$ maps Y_k to X_k; on $[0]$ it is sc-smooth as the pairing of the projection with $(c,M(b)y)$, the latter the composite of the sc-operator $((b,z),(c,y)) \mapsto (b,y)$ Chapter3/BasicGermCriterionProof · paramFiber0p with $(b,y) \mapsto M(b)y$ Chapter3/BasicGermCriterionProof · fiberIsoMap_isScSmooth; on $[1]$, $\Phi_M[1]$ is Φ_{M^1} for the shifted family $M^1(b) = M(b) : Y^1 \to X^1$ Chapter3/BasicGermCriterionProof · shiftFam Chapter3/BasicGermCriterionProof · shiftFiber_fiberIsoMap, and $(b,y) \mapsto M^1(b)y$ is sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter3/BasicGermCriterionProof · famMap_shiftFam_isScSmooth — the text's "if we raise the index by one, the map is also sc-smooth by Proposition 1.17". With $M = L^{-1}$ (Step 2) and $M = L$ (Step 1) the two maps are inverse to each other, so $\Phi = \Phi_{L^{-1}}$ is a strong bundle isomorphism Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleIso.

Step 4: the push-forward and the contraction map. With $h = f - f(0)$ Chapter3/BasicGermCriterionProof · shiftedGerm, $\Phi_* h = k$, $k(b,x) = (h_1(b,x), L(b)^{-1}h_2(b,x))$ Chapter3/BasicGermCriterionProof · pushGerm358, and the X-component of k is $x - H(b,x)$ with $H(b,x) = x - L(b)^{-1}h_2(b,x)$ Chapter3/BasicGermCriterionProof · contrMap358 Chapter3/BasicGermCriterionProof · pushGerm358_snd. Both are sc-smooth on U': sums, pairings and the composite of $(b,y) \mapsto L(b)^{-1}y$ with $(b,x) \mapsto (b,h_2(b,x))$. On level m, $H_m(b,w) = w - L(b)_m^{-1}((f_m(b,w))_2 - (f_m(0))_2)$ Chapter3/BasicGermCriterionProof · levelMap_contrMap358, and $|L(b)_m^{-1}| \le c_m^{-1}$ Chapter3/BasicGermCriterionProof · norm_symm_le.

Step 5: the contraction estimate Chapter3/BasicGermCriterionProof · contrMap358_contraction. For every m and $\varepsilon > 0$ there is $r > 0$ such that $(a,w) \in U'_m$ and $|H(a,w) - H(a,w')|_m \le \varepsilon |w - w'|_m$ for $a \in \widetilde{C}$, $|a| < r$, $w, w' \in X_m$ small on level m. Choose r below the radius r_3 of (3) for εc_m, below the radius of a ball around 0 in V_o, and so small that $(a,z) \in U_m$ whenever $|a| < r$ and $|z|_m < r$ ($U = V \cap C$ with V open; the level-m norm of z controls its level-0 norm through the norm of the inclusion) Chapter3/BasicGermCriterionProof · mem_lvlSet_of_small. First let $w_1, w_1' \in X_{m+1}$ with $|w_1|_m, |w_1'|_m < r$. On the convex set $S = \{z \in X_{m+1} : |z|_m < r\}$ the map $g(z) = f_m(a, \iota z)$ has the derivative $Df(a,z)(0,\iota\,\cdot)$ (Proposition 1.2.1 (preprint 1.16) on level m) Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level, and by (3) $|Df(a,z)(0,v) - Df(a,0)(0,v)|_m \le \varepsilon c_m |v|_m$. **The mean value inequality with the derivative bound in the weaker norm** Chapter3/BasicGermCriterionProof · norm_sub_sub_le_of_hasFDerivWithinAt_segment — the one-dimensional mean value inequality for $t \mapsto g(w_1' + t(w_1 - w_1')) - T\iota(w_1' + t(w_1 - w_1'))$ on $[0,1]$, which measures the increment in the norm of X_m rather than of X_{m+1}, as condition (3) requires — gives

$$|g(w_1) - g(w_1') - T\iota(w_1 - w_1')|_m \le \varepsilon c_m |\iota(w_1 - w_1')|_m, \qquad T = Df(a,0)(0,\cdot).$$

The Y-component of T is $L(a)_m$ Chapter3/BasicGermCriterionProof · snd_levelDeriv_eq_L (the level-m operator of Proposition 1.2.1 (preprint 1.16) restricts the level-0 one). Then $H(a,w_1) - H(a,w_1') = L(a)^{-1}[L(a)(w_1 - w_1') - (f(a,w_1) - f(a,w_1'))_2]$, whose norm is at most $c_m^{-1}\varepsilon c_m |w_1 - w_1'|_m$. This is the text's integral $\int_0^1 D_2 f(b, tx + (1-t)x')(x - x')\,dt$, made precise. For general $w, w' \in X_m$: approximate them by sequences from X_{m+1} (dense in X_m), small on level m for large indices, and pass to the limit with the continuity of H_m on U'_m — the text's "by the density of X_{m+1} in X_m".

Step 6: the basic germ and the conjugation. k is a basic germ on the germ $\{U'\}$ with contraction map H (Definition 3.1.7 (preprint 3.7) as formalized Chapter3/LocalCompactness · IsBasicGerm): it is an sc-smooth germ (Step 4), $k(0) = 0$, its X-component is $x - H(b,x)$, and H is an sc^0-contraction germ — sc^0, $H(0) = 0$

since $L(0)^{-1}0 = 0$, and the contraction estimate on every level with every ε (Step 5). The conjugation identity $(\Phi(u, h(u)))_2 = k(u)$ holds on all of U', because Φ covers the identity; with Φ from Step 3 these are the data of Definition 3.1.11 (preprint 3.11) for $h = f - f(0)$.

Second half: the family from conditions (1) and (2).

The family. At the smooth point $(b, 0)$ the linearization $Df(b, 0)$ is an sc-operator `Chapter3/BasicGermCriterionIndex · derivParamOp` `Chapter1/Properties · IsSc1With.scOperator_deriv` whose level maps are the level operators of Proposition 1.2.1 (preprint 1.16) `Chapter3/BasicGermCriterionIndex · derivParamOp_lvl`; $L(b) = \mathrm{pr}_Y \circ Df(b, 0) \circ \mathrm{inr}_X$ is an sc-operator $X \to Y$ (set to 0 where $(b, 0) \notin U$) `Chapter3/BasicGermCriterionIndex · derivFamOp`; and $(b, h) \mapsto L(b)h$ is jointly continuous on $O \times X_m$ for parameters with $(b, 0) \in U$, by the continuity of the level operators (Proposition 1.2.1 (2) (preprint 1.16)) `Chapter3/BasicGermCriterionIndex · continuousOn_levelDeriv_param` `Chapter3/BasicGermCriterionIndex · continuousOn_derivFamOp`.

The compactness property of the family `Chapter3/BasicGermCriterionIndex · familyCompactness_derivFamOp`. Let O be a set of parameters in $\widetilde{C}$ with $(b, 0) \in U$ and $|b| < r$ (r from (2)), $b_j \to b_0$ in O, (h_j) bounded in X_m, $L(b_j)h_j \to 0$. Write $Df(b_j, 0)(0, h_j) = (0, L(b_j)h_j) + ((Df(b_j, 0)(0, h_j))_Q, 0)$. The first summand tends to 0 in F_m. The second lies in $Q \oplus 0$, where all levels coincide, and is bounded: a jointly continuous family of operators is uniformly bounded near a parameter `Chapter3/BasicGermCriterionIndex · exists_bound_of_continuousOn_clm` (continuity at $(b_0, 0)$, then scaling), so $|Df(b_j, 0)(0, \cdot)|$ is bounded for j large. Condition (2) applied to the tails of the sequences gives a convergent subsequence: this is the compactness property of the abstract Lemmas 3.7.2, 3.7.3 (preprint 3.59, 3.60) (§III.25) `Chapter3/BasicGermCriterion · FamilyCompactness`. The text's "$z_j = Df(b_j)h_j \in A$ is bounded on every level since A is a smooth subspace" omits the uniform operator bound.

Lemma 3.7.2 (preprint 3.59). Let $V' = \{b : (b, 0) \in V\}$ ($U = V \cap C$) and $O = V' \cap B(0, r) \cap \widetilde{C}$, which contains 0. $L(0)$ is injective on X_0 by (2), so Lemma 3.7.2 (preprint 3.59) `Chapter3/BasicGermCriterion · lemma_3_59` gives $r_1 > 0$ with $L(b)$ injective on X_0 for $b \in O$, $|b| < r_1$, hence on every level (the inclusions are injective and commute with $L(b)$) `Chapter3/BasicGermCriterionIndex · ScOperator.injective_lvl`.

Lemma 3.7.3 (preprint 3.60). Take a closed ball $\bar{B}(0, r_1')$ inside $V' \cap B(0, r) \cap B(0, r_1)$ and $K = \bar{B}(0, r_1') \cap \widetilde{C}$, compact (since $\widetilde{C}$ is closed) and inside O. On K every $L(b)_m$ is injective, the family is jointly continuous and has the compactness property (restricted from O `Chapter3/BasicGermCriterionIndex · FamilyCompactness.mono`), so Lemma 3.7.3 (preprint 3.60) `Chapter3/BasicGermCriterion · lemma_3_60` gives $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ for $b \in K$.

The index argument. The text: "$P \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism". The formal proof uses a block computation instead. **The index of a block operator** `Chapter3/BasicGermCriterionIndex · fredholmIndex_block`: for a Fredholm $T : P \oplus X \to Q \oplus Y$ with P, Q finite-dimensional and the block $L = \mathrm{pr}_Y \circ T \circ \mathrm{inr}_X$ of finite index, $\mathrm{ind}\, T = \dim P + \mathrm{ind}\, L - \dim Q$. Indeed $T - \mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is $(p, x) \mapsto T(p, 0) + ((T(0, x))_Q, 0)$, of finite rank, hence compact, so $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is Fredholm with the index of T `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator` `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`; and the indices of pr_X and inr_Y are $\dim P$ and $-\dim Q$ `Chapter3/BasicGermCriterionIndex · hasFiniteIndex_snd_and_linIndex` `Chapter3/BasicGermCriterionIndex · hasFiniteIndex_inr_and_linIndex`, which add under composition `Chapter1/FredholmIndex · linIndex_comp`. At $b = 0$, $L(0)$ is bijective on level 0, of index 0, so $\mathrm{ind}\, Df(0) = \dim P - \dim Q$ on level 0, which is the sc-index (an sc-Fredholm operator has the same index on every level) `Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl`. **Surjectivity from the index** `Chapter3/BasicGermCriterionIndex · surjective_of_fredholmIndex_block`: if $\mathrm{ind}\, T = \dim P - \dim Q$ and the block L is injective, then L has finite index — its cokernel is a quotient of the cokernel of the Fredholm operator $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ — and index 0 by the block formula, so it is surjective. For $b \in K$ and every m, $Df(b, 0)_m$ is Fredholm of index $\mathrm{ind}\, Df(0) = \dim P - \dim Q$ by (1) and the level-wise index, and $L(b)_m$ is injective; so $L(b)_m$ is bijective, a topological isomorphism by the open mapping theorem,

and the $L(b)$, $b \in K$, are sc-isomorphisms. The sc-Fredholm property of the projection P and the composition rule for sc-Fredholm operators are not needed.

Conclusion. Extend $b \mapsto L(b)$ outside K by $L(0)$, put $V_o = B(0, r_1')$ (so $V_o \cap \widetilde{C} \subseteq K$), and apply the first half with the lower bounds of Lemma 3.7.3 (preprint 3.60). The neighbourhood replacement $U' = U \cap (V_o \oplus X)$ is the text's "replacing, if necessary, U by a smaller neighbourhood".

III.39 What the addendum taught us

1. **The global implicit function theorems hold with the text's hypotheses.** Theorem 3.6.11 (preprint 3.57), Corollary 3.1.27 (preprint 3.24) and Theorem 3.6.10 (preprint 3.56) are proved for an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with good position at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_57` `Chapter3/GlobalIFTTheoremsG · corollary_3_24` `Chapter3/GlobalIFTTheoremsG · theorem_3_56` (§III.37). Theorem 3.6.10 (preprint 3.56) reads "$f'(x)$ surjective" as good position, which the text's Definition 3.6.6 (preprint 3.53) makes the same thing at points with $d_X(x) = 0$; that the kernel condition is void in the chart form is not formalized.
2. **Two steps of the text's proof had to be replaced, and both repairs are specific to solution sets.** The regularity of the solution germ away from the centre, which the proof of Theorem 3.2.3 (preprint 3.27) asserts for any set and which fails in general (§III.24), follows for two graph descriptions of one set from the C^1-property of the first and the level-wise lifts of the second, by the re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` (§III.35). The relative openness of Σ, false for an arbitrary good complement, holds for a complement whose projection maps the partial quadrant into itself, and the proof of Theorem 3.6.8 (preprint 3.54) can produce such a complement from the start `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto` (§III.29). Read off the statement of the regularity theorem `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`, the first repair works for every set with the tangent germ property whose data satisfy the second regularity condition (§III.36).
3. **The proof of the local Theorem 3.6.8 (1) (preprint 3.54) has a gap, and it is the same gap as in the book's arXiv version.** At a nearby zero, Corollary 3.5.3 (preprint 3.41) gives the surjectivity of the filled linearization; the surjectivity of $f'(y)$ needs condition (3) of Definition 3.1.4 (preprint 3.4) at that zero, which the definition imposes at the centre only `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`. The local theorems were therefore first proved under that condition at the nearby zeros `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled` `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one` (§III.33); the kernel half needs nothing (§III.32). The global theorems do not need it, because their hypothesis already gives good position at every zero. The second addendum proves the surjectivity with Definition 3.1.4 (preprint 3.4) as stated (§§III.41–III.43).
4. **Definition 3.6.6 (preprint 3.53) presupposes a chart independence the text does not prove.** It holds at zeros, because only the fibre part of a chart transition survives there `Chapter3/LinearizationCharts · linAt_chart_change` (§III.31), and it is what lets the good position found in the chart at x_0 be transported to the chart centred at a nearby zero.
5. **Three silent steps are theorems of their own**: tameness is carried by sc-diffeomorphisms and by open subsets `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image` (§III.27); the kernel of the linearization of a basic germ plus an sc$^+$-germ consists of smooth points `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth` (§III.34); and the index raise of a strong bundle isomorphism is one `Chapter3/BasicClassComposition · IsStrongBundleIso.shift` (§III.28).
6. **The dimension of the solution set is locally constant, not constant**, and the dimension count that proves the re-centring is also what shows it (Sard's lemma in fixed dimension `Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty`, §III.35). Theorem 3.2.3 (preprint 3.27) is therefore also proved with a dimension that may vary `Chapter3/GlobalIFTTheorems · theorem_3_27_loc` (§III.36), and the solution set of Theorem 3.6.11 (preprint 3.57) splits into open and closed parts of fixed dimension `Chapter3/GlobalIFTTheoremsG · theorem_3_57_dim`.

7. **Proposition 3.6.5's (preprint 3.52) index needs the conjugation as data.** With the witnesses of the conjugation named, the index of $Tf(x_0)$ is the $n - N$ of every conjugating basic germ, and that number does not depend on the conjugation `Chapter3/FredholmGermIndexG · proposition_3_52_index` `Chapter3/FredholmGermIndexG · ConjugationData.index_eq` (§III.30).
8. **The quadrant inverse function theorem is not the right tool away from the centre**: it needs the degeneracy index to be preserved, which for the solution set is unknown; Lemma 3.8.21's (preprint 3.77) bootstrapping with the hypotheses its proof uses suffices `Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv` (§III.35).
9. **The basic germ criterion is proved in the coordinates of its proof**, with condition (1) needed only along the finite-dimensional directions and the text's index argument replaced by a block index formula `Chapter3/BasicGermCriterionIndex · fredholmIndex_block` `Chapter3/BasicGermCriterionIndex · theorem_3_58` (§III.38).
10. **What was left of Chapter 3** after this addendum: the surjectivity in Theorem 3.6.8 (1) (preprint 3.54) with Definition 3.1.4 (preprint 3.4) as stated; in Theorem 3.1.22 (preprint 3.19), the dimension as the Fredholm index and $\ker f'(y) = T_yS$; Proposition 3.6.5 (preprint 3.52) as an sc-Fredholm operator on T_xX; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting; Theorem 3.1.21 (preprint 3.18); Lemma 3.8.13 (preprint 3.70) for a general presentation of the partial quadrant; and Theorem 3.2.3 (preprint 3.27) for sets whose data violate the second regularity condition. The second addendum (§§III.40–III.46) settles the first three items, the third in charts, and Theorem 3.1.21 (preprint 3.18).

III.40 Theorem 3.1.21 (preprint 3.18): parametrized perturbations

The text states Theorem 3.1.21 (preprint 3.18) after a five-line argument: the section $\tilde{f}(r,x) = ((r,x), f(x))$ of the pull-back bundle $\pi^*P \to \mathbb{R}^n \times X$, $\pi(r,x) = x$, "is a sc-Fredholm section as is readily verified", $\tilde{s}(r,x) = ((r,x), \sum_i r_i s_i(x))$ is an sc$^+$-section, and Theorem 3.1.18 (preprint 3.15) adds them. The formalization follows this route. What it has to build first is everything the five lines take for granted: $\mathbb{R}^n$ as an M-polyfold with enough charts, the pull-back bundle as a strong bundle, the sc-calculus for a scalar factor, the product chart with its filler, and the extension of a basic germ by n inert parameters (Findings 160–167). Tameness is not used anywhere, as for Theorem 3.1.18 (preprint 3.15).

$\mathbb{R}^n$ and $\mathbb{R}^n \times X$. $\mathbb{R}^n$ carries the constant sc-structure `Chapter3/ParamPerturbation · rnSc` and the local model $(\mathbb{R}^n, \mathbb{R}^n, \mathbb{R}^n)$ `Chapter3/ParamPerturbation · rnModel`. For every a the translation $r \mapsto r - a$ is a chart `Chapter3/ParamPerturbation · rnChart`; two translation charts are compatible, because their transition maps are translations and a C^∞ map between finite-dimensional spaces is sc-smooth for the constant structures `Chapter3/ParamPerturbation · isScSmoothRetractMap_const_of_contDiff` `Chapter3/ParamPerturbation · rnChart_compatible`. The M-polyfold $\mathbb{R}^n$ has the identity chart as its atlas `Chapter3/ParamPerturbation · rnPolyfold`, and every translation chart is a chart of it `Chapter3/ParamPerturbation · isChart_rnChart`. These charts are needed: Definition 3.1.11 (preprint 3.11) asks for a chart *centred* at the smooth point (r_0, x), and the product of the identity of $\mathbb{R}^n$ with a chart of X is centred only at $r_0 = 0$. $\mathbb{R}^n$ is tame `Chapter3/ParamPerturbation · rnPolyfold_isTame` and every point lies on every level `Chapter3/ParamPerturbation · rnPolyfold_levelSet`; so (r,x) is on level m of the product iff x is `Chapter3/ParamPerturbation · prod_rn_mem_levelSet_iff`, smooth iff x is `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, and $\mathbb{R}^n \times X$ is tame when X is `Chapter3/ParamPerturbation · prod_rn_isTame` — which the text needs, since Definition 3.1.16 (preprint 3.14) is stated for tame bundles.

Two rules of the sc-calculus. Scalar multiplication $\mathbb{R} \oplus F \to F$ is continuous and bilinear on every level, hence sc-smooth by Proposition 1.2.4 (preprint 1.19) `Chapter3/ParamPerturbation · isScSmooth_smul`; composed with a pairing, $x \mapsto a(x)v(x)$ is sc-smooth for sc-smooth $a : U \to \mathbb{R}$ and $v : U \to F$ `Chapter3/ParamPerturbation · IsScSmooth.smul`; finite sums of sc-smooth maps are sc-smooth `Chapter3/ParamPerturbation · IsScSmooth.finsetSum`. The coordinate $r \mapsto r_i$ is an sc-operator `Chapter3/ParamPerturbation · rnCoordOp`. Chapter 1 states neither rule.

The pull-back bundle. Proposition 2.6.8 (preprint 2.54) is formalized as an existence statement `Chapter2/StrongBundlePullback · proposition_2_54`; here it is made a structure, the strong bundle π^*P with the

atlas of all pulled-back charts Chapter3/ParamPerturbation · StrongBundle.pullback, whose total space is paracompact by Theorem 2.3.10 (preprint 2.16) Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace. A pulled-back chart is made from a chart ψ of the base Z around z and a strong bundle chart c of P around $\pi(z)$, with π mapping the domain of ψ into that of c Chapter3/ParamPerturbation · exists_pullbackChartData; such a chart is a chart of π^*P as soon as c is merely compatible with the atlas of P Chapter3/ParamPerturbation · PullbackChartData.isChart_toChart. A map $\tau : Z \to Y$ over π is a section $z \mapsto (z, \tau(z))$ of π^*Y Chapter3/ParamPerturbation · pullbackLift Chapter3/ParamPerturbation · isSection_pullbackLift, whose value in a pulled-back chart is $(\psi(z), \mathrm{pr}_F \Phi(\tau(z)))$ Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift. For $\tau = \sigma \circ \pi$ the representative is $w \mapsto (w, \hat{\sigma}(g(w)))$ with $g = \varphi \circ \pi \circ \psi^{-1}$, which maps the retract of ψ into that of φ Chapter3/ParamPerturbation · PullbackChartData.g_mem_0 Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift_comp; so the pull-back of an sc-smooth section along an sc-smooth map is sc-smooth Chapter3/ParamPerturbation · StrongBundle.IsScSmoothSection.pullback (Finding 167).

The three sections. $F(r, x) = f(x) + \sum_i r_i s_i(x)$ is computed in the fibre $P^{-1}(x)$, from the fibre combination $\sum_i r_i s_i(x)$ Chapter3/ParamPerturbation · fiberComb, and read in the fibre of π^*Y over (r, x) Chapter3/ParamPerturbation · paramSection; $\tilde{f}$ Chapter3/ParamPerturbation · paramLift and $\tilde{s}$ Chapter3/ParamPerturbation · paramComb are the text's sections, and $F = \tilde{f} + \tilde{s}$ in the fibres of π^*Y Chapter3/ParamPerturbation · paramSection_eq_sectionAdd.

$\tilde{s}$ is an sc⁺-section Chapter3/ParamPerturbation · StrongBundle.isScPlusSection_paramComb. The atlas of π^*P contains the charts pulled back through *every* chart ψ of $\mathbb{R}^n \times X$, not only product charts, so the check must work for an arbitrary ψ. In a pulled-back chart, by the linearity of Φ on the fibre over $x = \pi(z)$, $\tilde{s}(z)$ has the value $(\psi(z), \sum_i r_i \hat{s}_i(x))$ Chapter3/ParamPerturbation · PullbackChartData.toChart_paramComb, so the principal part is $w \mapsto \sum_i \lambda_i(w) \hat{s}_i(g(w))$ with $\lambda(w) = (\psi^{-1}(w))_1$. The map λ is sc-smooth on the retract of ψ: it is the projection $\mathbb{R}^n \times X \to \mathbb{R}^n$ (an sc-smooth map of M-polyfolds) read in ψ and in the identity chart of $\mathbb{R}^n$ Chapter3/ParamPerturbation · isScSmooth_chart_coord. Each $\hat{s}_i \circ g$ is sc-smooth into F^1, s_i being sc⁺. The product rule and the sum rule conclude.

The product chart and its filler. For a strong bundle chart c of P and $r_0 \in \mathbb{R}^n$, the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd is $\psi = (r \mapsto r - r_0) \times \varphi$ on $\mathbb{R}^n \times V$, with the retraction $1 \times r$ of $\mathbb{R}^n \times U$ onto $\mathbb{R}^n \times O$; there $g = \varphi \circ \pi \circ \psi^{-1}$ is the projection $(a, u) \mapsto u$. *Filler conditions.* If g fills $\hat{f}$ in c, then $(a, u) \mapsto g(u)$ fills $(a, u) \mapsto \hat{f}(u)$ in the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd_isFillerOf. Conditions (1) and (2) are those of g, because the fibre projection of the product retraction at (a, u) is $\rho(r(u))$ — which uses $\varphi(\varphi^{-1}(r(u))) = r(u)$ on O. Condition (3): $D(1 \times r)(0) = 1 \times Dr(0)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_scDeriv_r, so its kernel is $0 \times \ker Dr(0)$; the fibre projection at 0 is $\rho(0)$ (from $r(0) = 0$); and $[1 - \rho(r(u))]g(u)$ does not depend on a, so its derivative at 0 on $0 \times \ker Dr(0)$ is that of g. *For a filling germ* g on G the same holds on the product germ $\mathbb{R}^n \times G$, with levels $\mathbb{R}^n \times G_m$, unbounded in the parameter direction, which Definition 3.1.1 (preprint 3.1) allows Chapter3/GermCalculusParam · ScNbhdGerm.univProd: $(a, u) \mapsto g(u)$ is a germ on it with the derivative family $Dg(u) \circ \mathrm{pr}_2$ Chapter3/GermCalculusParam · IsSc1GermWith.univProd, and the fill defect in the product chart is the pull-back along pr_2 of that of g, so condition (3) is the kernel identification just made, without a uniqueness argument Chapter3/GermCalculusParam · PullbackChartData.ofProd_isFillingGerm. *Principal parts* ($a = r - r_0$ the chart coordinate): $\tilde{s}$ has $(a, u) \mapsto \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramComb and F has $(a, u) \mapsto \hat{f}(u) + \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramSection, the text's $(r, u) \mapsto \hat{f}(u) + \sum_i r_i \hat{s}_i(u)$.

Inert parameters. Let Λ be finite-dimensional. The product $1_\Lambda \times \Phi$, $((a, u), k) \mapsto ((a, \varphi(u)), \Gamma(u, k))$, of a map Φ of $U \lhd F$ with the identity of Λ Chapter3/ParamPerturbation · paramMap is a strong bundle map when Φ is: on $[0]$ and $[1]$ it is $1 \times \Phi$ up to the reassociations of the factors, which are sc-operators Chapter3/ParamPerturbation · ScBanach.assocOp Chapter3/ParamPerturbation · ScBanach.assocInvOp Chap-

ter3/ParamPerturbation · IsStrongBundleMap.paramMap; and a strong bundle isomorphism, with inverse $1_\Lambda \times \Phi'$, when Φ is Chapter3/ParamPerturbation · IsStrongBundleIso.paramMap. The Λ-extension of a germ of neighbourhoods is $U''_m = \{((a,\lambda),w) \mid (a,w) \in U_m\}$ Chapter3/ParamPerturbation · ScNbhdGerm.paramExtend, unbounded in λ, which Definition 3.1.1 (preprint 3.1) allows. *The Λ-extension of a basic germ is basic* Chapter3/ParamPerturbation · IsBasicGerm.paramExtend: if $h(a,w) = (h_1(a,w), w - B(a,w))$ is basic on $\widetilde{C} \oplus W$, then $h'((a,\lambda),w) = h(a,w)$ is basic on $(\widetilde{C} \times \Lambda) \oplus W$, with $B'((a,\lambda),w) = B(a,w)$: it is the composite of h with the projection forgetting λ Chapter3/ParamPerturbation · paramForgetOp, and the contraction estimate of Definition 3.3.1 (preprint 3.28) holds with the same radius since $|a| \le |(a,\lambda)|$. *The Λ-extension of a conjugation* Chapter3/ParamPerturbation · IsConjugateToBasicGerm.paramExtend: if k is conjugated to h by Ψ, then $(\lambda,u) \mapsto k(u)$ is conjugated to h' by $\Psi''((\lambda,u),k) = ((((\psi(u))_1,\lambda),(\psi(u))_2),\Gamma(u,k))$, the composite Chapter3/BasicClassComposition · IsStrongBundleIso.comp of $1_\Lambda \times \Psi$ with the linear reassociation $(\lambda,(a,w)) \mapsto ((a,\lambda),w)$ Chapter3/ParamPerturbation · paramSwapOp Chapter3/ParamPerturbation · paramSwapInvOp, a strong bundle isomorphism as a pair of sc-isomorphisms Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap. The parameter quadrant is $\widetilde{C} \times \Lambda$ in $P \times \Lambda$, put second so as to reuse the product of a partial quadrant with a whole space; the order is immaterial (Finding 163).

$\tilde{f}$ is an sc-Fredholm section Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSection.param. *Sc-smooth*: it is the pull-back of f along the projection. *Regularizing* Chapter3/ParamPerturbationG · StrongBundle.regularizing_paramLift: if (r,x) is on level m and $\tilde{f}(r,x)$ on level $(m,m+1)$ of π^*Y, then $x \in X_m$ and $f(x) \in Y_{m,m+1}$ — tested in the pulled-back charts, whose fibre parts are those of P — so $x \in X_{m+1}$ because f is regularizing, and (r,x) is on level $m+1$. *The germ at a smooth point* (r_0,x) Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGerm.param: x is smooth, so Definition 3.1.11 (preprint 3.11) at x gives a centred chart c, a filling germ g on G_g, an sc⁺-germ t_0 on G_{t_0} with $t_0(0) = g(0)$ and a conjugation of $g - t_0$ to a basic germ. In the product chart at (r_0,x), which is centred there, the principal part of $\tilde{f}$ is $(a,u) \mapsto \hat{f}(u)$, its filling germ $(a,u) \mapsto g(u)$ on $\mathbb{R}^n \times G_g$ (the filler conditions above), the sc⁺-germ to subtract $(a,u) \mapsto t_0(u)$ on $\mathbb{R}^n \times G_{t_0}$ Chapter3/GermCalculusParam · IsScSmoothGerm.univProd, and $(a,u) \mapsto (g - t_0)(u)$ is conjugated to the $\mathbb{R}^n$-extension of the basic germ; the conjugation concerns $g - t_0$ near 0 in U, and $1 \times \Psi$ and the extended basic germ are whole-chart objects, so no germ calculus enters it. This is the "readily verified".

Theorem 3.1.21 (preprint 3.18) Chapter3/ParamPerturbationG · theorem_3_18. *Statement.* For a strong bundle $P : Y \to X$, an sc-Fredholm section f and sc⁺-sections $s_1, \dots, s_n$, the section F is an sc-Fredholm section of π^*P. *Proof.* $F = \tilde{f} + \tilde{s}$; $\tilde{f}$ is sc-Fredholm and $\tilde{s}$ is sc⁺; Theorem 3.1.18 (preprint 3.15) Chapter3/FredholmSectionsG · theorem_3_15. The paracompactness of $\mathbb{R}^n \times X$ and of π^*Y (Definitions 2.3.4, 2.6.5 (preprint 2.13, 2.51)) enters as instance arguments; the version without them supplies both by Theorem 2.3.10 (preprint 2.16) Chapter3/ParamPerturbationG · theorem_3_18' Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The text's sketch of the germ of F — filled version $(r,u) \mapsto g(u) + \sum_i r_i \hat{s}_i(r(u))$, sc⁺-part $t_0(u) + \sum_i r_i \hat{s}_i(r(u))$, conjugation $\Psi'(r,u,k) = (r - r_0, \Psi(u,k))$ — is exactly what Theorem 3.1.18 (preprint 3.15) produces from the germ data of $\tilde{f}$ and the principal part of $\tilde{s}$ Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd; the translation $r - r_0$ sits in the chart rather than in the conjugation (Finding 164).

The linearization (after Theorem 3.1.22 (preprint 3.19) in the text, used for the finite-dimensional reduction and for transversality). In the product chart at (r_0,x), for a chart c with $0 \in O$, the derivative of $\hat{F} \circ (1 \times r)$ at 0 is

$$(\delta a, \delta u) \mapsto D(\hat{G} \circ r)(0)\delta u + \sum_i \delta a_i \hat{s}_i(0), \qquad \hat{G} = \hat{f} + \sum_i r_{0,i} \hat{s}_i$$

Chapter3/ParamPerturbationG · scDeriv_paramSection: the product rule at $a = 0$, where the terms $\sum_i a_i D(\hat{s}_i \circ r)$ vanish. The tangent space of the product retract at 0 lies in $\mathbb{R}^n \times T_0O$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_mem_tangent, and in the chart form of the linearization (§III.31)

$\mathrm{Lin}_F(0)(\delta a,\delta u) = \mathrm{Lin}_{f+\sum r_{0,i}s_i}(0)\delta u + \sum_i \delta a_i \widehat{s}_i(0)$ `Chapter3/ParamPerturbationG · linAt_paramSection`. At $r_0 = 0$ this is the text's formula $\widetilde{f}'(0,x)(h,u) = f'(x)u + \sum_i h_i e_i$ `Chapter3/ParamPerturbationG · linAt_paramSection_zero`. The formula is the derivative at the centre of a chart and does not use $F(r_0,x)=0$; at a general (r_0,x) its first term is the linearization of $f+\sum r_{0,i}s_i$, not $f'(x)$ (Finding 165). The text's "sc¹-sections s_i" (p. 101 (preprint p. 75)) must be sc$^+$ for Theorem 3.1.21 (preprint 3.18) to apply (Finding 166).

III.41 Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses: the plan and the elementary lemmas

§III.33 showed what the text's argument for property (4) of Theorem 3.6.8 (preprint 3.54) leaves open: at a zero $z_0=\varphi(y)$ near 0, in the chart at x_0, the filled operator $Dg(z_0)$ is onto (Corollary 3.5.3 (preprint 3.41)), but $f'(y) = Dg(z_0)|_{T_{z_0}O}$ is onto only if condition (3) of the filler holds at z_0, which Definition 3.1.4 (preprint 3.4) does not ask; and the perturbation argument of the text's Proposition 6.2.2 (preprint 6.2) gives $\rho(z_0)\Lambda_0 \subseteq \text{range } f'(y)$ only on a neighbourhood depending on the finite-dimensional Λ_0. This section and the next two prove the surjectivity anyway, with the text's hypotheses, by a new argument found while formalizing (Findings 180–189). The text's statement is true; its proof, and that of the book's arXiv version, is incomplete.

The plan. By the assembly step of §III.31 it suffices to show, in the chart at x_0 and for every zero z_0 of g in O near 0, that $\rho(z_0)k \in Dg(z_0)(T_{z_0}O)$ for every smooth $k \in F_\infty$; the closed range of $f'(y)$ (Proposition 3.6.5 (preprint 3.52) at y) does the rest. Fix z_0 and k. Suppose $u(\mu)$, $0 \le \mu < \mu_0$, is a curve through $u(0)=z_0$ of zeros of the *perturbed filler* $g+\mu\rho(r(\cdot))k$, right differentiable at 0 into E_1 with velocity ξ. The curve lies in O (the first lemma below), so $r(u(\mu))=u(\mu)$ and $Dr(z_0)\xi=\xi$, i.e. $\xi \in T_{z_0}O$; differentiating $g(u(\mu)) = -\mu\rho(r(u(\mu)))k$ at $\mu=0$ gives $Dg(z_0)\xi = -\rho(z_0)k$. So one curve for each pair (z_0,k) suffices. It is produced by an implicit function theorem *at the nearby zero*, in the normal form of Theorem 3.4.2 (preprint 3.38): its infinite-dimensional part is Theorem 3.3.3 (preprint 3.30) for a contraction germ extended by the parameter μ, applied at the centre, whose regions in the parameter a do not depend on k (§III.42); its finite-dimensional part is a one-sided implicit function theorem at the image of z_0, in the direction of an interior vector of the partial quadrant that good position provides (this section and §III.43). The set of zeros to which the argument applies is cut out by conditions on the data at x_0 alone. This section proves the elementary lemmas, with no sc-structure involved.

Zeros of the perturbed filler lie in O `Chapter3/GlobalIFTNearbyOntoLemmas · StrongRetr.mem_O_of_perturbed_zero`. *Statement.* Let R be a strong bundle retraction on $U \lhd F$, $u \in U$, $\mu \in \mathbb{R}$, $k \in F$, and let condition (2) of Definition 3.1.4 (preprint 3.4) hold at u ($g(u)=\rho(r(u))g(u)$ implies $u \in O$). If $g(u)+\mu\rho(r(u))k = 0$, then $u \in O$. *Proof.* $r(u) \in U$ `Chapter3/Filler · StrongRetr.r_mem` lies in O, $\rho(r(u))$ is linear `Chapter3/Filler · StrongRetr.ρ_isLinear` and idempotent on the fibre over a point of O `Chapter3/Filler · StrongRetr.ρ_ρ`; so $\rho(r(u))g(u) = -\mu\rho(r(u))\rho(r(u))k = -\mu\rho(r(u))k = g(u)$, and condition (2) applies. □

An inward direction from good position `Chapter3/GlobalIFTNearbyOntoLemmas · exists_inward_of_isInGoodPosition`. *Statement.* Let $C = \{x : \ell_i(x) \ge 0\}$ be a partial quadrant presented by an sc-isomorphism T, with coordinate functionals $\ell_i = (T\cdot)_i$, $z \in C$, and $N \ne 0$ a subspace in good position to the partial quadrant $C_z = \{v : \ell_i(v) \ge 0 \text{ for the } i \text{ with } \ell_i(z)=0\}$. Then there are $v \in N$ and $\eta, t_0 > 0$ with $z + t(v+\zeta) \in C$ for all $t \in [0,t_0]$ and $|\zeta| < \eta$. *Proof.* Let v be an interior point of $N \cap C_z$ in N (good position). *For an active index i, $\ell_i(v) > 0$.* Otherwise $\ell_i(v)=0$; for $u \in N$ and small t, $v \pm tu \in C_z$, so $\pm t\ell_i(u) \ge 0$ and ℓ_i vanishes on N. Let P be a good complement with constant ε and $p \in P$; choose $n_0 \ne 0$ in $N \cap C_z$ (it has interior and $N \ne 0$) and scale it so that $|p| \le \varepsilon|n_0|$. Good position gives $n_0 \pm p \in C_z$, so $\ell_i(\pm p) = \ell_i(n_0 \pm p) \ge 0$ and $\ell_i(p)=0$. Then ℓ_i vanishes on $N \oplus P = E$, which contradicts the surjectivity of the coordinates of a presentation. *The estimates.* A finite family of positive numbers has a positive lower bound `Chapter3/GlobalIFTNearbyOntoLemmas · exists_pos_le_of_forall_pos_fin`: $\ell_i(v) \ge m$ for the active i and $\ell_i(z) \ge m$ for the others. For the active

i, $\ell_i(z+t(v+\zeta)) = t(\ell_i(v)+\ell_i(\zeta)) \geq t(m - \|\ell_i\|\,|\zeta|) \geq 0$ when $|\zeta|$ is small; for the others, $\ell_i(z+t(v+\zeta)) \geq m - t\|\ell_i\|(|v|+|\zeta|) \geq 0$ when t is small. □ The lemma needs $N \neq 0$: for the zero subspace good position is an empty condition, and the case $\dim K = 0$ of Theorem 3.6.8 (1) (preprint 3.54) is treated separately (§III.43). The note behind the formalization phrased it as "N meets the interior of C_z"; the cone condition is what transports along the linear coordinate change (Finding 185).

Inward directions absorb any vector after rescaling `Chapter3/GlobalIFTNearbyOntoLemmas · inward_add_smul`. If $a_0 + t(\alpha+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$ and $|\beta| < \eta$, then for every a_1 and $s \geq (|a_1|+1)/\eta$ the vector $\alpha_0 = a_1 + s\alpha$ satisfies $a_0 + t(\alpha_0+\beta) \in \widetilde{C}$ for $t \in [0, t_0/s]$ and $|\beta| < 1$: indeed $a_0 + t(\alpha_0+\beta) = a_0 + ts(\alpha + (a_1+\beta)/s)$ with $ts \leq t_0$ and $|(a_1+\beta)/s| < (|a_1|+1)/s \leq \eta$.

A one-sided curve of zeros `Chapter3/GlobalIFTNearbyOntoLemmas · exists_curve_of_inward` (the finite-dimensional implicit function theorem at a boundary point, in an inward direction). *Statement.* Let $\widetilde{C} \subseteq P$ be convex, $\Omega \subseteq \mathbb{R}\times P$ open, $D = \Omega \cap (\mathbb{R}\times\widetilde{C})$, $a_0 \in \widetilde{C}$ with $(0,a_0) \in \Omega$, $H : \mathbb{R}\times P \to Q$ of class C^1 on D, $H(0,a_0) = 0$, $D_aH(0,a_0)$ onto (the derivative within D, restricted to $0\times P$), and α_0 with $DH(0,a_0)(1,\alpha_0) = 0$ and $a_0 + t(\alpha_0+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$, $|\beta| < \eta$. Then there are $\mu_0 > 0$ and $a : \mathbb{R} \to P$ with $a(0) = a_0$, right derivative α_0 at 0, and $(\mu, a(\mu)) \in D$, $H(\mu,a(\mu)) = 0$ for $0 \leq \mu < \mu_0$. *Proof. Unique derivatives.* The inward property puts a ball around $a_0 + t_0\alpha_0$ into $\widetilde{C}$, so $\mathbb{R}\times\widetilde{C}$ is convex with nonempty interior and D has unique derivatives; derivatives within D are well defined. *The contraction.* Let S be a right inverse of $M_0 = D_aH(0,a_0)$ and Z its range; SM_0 is the identity on Z. For $\mu \geq 0$ and $\zeta \in Z$ with $|\zeta| \leq c\mu$, $c = \eta/2$, the point $(\mu, a_0 + \mu\alpha_0 + \zeta)$ lies in D and close to $(0,a_0)$ `Chapter3/GlobalIFTNearbyOntoLemmas · dist_curve_le`, and $a_0 + \mu\alpha_0 + \zeta = a_0 + \mu(\alpha_0 + \zeta/\mu)$ is in $\widetilde{C}$ by the inward property ($|\zeta/\mu| \leq c < \eta$). Put $T_\mu(\zeta) = \zeta - SH(\mu, a_0+\mu\alpha_0+\zeta)$. Its derivative in ζ is $1 - SD_aH(\mu, a_0+\mu\alpha_0+\zeta)|_Z$, which tends to $1 - SM_0|_Z = 0$ as $(\mu,\zeta) \to 0$ by the continuity of DH at $(0,a_0)$; the mean value inequality on the convex ball makes T_μ a $\frac{1}{2}$-contraction for small μ. The differentiability of H at $(0,a_0)$ in the direction $(1,\alpha_0)$, with $DH(0,a_0)(1,\alpha_0) = 0$, gives $|T_\mu(0)| = |SH(\mu, a_0+\mu\alpha_0)| = o(\mu)$; so $|T_\mu(0)| \leq c\mu/2$ for small μ and T_μ maps the ball of radius $c\mu$ into itself. *The curve.* The fixed point $\zeta(\mu)$ has $SH(\mu, a_0+\mu\alpha_0+\zeta(\mu)) = 0$, hence $H = 0$ there (S is injective), and $|\zeta(\mu)| \leq 2|T_\mu(0)| = o(\mu)$; so $a(\mu) = a_0 + \mu\alpha_0 + \zeta(\mu)$ has right derivative α_0 at 0. □

Two facts on right derivatives. If k is continuous at 0 from the right, $\mu \mapsto \mu k(\mu)$ has the right derivative $k(0)$ at 0 `Chapter3/GlobalIFTNearbyOntoLemmas · hasDerivWithinAt_smul_right_of_continuousWithinAt`; and right derivatives at 0 on $[0,\infty)$ are unique `Chapter3/GlobalIFTNearbyOntoLemmas · eq_of_hasDerivWithinAt_Ici`.

III.42 The λ-extended contraction germ and the perturbation in the normal form

Throughout, the normal form is the data of Theorem 3.4.2 (preprint 3.38) from §III.20 `Chapter3/LocalSolvability · ShiftedBasicData`: a germ $g : O(\widetilde{C}\oplus W, 0) \to Q\oplus W$, sc-smooth, whose index raise $g^1 = (G_1, w - B(a,w))$ is a basic germ on the shifted structures; at the chart at x_0 it is obtained from the stability data of the graph description by Theorem 3.4.2 (preprint 3.38) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`. (In this section g and e denote the normal form and its perturbation; the filler of the chart at x_0 returns in §III.43.) The norms of w below are those of W_{m+1}, the shifted level m.

The perturbation on the shifted structure. For an sc$^+$-perturbation $e : O(\widetilde{C}\oplus W, 0) \to (Q\oplus W)^1$, its W-component on level 1 `Chapter3/GlobalIFTNearbyOntoGerm · perturbEW` is sc-smooth on the index-raised germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_isScSmoothGerm`, by Proposition 1.2.2 (preprint 1.17) for germs applied to the level-raised map `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelMap_e_isScSmoothGerm` `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_eq_on`. Its level maps are those of e read on the shifted levels `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_eW`.

Contraction regions that do not depend on the perturbation Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii. A system of regions consists of radii ν_m in P and ρ_m in W_{m+1} such that for $a \in \widetilde{C}$, $|a| < \nu_m$ and $|w|, |w'| < \rho_m$: (a, w) lies in the domain of g^1 on level m, $|B(a,w) - B(a,w')| \le \frac{1}{4}|w - w'|$, and $|B(a,0)| \le \rho_m/8$; ν decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii.ν_anti_le and the ρ_m compatible with the inclusions. *They exist below any prescribed radii* δ_m Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii: on each level, Definition 3.3.1 (preprint 3.28) with $\varepsilon = \frac{1}{4}$ gives a radius on which $B(a,\cdot)$ is a $\frac{1}{4}$-contraction, and since $B(0,0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_B_zero and B is continuous at 0 on each level, $|B(a,0)| \le \rho_m/8$ for $|a|$ small; the radii are then shrunk level by level to be monotone and below δ_m. Nothing here involves a perturbation.

The λ-extended germ. On $(\mathbb{R} \times \widetilde{C}) \oplus W^1$ take the germ of neighbourhoods $\{((\mu, a), w) \mid (a, w) \in U^1_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbGerm, with the projections forgetting μ and keeping μ as sc-operators Chapter3/GlobalIFTNearbyOntoGerm · perturbProj Chapter3/GlobalIFTNearbyOntoGerm · perturbMuProj and $\mathbb{R} \times \widetilde{C}$ quadrant-like Chapter3/GlobalIFTNearbyOntoGerm · isQuadrantLike_perturb, and put

$$B_\mu(a, w) = B(a, w) - \mu\, Pe(a, w)$$

Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB, P the projection onto W, with $B_\mu(0,0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_zero. It is sc-smooth on the germ: B composed with the projection, minus the scalar product of the μ-coordinate with Pe composed with the projection, by the product rule with a scalar factor of §III.40 Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_isScSmoothGerm Chapter3/ParamPerturbation · IsScSmooth.smul; its level maps are $B_m - \mu(Pe)_m$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_perturbB. Given constants $c_m \ge 0$ with $|Pe(a,w)| \le c_m$ and $|Pe(a,w) - Pe(a,w')| \le c_m|w - w'|$ on the regions, define the μ-radii $\mu_m = \min_{j \le m} \min(\frac{1}{4}, \rho_j/8)/(c_j + 1)$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu, positive and decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_pos Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_anti, with $\mu_m c_m \le \frac{1}{4}$ and $\mu_m c_m \le \rho_m/8$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_mul_le.

Lemma: B_μ is a contraction germ in the working form Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction, with the parameter regions $(-\mu_m, \mu_m) \times \{|a| < \nu_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.mem_perturbContraction_N_iff, the open balls $\{|w| < \rho_m\}$ and the closed balls $\{|w| \le \rho_m/2\}$ (Definition 3.A, §III.7). *The contraction*: $|B_\mu(a,w) - B_\mu(a,w')| \le (\frac{1}{4} + |\mu|c_m)|w - w'| \le \frac{1}{2}|w - w'|$. *The closed balls are invariant*: for $|w| \le \rho_m/2$, $|B_\mu(a,w)| \le |B(a,w) - B(a,0)| + |B(a,0)| + |\mu||Pe(a,w)| \le \frac{1}{4} \cdot \frac{\rho_m}{2} + \frac{\rho_m}{8} + |\mu|c_m \le \frac{\rho_m}{2}$. The regions in a and w are those of B and do not depend on e; only the μ-intervals do. The general construction of a working form from Definition 3.3.1 (preprint 3.28) Chapter3/GermIFT · toContractionGerm is *not* used: its radii in a would depend on e, and the finite-dimensional step needs the nearby zero inside the level-3 region for every perturbation (Finding 182).

Theorem 3.3.3 (preprint 3.30) for B_μ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_isSc1GermWith. The solution germ $\delta_\mu(\mu, a)$ is sc¹ on the germ re-indexed by one level, $((-\mu_{m+1}, \mu_{m+1}) \times \{|a| < \nu_{m+1}\}) \cap (\mathbb{R} \times \widetilde{C})$, with the derivative family of the proof of Theorem 3.3.3 (preprint 3.30) Chapter3/ContractionGerm · theorem_3_30. *At* $\mu = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_sol_zero: a fixed point $w \in W_1$ of $B(a,\cdot)$ with $|w| < \rho_0$, $|a| < \nu_0$, $a \in \widetilde{C}$, is $\delta_\mu(0, a)$, by the uniqueness of the fixed point in the open ball of level 0.

The perturbation in the normal form. Now the chart at x_0 returns: the strong bundle retraction R on $U \lhd F$, the conjugation Ψ covering ψ with inverse Ψ', and the linear maps glue, split, fiber of the coordinate change of Theorem 3.4.2 (preprint 3.38) (§III.18). The perturbation $u \mapsto \rho(r(u))k$ of the filler,

pushed to the normal form, is $\hat{e}_k(y) = \Xi(y,k)$ with the *joint* map

$$\Xi(y,\kappa) = \operatorname{fiber}\Big(\Psi[1](\psi^{-1}(\operatorname{glue} y),\ (R[1](r(\psi^{-1}(\operatorname{glue} y)),\kappa))_F)\Big)_{F'}$$

Chapter3/GlobalIFTNearbyOntoXi · xiMap on $\psi(U) \oplus F^1$ Chapter3/GlobalIFTNearbyOntoXi · xiDom Chapter3/GlobalIFTNearbyOntoXi · mem_xiDom, a relatively open set ($\operatorname{glue}^{-1}(U')$ is relatively open in the quadrant of the normal form) Chapter3/GlobalIFTNearbyOntoXi · glue₀_preimage_relOpen Chapter3/GlobalIFTNearbyOntoXi · xiDom_relOpen. Ξ *is sc-smooth into the index-raised target* Chapter3/GlobalIFTNearbyOntoXi · xiMap_isScSmooth: it is the composite of the linear glue, $\psi^{-1} \times 1$, the map $(u,\kappa) \mapsto (u,(R[1](r(u),\kappa))_F)$, the strong bundle map $\Psi[1]$ on the index-raised bundle (Definition 2.6.1 (preprint 2.47)), a projection and the linear fiber. Ξ *is linear in* κ Chapter3/GlobalIFTNearbyOntoXi · xiMap_smul Chapter3/GlobalIFTNearbyOntoXi · xiMap_zero, also on every level Chapter3/GlobalIFTNearbyOntoXi · levelMap_xiMap_smul, since R and Ψ are linear on the fibres. For a smooth k with level-1 representative k_1, $\hat{e}_k = \Xi(\cdot,k_1)$ Chapter3/GlobalIFTNearbyOntoXi · ehat is sc-smooth into the index-raised target Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmooth (a constant map with smooth value is sc-smooth Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth), an sc⁺-perturbation on the germ of Theorem 3.4.2 (preprint 3.38) Chapter3/GlobalIFTNearbyOntoXi · pushGerm₀_U_subset Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmoothGerm, with level maps $(\hat{e}_k)_m(y) = \Xi_m(y,k_{m+1})$ Chapter3/GlobalIFTNearbyOntoXi · levelMap_ehat.

Lemma: bounds uniform in k Chapter3/GlobalIFTNearbyOntoXi · exists_xi_bounds. *For every level* m *there are* $\delta > 0$ *and* $M \geq 0$ *such that for* $|a| < \delta$, $|\kappa|_{m+2} < \delta$ *and a convex set* S *of* $w \in W_{m+1}$ *with* $|w| < \delta$ *on* S *(inside the domain of* Ξ *on level* $m+1$*):* $|\Xi_m((a,w),\kappa)| \leq 1$ *and* $|\Xi_m((a,w),\kappa) - \Xi_m((a,w'),\kappa)| \leq M|w-w'|$ *on* S*. Proof.* By Proposition 1.2.3 (preprint 1.18) Chapter1/Levels · IsScSmooth.contDiffOn_levels, Ξ is C^1 from level $m+1$ to level m of its index-raised target, which is the shifted level m of the normal form: on the shifted structure the perturbation is C^1 on every level, which is what contraction estimates on the shifted levels need. *This is where the index raise of Theorem 3.4.2 (preprint 3.38) is used* (Finding 183). Ξ_m vanishes at $\kappa = 0$, so by continuity of Ξ_m and of its derivative at 0 both are small on a small ball, and the mean value inequality on the convex S gives the Lipschitz bound. □ *The constants for* $\hat{e}_k$ Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds: there are radii $\delta_m > 0$, depending on Ξ only, such that for every system of regions with $\nu_m, \rho_m \leq \delta_m$ and every smooth k, the constants $c_m = (2|k|_{m+2}/\delta_m)(M_m+1)$ bound $(P\hat{e}_k)_m$ and its Lipschitz constant in w on the regions. The proof is the *scaling* $\Xi_m(y,\kappa) = (2|\kappa|/\delta_m)\,\Xi_m(y,(\delta_m/2|\kappa|)\kappa)$, which needs only the linearity of Ξ in κ; the note behind the formalization used the linearity of $D_y\Xi$ in κ instead.

III.43 The curve of perturbed zeros, the surjectivity in the chart at x_0, and Theorem 3.6.8 (1) (preprint 3.54)

The reduction at a nearby zero (in the normal form, with a perturbation e, regions R and constants c as in §III.42). Let $z = (a_0,w_0)$ be a level-1 zero of g in the domain of the germ with $a_0 \in \widetilde{C}$, $|a_0| < \nu_3$ and $|w_0|_1 < \rho_0$. Then w_0 is a fixed point of $B(a_0,\cdot)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.B_eq_of_zero (because $g^1 = (G_1, w - B)$), so $w_0 = \delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.sol_zero_eq. On the domain $D_3 = (-\mu_3,\mu_3) \times (\{|a| < \nu_3\} \cap \widetilde{C})$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDom, which has unique derivatives Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.uniqueDiffOn_curveDom, put $(\mu,a) \mapsto (a,\delta_\mu(\mu,a)) \in E_1$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift_mem and

$$\widetilde{G}(\mu,a) = (g + \mu\,\iota e)(a,\delta_\mu(\mu,a))$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap. *Its* W*-component vanishes* Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap_snd: $\delta_\mu(\mu,a)$ is the fixed point of $B_\mu(a,\cdot)$, and the W-component of $g + \mu\iota e$ is $w - B_\mu(a,w)$ (read on level 1, Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.

snd_g_toBase). So the whole equation is carried by $H = (\widetilde{G})_Q$. $\widetilde{G}$ *is* C^1 on D_3 Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.contDiffOn_curveMap: δ_μ is C^1 into W_2 by Theorem 3.3.3 (preprint 3.30) on the re-indexed germ (Proposition 1.2.1 (preprint 1.16) on level 1), and g, e are C^1 from level 1 to level 0. *Its derivative at* $(0, a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_curveMap is

$$D\widetilde{G}(0,a_0)(\dot\mu,\alpha) = Dg(z)(\alpha,\ \iota D\delta_\mu(0,a_0)(\dot\mu,\alpha)) + \dot\mu\,\iota e(z)$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solD Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv: the chain rule, with the derivative of the solution germ on level 1 Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level, the derivative of g at z Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_Dg', and the product rule for $\mu\,\iota e$, whose derivative at $\mu = 0$ in the base directions vanishes. Its W-component vanishes too Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_snd.

Lemma: $D_aH(0,a_0)$ is onto, by a dimension count Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq. *Statement.* Let moreover $\ker Dg(z)$ be finite-dimensional of dimension $\dim P - \dim Q$. Then $M_0 = D_aH(0,a_0) : P \to Q$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.DH is onto, and $\ker Dg(z)$ is the graph of $L = \iota D_a\delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solL over $\ker M_0$. *Proof.* At $\dot\mu = 0$ the derivative formula reads $Dg(z)(\alpha, L\alpha) = (M_0\alpha, 0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_inr Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.Dg'_graph, so the graph of L over $\ker M_0$ lies in $\ker Dg(z)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.map_graph_le_ker. The graph map is injective, so $\dim\ker M_0 \le \dim\ker Dg(z) = \dim P - \dim Q$; rank–nullity gives $\operatorname{rank} M_0 \ge \dim Q$, so M_0 is onto, and then $\dim\ker M_0 = \dim\ker Dg(z)$ and the inclusion is an equality. □ The obvious route, eliminating the W-block with $(1 - D_2B(z))^{-1}$, needs $\|D_2B(z)\| \le \frac12$ on the level where $Dg(z)$ is known to be onto, i.e. the surjectivity on the shifted structure; the dimension count uses only the kernel dimension on level 0 (Finding 184).

Proposition: the curve of perturbed zeros Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.exists_perturbed_curve. *Statement.* Under the hypotheses of the lemma, let $v \in \ker Dg(z)$ with $a_0 + t(v_1+\beta) \in \widetilde{C}$ for $t \in [0,t_0]$, $|\beta| < \eta$ (v_1 the P-component). Then there are $\mu_0 > 0$ and a curve $u'(\mu) = (a(\mu), \delta_\mu(\mu, a(\mu)))$, $0 \le \mu < \mu_0$, on level 2, in the domain of the germ with $a(\mu) \in \widetilde{C}$, with $u'(0) = z$ (on level 1), right differentiable at 0 as a map into E_2, and $(g + \mu\iota e)(u'(\mu)) = 0$. *Proof.* (i) By the lemma, M_0 is onto and $v_1 \in \ker M_0$ (as $v = (v_1, Lv_1)$). (ii) Choose a_1 with $M_0a_1 = -D_\mu H(0,a_0)$, and $\alpha_0 = a_1 + sv_1$ with s from the rescaling lemma of §III.41: α_0 is inward at a_0 (with $|\beta| < 1$) and $DH(0,a_0)(1,\alpha_0) = D_\mu H(0,a_0) + M_0a_1 + sM_0v_1 = 0$. (iii) H is C^1 on D_3 with $H(0,a_0) = 0$, and $\widetilde{C}$ is convex; the one-sided curve lemma (§III.41) on $\$\square = \$$ the open region of level 3 gives $a(\mu)$ with right derivative α_0 and $H(\mu, a(\mu)) = 0$. (iv) The level-1 map $(\mu, a) \mapsto \delta_\mu(\mu,a) \in W_2$ is differentiable at $(0,a_0)$ within D_3 (Theorem 3.3.3 (preprint 3.30), Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level), so by the chain rule for $\mu \mapsto (\mu, a(\mu))$ the curve u' is right differentiable at 0 into E_2. (v) At $\mu = 0$, $\delta_\mu(0,a_0) = w_0$; and $(g + \mu\iota e)(u'(\mu)) = \widetilde{G}(\mu, a(\mu)) = (H(\mu,a(\mu)), 0) = 0$. □

The normal form at a nearby point. For the chart at x_0: at every level-1 point y of $\psi(U_1)$ in the germ, $Dg(y) =$ fiber $\circ\, D(h+t)(\operatorname{glue} y) \circ$ glue for the normal form g of $h + t$ Chapter3/GlobalIFTNearbyOnto · StabilityData.scDerivGerm_pushMap₀_at (the coordinate change is linear), and the kernel of fiber $\circ\, T \circ$ glue is $\operatorname{split}(\ker T)$ Chapter3/GlobalIFTNearbyOnto · StabilityData.ker_fiber₀_comp_glue₀, split being injective Chapter3/GlobalIFTNearbyOnto · StabilityData.split₀_injective. Corollary 3.4.3 (preprint 3.39) is not used: the facts at the nearby zero come from the chart at x_0 through these linear maps (Finding 187).

Theorem A: the surjectivity in the chart at x_0 Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_chart_onto. *Statement.* Take the data of the graph description in the chart at x_0 (§III.29, Chapter3/GlobalIFTNearbyGraphG · theorem_3_54_chart_graph): the strong bundle retraction $R = (r, \rho)$ on U, a

filling germ g of $\mathbf{f}$ on G_g with a derivative family Dg, $\mathbf{f}(0)=0$, the conjugation Ψ covering ψ, $\psi(0)=0$, the stability data on a germ that ψ' maps into G_g level by level (inside G'', §III.48) with the basic germ h', $D(h'+\iota t)(0)$ onto with kernel K in good position, a closed good complement Y of K, the C^1 graph $\sigma : V' \to Y$ with $\sigma(0)=0$, $D\sigma(0)=0$ and its C^k-lifts, an open W_o on which $h'+\iota t$ is the push-forward of g, an open $N_2 \ni 0$ on which ψ maps into the germ and into $U' \cap W_o$, and on which ψ maps the zeros of g in O onto the graph. Assume condition (2) of Definition 3.1.4 (preprint 3.4) on $U \cap N_2$ and $K \neq 0$. Then there is an open $N_4 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_4$ of g, every level-1 point $z_{0,1}$ over z_0 and every smooth k, there is $x \in T_{z_0}O = Dr(z_{0,1})E_0$ with $Dg(z_{0,1})x = \rho(z_0)k$. *Proof*, in the order of the Lean:

1. *Data at x_0, independent of k.* Lemma 3.6.9 (preprint 3.55) on the graph with its radius ρ_2 `Chapter3/GlobalIFTFull · lemma_3_55_relOpen` (§III.32); the kernel at the nearby zeros on an open $N_3 \subseteq N_2$ `Chapter3/GlobalIFTNearbyKernelG · theorem_3_54_chart_kernel` (§III.32); the normal form $\mathcal{D}$ of Theorem 3.4.2 (preprint 3.38) `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`; the radii δ_m of the bounds of Ξ `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds` and contraction regions below them `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii`; and the radius r_{42} of Lemma 3.5.5 (preprint 3.42) for the bound ρ_0 `Chapter3/LocalSolvability · StabilityData.lemma_3_42`: zeros of the normal form on level 0 smaller than r_{42} have their W-component on level 1, smaller than ρ_0 there.
2. *The neighbourhood.* Let π be the projection onto K along Y (a topological complement: Y closed, K finite-dimensional), $c_s = \|\mathrm{split}\|+1$, and $r' = \min(\rho_2/(\|\pi\|+1), r_{42}/c_s, \nu_3/c_s)$. By the continuity of ψ on U, $\psi^{-1}(B(0,r')) \cap U = O_4 \cap U$ for an open $O_4 \ni 0$; put $N_4 = N_3 \cap O_4$. Nothing here depends on k.
3. *At a zero $z_0 \in O \cap N_4$.* The kernel theorem at z_0 gives $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point x_1 over it, $D(h'+\iota t)(x_1)$ onto and Fredholm of index $\dim P - \dim Q$, its kernel N_v, and $\ker Dg(z_{0,1}) = D\psi^{-1}(x_1)N_v \subseteq T_{z_0}O$. Since $|v| = |\pi(\psi(z_0))| \le \|\pi\| r' < \rho_2$, Lemma 3.6.9 (preprint 3.55) at v gives $x_1 \in C'$ and N_v in good position to C'_{x_1}.
4. *The zero of the normal form.* $z = \mathrm{split}(x_1)$ lies in the germ on level 1, with $a_0 = (x_1)_P \in \widetilde{C}$; $|\mathrm{split}\,\psi(z_0)| < c_s r'$, so $|a_0| < \nu_3$, and Lemma 3.5.5 (preprint 3.42) gives $|w_0|_1 < \rho_0$; g vanishes at z because $h'+\iota t$ vanishes at the graph point.
5. *Its kernel.* $Dg(z) = \mathrm{fiber} \circ D(h'+\iota t)(x_1) \circ \mathrm{glue}$, so $\ker Dg(z) = \mathrm{split}(N_v)$, finite-dimensional of dimension $\dim\ker D(h'+\iota t)(x_1) = \dim P - \dim Q$ (for a surjective Fredholm operator the kernel dimension is the index, `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`).
6. *The inward direction.* $N_v \neq 0$: its dimension is $\dim P - \dim Q$, which is also $\dim K$ ($D(h'+\iota t)(0)$ is onto and Fredholm of that index, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`), and $K \neq 0$. The inward lemma at x_1 (§III.41) gives $v_v \in N_v$ with $x_1 + t(v_v+\zeta) \in C'$; then split $v_v \in \ker Dg(z)$, and its P-component is inward at a_0 in the quadrant of the normal form for $|\beta| < \eta/(\|\mathrm{glue}\|+1)$, since $\mathrm{glue}(\beta,0)$ has norm at most $\|\mathrm{glue}\|\,|\beta|$.
7. *The perturbation and the curve.* Take the level-1 representative k_1 of k; $\hat{e}_k$ is an sc$^+$-perturbation (§III.42) with constants c_m on the regions; the proposition above gives μ_0 and the curve u' of zeros of $g + \mu\iota\hat{e}_k$ through z.
8. *Back in the chart.* $u(\mu) = \psi^{-1}(\mathrm{glue}\,u'(\mu))$ on level 1: ψ^{-1} is C^1 from level 2 to level 1 (Proposition 1.2.3 (preprint 1.18), `Chapter1/Levels · IsScSmooth.contDiffOn_levels`), so u is right differentiable at 0 with some velocity ξ_1, $u(0) = z_{0,1}$, and its base points $u_\mu = \psi^{-1}(\mathrm{glue}\,\iota u'(\mu))$ lie in U.
9. *The perturbed equation.* By continuity at $\mu = 0$, $\mathrm{glue}\,\iota u'(\mu) \in W_o$ and $u_\mu \in N_2$ for $0 \le \mu < \mu_2$ `Chapter3/GlobalIFTNearbyOnto · exists_Ico_of_eventually_nhdsWithin_Ici`. There the push-forward identity of g, the description of $\hat{e}_k$ as the Ψ-fibre image of $\rho(r(u_\mu))k$ transported by the linear fiber, and the injectivity of the fibre maps of Ψ turn $(g+\mu\iota\hat{e}_k)(u'(\mu)) = 0$ into $g(u_\mu) + \mu\rho(r(u_\mu))k = 0$.
10. *Tangency.* By the first lemma of §III.41 and condition (2) on N_2, $u_\mu \in O$.
11. *Differentiation.* r is sc^1 on U, and g is an sc^1-germ on G_g; the curve $u(\mu)$ stays on level 1 of G_g, because $\mathrm{glue}\,\iota u'(\mu)$ lies on level 1 of the germ of the stability data and ψ'_1 maps that into $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_inclK_mem_of_subset_shrinkGerm`. So both are differen-

tiable from level 1 to level 0 at $z_{0,1}$ along the curve, g by the chain rule within $(G_g)_1$ with derivative $Dg(z_{0,1})$ (with g sc-smooth on U no such condition was needed). Along the curve, $r(u_\mu) = u_\mu$, so the right derivatives at 0 agree, $Dr(z_{0,1})\xi_0 = \xi_0$ ($\xi_0 = \iota\xi_1$; uniqueness of right derivatives, §III.41): $\xi_0 \in T_{z_0}O$. The map $\mu \mapsto \rho(r(u_\mu))k$ is continuous at 0 (R is continuous on level 0), so $\mu \mapsto \mu\rho(r(u_\mu))k$ has the right derivative $\rho(r(z_0))k = \rho(z_0)k$; differentiating $g(u_\mu) = -\mu\rho(r(u_\mu))k$ gives $Dg(z_{0,1})\xi_0 = -\rho(z_0)k$. Take $x = -\xi_0$. □

Where the hypotheses enter. Tameness and good position at x_0 through the graph description, Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55) (steps 1, 3) and the inward direction at boundary zeros (step 6); condition (2) of the filler through the tangency (step 10); $K \neq 0$ for the inward direction. The sc-Fredholm property at y is not used here; it enters only through the closed range in the assembly step. Condition (3) of the filler at the nearby zeros is not used. *Why there is no uniformity problem.* The only k-dependent quantities are the constants c_m and the μ-intervals; N_4 is fixed in step 2 before k is chosen. The perturbation argument of §III.33 applied the implicit function theorem at x_0 to the Λ_0-extended problem, whose neighbourhood depends on Λ_0; here it is applied at the nearby zero (Finding 181).

Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'`. *Statement.* Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position. Then there is an open $V_x \ni x_0$ such that (f, y) is in good position at every zero $y \in V_x$. *Proof.* Destructure the good position at x_0: the centred chart c, the filling germ g with its derivative family, the sc⁺-germ, the conjugation, shrunk to G'' (§III.48), and the good position in the chart. Tameness gives $(1 - Dr(0))E \subseteq E_0$ by Proposition 2.5.3 (preprint 2.35) `Chapter2/TameRetracts · proposition_2_35`; the graph description in the chart `Chapter3/GlobalIFTNearbyGraphG · theorem_3_54_chart_graph` and the kernel at the nearby zeros `Chapter3/GlobalIFTNearbyKernelG · theorem_3_54_chart_kernel` give $N_1 \supseteq N_2 \supseteq N_3$. A zero y of f in the chart domain with $\varphi(y) \in N_2$ is a zero of g ($\mathbf{f} = g$ on $O \cap N_1$). *Case $K = 0$.* Put $V_x = \varphi^{-1}(N_2)$ in the chart domain. A zero $y \in V_x$ maps by $\psi\varphi$ onto a graph point $v + \sigma(v)$ with $v \in K = 0$, so $\psi(\varphi(y)) = \sigma(0) = 0 = \psi(0)$, $\varphi(y) = 0$ (as $\psi^{-1}\psi = 1$ on U), and $y = x_0$ by the injectivity of the chart: the only zero is x_0, in good position by hypothesis. *Case $K \neq 0$.* Theorem A gives N_4 (condition (2) on $U \cap N_2$ is the filler's, §III.29); put $V_x = \varphi^{-1}(N_3 \cap N_4)$ in the chart domain. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O$ is a zero of g; the kernel theorem gives the level-1 point $z_{0,1}$, $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$ and its good position inside $T_{z_0}O$; y is smooth `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so (f, y) is an sc-Fredholm germ; $\mathbf{f} = g$ on O near z_0. The assembly step in its germ form `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense` (§III.31; whole-chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`) applies at $z_{0,1} \in (G_g)_1$ with (a) from Theorem A — $\rho(z_0)\hat{k}$ in the range of $Dg(z_{0,1})|_{T_{z_0}O}$ for every smooth $\hat{k}$ — and (b) from the kernel theorem, $\ker Dg(z_{0,1}) \cap T_{z_0}O = \ker Dg(z_{0,1})$. □ **No hypothesis is added**: the statement has exactly the hypotheses of the kernel theorem of §III.32 plus the sc-Fredholm property (Finding 189).

Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) with the text's hypotheses. Theorem 3.6.8 (1) and (2) (preprint 3.54) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54'`: (1) as above on V_x, and (2), the regular data of Definition 3.2.1 (preprint 3.25) at x_0 of a dimension d that every nearby datum shares, by §III.37 `Chapter3/GlobalIFTTheoremsG · theorem_3_54_two` fed with (1). Theorem 3.1.26 (preprint 3.23) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_23'`: (1) supplies the open set of nearby good position that the boundary case from the nearby good position needs `Chapter3/GlobalIFTTheoremsG · theorem_3_23_of_nearby`, and every zero of the resulting U is in good position. Theorem 3.1.22 (preprint 3.19) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_19'`: Theorem 3.1.26 (preprint 3.23) and Lemma 2.4.6 (preprint 2.26), $d_S \leq d_X = 0$ `Chapter2/SubMPolyfold · lemma_2_26`. The first forms under the filler condition at the nearby zeros (§§III.33, III.37) are kept; that condition is now a theorem (§III.45).

III.44 The dimension of the solution set is the Fredholm index, and the index is locally constant

Theorem 3.1.22 (preprint 3.19) says that the solution set is a manifold "whose dimension agrees with the Fredholm index", and the proof of Theorem 3.6.8 (preprint 3.54) takes $d = \dim\ker f'(x)$. In the statements of §§III.34–III.37 the dimension was existential. This section identifies it, at every zero and in every chart, and deduces the local constancy of the index along the solution set (Findings 170–171).

"$f'(y)$ is onto, Fredholm, of index d", chart-intrinsically `Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex`: in every strong bundle chart around y compatible with the atlas, at the level-1 point z over $\varphi(y)$, every corestriction $L : T_zO \to K_z$ of the linearization $\mathrm{Lin}(z)$ (§III.31) is onto and Fredholm with $\mathrm{ind}\, L = \dim\ker L = d$. "$f'(y)$ is onto" alone `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt`: in *some* chart, $\mathrm{Lin}(z)$ maps onto the fibre $\{k \mid (\varphi(y), k) \in K\}$; by the chart independence it is then onto in every chart.

Proposition 3.6.5 (preprint 3.52) in every chart `Chapter3/GlobalIFTIndexG · StrongBundle.linAt_index_of_conjugationData`. *Statement.* Let f be an sc-smooth section, x_0 a zero, c a strong bundle chart centred at x_0 with a filling germ g of **f** and a derivative family Dg on a germ inside U, an sc⁺-germ s with $s(0) = g(0)$, and conjugation data d_c of $g - s$ to a basic germ `Chapter3/FredholmGermIndex · ConjugationData` with finite-dimensional spaces P_n, Q. Then in every strong bundle chart c' around x_0, at the level-1 point z' over $\varphi'(x_0)$, every corestriction L' of $\mathrm{Lin}'(z')$ is Fredholm of index $\dim P_n - \dim Q$, and onto if the germ is in good position at $Dg(0)$ in c. *Proof.* Proposition 3.6.5 (preprint 3.52) in c (§III.30, `Chapter3/FredholmGermIndexG · proposition_3_52_local_index`): $Tf(0)$ is Fredholm of index $\dim P_n - \dim Q$. $Tf(0) = Dg(0)|_{T_0O}$ is $\mathrm{Lin}(0)$ by the bridge for germs at the centre, which lies in every germ `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt` (whole-chart: `Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt`), up to the identification of the fibres $\rho(0)F$ and $\rho(\iota 0)F$, which are equal subspaces but different types in Lean `Chapter3/GlobalIFTIndex · exists_rangeS_pCLM_equiv`. The chart independence (§III.31, `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`) carries the Fredholm property, the index and the surjectivity from c at 0 to c' at z'; the four spaces are complete as ranges of idempotents `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`. □ At a zero in good position this gives $\mathrm{IsSurjLinOfIndex}(f, x_0, \dim P_n - \dim Q)$ `Chapter3/GlobalIFTIndexG · StrongBundle.isSurjLinOfIndex_of_conjugationData`, because for a surjective Fredholm operator the kernel dimension is the index `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`; and the d with $\mathrm{IsSurjLinOfIndex}(f, x_0, d)$ equals $\dim P_n - \dim Q$ for every conjugation in the chart of Definition 3.6.6 (preprint 3.53) `Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.exists_isSurjLinOfIndex` (a conjugation exists, `Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty`). The first clause of Theorem 3.1.22 (preprint 3.19) with the index `Chapter3/GlobalIFTIndexG · theorem_3_19_linearization_index`: for an sc-Fredholm section and a zero in good position, in the chart of Definition 3.6.6 (preprint 3.53), $Dg(0)$ and $Tf(x_0)$ are Fredholm, $Tf(x_0)$ is onto, and $\mathrm{ind}\, Dg(0) = \mathrm{ind}\, Tf(x_0) = \dim\ker Tf(x_0) = \dim d.P_n - \dim d.Q$ for every conjugation d — the old formulation's "$n - N$ for some natural numbers" made precise.

The conjugation exposed. The graph description and the tangent germ data at the centre are built from the chart of Definition 3.6.6 (preprint 3.53) by destructuring the existential good position, so their statements did not record which conjugation they used. Their proofs now prove versions that expose it `Chapter3/GlobalIFTG · theorem_3_54_graph_proj_conj` `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart_proj_conj`: the filling germ with its germ and derivative family Dg, the sc⁺-germ, the good position at $Dg(0)$ in the chart and conjugation data d_c, with $\dim K = \dim d_c.P_n - \dim d_c.Q$ for $K = \ker D(h+t)(0)$, because $D(h+t)(0)$ is onto and Fredholm of index $n - N$ (§III.18, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`). The earlier statements are corollaries.

Dimension = index. *At the centre* `Chapter3/GlobalIFTIndexG · theorem_3_54_tangentGermChart_index`: for an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero x_0 in good position, the data of Definition 3.2.1 (preprint 3.25) at x_0 for $\{y \in V_x : f(y) = 0\}$ (with $\pi(O) \subseteq C$)

have a dimension d with IsSurjLinOfIndex(f, x_0, d): the exposed $d = \dim K = \dim d_c.P_n - \dim d_c.Q$ and the previous paragraph. The same for the global solution set `Chapter3/GlobalIFTIndexG · theorem_3_54_tangentGermChart_solutionSet_index` (restriction of the chart, §III.37) and for the regular data of Theorem 3.6.8 (2) (preprint 3.54) `Chapter3/GlobalIFTIndexG · theorem_3_54_two_index` (the re-centring keeps the dimension, `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`). *At a nearby zero* `Chapter3/GlobalIFTIndexG · isSurjLinOfIndex_of_forall_dim`: if y is a zero in good position at which every datum of Definition 3.2.1 (preprint 3.25) for the solution set has dimension d, then IsSurjLinOfIndex(f, y, d) — the data at y from the centre theorem at y have dimension ind $f'(y)$, and that dimension is d.

The index is locally constant along the solution set `Chapter3/GlobalIFTIndexG · theorem_3_54_index_locally_constant`. *Statement.* For an sc-Fredholm section of a strong bundle whose charts have tame base retractions, a zero x_0 in good position: there are an open $V_x \ni x_0$ and d with IsSurjLinOfIndex(f, y, d) at every zero $y \in V_x$; in particular ind $f'(y) =$ ind $f'(x_0)$. *Proof.* Theorem 3.6.8 (1) (preprint 3.54) (§III.43) gives an open $V_{x,1}$ of nearby good position; Theorem 3.6.8 (2) (preprint 3.54) with the index on it gives regular data D of dimension d with source in $V_{x,1}$ along which every datum has dimension d; on the source of D, every zero is in good position with all its data of dimension d, and the previous paragraph applies. □ The proof does not repeat the kernel computation of the matrix argument (Finding 188): "the dimension is locally constant" (from the re-centring and Sard's lemma in fixed dimension, §III.35) and "the dimension is the index" give "the index is locally constant". Along the solution set this is the index stability of the text's Proposition 6.2.2 (preprint 6.2), whose proof needs exactly the surjectivity of §III.43. *Theorem 3.6.8 (preprint 3.54) with the index* `Chapter3/GlobalIFTIndexG · theorem_3_54_index`: on an open $V_x \ni x_0$ every zero is in good position with IsSurjLinOfIndex(f, y, d), and the solution set has at x_0 regular data of dimension d whose chart domain *is* V_x, every datum at a point of the solution set in V_x having dimension d.

Theorems 3.1.26, 3.1.22 and 3.6.11 (preprint 3.23, 3.19 and 3.57) with the index. Theorem 3.1.26 (preprint 3.23) `Chapter3/GlobalIFTIndexG · theorem_3_23_index`: for an sc-Fredholm section with tame base retractions and a zero x in good position, an open $U \ni x$ and d such that every zero $y \in U$ is in good position with IsSurjLinOfIndex(f, y, d), the zeros in U are smooth, and $S(f, U)$ is a sub-M-polyfold with tame induced structure equivalent to a d-dimensional manifold with boundary with corners — the proof of the boundary case from the nearby good position (§III.37) with Theorem 3.6.8 (2) (preprint 3.54) with the index. Theorem 3.1.22 (preprint 3.19) `Chapter3/GlobalIFTIndexG · theorem_3_19_index`: for $d_X \equiv 0$ the same, with degeneracy index 0 everywhere (Lemma 2.4.6 (preprint 2.26)): "a smooth manifold whose dimension agrees with the Fredholm index", now with the tameness of the induced structure. Theorem 3.6.11 (preprint 3.57) `Chapter3/GlobalIFTIndexG · theorem_3_57_index`: under the hypotheses of §III.37 (good position at every zero, $S \neq \emptyset$), at every $x \in S$ there are regular data of a dimension d, shared by every datum at the points of S in their chart domain, with IsSurjLinOfIndex(f, y, d) at every zero y of that domain.

III.45 The kernel as tangent space, good position at interior zeros, and the filler condition as a theorem

"ker $f'(y) = T_y S(f, U)$", in a chart `Chapter3/GlobalIFTIndexG · theorem_3_54_ker_eq_tangent`. *Statement.* For an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero x_0 in good position, there are the strong bundle chart c centred at x_0 of Definition 3.6.6 (preprint 3.53), the conjugating sc-diffeomorphism $\psi : U \to U' \subseteq C'$ with inverse ψ' and $\psi(0) = 0$, the finite-dimensional $K = \ker D(h + t)(0)$, the solution graph $\sigma : V' \to E'$ (C^1, $\sigma(0) = 0$, $D\sigma(0) = 0$, V' relatively open in $K \cap C'$), an open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$, and an open $V_{x,2} \subseteq V_x$ containing x_0, such that at every zero $y \in V_{x,2}$, with $\psi(\varphi(y)) = v + \sigma(v)$, and every level-1 point z over $\varphi(y)$:

$$D\psi(z)(\ker \mathrm{Lin}(z)) = N_v = \{n + D\sigma(v)n \mid n \in K\},$$

the tangent space at v of the graph, which is the solution set in the chart $\psi \circ \varphi$ of its tangent germ data. *Proof.* From the kernel statement at the nearby zeros (§III.32, `Chapter3/GlobalIFTNearbySurjG · theorem_`

3_54_nearby_kernel): z lies on level 1 of the germ of g, $\ker Dg(z) = D\psi'(\psi z)N_v$, $\ker Dg(z) \subseteq T_zO$, and $\mathrm{Lin}(z) = Dg(z)$ on T_zO (the bridge for germs, which needs exactly z in the germ and $0 \in O$, Chapter3/LinearizationChartsG · linAt_eq_germDeriv; whole-chart Chapter3/LinearizationCharts · linAt_eq_scDeriv), so $\ker \mathrm{Lin}(z) = \ker Dg(z)$; and $D\psi(z)D\psi'(\psi z) = 1$ on level 0 by the chain rule for $\psi \circ \psi' = 1$ on U'. □ The intrinsic statement, with T_yS as a subspace of T_yX (Definition 2.5.8 (preprint 2.38)), is not formalized: T_yX has no linear structure in the formalization (Chapter 2 gap G.13); the chart form is the statement (Finding 172).

Good position at a zero with $d_X = 0$ is surjectivity. *Linear algebra* Chapter3/GlobalIFTIndex · isInGoodPositionIn_of_subset: a subspace N of $T \subseteq E_0$ is in good position inside T to every cone C containing T — $N \cap C = N$ has interior in N, and any algebraic complement of N in T is a good complement with any constant, both sides of the condition being true. *The statement* Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_linAt_surj_of_degIndex_eq_zero: let f be an sc-smooth section, y a zero with (f, y) an sc-Fredholm germ and $d_X(y) = 0$, c a strong bundle chart around y with tame base retraction, and $\mathrm{Lin}(z)$ onto K_z at the level-1 point z over $\varphi(y)$; then (f, y) is in good position. *Proof.* The model of c is a tame retract, so $d_C(\varphi(y)) = d_X(y) = 0$ by the strong form of Proposition 2.5.6 (preprint 2.36) Chapter2/TameRetracts · degIndex_eq_chartDegIndex_of_isTameRetract; no coordinate of the presentation vanishes at $\varphi(y)$, so the partial quadrant at $\varphi(y)$ is the whole space, and the kernel condition of the assembly step holds by the linear algebra. The assembly step with all of K_z in the range Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj concludes. □ For an sc-Fredholm section with tame base retractions, "$f'(y)$ onto in some chart" therefore gives good position at a zero with $d_X(y) = 0$ Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt (y is smooth, so the germ is sc-Fredholm); conversely good position gives "$f'(y)$ onto" in the chart of Definition 3.6.6 (preprint 3.53) Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.isLinSurjAt; so at such a zero the two are equivalent Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt — the text's remark after Theorem 3.6.8 (preprint 3.54). In a non-tame chart $d_C(\varphi(y))$ can exceed $d_X(y)$, and the cone would have to be transported from a chart realizing $d_X(y)$; the standing tameness hypothesis makes that unnecessary (Finding 173).

Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) with "$f'(x)$ surjective". Theorem 3.1.22 (preprint 3.19) Chapter3/GlobalIFTIndexG · theorem_3_19_of_surjective: for $d_X \equiv 0$, an sc-Fredholm section with tame base retractions, $f(x_0) = 0$ and $f'(x_0)$ onto in some chart: the equivalence gives good position at x_0, and Theorem 3.1.22 (preprint 3.19) with the index (§III.44) applies. Theorem 3.6.10 (preprint 3.56) with the tameness and the regular data Chapter3/GlobalIFTIndexG · theorem_3_56_tame: for $d_X \equiv 0$ and good position at every zero, $S \neq \emptyset$: Theorem 3.6.11 (preprint 3.57) with the index (§III.44) and Lemma 2.4.6 (preprint 2.26) — the solution set is a sub-M-polyfold with tame induced structure, a manifold of locally constant dimension without boundary, with regular data at every zero of a dimension equal to the index at the zeros of their chart domain. With "$f'(x)$ surjective at every zero" Chapter3/GlobalIFTIndexG · theorem_3_56_of_surjective: the equivalence at every zero, then the previous statement.

The converse of the matrix argument, and the filler condition as a theorem. *The converse* Chapter3/GlobalIFTIndex · normal_injective_of_exists_range_scDeriv_eq_ρ. Let z be a level-1 point over a point z_0 of the retract with $\ker Dg(z) \subseteq T_{z_0}O$, and suppose $\rho(z_0)k \in Dg(z)(T_{z_0}O)$ for every $k \in F$. Then $Dr(z)b = 0$ and $\rho(z_0)Dg(z)b = Dg(z)b$ imply $b = 0$: choose $a \in T_{z_0}O$ with $Dg(z)a = \rho(z_0)Dg(z)b = Dg(z)b$; then $a - b \in \ker Dg(z) \subseteq T_{z_0}O$, so $b \in T_{z_0}O$; writing $b = Dr(z)w$, idempotence gives $b = Dr(z)b = 0$ Chapter3/LinearizationCharts · scDeriv_r_idem_at. The surjectivity of $Dg(z)$ is not needed for this direction. With the direction of §III.33 Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective, under $Dg(z)$ onto and $\ker Dg(z) \subseteq T_{z_0}O$ (and g agreeing with **f** on O near a zero z_0 of **f**): $\rho(z_0)F \subseteq Dg(z)(T_{z_0}O)$ iff condition (3) of Definition 3.1.4 (preprint 3.4) holds at z_0 Chapter3/GlobalIFTIndex · exists_range_scDeriv_eq_ρ_iff_normal_injective — the "iff" of Finding 100, both directions formalized. Both are linear algebra in the operator $Dg(z)$ and hold for a germ at the points of its germ Chapter3/GlobalIFTIndexG · normal_injective_of_exists_range_deriv_eq_ρ_germ Chapter3/GlobalIFTIndexG · exists_range_deriv_eq_ρ_iff_normal_injective_germ. *The filler condition* Chap-

ter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm: for an sc-Fredholm section with tame base retractions and a zero x_0 in good position, (f, x_0) is in good position with the filler condition at the nearby zeros Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled. *Proof.* Take the chart and filler of the kernel statement (§III.32) and the open $V_{x,1}$ of nearby good position (§III.43); let N be an open neighbourhood of 0 on which $\mathbf{f} = g$ on O and $O \cap N = \varphi(V_{x,2} \cap V_{x,1})$ (Chapter2/MPolyfold · exists_isOpen_image_eq). At a level-1 zero z_1 of g over a point of $O \cap N$, on level 1 of the germ of g (the condition asks no more, §III.33): $z_0 = \varphi(y)$ for a zero y of f in $V_{x,2} \cap V_{x,1}$; y is in good position, so $\mathrm{Lin}(z_1)$ is onto K_{z_0} in the chart c (§III.44); the kernel statement gives $\ker Dg(z_1) \subseteq T_{z_0}O$; and the converse gives condition (3) at z_0. □ So the added hypothesis of the first forms of Theorems 3.6.8, 3.1.26, 3.1.22 (preprint 3.54, 3.23, 3.19) (§§III.33, III.37) holds whenever their other hypotheses do: it was never a restriction, but proving that needed the surjectivity of §III.43 (Finding 174).

III.46 What the second addendum taught us

1. **The text's Theorem 3.6.8 (1) (preprint 3.54) is true, and the gap of its proof can be closed.** The surjectivity of $f'(y)$ at the zeros near a zero in good position does not follow from Corollary 3.5.3 (preprint 3.41) and Lemma 3.6.9 (preprint 3.55), as the text (and the book's arXiv version) says; it follows from an implicit function theorem *at the nearby zero* for the perturbed filler $g + \mu\rho(r(\cdot))k$, one smooth direction k at a time, whose curve of zeros is tangent to the retract by condition (2) of the filler Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_chart_onto Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one' (§§III.41–III.43). With it, Theorems 3.6.8, 3.1.26 and 3.1.22 (preprint 3.54, 3.23 and 3.19) hold with the text's hypotheses.
2. **The uniformity problem disappears when the contraction regions are written down.** Theorem 3.3.3 (preprint 3.30) is applied at the centre to the k-extended contraction germ, whose regions in a are those of the unperturbed germ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction; only the μ-interval depends on k, and the bounds of the perturbation scale linearly in k Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds. The black-box use of Theorem 3.5.11 (preprint Corollary 3.47) for an extended problem gives neighbourhoods depending on the extension.
3. **The index raise of Theorem 3.4.2 (preprint 3.38) is what makes the perturbation C^1 on the shifted levels** (§III.42), and the surjectivity of the reduced finite-dimensional problem comes from a dimension count on level 0 Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq, not from inverting $1 - D_2B$.
4. **The index is locally constant along the solution set** Chapter3/GlobalIFTIndexG · theorem_3_54_index_locally_constant, the special case of the text's Proposition 6.2.2 (preprint 6.2), and the dimension of the solution set is the Fredholm index at every zero, in every chart Chapter3/GlobalIFTIndexG · theorem_3_19_index (§III.44); $\ker f'(y)$ is the tangent space of the solution set in the chart of the proof Chapter3/GlobalIFTIndexG · theorem_3_54_ker_eq_tangent (§III.45).
5. **The filler condition at the nearby zeros was never a restriction** Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm, and good position at a zero with $d_X = 0$ is surjectivity for a tame strong bundle Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt (§III.45), so Theorems 3.1.22 and 3.6.10 (preprint 3.19 and 3.56) hold with "$f'(x)$ surjective".
6. **Theorem 3.1.21's (preprint 3.18) "readily verified" is a construction**: translation charts of $\mathbb{R}^n$, the pull-back bundle as a structure, the product rule for a scalar factor, and the extension of a basic germ by inert parameters Chapter3/ParamPerturbation · IsBasicGerm.paramExtend Chapter3/ParamPerturbationG · theorem_3_18 (§III.40).
7. **What is left of Chapter 3**: Proposition 3.6.5 (preprint 3.52) as an sc-Fredholm operator on T_xX; $\ker f'(y) = T_yS$ with $T_yS \subseteq T_yX$; Theorem 3.7.1 (preprint 3.58) for a general sc-splitting; Lemma 3.8.13 (preprint 3.70) for a general presentation of the partial quadrant; Theorem 3.2.3 (preprint 3.27) for sets whose data violate the second regularity condition; and Proposition 6.2.2 (preprint 6.2) beyond solution sets (since proved, with the text's hypotheses for paths on $\mathbb{R}$, §III.49). The

only added hypotheses are the closedness of good complements and the regularity conditions of Theorem 3.2.3 (preprint 3.27) for general subsets.

III.47 The tameness hypothesis repaired

The implicit function theorems of §§III.23 and III.32–III.45 were first formalized with the hypothesis, `htame` in the Lean, that the base retraction of **every** strong bundle chart compatible with the atlas is tame; the phrase "a strong bundle whose charts have tame base retractions" in those sections describes it. No strong bundle over a nonempty base satisfies it, so the statements that assumed it were vacuously true (Finding 190). This section records why, and what replaced it. The proofs told above are unchanged; the hypotheses and the charts of the conclusions changed, and the names of the theorems did not.

Why the hypothesis fails. Compatibility of strong bundle charts (Definition 2.6.4 (preprint 2.50)) constrains the transition maps between the retracts K, not the retraction whose image K is. From a compatible chart with base retraction r on $U \subseteq C$, the chart $(1, \varphi)$ onto $\{1\} \times O$, with model $\mathbb{R} \oplus E$, quadrant $[0,\infty) \times C$ and base retraction $r'(a,u) = (1, r(u))$ on $[0,\infty) \times U$, has the transition maps of the first chart composed with affine maps, so it is compatible with every chart the first one is; but r' lowers the degeneracy index at $(0,u)$. The last point is machine-checked. The constant retraction $(a,w) \mapsto (1,0)$ of $[0,\infty) \oplus W$ `Chapter3/TameVacuity · ScRetraction.halfLineConst` has retract $\{(1,0)\}$ `Chapter3/TameVacuity · ScRetraction.halfLineConst_retract`, and its product with any sc-smooth retraction r of a nonempty U is not tame `Chapter3/TameVacuity · ScRetraction.halfLineConst_prod_not_isTame`: the degeneracy index of a product is the sum of the indices `Chapter2/FibredProduct · degIndex_prod`, so it is $1 + d_C(u)$ at $((0,0),u)$, $u \in r(U)$, and $d_C(u)$ at the image $((1,0),u)$, against condition (1) of Definition 2.5.2 (preprint 2.34). That the chart $(1,\varphi)$ is compatible is not formalized.

The new hypotheses. The text asks for the tameness of particular charts: the chart of an sc-Fredholm germ has "a tame strong local bundle" $K \to O$ (Definitions 3.1.6 and 3.6.3 (preprint 3.6 and 3.50)), and the proof of Proposition 3.6.5 (preprint 3.52) starts from "a tame strong bundle chart". A *tame chart* is a compatible strong bundle chart whose base retraction is tame `Chapter3/TameGerms · StrongBundle.IsTameChart`. *Tame germ data* at x are the data of Definition 3.1.11 (preprint 3.11) (§III.6) in a tame chart `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame`; *good position in a tame chart* adds Definition 3.6.6 (preprint 3.53) in that chart (§III.21) `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame`; and an sc-Fredholm section *with tame germ data* has them at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionTame`. Each implies the notion without tameness `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame.isScFredholmGerm` `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame.isGoodPositionGerm`. They can be met: tameness depends only on the domain of a retraction and its values there `Chapter3/TameGerms · ScRetraction.IsTame.congr`, and the identity retraction is tame `Chapter2/Tame · ScRetraction.id_isTame`, so a compatible chart whose base retraction is the identity of U is tame `Chapter3/TameGerms · StrongBundle.isTameChart_of_eqOn_id`.

The restated theorems. The proofs used the tameness for three charts only (Finding 191): the chart of the germ data at x_0, for $(1 - Dr(0))E \subseteq E_0$ in the linear part (§III.23) `Chapter3/GlobalIFTG · theorem_3_54_graph_proj_conj`; the chart $\psi \circ \varphi$ built from it, whose model is tame because r is (§§III.27, III.34) `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart_proj_conj`; and the chart in which $f'(y)$ is onto at a zero with $d_X(y) = 0$ (§III.45). So each theorem now takes (f, x_0) in good position in a tame chart in place of good position and `htame`; where it concludes something at the other zeros it takes an sc-Fredholm section with tame germ data; its conclusions are good position in tame charts, and the charts it produces are tame. This applies to Theorem 3.6.8 (1) (preprint 3.54) with the text's hypotheses `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'`, to its first form, whose filler condition is now stated in a tame chart `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled` `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one`, to Theorems 3.1.26 and 3.1.22 (preprint 3.23 and 3.19) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_23'` `Chapter3/GlobalIFTNearbyOntoG · theorem_3_19'`, to the global theorems, with good position in a tame chart at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_57` `Chapter3/GlobalIFTTheoremsG · theorem_3_56`

Chapter3/GlobalIFTTheoremsG · corollary_3_24, to the forms with the index Chapter3/GlobalIFTIndexG · theorem_3_54_index, and to the filler condition as a theorem Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm.

Reading the conclusion in the tame chart. At a nearby zero y the good position is proved in the chart at x_0 and carried to a chart centred at y (§III.31). That chart was the one Definition 3.1.11 (preprint 3.11) supplies at y, of which nothing says that it is tame. The chart change now gives the good position in **every** chart carrying germ data at y Chapter3/LinearizationCharts · StrongBundle.isGoodPositionAt_of_chart_dense, the statement of §III.31 being its corollary Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense. A good position given in any chart therefore holds in the tame chart of the germ data Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame: in the chart of the given good position $Tf(0)$ maps onto $K_0 \supseteq \rho(0)F_\infty$, $\ker Dg(0) \subseteq T_0O$ and $C_0 = C$, which are the hypotheses of the chart change at $z = 0$; in the germ reading the point $z = 0$ lies in every germ, $Tf(0)$ is onto all of K_0, and the form of the chart change with all of K_0 in the range needs no Proposition 3.6.5 (preprint 3.52) (§III.31) Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart. The same step gives good position from "$f'(y)$ surjective" at a zero with $d_X(y) = 0$: the surjectivity, given in one chart, holds in every chart Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective, in particular in the tame chart of the germ data at y, where $d_C = d_X = 0$ Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt.

Theorems 3.1.18 and 3.1.21 (preprint 3.15 and 3.18) keep tame germ data. The proof of Theorem 3.1.18 (preprint 3.15) (§III.6) does not change the chart, so $f + s$ has germ data in the tame charts of f Chapter3/TameGermsAddG · StrongBundle.IsScFredholmGermTame.sectionAdd Chapter3/TameGermsAddG · theorem_3_15_tame. For Theorem 3.1.21 (preprint 3.18) (§III.40) the germ of $\tilde{f}$ at (r_0, x) is taken in the product chart $(r \mapsto r - r_0) \times \varphi$, whose base retraction $1 \times r$ on $\mathbb{R}^n \times U$ is tame when r is: the identity retraction of $\mathbb{R}^n$ is tame, so $d(a, u) = d_C(u)$, and the minimal subspaces and tangent spaces of the product are products Chapter2/FibredProduct · ScRetraction.IsTameWith.prod Chapter3/ParamPerturbation · PullbackChartData.ofProd_isTame. Hence $\tilde{f}$ has tame germ data Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGermTame.param Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSectionTame.param, and so has $F(r, x) = f(x) + \sum_i r_i s_i(x)$ Chapter3/ParamPerturbationG · theorem_3_18_tame, which is what the implicit function theorems need when applied to F.

Not done (Finding 199): the compatibility of the chart $(1, \varphi)$, that is, the vacuity of `htame` at the level of charts; and the passage from an atlas of tame charts to tame germ data (Definition 3.1.11 (preprint 3.11) □ Definition 3.6.3 (preprint 3.50)). The text builds the tame chart into Definition 3.6.3 (preprint 3.50), and so do the theorems.

Germ data and the charts of a tame atlas (Findings 350–353, fourth addendum). The passage was to be made by re-centring a tame chart of the atlas at the point and transporting the filler and the conjugation to it. That strategy fails, because a filling is bound to its chart. *Condition (3) of Definition 3.1.4 (preprint 3.4) is a condition on the chart*: a filling germ in a chart with strong bundle retraction $R = (r, \rho)$ gives a topological isomorphism $\ker Dr(0) \cong \ker \rho(0)$, the linearization of the fill defect $y \mapsto [1 - \rho(r(y))]g(y)$ on $\ker Dr(0)$ Chapter3/TameGermData · IsFillingGerm.nonempty_kerEquiv; so in a chart with $\ker Dr(0) \neq 0$ and $\rho(0)$ injective no map fills anything Chapter3/TameGermData · not_isFillingGerm_of_ker. *A tame chart with no filling.* For the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ take the model $\mathbb{R} \oplus \mathbb{R}^n$ without boundary, the base retraction $r(a, u) = (0, u)$, the product of the constant retraction of $\mathbb{R}$ onto 0 with the identity Chapter3/TameGermData · TrivialBundle.Reven, the chart $x \mapsto (0, x)$, and $R((a, u), h) = ((0, u), h)$ Chapter3/TameGermData · TrivialBundle.evenBundleChart. It is compatible with the translation charts Chapter3/TameGermData · TrivialBundle.isChart_evenBundleChart and tame — the degeneracy index of $\mathbb{R} \oplus \mathbb{R}^n$ is 0 everywhere and $\mathbb{R} \oplus 0$ is an sc-complement of $T_xO = 0 \oplus \mathbb{R}^n$ Chapter3/TameGermData · TrivialBundle.Reven_isTame Chapter3/TameGermData · TrivialBundle.isTameChart_evenBundleChart — but $\ker Dr(0) = \mathbb{R} \oplus 0$ while $\rho(0) = 1$, so no map fills the principal part of any section in it Chapter3/TameGermData · TrivialBundle.not_isFillingGerm_evenBundleChart. *A tame atlas without germ data in its chart.* With the one-chart atlas $\{\text{this chart}\}$ Chapter3/TameGermData · TrivialBundle.bundleEven every chart of the atlas is tame Chapter3/TameGermData · TrivialBundle.bundleEven_atlas_isTameChart, the graph section of a C^∞ map has tame germ data — in the compati-

ble translation charts `Chapter3/TameGermData · TrivialBundle.isScFredholmSectionTame_graphSection_even` — and in the chart of the atlas no section has a filling `Chapter3/TameGermData · TrivialBundle.tame_atlas_no_germData_in_atlas_chart`. So Definition 3.6.3 (preprint 3.50) must be the existential over all compatible tame charts, as formalized; read as "in a chart of the tame atlas" it would make every section of this bundle non-Fredholm. Whether germ data in some compatible chart give germ data in some compatible tame chart, for a tame atlas, stays open: the transition between two charts is defined on the retracts only, while the filling lives off them, so it cannot simply be pushed forward.

III.48 Definition 3.1.11 (preprint 3.11) on germs

The text states Definitions 3.1.4, 3.1.6 and 3.1.11 (preprint 3.4, 3.6 and 3.11) for germs: the filled version is "a germ $(g, 0)$", the sc$^+$-section a local one, and a germ of neighbourhoods (Definition 3.1.1 (preprint 3.1)) is a sequence U_m that may shrink with m. The first formalization asked g and s to be sc-smooth on the whole chart domain U, on the ground that a chart can be restricted. This section records why that was narrower, and the two devices on which the proofs in the text's germ reading rest: the calculus of germs with whole-chart maps, and the shrinking of the basic germ (Findings 260–334).

Germs are not restrictions. On $\mathbb{R}$ with the constant structure, $f(x) = \sum_n c_n(x - a_n)^{n+1}|x - a_n|$, $a_n = 1/(n + 2)$, $c_n = (4(n + 2))^{-(n+2)}$, is C^k on $U_k = (-1/(k + 2), 1/(k + 2))$ — the terms $n \le k$ have their singular points outside, the tail is C^{k+2} — and C^{n+2} on no open set containing a_n. So f is an sc-smooth germ on (U_m) and sc-smooth on no open $U \ni 0$ `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth`. Restricting a chart on level 0 never makes the levels shrink, so the whole-chart reading is a restriction of the class of sections; for a section sc-smooth on X the two readings of Definition 3.1.11 (preprint 3.11) differ exactly in the regularity of g and s away from their germs (§III.6). Whether every germ datum can be traded for a whole-chart one is open: replacing the sc$^+$-germ by a whole-chart sc$^+$-section costs one level (the index raise of Theorem 3.4.2 (preprint 3.38)), and on level 0 no argument was found in either direction (Finding 264). No theorem depends on the answer, since the theorems are proved in the germ reading.

The calculus with whole-chart maps. The text forms $g + s$, $g - s$, $[1 - \rho(r(y))]g(y)$ and $\Psi_*(g)$ for germs without comment. Sums, differences, scalar multiples and compositions with sc-operators of sc^1-germs are sc^1-germs with the explicit derivative families, on $G_1 \cap G_2$ for germs on two germs `Chapter3/GermCalculus · IsSc1GermWith.add` `Chapter3/GermCalculus · IsScSmoothGerm.add_inter`. The chain rule of §III.1 composes two germs *at* 0, but R and Ψ are applied at $(r(y), g(y))$, which is $(0, 0)$ only when $g(0) = 0$ and $0 \in O$; so the chain rule is proved again with the outer map sc^1 on a set V and the inner one a germ mapping G_0 into V, with no condition at a base point `Chapter3/GermCalculus · IsSc1With.comp_germ` `Chapter3/GermCalculus · IsScSmooth.comp_germ`, and with a whole-chart inner map ψ, $\psi(0) = 0$, on the preimage germ `Chapter3/GermCalculus · IsScSmoothGerm.comp_preimage`. Hence the fill defect of a germ is a germ on the same germ (§III.4); the push-forward $\Psi_*(g)$ is a germ on the germ of the points that ψ^{-1} maps into the germ of g, with $D(\Psi_*g)(0) = \Gamma(0)Dg(0)D\psi(0)^{-1}$ (§III.23); a germ plus a whole-chart map is a germ on the same germ (Theorem 3.1.18 (preprint 3.15), §III.6); and $(a, u) \mapsto g(u)$ is a germ on $\mathbb{R}^n \times G$ (Theorem 3.1.21 (preprint 3.18), §III.40). The strong bundle isomorphism Ψ and the basic germ h remain what they were.

A basic germ restricts. Definitions 3.1.7 and 3.3.1 (preprint 3.7 and 3.28) pass to every smaller germ with the same h and B: the contraction field after shrinking its radius, since every level of a germ is a relatively open neighbourhood of 0, and the sc-smoothness by the uniqueness of derivatives in the quadrant `Chapter3/BasicGermShrink · IsBasicGerm.mono`.

The shrunk basic germ G''. The text does not relate the germ G of the basic germ to the germs G_g, G_s of g and s, and nothing forces $\psi^{-1}(G_m) \subseteq (G_g)_m$. For conjugation data with base map ψ, inverse ψ' and basic germ on G, let $G''_m = \{x \in G_m \mid \psi'_m(x) \in (G_g)_m \cap (G_s)_m\}$, the preimage germ along the sc^0-germ $\psi'|_G$ (ψ' is sc-smooth on all of U' and $\psi'(0) = 0$) `Chapter3/BasicGermShrink · ConjugationData.`

shrinkGerm. The conjugation data shrunk to G'' have the same Ψ, h, B and conjugation identity Chapter3/BasicGermShrink · ConjugationData.shrink Chapter3/BasicGermShrink · ConjugationData.basic_shrinkGerm; a point of G''_m is sent by ψ'_m into $(G_g)_m \cap (G_s)_m$ Chapter3/BasicGermShrink · ConjugationData.levelMap_mem_inter_of_mem_shrinkGerm, and read on the E-side, $u \in U_m$ with $\psi_m(u) \in G''_m$ lies in both germs Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem. *Corollary 3.5.3 (preprint 3.41) on G''*: every zero x of the normal form with $|x|_0$ small is ιx_1 with x_1 on level 1 of G'' — it is small on level 1 (Theorem 3.5.1 (preprint 3.40)), and the contraction field puts a level-1 ball around 0 inside G''_1 — so $\psi'_1(x_1)$ lies on level 1 of G_g and G_s Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_41_shrink. This is the device of every argument at the nearby zeros: the level-1 lift $z_{0,1}$ of a zero near x_0 lies in the germ of g, where $Dg(z_{0,1})$ is defined (§§III.32, III.33, III.43, III.45). It relies on the lifts being small on level 1; a point small on level 0 only need not lie in G''_1. At the centre no shrinking is needed: 0 lies in every germ, and $Dg(0)$ is the same for all germs and families (§III.4).

What the proofs use of g and s. At 0: $Dg(0)$ and $Ds(0)$ (Propositions 3.1.5 and 3.6.5 (preprint 3.5 and 3.52), Definition 3.6.6 (preprint 3.53), the linear part of Theorem 3.6.8 (preprint 3.54); §§III.4, III.21, III.23, III.30). On level 0 near 0: continuity of g and of the sc⁺-part (Theorem 3.1.13 (preprint 3.12), §III.13; the local step of Theorem 5.2.2 (preprint 5.5), §V.6). At the level-1 lifts of the nearby zeros: Dg, through G'' (Theorem 3.6.8 (1) (preprint 3.54), §§III.32–III.33, III.43, III.45). Nowhere g or s on $U \cap E_m$ away from their germs. So every theorem of Chapter 3 holds in the germ reading with the text's hypotheses and its former conclusions, and no hypothesis is added; the statements that named the derivative of g on U at 0 (Proposition 3.6.5 (preprint 3.52), the linearization statements of Theorems 3.1.22 and 3.6.8 (preprint 3.19 and 3.54)) now name the germ, a derivative family Dg and $Dg(0)$, and Corollary 5.2.5 (preprint 5.8) in a chart has an sc⁺-germ on the shifted germ for its sc⁺-part (§V.7).

The switch. The canonical names now denote the germ reading: Definitions 3.1.11, 3.1.16, 3.6.3, 3.6.4 and 3.6.6 (preprint 3.11, 3.14, 3.50, 3.51 and 3.53) with the tame, filled and sc variants of good position Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm Chapter3/TameGermsG · StrongBundle.IsScFredholmSection Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm, and the theorems built on them. The whole-chart notions and theorems keep their statements under the names …`WholeChart` and …`_wholeChart`, and each whole-chart notion implies its germ twin Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionWholeChart.isScFredholmSection Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGermWholeChart.isGoodPositionGerm. The non-vacuity witnesses are whole-chart data, hence germ data. Chapter 5 is stated with the germ notions: the local step of Theorem 5.2.2 (preprint 5.5) and the spanning sections changed locally, and Corollary 5.2.5 (preprint 5.8) in a chart changed its statement (§§V.6, V.12, V.7).

III.49 Definition 6.2.1 (preprint 6.1) and Proposition 6.2.2 (preprint 6.2): the index along paths

The text's Chapter 6 opens with the index of an sc-Fredholm germ at a smooth point which need not be a zero (Definition 6.2.1 (preprint 6.1), pp. 237–240 (preprint pp. 200–201)) and proves that it is constant along sc-smooth paths of smooth points (Proposition 6.2.2 (preprint 6.2), p. 240 (preprint p. 201)). Its proof is the device that §III.44 used along the zero set: stabilize by finitely many sc⁺-sections and apply the implicit function theorem. This section retells both: first with sc⁺-sections on all of X, which needs bump functions (Findings 340–348), then as the text does it, with locally defined sections on the restriction of the bundle to a chart domain and without bump functions (Findings 349, 360–367).

Definition 6.2.1 (preprint 6.1) Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt. *f has index i at x* if there is an sc⁺-section s of P with $(f+s)(x) = 0$ and $(f+s)'(x)$ Fredholm of index i in every strong bundle chart around x (the chart-intrinsic statement of §III.44). The text's $f - s$ with $s(x) = f(x)$ is $f + (-s)$; the text's section is local, this one global; the reading with local sections, and the agreement of the two readings under bump functions, close this section. *Well defined.* At a zero of an sc-Fredholm section the index in `IsLinOfIndex` is unique, because in a chart $\mathrm{Lin}(z)$ is a bounded operator into K_z (§III.44), so the quantification over corestrictions is not vacuous Chapter3/IndexAlongPaths · StrongBundle.IsLinOfIndex.

unique. Adding an sc⁺-section s with $s(x) = 0$ at a zero x of f does not change the index: the germ datum of f at x, with conjugation data dc, gives the index $\dim P_n - \dim Q$ of $f'(x)$ (Proposition 3.6.5 (preprint 3.52)), and Theorem 3.1.18 (preprint 3.15) gives germ data of $f + s$ in the same chart whose conjugation data are dc up to their values on U, since $(g + \hat{s} \circ r) - (\hat{s}_1 \circ r + t_0) = g - t_0$ Chapter3/IndexAlongPaths · StrongBundle.isLinOfIndex_sectionAdd_of_isZeroAt. For two sections s, s' of the definition, $s' - s$ is an sc⁺-section vanishing at x Chapter3/IndexAlongPaths · isZeroAt_sectionDiff_of_isZeroAt_sectionAdd and $(f + s) + (s' - s) = f + s'$ Chapter3/IndexAlongPaths · sectionAdd_sectionAdd_sectionDiff, so the indices agree Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.unique. The text argues instead that $(t - s)'(x)$ is an sc⁺-operator, hence compact on every level, and invokes Proposition 1.1.11 (preprint 1.10); the project has no lemma that the linearization of an sc⁺-section is compact, and the germ data make it unnecessary. At a zero the index of the germ is that of $f'(x)$ Chapter3/IndexAlongPaths · StrongBundle.isIndexAt_of_isZeroAt Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.isLinOfIndex_of_isZeroAt. *Existence.* If X admits bump functions, every smooth point has an index: $f(x)$ is a smooth point of Y, Lemma 5.3.3 (preprint 5.11) gives an sc⁺-section s_0 with $s_0(x) = f(x)$, and for $s = -s_0$ the germ data of the sc-Fredholm section $f + s$ at its zero x (Theorem 3.1.18 (preprint 3.15)) give the index Chapter3/IndexAlongPaths · StrongBundle.exists_isIndexAt.

Along the zero set, without good position (Findings 240–242). Let X admit bump functions, f be an sc-Fredholm section with tame germ data and $f(x_0) = 0$; neither good position nor surjectivity at x_0 is assumed. In a chart centred at x_0, finitely many smooth vectors span a complement of the range of $\mathrm{Lin}(0)$, closed of finite codimension by Proposition 3.6.5 (preprint 3.52), and Lemma 5.3.3 (preprint 5.11) realizes them as allowable sc⁺-sections $s_1, \dots, s_k$ Chapter3/IndexLocallyConstantG · exists_spanning_sections_germ; with further sections that tilt the kernel into good position at a corner (the conditions (i)–(ii) of §V.19), the parametrized section $F(\lambda, x) = f(x) + \sum_i \lambda_i s_i(x)$ of Theorem 3.1.21 (preprint 3.18) is in good position at $(0, x_0)$ in a tame chart Chapter3/IndexLocallyConstantG · exists_sections_isParamGP_germ Chapter3/IndexLocallyConstantG · isGoodPositionGermTame_paramSection_of_isParamGP. So the index of F' is one integer d at the zeros of F near $(0, x_0)$ (§III.44), among them $(0, y)$ for every zero y of f near x_0. At $(0, y)$, in the product chart, $\mathrm{Lin}_F(0)(\delta\lambda, \delta u) = \mathrm{Lin}_f(0)\delta u + \sum_i \delta\lambda_i \hat{s}_i(0)$, a finite-rank extension, so $d = k + \mathrm{ind}\, f'(y)$ Chapter3/IndexLocallyConstant · fredholmIndex_eq_of_param Chapter3/IndexLocallyConstantG · StrongBundle.isLinOfIndex_of_paramSection_isLinOfIndex. Hence $f'(y)$ has index $d - k$ in every chart at every zero y near x_0 Chapter3/IndexLocallyConstantG · index_locally_constant_of_admitsBumpFunctions, also as an sc-Fredholm operator on every level Chapter3/IndexLocallyConstantG · theorem_3_54_index_locally_constant_general. This is the text's device for Proposition 6.2.2 (preprint 6.2) in the case of a zero.

The local form. For X with bump functions, f with tame germ data and an sc⁺-section s with $(f + s)(x_0) = 0$, the index of the germ is one integer at the zeros of $f + s$ near x_0: $f + s$ has tame germ data (§III.47) and the previous paragraph applies to it Chapter3/IndexAlongPaths · isIndexAt_locally_constant_of_isZeroAt_sectionAdd. Hence along a path continuous at t_0 on which $f + s$ vanishes near t_0 the index is constant near t_0 Chapter3/IndexAlongPaths · isIndexAt_eventually_const_of_path, and on a preconnected parameter space a locally constant index is constant: by uniqueness the index is a locally constant function of t Chapter3/IndexAlongPaths · isIndexAt_const_of_forall_eventually. What remains is a section s, depending on t_0, that vanishes along the path — the text's $s(t, x)$ on $[0, 1] \times X$ — and the passage from the product back to X.

Bump functions on $\mathbb{R}^n \times X$. The device works on the product of Theorem 3.1.21 (preprint 3.18) (§III.40), which must admit bump functions when X does. A C^∞ function on $\mathbb{R}^n$ is sc-smooth on the M-polyfold $\mathbb{R}^n$ Chapter3/IndexAlongPaths · isScSmoothMap_rn_real_of_contDiff, products of sc-smooth functions are sc-smooth by the product rule in every chart Chapter3/IndexAlongPaths · MPolyfold.IsScSmoothMap.mul, and $(r, x) \mapsto \beta_1(r)\beta_2(x)$ has support in $\mathrm{supp}\,\beta_1 \times \mathrm{supp}\,\beta_2$ Chapter3/IndexAlongPaths · tsupport_prod_mul_subset; so $\mathbb{R}^n \times X$ admits bump functions Chapter3/IndexAlongPaths · MPolyfold.admitsBumpFunctions_prod_rn. Then for the pull-back $\tilde{f} = f \circ \pi$ and an sc⁺-section S of $\pi^* P$ with $\tilde{f} + S$ vanishing along the graph $t \mapsto (t, \varphi(t))$ near t_0, the index of $(\tilde{f} + S)'(t, \varphi(t))$ in every chart is one integer for t near t_0: $\tilde{f} + S$ has tame germ data (§§III.40, III.47), and the local constancy along its zero set applies Chapter3/IndexAlongPaths · isLinOfIndex_paramLift_sectionAdd_eventually_const. The proposition thereby reduces to two steps, the section along the

path (E) and the index count on a slice (S) Chapter3/IndexAlongPaths · proposition_6_2_of_sections_of_slice.

(E): the text's section along the path Chapter3/IndexAlongPathsDevice · exists_isScPlusSection_pullback_of_path. For an sc-smooth section f (sc-Fredholm is not needed), an sc-smooth path $\varphi : \mathbb{R}^n \to X$ and t_0, there is an sc$^+$-section S of π^*P with $\tilde{f} + S = 0$ along the graph near t_0. Take a strong bundle chart c around $\varphi(t_0)$. First the path is reparametrized by a C^∞ map σ of $\mathbb{R}^n$ equal to the identity near t_0 with values in $\varphi^{-1}(\operatorname{dom}\varphi_c)$, $\sigma(t) = t_0 + b(t-t_0)(t-t_0)$ for a smooth bump b Chapter3/IndexAlongPathsDevice · exists_contDiff_eq_id_near_mapsTo; C^∞ self-maps of $\mathbb{R}^n$ are sc-smooth Chapter3/IndexAlongPathsDevice · isScSmoothMap_rn_of_contDiff, so $\psi = \varphi \circ \sigma$ is an sc-smooth path that stays in the chart. The representative $t \mapsto \Phi(f(\psi(t)))$ is sc-smooth from the constant structure $\mathbb{R}^n$ into $E \oplus F$; Proposition 1.2.2 (preprint 1.17) makes it sc-smooth from $(\mathbb{R}^n)^1 = \mathbb{R}^n$ into $(E \oplus F)^1 = E^1 \oplus F^1$, and its F^1-component $\widehat{e}_1$ is the curve of smooth points that the text's $s(t,x) = R(x, f(\varphi(t)))$ needs in F^1 — no new analysis is required. In the product chart $(r \mapsto r - t_0) \times \varphi_c$ of π^*P the local section is $\tau(a,u) = -R[1](u, \widehat{e}_1(a + t_0))_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is cut off by a bump of X equal to 1 near $\varphi(t_0)$ with support in the chart domain (Proposition 5.5.3 (preprint 5.30)) and extended by 0 (§V.9). Along the graph $\Phi(S(t,\varphi(t)))_2 = -R(\varphi_c(\varphi(t)), \Phi(f(\varphi(t)))_2)_2 = -\Phi(f(\varphi(t)))_2$, since $\Phi(f(\varphi(t))) \in K$ is fixed by R, so $\tilde{f} + S$ vanishes there for t near t_0, where $\sigma(t) = t$ and the bump is 1. The bump function and the reparametrization make the text's local section global; the text needs neither.

(S): the slice. The slice $S(t,\cdot)$ of a section of π^*P, read in Y through the fibres $(\pi^*Y)_{(t,x)} = P^{-1}(x)$, is a section of P Chapter3/IndexAlongPathsDevice · pullbackSlice Chapter3/IndexAlongPathsDevice · IsSection.pullbackSlice; $G(t,x) = 0$ iff the slice vanishes at x Chapter3/IndexAlongPathsDevice · isZeroAt_pullback_iff_pullbackSlice, and the slice of $\tilde{f} + S$ is $f + S(t,\cdot)$ Chapter3/IndexAlongPathsDevice · pullbackSlice_sectionAdd_paramLift. *The slice of an sc$^+$-section is sc$^+$* Chapter3/IndexAlongPathsDevice · StrongBundle.IsScPlusSection.pullbackSlice: for a chart c of the atlas of P, the product chart $(r \mapsto r - t) \times \varphi_c$ is a chart of the atlas of π^*P, and the local representative of $S(t,\cdot)$ is that of S composed with the sc-operator $u \mapsto (0,u)$ Chapter3/IndexAlongPathsDevice · ScBanach.prodSndLeftOp and projected. *The linearization on the slice direction*: in the product chart at (t,x), whose base retraction is $(a,u) \mapsto (a, r(u))$ Chapter3/IndexAlongPathsDevice · PullbackChartData.r_ofProd, $\mathrm{Lin}_G(0)(0,\delta u) = \mathrm{Lin}_{G(t,\cdot)}(0)\delta u$, by the chain rule for $u \mapsto \hat{G}(0, r(u))$ Chapter3/IndexAlongPathsDevice · StrongBundle.linAt_ofProd_slice. *The index count*: if G is an sc-Fredholm section of π^*P whose slice $g = G(t,\cdot)$ is an sc-Fredholm section with $g(x) = 0$, and $G'(t,x)$ is Fredholm of index i in every chart, then $g'(x)$ is Fredholm of index $i - n$ in every chart: $\mathrm{Lin}_G(0)(\delta a, \delta u) = \mathrm{Lin}_g(0)\delta u + \mathrm{Lin}_G(0)(\delta a, 0)$ by linearity and the slice identity, the last term has rank at most n, and the parameter count of the zero-set case applies, the chart independence coming from the germ datum of g at x Chapter3/IndexAlongPathsDevice · StrongBundle.isLinOfIndex_pullbackSlice_of_isLinOfIndex. For $G = \tilde{f} + S$ ($n = 1$) this is $\mathrm{ind}(f,x) = i - 1$, with $S(t,\cdot)$ as the section of Definition 6.2.1 (preprint 6.1) Chapter3/IndexAlongPathsDevice · StrongBundle.isIndexAt_of_pullback_isLinOfIndex.

Proposition 6.2.2 (preprint 6.2) Chapter3/IndexAlongPathsDevice · proposition_6_2. *Let X admit sc-smooth bump functions, f be an sc-Fredholm section with tame germ data and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* $\mathrm{ind}(f, \varphi(t))$ *is one integer for all t*; in particular $\mathrm{ind}(f,\varphi(0)) = \mathrm{ind}(f,\varphi(1))$. For each t_0, (E) gives S; the index of $\tilde{f} + S$ is constant near t_0 along the graph; (S) turns it into $\mathrm{ind}(f,\varphi(t)) + 1$; and $\mathbb{R}$ is connected. The paracompactness of $\mathbb{R} \times X$ and of π^*Y is supplied by Theorem 2.3.10 (preprint 2.16) Chapter3/IndexAlongPathsDevice · proposition_6_2'. *Against the text.* (i) The path is defined on $\mathbb{R}$, not on $[0,1]$: $\mathbb{R} \times X$ is the product of Theorem 3.1.21 (preprint 3.18) and has no boundary, and the text's "tame M-polyfold $[0,1] \times X$" is not used; a path on $[0,1]$ that extends sc-smoothly to $\mathbb{R}$ is covered, and by the reparametrization in (E) only a neighbourhood of each t_0 matters. (ii) The hypotheses are tame germ data, the reading of "tame strong bundle" of §III.47, and bump functions, which the text does not assume ("we do not need a sc-smooth partition of unity"): they are used for the stabilizing sections (Lemma 5.3.3 (preprint 5.11)) and for the cut-off of the section along the path, both globalizations of the text's locally defined sections. Without them the device runs on the restriction of P to a chart domain, as the text intends; that is the rest of this section. (iii) Definition 6.2.1 (preprint 6.1) is proved well defined without Proposition 1.1.11 (preprint 1.10). (iv) The two points the text's proof does not address — the tilting sections for the

kernel condition at a corner and the uniformity in the zero — are those of the zero-set case above.

Without bump functions: the restriction to an open set (Findings 349, 360). The text restricts the bundle to a chart domain without comment ("in a chart, $s(y) = R(y, f(x))$"; "we do not need a sc-smooth partition of unity", preprint p. 202). For an open $V \subseteq X$, a chart of X whose domain lies in V is transported to a chart of the subtype V with the same model `Chapter3/RestrictOpen · MChart.restrictTo`, compatibility is transported `Chapter3/RestrictOpen · MChart.Compatible.restrictTo`, and the charts of X inside V form the atlas of the M-polyfold V `Chapter3/RestrictOpen · MPolyfold.restrictOpen`, whose levels are the traces of those of X `Chapter3/RestrictOpen · MPolyfold.mem_levelSet_restrictOpen_iff`; an sc-smooth map with values in V is sc-smooth into V `Chapter3/RestrictOpen · MPolyfold.IsScSmoothMap.codRestrict_restrictOpen`. The restriction $P|V$ lives on the pull-back $\iota^*Y = \{(v, y) \mid P(y) = v\}$ of Definition 2.6.7 (preprint 2.53), as the restricted sections of Definition 5.3.13 (preprint 5.20) do (§V.23), and its fibres are those of P. A strong bundle chart of P with base domain inside V is transported with the same F, U, R, K `Chapter3/RestrictOpen · StrongBundleChart.restrictTo` `Chapter3/RestrictOpen · StrongBundleChart.Compatible.restrictTo`, and these charts form the atlas of $P|V$ `Chapter3/RestrictOpen · StrongBundle.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.restrictTo_mem_restrictOpen_atlas`; every chart of P restricts to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpen · StrongBundleChart.restrictInter`. The restriction $f|_V$ of an sc-smooth, sc$^+$ or regularizing section is again one `Chapter3/RestrictOpen · StrongBundle.IsScSmoothSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.IsScPlusSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.regularizing_restrictOpen`. Two choices of the formalization: the paracompactness of V and of ι^*Y is supplied by Theorem 2.3.10 (preprint 2.16), and V is nonempty, for the junk values of the transported inverse charts.

Germ data and the index on $P|V$ (Findings 361, 362). The data of Definition 3.1.11 (preprint 3.11) on a relatively open U restrict to every relatively open $U_0 \subseteq U$ containing 0: the germs of the filling and of the sc$^+$-part are intersected with U_0, and $Dr(0)$ and $\rho(0)$ do not change, so the kernel isomorphism of condition (3) is the same `Chapter3/RestrictOpenGerms · IsFillingGerm.of_subset` `Chapter3/RestrictOpenGerms · IsScPlusGerm.of_subset`; the strong bundle isomorphism Ψ of the conjugation is restricted to $U_0 \lhd F \to \psi(U_0) \lhd F'$, where $\psi(U_0) = U' \cap \psi'^{-1}(U_0)$ is relatively open by the continuity of ψ', and the basic germ is shrunk to $G \cap \psi'^{-1}(U_0)$ as for G'' (§III.48), with the same P_n, Q, h, B `Chapter3/RestrictOpenGerms · ConjugationData.ofSubset` `Chapter3/RestrictOpenGerms · IsConjugateToBasicGerm.of_subset`. So a germ datum at $x \in V$ survives in its chart restricted to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpenGerms · StrongBundleChart.germData_restrictInter`, which is tame when the chart is `Chapter3/RestrictOpenGerms · StrongBundle.IsTameChart.restrictInter`, and an sc-Fredholm section with tame germ data restricts to one on $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsScFredholmSectionTame.restrictOpen`. *The index.* At a zero $x \in V$, $f'(x)$ has index i in every chart of P iff $(f|_V)'(x)$ has index i in every chart of $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.restrictOpen` `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.of_restrictOpen`. The charts are not compared — a chart of $P|V$ need not come from one of P —: in either bundle the index is $\dim P_n - \dim Q$ for the germ datum (§III.44), whose restriction keeps P_n and Q `Chapter3/RestrictOpenGerms · StrongBundle.isLinOfIndex_restrictOpen_of_conjugationData`, and uniqueness identifies it with i. The same argument passes from $P|V$ to $P|W$ for an open $W \subseteq V$, where the chart restricted to $\operatorname{dom}\varphi \cap W$ is common to both and the two linearizations agree in it `Chapter3/IndexAlongPathsLocal · sectionRestrict` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsScPlusSection.restrictOpen_of_subset` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsLinOfIndex.restrictOpen_of_subset`; Definition 6.2.1 (preprint 6.1) restricts accordingly `Chapter3/RestrictOpenGerms · StrongBundle.IsIndexAt.restrictOpen` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.restrictOpen_of_subset`.

The stabilizing sections of a chart, without cut-off (Finding 363). In the zero-set argument above, bump functions enter only through Lemma 5.3.3 (preprint 5.11), as sc$^+$-sections through given smooth points of the fibres. That property — through every smooth point e of the fibre over a smooth point x there is an sc$^+$-section s with $s(x) = e$ — is isolated `Chapter3/IndexLocallyConstantLocal · StrongBundle.HasScPlusSectionsThrough`. Bump functions give it by Lemma 5.3.3 (preprint 5.11) `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`, and so does a chart of the atlas over the whole base, with no cut-off `Chapter3/IndexLocallyConstantLocal · StrongBundle.hasScPlusSectionsThrough_of_chart`: $\hat{e} = \Phi(e)_2$ is a smooth point of F, with a lift $\hat{e}_1 \in F^1$, and the section

$y \mapsto \Phi^{-1}(\varphi(y), R(\varphi(y), \hat{e}))$ — the text's $R(\cdot, e)$ — has the local representative $z \mapsto R[1](z, \hat{e}_1)_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is the cut-off section of Lemma 5.3.3 (preprint 5.11) with $\beta = 1$ (§V.9), and its value at x is e since $\Phi(e) \in K$ is fixed by R. The restriction $P|V$ to a chart domain V has such a chart: the chart itself, transported to V. The three steps of the zero-set argument — the spanning sections, the tilting sections that put the parametrized section in good position, the local constancy — are repeated with this property in place of bump functions `Chapter3/IndexLocallyConstantLocal · exists_spanning_sections_of_sections` `Chapter3/IndexLocallyConstantLocal · exists_sections_isParamGP_of_sections` `Chapter3/IndexLocallyConstantLocal · index_locally_constant_of_sections`; the support and norm bounds of Lemma 5.3.3 (preprint 5.11), which the index argument does not use, are dropped. *The index along the zero set with the text's hypotheses* `Chapter3/IndexLocallyConstantLocal · index_locally_constant_tame`: *let f be an sc-Fredholm section with tame germ data of any strong bundle and x_0 a zero; then there are an open $V \ni x_0$ and $i \in \mathbb{Z}$ with $f'(y)$ Fredholm of index i in every chart at every zero $y \in V$.* Take a chart around x_0 with domain V: $f|_V$ has tame germ data on $P|V$, which has sc$^+$-sections through every smooth fibre point, so the index of $f|_V$ is one integer at its zeros near x_0, and at each of them it is the index on P. In sc form, with the sc-Fredholm property of $\mathrm{Lin}(z)$ at every zero in every chart `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`, $\mathrm{Lin}(z)$ is sc-Fredholm of index i on every level `Chapter3/IndexLocallyConstantLocal · theorem_3_54_index_locally_constant_general_tame`. This is the localization that Finding 343 described; the gap ind $C(z) = 0$ of Finding 243 remains only without tame germ data.

Definition 6.2.1 (preprint 6.1) with local sections (Finding 364) `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal`. *f has index i at x* if there are an open $V \ni x$ and an sc$^+$-section s of $P|V$ with $(f|_V + s)(x) = 0$ and $(f|_V + s)'(x)$ Fredholm of index i in every chart of $P|V$: the global definition at the beginning of this section, on $P|V$. *Well defined.* Two local sections, on V and on V', both restrict to $V \cap V'$ (the restriction of Definition 6.2.1 (preprint 6.1) to a smaller open set, above), where the uniqueness on $P|V \cap V'$ applies `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal.unique`. A global section gives a local one, with $V = X$ `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.isIndexAtLocal`, and at a zero the index is that of $f'(x)$ `Chapter3/IndexAlongPathsLocal · StrongBundle.isIndexAtLocal_of_isZeroAt`. *Existence, without bump functions.* On the domain V of a chart around x, the section through $f(x)$ of the previous paragraph, negated, is an sc$^+$-section s of $P|V$ with $(f|_V + s)(x) = 0$, and Theorem 3.1.18 (preprint 3.15) gives the germ datum of $f|_V + s$ at x, hence an index `Chapter3/IndexAlongPathsLocal · StrongBundle.exists_isIndexAtLocal`. As for the global reading, a locally constant local index along a map from a preconnected space is constant `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_const_of_forall_eventually`.

Proposition 6.2.2 (preprint 6.2) with the text's hypotheses (Finding 365) `Chapter3/IndexAlongPathsLocal · proposition_6_2_local`. *Let f be an sc-Fredholm section with tame germ data of a strong bundle $P : Y \to X$ — no bump functions — and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* ind$(f, \varphi(t))$*, read with local sections, is one integer for all t*; any two values of the index at $\varphi(0)$ and at $\varphi(1)$ agree `Chapter3/IndexAlongPathsLocal · proposition_6_2_local'`. The proof is that of (E), (S) and Proposition 6.2.2 (preprint 6.2) above, run near each t_0 on $P|V$, V the domain of a chart c around $\varphi(t_0)$ `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_eventually_const_of_path`. The path is reparametrized into V as in (E), and is then an sc-smooth path of the M-polyfold V. The section along the path is the one of (E) with the cut-off as a datum `Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_bump`, taken equal to 1, which is possible because the chart $c|_V$ of $P|V$ covers its base `Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_chart`. The product chart of $c|_V$ covers $\mathbb{R} \times V$, so the pull-back has sc$^+$-sections through every smooth fibre point without cut-off `Chapter3/IndexAlongPathsLocal · hasScPlusSectionsThrough_pullback_rn_of_chart`, and the index of $\tilde{f} + S$ is constant along the graph near t_0 `Chapter3/IndexAlongPathsLocal · isLinOfIndex_paramLift_sectionAdd_eventually_const_of_sections`. The slice relation (S), applied to $P|V$ unchanged, turns it into ind$(f|_V, \varphi(t)) = i - 1$ near t_0, a local index of f, and the connectedness of $\mathbb{R}$ ends the proof. *Against the text*, what remains is the domain of the path: it is $\mathbb{R}$, and a path sc-smooth on $[0, 1]$ only is not covered, the project having no M-polyfold $[0, 1]$. The hypotheses on f are the text's, tame germ data being the reading of "tame strong

bundle" of §III.47.

The two readings agree under bump functions (Finding 367) `Chapter3/IndexAlongPathsLocalBump · StrongBundle.isIndexAtLocal_iff_isIndexAt`. If X admits bump functions, a local section of Definition 6.2.1 (preprint 6.1) gives a global one with the same index `Chapter3/IndexAlongPathsLocalBump · StrongBundle.IsIndexAtLocal.isIndexAt`: the local section s on V is restricted to $W = \operatorname{dom}\varphi \cap V$ for a chart φ around x, read in that chart as $\tau : O \to F^1$, and cut off by a bump β supported in W and equal to 1 near x; the cut-off of Lemma 5.3.3 (preprint 5.11) is made here in a chart of P that need not belong to the atlas `Chapter3/IndexAlongPathsLocalBump · StrongBundle.cutoffSection_mem_oneSet_of_isChart Chapter3/IndexAlongPathsLocalBump · StrongBundle.isScPlusSection_cutoffSection_of_isChart`. Then $f + \beta\tau$ vanishes at x, and its linearization there has the index of $(f|_W + s)'(x)$: the two sections agree where $\beta = 1$, so the two linearizations coincide in the chart restricted to that set, and the germ datum of $f + \beta\tau$ at x carries the index to every chart. So the global definition of the beginning of this section and the text's local one are the same notion whenever the hypotheses of Chapter 5 hold, and only the text's is available without them.

III.50 The 2021 edition: the filling without tameness (Definitions 3.1.4, 3.6.1 (preprint 3.4, 3.48)) and a corner example

The 2021 filling. Definition 3.1.4 (p. 91) and its repetition Definition 3.6.1 (p. 134) of the Springer edition ask the strong bundle retraction $R(u,h) = (r(u), \rho(u)h)$ to cover an *sc-smooth* retraction r with $r(0) = 0$. The 2017 text and the preprint ask for a *tame* one. Otherwise the conditions are those of §III.3: - (1) $\widehat{f} = g$ on O near 0; - (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; - (3) the linearization at 0 of $k(y) = [1 - \rho(r(y))]g(y)$, restricted to $\ker Dr(0)$, is a topological linear isomorphism onto $\ker \rho(0)$.

The formal definition is a structure with $r(0) = 0$ as a field and (1)–(3) in the germ reading of §III.48 `Chapter3/Book2021Germs · IsFilling2021 Chapter3/Book2021Germs · HasFilling2021`. The 2017 filling adds the tameness of r `Chapter3/Book2021Germs · IsFilling2017 Chapter3/Book2021Germs · isFilling2017_iff`.

The project's filling is the 2021 one. The germ filling of §III.3 never asked r to be tame `Chapter3/GermFilling · IsFillingGerm`. A 2021 filling is exactly a germ filling with $r(0) = 0$ `Chapter3/Book2021Germs · isFilling2021_iff Chapter3/Book2021Germs · IsFilling2021.isFillingGerm`. Conversely, a germ filling is a 2021 filling as soon as $0 \in O$, because points of O are fixed by r `Chapter3/Book2021Germs · IsFillingGerm.isFilling2021`; so is a whole-chart filled version `Chapter3/Book2021Germs · IsFilledVersion.isFilling2021`. In a chart centred at a point of X, $0 = \varphi(x_0)$ lies in O, so the text's standing assumption $r(0) = 0$ costs nothing there. A 2017 filling is a 2021 filling `Chapter3/Book2021Germs · IsFilling2017.isFilling2021`. Both notions are satisfiable: over the identity strong bundle retraction, which is tame, every sc-smooth g fills itself `Chapter3/Book2021Germs · isFilling2017_id`, for instance $g = 0$ on $\mathbb{R}$ `Chapter3/Book2021Germs · isFilling2017_rn`.

A 2021 filling that is not a 2017 filling. Take the following data: - $E = \mathbb{R} \oplus \mathbb{R}$, $C = U = [0,\infty) \times \mathbb{R}$, $F = \mathbb{R}$; - $R((a,b),h) = ((0,b),0)$ `Chapter3/Book2021Germs · CornerFilling.Rc`, a strong bundle retraction `Chapter3/Book2021Germs · CornerFilling.isStrongBundleRetraction_Rc`, so $r(a,b) = (0,b)$, $O = \{0\} \times \mathbb{R}$ and $\rho \equiv 0$ `Chapter3/Book2021Germs · CornerFilling.ρCLM_Rc`; - the zero section $\widehat{f} = 0$ of $K \to O$, and $g(a,b) = a$ `Chapter3/Book2021Germs · CornerFilling.gc Chapter3/Book2021Germs · CornerFilling.isScSmooth_gc`.

The derivatives at 0 are $Dr(0)(a,b) = (0,b)$ and $Dg(0)(a,b) = a$ `Chapter3/Book2021Germs · CornerFilling.scDeriv_r Chapter3/Book2021Germs · CornerFilling.scDeriv_g`. Since $\rho \equiv 0$, the fill defect k is g itself `Chapter3/Book2021Germs · CornerFilling.fillDefect_gc`. Now check the three conditions: - (1) $g = 0$ on O; - (2) $g(y) = \rho(r(y))g(y) = 0$ forces $y \in O$; - (3) $\ker Dr(0) = \mathbb{R} \oplus 0$ `Chapter3/Book2021Germs · CornerFilling.mem_ker_scDeriv_r`, and $Dk(0)$ maps it by $(a,0) \mapsto a$ onto $\ker \rho(0) = \mathbb{R}$ `Chapter3/Book2021Germs · CornerFilling.kerMap Chapter3/Book2021Germs · CornerFilling.kerMap_bijective`.

So g is a filled version `Chapter3/Book2021Germs · CornerFilling.isFilledVersion_gc`. With $0 \in O$ `Chapter3/`

Book2021Germs · CornerFilling.zero_mem_0, it is a 2021 filling Chapter3/Book2021Germs · CornerFilling.isFilling2021_gc. But r is not tame: $d_C(1,0) = 0$, while $d_C(r(1,0)) = d_C(0,0) = 1$, against Definition 2.5.2 (1) (preprint 2.33 (1)) Chapter3/Book2021Germs · CornerFilling.not_isTame_r. So g is not a 2017 filling Chapter3/Book2021Germs · CornerFilling.not_isFilling2017_gc, and the 2021 notion is strictly more general Chapter3/Book2021Germs · exists_isFilling2021_not_isFilling2017.

Sc-Fredholm germs and sections. With the 2021 filling in place of the filling of §III.6: - the 2021 sc-Fredholm germ (Definition 3.1.11 = 3.6.3) Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2021; - the 2021 sc-Fredholm section (Definition 3.1.16 = 3.6.4) Chapter3/Book2021Germs · StrongBundle.IsScFredholmSection2021; - the 2017 germ, with the 2017 filling Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2017.

In a chart centred at x_0, $0 \in O$ Chapter3/Book2021Germs · StrongBundle.zero_mem_0_of_chart. So the 2021 germ is the project's germ Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2021_iff Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm, and the 2021 section the project's section Chapter3/Book2021Germs · StrongBundle.isScFredholmSection2021_iff Chapter3/TameGermsG · StrongBundle.IsScFredholmSection. The 2017 germ is the germ with tame germ data of §III.47 Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2017_iff Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame, and 2017 implies 2021 Chapter3/Book2021Germs · StrongBundle.IsScFredholmGerm2017.isScFredholmGerm2021. The graph section of a C^∞ map $\mathbb{R}^n \to \mathbb{R}^k$ satisfies both Chapter3/Book2021Germs · isScFredholmSection2021_graphSection Chapter3/Book2021Germs · isScFredholmGerm2017_graphSection. Whether every 2021 germ is a 2017 germ is the question of Finding 353, the passage from germ data in some compatible chart to germ data in a tame one (gap G.79).

III.51 The 2021 edition: the theorems whose proofs do not use tameness

The project's proofs of the following results never used the tameness of r; each 2021 statement is the earlier theorem transported through the equivalences of §III.50: - **Proposition 3.1.5 (preprint 3.5)** (§III.4). If g is a 2021 filling of $\hat{f}$ with $\hat{f}(0) = 0$, then $0 \in O$ by $r(0) = 0$, and $g(0) = \hat{f}(0) = 0$ by (1). So the germ form of the proposition applies Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_5. $Dg(0)$ maps $T_0O = \operatorname{range} Dr(0)$ into $K_0 = \operatorname{range} \rho(0)$. The restriction $T\hat{f}(0) : T_0O \to K_0$ is onto iff $Dg(0)$ is, and is Fredholm iff $Dg(0)$ is, with the same index Chapter3/Book2021Germs · proposition_3_1_5_2021. - **Theorem 3.1.13 and Corollary 3.1.15 (preprint 3.12 and 3.13)** (§III.13) Chapter3/SectionCompactnessG · theorem_3_12 Chapter3/SectionCompactnessG · corollary_3_13 Chapter3/Book2021Germs · theorem_3_1_13_2021 Chapter3/Book2021Germs · corollary_3_1_15_2021. - **Theorem 3.1.18 (preprint 3.15)** (§III.6) Chapter3/FredholmSectionsG · theorem_3_15 Chapter3/Book2021Germs · theorem_3_1_18_2021. - **Theorem 3.1.21 (preprint 3.18)** (§III.40) Chapter3/ParamPerturbationG · theorem_3_18 Chapter3/Book2021Germs · theorem_3_1_21_2021. - **Proposition 3.6.5 (preprint 3.52)** (§§III.21, III.30) Chapter3/FredholmGermLinearizationG · proposition_3_52 Chapter3/Book2021Germs · proposition_3_6_5_2021.

So the 2021 change of the filling does not affect these statements, and the 2021 text's version of them is verified as printed.

III.52 The 2021 edition: Theorem 3.6.8 (preprint 3.54), the step on p. 138

The proof of Theorem 3.6.8 (pp. 137–138) starts from a germ in good position (Definition 3.6.6, §III.21). There $\ker T\tilde{f}(0)$ is in good position to the partial cone $C_0O = T_0O \cap C$ in T_0O. The proof passes to the good position of $\ker Dg(0)$ to C in E, the hypothesis of the implicit function theorem in the quadrant (§III.17). The passage uses an sc-complement of T_0O inside the subspace E_0 of C at 0, namely $\ker Dr(0)$. Proposition 2.5.3 (preprint 2.35) supplies it for a tame retract. In the linear part of §III.23 this is the

hypothesis $(1 - Dr(0))E \subseteq E_0$ `Chapter3/GoodPositionGermG · theorem_3_54_linear`. The 2021 text keeps the sentence that the retract is tame by assumption, while its Definition 3.6.1 no longer assumes it.

The step does not go through without tameness. In the example of §III.50, every other hypothesis of the linear part holds: - g is a 2021 filling of $\hat{f} = 0$, $0 \in O$, $\hat{f}(0) = 0$, and $\hat{f}$ is a section of K; - g is sc^1 with derivative $Dg(0)$, and $Dg(0)$ is onto; - the germ is in good position in the chart `Chapter3/GoodPositionGermG · IsGoodPositionAtGerm` `Chapter3/Book2021Germs · CornerFilling.isGoodPositionAtGerm_gc`: $T\tilde{f}(0) : T_0O \to K_0 = 0$ is onto, and its kernel $T_0O = 0 \oplus \mathbb{R}$ is all of $T_0O \cap C$. Here $\ker Dg(0) = \operatorname{range} Dr(0)$ `Chapter3/Book2021Germs · CornerFilling.ker_Lg_eq_range_Lr`.

But $(1 - Dr(0))(1, 0) = (1, 0)$ is not in $E_0 = 0 \oplus \mathbb{R}$ `Chapter3/Book2021Germs · CornerFilling.not_range_le_minimalSubspace`. And $\ker Dg(0) = 0 \oplus \mathbb{R}$ is not in good position to C in E `Chapter3/Book2021Germs · CornerFilling.not_isInGoodPosition_gc`, for two reasons: - a complement P contains a point p with first coordinate 1, from writing $(1, 0) = n + p$; - the first coordinate vanishes at $(0, 1) \in \ker Dg(0) \cap C$, so by Lemma 3.8.8 (preprint 3.65) every good complement lies in $\{a = 0\}$ `Chapter3/GoodPosition · lemma_3_65`.

The whole is one statement `Chapter3/Book2021Germs · theorem_3_6_8_tame_step_2021`. So this step of the proof does not hold as we read it for the 2021 definitions. The conclusion of the theorem is not refuted by the example: its solution set $O \cong \mathbb{R}$ is a manifold. Since $X = \mathbb{R}$ there has no boundary, even the proof of Theorem 3.6.10 (preprint 3.56), where $\partial X = \emptyset$, passes through this step.

The project states Theorem 3.6.8 and what rests on it (Theorems 3.1.22, 3.1.26, Corollary 3.1.27, Theorems 3.6.10 and 3.6.11 (preprint 3.19, 3.23, 3.24, 3.56 and 3.57)) with tame germ data, the 2017 notion (§III.47). For the untamed 2021 data they are open (gap G.79). One way to close the gap would answer Finding 353: show that germ data in a compatible chart can be traded for germ data in a tame one.

III.53 The 2021 edition: Theorem 3.5.1 (preprint 3.40) with the sequential conclusion

The 2021 statement of Theorem 3.5.1 (p. 118) adds a second conclusion. If points (a_k, w_k) of the solution set S converge to 0 in E_0, then for every m they eventually lie in C_m and converge to 0 there. The formal statement has both conclusions `Chapter3/Book2021Germs · theorem_3_5_1_2021`, in the following setting: - h a basic germ on an sc-germ G of neighbourhoods of 0 in $C = \widetilde{C} \times W$, with $\widetilde{C}$ closed; - $U = G_0$, s an sc$^+$-germ on G with $s(0) = 0$, and $S = \{x \in U : h(x) + s(x) = 0\}$.

The neighbourhoods. Take the open sets $O'(m)$ of the local Theorem 3.1.13 of §III.12 `Chapter3/LocalCompactness · theorem_3_12_local` and put $O(m) = O'(m) \cap G_0$. These are nested, relatively open in C, contain 0 and lie in U. The closure of $S \cap O(m)$ is taken in E_0. It lies in C, which is closed and contains G_0, and it lies in the image of E_m, because it is the image of the compact set K_m of §III.12.

The sequences. A zero of $h + s$ solves $w = B(a, w) - \mathbf{s}(a, w)$ with $\mathbf{s}$ the W-component of s `Chapter3/Book2021Germs · IsBasicGerm.solves`. $\mathbf{s}$ is an sc^0-germ into W^1 with $\mathbf{s}(0) = 0$. So the local Corollary 3.1.15 of §III.12 applies `Chapter3/LocalCompactness · corollary_3_13_local`. For every m there are $y_k \in E_m$ with $\iota_m(y_k) = x_k$ for k large and $y_k \to 0$ in E_m; these are the lifts $L_m x_k$. "Converges on level m" is thus stated with level-m lifts, as in Corollary 3.1.15 (preprint 3.13).

III.54 The 2021 edition: cones and Lemmas 3.8.4–3.8.6, Corollary 3.8.7 (Appendix 3.8.1.3)

The appendix on "cones and partial quadrants in infinite dimensions" (pp. 145–149) is new in the 2021 edition. Its notions: - a *closed convex cone* K in a finite-dimensional space is a closed convex set with

$K \cap (-K) = \{0\}$ and $tK \subseteq K$ for $t \geq 0$ Chapter3/Book2021Items · IsClosedConvexCone; - *a generates an extreme ray* if $a \in K \setminus \{0\}$, and $y \in K$, $a - y \in K$ imply $y \in \mathbb{R}_+ a$ Chapter3/Book2021Items · IsExtremeRayGen.

A closed convex cone is pointed Chapter3/Book2021Items · IsClosedConvexCone.eq_zero. $[0, \infty)$ is one, generated by the extreme ray of 1 Chapter3/Book2021Items · isClosedConvexCone_Ici Chapter3/Book2021Items · isExtremeRayGen_one.

The setting is the good position of §III.14: $C = T^{-1}([0,\infty)^n \oplus W)$ for a presentation T (the text's case is $T = \mathrm{id}$), and N a finite-dimensional subspace in good position. Within it: - $N \cap C$ is the cone of N Chapter3/GoodPosition · quadCone; - Γ, the union of the sets $\sigma_a = \{i : a_i = 0\}$ over $0 \neq a \in N \cap C$, is the vanishing index set Chapter3/GoodPosition · vanIdx; - $P : N \to \mathbb{R}^\Gamma$ is the coordinate projection Chapter3/GoodPosition · sigmaCoord, whose kernel is $N \cap (\mathbb{R}^{\Gamma^c} \oplus W)$.

The quadrant form of a pointed cone. If $N \cap C$ is pointed, a linear isomorphism $e : N \cong \mathbb{R}^{\dim N}$ carries $N \cap C$ onto $[0,\infty)^{\dim N}$ Chapter3/Book2021Items · exists_quadrantEquiv_of_pointed. It comes from the normal form of §III.14, $N \cap C \cong [0,\infty)^k \times N_0$ Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm, in which pointedness forces $N_0 = 0$. The vectors $e^{-1}(e_j)$ generate extreme rays Chapter3/Book2021Items · isExtremeRayGen_symm_single. A nonzero $a \in N \cap C$ has a coordinate $e(a)_j > 0$ with $a - e(a)_j e^{-1}(e_j) \in N \cap C$ Chapter3/Book2021Items · exists_sub_single_mem. As in §III.14, this replaces the Krein–Milman-type Lemmas 3.8.1–3.8.3 (preprint 3.62–3.64): the cone is a quadrant at once.

Lemma 3.8.4 Chapter3/Book2021Items · lemma_3_8_4_2021. Let $N^\perp$ be a good complement and $N \cap C$ a closed convex cone. Parts (1) and (3) say that $y_i = 0$ for $y \in N^\perp$ and $i \in \sigma_a$, a a generator, respectively $i \in \Gamma$. They hold without the cone hypothesis, because they are Lemma 3.8.8 (preprint 3.65) (§III.14) Chapter3/GoodPosition · lemma_3_65. For (2), $\Gamma = \bigcup_{a \text{ generator}} \sigma_a$, the inclusion $\supseteq$ is the definition. For $\subseteq$, let $0 \neq a \in N \cap C$ with $a_i = 0$, and take j and $b = e^{-1}(e_j)$ as above. Then $0 \leq (a - e(a)_j b)_i = -e(a)_j b_i \leq 0$ forces $b_i = 0$, so $i \in \sigma_b$. The text writes a as a sum of extreme-ray generators (Lemma 3.8.1); here only pointedness is used.

Lemma 3.8.5 Chapter3/Book2021Items · lemma_3_8_5_2021. The text's statement is: either $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$, in which case $\dim N = 1$ and $\Gamma = \emptyset$, or $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$. The proof assumes $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) \neq \{0\}$, and the formal statement is the dichotomy it shows. P is onto Chapter3/GoodPosition · sigmaCoord_surjective. - $\Gamma \neq \emptyset$. A point x of N with $x_i = 0$ for $i \in \Gamma$ lies in $N \cap C$, and so does $-x$ Chapter3/GoodPosition · mem_quadCone_of_vanIdx, so $x = 0$. - $\Gamma = \emptyset$ *and* $N \neq 0$. The kernel is all of N. If $\dim N \geq 2$, the generators b_0, b_1 have all coordinates positive (no index vanishes on a nonzero point of the cone). Then $b_0 - t b_1 \in N \cap C$ for small $t > 0$, but its e-coordinate j_1 is $-t < 0$. - $N = 0$. The kernel is $\{0\}$, and the first alternative holds.

Read literally, the first alternative does not hold as we read it for $N = 0$ Chapter3/Book2021Items · lemma_3_8_5_not_literal_2021. In $E = \mathbb{R}$, $C = [0,\infty)$, the subspace $N = 0$ is in good position, $N \cap C = \{0\}$ is a closed convex cone, and $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$; but $\dim N = 0$ and $N \ncong \mathbb{R}$. The dichotomy holds there through its other alternative.

Lemma 3.8.6. (1) If $N \cap (\mathbb{R}^{\Gamma^c} \oplus W) = \{0\}$, then P is a linear isomorphism carrying $N \cap C$ onto $[0,\infty)^\Gamma$ Chapter3/Book2021Items · lemma_3_8_6_2021. This needs no cone hypothesis: P is injective by hypothesis and onto. A point whose Γ-coordinates are ≥ 0 is in $N \cap C$ by the segment argument of §III.14 when $\Gamma \neq \emptyset$; when $\Gamma = \emptyset$, $N = 0$. (2) If $N \cap C$ is a closed convex cone, $N \subseteq \mathbb{R}^{\Gamma^c} \oplus W$ and $N \neq 0$, then $\dim N = 1$ by Lemma 3.8.5. A linear isomorphism $N \cong \mathbb{R}$ then carries $N \cap C$ onto $[0,\infty)$ Chapter3/Book2021Items · lemma_3_8_6_two_2021. As printed, (2) does not hold as we read it for $N = 0$, by the same example. With $N \neq 0$ its hypotheses are met by $N = E = \mathbb{R}$, $C = [0,\infty)$ Chapter3/Book2021Items · halfLine_witness_2021.

Corollary 3.8.7. If $N \cap C$ is a closed convex cone, it is a partial quadrant in N, and $(N, N \cap C) \cong (\mathbb{R}^{\dim N}, [0,\infty)^{\dim N})$ Chapter3/Book2021Items · corollary_3_8_7_2021. The first part is Proposition 3.1.25 (preprint 3.22) (§III.14), which needs no cone hypothesis Chapter3/GoodPosition · proposition_3_22. The second is the quadrant form above, which the text also derives on p. 149.

III.55 What the 2021 additions taught us

- **The 2021 filling is the formalization's.** The germ filling never asked for tameness. The 2021 text drops it, so the formal definitions and the printed ones now agree `Chapter3/Book2021Germs · isFilling2021_iff` `Chapter3/Book2021Germs · StrongBundle.isScFredholmGerm2021_iff`. The 2017 notion is strictly narrower `Chapter3/Book2021Germs · exists_isFilling2021_not_isFilling2017`.
- **One proof step needs the dropped hypothesis.** All the theorems of Chapter 3 that the project proved without tameness hold for the 2021 definitions as printed. The passage of Theorem 3.6.8 from good position in T_0O to good position in E (p. 138) does not hold as we read it without a tame chart `Chapter3/Book2021Germs · theorem_3_6_8_tame_step_2021`. The implicit function theorems remain verified with tame germ data, and open for untamed data (G.79).
- **Krein–Milman is not needed.** A pointed cone in good position is a quadrant `Chapter3/Book2021Items · exists_quadrantEquiv_of_pointed`. Lemma 3.8.4 (2) follows by one subtraction, and (1) and (3) need no cone hypothesis.
- $N = 0$. Lemma 3.8.5, read as an implication, and Lemma 3.8.6 (2) need $N \neq 0$ `Chapter3/Book2021Items · lemma_3_8_5_not_literal_2021`. The dichotomy the proof shows holds for all N.

Part IV. Manifolds with boundary with corners (HWZ Chapter 4, §§4.1–4.4)

Chapter 4 asks when an M-polyfold — a space glued from images of sc-smooth retractions — is secretly an ordinary manifold with boundary with corners, and why the fibres of a family of such manifolds are good for almost every parameter. The pivot is the *sc⁺ condition*: a retraction that gains a level. At a smooth point its derivative is compact, so its retract has finite-dimensional tangent spaces (Lemma 4.1.7 (preprint 4.7)), and near a point where the tangent space is in good position to the partial quadrant the retract is the graph of a map over that tangent space (Proposition 4.1.9 (preprint 4.9)). Theorem 4.1.2 (preprint 4.2) characterizes the manifolds with corners among M-polyfolds by these two properties; Theorem 4.2.7 (preprint 4.18) applies the characterization to submanifolds in good position; Theorem 4.2.11 (preprint 4.22) describes a submanifold near a point of general position; Theorem 4.3.1 (preprint 4.23) combines that description with Sard's theorem on the strata of the corners. This Part retells: finite-dimensional sc-spaces and corner charts (§IV.1); sc⁺-maps (§IV.2); sc⁺-retractions, Lemmas 4.1.5 and 4.1.7 (preprint 4.5 and 4.7) (§IV.3); Proposition 4.1.8 (preprint 4.8) with its chart form and Remark 4.1.3 (preprint 4.3) (§IV.4); charts centred at a smooth point, which the text uses and never constructs (§IV.5); Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), by a direct proof (§§IV.6–IV.9); Theorem 4.1.2 (preprint 4.2) (§IV.10); smooth finite-dimensional submanifolds and Remark 4.2.2 (preprint 4.14) (§IV.11); Proposition 4.2.3 (preprint 4.15) (§IV.12); good and general position in charts and their chart independence (§IV.13); Lemma 4.2.8 (preprint 4.19) and Theorem 4.2.7 (preprint 4.18) (§IV.14); Remark 4.2.9 (preprint 4.20) (§IV.15); the linear algebra at the corner of a centred chart (§IV.16); Theorem 4.2.11 (preprint 4.22) (§§IV.17–IV.18) and Remark 4.3.2 (preprint 4.24) (§IV.19); Sard's theorem (§IV.20); the finite-dimensional core of Theorem 4.3.1 (preprint 4.23) (§§IV.21–IV.22); the fibre over a regular value (§IV.23); Theorem 4.3.1 (preprint 4.23) (§IV.24); what is not formalized (§IV.25); and what the formalization taught us (§IV.26).

What is proved, and what is not. Theorems 4.1.2, 4.2.7, 4.2.11 and 4.3.1 (preprint 4.2, 4.18, 4.22 and 4.23) are proved with the text's hypotheses; Theorem 4.1.2 (preprint 4.2) needs no tameness at all. Proposition 4.1.9 (preprint 4.9) is proved with two hypotheses the text leaves implicit — the good complement Y is closed, and p is the projection onto T_0O *along* Y rather than $Dt(0)$ — and by a direct argument (§§IV.6–IV.9), because the text's route through Lemma 4.1.11 (preprint 4.10) and Theorem 3.5.11 (preprint Corollary 3.47) yields the graph map only as a germ over a complement of its own choosing; where Proposition 4.1.9 (preprint 4.9) is applied (Theorems 4.1.2, 4.2.7, 4.2.11 (preprint 4.2, 4.18, 4.22), Remark 4.2.9 (preprint 4.20)), the closed complement is chosen, so nothing is added there. One step of Lemma 4.2.8 (preprint 4.19) is false and is repaired (§IV.14). Theorem 4.2.11 (4) (preprint 4.22), which the text leaves to "the previous arguments", is proved (§IV.18). In Theorem 4.3.1 (preprint 4.23) "regular value" has to be read as the stratified notion of its proof, and the finite cover of its proof is replaced by a countable one (§IV.24). Proved although the text does not construct it: charts and strong bundle charts centred at a smooth point (§IV.5), an sc⁺-retraction onto the fibre A_λ (§IV.23), Theorem 3.8.22 (preprint 3.78) in C^∞ form (§IV.21), and Sard's theorem in all dimensions (§IV.20). Remark 4.1.3's (preprint 4.3) "$X^1 \neq X$ is possible", for which the text gives no example, is proved by one, an M-polyfold whose retraction loses

a level in its first derivative, with a new criterion for sc-smoothness (§IV.27). Not formalized: Lemma 4.1.11 (preprint 4.10) (no longer needed), the whole of §4.4 on sc-differential forms, the open question after Proposition 4.2.3 (preprint 4.15), and the chart independence of stratified regularity (§IV.25).

Five structural choices. (1) *sc⁺ is a property of maps.* The text defines it for retractions and operators and uses it for maps; here a map is sc⁺ when its lift into F^1 is sc-smooth, and the calculus of such maps is proved first (§IV.2). (2) *Tangent spaces of X are read in charts.* T_xX carries no linear structure in the formalization (gap G.7 of Chapter 2), so T_aA, good position, general position and stratified regularity are stated in a chart, and — except for stratified regularity — proved independent of the chart (§IV.13). (3) *Centred charts are re-centred charts.* A chart with $\psi(x) = 0$ is the chart $\varphi - \varphi(x)$ on the set where the partial quadrant C agrees with $C_{\varphi(x)}$, which exists only at smooth points (§IV.5). (4) *Proposition 4.1.9 (preprint 4.9) on fixed sets.* The sc-diffeomorphism $p : O \cap W \to V$ is proved with $\Gamma = p^{-1}$ sc-smooth on a *fixed* relatively open V, by contraction arguments on level 1, a frame formula for $D\Gamma$ and a C^k-bootstrap along the finite-dimensional tangent space. (5) *Stratified Sard.* Sard's theorem is applied stratum by stratum in the chart of Theorem 4.2.11 (preprint 4.22), and the null sets of critical values of countably many charts are collected.

Map.

- `Chapter4/ScPlus`: sc⁺-maps, Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7), Proposition 4.1.8 (preprint 4.8) and its chart form, Remark 4.1.3 (preprint 4.3), finite-dimensional sc-spaces.
- `Chapter4/Recentre`: the partial quadrant C_e and its neighbourhood $G(e)$, translations, re-centred retracts, charts and strong bundle charts.
- `Chapter4/ScPlusGraph`: the mean value estimate for sc⁺-retractions, Proposition 4.1.9 (preprint 4.9) with Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12), the frame formula and the bootstrap.
- `Chapter4/Characterization`: Definition 4.1.1 (preprint 4.1), corner charts, Theorem 4.1.2 (preprint 4.2).
- `Chapter4/Submanifolds`: Definitions 4.2.1, 4.2.5, 4.2.6, 4.2.10 (preprint 4.13, 4.16, 4.17, 4.21), Remark 4.2.2 (preprint 4.14), Proposition 4.2.3 (preprint 4.15), Theorem 4.2.7 (preprint 4.18), Lemma 4.2.8 (preprint 4.19), Remark 4.2.9 (preprint 4.20), the chart independence of good and general position.
- `Chapter4/LocalStructure`: Theorem 4.2.11 (1)–(4) (preprint 4.22), Remark 4.3.2 (preprint 4.24), the linear algebra of a centred chart, Lemmas 7.A and 7.B of the exposition.
- `Vendor/MilnorSard/…`: Sard's theorem, copied from our Milnor project (`math_faithful`) with the namespace renamed.
- `Chapter4/FamiliesLocal`, `Chapter4/Families`: the strata and Sard on them, Theorem 3.8.22 (preprint 3.78) in C^∞ form, the fibre retraction, the sc⁺-retraction onto A_λ and its general position, Theorem 4.3.1 (preprint 4.23).
- Added in the fourth addendum (§IV.27): `Chapter4/RemarkFourThreeExample` (parameter families of level operators with one level of loss, their sc-smoothness, the reduction of Remark 4.1.3's (preprint 4.3) example), `Chapter4/RemarkFourThreeFamily` (the family of moving projections, the M-polyfold of a retract, the example).

Throughout, $C = T^{-1}([0,\infty)^n \oplus W)$ is a partial quadrant presented by an sc-isomorphism $T : E \cong \mathbb{R}^n \oplus W$, with coordinates $\lambda_1, \dots, \lambda_n$; $I(e) = \{i : \lambda_i(e) = 0\}$, $d_C(e) = \#I(e)$, $C_e = \{\lambda_i \geq 0, i \in I(e)\}$ and $E_e = \{\lambda_i = 0, i \in I(e)\}$ (Definition 2.5.1 (preprint 2.33), §II.12). For a retract $O = r(U)$ and a level-1 point z, $T_zO = Dr(z)E_0$ is the tangent space of Definition 2.3.12 (preprint 2.17) and $T_z^RO = T_zO \cap E_z$ the reduced tangent space.

IV.1 Finite-dimensional sc-spaces and corner charts (Definition 4.1.1 (preprint 4.1))

Finite-dimensional sc-spaces. Definition 4.1.1 (preprint 4.1) models its charts on finite-dimensional spaces "with the constant sc-structure"; the text remarks after Definition 1.1.1 (preprint 1.1) that a finite-dimensional space has only that one. This is proved. If E_0 is finite-dimensional, every level E_m is, since it injects into E_0 `Chapter4/ScPlus · ScBanach.finiteDimensional_lvl`. The range of $E_m \to E_0$ is then a finite-dimensional, hence closed, subspace; it contains the smooth points, which are dense (Definition 1.1.1 (2) (preprint 1.1)); so it is all of E_0 `Chapter4/ScPlus · ScBanach.toBase_surjective`. The inclusions are therefore bijective, and they form an sc-isomorphism of E onto E_0 with the constant structure `Chapter4/ScPlus · ScBanach.scIsoConst`; in particular every point of E_0 is smooth `Chapter4/Characterization · ScBanach.mem_smooth_of_finiteDimensional`. The density axiom is what makes this work: $E_0 = \mathbb{R}$, $E_m = 0$ for $m \geq 1$ has compact injective inclusions and levels that differ (Findings, Chapter 4, item 6).

Definition 4.1.1 (preprint 4.1). A chart of X is a *corner chart* if its local model is (U, C, V) with V a finite-dimensional real normed space carrying the constant sc-structure, C a partial quadrant in V, U relatively open in C, and the identity retraction `Chapter4/Characterization · MChart.IsCornerChart`. Its model is then relatively open in a partial quadrant `Chapter4/Characterization · MChart.IsCornerChart.isRelOpenInQuadrant`, finite-dimensional `Chapter4/Characterization · MChart.IsCornerChart.finiteDimensional`, a tame retract — the identity retraction is tame `Chapter2/Tame · ScRetraction.id_isTame Chapter4/Characterization · MChart.IsCornerChart.isTameRetract` — and an sc$^+$-retract by Remark 4.1.3 (preprint 4.3) (§IV.4) `Chapter4/Characterization · MChart.IsCornerChart.isScPlusRetract`; every point of its model space is smooth `Chapter4/Characterization · MChart.IsCornerChart.mem_smooth`. *X has a corner structure* if it has an atlas of corner charts equivalent to its own atlas `Chapter4/Characterization · MPolyfold.HasCornerStructure`. The dimension of V may vary from chart to chart, as in the text. The text's "sc-smoothly compatible charts", "φ an sc-diffeomorphism", is read as compatibility with the atlas of X, as for tameness (Definition 2.5.7 (preprint 2.37)).

The definition is local `Chapter4/Characterization · MPolyfold.hasCornerStructure_iff`: X has a corner structure iff every point lies in a corner chart that is a chart of the M-polyfold structure. One direction takes a chart of the equivalent atlas through the point. For the other, the corner charts that are charts of X form an atlas: they cover X by hypothesis, any two charts of X are compatible `Chapter2/MPolyfold · MPolyfold.IsChart.compatible`, and the atlas is equivalent to that of X because each of its members is a chart of X. *A corner structure makes X tame* `Chapter4/Characterization · MPolyfold.HasCornerStructure.isTame`: the same atlas is an atlas of tame charts.

The transition maps are C^∞. For a chart whose model is relatively open in a partial quadrant, every overlap $\chi(V \cap V')$ is relatively open `Chapter4/Characterization · MChart.isRelOpenInQuadrant_overlap`. For compatible corner charts, the transition map is sc-smooth between the overlap retracts; composed with the identity retraction of the overlap it is sc-smooth on a relatively open subset of a partial quadrant in a finite-dimensional space with the constant structure, and such maps are exactly the C^∞ maps with derivatives within the set `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const Chapter4/Characterization · MChart.IsCornerChart.contDiffOn_transition`. That is the text's "classically smooth"; "the atlas defines the structure of a smooth manifold with boundary with corners" is not formalized beyond it, because Mathlib's manifolds with corners have a single model space and here the partial quadrants vary from chart to chart.

Arbitrary finite-dimensional models. The constant structure is no restriction. A partial quadrant of a finite-dimensional E is one of E_0 with the constant structure `Chapter4/Characterization · IsPartialQuadrant.toConst Chapter4/Characterization · IsRelOpenInQuadrant.toConst`, and the identity of E_0 is an sc-diffeomorphism between the two models `Chapter4/Characterization · IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst`. Composing a chart with it (the construction `compDiffeo` of Chapter 3) `Chapter3/GlobalIFTCenter · MChart.compDiffeo` gives, with the same map and domain, a corner chart of the same dimension `Chapter4/Characterization · MChart.exists_isCornerChart_of_finiteDimensional`. Hence X has a corner structure iff it has an equivalent atlas of charts onto relatively open subsets of par-

tial quadrants in sc-Banach spaces with finite-dimensional E_0 Chapter4/Characterization · MPolyfold.hasCornerStructure_iff_finiteDimensional — verbatim the definition of the manifolds with corners of locally constant dimension of Theorem 3.2.3 (preprint 3.27) (§III.36) — and the manifolds with corners of dimension n of Theorem 3.2.3 (preprint 3.27) are the corner structures all of whose charts have dimension n Chapter4/Characterization · MPolyfold.isManifoldWithCorners_iff Chapter4/Characterization · MPolyfold.IsManifoldWithCorners.hasCornerStructure.

IV.2 Sc⁺-maps

The text uses "sc⁺-map" in the proof of Lemma 4.1.7 (preprint 4.7) and composes sc⁺-retractions with sc-smooth maps in Lemma 4.1.5 (preprint 4.5), but defines sc⁺ only for retractions (Definition 4.1.4 (preprint 4.4)) and operators (Definition 1.1.5 (preprint 1.4)). The formalization defines it for maps on any set, as the neighbouring notions of the project are (sc⁺-operators are sc-operators into F^1, sc⁺-sections sc-smooth maps into $K[1]$).

The lift. For $x \in E_1 \subseteq E_0$, $1_0^1 x$ is the point of E_1 over x, with a junk value off E_1 Chapter4/ScPlus · ScBanach.liftOne Chapter4/ScPlus · ScBanach.toBase_liftOne Chapter4/ScPlus · ScBanach.liftOne_toBase.

Definition. f is *sc⁺ on* U if $f(U) \subseteq F_1$ and the lift $x \mapsto 1_0^1 f(x)$, a map $U \to F^1$, is sc-smooth Chapter4/ScPlus · IsScPlus. Then $f(U_m) \subseteq F_{m+1}$: the lift is sc⁰, so it maps U_m into $(F^1)_m = F_{m+1}$ Chapter4/ScPlus · IsScPlus.mem_range_succ. On an sc-domain the canonical lift can be replaced by any sc-smooth $g : U \to F^1$ with $\iota \circ g = f$, which is how the text phrases it Chapter4/ScPlus · IsScPlus.iff_exists. The calculus: an sc⁺-map is sc-smooth, being $\iota\circ$ (lift) with the sc-operator $\iota : F^1 \to F$ Chapter4/ScPlus · IsScPlus.isScSmooth; the property depends only on the values on U Chapter4/ScPlus · IsScPlus.congr and passes to smaller sc-domains Chapter4/ScPlus · IsScPlus.mono; *sc⁺ ∘ sc-smooth is sc⁺*, the lift of $g \circ f$ being (lift of g) $\circ f$, sc-smooth by the chain rule Chapter1/ChainRule · IsScSmooth.comp Chapter4/ScPlus · IsScPlus.comp_isScSmooth; *sc-smooth ∘ sc⁺ is sc⁺*, the lift of $g \circ f$ being $g^1\circ$ (lift of f) with $g^1 : V^1 \to G^1$ the map g induces on level 1, sc-smooth by Proposition 1.2.2 (preprint 1.17) Chapter1/Levels · IsScSmooth.shift Chapter4/ScPlus · IsScSmooth.comp_isScPlus; and an sc⁺-operator is an sc⁺-map Chapter4/ScPlus · ScPlus.isScPlus.

The derivative at a smooth point Chapter4/ScPlus · IsScPlus.exists_scPlus_deriv. Let f be sc⁺ on an sc-domain U and $x \in U_1$ smooth. The lift g is sc-smooth into F^1, so its derivative at the smooth point x is an sc-operator $E \to F^1$ (the remark after Proposition 1.2.1 (preprint 1.16)), that is an sc⁺-operator $S : E \to F$. Since $f = \iota \circ g$ on U, the chain rule and the uniqueness of the derivative on an sc-domain give $Df(x) = \iota\circ Dg(x)$, the level-0 operator of S. This is the text's step "$Dr(o)$ is a sc⁺-operator, since r is a sc⁺-map". Consequently $Df(x)$ is compact, and so is every level operator of S Chapter4/ScPlus · IsScPlus.isCompactOperator_deriv.

A compact projection has finite-dimensional range Chapter4/ScPlus · finiteDimensional_range_of_isCompactOperator_of_idem: if $P = P \circ P$ is compact, its range is the closed set $\{v : Pv = v\}$, P restricts to a compact operator of the range which is the identity, and a normed space with compact identity is finite-dimensional (Riesz). The text takes this for granted.

IV.3 Sc⁺-retractions and sc⁺-retracts (Definitions 4.1.4, 4.1.6 (preprint 4.4, 4.6), Lemmas 4.1.5, 4.1.7 (preprint 4.5, 4.7))

Definition 4.1.4 (preprint 4.4). An sc-smooth retraction $r : U \to U$ is an *sc⁺-retraction* if it is an sc⁺-map on U Chapter4/ScPlus · ScRetraction.IsScPlus. The text's condition $r(U_m) \subseteq U_{m+1}$ is then a consequence, not a hypothesis Chapter4/ScPlus · ScRetraction.IsScPlus.mem_lvlSet_succ. On an open subset V of an M-polyfold, an sc⁺-retraction is a map s, sc-smooth on V, with $s(V) \subseteq V$, $s\circ s = s$ on V, $s(V) \subseteq X_1$, and $1_0^1 \circ s : V \to X^1$ sc-smooth Chapter4/ScPlus · MPolyfold.IsScPlusRetractionOn; the junk value of 1_0^1 off X_1 is quantified over and plays no role.

Lemma 4.1.5 (preprint 4.5) Chapter4/ScPlus · lemma_4_5. *If t is an sc⁺-retraction onto O and s an sc-smooth retraction (of a possibly different relatively open set) onto O, then s is sc⁺.* The text's proof: s maps its domain into O, which lies in the domain of t, and t fixes O, so $s = t \circ s$, which is sc⁺ □ sc-smooth. (The text writes "$s(V) = 0$" for $s(V) = O$.)

Definition 4.1.6 (preprint 4.6). (O, C, E) is an *sc⁺-retract* if it is the retract of an sc⁺-retraction Chapter4/ScPlus · IsScPlusRetract; it is then a retract Chapter4/ScPlus · IsScPlusRetract.isScRetract, every sc-smooth retraction onto it is sc⁺ by Lemma 4.1.5 (preprint 4.5) Chapter4/ScPlus · IsScPlusRetract.isScPlus, and so the property is intrinsic Chapter4/ScPlus · isScPlusRetract_iff_forall.

Lemma 4.1.7 (preprint 4.7) (i) Chapter4/ScPlus · lemma_4_7_i. *The retract of an sc⁺-retraction consists of smooth points.* By induction on m: a point $y \in O$ lies in E_m; then $y \in U_m$, and $y = r(y) \in E_{m+1}$ by the level gain. (ii) Chapter4/ScPlus · lemma_4_7_ii. *At a level-1 point o over a point of O, $Dr(o)E_0$ is finite-dimensional.* o is smooth by (i), so $Dr(o)$ is compact (§IV.2); it is a projection because $r \circ r = r$ gives $Dr(o) \circ Dr(o) = Dr(o)$ at points of O Chapter2/Retracts · ScRetraction.scDeriv_comp_self; and a compact projection has finite-dimensional range. In the form with the tangent space of Definition 2.3.12 (preprint 2.17) Chapter4/ScPlus · lemma_4_7_ii_tangentSpace: every $o \in O$ has a smooth lift $o_1 \in E_1$ Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth, and $T_{o_1}O$ is the range of $Dr(o_1)$ Chapter2/Retracts · tangentSpace_eq_coe_submodule, finite-dimensional. Tangent spaces are indexed by level-1 points, which is why the statement mentions the lift.

IV.4 Proposition 4.1.8 (preprint 4.8), its chart form, and Remark 4.1.3 (preprint 4.3)

Proposition 4.1.8 (preprint 4.8) Chapter4/ScPlus · proposition_4_8. *A retract (O, C, E) is an sc⁺-retract iff $O \subseteq E_\infty$ and the lift 1^1_0 is sc-smooth from (O, C, E) to (O_1, C_1, E^1)* in the sense of Definition 2.1.14 (preprint 2.7). (□) $O \subseteq E_\infty$ is Lemma 4.1.7 (i) (preprint 4.7); and by the criterion of Definition 2.1.14 (preprint 2.7) through one retraction Chapter2/Retracts · isScSmoothRetractMap_iff_exists, the lift is sc-smooth between the retracts because $1^1_0 \circ r$ is the lift of the sc⁺-map r. (□) For a retraction r onto O, the composite $U \to O \to O_1 \to E^1$ is sc-smooth (Definition 2.1.14 (preprint 2.7), the field `comp_isScSmooth` of Chapter2/Retracts · IsScSmoothRetractMap) and is the lift of r, whose values lie in E_1 because $O \subseteq E_\infty$. The text's proof is complete; it ends "the lemma is proved".

The chart form. Theorem 4.1.2 (preprint 4.2) needs the statement for the charts of an M-polyfold, which the text does not state. The map $1^1_0 : X \to X^1$ is defined on X_1, with a junk value x_0 off it Chapter4/ScPlus · MPolyfold.toLevelOne Chapter4/ScPlus · MPolyfold.levelVal_toLevelOne Chapter4/ScPlus · MPolyfold.toLevelOne_levelVal. In the chart φ of X and the level chart $(V_1, \varphi, (O_1, C_1, E^1))$ of X^1 Chapter2/LevelPolyfold · MPolyfold.levelChart it is the lift $O \to O_1$ Chapter4/ScPlus · MPolyfold.levelChart_toLevelOne. A point of a chart domain is smooth iff its image is Chapter4/ScPlus · MPolyfold.mem_smoothSet_iff. So *the model of a chart $\varphi : V \to O$ is an sc⁺-retract iff $V \subseteq X_\infty$ and 1^1_0 is sc-smooth on V* Chapter4/ScPlus · MPolyfold.proposition_4_8_chart: (□) continuity on V comes from the lift read in the charts, and sc-smoothness is checked in the pair of charts φ, `levelChart` φ Chapter2/LevelPolyfold · IsScSmoothOn.of_exists_charts, where it is Proposition 4.1.8 (preprint 4.8); (□) the same pair of charts turns the sc-smoothness of 1^1_0 on V into that of the lift on O. Hence "$X_0 = X_\infty$ and 1^1_0 sc-smooth" holds iff every chart of X is modelled on an sc⁺-retract Chapter4/ScPlus · MPolyfold.isScSmoothMap_toLevelOne_iff, and already if some cover of X by such charts exists Chapter4/Characterization · MPolyfold.isScSmoothMap_toLevelOne_of_cover — sc-smoothness being local.

Remark 4.1.3 (preprint 4.3). The inclusion $\iota : (O_1, C_1, E^1) \to (O, C, E)$ is sc-smooth between retracts Chapter4/ScPlus · isScSmoothRetractMap_toBase_one, so $1^0_1 : X^1 \to X$ is always sc-smooth, being ι in the charts `levelChart` φ and φ Chapter4/ScPlus · MPolyfold.remark_4_3_levelVal. If every point lies in X_1 and 1^1_0 is sc-smooth, it is an sc-diffeomorphism with inverse 1^0_1 Chapter4/ScPlus · MPolyfold.remark_4_3_isScDiffeo. For a finite-dimensional V with the constant structure, $V^1 = V$ definitionally, the lift is the identity, and every

retract in V is an sc$^+$-retract, its retraction being its own lift `Chapter4/ScPlus · remark_4_3_const`. The remark's "it is possible that $X^1 \neq X$ as M-polyfolds although the sets agree" is given without an example; one is constructed in §IV.27.

IV.5 Charts centred at a smooth point

Theorem 4.1.2 (ii) (preprint 4.2) asks for charts with $\psi(x) = 0$, the proofs of Theorems 4.2.7 and 4.2.11 (preprint 4.18 and 4.22) and of Lemma 5.3.3 (preprint 5.11) use them, and the text never says where they come from. Translating a chart $\varphi : V \to O \subseteq C$ by $e = \varphi(x)$ fails twice: translation by e preserves every level only if e is smooth, and $C - e$ is not a partial quadrant unless $e \in C \cap (-C)$. The construction (Findings, items 1, 2) uses the partial quadrant C_e and the neighbourhood on which it agrees with C.

The partial quadrant C_e and the neighbourhood $G(e)$. Let $G(e) = \{v : \lambda_i(v) > 0 \text{ whenever } \lambda_i(e) \neq 0\}$ `Chapter4/Recentre · recentreNbhd`, open `Chapter4/Recentre · isOpen_recentreNbhd` and containing e when $e \in C$ `Chapter4/Recentre · mem_recentreNbhd`. On it C and C_e agree: $G(e) \cap C = G(e) \cap C_e$, because on $G(e)$ the coordinates not vanishing at e are positive anyway `Chapter4/Recentre · recentreNbhd_inter_eq`. C_e is invariant under translation by e `Chapter4/Recentre · add_mem_partialQuadrantAtWith_iff`, so $C_e - e = C_e$ `Chapter4/Recentre · image_sub_partialQuadrantAtWith`. *C_e is a partial quadrant* — Definition 2.5.1 (preprint 2.33) asserts it without proof: the vanishing indices $I(e)$ `Chapter4/Recentre · zeroIdx`, $d = \#I(e)$ of them `Chapter4/Recentre · recentreDim`, are put first, $\mathbb{R}^n \cong \mathbb{R}^{I(e)} \times \mathbb{R}^{\text{rest}}$ `Chapter4/Recentre · splitCoordsL`, which gives an sc-isomorphism $\mathbb{R}^n \oplus W \cong \mathbb{R}^{I(e)} \oplus (\mathbb{R}^{\text{rest}} \oplus W)$ `Chapter4/Recentre · splitCoordsIso` and the presentation T_e of C_e `Chapter4/Recentre · recentreIso Chapter4/Recentre · partialQuadrantAtWith_eq_preimage Chapter4/Recentre · isPartialQuadrant_partialQuadrantAtWith`: the coordinates not vanishing at e join the linear factor. *Re-centring keeps the tame data near e.* For $v \in G(e)$ the vanishing coordinates of $v - e$ for T_e are those of v for T `Chapter4/Recentre · recentreIso_fst_sub Chapter4/Recentre · zero_of_mem_recentreNbhd`, so $d_{C_e}(v-e) = d_C(v)$ `Chapter4/Recentre · degIndexWith_recentreIso Chapter4/Recentre · degIndex_sub_recentre`, and the minimal subspaces and partial quadrants agree: $E^{T_e}_{v-e} = E^T_v$ `Chapter4/Recentre · minimalSubspaceWith_recentreIso`, $(C_e)_{v-e} = C_v$ `Chapter4/Recentre · partialQuadrantAtWith_recentreIso`.

Translations. Translation by a smooth vector a is sc-smooth on every sc-domain: its level maps are $x \mapsto x + a_m$, a_m the lift of a to E_m, and the criterion of Proposition 1.2.4 (preprint 1.19) applies `Chapter1/Levels · IsScSmooth.of_contDiffOn_levels Chapter4/Recentre · isScSmooth_add_const Chapter4/Recentre · isScSmooth_sub_const`. For U relatively open in C with $U \subseteq G(e)$, the translate $U - e$ is relatively open in C_e `Chapter4/Recentre · relOpen_translate` and, for smooth e, $v \mapsto v - e$ is an sc-diffeomorphism $U \to U - e$ `Chapter4/Recentre · isScDiffeoQuadrant_translate`.

Re-centred retracts. A retraction r restricted to $U \cap r^{-1}(W) \cap W$, W open, is a retraction whose domain lies in W `Chapter4/Recentre · ScRetraction.restrictIn`, onto $r(U) \cap W$ `Chapter4/Recentre · ScRetraction.restrictIn_retract`, with the same derivative `Chapter4/Recentre · ScRetraction.scDeriv_restrictIn`, and tame if r is `Chapter4/Recentre · ScRetraction.IsTameWith.restrictIn Chapter4/Recentre · ScRetraction.restrictIn_isTame`. Transporting the restriction to $G(e)$ along the translation `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport` shows that $((O \cap G(e)) - e, C_e, E)$ is a retract, with retraction $u \mapsto r(u + e) - e$ `Chapter4/Recentre · isScRetract_recentre`, and tame if (O, C, E) is `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport Chapter4/Recentre · isTameRetract_recentre` — the transport of tameness uses exactly the invariance of d and of the minimal subspaces above.

Re-centred charts. For a chart φ, a presentation of its partial quadrant and a smooth e, the domain $V \cap \varphi^{-1}(G(e))$ `Chapter4/Recentre · MChart.recentreSource` is open, and the chart $\varphi_e = \varphi - e$ on it is φ restricted `Chapter2/MPolyfold · MChart.restrict` and composed with the translation `Chapter4/Recentre · MChart.recentre`, modelled on $((O \cap G(e)) - e, C_e, E)$. It is a chart of the M-polyfold, its transition maps being those of φ composed with translations `Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart Chapter4/Recentre · MChart.recentre_isChart`; $\varphi_e(x) = 0$ when $\varphi(x) = e$ `Chapter4/Recentre · MChart.recentre_apply_self`; its model is tame when that of φ is `Chapter4/Recentre · MChart.isTameRetract_recentre`; and it computes the same degeneracy in-

dex Chapter4/Recentre · MChart.chartDegIndex_recentre. So *every smooth point x of a chart domain is the centre of a chart of X* inside that domain, tame if the original is Chapter4/Recentre · MPolyfold.exists_centredChart, and *in a tame M-polyfold every smooth point is the centre of a tame chart* Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart Chapter4/Recentre · MPolyfold.IsTame.exists_centredChart. At the centre every coordinate of C_e vanishes, so the minimal subspace there is $\{\lambda = 0\}$, the text's "$\{0\} \oplus W$", which is what §§IV.14, IV.16 and IV.17 use.

Re-centred strong bundle charts (for Lemma 5.3.3 (preprint 5.11)). Only the base is translated: $(u, h) \mapsto (u \mp e, h)$ Chapter4/Recentre · subFst Chapter4/Recentre · addFst is sc-smooth on $E \oplus F$ and on $E \oplus F^1$ because $(e, 0)$ is smooth in both Chapter4/Recentre · mem_smooth_prod_zero Chapter4/Recentre · isScSmooth_addFst Chapter4/Recentre · isScSmooth_subFst. A strong bundle retraction restricts to $(U \cap r^{-1}(G) \cap G) \lhd F$ Chapter4/Recentre · IsStrongBundleRetraction.restrictIn and translates to $R_e(u, h) = (r(u+e) - e, \Gamma(u+e, h))$ Chapter4/Recentre · IsStrongBundleRetraction.translate; with $\Phi_e(y) = (\Phi(y)_1 - e, \Phi(y)_2)$ this is the re-centred chart Chapter4/Recentre · StrongBundleChart.recentre. Its overlaps are translates of open parts of the old ones Chapter4/Recentre · StrongBundleChart.recentre_overlap Chapter4/Recentre · StrongBundleChart.overlap_recentre_eq, its transition maps on $[0]$ and $[1]$ are the old ones composed with translations by $(e, 0)$, so compatibility passes to it Chapter4/Recentre · StrongBundleChart.Compatible.recentre, and it is a strong bundle chart Chapter4/Recentre · StrongBundle.IsChart.recentre. Every smooth point of the base is the centre of one Chapter4/Recentre · StrongBundle.IsChart.exists_centred. No hypothesis beyond the smoothness of the point is needed.

IV.6 Proposition 4.1.9 (preprint 4.9): the data, and the mean value estimate

The text. Let $t : U \to U$ be an sc⁺-retraction onto $O \ni 0$ with T_0O in good position to C, Y a good complement and $p = Dt(0)$ "the sc-projection". Then p maps a neighbourhood of 0 in O sc-diffeomorphically onto one in $T_0O \cap C$. The proof: the section $\mathbf{f}(u) = (1-p)(u - t(u))$ of $U \lhd Y$ is sc-Fredholm (Lemma 4.1.11 (preprint 4.10)), Theorem 3.5.11 (preprint Corollary 3.47) describes its zeros near 0 as a graph $\{a + \delta(a)\}$, Lemma 4.1.12 (preprint 4.11) says that every small zero is on the graph, Lemma 4.1.13 (preprint 4.12) that the graph lies in O.

What changes, and why. (a) $Dt(0)$ is a projection onto T_0O along $\ker Dt(0)$, which nothing relates to Y; Lemma 4.1.13 (preprint 4.12) writes $t(a + \delta(a)) = p\,t(a + \delta(a)) + \delta(a)$ with $\delta(a) \in Y$, which needs the projection *along* Y. (b) Y must be closed, or that projection is not continuous. (c) Theorem 3.5.11 (preprint Corollary 3.47) gives δ over a complement of its own choosing and sc-smooth only as a germ — C^k into E_m on neighbourhoods shrinking with m and k (§III.20) — while the claim is an sc-diffeomorphism between fixed open sets. The formalization therefore takes p to be the projection along a closed Y and proves the proposition directly, with the sc⁺ property doing the work of the Fredholm theory (Findings, items 20–28).

The data Chapter4/ScPlusGraph · GraphData: the sc⁺-retraction t with $0 \in O$; N with $N = T_0O$, which equals $Dt(0)E_0$ Chapter4/ScPlusGraph · GraphData.N_eq_range and is finite-dimensional by Lemma 4.1.7 (preprint 4.7) Chapter4/ScPlusGraph · GraphData.finiteDimensional; N in good position; a good complement Y, closed; and $p : E_0 \to N$ continuous linear with $p = 1$ on N, $p = 0$ on Y. The tangent space consists of smooth points: $Dt(0)$ is an sc⁺-operator and N its fixed vectors Chapter4/ScPlusGraph · GraphData.smooth. $Y = \ker p$ Chapter4/ScPlusGraph · GraphData.mem_Y_iff. Because $N \subseteq E_\infty$ is finite-dimensional, it lifts continuously to every level Chapter3/TangentGerm · finSmoothLift Chapter4/ScPlusGraph · GraphData.liftN, and p, $q = 1 - p$ act on every level E_m Chapter4/ScPlusGraph · GraphData.pL Chapter4/ScPlusGraph · GraphData.qL. The identities the text uses hold for this p on every level: $(1-p)Dt(0) = 0$, because the range of $Dt(0)$ is N Chapter4/ScPlusGraph · GraphData.qL_lvlDeriv_zero, and $Dt(0) = 1$ on N Chapter4/ScPlusGraph · GraphData.lvlDeriv_zero_of_mem_N. The good complement has a constant ε: for $n \in N$, $y \in Y$, $|y|_0 \le \varepsilon|n|_0$, $n \in C \iff n + y \in C$ Chapter4/ScPlusGraph · GraphData.eps Chapter4/ScPlusGraph · GraphData.eps_spec. Its level-1 version is $\varepsilon' = \varepsilon/(\|\iota_1\|\|\mathrm{lift}_N\| + 1)$ Chapter4/ScPlusGraph · GraphData.eps': $|y|_1 \le \varepsilon'|a|_1$ implies $|y|_0 \le \varepsilon|a|_0$ for $a \in N$

Chapter4/ScPlusGraph · GraphData.norm_toBase_le_eps_mul.

The level data of an sc⁺-retraction. The lift $t^1 : U \to E^1$ Chapter4/ScPlusGraph · ScRetraction.liftOne is sc-smooth Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isScSmooth_liftOne, sc¹ with derivative family Dt^1 Chapter4/ScPlusGraph · ScRetraction.liftDeriv Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isSc1With_liftOne. By Proposition 1.2.1 (preprint 1.16) for t^1, the map $t : U_{k+1} \to E_{k+1}$ Chapter4/ScPlusGraph · ScRetraction.lvlMap is differentiable within U_{k+1} with a derivative $Dt(x) \in \mathcal{L}(E_{k+1})$ Chapter4/ScPlusGraph · ScRetraction.lvlDeriv Chapter4/ScPlusGraph · ScRetraction.IsScPlus.hasFDerivWithinAt_lvlMap that depends continuously on x in the operator norm Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlDeriv; t is continuous on U_{k+1} Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlMap, fixes O_{k+1} Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlMap_eq_self, is t itself through the inclusions Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlMap, and its level derivatives are restrictions of Dt Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_liftDeriv Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlDeriv, projections at points of O Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlDeriv_lvlDeriv.

The mean value estimate (the text's (76)–(77)). The abstract form Chapter4/ScPlusGraph · exists_norm_sub_sub_le_of_hasFDerivWithinAt: if s is differentiable within a convex set S with derivative D continuous at $u_0 \in S$, then for every $\theta > 0$ there is $r > 0$ with $|s(z')-s(z)-D(u_0)(z'-z)| \le \theta|z'-z|$ for $z, z' \in S \cap B(u_0, r)$ — the mean value inequality for $s - D(u_0)$. For an sc⁺-retraction Chapter4/ScPlusGraph · ScRetraction.IsScPlus.exists_mvt: at $u_0 \in U_{k+1}$ a ball meets C_{k+1} inside U_{k+1} Chapter4/ScPlusGraph · ScRetraction.exists_ball_inter_subset Chapter4/ScPlusGraph · ScRetraction.lvlSet_U_subset_lvlSet_C, the set $C_{k+1} \cap B(u_0, r_0)$ is convex, and the abstract form and the continuity of Dt give r with $C_{k+1} \cap B(u_0, r) \subseteq U_{k+1}$, $\|Dt(z) - Dt(u_0)\| \le \theta$ and the estimate with constant θ there. The text integrates along segments in U_{k+1}, which need not be convex; the convex set $C_{k+1} \cap B$ is the repair.

A shortcut does not work: inverting $p \circ t$ on the finite-dimensional quadrant $N \cap C$ by Proposition 3.8.17 (preprint 3.73). In $\mathbb{R}^3$ with $C = [0,\infty)^2 \times \mathbb{R}$, $\Gamma(a) = (a_1, a_2, a_2^2)$ and $t(x) = \Gamma(x_1 + (x_3 - x_2^2)^2, x_2)$, t is a retraction with $N = \mathbb{R}^2 \times 0$, but $p\,t(x_1, x_2, 0) = (x_1 + x_2^4, x_2)$ misses $(0, y_2)$ for $y_2 > 0$.

IV.7 Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12): uniqueness and existence on level 1

Lemma 4.1.12 (preprint 4.11), the contraction estimate Chapter4/ScPlusGraph · GraphData.exists_contraction_radius. For $\theta > 0$ the mean value estimate at $u_0 = 0$ gives $r > 0$ with the estimates of §IV.6 on $C_{k+1} \cap B(0, r)$ and, for $u, u' \in O_{k+1}$ with norms below r, $|(1-p)(u'-u)|_{k+1} \le \|1-p\|\theta|u'-u|_{k+1}$: since $u = t(u)$, $u' = t(u')$ and $(1-p)Dt(0) = 0$, $(1-p)(u'-u) = (1-p)[t(u') - t(u) - Dt(0)(u'-u)]$. **Uniqueness** Chapter4/ScPlusGraph · GraphData.eq_of_p_eq: if $p(u) = p(u')$ and $|(1-p)(u'-u)| \le c|u'-u|$ with $c < 1$, then $u' - u \in Y$, so $(1-p)(u'-u) = u'-u$ and $u' = u$. This is what the text's Lemma 4.1.12 (preprint 4.11) is used for; its statement, "$y = \delta(a)$", presupposes the δ of Theorem 3.5.11 (preprint Corollary 3.47).

Lemma 4.1.13 (preprint 4.12), existence by a contraction Chapter4/ScPlusGraph · GraphData.exists_graph_point. Suppose the estimates hold on $C_1 \cap B(0, r)$ with $\theta < 1$, $\|1-p\|\theta < 1$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$. As $t(0) = 0$ and $t : U_1 \to E_1$ is continuous at 0, there is δ with $|t(u)|_1 < r$ for $|u|_1 < \delta$; put $r' = \min(r, \delta)$ and $\rho = r'/(1+\varepsilon')$. For $a \in N \cap C$ with $|a|_1 < \rho$:

1. The set $B_a = \{y \in Y_1 : |y|_1 \le \varepsilon'|a|_1\}$ is closed, hence complete, because Y is closed — the first use of that hypothesis.
2. For $y \in B_a$, $a + y \in C_1$ by the good complement property, $|y|_0 \le \varepsilon|a|_0$ coming from $|y|_1 \le \varepsilon'|a|_1$; and $|a+y|_1 \le (1+\varepsilon')|a|_1 < r'$. So $a + y$ lies in U_1, where t is defined — a point the text does not check.
3. $\Phi_a(y) = (1-p)t(a+y)$ maps B_a into itself: $\Phi_a(y) = (1-p)[t(a+y) - t(0) - Dt(0)(a+y)]$, of norm at most $\|1-p\|\theta(1+\varepsilon')|a|_1 \le \varepsilon'|a|_1$, and it lies in Y.
4. Φ_a is a contraction with constant $\|1-p\|\theta < 1$, by the estimate between $a+y$ and $a+y'$.

5. Its fixed point y (Banach's theorem on the complete B_a) gives $u = a + y$ with $(1-p)(t(u)-u) = 0$, i.e. $w - u \in N$ for $w = t(u)$. Now $w \in O$, $|w|_1 < r$ by the choice of δ, $t(w) = w$, and $Dt(0)(w-u) = w-u$. The estimate between u and w reads $|t(w) - t(u) - Dt(0)(w-u)|_1 = |w-u|_1 \le \theta|w-u|_1$, so $w = u$: $u = t(u) \in O$ — the text's argument "$a_1 = a$".

So $u \in O_1$, $p(u) = a$, $u - a \in Y$, $|u-a|_1 \le \varepsilon'|a|_1$ and $|u|_1 < r$. The text's "$|a(\tau)|_1 \le \varepsilon_1$ if $|a|_0 < \varepsilon$" is the level-1 continuity of t at 0 used in step 5.

IV.8 The step $p(O \cap W) \subseteq C$, and the neighbourhoods

p maps points of O near 0 into C `Chapter4/ScPlusGraph · GraphData.p_mem_C`, which the conclusion "$p : U_0 \to V \subseteq T_0O \cap C$" needs and the text never checks. If $u \in O_1$ has $|(1-p)u|_1 \le \kappa|u|_1$ with $\kappa \le \frac{1}{2}$, then $|u|_1 \le |pu|_1 + \frac{1}{2}|u|_1$, so $|u|_1 \le 2|pu|_1 \le 2\|\mathrm{lift}_N\||pu|_0$, and $|(1-p)u|_0 \le \|\iota_1\|\kappa|u|_1 \le 2\|\iota_1\|\kappa\|\mathrm{lift}_N\||pu|_0$; if that constant is at most ε, the good complement property gives $p(u) \in C \iff p(u) + (1-p)u = u \in C$, and $u \in O \subseteq C$. The hypothesis on u is the contraction estimate between 0 and u.

The neighbourhoods `Chapter4/ScPlusGraph · GraphData.Nbhd` `Chapter4/ScPlusGraph · GraphData.exists_nbhd`. For every $\theta_0 > 0$ the proof chooses, in this order:

1. $\theta = \min(\theta_0, \theta_1, \theta_2)$ with $\theta_1 = \varepsilon'/(2(\|1-p\|+1)(1+\varepsilon'))$ (so that $\|1-p\|\theta \le \frac{1}{2}$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$, for Lemmas 4.1.12 and 4.1.13 (preprint 4.11 and 4.12)) and $\theta_2 = \varepsilon/(2\|\iota_1\|\|\mathrm{lift}_N\|(\|1-p\|+1)+1)$ (for $p(O \cap W) \subseteq C$);
2. r from the contraction estimate on level 1, and ρ and the graph points $u(a)$ from Lemma 4.1.13 (preprint 4.12);
3. δ_0 from the continuity of the lift $t^1 : U \to E_1$ at 0 on level 0: $|t(u)|_1 < r$ for $u \in U$, $|u|_0 < \delta_0$;
4. $\rho' = \min(\rho, r/(1+\varepsilon'), \delta_0/(\|\iota_1\|(1+\varepsilon')+1))$, the open $V' = \{a \in N : |a|_1 < \rho'\}$ of N, $V = V' \cap C$, and $W = B_0(0,\delta_0) \cap p^{-1}(V')$, open in E_0;
5. $\Gamma(a) = u(a)$ for $a \in V$.

Then $\Gamma(a) \in O \cap W$ (on level 0, $|\Gamma(a)|_0 \le \|\iota_1\|(1+\varepsilon')|a|_1 < \delta_0$), $p\Gamma(a) = a$, $\Gamma(a) - a \in Y$, $|\Gamma(a)|_1 < r$. A point $u_0 \in O \cap W$ satisfies $u_0 = t(u_0)$, hence $|u_0|_1 < r$ by step 3 — *this is how a neighbourhood open on level* 0 *is controlled on level* 1, the sc$^+$ property a second time; the contraction estimate between 0 and u_0 and the first constraint on θ then give $p(u_0) \in C$, so $p(u_0) \in V$, and uniqueness gives $\Gamma(p(u_0)) = u_0$. Also $\|Dt(z) - Dt(0)\|_{\mathcal{L}(E_1)} \le \theta_0$ on $C_1 \cap B(0,r)$, and $|(1-p)(u'-u)|_1 \le \frac{1}{2}|u'-u|_1$ on $O_1 \cap B_1(0,r)$. Consequently $\Gamma(V) = O \cap W$ `Chapter4/ScPlusGraph · GraphData.Nbhd.image_V`, $\Gamma(0) = 0$ `Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_zero`, and p is injective on $O \cap W$ `Chapter4/ScPlusGraph · GraphData.Nbhd.injOn_p`.

IV.9 The regularity of Γ on the fixed V: the frame formula and the bootstrap

Continuity into every level. $\Gamma(V) \subseteq O \subseteq E_\infty$ `Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_mem_smooth`, so Γ has lifts $\Gamma_m : V \to E_m$ `Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl` with values in O_m `Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_mem_retract`. Γ_1 is Lipschitz: with $u = \Gamma(a)$, $u' = \Gamma(a')$, $|u'-u|_1 \le |p(u'-u)|_1 + |(1-p)(u'-u)|_1 \le \|\mathrm{lift}_N\||a'-a| + \frac{1}{2}|u'-u|_1$, so $|u'-u|_1 \le 2\|\mathrm{lift}_N\||a'-a|$ `Chapter4/ScPlusGraph · GraphData.Nbhd.norm_Γ_lvl_one_sub_le` `Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl_one`. And $\Gamma_{k+1} = t^1 \circ \Gamma_k$ on level k, because $\Gamma(a) = t(\Gamma(a))$ `Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_succ`; the lift t^1 is continuous from U_k into E_{k+1}, so by induction Γ is continuous into every level `Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl` `Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ`.

The frame at a point $u \in O$ near 0. Let $A(u) = p \circ Dt(u)|_N : N \to N$ `Chapter4/ScPlusGraph · GraphData.A`, $M(u) = A(u)^{-1}$ `Chapter4/ScPlusGraph · GraphData.M` and $B_u = Dt(u) \circ M(u)$, the *frame* `Chapter4/`

ScPlusGraph · GraphData.frame, with its level-$(k+1)$ version Chapter4/ScPlusGraph · GraphData.frameL. For $n \in N$, $Dt(u)n - n = \iota_1(Dt_1(u) - Dt_1(0))\mathrm{lift}(n)$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_eq. Choose θ_0 with $\|p\|\|\iota_1\|\|\mathrm{lift}_N\|\theta_0 \le \frac{1}{2}$ and $\|1-p\|\theta_0 < 1$ Chapter4/ScPlusGraph · GraphData.FrameSmall Chapter4/ScPlusGraph · GraphData.exists_frameSmall, and let $\|Dt_1(u) - Dt_1(0)\| \le \theta_0$. Then (L1) $|A(u)n - n| \le \frac{1}{2}|n|$, so $A(u)$ is injective Chapter4/ScPlusGraph · GraphData.A_injective, hence an isomorphism of the finite-dimensional N Chapter4/ScPlusGraph · GraphData.Aeq, with $A(u)M(u) = 1$ Chapter4/ScPlusGraph · GraphData.A_M and $p \circ B_u = 1$ Chapter4/ScPlusGraph · GraphData.p_frame. (L2) $T_uO \cap Y = 0$: a vector $v = Dt(u)v$ in Y satisfies $v = (1-p)v = (1-p)(Dt(u) - Dt(0))v$ on level 1, so $|v|_1 \le \|1-p\|\theta_0|v|_1$ Chapter4/ScPlusGraph · GraphData.eq_zero_of_mem_Y_of_scDeriv_eq. Hence for $u \in O_1$: $Dt(u)x - B_u(p\,Dt(u)x)$ lies in $T_uO \cap Y$ and vanishes, $(1 - B_up)Dt(u) = 0$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_frame; $Dt(u)B_u = B_u$ Chapter4/ScPlusGraph · GraphData.scDeriv_frame; and both hold on every level Chapter4/ScPlusGraph · GraphData.p_toBase_frameL Chapter4/ScPlusGraph · GraphData.lvlDeriv_sub_frameL. On the points of $\Gamma(V)$ the derivative bound holds by the choice of r Chapter4/ScPlusGraph · GraphData.Nbhd.deriv_bound_Γ.

The abstract step. On a set S of a normed space, let s satisfy the mean value estimate with derivative D_0 at u_0 and constant θ, let q, B be continuous linear with $q \circ B = 1$ and $(1-Bq)D_0 = 0$, and $\theta\|1-Bq\| \le \frac{1}{2}$. For fixed points u, u' of s in S, $(1-Bq)(u'-u) = (1-Bq)[s(u') - s(u) - D_0(u'-u)]$, of norm at most $\frac{1}{2}|u'-u|$, so $|u'-u| \le 2\|B\||q(u'-u)|$ Chapter4/ScPlusGraph · norm_sub_le_of_projection. And *every continuous right inverse Γ of q with values in the fixed points of s in S, through u_0, is differentiable at $z_0 = q(u_0)$ with derivative B* Chapter4/ScPlusGraph · hasFDerivWithinAt_of_projection: $\Gamma(z) - \Gamma(z_0) - B(z-z_0) = (1 - Bq)(\Gamma(z) - u_0) = (1-Bq)[(\Gamma(z) - u_0) - D_0(\Gamma(z) - u_0)]$, which is $o(|\Gamma(z) - u_0|)$ by the differentiability of s at u_0, and $|\Gamma(z) - u_0| \le 2\|B\||z - z_0|$ by the Lipschitz estimate.

The frame formula Chapter4/ScPlusGraph · GraphData.Nbhd.hasFDerivWithinAt_Γ_lvl. At $a_0 \in V$ and on level $k+1$, apply the abstract step with $s = t$, $u_0 = \Gamma_{k+1}(a_0)$, $q = p \circ \iota_{k+1}$, B the level-$(k+1)$ frame at u_0, $\theta = 1/(2(\|1-Bq\|+1))$, and $S = C_{k+1} \cap B(u_0, r)$ from the mean value estimate *at* u_0; Γ_{k+1} stays in $B(u_0, r)$ near a_0 by continuity. So

$$D\Gamma(a) = Dt(\Gamma(a))|_N \circ \left(p \circ Dt(\Gamma(a))|_N\right)^{-1} \quad \text{into every level } E_{k+1}.$$

The estimate is used at $\Gamma(a)$ with the projection $1 - B_{\Gamma(a)}p$, not at 0 with $1-p$: this is why the regularity holds at every point of the fixed V.

The C^k-bootstrap. By Proposition 1.2.3 (preprint 1.18) for the tangent map of the lift t^1, $(x,w) \mapsto Dt_{j+1}(x)w$ is C^k on $U_{j+k+1} \times E_{j+k+1}$, read through the inclusions into level $j+1$ Chapter4/ScPlusGraph · GraphData.contDiffOn_lvlDeriv_apply. If Γ_{j+k+1} and $w : V \to N$ are C^k, then so is $a \mapsto Dt_j(\Gamma_{j+1}(a))\,\mathrm{lift}(w(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_lvlDeriv_Γ; in particular $a \mapsto A(\Gamma(a))$ is C^k when Γ_{k+1} is Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_A_Γ, and so is its inverse, inversion being smooth on the invertible operators Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_M_Γ. Now induct on k, for all levels at once Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl: Γ_{j+1} is C^0 (continuity); if every $\Gamma_{j'+1}$ is C^k, the derivative $a \mapsto \mathrm{frame}_j(\Gamma_{j+1}(a))$ applied to n is $Dt_j(\Gamma_{j+1}(a))\mathrm{lift}(M(\Gamma_1(a))n)$, which is C^k because Γ_{j+k+1} is (the hypothesis at $j' = j+k$) and $M(\Gamma_1(\cdot))n$ is (the hypothesis at $j' = k$, through A); so Γ_{j+1} is C^{k+1}. Derivatives of t are only ever applied to the smooth vectors of the finite-dimensional N, so the loss of levels in Tt is absorbed by Γ being C^k into *every* level and does not accumulate. Hence Γ is C^∞ into every level Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_succ_infty Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_infty and sc-smooth on V by the criterion of Proposition 1.2.4 (preprint 1.19) for a finite-dimensional source Chapter4/ScPlusGraph · GraphData.Nbhd.isScSmooth_Γ.

The sc-diffeomorphism. $(O \cap W, C, E)$ is a retract (Proposition 2.3.1 (1) (preprint 2.10)) Chapter2/Retracts · proposition_2_10_1 Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_inter_W; $(V, N \cap C, N)$ is a local model, $N \cap C$ being a partial quadrant of N (Proposition 3.1.25 (preprint 3.22)) and V relatively open in it Chapter4/ScPlusGraph · GraphData.Nbhd.V_isRelOpenInQuadrant Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_

V. p is sc-smooth $E \to N$ as a continuous linear map into a finite-dimensional space (Corollary 1.2.5 (preprint 1.20)) `Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional` `Chapter4/ScPlusGraph · GraphData.p_isScSmooth`, so p and Γ are sc-smooth between the retracts `Chapter4/ScPlusGraph · GraphData.Nbhd.p_isScSmoothRetractMap` `Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_isScSmoothRetractMap`.

Proposition 4.1.9 (preprint 4.9) `Chapter4/ScPlusGraph · proposition_4_9`. *Statement.* Let t be an sc⁺-retraction onto $O \ni 0$, $N = T_0O$ in good position to C, Y a closed good complement of N and $p : E_0 \to N$ the projection along Y. There are an open $W \ni 0$ in E_0, a relatively open $V \ni 0$ of $N \cap C$ and $\Gamma : N \to E_0$, sc-smooth on V, with $p \circ \Gamma = 1$ on V, $\Gamma(a) - a \in Y$, $\Gamma(V) = O \cap W$, $\Gamma \circ p = 1$ on $O \cap W$, $\Gamma(0) = 0$, and p, Γ sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *Proof.* θ_0 from the frame condition, the neighbourhoods of §IV.8 for it, and the regularity above. The finite dimension of N, an instance argument needed to form N as an sc-Banach space, is Lemma 4.1.7 (preprint 4.7) `Chapter4/ScPlusGraph · finiteDimensional_tangentSpace_zero`.

IV.10 Theorem 4.1.2 (preprint 4.2): the characterization

Statement `Chapter4/Characterization · MPolyfold.theorem_4_2_general`. For every M-polyfold X and every junk value $x_0 \in X_1$ of 1^1_0, the following are equivalent. (i) X has a corner structure. (ii) Every point of X is smooth, $1^1_0 : X \to X^1$ is sc-smooth, and every x lies in a chart ψ of X with $\psi(x) = 0$ whose model is a tame retract and whose tangent space $N = T_0O$ is finite-dimensional and in good position to C. The text's form, for tame X, is the same statement with an unused hypothesis `Chapter4/Characterization · MPolyfold.theorem_4_2`. Tameness is used neither in the text's proof nor here, and (i) implies it (§IV.1).

Charts composed with an sc-diffeomorphism between retracts. The proof of (ii) □ (i) composes a chart with p, which is an sc-diffeomorphism between retracts, not between relatively open sets, so the construction of Chapter 3 does not apply. For a chart φ onto (O, C, E) and mutually inverse f, g sc-smooth between (O, C, E) and (O', C', E'), $f \circ \varphi$ is a chart onto (O', C', E') `Chapter4/Characterization · MChart.compRetract`, and a chart of the M-polyfold when φ is `Chapter4/Characterization · MChart.compRetract_isChart`: for a chart φ' of X the transition maps are $(\varphi' \circ \varphi^{-1}) \circ g$ and $f \circ (\varphi \circ \varphi'^{-1})$, compositions of maps sc-smooth between retracts (Theorem 2.1.15 (preprint 2.8)) `Chapter2/Retracts · IsScSmoothRetractMap.comp`. This is the compatibility the text's "we obtain a smoothly compatible atlas" skips (Findings, item 55).

(i) □ (ii). Corner charts are modelled on sc⁺-retracts, so every point is smooth and 1^1_0 is sc-smooth by Proposition 4.1.8 (preprint 4.8) on a cover (§IV.4). The text stops here; (ii) also asks for a centred tame chart with T_0O finite-dimensional and in good position (Findings, item 53). Let x lie in a corner chart χ. Its image is smooth (§IV.1), so the re-centred chart $\psi = \chi - \chi(x)$ exists; its model is a relatively open subset of $C_{\chi(x)}$ in the same finite-dimensional space `Chapter4/Characterization · image_sub_const_eq_preimage_add'` `Chapter4/Characterization · MChart.IsCornerChart.exists_centred`. With the identity retraction it is a tame retract; the tangent space of a relatively open subset of a partial quadrant is the whole space, because the derivative of the identity retraction is the identity `Chapter4/Characterization · tangentSpace_relOpen`; and the whole space is in good position to every partial quadrant, with the complement 0, the interior of a partial quadrant being nonempty `Chapter4/Characterization · isInGoodPosition_top` `Chapter4/Characterization · MChart.IsCornerChart.exists_centred_tame`.

(ii) □ (i). *The data of the corner chart* `Chapter4/Characterization · CornerChartData` for a chart ψ and a finite-dimensional N: a closed good complement Y, the projection p along it, W, V, Γ with p, Γ mutually inverse and sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *They exist* when the model of ψ is an sc⁺-retract containing 0 and $N = T_0O$ is in good position `Chapter4/Characterization · exists_cornerChartData`: here no complement is prescribed, so one can be chosen — the normal form of Proposition 3.1.25 (preprint 3.22) gives a continuous projection onto N whose kernel is a good complement, closed, even mapping C into C `Chapter3/GoodComplementProjection · exists_proj_goodComplement` — and Proposition 4.1.9 (preprint 4.9) is applied to an sc⁺-retraction onto O (Definition 4.1.6 (preprint 4.6)). The issue "$p = Dt(0)$" of §IV.6 does not arise, and nothing is added to Theorem 4.1.2 (preprint

4.2) (Findings, item 54). *The chart* $p \circ \psi$ on $\psi^{-1}(W)$ Chapter4/Characterization · CornerChartData.chart is ψ restricted to $\psi^{-1}(W)$ composed with p by `compRetract`, modelled on $(V, N \cap C, N)$ Chapter4/Characterization · CornerChartData.model; it is a corner chart Chapter4/Characterization · CornerChartData.chart_isCornerChart, a chart of X Chapter4/Characterization · CornerChartData.chart_isChart, contains x when $\psi(x) = 0$ Chapter4/Characterization · CornerChartData.mem_chart_source and sends it to 0 Chapter4/Characterization · CornerChartData.chart_apply_eq_zero. *At every point* Chapter4/Characterization · MPolyfold.theorem_4_2_chart: if all points are smooth and 1^1_0 is sc-smooth, the model of every chart is an sc⁺-retract (§IV.4), so for a chart ψ with $\psi(x) = 0$ and $N = T_0O$ finite-dimensional and in good position the corner chart $p \circ \psi$ exists. With such a chart at every point, X has a corner structure by the local form of Definition 4.1.1 (preprint 4.1) Chapter4/Characterization · MPolyfold.hasCornerStructure_of_centred — whether or not the charts ψ are tame.

IV.11 Smooth finite-dimensional submanifolds; the chart retraction; Remark 4.2.2 (preprint 4.14)

Definition 4.2.1 (preprint 4.13). $A \subseteq X$ is a *smooth finite-dimensional submanifold* if every $a \in A$ has an open $V \ni a$ and an sc⁺-retraction s on V with $s(V) = A \cap V$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold; the data at a point are a structure Chapter4/Submanifolds · MPolyfold.SubmanifoldData Chapter4/Submanifolds · MPolyfold.isSmoothSubmanifold_iff. An sc⁺-retraction is an sc-smooth retraction, so the data are those of Definition 2.3.18 (preprint 2.20) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.toSubData, A is a sub-M-polyfold Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold and carries the induced structure of Proposition 2.3.19 (preprint 2.21) Chapter2/SubMPolyfold · MPolyfold.subPolyfold. Its points lie on level 1, as $a = s(a) \in X_1$ Chapter4/Submanifolds · MPolyfold.SubmanifoldData.mem_levelSet_one.

The sc⁺ condition read in a chart (Lemma 6.A of the exposition). The text never reads the sc⁺ condition in charts, and all of §§4.2–4.3 rests on it (Findings, item 60). Let φ be a chart of X and $W = V \cap \operatorname{dom}\varphi \cap s^{-1}(V \cap \operatorname{dom}\varphi)$, the open set of the proof of Proposition 2.3.19 (preprint 2.21) Chapter2/SubMPolyfold · MPolyfold.SubData.shrink. (a) The lift of $\varphi \circ s \circ \varphi^{-1}$ into E^1 is sc-smooth from $(\varphi(W), C, E)$ to (O_1, C_1, E^1) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScSmoothRetractMap_lift: it is the sc-smoothness of $1^1_0 \circ s$ read in the charts φ of X and `levelChart` φ of X^1, where 1^1_0 is the lift. (b) For every retraction r onto O: $\varphi(W) = O \cap G$ for an open G; $\rho = \varphi \circ s \circ \varphi^{-1}$ is sc-smooth on the retract $\varphi(W)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth and idempotent there Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem, with image $\varphi(W \cap A)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image; so $t = \rho \circ r$ on $U \cap r^{-1}(G)$ — the restriction of r Chapter2/Retracts · ScRetraction.restrict composed with ρ Chapter2/Metrizable · ScRetraction.compRetraction — is a retraction onto $\varphi(W \cap A)$, and it is sc⁺: its values $\varphi(s(x))$ lie in E_1 because $s(x) \in X_1$, and its lift is the map of (a) composed with r Chapter4/Submanifolds · MPolyfold.SubmanifoldData.exists_chartRetraction. This is the text's $t = \varphi \circ s \circ \varphi^{-1} \circ r$ of the proof of Theorem 4.2.7 (preprint 4.18), with W in place of "$V = \varphi^{-1}(O)$". Hence the model $\varphi(W \cap A)$ of the chart of A induced by φ is an sc⁺-retract Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScPlusRetract_image, and $\varphi(a)$ is smooth for $a \in A \cap V \cap \operatorname{dom}\varphi$ (Lemma 4.1.7 (i) (preprint 4.7)) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.apply_mem_smooth.

Tangent spaces of sub-retracts. The tangent space of a retract is a germ: for an open K, $T_z(Q \cap K) = T_zQ$ Chapter4/Submanifolds · tangentSpace_inter_isOpen. For retracts $Q \subseteq O = r(U)$, $T_zQ \subseteq T_zO$: a retraction s onto Q satisfies $s = r \circ s$, so $Ds(z) = Dr(z)Ds(z)$ Chapter4/Submanifolds · tangentSpace_subset_of_subset. And if $F \circ r$ is sc-smooth and F maps the sub-retract Q into a retract Q', then $D(F \circ r)(z)$ maps T_zQ into $T_{F(z)}Q'$ (Proposition 2.1.13 (3) (preprint 2.6) for $F|_Q$) Chapter4/Submanifolds · scDeriv_mem_tangentSpace_of_subset.

Remark 4.2.2 (preprint 4.14) in a chart Chapter4/Submanifolds · MPolyfold.SubmanifoldData.remark_4_14. The text: $T_aA = Ts(a)(T_aX)$ is the image of the sc⁺-operator $Ts(a)$, hence finite-dimensional. In a chart, at $a \in A \cap V \cap \operatorname{dom}\varphi$ and the level-1 point z over $\varphi(a)$: z is smooth (Lemma 4.1.7 (i) (preprint 4.7) for the chart retraction t); the tangent map of ρ at z maps $T_zO = T_z(\varphi(W))$ onto $T_z(\varphi(W \cap A))$ (Proposition

2.3.19 (4) (preprint 2.21) for ρ) Chapter2/Metrizable · proposition_2_21_4; $T_z(\varphi(W \cap A)) \subseteq T_z O$; and it is the range of $Dt(z)$, finite-dimensional by Lemma 4.1.7 (ii) (preprint 4.7). The sc⁺-operator is $Dt(z)$, not $Ts(a)$.

Points with $d_X = 0$. The remark adds that near points with $d_X(a) = 0$, A is a classical manifold. Every subspace is in good position to the whole space, any algebraic complement being good Chapter4/Submanifolds · isInGoodPosition_univ, and at a point with $d_C(e) = 0$ the partial quadrant C_e is the whole space Chapter4/Submanifolds · partialQuadrantAtWith_eq_univ_of_degIndex_eq_zero. So at $a \in A$ with $d_X(a) = 0$, in a chart realizing $d_X(a) = 0$ Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq, A is in good position (Definition 4.2.5 (preprint 4.16), §IV.13) Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_of_degIndex_eq_zero; and if $d_X = 0$ on all of A, re-centring at each point gives charts modelled in $C_{\varphi(a)} = E$, in which $T_0(\psi(W \cap A))$ is in good position, so the induced structure has a corner structure by the core of Theorem 4.2.7 (preprint 4.18) (§IV.14) — without tameness of X Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_degIndex_eq_zero. The local statement for the open piece $A \cap \{d_X = 0\}$ of an arbitrary A is not formalized.

The open question after Proposition 4.2.3 (preprint 4.15) — is a sub-M-polyfold with $A = A_\infty$ and finite, locally constant $\dim T_a A$ a smooth finite-dimensional submanifold; is the "T" $\{(x, 0) : |x| < 1\} \cup \{(0, y) : 0 \le y < 1\}$ homeomorphic to a retract $O = O_\infty$ — is a question and is not formalized. Its last sentence, that this is impossible for sc⁺-retracts, is for a retract (O, E, E) without boundary the statement just proved with $A = O$: an sc⁺-retract in E is a manifold; that the "T" is not a topological manifold is not formalized.

IV.12 Proposition 4.2.3 (preprint 4.15)

(1) $A \subseteq X_\infty$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.subset_smoothSet: in a chart at a, $\varphi(a)$ lies in the retract of the sc⁺ chart retraction (Lemma 4.1.7 (i) (preprint 4.7)). The text does not distinguish this from "$A = A_\infty$ for the induced structure", which holds because the levels of the induced structure are the traces of those of X, computed in the induced charts Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.mem_levelSet_subPolyfold_iff Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_1.

(2) *A is locally closed* Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_2. The text's definition is garbled — "a point $b \in V(a)$ belongs to A if $U \cap A \neq \emptyset$ for all open neighbourhoods U of b", with U also the letter of the domain — and is read as $V(a) \cap \overline{A} \subseteq A$, one of the equivalent forms of Mathlib's `IsLocallyClosed` (the intersection of an open and a closed set). Proof: for $b \in V \cap \overline{A}$, b is in the closure of $A \cap V$ (V open), so the filter of neighbourhoods of b within $A \cap V$ is nontrivial; s is continuous at b and equal to the identity on $A \cap V$, so $s(b)$ and b are both limits of the identity along that filter, and $s(b) = b$ (X Hausdorff); hence $b \in s(V) \subseteq A$. A filter replaces the text's sequence $a_k \to b$, so no metrizability is needed.

(3) $\partial A = \{d_A \ge 1\} \subseteq \partial X$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_3, by $d_A \le d_X$ (Lemma 2.4.6 (preprint 2.26)) Chapter2/SubMPolyfold · lemma_2_26. The text's argument for $d_X(a) = 0$ is the proof of that lemma; it needs the chart re-centred at a, since only $C_{\varphi(a)}$, not C, is the whole space.

(4) $1^1_0 : A \to A^1$ *is sc-smooth* Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_4. The text says only that "the postulated sc⁺-retraction is the identity on the retract A". The proof: every point of A lies in an induced chart Chapter2/SubMPolyfold · MPolyfold.subChart whose model $\varphi(W \cap A)$ is an sc⁺-retract (§IV.11), and Proposition 4.1.8 (preprint 4.8) on a cover applies (§IV.4).

IV.13 Good and general position in charts, and their chart independence (Definitions 4.2.5, 4.2.6, 4.2.10 (preprint 4.16, 4.17, 4.21))

Definition 4.2.5 (preprint 4.16) in a chart. For retracts $Q \subseteq O$, a presentation T and a level-1 point z: T_zQ is *in good position to C_z inside T_zO* `Chapter4/Submanifolds · IsGoodPositionRetractAt` — in the sense of Definition 3.1.24 (preprint 3.21) inside a subspace `Chapter3/GoodPositionGerm · IsInGoodPositionIn`: $T_zQ \subseteq T_zO$, $T_zQ \cap C_z$ has interior points in T_zQ, and T_zQ has a complement inside T_zO with a constant ε. The condition depends only on the two tangent spaces `Chapter4/Submanifolds · IsGoodPositionRetractAt.congr`. For the data d of Definition 4.2.1 (preprint 4.13), a chart φ and $Q = \varphi(W \cap A)$ it is the chart condition `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGoodPositionChart`, and A *is in good position at a* if it holds for some data at a, some chart at a, some presentation and the level-1 point over $\varphi(a)$ `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt`. Explicitly, for retractions r onto O and t onto $\varphi(W \cap A)$, the condition says that $Dt(z)E$ is in good position to C_z inside $Dr(z)E$ `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isGoodPositionChart_iff`. The text states Definition 4.2.5 (preprint 4.16) for tame X so that C_aX is a partial quadrant (Proposition 2.5.6 (preprint 2.36)); the chart form does not need it.

Definition 4.2.10 (preprint 4.21) in a chart. *T_zQ has in T_zO an sc-complement contained in T^R_zO* `Chapter4/Submanifolds · IsGeneralPositionRetractAt`: an sc-subspace $B \subseteq T_zO \cap E_z$ of E with $(T_zO)_m = (T_zQ)_m \oplus B_m$ on every level, the sc-structure of T_zO being the one induced from E; again it depends only on the tangent spaces `Chapter4/Submanifolds · IsGeneralPositionRetractAt.congr`. The chart condition `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGeneralPositionChart` and *general position at a* `Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt` are defined as for good position. **Definition 4.2.6 (preprint 4.17)**: A is *tame* if its induced structure is `Chapter4/Submanifolds · MPolyfold.IsTameSubmanifold`.

Transport by a germ of sc-diffeomorphisms. Let F be a germ of sc-diffeomorphisms between retracts $O = r(U)$ and $O' = r'(U')$ with inverse G `Chapter2/Boundary · IsDiffeoGerm`, mapping sub-retracts Q, Q' into each other. At $z \in Q_1$, $\mathcal{A} = D(F \circ r)(z)$ restricts to an isomorphism $T_zO \cong T_{F(z)}O'$ with inverse $D(G \circ r')(F(z))$, carrying the partial cones onto each other (Proposition 2.4.10 (preprint 2.30)) `Chapter2/Boundary · proposition_2_30_cone` and T_zQ onto $T_{F(z)}Q'$ (§IV.11 in both directions). Good position inside a subspace is transported by such an isomorphism (Lemma 3.5.9 (preprint 3.45)) `Chapter3/LinearizationCharts · IsInGoodPositionIn.transport`, so *good position of Q at z gives good position of Q' at $F(z)$* `Chapter4/Submanifolds · IsGoodPositionRetractAt.transport`. For general position z must be smooth: then $\mathcal{A}$ is an sc-operator, it carries T^R_zO onto $T^R_{F(z)}O'$ (Proposition 2.4.10 (preprint 2.30)) `Chapter2/Boundary · proposition_2_30_reduced`, and the image of the sc-complement B is an sc-complement inside $T^R_{F(z)}O'$, level by level `Chapter4/Submanifolds · IsGeneralPositionRetractAt.transport`.

The chart change `Chapter4/Submanifolds · MPolyfold.chart_change_of_transport`. Let Φ be a property of $(Q \subseteq O, T, z)$ that depends only on the tangent spaces and is carried by germs of sc-diffeomorphisms at smooth points. If Φ holds for $\varphi(G \cap A) \subseteq O$ at z (smooth), it holds for $\varphi'(G' \cap A) \subseteq O'$ at z', for any presentations: on $H = G \cap G'$ the transition $\varphi' \circ \varphi^{-1}$ is a germ of sc-diffeomorphisms between the overlap retracts `Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm` carrying $\varphi(H \cap A)$ onto $\varphi'(H \cap A)$ `Chapter4/Submanifolds · MChart.image_inter_eq_inter`, and the tangent spaces at z are those of the open pieces (the germ property). Applied to the two properties `Chapter4/Submanifolds · MPolyfold.isGoodPositionRetractAt_chart_change` `Chapter4/Submanifolds · MPolyfold.isGeneralPositionRetractAt_chart_change`, with G, G' the open sets of two submanifold charts: *if A is in good position at a, the chart condition holds for every data, chart, presentation and lift* `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isGoodPositionChart`, and good position is equivalent to the condition for all of them `Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_iff_forall`; likewise for general position `Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart` `Chapter4/Submanifolds · MPolyfold.isGeneralPositionAt_iff_forall`, the smoothness of the point holding on A. The text takes this independence for granted (Findings, item 66).

IV.14 Lemma 4.2.8 (preprint 4.19) and Theorem 4.2.7 (preprint 4.18)

Lemma 4.2.8 (preprint 4.19), the complement kept explicit `Chapter4/Submanifolds · isGoodComplement_sup_range`. Let π be a bounded idempotent with $(1-\pi)E_0$ inside the minimal subspace at 0, $E_0^{\min} = \{\lambda = 0\}$, and $Y \subseteq \pi E_0$ a complement of N inside πE_0 with constant ε. Then $Y \oplus (1-\pi)E_0$ is a good complement of N in E_0 with constant $\varepsilon/(\|\pi\|+1)$: for $y \in Y$, $k \in (1-\pi)E_0$ with $|y+k| \le \frac{\varepsilon}{\|\pi\|+1}|n|$, $y = \pi(y+k)$, so $|y| \le \varepsilon|n|$ and $n \in C \iff n+y \in C$; and adding $k \in E_0^{\min}$ changes no coordinate, so $n+y \in C \iff n+y+k \in C$. The splitting: $N \oplus Y = \pi E_0$ and $\pi E_0 \oplus (1-\pi)E_0 = E_0$.
Lemma 4.2.8 (preprint 4.19) `Chapter4/Submanifolds · lemma_4_19`: *in the situation of the lemma, if N is in good position to C inside πE_0, then N is in good position to C in E_0, with good complements $Y \oplus (1-\pi)E_0$.* The good-position half is the transport lemma of the proof of Theorem 3.6.8 (preprint 3.54) `Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition` (§III.23).

The text's proof makes a false step. It concludes $n+y \in C$ from $n+y+z \in C$ "since $z \in Z \subset C$". This needs $-z \in C$: in $C = [0,\infty)$, $z = 1$ and $n+y = -\frac{1}{2}$ give $n+y+z \in C$, $n+y \notin C$. What holds, and what is used, is $Z \subseteq E_0^{\min} \subseteq C \cap (-C)$ — Proposition 2.5.3 (preprint 2.35) at the corner of the centred chart. The text's norm $|y|_0 + |z|_0$ on $Y \oplus Z$ is not the norm of E; the constant $\varepsilon/(\|\pi\|+1)$ compares them (Findings, item 69).

Good position read in a tame centred chart `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isInGoodPosition_of_tame`. If A is in good position at a and ψ is a chart of X with $\psi(a) = 0$ modelled on a tame retract $O = r(U)$, then $N = T_0(\psi(W \cap A))$ is in good position to C in E: by chart independence the chart condition holds in ψ, i.e. N is in good position to C_0 inside $T_0O = \operatorname{range} Dr(0)$; at the corner of the centred chart $C_0 = C$ `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero`; Proposition 2.5.3 (preprint 2.35) for the tame r at the smooth point 0 gives $(1-Dr(0))E_0 \subseteq E_0^{\min}$ `Chapter2/TameRetracts · proposition_2_35`; and $Dr(0)$ is idempotent. Lemma 4.2.8 (preprint 4.19) with $\pi = Dr(0)$ concludes. Tameness of X enters Theorem 4.2.7 (preprint 4.18) here and nowhere else.

The corner charts of A `Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_centred_isInGoodPosition`. If every $a \in A$ lies in a chart ψ of X with $\psi(a) = 0$ in which $T_0(\psi(W \cap A))$ is in good position to C in E, the induced structure has a corner structure. Proof (for nonempty A): every point of A is smooth and $1_0^1 : A \to A^1$ is sc-smooth (Proposition 4.2.3 (4) (preprint 4.15)); the chart of A induced by ψ is a chart of the induced structure, centred at a, modelled on $\psi(W \cap A)$; its tangent space at 0 is the range of $Dt(0)$ for the sc⁺ chart retraction, finite-dimensional (Lemma 4.1.7 (preprint 4.7)), and in good position by hypothesis; Theorem 4.1.2 (ii) (preprint 4.2) □ (i) for the M-polyfold A (§IV.10) concludes.

Theorem 4.2.7 (preprint 4.18) `Chapter4/Submanifolds · MPolyfold.theorem_4_18`. *Statement.* Let X be tame and A a smooth finite-dimensional submanifold in good position at every point. Then the induced structure on A has a corner structure, and A is tame. *Proof* (the text's route through Theorem 4.1.2 (preprint 4.2)): at $a \in A$ take a tame chart centred at a (§IV.5) — the "sc-smooth tame chart $\varphi : (V,a) \to (O,0)$" the text takes for granted; good position read in it (above) gives the hypothesis of the previous paragraph; and a corner structure makes A tame (§IV.1). What the text skips, besides the false step of Lemma 4.2.8 (preprint 4.19): the centred chart; the set W in place of "$V = \varphi^{-1}(O)$" (§IV.11); the sc⁺ condition in charts; the closed complement and the projection along it in place of "$p = Dt(0)$" (§IV.10); and the compatibility of the charts $p \circ \varphi$ (§IV.10). The text's "$V(a) = \varphi^{-1}(V_0) \cap A$" should be $(p\circ\varphi)^{-1}(V_0) \cap A$; its "Proposition 4.2" is Theorem 4.1.2 (preprint 4.2) (Findings, item 68). No hypothesis is added.

IV.15 Remark 4.2.9 (preprint 4.20): good position is an open condition

Statement `Chapter4/Submanifolds · MPolyfold.remark_4_20`. For tame X, if A is in good position at a, there is an open $U \ni a$ with A in good position at every point of $U \cap A$. The text omits "tame" (Definition 4.2.5 (preprint 4.16) is for tame X, and Lemma 4.2.8 (preprint 4.19) is used) and argues "since the tangent spaces of A move only slowly, Lemma 3.55 implies…". The formal proof makes this precise.

The tangent spaces along a graph Chapter4/Submanifolds · tangentSpace_eq_range_graph. If the retract Q is the image of $\Gamma : (V, N \cap C, N) \to (Q, C, E)$, sc-smooth between retracts with inverse p, then at $v \in V$ and the level-1 point y over $\Gamma(v)$: Γ is differentiable within V with derivative its sc-derivative Chapter4/Submanifolds · hasFDerivWithinAt_scDeriv_const, $T_y Q = D\Gamma(v)N$, and $p \circ D\Gamma(v) = 1$ — the tangent maps of the germ of sc-diffeomorphisms Γ, p.

Proof of Remark 4.2.9 (preprint 4.20). Take a tame chart ψ centred at a with tame retraction r, the sc$^+$ chart retraction t onto $\psi(W \cap A)$, and $N = T_0$, in good position to C in E (§IV.14). The normal form of Proposition 3.1.25 (preprint 3.22) gives a continuous projection π onto N with closed kernel a good complement Chapter3/GoodComplementProjection · exists_proj_goodComplement, and Proposition 4.1.9 (preprint 4.9) gives W_1, V, Γ over $N \cap C$. Write $\Gamma(v) = v + \sigma(v)$ with $\sigma(v) \in \ker \pi$. Then σ is C^1 on V, $\sigma(0) = 0$, and $D\sigma(0) = 0$: $D\Gamma(0)m$ lies in $T_0 = N$, so $D\Gamma(0)m = \pi D\Gamma(0)m = m$. The tangent space at $\Gamma(v)$ is the graph $\{n + D\sigma(v)n\}$ Chapter3/GoodPositionOpen · graphSpace. Lemma 3.6.9 (preprint 3.55), with the closed complement $\ker \pi$ Chapter3/GlobalIFTFull · lemma_3_55_relOpen, gives $\rho > 0$ such that for $|v| < \rho$ this graph is in good position to $C_{\Gamma(v)}$ in E. It lies in $T_{\Gamma(v)}O$ (Remark 4.2.2 (preprint 4.14) in the chart), and good position in E of a subspace of $T_{\Gamma(v)}O$ gives good position inside $T_{\Gamma(v)}O$, with the complement cut down by the modular law Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le. That is Definition 4.2.5 (preprint 4.16) at $b = \psi^{-1}(\Gamma(v))$, read in ψ. The neighbourhood is $U = W \cap \psi^{-1}(W_1 \cap \{|\pi u| < \rho\})$.

IV.16 Linear algebra at the corner of a centred chart; Lemmas 7.A and 7.B of the exposition

In a chart centred at 0, every coordinate vanishes at 0, so the minimal subspace there is $E_0^{\min} = \{\lambda = 0\}$, the text's "$\{0\} \oplus W$" Chapter4/LocalStructure · mem_minimalSubspaceWith_zero, and it lies in every E_x Chapter4/LocalStructure · minimalSubspaceWith_zero_le. Adding a vector of $E_0^{\min}$ changes no coordinate Chapter4/LocalStructure · coord_add_of_mem_zero, hence neither the degeneracy index Chapter4/LocalStructure · degIndexWith_add_of_mem_zero nor membership in C Chapter4/LocalStructure · add_mem_iff_of_mem_zero.

Closed complements. If N is finite-dimensional, K closed and $N + K$ the whole space, N has a closed complement $Y \subseteq K$: take a continuous projection q onto the finite-dimensional $N \cap K$ and $Y = K \cap \ker q$ Chapter4/LocalStructure · exists_isClosed_isCompl_le. The projection along a closed complement of a finite-dimensional subspace is continuous, with kernel exactly the complement Chapter4/LocalStructure · exists_proj_of_isCompl.

Good position from $N + E_0^{\min} = E_0$. Let N be finite-dimensional with $N + E_0^{\min} = E_0$. The coordinates restricted to N Chapter4/LocalStructure · coordsOn map N onto $\mathbb{R}^n$ Chapter4/LocalStructure · coordsOn_surjective, so $N \cap C$ has interior points, those with all coordinates positive Chapter4/LocalStructure · interior_nonempty_of_sup_eq_top. A complement inside $E_0^{\min}$ is good with every ε Chapter4/LocalStructure · isGoodComplement_of_le_zero. Hence N is in good position, with a closed good complement $Y \subseteq E_0^{\min}$ Chapter4/LocalStructure · exists_goodComplement_le_zero. **The degeneracy index on the slice** Chapter4/LocalStructure · degIndex_slice: $N \cap C$ is the partial quadrant of N presented by $v \mapsto (\lambda(v), v - s(\lambda(v)))$, s a linear section of $\lambda|_N$, and $d_{N \cap C}(v) = d_C(v)$ — both count the vanishing coordinates. This replaces the text's splitting $T_0 A = N \oplus (T_0 A \cap W)$, $E = N \oplus W$ (Findings, item 83).

Lemma 7.A: an sc-complement inside an sc-subspace Chapter4/LocalStructure · exists_scComplement_le. Let $N \subseteq E_\infty$ be finite-dimensional, R an sc-subspace and $M = N + R$. Then N has in M an sc-complement $B \subseteq R$: $K = N \cap R$ is finite-dimensional and smooth, so it has an sc-complement G in E (Proposition 1.1.7 (preprint 1.6)) Chapter1/ScBanach · ScBanach.proposition_1_6; put $B = R \cap G$. Then $R = K \oplus B$, $N + R = N + B$, $N \cap B \subseteq K \cap G = 0$, and $M_m = N \oplus B_m$ on every level because $N \subseteq E_\infty$. B is closed with $B_m = (1 - P)R_m$, P the sc-projection onto K, so the smooth points of B are dense in every level: for $x \in B_m$ and smooth $u \in R_m$ near x, $u - Pu \in B$ is smooth and near $x - Px = x$.

General position from $T_yQ + E_0^{\min} = E_0$ `Chapter4/LocalStructure · isGeneralPositionRetractAt_of_sup_eq_top`. Let r be tame, $Q \subseteq O$ a retract, $y \in Q_1$ smooth, and T_yQ finite-dimensional, smooth, with $T_yQ + E_0^{\min} = E_0$. As $E_0^{\min} \subseteq E_y$, intersecting with $T_yO \supseteq T_yQ$ gives $T_yO = T_yQ + (T_yO \cap E_0^{\min}) \subseteq T_yQ + T_y^RO$ (modular law). T_y^RO is an sc-subspace by Proposition 2.5.6 (preprint 2.36) for the tame r `Chapter2/PartialCone · proposition_2_36_partialCone`, and Lemma 7.A gives the sc-complement inside it. This is the step of Theorem 4.2.11 (3) (preprint 4.22) "implying that T_aA has in T_aO a sc-complement contained in T_a^RO" (Findings, item 85).

Lemma 7.B: general position implies good position `Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt`. For r tame and a smooth $z \in O_1$: if T_zQ has in T_zO an sc-complement $B \subseteq T_z^RO$, then T_zQ is in good position to C_z inside T_zO. $B \subseteq E_z$ changes none of the coordinates λ_i, $i \in I(z)$, so it is a good complement with every ε. The coordinates $\lambda_{I(z)}$ restricted to T_zO have kernel T_z^RO, whose codimension is $d_C(z) = \#I(z)$ (Proposition 2.5.6 (preprint 2.36)) `Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq`, so they map T_zO onto $\mathbb{R}^{I(z)}$; they kill B, so they map T_zQ onto $\mathbb{R}^{I(z)}$, and the preimage of the open positive orthant gives interior points of $T_zQ \cap C_z$. The text never states this step, which Theorem 4.2.11 (4) (preprint 4.22) and Theorem 4.3.1 (preprint 4.23) need for tameness and corner structures.

IV.17 Theorem 4.2.11 (1)–(3) (preprint 4.22): local structure in general position

$T_0Q + E_0^{\min} = E_0$ from general position at the centre `Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top`. If T_0Q has in T_0O an sc-complement $B \subseteq T_0^RO \subseteq E_0^{\min}$ and r is tame, then $E_0 = T_0O + (1 - Dr(0))E_0$ with $(1 - Dr(0))E_0 \subseteq E_0^{\min}$ (Proposition 2.5.3 (preprint 2.35)), so $E_0 = T_0Q + B + (1 - Dr(0))E_0 \subseteq T_0Q + E_0^{\min}$. (The text: "$E = T_0O \oplus Z = T_0A \oplus \widetilde{A} \oplus Z$, so T_0A has in E an sc-complement Y contained in W"; only $T_0A + W = E$ is used afterwards.)

$T_yQ + E_0^{\min} = E_0$ along the graph `Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top_graph`. If $\Gamma : V \to Q \cap W$ is the graph map of Proposition 4.1.9 (preprint 4.9) for a projection p onto N whose kernel lies in $Y \subseteq E_0^{\min}$, and $N + E_0^{\min} = E_0$, then at y over $\Gamma(v)$, $T_yQ = D\Gamma(v)N$ (§IV.15) and $D\Gamma(v)n - n \in \ker p \subseteq E_0^{\min}$ because $p \circ D\Gamma(v) = 1$; so $T_yQ + E_0^{\min} = N + E_0^{\min} = E_0$. This is the text's "by means of the particular form of the map $\alpha : h \mapsto h + D\delta(v)h$, also $T_aA + W = E$", which needs $D\delta(v)h \in W$.

Theorem 4.2.11 (preprint 4.22) in a tame centred chart `Chapter4/LocalStructure · theorem_4_22_chart`. Let r be tame onto $O \ni 0$, t an sc⁺-retraction onto $Q \subseteq O$ with $0 \in Q$, $N = T_0Q$, and suppose Q is in general position at 0. Then: $N + E_0^{\min} = E_0$ (above); N is in good position with a closed good complement $Y \subseteq E_0^{\min}$ (§IV.16) and the projection p along it, with kernel exactly Y; Proposition 4.1.9 (preprint 4.9) for this Y gives W, V, Γ; (2) $d_{N\cap C}(v) = d_C(\Gamma(v))$ for $v \in V$, because $d_{N\cap C}(v) = d_C(v)$ on the slice and $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate; (3) at every level-1 point y over $Q \cap W$, Q is in general position: y is smooth (Lemma 4.1.7 (i) (preprint 4.7)), T_yQ is the range of $Dt(y)$, finite-dimensional (Lemma 4.1.7 (ii) (preprint 4.7)) and smooth — the range of $Dt(y)$ is an sc-subspace at a smooth point `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`, finite-dimensional sc-subspaces are smooth `Chapter1/ScBanach · ScBanach.proposition_1_6_iff` — and $T_yQ + E_0^{\min} = E_0$ along the graph gives general position (§IV.16).

Theorem 4.2.11 (preprint 4.22) `Chapter4/LocalStructure · MPolyfold.theorem_4_22`. *Statement.* Let X be tame, A a smooth finite-dimensional submanifold, in general position at $x \in A$. There are a chart ψ of X with $\psi(x) = 0$, a tame retraction r onto its model for a presentation with n half-line coordinates, and an open $U \ni x$ inside the domain of ψ, such that: (1) $d_X(x) = n$, and $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O; (2) $d_X(a) = d_A(a)$ for $a \in A \cap U$; (3) A is in general position at every point of $A \cap U$; (4) for every $\sigma \subseteq \{1, \dots, n\}$ the intersection of local faces F_σ is a tame sub-M-polyfold inside U with

$d_{F_\sigma} + \#\sigma = d_X$, and $A \cap F_\sigma$ is a tame smooth finite-dimensional submanifold of F_σ, in general position in F_σ at each of its points. *Proof.* x is smooth (Proposition 4.2.3 (1) (preprint 4.15)), so there is a tame chart ψ centred at x (§IV.5), with a tame retraction r and a presentation T; the chart retraction t of A in ψ is sc$^+$ (§IV.11), and general position at x read in ψ (chart independence, §IV.13) is general position of $t(U)$ at 0. Theorem 4.2.11 (preprint 4.22) in the chart gives Y, p, W_1, V, Γ; put $U = W \cap \psi^{-1}(W_1)$, W the open set of the submanifold chart.

1. *(1) is a statement about the chart.* At the centre all n coordinates vanish, so $d_C(0) = n$; the tame chart realizes d_X `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so $d_X(x) = n$. The face count is Corollary 2.5.13 (preprint 2.41) for the M-polyfold O `Chapter2/TameRetracts · isFaceStructured_toMPolyfold` `Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith`. "Precisely d local faces contained in U" is read for the faces $O \cap \{\lambda_i = 0\}$ of the chart model; their identification with global faces of X is not formalized (Findings, item 86).
2. *(2).* The text: "$p : A \to (T_0A \cap C) \cap V$ is a sc-diffeomorphism … by Proposition 2.24, $d_A(a) = d_{T_0A\cap C}(p(a))$". The sc-diffeomorphism is the corner chart $p \circ \psi$ of A: the data of §IV.10 for the chart of A induced by ψ, with the complement $Y \subseteq E_0^{\min}$, give a chart of the induced structure modelled on $(V, N \cap C, N)$ with the identity retraction. That model is tame, so the chart realizes d_A `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`; ψ realizes d_X; and $d_{N\cap C}(p\psi(a)) = d_C(\Gamma(p\psi(a))) = d_C(\psi(a))$ by (2) of the chart form (Findings, item 84).
3. *(3)* is (3) of the chart form, read back through the chart: at $b \in U \cap A$ the level-1 point over $\psi(b)$ is smooth, and the chart condition in ψ is general position at b.
4. *(4)* is §IV.18.

The text's further claims "$\delta(0) = 0$ and $D\delta(0) = 0$" hold (§IV.15) but are not used.

IV.18 Theorem 4.2.11 (4) (preprint 4.22): the faces, and $A \cap F_\sigma$

The text: "in view of the arguments in the proof of Proposition 2.43, the sets F_σ are tame M-polyfolds. Moreover, $A \cap F_\sigma$ are smooth submanifolds of F_σ. The previous arguments, but now applied to $A \cap F_\sigma$ conclude the proof." Four things are missing (Findings, item 89): Proposition 2.5.15 (preprint 2.43) is about one face, and intersections need Proposition 2.5.16 (preprint 2.44) (second sentence); "smooth submanifold of F_σ" needs an sc$^+$-retraction *in the M-polyfold* F_σ; general position in F_σ needs an argument; and tameness of $A \cap F_\sigma$ needs Theorem 4.2.7 (preprint 4.18) in F_σ and good position.

Killing coordinates. For $\sigma \subseteq \{1, \dots, n\}$, $a \mapsto a \cdot 1_\sigma$ keeps the coordinates in σ `Chapter4/LocalStructure · maskL`, and for a linear section s of $\lambda|_N$, $\kappa_\sigma(v) = v - s(\lambda(v)1_\sigma)$ `Chapter4/LocalStructure · killOn` has $\lambda_i(\kappa_\sigma v) = 0$ for $i \in \sigma$ and $\lambda_j(\kappa_\sigma v) = \lambda_j(v)$ otherwise `Chapter4/LocalStructure · coordsOn_killOn`, fixes the vectors of the stratum $f_\sigma = \{\lambda_i = 0, i \in \sigma\}$ `Chapter4/LocalStructure · killOn_eq_self`, is idempotent `Chapter4/LocalStructure · killOn_idem` and maps $N \cap C$ into itself `Chapter4/LocalStructure · killOn_mem`. A continuous linear map into a finite-dimensional space is an sc-operator for the constant structure `Chapter4/LocalStructure · opToConst`.

The sc$^+$-retraction onto $A \cap F_\sigma$ in the chart `Chapter4/LocalStructure · exists_scPlus_retraction_corner`. In the situation of the chart form of Theorem 4.2.11 (preprint 4.22), for every open $B \ni 0$ there are an open $G \ni 0$, $G \subseteq W \cap B$, and an sc$^+$-retraction $t_\sigma = \Gamma \circ \kappa_\sigma \circ p \circ t$ on $U \cap t^{-1}(G)$ onto $Q_\sigma = Q \cap G \cap f_\sigma$: (a) it is sc$^+$, as sc-smooth $\square$ sc$^+$ (§IV.2), p, κ_σ and Γ being sc-smooth; (b) it is a retraction, because $t\Gamma = \Gamma$, $p\Gamma = 1$ and $\kappa_\sigma^2 = \kappa_\sigma$; (c) its image is $Q \cap G \cap f_\sigma$, because $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate, so $\lambda(\Gamma(v)) = \lambda(v)$; (d) at a point y of Q_σ the tangent vectors of Q lying in f_σ are tangent to Q_σ: for such h, $k = ph$ has $\lambda(k) = \lambda(h)$, so $\kappa_\sigma k = k$ and $Dt_\sigma(y)h = D\Gamma(w)\kappa_\sigma pDt(y)h = D\Gamma(w)k = h$, $w = p(y)$, by the chain rule and $T_yQ = D\Gamma(w)N$.

The corner model `Chapter4/LocalStructure · exists_cornerModel_retraction`. For a retraction S with retract in f_σ, $\pi_\sigma \circ S \circ \iota_\sigma$ on $\iota_\sigma^{-1}(S.U)$, with the corner inclusion and projection of Chapter 2 `Chapter2/`

CornerGerms · cornerIncl Chapter2/CornerGerms · cornerProj, is a retraction of $[0,\infty)^{n-\#\sigma} \oplus W$ onto $\pi_\sigma(S(U))$, sc$^+$ if S is.

Theorem 4.2.11 (4) (preprint 4.22) in the chart Chapter4/LocalStructure · theorem_4_22_four_chart. For every σ and every open $B_0 \ni 0$ there are faces $F_i \ni 0$ of O ($i \in \sigma$) and an open $G \ni 0$, $G \subseteq W \cap B_0$, such that: *F_σ near 0 is a tame retract* — $O \cap G \cap f_\sigma$ is the retract of the face retraction of Proposition 2.5.16 (preprint 2.44) Chapter2/CornerGerms · corner_retraction restricted to G, whose corner model is the retract of a tame retraction r' with $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ Chapter2/CornerGerms · corner_tameModel; *$A \cap F_\sigma$ near 0 is an sc$^+$-retract* — $Q \cap G \cap f_\sigma$ is the retract of t_σ, and its corner model that of the sc$^+$-retraction t', inside the retract of r'; *general position in F_σ* — at a level-1 point y' of $\pi_\sigma(Q \cap G \cap f_\sigma)$, with $y = \iota_\sigma(y')$: $T_yQ + E_0^{\min} = E_0$ (§IV.17) gives for every $b \in \mathbb{R}^{n-\#\sigma}$ a $v \in T_yQ$ with $\lambda(v) = \iota(b)$, zero on σ; $v \in f_\sigma$, so v is tangent to Q_σ, and $\pi_\sigma v$ is tangent to $\pi_\sigma(Q_\sigma)$ at y' (§IV.11), with coordinates b; so the tangent space of the corner model plus its minimal subspace at 0 is everything, and general position follows (§IV.16), in the corner model with the identity presentation.

sc$^+$-retractions in a chart are sc$^+$-retractions on open subsets Chapter4/LocalStructure · MPolyfold.isScPlusRetractionOn_of_chart — the converse of §IV.11, which the text does not state. For a chart χ and an sc$^+$-retraction τ of its model with $\tau(U) \subseteq O$, $s = \chi^{-1} \circ \tau \circ \chi$ is an sc$^+$-retraction on the open set $\chi^{-1}(\tau.U)$, with image $\chi^{-1}(\tau(U))$: sc-smoothness of s and of $1_0^1 \circ s$ are read in the charts χ of X and `levelChart` χ of X^1, where they are τ and its lift; the values lie in X_1 since $\tau(U) \subseteq E_\infty$ (Lemma 4.1.7 (i) (preprint 4.7)).

(4) on the M-polyfold, for one σ Chapter4/LocalStructure · MPolyfold.theorem_4_22_four_of_chart. In the tame chart ψ centred at x with the chart data of §IV.17: the faces F_i and G of the chart form, with G inside the domain of the submanifold chart, give $F_\sigma = \psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ Chapter4/LocalStructure · MChart.cornerPreimage, which contains x and lies in W. It is a sub-M-polyfold, a local property Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local; its face chart $\pi_\sigma \circ \psi$, compatible with the induced charts Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart, is a chart of the induced structure Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible modelled on the tame corner model, so F_σ is tame Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart with $d_{F_\sigma} + \#\sigma = d_X$. $A \cap F_\sigma$ is, in the face chart, the retract of the sc$^+$-retraction t', which the previous paragraph turns into an sc$^+$-retraction on an open subset of F_σ: it is a smooth finite-dimensional submanifold of F_σ. It is in general position at each of its points by the chart form read in the face chart. Its tameness is Theorem 4.2.7 (preprint 4.18) in the tame M-polyfold F_σ, good position following from general position (Lemma 7.B). F_σ is a germ, chosen inside U — the text's "contained in U" — and G depends on σ. Paracompactness of F_σ and $A \cap F_\sigma$, instance arguments of the induced structures, holds for every subset Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype.

The proof of Theorem 4.3.1 (preprint 4.23) uses $\dim(A \cap F_\sigma) = \dim A - \#\sigma$, which Theorem 4.2.11 (preprint 4.22) does not state; it is proved in the model in §IV.21.

IV.19 Remark 4.3.2 (preprint 4.24)

If $T_zO = T_zQ + B$ with $B \subseteq K$, the image of T_zQ spans T_zO/K, so $\dim(T_zO/K) \le \dim T_zQ$ Chapter4/LocalStructure · quotDim_le_finrank_of_eq_sup. **In a chart** Chapter4/LocalStructure · remark_4_24_chart: at a smooth point z of a tame retract, $\dim(T_zO/T_z^RO) = d_C(z)$ (Proposition 2.5.6 (preprint 2.36)), and if T_zQ has in T_zO an sc-complement inside T_z^RO, then $d_C(z) \le \dim T_zQ$. **On the M-polyfold**: for tame X and smooth x, in a tame chart at x the reduced tangent space has codimension $d_X(x)$ Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart Chapter4/LocalStructure · MPolyfold.remark_4_24_codim — the remark's first sentence, which needs X tame and x smooth; if A is in general position at a, then $d_X(a) \le \dim T_aA$ in any tame chart at a Chapter4/LocalStructure · MPolyfold.remark_4_24; and a submanifold in general position with $T_aA = 0$ does not meet ∂X at a Chapter4/LocalStructure · MPolyfold.remark_4_24_not_mem_boundary. The remark's "$x \in A_\lambda$ can only belong to a corner if $d_X(x) \le \dim A_\lambda$" holds for every sub-

manifold in general position, so it applies to the A_λ of Theorem 4.3.1 (preprint 4.23) once their general position is proved (§IV.24).

IV.20 Sard's theorem, vendored

The proof of Theorem 4.3.1 (preprint 4.23) applies "Sard's theorem" to maps $A \cap F^\circ_\sigma \to \mathbb{R}^m$ between spaces of arbitrary dimensions. Mathlib has the cases $\dim E \le \dim F$ only. The Milnor project of this repository proves the general theorem by Milnor's induction; its import closure is copied into `Poly–fold/Vendor/MilnorSard/` with the namespace renamed and no statement or proof changed (Findings, item 100). The proof, in Milnor's steps:

- *The objects.* Crit $= \{x \in U : Df(x) \text{ not onto}\}$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet`, and C_k, the points where all derivatives of orders $1, \dots, k$ vanish `Vendor/MilnorSard/Basic · MilnorSard.Sard.vanishSet`, with Crit $\subseteq$ (Crit $\setminus C_1$) $\cup \bigcup_{1\le i<k}(C_i \setminus C_{i+1}) \cup C_k$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet_subset_union`. The statement in dimension n `Vendor/MilnorSard/Basic · MilnorSard.Sard.SardStatement`.
- *Straightening.* A smooth w with $Dw(\bar{x}) \neq 0$ is straightened by $H(x) = (w(x), Px)$, a local diffeomorphism by the inverse function theorem `Vendor/MilnorSard/InverseFunction · MilnorSard.isLocalDiffeomorphAt_of_isInvertible_mfderiv` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.Straightening` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.exists_straightening`.
- *Step 1,* $f(\text{Crit} \setminus C_1)$ null, locally `Vendor/MilnorSard/Step1 · MilnorSard.Sard.step1_local`: in straightened coordinates $g = f \circ H^{-1}$ maps hyperplanes to hyperplanes, the critical points of g on a hyperplane are those of its restriction, the induction hypothesis in dimension $n-1$ applies to the restrictions, and Fubini concludes.
- *Step 2,* $f(C_k \setminus C_{k+1})$ null, locally `Vendor/MilnorSard/Step2 · MilnorSard.Sard.step2_local`: a k-th derivative w vanishes on C_k with $Dw \neq 0$; after straightening, C_k lies in a hyperplane, on which every point of C_k is critical, and the induction hypothesis applies. Local null sets glue along a countable cover `Vendor/MilnorSard/Step2 · MilnorSard.Sard.measure_image_eq_zero_of_locally`.
- *Step 3,* $f(C_k \cap K)$ null for $(k+1)\dim F > \dim E$ and K compact convex `Vendor/MilnorSard/Step3 · MilnorSard.Sard.measure_image_vanishSet_inter_eq_zero`: Taylor's bound $|f(y) - f(x)| \le c|y - x|^{k+1}$ for $x \in C_k$ `Vendor/MilnorSard/Step3 · MilnorSard.Sard.norm_sub_le_of_iteratedFDeriv_eq_zero` makes f Hölder of exponent $k+1$ there, so $f(C_k \cap K)$ has Hausdorff dimension at most $\dim E/(k+1) < \dim F$, hence measure zero `Vendor/MilnorSard/Basic · MilnorSard.Sard.addHaar_eq_zero_of_dimH_lt_finrank` — Milnor's cube count is the standard proof of that estimate.
- *The induction* `Vendor/MilnorSard/Sard · MilnorSard.Sard.sardStatement`, with the zero-dimensional target treated separately (no critical points) `Vendor/MilnorSard/Sard · MilnorSard.Sard.critSet_eq_empty_of_subsingleton`.

Sard's theorem `Vendor/MilnorSard/Sard · MilnorSard.sard`. *For E, F finite-dimensional real normed spaces in the same universe, a Haar measure μ on F, $U \subseteq E$ open and f of class C^∞ on U, $\mu(f(\text{Crit})) = 0$;* and the complement of the critical values is dense `Vendor/MilnorSard/Sard · MilnorSard.dense_compl_image_critSet`.

IV.21 The strata, Sard on the strata, and Theorem 3.8.22 (preprint 3.78) in C^∞ form

The finite-dimensional model of the proof of Theorem 4.3.1 (preprint 4.23): a finite-dimensional V (in the application T_0A of the chart of Theorem 4.2.11 (preprint 4.22)) with linear coordinates $\ell : V \to \mathbb{R}^n$ (those of the tame chart, onto), the partial quadrant $C = \{\ell \ge 0\}$ `Chapter4/FamiliesLocal · quadC`, a relatively open $U = O \cap C$ and a map f of class C^∞ on U (in the application the projection to $\mathbb{R}^m$ composed with the graph map Γ).

The strata. The stratum subspace of σ is $\{\ell_i = 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · stratSub Chapter4/`

FamiliesLocal · mem_stratSub, the tangent space of $A \cap F_\sigma$; $\sigma(v) = \{i : \ell_i(v) = 0\}$ is the set of faces through v Chapter4/FamiliesLocal · zeroSetOf, and v lies in the stratum subspace of $\sigma(v)$ Chapter4/FamiliesLocal · mem_stratSub_zeroSetOf. **The dimension count** Chapter4/FamiliesLocal · finrank_stratSub: if ℓ is onto, the stratum subspace of σ has dimension $\dim V - \#\sigma$ — it is the kernel of $v \mapsto (\ell_i(v))_{i\in\sigma}$, which is onto $\mathbb{R}^\sigma$. This is the count "$A \cap F_\sigma^\circ$ is a smooth manifold … having dimension $\dim(A) - \#\sigma$", which the text attributes to "Theorem 4.23" (it means Theorem 4.2.11 (preprint 4.22), which does not state it).

Stratified critical points Chapter4/FamiliesLocal · stratCritSet: the $v \in U$ at which $Df(v)$ restricted to the stratum subspace of $\sigma(v)$ is not onto — the critical points of the text's $p_\sigma : A \cap F_\sigma^\circ \to \mathbb{R}^m$ for $\sigma = \sigma(v)$. **Sard on the strata** Chapter4/FamiliesLocal · measure_image_stratCritSet: *the stratified critical values form a null set for every Haar measure.* The open stratum of σ, $\{k \in O : \ell_i(k) = 0\ (i \in \sigma),\ \ell_j(k) > 0\ (j \notin \sigma)\}$, is an open subset of the stratum subspace and lies in U; f restricted to it is C^∞, with derivative $Df(v)$ restricted to the stratum subspace; so a stratified critical point v with $\sigma = \sigma(v)$ is a critical point of f on the open stratum of σ, and Sard's theorem applies to each of the finitely many strata. The union of the open strata is U, and a point is stratified-regular iff it is a regular point of the p_σ for its own σ; so the stratified critical values are the union of the critical values of all the p_σ.

Theorem 3.8.22 (preprint 3.78) in C^∞ form. Theorem 3.8.22 (preprint 3.78) (§III.17) is stated for one j, with j-dependent constants in its proof; the fibre retraction below needs one solution that is C^∞. In the normalized form Chapter4/FamiliesLocal · ContractionData.exists_solution_smooth: if the contraction data of the proof hold for every $j \geq 1$, the radii of Lemmas 3.8.23–3.8.24 (preprint 3.79–3.80) Chapter3/QuadrantIFT · ContractionData.exists_radii and the fixed point $\tau(k)$ are those for $j = 1$, independent of j, and only the regularity statement depends on j Chapter3/QuadrantIFT · Radii.contDiffOn_τ; so one τ is C^j for every j. **Theorem 3.8.22 (preprint 3.78), C^∞** Chapter4/FamiliesLocal · theorem_3_78_smooth: for U relatively open in a partial quadrant C of a finite-dimensional V, $0 \in U$, f of class C^∞ on U with $f(0) = 0$ and $Df(0)$ onto, $K = \ker Df(0)$ in good position to C with good complement Y: there are a relatively open $V' \ni 0$ of $K \cap C$ and $\tau : K \to Y$, C^∞ on V', with $\tau(0) = 0$, $D\tau(0) = 0$, $k + \tau(k) \in U$ and $f(k + \tau(k)) = 0$, and the uniqueness of $\tau(k)$ among small y — the proof of Theorem 3.8.22 (preprint 3.78) Chapter3/QuadrantIFT · theorem_3_78 (the splitting $V \cong K \times Y$, $L = Df(0)$ an isomorphism $Y \to F$, $B(k, y) = y - L^{-1} f(k + y)$) with the C^∞ normalized form in place of the C^j one.

IV.22 The fibre over a stratified regular point

Partial quadrants of the model. $\{\ell_i \geq 0, i \in \sigma\}$ Chapter4/FamiliesLocal · quadAt is, for $\sigma = \sigma(v_0)$, the partial quadrant C_{v_0} of Definition 2.5.1 (preprint 2.33). With a linear section s of ℓ, $v \mapsto (\ell(v), v - s(\ell(v)))$ is a linear isomorphism $V \cong \mathbb{R}^n \times \ker \ell$ Chapter4/FamiliesLocal · sliceEquiv, which presents both as partial quadrants Chapter4/FamiliesLocal · isPartialQuadrant_quadC Chapter4/FamiliesLocal · isPartialQuadrant_quadAt; C is convex Chapter4/FamiliesLocal · convex_quadC with nonempty interior Chapter4/FamiliesLocal · interior_quadC_nonempty, so derivatives within $O \cap C$ are unique Chapter4/FamiliesLocal · uniqueDiffWithinAt_quadC.

The kernel at a stratified regular point is in good position Chapter4/FamiliesLocal · exists_goodComplement_stratSub — the argument of Lemma 5.26 (preprint; not in the 2021 edition), which the text needs here and does not give. Let $K + \{\ell_i = 0, i \in \sigma\} = V$ (for $K = \ker Df(v_0)$, $\sigma = \sigma(v_0)$, this is stratified regularity). The stratum subspace is closed, so K has a closed complement Y inside it (§IV.16). Adding a vector of Y changes none of the coordinates ℓ_i, $i \in \sigma$, so Y is a good complement with $\varepsilon = 1$. For the interior points: choose x with $\ell(x) = (1, \dots, 1)$ and write $x = k + y$; then $\ell_i(k) = 1$ for $i \in \sigma$, and the open set $\{k' \in K : \ell_i(k') > 0, i \in \sigma\}$ contains k and lies in $K \cap \{\ell_i \geq 0, i \in \sigma\}$.

The fibre retraction Chapter4/FamiliesLocal · exists_fibre_retraction. *Let ℓ be onto, f C^∞ on $O \cap C$ and $v_0 \in O \cap C$ not stratified-critical. There are an open $O' \ni v_0$, $O' \subseteq O$, and ρ, C^∞ on $O' \cap C$, mapping $O' \cap C$ into itself and into $f^{-1}(f(v_0))$, equal to the identity on the fibre there, with $D\rho(v_0)$ of range* $\ker Df(v_0)$ *and equal to the identity on it.* Proof:

1. Stratified regularity gives $Df(v_0)$ onto (its restriction to the stratum already is) and $K +$ stratum $=$

V for $K = \ker Df(v_0)$; so K is in good position to C_{v_0} with a closed good complement Y in the stratum subspace.
2. Near 0, $v_0 + C_{v_0}$ and C agree: on $G_0 = \{w : \ell_j(v_0+w) > 0, j \notin \sigma\}$, a point $v_0 + w$ is in C iff $w \in C_{v_0}$, since $\ell_i(v_0) = 0$ for $i \in \sigma$ — the re-centring of §IV.5 in the model. So $U_1 = (O - v_0) \cap G_0 \cap C_{v_0}$ is relatively open in C_{v_0}, contains 0, and $w \mapsto v_0 + w$ maps it into U.
3. $g(w) = f(v_0+w) - f(v_0)$ is C^∞ on U_1, $g(0) = 0$, $Dg(0) = Df(v_0)$ onto; Theorem 3.8.22 (preprint 3.78) in C^∞ form gives the fibre near v_0 as the graph $\{v_0 + k + \tau(k)\}$ of a C^∞ map $\tau : V' \to Y$, $V' \subseteq K \cap C_{v_0}$, unique for small k and y.
4. With $q(v)$ the projection of $v - v_0$ onto K along Y `Chapter4/LocalStructure · exists_proj_of_isCompl`, put $\rho(v) = v_0 + q(v) + \tau(q(v))$ on $O' \cap C$, O' a small ball around v_0 chosen so that q maps it into the uniqueness region and $\rho(O' \cap C) \subseteq O'$. The projection along Y keeps the coordinates in σ, so q maps C near v_0 into C_{v_0}; ρ lands on the fibre; and on the fibre $\rho(v) = v$ by uniqueness.
5. $D\tau(0) = 0$ gives $D\rho(v_0) = q$, whose range is K and which is the identity on K.

IV.23 Stratified regularity, the sc⁺-retraction onto A_λ, and its general position

The family. For tame X, $\mathbb{R}^m \times X$ with the product charts is tame `Chapter3/ParamPerturbation · prod_rn_isTame` (§III.40 for $\mathbb{R}^m$ as an M-polyfold `Chapter3/ParamPerturbation · rnPolyfold` with its translation charts `Chapter3/ParamPerturbation · rnChart`). The fibre of $A \subseteq \mathbb{R}^m \times X$ over λ is $A_\lambda = \{x : (\lambda, x) \in A\}$ `Chapter4/Families · fibreSet`. $\mathbb{R}^m$ is presented as a partial quadrant without half-line coordinates `Chapter4/Families · rnPres` `Chapter4/Families · rnPres_eq`, and the product chart $\chi = (r \mapsto r - a) \times \varphi$ has the product presentation `Chapter2/FibredProduct · prodPresentation`, whose minimal subspace at z is $E_z = \mathbb{R}^m \times (E_X)_{\varphi(z_2)}$.

Stratified regular points `Chapter4/Families · IsStratRegularPoint`. $z \in A$ is a stratified regular point if, for some data of Definition 4.2.1 (preprint 4.13) at z, some $a \in \mathbb{R}^m$, some chart φ of X at z_2, some presentation of its partial quadrant and the level-1 point y over $\chi(z)$, every $\mu \in \mathbb{R}^m$ is the first component of a tangent vector v of the chart model of A at y with $v_2 \in E_{\varphi(z_2)}$: the projection maps $T_z A \cap E_z$ — the tangent space of the stratum through z — onto $\mathbb{R}^m$. λ is a *stratified regular value* if every point of A over λ is a stratified regular point `Chapter4/Families · IsStratRegularValue`. This is the text's "regular value of every p_σ". The chart independence is not formalized (§IV.25). *The naive notion does not suffice*: in $\mathbb{R} \times [0, \infty)$, $A = \{(x, x) : 0 \le x < 1\} \cup (-1, 1) \times \{5\}$ and $\varepsilon = \frac{1}{2}$ satisfy the hypotheses of Theorem 4.3.1 (preprint 4.23) (at $(0,0)$, $T_z A = \mathbb{R}(1,1)$ has the complement $\mathbb{R} \times 0 = T_z^R Z$), Dp is onto everywhere, but $A_0 = \{0, 5\}$ is zero-dimensional and meets ∂X, so it is not in general position (Remark 4.3.2 (preprint 4.24)); $(0,0)$ is stratified-critical, $T_z A \cap E_z = 0$.

The slice $e \mapsto (c, e)$: the inclusion $h \mapsto (0, h)$ `Chapter4/Families · ScBanach.prodInrOp` followed by the translation by the smooth vector $(c, 0)$ `Chapter4/Families · mem_smooth_rn_zero` is sc-smooth `Chapter4/Families · isScSmooth_slice` with derivative $h \mapsto (0, h)$ `Chapter4/Families · scDeriv_slice`. On a finite-dimensional source the sc-derivative is the classical one `Chapter4/Families · hasFDerivWithinAt_scDeriv_const'`.

The sc⁺-retraction onto A_λ `Chapter4/Families · exists_slice_retraction` — Definition 4.2.1 (preprint 4.13) requires it and the text does not construct it. In the model $\mathbb{R}^m \oplus E$ of a product chart, let t be an sc⁺-retraction (the chart model of A), described near a point as the graph $\Gamma : V \to t(U) \cap W$ with inverse p (Proposition 4.1.9 (preprint 4.9)), and ρ a C^∞ retraction of $O' \cap (\mathbb{R}^m \times C) \subseteq V$ onto the points v with $\Gamma(v)_1 = c$ (the fibre retraction of §IV.22 for $f = (\Gamma)_1$). Then

$$\tau(e) = \big(\Gamma(\rho(p(t(c, e))))\big)_2$$

is an sc⁺-retraction of E (partial quadrant C) onto $\{e : (c, e) \in t(U) \cap W,\ p(c, e) \in O'\}$: it is sc⁺ because t is sc⁺ and the slice, p, ρ (sc-smooth as a C^∞ map on a finite-dimensional set, Corollary 1.2.5 (preprint 1.20)), Γ and the second projection `Chapter2/StrongBundle · ScBanach.prodSndOp` are sc-smooth (§IV.2); and it

is a retraction, because for $e' = \tau(e)$, $\Gamma(w) = (c, e')$ with $w = \rho(\cdot)$ on the fibre, so $t(c, e') = \Gamma(w)$, $p\Gamma(w) = w = \rho(w)$ and $\tau(e') = e'$. *Its tangent space*: if $D\rho(v_0) = 1$ on the h with $(D\Gamma(v_0)h)_1 = 0$, then at the level-1 point over $\Gamma(v_0)_2$ the tangent space of the retract of τ contains $(D\Gamma(v_0)h)_2$ for all such h, by the chain rule $D\tau(y)w = (D\Gamma(v_0)D\rho(v_0)p\, Dt(c,y)(0,w))_2$, Dt fixing $T(t(U)) = D\Gamma(v_0)N$ and $p \circ D\Gamma(v_0) = 1$.

General position of the fibre `Chapter4/Families · isGeneralPositionRetractAt_slice`. Let r_X be tame, the chart model of A in general position at $z = (c, y)$ with respect to id $\times\, r_X$ and the product presentation, z regular on its stratum (every μ is the first component of some $a \in T_zA \cap E_z$), and τ an sc$^+$-retraction onto $Q \subseteq O$ whose tangent space at the smooth y contains the second components of the tangent vectors of A at z with first component 0. Then Q is in general position at y. For $w \in T_yO$ write $(0, w) = a + b$ with $a \in T_zA$ and $b \in T_z^RZ = \mathbb{R}^m \times T_y^RO$ (general position of A at z); correct a by $a_2 \in T_zA \cap E_z$ with the same first component as a (regularity on the stratum); then $(a - a_2)_2 \in T_yQ$ and $(b + a_2)_2 \in T_y^RO$, a_2 lying in E_z, so $w \in T_yQ + T_y^RO$. Proposition 2.5.6 (preprint 2.36) makes T_y^RO an sc-subspace and Lemma 7.A gives the sc-complement (§IV.16). This is the text's "$T_{z'}(A \cap F^\circ_\sigma) = T_{z'}((\{\lambda\} \times A_\lambda) \cap F^\circ_\sigma) \oplus \xi_{z'}$ … in view of the assumption (2)", made precise: regularity on the stratum is what puts a_2 in E_z.

IV.24 Theorem 4.3.1 (preprint 4.23): families and Sard's theorem

The conclusions at a point `Chapter4/Families · FibreLocal`: $z = (\lambda, x)$ is a stratified regular point, A_λ is near x the image of an sc$^+$-retraction on an open subset of X, and A_λ is in general and in good position at x.

The local step `Chapter4/Families · exists_local_family`. *For X tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold in general position at every point, every $z_0 \in A$ has an open neighbourhood U and a Lebesgue-null* CV $\subseteq \mathbb{R}^m$ *such that the conclusions hold at every $z \in A \cap U$ with $z_1 \notin$ CV.* Proof:

1. *The chart.* $z_{0,2}$ is smooth `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, so there is a tame chart φ of X centred at it (§IV.5); the product chart $\chi = (r \mapsto r - \lambda_0) \times \varphi$ is a chart of $\mathbb{R}^m \times X$ `Chapter3/ParamPerturbation · isChart_rnChart` `Chapter2/FibredProduct · MPolyfold.IsChart.prod` centred at z_0, with the tame retraction id $\times\, r_X$ `Chapter2/FibredProduct · ScRetraction.prod` `Chapter2/FibredProduct · ScRetraction.IsTameWith.prod` and coordinates those of φ ($\mathbb{R}^m$ has none).
2. *Theorem 4.2.11 (preprint 4.22) in the chart.* The chart retraction t of A in χ is sc$^+$ (§IV.11), and general position at z_0 read in χ is general position at 0; Theorem 4.2.11 (preprint 4.22) in the chart (§IV.17) gives $N = T_0(\chi A)$ with $N + E_0^{\min} = E_0$, so the coordinates ℓ are onto on N, a closed good complement $Y \subseteq E_0^{\min}$, the projection p, and $\Gamma : V \to \chi(A)$ over $V = O_V \cap \{\ell \geq 0\} \subseteq N$, with $\ell \circ \Gamma = \ell$.
3. *Sard.* $f(v) = \Gamma(v)_1 + \lambda_0$, the projection P read in the chart, is sc-smooth on V as the first projection `Chapter2/StrongBundle · ScBanach.prodFstOp` composed with Γ, hence C^∞ (§IV.1); CV $= f($stratified critical points of $f)$ is null (§IV.21). $U = W_A \cap \chi^{-1}(W)$, with W_A the open set of the submanifold chart (§IV.11) and W that of Proposition 4.1.9 (preprint 4.9).
4. *A point $z \in A \cap U$ with $z_1 \notin$ CV.* $\chi(z) \in t(U) \cap W$, so $v_0 = p(\chi z) \in V$ with $\Gamma(v_0) = \chi(z)$ and $f(v_0) = z_1$; hence v_0 is not stratified-critical. Since $\ell(D\Gamma(v)h) = \ell(h)$, $D\Gamma(v_0)$ maps the stratum subspace of $\sigma(v_0)$ onto $T_yA \cap E_y$, $y = \Gamma(v_0)$; so z is a stratified regular point, in the chart χ.
5. *The fibre.* The fibre retraction ρ of f at v_0 (§IV.22), the sc$^+$-retraction τ onto the chart image of A_{z_1} (§IV.23), the sc$^+$-retraction $\varphi^{-1} \circ \tau \circ \varphi$ on an open subset of X (§IV.18), whose image is A_{z_1} near z_2; the tangent space of τ's retract contains the vertical tangent vectors of A, since $D\rho(v_0) = 1$ on $\ker Df(v_0)$; general position of A_{z_1} at z_2 (§IV.23); good position from general position (Lemma 7.B).

The cover `Chapter4/Families · theorem_4_23_of_subset`. *For X tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with compact closure, in general position at every point, and any set $S \subseteq \mathbb{R}^m$ of parameters, there is $\Sigma \subseteq S$ with $S \setminus \Sigma$ null, consisting of stratified regular values, with the conclusions at every point of every A_λ, $\lambda \in \Sigma$.* The text covers $\{z \in A : |p(z)| \leq \varepsilon\}$ "with the finitely many such neighborhoods $U(z_1), \dots, U(z_k)$".

That set need not be compact: in $\mathbb{R}\times\mathbb{R}$, $A=(-2\varepsilon,2\varepsilon)\times\{2\}\cup\{(\varepsilon(1-x),x):0<x<1\}$ has compact closure and compact non-empty fibres over $|\lambda|\le\varepsilon$, but $\{|p|\le\varepsilon\}$ contains the open segment, which accumulates at $(\varepsilon,0)\notin A$, and neighbourhoods of radius less than half the distance to $(\varepsilon,0)$ have no finite subfamily covering it. A countable subcover suffices: $\mathbb{R}^m\times X$ is metrizable (Theorem 2.3.10 (preprint 2.16)) `Chapter2/Metrizable · MPolyfold.metrizableSpace`, so the compact $\overline{A}$, and with it A, is separable, hence second countable; countably many of the neighbourhoods of the local step cover A, the countable union of their null sets is null, and $\Sigma=S\setminus\bigcup_i \mathrm{CV}(z_i)$ (Findings, item 103). Some countability is necessary: an uncountable discrete A can have critical values of positive measure; here it comes from the compactness of $\overline{A}$, which the text assumes.

Theorem 4.3.1 (preprint 4.23) `Chapter4/Families · theorem_4_23`. *Statement.* Let X be tame, $A\subseteq\mathbb{R}^m\times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon>0$, and (1) for $|\lambda|\le\varepsilon$ ($|\lambda|^2=\sum\lambda_i^2$) the set $A\cap\{z_1=\lambda\}$ is compact and non-empty, (2) A is in general position at every point. Then there is $\Sigma\subseteq\{|\lambda|<\varepsilon\}$ with null complement in $\{|\lambda|<\varepsilon\}$, consisting of stratified regular values, such that for $\lambda\in\Sigma$ the fibre A_λ is compact, a smooth finite-dimensional submanifold of X, in general position at each of its points, with a corner structure (Definition 4.1.1 (preprint 4.1)), and tame. *Proof.* The cover for $S=\{|\lambda|<\varepsilon\}$. A_λ is the image of the compact $A\cap\{z_1=\lambda\}$ under the projection to X, hence compact. It is a smooth finite-dimensional submanifold and in general position by the conclusions at its points; it is in good position at every point (Lemma 7.B), so Theorem 4.2.7 (preprint 4.18) gives the corner structure and tameness. Hypothesis (2), misprinted in the text ("there exists a sc-complement T_zA in T_zZ which is contained in T^R_zZ"), is Definition 4.2.10 (preprint 4.21), as the proof uses it. Not used: the tameness of the induced structure of A, the non-emptiness in (1), $\varepsilon>0$ (Findings, item 101). Every stratified regular value is a regular value — $T_zA\cap E_z\subseteq T_zA$ — so Σ consists of regular values in the text's sense; but the conclusion is not true at every ordinary regular value (§IV.23).

IV.25 What is not formalized

- **Lemma 4.1.11 (preprint 4.10)** (the section $u\mapsto(1-p)(u-t(u))$ of $U\lhd Y$ is sc-Fredholm): no longer used, since Proposition 4.1.9 (preprint 4.9) is proved directly. It would need $U\lhd Y$ as a strong bundle, Y and $Y\cap F$ as sc-subspaces, a complement B of $Y\cap F$ in Y chosen inside Y_∞ (an arbitrary algebraic complement does not give an sc-splitting), and at each smooth point the re-centred strong bundle chart (§IV.5), the trivial filler, the sc$^+$-section $s(v)=\mathbf{f}(u)+(1-p)[t(u)-t(u+v)]$ and the conjugation of $g-s=(1-p)v$ to a basic germ with contraction part 0. The text's proof has a sign slip in $D\mathbf{f}(u)$, calls $D\mathbf{f}(u)$ the sc$^+$-operator where $Dt(u)$ is meant, argues "a perturbation of the identity" for an operator into Y, writes $L(T_0O)$ for $L(Y)$, and omits the interior points of $(T_0O\oplus B)\cap C$ (Findings, item 57).
- **§4.4** (Definitions 4.4.1, 4.4.6 (preprint 4.25, 4.27), the exterior derivative, the relative complex, Theorem 4.4.7 (preprint 4.28); Proposition 4.4.5 (preprint 4.26) is proved in the 2021 edition and formalized, §IV.29): it needs $\oplus^kTX$ as an M-polyfold and a linear T_xX (gap G.7), the M-polyfolds X^i for $i\ge2$ with $(TX)^i\cong T(X^i)$ (gap G.6, now closed, §IV.29), and integration on manifolds with corners of varying dimension. The text's $d:\Omega^k(X^{i+1})\to\Omega^{k+1}(X^i)$ should presumably run $\Omega^k(X^i)\to\Omega^{k+1}(X^{i+1})$, the bracket of Proposition 4.4.5 (preprint 4.26) living one level up; $d^2=0$ and $dj^*=j^*d$ are asserted; forms on ∂X are not defined; and Theorem 4.4.7 (preprint 4.28) has no proof (Findings, items 120–124).
- **The open question** after Proposition 4.2.3 (preprint 4.15) (§IV.11). For **Remark 4.1.3 (preprint 4.3)** the example is formalized (§IV.27); the argument that an example cannot come from a retraction whose level maps are C^1 is not.
- **The chart independence of stratified regularity** (§IV.23), true by the transport argument of §IV.13, since transition maps of product charts are id $\times\,\tau$ up to translation; Theorem 4.3.1 (preprint 4.23) does not need it.
- The local form of Remark 4.2.2 (preprint 4.14) near $\{d_X=0\}$ (§IV.11), the identification of the local faces of a chart with global faces in Theorem 4.2.11 (1) (preprint 4.22) (§IV.17), and the classical

manifold structure behind Definition 4.1.1 (preprint 4.1) beyond the C^∞ transitions (§IV.1).

IV.26 What the formalization taught us

1. *Centred charts are a construction.* They exist only at smooth points and live in the partial quadrant $C_{\varphi(x)}$, on the set where it agrees with C; four theorems of the chapter and Lemma 5.3.3 (preprint 5.11) rely on them, and the text never builds them (§IV.5).
2. *Proposition 4.1.9 (preprint 4.9) is a statement about fixed sets, and its proof is not the implicit function theorem.* The text's route, as we read it, gives a germ over a different complement from the one needed. The direct proof uses only the sc⁺ property — a mean value estimate on every level and the level-1 continuity of t — and its regularity argument rests on two observations: the estimate is applied at $\Gamma(a)$ with the frame $B_{\Gamma(a)}$, not at 0, and derivatives of t only meet the finite-dimensional smooth tangent space, so the loss of levels does not accumulate (§§IV.6–IV.9). The projection must be the one along the closed good complement, not $Dt(0)$.
3. *Theorem 4.1.2 (preprint 4.2) does not need tameness*, and its compatibility claim needs charts composed with sc-diffeomorphisms between retracts (§IV.10).
4. *The sc⁺ condition has to be read in charts*, in both directions: a submanifold's chart retraction is sc⁺ (§IV.11), and an sc⁺-retraction in a chart is one on an open set (§IV.18). Every statement of §§4.2–4.3 passes through these two facts.
5. *One step of Lemma 4.2.8 (preprint 4.19) is false*, and the repair is the observation that the complement of T_0O lies in $C \cap (-C)$ at the corner of a centred chart (§IV.14).
6. *General position implies good position* (§IV.16), a step the text uses in Theorem 4.2.11 (4) (preprint 4.22) and Theorem 4.3.1 (preprint 4.23) without stating it.
7. *Theorem 4.3.1 (preprint 4.23) needs stratified regular values and a countable cover*, and the fibre's manifold structure needs the implicit function theorem at corner points with the kernel in good position — the argument of Lemma 5.26 (preprint) — and Theorem 3.8.22 (preprint 3.78) in C^∞ form (§§IV.21–IV.24). Sard's theorem had to be supplied in all dimensions (§IV.20).

IV.27 Remark 4.1.3's (preprint 4.3) example: a retraction that loses a level

Remark 4.1.3 (preprint 4.3) says that "$X^1 \neq X$" is possible as M-polyfolds although the underlying sets agree, and gives no example. By Proposition 4.1.8 (preprint 4.8) in charts (§IV.4) an example is an sc-retract $O = r(U)$ with $O = O_\infty$ that is not an sc⁺-retract. It cannot come from a retraction whose level maps $U_m \to E_m$ are C^1 — the maps that Proposition 1.2.4 (preprint 1.19), Corollary 1.2.5 (preprint 1.20) and their composites produce: then $T_oO = Dr(o)E_0$ is finite-dimensional by a Baire category argument (O is a countable union of the compact sets $\mathrm{cl}_0\{x \in O : |x|_1 \leq n\}$), and the topologies of O in E_0 and E_1 agree near o (Findings 141; an argument, not formalized). So the retraction must lose a level in its first derivative, as the splicings of HWZ do, and its sc-smoothness needs a criterion the project did not have (Findings 140–146).

A criterion with loss. Let V be finite-dimensional and $L(p)_m : Z_m \to W_m$, $p \in V$, be bounded operators compatible with the inclusions, such that $(p,z) \mapsto L(p)_m z$ is continuous on $V \times Z_m$ — strong continuity, not continuity of $p \mapsto L(p)_m$ in the operator norm — and $p \mapsto A_m(p) = L(p)_m \circ \iota$ is C^∞ as a map into the operators $Z_{m+1} \to W_m$: one level of loss buys all the p-derivatives `Chapter4/RemarkFourThreeExample · IsLossFamily` `Chapter4/RemarkFourThreeExample · famLoss` `Chapter4/RemarkFourThreeExample · famDeriv`. *Then $(p,z) \mapsto L(p)z$ is sc-smooth on $V \oplus Z$* `Chapter4/RemarkFourThreeExample · lossFamMap` `Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth`. The map is sc⁰ by the strong continuity `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc0`, and sc¹ with derivative $(\pi,\zeta) \mapsto \partial_p A_0(p)[\pi]z + L(p)\zeta$ at a level-1 point (p,z) `Chapter4/RemarkFourThreeExample · IsLossFamily.differentiable_famLoss` `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc1With`: the increment in p is

taken in the operator norm of A_0, which needs $z \in Z_1$, and the derivative family is continuous on the tangent levels by the C^∞ dependence of $\partial_p A_m$ Chapter4/RemarkFourThreeExample · IsLossFamily.continuous_famDeriv. The tangent map is again a map of the same kind: after the sc-isomorphism that reorders (p,z,π,ζ) as $((p,\pi),(z,\zeta))$ Chapter4/RemarkFourThreeExample · famSwap it is the map of the *tangent family* $\tilde{L}(p,\pi)(z,\zeta) = (L(p)z, \partial_p A(p)[\pi]z + L(p)\zeta)$ on $V \times V$ Chapter4/RemarkFourThreeExample · famTangent Chapter4/RemarkFourThreeExample · tangentMapWith_lossFamMap_eq, and $\tilde{L}$ is a loss family Chapter4/RemarkFourThreeExample · IsLossFamily.tangent. Induction on k gives sc^k for every k Chapter4/RemarkFourThreeExample · isSck_lossFamMap. Only one level of loss is allowed: maps whose j-th derivative costs j levels, such as the map $(t,x) \mapsto (\cos(kt)x_k)_k$ of the example after Definition 1.1.15 (preprint 1.14), are not covered.

The reduction. For a loss family of projections ($L(p)_0^2 = L(p)_0$), $r(p,z) = (p, L(p)z)$ is an sc-smooth retraction of $V \oplus Z$ Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth_famRetr Chapter4/RemarkFourThreeExample · IsLossFamily.retraction; if the ranges of the $L(p)_0$ are smooth points, $O = O_\infty$ Chapter4/RemarkFourThreeExample · IsLossFamily.retract_subset_smooth; and if the lift $O \to E_1$ is discontinuous, Proposition 4.1.8 (preprint 4.8) says that O is not an sc^+-retract Chapter4/RemarkFourThreeExample · IsLossFamily.not_isScPlusRetract. These are the data of the example Chapter4/RemarkFourThreeExample · RemarkFourThreeData Chapter4/RemarkFourThreeExample · exists_retract_smooth_not_scPlus_of_data.

The family. On the weighted sequence space X of the example after Definition 1.1.15 (preprint 1.14) ($|x|_m = \sup_k (k+1)^m |x_k|$; Chapter1/ScOneNotNormContinuous · ScOneExample.X) the moving vector must pass between coordinates of comparable weight, or the level-m operators are unbounded, so the transitions must be fast and the gain per step geometric: the indices are $n_k = 2^k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.nk. With nodes $1/(k+1)$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.node and angles $\theta_k(t)$ turning from 0 (for $t \geq 1/(k+1)$) to $\pi/2$ (for $t \leq 1/(k+2)$) through Mathlib's smooth transition, flat at both ends Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition, the coefficients $b_k = \sin\theta_{k-1}\cos\theta_k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb vanish except $b_k = \cos\theta_k$ and $b_{k+1} = \sin\theta_k$ at the active block $k = \lfloor 1/t \rfloor - 1$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_eq_zero_of_pos Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt_succ, and all vanish for $t \leq 0$. $L(t) = \sum_k$ (block k) Chapter4/RemarkFourThreeFamily · RemarkFourThree.blk Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam is a sum with two non-zero terms, which on level 0 is the projection onto $b_k e_{n_k} + b_{k+1} e_{n_{k+1}}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_of_pos; on level m its entries carry the weight ratios $((a+1)/(b+1))^m \leq 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.elemL. It is a loss family Chapter4/RemarkFourThreeFamily · RemarkFourThree.isLossFamily_fam: compatible with the inclusions Chapter4/RemarkFourThreeFamily · RemarkFourThree.toBase_fam; uniformly bounded, $\|L(t)_m\| \leq 3 + 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.norm_fam_le; strongly continuous, from the bound and the norm continuity after the dense inclusion by an $\varepsilon/3$ argument Chapter4/RemarkFourThreeFamily · RemarkFourThree.continuous_apply_of_bound; and $t \mapsto L(t)_m \circ \iota$ is C^∞ in the operator norm, by Mathlib's differentiation of series Mathlib/Analysis/Calculus/SmoothSeries · contDiff_tsum: j derivatives of block k are bounded by a polynomial in k of degree $2j$ (Leibniz's rule and the derivatives of the flat transition), while one level of loss gains a factor $(2+2^m)2^{-k}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.contDiff_fam_comp. Every $L(t)_0$ is a projection Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_idem with finitely supported, hence smooth, range Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_mem_smooth. The lift is discontinuous: the points $o_k = (1/(k+1), (1/(k+1))e_{n_k})$ lie in O Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_node_smul_δv and tend to 0 in E_0, while their level-1 coordinates $(2^k+1)/(k+1)\, e_{n_k}$ have norm at least 1 Chapter4/RemarkFourThreeFamily · RemarkFourThree.not_continuousOn_liftOne. So the data exist Chapter4/RemarkFourThreeFamily · RemarkFourThree.remarkFourThreeData, and there is an sc-retract $O = O_\infty$ that is not an sc^+-retract Chapter4/RemarkFourThreeFamily · remark_4_3_retract_example.

The M-polyfold. A retract with its single chart $O \to O$ is an M-polyfold Chapter4/RemarkFourThreeFamily · retractChart Chapter4/RemarkFourThreeFamily · retractPolyfold; if $O \subseteq E_\infty$ and O

is not an sc$^+$-retract, every point of it is smooth and $1^1_0 : X \to X^1$ is not sc-smooth, for every junk value (Proposition 4.1.8 (preprint 4.8) in charts, §IV.4) `Chapter4/RemarkFourThreeFamily · retractPolyfold_smooth_not_toLevelOne`. *Remark 4.1.3's (preprint 4.3) example* `Chapter4/RemarkFourThreeFamily · remark_4_3_example`: there is an M-polyfold with $X_0 = X_\infty$ for which 1^1_0 is not sc-smooth, hence (by the remark's first part, §IV.4) not an sc-diffeomorphism: $X^1 \neq X$ as M-polyfolds. In this example X and X^1 even carry different topologies on the same set, since $o_k \to 0$ in X and not in X^1. "$X^1 \neq X$" is read as "1^1_0 is not sc-smooth"; the M-polyfold is a single retract, which is all the remark needs; its parameter space is $\mathbb{R}$ and its partial quadrant all of $\mathbb{R} \oplus X$, so there is no boundary.

IV.28 The Springer 2021 edition: Theorems 4.2.11 and 4.3.1, vector fields and the Lie bracket

The 2021 edition changes two statements of §§4.2–4.3 and rewrites §4.4, adding sc-smooth vector fields (Definition 4.4.3, Lemma 4.4.4), a proof of Proposition 4.4.5 (preprint 4.26) in a new Appendix 4.5, and a Poincaré lemma. The files are `Chapter4/Book2021` (the two theorems), `Chapter4/VectorFields` (vector fields and the bracket) and `Chapter4/BracketLevel` (an example on the level of the bracket). The forms of §4.4 remain unformalized (gap G.80, continuing G.37–G.40; Findings, Chapter 4, items 150–159).

Theorem 4.2.11 (preprint 4.22), 2021 form `Chapter4/Book2021 · MPolyfold.theorem_4_2_11_2021`. *Statement.* As in §IV.17–§IV.18, with (1) strengthened: the $d = d_X(x)$ local faces lie in U *and contain* x. Formally: a tame chart ψ centred at x with tame retraction R and presentation $T : E \cong \mathbb{R}^n \oplus W$, and an open $U \ni x$ in its domain, such that $d_X(x) = n$, $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O, and for every $i < n$ there are a face $F_i \ni 0$ of the retract in the i-th coordinate hyperplane and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap F_i) \subseteq U$; (2)–(4) unchanged. *Proof.* Everything except the new clause is `Chapter4/LocalStructure · MPolyfold.theorem_4_22`, which already counts the faces *through* $\psi(x)$ — the 2021 reading of (1). For the new clause apply (4) to $\sigma = \{i\}$: it gives a family $(F_j)_{j\in\{i\}}$ of faces through 0 in the coordinate hyperplanes and an open $G \ni 0$ with $x \in \psi^{-1}(G \cap \bigcap_j F_j) \subseteq U$; since the index set has one element, the intersection over it is F_i itself (an extensionality step on the preimage set). The local faces remain those of the chart model (gap G.35).

Theorem 4.3.1 (preprint 4.23), 2021 form `Chapter4/Book2021 · theorem_4_3_1_2021`. *Statement.* X tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, (1) $A \cap \{z_1 = \lambda\}$ compact for $|\lambda| \leq \varepsilon$ — no longer "non-empty" — and (2) general position at every point. Then there is a *measurable* $\Sigma \subseteq B = \{|\lambda| < \varepsilon\}$ with $B \setminus \Sigma$ null, of stratified regular values `Chapter4/Families · IsStratRegularValue`, such that each A_λ, $\lambda \in \Sigma$, is compact, a smooth finite-dimensional submanifold in general position at its points, with a corner structure, and tame. *Proof.* The cover of §IV.24 for $S = B$ `Chapter4/Families · theorem_4_23_of_subset` gives $\Sigma_0 \subseteq B$ with $B \setminus \Sigma_0$ null and the local conclusions at every point of every fibre. B is measurable (a strict sublevel set of a continuous function). Let N be a measurable null set containing $B \setminus \Sigma_0$ (Mathlib's measurable hull `Mathlib/MeasureTheory/Measure/MeasureSpaceDef · MeasureTheory.toMeasurable`) and $\Sigma = B \setminus N$: measurable, inside Σ_0, with $B \setminus \Sigma \subseteq N$ null. For $\lambda \in \Sigma$, the fibre A_λ is the image of the compact set $A \cap \{z_1 = \lambda\}$ (hypothesis (1), $|\lambda| < \varepsilon$) under the second projection, hence compact; it is a submanifold and in general and good position by the local conclusions; Theorem 4.2.7 (preprint 4.18) `Chapter4/Submanifolds · MPolyfold.theorem_4_18` gives the corner structure and tameness. Neither non-emptiness, nor the tameness of A, nor $\varepsilon > 0$ is used.

Vector fields in a local model (Definition 4.4.3). For a local model (O, C, E) a map $A : E_1 \to E_0$ is a *vector field* if $x \mapsto (x, A(x))$ is an sc-smooth map between retracts from (O^1, C^1, E^1) to the tangent (TO, TC, TE) `Chapter4/VectorFields · IsRetractVectorField` — the text's sc-smooth section of $TX \to X^1$, read in a chart: its domain is X^1. The zero map is one on every sc-smooth retract: $(x, 0)$ lies in TO `Chapter2/TangentBundle · mk_zero_mem_retractTangent` and $x \mapsto (x, 0)$ is the pair of the retraction and the zero operator `Chapter4/VectorFields · isRetractVectorField_zero`. Consequences: $(x, A(x)) \in TO$ for $x \in O_1$ `Chapter4/VectorFields · IsRetractVectorField.mem_retractTangent`; for a retraction r onto O, $\tilde{A} = A \circ r^1$ is sc-

smooth on U^1, as the second component of the section composed with the shifted retraction Chapter4/VectorFields · IsRetractVectorField.isScSmooth_comp; and $Dr(x)A(x) = A(x)$ for $x \in O_1$, formula (4.8), because T_xO is the fixed space of $Dr(x)$ Chapter4/VectorFields · IsRetractVectorField.scDeriv_apply.

Vector fields on an M-polyfold (Definition 4.4.3). $A : X^1 \to TX$ is an sc-smooth vector field if it is sc-smooth for the M-polyfold structures of X^1 and of TX and $p(A(x)) = x$ Chapter4/VectorFields · MPolyfold.IsScVectorField. The zero section takes at x the zero tangent vector in a chart chosen around x Chapter4/VectorFields · MPolyfold.zeroTangentPre Chapter4/VectorFields · MPolyfold.zeroVectorField; the tangent transition maps are linear in the fibre, so in every chart ψ around x it reads $(\psi(x), 0)$ Chapter4/VectorFields · MPolyfold.zeroTangentPre_transport Chapter4/VectorFields · MPolyfold.tangentChartMap_zeroVectorField. Hence it is continuous (the preimage of a basic open set of TX is a basic open set of X^1) Chapter4/VectorFields · MPolyfold.continuous_zeroVectorField and, in each chart, the sc-smooth map $z \mapsto (z, 0)$: the zero section is an sc-smooth vector field Chapter4/VectorFields · MPolyfold.isScVectorField_zero.

Raising the index (Lemma 4.4.4, local form, $i = 1$). *For a retraction r onto O, $T(O^1) = (TO)^1$* Chapter4/VectorFields · retractTangent_shift: a point (x, h) of $E_2 \oplus E_1$ lies in the tangent of the shifted retract iff $Tr^1(x, h) = (x, h)$, and the base point of $Tr^1(x, h)$ in $E_1 \oplus E_0$ is $Tr(x, h)$, because the derivative of the shifted retraction is the level-1 restriction of Dr Chapter1/ScSmooth · IsSc1.scDeriv_eqOn; the inclusion of levels being injective, the two fixed-point conditions agree. *If A is a vector field on (O, C, E), its level map $A_1 : E_2 \to E_1$ is a vector field on (O^1, C^1, E^1)* Chapter4/VectorFields · IsRetractVectorField.shift: shifting the sc-smooth map $x \mapsto (x, A(x))$ gives an sc-smooth map from O^2 into $(TO)^1$ Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift, which by the identity just proved is $T(O^1)$, and its level-1 values are $(x, A_1(x))$ Chapter1/ScSmooth · toBase_levelMap. Larger i by iteration; the M-polyfold form, through the identification $T(X^1) \cong (TX)^1$ of Proposition 2.3.17 (gap G.6, closed), is in §IV.29, for $i = 1$ and for every i.

The second derivative and its symmetry. For f sc² on U, $z \in U_2$ and $a, b \in E_1$, $D^2f(z)(a, b)$ is defined as the second component of $D(Tf)(z, a)(b, 0)$ — the derivative of $z \mapsto Df(z)a$ in the direction b Chapter4/VectorFields · scSecondDeriv. (The text writes $D^2r(x)(A(x), B(x))$ with $A(x), B(x) \in E_1$, where $r : U_2 \to E_0$ has a second derivative only in directions of E_2.) The vertical part of $D(Tf)$ is $D(Tf)(z, a)(0, l) = (0, Df(z)l)$ Chapter4/VectorFields · scDeriv_tangentMap_inr. *Symmetry* Chapter4/VectorFields · scSecondDeriv_symm: for U relatively open in a partial quadrant, $D^2f(z)(a, b) = D^2f(z)(b, a)$. (i) On E_2: $f \circ \iota : U_2 \to F_0$ is C^2 within U_2 (sc² gives C^2 on level 2), every point of U_2 lies in the closure of the interior of U_2 Chapter4/VectorFields · IsRelOpenInQuadrant.mem_closure_interior_lvlSet, and Mathlib's symmetry of second derivatives within a set applies Mathlib/Analysis/Calculus/FDeriv/Symmetric · ContDiffWithinAt.isSymmSndFDerivWithinAt; for $v, w \in E_2$ the iterated derivative within U_2 is $D^2f(z)(\iota v, \iota w)$ Chapter4/VectorFields · fderivWithin_fderivWithin_eq, by the derivative of Df along level 2 Chapter4/VectorFields · hasFDerivWithinAt_two Chapter4/VectorFields · hasFDerivWithinAt_scDeriv_apply. (ii) On E_1: $(a, b) \mapsto D^2f(z)(a, b)$ is continuous ($D(Tf)$ is continuous on level 0), $E_2 \times E_2$ is dense in $E_1 \times E_1$, and two continuous functions equal on a dense set are equal.

The bracket in a local model (§4.5.1). For a retraction r onto O and $x \in E_2$, $[A, B](x) = D\tilde{A}(x)B_1(x) - D\tilde{B}(x)A_1(x)$ with $\tilde{A} = A \circ r^1$, $\tilde{B} = B \circ r^1$ Chapter4/VectorFields · retractLieBracket; $[A, B] = -[B, A]$ Chapter4/VectorFields · retractLieBracket_antisymm. On O^2, $(A \circ r^1)_1 = A_1$, since r^1 fixes O^1 Chapter4/VectorFields · levelMap_comp_shift_eq. *The differentiated identity* Chapter4/VectorFields · IsRetractVectorField.diff_identity: for $x \in O_2$ and b with $Dr^1(x)b = b$, $D^2r(x)(A(x), b) + Dr(x)D\tilde{A}(x)b = D\tilde{A}(x)b$. The map $\alpha : y \mapsto (r^1(y), \tilde{A}(y))$ is sc-smooth on U^1 with values in TO (as $r^1(y) \in O^1$), so $Tr \circ \alpha = \alpha$ on all of U^1, TO being the fixed set of Tr. Differentiate at x in the direction b by the chain rule Chapter1/ChainRule · theorem_1_21'; the derivative of α is $(Dr^1(x)b, D\tilde{A}(x)b) = (b, D\tilde{A}(x)b)$, and $\alpha_1(x) = (x, A_1(x))$ since r^1 fixes x. Split $(b, D\tilde{A}(x)b) = (b, 0) + (0, D\tilde{A}(x)b)$: the first part gives $D^2r(x)(A(x), b)$ by definition, the second the vertical part $Dr(x)D\tilde{A}(x)b$; compare second components. *Tangency* Chapter4/VectorFields · retractLieBracket_mem_tangent: by Lemma 4.4.4, $B_1(x)$ and $A_1(x)$ are fixed by $Dr^1(x)$; apply the identity to $\tilde{A}$ in the direction $B_1(x)$ and to $\tilde{B}$ in the direction $A_1(x)$, subtract, and

cancel the second derivatives by symmetry: $Dr(x)[A,B](x) = [A,B](x)$.

Proposition 4.4.5 (preprint 4.26), local form `Chapter4/VectorFields · proposition_4_4_5_local_2021`. *Statement.* For vector fields A, B on (O,C,E) with retraction r: $(x,[A,B](x)) \in TO$ for $x \in O_2$, and $x \mapsto (x,[A,B](x))$ is an sc-smooth map from (O^2,C^2,E^2) to (TO,TC,TE). *Proof.* The first clause is tangency, T_xO being the fixed space of $Dr(x)$ at $x \in O_1$. For sc-smoothness between retracts it suffices to give an sc-smooth extension to U^2 that agrees on O^2 `Chapter2/Retracts · isScSmoothRetractMap_iff_exists`. First component: the inclusion $E^2 \to E^1$ composed with r^2. Second component: $y \mapsto D\tilde{A}(r^2y)(\tilde{B})_1(y) - D\tilde{B}(r^2y)(\tilde{A})_1(y)$. Each term is the second component of the tangent map $T\tilde{A}$, which is sc-smooth because $\tilde{A}$ is, composed with the sc-smooth pair $(r^2,(\tilde{B})_1)$ `Chapter4/VectorFields · isScSmooth_scDeriv_apply`; and $B_1 \circ r^2 = (\tilde{B})_1$ on U^2 `Chapter4/VectorFields · levelMap_shift_shift_eq`, so the extension is $[A,B]\circ r^2$. The statement on an M-polyfold, with Lemma 4.5.1, whose proof the text leaves to the reader, is in §IV.29 (gap G.38, closed).

The level of the bracket: an example `Chapter4/BracketLevel · lieBracket_level_counterexample_2021`. The text concludes from tangency that the bracket lies in "the fiber $(T_xO)_1$ whenever $x \in O_2$" (p. 200) and that for $x \in O_2$ it "belongs to the level 1" (p. 201). Read as $[A,B](x) \in E_1$, this does not hold as we read it. *Generalities.* For $O = E$ (no boundary, identity retraction `Chapter4/BracketLevel · idRetraction`), $TO = TE$ `Chapter4/BracketLevel · retractTangent_univ`; every sc-operator $S : E^1 \to E$ is a vector field `Chapter4/BracketLevel · isRetractVectorField_scOperator`, and the bracket of two is the commutator $S_0T_1 - T_0S_1$, the derivative of a linear map being itself `Chapter4/BracketLevel · retractLieBracket_scOperator`. *The space.* E is the weighted sequence space with $|x|_m = \sup_k (k+1)^m|x_k|$ `Chapter1/ScOneNotNormContinuous · ScOneExample.X`, level m being c_0 in the coordinates $(k+1)^m x_k$. *The operators.* $(Sx)_k = (-1)^k(k+1)x_k$ `Chapter4/BracketLevel · BracketLevel.opS`, on every level the diagonal $\mathrm{diag}((-1)^k)$ in the weighted coordinates; $(Tx)_k = (k+1)x_{k+1}$ `Chapter4/BracketLevel · BracketLevel.opT`, on level m the weighted shift with weights $((k+1)/(k+2))^{m+1} \le 1$ `Chapter4/BracketLevel · BracketLevel.wshift` `Chapter4/BracketLevel · BracketLevel.tw`; both are vector fields `Chapter4/BracketLevel · BracketLevel.isRetractVectorField_opS` `Chapter4/BracketLevel · BracketLevel.isRetractVectorField_opT`. *The computation.* $[S,T]x = ((-1)^k(k+1)(2k+3)x_{k+1})_k$ — the sign $(-1)^k$ makes the two terms add instead of cancelling; with polynomial weights the commutator of a non-oscillating diagonal operator and a shift would lose only one level. For $x_k = (k+1)^{-3}$, a point of E_2 `Chapter4/BracketLevel · BracketLevel.xPt`, the level-1 coordinates $(k+1)[S,T](x)_k$ have absolute value $P^3 + P^2 \ge 1/4$, $P = (k+1)/(k+2) \ge 1/2$, so they do not tend to 0 and $[S,T]x \notin E_1$ `Chapter4/BracketLevel · BracketLevel.bracket_not_mem_level_one`. The universal statement therefore fails `Chapter4/BracketLevel · not_forall_lieBracket_mem_level_one_2021`. Here $T_xO = E_0$, so $(T_xO)_1 = E_1$ and the sentence of p. 200 is covered too. What holds, and what Proposition 4.4.5 states, is the shifted statement $x \in O_{m+2} \Rightarrow [A,B](x) \in E_m$: $[A,B]$ is a tangent vector at the point x of level 1 of X^1 — the local form above. A first example on a weighted ℓ^2 space ($E_m = \ell^2$ with weights 4^{mk}, $S = \mathrm{diag}(2^k)$, $(Tx)_k = 2^k x_{k+1}$, $x_j = 4^{-j}/j$) is sketched in the header of `Chapter4/VectorFields` and not formalized, the project having no weighted ℓ^2 sc-structure.

What remains. (Lemma 4.4.4 and Proposition 4.4.5 on an M-polyfold and Lemma 4.5.1 are now §IV.29.) Definitions 4.4.1–4.4.2, d and $d \circ d = 0$, Definition 4.4.6, Theorem 4.4.7, Lemmas 4.4.8 and 4.5.2 (gap G.80); and the example the 2021 edition adds to Remark 4.1.3 (preprint 4.3), the space of its Subsection 1.4.4, for which 1^1_0 is not sc-smooth because Theorem 4.1.2 (preprint 4.2) would make it a manifold of constant local dimension (gap G.31; the formalized example of §IV.27 is a different one).

IV.29 The Springer 2021 edition: vector fields and the bracket on an M-polyfold (Lemmas 4.4.4, 4.5.1, Proposition 4.4.5)

The local forms of §IV.28 are carried to M-polyfolds in `Chapter4/VectorFieldsGlobal`, on top of the identification $\Phi : T(X^1) \cong (TX)^1$ of Part I's Proposition 2.3.17 `Chapter2/TangentLift · MPolyfold.tangentShiftEquiv` `Chapter2/TangentLift · proposition_2_3_17_2021`. The plan: Lemma 4.5.1 in a local model,

by the computation of §4.5.2; principal parts in charts and the independence of the chart; the bracket $X^2 \to TX$; Lemma 4.5.1 on M-polyfolds; Lemma 4.4.4 for $i = 1$ and for every i (gap G.38 closed; Findings, Chapter 4, items 160–164).

Lemma 4.5.1, local form (the computation of §4.5.2) `Chapter4/VectorFieldsGlobal · retractLieBracket_natural`. *Statement.* Let r, r' be retractions onto O, O', f an sc-smooth map from the retract O to the retract O', $g = f \circ r$, and A, B vector fields on O, A', B' on O' with $Dg(x)A(x) = A'(f(x))$ and $Dg(x)B(x) = B'(f(x))$ for $x \in O_1$. Then $Dg(x)[A,B](x) = [A',B'](f(x))$ for $x \in O_2$. The text states it for a local sc-diffeomorphism $\varphi : X \to X$ with the relations at points of X_2, and leaves the proof to the reader (p. 201); §4.5.2 is the computation, and it differentiates the relations, so they are assumed on the open set O_1 of level 1. *Proof.* (i) *The relation differentiated* `Chapter4/VectorFieldsGlobal · IsRetractVectorField.diff_related`: the map $y \mapsto Dg(r^1 y)\tilde{A}(y) - \tilde{A}'(g_1(y))$, $\tilde{A} = A \circ r^1$, $\tilde{A}' = A' \circ r'^1$, vanishes on U^1 (since $r^1 y \in O_1$ and $g \circ r = g$); differentiating at $x \in O_2$ in a direction b with $Dr^1(x)b = b$, as in §IV.28's differentiated identity, gives $D^2g(x)(A(x),b) + Dg(x)D\tilde{A}(x)b = D\tilde{A}'(g_2(x))Dg^1(x)b$. (ii) *The relation on level* 1 `Chapter4/VectorFieldsGlobal · IsRetractVectorField.related_shift`: the derivative of the shifted map g^1 at $x \in O_2$ is the level-1 restriction of $Dg(x)$ `Chapter1/ScSmooth · IsSc1.scDeriv_eqOn`, so $Dg^1(x)B_1(x) = B_1'(g_2(x))$; and $g_2 = f_2$ on O_2 `Chapter4/VectorFieldsGlobal · levelMap_levelMap_comp_eq`. (iii) Apply (i) to A with $b = B_1(x)$ and to B with $b = A_1(x)$ (both fixed by $Dr^1(x)$, §IV.28), rewrite the right sides by (ii), subtract, and cancel $D^2g(x)(A(x),B(x)) - D^2g(x)(B(x),A(x))$ by symmetry `Chapter4/VectorFields · scSecondDeriv_symm`: $Dg(x)[A,B](x) = D\tilde{A}'(f_2x)B_1'(f_2x) - D\tilde{B}'(f_2x)A_1'(f_2x) = [A',B'](f(x))$. §4.5.2 ends with "The proof of Proposition 4.5.1 is complete"; there is no Proposition 4.5.1, a possible misprint for Lemma 4.5.1.

On an open part. A transition map of charts is defined only on an open part $O \cap W$ of a model. A vector field on O is one on $O \cap W = r'(U')$ for the restricted retraction r' `Chapter4/VectorFieldsGlobal · IsRetractVectorField.restrict`, and at points of $(O \cap W)_2$ the two brackets agree, since r' is r on the smaller open domain U' and a derivative on an open subdomain is the restriction of the derivative `Chapter4/VectorFieldsGlobal · retractLieBracket_restrict`. So for f sc-smooth on $O \cap W$ with $Tf(z,A(z)) = (f(z),A'(f(z)))$ and $Tf(z,B(z)) = (f(z),B'(f(z)))$ on $(O \cap W)_1$, also $Tf(x,[A,B](x)) = (f(x),[A',B'](f(x)))$ on $(O \cap W)_2$ `Chapter4/VectorFieldsGlobal · retractLieBracket_natural_inter`.

Principal parts and the independence of the chart. Tangent charts change by $T\psi = T(\psi \circ \varphi^{-1}) \circ T\varphi$ over the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_change` (formula (2.9)). Choose a retraction onto the model of each chart `Chapter4/VectorFieldsGlobal · MPolyfold.chartRetraction` `Chapter4/VectorFieldsGlobal · MPolyfold.chartRetraction_retract`. The *principal part* of a vector field A in φ is the second component of $T\varphi(A(\varphi^{-1}z))$ `Chapter4/VectorFieldsGlobal · MPolyfold.principalPart`, so $T\varphi(A(y)) = (\varphi(y), A_\varphi(\varphi(y)))$ for y of level 1 in the domain of φ `Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_eq_principalPart`; read through the charts φ^1 of X^1 and $T\varphi$ of TX, the sc-smoothness of A says that A_φ is a vector field on the model `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.isRetractVectorField_principalPart`. The principal parts in two charts are related by the transition map on level 1 of the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.principalPart_related`. The bracket of the principal parts `Chapter4/VectorFieldsGlobal · MPolyfold.chartLieBracket` satisfies the local Proposition 4.4.5 `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.chartLieBracket_spec`, and — the use of Lemma 4.5.1 that p. 201 names — the brackets in φ and ψ are related by $T(\psi\varphi^{-1})$ on level 2 of the overlap `Chapter4/VectorFieldsGlobal · MPolyfold.IsScVectorField.chartLieBracket_change`: the overlap is $O \cap W$ for an open W, and the local form on an open part applies.

Proposition 4.4.5 (preprint 4.26) on an M-polyfold `Chapter4/VectorFieldsGlobal · proposition_4_4_5_2021`. *Statement.* For sc-smooth vector fields A, B on X, the bracket $[A,B] : X^2 \to TX$ is sc-smooth and $p([A,B](x)) = x$. The text's "sc-smooth section of $TX|X^2 \to X^2$" is read as an sc-smooth map $X^2 \to TX$ covering the inclusion $X^2 \to X^1$; the pull-back bundle itself is not built (gap G.92). *Construction.* At $x \in X^2$ take the chart φ chosen around x; the level lift of x in the chart $(\varphi^1)^1$ of X^2 lies in O_2 and is $\varphi(x)$ on level 1 `Chapter4/VectorFieldsGlobal · MPolyfold.levelLift_levelChart_spec`, so $(\varphi(x),[A_\varphi,B_\varphi](\varphi(x)))$ lies in TO and defines a tangent vector at x `Chapter4/VectorFieldsGlobal · MPolyfold.lieBracketPre` `Chapter4/`

VectorFieldsGlobal · MPolyfold.lieBracket over x Chapter4/VectorFieldsGlobal · MPolyfold.tangentProj_lieBracket. *Independence of the chart* Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_lieBracket: in every chart ψ around x, $T\psi([A,B](x)) = (\psi(x), [A_\psi, B_\psi](\psi(x)))$, by the change of tangent charts and the relation of the brackets in φ and ψ. *Proof of the statement.* On the domain of $(\psi^1)^1$, $T\psi \circ [A,B]$ is the section $z \mapsto (z, [A_\psi, B_\psi](z))$ composed with the chart Chapter4/VectorFieldsGlobal · MPolyfold.tangentChartMap_lieBracket_eq. That section is continuous and sc-smooth by the local proposition, so $[A,B]$ is continuous (the preimage of a basic open set of TX is open in each chart domain of X^2) Chapter4/VectorFieldsGlobal · MPolyfold.continuous_lieBracket, and sc-smooth in the charts $(\psi^1)^1$ of X^2 and $T\psi$ of TX.

Lemma 4.5.1 on M-polyfolds Chapter4/VectorFieldsGlobal · lemma_4_5_1_2021. *Statement.* Let $f : X \to Y$ be sc-smooth, with level maps $f_1 : X^1 \to Y^1$ Chapter4/VectorFieldsGlobal · MPolyfold.liftLevelOneMap and $f_2 : X^2 \to Y^2$ Chapter4/VectorFieldsGlobal · MPolyfold.levelTwoMap, and A, B on X, A', B' on Y sc-smooth vector fields with $Tf \circ A = A' \circ f_1$ and $Tf \circ B = B' \circ f_1$ on X^1. Then $Tf \circ [A,B] = [A',B'] \circ f_2$ on X^2. This is more general in the map than the text's local sc-diffeomorphism $X \to X$; one between open subsets of X is covered only through such an f, open subsets of an M-polyfold not being built as M-polyfolds (gap G.92). *Proof.* Fix $x \in X^2$, a chart φ around x and ψ around $f(x)$. Both sides lie over $f(x)$; in $T\psi$ the right side is $(\psi f(x), [A'_\psi, B'_\psi](\psi f(x)))$ by the independence of the chart, and on level 1, $(\psi f \varphi^{-1})_1(\varphi(x)) = \psi(f(x))$ Chapter4/VectorFieldsGlobal · MPolyfold.levelMap_coord_levelLift. The left side, through φ, is $T(\psi f \varphi^{-1})(\varphi(x), [A_\varphi, B_\varphi](\varphi(x)))$. The map $\psi f \varphi^{-1}$ is sc-smooth on the open part of the model of φ that $\varphi(f^{-1}(\text{domain of } \psi))$ cuts out, and the hypotheses read in the charts say that it relates A_φ, B_φ to A'_ψ, B'_ψ on level 1 there; the local form on an open part gives the equality.

Lemma 4.4.4 on an M-polyfold, $i = 1$ Chapter4/VectorFieldsGlobal · lemma_4_4_4_2021. *Statement.* For an sc-smooth vector field A, $A^{(1)} = \Phi^{-1} \circ A_1 : X^2 \to T(X^1)$ Chapter4/VectorFieldsGlobal · MPolyfold.raiseIndex, $A_1 : X^2 \to (TX)^1$ the level-1 map of A Chapter4/VectorFieldsGlobal · MPolyfold.vectorFieldLevelOne, is an sc-smooth vector field on the M-polyfold X^1, and $\Phi(A^{(1)}(x)) = A(x)$ in $(TX)^1$. *Proof.* A_1 is sc-smooth, as the level-1 map of the sc-smooth A; Φ^{-1} is sc-smooth (Proposition 2.3.17), so $A^{(1)}$ is; Φ commutes with the projections to X^2, so $p(A^{(1)}(x)) = x$ follows from $p(A(x)) = x$; and $\Phi \circ \Phi^{-1}$ is the identity. Every i follows by applying the lemma to X^1, X^2, …, as the next paragraph does. A witness for the hypotheses of the section: the zero section Chapter4/VectorFields · MPolyfold.isScVectorField_zero.

Lemma 4.4.4 on an M-polyfold, every i Chapter2/TangentLiftAll · lemma_4_4_4_all_2021. To state the lemma once for all i, the M-polyfolds are bundled with their spaces Chapter2/TangentLiftAll · BundledMPolyfold, so that the lifts $X^0 = X$, $X^{i+1} = (X^i)^1$ form a family Chapter2/TangentLiftAll · BundledMPolyfold.lift. Part I's Proposition 2.3.17 for every i Chapter2/TangentLiftAll · proposition_2_3_17_all_2021 gives the sc-diffeomorphisms $\Phi_i : T(X^i) \to (TX)^i$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv and $J_i : (X^1)^i \to (X^i)^1$ Chapter2/TangentLiftAll · BundledMPolyfold.shiftLiftEquiv, and Φ_i commutes with the inclusions into TX. *Statement.* With $A^{(0)} = A$ and $A^{(i+1)} = (A^{(i)})^{(1)}$, the case $i = 1$ on X^i Chapter2/TangentLiftAll · BundledMPolyfold.raiseIndexAll: for every i, $A^{(i)}$ is an sc-smooth vector field on X^i; its image in TX at $x \in X^{i+1}$ is $A(x)$ Chapter2/TangentLiftAll · BundledMPolyfold.tangentIncl_raiseIndexAll; and $\Phi_i \circ A^{(i)} \circ J_i$ is the i-th index raise of A, viewed as an sc-smooth map $X^1 \to TX$ Chapter2/TangentLiftAll · BundledMPolyfold.vectorFieldScMap Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv_raiseIndexAll — the diagram printed before the lemma. *Proof.* The first claim is the case $i = 1$, applied i times. The second by induction: the case $i = 1$ for X^i says that $A^{(i+1)}(x)$ and $A^{(i)}(x)$ agree after $T(X^{i+1}) \to T(X^i)$. For the third, both sides are compared after the inclusion $(TX)^i \to TX$, which is injective: on the left it is the inclusion $T(X^i) \to TX$ after Φ_i, so the second claim applies; on the right it is A on the underlying point. *Witness:* the zero vector field, for every i Chapter2/TangentLiftAll · BundledMPolyfold.lemma_4_4_4_all_2021_zero (Findings, Chapter 4, item 164).

What remains. The pull-back bundle $TX|X^2 \to X^2$ as a bundle object, and Lemma 4.5.1 for a local sc-diffeomorphism between open subsets (gap G.92); and the forms of §4.4 (gap G.80, §IV.28).

Part V. The Fredholm package for M-polyfolds (HWZ Chapter 5, §§5.1–5.3, 5.5 (preprint §§5.1–5.4))

Chapter 5 turns the implicit function theorems of Chapter 3 into the working tools of a Fredholm theory: a way to measure perturbations (auxiliary norms, §5.1), the compactness that keeps the solution sets of small perturbations under control (§5.2), a supply of perturbations built from sc-smooth bump functions (the start of §5.3), and the perturbation theorems themselves — transversality when there is no boundary (Theorem 5.3.5 (preprint 5.13)), general position (Theorem 5.3.10 (preprint 5.18)), good position relative to the boundary (Theorem 5.3.12 (preprint 5.19) and Theorem 5.25 (preprint; not in the 2021 edition)), and the lemmas of the Morse-type Theorem 5.3.15 (preprint 5.22). §5.2 (preprint Appendix 5.4.1) proves Theorem 5.2.2 (preprint 5.5); §5.5 (preprint Appendix 5.4.2) relates sc-smooth bump functions and partitions of unity to the geometry of Banach spaces. This Part retells §5.1 (§§V.1–V.3); §5.2, with the proof that the preprint gives in Appendix 5.4.1 (§§V.4–V.7), the longest retelling being Theorem 5.2.2 (preprint 5.5) (§§V.5–V.6); the start of §5.3 — bump functions, the calculus of sections, Lemma 5.3.3 (preprint 5.11), allowable sections (§§V.8–V.10); Theorem 5.3.5 (preprint 5.13) (§§V.11–V.15); general position, Lemma 5.26 (preprint; not in the 2021 edition) and Theorem 5.3.10 (preprint 5.18) (§§V.16–V.20); Theorem 5.3.12 (preprint 5.19) and Theorem 5.25 (preprint) (§§V.21–V.22); the homotopy lemmas (§V.23); §5.5 (preprint Appendix 5.4.2) (§§V.24–V.26); what is not formalized (§V.27); and what the formalization taught us (§V.28). A fourth addendum adds Corollary 5.2.5 (preprint 5.8) on the shifted strong bundle (§V.29), the metric space of the allowable sections (§V.30) and weak against pointwise bump functions (§V.31).

What is proved, and what is not. Theorems 5.2.2, 5.2.3, 5.2.4, 5.3.5, 5.3.10, 5.3.12, 5.5.7, 5.5.10 (preprint 5.5, 5.6, 5.7, 5.13, 5.18, 5.19, 5.34, 5.37) and Theorem 5.25 (preprint) are proved, Theorems 5.2.2–5.2.4 (preprint 5.5–5.7) without tameness; so are Lemmas 5.1.2, 5.3.3, 5.3.16, 5.3.17 (preprint 5.2, 5.11, 5.23, 5.24) and Lemma 5.26 (preprint), Propositions 5.1.3, 5.3.2, 5.5.3, 5.5.8, 5.5.11, 5.5.15 (preprint 5.3, 5.10, 5.30, 5.35, 5.38, 5.42), Remark 5.3.4 (preprint 5.12) and Corollary 5.5.17 (preprint 5.44), the last with Theorem 5.5.16 (preprint 5.43) as an explicit hypothesis. Every definition of the chapter is formalized. Corollary 5.2.5 (preprint 5.8) is proved on the shifted strong bundle $P^1 : Y^1 \to X^1$, which the text uses without defining it, and by iteration for every fixed i, with the filler condition on level 1 that its "it is a priori clear" hides (§V.29). The perturbation theorems take f with *tame germ data* (Definition 3.6.4 (preprint 3.51)), the reading of the tame strong bundle with which §5.3 opens and the hypothesis under which the §3.6 implicit function theorems are proved since their repair (Part III, Findings of Chapter 3, items 190–199); Theorem 5.3.10 (preprint 5.18) also assumes X tame, which its proof uses through Theorems 4.2.11 and 4.3.1 (preprint 4.22 and 4.23). The metric space $(\Gamma_U^{+,1}(P), \rho)$ is built, and the text's "not complete" turns out to need a hypothesis (§V.30). Not formalized: Theorem 5.3.15 (preprint 5.22) and Theorem 5.27 (preprint; not in the 2021 edition), Remarks 5.3.6, 5.3.8, 5.3.14 (preprint 5.14, 5.16, 5.21), the cited Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43), the Sobolev remark, a dense form of Theorem 5.3.12 (preprint 5.19), and Corollary 5.2.5 (preprint 5.8) as one statement for a family indexed by i (§V.27).

Four structural choices. (1) *The linearization is read in charts.* T_xX carries no linear structure in the formalization (gap G.7 of Chapter 2), so $f'(x)$ is the operator $\mathrm{Lin}(z) : T_zO \to K_z$ of a strong bundle chart (Part III), transversality is surjectivity of Lin in a chart, and general position is stated in a tame chart; its chart independence is proved where it matters (§V.18). (2) *"Generic" is "off a null set".* Every perturbation theorem is proved first in a generic form — for λ outside a Lebesgue null set of a small ball, $s_0 + \sum \lambda_i t_i$ does the job — and the text's dense-set form is derived from it. (3) *The parametrized solution set is not built as a manifold.* The Sard step is local: at each zero of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in good position, Theorem 3.6.8 (2) (preprint 3.54) gives a regular tangent germ chart, Sard's theorem is applied on its finite-dimensional model, and finitely many null sets are collected (§V.14). Where Theorem 4.3.1 (preprint 4.23) needs a submanifold, the *bridge lemma* supplies it (§V.17). (4) *Density is pointwise.* "Within ε" is $N(s(x) - s'(x)) \le \varepsilon$ for all x; this is $\rho(s, s') \le \varepsilon$ for the metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$, which the perturbation theorems never needed and which was built afterwards (§V.30).

Map.

- `Chapter5/AuxNorm`: the charts $\Phi[1]$, the fibre $(Y_{0,1})_x$ and its chart norms, Definition 5.1.1 (preprint 5.1), Lemma 5.1.2 (preprint 5.2), Proposition 5.1.3 (preprint 5.3).
- `Chapter5/Compactness`: Definition 5.2.1 (preprint 5.4), Theorems 5.2.2–5.2.4 (preprint 5.5–5.7), Corollary 5.2.5 (preprint 5.8) in a chart, Remark 5.3.4 (preprint 5.12).
- `Chapter5/BumpFunctions`: $\mathbb{R}$ as an M-polyfold, Definitions 5.3.1, 5.5.2, 5.5.6 (preprint 5.9, 5.29, 5.33), Propositions 5.3.2, 5.5.3, 5.5.8 (preprint 5.10, 5.30, 5.35), Theorem 5.5.7 (preprint 5.34), the calculus of sections, Lemma 5.3.3 (preprint 5.11), $\Gamma^{+,1}_U(P)$.
- `Chapter5/Transversality`: Definitions 5.3.7, 5.3.9 (preprint 5.15, 5.17), Lemma 5.26 (preprint), the parametrized section, the Sard step, Theorem 5.3.5 (preprint 5.13).
- `Chapter5/GeneralPosition`: the bridge lemma, $T_yS = \ker f'(y)$, chart independence and openness of general position, properties (i)–(ii), Theorems 5.3.10, 5.3.12 (preprint 5.18, 5.19) and Theorem 5.25 (preprint).
- `Chapter5/Homotopy`: Definition 5.3.13 (preprint 5.20), Lemmas 5.3.16, 5.3.17 (preprint 5.23, 5.24).
- `Chapter5/PartitionsOfUnity`: the algebra of sc-smooth functions, Definitions 5.5.1, 5.5.4, 5.5.9, 5.5.12, 5.5.14 (preprint 5.28, 5.31, 5.36, 5.39, 5.41), Theorem 5.5.10 (preprint 5.37), Propositions 5.5.11, 5.5.15 (preprint 5.38, 5.42), Corollary 5.5.17 (preprint 5.44), the statements of Theorems 5.5.13, 5.5.16 (preprint 5.40, 5.43).
- Added in the fourth addendum (§§V.29–V.31): `Chapter5/StrongBundleShift` (the shifted strong bundle P^1, raising the index of sc-smooth maps and sections), `Chapter5/Corollary58` (the filler condition on every level, Corollary 5.2.5 (preprint 5.8)), `Chapter5/AllowableSectionsMetric` (the metric ρ, completeness and its failure), `Chapter5/BumpWeakPointwise` (bump points, an sc-Banach space as an M-polyfold).

Throughout, $P : Y \to X$ is a strong bundle over an M-polyfold (both paracompact and Hausdorff), a section is a map $s : X \to Y$ with $P \circ s = \mathrm{id}$, 0_x is the zero of the fibre over x, and $\operatorname{supp} s$ is the closure of $\{s \neq 0\}$ `Chapter5/BumpFunctions · sectionSupport`. A strong bundle chart Φ has base chart $\varphi : V \to O$, $O = r(U)$ a retract, and local bundle $K \to O$; the *principal part* of f is $\hat{f}(u) = \Phi(f(\varphi^{-1}(u)))_2$, and $\mathrm{Lin}(z)$ is its linearization at a level-1 point z (Part III). $|\lambda|$ is the maximum norm on $\mathbb{R}^m$.

V.1 The fibre $Y_{0,1}$ and the charts $\Phi[1]$

§5.1 works on the part $Y_{0,1}$ of biregularity $(0, 1)$, the M-polyfold $Y[1]$ of §2.6 (preprint §2.5) (Part II), and assumes two things the project did not have: charts $\Phi[1]$ for *every* strong bundle chart, and a Banach structure on the fibres of $Y_{0,1}$.

$Y_{0,1}$ in any chart. For a strong bundle chart Φ compatible with the atlas — the text's "strong bundle isomorphism $\Phi : Y|V \to K$" — a point y over $\operatorname{dom}\varphi$ lies in $Y_{0,1}$ iff $\Phi(y)_2 \in F_1$: compare with a chart of the atlas through the transition map, which is a strong bundle map and so preserves biregularity `Chap-`

ter5/AuxNorm · StrongBundle.mem_oneSet_iff_of_isChart. On $P^{-1}(V) \cap Y_{0,1}$ the map $w \mapsto \Phi(w)$ lifted to $K[1]$ is then a partial homeomorphism Chapter5/AuxNorm · StrongBundle.oneChartOf, compatible with every chart $\Phi'[1]$ of the atlas because the transition is the $[1]$-transition of Definition 2.6.4 (preprint 2.50) Chapter5/AuxNorm · StrongBundle.oneChartOf_transition Chapter5/AuxNorm · StrongBundle.oneChartOf_compatible, hence a chart of the M-polyfold $Y[1]$ Chapter5/AuxNorm · StrongBundle.isChart_oneChartOf. For a chart of the atlas it is the chart of Part II Chapter5/AuxNorm · StrongBundle.oneChart_eq_oneChartOf.

The fibre. The fibres $P^{-1}(x)$ carry linear structures; the points of $P^{-1}(x)$ in $Y_{0,1}$ form a linear subspace $(Y_{0,1})_x$, because the charts are linear on the fibres and $F_1 \subseteq F_0$ is a subspace Chapter5/AuxNorm · StrongBundle.oneFiber. Its points are points of $Y[1]$ and conversely Chapter5/AuxNorm · StrongBundle.toOne Chapter5/AuxNorm · StrongBundle.ofOne; its zero is the point 0_x of $Y[1]$ Chapter5/AuxNorm · StrongBundle.zeroOne Chapter5/AuxNorm · StrongBundle.fiberZero_mem_oneSet.

The chart norm $|h|_1$. In a chart around x, the fibre component of $\Phi[1]$ is a linear map $(Y_{0,1})_x \to F_1$ Chapter5/AuxNorm · StrongBundle.liftOneFiber, injective because Φ is injective on the fibre Chapter5/AuxNorm · StrongBundle.liftOneFiber_injective. Its image is the slice $\{h \in F_1 \mid (\varphi(x), h) \in K[1]\}$ Chapter5/AuxNorm · StrongBundle.mem_range_liftOneFiber, the fixed point set of the continuous map $h \mapsto R[1](\varphi(x), h)$, hence closed in F_1 Chapter5/AuxNorm · StrongBundle.isClosed_slice. So a sequence of the fibre that is Cauchy for $|\Phi[1](\cdot)_2|_1$ converges: its images converge in F_1, the limit stays in the slice, and it is the image of a point of the fibre Chapter5/AuxNorm · StrongBundle.complete_liftOneFiber. This is the text's "the fibers Y_x have the structure of Banach spaces". The norm depends on the chart, but only up to equivalence, and locally uniformly: by Lemma 5.1.2's (preprint 5.2) upper bound (§V.2) applied to $g = |\Phi_1[1](\cdot)_2|_1$, the norms of two charts are comparable with one constant on the fibres over a neighbourhood of x Chapter5/AuxNorm · StrongBundle.exists_norm_liftOne_le_mul, and so are finitely many charts and one more Chapter5/AuxNorm · StrongBundle.exists_uniform_comparison.

V.2 Auxiliary norms and Lemma 5.1.2 (preprint 5.2)

Definition 5.1.1 (preprint 5.1). A function $N : Y[1] \to \mathbb{R}$ is a *complete norm on the fibre over x* if $N(v) = 0$ only for $v = 0$, $N(tv) = |t|N(v)$, $N(v + w) \le N(v) + N(w)$, and every N-Cauchy sequence of $(Y_{0,1})_x$ converges Chapter5/AuxNorm · StrongBundle.IsCompleteFiberNorm. An *auxiliary norm* is a continuous such N, complete on every fibre, with condition (2): if $P(w_k) \to x$ and $N(w_k) \to 0$, then $w_k \to 0_x$ in $Y[1]$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm. It is non-negative Chapter5/AuxNorm · StrongBundle.IsAuxNorm.nonneg and homogeneous on the fibres Chapter5/AuxNorm · StrongBundle.IsAuxNorm.smul.

Lemma 5.1.2 (preprint 5.2), the upper bound. Let $g : Y[1] \to \mathbb{R}$ be continuous at 0_x and positively homogeneous on the fibres near x, and Φ a chart around x. By continuity at 0_x — read through the chart $\Phi[1]$, whose neighbourhoods of $(\varphi(x), 0)$ are the sets $|p - \varphi(x)|_0 + |h|_1 < \varepsilon$ — $g < 1$ where $|p - \varphi(x)|_0 < \varepsilon/2$ and $|h|_1 \le \varepsilon/2$; homogeneity scales every vector of a nearby fibre to $|h|_1 = \varepsilon/2$, which gives $g(w) \le (2/\varepsilon)|h|_1$ on the fibres over a neighbourhood V of x Chapter5/AuxNorm · StrongBundle.exists_le_mul_norm_liftOne. This is the text's first half of "(i) □ (ii)".

Lemma 5.1.2 (preprint 5.2), the lower bound. For an auxiliary norm, $|h|_1 \le A\,N(w)$ near x Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le. Suppose not: then for every k there is w_k over $\varphi^{-1}(B(\varphi(x), 1/(k+1)))$ with $|h_k|_1 > (k+1)N(w_k)$; rescale to $|h_k|_1 = 1$. Then $N(w_k) \to 0$ and $P(w_k) \to x$ — the points of the shrinking balls converge to x because φ^{-1} is continuous on O, so no metrizability of X is needed — hence $w_k \to 0_x$ by condition (2), and $\Phi[1](w_k) \to (\varphi(x), 0)$ by the continuity of $\Phi[1]$, contradicting $|h_k|_1 = 1$. Together: **in every chart around** x there are a neighbourhood V and $0 < a < A$ with $aN(w) \le |h|_1 \le AN(w)$ over V Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_bounds.

Lemma 5.1.2 (preprint 5.2). For N continuous and a complete norm on every fibre, N is an auxiliary norm iff every x has a chart and such two-sided bounds near x Chapter5/AuxNorm · lemma_5_2. "(i) □ (ii)" is the previous paragraph. "(ii) □ (i)", "obvious" in the text: if $P(w_k) \to x$ and $N(w_k) \to 0$, the bound

gives $|h_k|_1 \to 0$, so $\Phi[1](w_k) \to (\varphi(x), 0)$, and $w_k \to 0_x$ because $\Phi[1]^{-1}$ is continuous on $K[1]$ (§V.1). In the statement V is a neighbourhood of x on which the bounds hold; the text's V is the domain of Φ, but its own proof gives them "for $P(w)$ close to x".

V.3 Proposition 5.1.3 (preprint 5.3)

Existence. For the charts Φ_i of the atlas and a partition of unity (ρ_i) subordinate to their domains — it exists since X is paracompact and Hausdorff, hence normal `Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate` — the *glued norm* is $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)_2|_1$ `Chapter5/AuxNorm · StrongBundle.gluedNorm` `Chapter5/AuxNorm · StrongBundle.gluedNorm_eq_sum`. It is continuous on $Y[1]$: each term vanishes off the closed set $P^{-1}(\operatorname{supp}\rho_i)$, which lies in the open domain of $\Phi_i[1]$, where the term is continuous, and the sum is locally finite `Chapter5/AuxNorm · StrongBundle.continuous_gluedNorm`. Near a point x only finitely many ρ_i are non-zero, all with x in the domain of Φ_i and summing to 1 `Chapter5/AuxNorm · StrongBundle.sum_partition_eq_one`; their chart norms are uniformly comparable with the norm of any one chart Φ' around x (§V.1), so N is two-sided comparable with $|\Phi'[1](\cdot)_2|_1$ near x `Chapter5/AuxNorm · StrongBundle.gluedNorm_bounds`. On each fibre N is a convex combination of norms, hence a norm, and comparable with a complete norm, hence complete `Chapter5/AuxNorm · StrongBundle.complete_of_bounds` `Chapter5/AuxNorm · StrongBundle.isCompleteFiberNorm_gluedNorm`. Lemma 5.1.2 (preprint 5.2) "(ii) □ (i)" concludes `Chapter5/AuxNorm · proposition_5_3_exists`. The text gives this as one sentence.

Comparison. For auxiliary norms N_0, N_1, Lemma 5.1.2 (preprint 5.2) "(i) □ (ii)" for both in the *same* chart around y — which is why "(i) □ (ii)" was proved in every chart — gives $r_y > 0$ with $r_y N_0 \le N_1$ and $r_y N_1 \le N_0$ over a neighbourhood of y. At each point the admissible constants r form a convex set (r admissible and $0 < r' \le r$ give r' admissible), so a partition of unity glues the local constants into a continuous $f > 0$ with values in these sets `Mathlib/Analysis/Convex/PartitionOfUnity · exists_continuous_forall_mem_convex_of_local_const`, and $f(x)N_0 \le N_1 \le f(x)^{-1}N_0$ on the fibre over x `Chapter5/AuxNorm · proposition_5_3_compare`. Tameness plays no role in §5.1.

V.4 Definition 5.2.1 (preprint 5.4) and proper sections

"$N(y) \le r$" for a point y of Y is read with the text's convention $N(y) = \infty$ off $Y_{0,1}$: it means $y \in Y_{0,1}$ and $N(y) \le r$ `Chapter5/Compactness · StrongBundle.AuxLe`, a property of the point only `Chapter5/Compactness · StrongBundle.AuxLe.congr`. A section f is *weakly proper* if $f^{-1}(0)$ is compact and for every auxiliary norm N there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc⁺-section s with support in U and $N(s) \le 1$ `Chapter5/Compactness · StrongBundle.IsWeaklyProper`; *proper* if instead $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le 1\}$ is compact `Chapter5/Compactness · StrongBundle.IsProper`.

Proper implies weakly proper for a continuous f `Chapter5/Compactness · StrongBundle.IsProper.isWeaklyProper`. Take the U of properness. A solution of $f(x) = s(x)$ outside U has $s(x) = 0$, so $f(x) = 0$ and $x \in f^{-1}(0) \subseteq U$ after all; inside U, $f(x) = s(x)$ gives $N(f(x)) = N(s(x)) \le 1$. So $\{f = s\}$ lies in the compact closure, and it is closed because f and s are continuous and Y is Hausdorff. Weakly proper sections have compact solution sets by definition `Chapter5/Compactness · StrongBundle.IsWeaklyProper.isCompact_solutionSet`. The text's "obviously" hides the continuity of f.

V.5 Theorem 5.2.2 (preprint 5.5): the local form for basic germs

Theorem 5.2.2 (preprint 5.5) says that for sc-Fredholm sections a compact solution set already forces properness. The appendix proves it in the coordinates of a basic germ, and that local statement is the heart of the matter.

The setting. A basic germ (Definition 3.1.7 (preprint 3.7), Part III) is $h : \tilde{C} \oplus W \supseteq U_0 \to \mathbb{R}^N \oplus W$ whose W-component is $w - B(a,w)$, where B is an sc^0-contraction germ: on small balls, $w \mapsto B(a,w)$ contracts with constant $\leq 1/2$ on level 0, uniformly in a, and $B(0,0) = 0$. Part III built the *parametrized fixed point* $\delta(a,z)$, the unique solution w in a closed ball $\bar{B}(r)$ of $w = B(a,w) - z$ for (a,z) in a set Σ of small parameters, continuous in (a,z) on level 0 `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_continuousOn` `Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero`.

The local form `Chapter5/Compactness · IsSc0ContractionGerm.exists_isCompact_bdd`. Assume $\tilde{C}$ closed. There is an open $O \ni 0$, *independent of c*, such that for every c the points $x = (a,w) \in U_0 \cap O$ with $w - B(a,w) \in W_1$ and $|w - B(a,w)|_1 \leq c$ lie in a compact set. The proof:

1. *A small ball.* $B(a,w) - w$ is continuous at 0 within U_0 and vanishes there, so for $|(a,w)|_0 < \rho_1$ it has norm $< r/8$. Let $\rho = \min(\rho_1, \rho'/2, r/2)$, with ρ' the radius of the parameter set, and $O = B(0,\rho)$.
2. *The compact set of parameters.* The inclusion $W_1 \to W_0$ is compact, so the image of the closed level-1 ball of radius c lies in a compact $K_c \subseteq W_0$. The set $L = (\tilde{C} \cap \bar{B}(\rho)) \times (K_c \cap \bar{B}(r/8))$ is compact (here $\tilde{C}$ closed is used) and lies in Σ.
3. *The candidate.* The compact set is the image of L under the continuous map $(a,z) \mapsto (a, \delta(a,z))$.
4. *Every bounded point is hit.* Let $x = (a,w) \in U_0 \cap O$ with $w - B(a,w) = \iota(z_1)$, $|z_1|_1 \leq c$. Put $p = (a, B(a,w) - w)$. Then $a \in \tilde{C}$ with $|a| < \rho$; $B(a,w) - w = \iota(-z_1)$ lies in K_c and has norm $< r/8$ by step 1; so $p \in L$. And w lies in the ball $\bar{B}(r)$ and solves $w = B(a,w) - (B(a,w) - w)$, so $w = \delta(p)$ by uniqueness `Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero`, i.e. x is the image of p.

Only the level-0 contraction is used. For a basic germ f itself the W-component of $f(x)$ is $w - B(a,w)$, so the same O works with the level-1 bound $|f(x)|_1 \leq c$ `Chapter5/Compactness · IsBasicGerm.exists_isCompact_bdd`. This is the appendix's (82)–(90), where the text writes "$a \in \mathbb{R}^n$" for $a \in \tilde{C}$ (B is defined on $\tilde{C} \oplus W$ only), "$\mathbb{R}^n \oplus Y_1$" for $\mathbb{R}^N \oplus W_1$, and in (89) mixes the $\mathbb{R}^N$- and W-components; the t of its "$h + t$" and its estimates (83), (85) disappear in the next step.

V.6 Theorem 5.2.2 (preprint 5.5): the local step and the covering

Uniform fibre bounds. A strong bundle map Ψ (Definition 2.6.1 (preprint 2.47)) is linear on the fibres and $\Psi[1]$ is continuous; at $(u_0, 0)$, where $\Psi[1](u_0,0)_2 = 0$ `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_zero`, continuity gives $|\Psi[1](u,k)_2|_1 \leq 1$ for $|u - u_0|_0 < \delta$, $|k|_1 \leq \eta$, and fibre linearity `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_smul` turns this into $|\Psi[1](u,k)_2|_1 \leq \eta^{-1}|k|_1$ for all $k \in F_1$ `Chapter5/Compactness · IsStrongBundleMap.exists_fiber_bound_one`. The text uses this bound silently when it replaces $N(f(x)) \leq 1$ by a bound in basic germ coordinates.

The local step `Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd`. Let (f, x_0) be an sc-Fredholm germ (Definition 3.1.11 (preprint 3.11)) — x_0 need not be a zero. Its data, in the text's germ reading (§III.6): a strong bundle chart Φ centred at x_0, a filling germ g of the principal part $\hat{f}$ on a germ inside the domain U of the retraction, an sc$^+$-germ t_0 on a germ G_s, and a strong bundle isomorphism Ψ with base map ψ conjugating $g - t_0$ to a basic germ h: $h(\psi(u)) = \Psi(u, g(u) - t_0(u))_2$ for $u \in U$ near 0. The claim is an open $V \ni x_0$, independent of C, such that for every C the closure in X of $Z_C = \{x \in V \mid |\Phi(f(x))_2|_1 \leq C\}$ is compact.

1. *The neighbourhood.* Near 0 the filler agrees with the principal part on O and the conjugation identity holds (a set N_1); t_0 is continuous into F_1 on level 0 of its germ, $(G_s)_0 = O_s \cap C$ with O_s open — not on U — so the neighbourhood is also cut with O_s, and $|t_0(u)|_1 \leq |t_0(0)|_1 + 1$ for $u \in U \cap O_s$, $|u|_0 < \delta_2$ (with t_0 sc-smooth on U the cut was not needed); the fibre bound of Ψ holds for $|u|_0 < \delta_3$ with constant M. On the side of the basic germ, the local form (§V.5) gives an open O'; ψ maps a neighbourhood of 0 into a closed ball $\bar{B}(r_0) \cap C'$ inside the domain U' of ψ^{-1} (relative openness of U'). V is the set of points of the chart domain whose image lies in all of these.

2. *The bound transported.* For $x \in Z_C$, $u = \varphi(x) \in O$, $g(u) = \hat{f}(u)$ lies in F_1 with $|g(u)|_1 \le C$, so $|g(u) - t_0(u)|_1 \le C + |t_0(0)|_1 + 1$, and $|h(\psi(u))|_1 = |\Psi[1](u, g(u) - t_0(u))_2|_1 \le M(C + |t_0(0)|_1 + 1) =: c'$. *Subtracting the sc⁺-part before the conjugation* is what makes the text's t and its estimates unnecessary, and the neighbourhood independent of C.
3. *Compactness on the side of the basic germ.* So $\psi(\varphi(Z_C))$ lies in the compact set $K_{c'}$ of the local form, intersected with the closed ball $\bar{B}(r_0) \cap C'$; its closure A is compact and inside U'.
4. *Back to U.* ψ^{-1} is continuous on U', so $T = \psi^{-1}(A)$ is compact, contains $\varphi(Z_C)$ and lies in U.
5. *Back to O.* The closure of $\varphi(Z_C)$ lies in $T \subseteq U$ and in the closure of O; since O is the fixed point set of the continuous retraction r on U, it is closed in U, so the closure lies in O.
6. *Back to X.* φ^{-1} is continuous on O, so φ^{-1} of that closure is compact and contains Z_C, hence its closure.

The text's "it suffices to find in local coordinates an open neighborhood" leaves steps 4–6 implicit: its set $\{(a, w) \in U \mid |(h + t)(a, w)|_1 \le c\}$ lives in the chart domain U of the filling, which contains points not in the retract.

Theorem 5.2.2 (preprint 5.5) `Chapter5/Compactness · theorem_5_5`. Let f be sc-Fredholm with compact solution set, and N an auxiliary norm. At each zero x the germ (f, x) is sc-Fredholm, so the local step gives a chart Φ and an open $V(x)$; Lemma 5.1.2's (preprint 5.2) lower bound in that chart `Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le` gives $V_1(x)$ and A with $|\Phi(w)_2|_1 \le A\,N(w)$ over $V_1(x)$, so on $V(x) \cap V_1(x)$ the condition $N(f(y)) \le 1$ implies $|\Phi(f(y))_2|_1 \le A$, and $\mathrm{cl}\{y \in V(x) \cap V_1(x) \mid N(f(y)) \le 1\}$ is a closed subset of a compact set. Finitely many of these neighbourhoods cover the compact $f^{-1}(0)$; their union is U, and the closure of $\{y \in U \mid N(f(y)) \le 1\}$ lies in the finite union of the compact closures. That is the text's last sentence, "since the solution set of f is compact, the proof is complete". Tameness is not used, as in the appendix's statement of the theorem.

The equivalence. For an sc-Fredholm section, which is continuous, proper □ weakly proper (§V.4) □ compact solution set □ proper (Theorem 5.2.2 (preprint 5.5)) `Chapter5/Compactness · theorem_5_5_iff`.

V.7 Theorems 5.2.3, 5.2.4 (preprint 5.6, 5.7) and Corollary 5.2.5 (preprint 5.8)

Theorem 5.2.3 (preprint 5.6) is Theorem 3.1.13 (preprint 3.12) at a zero: a zero of an sc-Fredholm section is a smooth point (regularizing), where (f, x) is an sc-Fredholm germ, and Theorem 3.1.13 (preprint 3.12) gives nested open $U(i) \ni x$ with $\mathrm{cl}\{y \in U(i) \mid f(y) = 0\}$ the image of a compact subset of X_i `Chapter3/SectionCompactnessG · theorem_3_12` `Chapter5/Compactness · theorem_5_6`.

The space X_∞. The text says "X_∞ is a metric space" but never gives it a topology. It is the set of smooth points with the initial topology of the inclusions $X_\infty \to X_m$ `Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq`, metrizable by Theorem 2.3.10 (preprint 2.16) `Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel`, and a sequence converges in it iff it converges on every level `Chapter5/Compactness · MPolyfold.tendsto_smoothLevel_iff`; the inclusions into X and into each X_i are continuous `Chapter5/Compactness · MPolyfold.continuous_smoothVal` `Chapter5/Compactness · MPolyfold.continuous_smoothIncl`.

Theorem 5.2.4 (preprint 5.7) `Chapter5/Compactness · theorem_5_7`. $S = f^{-1}(0)$ lies in X_∞ because f is regularizing. Take a sequence (x_k) in S; a subsequence converges in X to some x. The limit is a zero, since a limit of zeros of a continuous section is a zero `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`, so it is smooth and (f, x) is an sc-Fredholm germ. Now *Corollary 3.1.15 (preprint 3.13) at the point x* gives the convergence $x_k \to x$ on every level `Chapter3/Chapter5BridgeG · corollary_3_13_section`, hence in X_∞. The text cites Theorem 5.2.3 (preprint 5.6), which gives compactness in X_i, not the convergence of a given sequence. Tameness is not used.

Corollary 5.2.5 (preprint 5.8), the compact solution set. The zeros in X_i are the image of the compact subset of X_∞ of Theorem 5.2.4 (preprint 5.7) under the continuous inclusion $X_\infty \to X_i$ `Chapter5/`

Compactness · isCompact_solutionSet_level.

Corollary 5.2.5 (preprint 5.8) in a chart. "It is a priori clear that f^i is sc-Fredholm" is not clear, and $Y^i \to X^i$ is not defined in the text. What is formalized is the shift of the germ data. A basic germ f on $\tilde{C} \oplus W$ shifts to a basic germ f^1 on $\tilde{C} \oplus W^1$ with contraction germ B^1: the shifted neighbourhood germ Chapter5/Compactness · ScNbhdGerm.shiftParam, the contraction estimates on the shifted levels Chapter5/Compactness · IsSc0ContractionGerm.shift, and sc-smoothness of f^1 by Proposition 1.2.2 (preprint 1.17) for germs, which needs $\tilde{C}$ a partial quadrant Chapter5/Compactness · IsBasicGerm.shift. If k is conjugated to h by Ψ, then k^1 is conjugated to h^1 by the index raise Ψ^1 Chapter3/BasicClassComposition · IsStrongBundleIso.shift Chapter5/Compactness · IsConjugateToBasicGerm.shift. Hence, for an sc-Fredholm section with compact solution set, the zeros in X_1 are compact, and at every smooth point, in the chart of the germ, $(g - s)^1$ is conjugated to a basic germ and s^1 is an sc$^+$-germ on the shifted germ G^1_s (Proposition 1.2.2 (preprint 1.17) for germs, Chapter3/GermLevels · IsSckGerm.shift), the text's setting Chapter5/Compactness · corollary_5_8_chart. In the germ reading of Definition 3.1.11 (preprint 3.11) (§III.48) g is a filling germ and $g - s$ only an sc^0-germ on the intersection of the two germs; the level shift of the conjugation needs no more, since the conjugation identity is used near 0 only Chapter5/Compactness · IsConjugateToBasicGerm.shift_germ. The statement changed with the switch: in the whole-chart reading s^1 was an sc$^+$-section of all of $U^1 \lhd F^1$. The filler of g^1 was the gap: its condition (3) on level 1 does not follow from level 0. It is closed, together with the shifted strong bundle and the whole corollary, in §V.29.

V.8 $\mathbb{R}$ as an M-polyfold, and bump functions

$\mathbb{R}$. "An sc-smooth map $f : X \to \mathbb{R}$" needs $\mathbb{R}$ as an M-polyfold: the constant sc-structure Chapter5/BumpFunctions · realSc and the single chart id Chapter5/BumpFunctions · realChart Chapter5/BumpFunctions · realPolyfold Chapter5/BumpFunctions · isChart_realChart. Sc-smoothness of a map between M-polyfolds can be checked in one pair of charts around each point, continuity included, since the local representatives determine the map near the point Chapter5/BumpFunctions · MPolyfold.isScSmoothMap_of_local Chapter5/BumpFunctions · isScSmoothMap_real_of_local; an sc-smooth function read in any chart is sc-smooth on the retract Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.real_chart; constants are sc-smooth Chapter5/BumpFunctions · isScSmoothMap_real_const; a C^∞ function $\sigma : \mathbb{R} \to \mathbb{R}$ is an sc-smooth self-map of $\mathbb{R}$ (Corollary 1.2.5 (preprint 1.20) in the constant structure) Chapter5/BumpFunctions · isScSmoothMap_real_of_contDiff, so $\sigma \circ f$ is sc-smooth Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.comp_contDiff.

Definitions 5.3.1 and 5.5.2 (preprint 5.9 and 5.29). *X admits sc-smooth bump functions* if every x and open $U \ni x$ have an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions. The text has two definitions: Definition 5.3.1 (preprint 5.9) asks "$f \neq 0$", Definition 5.5.2 (preprint 5.29) "not identically zero". The proof of Proposition 5.5.3 (preprint 5.30) starts from "a bump function g with $g(0) = 1$", i.e. $g(x) \neq 0$ at the given point; on an M-polyfold there are no translations to move a bump to x (even on an sc-Banach space, translation by a non-smooth vector does not preserve the levels). So Definition 5.3.1 (preprint 5.9) is read with $f(x) \neq 0$; Definition 5.5.2 (preprint 5.29) as written Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctionsWeak is implied Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions.weak. How far apart the two readings are is §V.31.

Propositions 5.3.2 and 5.5.3 (preprint 5.10 and 5.30). Given x and U, take a bump g with $g(x) \neq 0$ and let $f = \tau(3g/g(x) - 1)$, τ the smooth transition from 0 on $(-\infty, 0]$ to 1 on $[1, \infty)$ Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition. Then f is sc-smooth, $0 \leq f \leq 1$, $f(x) = \tau(2) = 1$, $f \equiv 1$ on the open set $\{g/g(x) > 2/3\} \ni x$, and $f \neq 0$ only where $g/g(x) > 1/3$, so $\operatorname{supp} f \subseteq \operatorname{supp} g \subseteq U$ Chapter5/BumpFunctions · proposition_5_30 Chapter5/BumpFunctions · proposition_5_10. The text's σ ("0 for $s \leq 1$ and 1 for $s \geq 1$") is not a function.

V.9 The calculus of sections, and Lemma 5.3.3 (preprint 5.11)

$\Gamma^+(P)$ and its algebra. $\Gamma^+(P)$ is the set of sc⁺-sections (Definition 2.6.6 (preprint 2.52)) `Chapter5/BumpFunctions · StrongBundle.scPlusSections`; their values lie in $Y_{0,1}$, because in a chart their principal parts take values in F_1 `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.mem_oneSet`. The text uses, and does not state, the rules that make $\Gamma^+(P)$ a module over the sc-smooth functions. The zero section is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_fiberZero`. *The product rule:* for f sc-smooth and s sc⁺, the fibrewise product $f \cdot s$ `Chapter5/BumpFunctions · sectionSmul` is sc⁺: in a chart its principal part on level 1 is $(f \circ \varphi^{-1}) \cdot \hat{s}$, the product of an sc-smooth function and an sc-smooth map into F^1, sc-smooth by the product rule of the sc-calculus `Chapter3/ParamPerturbation · IsScSmooth.smul Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.smul_left`. Sums are sc⁺, since principal parts add in a chart `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.add`, and by induction finite sums `Chapter5/BumpFunctions · sectionSum Chapter5/BumpFunctions · StrongBundle.isScPlusSection_sectionSum`.

Cut-offs. Lemma 5.3.3 (preprint 5.11) builds a section in one chart and "extends it by 0". For a chart Φ of the atlas, $\tau : O \to F_1$ with $z \mapsto (z, \tau(z))$ sc-smooth into $K[1]$, and β sc-smooth with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$, the section $y \mapsto \Phi^{-1}(\varphi(y), \beta(y)\tau(\varphi(y)))$ on $\operatorname{dom}\varphi$, 0_y outside `Chapter5/BumpFunctions · StrongBundle.cutoffSection`, is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_cutoffSection`: on $\operatorname{dom}\varphi$ its representative in $\Phi[1]$ is $z \mapsto (z, \beta(\varphi^{-1}(z))\tau(z))$, sc-smooth by the product rule; on the open set $X \setminus \operatorname{supp}\beta$ it is the zero section; the two open sets cover X.

Lemma 5.3.3 (preprint 5.11) `Chapter5/BumpFunctions · lemma_5_11`. Let X admit bump functions, N be an auxiliary norm, x smooth, e a smooth point of Y over x, $\varepsilon > 0$ and $U \ni x$ open. Take any chart Φ of the atlas around x — neither a centred chart nor tameness is needed — and $\hat{e} = \Phi(e)_2$, a smooth point of F because e is smooth. The constant $\hat{e}$ is sc-smooth into F^1 `Chapter5/BumpFunctions · isScSmooth_const_of_mem_smooth`, and $R[1]$ is sc-smooth, so $t(z) = R(z, \hat{e})$ gives an sc⁺-section near x with $t(x) = e$. Since N is continuous on $Y[1]$ and t is continuous into $Y[1]$, the set $\{y \mid N(t(y)) < N(e) + \varepsilon\}$ is an open neighbourhood of x. Proposition 5.5.3 (preprint 5.30) gives $\beta : X \to [0,1]$ with $\beta(x) = 1$ and support in that set, in U and in the chart domain, and the cut-off $s = \beta \cdot t$ is sc⁺ with $s(x) = e$ and $\operatorname{supp} s \subseteq U$. Finally, at every y, $N(s(y)) = \beta(y)N(t(y)) \le N(t(y)) < N(e) + \varepsilon$, because N is a norm on the fibre and $0 \le \beta \le 1$; off the support, $N(s(y)) = N(0_y) = 0$. The text states the bound only at x, where it is trivial; its proof gives, and §5.3 needs, the bound everywhere.

V.10 Allowable sections and Remark 5.3.4 (preprint 5.12)

$\Gamma^{+,1}_U(P)$. For an auxiliary norm N and an open U, the allowable sections are the sc⁺-sections with support in U and $N(s(x)) \le 1$ at every point `Chapter5/BumpFunctions · StrongBundle.allowableSections` — a condition at every point because sc⁺-sections take values in $Y_{0,1}$. The text's metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$ is used through its pointwise form: $N(s(x) - s'(x))$, the difference taken in the fibre $(Y_{0,1})_x$ `Chapter5/Transversality · StrongBundle.sectionDistAt`. The metric space itself is §V.30.

Remark 5.3.4 (preprint 5.12) `Chapter5/Compactness · remark_5_12`. Let $S_f \subseteq U$ with $\operatorname{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact (the U of properness), and let s vanish outside U with $N(s(x)) \le 1$ everywhere — properties that pass to uniform limits of allowable sections, which is all the remark uses of the completion. At a zero x of $f + s$: if $x \notin U$, then $s(x) = 0$ and $f(x) = 0$, so $x \in S_f \subseteq U$, a contradiction; so $x \in U$, and $f(x) = -s(x)$ gives $N(f(x)) = N(s(x)) \le 1$ `Chapter5/Compactness · StrongBundle.IsAuxNorm.auxLe_of_sectionAdd`. So $S_{f+s} \subseteq \{x \in U \mid N(f(x)) \le 1\}$, whose closure is compact. The remark claims the solution set compact, which needs it closed; for an allowable s this holds because $f + s$ is sc-Fredholm (Theorem 3.1.18 (preprint 3.15)) `Chapter3/FredholmSectionsG · theorem_3_15` and solution sets of continuous sections are closed `Chapter5/Transversality · isClosed_solutionSet`. This is how part (1) of Theorems 5.3.5, 5.3.10, 5.3.12 (preprint 5.13, 5.18, 5.19) and Theorem 5.25 (preprint) is proved: U from Theorem 5.2.2 (preprint 5.5), then Remark 5.3.4 (preprint 5.12) and closedness.

V.11 Transversality and the parametrized section

Definition 5.3.7 (preprint 5.15). g is transversal if at every zero its linearization is onto in a chart `Chapter5/Transversality · StrongBundle.IsTransversal`, hence in every chart `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. The kernel of $\mathrm{Lin}(z)$ in a chart is a subspace of E_0 inside T_zO `Chapter5/Transversality · StrongBundleChart.linKer`.

The parametrized section. For a section g and sections $t_1, \dots, t_m$, Theorem 3.1.21 (preprint 3.18) (Part III) makes $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ a section of the pull-back of P to $\mathbb{R}^m \times X$, sc-Fredholm when g is and the t_i are sc$^+$, and with tame germ data when g has them `Chapter3/ParamPerturbationG · theorem_3_18_tame`. Its *slice* at λ is the section $g + \sum \lambda_i t_i$ of P `Chapter5/Transversality · paramSlice Chapter5/Transversality · paramSlice_eq_sectionAdd`, and (λ, y) is a zero of F iff y is a zero of the slice `Chapter5/Transversality · isZeroAt_paramSection_iff`. The slices are sc-Fredholm, since $\sum \lambda_i t_i$ is sc$^+$ (§V.9, Theorem 3.1.18 (preprint 3.15)) `Chapter5/Transversality · isScFredholmSection_paramSlice`; slices of sc$^+$-sections are sc$^+$ `Chapter5/Transversality · isScPlusSection_paramSlice` with support in the union of the supports `Chapter5/Transversality · sectionSupport_paramSlice_subset` and vanish off it `Chapter5/Transversality · isZeroAt_paramSlice_of_not_mem_support`. *Allowability:* if $N(s_0) \le 1-\varepsilon$ and $N(t_i) \le 1$, the triangle inequality on the fibre gives $N(s_0 + \sum \lambda_i t_i) \le 1 - \varepsilon + m|\lambda|$ `Chapter5/Transversality · StrongBundle.IsAuxNorm.toOne_sum_le`, which is ≤ 1 when $m|\lambda| \le \varepsilon$ `Chapter5/Transversality · auxLe_paramSlice` — the text's "$N(\sum \lambda_j t_j(y)) < \varepsilon/2$", with $|\lambda|$ the maximum norm.

The linear algebra of "regular value ⇒ transversal". Let $M(a, u) = Lu + \sum a_i e_i : \mathbb{R}^m \oplus T \to K$ be onto. If the projection $\ker M \to \mathbb{R}^m$ is onto, L is onto: given $k = Lu + \sum a_i e_i$, choose u' with $(a, u') \in \ker M$, so $k = L(u - u')$ `Chapter5/Transversality · surjective_of_forall_exists_ker_fst`. Conversely, if L is onto, every a lifts to $\ker M$ `Chapter5/Transversality · forall_exists_ker_fst_of_surjective`. In the product chart $(r \mapsto r - \lambda) \times \Phi$ of a chart Φ centred at y, the linearization of F at (λ, y) is $\mathrm{Lin}_F(0)(a, u) = \mathrm{Lin}_{g_\lambda}(0)u + \sum a_i \hat{t}_i(0)$ `Chapter3/ParamPerturbationG · linAt_paramSection`, the fibre of the product chart over its centre being the fibre K_0 of Φ `Chapter5/Transversality · mem_K_ofProd_zero_iff`. So if Lin_F is onto at (λ, y) and its kernel projects onto $\mathbb{R}^m$, the slice g_λ is transversal at y `Chapter5/Transversality · isLinSurjAt_paramSlice_of_ker_fst`. In the other direction, if in a chart centred at a zero y of g the range of $\mathrm{Lin}_g(0)$ and the $\hat{t}_i(0)$ span K_0, then F is transversal at $(0, y)$ `Chapter3/ParamPerturbationG · linAt_paramSection_zero Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_spanning`; and adding sections keeps this, because a combination over a subfamily is one over the family `Chapter5/Transversality · exists_sum_comp_eq Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_comp`. Only "regular value ⇒ transversal" is needed in Theorem 5.3.5 (preprint 5.13).

V.12 Spanning sections

"For every $x \in S_{f+s_0}$ we find finitely many allowable sc$^+$-sections such that the range of $(f + s_0)'(x)$ together with the sections span Y_x": the text does not say how. Two ingredients.

Linear algebra `Chapter5/Transversality · exists_fin_mem_of_dense_of_closedComplemented`. Let $R \subseteq K$ be closed, complemented and of finite codimension, and $D \subseteq K$ a dense subspace. With π a continuous projection onto R, $Q = \ker \pi$ is finite-dimensional, and $(1 - \pi)(D)$ is a dense subspace of Q, hence closed, hence all of Q; a basis of Q lifts to finitely many $d_i \in D$, and $K = R + \mathrm{span}\{d_i\}$.

The sections `Chapter5/Transversality · exists_spanning_sections`. Let g be sc-Fredholm, $g(y) = 0$, $U \ni y$ open. In the chart Φ of the germ, centred at y, $\mathrm{Lin}(0) : T_0O \to K_0$ is $Dg(0)|_{T_0O}$ for a derivative family Dg of the filling germ on its germ (the bridge for germs at the centre, `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt`), hence Fredholm (Proposition 3.6.5 (preprint 3.52) for germs, which exposes the germ and Dg) `Chapter3/FredholmGermLinearizationG · proposition_3_52`, so its range R is closed, complemented and of finite codimension in $K_0 = \rho(0)F$. The smooth vectors $\rho(0)F_\infty$ are dense in K_0 — $\rho(0)$ is continuous and preserves the levels at the smooth point 0, and F_∞ is dense — so finitely many

smooth k_i span a complement. Each k_i is $\Phi(e_i)_2$ for a smooth point e_i of the fibre over y `Chapter5/Transversality · mem_smoothSet_toMPolyfold0_of_chart`; Lemma 5.3.3 (preprint 5.11) gives s_i with $s_i(y) = e_i$, support in U and $N(s_i) < N(e_i) + 1$, and $s_i/(N(e_i)+1)$ is allowable, with support unchanged `Chapter5/Transversality · sectionSupport_sectionSmul_const_subset`, N scaled `Chapter5/Transversality · StrongBundle.IsAuxNorm.oneMk_sectionSmul_const` and principal part scaled `Chapter5/Transversality · principal_sectionSmul_const`. The density of the smooth vectors is the point the text omits: Lemma 5.3.3 (preprint 5.11) only reaches smooth points of the fibre.

V.13 Theorem 5.3.5 (preprint 5.13): openness, and the omitted compactness step

No boundary on $\mathbb{R}^m \times X$. "The interior case, Theorem 3.19" is applied to F on $\mathbb{R}^m \times X$, so $\partial(\mathbb{R}^m \times X) = \emptyset$ is needed: $d_{\mathbb{R}^m} \equiv 0$ `Chapter5/Transversality · rnPolyfold_degIndex` and $d_{\mathbb{R}^m \times X} \le d_{\mathbb{R}^m} + d_X$ `Chapter2/FibredProduct · MPolyfold.degIndex_prod_le` `Chapter5/Transversality · isWithoutBoundary_prod_rn`. At a point with $d = 0$ good position is surjectivity `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`.

Transversality spreads. If g has tame germ data and F is transversal at $(0,x)$, x a zero of g, then $(F,(0,x))$ is in good position, and Theorem 3.6.8 (1) (preprint 3.54) gives an open $V \ni (0,x)$ all of whose zeros are in good position, hence transversal; $U(x) = \{y \mid (0,y) \in V\}$ is the text's neighbourhood `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'` `Chapter5/Transversality · exists_isOpen_isLinSurjAt_zero_near`. Likewise, if F is transversal at every $(0,y)$, $y \in S_g$, the union of these sets is an open $V \supseteq \{0\} \times S_g$ all of whose zeros of F are in good position in a tame chart `Chapter5/Transversality · exists_isOpen_goodPosition_of_isLinSurjAt`.

The omitted step. The text passes from the neighbourhood U of $\{0\} \times S_{f+s_0}$ to "$U \subset \{|\lambda| < \delta_0\} \oplus X$" without argument; what is needed is that *every* zero of F with $|\lambda|$ small lies in V. The topological core: if $Z \subseteq \mathbb{R}^m \times X$ is compact and the open V contains $Z \cap \{\lambda = 0\}$, then $Z \setminus V$ is compact, its projection to $\mathbb{R}^m$ is compact and misses 0, and any δ below its distance to 0 works `Chapter5/Transversality · exists_pos_forall_mem_of_isCompact`. The compact Z is the set of zeros of F with $|\lambda| \le \delta_1$: for such λ the perturbation $s_0 + \sum \lambda_i t_i$ is allowable (§V.11), so by Remark 5.3.4 (preprint 5.12) its zeros lie in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ of Theorem 5.2.2 (preprint 5.5) — Z is a closed subset of $\bar{B}(\delta_1)\times$ (that compact set), closed because F is continuous.

V.14 Theorem 5.3.5 (preprint 5.13): the Sard step

"$\tilde{S} := F^{-1}(0)$ is a smooth manifold"; "taking a regular value λ_0 for the projection $\tilde{S} \to \mathbb{R}^m$, it is easily verified that $f + s_0 + s(\lambda_0, \cdot)$ is transversal". The formalization does not build $\tilde{S}$; it proves the needed statement near each zero `Chapter5/Transversality · exists_null_isLinSurjAt_paramSlice_near`.

Claim. Let g have tame germ data, t_i be sc$^+$, and z_0 a zero of F in good position in a tame chart. There are an open $W \ni z_0$ and a null set $CV \subseteq \mathbb{R}^m$ such that for every zero $(\lambda, y) \in W$ with $\lambda \notin CV$ the slice g_λ is transversal at y.

1. *The chart of $\tilde{S}$.* Theorem 3.6.8 (2) (preprint 3.54) gives at z_0 a regular tangent germ chart D of the solution set of F (Definition 3.2.1 (2) (preprint 3.25)): a chart $D.\varphi$ of $\mathbb{R}^m \times X$ with domain W, a finite-dimensional N, a relatively open $V \subseteq N \cap C$ and an sc-smooth graph map Γ with $D.\varphi^{-1} \circ \Gamma$ parametrizing $\tilde{S} \cap W$ `Chapter3/GlobalIFTTheoremsG · theorem_3_54_two`.
2. *The projection.* $q = p \circ D.\varphi^{-1} \circ \Gamma : V \to \mathbb{R}^m$, p the projection to the parameter, is sc-smooth on a finite-dimensional space with the constant structure, hence C^∞ `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`.

3. *Sard on the strata.* V is relatively open in a partial quadrant of N; Sard's theorem on each stratum of the quadrant (Part IV) makes the set CV of stratified critical values of q null `Chapter4/FamiliesLocal · measure_image_stratCritSet`. Since $\partial X = \emptyset$ the points over the zeros are interior and classical Sard would do, but proving that needs the degeneracy index of the induced structure; the stratified form makes the question unnecessary.
4. *The point.* A zero $(\lambda, y) \in W$ is $D.\varphi^{-1}(\Gamma(v))$ for some $v \in V$ with $q(v) = \lambda$. Let Φ be a chart centred at y and $\psi = (r \mapsto r - \lambda) \times \varphi$ the product chart. On a relatively open $V' \subseteq V$ around v, $\mathrm{gmap} = \psi \circ D.\varphi^{-1} \circ \Gamma$ is sc-smooth (the transition between the two charts of $\mathbb{R}^m \times X$ restricted to the overlap), and $q = \mathrm{pr}_1 \circ \mathrm{gmap} + \lambda$ there.
5. *Kernel vectors.* If $\lambda \notin CV$, $Dq(v)$ maps the stratum through v onto $\mathbb{R}^m$. The values of $D\,\mathrm{gmap}(v)$ lie in the tangent space T_0O of the product chart, and in $\ker \mathrm{Lin}_F(0)$: the principal part of F vanishes along $r \circ \mathrm{gmap}$, and the chain rule gives $\mathrm{Lin}_F(0) \circ D\,\mathrm{gmap}(v) = 0$. So for every $a \in \mathbb{R}^m$ there is a kernel vector with first component a.
6. *Conclusion.* F is transversal at (λ, y) (good position near z_0), and §V.11 gives the transversality of the slice at y.

Only range $D\,\mathrm{gmap}(v) \subseteq \ker \mathrm{Lin}_F$ is used, not "$T\tilde{S} = \ker F'$".

V.15 Theorem 5.3.5 (preprint 5.13): the assembly

The generic form `Chapter5/Transversality · theorem_5_13_generic`. Let $\partial X = \emptyset$, X admit bump functions, f have tame germ data and compact solution set, N be an auxiliary norm.

1. *U and part (1).* Theorem 5.2.2 (preprint 5.5) gives $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact `Chapter5/Compactness · theorem_5_5`; for allowable s, Remark 5.3.4 (preprint 5.12) and closedness give S_{f+s} compact (§V.10).
2. *The section $g = f + s_0$.* For s_0 sc$^+$ with $\mathrm{supp}\, s_0 \subseteq U$ and $N(s_0) \le 1 - \varepsilon$, g has tame germ data `Chapter3/TameGermsAddG · theorem_3_15_tame`, a compact solution set (part (1)), and $S_g \subseteq U$ (Remark 5.3.4 (preprint 5.12)).
3. *Local families.* At every zero x of g, spanning sections $s_1^x, \dots, s_{k_x}^x$ supported in U (§V.12) make F_x transversal at $(0, x)$, and at $(0, y)$ for the zeros y in an open $U(x)$ (§V.13).
4. *One family.* Finitely many $U(x_j)$ cover the compact S_g; the families are united through an equivalence $\Sigma_j \mathrm{Fin}(k_{x_j}) \simeq \mathrm{Fin}(m)$ into $t_1, \dots, t_m$, allowable and supported in U; adding sections keeps transversality (§V.11), so F, whose slice at $\lambda = 0$ is g `Chapter5/Transversality · paramSlice_zero`, is transversal at every $(0, y)$, $y \in S_g$.
5. *Good position near the zero slice.* An open $V \supseteq \{0\} \times S_g$ of zeros of F in good position (§V.13).
6. *Allowability and compactness.* For $|\lambda| \le \delta_1 = \varepsilon/(m+1)$ the slices $s_0 + \sum \lambda_i t_i$ are allowable, and the zeros of F over $|\lambda| \le \delta_1$ form a compact Z (§V.13).
7. *The omitted step.* $Z \cap \{|\lambda| < \delta_0\} \subseteq V$ for some $\delta_0 > 0$.
8. *Sard.* With $\delta = \min(\delta_0, \delta_1)$, the zeros Z_0 of F over $|\lambda| \le \delta/2$ are compact and lie in V; the Sard step (§V.14) at each point of Z_0 gives an open W_z and a null CV_z; finitely many W_z cover Z_0, and $CV = \bigcup CV_z$ is null.
9. *The conclusions.* For $|\lambda| < \delta/2$, $\lambda \notin CV$: $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable; every zero y of $f + s_\lambda$ gives a zero $(\lambda, y) \in Z_0$, inside some W_z, so $f + s_\lambda$ is transversal at y; S_{f+s_λ} is compact by part (1); and if it is non-empty, Theorem 3.6.10 (preprint 3.56) for a transversal section over X without boundary, applied to $f + s_\lambda$ with tame germ data, makes it a sub-M-polyfold whose induced structure is tame, a manifold without boundary of locally constant dimension, with regular tangent germ data of dimension the Fredholm index at every zero `Chapter3/GlobalIFTIndexG · theorem_3_56_of_surjective` `Chapter5/Transversality · StrongBundle.SolutionSetIsManifold`.

The dense form `Chapter5/Transversality · theorem_5_13`. Given an allowable s' and $\varepsilon > 0$, let $\varepsilon' = \min(\varepsilon, 1)$ and $s_0 = (1 - \varepsilon'/2)s'$, so $N(s_0) \le 1 - \varepsilon'/2$. The generic form gives t, δ, CV; since CV is null and the ball

$\{|\lambda| < \min(\delta, \varepsilon'/(2m+2))\}$ has positive measure, some λ in it avoids CV. Then $s = s_0 + \sum \lambda_i t_i$ has $N(s(x) - s'(x)) \le (\varepsilon'/2)N(s'(x)) + \sum |\lambda_i| N(t_i(x)) \le \varepsilon'$. The text's "$N(s_0(y) + s(\lambda_0, y)) \le 1 - \varepsilon/2$" is a slip for ≤ 1, its "$N(f(x)) \le 1$" in the statement is $N(s(x)) \le 1$, and its "the germ (f, x) is in good position" is $(f + s, x)$.

V.16 Definition 5.3.9 (preprint 5.17) and Lemma 5.26 (preprint)

Definition 5.3.9 (preprint 5.17), in a tame chart `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm`. (f, y), $f(y) = 0$, is in general position if in some tame strong bundle chart around y (Definition 3.6.3 (preprint 3.50)) `Chapter3/TameGerms · StrongBundle.IsTameChart`, at the level-1 point z over $\varphi(y)$, $\mathrm{Lin}(z)$ is onto K_z, and for every presentation T of the partial quadrant the kernel has an sc-complement B in T_zO inside the reduced tangent space $T_z^R O = T_zO \cap E_z$ `Chapter2/Tame · reducedTangentSpaceWith`, on every level: $(T_zO)_m = (\ker \mathrm{Lin}(z))_m \oplus B_m$. The chart is tame because $T_x^R X$ is the reduced tangent space of a tame M-polyfold and the dimension count below needs it. General position includes transversality `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isLinSurjAt`.

The linear algebra of Lemma 5.26 (preprint) `Chapter5/Transversality · isInGoodPositionIn_of_scComplement_le_reduced`. Let r be a tame retraction, z a smooth point of the retract, and $N, B \subseteq T_zO$ with $T_zO = N \oplus B$ and $B \subseteq T_z^R O$. Then N is in good position to the cone C_z inside T_zO (Definition 3.1.24 (preprint 3.21)): B is a good complement, because adding a vector of E_z does not change membership in C_z (the coordinates λ_i, $i \in I(z)$, vanish on E_z); and $N \cap C_z$ has non-empty interior in N, because the coordinates λ_i, $i \in I(z)$, map T_zO onto $\mathbb{R}^{I(z)}$ — $\dim T_zO/T_z^R O = d_C(z)$ in a tame retract (Proposition 2.5.6 (preprint 2.36)) — and vanish on B, so they map N onto $\mathbb{R}^{I(z)}$, and a vector of N with all these coordinates equal to 1 is interior. The text omits the interior condition; this is the argument of Part IV for tangent spaces `Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt`, for a subspace not given as one.

Lemma 5.26 (preprint) `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGerm`. For f sc-Fredholm and (f, y) in general position: in the chart of Definition 5.3.9 (preprint 5.17) $\mathrm{Lin}(z)$ is onto and its kernel is in good position by the linear algebra; Definition 3.6.6 (preprint 3.53) then follows `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`. For f with tame germ data, good position holds in the tame chart of the germ data `Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame` `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame` — the form the §3.6 theorems take. X itself need not be tame. (The text places the lemma after Theorem 5.25 (preprint) and calls it "clear" after Theorem 5.3.10 (preprint 5.18); it is used in Theorem 5.3.10 (preprint 5.18) and Theorem 5.25 (preprint) here.)

V.17 The bridge lemma, and $T_yS = \ker f'(y)$

Theorem 5.3.10 (preprint 5.18) applies Theorem 4.3.1 (preprint 4.23) to the parametrized solution set $\tilde{S}$, which needs $\tilde{S}$ to be a smooth submanifold in the sense of Definition 4.2.1 (preprint 4.13) — locally the image of an sc⁺-retraction. The implicit function theorem gives tangent germ data (Definition 3.2.1 (preprint 3.25)), and the text never connects the two.

sc⁺ on a finite-dimensional domain `Chapter5/GeneralPosition · isScPlus_of_isScSmooth_const`. On a finite-dimensional space with the constant structure all levels of the domain coincide, so an sc-smooth f is C^∞ into every F_m, and its lift into F^1 is sc-smooth: f is sc⁺.

The bridge lemma. Let D be regular tangent germ data for M at x (Definition 3.2.1 (2) (preprint 3.25)), with graph map $\Gamma : V \to E$ over the finite-dimensional N, projection π onto N, and r a retraction onto the chart model. Then Γ is sc⁺ on V `Chapter5/GeneralPosition · MPolyfold.TangentGermChart.Γ_isScPlus`, so $\tau = \Gamma \circ \pi \circ r$, on the set of points whose image under $\pi \circ r$ lies in V, is an sc-smooth retraction (as in the proof of Theorem 3.2.3 (preprint 3.27)) `Chapter3/TangentGerm · theorem_3_27`

which is sc$^+$ (sc$^+$ $\square$ sc-smooth), with $\tau \circ \Gamma = \Gamma$ on V and retract the chart image of M Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_scPlusRetraction. Transported to X it is an sc$^+$-retraction on an open set with image $M \cap V$ Chapter4/LocalStructure · MPolyfold.isScPlusRetractionOn_of_chart Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_isScPlusRetractionOn. Hence a set with regular tangent germ data at each point is a smooth finite-dimensional submanifold Chapter5/GeneralPosition · MPolyfold.isSmoothSubmanifold_of_isRegular. For solution sets: if all zeros of f in an open W are in good position in a tame chart, Theorem 3.6.8 (2) (preprint 3.54) gives regular data at each, on charts inside W, and $S_f \cap W$ is a smooth submanifold Chapter5/GeneralPosition · StrongBundle.isSmoothSubmanifold_solutionSet_inter.

$T_y S \subseteq \ker f'(y)$ Chapter5/GeneralPosition · tangentSpace_subset_linKer. In a chart, let $Q \subseteq O$ be a retract on which the principal part $\hat{f}$ vanishes, S a retraction onto Q, z a level-1 point of Q. Then $T_z Q = \operatorname{range} DS(z) \subseteq T_z O$, and $\hat{f} \circ r \circ S = 0$ gives $D(\hat{f} \circ r)(z) \circ DS(z) = 0$ by the chain rule, i.e. $T_z Q \subseteq \ker \mathrm{Lin}(z)$ Chapter5/GeneralPosition · StrongBundleChart.mem_linKer_iff.

Equality, by counting Chapter5/GeneralPosition · exists_tangentSpace_eq_linKer. Let f have tame germ data and (f, x_0) be a zero in good position in a tame chart. The index form of Theorem 3.6.8 (preprint 3.54) gives an open $V \ni x_0$ whose zeros are in good position, and regular data D of a fixed dimension d at each zero, with $\dim \ker \mathrm{Lin}(z) = d$ in *every* chart Chapter3/GlobalIFTIndexG · theorem_3_54_index Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex. On the other side, the tangent space of the bridge retraction τ at the point over $\Gamma(v)$ contains the d vectors $D\Gamma(v)b_i$, b_i a basis of N: $D\tau \circ D\Gamma(v) = D(\tau \circ \Gamma)(v) = D\Gamma(v)$, and $\pi \circ D\Gamma(v) = 1$ makes them independent Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_linearIndependent_tangentSpace. "Having d independent tangent vectors" Chapter5/GeneralPosition · HasIndepTangent is carried by germs of sc-diffeomorphisms, since $D(F \circ r)(z)$ maps $T_z Q$ into $T_{F(z)} Q'$ and is injective on $T_z O$ Chapter5/GeneralPosition · HasIndepTangent.transport, so it passes from the chart of D to any chart Φ through the transition maps Chapter4/Submanifolds · MPolyfold.chart_change_of_transport. A d-dimensional subspace of the d-dimensional $\ker \mathrm{Lin}(z)$ — tangent spaces of retracts are germs, so the retract $\varphi(G \cap S_f \cap V)$ of any small open G has the same tangent space — is all of it.

V.18 General position: chart independence and openness

Definition 5.3.9 (preprint 5.17) is Definition 4.2.10 (preprint 4.21) Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff. Near a zero x_0 in good position (with f of tame germ data) there is an open V such that every zero in V is in good position, $S = S_f \cap V$ is a smooth submanifold (§V.17), and at every zero $y \in V$: (f, y) is in general position iff S is in general position at y (Definition 4.2.10 (preprint 4.21), Part IV). Since $T_z S = \ker \mathrm{Lin}(z)$ in every chart, an sc-complement of $\ker \mathrm{Lin}(z)$ inside $T_z^R O$ *is* an sc-complement of $T_z S$ inside $T_z^R O$. Definition 4.2.10 (preprint 4.21) is chart independent (Part IV) Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart, so Definition 5.3.9 (preprint 5.17) is too, at these zeros.

From submanifold general position to good position Chapter5/GeneralPosition · isGoodPositionGermTame_of_isGeneralPositionAt. If $f'(y)$ is onto and a smooth submanifold $A \subseteq S_f$ is in general position at y, then in the tame chart of the germ data $T_z A \subseteq \ker \mathrm{Lin}(z)$ (§V.17) Chapter5/GeneralPosition · StrongBundleChart.linKer_le and $T_z A + T_z^R O = T_z O$; so $\ker \mathrm{Lin}(z) + T_z^R O = T_z O$, the kernel has an algebraic complement inside $T_z^R O$ Chapter5/GeneralPosition · exists_compl_le_of_sup_eq, and §V.16 gives good position.

Openness Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm. Let X be tame, f with tame germ data, (f, x_0) in general position. By Lemma 5.26 (preprint) it is in good position; the previous paragraph turns its general position into general position of the submanifold $S_f \cap V$ at x_0; Theorem 4.2.11 (3) (preprint 4.22) for tame X (Part IV) makes that open along the submanifold Chapter4/LocalStructure · MPolyfold.theorem_4_22; Definition 4.2.10 (preprint 4.21) depends only on the germ of the submanifold, since tangent spaces of chart models agree on open sets Chapter5/GeneralPosition · MPolyfold.IsGeneralPositionAt.congr_nhds; and the equivalence turns it back into general position of (f, y) at the nearby zeros. The text uses this without proof ("they will also hold at $(0, y)$ for $y \in S_f$ close to x";

"the properties (i)–(ii) hold for all $(\lambda, y) \in \tilde{S}$").

V.19 Properties (i)–(ii)

In a chart centred at y, with a presentation T, the properties of the proof of Theorem 5.3.10 (preprint 5.18) are `Chapter5/GeneralPosition · StrongBundleChart.IsParamGP`: (i) $\mathrm{Lin}_g(0)(T_0O) + \mathrm{span}\{\hat{t}_i(0)\} = K_0$ — $F'(0,y)$ onto; (ii) every $w \in T_0O$ is $u + e$ with $e \in E_0$ and $\mathrm{Lin}_g(0)u + \sum a_i\hat{t}_i(0) = 0$ for some a — $\ker F'(0,y)$ transversal to $\mathbb{R}^m \oplus T^R$. Adding sections keeps both, by linear algebra `Chapter5/GeneralPosition · StrongBundleChart.IsParamGP.comp`.

(i)–(ii) give general position `Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP`. In the tame product chart $(r \mapsto r) \times \Phi$ of a tame chart centred at y `Chapter5/GeneralPosition · pc0`, whose tangent space at the centre is $\mathbb{R}^m \times T_0O$ `Chapter5/GeneralPosition · ofProd_mem_rangeS` and whose reduced tangent space is $\mathbb{R}^m \times T_0^RO$, the minimal subspace of $\mathbb{R}^m$ being everything `Chapter5/GeneralPosition · minimalSubspaceWith_rnPres`: (i) is the surjectivity of $F'(0,y)$; (ii) says $\ker F'(0,y) + T^R = T(\mathbb{R}^m \times O)$ on level 0, so the kernel has an algebraic complement inside T^R, and §V.16 gives good position of $(F,(0,y))$. *Then* $\ker F'(0,y) = T_{(0,y)}\tilde{S}$ (§V.17), which is finite-dimensional and smooth; a finite-dimensional smooth subspace has an sc-complement inside any sc-subspace it complements algebraically `Chapter4/LocalStructure · exists_scComplement_le`, which gives Definition 5.3.9 (preprint 5.17) for F. The text's "at this point the combined system of sections already satisfies (i) and (ii)" compresses these two steps.

General position gives (i)–(ii) in every tame chart centred at y `Chapter5/GeneralPosition · isParamGP_of_isGeneralPositionGerm`: (i) is surjectivity read in the product chart; for (ii), $\tilde{S}$ is in general position at $(0,y)$ (§V.18), and in the product chart its tangent space is $\ker F'(0,y)$, which therefore has a complement inside $\mathbb{R}^m \times T^R$.

Achieving (i)–(ii) `Chapter5/GeneralPosition · exists_sections_isParamGP`. Spanning sections give (i) (§V.12) `Chapter5/GeneralPosition · spanning_of_isLinSurjAt`. For (ii), "take a linear subspace L complementing T_x^RX in T_xX and add sections which at x span $f'(x)L$": the sections must come from Lemma 5.3.3 (preprint 5.11), which needs *smooth* points of the fibre, so L must consist of smooth vectors. The coordinates $\lambda_{I(0)}$ map T_0O onto $\mathbb{R}^{I(0)}$ with kernel T_0^RO (Proposition 2.5.6 (preprint 2.36)), and already map the dense smooth vectors of T_0O onto it (a dense subspace of a finite-dimensional image is all of it); so there are smooth $l_i \in T_0O$ with $\lambda(l_i) = e_i$. $\mathrm{Lin}_g(0)l_i$ is smooth (Lin is an sc-operator at the smooth point), Lemma 5.3.3 (preprint 5.11) gives allowable sections through $\Phi^{-1}(0, \mathrm{Lin}_g(0)l_i)$ supported in U, and with these added every $w \in T_0O$ is $u + e$: subtract $\sum \lambda_i(w)l_i$, whose image under $\mathrm{Lin}_g(0)$ is cancelled by the new sections.

V.20 Theorem 5.3.10 (preprint 5.18)

From a family `Chapter5/GeneralPosition · theorem_5_18_of_family`. Let X be tame, f with tame germ data and compact solution set, N auxiliary, U from Theorem 5.2.2 (preprint 5.5). Fix s_0 with $\mathrm{supp}\, s_0 \subseteq U$, $N(s_0) \le 1 - \varepsilon$, and allowable $t_1, \dots, t_m$ with F in general position at every $(0,x)$, $x \in S_g$, $g = f + s_0$.

1. *General position near the zero slice.* $\mathbb{R}^m \times X$ is tame `Chapter3/ParamPerturbation · prod_rn_isTame` and F has tame germ data `Chapter3/ParamPerturbationG · theorem_3_18_tame`, so openness (§V.18) gives, around each $(0,x)$, an open set on which the zeros of F are in good and general position; their union $W \supseteq \{0\} \times S_g$.
2. *Compactness and the omitted step* as in §V.13: the zeros Z of F over $|\lambda| \le \delta_1$ are compact, and $Z \cap \{|\lambda| < \delta_0\} \subseteq W$.

3. *The submanifold.* $A = \tilde{S} \cap W \cap \{|\lambda| < \delta\}$ is a smooth finite-dimensional submanifold of $\mathbb{R}^m \times X$ (the bridge lemma, §V.17), in general position at each of its points (Definition 4.2.10 (preprint 4.21), by §V.18 and locality), with compact closure (inside Z).
4. *Theorem 4.3.1 (preprint 4.23).* In its countable-cover form (Part IV) `Chapter4/Families · theorem_4_23_of_subset`, it gives a set Σ of parameters with null complement over which the fibre A_λ is a smooth submanifold of X in general position.
5. *Transversality.* The Sard step of §V.14 on the compact zeros over $|\lambda| \le \delta/2$ gives a further null set off which the slices are transversal at their zeros.
6. *The conclusions for λ.* For $|\lambda| < \delta/2$ off both null sets, with $h = f + s_\lambda$: the fibre of A over λ is S_h; h is transversal; S_h is a submanifold in general position, so (h, y) is in good position (§V.18), and then in general position by the equivalence of §V.18 applied at y together with the locality of Definition 4.2.10 (preprint 4.21); Theorem 4.2.7 (preprint 4.18) gives the corner structure and tameness of S_h `Chapter4/Submanifolds · MPolyfold.theorem_4_18`, and Theorem 4.2.11 (2) (preprint 4.22) gives $d_{S_h} = d_X$ on S_h `Chapter4/LocalStructure · MPolyfold.theorem_4_22`. These are the conclusions collected in `Chapter5/GeneralPosition · StrongBundle.SolutionSetInGeneralPosition`.

The generic form `Chapter5/GeneralPosition · theorem_5_18`. At every zero x of g choose a tame chart centred at x with a presentation; sections with (i)–(ii) there (§V.19) make F_x general at $(0, x)$, hence (openness, §V.18) at $(0, y)$ for the zeros y in an open $U(x)$. Finitely many $U(x_j)$ cover S_g; the united family t keeps (i)–(ii) at every $(0, y)$ — read (i)–(ii) off general position in the chart of x_j, add sections, go back — so F is in general position at every $(0, y)$, and the previous theorem applies. "Replacing $f + s_0$ by f and N by cN" is not needed; the proof keeps s_0.

The dense form `Chapter5/GeneralPosition · theorem_5_18_dense` is derived exactly as for Theorem 5.3.5 (preprint 5.13) (§V.15).

The text's $\tilde{S}$ omits the equation $F = 0$, and its "$p^{-1}(\{|\lambda| \le \varepsilon\})$ is compact" is the omitted step of §V.13.

V.21 Theorem 5.25 (preprint)

`Chapter5/GeneralPosition · theorem_5_25`. Let X be tame with bump functions, f proper with tame germ data. Properness gives a compact solution set; Theorem 5.3.10 (preprint 5.18) with $s_0 = 0$ gives U, t, δ and a null CV `Chapter5/GeneralPosition · StrongBundle.IsAuxNorm.fiberZero_eq` `Chapter5/GeneralPosition · sectionAdd_fiberZero`; the ball $\{|\lambda| < \delta\}$ has positive measure, so some λ avoids CV, and $s = \sum \lambda_i t_i$ is allowable with $f + s$ in general position at every zero and all the conclusions of Theorem 5.3.10 (preprint 5.18); Lemma 5.26 (preprint) turns general into good position in the tame chart of the germ data (§V.16). The text's proof says "in view of Theorem 3.54 … in general position", which is the openness of §V.18 (Theorem 3.6.8 (preprint 3.54) is about good position), and writes $x \in S_f$ for $x \in S_{f+s_0}$. Here U is the neighbourhood Theorem 5.3.10 (preprint 5.18) produces; the text lets the reader choose "an associated" one.

V.22 Theorem 5.3.12 (preprint 5.19)

`Chapter5/GeneralPosition · theorem_5_19`. Let X admit bump functions, f have tame germ data and compact solution set, and (f, x) be in good position at the zeros $x \in \partial X$. The text's "as before" applies the implicit function theorem to F at the boundary zeros $(0, x)$, where all t_i vanish and $\ker F'(0, x) = \mathbb{R}^m \times \ker f'(x)$; this is in good position to $\mathbb{R}^m \times C_x$ only if $\ker f'(x) + E_x \supseteq T_x X$, a general position condition that good position does not imply ($\mathbb{R}(1, 1)$ in $[0, \infty)^2$; checked on paper only). The formal proof avoids those points:

1. *Near the boundary zeros.* Theorem 3.6.8 (1) (preprint 3.54) at each zero of f on ∂X gives an open set of zeros in good position `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'`; their union V_{bd}, and an open V_1 with $S_f \cap \partial X \subseteq V_1$ and $\operatorname{cl} V_1 \subseteq V_{bd}$.

2. *Away from them.* The zeros of f outside V_1 are not on ∂X, so interior ($d = 0$). Spanning sections at these zeros are taken with supports in $U \setminus (\partial X \cup \operatorname{cl} V_1)$, an open set containing them; finitely many families cover the compact $S_f \setminus V_1$, and F is onto at every $(0, y)$, $y \in S_f$ outside V_1.
3. *Good position of F off the boundary.* At the interior points good position is surjectivity `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`; Theorem 3.6.8 (1) (preprint 3.54) for F gives an open $W \supseteq \{0\} \times (S_f \setminus V_1)$ of zeros in good position, off $\mathbb{R}^m \times \partial X$.
4. *Compactness and Sard* as in §V.15 for the zeros of F over small λ outside $\mathbb{R}^m \times V_1$.
5. *The conclusions.* For small λ off the null set, $h = f + s_\lambda$. On V_1 all t_i vanish, so $h = f$ there, and the good position of (f, y) at the zeros in V_1 passes to (h, y) because good position depends only on the germ of the section `Chapter3/Chapter5BridgeG · StrongBundle.IsGoodPositionGermTame.congr_nhds`. Off V_1 the zeros of h are interior, h is transversal there by the Sard step, hence in good position. Theorem 3.6.11 (preprint 3.57) makes S_h, if non-empty, a sub-M-polyfold whose induced structure is tame and a manifold with boundary with corners `Chapter3/GlobalIFTTheoremsG · theorem_3_57`.

$s_\lambda = \sum \lambda_i t_i$ has support disjoint from ∂X, so it vanishes on the open neighbourhood $X \setminus \operatorname{supp} s_\lambda$ of ∂X, and it is "arbitrarily small" as $\lambda \to 0$. The text's tameness of X is not used, and its "(f, x) in good position for every $x \in S_{f+s}$" is $(f + s, x)$. No dense form is stated.

V.23 Definition 5.3.13 (preprint 5.20), and Lemmas 5.3.16 and 5.3.17 (preprint 5.23 and 5.24)

Definition 5.3.13 (preprint 5.20) `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy`. X tame, f with tame germ data and compact solution set, $t : X \to \mathbb{R}$ sc-smooth with $t(X) = [0, 1]$, and for every $\tau \in [0, 1]$: $X_\tau = t^{-1}(\tau)$ a sub-M-polyfold (Definition 2.5.7 (2017: 2.19)) with tame induced structure, on which the restriction $f|X_\tau$ — the section $a \mapsto (a, f(a))$ of the pull-back of P along the inclusion `Chapter5/Homotopy · restrictSection Chapter3/ParamPerturbation · StrongBundle.pullback`, the inclusion being sc-smooth (Proposition 2.3.19 (preprint 2.21)) — has tame germ data. The text never says what $Y|X_t$ is; the pull-back of Proposition 2.6.8 (preprint 2.54) is the strong bundle structure available. The end problems are the restrictions to $t^{-1}(0)$ and $t^{-1}(1)$; their solution sets are $S_f \cap X_\tau$ `Chapter5/Homotopy · isZeroAt_restrictSection_iff`, closed in the compact S_f, hence compact `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict`.

Lemma 5.3.16 (preprint 5.23) and the hidden hypothesis. The proof of Theorem 5.3.15 (preprint 5.22) fixes a critical point x of $t|S$ and says "since x is a smooth point, we find a one-dimensional smooth linear sc-subspace $Z \subset T_x X$ such that $T_x X = Z \oplus \ker(dt(x))$". For $dt(x) = 0$, $\ker dt(x) = T_x X$ and the only complement is 0: a non-zero complement exists only if $\ell \neq 0$ `Chapter5/Homotopy · ne_zero_of_isCompl_ker`. Theorem 5.3.15 (preprint 5.22) does not assume that t is a submersion at the critical points of $t|S$, so this is a hidden hypothesis. When $\ell \neq 0$, some vector e of any spanning set (e.g. the dense smooth vectors) has $\ell(e) \neq 0$, and $\mathbb{R}e$ complements $\ker \ell$ `Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero`. Lemma 5.3.16 (preprint 5.23) is then linear algebra: $\ell|_Z$ is an isomorphism onto $\mathbb{R}$, so $b = (\ell|_Z)^{-1} \circ \tau$ is the unique $b : S \to Z$ with $\ell \circ b = \tau$, and adding a with values in $\ker \ell$ does not change $\ell \circ b$ `Chapter5/Homotopy · lemma_5_23`.

The calculus. At a level-1 point z, $D(a \cdot w)(z)v = Da(z)v \cdot w(z) + a(z)Dw(z)v$ for sc-smooth $a : U \to \mathbb{R}$, $w : U \to F$ `Chapter5/Homotopy · scDeriv_smul_apply`, and $D(\ell(\rho(\cdot) - p))(z) = \ell \circ D\rho(z)$ `Chapter5/Homotopy · scDeriv_clm_sub_apply`. The function $g = \beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ on $\operatorname{dom}\varphi$, 0 outside `Chapter5/Homotopy · chartLinCutoff`, is sc-smooth when β is with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$ `Chapter5/Homotopy · isScSmoothMap_chartLinCutoff`.

Lemma 5.3.17 (preprint 5.24). *The construction* `Chapter5/Homotopy · exists_scPlusSection_linAt`: for a chart Φ around the smooth point x, a smooth point e over x, a continuous linear ℓ on the model space and $U \ni x$, let s_e be the sc$^+$-section through e of Lemma 5.3.3 (preprint 5.11), supported in $U \cap \operatorname{dom}\varphi$, β a bump with $\beta \equiv 1$ near x and support in the chart domain, and $s = g \cdot s_e$ with g as above. Then s is sc$^+$ (§V.9),

$s(x) = g(x)e = 0$ `Chapter5/Homotopy · principal_sectionSmul`, $\operatorname{supp} s \subseteq U$ `Chapter5/Homotopy · sectionSupport_sectionSmul_subset`, and in the chart, by the product rule, $\mathrm{Lin}_s(z)v = D(g \circ \varphi^{-1} \circ r)(z)v \cdot \hat{s}_e(\varphi(x)) + g(x)\mathrm{Lin}_{s_e}(z)v = \ell(Dr(z)v)\Phi(e)_2 = \ell(v)\Phi(e)_2$ for $v \in T_zO$, since $g \circ \varphi^{-1} = \ell(\cdot - \varphi(x))$ near $\varphi(x)$ `Chapter5/Homotopy · chartLinCutoff_of_mem` and $Dr(z) = 1$ on T_zO. *The lemma* `Chapter5/Homotopy · lemma_5_24`: for a finite-dimensional $K \subseteq T_zO$ (the text's $T_xS = \ker f'(x)$) and a linear $b : K \to T_zO$ with $\mathrm{Lin}_f(z)(bv) = \lambda(v)\Phi(e)_2$, extend $-\lambda$ (linear on the finite-dimensional K) to a continuous linear functional ℓ on the model space (Hahn–Banach) and apply the construction: $\mathrm{Lin}_f(z)(bv) + \mathrm{Lin}_s(z)v = \lambda(v)\Phi(e)_2 - \lambda(v)\Phi(e)_2 = 0$. The text's graph of S over T_xS, its complement L with $j : L \to \mathbb{R}$, and its case "if this operator is the zero operator we can take $s = 0$" are not needed. That $f'(x) \circ b_\tau$ has its image in the line of a *smooth* point is a hypothesis; the text's justification ("the image of any vector in T_xS belongs to Y_∞") is not formalized.

V.24 Bump functions in local models: Theorem 5.5.7 (preprint 5.34) and Proposition 5.5.8 (preprint 5.35)

Definition 5.5.6 (preprint 5.33) `Chapter5/BumpFunctions · LocalModel.HasBumpProperty`. For $x \in O$ and W open in E_0 with $\mathrm{cl}(O \cap W) \subseteq O$, an sc-smooth f on O with $f(x) \neq 0$ (as in Definition 5.3.1 (preprint 5.9)) and support in O inside W.

A retract is closed near its points `Chapter5/BumpFunctions · closure_inter_ball_subset`. If $B(z,r) \subseteq W'$ and $U = C \cap W'$ is the domain of a retraction ρ onto O, then $O \cap \mathrm{cl}\, B(z, r/2) \subseteq O$: a limit of points of O in the closed half ball lies in C (C closed) and in $B(z,r)$, hence in U, where ρ is continuous, and it is a fixed point of ρ.

Theorem 5.5.7 (preprint 5.34) `Chapter5/BumpFunctions · theorem_5_34`. "□ ": every chart's model has the property — given x, W, a bump of X at $\varphi^{-1}(x)$ supported in $\varphi^{-1}(O \cap W)$, read in the chart. "□ ", "clearly" in the text, hides a support condition: a local bump f_0 with support in $U(x)$, $\mathrm{cl}_E(U(x)) \subseteq O$, gives $f = f_0 \circ \varphi$ on the chart domain, but $\mathrm{cl}_X\{f \neq 0\}$ may leave the domain. For $X = E$ a Hilbert space, the chart $\varphi(u) = u/(1 - |u|_0^2)$ from the open unit ball onto $O = E$, $U(x) = O$ and $f_0 \equiv 1$, the extension by 0 is the indicator of the ball. The proof chooses $U_1 \ni x$ with $\mathrm{cl}_X U_1 \subseteq U \cap \operatorname{dom}\varphi$ (regularity of X) and r with $B(\varphi(x), r) \cap O \subseteq \varphi(U_1)$, takes the local bump in $W = B(\varphi(x), r/2)$ — admissible by the previous paragraph — and extends $f_0 \circ \varphi$ by 0: its support lies in $\mathrm{cl}_X U_1 \subseteq U$, and it is sc-smooth because it vanishes on the open complement of its support and is sc-smooth in the chart (§V.8).

Proposition 5.5.8 (preprint 5.35). If $\kappa\|v\|^2 \leq b(v,v)$ for a continuous bilinear b on E_0, then $f(v) = \tau(1 - b(v-x, v-x)/\rho)$ is C^∞ on E_0, equal to 1 at x and supported in a small ball; it is sc-smooth on E by Corollary 1.2.5 (preprint 1.20) from the base level `Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional`, and so is its composition with any retraction onto O `Chapter5/BumpFunctions · LocalModel.hasBumpProperty_of_quadratic`. "E_0 is a Hilbert space" is stated as $\|v\|^2 = b(v,v)$ `Chapter5/BumpFunctions · proposition_5_35`: an inner product structure on E_0 would be a second normed structure next to that of the sc-scale. Translating by a non-smooth x is legitimate here because f is C^∞ on E_0 itself; for a merely sc-smooth function it would not be. With Theorem 5.5.7 (preprint 5.34): an M-polyfold covered by charts on sc-Hilbert spaces admits sc-smooth bump functions, the remark before Definition 5.3.1 (preprint 5.9) `Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert`.

V.25 Partitions of unity: Theorem 5.5.10 (preprint 5.37) and Proposition 5.5.11 (preprint 5.38)

The algebra of sc-smooth functions. Sums, differences and products of sc-smooth functions are sc-smooth, on retracts `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_add` `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_mul` and on X `Chapter5/PartitionsOfUnity · MPolyfold.`

IsScSmoothMap.real_add Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_sub Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_mul, and so are finite sums and products Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetSum Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetProd. Sc-smoothness is local Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_eventuallyEq, so a locally finite sum, which near each point is a finite sum, is sc-smooth Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsum; and a function supported in a chart domain with sc-smooth representative is sc-smooth on X Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_chart.

Definitions 5.5.1 and 5.5.9 (preprint 5.28 and 5.36). A partition of unity is Mathlib's (continuous $\rho_i \geq 0$, locally finite supports, $\sum \rho_i = 1$), sc-smooth if every ρ_i is Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity; X admits sc-smooth partitions of unity if every open cover has a subordinate one Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity. The approximation property: for $V = O \cap W$ with $\mathrm{cl}\, V \subseteq O$, continuous $f : O \to [0,1]$ supported in V and $\varepsilon > 0$, an sc-smooth $g : O \to [0,1]$ supported in V with $|f-g| \leq \varepsilon$ Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty. Partitions of unity give bump functions: the function attached to U of a partition subordinate to $\{U, X\setminus\{x\}\}$ is 1 at x Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity.admitsBumpFunctions — the easy half of the open problem the text states after Definition 5.5.2 (preprint 5.29).

"(i) □ (ii)" Chapter5/PartitionsOfUnity · hasApproxProperty_restrict. The text calls $f\circ\varphi$ "sc-smooth" (f is only continuous) and asserts $\mathrm{cl}_X(\varphi^{-1}(V)) \subseteq U$; with the chart of the unit ball above, $V = O$ and $f \equiv 1$, the extension by 0 is not continuous. Since (ii) asks only for *some* atlas, the proof restricts each chart to an open s with $\mathrm{cl}_X(s) \subseteq \mathrm{dom}\,\varphi$. Then for closed $A \subseteq \varphi(s)$ the set $\varphi^{-1}(A)$ is closed in X Chapter5/PartitionsOfUnity · MChart.isClosed_inter_preimage, so $F = f \circ \varphi$ extended by 0 is continuous. Cover X by $\{F < \varepsilon/2\}$ and the sets $U_x = \{|F - F(x)| < \varepsilon/2\} \cap \varphi^{-1}(V)$, take a subordinate sc-smooth partition of unity (β_0, β_x), and let $\hat{g} = \sum_x \beta_x F(x)$: a locally finite sum of sc-smooth functions with values in $[0,1]$ Chapter5/PartitionsOfUnity · PartitionOfUnity.finsum_mul_mem_Icc, supported in $\varphi^{-1}(V)$ Chapter5/PartitionsOfUnity · PartitionOfUnity.tsupport_finsum_mul_subset, and within $\varepsilon/2$ of F, since every $F(x)$ with $\beta_x(y) \neq 0$ is Chapter5/PartitionsOfUnity · PartitionOfUnity.abs_sub_finsum_mul_le. Whether *every* chart has the property is not settled.

"(ii) □ (i)". Every point of a chart domain has small open neighbourhoods N with $\mathrm{cl}_X N \subseteq \mathrm{dom}\,\varphi$ and $\mathrm{cl}\,\varphi(N) \subseteq O$ — half a ball inside the domain of a retraction, closed by §V.24 Chapter5/PartitionsOfUnity · MChart.exists_small_nhds — and the approximation property transports to continuous functions on X supported in such N Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty.exists_approx. *Sc-smooth Urysohn functions* Chapter5/PartitionsOfUnity · exists_isScSmoothMap_urysohn: for disjoint closed s, t, cover t by small neighbourhoods W_x inside $X \setminus s$, take continuous $f_x = 1$ near x supported in W_x, a locally finite refinement, approximations $\hat{g}_x$ within $1/2$ — so $\hat{g}_x \geq 1/2$ where $f_x = 1$ — and $F = \tau(4\sum \hat{g}_x - 1)$, sc-smooth, 0 on s and 1 on t. Then Mathlib's bump coverings produce a covering indexed by the given cover with sc-smooth functions Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop, and $\rho_i = f_i \prod_{j<i}(1 - f_j)$, locally a finite product, is the partition of unity Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx. This replaces the text's refinement "with the same index set", which does not exist in general (the cover $\{X\}$); metrizability is not needed.

Theorem 5.5.10 (preprint 5.37) Chapter5/PartitionsOfUnity · theorem_5_37 combines the two directions, "atlas" read as a family of charts covering X.

Proposition 5.5.11 (preprint 5.38) Chapter5/PartitionsOfUnity · proposition_5_38 is not "an immediate consequence of Theorem 5.37": that theorem produces some atlas, not the given model, and applied to O would need partitions of unity on an open subset of C. The proof repeats "(i) □ (ii)" in E_0: $A = \{x \in O \mid f(x) \geq \varepsilon/2\}$ is closed in E_0 because $\mathrm{cl}\, V \subseteq O$; $E_0 \setminus A$ and balls $B(x, r_x) \subseteq W$, $x \in V$, on which f varies by less than $\varepsilon/2$, cover E_0; a C^∞ partition of unity of E_0 gives $g = \sum \beta_x f(x)$, C^∞ on E_0, hence sc-smooth (Corollary 1.2.5 (preprint 1.20)), composed with a retraction onto O.

V.26 Banach space geometry: Definitions 5.5.4, 5.5.12, 5.5.14 (preprint 5.31, 5.39, 5.41), Proposition 5.5.15 (preprint 5.42), Corollary 5.5.17 (preprint 5.44)

A Banach space admits a C^k-bump function if some C^k function, not identically zero, has bounded support `Chapter5/PartitionsOfUnity · HasContDiffBump`, and C^k partitions of unity if every open cover has a subordinate C^k one `Chapter5/PartitionsOfUnity · AdmitsContDiffPartitionsOfUnity`. $c_0(\Gamma)$ is the subspace of $\ell^\infty(\Gamma)$ of functions tending to 0 along the cofinite filter `Chapter5/PartitionsOfUnity · cZero`, equivalently with $\{|f(\gamma)| > \varepsilon\}$ finite for every ε `Chapter5/PartitionsOfUnity · mem_cZero_iff`, normed by the supremum `Chapter5/PartitionsOfUnity · norm_cZero`; coordinate-wise C^k embeddings are as in the text `Chapter5/PartitionsOfUnity · IsCoordinatewiseContDiffEmbedding`, and Toruńczyk's Theorem 5.5.13 (preprint 5.40) is recorded as a statement, not proved and not used `Chapter5/PartitionsOfUnity · TorunczykStatement`. WCG means a weakly compact set with dense span `Chapter5/PartitionsOfUnity · IsWCG`; reflexive, used in Proposition 5.5.15 (preprint 5.42) and not defined in the text, means $V \to V''$ onto `Chapter5/PartitionsOfUnity · IsReflexive`.

Proposition 5.5.15 (preprint 5.42). (i) For reflexive V the weak closure of the closed unit ball is weakly compact — Kakutani's theorem, in Mathlib as Banach–Alaoglu in V'' through the embedding of (V, weak) into $(V'', \text{weak-*})$ `Mathlib/Analysis/Normed/Module/DoubleDual · NormedSpace.isCompact_closure_of_isBounded` — and its span contains the ball, hence is V `Chapter5/PartitionsOfUnity · span_closedBall_eq_top` `Chapter5/PartitionsOfUnity · proposition_5_42_reflexive`. (ii) For separable V, a dense sequence (y_n) scaled by $1/((n+1)(|y_n|+1))$ is a null sequence; with 0 it is norm compact, hence weakly compact, and its span contains every y_n `Chapter5/PartitionsOfUnity · proposition_5_42_separable`. The text's "$\{0\} \cup \{x_n/n\}$" has its index range missing.

Corollary 5.5.17 (preprint 5.44) `Chapter5/PartitionsOfUnity · corollary_5_44`. Let $\|v\|^2 = b(v,v)$ on E_0, and assume Theorem 5.5.16 (preprint 5.43) for E_0 and $k = \infty$, recorded as a statement `Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement`. A Hilbert space is reflexive: the parallelogram law holds, so the norm comes from an inner product `Mathlib/Analysis/InnerProductSpace/OfNorm · InnerProductSpace.ofNorm`, and every $\phi \in V''$ is evaluation at the Riesz representative of $x \mapsto \phi(\langle x, \cdot\rangle)$ `Chapter5/PartitionsOfUnity · isReflexive_of_hilbert`; hence WCG by Proposition 5.5.15 (i) (preprint 5.42). It has the C^∞-bump $\tau(1 - b(v,v))$ `Chapter5/PartitionsOfUnity · hasContDiffBump_of_hilbert`. Theorem 5.5.16 (preprint 5.43) gives C^∞ partitions of unity of E_0, and Proposition 5.5.11 (preprint 5.38) — not Theorem 5.5.10 (preprint 5.37), as the text says — the approximation property. With Theorem 5.5.10 (preprint 5.37), an M-polyfold on such charts admits sc-smooth partitions of unity `Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert`.

V.27 What is not formalized

- **Corollary 5.2.5 (preprint 5.8) as one statement for all i.** It is proved for each fixed i (§V.29; gap G.41 closed in the fourth addendum). Not formalized: a family P^i indexed by i, and the tame form of the conclusion, which needs the tameness of the index raises of tame charts. The identification of the iterated base $(X^1)^1$ with X_2 as a topological space is proved since (§II.32; gap G.6, closed). Condition (3) of a filling on level 1 does not follow from level 0 alone — on $E_m = \{x \mid \sum e^{2mk} x_k^2 < \infty\}$ the right shift T makes $1 - T/2$ invertible on E_0 while $(1 - T/2)^{-1} e_0 = \sum 2^{-n} e_n \notin E_1$ — so the corollary's proof needs the regularity argument of §V.29, through Lemma 3.1.20 (preprint 3.17) `Chapter3/Filler · lemma_3_17`.
- **The non-completeness of $(\Gamma_U^{+,1}(P), \rho)$ in the text's setting** (§V.30), and the identification of the completion of Remark 5.3.4 (preprint 5.12); **Remarks 5.3.6, 5.3.8, 5.3.14 (preprint 5.14, 5.16, 5.21)** (gaps G.43, G.44, G.47), and **a dense form of Theorem 5.3.12 (preprint 5.19)** (gap G.45).
- **Weak against pointwise bump functions at a non-smooth point** (§V.31).

- **Finding 88** (gap G.46): the failure of good position of F at the boundary zeros (§V.22) is checked on paper only.
- **Theorem 5.3.15 (preprint 5.22)** (gap G.48): it needs finite-dimensional Morse theory on manifolds with boundary, the cotangent bundle of the solution set, a parametric Sard theorem and a trivialization of the family S_λ, none in Mathlib or the project, and the hidden hypothesis $dt(x) \neq 0$ (§V.23). **Theorem 5.27 (preprint)** (gap G.49): "the proof is left to the reader".
- **Theorems 5.5.5, 5.5.13, 5.5.16 (preprint 5.32, 5.40, 5.43) and the Sobolev remark** (gaps G.50–G.53): deep Banach space geometry cited from [9], [12], and Sobolev spaces with their sc-structures (gap G.1). Nothing on the main line uses partitions of unity.

V.28 What the formalization taught us

The tameness hypothesis. The first formal versions of Theorem 5.3.5 (preprint 5.13) and Lemma 5.26 (preprint) assumed, as the §3.6 theorems then did, that the base retraction of *every* compatible strong bundle chart — and of every chart of the pull-backs to $\mathbb{R}^k \times X$ — is tame. No strong bundle over a nonempty base satisfies this (a compatible chart $(1, \varphi)$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ is never tame), so those statements were vacuous. The repair is the text's own Definitions 3.6.3, 3.6.4, 3.6.6 (preprint 3.50, 3.51, 3.53) in a tame chart: the perturbation theorems take f with tame germ data, the parametrized sections inherit them through the tame product chart, and Definition 5.3.9 (preprint 5.17) lives in a tame chart, where Lemma 5.26's (preprint) count $\dim T_zO/T_z^RO = d_C(z)$ holds.

What "it is easily verified" costs. The perturbation theorems of §5.3 are sketches. Their formal proofs needed the density of smooth vectors (§V.12), the compactness of the zeros over a small parameter ball (§V.13), a local, stratified Sard step (§V.14), the bridge from tangent germ data to Definition 4.2.1 (preprint 4.13) (§V.17), $T_yS = \ker f'(y)$ by a dimension count, which gives the chart independence of Definition 5.3.9 (preprint 5.17) and the openness of general position (§§V.17–V.18), and, in Theorem 5.3.12 (preprint 5.19), a route around the implicit function theorem at boundary points where its hypothesis fails (§V.22).

V.29 Corollary 5.2.5 (preprint 5.8) on the shifted strong bundle

Corollary 5.2.5 (preprint 5.8) says that, for an sc-Fredholm section f with compact solution set, f^i is an sc-Fredholm section of $P^i : Y^i \to X^i$ with compact solution set, and that "it is a priori clear that f^i is sc-Fredholm". The text defines $Y[i]$ for $i = 0, 1$ only, never Y^i, and "a priori clear" hides the filler condition on level 1 (§V.7). Both are supplied here (Findings, Chapter 5, items 140–146).

The reading of Y^1. $Y^1 = Y_{1,1}$ with $(Y^1)_{m,k} = Y_{m+1,k+1}$ — the convention of E^1 and of $(U \lhd F)^1 = U^1 \lhd F^1$ — not $Y[1] = Y_{0,1}$ and not $Y_{1,2}$. It is forced by "f^i is obtained by raising the index": an sc-smooth section maps X_1 into $Y_{1,1}$, and in general not into $Y_{1,2}$.

Fibres of a strong bundle. Through a chart, $P^{-1}(x)$ is linearly homeomorphic to $K_{\varphi(x)} = \{h \mid \Gamma(\varphi(x), h) = h\}$ `Chapter5/StrongBundleShift · StrongBundleChart.fiberK` `Chapter5/StrongBundleShift · StrongBundleChart.fiberCLE`, the fixed points of a continuous projection, a closed subspace of the fibre model `Chapter5/StrongBundleShift · StrongBundleChart.isClosed_fiberK`; so every fibre is a Banach space `Chapter5/StrongBundleShift · StrongBundleChart.isBanachable_fiber`, which Definition 2.6.5 (preprint 2.51) asks of a strong bundle.

The levels of X^1, and raising the index of maps. $(X^1)_m = X_{m+1}$ and $(X^1)_\infty = X_\infty$ as sets `Chapter5/StrongBundleShift · MPolyfold.mem_shift_levelSet_iff` `Chapter5/StrongBundleShift · MPolyfold.mem_shift_smoothSet_iff` (their topologies agree too, §II.32; gap G.6, closed). An sc-smooth map $f : X \to Z$ between M-polyfolds induces an sc-smooth $f_1 : X^1 \to Z^1$: continuity, and in the level charts the local representatives are the level-1 maps of those of f, sc-smooth by Proposition 1.2.2 (preprint 1.17) for

retracts Chapter5/StrongBundleShift · MPolyfold.levelChart_apply_levelOne Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne.

The shifted strong bundle Chapter5/StrongBundleShift · StrongBundle.shift. Its total space is $Y_{1,1}$, the level 1 of the M-polyfold $Y[0]$ with its level topology, Hausdorff and paracompact by Theorem 2.3.10 (preprint 2.16) Chapter5/StrongBundleShift · StrongBundle.ShiftSpace; the projection is $P^1 = P|_{Y_{1,1}}$ Chapter5/StrongBundleShift · StrongBundle.shiftProj, continuous and onto Chapter5/StrongBundleShift · StrongBundle.continuous_shiftProj Chapter5/StrongBundleShift · StrongBundle.surjective_shiftProj; the fibre $P^{-1}(x) \cap Y_{1,1}$ is a linear subspace — in a chart it is the preimage of F_1 under the linear fibre map Chapter5/StrongBundleShift · StrongBundle.shiftFiberSubmodule — carries the topology of F_1, not that of $P^{-1}(x)$, and is a Banach space, a closed subspace of F_1 through a chart Chapter5/StrongBundleShift · StrongBundle.banachable_shift. The atlas consists of the *index raises* of the strong bundle charts Chapter5/StrongBundleShift · StrongBundleChart.shift Chapter5/StrongBundleShift · StrongBundle.shiftAtlas: base chart the level chart $\varphi^1 : V_1 \to O_1$, fibre model F^1, retraction R on $U^1 \lhd F^1$ Chapter5/StrongBundleShift · IsStrongBundleRetraction.shift, bundle chart the level chart of Φ in $Y[0]$. Compatible charts have compatible index raises Chapter5/StrongBundleShift · StrongBundleChart.Compatible.shift: on $[0]$ the transitions are level charts of compatible charts, on $[1]$ the transition $(\Phi'^1 \circ (\Phi^1)^{-1})[1]$ is the level-1 map of $(\Phi' \circ \Phi^{-1})[1]$ (Proposition 1.2.2 (preprint 1.17) for retracts) Chapter5/StrongBundleShift · StrongBundleChart.shift_transition_one, and the F_1-condition of Definition 2.6.4 (preprint 2.50) holds because the $[1]$-transition is sc^0 Chapter5/StrongBundleShift · StrongBundleChart.shift_compatible_range. Checks: $(Y^1)_{m,k} = Y_{m+1,k+1}$ Chapter5/StrongBundleShift · StrongBundle.mem_shift_doubleLevel_iff, the M-polyfold $Y^1[0]$ is $Y[0]^1$ Chapter5/StrongBundleShift · StrongBundle.isChart_shift_toMPolyfold0_iff, and the index raise of every chart of P is a chart of P^1 Chapter5/StrongBundleShift · StrongBundle.isChart_shift.

The section f^1. $f^1 = f|_{X_1}$ Chapter5/StrongBundleShift · StrongBundle.sectionShift is a section of P^1 Chapter5/StrongBundleShift · StrongBundle.isSection_sectionShift, sc-smooth by the raising of the index of maps Chapter5/StrongBundleShift · StrongBundle.IsScSmoothSection.shift, with the zeros of f in X_1 as its zeros Chapter5/StrongBundleShift · StrongBundle.solutionSet_sectionShift, and regularizing, from $(X^1)_m = X_{m+1}$ and $(Y^1)_{m,m+1} = Y_{m+1,m+2}$ Chapter5/StrongBundleShift · StrongBundle.regularizing_sectionShift. In the index raise of a chart the principal part of f^1 is the level-1 map of that of f Chapter5/Corollary58 · StrongBundleChart.principal_sectionShift, and the raised chart is centred where the chart is Chapter5/Corollary58 · StrongBundleChart.shift_φ_eq_zero.

Condition (3) on every level, at every centre Chapter5/Corollary58 · IsFillingGerm.mem_range_toBase_of_fillDefect. *Let (g, s) be a datum of Definition 3.1.11 (preprint 3.11) in a chart — g a filling germ of $\hat{f}$ with fill defect $k = [1 - \rho(r)]g$ and derivative family Dk, s an sc$^+$-germ with $s(0) = g(0)$, $g - s$ conjugated to a basic germ. If $v \in \ker Dr(0)$ and $Dk(0)v \in F_m$, then $v \in E_m$* — at the centre of the datum, which need not be a zero. At a zero this is the regularity of §III.45 Chapter3/LinearizationChartsScG · IsFillingGerm.mem_range_toBase_of_scDeriv_r_eq_zero. The reduction to a zero needs neither a global sc$^+$-section nor a bump function: $g(0) = \iota s(0)$ is a smooth point of F Chapter5/Corollary58 · IsScSmoothGerm.toBase_zero_mem_smooth; $\hat{s}(z) = \Gamma(z, -g(0))$ is sc-smooth on U Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth and a section of $K \to O$, so by Lemma 3.1.20 (preprint 3.17) $g' = g + \hat{s} \circ r$ fills $\hat{f} + \hat{s}$ with the *same* fill defect Chapter3/GermCalculus · IsFillingGerm.add_section, and $g'(0) = g(0) - \Gamma(0, g(0)) = 0$. With the sc$^+$-germ $s' = s + \Gamma^1(r(\cdot), -s(0))$, $g' - s' = g - s$ on U, so Proposition 3.6.5 (preprint 3.52) in sc form makes $Dg'(0)$ sc-Fredholm Chapter3/FredholmGermLinearizationScG · proposition_3_52_local_sc; and $Dk(0) = (1 - \rho(0))Dg'(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, so the case of a zero applies to g'.

The filling on level 1 Chapter5/Corollary58 · IsFillingGerm.germ_shift. The fill defect of g^1 is the level-1 map of k, an sc^1-germ with derivative family $(Dk)_1$ by Proposition 1.2.2 (preprint 1.17) for germs Chapter3/GermLevels · IsSc1GermWith.shift; $\iota\, Dr^1(0) = Dr(0)\, \iota$, by the uniqueness of the derivative of $r^1 = r_1$ on the shifted germ Chapter5/Corollary58 · toBase_scDeriv_r_shift, and $\iota\, \rho^1(0) = \rho(0)\, \iota$ Chapter5/Corollary58 · toBase_ρCLM_shift. So $Dk^1(0) = Dk(0)|_{E_1}$ maps $\ker Dr^1(0)$ to $\ker \rho^1(0)$; it is injective by condition (3) on level 0, onto by the regularity just proved with $m = 1$, and a topological isomorphism by the open mapping

theorem. Conditions (1) and (2) of the filling shift directly Chapter5/Corollary58 · StrongBundleChart.shift_filling_eq_on Chapter5/Corollary58 · StrongBundleChart.shift_filling_mem_of_fixed, and g^1 is a filling germ of the principal part of f^1 in Φ^1 Chapter5/Corollary58 · StrongBundleChart.shift_isFillingGerm.

Corollary 5.2.5 (preprint 5.8). At every smooth point of X^1, the datum (g, s) of f in a chart Φ gives the datum (g^1, s^1) of f^1 in Φ^1: g^1 fills, s^1 is an sc⁺-germ on the shifted germ, $s^1(0) = g^1(0)$, and $g^1 - s^1 = (g - s)_1$ near 0 is conjugated to a basic germ by the index raise of the conjugation (§V.7) Chapter5/Corollary58 · isScFredholmGerm_sectionShift. *For an sc-Fredholm section f, f^1 is an sc-Fredholm section of P^1* Chapter5/Corollary58 · corollary_5_8_isScFredholmSection; *if f has compact solution set, so has f^1*, by the compactness on level 1 (§V.7) Chapter5/Corollary58 · corollary_5_8. Since this holds for every strong bundle, it applies to P^1 and f^1: $f^2 = (f^1)^1$ is an sc-Fredholm section of $P^2 = (P^1)^1$ with compact solution set Chapter5/Corollary58 · corollary_5_8_two, and so on for every fixed i. The unpacked data in the shifted chart are kept for reference Chapter5/Corollary58 · corollary_5_8_data. No hypothesis beyond the text's is added, and tameness is not used. *Not done*: a family P^i indexed by i (a dependent type of strong bundles), the identification of $(X^1)^1$ with X_2 as topological spaces, and the tame form of the conclusion — germ data of f^1 in tame charts of P^1 — which needs the presentation independence of the degeneracy index for C^1 and that $A \cap E_1$ is an sc-complement of T_xO^1 when A is one of T_xO; there is no counterexample either.

V.30 The metric ρ on $\Gamma^{+,1}_U(P)$

The text: $\Gamma^{+,1}_U(P)$ "becomes a metric space" for $\rho(s, s') = \sup_x N(s(x) - s'(x))$, and "is not complete" (p. 217 (preprint p. 174)). The first is proved here; the second turns out to need a hypothesis (Findings, Chapter 5, items 150–153).

The metric. $\rho(s, s')$ is the supremum of the pointwise distances of §V.10 Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist. For an auxiliary norm the fibre norm is symmetric and subadditive (Definition 5.1.1 (1) (preprint 5.1)) Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.neg Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_le Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_triangle, so for allowable sections $N(s(x) - s'(x)) \le N(s(x)) + N(s'(x)) \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.sectionDistAt_le_two: the supremum is finite and $\rho \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_two. $\rho(s, s) = 0$, symmetry and the triangle inequality come from the fibre norms Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_self Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_comm Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_triangle, and $\rho(s, s') = 0$ forces $s = s'$ because N vanishes only at 0_x Chapter5/AllowableSectionsMetric · StrongBundle.eq_of_allowableDist_eq_zero. So ρ is a metric Chapter5/AllowableSectionsMetric · StrongBundle.allowableMetric, carried by a type synonym of $\Gamma^{+,1}_U(P)$, since the sections $X \to Y$ carry the product topology Chapter5/AllowableSectionsMetric · StrongBundle.AllowableSpace. Only Definition 5.1.1 (1) (preprint 5.1) is used. The density statements of Theorems 5.3.5 and 5.3.10 (preprint 5.13 and 5.18) (§§V.15, V.20) are $\rho(s, s') \le \varepsilon$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_iff.

Not complete, by an example Chapter5/AllowableSectionsMetric · TrivialBundle.not_completeSpace_allowableSections. Take the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ of the non-vacuity witnesses of Chapter 3, the fibre norm $N(x, v) = |v|$ — an auxiliary norm, including the condition on the topology of $Y[1]$ Chapter5/AllowableSectionsMetric · TrivialBundle.fibNorm Chapter5/AllowableSectionsMetric · TrivialBundle.isAuxNorm_fibNorm — and $U = \mathbb{R}$. Graphs of C^∞ maps are sc⁺-sections Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_graphSection, so $s_j(x) = (x, \sqrt{x^2 + 1/(j+1)^2}/(2\sqrt{1+x^2}))$ is allowable, with $N \le 1/2$ Chapter5/AllowableSectionsMetric · TrivialBundle.approx Chapter5/AllowableSectionsMetric · TrivialBundle.graphSection_approx_mem, and $\rho(s_i, s_j) \le (\varepsilon_i + \varepsilon_j)/2$ with $\varepsilon_j = 1/(j+1)$, by $|\sqrt{x^2 + a^2} - |x|| \le a$ Chapter5/AllowableSectionsMetric · TrivialBundle.abs_sqrt_sub_abs_le Chapter5/AllowableSectionsMetric · TrivialBundle.abs_approx_sub_limitFn. A ρ-limit s would have, by pointwise convergence, the fibre part $|x|/(2\sqrt{1+x^2})$ Chapter5/AllowableSectionsMetric · TrivialBundle.limitFn, which is not differentiable at 0 Chapter5/AllowableSectionsMetric · TrivialBundle.not_differentiableAt_limit; but the fibre part of an sc⁺-section

of the trivial bundle is C^1, by Proposition 1.2.3 (preprint 1.18) in the chart $\Phi = \mathrm{id}$ `Chapter5/AllowableSectionsMetric · TrivialBundle.differentiable_snd_of_isScPlusSection`.

Complete, in two cases. For $U = \emptyset$ every allowable section is the zero section, so the space is a point, for every bundle and auxiliary norm `Chapter5/AllowableSectionsMetric · StrongBundle.subsingleton_allowableSections_empty` `Chapter5/AllowableSectionsMetric · StrongBundle.completeSpace_allowableSections_empty`. Over the point $\mathbb{R}^0$ with fibre $\mathbb{R}^k$ every section is the graph of a constant, hence sc^+ `Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_of_isSection_point`, and the space is the closed unit ball of $\mathbb{R}^k$ (or $\{0\}$), complete for every U `Chapter5/AllowableSectionsMetric · TrivialBundle.completeSpace_allowableSections_point`. So the text's "not complete" is false without a hypothesis and is read as "not complete in general". In the text's setting — U a neighbourhood of the solution set of an sc-Fredholm section on an M-polyfold with bump functions — one expects non-completeness as soon as U contains a non-isolated point over which $Y_{0,1}$ is nonzero, by the same mechanism (bump functions times a non-smooth uniform limit); this is not formalized, and neither is the identification of the metric completion with the sections of Remark 5.3.4 (preprint 5.12) (§V.10).

V.31 Weak and pointwise bump functions

§V.8 reads Definition 5.3.1 (preprint 5.9) pointwise, $f(x) \neq 0$, because Proposition 5.5.3 (preprint 5.30) needs it, and keeps Definition 5.5.2's (preprint 5.29) "not identically zero" as the weak reading. How far apart are they? (Findings, Chapter 5, items 160–163.)

Bump points. x is a *bump point* if every open $U \ni x$ carries an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ `Chapter5/BumpWeakPointwise · MPolyfold.HasBumpAt`. The pointwise reading says that every point is a bump point `Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctions_iff_forall_hasBumpAt`, and the weak reading holds as soon as the bump points are dense: a bump at a bump point inside U is not identically zero and is supported in U `Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctionsWeak_of_dense`. (The converse, weak ⇒ dense bump points, is a nested-neighbourhoods argument with Cantor's theorem in a chart; not formalized.)

On an sc-Banach space. An sc-Banach space E is an M-polyfold with the identity chart and model (E, E, E) `Chapter5/BumpWeakPointwise · scChart` `Chapter5/BumpWeakPointwise · scPolyfold`, whose sc-smooth functions are the sc-smooth maps $E \to \mathbb{R}$ `Chapter5/BumpWeakPointwise · isScSmoothMap_scPolyfold_iff`. *There the weak reading gives a bump point at every smooth point* `Chapter5/BumpWeakPointwise · MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth`: given $x \in E_\infty$ and $B(x, \varepsilon) \subseteq U$, a weak bump supported in $B(x, \varepsilon/2)$ is non-zero on a non-empty open set, which contains a smooth point y because E_∞ is dense; $a = y - x$ is smooth, so $v \mapsto f(v + a)$ is sc-smooth `Chapter4/Recentre · isScSmooth_add_const`, non-zero at x, with support in $B(x, \varepsilon) \subseteq U$. The argument stops at E_∞: translation by a vector $a \notin E_m$ is not even sc^0, since it sends $0 \in E_m$ out of E_m `Chapter5/BumpWeakPointwise · not_isSc0_add_const`. On M-polyfolds built on sc-Hilbert spaces both readings hold at every point `Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert`. *Open*: whether the weak reading gives a bump function non-zero at a non-smooth point, on an sc-Banach space or an M-polyfold; a counterexample would need an E_0 without C^1 bump functions.

V.32 The Springer 2021 edition: Definition 5.2.1, Theorems 5.2.2 and 5.3.5, Definition 5.3.13

The 2021 edition adds two compactness properties to Definition 5.2.1 (preprint 5.4) and extends Theorem 5.2.2 (preprint 5.5) to them, concludes in Theorem 5.3.5 (preprint 5.13) that the perturbed solution sets are compact sub-M-polyfolds, and changes Definition 5.3.13 (preprint 5.20): t need not be onto, the fibres X_τ are no longer asked to be Fredholm problems, and $dt(x)$ is asked onto at every smooth point. Chapter 5 has no item that is new in the 2021 edition. The file is `Chapter5/Book2021.lean` (Findings, Chapter 5, items 170–179).

Definition 5.2.1, 2021 form Chapter5/Book2021 · StrongBundle.IsWeaklyProperOnU Chapter5/Book2021 · StrongBundle.IsWeaklyProperLtInfty Chapter5/Book2021 · StrongBundle.IsProperLtInfty. *Statement.* (2) weakly proper: compact solution set, and for every auxiliary norm N an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc$^+$-section s supported in U with $N(s(y)) \le 1$ for $y \in U$; (4) weakly proper$_{<\infty}$: the same for every such s with $N(s) \le C$ on U for some constant C; (5) proper$_{<\infty}$: compact solution set, and for every N an open $U \supseteq f^{-1}(0)$ with $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le C\}$ compact for every $C > 0$, where $N(f(x)) \le C$ includes $f(x) \in Y_{0,1}$ Chapter5/Compactness · StrongBundle.AuxLe. *The bound on U is the old bound everywhere* Chapter5/Book2021 · StrongBundle.isWeaklyProperOnU_iff: a section supported in U vanishes off U Chapter5/Book2021 · StrongBundle.isZeroAt_of_not_mem_of_support_subset, and the zero of a fibre has $N \le r$ for every $r \ge 0$ Chapter5/Book2021 · StrongBundle.IsAuxNorm.auxLe_fiberZero; so (2) is the `IsWeaklyProper` of §V.4 Chapter5/Compactness · StrongBundle.IsWeaklyProper, and (1), (3) are unchanged Chapter5/Compactness · StrongBundle.IsProper.

The a priori dependencies (p. 209). Proper$_{<\infty}$ gives proper with $C = 1$ Chapter5/Book2021 · StrongBundle.IsProperLtInfty.isProper; weakly proper$_{<\infty}$ gives weakly proper, since the bound 1 is a constant Chapter5/Book2021 · StrongBundle.IsWeaklyProperLtInfty.isWeaklyProperOnU; weakly proper includes a compact solution set Chapter5/Book2021 · StrongBundle.IsWeaklyProperOnU.isCompact_solutionSet. For a continuous f, proper gives weakly proper by §V.4 Chapter5/Book2021 · StrongBundle.IsProper.isWeaklyProperOnU, and proper$_{<\infty}$ gives weakly proper$_{<\infty}$ Chapter5/Book2021 · StrongBundle.IsProperLtInfty.isWeaklyProperLtInfty: let $N(s) \le C$ on U. A solution x of $f = s$ lies in U — otherwise $s(x) = 0$, so $x \in f^{-1}(0) \subseteq U$ — and there $N(f(x)) = N(s(x)) \le \max(C, 1)$; so $\{f = s\}$, closed because f and s are continuous, lies in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le \max(C, 1)\}$ Chapter5/Book2021 · StrongBundle.AuxLe.mono. The text's remark that the notions differ for general sc-smooth sections is not formalized.

Theorem 5.2.2 (preprint 5.5), 2021 form Chapter5/Book2021 · theorem_5_2_2_2021. *Statement.* An sc-Fredholm section with compact solution set is proper$_{<\infty}$. *Proof.* As for §V.6 Chapter5/Compactness · theorem_5_5. At a zero x, the local step gives a chart c, an open $V \ni x$ and for every bound b a compact set containing the closure of the points of V whose fibre part has level-1 norm $\le b$ Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd; near x the chart norm is at most $A \cdot N$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le. So on a smaller V the closure of $\{N(f) \le C\}$ lies in the compact set for $b = A \cdot C$, and V does not depend on C. Finitely many such V cover $f^{-1}(0)$, and the closure of the finite union lies in the finite union of the compact closures. □ The text's first display writes $N(f(x)) \le d$ for the points z of the set, a possible misprint for $N(f(z)) \le d$; the later display has y throughout. *The five properties are equivalent* for sc-Fredholm sections Chapter5/Book2021 · theorem_5_2_2_2021_tfae: the cycle (5) □ (3) □ (2) □ (1) □ (5) and (5) □ (4) □ (2), with f continuous because it is sc-smooth.

Theorem 5.3.5 (preprint 5.13), 2021 form Chapter5/Book2021 · theorem_5_3_5_2021. *Statement.* The conclusions of §V.15 Chapter5/Transversality · theorem_5_13, and moreover S_{f+s} is a compact sub-M-polyfold for the transversal perturbations s. *Proof.* `theorem_5_13` gives S_{f+s} compact and, if non-empty, a manifold, a sub-M-polyfold in particular; the empty set is a sub-M-polyfold Chapter5/Book2021 · isSubMPolyfold_solutionSet_of_manifold. □ The perturbations are the allowable sections, bounded by $N(s) \le 1$ Chapter5/BumpFunctions · StrongBundle.allowableSections; the printed "$N(f(x)) \le 1$ for all x" is read as $N(s(x)) \le 1$, a possible misprint, as in §V.15.

Definition 5.3.13 (preprint 5.20), 2021 form Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy. *Statement.* X tame; f an sc-Fredholm section with tame germ data and compact solution set; $t : X \to \mathbb{R}$ sc-smooth with values in $[0, 1]$; for $i = 0, 1$ the end $t^{-1}(i)$ a sub-M-polyfold with tame induced structure over which the restriction of f to the pull-back bundle has tame germ data Chapter5/Book2021 · StrongBundle.IsCompactFredholmRestriction Chapter5/Homotopy · restrictSection; and (3) $dt(x)$ onto at every smooth x. Since T_xX has no linear structure in the project, (3) is read in a chart φ around x with a retraction R onto its model and the level-1 point z over $\varphi(x)$: every $a \in \mathbb{R}$ is the derivative of $t \circ \varphi^{-1} \circ R$ at z on some $DR(z)v \in T_zO$ Chapter5/Book2021 · MPolyfold.HasSurjDerivAt. *Relation to §V.23.* The data of the earlier reading Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy together with (3) give the new data Chapter5/Book2021 · StrongBundle.IsGenCompactHomotopy.isSubmersiveGenCompactHomotopy (t onto $[0, 1]$ gives values in $[0, 1]$,

and the fibre conditions at $\tau = 0, 1$ are the end conditions). Every fibre $t^{-1}(\tau)$ still has a compact solution set: it is closed, and the solution set of the restriction is the trace of $f^{-1}(0)$ Chapter5/Book2021 · StrongBundle.IsSubmersiveGenCompactHomotopy.isCompact_solutionSet_restrict.

A witness, and the converse. Over an empty end the end conditions hold: $\emptyset$ is a tame sub-M-polyfold whose atlas has no charts, and every section over it is tame sc-Fredholm Chapter5/Book2021 · StrongBundle.isCompactFredholmRestriction_of_eq_empty Chapter5/Book2021 · isScSmoothRetractMap_of_domain_empty. On the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ take the graph section of the identity, with solution set $\{0\}$ Chapter5/Book2021 · GenCompactHomotopyWitness.solutionSet_graphSection_id and tame germ data Chapter3/NonVacuityG · TrivialBundle.isScFredholmSectionTame_graphSection, and $t(x) = 1/2 + \arctan(x)/\pi$ Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime, a C^∞ function with values in $(0, 1)$ Chapter5/Book2021 · GenCompactHomotopyWitness.contDiff_arctanTime Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime_mem_Ioo. Both ends are empty Chapter5/Book2021 · GenCompactHomotopyWitness.arctanTime_preimage_eq_empty, and $t'(x) = 1/(\pi(1+x^2)) \neq 0$ Chapter5/Book2021 · GenCompactHomotopyWitness.hasFDerivAt_arctanTime, so $dt(x)$ is onto in the identity chart Chapter5/Book2021 · GenCompactHomotopyWitness.hasSurjDerivAt_arctanTime. So the 2021 definition is satisfiable Chapter5/Book2021 · GenCompactHomotopyWitness.isSubmersiveGenCompactHomotopy_witness; the same data are not a generalized compact homotopy in the earlier sense, as 0 is not a value of t Chapter5/Book2021 · GenCompactHomotopyWitness.not_isGenCompactHomotopy_witness.

Lemma 5.3.16 (preprint 5.23) under (3) Chapter5/Book2021 · lemma_5_3_16_2021. *Statement.* If $dt(x)$ is onto, then in a chart the derivative ℓ on $T_z O$ is non-zero, $\ker \ell$ has a one-dimensional complement Z, and the conclusions of Lemma 5.3.16 hold. *Proof.* Take v with $\ell(DR(z)v) = 1$; so $\ell \neq 0$. A vector on which ℓ is non-zero spans a complement of $\ker \ell$ Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero, and §V.23 applies Chapter5/Homotopy · lemma_5_23. □ This is the hypothesis $dt(x) \neq 0$ that §V.23 found missing from Theorem 5.3.15 (preprint 5.22): the 2021 condition (3) supplies it. That Z can be chosen smooth, and Theorem 5.3.15 itself, are not formalized (gap G.48).

Theorem 5.5.5 (preprint 5.32). The 2017 and 2021 statements print the exponent p that the preprint omits in (1) and (2); the theorem is still cited, not stated (gap G.50).

Part VI. Linearizations, orientations and invariants (HWZ Chapter 6, §§6.2–6.8 (preprint §§6.1–6.7))

Chapter 6 builds the orientation theory of sc-Fredholm sections. At a smooth point x it collects the linearizations of f into a convex set $\mathrm{Lin}(f,x)$ of Fredholm operators (§6.2 (preprint §6.1)); it fixes the conventions for determinant lines and for the isomorphism attached to a four-term exact sequence (§6.3 (preprint §6.2)); it defines the determinant $\det(T) = \lambda(\ker T) \otimes \lambda^*(\operatorname{coker} T)$ of a Fredholm operator, compares it with the determinants of PT for good left-projections P and of the stabilizations T_φ, and builds the determinant line bundle $\mathrm{DET}(E,F)$ over the Fredholm operators (§§6.4–6.5 (preprint §§6.3–6.4)); it then restricts that bundle to $\mathrm{Lin}(f,x)$, propagates its orientations along sc-smooth paths of X, first in small neighbourhoods (Theorem 6.6.11 (preprint 6.35)) and then globally, as a functor on the path category (Theorem 6.6.13 (preprint 6.37)), and defines orientability and orientations of f, orients the solution set (Theorem 6.6.18 (preprint 6.41)) and the sections $f+s$ over $X \times [0,1]$ (Remark 6.6.19 (preprint 6.42)) (§6.6 (preprint §6.5)); §6.7 (preprint §6.6) states the invariant $\Psi_{(f,o)}$ (Theorem 6.7.1 (preprint 6.43)), and the appendix §6.8 (preprint §6.7) proves Lemma 6.3.3 (preprint 6.5) and Proposition 6.4.11 (preprint 6.12). This Part retells §6.2 (preprint §6.1) (§VI.1); §6.3 (preprint §6.2) — determinant lines, Definition 6.3.1 (preprint 6.3) and Proposition 6.3.2 (preprint 6.4), the isomorphism Φ_E, Lemma 6.3.3 (preprint 6.5) and Proposition 6.3.5 (preprint 6.6) (§§VI.2–VI.3); §6.4 (preprint §6.3) — Definitions 6.4.1–6.4.3 (preprint 6.7–6.9), Lemma 6.10 (preprint; not in the 2021 edition), the isomorphisms γ_T^P, the stabilization and the direct sums (§§VI.4–VI.6); §6.5 (preprint §6.4) — the local determinant bundles and $\mathrm{DET}(E,F)$ (§§VI.7–VI.8); §6.6 (preprint §6.5) — orientations over $\mathrm{Lin}(f,x)$ (§VI.9), Proposition 6.6.3 (preprint 6.28) (§VI.10), Lemmas 6.6.4–6.6.6 (preprint 6.29–6.31) (§§VI.11–VI.13), the push-forward (§§VI.14–VI.15), Lemma 6.6.8 (preprint 6.33), Lemma 6.6.10 (preprint 6.34) and Proposition 6.6.7 (preprint 6.32) (§§VI.16–VI.17), Theorem 6.6.11 (preprint 6.35) and Definition 6.6.12 (preprint 6.36) (§§VI.18–VI.19), the path category (§VI.20), the comparison of charts (§VI.21), the transport and Theorem 6.6.13 (preprint 6.37) (§§VI.22–VI.23), Definitions 6.6.14–6.6.16 (preprint 6.38–6.40) (§VI.24), Theorem 6.6.18 (preprint 6.41) (§VI.25) and Remark 6.6.19 (preprint 6.42) (§VI.26); §6.7 (preprint §6.6) and Theorem 6.7.1 (preprint 6.43), which are not formalized (§VI.27); the appendix §6.8 (preprint §6.7) (§§VI.28–VI.29); what is not formalized (§VI.30); and what the formalization taught us (§VI.31). Proposition 6.2.2 (preprint 6.2) was formalized with Chapter 3 and is retold in §III.49. Each item carries the number of its comparison item (V6.1–V6.64) in `Exposition/verification/ch06.md`.

What is proved, and what is not. Theorems 6.6.11 and 6.6.13 (preprint 6.35 and 6.37) are proved; so are Propositions 6.3.2, 6.3.5, 6.4.11, 6.4.15, 6.5.4, 6.5.7, 6.5.8, 6.6.1, 6.6.7 (preprint 6.4, 6.6, 6.12, 6.15, 6.20, 6.23, 6.24, 6.26, 6.32), Lemma 6.3.3, Proposition 6.4.10, Lemma 6.4.14, Lemma 6.5.1, Lemma 6.5.2, Lemma 6.5.5, Lemma 6.5.9, Lemma 6.6.6, Lemma 6.6.8, Lemma 6.6.10 (preprint Lemmas 6.5, 6.11, 6.14, 6.17, 6.18, 6.21, 6.25, 6.31, 6.33, 6.34) and Lemma 6.10 (preprint), Corollary 6.4.12 (preprint 6.13), Lemma 6.6.4 (preprint 6.29) in the form the text uses afterwards, the kernel statement of Lemma 6.6.5 (preprint 6.30), Proposition 6.6.3 (preprint 6.28) with (1′) in place of (1), the equivalence of Definitions 6.6.14 and 6.6.15 (preprint

6.38 and 6.39), Theorem 6.6.18 (preprint 6.41) in the form of tangent orientations that vary continuously along the paths of zeros in a chart, and Remark 6.6.19 (preprint 6.42) at a point. Every definition of the chapter is formalized. Three places needed more than a reading. The projection of the proof of Lemma 6.10 (preprint) does not lie below the two given projections as we read it, and the lemma is proved by another construction, without the hypothesis that T is Fredholm (§VI.4). The isomorphism Φ_E of formula (94) does not vary continuously in the topology of DET where $\dim \operatorname{coker} T$ changes parity, as we read it, so the propagation uses $(-1)^{\dim \operatorname{coker} T}\Phi_E$ (§VI.14); the map $h \mapsto \tilde{h}$ of Remark 6.6.19 (preprint 6.42) gets the same sign (§VI.26). Proposition 6.6.3 (1) (preprint 6.28), the sc-smooth contractibility of the neighbourhood, is not established by its proof as we read it; what Theorem 6.6.11 (preprint 6.35) uses is (1′), the homotopy of two paths with the same end points inside a larger region, and that is proved (§§VI.10, VI.18). Five proofs the text omits are supplied: the orientability of a line bundle over a convex set (§VI.9), the general position of Lemma 6.6.4 (preprint 6.29) (§VI.11), the "simple computation" of Lemma 6.6.8 (preprint 6.33) (§VI.16), the covering argument of Theorem 6.6.13 (preprint 6.37) (§VI.22) and the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) (§VI.24). Not formalized: Theorem 6.7.1 (preprint 6.43) and the admissible sections of §6.7 (preprint §6.6) (gaps G.54, G.55), Proposition 6.6.3 (1) (preprint 6.28) (gap G.61), the orientation of the solution set as an abstract manifold, its chart independence and the perturbation paragraph after Theorem 6.6.18 (preprint 6.41) (gaps G.67–G.69), Remark 6.6.19 (preprint 6.42) beyond a point (gap G.66), the general Lemma 6.5.3 (preprint 6.19), several remarks, and the facts on paths recalled on preprint p. 201 (§VI.30).

Six structural choices. (1) *Chart points.* T_xX carries no linear structure in the formalization (gaps G.6, G.7, G.13 of Chapter 2), so $\mathrm{Lin}(f,x)$, $\mathrm{DET}(f,x)$ and Or_x^f are made at a *chart point* (c,z) of x, with operators $T_zO = Dr(z)E \to K_z = \rho(z)F$ on level 0, and they are identified between charts by explicit transition maps (§VI.21); the text's $f-s$ with $s(x)=f(x)$ is $f+s$ with $(f+s)(x)=0$. (2) *Global sections, from bump functions.* The text assumes sc-smooth partitions of unity from §6.3 (preprint §6.2) on, "to simplify the presentation" (p. 241 (preprint p. 203)). The formalization assumes sc-smooth bump functions, which partitions of unity give (§V.25), and uses them only to make global sc$^+$-sections (Findings, Chapter 6, item 43). (3) *Families on* $\mathbb{R}^n$. Paths, homotopies and families are sc-smooth maps on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ whose conditions are imposed on $[0,1]$, $[0,1]^2$, $[0,1]^n$, and $X \times [0,1]$ is read as $\mathbb{R}^n \times X$ with the pull-back π^*P of the bundle along the projection (decision D2, as for Proposition 6.2.2 (preprint 6.2) in §III.49; gap G.65; Findings, Chapter 6, item 102). (4) *The sign* $\alpha(T) = (-1)^{\dim \operatorname{coker} T}$ at both end points of every propagation (§VI.14). (5) *(1′), and orientable neighbourhoods as pairs* $V \subseteq U$, since Proposition 6.6.3 (preprint 6.28) gives (1′) rather than contractibility (§§VI.10, VI.19). (6) *Dyadic subdivisions* in place of the finite coverings of the proof of Theorem 6.6.13 (preprint 6.37) (§VI.22). An orientation of a line is a ray (Mathlib's `Module.Ray`); an orientation of a topological line bundle is a choice of rays given near every point by a continuous nowhere-zero section.

Map.

- `Chapter6/LinSet`: chart points, T_zO, K_z, sc$^+$-operators, Definition 6.2.1 (preprint 6.1), the index and convexity of $\mathrm{Lin}(f,x)$.
- `Chapter6/DetLine`: $\lambda(E)$, $\lambda^*(E)$, wedges, v^*, $\lambda(\Phi)$, Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4).
- `Chapter6/ExactSequence`: four-term exact sequences, the choices of (91), Φ_E, Lemma 6.3.3 (preprint 6.5) with the proof of §6.8.1 (preprint §6.7.1), (92), Proposition 6.3.5 (preprint 6.6).
- `Chapter6/FredholmDet`: Definitions 6.4.1, 6.4.2 (preprint 6.7, 6.8).
- `Chapter6/GoodProjections`: Definition 6.4.3 (preprint 6.9), the order on Π_T, the facts after it, Lemma 6.10 (preprint) and the example on $\mathbb{R}^2$.
- `Chapter6/ProjectionSequence`: Proposition 6.4.10 (preprint Lemma 6.11), γ_T^P, Proposition 6.4.11 (preprint 6.12) with the proof of §6.8.2 (preprint §6.7.2), Corollary 6.4.12 (preprint 6.13).
- `Chapter6/Stabilization`: Lemma 6.4.14 (preprint 6.14), formula (94), Proposition 6.4.15 (preprint 6.15).
- `Chapter6/DirectSumDet`: Definition 6.4.16 (preprint 6.16).
- `Chapter6/LocalDetBundle`: Lemmas 6.5.1, 6.5.2 (preprint 6.17, 6.18), the frames of Proposition 6.5.4

(preprint 6.20), Lemma 6.5.5 (preprint 6.21).

- `Chapter6/DetBundle`: $\mathrm{DET}(E,F)$ as a vector prebundle, Definition 6.5.6 (preprint 6.22), Propositions 6.5.7, 6.5.8 (preprint 6.23, 6.24), Lemma 6.5.9 (preprint 6.25), Proposition 6.6.1 (preprint 6.26).
- `Chapter6/LineBundleOrientation`: orientations of line bundles, the orientation double cover.
- `Chapter6/LinOrientation`: Definition 6.6.2 (preprint 6.27).
- `Chapter6/OrientableNbhd`: sc-smooth homotopies, Proposition 6.6.3 (preprint 6.28).
- `Chapter6/Lemma629`: Lemma 6.6.4 (preprint 6.29).
- `Chapter6/KernelFamily`: the kernels of Lemma 6.6.5 (preprint 6.30).
- `Chapter6/SubspaceBundle`: continuous families of subspaces, their determinant lines, continuation over $[0,1]$.
- `Chapter6/StabilizationContinuity`: families of kernels and their images, the stabilization sign, the example without the sign.
- `Chapter6/ParamKernel`: the stabilized section in a product chart, Lemma 6.6.6 (preprint 6.31).
- `Chapter6/Continuation`: continuation over a simply connected base, the propagation $\Phi_{E_1}^{-1}\circ$ continuation $\circ\,\Phi_{E_0}$.
- `Chapter6/Propagation`: stabilized kernels and the push-forward φ_*; Lemmas 6.6.8 and 6.6.10 (preprint 6.33 and 6.34) for it.
- `Chapter6/StabKernelsExist`: the stabilized kernels exist.
- `Chapter6/PushData`: push-forwards along paths, their existence, halves and reparametrizations.
- `Chapter6/AddSections`: the linear algebra of Lemma 6.6.8 (preprint 6.33).
- `Chapter6/FamilySections`: sections over parameter families.
- `Chapter6/PropagationWellDefined`: Proposition 6.6.7 (preprint 6.32), the homotopy argument, Theorem 6.6.11 (preprint 6.35).
- `Chapter6/ScPaths`: sc-smooth paths in a chart, concatenation.
- `Chapter6/ContinuousOrientation`: Definition 6.6.12 (preprint 6.36), continuous orientations.
- `Chapter6/ScHomotopy`: reparametrization homotopies.
- `Chapter6/PathCategory`: homotopy classes and the path category PX.
- `Chapter6/ChartTransfer`: conjugation of determinants, the transfer between chart points.
- `Chapter6/PushTransfer`: the push-forward does not depend on the chart.
- `Chapter6/SectionOrientation`: orientable neighbourhoods as structures, Or_x^f in a chosen chart, Definitions 6.6.15, 6.6.16 (preprint 6.39, 6.40).
- `Chapter6/PathTransport`: transport along paths from local push-forwards.
- `Chapter6/OrientationFunctor`: Theorem 6.6.13 (preprint 6.37), Definitions 6.6.14–6.6.16 (preprint 6.38–6.40) and their equivalences.
- `Chapter6/SolutionOrientation`: Theorem 6.6.18 (preprint 6.41).
- `Chapter6/ProductOrientation`: Remark 6.6.19 (preprint 6.42).

Throughout, E and F are real Banach spaces, $T : E \to F$ is bounded, $R(T)$ is its range and $\operatorname{coker} T = F/R(T)$; T is Fredholm if $\ker T$ and $\operatorname{coker} T$ are finite-dimensional, with index $\dim\ker T - \dim\operatorname{coker} T$. A *line* is a one-dimensional real space. $P : Y \to X$ is a strong bundle over an M-polyfold, both paracompact and Hausdorff, and from §VI.10 on X is tame with sc-smooth bump functions and f is an sc-Fredholm section with tame germ data (Definition 3.6.4 (preprint 3.51)), the reading of Chapter 5 (Part V). A strong bundle chart c has base chart φ, retraction r on an open set U of a partial quadrant C, retract $O = r(U)$, and fibre projections $\rho(u)$; the *principal part* of a section is its fibre coordinate in the chart. $\mathbb{R}^1$ is the M-polyfold of the parameter t, with t the point of $\mathbb{R}^1$ `Chapter6/OrientableNbhd · pt1`, and (τ,t) the point of $\mathbb{R}^2$ `Chapter6/OrientableNbhd · pt2`; κ is Mathlib's smooth transition function, 0 on $(-\infty,0]$, 1 on $[1,\infty)$.

VI.1 Definition 6.2.1 (preprint 6.1): the space $\mathrm{Lin}(f,x)$

§6.2 (preprint §6.1) (pp. 237–240 (preprint pp. 200–201)) works at a smooth point x. Since $f(x)$ is a smooth point of the fibre, there is an sc$^+$-section s with $s(x) = f(x)$ — in a chart, $s(y) = R(y, f(x))$ — and $(f-s)'(x) : T_xX \to Y_x$ is sc-Fredholm. Definition 6.2.1 (preprint 6.1) collects these linearizations,

perturbed by sc⁺-operators, into $\mathrm{Lin}(f,x)$; the index is the index of any member, and Proposition 6.2.2 (preprint 6.2) says that it is constant along sc-smooth paths.

The chart point (V6.1). A chart point of x is a strong bundle chart c of the atlas whose base chart contains x, with the level-1 point z of the model over $\varphi(x)$ `Chapter6/LinSet · ChartPoint`. In it T_xX is $T_zO = Dr(z)E$, the range of the derivative of the retraction `Chapter6/LinSet · ChartPoint.T`, and Y_x is $K_z = \rho(z)F$, the range of the fibre projection over z `Chapter6/LinSet · ChartPoint.K`; both are closed, as ranges of continuous projections, hence Banach spaces. An sc⁺-operator $T_xX \to Y_x$ is level 0 of an sc⁺-operator between these sc-subspaces `Chapter6/LinSet · ChartPoint.IsScPlusOp`.

Definition 6.2.1 (preprint 6.1) (V6.1). $\mathrm{Lin}(f,x)$, at the chart point p, is the set of the operators $(f+s)'(x)+a : T_zO \to K_z$ for an sc⁺-section s with $(f+s)(x)=0$ and an sc⁺-operator a `Chapter6/LinSet · ChartPoint.linSet` `Chapter6/LinSet · ChartPoint.mem_linSet_iff`. The linearization of $f+s$ at its zero x is the derivative at z of the principal part composed with r, restricted to T_zO — the $\mathrm{Lin}(z)$ of Part III. Replacing the text's s by $-s$ turns $f-s$ with $s(x)=f(x)$ into $f+s$ with $(f+s)(x)=0$.

Two sections differ by an sc⁺-operator. For two sections s, t with $(f+s)(x)=(f+t)(x)=0$, the principal parts of $f+s$ and $f+t$ differ by that of the sc⁺-section $s-t$, whose derivative at z, followed by $\rho(z)$, is level 0 of an sc⁺-operator: the derivative of an sc⁺-map is sc⁺ on the shifted levels. So $\mathrm{Lin}(f,x)$ is $(f+s)'(x)$ plus all the sc⁺-operators, for any one s `Chapter6/LinSet · ChartPoint.exists_isScPlusOp_linAt_eq`, and two members differ by an sc⁺-operator `Chapter6/LinSet · ChartPoint.isScPlusOp_sub_of_mem_linSet`. The paragraph before the definition puts this as "$(f-s)'(x)$ is an sc-Fredholm operator if and only if $(f-s)'(x)$ is"; it is read with $(f-t)'(x)$ in the second place (possible misprint; Findings, Chapter 6, item 41).

Not empty. The text makes its section locally, in a chart. The formalization uses global sc⁺-sections: $\mathrm{Lin}(f,x)$ is not empty when P has sc⁺-sections through the smooth points of its fibres, one through $-f(x)$ giving a member `Chapter6/LinSet · ChartPoint.linSet_nonempty`, and bump functions give such sections `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. This is the hidden hypothesis of the definition: an sc⁺-section through $f(x)$.

Fredholm, of one index. Every member is Fredholm on level 0, with the index $\mathrm{ind}(f,x)$ of the germ at x (§III.49) `Chapter6/LinSet · ChartPoint.isFredholm_isIndexAt_of_mem_linSet` `Chapter6/LinSet · ChartPoint.exists_index_linSet` `Chapter6/LinSet · ChartPoint.fredholmIndex_of_mem_linSet`. The proof: $f+s$ has germ data at x (an sc-Fredholm germ plus an sc⁺-germ is one), so its linearization at the zero x is Fredholm of the index of the germ, through the conjugation to a basic germ (Proposition 3.6.5 (preprint 3.52), §III.30); adding an sc⁺-operator, compact on level 0, keeps the Fredholm property and the index; and the index of the germ does not depend on the choices. That the members are sc-Fredholm on every level, as the text says, is not stated: the determinant lines use level 0 only (gap G.64; Findings, Chapter 6, item 40).

Convex `Chapter6/LinSet · ChartPoint.convex_linSet`. $(1-\tau)(L+a)+\tau(L+b) = L+((1-\tau)a+\tau b)$, and the sc⁺-operators form a vector space. The text notes this on preprint p. 219, where Proposition 6.6.1 (preprint 6.26) needs it.

Proposition 6.2.2 (preprint 6.2) (V3.98). The constancy of the index along sc-smooth paths was formalized with Chapter 3 and is retold in §III.49: for paths on $\mathbb{R}$, with the text's hypotheses — tame germ data, no bump functions, the locally defined sc⁺-sections of Definition 6.2.1 (preprint 6.1). It is not repeated here. The facts on paths recalled before it on preprint p. 201 — an M-polyfold is locally path connected, and two smooth points in one path component are joined by an sc⁺-smooth path — are formalized only inside a chart (§VI.18; gap G.56; Findings, Chapter 6, item 42).

VI.2 Determinant lines, Definition 6.3.1 (preprint 6.3), Proposition 6.3.2 (preprint 6.4)

$\lambda(E)$ **and** $\lambda^*(E)$ (V6.2). For a finite-dimensional real space E, $\lambda(E) = \Lambda^{\max}E$ is the top exterior power `Chapter6/DetLine · maxPow` and $\lambda^*(E)$ its dual `Chapter6/DetLine · maxPowDual`; both are lines `Chapter6/DetLine · finrank_maxPow`, $\lambda(\{0\}) = \mathbb{R}$ being the zeroth exterior power. A family $a_1, \dots, a_n$ with $n = \dim E$ has its wedge $a_1 \wedge \cdots \wedge a_n$ `Chapter6/DetLine · wedge`. Two rules carry every computation of the chapter. A change of basis multiplies the wedge by its determinant `Chapter6/DetLine · wedge_eq_det_smul`, also when only one block of a two-block wedge $a \wedge b$ is changed `Chapter6/DetLine · wedge_append_basis_left` `Chapter6/DetLine · wedge_append_basis_right`; and adding to the vectors of one block vectors of the span of the other does not change the wedge `Chapter6/DetLine · wedge_append_add_left` `Chapter6/DetLine · wedge_append_add_right`. For $v \neq 0$ in a line, v^* is the functional with $v^*(v) = 1$ `Chapter6/DetLine · dualVec` `Chapter6/DetLine · dualVec_self`, and that value determines it `Chapter6/DetLine · eq_dualVec_of_apply_eq_one`. For $\dim E = \dim F$ and $\Phi : E \to F$, $\lambda(\Phi)(a_1 \wedge \cdots \wedge a_n) = \Phi a_1 \wedge \cdots \wedge \Phi a_n$ `Chapter6/DetLine · maxPowMap` `Chapter6/DetLine · maxPowMap_wedge`, with $\lambda(\Psi\Phi) = \lambda(\Psi)\lambda(\Phi)$ `Chapter6/DetLine · maxPowMap_comp`, and $\lambda(\Phi) \neq 0$ iff Φ is bijective `Chapter6/DetLine · maxPowMap_ne_zero_iff`; $\lambda^*(\Phi)$ is its dual `Chapter6/DetLine · maxPowDualMap` (pp. 241–243 (preprint pp. 203–204)).

Definition 6.3.1 (preprint 6.3) (V6.3). $\iota : \lambda(E^*) \to \lambda^*(E)$ is $\iota(e_1^* \wedge \cdots \wedge e_n^*)(a_1 \wedge \cdots \wedge a_n) = \det(e_i^*(a_j))$ `Chapter6/DetLine · iotaDual` `Chapter6/DetLine · iotaDual_wedge_wedge`. The formula is alternating and multilinear in both families, so it descends to the exterior powers; that is the text's well-definedness. ι is a nonzero map of lines `Chapter6/DetLine · iotaDual_ne_zero`, and for a basis with its dual basis $\iota(e_1^* \wedge \cdots \wedge e_n^*) = (e_1 \wedge \cdots \wedge e_n)^*$ `Chapter6/DetLine · iotaDual_wedge_dualBasis`, since the matrix is the identity. The text's matrix $(e_i^*(a_j))$ is the transpose of the one of Mathlib's pairing, with the same determinant (preprint p. 204).

Proposition 6.3.2 (preprint 6.4) (V6.4) `Chapter6/DetLine · proposition_6_4`. For an isomorphism $\Phi : E \to F$, $\lambda^*(\Phi) \circ \iota = \iota \circ \lambda(\Phi^*)$. Both sides, evaluated at $f_1^* \wedge \cdots \wedge f_n^*$ and $a_1 \wedge \cdots \wedge a_n$, give $\det(f_i^*(\Phi a_j))$: the left side by the definition of $\lambda^*(\Phi)$, the right side because $(\Phi^* f_i^*)(a_j) = f_i^*(\Phi a_j)$ (p. 243 (preprint p. 205)).

VI.3 Four-term exact sequences: Φ_E, Lemma 6.3.3 (preprint 6.5), Proposition 6.3.5 (preprint 6.6)

The data of (91) (V6.5). Let $\mathbf{E} : 0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\gamma} D \to 0$ be an exact sequence of finite-dimensional spaces `Chapter6/ExactSequence · FourTermExact`. A *choice* `Chapter6/ExactSequence · FourTermExact.Choice` consists of a complement Z of $\alpha(A)$ in B, a complement V of $\beta(B)$ in C, bases a of A, d of D, b' of Z, and the γ-preimages c in V of the basis d. Then $\alpha(a), b'$ is a basis of B, and $c, \beta(b')$ is one of C, because β maps Z isomorphically onto $\beta(B)$ and γ maps V isomorphically onto D. Formula (91) assigns to the choice the value $(\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}) \otimes (c_1 \wedge \cdots \wedge c_l \wedge \beta(b'_1) \wedge \cdots \wedge \beta(b'_{m-n}))^*$ as the image of $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, with the text's two ordering conventions `Chapter6/ExactSequence · FourTermExact.Choice.value`. $\Phi_E : \lambda(A) \otimes \lambda^*(D) \to \lambda(B) \otimes \lambda^*(C)$ is the linear map of lines defined by one fixed choice `Chapter6/ExactSequence · FourTermExact.canonicalChoice` `Chapter6/ExactSequence · FourTermExact.phi`; it is nonzero `Chapter6/ExactSequence · FourTermExact.phi_ne_zero`, hence an isomorphism `Chapter6/ExactSequence · FourTermExact.phiEquiv` (preprint p. 206).

Lemma 6.3.3 (preprint 6.5) (V6.6) `Chapter6/ExactSequence · FourTermExact.lemma_6_5`. The text: Φ_E does not depend on the choices. The formalization proves the strongest form: Φ_E, defined from one choice, maps the source element $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$ of *any other* choice to the right-hand side of (91) computed with that choice. This is the form every later computation uses — γ_T^P is always evaluated in bases adapted to the problem at hand (Findings, Chapter 6, item 1). The proof is §6.8.1 (preprint §6.7.1) (§VI.28).

Proposition 6.3.5 (preprint 6.6) (V6.7). With the contraction $\bar{\iota}(v \otimes b \otimes \varphi) = \varphi(v)\, b$ `Chapter6/`

ExactSequence · contractDual, (92) defines $\Psi_E(c \otimes h) = \bar{\iota}(c \otimes \Phi_E(h))$ Chapter6/ExactSequence · FourTermExact.psi. The proposition Chapter6/ExactSequence · FourTermExact.proposition_6_6: Ψ_E maps the wedge of the basis $c, \beta(b')$ of C, tensored with $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, to $\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}$. By Lemma 6.3.3 (preprint 6.5) in its strong form, Φ_E of the source is the value of (91) for this choice, and the dual wedge evaluated on the wedge is 1. The wedge in C is taken in the order of (91); the statement exchanges the names of its two blocks (possible misprint; Findings, Chapter 6, item 2) (preprint p. 207).

VI.4 Determinants of Fredholm operators, good left-projections, and Lemma 6.10 (preprint)

Definition 6.4.1 (preprint 6.7) (V6.8). $\det(T) = \lambda(\ker T) \otimes \lambda^*(F/R(T))$ Chapter6/FredholmDet · linDet Chapter6/FredholmDet · fredholmDet, a line for T Fredholm Chapter6/FredholmDet · finrank_fredholmDet. Equal operators have identified determinants Chapter6/FredholmDet · linDetEquivOfEq: PQT and PT are the same operator when $P \leq Q$, but not the same expression, and Proposition 6.4.11 (preprint 6.12) compares their determinants. The alternative definition $\lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ mentioned after the definition is not formalized (gap G.57) (preprint p. 208).

Definition 6.4.2 (preprint 6.8) (V6.9). An orientation of T is a ray of $\det(T)$ Chapter6/FredholmDet · FredholmOrientation; there is one Chapter6/FredholmDet · nonempty_fredholmOrientation, and exactly two Chapter6/FredholmDet · exists_two_fredholmOrientation (preprint p. 208).

Definition 6.4.3 (preprint 6.9) and the facts after it (V6.10). A *good left-projection* for T is a bounded projection $P : F \to F$ with $\dim F/R(P) < \infty$ and $R(PT) = R(P)$ Chapter6/GoodProjections · IsGoodLeftProj; Π_T is their set Chapter6/GoodProjections · goodLeftProjs. $P \leq Q$ iff $P = PQ = QP$ Chapter6/GoodProjections · ProjLE, a partial order Chapter6/GoodProjections · ProjLE.refl Chapter6/GoodProjections · ProjLE.antisymm Chapter6/GoodProjections · ProjLE.trans. A good left-projection is Fredholm of index 0 — $\ker P$ is a complement of $R(P)$, of dimension $\dim F/R(P)$ Chapter6/GoodProjections · IsGoodLeftProj.isFredholm Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_eq_zero — so PT is Fredholm with $\mathrm{ind}(PT) = \mathrm{ind}(T)$ when T is, by the index of a composite Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp. For $P, P' \in \Pi_T$, $R(P) \cap R(P')$ has finite codimension Chapter6/GoodProjections · IsGoodLeftProj.finiteDimensional_quotient_inf. And $\Pi_T \neq \emptyset$ for T Fredholm Chapter6/GoodProjections · exists_isGoodLeftProj — the projection onto $R(T)$ along a finite-dimensional complement is one — which the text uses from Proposition 6.4.10 (preprint Lemma 6.11) on without stating it (Findings, Chapter 6, item 3) (preprint p. 208).

Lemma 6.10 (preprint) (V6.11) Chapter6/GoodProjections · lemma_6_10. *For $P, P' \in \Pi_T$ there is $P'' \in \Pi_T$ with $P'' \leq P$ and $P'' \leq P'$*, for any bounded T. The text's P'' projects onto $R(P) \cap R(P')$ along a complement assembled from $\ker P$, $\ker P'$ and a further piece, and $P'' = P''P$ needs $\ker P \subseteq \ker P''$. As we read it, this does not hold in general: on $\mathbb{R}^2$ with $T = 1$, let P project onto the x-axis along the y-axis and P' along the diagonal; $R(P) \cap R(P')$ is the x-axis, and a projection onto it below both would equal both Chapter6/GoodProjections · lemma_6_10_text_reading. The lemma holds with a smaller range. *Step 1.* $K = \ker P + \ker P'$ is finite-dimensional, each kernel having the dimension of a finite-dimensional quotient. *Step 2.* K has a closed complement N (finite-dimensional subspaces are complemented, by Hahn–Banach), and $R'' = R(P) \cap R(P') \cap N$ is closed, of finite codimension, and meets K only in 0. *Step 3.* R'' has a complement $C \supseteq K$, topological because the quotient by R'' is finite-dimensional; P'' is the projection onto R'' along C. *Step 4.* $\ker P \subseteq C = \ker P''$ gives $P''P = P''$, and $R'' \subseteq R(P)$ gives $PP'' = P''$; the same for P'. *Step 5.* $R(P''T) = P''(R(PT)) = P''(R(P)) = R''$, since P'' is the identity on $R'' \subseteq R(P)$. Nothing uses that T is Fredholm. The 2017 version drops the lemma and remarks that a common lower bound need not exist; with the order of Definition 6.4.3 (preprint 6.9), used by both versions, it always exists, as we read it, so the preprint's route to Lemma 6.5.9 (preprint 6.25) is sound and is the one formalized (Findings, Chapter 6, item 4) (preprint pp. 208–209).

VI.5 Proposition 6.4.10 (preprint Lemma 6.11), γ_T^P, Proposition 6.4.11 (preprint 6.12) and Corollary 6.4.12 (preprint 6.13)

Proposition 6.4.10 (preprint Lemma 6.11) (V6.12). For $P \in \Pi_T$, the sequence $\mathbf{E}(T,P) : 0 \to \ker T \to \ker(PT) \to F/R(P) \to \operatorname{coker} T \to 0$, with the inclusion `Chapter6/ProjectionSequence · projSeqJ`, $x \mapsto Tx + R(P)$ `Chapter6/ProjectionSequence · projSeqPhi` and $y + R(P) \mapsto (I-P)y + R(T)$ `Chapter6/ProjectionSequence · projSeqPi`, is exact `Chapter6/ProjectionSequence · lemma_6_11`, a four-term exact sequence of §VI.3 `Chapter6/ProjectionSequence · projSeq`. The four checks use only $P^2 = P$ and $R(PT) = R(P)$. At $\ker(PT)$: if $PTx = 0$ and $Tx \in R(P)$, then $Tx \in R(P) \cap \ker P = 0$. At $F/R(P)$: the last map is well defined because $(I-P)R(P) = 0$; it kills $[Tx]$ for $x \in \ker(PT)$, since $(I-P)Tx = Tx$; and if $(I-P)y = Tx$, then $PTx = 0$ and $y - Tx = Py \in R(P)$. Onto: $Py = PTe$ for some e, and then $(I-P)(y-Te) = y - Te$, which has the class of y in $\operatorname{coker} T$ (preprint p. 209).

The isomorphism γ_T^P (V6.13). $\gamma_T^P : \det(T) \to \det(PT)$ is $\Phi_{\mathbf{E}(T,P)}$ followed by the identification of $\lambda^*(F/R(P))$ with $\lambda^*(F/R(PT))$ `Chapter6/ProjectionSequence · gammaProj` `Chapter6/ProjectionSequence · gammaProj_tmul` — the text's "recalling $R(P) = R(PT)$" (Findings, Chapter 6, item 5). By Lemma 6.3.3 (preprint 6.5) it is given on wedges by (91) for any choice `Chapter6/ProjectionSequence · gammaProj_wedge_tmul` (preprint p. 210).

Proposition 6.4.11 (preprint 6.12) (V6.14). *For $P \le Q$ in Π_T, QT is Fredholm, $P \in \Pi_{QT}$, and $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$,* up to the identification of $\det(P(QT))$ with $\det(PT)$ `Chapter6/ProjectionSequence · proposition_6_12`. $P \in \Pi_{QT}$ because $R(PQT) = R(PT) = R(P)$ `Chapter6/ProjectionSequence · IsGoodLeftProj.of_projLE`. The identity is a computation in adapted bases `Chapter6/ProjectionSequence · comp_comp_eq_of_projLE`, the proof of §6.8.2 (preprint §6.7.2), retold in §VI.29 (preprint p. 211).

Corollary 6.4.12 (preprint 6.13) (V6.15) `Chapter6/ProjectionSequence · corollary_6_13`. For $P \le S$ and $P \le R$ in Π_T, $(\gamma_{RT}^P)^{-1} \circ \gamma_{ST}^P = \gamma_T^R \circ (\gamma_T^S)^{-1}$. Proposition 6.4.11 (preprint 6.12) twice gives $\gamma_{ST}^P \gamma_T^S = \gamma_T^P = \gamma_{RT}^P \gamma_T^R$; rearranging gives the claim. So the left side, the comparison of two local frames through a common lower bound, does not depend on the lower bound (preprint p. 211).

VI.6 The stabilization: Lemma 6.4.14 (preprint 6.14), formula (94), Proposition 6.4.15 (preprint 6.15), Definition 6.4.16 (preprint 6.16)

Lemma 6.4.14 (preprint 6.14) (V6.16). For T Fredholm and $\varphi : \mathbb{R}^n \to F$ with $T_\varphi(e,r) = Te + \varphi r$ onto, the sequence $0 \to \ker T \xrightarrow{j} \ker T_\varphi \xrightarrow{\pi} \mathbb{R}^n \xrightarrow{[\varphi]} F/R(T) \to 0$, with $j(e) = (e,0)$ `Chapter6/Stabilization · stabJ`, $\pi(e,r) = r$ `Chapter6/Stabilization · stabPi` and $[\varphi](r) = \varphi(r) + R(T)$ `Chapter6/Stabilization · stabClass`, is exact `Chapter6/Stabilization · lemma_6_14` `Chapter6/Stabilization · stabSeq`. j is injective; $\ker\pi \cap \ker T_\varphi$ is $\{(e,0) \mid Te = 0\} = j(\ker T)$; for $(e,r) \in \ker T_\varphi$, $\varphi r = -Te \in R(T)$, and conversely $\varphi r = -Te$ puts (e,r) in $\ker T_\varphi$; $[\varphi]$ is onto because T_φ is. A stabilizing φ exists for every Fredholm T `Chapter6/Stabilization · exists_surjective_coprod`: send a basis of $\mathbb{R}^n$ to representatives of a basis of the cokernel. The proof writes $j(e) = (0,e)$, read as $(e,0)$ (possible misprint; Findings, Chapter 6, item 8) (pp. 251–256 (preprint pp. 211–212)).

Formula (94) (V6.17). $\psi : \lambda^*(\mathbb{R}^n) \to \lambda^*(\{0\})$ maps $(e_1 \wedge \cdots \wedge e_n)^*$ to 1^* for the standard basis `Chapter6/Stabilization · stdWedge` `Chapter6/Stabilization · stabOne` `Chapter6/Stabilization · stabPsi`. Then $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c_1,\dots,c_n]^{-1}\, 1^*$ for every basis `Chapter6/Stabilization · stabPsi_dualVec`: $c_1 \wedge \cdots \wedge c_n = \det[c]\, e_1 \wedge \cdots \wedge e_n$, and the dual vectors scale inversely (p. 256 (preprint p. 212)).

Proposition 6.4.15 (preprint 6.15) (V6.18). Since T_φ is onto, $\det(T_\varphi) = \lambda(\ker T_\varphi) \otimes \lambda^*(\{0\})$, and $\iota_\varphi = (1 \otimes \psi) \circ \Phi_{\mathbf{E}}$, for the sequence $\mathbf{E}$ of Lemma 6.4.14 (preprint 6.14), is an isomorphism $\det(T) \to \det(T_\varphi)$ `Chapter6/Stabilization · stabIota` `Chapter6/Stabilization · stabIota_bijective`. The formula `Chapter6/`

Stabilization · proposition_6_15: for a basis a of $\ker T$, vectors $\bar{c}_1, \dots, \bar{c}_m$ with $[\varphi](\bar{c}_i)$ a basis of $F/R(T)$, and $\bar{b}_i = (b_i, r_i)$ completing $(a, 0)$ to a basis of $\ker T_\varphi$, $\iota_\varphi(h) = \det(\bar{c}, r)^{-1}\,((a_1, 0) \wedge \cdots \wedge (a_k, 0) \wedge \bar{b}_1 \wedge \cdots \wedge \bar{b}_l) \otimes 1^*$. These data are a choice of (91) for $\mathbf{E}$ — Z the span of the $\bar{b}_i$, V the span of the $\bar{c}_j$ — so Lemma 6.3.3 (preprint 6.5) gives $\Phi_{\mathbf{E}}(h) = ((a, 0) \wedge \bar{b}) \otimes (\bar{c} \wedge r)^*$, and (94) turns the second factor into $\det(\bar{c}, r)^{-1} 1^*$. To form $\det(\bar{c}, r)$ one needs $m + l = n$: the statement takes it as a hypothesis, and it is proved to hold always Chapter6/Stabilization · stab_card_eq; that the r_i and $\bar{c}_j$ form a basis of $\mathbb{R}^n$, which the text notices, is proved — the r_i span $\pi(\ker T_\varphi) = \ker[\varphi]$, and the $\bar{c}_j$ a complement of it Chapter6/Stabilization · isCompl_range_stabPi_span. The statement calls ψ "φ" (possible misprint). The identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after the proposition are not formalized; the formalized statements use only (94) (gap G.58; Findings, Chapter 6, items 9, 10) (preprint p. 213).

Definition 6.4.16 (preprint 6.16) (V6.19). For Fredholm T, S, the operator $T \oplus S$ has $\ker(T \oplus S) \cong \ker T \times \ker S$ and $\operatorname{coker}(T \oplus S) \cong \operatorname{coker} T \times \operatorname{coker} S$, through the inclusions of the factors Chapter6/DirectSumDet · kerInl Chapter6/DirectSumDet · kerInr Chapter6/DirectSumDet · cokerInl Chapter6/DirectSumDet · cokerInr Chapter6/DirectSumDet · kerProdEquiv Chapter6/DirectSumDet · cokerProdEquiv; the dimensions add Chapter6/DirectSumDet · finrank_ker_prodMap Chapter6/DirectSumDet · finrank_coker_prodMap, so $T \oplus S$ is Fredholm Chapter6/DirectSumDet · isFredholm_prodMap. The map $\det(T) \otimes \det(S) \to \det(T \oplus S)$, $h \otimes h' \mapsto (a \wedge a') \otimes (d \wedge d')^*$, is defined by the formula in one choice of bases Chapter6/DirectSumDet · directSumVec Chapter6/DirectSumDet · directSumDet, satisfies it for every choice Chapter6/DirectSumDet · directSumVec_eq_smul Chapter6/DirectSumDet · definition_6_16 — a change of bases multiplies source and target by the same product of determinants, the block wedges by the block determinants — and is an isomorphism Chapter6/DirectSumDet · directSumDet_bijective. "Natural" is read as the formula for every choice, as in Lemma 6.3.3 (preprint 6.5) (Findings, Chapter 6, item 11) (p. 258 (preprint p. 214)).

VI.7 Local determinant bundles: Lemmas 6.5.1–6.5.3 (preprint 6.17–6.19), Proposition 6.5.4 (preprint 6.20), Lemma 6.5.5 (preprint 6.21)

Lemma 6.5.1 (preprint 6.17) (V6.20). The Fredholm operators form a subspace of $L(E, F)$ Chapter6/LocalDetBundle · FredholmOp. *For a bounded T, $P \in \Pi_T$ and a topological complement Y of* $\ker(PT)$*, there is $\varepsilon > 0$ such that for $\|S - T\| < \varepsilon$, $P \in \Pi_S$ and $PS|_Y : Y \to R(P)$ is a topological isomorphism* Chapter6/LocalDetBundle · lemma_6_17. *Step 1.* $PT|_Y$ is injective, since $Y \cap \ker(PT) = 0$, and onto $R(P)$, since $PT(E) = PT(Y)$ and $R(PT) = R(P)$; as a bijection between the Banach spaces Y and $R(P)$ (both closed) it is an isomorphism by the open mapping theorem. *Step 2.* $\|PS|_Y - PT|_Y\| \le \|P\|\,\|S - T\|$ Chapter6/LocalDetBundle · projRestrict, and the invertible operators form an open set. *Step 3.* $R(PS) \supseteq PS(Y) = R(P)$, so $P \in \Pi_S$. That T is Fredholm is not used (Findings, Chapter 6, item 20); a topological complement is an algebraic complement with a continuous projection (p. 258 (preprint p. 214)).

Lemma 6.5.2 (preprint 6.18) (V6.21). A *local determinant chart* is the data (T, P, Y, ε) of Lemma 6.5.1 (preprint 6.17) with its conclusions Chapter6/LocalDetBundle · LocalDetChart. For $k \in \ker(PT)$, $y(k, S) = -(PS|_Y)^{-1} PSk$ Chapter6/LocalDetBundle · LocalDetChart.ySol is continuous in (k, S), inversion being continuous, and $k + y(k, S) \in \ker(PS)$. The map $(S, k) \mapsto (S, k + y(k, S))$ Chapter6/LocalDetBundle · LocalDetChart.kerMap is onto $\bigcup_{\|S-T\|<\varepsilon} \{S\} \times \ker(PS)$, inverted by the continuous projection onto $\ker(PT)$ along Y Chapter6/LocalDetBundle · LocalDetChart.kerProj: for $w \in \ker(PS)$ write $w = k + y$ with $y \in Y$; then $PSy = -PSk$, so $y = y(k, S)$ Chapter6/LocalDetBundle · LocalDetChart.lemma_6_18. "Is a trivial vector bundle" is stated as this trivialization and its inverse (Findings, Chapter 6, item 21) (preprint p. 215).

Lemma 6.5.3 (preprint 6.19) and Proposition 6.5.4 (preprint 6.20) (V6.22). For a basis k of $\ker(PT)$, $k_i(S) = k_i + y(k_i, S)$ is a continuous frame of the kernels Chapter6/LocalDetBundle · LocalDetChart.kerFrame, and with a constant basis e of $F/R(P) = F/R(PS)$, $\omega(S) = (k_1(S) \wedge \cdots \wedge k_n(S)) \otimes (e_1 \wedge \cdots \wedge e_l)^*$ Chapter6/LocalDetBundle · LocalDetChart.frame is a nowhere-zero frame Chapter6/LocalDetBundle · LocalDetChart.frame_ne_zero of $\mathrm{DET}(T, P, \varepsilon) = \bigcup_S \det(PS)$. The frames of two charts with the same P differ by a

continuous nowhere-zero factor `Chapter6/LocalDetBundle · proposition_6_20`: the determinant of the matrix expressing one kernel frame in the other, continuous because both frames are continuous in S, times the constant change of the bases of $F/R(P)$. Lemma 6.5.3 (preprint 6.19) — exterior powers, duals, sums and tensor products of topological vector bundles — is not formalized in general (gap G.59): it is applied only to trivial bundles, and Proposition 6.5.4 (preprint 6.20) is proved with the explicit frames; "in a natural way" is read as this relation between frames (Findings, Chapter 6, item 22) (preprint p. 216).

Lemma 6.5.5 (preprint 6.21) (V6.23). *For $Q \le P$, γ^Q_{PS} maps the P-frame at S to a continuous nowhere-zero multiple of the Q-frame, on the common ball of any two local determinant charts* `Chapter6/LocalDetBundle · LocalDetChart.gammaProj_frame_eq Chapter6/LocalDetBundle · lemma_6_21`. The text's sketch — extend the kernel sections of PS to sections spanning $\ker(QS)$ — is the proof. For a basis $f_1, \dots$ of $R(P) \cap \ker Q$, $z_m(S) = (PS|_Y)^{-1} f_m$ is continuous and $PS z_m(S) = f_m$; the $k_i(S)$ and $z_m(S)$ span $\ker(QPS) = \ker(QS)$; and in the choice of (91) for $\mathbf{E}(PS, Q)$ these vectors go with the *constant* classes $[f_m]$. So $\gamma^Q_{PS}(\omega_P(S))$ is $(k(S) \wedge z(S)) \otimes (e \wedge [f])^*$, continuous in S, and its ratio to the Q-frame of the other chart is the determinant of a continuous change of basis. The lemma is stated for two charts with different centres and radii, the form Lemma 6.5.9 (preprint 6.25) uses (Findings, Chapter 6, item 23) (pp. 259–261 (preprint pp. 216–217)).

VI.8 $\mathrm{DET}(E, F)$: Definition 6.5.6 (preprint 6.22), Propositions 6.5.7 and 6.5.8 (preprint 6.23 and 6.24), Lemma 6.5.9 (preprint 6.25)

The prebundle (V6.24). The fibre over a Fredholm S is $\det(S)$ `Chapter6/DetBundle · DetFiber`. A local determinant chart gives the pretrivialization $(S, h) \mapsto (S, \omega(S)^*(\gamma_S^P h))$ over its ball, with model fibre $\mathbb{R}$ `Chapter6/DetBundle · LocalDetChart.pretriv Chapter6/DetBundle · LocalDetChart.coord` — the text's $\gamma_{(T,P,\varepsilon)}$ read in the frame of Proposition 6.5.4 (preprint 6.20) (Findings, Chapter 6, item 24). Every Fredholm operator is the centre of one `Chapter6/DetBundle · exists_localDetChart`: $\Pi_T \neq \emptyset$ (§VI.4) and the finite-dimensional $\ker(PT)$ has a topological complement. The charts form a Mathlib vector prebundle `Chapter6/DetBundle · detPrebundle`, which needs exactly that the coordinate changes be continuous, Lemma 6.5.9 (preprint 6.25) below; its topology and bundle structure are those of $\mathrm{DET}(E, F)$ (preprint p. 217).

Lemma 6.5.9 (preprint 6.25) (V6.28), proved first. *The transition between two charts is $(S, t) \mapsto (S, g(S)t)$ with g continuous and nowhere zero on the common ball* `Chapter6/DetBundle · lemma_6_25_local Chapter6/DetBundle · lemma_6_25`. The proof is the text's: a common lower bound $Q \le P, P'$ (Lemma 6.10 (preprint), §VI.4); Corollary 6.4.12 (preprint 6.13) to pass from γ^P to $\gamma^{P'}$ through Q; and Lemma 6.5.5 (preprint 6.21) for each chart, which turns both frames into multiples of Q-frames (preprint p. 218).

Definition 6.5.6 (preprint 6.22), Propositions 6.5.7 and 6.5.8 (preprint 6.23 and 6.24). $\mathcal{B}$ is the collection of sets lying over the ball of some chart whose image in ball $\times \mathbb{R}$ is open (V6.25) `Chapter6/DetBundle · detBasis`. It is a basis of the topology of the prebundle (V6.26) `Chapter6/DetBundle · proposition_6_23`: its members are open, the charts being homeomorphisms onto ball $\times \mathbb{R}$, and every open set is the union of its intersections with chart domains, which are members; so the topology $\mathcal{B}$ generates is that one. $\mathrm{DET}(E, F)$ is a topological vector bundle with model fibre $\mathbb{R}$, and every $\gamma_{(T,P,\varepsilon)}$, read in its frame, is a trivialization (V6.27) `Chapter6/DetBundle · proposition_6_24` (preprint p. 218). The remark that $\mathrm{DET}(E, F)$ is not orientable and the classical case at the opening of §6.6 (preprint §6.5) are not formalized (gap G.60; Findings, Chapter 6, item 26).

VI.9 Orientations of line bundles, Proposition 6.6.1 (preprint 6.26), Definition 6.6.2 (preprint 6.27)

Orientations of line bundles (V6.29). An orientation of a topological line bundle is a choice of rays of the fibres given near every point by a continuous nowhere-zero section `Chapter6/`

LineBundleOrientation · IsLineOrientation Chapter6/LineBundleOrientation · LineOrientation. Two orientations that agree on one fibre agree over a preconnected base Chapter6/LineBundleOrientation · LineOrientation. eq_of_apply_eq: the set where they agree is open and closed. The rays of the fibres form a two-sheeted covering Chapter6/LineBundleOrientation · OrCover Chapter6/LineBundleOrientation · isCoveringMap_orCover, and an orientation is a continuous section of it; over a simply connected, locally path-connected base the lifting criterion extends every orientation of one fibre Chapter6/LineBundleOrientation · exists_ lineOrientation_apply_eq, and over a nonempty such base there are exactly two orientations Chapter6/ LineBundleOrientation · exists_two_lineOrientation. The text gives "since the base space is contractible" as the reason, without the theorem it needs (Findings, Chapter 6, item 25) (p. 263 (preprint p. 219)).

Proposition 6.6.1 (preprint 6.26) (V6.30). Over a nonempty convex set C of Fredholm operators Chapter6/ DetBundle · fredholmIncl, the restriction $\mathrm{DET}(E,F)|_C$ Chapter6/DetBundle · DetOn is a topological line bundle with exactly two orientations Chapter6/DetBundle · proposition_6_26: a nonempty convex set of a normed space is contractible, hence simply connected, and locally path-connected. The proposition is stated for every nonempty convex C; $\mathrm{Lin}(f,x)$ is one (§VI.1) (preprint p. 219).

Definition 6.6.2 (preprint 6.27) (V6.31). At a chart point the members of $\mathrm{Lin}(f,x)$ are Fredholm Chapter6/LinOrientation · ChartPoint.isFredholm_of_mem_linSet; $\mathrm{DET}(f,x)$ is the pull-back of $\mathrm{DET}(T_zO,K_z)$ to $\mathrm{Lin}(f,x)$ Chapter6/LinOrientation · ChartPoint.detLin, and Or_x^f is the set of its orientations Chapter6/ LinOrientation · ChartPoint.Or, with exactly two elements Chapter6/LinOrientation · ChartPoint.exists_two_or. For each $L \in \mathrm{Lin}(f,x)$, $o \mapsto o(L)$ is a bijection onto the two orientations of $\det(L)$ Chapter6/ LinOrientation · ChartPoint.orEquivRay: injective because an orientation is determined by one value over the connected base, onto because each ray extends. The category $\mathcal{O}_f$ is read through its objects, the two-element sets, with bijections as morphisms; between two two-element sets there are bijections Chapter6/ OrientationFunctor · nonempty_equiv_of_two, each determined by one value Chapter6/OrientationFunctor · equiv_ eq_of_apply_eq_of_two. No negation is set up on the tensor products, so $\{o_x,-o_x\}$ is stated as "exactly two elements" (Findings, Chapter 6, item 44) (preprint pp. 219–220). The comparison of Or_x^f in two charts is §VI.21.

VI.10 Proposition 6.6.3 (preprint 6.28): the neighbourhoods of the propagation

The text: every smooth point x has an open neighbourhood U which (1) is sc-smoothly contractible, (2) has $\{y \in \mathrm{cl}(U) \mid f(y)=0\}$ compact, (3) carries, for every sc-smooth path $\varphi:[0,1]\to U$, an sc$^+$-section $s: U\times[0,1]\to Y$ with $s(\varphi(t),t)=f(\varphi(t))$, and (4) carries sc$^+$-sections $q: U\times[0,1]\to Y$ through every smooth point of every fibre over U_∞, at every t_0. The proof takes local strong bundle coordinates, $P: K\to O$ with $x=0$.

The statement formalized (V6.32) Chapter6/OrientableNbhd · proposition_6_28. *With sc-smooth bump functions, every smooth x has open $U\subseteq W$ in the domain of one chart of the atlas with* (1′) *any two sc-smooth paths in U with the same end points are sc-smoothly homotopic in W with the end points fixed* Chapter6/ OrientableNbhd · IsScHomotopyRel; (2) $\{y\in\mathrm{cl}(U)\mid f(y)=0\}$ *is compact*; (3) *for every sc-smooth family $\Phi:\mathbb{R}^n\to X$ with $\Phi([0,1]^n)\subseteq W$ there is an sc$^+$-section S of $\pi^*P\to\mathbb{R}^n\times X$ with $(f+S)(q,\Phi(q))=0$ for q near $[0,1]^n$, depending on $q\in[0,1]^n$ only through $\Phi(q)$*; (4) *π^*P has sc$^+$-sections through all smooth fibre points.* A path "in U" is one with $\gamma([0,1])\subseteq U$.

The neighbourhoods. Take a chart c of the atlas at x, $u_0=\varphi(x)$, the retraction r on $U_c\subseteq C$ with $O=r(U_c)$. Choose δ with $B_\delta\cap C\subseteq U_c$ (the domain of the retraction is relatively open); δ_2 with $r(B_{\delta_2}\cap U_c)\subseteq B_\delta$ (continuity of r at u_0 on level 0, $r(u_0)=u_0$); a neighbourhood N of x whose closure meets the zeros of f in a compact set Chapter6/OrientableNbhd · exists_nhds_isCompact_zeros — at a zero by the local compactness of Theorem 5.2.3 (preprint 5.6) (§V.7), elsewhere because the zero set is closed and misses a neighbourhood of x; a closed neighbourhood $T\subseteq N$ of x and δ_3 with $\varphi^{-1}(B_{\delta_3}\cap O)\subseteq T$. With $\delta'=\min(\delta,\delta_2,\delta_3)$, $W=\varphi^{-1}(B(u_0,\delta)\cap O)$ and $U=\varphi^{-1}(B(u_0,\delta')\cap O)$.

(1′). Let γ_0, γ_1 be sc-smooth paths in U with the same end points. Since paths are maps on $\mathbb{R}$ that lie in U only for $t \in [0,1]$, first take a C^∞ map $\sigma : \mathbb{R} \to \mathbb{R}$ equal to the identity near $[0,1]$ with values in the open set $\gamma_0^{-1}(U) \cap \gamma_1^{-1}(U)$ `Chapter6/OrientableNbhd · exists_contDiff_eq_id_near_Icc_mapsTo`. Then $\Phi(\tau,t) = \varphi^{-1}(r((1-\kappa(\tau))\,\varphi(\gamma_0(\sigma(t))) + \kappa(\tau)\,\varphi(\gamma_1(\sigma(t)))))$. The convex combination lies in the convex set $B_{\delta'} \cap C$, hence in U_c; r maps it into $B_\delta \cap O$, so Φ takes values in W; it is sc-smooth as a composite of sc-smooth maps in the chart; at $\tau = 0, 1$ it is γ_0, γ_1 on $[0,1]$, because σ is the identity there and r fixes O; and the end points stay fixed, because γ_0 and γ_1 agree there. The text's contraction $\widehat{r}(t,y) = r(ty)$ contracts $r(V)$ for a convex V; for $U = r(V')$ it needs $r(V') \subseteq V'$, which is not stated, and a convex domain of the retraction is not available in general. As we read it, (1) is not established for U (gap G.61; Findings, Chapter 6, item 45). What Theorem 6.6.11 (preprint 6.35) uses is (1′).

(2). $\mathrm{cl}(U) \subseteq T \subseteq \mathrm{cl}(N)$, since $U \subseteq \varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$ and T is closed; so the zeros in $\mathrm{cl}(U)$ are a closed subset of a compact set.

(3), sections along families `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_family`. This is the text's $s(y) = R(y, f(x))$ made to depend on the parameter. An sc-smooth map on the finite-dimensional $\mathbb{R}^n$ takes values in the smooth points, and the principal part $q \mapsto \widehat{f}(\Phi(q))$ lifts to a C^∞ map into the level-1 fibre model. In the chart, the section over (q,y) whose fibre coordinate is the projection onto $K_{\varphi(y)}$ of that lift of $-\widehat{f}(\Phi(q))$ — after a reparametrization of the parameter, equal to the identity near $[0,1]^n$, that keeps the family in the chart domain — is an sc$^+$-section of the pull-back over the chart domain, and at $y = \Phi(q)$ it is $-f(\Phi(q))$. Cut off by an sc-smooth function on X supported in the chart domain and equal to 1 near a compact set containing $\Phi([0,1]^n)$ `Chapter6/OrientableNbhd · exists_isScSmoothMap_eq_one_nhdsSet`, it becomes a global sc$^+$-section, with $f + S = 0$ at the points $(q, \Phi(q))$ where the cut-off is 1 `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_mapsTo_of_bump`. For $q \in [0,1]^n$ it is the projection of $-\widehat{f}(\Phi(q))$, so it depends on q only through $\Phi(q)$. The text's (3) is for paths; it is stated for families because the homotopies of Theorem 6.6.11 (preprint 6.35) need it (Findings, Chapter 6, item 46).

(4) `Chapter6/OrientableNbhd · hasScPlusSectionsThrough_pullback_rn_of_bump`. Products of bump functions on $\mathbb{R}^n$ and on X are bump functions on $\mathbb{R}^n \times X$, and they give the sections. $\mathbb{R}^n \times X$ `Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod` and the total space of π^*P `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` are paracompact, as the project's M-polyfolds and strong bundles must be. *Hidden hypotheses:* bump functions, for the global sections of (3) and (4), where the text assumes partitions of unity from §6.3 (preprint §6.2) on. The construction uses neither that x is smooth nor that X is tame (pp. 263–264 (preprint pp. 220–221)).

VI.11 Lemma 6.6.4 (preprint 6.29): general position along a family

The text (preprint pp. 221–222): along a path φ with a section $s(\cdot,t)$, the section $F(t,y,\lambda) = f(y) - s(y,t) + \sum \lambda_i s_i(y,t)$ of the bundle over $[0,1] \times \mathbb{R}^k \times X$; Lemma 6.6.4 (preprint 6.29) asserts that there are $s_1, \dots, s_k$ with property (P): for every t, the slice F_t is in general position to the boundary of $X \times \mathbb{R}^k$ at $(\varphi(t), 0)$. The proof is the sentence "using the results from the transversality theory".

The statement formalized (V6.33) `Chapter6/Lemma629 · lemma_6_29`. *Over a tame X with bump functions, for f with tame germ data, an sc-smooth $\Phi : \mathbb{R}^n \to X$ and an sc$^+$-section S of π^*P with $(f+S)(q,\Phi(q)) = 0$ near $[0,1]^n$, there are finitely many sc$^+$-sections $s_1, \dots, s_k$ of π^*P such that $F(\lambda,(q,y)) = f(y) + S(q,y) + \sum \lambda_i s_i(q,y)$ is in general position at $(0,(q,\Phi(q)))$ for all q near $[0,1]^n$.*

The proof is that of Theorem 5.3.10 (preprint 5.18) (§V.20). The points $(0,(q,\Phi(q)))$ are zeros of F `Chapter6/Lemma629 · isZeroAt_paramSection_zero`, and for q in a compact neighbourhood of $[0,1]^n$ they form a compact set. At each of them finitely many sections make the parametrized section satisfy properties (i)–(ii) of §V.19; general position is open (§V.18); finitely many neighbourhoods cover the compact set, and the sections of all of them together do the job `Chapter6/Lemma629 · exists_sections_isGeneralPositionGerm_of_isCompact`. *Hidden hypotheses:* X tame, bump functions, tame germ data, the standing hypotheses of

§6.6 (preprint §6.5) and of Theorem 5.3.10 (preprint 5.18). The conclusion is the general position of F at the points $(t, \varphi(t), 0)$, which the text derives from (P) and which Lemmas 6.6.5 and 6.6.6 (preprint 6.30 and 6.31) use. (P) itself, the general position of each slice F_t, is not stated (gap G.63); as we read it, it follows, since the tangent vectors of the family of zeros lie in $\ker F'$ and $\mathbb{R}^n$ has no boundary, but this chart computation is not carried out (Findings, Chapter 6, item 47).

VI.12 Lemma 6.6.5 (preprint 6.30): kernels as continuous families of subspaces

The text (preprint p. 222): near $(t, \varphi(t), 0)$ the solution set of F is a manifold with boundary with corners, $\pi(t, y, \lambda) = t$ restricted to it is a submersion, and the kernels L_t form a smooth vector bundle L over $[0, 1]$. What the propagation needs is less: the kernels form a continuous family, so that an orientation of one of them can be continued to the others.

The kernels (V6.34). At a zero with surjective linearization and germ data of index d, $\dim\ker = d$ `Chapter6/KernelFamily · StrongBundle.IsSurjLinOfIndex.finrank_linKer`. Near a zero in good position the kernel is the tangent space of the solution set (Theorem 3.6.8 (preprint 3.54), §III.45), the range of the derivative of an sc-retraction onto the solution set read in the chart. So along a continuous family y_q of zeros in good position, read in one chart at continuously varying level-1 points z_q, near every q_0 the vectors $DS(z_q)b_i$, for a basis b of the kernel at q_0 and that retraction S, are a continuous frame of the kernels `Chapter6/KernelFamily · exists_local_frame_linKer` — continuous because $(z, v) \mapsto DS(z)v$ is, independent near q_0 because independence is open, and spanning because the dimension is constant. The kernels form a continuous family of subspaces of the model space, of dimension the index `Chapter6/KernelFamily · exists_subspaceFamily_linKer`. For an sc-smooth family of zeros $\mathbb{R}^n \to X$ read in a chart `Chapter6/KernelFamily · hasFDerivAt_scDeriv_rn`, differentiating $\hat{f}(\Phi(q)) = 0$ shows that the derivative lies in the kernel of the linearization `Chapter6/KernelFamily · fderiv_mem_linKer`. The bundle L is a continuous family of subspaces; the solution set as a manifold with boundary with corners near $(t, \varphi(t), 0)$, the map π and the smooth structure of L are not formalized (gap G.62; Findings, Chapter 6, item 48).

Continuous families of subspaces (V6.35). A family of n-dimensional subspaces L_b of a normed space, spanned near every point by a continuous frame `Chapter6/SubspaceBundle · SubspaceFamily`, has the line bundle $b \mapsto \lambda(L_b)$ `Chapter6/SubspaceBundle · SubspaceFamily.det`, with the wedges of the frames as trivializations `Chapter6/SubspaceBundle · SubspaceFamily.Frame.pretriv`; the coordinate change between two frames is the determinant of the matrix expressing one in the other, continuous because that matrix solves a linear system with continuous coefficients `Chapter6/SubspaceBundle · SubspaceFamily.Frame.coordChange_continuousOn`. Families and orientations restrict along continuous maps `Chapter6/SubspaceBundle · SubspaceFamily.comp` `Chapter6/SubspaceBundle · LineOrientation.comp`. Over $[0, 1]$ an orientation of $\lambda(L_0)$ extends uniquely to an orientation of $\lambda(L)$ (§VI.9), and its value at 1 is the continuation `Chapter6/SubspaceBundle · SubspaceFamily.transport` `Chapter6/SubspaceBundle · SubspaceFamily.transport_eq`, injective `Chapter6/SubspaceBundle · SubspaceFamily.transport_injective` and compatible with restriction `Chapter6/SubspaceBundle · SubspaceFamily.transport_comp`; over a simply connected, locally path-connected base the continuation along a path depends only on its end points. This is the text's "an orientation of the lines … determines by continuation an orientation of all the other lines" (pp. 264–266 (preprint pp. 222–223)), and the bundles $\tilde{L}$ of pp. 266–273 (preprint pp. 224–227). "Vector bundle" is read as a family of subspaces of one model space with continuous local frames.

Families of kernels and their images (V6.36). The kernels of a norm-continuous family of surjective Fredholm operators over a connected base form a continuous family `Chapter6/StabilizationContinuity · exists_subspaceFamily_ker`: their dimension is the index, and near q_0 the kernel of A_q is the graph over $\ker A_{q_0}$ given by Lemma 6.5.2 (preprint 6.18) with $P = 1$. The images of a continuous family under a jointly continuous family of linear maps injective on its members form one `Chapter6/StabilizationContinuity · SubspaceFamily.exists_map`, and continuous orientations are carried along `Chapter6/`

StabilizationContinuity · SubspaceFamily.isLineOrientation_map. Cutting a continuous family by the kernel of a fixed ℓ that is onto on every member gives one Chapter6/StabilizationContinuity · SubspaceFamily.exists_inf_ker. "As τ varies, the bundle varies continuously" (pp. 265–272 (preprint pp. 223–226)) is read in the operator norm, and for maps into a model space as joint continuity.

VI.13 The stabilized section in a product chart, and Lemma 6.6.6 (preprint 6.31)

The text (pp. 264–266 (preprint pp. 222–223)): with $T = (f - s(\cdot,t))'(\varphi(t))$, the kernel L_t of $F_t'(\varphi(t),0)$ fits into the sequence (96) $0 \to \ker T \to L_t \to \mathbb{R}^k \to \operatorname{coker} T \to 0$, and Lemma 6.6.6 (preprint 6.31) says that it is exact.

The product chart (V6.37). The parametrized section F lives on $\mathbb{R}^k \times (\mathbb{R}^n \times X)$; it is read in the product chart $(\lambda \mapsto \lambda) \times ((q \mapsto q) \times \varphi)$ at $(0,(q,z))$, whose tangent space and fibre are products Chapter6/ParamKernel · PullbackChartData.ofProd_scDeriv_r_apply Chapter6/ParamKernel · PullbackChartData.ofProd_mem_tangent_iff Chapter6/ParamKernel · PullbackChartData.mem_K_ofProd_iff. There the linearization of F is $(\lambda,(\sigma,v)) \mapsto D(\hat g_q \circ r)(z)v + \sum \lambda_i \hat s_i(q,\cdot)(z) + D_q\sigma$, with $g_q = f + S(q,\cdot)$ and D_q the derivative in the parameter Chapter6/ParamKernel · StrongBundle.scDeriv_ofProd_slice_apply Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_apply Chapter6/ParamKernel · StrongBundleChart.paramDerivAt Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_ofProd_ofProd. With the projection to σ and the inclusion of the slice $\sigma = 0$ Chapter6/ParamKernel · sliceProj Chapter6/ParamKernel · sliceIncl, the kernel cut by $\sigma = 0$ is the image of $\ker(T \oplus \varphi)$ Chapter6/ParamKernel · linKer_inf_ker_sliceProj_eq_map Chapter6/ParamKernel · exists_sliceOps, where $T = g_q'(x)$ lies in $\mathrm{Lin}(f,x)$ Chapter6/ParamKernel · mem_linSet_sectionAdd_pullbackSlice and $\varphi(\lambda) = \sum \lambda_i s_i(q,x)$. *$T \oplus \varphi$ is onto when F is in general position at* $(0,(q,\Phi(q)))$ Chapter6/ParamKernel · surjective_coprod_of_isLinSurjAt: given w, the surjectivity of F' gives (λ,σ,v) with image w; the derivative of the curve of zeros, $(0,\sigma,D\Phi\,\sigma)$, lies in $\ker F'$ (§VI.12); subtracting it gives $(\lambda,0,v - D\Phi\,\sigma)$ with the same image, a preimage under $T \oplus \varphi$. The kernel is then a closed subspace of the product with the right inclusion Chapter6/ParamKernel · exists_sliceKernel. $\ker F_t'(\varphi(t),0)$ is read as the kernel of the linearization of the whole F cut by $\sigma = 0$ (Findings, Chapter 6, item 49) (pp. 263–266 (preprint pp. 221–223)).

Lemma 6.6.6 (preprint 6.31) (V6.38) Chapter6/ParamKernel · lemma_6_31. With these operators, (96) $0 \to \ker T \to \ker(T \oplus \varphi) \to \mathbb{R}^k \to K_z/R(T) \to 0$ is the stabilization sequence of T and φ (Lemma 6.4.14 (preprint 6.14), §VI.6), hence exact. The text has c onto by (P); here it is onto by the general position of F (§VI.11) and the derivative of the family of zeros, which absorbs the $\mathbb{R}^n$-directions (preprint p. 223).

VI.14 Continuation, the sign, and the push-forward

The text (pp. 265–269 (preprint pp. 223–224)): Φ_{E_t} of (94) identifies $\det(T_t)$ with $\lambda(L_t)$; an orientation o_0 of $\det(T_0)$ gives one of $\lambda(L_0)$, continuation carries it to $\lambda(L_1)$, and $\Phi_{E_1}^{-1}$ carries it to $\det(T_1)$: this defines $o_1 = \varphi_* o_0$.

Continuation (V6.39). Over a simply connected, locally path-connected base the continuation of a ray of $\lambda(L_{q_0})$ to $\lambda(L_{q_1})$ is the value at q_1 of the unique orientation extending it Chapter6/Continuation · SubspaceFamily.cont Chapter6/Continuation · SubspaceFamily.cont_eq. It is the identity from a point to itself Chapter6/Continuation · SubspaceFamily.cont_self, composes Chapter6/Continuation · SubspaceFamily.cont_trans, is a bijection of rays Chapter6/Continuation · SubspaceFamily.contEquiv, is compatible with pull-back Chapter6/Continuation · SubspaceFamily.cont_comp, and along $[0,1]$ it is the continuation of §VI.12 Chapter6/Continuation · SubspaceFamily.cont_eq_transport. For a stabilization $T \oplus \varphi$ that is onto, $\det(T \oplus \varphi) \cong \lambda(\ker(T \oplus \varphi)) \otimes \lambda^*(\{0\}) \cong \lambda(\ker(T \oplus \varphi))$ Chapter6/Continuation · detEquivOfSurjective, and through ι_φ (§VI.6) and an identification of the kernel with a member of L the rays of $\det(T)$ and of $\lambda(L)$ correspond Chapter6/Continuation · stabRayEquiv. The propagation is Φ_{E_0}, then continuation,

then $\Phi_{E_1}^{-1}$ on rays `Chapter6/Continuation · propagate`, with the identity on a constant family `Chapter6/Continuation · propagate_self`, composition `Chapter6/Continuation · propagate_trans` and pull-back `Chapter6/Continuation · propagate_comp`. It is stated over any simply connected, locally path-connected base, so that one statement serves the paths and the squares; the fixed identification $\lambda^*(\mathbb{R}^k) \cong \lambda^*(\{0\})$ of (94) occurs at both ends.

The sign (V6.40). In a local determinant chart with projection P, the chart coordinate of $\iota_\psi(x) \in \det(S \oplus \psi)$ is $(-1)^{k \dim F/R(P) + \dim \operatorname{coker} S}\, D(S,\psi)$ times that of x `Chapter6/StabilizationContinuity · coord_stabIota`, with D `Chapter6/StabilizationContinuity · stabD` continuous `Chapter6/StabilizationContinuity · continuousOn_stabD` and nowhere zero `Chapter6/StabilizationContinuity · stabD_ne_zero` near the centre. The first summand of the exponent is constant on a chart, the second is not. So where $\dim\operatorname{coker} T$ changes parity along a family, the rays carried by Φ_E jump. With $\alpha(T) = (-1)^{\dim\operatorname{coker} T}$ `Chapter6/StabilizationContinuity · stabSign`: *for a continuous orientation o of* DET*, a norm-continuous family T_q and a continuous stabilization φ_q with $T_q \oplus \varphi_q$ onto, the rays $\alpha(T_q)\,\Phi(o(T_q))$ of $\lambda(\ker(T_q \oplus \varphi_q))$ form an orientation* `Chapter6/StabilizationContinuity · isLineOrientation_stabRay`. Without the sign they need not `Chapter6/StabilizationContinuity · not_isLineOrientation_stabRay_uncorrected`: $E = F = \mathbb{R}$, $k = 1$, $T_q = q/(1+|q|)$ and $\varphi(r) = r_0$, where $\dim\operatorname{coker} T_q$ is 1 at $q = 0$ and 0 elsewhere. As we read it, the orientations carried by Φ_E of (94) along a continuous family are not continuous where $\dim\operatorname{coker} T$ changes parity, and Lemmas 6.6.8, 6.6.10 (preprint 6.33, 6.34) and the homotopy argument before Theorem 6.6.11 (preprint 6.35) use that they are. The propagation uses $\alpha(T)\Phi_E$, which on each operator carries the text's orientation or its opposite (Findings, Chapter 6, item 50) (p. 256 (preprint p. 212), as used on pp. 265–272 (preprint pp. 223–226)).

The push-forward (V6.41). At a chart point p of a point y `Chapter6/Propagation · chartPt`, a section S with $(f+S)(y) = 0$ and sections $s_1, \dots, s_k$ give $T = (f+S)'(y) \in \operatorname{Lin}(f, y)$ `Chapter6/Propagation · ChartPoint.stabT` `Chapter6/Propagation · ChartPoint.stabT_mem_linSet` and $\varphi(\lambda) = \sum \lambda_i \hat{s}_i(z)$ `Chapter6/Propagation · ChartPoint.stabPhi`, with appended sections giving the appended stabilization `Chapter6/Propagation · ChartPoint.stabPhi_finAppend`; $T \oplus \varphi$ is Fredholm `Chapter6/Propagation · isFredholm_stabCoprod`, and the sign is an involution `Chapter6/Propagation · stabSign_inv` `Chapter6/Propagation · toPerm_stabSign_symm`. A family of *stabilized kernels* `Chapter6/Propagation · IsStabKernels` over a parameter space Q consists of a global sc$^+$-section S of the pull-back with $f + S = 0$ at the points $(q, y(q))$, sections $s_1, \dots, s_k$, a continuous lift of $q \mapsto \varphi(y(q))$ to level 1 of the chart, and a continuous family Λ of subspaces of $E_0 \times \mathbb{R}^k$ such that at every chart point of $y(q)$ in the chart $T_q \oplus \varphi_q$ is onto and Λ_q is the image of its kernel. It restricts along continuous maps `Chapter6/Propagation · IsStabKernels.comp`. Its *push-forward* between two parameters a, b `Chapter6/Propagation · IsStabKernels.push`: $o \in \mathrm{Or}^f_{y(a)}$ gives the ray $o(T_a)$, then $\alpha(T_a)\Phi_{E_a}(o(T_a))$, a ray of $\lambda(\Lambda_a)$; continuation carries it to $\lambda(\Lambda_b)$; $\alpha(T_b)\Phi_{E_b}^{-1}$ carries it to $\det(T_b)$; and $\mathrm{Or}^f_{y(b)} \cong \{\text{rays of } \det(T_b)\}$ (§VI.9) gives the image. It depends only on the data `Chapter6/Propagation · IsStabKernels.push_congr`. With $\mathrm{orRay}_q(o)$, the ray of $\lambda(\Lambda_q)$ read from o at q `Chapter6/Propagation · IsStabKernels.orRay`, $\varphi_* = \mathrm{orRay}_b^{-1} \circ \mathrm{cont}_{ab} \circ \mathrm{orRay}_a$ `Chapter6/Propagation · IsStabKernels.push_eq_orRay`. *Along a path with continuously varying data and a fixed point, continuation carries the rays read at one parameter to those read at the next* `Chapter6/Propagation · IsStabKernels.cont_orRay_path`: the operators T_q, φ_q vary continuously; their kernels form a continuous family (§VI.12); the sign makes the rays $\alpha\Phi(o(T_q))$ an orientation; and the inclusion into $E_0 \times \mathbb{R}^k$ carries that orientation to $\lambda(\Lambda)$. The push-forward is compatible with pull-back `Chapter6/Propagation · IsStabKernels.push_comp` and with concatenation `Chapter6/Propagation · IsStabKernels.push_trans`. For given end-point operators the result differs from the text's by $\alpha(T_a)\alpha(T_b)$; the sections enter pointwise, and the continuity in the parameter is that of Λ (Findings, Chapter 6, item 50) (preprint p. 224).

VI.15 The stabilized kernels exist; push-forwards along paths

Lemma 6.6.5 (preprint 6.30) for the push-forward (V6.42) `Chapter6/StabKernelsExist · exists_isStabKernels`. *For an sc-smooth $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n)$ in the domain of a chart, an sc$^+$-section S with $f + S = 0$*

along Φ near $[0,1]^n$, and sections $s_1, \dots, s_k$ as in Lemma 6.6.4 (preprint 6.29), the stabilized kernels of the slices form a continuous family over $[0,1]^n$. The family $q \mapsto \varphi(\Phi(q))$ has a continuous lift to level 1 `Chapter6/StabKernelsExist · exists_continuousOn_lift_chart`, the values of an sc-smooth map on $\mathbb{R}^n$ being smooth points. The zeros $(0,(q,\Phi(q)))$ of F are in general position, hence in good position; their kernels form a continuous family (§VI.12); cutting it by $\{d\sigma = 0\}$, a fixed functional onto on every member because the derivative of the family of zeros lies in the kernel, keeps it continuous (§VI.12); and §VI.13 identifies the cut with $\ker(T_q \oplus \varphi_q)$. The deviation is that of §VI.12 (gap G.62).

Push-forwards along paths (V6.43). The parameter points of $[0,1]$ `Chapter6/PushData · pt1_mem_Icc`, level-1 lifts at smooth points `Chapter6/PushData · exists_lift_of_smooth` and the chart points they give `Chapter6/PushData · chartPtOf` are the bookkeeping. *$e : \mathrm{Or}^f_{\gamma(0)} \to \mathrm{Or}^f_{\gamma(1)}$ is a push-forward along an sc-smooth path γ in a chart* `Chapter6/PushData · IsPush` if there are global sc⁺-sections $S, s_1, \dots, s_k$ of $\pi^*P \to \mathbb{R} \times X$ with $(f+S)(t,\gamma(t)) = 0$ for t near $[0,1]$, and a family of stabilized kernels of their slices along γ whose push-forward is e. Equal data give equal push-forwards `Chapter6/PushData · IsStabKernels.push_eq_of_eq`. Pulled back along an affine reparametrization the sections still vanish along the reparametrized path `Chapter6/PushData · isZeroAt_sectionAdd_reparamSection` and give a push-forward along it `Chapter6/PushData · isPush_reparam`; so a push-forward along γ is the composite of push-forwards along its two halves, each reparametrized to $[0,1]$ `Chapter6/PushData · IsPush.split`. *Along every sc-smooth path in a set with property (3) of Proposition 6.6.3 (preprint 6.28) there is a push-forward* `Chapter6/PushData · exists_isPush`: (3) gives S, Lemma 6.6.4 (preprint 6.29) gives $s_1, \dots, s_k$, and the stabilized kernels exist. Global sections of the pull-back stand in place of the text's sections over $U \times [0,1]$, and (P) is used only to construct a push-forward (preprint p. 224).

VI.16 Lemma 6.6.8 (preprint 6.33) and sections over families

Lemma 6.6.8 (preprint 6.33) (V6.45) `Chapter6/Propagation · IsStabKernels.push_append`. *Appending sc⁺-sections $\bar{s}_1, \dots, \bar{s}_d$ after $s_1, \dots, s_k$ does not change the push-forward.* The text proves it with a homotopy $\tilde{F}(\tau,\cdot)$ and "one verifies by a simple computation". The computation is this. For $\ell_1, \dots, \ell_d \in \ker(T \oplus (\varphi,\psi))$ whose ψ-coordinates are the standard basis of $\mathbb{R}^d$, the map $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$ `Chapter6/AddSections · wedgeExt` `Chapter6/AddSections · wedgeExt_wedge` does not depend on the ℓ_j — two choices differ by vectors of the image of $\ker(T \oplus \varphi)$, the span of the first block `Chapter6/AddSections · wedgeExt_congr` — and is bijective `Chapter6/AddSections · wedgeExt_bijective`. With the appended stabilization `Chapter6/AddSections · appendStab` and the kernel inclusion $(v,\lambda) \mapsto (v,\lambda,0)$ `Chapter6/AddSections · kerAppend`, *the stabilization isomorphism of (φ,ψ) is that of φ followed by $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$, exactly, with no sign* `Chapter6/AddSections · detEquiv_stabIota_append`: a choice of Proposition 6.4.15 (preprint 6.15) for (φ,ψ) is one for φ with the ℓ_j appended to the kernel basis and the standard basis vectors to the r_i, and with the appended sections last the matrix $(\bar{c}, r)$ becomes block triangular with an identity block, so the determinant in (94) is unchanged. Along a family the ℓ_j are chosen continuously `Chapter6/AddSections · SubspaceFamily.exists_continuous_lifts` — the larger kernels project onto $\mathbb{R}^d$ because $T \oplus \varphi$ is already onto — and the wedge extension carries continuous orientations along `Chapter6/AddSections · isLineOrientation_wedgeExt`, through the kernel inclusion `Chapter6/AddSections · kerEquivMap`. So the stabilized rays `Chapter6/AddSections · stabRayEquiv_append` and the propagations `Chapter6/AddSections · propagate_append` agree, and with them the push-forwards. The homotopy of the text's proof is not needed (Findings, Chapter 6, item 51) (pp. 266–271 (preprint pp. 224–225)).

Sections over families (V6.46). For g affine, $(q,y) \mapsto S(g(q),y)$ `Chapter6/FamilySections · reparamSection` is again an sc⁺-section of the pull-back `Chapter6/FamilySections · StrongBundle.IsScPlusSection.reparam`, through the sc-smooth product map $g \times 1$ `Chapter6/FamilySections · rnProdMapOp`, with the expected slices `Chapter6/FamilySections · pullbackSlice_reparamSection`. Sc⁺-sections may be checked in one chart around each point `Chapter6/FamilySections · StrongBundle.isScPlusSection_of_local`. Adding sections keeps the general position of F at $(0,y)$ `Chapter6/FamilySections · StrongBundle.isGeneralPositionGerm_paramSection_of_comp`: the linearization only gains columns, and the properties of Definition 5.3.9 (preprint 5.17) survive. These are the sections $s(y,\tau,t)$ over a square and the text's "adding more sections, the properties will still hold" (pp. 266–

272 (preprint pp. 224–226)).

VI.17 Lemma 6.6.10 (preprint 6.34) and Proposition 6.6.7 (preprint 6.32)

Lemma 6.6.10 (preprint 6.34) (V6.47) `Chapter6/Propagation · IsStabKernels.push_square`. *For a family of stabilized kernels over the square $[0,1]^2$ with the end points x_0, x_1 fixed and the end-point operators varying continuously, the push-forwards along the edges $\tau = 0$ and $\tau = 1$ agree.* The edges are the restrictions to affine segments of the square `Chapter6/PropagationWellDefined · sqEdge Chapter6/PropagationWellDefined · sqPath`. On the edge $t = 0$ the point is x_0 and the operators vary continuously, so by §VI.14 continuation along it carries $\mathrm{orRay}_{(0,0)}(o)$ to $\mathrm{orRay}_{(1,0)}(o)$; the same on $t = 1$. The square is simply connected, so continuation from $(0,0)$ to $(1,1)$ is the same through $(0,1)$ and through $(1,0)$. Writing both push-forwards as $\mathrm{orRay}^{-1} \circ \mathrm{cont} \circ \mathrm{orRay}$ gives the claim. The text's convex homotopy s^τ is part of the square of Proposition 6.6.7 (preprint 6.32) below, and the continuity of the end-point orientations along τ is that of §VI.14, with the sign (preprint p. 226).

Proposition 6.6.7 (preprint 6.32) (V6.44) `Chapter6/PropagationWellDefined · proposition_6_32`. *Two push-forwards along the same sc-smooth path in the domain of a chart agree.* Let the data be $(S^0; s^0_1, \dots, s^0_{k^0})$ and $(S^1; s^1_1, \dots, s^1_{k^1})$. The text combines Lemma 6.6.8 (preprint 6.33) (more sections) and Lemma 6.6.10 (preprint 6.34) (a homotopy of the data); the formalization does both over one square (Findings, Chapter 6, item 52). *Step 1, the square.* $\Phi(\tau, t) = \gamma(t)$, constant in τ, and $S^\tau = (1-\tau)S^0 + \tau S^1$, a section of the pull-back over $\mathbb{R}^2 \times X$ by §VI.16; $f + S^\tau$ vanishes along Φ since both $f + S^0$ and $f + S^1$ do and the fibres are linear `Chapter6/PropagationWellDefined · isZeroAt_sectionAdd_combo`. *Step 2, the sections.* With $\widehat{s}^i_j = s^i_j$ for $j < k^i$ and 0 otherwise, the sections $(1-\tau)\widehat{s}^0_j + \tau\widehat{s}^1_j$, followed by the sections that Lemma 6.6.4 (preprint 6.29) gives on the square. The parametrized section is in general position along the square — adding sections keeps it (§VI.16) — so the stabilized kernels exist over $[0,1]^2$ (§VI.15). *Step 3, the edges $\tau = 0, 1$.* At $\tau = 0$ the sections s^0 are the first k^0 of the family and at $\tau = 1$ the sections s^1 are the first k^1, so Lemma 6.6.8 (preprint 6.33) compares each datum with the edge of the square without reordering `Chapter6/PropagationWellDefined · IsStabKernels.push_eq_of_prefix`, the edges being data of the same kind `Chapter6/PropagationWellDefined · IsStabKernels.congr_data Chapter6/PropagationWellDefined · IsStabKernels.push_congr_data`. *Step 4, the edges $t = 0, 1$.* There the point is fixed and the operators are affine in τ `Chapter6/PropagationWellDefined · linAt_principal_combo`, hence continuous `Chapter6/PropagationWellDefined · continuous_principal_pullbackSlice`. *Step 5.* Lemma 6.6.10 (preprint 6.34) equates the push-forwards along the edges $\tau = 0$ and $\tau = 1$, and Step 3 identifies them with the two given ones. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) (X tame, bump functions, tame germ data). "Does not depend on the choices" is read as the agreement of any two data (pp. 266–272 (preprint pp. 224–226)).

VI.18 Theorem 6.6.11 (preprint 6.35), and paths in a chart

Theorem 6.6.11 (preprint 6.35) (V6.48). U has the *local propagation property* in a chart c `Chapter6/PropagationWellDefined · HasLocalPropagation` if U lies in the domain of c, along every sc-smooth path in U there is a push-forward, and any two push-forwards along paths in U with the same end points agree. `Chapter6/PropagationWellDefined · theorem_6_35`: *over a tame X with bump functions, for f with tame germ data, every smooth point x has an open $U \ni x$ and a chart c with the local propagation property.* The statement names the bundle and the point but not the section (possible omission); it is read for an sc-Fredholm section with tame germ data (Findings, Chapter 6, item 53). *Step 1. $U \subseteq W$ and c* from Proposition 6.6.3 (preprint 6.28); push-forwards exist along paths in W, a fortiori in U, by (3) (§VI.15). *Step 2, homotopy invariance* `Chapter6/PropagationWellDefined · push_eq_of_homotopy`. For an sc-smooth $\Phi : \mathbb{R}^2 \to X$ with $\Phi([0,1]^2) \subseteq W$ and the end points fixed, (3) for $n = 2$ gives S, Lemma 6.6.4 (preprint 6.29) gives sections, and the stabilized kernels exist over the square. The edge $t \mapsto (\tau, t)$ is an affine map, so the restriction to it is a push-forward datum along $\Phi(\tau, \cdot)$ (§§VI.14–VI.15). Along $t = 0, 1$ the point is constant and so is the section $S(q, \cdot)$, because it depends on q only through $\Phi(q)$; the end-point operators are then constant, and

Lemma 6.6.10 (preprint 6.34) applies. With Proposition 6.6.7 (preprint 6.32), any push-forward along $\Phi(0,\cdot)$ equals any along $\Phi(1,\cdot)$. *Step 3, the reparametrization.* The homotopy of (1′) is γ_0 and γ_1 at its edges only for $t \in [0,1]$, while a push-forward datum asks for zeros of $f+S$ for t near $[0,1]$. The proof therefore passes through $\gamma_i \circ \kappa$, which maps all of $\mathbb{R}$ into $\gamma_i([0,1])$, with three homotopies whose edges are exact for all t: $H(\sigma,t) = \gamma_0((1-\sigma)t + \sigma\kappa(t))$ from γ_0 to $\gamma_0 \circ \kappa$; the homotopy $(\tau,t) \mapsto \Phi(\tau,\kappa(t))$ from $\gamma_0 \circ \kappa$ to $\gamma_1 \circ \kappa$; and the analogue of H for γ_1. Step 2 applied three times gives the agreement. U is not shown to be sc-smoothly contractible (gap G.61); what the proof uses is (1′) (pp. 271–273 (preprint pp. 226–227)).

Paths in a chart (V6.49). Sc-smooth maps that agree near every point of a set are sc-smooth together `Chapter6/ScPaths · MPolyfold.isScSmoothMap_of_eventuallyEq`; C^∞ maps of the parameter compose `Chapter6/ScPaths · pt1_apply` `Chapter6/ScPaths · contDiff_pt1_comp`. *For an open U in the domain of a chart and $x \in U$ there is an open $V \ni x$ in U whose smooth points are pairwise joined by sc-smooth paths in U* `Chapter6/ScPaths · exists_nhds_scPath`: in the chart, $t \mapsto \varphi^{-1}(r(u_1 + \kappa(t)(u_2 - u_1)))$, for u_1, u_2 in a small ball around $\varphi(x)$, as in (1′). The concatenation `Chapter6/ScPaths · concatPath` is $\gamma_1(\kappa(3s))$ for $s \le 1/2$ and $\gamma_2(\kappa(3s-2))$ for $s > 1/2$ `Chapter6/ScPaths · concatPath_of_le` `Chapter6/ScPaths · concatPath_of_gt`. It is constant on $[1/3, 2/3]$, with value $\gamma_1(1) = \gamma_2(0)$, so the two pieces glue sc-smoothly `Chapter6/ScPaths · isScSmoothMap_concatPath`; it has the right end points `Chapter6/ScPaths · concatPath_zero` `Chapter6/ScPaths · concatPath_one` and halves `Chapter6/ScPaths · concatPath_half₁` `Chapter6/ScPaths · concatPath_half₂`. Reparametrizing by a C^∞ map keeps sc-smoothness `Chapter6/ScPaths · isScSmoothMap_reparam`, and push-forwards compose along concatenations `Chapter6/ContinuousOrientation · HasLocalPropagation.eq_trans`. Paths are joined only locally, inside a chart; the global facts recalled on preprint p. 201 are not formalized (gap G.56) (pp. 272–274 (preprint pp. 227–228)).

VI.19 Definition 6.6.12 (preprint 6.36): orientable neighbourhoods and continuous orientations

Definition 6.6.12 (preprint 6.36) (V6.50). The text: an orientable neighbourhood of (f,x) is an open, sc-smoothly contractible U with the local propagation property, and a continuous orientation over it is a choice $y \mapsto o_y$ carried into itself by the propagation. Since Proposition 6.6.3 (preprint 6.28) gives (1′), an orientable neighbourhood is a pair $V \subseteq U$ `Chapter6/SectionOrientation · IsOrientableNbhd`: U open with the local propagation property in a chart, $V \ni x$ open with any two smooth points joined by an sc-smooth path in U. A *continuous orientation* is a choice $y \mapsto o_y$ on V_∞, read in the chart, carried into itself by the push-forwards along the paths in U `Chapter6/ContinuousOrientation · IsContinuousOrientation`. *There are exactly two* `Chapter6/ContinuousOrientation · exists_two_isContinuousOrientation`: given o_x, set o_y to be the push-forward of o_x along any path in U from x to y, well defined by the local propagation property; push-forwards compose along concatenations, so this is a continuous orientation; the two values of o_x give two different ones; and a continuous orientation is determined by its value at x. *Every smooth point has an orientable neighbourhood* `Chapter6/ContinuousOrientation · exists_orientableNbhd` `Chapter6/SectionOrientation · exists_isOrientableNbhd`, by Theorem 6.6.11 (preprint 6.35) and §VI.18. *Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5) (Findings, Chapter 6, item 54) (preprint p. 227).

VI.20 The path category PX

Reparametrization homotopies (V6.51). The values of an sc-smooth path at parameters are smooth points `Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap`, and C^∞ maps of $\mathbb{R}^2$ compose `Chapter6/ScHomotopy · contDiff_pt2_comp`. *For a C^∞ θ with $\theta(0) = 0$, $\theta(1) = 1$, $\gamma \circ \theta$ is sc-smoothly homotopic to γ with the end points fixed* `Chapter6/ScHomotopy · isScHomotopyRel_reparam`, by $H(\tau,t) = \gamma((1-\tau)t + \tau\theta(t))$, and inside every set containing $\gamma([0,1])$ when $\theta([0,1]) \subseteq [0,1]$ `Chapter6/ScHomotopy · reparam_mem`. With the halves of a path `Chapter6/ScHomotopy · leftHalf` `Chapter6/ScHomotopy · rightHalf`, the concatenation of the two halves of γ is such a reparametrization `Chapter6/ScHomotopy · halvesReparam` `Chapter6/ScHomotopy · concatPath_halves` (preprint p. 228).

The category (V6.52). The tools are the value $\kappa(1/2)$ Chapter6/PathCategory · smoothTransition_half, constant maps at smooth points Chapter6/PathCategory · isScSmoothMap_const_of_mem_smoothSet, maps defined by cases on closed pieces that agree near the seams Chapter6/PathCategory · isScSmoothMap_ite Chapter6/PathCategory · isScSmoothMap_ite₃ and the middle third of a concatenation Chapter6/PathCategory · concatPath_of_le_two_thirds Chapter6/PathCategory · concatPath_of_one_third_le. Sc-smooth homotopy with the end points fixed Chapter6/PathCategory · ScHomotopic is an equivalence relation on the sc-smooth paths from x to y Chapter6/PathCategory · ScHomotopic.refl Chapter6/PathCategory · ScHomotopic.symm Chapter6/PathCategory · ScHomotopic.trans. Transitivity needs care: a homotopy meets its edge conditions only on $[0,1]$ (decision D2), and two homotopies glued in τ need not agree off $[0,1]$, so all paths and homotopies are first reparametrized by κ in t (Findings, Chapter 6, item 71). The paths with smooth end points Chapter6/PathCategory · ScPath Chapter6/PathCategory · ScPath.source_mem_smoothSet Chapter6/PathCategory · ScPath.target_mem_smoothSet have the constant paths Chapter6/PathCategory · ScPath.const, concatenation Chapter6/PathCategory · ScPath.concat and reversal Chapter6/PathCategory · ScPath.reverse. Concatenation is well defined on classes Chapter6/PathCategory · ScPath.concat_congr, has the constant paths as units Chapter6/PathCategory · ScPath.const_concat Chapter6/PathCategory · ScPath.concat_const, is associative Chapter6/PathCategory · ScPath.concat_assoc — the two bracketings differ by an explicit reparametrization Chapter6/PathCategory · assocReparam, joined by a three-piece homotopy Chapter6/PathCategory · isScHomotopyRel_triHomotopy — and the reversed paths are inverses Chapter6/PathCategory · ScPath.concat_reverse Chapter6/PathCategory · ScPath.reverse_concat. The classes Chapter6/PathCategory · ScPathClass with their composition Chapter6/PathCategory · ScPathClass.comp make the smooth points a category PX Chapter6/PathCategory · PathCat Chapter6/PathCategory · PathCat.category, indeed a groupoid Chapter6/PathCategory · PathCat.groupoid, with the morphism of a path Chapter6/PathCategory · PathCat.homMk, composition and identities on representatives Chapter6/PathCategory · PathCat.comp_mk Chapter6/PathCategory · PathCat.id_eq_mk, equality of morphisms as homotopy Chapter6/PathCategory · PathCat.mk_eq_mk_iff, every morphism represented Chapter6/PathCategory · PathCat.exists_eq_mk and inverses by reversal Chapter6/PathCategory · PathCat.inv_mk. The text does not name the identities; they are the classes of the constant paths. The second piece $\psi(\beta(s))$ of the composition on preprint p. 228 leaves $[0,1]$, since $\beta([1/2,1]) = [1,2]$; it is read as $\psi(\beta(s)-1)$ (possible misprint; Findings, Chapter 6, item 55). Mathlib's $[\varphi] \gg [\psi]$ is the text's $[\psi] * [\varphi]$ (pp. 272–274 (preprint pp. 227–228)).

VI.21 Comparing charts: the transfer, and Or_x^f in a chosen chart

The proof of Theorem 6.6.13 (preprint 6.37) carries an orientation along orientable neighbourhoods of different points of a path, which may lie in different charts. The text takes T_xX, Y_x, $\mathrm{Lin}(f,x)$ and Or_x^f to be the same whichever chart is used (preprint p. 219); here they are made at chart points (§VI.1), so the charts have to be compared (Findings, Chapter 6, item 70).

Conjugation of determinants (V6.53). For isomorphisms A, Γ with $S'A = \Gamma S$, A maps $\ker S$ onto $\ker S'$ Chapter6/ChartTransfer · conjKerEquiv and Γ induces $\operatorname{coker} S \cong \operatorname{coker} S'$ Chapter6/ChartTransfer · conjCokerEquiv; together they give $\det(S) \cong \det(S')$ Chapter6/ChartTransfer · conjDet, on wedges $(a \wedge \dots) \otimes (d \wedge \dots)^* \mapsto (Aa \wedge \dots) \otimes ([\Gamma d] \wedge \dots)^*$ Chapter6/ChartTransfer · conjDet_wedge_tmul, compatible with composition, equal data and the identity Chapter6/ChartTransfer · conjDet_trans Chapter6/ChartTransfer · conjDet_congr Chapter6/ChartTransfer · conjDet_refl. Good left-projections are conjugated, $\Gamma P \Gamma^{-1} \in \Pi_{S'}$ Chapter6/ChartTransfer · IsGoodLeftProj.conj, and the γ^P commute with the conjugation Chapter6/ChartTransfer · gammaProj_conjDet; read in local determinant charts the conjugation has a continuous nowhere-zero coordinate Chapter6/ChartTransfer · coord_conjDet_coordInv, so continuous orientations of DET are carried along Chapter6/ChartTransfer · isLineOrientation_conj Chapter6/ChartTransfer · LineOrientation.conj Chapter6/ChartTransfer · LineOrientation.conj_apply, with the same rules Chapter6/ChartTransfer · LineOrientation.conj_refl Chapter6/ChartTransfer · LineOrientation.conj_trans Chapter6/ChartTransfer · LineOrientation.conj_congr.

The transfer between two chart points (V6.53). For chart points p, p' of x, A is the derivative at z of the transition of the base charts Chapter6/ChartTransfer · ChartPoint.baseDeriv, with the cocycle rules Chapter6/

ChartTransfer · ChartPoint.baseDeriv_self Chapter6/ChartTransfer · ChartPoint.baseDeriv_trans, an sc-operator Chapter6/ChartTransfer · ChartPoint.isScOperator_baseDeriv; Γ is the fibre part of the transition of the strong bundle charts Chapter6/ChartTransfer · ChartPoint.fiberMap Chapter6/ChartTransfer · ChartPoint.fiberMap_self Chapter6/ChartTransfer · ChartPoint.fiberMap_trans Chapter6/ChartTransfer · ChartPoint.isScOperator_fiberMap. They restrict to isomorphisms $T_zO \cong T_{z'}O'$ and $K_z \cong K_{z'}$ Chapter6/ChartTransfer · ChartPoint.transferT Chapter6/ChartTransfer · ChartPoint.transferK, again with the cocycle rules Chapter6/ChartTransfer · ChartPoint.transferT_self_apply Chapter6/ChartTransfer · ChartPoint.transferT_trans_apply. At a zero the linearizations in the two charts are related by them, the chain rule applied to the principal parts Chapter6/ChartTransfer · ChartPoint.scDeriv_principal_baseDeriv, and sc⁺-operators conjugate to sc⁺-operators Chapter6/ChartTransfer · ChartPoint.IsScPlusOp.conj; so $L \in \mathrm{Lin}_p(f,x)$ iff $\Gamma L A^{-1} \in \mathrm{Lin}_{p'}(f,x)$ Chapter6/ChartTransfer · ChartPoint.mem_linSet_transfer_iff. Conjugation of orientations over this bijection of the bases is the transfer $\mathrm{Or}_x^f(p) \cong \mathrm{Or}_x^f(p')$ Chapter6/ChartTransfer · ChartPoint.orTransfer Chapter6/ChartTransfer · ChartPoint.orTransfer_apply, with the cocycle rules Chapter6/ChartTransfer · ChartPoint.orTransfer_self Chapter6/ChartTransfer · ChartPoint.orTransfer_trans.

The push-forward does not depend on the chart (V6.54). The sign is invariant, conjugation preserving $\dim\operatorname{coker}$ Chapter6/PushTransfer · stabSign_conj; the stabilization is natural — $(\Gamma S A^{-1}) \oplus \Gamma\varphi$ is the conjugate of $S \oplus \varphi$, with the conjugated kernel, onto iff $S \oplus \varphi$ is Chapter6/PushTransfer · coprod_conj Chapter6/PushTransfer · ker_coprod_conj Chapter6/PushTransfer · surjective_coprod_conj — and so is its isomorphism on rays Chapter6/PushTransfer · conj_stab_key Chapter6/PushTransfer · stabRayEquiv_conjDet; continuation is natural under jointly continuous families of injective linear maps Chapter6/PushTransfer · SubspaceFamily.cont_map. The principal parts of the sections transform by Γ Chapter6/PushTransfer · ChartPoint.fiberMap_principal, so the operators T and φ of a datum in one chart are the conjugates of those in the other Chapter6/PushTransfer · ChartPoint.conj_of_linAt Chapter6/PushTransfer · ChartPoint.conj_of_principal, and A varies continuously along a path Chapter6/PushTransfer · continuous_baseDeriv. A family of stabilized kernels in one chart is therefore carried by $A \times 1$ to one in the second chart, with the same sections Chapter6/PushTransfer · IsStabKernels.exists_transfer, the rays read from an orientation corresponding under the transfer Chapter6/PushTransfer · IsStabKernels.orRay_transfer; a push-forward in one chart, conjugated by the transfers at the end points, is a push-forward in the other Chapter6/PushTransfer · IsPush.transfer, and the local propagations of two charts agree Chapter6/PushTransfer · HasLocalPropagation.push_compat. The text does not compare charts (pp. 264–274 (preprint pp. 222–228)).

Or_x^f in a chosen chart (V6.55). A chart of the atlas at x is chosen for every x Chapter6/SectionOrientation · orChart Chapter6/SectionOrientation · isChart_orChart Chapter6/SectionOrientation · mem_orChart, and Or_x^f is the set of the orientations read there Chapter6/SectionOrientation · Orx, with two elements Chapter6/OrientationFunctor · exists_two_orx. It is identified with the orientations read in any other chart at x by the transfer Chapter6/SectionOrientation · toChartOr Chapter6/SectionOrientation · toChartOr_orChart, compatibly with composition Chapter6/OrientationFunctor · toChartOr_trans, and a map $y \mapsto o_y$ is read in the chart of an orientable neighbourhood through it Chapter6/SectionOrientation · restrictOr. A chosen chart stands for the text's chart-free Or_x^f (preprint pp. 220, 228).

VI.22 Transport along paths from local push-forwards

The proof of Theorem 6.6.13 (preprint 6.37) (p. 273 (preprint p. 228)): for a path φ, "using the compactness of $[0,1]$ and a finite covering" by orientable neighbourhoods, φ_* is the composite of the local propagations; "by a covering argument" it depends only on $[\varphi]$; and "it is obvious that Γ_f is unique". The formalization separates this argument from orientations: it is proved for any family of types over X_∞ (Findings, Chapter 6, item 72).

Local transport data (V6.56) Chapter6/PathTransport · LocalTransport: a family of *good* open sets covering X_∞, all fibres in bijection, and for every good set U a push-forward along every sc-smooth path in U, such that push-forwards in one good set compose and do not depend on the good set containing the path. Then the push-forward along a loop in a good set is the identity Chapter6/PathTransport · LocalTransport.

push_eq_refl, in a good set it depends only on the end points Chapter6/PathTransport · LocalTransport.push_eq_push, and only on the path on $[0,1]$ Chapter6/PathTransport · LocalTransport.push_congr.

The dyadic transport. At level 0 it is the push-forward in a chosen good set containing $\gamma([0,1])$ (an arbitrary bijection if there is none); at level $n+1$ it is the composite of the transports at level n along the two halves Chapter6/PathTransport · LocalTransport.transportN. On a path in a good set every level gives the push-forward Chapter6/PathTransport · LocalTransport.transportN_eq_push, by the composition rule. The pieces of level N lie in good sets Chapter6/PathTransport · LocalTransport.IsFine for some N: the preimages of the good sets cover the compact $[0,1]$, so there is a Lebesgue number δ Chapter6/PathTransport · LocalTransport.IsLebesgueNumber Chapter6/PathTransport · LocalTransport.exists_isLebesgueNumber, and $2^N\delta > 1$ suffices Chapter6/PathTransport · LocalTransport.isFine_of_isLebesgueNumber. From such an N on the levels agree Chapter6/PathTransport · LocalTransport.transportN_eq_of_isFine, since at the next level each piece is split inside its good set. The *transport* is the value at such a level Chapter6/PathTransport · LocalTransport.transport.

Its properties. It is the push-forward on a path in a good set Chapter6/PathTransport · LocalTransport.transport_eq_push — property (□) — depends only on the path on $[0,1]$ Chapter6/PathTransport · LocalTransport.transport_congr, is the identity on paths constant on $[0,1]$ Chapter6/PathTransport · LocalTransport.transport_const, and is the composite along the two halves Chapter6/PathTransport · LocalTransport.transport_halves. *Homotopy invariance* Chapter6/PathTransport · LocalTransport.transport_eq_of_isScHomotopyRel is the covering argument. For a homotopy $\Phi : \mathbb{R}^2 \to X$ the preimages of the good sets cover $[0,1]^2$, with a Lebesgue number δ Chapter6/PathTransport · LocalTransport.exists_isSqLebesgueNumber. *The square lemma* Chapter6/PathTransport · LocalTransport.transportN_square: for a rectangle of $[0,1]^2$ with sides $< 2^n\delta$, the transports at level n along its vertical, horizontal and diagonal segments Chapter6/PathTransport · vSeg Chapter6/PathTransport · hSeg Chapter6/PathTransport · dSeg commute around its boundary — left side then top equals bottom then right side. By induction on n: for $n = 0$ the rectangle lies in one good set, and both composites are the push-forward along the diagonal; for $n+1$ the rectangle is cut into four quadrants and the four commutative squares are pasted, the interior edges cancelling Chapter6/PathTransport · equiv_paste_squares. Applied to $[0,1]^2$ at a level with $2^n\delta > 1$, with the sides $\tau \mapsto \Phi(\tau,0)$, $\tau \mapsto \Phi(\tau,1)$ constant on $[0,1]$ and of trivial transport, it gives the invariance. The transport is invariant under reparametrization Chapter6/PathTransport · LocalTransport.transport_reparam (§VI.20) and takes concatenations to composites Chapter6/PathTransport · LocalTransport.transport_concatPath, the halves of a concatenation being reparametrizations of the two paths. *Uniqueness* Chapter6/PathTransport · LocalTransport.transport_unique: a family of bijections along the sc-smooth paths that is homotopy invariant, takes concatenations to composites and is the push-forward on paths in good sets, is the transport — $G(\gamma) = G(\text{left half}) \circ G(\text{right half})$ through the reparametrization, and induction on the dyadic level. Dyadic subdivisions of $[0,1]$ and of $[0,1]^2$ stand in place of arbitrary finite coverings (preprint p. 228).

VI.23 Theorem 6.6.13 (preprint 6.37): the functor Γ_f

The text: *for a strong bundle over a tame M-polyfold and an sc-Fredholm section f there is a uniquely determined functor $\Gamma_f : PX \to \mathcal{O}_f$, $x \mapsto \mathrm{Or}^f_x$, with (□): for every smooth x, every sc-smoothly contractible open $U \ni x$ on which f has the local propagation property, and every sc-smooth path $\varphi : [0,1] \to U$, the propagation φ_* and $\Gamma_f([\varphi])$ coincide.*

The local data of f (V6.57). U is a propagation neighbourhood if it is open with the local propagation property in some chart Chapter6/OrientationFunctor · IsPropagationNbhd. Along a path in such a U the push-forward, read in the chosen charts of the end points through the transfer (§VI.21), is orPush Chapter6/OrientationFunctor · orPush; it equals every push-forward along the path in every chart in which the path lies in a set with the local propagation property Chapter6/OrientationFunctor · orPush_eq — by the chart independence of §VI.21 and the cocycle rule of the transfer — it does not depend on the neighbourhood Chapter6/OrientationFunctor · orPush_compat, and it composes Chapter6/OrientationFunctor · orPush_trans. With Theorem 6.6.11 (preprint 6.35), which makes the propagation

neighbourhoods cover X_∞, and §VI.9, which puts every Or_x^f in bijection with every other, these are local transport data `Chapter6/OrientationFunctor · orLocalTransport`, and their transport satisfies (□) `Chapter6/OrientationFunctor · orLocalTransport_transport_eq`.

The functor. $\Gamma_f([\varphi])$ is the transport along φ, well defined on classes by homotopy invariance `Chapter6/OrientationFunctor · orFunctorMap` `Chapter6/OrientationFunctor · orFunctorMap_mk`; it takes identities to identities `Chapter6/OrientationFunctor · orFunctorMap_id` and compositions to composites `Chapter6/OrientationFunctor · orFunctorMap_comp`, by the constancy and concatenation rules of §VI.22; and it is a functor $PX \to \mathrm{Type}$ `Chapter6/OrientationFunctor · orFunctor`. Property (□) for a family G on the morphisms of PX, on the members of a family of open sets `Chapter6/OrientationFunctor · IsStarPropertyOn` and on all of them `Chapter6/OrientationFunctor · IsStarProperty`, reads $G([\varphi]) = \mathrm{toChartOr}_{\varphi(1)}^{-1} \circ e \circ \mathrm{toChartOr}_{\varphi(0)}$ for every chart c, every open U with the local propagation property in c, every sc-smooth path φ in U with smooth end points and every push-forward e along φ in c. Γ_f has it `Chapter6/OrientationFunctor · orFunctorMap_isStarProperty`, so the property is satisfiable. *Uniqueness*: a family with the rules for identities and composites and with (□) on the members of a family of such open sets covering X_∞ is Γ_f `Chapter6/OrientationFunctor · orFunctorMap_unique_of_cover` — restrict the local data to the cover `Chapter6/OrientationFunctor · LocalTransport.restrict` and apply the uniqueness of the transport — in particular with (□) everywhere `Chapter6/OrientationFunctor · orFunctorMap_unique`.

Theorem 6.6.13 (preprint 6.37) `Chapter6/OrientationFunctor · theorem_6_37`. *Over a tame X with bump functions, for f with tame germ data and P with sc⁺-sections through the smooth fibre points, there is exactly one family of bijections $\Gamma_f([\varphi]) : \mathrm{Or}_x^f \to \mathrm{Or}_y^f$ on the morphisms of PX taking identities to identities and compositions to composites, with property (□). Hidden hypotheses:* the standing ones of §6.6 (preprint §6.5), and sc⁺-sections through every smooth fibre point, so that every Or_x^f has two elements (bump functions give them). (□) is asked for every open set with the local propagation property, not only for sc-smoothly contractible ones (gap G.61), and the uniqueness needs it only on a covering family — for instance on the sc-smoothly contractible ones, wherever they cover X_∞. $\mathcal{O}_f$ is read as types and bijections, and "uniquely determined" for the maps on morphisms: on objects there is no choice (preprint p. 228).

VI.24 Definitions 6.6.14–6.6.16 (preprint 6.38–6.40): orientability and orientations

Definition 6.6.14 (preprint 6.38) (V6.58). *f is orientable if for every pair of smooth points x, y in the same path component, $\Gamma_f([\varphi])$ does not depend on the class $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · IsOrientableSection`. "In the same path component" is read as "joined by an sc-smooth path" `Chapter6/OrientationFunctor · scPathSetoid`; for points not so joined there is nothing to check (Findings, Chapter 6, item 73). *Equivalently there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for every $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · isOrientableSection_iff_exists`: from left to right, choose a base point in each class of points joined by sc-smooth paths and set $o_y = \Gamma_f([\varphi])o_{\text{base}}$ for any $[\varphi]$, which does not depend on $[\varphi]$; from right to left, $\Gamma_f([\varphi])$ and $\Gamma_f([\psi])$ are bijections between two-element sets with the same value at o_x, hence equal (§VI.9) (preprint p. 229).

Definition 6.6.15 (preprint 6.39) (V6.59). *f is orientable if there is $y \mapsto o_y$ on X_∞ such that every smooth point has an orientable neighbourhood over which it is one of the two continuous orientations* `Chapter6/SectionOrientation · IsOrientableSection'`. The definition writes "$y \in U_\infty$" before any U is introduced; it is read as $y \in X_\infty$ (possible misprint; Findings, Chapter 6, item 74). *This holds iff there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi]$* `Chapter6/OrientationFunctor · isOrientableSection'_iff_exists`. From right to left: on an orientable neighbourhood $V \subseteq U$ the push-forwards along paths in U are values of Γ_f, by (□), so the restriction is a continuous orientation. From left to right: elements carried into each other by the transport along the paths in the members of an open covering are carried into each other along every path `Chapter6/OrientationFunctor · LocalTransport.transport_apply_eq_of_local` — the transport is computed on

dyadic pieces inside orientable neighbourhoods. *So Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39) are equivalent* `Chapter6/OrientationFunctor · isOrientableSection_iff`. The text calls them equivalent without proof (Findings, Chapter 6, item 73) (preprint p. 229).

Definition 6.6.16 (preprint 6.40) (V6.60). *An orientation of f is a map $y \mapsto o_y$ on X_∞ whose restriction to every orientable neighbourhood of every smooth point is a continuous orientation* `Chapter6/SectionOrientation · IsSectionOrientation`. Equivalently $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi] : x \to y$ `Chapter6/OrientationFunctor · isSectionOrientation_iff`, by the argument of Definition 6.6.15 (preprint 6.39); so f has an orientation iff it is orientable `Chapter6/OrientationFunctor · exists_isSectionOrientation_iff`, two orientations that agree at a point agree on the smooth points joined to it by sc-smooth paths `Chapter6/OrientationFunctor · IsSectionOrientation.apply_eq_iff`, and there are exactly two on each such class `Chapter6/OrientationFunctor · IsSectionOrientation.exists_two`. An orientable neighbourhood is the pair of §VI.19 (Findings, Chapter 6, item 75) (preprint p. 229).

VI.25 Theorem 6.6.18 (preprint 6.41): the natural orientation of the solution set

The text: *over a tame X, let f be a proper sc-Fredholm section in general position to the boundary at every zero, and o^f an orientation of f; then $S = f^{-1}(0)$ is a smooth compact manifold with boundary with corners with a natural orientation.* The proof: S is such a manifold "from the transversality discussion"; $\ker f'(x) = T_xS$ and, $f'(x)$ being onto, T_xS is oriented by o^f_x; and "one verifies that the differential geometric local prolongation of the orientation on S is the same as the Fredholm one".

The first assertion (V6.61) `Chapter6/SolutionOrientation · theorem_6_41_solutionSet`. *For X tame and a proper f with tame germ data in general position (Definition 5.3.9 (preprint 5.17)) at every zero, S is compact, a smooth submanifold in general position with the induced structure of a tame manifold with boundary with corners* — the conclusions of Theorem 5.3.10 (preprint 5.18) (§V.20), with good position from general position by Lemma 4.2.8 (preprint 4.19) (§IV.14) `Chapter6/SolutionOrientation · MPolyfold.IsGeneralPositionAt.isGoodPositionAt`. In a chart, $T_xS = \ker f'(x)$ `Chapter6/SolutionOrientation · tangentSpace_solutionSet_eq_linKer` (§V.17). "Proper" is the text's hypothesis; the proof uses only the compactness of S, which for sc-Fredholm sections is equivalent to it (Theorem 5.2.2 (preprint 5.5), §V.6), so nothing of the hypothesis is lost (Findings, Chapter 6, item 90).

The tangent orientation. At a zero, $f'(x)$ — the linearization of $f + s$ with $s = 0$ `Chapter6/SolutionOrientation · ChartPoint.linZero` — lies in $\mathrm{Lin}(f, x)$ `Chapter6/SolutionOrientation · ChartPoint.linZero_mem_linSet` and is onto `Chapter6/SolutionOrientation · ChartPoint.linZero_surjective`, with kernel T_xS in the chart `Chapter6/SolutionOrientation · ChartPoint.map_ker_linZero`. So an element o_x of Or^f_x gives a ray of $\det f'(x) \cong \lambda(\ker f'(x)) = \lambda(T_xS)$, the *tangent orientation* `Chapter6/SolutionOrientation · ChartPoint.tangentOr`, read in a chart at a zero `Chapter6/SolutionOrientation · tangentOrAt`. The sign of a surjective operator is 1 `Chapter6/SolutionOrientation · stabSign_of_surjective`, and the stabilization by $\mathbb{R}^0$ of a surjective operator is $\det(T) \cong \lambda(\ker T)$ on rays `Chapter6/SolutionOrientation · stabRayEquiv_zero`, so the tangent orientation is the stabilized ray of §VI.14 for the empty family of sections `Chapter6/SolutionOrientation · ChartPoint.stabRayEquiv_eq_tangentOr`.

"The local prolongation is the Fredholm one." Along an sc-smooth path γ of zeros in the domain of a chart, the datum $S = 0$, $k = 0$ is a family of stabilized kernels `Chapter6/SolutionOrientation · isStabKernels_zero`, its kernels being those of $f'(\gamma_q)$ `Chapter6/SolutionOrientation · SubspaceFamily.congrL`; they form a continuous family of subspaces, by Lemma 6.6.5 (preprint 6.30) (§VI.12) `Chapter6/SolutionOrientation · exists_tangentFamily` `Chapter6/SolutionOrientation · tangentFamily` `Chapter6/SolutionOrientation · tangentFamily_L`, and the rays read from an orientation are the tangent orientations `Chapter6/SolutionOrientation · IsStabKernels.orRay_eq_tangentOr`. So along a path of zeros the push-forward is continuation between tangent orientations `Chapter6/SolutionOrientation · cont_tangentOr`, and *the tangent orientations given by an orientation of* DET *along the path form an orientation of the line bundle $\lambda(\Lambda)$* `Chapter6/SolutionOrientation · isLineOrientation_`

tangentOr. Intermediate parameters are reached by the paths $s \mapsto \gamma(t\kappa(s))$, which stay in $\gamma([0,t])$ for all $s \in \mathbb{R}$. For an orientation o of f and an sc-smooth path of zeros in an orientable neighbourhood $V \subseteq U$, the tangent orientations of o along it form an orientation of $\lambda(\Lambda)$ Chapter6/SolutionOrientation · isLineOrientation_tangentOr_of_isSectionOrientation, and together this is the theorem Chapter6/SolutionOrientation · theorem_6_41 (Findings, Chapter 6, item 91).

Readings and what is left. *Hidden hypotheses:* tame germ data, the standing hypothesis of Chapter 5; the orientation o exists iff f is orientable (§VI.24). "A natural orientation" is read as the tangent orientations with their continuity along the sc-smooth paths of zeros in a chart, and the prolongation sentence as the identity of continuation and push-forward along such paths. Not formalized: an orientation of S as an abstract manifold, of its tangent bundle in the charts of the induced structure (gap G.67); the compatibility of the tangent orientations read in two charts at one zero (gap G.68); and the paragraph after the theorem — for f not generic, a small sc$^+$-perturbation s supported near the zero set, the orientation of S_{f+s} from o^f, and $\mathrm{Lin}(f+s,x) = \mathrm{Lin}(f,x)$ for the orientation of $\det((f+s)'(x))$ (gap G.69) (Findings, Chapter 6, items 92, 93) (pp. 274–276 (preprint pp. 229–230)).

VI.26 Remark 6.6.19 (preprint 6.42): the orientation of $\hat{f}$ at a point

The text: for an oriented (f,o) and an sc$^+$-section s of $Y \to X \times [0,1]$, $\hat{f}(x,t) = f(x) + s(x,t)$ has a natural orientation from o and the standard orientation of $[0,1]$: for $L = (f-t)'(x)$ oriented by $h = (a_1 \wedge \cdots \wedge a_n) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, orient $\tilde{L}(a,b) = La$ by $\tilde{h} = (a_1 \wedge \cdots \wedge a_n \wedge e) \otimes (b_1 \wedge \cdots \wedge b_l)^*$, e the standard orientation of $[0,1]$; $\tilde{L}$ lies in the linearization space of $\hat{f}$ at (x,t) (p. 276 (preprint p. 230)).

The linear algebra (V6.62). For Fredholm L, $\tilde{L} = L \oplus 0 : E \times \mathbb{R} \to F$ Chapter6/ProductOrientation · extOp is Fredholm Chapter6/ProductOrientation · isFredholm_extOp with $\ker \tilde{L} = \ker L \times \mathbb{R}$ — one dimension more Chapter6/ProductOrientation · finrank_ker_extOp, the new direction $e = (0,1)$ Chapter6/ProductOrientation · extE — and the same range Chapter6/ProductOrientation · range_extOp and cokernel Chapter6/ProductOrientation · extCoker. The map $h \mapsto \tilde{h}$, $(a_1 \wedge \cdots \wedge a_n) \otimes d^* \mapsto ((a_1,0) \wedge \cdots \wedge (a_n,0) \wedge e) \otimes d^*$, is an isomorphism $\det(L) \cong \det(\tilde{L})$ Chapter6/ProductOrientation · extDet Chapter6/ProductOrientation · extDet_wedge_tmul, the wedge extension of §VI.16 with one vector. In local determinant charts its coordinate is $(-1)^{\dim F/R(P) + \dim \operatorname{coker} L}$ times a continuous nowhere-zero function Chapter6/ProductOrientation · coord_extDet. Over the convex $\mathrm{Lin}(f,x)$, where $\dim \operatorname{coker} L$ can change parity, the orientation defined by $\tilde{h}$ at one L could therefore differ from the one defined at another, as we read it. With the sign of §VI.14, $(-1)^{\dim \operatorname{coker} L}\, h \mapsto \tilde{h}$ Chapter6/ProductOrientation · extDetN has a coordinate continuous up to a sign constant on a chart Chapter6/ProductOrientation · coord_extDetN, agrees with the text's map when L is onto Chapter6/ProductOrientation · extDetN_eq_of_surjective, and pulls continuous orientations back along $L \mapsto \tilde{L}$ Chapter6/ProductOrientation · isLineOrientation_extPull Chapter6/ProductOrientation · LineOrientation.extPull (Findings, Chapter 6, item 95). Transported along isomorphisms A, Γ, $\hat{L} = \Gamma^{-1}(L \oplus 0)A$ Chapter6/ProductOrientation · extConjOp gets the isomorphism $h \mapsto \tilde{h}$ followed by the conjugation of §VI.21 Chapter6/ProductOrientation · extConjDet; and for a preconnected C and a simply connected, locally path-connected $\hat{C}$ containing the $\hat{L}$, an orientation o of DET over a nonempty C gives exactly one orientation $\hat{o}$ over $\hat{C}$ with $\hat{o}(\hat{L})$ the image of $o(L)$ for every $L \in C$ Chapter6/ProductOrientation · existsUnique_extConjOr.

In charts. $X \times [0,1]$ is read as $\mathbb{R} \times X$ (decision D2), with the parameter first; s is an sc$^+$-section S of π^*P, and $\hat{f} = \tilde{f} + S$ with $\tilde{f}(t,y) = f(y)$. At the product chart point (t,z) Chapter6/ProductOrientation · ChartPoint.prodPt the tangent space is $\mathbb{R} \times T_zO$ Chapter6/ProductOrientation · ChartPoint.mem_prodPt_T_iff and the fibre K_z Chapter6/ProductOrientation · ChartPoint.mem_prodPt_K_iff. The text's $\tilde{L}(a,b) = La$ puts the $\mathbb{R}$-factor second, so $\hat{L} = \Gamma^{-1}(L \oplus 0)A$ with A the swap of the factors and $\Gamma = 1$ Chapter6/ProductOrientation · ChartPoint.prodT Chapter6/ProductOrientation · ChartPoint.prodK Chapter6/ProductOrientation · ChartPoint.prodLin. Sc$^+$-operators lift to the product Chapter6/

ProductOrientation · ChartPoint.IsScPlusOp.prodPt, sections of P lift to t-constant sections of π^*P Chapter6/ProductOrientation · constLift, and so $\hat{L} \in \mathrm{Lin}(\hat{f},(t,x))$ for every $L \in \mathrm{Lin}(f,x)$ Chapter6/ProductOrientation · ChartPoint.prodLin_mem_linSet — the derivative in t of a t-constant lift vanishes, and S only contributes an sc⁺-operator. *An orientation o of* $\mathrm{DET}(f,x)$ *gives exactly one orientation* $\hat{o}$ *of* $\mathrm{DET}(\hat{f},(t,x))$ *with* $\hat{o}(\hat{L})$ *the image of* $o(L)$ *for every* L Chapter6/ProductOrientation · ChartPoint.existsUnique_prodOr Chapter6/ProductOrientation · ChartPoint.prodOr Chapter6/ProductOrientation · ChartPoint.prodOr_apply, and $o \mapsto \hat{o}$ is injective Chapter6/ProductOrientation · ChartPoint.prodOr_injective (Findings, Chapter 6, item 96). *Hidden hypotheses:* x smooth and $\mathrm{Lin}(f,x)$ nonempty (bump functions give it, §VI.1); only the orientation of $\mathrm{DET}(f,x)$ at the one point is used, not an orientation of f. That $(t,y)\mapsto\hat{o}$ is an orientation of $\hat{f}$ in the sense of Definition 6.6.16 (preprint 6.40) — continuity over the orientable neighbourhoods of $\mathbb{R}\times X$, independence of the chart of the pull-back bundle — built from an orientation of f, is not formalized (gap G.66; Findings, Chapter 6, item 97).

VI.27 §6.7 (preprint §6.6) and Theorem 6.7.1 (preprint 6.43): not formalized

§6.7 (preprint §6.6) (p. 276 (preprint p. 230)) recalls, for a proper oriented (f,o) over a tame X with sc-smooth bump functions and an auxiliary norm N, the neighbourhood U of S of Theorem 5.3.10 (preprint 5.18) and the sc⁺-perturbations s supported in U with $N(s)\le 1$ for which $f+s$ is in general position at its zeros; it calls them *admissible for the pair* (N,U). Theorem 5.3.10 (preprint 5.18) is formalized (§V.20) and the orientation of S_{f+s} is the subject of §VI.25, but the notion itself, used only in Theorem 6.7.1 (preprint 6.43), is not formalized (gap G.55; Findings, Chapter 6, item 101).

Theorem 6.7.1 (preprint 6.43) (p. 277 (preprint p. 231)): a well-defined linear map $\Psi_{(f,o)} : H^*_{dR}(X,\partial X)\to\mathbb{R}$, $\Psi_{(f,o)}([\omega,\tau]) = \int_{S_{f+s}}\omega - \int_{\partial S_{f+s}}\tau$ for every admissible s, with its proof (pp. 277–279 (preprint pp. 231–233)) and the case $\partial X=\emptyset$ (p. 279 (preprint p. 233)). It is not formalized (gap G.54): it needs the sc-differential forms and their cohomology of §4.4 (gaps G.37–G.40), and integration over oriented compact manifolds with boundary with corners with Stokes' theorem, which Mathlib does not have (decision D1). What the theorem rests on is formalized: the perturbations of Theorem 5.3.10 (preprint 5.18) (Part V) and the tangent orientations of Theorem 6.6.18 (preprint 6.41) (§VI.25) (Findings, Chapter 6, item 100).

VI.28 Appendix §6.8.1 (preprint §6.7.1): the proof of Lemma 6.3.3 (preprint 6.5)

The text's proof (preprint pp. 234–235) has three steps: a change of the basis of the complement of $\alpha(A)$ multiplies both wedges of (91) by its determinant; a change of the bases of A and D multiplies the source by $\det(\tau)/\det(\varepsilon)$; and a change of complements adds to the basis of the complement of $\alpha(A)$ vectors of $\alpha(A)$, and to the γ-preimages vectors of $\beta(B)$, which leaves the wedges unchanged.

The formalized proof (V6.63) does the three steps at once. For two choices χ, χ' there is a scalar μ with $\mathrm{source}(\chi')=\mu\,\mathrm{source}(\chi)$ and $\mathrm{value}(\chi')=\mu\,\mathrm{value}(\chi)$ Chapter6/ExactSequence · FourTermExact.Choice.exists_smul_source_value. *Step 1.* Project the basis b'' of the new complement Z' onto Z along $\alpha(A)$; the projections form a basis of Z, related to b' by some σ, and since $\beta\circ\alpha=0$ they have the same β-images as b''. *Step 2.* In B, $\alpha(a')\wedge b''=\alpha(a')\wedge p(b'')$, adding vectors of the span of the first block; with $a'=\tau a$ this is $\det\tau\,\det\sigma\;\alpha(a)\wedge b'$. *Step 3.* In C, the γ-preimages c' of $d'=\varepsilon d$ in V' differ from the γ-preimages in V by vectors of $\ker\gamma=\beta(B)$, the span of the second block, so the wedge of the basis of C Chapter6/ExactSequence · FourTermExact.Choice.wedgeC is $\det\varepsilon\,\det\sigma$ times that of χ. *Step 4.* So the value scales by $\det\tau\,\det\sigma/(\det\varepsilon\,\det\sigma)=\det\tau/\det\varepsilon$, as the source does, and Lemma 6.3.3 (preprint 6.5) follows from the fixed choice that defines Φ_E. The step "independent of the choices of the basis if

and only if $\det(\tau) = \det(\varepsilon)$" is read as this homogeneity of (91) in the bases of A and D, τ being the σ just introduced; the corrections p_i lie in $\beta(B)$, written $\beta(C)$, and are combinations of the β-images, whose names are exchanged with those of the γ-preimages in the last step (possible misprints; Findings, Chapter 6, item 1).

VI.29 Appendix §6.8.2 (preprint §6.7.2): the proof of Proposition 6.4.11 (preprint 6.12)

(V6.64.) Let $P \le Q$ in Π_T. *Step 1, adapted bases.* Take a basis a of $\ker T$ and classes $[d_i]$ forming a basis of $\operatorname{coker} T$ with $d_i \in \ker Q$, possible because $R(QT) = R(Q)$; complete a by b to a basis of $\ker(QT)$, and (a,b) by $\tilde{b}$ to a basis of $\ker(PQT) = \ker(PT)$. *Step 2, the three values.* By (91) with these choices (Lemma 6.3.3 (preprint 6.5), §VI.3), $\gamma_T^Q(h_0) = (a \wedge b) \otimes ([d]_Q \wedge [Tb]_Q)^*$; $\gamma_{QT}^P \gamma_T^Q(h_0) = (a \wedge b \wedge \tilde{b}) \otimes ([d]_P \wedge [Tb]_P \wedge [QT\tilde{b}]_P)^*$, the classes $[d]_P$, $[Tb]_P$ being preimages of $[d]_Q$, $[Tb]_Q$ because $Pd_i = PQd_i = 0$ and $PTb \in R(P) \subseteq R(Q)$; and $\gamma_T^P(h_0) = (a \wedge b \wedge \tilde{b}) \otimes ([d]_P \wedge [Tb]_P \wedge [T\tilde{b}]_P)^*$ `Chapter6/ProjectionSequence · gammaProj_wedge_tmul`. *Step 3, the spanning.* $[T\tilde{b}]_P - [QT\tilde{b}]_P = [(I-Q)T\tilde{b}]_P$, and $(I-Q)T\tilde{b} \in \ker Q$. If $Qg = 0$, then $g = \sum c_i d_i + Tx$ with $x \in \ker(QT)$ — write $[g] = \sum c_i [d_i]$ in $\operatorname{coker} T$ and use $Qd_i = 0$ — so the classes of the d_i and of the Tb_j span what the text calls $\ker \tilde{Q}$ `Chapter6/ProjectionSequence · mem_span_append_of_ker`. *Step 4.* The last block of the third value differs from that of the second by vectors of the span of the first two blocks, which does not change the wedge (§VI.2), and with the identification of $\det(P(QT))$ with $\det(PT)$ the two values agree `Chapter6/ProjectionSequence · comp_comp_eq_of_projLE`. The spanning is proved directly in F; the name α_q in the last part is read as α^P (possible misprint; Findings, Chapter 6, items 6, 7) (pp. 280–286 (preprint pp. 235–237)).

VI.30 What is not formalized

- **Theorem 6.7.1 (preprint 6.43)** (gap G.54): the invariant $\Psi_{(f,o)}$, its proof and the case $\partial X = \emptyset$ need the forms and cohomology of §4.4 (gaps G.37–G.40), integration and Stokes' theorem (decision D1) (§VI.27). **The admissible sections of §6.7 (preprint §6.6)** (gap G.55), used only there.
- **The facts on paths of preprint p. 201** (gap G.56): local path connectedness and sc$^+$-smooth paths within a path component; only the local form inside a chart is formalized (§VI.18), and "in the same path component" is read as "joined by an sc-smooth path" (§VI.24).
- **§§6.3–6.5 (preprint §§6.2–6.4) remarks**: the alternative determinant $\lambda(\ker T) \otimes \lambda(\operatorname{coker}(T)^*)$ (gap G.57, §VI.4); the identification (95) and the maps after Proposition 6.4.15 (preprint 6.15) (gap G.58, §VI.6); Lemma 6.5.3 (preprint 6.19) in general, applied only to trivial bundles (gap G.59, §VI.7); the non-orientability of $\mathrm{DET}(E,F)$ and the classical case at the opening of §6.6 (preprint §6.5) (gap G.60, §VI.8).
- **Proposition 6.6.3 (1) (preprint 6.28)** (gap G.61): the sc-smooth contractibility of U, and with it the word "contractible" in Theorem 6.6.11 (preprint 6.35), Definition 6.6.12 (preprint 6.36) and (□) of Theorem 6.6.13 (preprint 6.37); (1′) is proved instead and is what the proofs use (§§VI.10, VI.18, VI.19, VI.23).
- **Lemma 6.6.5 (preprint 6.30) as a statement about manifolds** (gap G.62): the solution set near $(t, \varphi(t), 0)$, the submersion π and the smooth bundle L; the kernels are a continuous family of subspaces (§§VI.12, VI.15). **Property (P) for the slices** (gap G.63); the general position of F that the text derives from it is proved (§VI.11).
- **Lin on every level** (gap G.64): the members of $\mathrm{Lin}(f,x)$ are Fredholm on level 0, not shown sc-Fredholm on every level (§VI.1).
- **Paths on** $[0,1]$ (gap G.65): paths, homotopies and families sc-smooth on $[0,1]$, $[0,1]^2$, $[0,1]^n$ only, and $X \times [0,1]$ as an M-polyfold with boundary; they are read on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ and $\mathbb{R}^n \times X$ (decision D2), and a path that does not extend sc-smoothly to $\mathbb{R}$ is not covered.

- **Remark 6.6.19 (preprint 6.42) beyond a point** (gap G.66, §VI.26).
- **Theorem 6.6.18 (preprint 6.41), the global parts** (§VI.25): the orientation of S as an abstract manifold (gap G.67), the compatibility of the tangent orientations read in two charts (gap G.68), and the perturbation paragraph after the theorem (gap G.69).
- **The chart-free objects** T_xX, Y_x, $\mathrm{Lin}(f,x)$, Or_x^f (gaps G.6, G.7, G.13 of Chapter 2): they are made in chart points and identified between charts (§VI.21).

VI.31 What the formalization taught us

The sign of (94). The main change of substance is a sign. The coordinate of Φ_E in a local determinant chart carries the factor $(-1)^{\dim \operatorname{coker} S}$, which jumps where the parity of the cokernel dimension changes; the orientations it carries along a continuous family are then not continuous, as we read it, and an example on $\mathbb{R}$ shows it (§VI.14). The continuity statements of pp. 265–272 (preprint pp. 223–226) are not written in coordinates in the text, and it is in coordinates that the sign appears. The normalized $\alpha(T)\Phi_E$ at both ends of every propagation changes nothing on surjective operators — the tangent orientations of Theorem 6.6.18 (preprint 6.41) are the text's — and the same sign is used in Remark 6.6.19 (preprint 6.42) (§VI.26).

Lemma 6.10 (preprint) and the 2017 remark. The common lower bound of two good left-projections exists, but not with the range $R(P)\cap R(P')$: a projection onto a smaller, finite-codimensional piece of it, along a complement containing both kernels, is below both (§VI.4). This keeps the preprint's construction of $\mathrm{DET}(E,F)$ — one lower bound, Corollary 6.4.12 (preprint 6.13), Lemma 6.5.5 (preprint 6.21) — intact.

Contractibility was never needed. Proposition 6.6.3 (1) (preprint 6.28) is not established for the neighbourhood its proof builds, as we read it, but every later use of contractibility is a use of (1′): two paths in U with the same end points are homotopic in a larger region where sections along families exist. Orientable neighbourhoods become pairs $V\subseteq U$, and property ( ) of Theorem 6.6.13 (preprint 6.37) holds on every set with the local propagation property (§§VI.10, VI.19, VI.23).

Paths on $\mathbb{R}$ cost reparametrizations. With paths and homotopies on $\mathbb{R}$ whose conditions hold on $[0,1]$ (decision D2), the homotopy of (1′) has the right edges only on $[0,1]$ while push-forward data need zeros near $[0,1]$; transitivity of homotopy, associativity of concatenation and Theorem 6.6.11 (preprint 6.35) all pass through the smooth transition function κ (§§VI.18, VI.20).

Charts must be compared. The text's chart-free Or_x^f becomes a set read in a chosen chart, and the invariance of the local propagation under a change of chart — the naturality of the stabilization isomorphism, of the sign and of continuation — is a file of its own (§VI.21).

"By a covering argument" and "one verifies". The covering argument of Theorem 6.6.13 (preprint 6.37) is a dyadic transport with a square lemma, proved for any family of types (§VI.22); the "simple computation" of Lemma 6.6.8 (preprint 6.33) is an exact identity of stabilization isomorphisms, with no sign, once the appended sections come last (§VI.16); and the equivalence of Definitions 6.6.14 and 6.6.15 (preprint 6.38 and 6.39), stated without proof, needs the transport computed on dyadic pieces inside orientable neighbourhoods (§VI.24).

VI.32 The Springer 2021 edition: Lemma 6.4.4, Proposition 6.4.13, Lemmas 6.6.8 and 6.6.9

The 2021 edition adds index equalities to Lemma 6.4.4 (2017: 6.3), makes the hypothesis of Proposition 6.4.13 (2017: 6.6) explicit, and splits preprint Lemma 6.33 (2017: 6.15) into Lemmas 6.6.8 and 6.6.9. Neither Lemma 6.4.4 nor Proposition 6.4.13 has a preprint counterpart. The file is `Chapter6/Book2021.lean` (Findings, Chapter 6, items 110–115).

Lemma 6.4.4 (2017: 6.3) Chapter6/Book2021 · lemma_6_4_4_2021. *Statement.* Let E, F be Banach spaces, $T_0 : E \to F$ Fredholm and $P_0 \in \Pi_{T_0}$ Chapter6/GoodProjections · IsGoodLeftProj. There is $\varepsilon > 0$ such that for every T and every projection P with $\|T - T_0\| < \varepsilon$, $\|P - P_0\| < \varepsilon$: T is Fredholm, $P \in \Pi_T$, and $\operatorname{ind}(P \circ T) = \operatorname{ind}(T) = \operatorname{ind}(T_0) = \operatorname{ind}(P_0 \circ T_0)$, all indices of maps $E \to F$ Chapter1/FredholmIndex · fredholmIndex. *Proof.* (a) With a quasi-inverse v of T_0, every T with $\|T - T_0\| < 1/(\|v\|+1)$ is Fredholm of the index of T_0 Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt. (b) Instead of the text's map $\Phi_{(P,T)}(x,h) = PTx + (I-P)h$, take the intertwiner $J(P) = PP_0 + (1-P)(1-P_0)$ Chapter6/Book2021 · projIntertwiner. It is continuous in P Chapter6/Book2021 · continuous_projIntertwiner, $J(P_0) = 1$ Chapter6/Book2021 · projIntertwiner_self, so $J(P)$ is a unit for P near P_0; and $J(P)P_0 = PJ(P)$ for every projection P Chapter6/Book2021 · projIntertwiner_mul, so $P = JP_0J^{-1}$. (c) $J^{-1}T$ is near T_0, so by Lemma 6.5.1 (preprint 6.17) $P_0 \in \Pi_{J^{-1}T}$ Chapter6/LocalDetBundle · lemma_6_17. Then $R(P) = JR(P_0)$ has finite codimension, $(1-P)F$ complements it, and $R(PT) = JR(P_0J^{-1}T) = JR(P_0) = R(P)$: $P \in \Pi_T$. (d) The indices of $P \circ T$ and $P_0 \circ T_0$ are those of T and T_0 Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp. □ As a map onto $R(P)$ the operator $P \circ T$ would have index $\dim\ker(PT)$; the text's equalities hold for the map $E \to F$. E and F are taken in the universe Type, that of (a).

A common complement Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq. *Statement.* Two finite-dimensional subspaces K_1, K_2 of the same dimension of a real vector space have a common algebraic complement. *Proof.* Induction on $n = \dim(K_1 + K_2) - \dim K_1$ Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq_aux. For $n = 0$, $K_1 = K_2$. Otherwise neither contains the other; take $a \in K_2 \setminus K_1$, $b \in K_1 \setminus K_2$. Then $a+b$ lies in neither, $K_i + \mathbb{R}(a+b)$ has dimension $\dim K_i + 1$ Chapter6/Book2021 · finrank_sup_span_singleton_of_notMem and their sum is still $K_1 + K_2$, so the induction gives a common complement X' of $K_1 + \mathbb{R}(a+b)$ and $K_2 + \mathbb{R}(a+b)$, and $X' + \mathbb{R}(a+b)$ complements K_1 and K_2 Chapter6/Book2021 · isCompl_sup_span_singleton. □ The text takes X closed; that is not needed for the algebraic part below, and a closed one is §VI.33.

Proposition 6.4.13 (2017: 6.6), the algebraic part Chapter6/Book2021 · proposition_6_4_13_2021. *Statement.* Let T be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$ (the 2021 hypothesis $\Pi_T \cap \mathcal{P}_H \neq \emptyset$ is the existence of such projections). There are linear isomorphisms $A_T : \ker(PT) \to \ker(QT)$, $B_T : F/R(P) \to F/R(Q)$ with $A_T \circ j_P = j_Q$, $\Phi_Q \circ A_T = B_T \circ \Phi_P$ and $\pi_Q \circ B_T = \pi_P$, for the maps of $E(T,P)$, $E(T,Q)$ (§VI.5) Chapter6/ProjectionSequence · projSeqJ Chapter6/ProjectionSequence · projSeqPhi Chapter6/ProjectionSequence · projSeqPi. *Proof (the text's).* PT and QT are Fredholm Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp of index $\operatorname{ind}(T)$, and $F/R(P) = F/R(Q)$, so $\dim\ker(PT) = \dim\ker(QT)$; let X be a common complement. QT maps X isomorphically onto $R(Q)$; for $e \in \ker(PT)$ let $x(e) \in X$ solve $QTx = QTe$ and put $A_T(e) = e - x(e) \in \ker(QT)$. It is injective ($A_T(e) = 0$ puts e in $X \cap \ker(PT) = 0$), the identity on $\ker T$, and onto by the dimensions. For $f \in F$ write $(1-P)f = Tx + \tilde{f}$ with $x \in X$, $\tilde{f} \in (1-Q)F$ (solve $QTx = Q(1-P)f$), and put $B_T(f + H) = \tilde{f} + H$. The three identities are direct computations; B_T is injective by a direct computation, hence an isomorphism, the two quotients having the same finite dimension (the text uses the five lemma). □ The last sentence, that A_T and B_T depend continuously on T, is §VI.33, with a closed X (gap G.81, closed).

Lemmas 6.6.8 and 6.6.9 Chapter6/Book2021 · lemma_6_6_8_2021 Chapter6/Book2021 · lemma_6_6_9_2021. *Statement.* For stabilized kernels of the data $(f, s_1, \dots, s_k)$ along a family Chapter6/Propagation · IsStabKernels and of the data with d sections appended, the push-forwards of orientations agree: for zero sections (Lemma 6.6.8, the parameters μ as dummy variables) and for arbitrary sections $\bar{s}_1, \dots, \bar{s}_d$ (Lemma 6.6.9). *Proof.* Both are §VI.16 Chapter6/Propagation · IsStabKernels.push_append, Lemma 6.6.8 with $\bar{s}_j = 0$. □ The statement of preprint Lemma 6.33 (2017: 6.15) is Lemma 6.6.9, and the first step of its proof is Lemma 6.6.8; the formalization of §VI.16 never needed that step.

Already covered. The other 2021 changes of the chapter concern statements verified before: Proposition 6.4.15 with its hypotheses made explicit (§VI.6) Chapter6/Stabilization · proposition_6_15, Proposition 6.6.3 for a tame X and sections valued in $Y \times [0,1]$ (§VI.10, without tameness) Chapter6/

OrientableNbhd · proposition_6_28, and Lemma 6.6.5 with the solution set near $(t, \varphi(t), 0)$ in the points (t, y, λ), whose kernel family (§VI.12) is unaffected Chapter6/KernelFamily · exists_subspaceFamily_linKer.

VI.33 The Springer 2021 edition, continued: Proposition 6.4.13 (last sentence) and Theorem 6.6.17

The last sentence of Proposition 6.4.13 (2017: 6.6), the continuity of A_T and B_T in T, is proved with a closed common complement. Theorem 6.6.17 is new in the 2021 edition: a topology on the orientation space O_f of pp. 274–275, whose proof the text leaves to the reader. It is proved with property (2) under a hypothesis on X, and without that hypothesis for properties (1) and (3). The files are `Chapter6/Book2021Continuity.lean` and `Chapter6/OrientationSpace.lean` (Findings, Chapter 6, items 116–121).

A common closed complement Chapter6/Book2021Continuity · exists_isClosed_isTopCompl_isTopCompl_of_finrank_eq. *Statement.* Two finite-dimensional subspaces K_1, K_2 of the same dimension of a real normed space E have a common closed complement X: $E = K_1 \oplus X = K_2 \oplus X$ topologically. *Proof.* Let X_0 be a common algebraic complement (§VI.32) Chapter6/Book2021 · exists_isCompl_isCompl_of_finrank_eq and Z a closed complement of the finite-dimensional $K_1 + K_2$. Then $X = Z + X_0 \cap (K_1 + K_2)$ complements K_i: inside $K_1 + K_2$, $X_0 \cap (K_1 + K_2)$ complements K_i, and Z complements $K_1 + K_2$ Chapter6/Book2021Continuity · isCompl_sup_inf_of_isCompl. X is closed, a closed subspace plus a finite-dimensional one, and an algebraic complement of a finite-dimensional subspace that is closed is a topological one. □

Proposition 6.4.13 (2017: 6.6), the last sentence Chapter6/Book2021Continuity · proposition_6_4_13_continuity_2021. *Statement.* Let T be Fredholm and $P, Q \in \Pi_T$ with $R(P) = R(Q) = H$. There are a closed $X \subseteq E$ and $\varepsilon > 0$ such that on $\|S - T\| < \varepsilon$ the maps $S \mapsto A_S = 1 - \xi_S \circ S$ Chapter6/Book2021Continuity · kerTransition and $S \mapsto B_S = (1 - P) - S\xi_S(1 - P)$ Chapter6/Book2021Continuity · cokerTransition, where $\xi_S = (QS|_X)^{-1} \circ Q$ Chapter6/Book2021Continuity · solveOp Chapter6/Book2021Continuity · projCod, are continuous into $L(E)$, $L(F)$ in the operator norm, and jointly with the vector; and for every such S: $P, Q \in \Pi_S$, $E = \ker(PS) \oplus X = \ker(QS) \oplus X$ topologically, $e - A_S e \in X$ and $QSA_S e = 0$ (6.15)–(6.16) Chapter6/Book2021Continuity · kerTransition_apply, $QB_S f = 0$ and $Sx + B_S f = (1 - P)f$ for some $x \in X$ (6.18)–(6.19) Chapter6/Book2021Continuity · cokerTransition_apply Chapter6/Book2021Continuity · solveOp_mem, and A_S, B_S induce isomorphisms $\ker(PS) \to \ker(QS)$, $F/R(P) \to F/R(Q)$ filling the diagram of $E(S, P)$, $E(S, Q)$ Chapter6/Book2021Continuity · diagram_of_solveOp. *Proof.* Take X from the previous statement for $\ker(PT)$, $\ker(QT)$, which have the same dimension (§VI.32). (a) By Lemma 6.5.1 (preprint 6.17) Chapter6/LocalDetBundle · lemma_6_17 there is ε with $P, Q \in \Pi_S$ and $QS|_X : X \to R(Q)$ an isomorphism for $\|S - T\| < \varepsilon$ Chapter6/LocalDetBundle · projRestrict. (b) $S \mapsto QS|_X$ is continuous Chapter6/LocalDetBundle · continuous_projRestrict and inversion is continuous at isomorphisms, X being complete; so ξ_S is continuous in S Chapter6/Book2021Continuity · continuousAt_solveOp, and so are A_S, B_S, which are built from ξ_S, S and P by composition. (c) $QS\xi_S = Q$ Chapter6/Book2021Continuity · apply_solveOp_of_projRestrict_eq gives (6.15)–(6.19). (d) $\ker(QS) \cap X = 0$ because $QS|_X$ is injective, and $\ker(QS) + X = E$ because $e - \xi_S Se \in \ker(QS)$ Chapter6/Book2021Continuity · disjoint_ker_comp_of_projRestrict_eq; the projection $1 - \xi_S S$ onto $\ker(QS)$ along X is continuous Chapter6/Book2021Continuity · isTopCompl_ker_comp_of_projRestrict_eq; the same for P, $R(P) = R(Q)$. (e) The diagram is §VI.32 for S with this X. □ *Satisfiable:* $T = 1$ on $\mathbb{R}$ and $P = Q$ any element of Π_T Chapter6/Book2021Continuity · proposition_6_4_13_continuity_2021_hyps. "Depend continuously" is read as the continuity of A_S on E and B_S on F; these determine the maps between the varying kernels $\ker(PS)$, $\ker(QS)$ and on F/H, and the joint continuity is continuity on the kernel bundle $\bigcup_S \{S\} \times \ker(PS)$ of Remark 6.4.5 (gap G.81, closed).

The orientation space Chapter6/OrientationSpace · OrSpace Chapter6/OrientationSpace · orProj. *Definition.* O_f is the set of pairs (x, o) with $x \in X_\infty$ and $o \in \mathrm{Or}_x^f$, read in the chosen chart (§VI.21) Chapter6/SectionOrientation · Orx, and $\pi_f(x, o) = x$; X_∞ carries the topology of Theorem 2.3.10 (§II.7) Chapter2/

Metrizable · MPolyfold.SmoothLevel. For an sc-smooth path φ (a map $\mathbb{R} \to X$, decision D2) the pull-back $\varphi^* O_f$ is the subspace of $[0,1] \times O_f$ of the (t,p) with $\varphi(t) = \pi_f(p)$ Chapter6/OrientationSpace · OrPullback, and a continuous section over an open U is a continuous $\sigma : U_\infty \to O_f$ with $\pi_f \circ \sigma = \mathrm{id}$ Chapter6/OrientationSpace · IsContSectionOver. The three properties of the theorem: (1) π_f is a local homeomorphism; (2) every $x \in X_\infty$ has an open U in X over whose smooth part there are exactly two continuous sections Chapter6/OrientationSpace · HasTwoContSections; (3) for every sc-smooth path φ and continuous section $t \mapsto (\varphi(t), o_t)$ of $\varphi^* O_f$, for every t_0 and t near t_0, o_t is the push-forward of o_{t_0} along the segment $\varphi|[t_0,t]$ Chapter6/OrientationSpace · segPath Chapter6/OrientationSpace · isScSmoothMap_segPath in a set with the local propagation property Chapter6/OrientationFunctor · IsPropagationNbhd Chapter6/OrientationFunctor · orPush, which does not depend on the set Chapter6/OrientationFunctor · orPush_compat Chapter6/OrientationSpace · IsPropagationContinuous. An *orientation topology* is a Hausdorff topology with (1)–(3) Chapter6/OrientationSpace · IsOrientationTopology. This is our reading of the text's "local propagation of o_{t_0}".

Sc-smooth paths in X_∞ Chapter6/OrientationSpace · continuous_smoothMk_comp Chapter6/OrientationSpace · eventually_exists_scPath_smoothLevel. *Statement.* (a) An sc-smooth $\Gamma : \mathbb{R}^n \to X$ is continuous into X_∞. (b) If N is a neighbourhood of x in X_∞, every $y \in X_\infty$ near x is joined to x by an sc-smooth path with all its points in N. *Proof.* (a) Its values are smooth points Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap; in a chart, Γ is continuous into every level m Chapter6/OrientationSpace · continuousOn_levelLift_comp Chapter6/OrientationSpace · rnChart_toBase, which is continuity into X_∞. (b) In a chart at x, take the retracted segment $t \mapsto r((1-t)u_0 + tu_1)$ Chapter6/OrientationSpace · chartSegPath, sc-smooth Chapter6/OrientationSpace · isScSmoothMap_chartSegPath; for y near x in X_∞ its points are near x in every level, hence in N. □ The neighbourhood version for open sets of X is Chapter6/OrientationSpace · exists_nhds_scPath_of_isOpen, from §VI.18 Chapter6/ScPaths · exists_nhds_scPath.

The topology Chapter6/OrientationSpace · orTopology. *Definition.* Fix local transport data L (§VI.22) Chapter6/PathTransport · LocalTransport. A *continuous orientation* over an open U is a family $y \mapsto o_y$ on U_∞ carried along by the transport of L along every sc-smooth path in U Chapter6/OrientationSpace · IsContOrOn; its sheet is $\{(y, o_y) : y \in U_\infty\}$ Chapter6/OrientationSpace · orSec Chapter6/OrientationSpace · orBasic. A set $W \subseteq O_f$ is open if $\{y \in U_\infty : (y, o_y) \in W\}$ is open in X_∞ for every continuous orientation o over an open U. *Properties, for every L.* (a) Two continuous orientations that agree at a point agree near it Chapter6/OrientationSpace · IsContOrOn.eventually_iff; so the sheets are open Chapter6/OrientationSpace · isOpen_orBasic and π_f is continuous Chapter6/OrientationSpace · continuous_orProj. (b) Every point of O_f lies on a sheet Chapter6/OrientationSpace · exists_isContOrOn: push the orientation from x along the paths of §VI.18 in a good set of L. (c) On a sheet π_f is a homeomorphism onto U_∞, with inverse $y \mapsto (y, o_y)$ Chapter6/OrientationSpace · continuousOn_orSec Chapter6/OrientationSpace · eq_orSec_of_mem_orBasic Chapter6/OrientationSpace · orSec_mem_orBasic Chapter6/OrientationSpace · orChartHomeo; with (b), property (1) Chapter6/OrientationSpace · isLocalHomeomorph_orProj. (d) Hausdorff Chapter6/OrientationSpace · t2Space_orTopology: points over different x are separated by π_f, X_∞ being metrizable Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel; the two points over one x lie on two sheets which, by (a), are disjoint over a neighbourhood of x. □ The text gives no construction ("in view of Proposition 6.6.3 and Theorem 6.6.11").

Property (2), under connected smooth neighbourhoods Chapter6/OrientationSpace · hasTwoContSections_orTopology. *Statement.* Suppose every smooth point has, inside every open $W \ni x$, an open $U \ni x$ with U_∞ preconnected in X_∞ Chapter6/OrientationSpace · MPolyfold.HasConnectedSmoothNbhds, and every Or_y^f has two elements. Then the topology has property (2) for every L, the two sections being $y \mapsto (y, \pm o_y)$. *Proof.* Take a sheet of o over $V \ni x$ (property (b)) and such a $U \subseteq V$. Every point over y is (y, o_y) or $(y, -o_y)$ Chapter6/OrientationSpace · eq_orSec_or_eq_orSec, and both sheets are open; so for a continuous section the sets where it equals (y, o_y) and where it equals $(y, -o_y)$ are open and complementary in U_∞, which is preconnected. □ The hypothesis holds for $\mathbb{R}^n$ (convex balls) Chapter6/OrientationSpace · rnPolyfold_hasConnectedSmoothNbhds and for every M-polyfold on a subsingleton Chapter6/OrientationSpace · MPolyfold.hasConnectedSmoothNbhds_of_subsingleton. Part I does not state it (gap G.90).

Theorem 6.6.17 `Chapter6/OrientationSpace · theorem_6_6_17_2021`. *Statement.* Let X be tame with bump functions and f an sc-Fredholm section with tame germ data and sc$^+$-sections through the smooth fibre points, the hypotheses under which Γ_f is built (§VI.23) `Chapter6/OrientationFunctor · theorem_6_37` `Chapter6/OrientationFunctor · orLocalTransport`. If X has connected smooth neighbourhoods, O_f carries exactly one Hausdorff topology with (1), (2), (3). Without that hypothesis, O_f carries exactly one Hausdorff topology with (1) and (3) `Chapter6/OrientationSpace · existsUnique_orientationTopology_2021`. *Proof.* Existence: the topology for the transport data of Γ_f. It has (1) and is Hausdorff (above), has (2) by the previous statement, every Or_y^f having two elements `Chapter6/OrientationFunctor · exists_two_orx`, and has (3) `Chapter6/OrientationSpace · isPropagationContinuous_orTopology`: near t_0 a continuous section of $\varphi^* O_f$ stays in one sheet `Chapter6/OrientationSpace · eq_orSec_iff` `Chapter6/OrientationSpace · eventually_segPath_mem`, and a sheet is carried along the segments of φ by the transport. Uniqueness: let t be any topology with (1) and (3) `Chapter6/OrientationSpace · eq_orTopology_of`. (i) A t-continuous local section s over an open $N \subseteq X_\infty$ that agrees with a continuous orientation o at y_0 agrees with it near y_0 `Chapter6/OrientationSpace · eventually_eq_orSec_of_continuousOn`: join y near y_0 to y_0 by a path in N (statement (b) above), which is continuous into X_∞ (statement (a)); then s along the path is a continuous section of the pull-back, and by (3) the set of parameters where it equals o is open and closed in $[0,1]$. (ii) Through every point a local inverse of π_f for t is such a section. By (i) the t-open sets and the open sets of the topology coincide. □ Neither Hausdorffness nor (2) is used in the uniqueness. The text's "uniquely defined" is read as "exactly one".

Not formalized. Property (2) without the hypothesis of connected smooth neighbourhoods (gap G.90): a continuous section over U_∞ is locally one of the two sheets, so (2) needs a U with connected smooth part; the paths of §VI.18 join the smooth points of a small V only inside a larger U. The proofs of §7.4 use that U_∞ is connected for connected open U (gap G.85); as we read it, that fact gives the hypothesis, M-polyfolds being locally path connected (Proposition 2.3.7). The remarks after the theorem (gap G.91): π_f as the orientation bundle, orientability as a global continuous section of π_f compared with Definitions 6.6.14–6.6.16 (§VI.24), the continuous $\mathbb{Z}_2$-action $(x,o) \mapsto (x,-o)$ and the line bundle $\det_f = \mathbb{R} \otimes_{\mathbb{Z}_2} O_f$.

Part VII. Ep-groupoids (HWZ Chapter 7, §§7.1–7.3, 7.5, 7.6 (2017 §§7.1–7.5))

Chapter 7 opens Part II of the book. Part II has no 2014 preprint, so its numbers and pages are those of the book (Springer 2021, pp. 287–356), with the 2017 text's in parentheses (2017 pp. 225–256; see `BookConcordance.md`). The chapter introduces ep-groupoids, the sc-smooth version of étale proper Lie groupoids. §7.1 defines groupoids, isotropy groups and ep-groupoids (Definitions 7.1.1–7.1.3 (2017: 7.1–7.3)), the local sc-diffeomorphism $\widehat{\varphi}$ of a morphism and its tangent (Definition 7.1.4 (2017: 7.4)), the levels and the degeneracy index of the orbit space (Definition 7.1.5 (2017: 7.5)), tame ep-groupoids and the category $\mathcal{EP}$ (Definitions 7.1.6, 7.1.7 (2017: 7.6, 7.7)). It proves that π is open, that $|X|$ has countable bases of neighbourhoods, the local closure relation and the finiteness of the isotropy groups (Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1)). It states the natural representation of G_x (Theorem 7.1.13 (2017: 7.1)), discusses the translation groupoid $G_x \ltimes U(x)$ and the action $\hat{\Phi}$ (pp. 293–297 (2017 pp. 231–233)), extends the natural representation to finite sets of isomorphic objects (Proposition 7.1.15 (2017: 7.5), Definition 7.1.18 (2017: 7.8)), identifies $|U(x)|$ with $G_x\backslash U(x)$ (Proposition 7.1.19 (2017: 7.6)) and defines local uniformizers and ep-subgroupoids (Definitions 7.1.20, 7.1.21 (2017: 7.9, 7.10), Proposition 7.1.22 (2017: 7.7)). §7.2 treats effective ep-groupoids, the subgroups T_x, the reduced morphism space $\mathbf{X}^R$ and the reduced ep-groupoid X_R (Definitions 7.2.1–7.2.15 (2017: 7.11–7.13), Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Propositions 7.2.4, 7.2.10 (2017: 7.8, 7.9) and Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14)). §7.3 proves that $|X|$ is locally metrizable, regular and Hausdorff (Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2), Definition 7.3.7 (2017: 7.14)) and lifts ep-groupoids to level 1 (Proposition 7.3.9 (2017: 7.11)). §7.5 (2017 §7.4) defines paracompact orbit spaces and sc-smooth partitions of unity (Definitions 7.5.1, 7.5.5 (2017: 7.15, 7.16), Remarks 7.5.2, 7.5.6 (2017: 7.1, 7.2)), proves Proposition 7.5.3 (2017: 7.12) and Theorems 7.5.4, 7.5.7 (2017: 7.3, 7.4), and characterizes the closed subsets of $|X|$ (Definition 7.5.8 (2017: 7.17), Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14)). The appendix §7.6 (2017 §7.5) proves Theorem 7.1.13 (2017: 7.1) as Theorem 7.6.1 (2017: 7.5) (Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9)) and Theorem 7.5.7 (2017: 7.4) as Theorem 7.6.4 (2017: 7.6) (Lemma 7.6.5 (2017: 7.10)), and ends with Corollary 7.6.6 (2017: 7.3). This Part retells §7.1 in §§VII.1–VII.12 — Definitions 7.1.1, 7.1.2 (2017: 7.1, 7.2) (§VII.1), Definition 7.1.3 (2017: 7.3) and properness (§VII.2), Definitions 7.1.4–7.1.6 (2017: 7.4–7.6) (§VII.3), Definition 7.1.7 (2017: 7.7) and the remark after it (§VII.4), Propositions 7.1.8–7.1.10 (2017: 7.1–7.3) (§VII.5), Lemma 7.1.11 (2017: 7.1) and Proposition 7.1.12 (2017: 7.4) (§VII.6), Theorem 7.1.13 (2017: 7.1) (§VII.7), pp. 293–297 (2017 pp. 231–233) (§VII.8), the uniqueness of natural representations (§VII.9), Proposition 7.1.15 (2017: 7.5) and Definition 7.1.18 (2017: 7.8) (§VII.10), Proposition 7.1.19 (2017: 7.6) and Definition 7.1.20 (2017: 7.9) (§VII.11), Definition 7.1.21 (2017: 7.10) and Proposition 7.1.22 (2017: 7.7) (§VII.12); §7.2 in §§VII.13–VII.17 — Definition 7.2.1 (2017: 7.11) and Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3) (§VII.13), Proposition 7.2.4 (2017: 7.8) (§VII.14), T_x and Definition 7.2.7 (2017: 7.12) (§VII.15), Proposition 7.2.10 (2017: 7.9) (§VII.16), X_R, Proposition 7.10 (2017; rewritten in 2021 as Theorem 7.2.14) and Definition 7.2.15 (2017: 7.13) (§VII.17); §7.3 in §§VII.18–VII.22 — Theorem 7.3.1 (2017: 7.2) and Lemmas 7.3.2, 7.3.3 (2017: 7.4, 7.5) (§VII.18), Lemmas 7.3.4, 7.3.5 (2017: 7.6, 7.7) and Corollary 7.3.6 (2017: 7.1) (§VII.19), Michael's theorem of Part I (§VII.20), Definition 7.3.7 (2017: 7.14) and Corollary 7.3.8 (2017: 7.2) (§VII.21), Proposition 7.3.9 (2017: 7.11) (§VII.22); §7.5 (2017 §7.4) in §§VII.23–VII.25

— Definition 7.5.1 (2017: 7.15), Remark 7.5.2 (2017: 7.1) and Proposition 7.5.3 (2017: 7.12) (§VII.23), Theorem 7.5.4 (2017: 7.3), Definition 7.5.5 (2017: 7.16), Remark 7.5.6 (2017: 7.2) and the statement of Theorem 7.5.7 (2017: 7.4) (§VII.24), Definition 7.5.8 (2017: 7.17) and Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14) (§VII.25); the appendix in §§VII.26–VII.29 — the proof of Theorem 7.1.13 (2017: 7.1) (§VII.26), Lemma 7.6.5 (2017: 7.10) (§VII.27), the proof of Theorem 7.5.7 (2017: 7.4) (§VII.28) and Corollary 7.6.6 (2017: 7.3) (§VII.29); the witnesses and counterexamples (§VII.30); what is not formalized (§VII.31); and what the formalization taught us (§VII.32). Each item carries the number of its comparison item (V7.1–V7.68) in `Exposition/verification/ch07.md`. The additions of the 2021 edition (V7.69–V7.104) are retold in §§VII.33–VII.43: §7.1 (§VII.33), §7.2 (§§VII.34–VII.35), Appendices 7.6.3 and 7.6.4 (§VII.36), §7.4, new in the 2021 edition (§§VII.37–VII.39), Appendix 7.6.5 (§VII.40), what is not formalized and what they taught us (§VII.41), Theorem 7.6.9 in full, the boundary ep-groupoid (§VII.42), and §7.4 on every lift X^i (§VII.43).

What is proved, and what is not. Theorems 7.1.13–7.6.4 (2017: 7.1–7.6) are proved, and so are Propositions 7.1.8–7.2.4 and 7.3.9–7.5.10 (2017: 7.1–7.8 and 7.11–7.14), Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) under the hypothesis that $\mathbf{X}^R$ is Hausdorff and paracompact, Lemmas 7.1.11–7.6.5 (2017: 7.1–7.10) and Corollaries 7.3.6–7.6.6 (2017: 7.1–7.3), Corollary 7.6.6 (2017: 7.3) with the theorem of Godefroy–Troyanski–Whitfield–Zizler as an explicit hypothesis, as in Part I. Every definition of the chapter is formalized; the tangent $T\varphi$ of Definition 7.1.4 (2017: 7.4) is a definition only. Five places needed more than a reading. The remark after Definition 7.1.7 (2017: 7.7), that a functor sc-smooth on objects is sc-smooth on morphisms, does not hold as we read it, and a counterexample on the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$ shows it; it holds for functors continuous on morphisms, which is what its argument uses (§VII.4). The proof of Proposition 7.2.10 (2017: 7.9) gives the charts of $\mathbf{X}^R$ and their sc-smooth transitions but, as we read it, neither Hausdorffness nor paracompactness, which become hypotheses, discharged for effective ep-groupoids (§VII.16). The natural representation of Theorem 7.1.13 (2017: 7.1) is taken on a connected neighbourhood, which the text chooses on 2017 p. 231 but does not put in the statement; connectedness makes Φ a homomorphism and Γ a functor, and the local path connectedness of M-polyfolds that supplies it is proved (§VII.7). The tameness of X^1 in Proposition 7.3.9 (2017: 7.11) needs that the index lift of a tame retraction is tame, which Part I left open; it is proved (§VII.22). And several omitted steps are supplied: the uniqueness of natural representations behind "adapting restrictions" (§VII.9), the conjugation $\varphi T_x \varphi^{-1} = T_y$ that the composition of X_R needs (§VII.15), the passage from compact to component compact sets in Corollary 7.3.8 (2017: 7.2) (§VII.21), the support bound that makes the sum in the proof of Theorem 7.5.7 (2017: 7.4) locally finite (§§VII.27–VII.28) and Michael's theorem on unions of closed paracompact sets (§VII.20). Proposition 7.1.15 (2017: 7.5) is proved by a construction without induction (§VII.10), Proposition 7.5.3 (2017: 7.12) by Smirnov's theorem pulled back to $|X^1|$ (§VII.23), Proposition 7.5.10 (2) (2017: 7.14) without Lemma 2.7.3 (2017: 2.7) (§VII.25), Lemma 7.3.4 (2017: 7.6) from the closedness of points and regularity (§VII.19), and the normalization of Theorem 7.5.7 (2017: 7.4) by bump coverings (§VII.28). Not formalized: the properties of $T\varphi$ (gap G.70), the set $|X_\infty|$ of the filtration of $|X|$ (gap G.72), the tangent facts (3), (4) recalled before Proposition 7.1.22 (2017: 7.7), the paracompactness of $\mathbf{X}^R$ when it is Hausdorff (gap G.71; Hausdorffness itself fails in general, §VII.16), the density of X_∞ in X_k (gap G.73), and the open question of 2017 p. 249 (§VII.31).

Seven structural choices. (1) *A groupoid is a structure of maps.* Objects and morphisms are types; s, t, u, ι are maps and the composition m is a total map whose value on a non-composable pair is a junk value never used; the groupoid laws are required on composable pairs. A bridge to Mathlib's groupoids makes "full" and "faithful" Mathlib's. (2) *Properness is Mathlib's.* Definition 7.1.3 (3) (2017: 7.3) is the properness, in Mathlib's sense, of t restricted to $s^{-1}(\mathrm{cl}_X V)$, for some open $V \ni x$, per point; on metrizable spaces it is equivalent to the sequential form the text uses (§VII.2). (3) *Condition (4) in two forms.* It is stated literally, on the fibred product of Part I's Proposition 2.5.20 (2017: 2.15; preprint 2.46, §II.17), and used in the equivalent *sheet form*: for every local inverse σ of s, the map $\psi \mapsto m(\sigma(t\psi), \psi)$ is sc-smooth. (4) *Natural representations are data.* "An open neighbourhood $U(x)$ equipped with the natural G_x-action" is always a datum (U, Φ, Γ) with U open, connected and containing x, never a hypothesis "for every such $U(x)$". (5) *$G \times U$ is a product with a discrete M-polyfold*, the finite disjoint union of copies of U, on which sc-smoothness is sc-smoothness on each slice (§VII.8). (6) *The reduced morphisms* $\mathbf{X}^R$ are a quotient whose M-polyfold structure is built under the hypotheses that it is Hausdorff and paracompact (§VII.16).

(7) *Michael's theorem without Hausdorff.* Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17) (gap G.15 there) are proved for regular spaces, and Mathlib's paracompactness, like Definition 7.5.1 (2017: 7.15), does not include Hausdorff (§VII.20).

Map.

- `Chapter7/Groupoid`: Definitions 7.1.1, 7.1.2 (2017: 7.1, 7.2); the orbit space, saturations, the conjugation of isotropy groups, regular objects, full subcategories, translation groupoids, functors and the bridge to Mathlib.
- `Chapter7/EpGroupoid`: Definition 7.1.3 (2017: 7.3) with the fibred-product and sheet forms of (4), Definition 7.1.4 (2017: 7.4), the levels, Definitions 7.1.5–7.5.8 (2017: 7.5–7.17) as structures and predicates, the trivial ep-groupoid and its witnesses.
- `Chapter7/ProperMaps`: proper maps between metrizable spaces, the sequential criterion, proper restrictions $t : s^{-1}(Q) \to X$.
- `Chapter7/LocalDiffeo`: local sc-diffeomorphisms, their levels, degeneracy index and tameness; Part I's partitions of unity on $\mathbb{R}^n$.
- `Chapter7/Basic`: the remark after Definition 7.1.7 (2017: 7.7), Propositions 7.1.8–7.1.12 (2017: 7.1–7.4), Lemma 7.1.11 (2017: 7.1), the conjugation $G_x \cong G_y$.
- `Chapter7/Discrete`: finiteness from proper locally injective maps, discrete M-polyfolds, $G \times X$ and its sheets.
- `Chapter7/LocConnected`: local path connectedness of retracts and M-polyfolds, uniqueness of sections on connected sets.
- `Chapter7/NaturalRep`: Theorem 7.1.13 (2017: 7.1) = Theorem 7.6.1 (2017: 7.5) with Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9).
- `Chapter7/NaturalAction`: the consequences of Theorem 7.1.13 (2017: 7.1) on pp. 293–297 (2017 pp. 231–233) and the uniqueness of natural representations.
- `Chapter7/NaturalRepFinite`: Proposition 7.1.15 (2017: 7.5) by the star construction, Definition 7.1.18 (2017: 7.8).
- `Chapter7/Uniformizer`: Proposition 7.1.19 (2017: 7.6), Definition 7.1.20 (2017: 7.9) and the claims after it.
- `Chapter7/SubGroupoid`: Definition 7.1.21 (2017: 7.10), the M-polyfold **A**, Proposition 7.1.22 (2017: 7.7).
- `Chapter7/Effective`: Definition 7.2.1's (2017: 7.11) consequences, Lemmas 7.2.2, 7.2.3 (2017: 7.2, 7.3), Proposition 7.2.4 (2017: 7.8), T_x.
- `Chapter7/Reduced`: Definition 7.2.7 (2017: 7.12) made precise, Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), Definition 7.2.15 (2017: 7.13).
- `Chapter7/OrbitTopology`: Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2–7.3.5 (2017: 7.4–7.7), Corollaries 7.3.6, 7.3.8 (2017: 7.1, 7.2).
- `Chapter7/Michael`: Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17), unions of closed paracompact sets.
- `Chapter7/Lift`: Proposition 7.3.9 (2017: 7.11), the map $|X^1| \to |X|$, Definition 7.5.1's (2017: 7.15) consequences, Remark 7.5.2 (2017: 7.1), Proposition 7.5.3 (2017: 7.12).
- `Chapter7/ProperSubsets`: Propositions 7.5.9, 7.5.10 (2017: 7.13, 7.14).
- `Chapter7/PartitionsOfUnity`: Theorem 7.5.4 (2017: 7.3), Lemma 7.6.5 (2017: 7.10), Theorem 7.5.7 (2017: 7.4) = Theorem 7.6.4 (2017: 7.6), Corollary 7.6.6 (2017: 7.3).
- `Chapter7/NonVacuity`: the translation groupoid of a finite group action, $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection, the trivial $\mathbb{Z}/3$-action, the counterexamples.
- `Chapter7/Book2021Basic`: the 2021 forms of the remark after Definition 7.1.7 and of Proposition 7.1.15, the data (7.3), Proposition 7.1.14, Lemma 7.1.16, Corollary 7.1.17 (§VII.33).
- `Chapter7/ReducedCriterion`: the 2021 §7.2 from Definition 7.2.5 on, LUC, Theorem 7.2.22, Appendices 7.6.3 and 7.6.4 (§§VII.34–VII.36).
- `Chapter7/Zhou`: §7.4, the index lifts, the Part I facts as hypotheses, (ZHOU), (SLUC), (VSLUC), (LR), (SLR) (§§VII.37–VII.39).
- `Chapter7/TameBoundaryEp`: Appendix 7.6.5, the boundary groupoid of a tame ep-groupoid, Theo-

rem 7.6.9 (topological part) (§VII.40).
- `Chapter7/TameBoundaryEpStructure`: Theorem 7.6.9 in full, on Part I's Theorem 2.7.13 (2)–(4): the boundary ep-groupoid and the ep-functor B (§VII.42).
- `Chapter7/ZhouLevels`: §7.4 on every lift X^i: Propositions 7.4.9, 7.4.16–7.4.18, and Corollary 7.4.11, Theorems 7.4.13, 7.4.24 for every X^i (§VII.43).

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid: X is the object M-polyfold and $\mathbf{X}$ the morphism M-polyfold, both Hausdorff and paracompact, as every M-polyfold of the project (Definition 2.3.4 (2017: 2.8), §II.5); $s, t : \mathbf{X} \to X$ are source and target, $m(\varphi, \psi) = \varphi \circ \psi$ the composition, $u(x) = 1_x$ the unit and $\iota(\varphi) = \varphi^{-1}$ the inverse. $|X|$ is the orbit space with the quotient topology and $\pi : X \to |X|$ the projection; G_x is the isotropy group of x, cl_X the closure in X, and $\mathbf{U}(x)$ the set of morphisms with source and target in $U(x)$. For a natural representation (U, Φ, Γ) at x, $g * y = \Phi(g)(y)$. A *sheet* is an open set $\mathbf{W} \subseteq \mathbf{X}$ on which s is an sc-diffeomorphism onto an open set, its inverse a *local inverse* of s. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses; the Lean names carry the preprint numbers (Theorem 2.3.10, 2017 Theorem 2.2, is preprint Theorem 2.16, §II.7; Lemma 2.7.3 and Proposition 2.7.4, 2017 Lemma 2.7 and Proposition 2.17, are `lemma_2_56` and `proposition_2_57`; Theorem 5.5.10, 2017 Theorem 5.12, is preprint Theorem 5.37, §V.25). Findings are those of `Exposition/Findings.md`, "Chapter 7".

VII.1 Definitions 7.1.1 and 7.1.2 (2017: 7.1 and 7.2): groupoids, the orbit space, isotropy

Definition 7.1.1 (2017: 7.1) (V7.1). A groupoid is a structure of maps s, t, m, u, ι on a type of objects and a type of morphisms `Chapter7/Groupoid · GroupoidStr`, with associativity, units and inverses required on composable pairs; the morphisms $x \to y$ are a set `Chapter7/Groupoid · GroupoidStr.mor`, and so are the morphisms between points of a set A `Chapter7/Groupoid · GroupoidStr.mor0n`. The composition of a non-composable pair is a junk value; no statement uses it. The text's list of consequences — $(\varphi^{-1})^{-1} = \varphi$, $1_x^{-1} = 1_x$, $(\varphi \circ \psi)^{-1} = \psi^{-1} \circ \varphi^{-1}$, uniqueness of inverses — is derived from the laws `Chapter7/Groupoid · GroupoidStr.inv_inv` `Chapter7/Groupoid · GroupoidStr.inv_unit` `Chapter7/Groupoid · GroupoidStr.inv_comp_rev` `Chapter7/Groupoid · GroupoidStr.eq_inv_of_comp_eq_unit`. Objects are isomorphic if there is a morphism between them `Chapter7/Groupoid · GroupoidStr.Isomorphic`, an equivalence relation `Chapter7/Groupoid · GroupoidStr.orbitSetoid` whose quotient is $|X|$ `Chapter7/Groupoid · GroupoidStr.OrbitSpace` with projection π `Chapter7/Groupoid · GroupoidStr.π`; $\pi(x) = \pi(y)$ iff $x \cong y$ `Chapter7/Groupoid · GroupoidStr.π_eq_π_iff`, $\pi(s\varphi) = \pi(t\varphi)$ `Chapter7/Groupoid · GroupoidStr.π_src_eq_π_tgt` and π is onto `Chapter7/Groupoid · GroupoidStr.surjective_π`. The saturation $\pi^{-1}(\pi(A))$, used from Definition 7.1.21 (2017: 7.10) on, is defined here `Chapter7/Groupoid · GroupoidStr.saturation`, with saturated sets `Chapter7/Groupoid · GroupoidStr.IsSaturated` `Chapter7/Groupoid · GroupoidStr.isSaturated_iff`. The bridge to Mathlib's groupoids is a type synonym of the objects `Chapter7/Groupoid · GroupoidStr.Ob` with the groupoid instance `Chapter7/Groupoid · GroupoidStr.instGroupoid`, whose isomorphism classes are those of the orbit relation `Chapter7/Groupoid · GroupoidStr.orbitSetoid_eq` (Findings, Chapter 7, item 1) (pp. 287–288 (2017 pp. 225–226)).

Definition 7.1.2 (2017: 7.2) (V7.2). G_x, the morphisms $x \to x$ `Chapter7/Groupoid · GroupoidStr.Isotropy`, is a group under composition with $gh = g \circ h$ `Chapter7/Groupoid · GroupoidStr.Isotropy.val_mul` `Chapter7/Groupoid · GroupoidStr.Isotropy.val_one` `Chapter7/Groupoid · GroupoidStr.Isotropy.val_inv`, the text's order. Mathlib's vertex group multiplies in the other order; it is identified with the opposite group `Chapter7/Groupoid · GroupoidStr.isotropyEquivVertexGroup`, and G_x with the automorphisms of x in the bridge `Chapter7/Groupoid · GroupoidStr.isotropyEquivAut` (Findings, Chapter 7, item 2) (p. 288 (2017 p. 226)).

VII.2 Definition 7.1.3 (2017: 7.3), and properness

Definition 7.1.3 (2017: 7.3) (V7.3) `Chapter7/EpGroupoid · EpGroupoid`. An ep-groupoid is a groupoid with M-polyfold structures on the objects and on the morphisms such that (1) s and t are local

sc-diffeomorphisms, (2) u and ι are sc-smooth, (3) every x has an open neighbourhood $V(x)$ with $t : s^{-1}(\mathrm{cl}_X V(x)) \to X$ proper, and (4) m is sc-smooth on $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$. "In particular all structure maps are sc-smooth": s and t are `Chapter7/EpGroupoid · EpGroupoid.isScSmoothMap_src` `Chapter7/EpGroupoid · EpGroupoid.isScSmoothMap_tgt`, and continuous `Chapter7/EpGroupoid · EpGroupoid.continuous_src` `Chapter7/EpGroupoid · EpGroupoid.continuous_tgt` `Chapter7/EpGroupoid · EpGroupoid.continuous_unit` `Chapter7/EpGroupoid · EpGroupoid.continuous_inv` (pp. 288–290 (2017 pp. 226–227)).

Condition (3). It is existential, one open $V(x)$ per point, as the text writes it. The universal reading, "for every open $V \ni x$", fails: in the pair groupoid of $\mathbb{Z}$ (objects $\mathbb{Z}$, one morphism between any two objects) t is proper over $s^{-1}(\{x\})$ but not over $s^{-1}(\mathbb{Z})$ `Chapter7/NonVacuity · Examples.pair_isProperMap` `Chapter7/NonVacuity · Examples.pair_not_isProperMap_univ` `Chapter7/NonVacuity · Examples.pair_not_forall_isProperMap`; every smaller open neighbourhood of x works as soon as one does (§VII.25) (Findings, Chapter 7, items 3, 103).

Condition (4). The text cites "Lemma 2.8 in [38]" for the M-polyfold structure of the fibred product; this is an external reference, and the statement used is Part I's Proposition 2.5.20 (2017: 2.15) (§II.17): since s is a local sc-diffeomorphism, $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$ is a sub-M-polyfold of $\mathbf{X} \times \mathbf{X}$ `Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct`, and (4) is stated on it `Chapter7/EpGroupoid · IsScSmoothComp`. The proofs use an equivalent *sheet form* `Chapter7/EpGroupoid · IsScSmoothCompSheet`: for every sc-diffeomorphism $s : \mathbf{W} \to V$ with inverse σ, the map $\psi \mapsto m(\sigma(t\psi), \psi)$ is sc-smooth on $t^{-1}(V)$. *Step 1.* The graph of $\sigma \circ t$ over $t^{-1}(V)$ is an open piece of the fibred product, and the second projection is an sc-diffeomorphism from it onto $t^{-1}(V)$, with inverse $\psi \mapsto (\sigma(t\psi), \psi)$. *Step 2.* So the literal (4) gives the sheet form by composition, and conversely the fibred product is covered by such graphs `Chapter7/EpGroupoid · isScSmoothComp_iff_sheet`. Every ep-groupoid satisfies the sheet form `Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet`. The "for every σ" of the sheet form carries no vacuity risk, being equivalent to the literal condition (Findings, Chapter 7, item 4).

Properness in sequential form (V7.4). The text uses properness through sequences: in Proposition 7.1.10 (2017: 7.3), Lemma 7.1.11 (2017: 7.1), Proposition 7.1.22 (2017: 7.7), Proposition 7.3.9 (2017: 7.11), Proposition 7.5.9 (2017: 7.13) and Lemma 7.6.2 (2017: 7.8). For a continuous map between metrizable spaces, properness is equivalent to: every sequence whose images converge has a convergent subsequence `Chapter7/ProperMaps · isProperMap_iff_exists_tendsto_subseq`. *Step 1, from properness.* The images and their limit form a compact set, whose preimage is compact, hence sequentially compact `Chapter7/ProperMaps · exists_tendsto_subseq_of_isProperMap`. *Step 2, to properness.* Preimages of compact sets are sequentially compact, hence compact in a metrizable space, and a continuous map with compact fibres of compacta into a compactly generated space is proper (Mathlib's characterization) `Chapter7/ProperMaps · isProperMap_of_exists_tendsto_subseq`. For the restriction of t to $s^{-1}(Q)$ the criterion reads: every sequence with $s(\varphi_k) \in Q$ and $t(\varphi_k)$ convergent has a subsequence converging to a morphism over Q `Chapter7/ProperMaps · isProperMap_domRestrict_iff`, over $\mathrm{cl}\,Q$ when Q is closed `Chapter7/ProperMaps · isProperMap_domRestrict_iff_of_isClosed` `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff`, and it is the compactness of $s^{-1}(Q) \cap t^{-1}(K)$ for compact K `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff_isCompact_preimage`. Three consequences recur. Properness passes to smaller closures, so any open $V' \subseteq V(x)$ around x will do `Chapter7/ProperMaps · isProperMap_domRestrict_closure_mono`; the image $t(s^{-1}(Q))$ is closed for closed Q, a proper map being closed `Chapter7/ProperMaps · isClosed_image_preimage_of_isProperMap_domRestrict`; and the fibres over points are finite when t is locally injective `Chapter7/ProperMaps · finite_setOf_of_isProperMap_domRestrict`. If t is proper over $s^{-1}(Q)$ and the units exist, Q is closed: a limit x of $x_n \in Q$ is the source of a limit of a subsequence of the units 1_{x_n} `Chapter7/ProperMaps · isClosed_of_isProperMap_domRestrict`. The hidden hypothesis is metrizability, which M-polyfolds have (Theorem 2.3.10 (2017: 2.2), §II.7) (Findings, Chapter 7, item 3).

VII.3 Definition 7.1.4 (2017: 7.4), the levels, Definitions 7.1.5 and 7.1.6 (2017: 7.5 and 7.6)

Definition 7.1.4 (2017: 7.4) (V7.5). For a morphism $\varphi : x \to y$ the text takes open $\mathbf{U}(\varphi)$, $U(x)$, $U(y)$ with $s : \mathbf{U}(\varphi) \to U(x)$ and $t : \mathbf{U}(\varphi) \to U(y)$ sc-diffeomorphisms and sets $\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} :$

$U(x) \to U(y)$. Such a sheet exists at every morphism `Chapter7/EpGroupoid · EpGroupoid.Sheet` `Chapter7/EpGroupoid · EpGroupoid.exists_sheet`: (1) gives open neighbourhoods of φ on which s, resp. t, is an sc-diffeomorphism onto an open set, and on their intersection both are, an sc-diffeomorphism restricting to one on every open subset. A sheet is chosen at every morphism `Chapter7/EpGroupoid · EpGroupoid.sheetAt` `Chapter7/EpGroupoid · EpGroupoid.mem_sheetAt`, and $\widehat{\varphi}$ is defined from it `Chapter7/EpGroupoid · EpGroupoid.Sheet.hat` `Chapter7/EpGroupoid · EpGroupoid.hat`: an sc-diffeomorphism `Chapter7/EpGroupoid · EpGroupoid.Sheet.isScDiffeoOn_hat` `Chapter7/EpGroupoid · EpGroupoid.isScDiffeoOn_hat`, a composite of sc-diffeomorphisms, with $\widehat{\varphi}(x) = y$ `Chapter7/EpGroupoid · EpGroupoid.hat_src`. It depends on the sheet only as a germ. $T\varphi = T\widehat{\varphi}(x)$ is read through the chain rule as $Tt \circ (Ts|_{T_\varphi \mathbf{X}})^{-1}$ with Part I's tangent maps `Chapter7/EpGroupoid · EpGroupoid.tangentMor`; that it is a linear isomorphism on level 0 for φ on level 1, and an sc-isomorphism for smooth φ, is not proved: T_xX carries no linear structure in Part I (gaps G.7, G.70). The display "$\widehat{\varphi} = t \circ (s|\mathbf{U}(\varphi))^{-1} \circ U(x) \to U(y)$" is read with : for the last $\circ$, and "$T\varphi : T_{s(\varphi)}X \to T_{t(\varphi)}$" with the missing X (possible misprints; Findings, Chapter 7, items 5, 110) (2017 p. 227).

The levels and the degeneracy index (V7.6). A local sc-diffeomorphism preserves the levels and the degeneracy index: in a sheet it is an sc-diffeomorphism, and Part I's Proposition 2.4.4 (2017: 2.7) (§II.11) applies `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.mem_levelSet_iff` `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.degIndex_eq`. Hence $\varphi \in \mathbf{X}_m$ iff $s(\varphi) \in X_m$ iff $t(\varphi) \in X_m$ `Chapter7/EpGroupoid · EpGroupoid.mem_levelSet_src_iff` `Chapter7/EpGroupoid · EpGroupoid.mem_levelSet_tgt_iff`, and $d_X(s\varphi) = d_{\mathbf{X}}(\varphi) = d_X(t\varphi)$ `Chapter7/EpGroupoid · EpGroupoid.degIndex_src` `Chapter7/EpGroupoid · EpGroupoid.degIndex_tgt`. The text's $ml(x)$, the largest m with $x \in X_m$, is the maximal level `Chapter7/LocalDiffeo · MPolyfold.maxLevel`, preserved too `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.maxLevel_eq`. $|X_k| = \pi(X_k)$ is defined `Chapter7/EpGroupoid · EpGroupoid.orbitLevel`; the set $|X_\infty|$ and the identity $|X_\infty| = \bigcap_k |X_k|$ are not stated (gap G.72; Findings, Chapter 7, items 6, 112) (pp. 289–291 (2017 pp. 227–228)).

Definition 7.1.5 (2017: 7.5) (V7.7). $d_{|X|}(|x|) = d_X(x)$ is well defined by the previous paragraph `Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit` `Chapter7/EpGroupoid · EpGroupoid.degIndexOrbit_π` (2017 p. 228).

Definition 7.1.6 (2017: 7.6) (V7.8). X is tame if the object M-polyfold is tame `Chapter7/EpGroupoid · EpGroupoid.IsTame`, "tame" being the existence of an equivalent tame atlas (Definition 2.5.7 (2017: 2.19), §II.14) `Chapter2/Tame · MPolyfold.IsTame` — not "every chart is tame", which would be unsatisfiable. *Then* $\mathbf{X}$ *is tame.* At a morphism, a sheet $s : \mathbf{W} \to V$ composed with a tame chart of X around $s(\varphi)$ is a tame chart of $\mathbf{X}$, since tameness is carried by sc-diffeomorphisms (§III.27) `Chapter7/LocalDiffeo · MPolyfold.IsScDiffeoOn.exists_tame_chart`; these charts are compatible with the atlas of $\mathbf{X}$ and form a tame atlas `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame`; conversely a surjective local sc-diffeomorphism transfers tameness down `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isTame_iff_of_surjective`. The witnesses are the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn` and the reflection groupoid of §VII.30 (Findings, Chapter 7, items 7, 106) (2017 p. 228).

VII.4 Definition 7.1.7 (2017: 7.7), and the remark after it

Definition 7.1.7 (2017: 7.7) (V7.9). A functor between groupoids is a pair of maps commuting with s, t, m on composable pairs and u `Chapter7/Groupoid · GroupoidFunctor`; it commutes with ι, by the uniqueness of inverses `Chapter7/Groupoid · GroupoidFunctor.mor_inv`, has identities and composites `Chapter7/Groupoid · GroupoidFunctor.id` `Chapter7/Groupoid · GroupoidFunctor.comp`, a Mathlib functor `Chapter7/Groupoid · GroupoidFunctor.toFunctor` and a map of orbit spaces `Chapter7/Groupoid · GroupoidFunctor.orbitMap`. An ep-functor is sc-smooth on objects and on morphisms `Chapter7/EpGroupoid · EpFunctor` `Chapter7/EpGroupoid · EpFunctor.id` `Chapter7/EpGroupoid · EpFunctor.comp` `Chapter7/EpGroupoid · EpFunctor.ext`, and $\mathcal{EP}$ is the category of bundled ep-groupoids and ep-functors `Chapter7/EpGroupoid · EP` (Findings, Chapter 7, item 8) (2017 p. 228).

The remark after Definition 7.1.7 (2017: 7.7) (V7.10). The text says that it suffices to ask sc-smoothness between the object spaces, because near φ one has $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s(\gamma)))$ for γ in a neighbour-

hood $\mathbf{U}(\varphi)$ with $f(\mathbf{U}(\varphi)) \subseteq \mathbf{U}(\psi)$, $\psi = f(\varphi)$. Choosing that neighbourhood uses the continuity of f on morphisms, and without it the claim does not hold as we read it. *The counterexample.* On the trivial action of $\mathbb{Z}/3$ on $\mathbb{R}$ (objects $\mathbb{R}$, morphisms $\mathbb{Z}/3 \times \mathbb{R}$, $s = t$ the projection), let f be the identity on objects and $(g, y) \mapsto (g, y)$ for $y \le 0$, $(g, y) \mapsto (g^2, y)$ for $y > 0$ `Chapter7/NonVacuity · Examples.remarkFunctor`. Since $g \mapsto g^2$ is an automorphism of $\mathbb{Z}/3$, this is a functor; it is sc-smooth on objects `Chapter7/NonVacuity · Examples.isScSmoothMap_remarkFunctor_obj` but not continuous on morphisms, jumping at $(g, 0)$ for $g \neq 1$ `Chapter7/NonVacuity · Examples.not_continuous_remarkFunctor_mor`, hence not sc-smooth there `Chapter7/NonVacuity · Examples.not_isScSmoothMap_remarkFunctor_mor`, and no ep-functor has these components `Chapter7/NonVacuity · Examples.not_exists_epFunctor_remarkFunctor` `Chapter7/NonVacuity · Examples.remark_definition_7_7_counterexample`. *The remark for functors continuous on morphisms.* Take a sheet $\mathbf{U}(\psi)$ of s at $\psi = f(\varphi)$, with inverse σ, and $\mathbf{U}(\varphi) = f^{-1}(\mathbf{U}(\psi))$, open by continuity. For $\gamma \in \mathbf{U}(\varphi)$, $f(\gamma)$ lies in $\mathbf{U}(\psi)$ and has source $f(s\gamma)$, so $f(\gamma) = \sigma(f(s\gamma))$ `Chapter7/Basic · EpGroupoid.mor_eq_srcInv_obj_src`, a composite of sc-smooth maps `Chapter7/Basic · EpGroupoid.isScSmoothMap_mor_of_continuous`. The sheet at φ and the neighbourhoods $U(s\varphi)$, $U(s\psi)$ of the text are not used. Since every ep-functor is continuous on morphisms `Chapter7/Basic · EpFunctor.continuous_mor`, the remark becomes an equivalence for functors sc-smooth on objects `Chapter7/Basic · EpGroupoid.isScSmoothMap_mor_iff_continuous`, and a functor sc-smooth on objects and continuous on morphisms is an ep-functor `Chapter7/Basic · EpFunctor.ofContinuous` (Findings, Chapter 7, items 9, 104) (pp. 290–292 (2017 pp. 228–229)).

VII.5 Propositions 7.1.8–7.1.10 (2017: 7.1–7.3): the topology of $|X|$

Proposition 7.1.8 (2017: 7.1) (V7.11) `Chapter7/Basic · EpGroupoid.proposition_7_1`. *π is open.* $|X|$ carries the quotient topology, so π is continuous `Chapter7/EpGroupoid · EpGroupoid.continuous_π`, and $\pi(U)$ is open iff its saturation $\pi^{-1}(\pi(U))$ is. *Step 1.* The saturation of U is $t(s^{-1}(U))$: $y \cong x \in U$ iff some φ has $s\varphi \in U$ and $t\varphi = y$ `Chapter7/Basic · EpGroupoid.preimage_π_image_eq`. *Step 2.* $s^{-1}(U)$ is open since s is continuous, and t is an open map, being a local sc-diffeomorphism `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.isOpenMap`. The text argues with the sheets $V(\varphi)$, which is the openness of t unpacked. So π is an open quotient map `Chapter7/Basic · EpGroupoid.proposition_7_1_isOpenQuotientMap` and carries neighbourhood filters to neighbourhood filters `Chapter7/Basic · EpGroupoid.proposition_7_1_map_nhds` (Findings, Chapter 7, item 10) (2017 p. 229).

Proposition 7.1.9 (2017: 7.2) (V7.12) `Chapter7/Basic · EpGroupoid.proposition_7_2`. *Every $z \in |X|$ has a countable basis of open neighbourhoods, chosen monotone.* X is metrizable (Theorem 2.3.10 (2017: 2.2)); for $z = \pi(x)$ the images of the balls $B(x, 1/(k+1))$ are open by Proposition 7.1.8 (2017: 7.1) and decreasing, and they form a basis because π maps the neighbourhood filter of x onto that of z `Chapter7/Basic · EpGroupoid.proposition_7_2_of_hasAntitoneBasis`. The statement is Mathlib's antitone basis indexed by $\mathbb{N}$ (radii $1/(k+1)$, the index starting at 0), and $|X|$ is first countable `Chapter7/Basic · EpGroupoid.proposition_7_2_firstCountableTopology` (Findings, Chapter 7, item 10) (2017 p. 229).

Proposition 7.1.10 (2017: 7.3) (V7.13) `Chapter7/Basic · EpGroupoid.proposition_7_3`. *Every x has an open U with $\pi(\mathrm{cl}_X A) = \mathrm{cl}_{|X|}\pi(A)$ for every $A \subseteq U$.* The text proves "$\supseteq$" with a sequence $\varphi_k : a_k \to b_k$, $b_k \to b$, $a_k \in A$, of which properness extracts a convergent subsequence. The formalization proves more, for every U with t proper over $s^{-1}(\mathrm{cl}_X U)$ and every $A \subseteq \mathrm{cl}_X U$, and without sequences `Chapter7/Basic · EpGroupoid.proposition_7_3_of_isProperMap`. *Step 1.* Since π is open and continuous, $\pi^{-1}(\mathrm{cl}\,\pi(A)) = \mathrm{cl}\,\pi^{-1}(\pi(A)) = \mathrm{cl}\,t(s^{-1}(A))$. *Step 2.* $t(s^{-1}(\mathrm{cl}A))$ is closed, as the image of the closed $s^{-1}(\mathrm{cl}A) \subseteq s^{-1}(\mathrm{cl}U)$ under a proper, hence closed, map (§VII.2), and contains $t(s^{-1}(A))$; so $\mathrm{cl}\,\pi(A) \subseteq \pi(t(s^{-1}(\mathrm{cl}A))) = \pi(\mathrm{cl}A)$. *Step 3.* "$\subseteq$" is the continuity of π. "For every z and every representative x" is "for every x" (Findings, Chapter 7, item 11) (pp. 291–293 (2017 pp. 229–230)).

VII.6 Lemma 7.1.11 (2017: 7.1), Proposition 7.1.12 (2017: 7.4), and the conjugation $G_x \cong G_y$

Lemma 7.1.11 (2017: 7.1) (V7.14) `Chapter7/Basic · EpGroupoid.lemma_7_1_1`. *If $t : s^{-1}(\mathrm{cl}_X U(x_0)) \to X$ is proper, only finitely many ψ have $s(\psi) = x$ and $t(\psi) \in U(x_0)$.* **The text's proof** argues by contradiction with distinct $\psi_n : s(\psi_n) \to x$, $s(\psi_n) \in U(x_0)$; properness gives a convergent subsequence, t is a local sc-diffeomorphism near the limit, and $t(\psi_n) = t(\psi_m) = x$ forces $\psi_n = \psi_m$. The proof thus treats the inverses of the morphisms of the statement, with s and t exchanged, and writes "$t : V(\psi) \to V(y)$" for $V(x)$ and "$s(\psi) \in s^{-1}(\mathrm{cl}_X(U(x_0)))$"; parentheses are unbalanced in the statement and the proof (possible misprints; Findings, Chapter 7, items 12, 96). **The formalized proof** does the proof's form first and has no sequences. *Step 1.* The fibre of the proper restriction of t over x is compact. *Step 2.* It is discrete, t being locally injective as a local sc-diffeomorphism `Chapter7/Discrete · MPolyfold.IsLocalScDiffeo.isLocallyInjective` `Chapter7/Discrete · IsLocallyInjective.isDiscrete_preimage_singleton`. *Step 3.* A compact discrete set is finite `Chapter7/Discrete · IsProperMap.finite_preimage_singleton_of_isLocallyInjective` `Chapter7/Discrete · finite_setOf_mem_of_isProperMap_domRestrict` `Chapter7/Discrete · MPolyfold.finite_setOf_mem_of_isLocalScDiffeo`. This gives the finiteness of $\{\psi \mid s\psi \in \mathrm{cl}_X U(x_0),\ t\psi = x\}$ `Chapter7/Basic · EpGroupoid.lemma_7_1_1_swap_of_isProperMap` `Chapter7/Basic · EpGroupoid.lemma_7_1_1_swap`. *Step 4.* The statement's form follows by ι, which exchanges s and t and is injective, and holds even for targets in $\mathrm{cl}_X U(x_0)$ `Chapter7/Basic · EpGroupoid.lemma_7_1_1_of_isProperMap`. The openness of $U(x_0)$ and $x_0 \in U(x_0)$ are kept as hypotheses but not used (2017 p. 230).

Proposition 7.1.12 (2017: 7.4) (V7.15) `Chapter7/Basic · EpGroupoid.proposition_7_4`. *G_x is finite.* Lemma 7.1.11 (2017: 7.1) with $x_0 = x$ and $U(x_0) = V(x)$ of Definition 7.1.3 (3) (2017: 7.3): G_x is contained in the finite set of morphisms with source x and target in $V(x)$ `Chapter7/Discrete · MPolyfold.finite_isotropy_of_isLocalScDiffeo`. The proposition is an instance, so the cardinality of G_x is its order as a finite group (Findings, Chapter 7, item 13) (2017 p. 230).

The conjugation (V7.16). For $\varphi : x \to y$, $g \mapsto \varphi \circ g \circ \varphi^{-1}$ is an isomorphism $G_x \to G_y$ `Chapter7/Groupoid · GroupoidStr.isotropyConj` `Chapter7/Groupoid · GroupoidStr.isotropyConj_val` `Chapter7/Basic · EpGroupoid.conjIsotropy` `Chapter7/Basic · EpGroupoid.conjIsotropy_val`, so isomorphic objects have isomorphic isotropy groups of the same order `Chapter7/Basic · EpGroupoid.nonempty_isotropy_mulEquiv` `Chapter7/Basic · EpGroupoid.nat_card_isotropy_eq`. "There is no canonical identification" is given its content: two choices φ, φ' give isomorphisms that differ by conjugation with $\varphi'\varphi^{-1} \in G_y$, an inner automorphism `Chapter7/Basic · EpGroupoid.isotropyConj_eq_isotropyConj_conj` (Findings, Chapter 7, item 13) (2017 p. 230).

VII.7 Theorem 7.1.13 (2017: 7.1), and local path connectedness

Theorem 7.1.13 (2017: 7.1) (V7.17) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1_1`. *For every object x and every open $V \ni x$ there are an open neighbourhood $U \subseteq V$ of x, a homomorphism $\Phi : G_x \to \mathrm{Diff}_{sc}(U)$ and an sc-smooth $\Gamma : G_x \times U \to \mathbf{X}$ with (1) $\Gamma(g, x) = g$, (2) $s(\Gamma(g, y)) = y$ and $t(\Gamma(g, y)) = \Phi(g)(y)$, (3) every morphism $h : y \to z$ between points of U is $\Gamma(g, y)$ for a unique g.* The data form a *natural representation* at x `Chapter7/EpGroupoid · EpGroupoid.NaturalRep`: U open, connected and containing x; $\Phi(g)$ an sc-diffeomorphism of U with inverse $\Phi(g^{-1})$ and $\Phi(gh) = \Phi(g) \circ \Phi(h)$ on U; Γ sc-smooth on each slice $\{g\} \times U$, which is sc-smoothness on $G_x \times U$ (§VII.8), with (1)–(3). The first consequences: $\Phi(g)$ maps U into U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.mapsTo_Φ`, $\Phi(1)$ is the identity on U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Φ_one`, every $\Phi(g)$ fixes x `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Φ_self`, $\Gamma(g, y)$ is a morphism between points of U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.Γ_mem_morOn`, $\Phi(g^{-1})\Phi(g) = 1$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_inv_Φ`, and (3) is a bijection $G_x \times U \to \mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.bijOn_Γ`. The action $g * y = \Phi(g)(y)$ is a group action on U `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.smul_val`. The proof is the text's Theorem 7.6.1 (2017: 7.5) in the appendix (§VII.26). *Hidden hypothesis: U* connected. The text chooses "the connected open neighborhood $U(x)$" on p. 293 (2017 p. 231) but does not put it into the statement; as we read it, the homomorphism property of Φ, the functoriality of Γ and the uniqueness of (Φ, Γ) all need it (Findings, Chapter 7, item

20) (2017 p. 231).

Local path connectedness (V7.18). Connected neighbourhoods exist because M-polyfolds are locally path connected, which the text uses without statement (2017 p. 231, Lemma 7.6.3 (2017: 7.9), Corollary 7.3.8 (2017: 7.2)). *Step 1, a retract.* For $o \in O = r(U)$, U open in a partial quadrant C, and a neighbourhood W of o, there is $\varepsilon > 0$ such that every $p \in O$ with $\|p - o\| < \varepsilon$ is joined to o inside $O \cap W$ by the path $\tau \mapsto r(o + \tau(p - o))$: the segment lies in the convex set $B_\varepsilon(o) \cap C \subseteq U$, r is continuous on level 0 and fixes o and p `Chapter7/LocConnected · ScRetraction.exists_ball_joinedIn`. So an sc-smooth retract is locally path connected `Chapter7/LocConnected · IsScRetract.locallyPathConnectedSpace`. *Step 2, an M-polyfold.* Through any chart around a point, path components of open sets are neighbourhoods `Chapter7/LocConnected · MChart.pathComponentIn_mem_nhds`, and X is locally path connected `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace`, hence locally connected `Chapter7/LocConnected · MPolyfold.locallyConnectedSpace`. *Step 3, consequences.* Open connected neighbourhoods form a basis `Chapter7/LocConnected · MPolyfold.hasBasis_isOpen_isConnected` `Chapter7/LocConnected · MPolyfold.hasBasis_isOpen_isPathConnected` `Chapter7/LocConnected · MPolyfold.exists_isOpen_isConnected_subset`; components, also of open sets, are open `Chapter7/LocConnected · MPolyfold.isOpen_connectedComponent` `Chapter7/LocConnected · MPolyfold.isOpen_connectedComponentIn`; an open set is connected iff path connected `Chapter7/LocConnected · MPolyfold.isConnected_iff_isPathConnected`; every quotient of X, in particular $|X|$, is locally path connected `Chapter7/LocConnected · MPolyfold.locallyPathConnectedSpace_quotient`; and finitely many points of a Hausdorff locally connected space have pairwise disjoint connected open neighbourhoods inside a given open set `Chapter7/LocConnected · exists_pairwiseDisjoint_isOpen_isConnected`. *Step 4, the tool for connectedness.* Two continuous sections of a locally injective map into a Hausdorff space that agree at one point of a preconnected set agree on all of it `Chapter7/LocConnected · eqOn_sections_of_isPreconnected`, by Mathlib's statement for separated maps `Mathlib/Topology/SeparatedMap · eqOn_of_comp_eqOn`. For s this says that a local inverse of s on a connected U is determined by one value; it is what makes Φ a homomorphism (§VII.26) and Γ a functor (§VII.8). The paths are continuous on level 0; sc-smooth paths between smooth points are Chapter 6's (§VI.18). This is the first fact of gap G.56 of Part I, now formalized (Findings, Chapter 7, items 21, 114).

VII.8 Pages 231–233: the sheets $\mathbf{U}(g)$, $G_x \ltimes U(x)$, the functor Γ, and $\hat{\Phi}$

Everything on these pages is derived from one natural representation (U, Φ, Γ); the text's "using the proof of Theorem 7.1 one verifies" is not needed. No possible misprint was found on pp. 293–297 (2017 pp. 231–233) (Findings, Chapter 7, item 22).

The splitting $\mathbf{U}(x) = \bigsqcup_g \mathbf{U}(g)$ (V7.19). Put $\mathbf{U}(g) = \Gamma(g, U)$. *Step 1, open.* The image of a continuous section of a local homeomorphism over an open set is open `Chapter7/NaturalAction · isOpen_image_of_section`; $\Gamma(g, \cdot)$ is a section of s by (2) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_image_Γ`. *Step 2, connected, disjoint, covering.* $\mathbf{U}(g)$ is connected as a continuous image of U `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isConnected_image_Γ`; two sheets are disjoint by the uniqueness in (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_inj` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.pairwise_disjoint_image_Γ`; they cover $\mathbf{U}(x)$ by the existence in (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.iUnion_image_Γ`, which is therefore open `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_morOn`. *Step 3, components.* Open, connected, disjoint and covering, they are the connected components of $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.connectedComponentIn_morOn`, and there are $\#G_x$ of them `Chapter7/NaturalAction · EpGroupoid.NaturalRep.injective_image_Γ`. *Step 4.* $s, t : \mathbf{U}(g) \to U$ are sc-diffeomorphisms, with inverses $\Gamma(g, \cdot)$ and $\Gamma(g, \cdot) \circ \Phi(g)^{-1}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_src_image` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_tgt_image`. Every $\varphi \in \mathbf{U}(x)$ has a sheet index h `Chapter7/NaturalAction · EpGroupoid.NaturalRep.index` with $\varphi = \Gamma(h, s\varphi)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_index` and $t\varphi = \Phi(h)(s\varphi)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.tgt_eq_Φ_index`, locally constant on $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.eventually_index_eq`. Once U is connected, the text's "choose $U(x)$ so small that $\mathbf{U}(x)$ splits into $\#G_x$-many components" needs no shrinking. "$\mathbf{U}(x)$ fits into an arbitrarily small open neighbourhood of G_x" belongs to the construction: Lemma

7.6.2 (2017: 7.8) holds for arbitrary open $N_g \ni g$ (§VII.26) `Chapter7/NaturalRep · EpGroupoid.lemma_7_8` (2017 p. 231).

The M-polyfold $G \times U$ (V7.20). "A natural M-polyfold structure as the finite disjoint union of copies of U" is the product with a discrete M-polyfold. *Step 1.* A discrete space is an M-polyfold with the charts $\{g\} \to \mathbb{R}^0$ `Chapter7/Discrete · discreteChart` `Chapter7/Discrete · discretePolyfold` `Chapter7/Discrete · isChart_discreteChart`, pairwise compatible because their transitions are the identity of the one-point space or the empty map `Chapter7/Discrete · discreteChart_compatible`; every point is smooth `Chapter7/Discrete · discretePolyfold_smoothSet`, and a map into it is sc-smooth iff continuous `Chapter7/Discrete · isScSmoothMap_discrete_iff`. *Step 2.* In $G \times X$ the slices $\{g\} \times X$ are open `Chapter7/Discrete · isOpen_sheet`, $x \mapsto (g, x)$ is sc-smooth `Chapter7/Discrete · isScSmoothMap_sheet` and the projection of a slice is an sc-diffeomorphism `Chapter7/Discrete · isScDiffeoOn_snd_sheet`, with $d_{G\times X}(g, x) = d_X(x)$ `Chapter7/Discrete · degIndex_prod_discrete`. *Step 3.* Hence a map on $G \times U$ is sc-smooth iff it is on each slice `Chapter7/Discrete · isScSmoothOn_prod_discrete_iff` `Chapter7/Discrete · isScSmoothOn_univ_prod_discrete_iff` `Chapter7/Discrete · isScSmoothMap_prod_discrete_iff`, the text's sheet-by-sheet reading of "Γ sc-smooth on $G \times U$". For a natural representation the objects are the open-subset M-polyfold U `Chapter7/NaturalAction · EpGroupoid.NaturalRep.objPolyfold`, the morphisms $G_x \times U$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.morPolyfold`, and $(g, y) \mapsto g*y$ is sc-smooth `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScSmoothMap_smul` (Findings, Chapter 7, item 23) (2017 p. 232).

The translation groupoid $G_x \ltimes U(x)$ is an ep-groupoid (V7.21). The groupoid has objects U, morphisms $G\times U$, $s(g, y) = y$, $t(g, y) = g*y$, $(g, h*z)\circ(h, z) = (gh, z)$ `Chapter7/Groupoid · GroupoidStr.translation` `Chapter7/EpGroupoid · EpGroupoid.NaturalRep.translationGroupoid`. *(1).* s is the projection, a local sc-diffeomorphism by Step 2 above `Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isLocalScDiffeo_src`; t is the projection after the shear $(g, y) \mapsto (g, g * y)$, an sc-diffeomorphism with inverse $(g, y) \mapsto (g, g^{-1} * y)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.shearHomeomorph` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isLocalScDiffeo_tgt`. *(3), for every V.* t is the shear followed by the projection $G \times U \to U$, proper because G_x is finite `Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isProperMap`. *(4), sheet form.* For a local inverse σ of s, $\sigma(t\psi) \circ \psi = (k(\psi)\psi_1, \psi_2)$ with k locally constant, sc-smooth on each slice `Chapter7/NaturalAction · EpGroupoid.NaturalRep.translation_isScSmoothCompSheet`. So $G_x \ltimes U(x)$ is an ep-groupoid `Chapter7/NaturalAction · EpGroupoid.NaturalRep.translationEpGroupoid`, built from (Φ, Γ) alone (2017 p. 232).

The functor $\Gamma : G_x \ltimes U(x) \to X$ (V7.22). *Step 1, composition.* $y \mapsto \Gamma(g, h*y)\circ\Gamma(h, y)$ and $y \mapsto \Gamma(gh, y)$ are continuous sections of s over U — the first by (4) in the sheet form, for the sheet $\mathbf{U}(g)$ — with the same value $g \circ h$ at x by (1); on the connected U they agree (§VII.7) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_mul`. *Step 2.* In the same way $\Gamma(1, y) = 1_y$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_one`, and $\Gamma(g^{-1}, g*y) = \Gamma(g, y)^{-1}$ by the uniqueness of inverses `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_inv`. *Step 3.* So Γ, the inclusion on objects, is a functor `Chapter7/NaturalAction · EpGroupoid.NaturalRep.functor`, full and faithful by (3) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.full` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.faithful`, onto the full subcategory on U `Chapter7/Groupoid · GroupoidStr.fullSub`, and an ep-functor `Chapter7/NaturalAction · EpGroupoid.NaturalRep.epFunctor`. *Hidden hypothesis: U* connected, without which the sheet index of $\Gamma(g, h*y)\circ\Gamma(h, y)$ is only locally constant (2017 p. 232).

The action $\hat{\Phi}$ on $\mathbf{U}(x)$ (V7.23). The text defines $\hat{\Phi}(g)(\varphi) = \Gamma(ghg^{-1}, g*s(\varphi))$ for $\varphi \in \mathbf{U}(h)$. The formalization defines the total map $\hat{\Phi}(g)(\varphi) = \Gamma(g, t\varphi)\circ\varphi\circ\Gamma(g, s\varphi)^{-1}$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat` and recovers the text's formula from Γ_{mul} and Γ_{inv} `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_Γ`, together with $\Phi(g)\Phi(h)\Phi(g)^{-1} = \Phi(ghg^{-1})$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_conj`. Then $\hat{\Phi}(g)$ maps $\mathbf{U}(h)$ onto $\mathbf{U}(ghg^{-1})$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_mem_image` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.image_Φhat` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.index_Φhat`; the two diagrams are $s \circ \hat{\Phi}(g) = \Phi(g) \circ s$ and $t \circ \hat{\Phi}(g) = \Phi(g) \circ t$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.src_Φhat` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.tgt_Φhat`; $\hat{\Phi}(1) = 1$ and $\hat{\Phi}(gh) = \hat{\Phi}(g)\hat{\Phi}(h)$ on $\mathbf{U}(x)$ `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_one` `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φhat_mul`; and $\hat{\Phi}(g)$ is an sc-diffeomorphism of $\mathbf{U}(x)$ with inverse

$\hat{\Phi}(g^{-1})$, sc-smooth by (4) in the sheet form and the sc-smoothness of ι Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScDiffeoOn_Φhat (pp. 294–297 (2017 pp. 232–233)).

VII.9 Uniqueness and restriction of natural representations

(V7.24.) The text uses without statement that natural representations can be shrunk and compared: "adapting the corresponding restrictions" (Lemma 7.2.3 (2017: 7.3)), "$U(x)$ can be chosen so small" (T_x, 2017 p. 241). Let $R = (U, \Phi, \Gamma)$ and $R' = (U', \Phi', \Gamma')$ be natural representations at x and C the connected component of x in $U \cap U'$, open and connected (§VII.7) Chapter7/NaturalAction · EpGroupoid.NaturalRep.isOpen_connectedComponentIn_inter Chapter7/NaturalAction · EpGroupoid.NaturalRep.isConnected_connectedComponentIn_inter. *Step 1.* $\Gamma(g, \cdot)$ and $\Gamma'(g, \cdot)$ are continuous sections of s over C with the value g at x, so they agree on C Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_eqOn; applying t, $\Phi(g) = \Phi'(g)$ on C Chapter7/NaturalAction · EpGroupoid.NaturalRep.Φ_eqOn. *Step 2.* C is invariant: $\Phi(g)(C)$ is connected, contains x and lies in $U \cap U'$ Chapter7/NaturalAction · EpGroupoid.NaturalRep.mapsTo_Φ_connectedComponentIn. *Step 3.* So a natural representation restricts to every invariant connected open neighbourhood of x inside U Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrict Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.restrict Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_restrict, in particular to C, where the restrictions of R and R' coincide Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrictInter Chapter7/NaturalAction · EpGroupoid.NaturalRep.restrictInter_eq, and natural representations exist inside every open $V \ni x$ Chapter7/NaturalRep · EpGroupoid.exists_naturalRep_shrink. *Step 4.* If $\Phi(g) = 1$ for one natural representation, then $\Phi'(g) = 1$ near x for every one Chapter7/NaturalAction · EpGroupoid.NaturalRep.eventually_Φ_eq_of_actsTrivially. Uniqueness holds on the component C; $U \cap U'$ itself need not be connected (Findings, Chapter 7, item 24).

VII.10 Proposition 7.1.15 (2017: 7.5) and Definition 7.1.18 (2017: 7.8): the star construction

Proposition 7.1.15 (2017: 7.5) (V7.25) Chapter7/NaturalRepFinite · EpGroupoid.proposition_7_5. *Let Y be a finite set of mutually isomorphic objects and V a neighbourhood of Y. There is a functor f assigning to every $y \in Y$ a connected open $U(y) \subseteq V$, mutually disjoint, and to every $\psi : y \to y'$ between points of Y an sc-diffeomorphism $f(\psi) : U(y) \to U(y')$, such that the morphisms between points of $\bigcup U(y)$ split into components $\mathbf{U}(\psi)$ with (1) the natural action of G_y on $U(y)$, (2) $s : \mathbf{U}(\psi) \to U(y)$, $t : \mathbf{U}(\psi) \to U(y')$ sc-diffeomorphisms, (3) $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$.* The datum is a structure Chapter7/EpGroupoid · EpGroupoid.NaturalRepFinite (Definition 7.1.18 (2017: 7.8), V7.26), "decomposes into the connected components" being read as: the $\mathbf{U}(\psi)$ are open, connected, pairwise disjoint and cover; they are then the components Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.connectedComponentIn_morOn (pp. 295–305 (2017 pp. 233–236)).

The text's proof (pp. 297–305 (2017 pp. 234–236)) is an induction on $\#Y$. $\#Y = 1$ is Theorem 7.1.13 (2017: 7.1). For $Y = \{y_1, y_2\}$ it fixes $\gamma_1 : y_1 \to y_2$ with a sheet and $\hat{\gamma}_1 = t \circ (s|\cdot)^{-1}$, takes a G_{y_1}-invariant connected $U(y_1)$ with $\hat{\gamma}_1(U(y_1))$ inside a G_{y_2}-invariant $W(y_2)$, puts $U(y_2) = \hat{\gamma}_1(U(y_1))$ and shows it invariant; the other morphisms $y_1 \to y_2$ are $g \circ \gamma_1$, $g \in G_{y_2}$, with components $\Phi_g(\mathbf{U}(\gamma_1))$, $\Phi_g(\varphi) = \Gamma(g, t\varphi) \circ \varphi$, and the morphisms $y_2 \to y_1$ get the inverse components. For $Y = Y' \cup \{y_{k+1}\}$ it fixes $\delta_j : y_{k+1} \to y_j$, shrinks as in the case of two objects, and adjusts the components of the morphisms of Y' to $(s|\mathbf{U}'(\gamma))^{-1}(U(y_i))$, showing $t(\mathbf{U}(\gamma)) = U(y_j)$.

The formalized proof (V7.25) is a *star*: everything is transported from one base object, and nothing is shrunk afterwards. *Step 1, transport along a sheet.* Let $R = (U, \Phi, \Gamma)$ be a natural representation at x and $\mathbf{S}_1, \mathbf{S}_2$ sheets of morphisms out of x with local inverses $a_i = (s|\mathbf{S}_i)^{-1}$ and $U \subseteq s(\mathbf{S}_i)$, $\hat{\gamma}_i = t \circ a_i$. The images $\hat{\gamma}_i(U)$ are open and connected Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isOpen_transportSet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isConnected_transportSet Chapter7/NaturalRepFinite · EpGroupoid.

NatRepFinite.isScDiffeoOn_hat_transportSet, $\hat\gamma_i^{-1}$ is the inverse there Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.back, and the conjugates $\hat\gamma_2 \circ \hat\Phi(g) \circ \hat\gamma_1^{-1}$ are sc-diffeomorphisms compatible with composition Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap_conjMap Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.conjMap_one. The transported morphisms are $\Theta_g(z) = a_2(\Phi(g)u) \circ \Gamma(g,u) \circ a_1(u)^{-1}$, $u = \hat\gamma_1^{-1}(z)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport, with source z and target $\hat\gamma_2(\Phi(g)u)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.src_transport Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.tgt_transport, equal to $\gamma_2 \circ g \circ \gamma_1^{-1}$, $\gamma_i = a_i(x)$, at the point $\hat\gamma_1(x)$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport_hat_self, sc-smooth by (4) in the sheet form, used twice, and the sc-smoothness of ι Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScSmoothOn_transport. *Step 2, the bijection.* A morphism k from $\hat\gamma_1(U)$ to $\hat\gamma_2(U)$ pulls back to the morphism $a_2(\cdot)^{-1} \circ k \circ a_1(\cdot)$ between points of U, which is $\Gamma(g,u)$ for a unique g by Theorem 7.1.13 (3) (2017: 7.1); so $k = \Theta_g(sk)$ for a unique g Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.existsUnique_transport Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transport_injective. *Step 3, a natural representation at y.* For $\gamma : x \to y$, $h \mapsto \gamma^{-1}h\gamma$ is an isomorphism $G_y \to G_x$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportConj, and $U(y) = \hat\gamma(U)$, $\Phi_y(h) = \hat\gamma \circ \Phi(\gamma^{-1}h\gamma) \circ \hat\gamma^{-1}$, $\Gamma_y(h,\cdot) = \Theta_{\gamma^{-1}h\gamma}$ is a natural representation at y Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_U Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_Φ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.transportRep_Γ, after shrinking R (same Φ, Γ) into s of a sheet of γ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.exists_spoke. *Step 4, the star.* Fix one x isomorphic to all of Y, morphisms $\gamma_y : x \to y$ with sheets $\mathbf{K}_y$, and by Hausdorff separation of the finite Y pairwise disjoint open $W_y \ni y$; take $U_0 = \bigcap_y (s(\mathbf{K}_y) \cap \hat\gamma_y^{-1}(W_y \cap V))$ and a natural representation inside U_0 (Theorem 7.1.13 (2017: 7.1)) Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.exists_star. Then $U(y) = \hat\gamma_y(U(x)) \subseteq W_y \cap V$ are pairwise disjoint. *Step 5, the functor.* $\psi \mapsto g_\psi = \gamma_{y'}^{-1} \circ \psi \circ \gamma_y \in G_x$ is a bijection from the morphisms $y \to y'$ onto G_x compatible with composition, units and inverses Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_comp Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_unit Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_inv Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.g_isotropy; $f(\psi) = \hat\gamma_{y'} \circ \Phi(g_\psi) \circ \hat\gamma_y^{-1}$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F is an sc-diffeomorphism Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.isScDiffeoOn_F and a functor, since $\psi \mapsto g_\psi$ is one and Φ is a homomorphism Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F_comp Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.F_unit. *Step 6, the components.* $\mathbf{U}(\psi) = \Theta_{g_\psi}(U(y))$ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.θ Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.sheet Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.mem_sheet are pairwise disjoint Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.pairwiseDisjoint_sheet and cover the morphisms between points of $\bigcup U(y)$, by Step 2 Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.morOn_eq; s and t are sc-diffeomorphisms on them, as images of sections Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_src_image_of_section Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.isScDiffeoOn_tgt_image_of_section, and $t \circ (s|\mathbf{U}(\psi))^{-1} = f(\psi)$. *Step 7, (1).* The natural representation of G_y on $U(y)$ is the transported one of Step 3, whose Φ_y is f on G_y Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.Star.toNaturalRepFinite; by §VII.9 it agrees near y with every other one. The text instead chooses $U(y)$ invariant under a natural representation of G_y given beforehand. The text's case of two objects already has this form ($U(y_2) = \hat\gamma_1(U(y_1))$); the adjustment of the components of Y' in the inductive step has no counterpart. "A neighbourhood V of Y" is a neighbourhood of the set Chapter7/NaturalRepFinite · EpGroupoid.proposition_7_5_of_mem_nhdsSet, reduced to an open one; $Y = \emptyset$ is allowed Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.emptyNaturalRepFinite, and enlarging V keeps a datum Chapter7/NaturalRepFinite · EpGroupoid.NatRepFinite.mono. "Mutually isomorphic" is used only through one x isomorphic to all of Y. Among the possible misprints of the proof (Findings, Chapter 7, item 26): "Φ_g" for $g \in G_{y_1}$ where $t(\varphi) \in U(y_2)$ requires G_{y_2}; "$\mathbf{U}(\gamma_g) := \Psi_g(\mathbf{U}(\gamma_1))$" for Φ_g; the induction step "let $k \geq 3$", which must start at $k \geq 2$ to reach $\#Y = 3$; "$\varphi = \Gamma(g, s(\varphi)) \circ \psi$" for

$\Gamma(g, t(\psi)) \circ \psi$; "$y : \mathbf{U}(\delta_j) \to U(y_j)$" for t; and "$g \in G_{y_i}$" for G_{y_j} on 2017 p. 236 (Findings, Chapter 7, items 25, 26).

VII.11 Proposition 7.1.19 (2017: 7.6) and Definition 7.1.20 (2017: 7.9): $|U(x)| = G_x\backslash U(x)$, local uniformizers

Proposition 7.1.19 (2017: 7.6) (V7.27) `Chapter7/Uniformizer · EpGroupoid.proposition_7_6`. *For a natural representation at x, $\tau : G_x\backslash U(x) \to |U(x)| = \pi(U(x))$, $G_x * y \mapsto |y|$, is a homeomorphism.* $G_x\backslash U(x)$ is the orbit space of the action $g * y$ with the quotient topology. **The text's proof**: τ is well defined and onto; it is injective because a morphism $\varphi : y \to y'$ is $\Gamma(g, y)$ by Theorem 7.1.13 (3) (2017: 7.1), so $y' = g * y$; $|U(x)|$ is open by Proposition 7.1.8 (2017: 7.1); and $\tau \circ \sigma = \pi|U(x)$ for the open quotient map σ, so τ is continuous and open. **The formalized proof.** *Step 1.* $|y| = |y'|$ iff $y' = g * y$ for some g `Chapter7/Uniformizer · EpGroupoid.NaturalRep.π_eq_π_iff_exists_smul`. *Step 2.* τ as a map into $|X|$ `Chapter7/Uniformizer · EpGroupoid.NaturalRep.orbitQuotientMap` is continuous by the quotient topology and injective by Step 1. *Step 3, open.* The image of an open Q is $\pi(\sigma^{-1}(Q))$, and $\sigma^{-1}(Q)$ is open in the open $U(x)$, so Proposition 7.1.8 (2017: 7.1) applies; the openness of σ is not needed. So τ is an open embedding into $|X|$ `Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_orbitQuotientMap` with range $\pi(U(x))$ `Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_orbitQuotientMap Chapter7/Uniformizer · EpGroupoid.proposition_7_6_apply_mk`. *The second form.* The functor $G_x \ltimes U(x) \to X$ of §VII.8 is fully faithful `Chapter7/Uniformizer · EpGroupoid.proposition_7_6_fullyFaithful`, and the same argument for the orbit space of the translation groupoid, whose classes are the same orbits, gives an open embedding with range $\pi(U(x))$ `Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_orbitMap Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_functor_orbitMap Chapter7/Uniformizer · EpGroupoid.proposition_7_6_orbitMap`, with $|U(x)|_{G_x \ltimes U(x)} \cong G_x\backslash U(x)$ `Chapter7/Uniformizer · EpGroupoid.translationOrbitHomeomorph` (Findings, Chapter 7, item 27) (pp. 299–306 (2017 pp. 236–237)).

Definition 7.1.20 (2017: 7.9) (V7.28). A local uniformizer around x is a fully faithful functor $\Psi_x : G_x \ltimes U(x) \to X$, the inclusion on objects, inducing a homeomorphism of $|U(x)|_{G_x \ltimes U(x)}$ onto the open $\pi(U(x))$ `Chapter7/EpGroupoid · EpGroupoid.LocalUniformizer`. "Embedding" is not a separate field: the injectivity on objects and morphisms, which the text notes after the definition, follows from the inclusion on objects and faithfulness `Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.injective_obj Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.injective_mor`; for the uniformizers built from natural representations it is also topological, Ψ_x being an open embedding on objects and on morphisms `Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_obj Chapter7/Uniformizer · EpGroupoid.NaturalRep.isOpenEmbedding_functor_mor Chapter7/Uniformizer · EpGroupoid.NaturalRep.range_functor_mor`. "There exists a local uniformizer around every object": every natural representation is one `Chapter7/Uniformizer · EpGroupoid.NaturalRep.toLocalUniformizer`, and one exists inside every open $V \ni x$ `Chapter7/Uniformizer · EpGroupoid.exists_localUniformizer`; its orbit map is a homeomorphism onto the open $|U(x)|$ `Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.isOpen_image Chapter7/Uniformizer · EpGroupoid.LocalUniformizer.orbitHomeomorph`. "$\Psi_x \circ G_x \ltimes U(x) \to X$" is read with :, and "a a local uniformizer" has a doubled article (possible misprints; Findings, Chapter 7, item 28) (2017 p. 237).

VII.12 Definition 7.1.21 (2017: 7.10), the M-polyfold A, and Proposition 7.1.22 (2017: 7.7)

Definition 7.1.21 (2017: 7.10) (V7.29). An ep-subgroupoid is the full subcategory on a set A of objects which is a sub-M-polyfold (Definition 2.3.18 (2017: 2.12), §II.10) and saturated `Chapter7/EpGroupoid · EpGroupoid.IsEpSubgroupoid`. Witnesses: X itself `Chapter7/EpGroupoid · EpGroupoid.isEpSubgroupoid_univ`, every saturated open set `Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_of_isOpen`, the saturation of every open set, open by Proposition 7.1.8 (2017: 7.1) `Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_`

saturation, and every sub-M-polyfold of a trivial ep-groupoid Chapter7/SubGroupoid · EpGroupoid.isEpSubgroupoid_trivial (Findings, Chapter 7, items 40, 105) (2017 p. 237).

The facts (1)–(4) on A (V7.30). By Part I's Proposition 2.3.19 (2017: 2.6) (§II.10), A has an M-polyfold structure Chapter2/SubMPolyfold · MPolyfold.subPolyfold Chapter7/SubGroupoid · EpGroupoid.subObjPolyfold with (1) the inclusion sc-smooth and a homeomorphism onto its image Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_val_subObj and (2) $i^{-1} \circ r$ sc-smooth for every sc-smooth retraction r onto $A \cap V$ Chapter7/SubGroupoid · EpGroupoid.isScSmoothOn_retractInto_subObj. The tangent facts (3), (4) are not restated, since $T_a A$ and $T_a X$ carry no linear structure here (gap G.7); in a chart they are Part I's Chapter2/Metrizable · proposition_2_21_3, and nothing below uses them (Findings, Chapter 7, item 40) (2017 pp. 237–238).

A is a sub-M-polyfold (V7.31). The text takes, for $\varphi \in \mathbf{A}$ with $a = s(\varphi)$, a sheet $s : \mathbf{U}(\varphi) \to U(a)$ and a retraction r of $U(a)$ onto $U(a) \cap A$, and shows that $\mathbf{r} = (s|\mathbf{U}(\varphi))^{-1} \circ r \circ s$ retracts $\mathbf{U}(\varphi)$ onto $\mathbf{U}(\varphi) \cap \mathbf{A}$. *Step 1, saturation.* For saturated A, $t(\varphi) \in A$ iff $s(\varphi) \in A$ Chapter7/SubGroupoid · EpGroupoid.src_mem_of_tgt_mem_of_isSaturated, so $\mathbf{A} = s^{-1}(A) = t^{-1}(A)$ Chapter7/SubGroupoid · EpGroupoid.morOn_eq_preimage_src Chapter7/SubGroupoid · EpGroupoid.morOn_eq_preimage_tgt. *Step 2, a more general statement.* The preimage of a sub-M-polyfold under a local sc-diffeomorphism is one: the retraction is pulled back along a sheet, after shrinking $U(a)$ to $(V \cap V_0) \cap r^{-1}(V \cap V_0)$ so that r maps it into itself Chapter7/SubGroupoid · isSubMPolyfold_preimage_of_isLocalScDiffeo Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_preimage_src Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_preimage_tgt Chapter7/SubGroupoid · EpGroupoid.isSubMPolyfold_morOn. So $\mathbf{A}$ is an M-polyfold Chapter7/SubGroupoid · EpGroupoid.subMorPolyfold with facts (1), (2) Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_val_subMor Chapter7/SubGroupoid · EpGroupoid.isScSmoothOn_retractInto_subMor; its paracompactness, which the project's M-polyfolds need, is that of a subspace of a metrizable space. The text's "$s(\mathbf{r}(\psi)) = r(s(\psi)) \in A$ and $t(\mathbf{r}(\psi)) \in U(a)$" is not used: the second claim does not hold in general as we read it, $t(\mathbf{r}(\psi))$ lying near $t(\varphi)$ rather than near a, and $s(\mathbf{r}(\psi)) \in A$ with saturation gives $t(\mathbf{r}(\psi)) \in A$ (possible misprint or extraction artefact for $s(\mathbf{r}(\psi)) \in U(a)$; Findings, Chapter 7, item 41) (2017 p. 238).

Proposition 7.1.22 (2017: 7.7) (V7.32) Chapter7/SubGroupoid · EpGroupoid.proposition_7_7. *For a saturated sub-M-polyfold A, the full subcategory on A is an ep-groupoid; A and $\mathbf{A}$ are sub-M-polyfolds, and all structure maps are the restrictions of those of X.* The text checks the conditions of Definition 7.1.3 (2017: 7.3) in a few lines; the formalization supplies the unstated step that *a map into a sub-M-polyfold which is sc-smooth into the ambient M-polyfold is sc-smooth into the sub-M-polyfold*: near a point it is $(i^{-1} \circ r) \circ i \circ f$, by fact (2) Chapter7/SubGroupoid · isScSmoothOn_subPolyfold_of_val. *(1).* A sheet of s maps $\mathbf{U}(\varphi) \cap \mathbf{A}$ onto $U(a) \cap A$; with the unstated step, s and t restrict to local sc-diffeomorphisms Chapter7/SubGroupoid · EpGroupoid.isLocalScDiffeo_fullSub_src Chapter7/SubGroupoid · EpGroupoid.isLocalScDiffeo_fullSub_tgt. *(2).* u and ι are restrictions Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_fullSub_unit Chapter7/SubGroupoid · EpGroupoid.isScSmoothMap_fullSub_inv. *(3).* The text uses a $U(a)$ with the natural G_a-action and t proper over $s^{-1}(\mathrm{cl}_X U(a))$ and notes that $U(a) \cap A$ is invariant; the invariance is not used, and any $V(a)$ of Definition 7.1.3 (3) (2017: 7.3) works with $V(a) \cap A$: a sequence with sources in $\mathrm{cl}_A(V(a) \cap A) \subseteq \mathrm{cl}_X V(a)$ and convergent targets in A has a subsequence converging in $\mathbf{X}$ (§VII.2), whose source lies in A by saturation and in $\mathrm{cl}_A(V(a) \cap A)$ since this set is closed in A Chapter7/SubGroupoid · EpGroupoid.exists_isProperMap_fullSub. *(4), literally.* The inclusion $\mathbf{A}\,{}_s\!\times_t \mathbf{A} \to \mathbf{X}\,{}_s\!\times_t \mathbf{X}$ is sc-smooth between the fibred products of Proposition 2.5.20 (2017: 2.15), by the unstated step applied twice, and m of A is m of X composed with it Chapter7/SubGroupoid · EpGroupoid.fullSub_comp_val Chapter7/SubGroupoid · EpGroupoid.isScSmoothComp_fullSub. So the full subcategory is an ep-groupoid Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid whose structure maps are the restrictions Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid_src_val Chapter7/SubGroupoid · EpGroupoid.subEpGroupoid_comp_val, and the inclusion is a full and faithful ep-functor Chapter7/SubGroupoid · EpGroupoid.subEpFunctor Chapter7/SubGroupoid · EpGroupoid.subEpFunctor_full Chapter7/SubGroupoid · EpGroupoid.subEpFunctor_faithful, which the text does not state. *Hidden hypotheses:* none; the paracompactness of A and $\mathbf{A}$ is proved (Findings, Chapter 7, item 42) (pp. 306–308 (2017 pp. 238–239)).

VII.13 Definition 7.2.1 (2017: 7.11), Lemmas 7.2.2 and 7.2.3 (2017: 7.2 and 7.3)

Definition 7.2.1 (2017: 7.11) (V7.33). X is *effective* if for every x, every $g \in G_x \setminus \{1_x\}$ and every open neighbourhood $\mathbf{U}(g)$ there is $\psi \in \mathbf{U}(g)$ with $s(\psi) \neq t(\psi)$ `Chapter7/EpGroupoid · EpGroupoid.IsEffective`; x is *regular* if $G_x = \{1_x\}$ `Chapter7/Groupoid · GroupoidStr.IsRegular`, and X_{reg} is the set of regular objects `Chapter7/Groupoid · GroupoidStr.regularSet`. "If $s(\varphi)$ is regular, so is $t(\varphi)$": conjugation with φ is an isomorphism of isotropy groups (§VII.6) `Chapter7/Groupoid · GroupoidStr.isRegular_tgt_iff` `Chapter7/Effective · EpGroupoid.isRegular_tgt_of_isRegular_src`, so X_{reg} is saturated `Chapter7/Effective · EpGroupoid.isSaturated_regularSet`. The definition is satisfiable non-vacuously, by $\mathbb{Z}/2$ acting on $\mathbb{R}$ by reflection (§VII.30) `Chapter7/NonVacuity · Examples.isEffective_reflection_nonvacuous` (Findings, Chapter 7, items 50, 101) (2017 p. 239).

Lemma 7.2.2 (2017: 7.2) (V7.34) `Chapter7/Effective · EpGroupoid.lemma_7_2`. *X_{reg} is open.* For regular x and a natural representation at x, every morphism $\varphi : a \to b$ between points of $U(x)$ is $\Gamma(1_x, a)$ by Theorem 7.1.13 (3) (2017: 7.1), and $\Gamma(1_x, a) = 1_a$ (§VII.8); so $b = a$, $\varphi = 1_a$, and G_a is trivial. The formalization proves the stronger statement that $U(x) \subseteq X_{reg}$ for *every* natural representation at a regular x `Chapter7/Effective · EpGroupoid.NaturalRep.subset_regularSet`. *Hidden hypothesis:* $\Gamma(1, a) = 1_a$, which needs $U(x)$ connected. "Examples show that the regular set can be empty" is the trivial $\mathbb{Z}/3$-action (§VII.30) (Findings, Chapter 7, item 51) (p. 308 (2017 p. 239)).

Lemma 7.2.3 (2017: 7.3) (V7.35) `Chapter7/Effective · EpGroupoid.lemma_7_3`. *Equivalent: (1) X is effective; (2) every open $V(x)$ contains a natural representation with $\Phi : G_x \to \mathrm{Diff}_{sc}(U(x))$ injective.* Injectivity is read as "$\Phi(g) = \mathrm{id}$ on U only for $g = 1$" `Chapter7/Effective · EpGroupoid.NaturalRep.injective_Φ_iff`, the triviality of the kernel of $G_x \to \mathrm{Perm}(U)$ `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ` `Chapter7/Effective · EpGroupoid.NaturalRep.mem_kerΦ` `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ_eq_bot_iff` `Chapter7/Effective · EpGroupoid.NaturalRep.kerΦ_eq_bot_iff_injective`. **The text's proof.** (1) ⇒ (2) by contradiction: if $\Phi(h) = 1$ with $h \neq 1$, effectiveness gives $\varphi_k \to h$ with $s(\varphi_k) \neq t(\varphi_k)$; $\varphi_k = \Gamma(h_k, s\varphi_k)$ with $h_k \neq h$, after a subsequence $h_k = g$, and in the limit $h = \Gamma(g, x) = g$. (2) ⇒ (1): $\Phi(g) \neq 1$ gives a with $\Phi(g)(a) \neq a$, and $\Gamma(g, a)$ has $s \neq t$; "by taking smaller and smaller neighborhoods $U(x)$ and adapting the corresponding restrictions of Γ" one finds such morphisms converging to g. **The formalized proof.** *(1) ⇒ (2), for every natural representation, without sequences* `Chapter7/Effective · EpGroupoid.IsEffective.eq_one`. If $\Phi(h) = 1$ on U, the sheet $\Gamma(h, U)$ is an open neighbourhood of h on which $s = t$, since $t(\Gamma(h, y)) = \Phi(h)(y) = y$; effectiveness forces $h = 1$. So effectiveness is the injectivity of every natural representation `Chapter7/Effective · EpGroupoid.isEffective_iff_forall_naturalRep`, and natural representations exist inside every $V(x)$ (§VII.9). *(2) ⇒ (1).* Fix one natural representation $R_0 = (U_0, \Phi_0, \Gamma_0)$, $g \neq 1$ and an open $\mathbf{U}(g)$. *Step 1.* Apply (2) once, to $V = \{y \in U_0 \mid \Gamma_0(g, y) \in \mathbf{U}(g)\}$, open and containing x. *Step 2.* The natural representation R that (2) gives inside V has $\Phi(g) \neq 1$, so some $a \in U$ has $\Phi(g)(a) \neq a$. *Step 3.* R need not be a restriction of R_0, as we read "adapting the corresponding restrictions"; it agrees with R_0 on the component of x in $U \cap U_0 = U$, i.e. on U (§VII.9) `Chapter7/Effective · EpGroupoid.NaturalRep.Γ_eqOn_of_subset` `Chapter7/Effective · EpGroupoid.NaturalRep.Φ_eqOn_of_subset`. *Step 4.* So $\Gamma_0(g, a) = \Gamma(g, a) \in \mathbf{U}(g)$ has source a and target $\Phi(g)(a) \neq a$. One application of (2) suffices (Findings, Chapter 7, item 52) (pp. 307–309 (2017 pp. 239–240)).

VII.14 Proposition 7.2.4 (2017: 7.8)

Proposition 7.2.4 (2017: 7.8) (V7.36) `Chapter7/Effective · EpGroupoid.proposition_7_8`. *If X is effective, X_{reg} is open and dense.* **The text's proof**: in a $U(x_0)$ take x with $\#G_x$ minimal; if $\#G_x > 1$, Lemma 7.2.3 (2017: 7.3) gives $U(x) \subseteq U(x_0)$ with Φ injective and, for $g \neq 1$, a point a with $\Phi(g)(a) \neq a$; G_a is isomorphic to $H = \{h \in G_x \mid \Phi(h)(a) = a\}$, so $\#G_x \leq \#G_a = \#H \leq \#G_x$, $H = G_x$, a contradiction. **The formalized proof.** *Step 1.* For a natural representation at x and $a \in U$, $h \mapsto \Gamma(h, a)$ is an isomorphism from the stabilizer of a onto G_a, by Theorem 7.1.13 (3) (2017: 7.1) and the functori-

ality of Γ Chapter7/Effective · EpGroupoid.NaturalRep.stabilizerEquivIsotropy. *Step 2.* Given a nonempty open O, take $x \in O$ minimizing $\#G_x$ over O itself, so the text's $U(x_0)$ is not needed. *Step 3.* If $G_x \neq 1$, take a natural representation inside O (§VII.9), injective by Lemma 7.2.3 (2017: 7.3), $g \neq 1$ and $a \in U$ with $\Phi(g)(a) \neq a$; then $\#G_a = \#\mathrm{Stab}(a) < \#G_x$, contradicting minimality. So O meets X_{reg}; openness is Lemma 7.2.2 (2017: 7.2). Only $\#G_a < \#G_x$ is used. *The converse* holds too, and is not in the text Chapter7/Effective · EpGroupoid.isEffective_of_dense_regularSet Chapter7/Effective · EpGroupoid.isEffective_iff_dense_regularSet: if X_{reg} is dense and $g \neq 1$, a regular a near x with $\Gamma(g,a)$ in a given $\mathbf{U}(g)$ has $\Gamma(g,a) \neq 1_a = \Gamma(1,a)$, so $s \neq t$ there (Findings, Chapter 7, item 53) (pp. 308–311 (2017 pp. 240–241)).

VII.15 The subgroups T_x, and Definition 7.2.7 (2017: 7.12)

T_x (V7.37). $g \in G_x$ *acts trivially* if $\Phi(g) = 1$ for some natural representation Chapter7/EpGroupoid · EpGroupoid.ActsTrivially, and T_x is the set of such g Chapter7/EpGroupoid · EpGroupoid.trivialSet Chapter7/EpGroupoid · EpGroupoid.one_mem_trivialSet. The text: since G_x is finite, $U(x)$ can be chosen so small that the kernel of Φ is T_x, a normal subgroup as a kernel. Since sc-smooth maps are not analytic, $\Phi(g) = \mathrm{id}$ near x does not give $\Phi(g) = \mathrm{id}$ on a connected $U(x)$, so the kernel can depend on $U(x)$; the text's "small" is made precise in three statements. *Step 1, for every representation.* $g \in T_x$ iff $\Phi(g) = \mathrm{id}$ near x, for any given natural representation (§VII.9, Step 4) Chapter7/Effective · EpGroupoid.NaturalRep.actsTrivially_iff. *Step 2, intrinsically.* $g \in T_x$ iff $s = t$ on a neighbourhood of g in $\mathbf{X}$ Chapter7/Effective · EpGroupoid.mem_trivialSet_iff_eventually: one direction is the sheet $\Gamma(g,U)$; for the other, a continuous section of s through g along which $s = t$ near x gives $\Phi(g) = \mathrm{id}$ near x Chapter7/Effective · EpGroupoid.mem_trivialSet_of_section. *Step 3, small neighbourhoods.* For one R_0, the common fixed set in U_0 of the finitely many elements of T_x is a neighbourhood N of x, and $\ker\Phi = T_x$ for *every* natural representation inside N Chapter7/Effective · EpGroupoid.exists_mem_nhds_trivialSet_iff Chapter7/Effective · EpGroupoid.exists_naturalRep_trivialSet_iff Chapter7/Effective · EpGroupoid.exists_naturalRep_kerΦ_eq. *Step 4.* So T_x is a kernel, a normal subgroup Chapter7/Effective · EpGroupoid.trivialSubgroup Chapter7/Effective · EpGroupoid.mem_trivialSubgroup Chapter7/Effective · EpGroupoid.trivialSubgroup_normal, its own normal closure Chapter7/Effective · EpGroupoid.normalClosure_trivialSet, and by Step 2 Definition 7.2.1 (2017: 7.11) says exactly $T_x = \{1_x\}$ for all x Chapter7/Effective · EpGroupoid.isEffective_iff_trivialSubgroup_eq_bot. *At nearby points.* For $g \in \ker\Phi$, $\Gamma(g,y) \in T_y$ for $y \in U$, which the proof of Proposition 7.2.10 (2017: 7.9) uses Chapter7/Effective · EpGroupoid.NaturalRep.mem_trivialSet_of_val_eq_Γ. *Along a morphism, not stated in the text.* For $\varphi : x \to y$ and $g \in T_x$, $z \mapsto \theta(z) \circ \Gamma(g, s\theta(z)) \circ \theta(z)^{-1}$, $\theta = (t|\mathbf{U}(\varphi))^{-1}$, is a continuous section of s through $\varphi g \varphi^{-1}$ — by (4) in the sheet form, twice — along which $s = t$ near y Chapter7/Effective · EpGroupoid.exists_conj_section; by Step 2, $\varphi g \varphi^{-1} \in T_y$ Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet_iff, and $\varphi T_x \varphi^{-1} = T_y$ Chapter7/Effective · EpGroupoid.map_isotropyConj_trivialSubgroup. This is what the descent of the composition to $\mathbf{X}^R$ needs. "Diff$_{sc}(U(x)$" lacks a parenthesis (possible misprint or extraction artefact). Witnesses: $T_x = 1$ for the trivial ep-groupoid Chapter7/Effective · EpGroupoid.trivialSubgroup_trivial and the reflection, $T_x = G_x \cong \mathbb{Z}/3$ for the trivial $\mathbb{Z}/3$-action (§VII.30) (Findings, Chapter 7, item 54) (2017 p. 241).

Definition 7.2.7 (2017: 7.12) (V7.38). $\mathbf{X}^T = \bigcup_x T_x$ Chapter7/EpGroupoid · EpGroupoid.nonEffectivePart; $\varphi \sim \varphi'$ iff $s(\varphi) = s(\varphi') = x$ and $\varphi = \varphi' \circ g$ with $g \in T_x$ Chapter7/EpGroupoid · EpGroupoid.ReducedRel; $\mathbf{X}^R$ is the set of classes Chapter7/EpGroupoid · EpGroupoid.ReducedMor. *Step 1,* $\mathbf{X}^T$ *intrinsically.* By §VII.15 Step 2, $\mathbf{X}^T$ is the interior of $\{\varphi \mid s\varphi = t\varphi\}$ Chapter7/Reduced · EpGroupoid.mem_nonEffectivePart_iff Chapter7/Reduced · EpGroupoid.nonEffectivePart_eq_interior Chapter7/Reduced · EpGroupoid.val_mem_nonEffectivePart_iff, open Chapter7/Reduced · EpGroupoid.isOpen_nonEffectivePart and closed under units, inverses, composition and conjugation Chapter7/Reduced · EpGroupoid.unit_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.inv_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.comp_mem_nonEffectivePart Chapter7/Reduced · EpGroupoid.conj_mem_nonEffectivePart. So $\varphi \sim \varphi'$ iff $\varphi = \varphi' \circ k$ with $k \in \mathbf{X}^T$ Chapter7/Reduced · EpGroupoid.reducedRel_iff, a form without dependent types. *Step 2, an equivalence relation,* because T_x is a subgroup Chapter7/Reduced · EpGroupoid.reducedRel_refl Chapter7/Reduced · EpGroupoid.ReducedRel.symm Chapter7/Reduced · EpGroupoid.ReducedRel.trans Chapter7/Reduced · EpGroupoid.reducedSetoid Chapter7/Reduced · EpGroupoid.reducedMor_eq_quotient. *Step 3, the left*

form. By the conjugation $\varphi T_x \varphi^{-1} = T_y$, $\varphi \sim \psi$ iff $\varphi = k \circ \psi$ with $k \in T_{t\psi}$ `Chapter7/Reduced · EpGroupoid.reducedRel_comp_left` `Chapter7/Reduced · EpGroupoid.ReducedRel.exists_comp_left`; the relation respects s, t, ι and composition on both sides `Chapter7/Reduced · EpGroupoid.ReducedRel.src_eq` `Chapter7/Reduced · EpGroupoid.ReducedRel.tgt_eq` `Chapter7/Reduced · EpGroupoid.ReducedRel.inv` `Chapter7/Reduced · EpGroupoid.ReducedRel.comp_left` `Chapter7/Reduced · EpGroupoid.ReducedRel.comp_right`. "The subset $\mathbf{X}^R$ of reduced morphisms" is a quotient, as the next sentence of the text says (possible misprint; Findings, Chapter 7, item 55) (2017 p. 241).

VII.16 Proposition 7.2.10 (2017: 7.9): the M-polyfold $\mathbf{X}^R$

Proposition 7.2.10 (2017: 7.9) (V7.39) `Chapter7/Reduced · EpGroupoid.proposition_7_9`. *$\mathbf{X}^R$ has a natural M-polyfold structure.* **The text's proof**: $\mathbf{X}^R$ carries the quotient topology; on a sheet $s : U(\varphi_0) \to U(s\varphi_0)$ the projection π' is continuous and injective; its image $U([\varphi_0])$ is open, by the map $\delta(\psi') = \psi' \circ \Gamma(g, s\psi')$ near an equivalent ψ, with $U(s\psi)$ chosen with $\ker \Phi = T_{s\psi}$; the same argument shows $\Psi_{\varphi_0} : U(\varphi_0) \to U([\varphi_0])$ open; and the transitions are, near φ with $\psi = \varphi \circ g$, the sc-smooth δ. **The formalized proof.** *Step 1, the topology.* The projection $\pi^R : \mathbf{X} \to \mathbf{X}^R$ `Chapter7/Reduced · EpGroupoid.reducedπ` `Chapter7/Reduced · EpGroupoid.reducedπ_eq_iff` `Chapter7/Reduced · EpGroupoid.surjective_reducedπ` is a quotient map `Chapter7/Reduced · EpGroupoid.isQuotientMap_reducedπ` `Chapter7/Reduced · EpGroupoid.continuous_reducedπ`. *Step 2, π^R is open* `Chapter7/Reduced · EpGroupoid.isOpenMap_reducedπ`. The text's δ composes on the right with a section of s; its continuity holds by (4) applied to inverses. The left form is used instead: for $\varphi \sim \psi$, $\varphi = \Gamma(g, t\psi) \circ \psi$ near ψ with $g \in T_{t\psi}$ and a natural representation with $\ker \Phi = T_{t\psi}$ (§VII.15); $\delta(\psi') = \Gamma(g, t\psi') \circ \psi'$ is sc-smooth by (4) in the sheet form `Chapter7/NaturalAction · EpGroupoid.NaturalRep.isScSmoothOn_comp_Γ_tgt` and $\psi' \sim \delta(\psi')$ since $\Gamma(g, t\psi') \in T_{t\psi'}$. So the saturation of an open set is open. *Step 3, sheets.* On every sheet $s : \mathbf{W} \to V$, π^R is injective — equivalent morphisms have the same source — and hence an open embedding `Chapter7/Reduced · EpGroupoid.injOn_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ_image_Γ`; "small" sheets are not needed. *Step 4, charts.* For a chart ψ of $\mathbf{X}$ with domain in a sheet $\mathbf{W}$, $\sigma = (s|\mathbf{W})^{-1}$, the chart of $\mathbf{X}^R$ is $[\varphi'] \mapsto \psi(\sigma(s[\varphi']))$ on $\pi^R(\operatorname{dom}\psi)$ `Chapter7/Reduced · EpGroupoid.reducedChart`; every point lies in one `Chapter7/Reduced · EpGroupoid.exists_chart_sheet`. *Step 5, transitions.* On an overlap, an element of the second sheet equivalent to φ' has the source $s(\varphi')$, so the text's δ equals $\sigma_2 \circ s$ and the transition is $\psi_2 \circ \sigma_2 \circ s \circ \psi_1^{-1}$ `Chapter7/Reduced · EpGroupoid.reducedChart_transition`, sc-smooth; what needs the text's argument is that the overlap is open, i.e. Step 2 `Chapter7/Reduced · EpGroupoid.overlap_reducedChart` `Chapter7/Reduced · EpGroupoid.reducedChart_compatible`. The charts form an atlas `Chapter7/Reduced · EpGroupoid.reducedCharts` `Chapter7/Reduced · EpGroupoid.reducedAtlas`, all without hypotheses. *Step 6, Hausdorff and paracompact.* An M-polyfold is a Hausdorff paracompact space (Definition 2.3.4 (2017: 2.8)). The proof gives neither as we read it, and $\mathbf{X}^R$ need not be Hausdorff (the example at the end of this paragraph, formalized); so the M-polyfold $\mathbf{X}^R$ is built under the hypotheses that $\mathbf{X}^R$ is Hausdorff and paracompact `Chapter7/Reduced · EpGroupoid.reducedPolyfold` `Chapter7/Reduced · EpGroupoid.isChart_reducedChart`, and then π^R is a local sc-diffeomorphism, with inverse $\sigma \circ s$ on $\pi^R(\mathbf{W})$ `Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedπ` `Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedπ` `Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedπ` (gap G.71, closed by the 2021 text, §VII.41). *The hypotheses are satisfiable.* For effective X, $\sim$ is equality `Chapter7/Reduced · EpGroupoid.reducedRel_iff_eq_of_isEffective`, π^R a homeomorphism `Chapter7/Reduced · EpGroupoid.reducedHomeomorph`, and $\mathbf{X}^R$ is Hausdorff, paracompact and regular `Chapter7/Reduced · EpGroupoid.t2Space_reducedMor_of_isEffective` `Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_of_isEffective` `Chapter7/Reduced · EpGroupoid.t3Space_reducedMor_of_isEffective`; this covers the trivial ep-groupoid of every M-polyfold `Chapter7/Reduced · EpGroupoid.t2Space_reducedMor_trivial` `Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_trivial`. Paracompactness also follows from regularity of $\mathbf{X}^R$ when $\mathbf{X}$ is second countable `Chapter7/Reduced · EpGroupoid.paracompactSpace_reducedMor_of_t3`; Hausdorffness alone with $\mathbf{X}$ second countable is not shown to suffice, a second countable, locally metrizable Hausdorff space need not be regular. Four possible misprints of the proof: "$\psi' = (s|U(\psi))^{-1}(s(\psi))$" for $s(\psi')$, "$U(\varphi) \subset \Psi_{\varphi_0}(U([\varphi_0]) \cap U([\psi_0]))$" for $\Psi_{\varphi_0}^{-1}$, "T_x agrees with the kernel of the $G_{s(\varphi)}$-action" for $T_{s(\varphi)}$, and a garbled sentence on the smoothness of the source and target maps (Find-

ings, Chapter 7, items 56, 57, 60, 111) (2017 pp. 242–243). *The Hausdorff hypothesis cannot be dropped* Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor. Let $\mathbb{Z}/2$ act by $(t,z)\mapsto(t,\pm z)$ on the retract O of Part I's Remark 4.2.2 (2017: 4.3) example, a point over each $t\le 0$ and a line over each $t>0$; the action is the restriction of an sc-operator Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.isScSmoothMap_smul. Over $t<0$ both elements act trivially, so $(-1,(t,0))\in\mathbf{X}^T$ Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.mem_nonEffectivePart; at $(0,0)$ points of O with $t>0$ tend to 0 and are moved by -1, so $(-1,(0,0))\notin\mathbf{X}^T$ Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_mem_nonEffectivePart. The classes of $(\pm1,(-1/(n+1),0))$ coincide and converge to both $[(-1,0)]\neq[(1,0)]$ Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor (Findings, Chapter 7, item 107; found on paper while reading, proved in the round-trip audit).

VII.17 The reduced ep-groupoid X_R, Proposition 7.10 (2017), Definition 7.2.15 (2017: 7.13)

X_R (V7.40). Objects X, morphisms $\mathbf{X}^R$, $s([\varphi])=s(\varphi), t([\varphi])=t(\varphi)$, with unit, inverse and composition descended by §VII.15 Chapter7/Reduced · EpGroupoid.reducedSrc Chapter7/Reduced · EpGroupoid.reducedTgt Chapter7/Reduced · EpGroupoid.reducedUnit Chapter7/Reduced · EpGroupoid.reducedInv Chapter7/Reduced · EpGroupoid.reducedComp Chapter7/Reduced · EpGroupoid.reducedComp_π, s and t onto Chapter7/Reduced · EpGroupoid.surjective_reducedSrc Chapter7/Reduced · EpGroupoid.surjective_reducedTgt, a groupoid Chapter7/Reduced · EpGroupoid.reducedGroupoidStr; the composition of a non-composable pair is a junk value, as in §VII.1 (2017 p. 243).

Proposition 7.10 (2017) (V7.40) Chapter7/Reduced · EpGroupoid.proposition_7_10. *Under the hypotheses of §VII.16, X_R is an ep-groupoid, and it is effective. (1).* For a sheet $\mathbf{W}$ of Definition 7.1.4 (2017: 7.4), $s,t:\pi^R(\mathbf{W})\to X$ are $s\circ(\pi^R|\mathbf{W})^{-1}$ and $t\circ(\pi^R|\mathbf{W})^{-1}$, the text's $s(\Psi_\varphi(\varphi'))=s(\varphi')$ Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedSrc Chapter7/Reduced · EpGroupoid.isScDiffeoOn_reducedTgt Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedSrc Chapter7/Reduced · EpGroupoid.isLocalScDiffeo_reducedTgt. *(2).* $u=\pi^R\circ u_X$ and $\iota=\pi^R\circ\iota_X\circ(\pi^R|\mathbf{W})^{-1}$ on $\pi^R(\mathbf{W})$ Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedUnit Chapter7/Reduced · EpGroupoid.isScSmoothMap_reducedInv. *(3), "proved similarly".* For the same $V(x)$, $t\circ\pi^R=t_X$ on $s^{-1}(\mathrm{cl}V(x))$, with π^R a continuous surjection onto the corresponding set of $\mathbf{X}^R$; a map whose composite with a continuous surjection is proper is proper (Mathlib) Chapter7/Reduced · EpGroupoid.exists_isProperMap_reduced. *(4), "reduced by our local charts",* in the sheet form: for an sc-diffeomorphism $s:\mathbf{W}'\to V'$ of $\mathbf{X}^R$ with inverse σ', $\kappa=\sigma_2\circ s\circ\sigma'$ is the inverse of s on an open set of $\mathbf{X}$ — a step the text leaves implicit — and $m(\sigma'(t\psi),\psi)=\pi^R(m(\kappa(t\psi),\psi))$ Chapter7/Reduced · EpGroupoid.isScSmoothCompSheet_reduced. So X_R is an ep-groupoid Chapter7/Reduced · EpGroupoid.reducedEpGroupoid and $X\to X_R$ an ep-functor Chapter7/Reduced · EpGroupoid.reductionFunctor. *Effective.* The text's point a with $\Phi(g)(a)\neq a$ is replaced by the intrinsic form of T_x: if $s=t$ near $[g]$ in $\mathbf{X}^R$, then $s=t$ near g in $\mathbf{X}$, π^R being continuous, so $g\in T_x$ and $[g]=[1_x]$ Chapter7/Reduced · EpGroupoid.isEffective_reducedEpGroupoid. For effective X, $X_R\cong X$ in $\mathcal{EP}$ Chapter7/Reduced · EpGroupoid.reductionInv Chapter7/Reduced · EpGroupoid.reductionIso (Findings, Chapter 7, items 58, 60) (2017 p. 244).

The isotropy of X_R, and Definition 7.2.15 (2017: 7.13) (V7.41). $g\mapsto[g]$ is a surjective homomorphism from G_x onto the isotropy of x in X_R Chapter7/Reduced · EpGroupoid.reducedIsotropyHom Chapter7/Reduced · EpGroupoid.surjective_reducedIsotropyHom with kernel T_x — the text's "$[1_x]=T_x$" Chapter7/Reduced · EpGroupoid.ker_reducedIsotropyHom — so that isotropy is G_x/T_x Chapter7/Reduced · EpGroupoid.quotientTrivialSubgroupEquiv. The effective isotropy group $G_x^{\mathrm{eff}}=G_x/T_x$ Chapter7/EpGroupoid · EpGroupoid.EffIsotropy is defined as the quotient by the normal closure of T_x, which is G_x/T_x by §VII.15 Chapter7/Effective · EpGroupoid.effIsotropyEquiv, and it is the isotropy group of X_R Chapter7/Reduced · EpGroupoid.effIsotropyEquivReduced. "For and object" (possible misprint; Findings, Chapter 7, item 59) (2017 p. 244).

VII.18 Theorem 7.3.1 (2017: 7.2), Lemmas 7.3.2 and 7.3.3 (2017: 7.4 and 7.5)

Theorem 7.3.1 (2017: 7.2) (V7.42) `Chapter7/OrbitTopology · EpGroupoid.theorem_7_2`. *$|X|$ is locally metrizable, regular and Hausdorff; if $|X|$ is paracompact, it is metrizable.* "Locally metrizable" is read as: every point has an open neighbourhood which is metrizable in the subspace topology. The three properties are Lemmas 7.3.2, 7.3.5 and 7.3.4 (2017: 7.4, 7.7 and 7.6). The metrizability `Chapter7/OrbitTopology · EpGroupoid.theorem_7_2_metrizable` is the text's "Nagata/Smirnov" in Part I's form (§II.7): a paracompact Hausdorff locally metrizable space is metrizable `Chapter2/Metrizable · metrizableSpace_of_locally`. The hypothesis "$|X|$ paracompact" is satisfiable, by the trivial ep-groupoid `Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial` (Findings, Chapter 7, item 70) (2017 p. 244).

Lemma 7.3.2 (2017: 7.4) (V7.43) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_4`. *Every $z \in |X|$ has an open metrizable neighbourhood; one can take $|U(x)|$ for a natural representation at x.* **The text's proof**: U is metrizable with a metric d; $\rho(y,y') = \max_g d(g*y, g*y')$ is an equivalent G_x-invariant metric; $\hat\rho([y],[y']) = \min\{\rho(g*y, h*y') \mid g,h \in G_x\}$ defines the quotient topology on $G_x\backslash U$; and Proposition 7.1.19 (2017: 7.6) carries it to $|U|$. **The formalized proof** follows it, for any finite group acting by homeomorphisms on a metrizable space `Chapter7/OrbitTopology · metrizableSpace_orbitRel_quotient`. *Step 1.* ρ is the maximum over the finite G_x `Chapter7/OrbitTopology · maxDist`, invariant `Chapter7/OrbitTopology · maxDist_smul`, and $d \le \rho$. *Step 2.* $\hat\rho$ is computed as $\min_h \rho(y, h*y')$, the same number by invariance `Chapter7/OrbitTopology · quotDist` `Chapter7/OrbitTopology · quotDist_mk`; it is symmetric and satisfies the triangle inequality by invariance, and $\hat\rho([y],[y']) = 0$ forces $y' \in G_x * y$, orbits of a finite group being closed. *Step 3, the topology.* A set of $G_x\backslash U$ is open in the quotient topology iff it contains a $\hat\rho$-ball around each of its points: $\{y' \mid \rho(y,y') < \varepsilon\} = \bigcap_g g^{-1}(B_d(g*y,\varepsilon))$ is open, and $d \le \rho$ gives the converse. This is Mathlib's construction of a metric space with a prescribed topology; ρ itself is never installed as a metric on U. So $G_x\backslash U(x)$ is metrizable `Chapter7/OrbitTopology · EpGroupoid.metrizableSpace_orbitRel_quotient_naturalRep` and so is $|U(x)|$, its homeomorphic image (§VII.11) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_4_naturalRep` (Findings, Chapter 7, item 71) (pp. 319–325 (2017 pp. 244–245)).

Lemma 7.3.3 (2017: 7.5) (V7.44) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_5`. *Every point of $|X|$ has a monotone countable basis of open neighbourhoods.* This is Proposition 7.1.9 (2017: 7.2) (§VII.5), proved there from the metrizability of X and the openness of π; the text derives it from Lemma 7.3.2 (2017: 7.4) (Findings, Chapter 7, item 72) (2017 p. 245).

VII.19 Lemmas 7.3.4 and 7.3.5 (2017: 7.6 and 7.7), Corollary 7.3.6 (2017: 7.1), subsets of $|X|$

The text proves Lemma 7.3.4 (2017: 7.6) before Lemma 7.3.5 (2017: 7.7); the formalization proves Lemma 7.3.5 (2017: 7.7) first and derives Lemma 7.3.4 (2017: 7.6) from it (Findings, Chapter 7, item 72).

Lemma 7.3.5 (2017: 7.7) (V7.46) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_7`. *$|X|$ is regular.* **The text's proof**: for closed $A \not\ni z = \pi(x)$ choose $O \ni x$ with $\pi(O) \cap A = \emptyset$ and t proper over $s^{-1}(\mathrm{cl}O)$, and $x \in O' \subseteq \mathrm{cl}O' \subseteq O$; every $y \in \pi^{-1}(A)$ has a $V(y)$ which no morphism from $\mathrm{cl}O'$ reaches, else properness gives φ with $s\varphi \in \mathrm{cl}O'$ and $t\varphi = y$; then $\pi(\bigcup V(y)) \supseteq A$ and $\pi(O')$ are disjoint. **The formalized proof** uses a closed neighbourhood basis `Mathlib/Topology/Separation/Regular · RegularSpace.of_exists_mem_nhds_isClosed_subset`. *Step 1.* For a neighbourhood N of $\pi(x)$, take a closed neighbourhood O' of x inside $\pi^{-1}(N) \cap V(x)$, X being regular and $V(x)$ from Definition 7.1.3 (3) (2017: 7.3). *Step 2.* $\pi(O')$ is closed by Proposition 7.1.10 (2017: 7.3), $O' \subseteq \mathrm{cl}_X V(x)$ `Chapter7/Basic · EpGroupoid.proposition_7_3_of_isProperMap`, a neighbourhood of $\pi(x)$ by Proposition 7.1.8 (2017: 7.1), and inside N. The same properness replaces the text's sets $V(y)$ (2017 p. 246).

Lemma 7.3.4 (2017: 7.6) (V7.45) `Chapter7/OrbitTopology · EpGroupoid.lemma_7_6`. *$|X|$ is Hausdorff.* **The text's proof** is by contradiction: $x_k \to x, x'_k \to x'$ with morphisms $\varphi_k : x_k \to x'_k$, of which properness extracts

a subsequence converging to a morphism $x \to x'$. **The formalized proof** is two lines. *Step 1, points are closed* `Chapter7/OrbitTopology · EpGroupoid.t1Space_orbitSpace`: the point $\{x\}$ is compact, hence a proper subset `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_of_isCompact`, and $\pi(\{x\})$ is closed by Proposition 7.5.10 (1) (2017: 7.14) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_14_one` (§VII.25), which uses the same properness as the text's sequence. *Step 2.* A regular T_1 space is Hausdorff, with Lemma 7.3.5 (2017: 7.7) (pp. 324–326 (2017 pp. 245–246)).

Corollary 7.3.6 (2017: 7.1) (V7.47) `Chapter7/OrbitTopology · EpGroupoid.corollary_7_1`. *A second countable $|X|$ is metrizable*: Urysohn's metrization theorem, an instance of Mathlib, $|X|$ being regular and Hausdorff by Lemmas 7.3.4, 7.3.5 (2017: 7.6, 7.7) (Findings, Chapter 7, item 73) (2017 p. 247).

Subsets of $|X|$ (V7.48). "The induced topology on a subset of $|X|$ is again locally metrizable, regular and Hausdorff": local metrizability passes to subspaces `Chapter7/OrbitTopology · EpGroupoid.exists_metrizable_nhds_subtype`, regularity and Hausdorffness are Mathlib's subspace instances, and so a paracompact subset of $|X|$ is metrizable, by Smirnov's theorem in the subspace `Chapter7/OrbitTopology · EpGroupoid.metrizableSpace_of_paracompactSpace_subtype` (Findings, Chapter 7, item 73) (2017 p. 247).

VII.20 Michael's theorem: Part I's Lemma 2.7.3 (2017: 2.7) and Proposition 2.7.4 (2017: 2.17)

(V7.50.) Corollary 7.3.8 (2017: 7.2) needs that a finite union of closed paracompact subsets of the regular $|X|$ is paracompact, and the text's route to Proposition 7.5.3 (2017: 7.12) uses Lemma 2.7.3 (2017: 2.7). Part I's Lemma 2.7.3 (2017: 2.7) (Michael: a regular Hausdorff space is paracompact iff every open cover has a locally finite closed refinement) and Proposition 2.7.4 (2017: 2.17) (a regular Hausdorff space that is the union of a locally finite family of closed paracompact subspaces is paracompact), 2017 p. 51, were gap G.15 of Part I; they are formalized here, for regular spaces and without Hausdorff. Michael's Lemma 1 has four conditions on a regular space: (a) paracompact; (b) every open cover has a σ-locally finite open refinement; (c) every open cover has a locally finite refinement by arbitrary sets; (d) every open cover has a locally finite closed refinement `Chapter7/Michael · michael_tfae`. *(d) □ (a), in every topological space* `Chapter7/Michael · paracompactSpace_of_closed_refinement`. Let (A_b) be a closed locally finite refinement of the cover (U_i), with $A_b \subseteq U_{f(b)}$; cover the space by open sets each meeting finitely many A_b and take a closed locally finite refinement (B_c) of that cover; then $V_b = (\bigcup\{B_c \mid B_c \cap A_b = \emptyset\})^c \cap U_{f(b)}$ is open (a locally finite union of closed sets is closed), contains A_b, and each B_c meets only the finitely many V_b with $B_c \cap A_b \neq \emptyset$; so (V_b) is a locally finite open refinement. *(c) □ (d) in a regular space* `Chapter7/Michael · closed_refinement_of_refinement`: refine first a cover by open sets whose closures lie in the cover, and take closures. *(b) □ (c), in every space* `Chapter7/Michael · refinement_of_sigma_refinement`: $t_{n,b} \setminus \bigcup_{k<n} \bigcup_{b'} t_{k,b'}$. Hence (c) □ (a) and (b) □ (a) in a regular space `Chapter7/Michael · paracompactSpace_of_refinement` `Chapter7/Michael · paracompactSpace_of_sigma_refinement`, and (a) □ (d) by Mathlib's shrinking lemma, with the same index and $v_i \subseteq u_i$ `Chapter7/Michael · exists_closed_precise_refinement` `Chapter7/Michael · exists_closed_precise_refinement_set`. **Lemma 2.7.3 (2017: 2.7)** is the equivalence of (a) and (d) `Chapter7/Michael · lemma_2_56`. **Proposition 2.7.4 (2017: 2.17)** `Chapter7/Michael · proposition_2_57` follows the text: closed locally finite refinements $(C^i_j)_j$ of the traces $(Y_i \cap U_j)_j$ by Lemma 2.7.3 (2017: 2.7) in each Y_i, which is regular because Y is; $C_j = \bigcup_i C^i_j$ is locally finite in Y because (Y_i) is and a locally finite family in a closed subspace is locally finite in the space `Chapter7/Michael · locallyFinite_image_val`; and (d) □ (a) in Y. The subspace forms are what Chapter 7 uses: a locally finite family of closed paracompact subsets has a paracompact union `Chapter7/Michael · paracompactSpace_iUnion`, so do finite families and pairs `Chapter7/Michael · paracompactSpace_iUnion_of_finite` `Chapter7/Michael · paracompactSpace_biUnion_of_finite` `Chapter7/Michael · paracompactSpace_union`, and a closed subset of a paracompact subspace is paracompact `Chapter7/Michael · paracompactSpace_of_isClosed_subset`. *The Hausdorff property is never used*; regularity enters through (c) □ (d) and (a) □ (d). Part I's 2017 p. 51 says "subcovering" in the definition of paracompactness, read as "refinement", as on 2017 p. 249 (Findings, Chapter 7, items 75, 114).

VII.21 Definition 7.3.7 (2017: 7.14) and Corollary 7.3.8 (2017: 7.2)

Definition 7.3.7 (2017: 7.14) (V7.49). $K \subseteq |X|$ is component compact if it meets every connected component of $|X|$ in a compact set `Chapter7/EpGroupoid · EpGroupoid.IsComponentCompact` (2017 p. 247).

Corollary 7.3.8 (2017: 7.2) (V7.51) `Chapter7/OrbitTopology · EpGroupoid.corollary_7_2`. *A component compact K has an open neighbourhood U with* $\mathrm{cl}_{|X|}U$ *paracompact; then* $\mathrm{cl}U$ *and U are metrizable.* **The text's proof**: work in a connected component of K, so that K may be assumed compact; every $z \in K$ has an open $U(z)$ with metrizable, hence paracompact, closure; finitely many $U(z_i)$ cover K; $\bar{U} = \bigcup \mathrm{cl}U(z_i)$ is paracompact, Hausdorff, regular and locally metrizable, hence metrizable, and so is $U = \bigcup U(z_i)$. **The formalized proof.** *Step 1, the compact case, inside a given open $W \supseteq K$* `Chapter7/OrbitTopology · EpGroupoid.corollary_7_2_of_isCompact`. Every z has an open $U(z)$ with $\mathrm{cl}U(z) \subseteq W$ metrizable, by Lemma 7.3.2 (2017: 7.4) and the regularity of $|X|$ `Chapter7/OrbitTopology · EpGroupoid.exists_isOpen_metrizableSpace_closure`; finitely many cover K; $U = \bigcup U(z_i)$ has $\mathrm{cl}U = \bigcup \mathrm{cl}U(z_i)$, paracompact by Michael (§VII.20), which the text does not cite, and metrizable by §VII.19. *Step 2, the components are open.* "We work in a connected component of K" is read in a component C of $|X|$, where $K \cap C$ is compact. This needs the components of $|X|$ to be open, which the text does not state: $|X|$ is a quotient of the locally path connected X (§VII.7), hence locally connected `Chapter7/OrbitTopology · EpGroupoid.locallyPathConnectedSpace_orbitSpace`. *Step 3, back to K, not in the text.* Step 1 in each component C, with $W = C$, gives $U_C \supseteq K \cap C$ with $\mathrm{cl}U_C \subseteq C$ paracompact. The components are open and pairwise disjoint, so $(\mathrm{cl}U_C)_C$ is locally finite, $\mathrm{cl}(\bigcup U_C) = \bigcup \mathrm{cl}U_C$, and this locally finite union of closed paracompact sets is paracompact (Proposition 2.7.4 (2017: 2.17), §VII.20) `Chapter7/Michael · paracompactSpace_iUnion`. The text's U names both $\bigcup \mathrm{cl}U(z_i)$ and $\bigcup U(z_i)$ (possible misprint); here U is the open one (Findings, Chapter 7, item 74) (2017 p. 247).

VII.22 Proposition 7.3.9 (2017: 7.11): index lifting

Proposition 7.3.9 (2017: 7.11) (V7.52) `Chapter7/Lift · EpGroupoid.proposition_7_11`. *If $X = (X, \mathbf{X})$ is an ep-groupoid, so is $X^1 = (X^1, \mathbf{X}^1)$; if X is tame, so is X^1.* X^1 is the M-polyfold of Part I with the levels shifted by one (§II.6). **The structure maps of X^1.** The text restricts s, t, u, ι, m to level 1 without comment. Sc-smooth maps preserve the levels, so s, t, u, ι restrict `Chapter7/LocalDiffeo · MPolyfold.levelOneMap` `Chapter7/Lift · EpGroupoid.liftSrc` `Chapter7/Lift · EpGroupoid.liftTgt` `Chapter7/Lift · EpGroupoid.liftUnit` `Chapter7/Lift · EpGroupoid.liftInv`; a composite of composable morphisms on level 1 is on level 1, since $ml(t(\varphi \circ \psi)) = ml(t\varphi) = ml(\varphi)$ (§VII.3) `Chapter7/Lift · EpGroupoid.comp_mem_levelSet` `Chapter7/Lift · EpGroupoid.liftComp`; the groupoid is `Chapter7/Lift · EpGroupoid.liftGroupoidStr`. **The text's proof** says that (1), (2) and (4) hold trivially, and proves (3) by sequences: for $x \in X_1$ and $V(x)$ with t proper over $s^{-1}(\mathrm{cl}_X V(x))$, $W(x) = V(x) \cap X_1$ is open in X_1 and $\mathrm{cl}_{X_1} W(x) \subseteq \mathrm{cl}_X V(x)$; for (φ_k) with $(t\varphi_k)$ in a compact subset of X_1, after a subsequence (a) $\varphi_k \to \varphi$ in $\mathbf{X}_0$, (b) s, t converge in X_0, (c) $t(\varphi_k) \to t(\varphi)$ in X_1, so (d) $\varphi \in \mathbf{X}_1$, and $\varphi_k = (t|\mathbf{U}(\varphi))^{-1}(t\varphi_k) \to \varphi$ in $\mathbf{X}^1$. **The formalized proof.** *(1).* The level-1 lift of a local sc-diffeomorphism is one: an sc-diffeomorphism on an open set lifts with its inverse `Chapter7/LocalDiffeo · MPolyfold.IsScDiffeoOn.levelOne` `Chapter7/LocalDiffeo · MPolyfold.IsLocalScDiffeo.levelOne` `Chapter7/Lift · EpGroupoid.isLocalScDiffeo_liftSrc` `Chapter7/Lift · EpGroupoid.isLocalScDiffeo_liftTgt`. *(2).* Part I's lifting of the index for sc-smooth maps, Proposition 1.2.2 (preprint 1.17) in charts (§I.15) `Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne` `Chapter7/Lift · EpGroupoid.isScSmoothMap_liftUnit` `Chapter7/Lift · EpGroupoid.isScSmoothMap_liftInv`. *(4).* The text's "trivially", as we read it, uses $(\mathbf{X}\,{}_s\!\times_t \mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t \mathbf{X}^1$ as M-polyfolds, which is not proved here; it is avoided through the sheet form (§VII.2). A local inverse σ_1 of s on level 1 agrees near each point with the level-1 lift of a local inverse σ of s, both inverting s near that point, and $\psi \mapsto m(\sigma(t\psi), \psi)$ lifts by the lifting of sc-smooth maps on open sets `Chapter7/LocalDiffeo · MPolyfold.IsScSmoothOn.levelOne` `Chapter7/Lift · EpGroupoid.isScSmoothCompSheet_lift` `Chapter7/Lift · EpGroupoid.isScSmoothComp_lift`. *(3), the text's argument* `Chapter7/Lift · EpGroupoid.exists_isProperMap_lift`. "$(t\varphi_k)$ in a compact subset of X_1" is replaced by "$t\varphi_k$ converges in X_1", the sequential criterion for a closed set (§VII.2) `Chapter7/ProperMaps · isProperMap_domRestrict_iff_of_isClosed`, X^1 and $\mathbf{X}^1$ being metrizable (Theorem 2.3.10 (2017: 2.2)) `Chapter2/Metrizable · MPolyfold.metrizableSpace_level`;

$\mathrm{cl}_{X_1} W(x) \subseteq \mathrm{cl}_X V(x)$ is the continuity of $X_1 \to X$; (a) is properness on level 0, (c) the hypothesis, (d) that t is a local sc-diffeomorphism and preserves levels; step (b) is not needed; and "$(t|\mathbf{U}(\varphi))^{-1}$ is sc^0" is used as the continuity on level 1 of its level-1 lift. So X^1 is an ep-groupoid `Chapter7/Lift · EpGroupoid.liftOne`, paracompact on both levels `Chapter2/Metrizable · MPolyfold.paracompactSpace_level`. The display (7.5), "$s^{-1}(\mathrm{cl}_{X_1}(W(x)) \to X_1$", has an unbalanced parenthesis (possible misprint; Findings, Chapter 7, items 80, 96).

Tameness of X^1 `Chapter7/Lift · EpGroupoid.isTame_liftOne`. The text calls it trivial. As we read it, it needs that the index lift of a tame retraction is tame, which Part I left open (Findings, Chapter 5, item 145); it is proved `Chapter7/Lift · MPolyfold.IsTame.levelOne`. *Step 1, the shifted presentation.* For a tame r with a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant C, the presentation $T^1 : E^1 \cong \mathbb{R}^n \oplus W^1$ of C^1 has the same $\mathbb{R}^n$-coordinates `Chapter7/Lift · shiftPres Chapter7/Lift · shift_presentation`, so $d_{C^1} = d_C$ on C^1 `Chapter7/Lift · degIndex_shift` and the subspace E_x for T^1 is $E_x \cap E_1$ `Chapter7/Lift · mem_minimalSubspaceWith_shift`. *Step 2, the tangent.* $Dr^1(x) = Dr(x)|_{E_1}$ (Proposition 1.2.2 (preprint 1.17)), whose range is $T_xO \cap E_1$ since $Dr(x)$ is a projection. *Step 3, the complement.* An sc-complement A of T_xO gives the sc-complement $A \cap E_1$ of T_xO^1, an sc-splitting shifting to one `Chapter7/Lift · isScSplitting_levelSub_one Chapter7/Lift · isScSubspace_levelSub_one`. *Step 4.* So r^1 is tame for T^1 `Chapter7/Lift · isTameWith_shift Chapter7/Lift · isTameRetract_shift`, hence tame by Part I's independence of the presentation `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`, and the level charts of tame charts form a tame atlas of X^1 (Findings, Chapter 7, item 81) (pp. 326–328 (2017 pp. 247–248)).

VII.23 Definition 7.5.1 (2017: 7.15), Remark 7.5.2 (2017: 7.1), and Proposition 7.5.3 (2017: 7.12)

Definition 7.5.1 (2017: 7.15) (V7.53). An ep-groupoid has paracompact orbit space if in addition (5) $|X|$ is paracompact `Chapter7/EpGroupoid · EpGroupoid.HasParacompactOrbitSpace`; the text stipulates that paracompact does not include Hausdorff, and Mathlib's notion does not. Since $|X|$ is Hausdorff and locally metrizable (Theorem 7.3.1 (2017: 7.2)), (5) is equivalent to the metrizability of $|X|$ `Chapter7/Lift · EpGroupoid.hasParacompactOrbitSpace_iff_metrizableSpace`, and it holds when $|X|$ is second countable (Corollary 7.3.6 (2017: 7.1)) `Chapter7/Lift · EpGroupoid.hasParacompactOrbitSpace_of_secondCountableTopology`. *The sentence before Remark 7.5.2 (2017: 7.1)*, "in case X has a second countable topology, (1)–(4) imply (5)", is read with X the object space `Chapter7/Lift · EpGroupoid.remark_7_1`: $|X|$ is then second countable, π being open and onto (Proposition 7.1.8 (2017: 7.1)), hence metrizable by Corollary 7.3.6 (2017: 7.1), hence paracompact. Remark 7.5.2 (2017: 7.1) itself, on the second countability required in [38], has no formal content, and "it is presently not known whether the property (5) is a consequence of the properties (1)–(4)" is not formalized (Findings, Chapter 7, items 82, 113) (p. 340 (2017 p. 249)).

The map $|X^1| \to |X|$. The text writes $|X_1|$ for the orbit space of X^1 and treats it as a subset of $|X|$ ("$A_x \cap |X_1|$", 2017 p. 250). Here $|X^1|$ keeps its own quotient topology: the inclusion $X^1 \to X$ is a functor `Chapter7/Lift · EpGroupoid.liftIncl` and induces $|X^1| \to |X|$ `Chapter7/Lift · EpGroupoid.orbitIncl Chapter7/Lift · EpGroupoid.orbitIncl_π`, continuous `Chapter7/Lift · EpGroupoid.continuous_orbitIncl` and injective, because a morphism between points of X_1 lies in $\mathbf{X}_1$ ($ml(s\varphi) = ml(t\varphi)$, §VII.3) `Chapter7/Lift · EpGroupoid.orbitIncl_injective`, with range $|X_1| = \pi(X_1)$ `Chapter7/Lift · EpGroupoid.range_orbitIncl`; the preimage of $\pi(U)$ is $\pi_1(U \cap X_1)$ `Chapter7/Lift · EpGroupoid.preimage_orbitIncl_image`.

Proposition 7.5.3 (2017: 7.12) (V7.54) `Chapter7/Lift · EpGroupoid.proposition_7_12`. *If X has paracompact orbit space, so has X^1.* **The text's proof**: by Lemma 2.7.3 (2017: 2.7) the cover $(|U(x)|)$ of $|X|$ has a locally finite closed refinement (A_x), $A_x \subseteq |U(x)|$; $A_{x,1} = A_x \cap |X_1|$ is a locally finite closed cover of $|X_1|$; G_x acts on $U(x)_1$ and, as in Lemma 7.3.2 (2017: 7.4), an invariant metric passes to $|U(x)|_1 = |U(x)_1|$, so $A_{x,1}$ is metrizable, hence paracompact, and $|X_1|$ is a locally finite union of closed paracompact sets (Proposition 2.7.4 (2017: 2.17)). **The formalized proof** proves more, that $|X^1|$ is metrizable `Chapter7/Lift · EpGroupoid.metrizableSpace_liftOne_orbitSpace`, by Smirnov's theorem

pulled back along $|X^1| \to |X|$ (Part I): a continuous injection into a paracompact Hausdorff space whose points have open neighbourhoods with metrizable preimages has a metrizable source `Chapter2/Metrizable · metrizableSpace_of_pullback`. *Step 1, the restricted action.* For a natural representation at x, G_x acts on $U(x)_1 = U(x) \cap X_1$ by the level-1 lifts of the $\Phi(g)$, homeomorphisms of $U(x)_1$ `Chapter7/Lift · EpGroupoid.NaturalRep.liftU` `Chapter7/Lift · EpGroupoid.NaturalRep.liftΦ` `Chapter7/Lift · EpGroupoid.NaturalRep.liftMulAction` `Chapter7/Lift · EpGroupoid.NaturalRep.continuous_liftSmul`. "$U(x)_1$ has the natural G_x-action" is read as this restricted action only: $U(x)_1$ need not be connected in X^1, so it is not a natural representation of X^1. *Step 2, the fibres.* Points of $U(x)_1$ have the same image in $|X^1|$ iff they are in one G_x-orbit, by Theorem 7.1.13 (3) (2017: 7.1) and §VII.3 `Chapter7/Lift · EpGroupoid.NaturalRep.liftOne_π_eq_iff`. *Step 3, the metric.* $G_x\backslash U(x)_1$ is metrizable by the text's $\widehat{\rho}$, the argument of Lemma 7.3.2 (2017: 7.4) repeated for a finite group acting by homeomorphisms on a metrizable space `Chapter7/Lift · lvMaxDist` `Chapter7/Lift · lvQuotDist` `Chapter7/Lift · metrizableSpace_orbitRel_quotient_of_finite`. *Step 4, Proposition 7.1.19 (2017: 7.6) on level* 1. $G_x\backslash U(x)_1 \to |X^1|$ is an open embedding `Chapter7/Lift · EpGroupoid.NaturalRep.liftOrbitQuotientMap` `Chapter7/Lift · EpGroupoid.NaturalRep.isOpenEmbedding_liftOrbitQuotientMap` onto the preimage of $|U(x)|$ `Chapter7/Lift · EpGroupoid.NaturalRep.range_liftOrbitQuotientMap`, so that preimage is metrizable `Chapter7/Lift · EpGroupoid.NaturalRep.metrizableSpace_preimage_orbitIncl`. *Step 5.* $|X|$ is paracompact by hypothesis and Hausdorff by Lemma 7.3.4 (2017: 7.6), and every point has the open neighbourhood $|U(x)|$; the pulled-back Smirnov theorem gives $|X^1|$ metrizable, hence paracompact (Stone). The regularity of $|X^1|$ and Michael's theorem are not needed on this route, though both are available (§§VII.19, VII.20) (Findings, Chapter 7, item 83) (pp. 340–342 (2017 pp. 249–250)).

VII.24 Theorem 7.5.4 (2017: 7.3), Definition 7.5.5 (2017: 7.16), Remark 7.5.6 (2017: 7.2), and the statement of Theorem 7.5.7 (2017: 7.4)

Theorem 7.5.4 (2017: 7.3) (V7.55) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_3`. *If $|X|$ is paracompact, every open cover (O_λ) of $|X|$ has a subordinate partition of unity (σ_λ), and every cover (U_λ) of X by saturated open sets has a subordinate continuous partition of unity (β_λ) taking equal values on isomorphic objects. Step 1.* The first statement is Mathlib's existence of subordinate partitions of unity on a paracompact normal space `Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate`. *Hidden hypothesis:* normality, which needs $|X|$ Hausdorff — Lemma 7.3.4 (2017: 7.6) — besides paracompact. *Step 2.* For the second, $O_\lambda = \pi(U_\lambda)$ is open (Proposition 7.1.8 (2017: 7.1)), and $\beta_\lambda = \sigma_\lambda \circ \pi$ `Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_apply` has $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ `Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_src`. *Step 3, supports.* $\operatorname{supp}(\sigma_\lambda \circ \pi) \subseteq \pi^{-1}(\operatorname{supp}\sigma_\lambda) \subseteq \pi^{-1}(\pi(U_\lambda)) = U_\lambda$, since U_λ is saturated `Chapter7/PartitionsOfUnity · EpGroupoid.tsupport_orbitPullback_subset` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitPullback_isSubordinate`. *Step 4, local finiteness.* If an open $V \ni \pi(x)$ meets finitely many supports of the σ_λ, the saturated open $\pi^{-1}(V)$ meets finitely many supports of the β_λ `Chapter7/PartitionsOfUnity · EpGroupoid.exists_isSaturated_nhds_finite_orbitPullback`. "Subordinate" is Mathlib's, $\operatorname{supp}\sigma_\lambda \subseteq O_\lambda$; "some of the $\sigma_\lambda \equiv 0$" says only that the index set is Λ (Findings, Chapter 7, item 90) (2017 p. 250).

Definition 7.5.5 (2017: 7.16) (V7.56). X has sc-smooth partitions of unity if every cover of X by saturated open sets has a subordinate sc-smooth partition of unity (β_λ) with (1) every point having a saturated open neighbourhood meeting finitely many supports and (2) $\beta_\lambda(s\varphi) = \beta_\lambda(t\varphi)$ `Chapter7/EpGroupoid · EpGroupoid.HasScPartitionsOfUnity`, the partitions being Part I's `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity` with supports the closed supports and index types in the universe of X. The hypothesis is satisfiable: for the trivial ep-groupoid it reduces to Part I's sc-smooth partitions of unity `Chapter7/EpGroupoid · EpGroupoid.hasScPartitionsOfUnity_trivial`, which $\mathbb{R}^n$ has `Chapter7/LocalDiffeo · rnPolyfold_admitsScPartitionsOfUnity` (Findings, Chapter 7, item 91) (pp. 341–343 (2017 pp. 250–251)).

Remark 7.5.6 (2017: 7.2) (V7.57). Apart from sc-Hilbert spaces, the existence of sc-smooth partitions of unity is "a zoo"; the remark has no formal content, and the sc-Hilbert case is Part I's (§VII.29) (2017

p. 251).

Theorem 7.5.7 (2017: 7.4) (V7.58) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_4`. *If $|X|$ is paracompact and the object M-polyfold admits sc-smooth partitions of unity, X has sc-smooth partitions of unity.* The proof is the text's Theorem 7.6.4 (2017: 7.6) (§§VII.27–VII.28). The hypotheses are satisfiable, by the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_4_hypotheses_rn` `Chapter7/PartitionsOfUnity · EpGroupoid.hasScPartitionsOfUnity_trivial_rn_of_theorem_7_4` (Findings, Chapter 7, items 94, 105) (2017 p. 251).

VII.25 Definition 7.5.8 (2017: 7.17), Propositions 7.5.9 and 7.5.10 (2017: 7.13 and 7.14)

Definition 7.5.8 (2017: 7.17) (V7.59). $Q \subseteq X$ is *proper* if $t : s^{-1}(Q) \to X$ is proper `Chapter7/EpGroupoid · EpGroupoid.IsProperSubset`, and an open U has the *properness property* if $\mathrm{cl}_X U$ is proper `Chapter7/EpGroupoid · EpGroupoid.HasPropernessProperty`; every x has such a U, by Definition 7.1.3 (3) (2017: 7.3) `Chapter7/EpGroupoid · EpGroupoid.exists_hasPropernessProperty`. In metrizable spaces Q is proper iff $s^{-1}(Q) \cap t^{-1}(K)$ is compact for every compact K `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iff_isCompact`, and for closed Q iff the sequential criterion holds `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iff_exists_tendsto_subseq`. Basic facts: a closed subset of a proper set is proper `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.of_subset`; $\emptyset$, finite unions and compact sets are proper `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_empty` `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.union` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_biUnion` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iUnion` `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_of_isCompact`; an open set inside one with the properness property has it `Chapter7/ProperSubsets · EpGroupoid.HasPropernessProperty.isProperSubset` `Chapter7/ProperSubsets · EpGroupoid.HasPropernessProperty.mono`, and every neighbourhood of x contains an open, also an open connected, neighbourhood of x with the properness property `Chapter7/ProperSubsets · EpGroupoid.exists_hasPropernessProperty_subset` `Chapter7/ProperSubsets · EpGroupoid.exists_hasPropernessProperty_isConnected_subset` (Findings, Chapter 7, item 92) (2017 p. 251).

Proposition 7.5.9 (2017: 7.13) (V7.60) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_13`. *A proper subset is closed.* The text: for $x_n \to x$ in Q, the units 1_{x_n} lie in the compact $(t|s^{-1}(Q))^{-1}(\{x_n\} \cup \{x\})$, a subsequence converges to some φ with $s(\varphi) \in Q$, and $s(\varphi) = x$. The formalization is this argument with the compact preimage (§VII.2) `Chapter7/ProperSubsets · EpGroupoid.IsProperSubset.isClosed` (2017 p. 251).

Proposition 7.5.10 (2017: 7.14) (V7.61). *(1) If Q is proper, $\pi(Q)$ is closed. (2) If $|X|$ is paracompact, $A \subseteq |X|$ is closed iff $A = \pi(Q)$ for a proper Q.* **(1)** `Chapter7/ProperSubsets · EpGroupoid.proposition_7_14_one`. The text uses sequences $\varphi_k : x_k \to y_k$ with $x_k \in Q$, $y_k \to y$. The formalization uses the closed-map property: a set of $|X|$ is closed iff its preimage is `Chapter7/ProperSubsets · EpGroupoid.isClosed_iff_preimage_π`, and $\pi^{-1}(\pi(Q)) = t(s^{-1}(Q))$ is the image of the closed $s^{-1}(Q)$ (Proposition 7.5.9 (2017: 7.13)) under the proper, hence closed, t `Chapter7/ProperSubsets · EpGroupoid.isClosed_saturation_of_isProperSubset`. The proof's "$x = s(\varphi) \in Q$" reuses the name x (possible misprint).

(2) `Chapter7/ProperSubsets · EpGroupoid.proposition_7_14_two`. One direction is (1). **The text's proof** of the other chooses for $z \in A$ a representative x_z and a neighbourhood $U(x_z)$ with the natural action whose closure Q_z is proper; by Lemma 2.7.3 (2017: 2.7) the closed, hence paracompact, A has a locally finite cover by closed $A_i \subseteq \pi(Q_{z_i})$; $P_i = (\pi|\mathrm{cl}_X U(x_{z_i}))^{-1}(A_i)$ and $Q = \bigcup P_i$ has $\pi(Q) = A$; it is proper because only finitely many A_i meet a neighbourhood of $\pi(y)$, so the sources lie in finitely many P_i. Lemma 2.7.3 (2017: 2.7) assumes a regular Hausdorff space. **The formalized proof** needs only the paracompactness of $|X|$. *Step 1.* Representatives x_z are chosen by the quotient, and $V(x_z)$ is any open neighbourhood with the properness property; the natural action is not used. *Step 2, an open refinement.* The open sets $\pi(V(x_z))$ cover the closed A; Mathlib gives an open locally finite precise refinement (v_z) of this cover of A, from paracompactness alone `Mathlib/Topology/Compactness/Paracompact · precise_refinement_set`. *Step 3, closures.* $P_z = \mathrm{cl}_X V(x_z) \cap \pi^{-1}(\mathrm{cl}(v_z) \cap A)$ is a closed subset of the proper $\mathrm{cl}_X V(x_z)$, hence proper,

with $\pi(P_z) \subseteq \mathrm{cl}(v_z) \cap A$; and $\pi(P_z) \supseteq v_z \cap A$, since $v_z \subseteq \pi(V(x_z))$. As the v_z cover A, $\pi(\bigcup P_z) = A$. *Step 4, properness of* $Q = \bigcup P_z$. The closures $\mathrm{cl}(v_z)$ of a locally finite family are locally finite, and a union of proper sets whose images form a locally finite family is proper: a compact K has compact $\pi(K)$, which meets only finitely many $\mathrm{cl}(v_z)$ `Chapter7/ProperSubsets · EpGroupoid.isProperSubset_iUnion_of_locallyFinite`. Neither Lemma 2.7.3 (2017: 2.7) nor the regularity of $|X|$ is used, and no version under a regularity hypothesis is needed; the closed refinement of Michael's theorem would give the text's route as well (§VII.20). The text writes $\mathrm{cl}_X(U_{x_{z_i}})$ for $\mathrm{cl}_X(U(x_{z_i}))$ (possible misprint; Findings, Chapter 7, item 92) (pp. 342–344 (2017 pp. 251–252)).

VII.26 Appendix §7.6.1 (2017 §7.5.1): the proof of Theorem 7.1.13 (2017: 7.1) (Theorem 7.6.1 (2017: 7.5), Lemmas 7.6.2 and 7.6.3 (2017: 7.8 and 7.9))

Theorem 7.6.1 (2017: 7.5) (V7.62) `Chapter7/NaturalRep · EpGroupoid.theorem_7_5` is Theorem 7.1.13 (2017: 7.1) with $\Phi(g) = t_g \circ s_g^{-1}$; it is stated as the same theorem. **The text's proof** (pp. 344–346 (2017 pp. 253–254)): for every $g \in G_{x_0}$ choose contractible open N_g^t, N_g^s on which t and s are sc-diffeomorphisms onto an open $U_0 \ni x_0$, disjoint for distinct g since G_{x_0} is finite; $N_g = N_g^t \cap N_g^s$, $s_g = s|N_g$, $t_g = t|N_g$; Lemmas 7.6.2 and 7.6.3 (2017: 7.8 and 7.9) give U; $\Phi(g) = t_g \circ s_g^{-1}$, a homomorphism "since the neighborhoods N_g are disjoint and since the structure maps are continuous"; $\Gamma(g, y) = s_g^{-1}(y)$. **The formalized proof.** *Step 1, the sheets* `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.exists_sheets`. G_{x_0} is finite (Proposition 7.1.12 (2017: 7.4)) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.finite_isotropy`; Hausdorff separation of the finite set $G_{x_0} \subseteq \mathbf{X}$ gives pairwise disjoint open $N_g \ni g$, shrunk so that s is an sc-diffeomorphism from N_g onto its open image with inverse $\sigma_g = s_g^{-1}$; t is one too, which is recorded but not used. Only connectedness will be needed, not contractibility. *Step 2,* $U \subseteq V$. The proof does not mention V; it starts at $U_0 = V \cap \bigcap_g s(N_g)$. *Step 3, Lemma 7.6.2 (2017: 7.8), then Lemma 7.6.3 (2017: 7.9)* (below). *Step 4, the homomorphism* `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isScSmoothOn_hat`. On the connected U, $y \mapsto \sigma_{gh}(y)$ and $y \mapsto \sigma_g(\Phi(h)y) \circ \sigma_h(y)$ are continuous sections of s — the second by (4) in the sheet form — agreeing at x_0, so they agree on U (§VII.7) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eqOn_σ_mul` and $\Phi(gh) = \Phi(g)\Phi(h)$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat_mul`. In the same way σ_1 is the unit section u on U `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eqOn_σ_one` and $\Phi(1) = \mathrm{id}$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.hat_one`, which the text does not mention and which makes $\Phi(g)$ an sc-diffeomorphism with inverse $\Phi(g^{-1})$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isScDiffeoOn_hat`. As we read it, "since the N_g are disjoint and the structure maps are continuous" shows only that the sheet index is locally constant; connectedness of U makes it constant. *Step 5, (3).* A morphism h between points of U lies in some $N_{g'}$ by Lemma 7.6.2 (2017: 7.8), and s is injective there, so $h = \sigma_{g'}(sh)$; disjointness of the N_g gives uniqueness `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.eq_of_mem_N`. *Step 6.* The natural representation is built on every open connected invariant $U \subseteq \bigcap_g s(N_g)$ with the conclusion of Lemma 7.6.2 (2017: 7.8) `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.mkRep`, and Lemmas 7.6.2, 7.6.3 (2017: 7.8, 7.9) provide one `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.exists_naturalRep_sheets`. "$\Phi : G_x \to \mathrm{Diff}_{sc}(U)$" is read with G_{x_0} (possible misprint; Findings, Chapter 7, item 29).

Lemma 7.6.2 (2017: 7.8) (V7.63) `Chapter7/NaturalRep · EpGroupoid.lemma_7_8`. *There is an open* $U_1 \subseteq U_0$ *of* x_0 *such that every morphism with source and target in* U_1 *lies in some* N_g — for any open $N_g \ni g$ and any neighbourhood U_0; the sheet property is not used. **The text's proof**: otherwise there are $h_k \notin \bigcup N_g$ with $s(h_k), t(h_k) \to x_0$; by properness $h_{k_l} \to h$, necessarily $h \in G_{x_0}$, so $h \in N_g$ for some g, a contradiction. **The formalized proof** follows it. *Step 1.* The sequence is taken along a countable decreasing basis of

x_0, X being metrizable. *Step 2.* "By the properness assumption" is Definition 7.1.3 (3) (2017: 7.3) at x_0, which applies to the h_k with $s(h_k) \in \mathrm{cl}V(x_0)$, i.e. to all large k (§VII.2). *Step 3.* "Necessarily $h \in G_{x_0}$": $t(h) = x_0$ comes with the properness, $s(h) = x_0$ from the continuity of s and the uniqueness of limits. *Step 4.* N_g is an open neighbourhood of the limit, so eventually $h_{k_l} \in N_g$. "The properness assumption of ep-polyfolds" is read as that of ep-groupoids (possible misprint; Findings, Chapter 7, item 30) (pp. 344–346 (2017 pp. 253–254)).

Lemma 7.6.3 (2017: 7.9) (V7.64) `Chapter7/NaturalRep · EpGroupoid.lemma_7_9`. *There is an open $U_2 \subseteq U_1$ of x_0 such that $U = \bigcup_g t_g s_g^{-1}(U_2)$ lies in U_1 and is invariant under all $t_g s_g^{-1}$.* **The text's proof**: choose U_2 so small that U and $t_g s_g^{-1}(U)$ lie in U_1; for $x = t_h s_h^{-1}(u)$, the point $v = t_g s_g^{-1} t_h s_h^{-1}(u) \in U_1$ is the target of a morphism $u \to v$, which by Lemma 7.6.2 (2017: 7.8) is $s_{g'}^{-1}(u)$, so $v = t_{g'} s_{g'}^{-1}(u) \in U$. **The formalized proof.** *Step 1.* Each $t_g s_g^{-1}$ is continuous at its fixed point x_0 and G_{x_0} is finite, so a U_2 with $t_h s_h^{-1}(U_2) \subseteq U_1$ and $t_g s_g^{-1} t_h s_h^{-1}(U_2) \subseteq U_1$ for all g, h exists; it is taken *connected* (§VII.7). *Step 2.* U is open, t being open and $s_g^{-1}(U_2) = N_g \cap s^{-1}(U_2)$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.unionHat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isOpen_image_hat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isOpen_unionHat` `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.mem_unionHat_iff`, and connected, a union of connected sets through $x_0 = t_g s_g^{-1}(x_0)$, so no passage to a component is needed `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.isConnected_unionHat`. *Step 3, invariance.* The morphism $u \to v$ is $\sigma_g(x) \circ \sigma_h(u)$, composable because $x \in U_1 \subseteq s(N_g)$; it lies in some $N_{g'}$ by Lemma 7.6.2 (2017: 7.8), and "it has necessarily the form $v = t_{g'} s_{g'}^{-1}(u)$" uses that s is injective on $N_{g'}$ `Chapter7/NaturalRep · EpGroupoid.NatRepConstruction.Sheets.image_σ` (Findings, Chapter 7, item 31) (2017 p. 254).

VII.27 Appendix §7.6.2 (2017 §7.5.2): Lemma 7.6.5 (2017: 7.10)

Theorem 7.6.4 (2017: 7.6) (V7.65) `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_6` is Theorem 7.5.7 (2017: 7.4), stated again (Findings, Chapter 7, item 95) (2017 p. 255).

Lemma 7.6.5 (2017: 7.10) (V7.66) `Chapter7/PartitionsOfUnity · EpGroupoid.lemma_7_10`. *Let $|X|$ be paracompact and X admit sc-smooth partitions of unity; let $U(x) \subseteq \mathrm{cl}_X U(x) \subseteq V(x)$ be open, $V(x)$ with the natural G_x-action and $U(x)$ invariant. A continuous $\beta : X \to [0,1]$ supported in $U(x)$ and invariant on $V(x)$ lies below an sc-smooth $f \geq 0$ supported in $V(x)$ and invariant on $V(x)$.* The formalization adds the conclusion $\mathrm{supp} f \subseteq \mathrm{cl}_X U(x)$, which the proof gives and Theorem 7.5.7 (2017: 7.4) needs, and omits the hypothesis $x \in U(x)$, which the proof does not use and Theorem 7.5.7 (2017: 7.4) cannot supply; the paracompactness of $|X|$ and the continuity of β are kept, unused. **The text's proof**: open $W(z)$, $z \in V(x)$, with numbers a_z — $\mathrm{cl}W(z) \subseteq U(x)$ and $\beta < a_z$ on $W(z)$ for $z \in U(x)$, otherwise $W(z) \subseteq V(x)$, $\beta|W(z) = 0$, $a_z = 0$; an sc-smooth partition of unity (σ_z) of $V(x)$ subordinate to them; $\gamma = \sum \sigma_z a_z \geq \beta$, supported in $\mathrm{cl}U(x)$; and $f(y) = |G_x|^{-1} \sum_g \gamma(g * y)$. **The formalized proof.** *Step 1, a cover of X.* The hypothesis is that X, not the open subset $V(x)$, admits sc-smooth partitions of unity, so the cover is completed by $X \setminus \mathrm{cl}U(x)$, where $\beta = 0$ and $a = 0$; and it is coarsened to two sets, $U(x)$ with $a = 1$ (as $\beta \leq 1$) and $X \setminus \mathrm{supp}\,\beta$ with $a = 0$. The text's $\mathrm{cl}W(z) \subseteq U(x)$ and $\beta < a_z$ are not needed. *Step 2, γ.* γ is the function attached to $U(x)$ of a partition of unity subordinate to $\{U(x), X \setminus \mathrm{supp}\,\beta\}$: sc-smooth, ≥ 0, supported in $U(x)$, and $\gamma = 1 \geq \beta$ on $\mathrm{supp}\,\beta$, where the other function vanishes. *Step 3, the average.* $f = |G_x|^{-1} \sum_g \gamma \circ \Phi(g)$ on $V(x)$ and 0 elsewhere. Each $\gamma \circ \Phi(g)$, extended by 0, is supported in $\Phi(g)^{-1}(\mathrm{cl}U(x)) = \mathrm{cl}U(x) \subseteq V(x)$ by the invariance of $U(x)$, hence sc-smooth on X `Chapter7/PartitionsOfUnity · isScSmoothMap_real_of_tsupport_subset` `Chapter7/PartitionsOfUnity · isScSmoothMap_real_of_isScSmoothOn_nhds`. *Step 4.* $\beta \leq f$ by the invariance of β: $\beta(y) = \beta(g*y) \leq \gamma(g*y)$ for every g. f is invariant by the reindexing $g \mapsto gh$, and $\mathrm{supp} f \subseteq \mathrm{cl}U(x)$ (Findings, Chapter 7, item 93) (2017 p. 255).

VII.28 Appendix §7.6.2 (2017 §7.5.2): the proof of Theorem 7.5.7 (2017: 7.4)

The text's proof (pp. 346–347 (2017 pp. 255–256)). For every x choose an open $V(x)$ with (1) an open $W(x)$ with the natural G_x-action, $V(x) \subseteq \mathrm{cl}V(x) \subseteq W(x)$ and $V(x)$ invariant; (2) $t : s^{-1}(\mathrm{cl}V(x)) \to X$ proper; (3) $V(x) \subseteq U_\lambda$ for some λ; (4) $(\pi^{-1}(|V(x)|) \setminus V(x)) \cap W(x) = \emptyset$. Take a locally finite refinement (V_i') of $(|V(x)|)$ with $V_i' \subseteq |V(x_i)|$, open $V_i'' \subseteq \mathrm{cl}V_i'' \subseteq V_i'$ covering $|X|$, $U_i'' = V(x_i) \cap \pi^{-1}(V_i'')$ and a partition of unity (β_i) subordinate to (V_i''). "Approximate $(\beta_i \circ \pi)|V(x_i)$ using Lemma 7.10" gives f_i supported in $V(x_i)$; averaging, extending to the saturation and by 0 gives h_i; $h = \sum h_i > 0$ is locally finite and (h_i/h) is the partition of unity.

The formalized proof (V7.67) builds the same functions h_i, but for a Urysohn lemma, and normalizes with Mathlib's bump coverings, as Part I's Theorem 5.5.10 (2017: 5.12) does (§V.25) `Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx`. The four points where the text needed more than a reading are marked (a)–(d).

Step 1, properties (1)–(3) `Chapter7/PartitionsOfUnity · EpGroupoid.exists_invariant_nhds_of_isOpen`. Given x and an open $O \ni x$, take $V_0 \subseteq O$ with the properness property (§VII.25), $W(x) = U$ of a natural representation inside V_0 (Theorem 7.1.13 (2017: 7.1)), an open $N \ni x$ with $\mathrm{cl}N \subseteq W(x)$, X being regular, and $V(x) = \bigcap_g \Phi(g)^{-1}(N)$: open, invariant, $x \in V(x)$, $\mathrm{cl}V(x) \subseteq \mathrm{cl}N \subseteq W(x)$, and proper over its closure by §VII.25. Elements of one orbit have one image in $|X|$ `Chapter7/PartitionsOfUnity · EpGroupoid.π_Φ_eq`.

Step 2, (4) follows from (1) and is not a further choice. The morphisms between points of $W(x)$ are the $\Gamma(g, \cdot)$ (Theorem 7.1.13 (3) (2017: 7.1)), so a point of $W(x)$ isomorphic to a point of the invariant $V(x)$ is $g * v$ for some $v \in V(x)$, hence lies in $V(x)$ `Chapter7/PartitionsOfUnity · EpGroupoid.mem_of_π_mem_image_of_invariant` `Chapter7/PartitionsOfUnity · EpGroupoid.saturation_diff_inter_eq_empty`.

Step 3, extension to the saturation. A G_x-invariant f on $W(x)$ defines F on $|X|$ by $F(|y|) = f(y)$ for $y \in W(x)$ and 0 elsewhere, well defined because points of $W(x)$ in one orbit differ by the natural action (§VII.11) `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension_π` `Chapter7/PartitionsOfUnity · EpGroupoid.orbitExtension_nonneg` `Chapter7/PartitionsOfUnity · EpGroupoid.support_orbitExtension_subset`. (d) *$F \circ \pi$ is sc-smooth* when f is sc-smooth with $\mathrm{supp} f \subseteq \mathrm{cl}V(x)$ `Chapter7/PartitionsOfUnity · EpGroupoid.isScSmoothMap_orbitExtension`: $F \circ \pi$ vanishes off the saturation of $\mathrm{supp} f$, which is closed because $\mathrm{supp} f \subseteq \mathrm{cl}V(x)$ is proper by (2) `Chapter7/ProperSubsets · EpGroupoid.isClosed_saturation_of_isProperSubset`; near a point $y \cong z$ of that saturation, with $\varphi : z \to y$, $F \circ \pi = f \circ s \circ (t|\mathbf{U}(\varphi))^{-1}$ (Definition 7.1.4 (2017: 7.4)). The text does not mention the closedness.

Step 4, the invariant Urysohn lemma `Chapter7/PartitionsOfUnity · EpGroupoid.exists_isScSmoothMap_orbit_urysohn`: *for disjoint closed $s', t' \subseteq |X|$ there is a continuous $F : |X| \to [0,1]$ with $F \circ \pi$ sc-smooth, $F = 0$ on s' and $F = 1$ on t'.* For every $x \in \pi^{-1}(t')$ Step 1 gives $V(x) \subseteq \mathrm{cl}V(x) \subseteq W(x) \subseteq \pi^{-1}(|X| \setminus s')$. The open sets $|V(x)|$ cover the closed t'; take an open locally finite precise refinement (V_x') of this cover of t' `Mathlib/Topology/Compactness/Paracompact · precise_refinement_set` and a bump covering (b_x) of t' subordinate to it `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate`, whose $\mathrm{supp}\, b_x \subseteq V_x'$ takes the place of the text's V_x''; put $U_x'' = V(x) \cap \pi^{-1}(V_x')$. (b) *The cut-off.* $c_x = b_x \circ \pi$ on $V(x)$, 0 elsewhere, is continuous and supported in U_x'': a point of $\mathrm{cl}V(x) \cap \pi^{-1}(\mathrm{supp}\, b_x)$ lies in $W(x)$ and is isomorphic to a point of $V(x)$, as $\mathrm{supp}\, b_x \subseteq |V(x)|$, hence lies in $V(x)$ by (4). (a) *Lemma 7.6.5 (2017: 7.10)* is applied with $W(x)$ in the role of $V(x)$ and U_x'' in the role of $U(x)$: its hypothesis $\mathrm{cl}U_x'' \subseteq W(x)$, with the natural action on $W(x)$, holds by (1), and U_x'' is invariant. The text's "approximate $(\beta_i \circ \pi)|V(x_i)$ using Lemma 7.10" reads as $V(x_i)$ in the role of $V(x)$ (possible misprint). It gives an invariant sc-smooth $f_x \geq c_x$ with $\mathrm{supp} f_x \subseteq \mathrm{cl}U_x''$, and F_x its extension (Step 3), using $\mathrm{cl}U_x'' \subseteq \mathrm{cl}V(x)$. (c) *Local finiteness.* $\mathrm{supp}(F_x \circ \pi) \subseteq \pi^{-1}(\mathrm{cl}V_x')$ by the support bound of Lemma 7.6.5 (2017: 7.10), and $(\mathrm{cl}V_x')$ is locally finite, so $G = \sum_x F_x$ is a locally finite sum. The text does not state this bound: Lemma 7.6.5 (2017: 7.10) as stated gives only $\mathrm{supp} f_x \subseteq W(x)$, and the saturations of the $W(x)$ need not be locally finite. *The*

function. $G \circ \pi$ is sc-smooth, $G = 0$ on s' since every $W(x)$ misses $\pi^{-1}(s')$, and $G \geq 1$ on t', since at every point of t' some b_x equals 1, and then $f_x \geq c_x = 1$ over it; $F = \tau \circ G$ with Mathlib's smooth transition function τ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`, 0 on $(-\infty, 0]$ and 1 on $[1, \infty)$.

Step 5, the normalization `Chapter7/PartitionsOfUnity · EpGroupoid.theorem_7_4`. Given a saturated open cover (U_λ), the $\pi(U_\lambda)$ cover $|X|$ `Chapter7/PartitionsOfUnity · EpGroupoid.iUnion_image_π_eq_univ`. Mathlib's existence of bump coverings subordinate to an open cover with all functions satisfying a predicate for which Urysohn's lemma holds `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop`, for the class of the g with $g \circ \pi$ sc-smooth and Step 4 as its input, gives (b_λ) on the paracompact normal $|X|$, subordinate to $(\pi(U_\lambda))$, each $b_\lambda \circ \pi$ sc-smooth. The partition $\rho_\lambda = b_\lambda \prod_{\mu<\lambda}(1 - b_\mu)$ `Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity` is locally a finite product of such functions, so $\rho_\lambda \circ \pi$ is sc-smooth, and $\beta_\lambda = \rho_\lambda \circ \pi$, Theorem 7.5.4's (2017: 7.3) pull-back (§VII.24), has the properties (1), (2) of Definition 7.5.5 (2017: 7.16). This replaces the text's (h_i/h) and the passage from the index set of the refinement back to Λ, which the text leaves implicit. "We average using the G_{x_i}-action": f_x is already invariant by Lemma 7.6.5 (2017: 7.10), and no second average is taken. "$t : s^{-1}(\mathrm{cl}_X(V(x)) \to X$" has an unbalanced parenthesis (possible misprint; Findings, Chapter 7, items 94, 96) (pp. 346–347 (2017 pp. 255–256)).

VII.29 Corollary 7.6.6 (2017: 7.3)

Corollary 7.6.6 (2017: 7.3) (V7.68) `Chapter7/PartitionsOfUnity · EpGroupoid.corollary_7_3`. *If $|X|$ is paracompact and the object M-polyfold has an atlas whose local models have Hilbert spaces E_0, every saturated open cover of X has a subordinate sc-smooth partition of unity of X.* The text cites Part I's Proposition 5.5.8 (2017: 5.6), Theorem 5.5.10 (2017: 5.12) and Corollary 5.5.17 (2017: 5.2), the ingredients of Part I's theorem that such an M-polyfold admits sc-smooth partitions of unity (§§V.25–V.26) `Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert`, and Theorem 7.5.7 (2017: 7.4) applies. "An atlas consisting of charts whose local models have Hilbert spaces" is existential, one chart per point; "every compatible chart has a Hilbert E_0" would be unsatisfiable. As in Part I, the theorem of Godefroy–Troyanski–Whitfield–Zizler (Part I's Theorem 5.5.16 (2017: 5.14)) is an explicit hypothesis `Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement` (gap G.53), which holds for finite-dimensional spaces `Chapter7/LocalDiffeo · godefroyTroyanskiWhitfieldZizlerStatement_of_finiteDimensional` `Chapter7/LocalDiffeo · admitsContDiffPartitionsOfUnity_of_finiteDimensional`. The hypotheses are satisfiable on $\mathbb{R}^1$ `Chapter7/LocalDiffeo · rnPolyfold_one_hilbert_hypothesis` `Chapter7/LocalDiffeo · rnPolyfold_one_admitsScPartitionsOfUnity_of_hilbert` `Chapter7/PartitionsOfUnity · EpGroupoid.corollary_7_3_hypotheses_rn` (Findings, Chapter 7, items 95, 106) (p. 347 (2017 p. 256)).

VII.30 Witnesses and counterexamples

Every hypothesis of the chapter that is a predicate on ep-groupoids or on their data is shown satisfiable, and the readings that fail are refuted (Findings, Chapter 7, items 100–107).

The trivial ep-groupoid of an M-polyfold Z, with only units, exists on every M-polyfold `Chapter7/EpGroupoid · EpGroupoid.trivial`. It is tame for tame Z and for $\mathbb{R}^n$ `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial` `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn`, effective with $X_{reg} = Z$ `Chapter7/EpGroupoid · EpGroupoid.isEffective_trivial` `Chapter7/EpGroupoid · EpGroupoid.regularSet_trivial`, every closed set is proper and every open set has the properness property `Chapter7/EpGroupoid · EpGroupoid.isProperSubset_trivial` `Chapter7/EpGroupoid · EpGroupoid.hasPropernessProperty_trivial`, $|X| \cong Z$ `Chapter7/EpGroupoid · EpGroupoid.trivialOrbitHomeomorph` is paracompact `Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial`, and it has natural representations on every connected open neighbourhood `Chapter7/EpGroupoid · EpGroupoid.trivialNaturalRep` `Chapter7/EpGroupoid · EpGroupoid.trivialNaturalRepFinite`. Its witnesses are vacuous for the isotropy, which the next ones are not.

The translation groupoid $K \ltimes U$ of a finite discrete group acting on an M-polyfold by sc-smooth maps, with morphisms the product with the discrete M-polyfold `Chapter7/NonVacuity · Examples.actionMorPolyfold`, is an ep-groupoid `Chapter7/NonVacuity · Examples.actionEpGroupoid`: s is a projection, t a projection after the shear `Chapter7/NonVacuity · Examples.actionShear` `Chapter7/NonVacuity · Examples.action_isLocalScDiffeo_src` `Chapter7/NonVacuity · Examples.action_isLocalScDiffeo_tgt`, (4) holds in the sheet form `Chapter7/NonVacuity · Examples.action_isScSmoothCompSheet` and t is proper on all of $\mathbf{X}$ `Chapter7/NonVacuity · Examples.action_isProperMap_tgt`. Its isotropy is the stabilizer `Chapter7/NonVacuity · Examples.isotropyEquivStabilizer` `Chapter7/NonVacuity · Examples.isRegular_action_iff` `Chapter7/NonVacuity · Examples.regularSet_action`, its orbit space U/K `Chapter7/NonVacuity · Examples.action_π_eq_π_iff` `Chapter7/NonVacuity · Examples.actionOrbitHomeomorph`, its proper sets the closed sets `Chapter7/NonVacuity · Examples.isProperSubset_action` `Chapter7/NonVacuity · Examples.isProperSubset_action_iff` `Chapter7/NonVacuity · Examples.hasPropernessProperty_action`, and its orbit space is paracompact for second countable U `Chapter7/NonVacuity · Examples.hasParacompactOrbitSpace_action`.

$\mathbb{Z}/2 \ltimes \mathbb{R}$ **by reflection** `Chapter7/NonVacuity · Examples.isScSmoothMap_units_smul` `Chapter7/NonVacuity · Examples.reflectionEpGroupoid` is effective `Chapter7/NonVacuity · Examples.isEffective_reflection` with a non-trivial isotropy element $(-1, 0)$ `Chapter7/NonVacuity · Examples.reflectionIsotropyNegOne`, so Definition 7.2.1 (2017: 7.11) is met non-vacuously `Chapter7/NonVacuity · Examples.isEffective_reflection_nonvacuous`; $X_{reg} = \mathbb{R} \setminus \{0\}$ `Chapter7/NonVacuity · Examples.regularSet_reflection`, $G_0 \cong \mathbb{Z}/2$ `Chapter7/NonVacuity · Examples.reflectionIsotropyZeroEquiv`, $T_x = 1$ `Chapter7/NonVacuity · Examples.trivialSubgroup_reflection`; it is tame and has paracompact orbit space `Chapter7/NonVacuity · Examples.isTame_reflection` `Chapter7/NonVacuity · Examples.hasParacompactOrbitSpace_reflection`.

The trivial action of $\mathbb{Z}/3$ **on** $\mathbb{R}$ `Chapter7/NonVacuity · Examples.trivialZ3EpGroupoid` has $G_x \cong \mathbb{Z}/3$ `Chapter7/NonVacuity · Examples.trivialZ3IsotropyEquiv`, $T_x = G_x$ `Chapter7/NonVacuity · Examples.trivialSubgroup_trivialZ3` `Chapter7/NonVacuity · Examples.trivialZ3TrivialSubgroupEquiv`, is not effective `Chapter7/NonVacuity · Examples.not_isEffective_trivialZ3`, and its regular set is empty `Chapter7/NonVacuity · Examples.regularSet_trivialZ3`: the text's "examples show that the regular set can be empty" (p. 308 (2017 p. 239)). It carries the counterexample to the remark after Definition 7.1.7 (2017: 7.7) (§VII.4).

The pair groupoid of $\mathbb{Z}$ `Chapter7/NonVacuity · Examples.pairGroupoidStr` `Chapter7/NonVacuity · Examples.pairEpGroupoid` is an ep-groupoid in which properness fails over all of $\mathbb{Z}$ `Chapter7/NonVacuity · Examples.pair_exists_not_isProperMap`, refuting the universal reading of Definition 7.1.3 (3) (2017: 7.3) (§VII.2). The abstract finiteness statements of §VII.6 are met by the trivial action of a finite discrete group, whose isotropy set has $|G|$ elements `Chapter7/Discrete · isotropy_snd_witness`.

$\mathbb{Z}/2$ **acting on a dimension-jumping retract** `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.X`, by $(t, z) \mapsto (t, \pm z)$ on the retract of Part I's Remark 4.2.2 (2017: 4.3) example, is a non-effective ep-groupoid whose reduced morphism space $\mathbf{X}^R$ is not Hausdorff `Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.exists_not_t2Space_reducedMor`: the hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017) cannot be dropped (§VII.16; Findings, Chapter 7, item 107).

VII.31 What is not formalized

- $T\varphi$ (Definition 7.1.4 (2017: 7.4), p. 289 (2017 p. 227); gap G.70): that it is well defined, equal to $T\hat{\varphi}(s(\varphi))$, a bounded linear isomorphism on level 0 for φ on level 1 and an sc-operator for smooth φ; $T\varphi$ is only defined, its linearity needing the linear structure of T_xX (gap G.7), and it is deferred to Chapter 8 (§VII.3; Findings, Chapter 7, item 110). The comparison with the classical properness of (s, t) on 2017 p. 227 has no formal content.
- $|X_\infty|$ (2017 p. 228; gap G.72): $|X_k| = \pi(X_k)$ is defined, $|X_\infty|$ and $|X_\infty| = \bigcap_k |X_k|$ are not stated (§VII.3; Findings, Chapter 7, item 112).
- **The density of** X_∞ **in** X_k (2017 p. 227; gap G.73): stated by the text without proof; Part I proves it only for sc-Banach spaces `Chapter1/ScBanach · ScBanach.denseRange_incl`, not for M-polyfolds (§VII.3; Findings, Chapter 7, item 116).

- **The remark after Definition 7.1.7 (2017: 7.7)** (pp. 290–292 (2017 pp. 228–229)): it does not hold as we read it; only the form for functors continuous on morphisms is proved (§VII.4; Findings, Chapter 7, item 9).
- **The tangent facts (3), (4)** before Proposition 7.1.22 (2017: 7.7) (2017 pp. 237–238; gap G.7): not restated for A or $\mathbf{A}$, since T_aA and T_aX carry no linear structure here; in a chart they are Part I's (§VII.12; Findings, Chapter 7, item 40).
- **$\mathbf{X}^R$ for a general ep-groupoid** (pp. 311–324 (2017 pp. 242–244); gap G.71): its Hausdorff and paracompact properties are hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017), discharged for effective X and the trivial ep-groupoid. Hausdorffness fails in general (§VII.16, the splicing example). Not formalized: paracompactness when $\mathbf{X}^R$ is Hausdorff, and whether Hausdorffness with $\mathbf{X}$ second countable suffices (Findings, Chapter 7, items 56, 60, 107, 111). *2021 edition:* paracompactness holds for every ep-groupoid (Proposition 7.2.12), and Theorem 7.2.14 assumes T_R Hausdorff, characterized in Theorem 7.2.22, so the question is settled (§§VII.34–VII.35; gap G.71, closed).
- **The open question of 2017 p. 249**, whether (5) follows from (1)–(4): not formalized and given no gap, since nothing depends on it; Remark 7.5.2 (2017: 7.1) and Remark 7.5.6 (2017: 7.2) have no formal content (§§VII.23, VII.24; Findings, Chapter 7, items 82, 113).
- **Contractibility** of N_g^t, N_g^s in the proof of Theorem 7.6.1 (2017: 7.5) (p. 344 (2017 p. 253)): not needed, connectedness suffices (§VII.26; Findings, Chapter 7, item 29). Nor is the identification $(\mathbf{X}\,{}_s\!\times_t\mathbf{X})^1 = \mathbf{X}^1\,{}_s\!\times_t\mathbf{X}^1$, which the sheet form avoids (§VII.22).
- **Corollary 7.6.6 (2017: 7.3)** holds with Part I's Theorem 5.5.16 (2017: 5.14) as an explicit hypothesis (gap G.53), which holds for finite-dimensional spaces (§VII.29; Findings, Chapter 7, item 95).
- **Gaps of Part I taken up**: G.15 (Lemma 2.7.3 (2017: 2.7), Proposition 2.7.4 (2017: 2.17)) is now formalized (§VII.20); of G.56, the local path connectedness of M-polyfolds is formalized (§VII.7), the sc$^+$-smooth paths inside a path component are not; the tameness of X^1, left open in Part I, is proved (§VII.22) (Findings, Chapter 7, item 114).

VII.32 What the formalization taught us

Continuity is the content of the remark after Definition 7.1.7 (2017: 7.7). The formula $f(\gamma) = (s|\mathbf{U}(\psi))^{-1}(f(s\gamma))$ is right wherever it applies; what it needs is a neighbourhood of φ mapped into the sheet at $f(\varphi)$, and for that the functor must be continuous on morphisms. A functor of the trivial $\mathbb{Z}/3$-action that swaps two isotropy elements on one side of 0 shows that sc-smoothness on objects alone does not give it, as we read the remark (§VII.4).

Connectedness is the hidden hypothesis of the natural representation. Every proof that Φ is a homomorphism, that Γ is a functor, that $\Gamma(1,a) = 1_a$ (Lemma 7.2.2 (2017: 7.2)), and that two natural representations agree compares two continuous sections of s that agree at x; without connectedness they agree only near x. The text chooses a connected $U(x)$ on 2017 p. 231 and never states it again. Making U connected a field of the datum, and proving that M-polyfolds are locally path connected, settles all of these at once (§§VII.7–VII.9, VII.26).

"Small" neighbourhoods and the kernel of Φ. Since sc-smooth maps are not analytic, the kernel of Φ can depend on $U(x)$, and the text's "for $U(x)$ small, $\ker\Phi = T_x$" needs a precise form: a neighbourhood N of x such that every natural representation inside N has kernel T_x, and the intrinsic description $g \in T_x$ iff $s = t$ near g. That description turns Definition 7.2.1 (2017: 7.11) into $T_x = \{1_x\}$, proves the effectiveness of X_R in one line, and gives the conjugation $\varphi T_x \varphi^{-1} = T_y$ that the composition of X_R needs and the text does not state (§§VII.15, VII.17).

The reduced charts do not give Hausdorffness. The proof of Proposition 7.2.10 (2017: 7.9) builds charts and sc-smooth transitions for $\mathbf{X}^R$ without any hypothesis; an M-polyfold also has to be Hausdorff and paracompact, and these do not follow from the charts as we read it. The formalization separates what is proved unconditionally from the two hypotheses, and shows them satisfiable (§VII.16).

A star instead of an induction. Proposition 7.1.15 (2017: 7.5) becomes a construction transported from one object along chosen morphisms, with nothing shrunk afterwards; the text's inductive adjustment of components, and most of its possible misprints, have no counterpart (§VII.10).

Properness does more than sequences. Where the text extracts convergent subsequences, the formalization mostly uses that a proper map is closed or that preimages of compact sets are compact: Proposition 7.1.10 (2017: 7.3), Proposition 7.5.10 (1) (2017: 7.14), the closedness of points behind Lemma 7.3.4 (2017: 7.6), and the closed saturations behind the sc-smoothness in Theorem 7.5.7 (2017: 7.4). With compact sets in place of sequences, Proposition 7.5.10 (2) (2017: 7.14) needs only paracompactness, no regularity (§§VII.5, VII.19, VII.25, VII.28).

Paracompact without Hausdorff. Definition 7.5.1 (2017: 7.15) stipulates that paracompact does not include Hausdorff, and Michael's theorem, which the text quotes for regular Hausdorff spaces, holds for regular spaces; one direction holds in every space. This closes gap G.15 of Part I and gives Corollary 7.3.8 (2017: 7.2) its finite and locally finite unions (§§VII.20, VII.21). For Proposition 7.5.3 (2017: 7.12) a shorter route proves more: $|X^1|$ is metrizable (§VII.23).

Lemma 7.6.5 (2017: 7.10) needs its support bound. As stated, Lemma 7.6.5 (2017: 7.10) bounds the support of f by $V(x)$; the proof of Theorem 7.5.7 (2017: 7.4) needs the bound $\mathrm{cl}\,U(x)$, which the lemma's own proof gives, for the sum $\sum h_i$ to be locally finite, and it applies the lemma to sets U_i'' that need not contain x_i. Stating the lemma with the stronger conclusion and the weaker hypothesis makes the proof of Theorem 7.5.7 (2017: 7.4) go through as written, with $W(x_i)$ in the role of $V(x)$ (§§VII.27–VII.28).

VII.33 The 2021 additions to §7.1: the remark after Definition 7.1.7, Propositions 7.1.14, 7.1.15, Lemma 7.1.16, Corollary 7.1.17

The Springer edition (2021) changes the remark after Definition 7.1.7 and Proposition 7.1.15 and adds Proposition 7.1.14, Lemma 7.1.16 and Corollary 7.1.17; Sections VII.33–VII.41 retell the additions of the 2021 edition, with its numbers and pages (V7.69–V7.98).

The remark after Definition 7.1.7, 2021 form (V7.69) `Chapter7/Book2021Basic · EpGroupoid.remark_7_1_7_2021`. *A functor sc-smooth on objects and continuous on morphisms is sc-smooth on morphisms, through* $f(\gamma) = (s|U(\psi))^{-1}(f(s(\gamma)))$ *on a neighbourhood* $U(\varphi)$ *mapped into a sheet* $U(\psi)$ *at* $\psi = f(\varphi)$. The 2021 text adds the continuity on morphisms, the hypothesis under which §VII.4 proved the 2017 remark; the continuity gives the neighbourhood $U(\varphi)$, the sheets give the formula, and sc-smoothness follows as in §VII.4. The functor of the counterexample of §VII.4 is sc-smooth on objects and not continuous on morphisms, so the 2021 hypothesis excludes it `Chapter7/Book2021Basic · EpGroupoid.remark_7_1_7_2021_remarkFunctor` (Findings, Chapter 7, item 117) (p. 291).

The data (7.3) and Proposition 7.1.14 (V7.70) `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_14_2021`. *For data (7.3) — an open* $W(x)$ *and sheets* $U^s(g)$, $U^t(g)$ *of* s *and* t *at each* $g \in G_x$ *onto* $W(x)$ *— and a representation* (Φ, Γ) *with the properties of Theorem 7.1.13, there is a* Φ*-invariant open* $V(x) \subseteq W(x) \cap U(x)$ *on which* $\Gamma(g, v) = s_g^{-1}(v)$ *and* $\Phi(g)(v) = t(s_g^{-1}(v))$. The data are a structure `Chapter7/Book2021Basic · EpGroupoid.IsotropySheetData` and exist at every object `Chapter7/Book2021Basic · EpGroupoid.exists_isotropySheetData`; "neighbourhoods $U^s(g)$, $U^t(g)$ for g" is read as neighbourhoods of g, without which $\Gamma(g, x) = g = s_g^{-1}(x)$ fails. *The proof.* On the open set N of the $v \in W(x) \cap U(x)$ with $\Gamma(g, v) \in U^s(g) \cap U^t(g)$ for all g, $\Gamma(g, v) = s_g^{-1}(s(\Gamma(g, v))) = s_g^{-1}(v)$; applying t gives the formula for $\Phi(g)$, and $\Gamma(g, v) = t_g^{-1}(\Phi(g)(v))$, the identity (7.4) of the proof, is part of the conclusion. $V(x) = \bigcap_g \Phi(g)^{-1}(N)$, a finite intersection, is open and invariant. Property (3) of Theorem 7.1.13 is a hypothesis as printed and is not used. The same holds for every natural representation `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_14_2021_of_naturalRep` (Findings, Chapter 7, item 118) (p. 296).

Proposition 7.1.15, 2021 form (V7.71) `Chapter7/Book2021Basic · EpGroupoid.proposition_7_1_15_2021`. *For a finite set* Y *of mutually isomorphic objects and* $V \in \mathcal{N}(Y)$ *there is a natural representation of* Y *in* V *whose sheets*

$\mathbf{U}(\psi) \ni \psi$ *are the components of the morphisms between points of* $\bigcup_y U(y)$*, with* $s : \mathbf{U}(\psi) \to U(s(\psi))$*,* $t : \mathbf{U}(\psi) \to U(t(\psi))$ *sc-diffeomorphisms,* $f(\psi) = t \circ (s|\mathbf{U}(\psi))^{-1}$ *and* $t(\psi') = f(\psi)(s(\psi'))$ *on* $\mathbf{U}(\psi)$*.* This is the star construction of §VII.10, whose structure already had the 2021 form of (2). *The 2017 form of (2).* As printed in 2017, $s, t : \mathbf{U}(\psi) \to U(y)$ with one y for both maps; with $y \in U(y)$, disjoint $U(y)$ and $\psi \in \mathbf{U}(\psi)$ this forces $s(\psi) = t(\psi)$ `Chapter7/Book2021Basic · EpGroupoid.src_eq_tgt_of_printed_2017`, and for $\mathbb{Z}/2 \ltimes \mathbb{R}$ by reflection with $Y = \{1, -1\}$ the hypotheses hold, a natural representation of the 2021 form exists, and no data of the printed 2017 form exist `Chapter7/Book2021Basic · EpGroupoid.not_printed_2017_reflection`. So the 2017 form does not hold as we read it, and the 2021 form is the one proved (Findings, Chapter 7, items 25, 119) (p. 297).

Lemma 7.1.16 (V7.72) `Chapter7/Book2021Basic · EpGroupoid.lemma_7_1_16_2021`. *If* $\psi'_k \in \mathbf{U}(\psi)$ *converge to* ψ' *and* $s(\psi'_k)$ *or* $t(\psi'_k)$ *converges to a point of* $U(s(\psi)) \cup U(t(\psi))$*, then* $\psi' \in \mathbf{U}(\psi)$ *and the sources and targets converge inside* $U(s(\psi))$*,* $U(t(\psi))$*.* The four cases reduce to two, since a limit lying in $U(b)$ of points of $U(a)$ lies in $U(a)$, the $U(y)$ being open and disjoint `Chapter7/Book2021Basic · EpGroupoid.NaturalRepFinite.mem_U_of_tendsto`; then s on $\mathbf{U}(\psi)$ is a homeomorphism onto $U(s(\psi))$ and the limits agree (Findings, Chapter 7, item 120) (p. 300).

Corollary 7.1.17 (4)–(7) (V7.73). *(4) For* $x \in Y$*, the natural representation of* G_x *on* $U(x)$ *has* $\Gamma(g, z) = (s|\mathbf{U}(g))^{-1}(z)$ *and* $f(g) = \Phi(g)$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_4_2021` — by a topological argument, the image of the connected $U(x)$ under $\Gamma(g, \cdot)$ lying in one component, since $f(\psi) = f(\psi')$ does not force $\psi = \psi'$ in a non-effective ep-groupoid. *(5)* $f(\varphi)$ is equivariant along the conjugation $g \mapsto \varphi \circ g \circ \varphi^{-1}$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_5_2021`. *(6)* The multiplication is an sc-diffeomorphism from $\mathbf{U}(\psi)\,{}_s\!\times_t \mathbf{U}(\varphi)$, an open subset of the fibred-product sub-M-polyfold of Part I's Proposition 2.5.20 (2017: 2.15), onto $\mathbf{U}(\psi \circ \varphi)$ `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_6_2021`: it is $(s|\mathbf{U}(\psi \circ \varphi))^{-1} \circ s \circ \mathrm{pr}_2$ once its image is known to lie in $\mathbf{U}(\psi \circ \varphi)$, which follows, as in the text, from the connectedness of the domain and the continuity of the multiplication, without sequences. *(7)* The $U(y)$ can be taken with the properness property `Chapter7/Book2021Basic · EpGroupoid.corollary_7_1_17_7_2021` (Findings, Chapter 7, item 120) (p. 301).

VII.34 The 2021 §7.2: T_x, the topology T_R, Lemma 7.2.11, Propositions 7.2.12, 7.2.13 and Theorem 7.2.14

Definition 7.2.5 and Lemma 7.2.6 (V7.74) `Chapter7/ReducedCriterion · EpGroupoid.definition_7_2_5_2021` `Chapter7/ReducedCriterion · EpGroupoid.lemma_7_2_6_2021`. *T_x is the maximal subgroup of G_x acting trivially on some small $W(x)$; conjugation by $\varphi : x \to y$ maps T_x onto T_y.* For a natural representation, T_x^V is the group of the g with $\Phi(g) = \mathrm{id}$ on $V \cap U(x)$ `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn` `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.mem_kerOn`; it grows as V shrinks `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_anti`, is normal for invariant V `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_normal` and lies in the intrinsic T_x of §VII.15 `Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.kerOn_le_trivialSubgroup`. "Maximal" is read as two statements: $T_x \supseteq T_x^W$ for every open $W \ni x$, with equality for every small W, by §VII.15 Step 3. Lemma 7.2.6 is the conjugation $\varphi T_x \varphi^{-1} = T_y$ that §VII.15 had proved for the composition of X_R, now an isomorphism `Chapter7/ReducedCriterion · EpGroupoid.trivialSubgroupEquiv_2021` `Chapter7/ReducedCriterion · EpGroupoid.trivialSubgroupEquiv_2021_apply` (Findings, Chapter 7, item 121) (p. 310).

Remark 7.2.8 and Lemma 7.2.9 (V7.75) `Chapter7/ReducedCriterion · EpGroupoid.remark_7_2_8_2021`. *T_R is the finest topology on $\mathbf{X}^R$ for which π is continuous; in general $(X, \mathbf{X}^R)$ cannot be made an ep-groupoid, but it can when T_R is Hausdorff.* The finest such topology is the quotient topology of §VII.16, and π is an open quotient map `Chapter7/ReducedCriterion · EpGroupoid.isOpenQuotientMap_reducedπ`; Lemma 7.2.9 is §VII.16 Steps 2 and 3 `Chapter7/Reduced · EpGroupoid.isOpenMap_reducedπ` `Chapter7/Reduced · EpGroupoid.isOpenEmbedding_domRestrict_reducedπ`. In Definition 7.2.7 the target map is printed "$s([\varphi]) = t_X(\varphi)$", a possible misprint for $t([\varphi])$ `Chapter7/Reduced · EpGroupoid.reducedTgt`. The remark's "in general not" is what the counterexample of §VII.16

shows (Findings, Chapter 7, items 122, 123) (p. 311).

Lemma 7.2.11 (V7.76). *If $\varphi_1, \dots, \varphi_k$ are the other members of the class of φ_0, there are neighbourhoods $U(\varphi_i)$ with disjoint closures, each mapped by π homeomorphically onto one $U([\varphi_0])$, which contain every ψ with $[\psi] \in U([\varphi_0])$.* The class is finite `Chapter7/ReducedCriterion · EpGroupoid.finite_reducedπ_preimage`, by properness. The conclusion is a predicate `Chapter7/ReducedCriterion · EpGroupoid.IsReducedEvenlyCovered`, with "homeomorphism onto $U([\varphi_0])$" read as: $\pi(U(\varphi_i)) = U([\varphi_0])$ and π an open embedding on $U(\varphi_i)$. *As we read it, the lemma needs T_R Hausdorff.* The text's proof of (3) takes a sequence ψ_k with $[\psi_k] \to [\varphi_0]$ and uses that the limit of $[\psi_k]$ is unique. Without that, (1)–(3) at every unit already make $\mathbf{X}^T$ closed `Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_isReducedEvenlyCovered`, and (1)–(3) at every morphism are equivalent to T_R Hausdorff `Chapter7/ReducedCriterion · EpGroupoid.isReducedEvenlyCovered_iff_t2Space`; in the ep-groupoid of §VII.16 they fail at $(1, o_0)$, whose class contains only $(1, o_0)$ while the morphisms $(-1, (-1/(n+1), 0))$ have classes converging to it `Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.reducedπ_preimage_one Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isReducedEvenlyCovered`. *Under T_R Hausdorff*, the setting of Theorem 7.2.14 where the lemma is used, every open $N \supseteq \pi^{-1}([\varphi_0])$ contains $\pi^{-1}(O)$ for an open $O \ni [\varphi_0]$, which gives (3) for sheets at the finitely many members of the class, chosen with disjoint closures `Chapter7/ReducedCriterion · EpGroupoid.exists_preimage_reducedπ_subset_of_t2Space Chapter7/ReducedCriterion · EpGroupoid.isReducedEvenlyCovered_of_t2Space`, and the lemma holds `Chapter7/ReducedCriterion · EpGroupoid.lemma_7_2_11_2021` (Findings, Chapter 7, item 124) (p. 315).

Proposition 7.2.12 (V7.77) `Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_12_2021`. *$(\mathbf{X}^R, T_R)$ is paracompact, for every ep-groupoid.* **The text's proof** refines an open cover through the neighbourhoods of Lemma 7.2.11, which as we read it need T_R Hausdorff. **The formalized proof** needs no hypothesis. *Step 1, descent.* Paracompactness descends along an open continuous surjection p if every point q has a compact K such that every open $N \supseteq K$ contains $p^{-1}(O)$ for some open $O \ni q$ `Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_of_isOpenMap`: the images of a locally finite precise refinement upstairs form one downstairs, with no separation axiom. *Step 2, the compact set.* For $q = [\varphi]$ and V the open set of Definition 7.1.3 (3) at $s(\varphi)$, $K = \{\psi \in s^{-1}(\operatorname{cl} V) \mid (s,t)(\psi) = (s,t)(\varphi)\}$ is compact by properness and contains the class of φ; (s,t) is proper, hence closed, on $s^{-1}(\operatorname{cl} V)$, so an open $N \supseteq K$ contains $\pi^{-1}(O)$ for an open $O \ni q$ `Chapter7/ReducedCriterion · EpGroupoid.exists_isCompact_preimage_reducedπ_subset`. So `ParacompactSpace X.ReducedMor` is an instance `Chapter7/ReducedCriterion · EpGroupoid.paracompactSpace_reducedMor`, and of the two hypotheses of §VII.16 only the Hausdorff one remains (gap G.71) (Findings, Chapter 7, items 111, 125) (p. 317).

Proposition 7.2.13 and Theorem 7.2.14 (V7.78) `Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_13_2021 Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_14_2021`. *The natural charts make $(\mathbf{X}^R, T_R)$ an M-polyfold iff T_R is Hausdorff; if T_R is Hausdorff, $(X, \mathbf{X}^R)$ is naturally an ep-groupoid.* An M-polyfold is Hausdorff and paracompact by definition, so Proposition 7.2.13 is read as the equivalence of "an M-polyfold structure with the atlas of §VII.16 exists" and T_R Hausdorff; one direction is the definition, the other is §VII.16 with Proposition 7.2.12. Theorem 7.2.14 is Proposition 7.10 (2017) of §VII.17 under exactly the hypothesis the formalization had taken: X_R `Chapter7/Reduced · EpGroupoid.reducedEpGroupoid`, with the M-polyfold $\mathbf{X}^R$ `Chapter7/Reduced · EpGroupoid.reducedPolyfold`, the groupoid structure of §VII.17 and the ep-functor $X \to X_R$; it is effective (Proposition 7.2.16) `Chapter7/Reduced · EpGroupoid.isEffective_reducedEpGroupoid`. The 2021 text thus agrees with the reading of §§VII.16–VII.17 (gap G.71, closed in the 2021 form) (Findings, Chapter 7, items 56, 107, 111, 126, 143) (p. 318).

VII.35 Proposition 7.2.17, the local unique continuation property, and Theorem 7.2.22

Proposition 7.2.17 (V7.79) `Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_17_2021`. *T_R is Hausdorff iff $\mathbf{X}^T$ is closed in $\mathbf{X}$. Step 1, the relation.* $[\varphi] = [\psi]$ iff $t\varphi = t\psi$ and $\psi^{-1} \circ \varphi \in \mathbf{X}^T$ `Chapter7/ReducedCriterion · EpGroupoid.reducedπ_eq_iff_comp_inv`; in particular $\varphi \in \mathbf{X}^T$ iff $[\varphi]$ is the class

of a unit Chapter7/ReducedCriterion · EpGroupoid.mem_nonEffectivePart_iff_reducedπ. *Step 2, closed relation.* $(\varphi,\psi)\mapsto\psi^{-1}\circ\varphi$ is continuous on $\{s\varphi=s\psi\}$ Chapter7/ReducedCriterion · EpGroupoid.continuous_comp_fibredProduct Chapter7/ReducedCriterion · EpGroupoid.continuousOn_comp_inv, so the relation is closed when $\mathbf{X}^T$ is Chapter7/ReducedCriterion · EpGroupoid.isClosed_reducedπ_eq_of_isClosed, and a quotient by an open quotient map is Hausdorff iff its relation is closed (Mathlib). *Step 3, conversely,* if T_R is Hausdorff, $\mathbf{X}^T=\{\varphi\mid[\varphi]=[1_{s(\varphi)}]\}$ is closed, the set where two continuous maps into a Hausdorff space agree Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_t2Space. Properness is not needed. Always, $\mathrm{cl}\,\mathbf{X}^T\subseteq\{s=t\}$ Chapter7/ReducedCriterion · EpGroupoid.isClosed_src_eq_tgt Chapter7/ReducedCriterion · EpGroupoid.closure_nonEffectivePart_subset (Findings, Chapter 7, item 127) (p. 320).

Definition 7.2.18 (= 7.4.1), Remarks 7.2.19 and 7.2.20 (V7.80). *(LUC): every object has a natural representation on which $\hat{g}=\mathrm{id}$ on a nonempty open subset forces $\hat{g}=\mathrm{id}$.* Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.HasLUC Chapter7/ReducedCriterion · EpGroupoid.IsLUC. The choice of $U(x)$ does not matter: if $\mathbf{X}^T$ is closed, every natural representation has the property Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.hasLUC_of_isClosed, so LUC holds iff it holds for all of them Chapter7/ReducedCriterion · EpGroupoid.isLUC_iff_forall_naturalRep, and Remark 7.2.20 (inside every $O(x)$) follows Chapter7/ReducedCriterion · EpGroupoid.remark_7_2_20_2021. Remark 7.2.19, some ep-groupoid is not LUC, is §VII.36 Chapter7/ReducedCriterion · EpGroupoid.Examples.remark_7_2_19_2021. Effective ep-groupoids are LUC, $\mathbf{X}^T$ being the closed set of units Chapter7/ReducedCriterion · EpGroupoid.isClosed_range_unit Chapter7/ReducedCriterion · EpGroupoid.nonEffectivePart_eq_range_unit_of_isEffective Chapter7/ReducedCriterion · EpGroupoid.isLUC_of_isEffective, and so are the trivial ep-groupoid, the reflection and the non-effective trivial $\mathbb{Z}/3$-action Chapter7/ReducedCriterion · EpGroupoid.isLUC_trivial Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_reflection Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_trivialZ3 (Findings, Chapter 7, items 128, 144) (p. 321).

Proposition 7.2.21, Theorem 7.2.22 and Proposition 7.2.23 (V7.81) Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_22_2021. *T_R Hausdorff, LUC and $\mathbf{X}^T$ closed are equivalent; under LUC, X_R is an ep-groupoid. Proposition 7.2.21, by connectedness.* In a natural representation, $\Gamma(g,y)\in\mathbf{X}^T$ iff $\Phi(g)=\mathrm{id}$ near y Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.eventually_Φ_eq_of_Γ_mem Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.Γ_mem_of_eventually. So the set of $y\in U(x)$ near which $\Phi(g)=\mathrm{id}$ is open, and closed in $U(x)$ when $\mathbf{X}^T$ is closed, hence empty or all of the connected $U(x)$: LUC. Conversely, under LUC, a morphism $\varphi\in\mathrm{cl}\,\mathbf{X}^T$ has $s\varphi=t\varphi=x$, so $g=\varphi\in G_x$; the open sheet $\Gamma(g,U(x))\ni\varphi$ meets $\mathbf{X}^T$ in some $\Gamma(g,y)$, so $\Phi(g)=\mathrm{id}$ near y, LUC gives $\Phi(g)=\mathrm{id}$ on $U(x)$, and $\varphi\in T_x$ Chapter7/ReducedCriterion · EpGroupoid.isClosed_nonEffectivePart_of_isLUC Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_21_2021. With Proposition 7.2.17 this is Theorem 7.2.22, and Proposition 7.2.23 is Theorem 7.2.14 under LUC Chapter7/ReducedCriterion · EpGroupoid.t2Space_reducedMor_of_isLUC Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_23_2021. For the trivial $\mathbb{Z}/3$-action, $\mathbf{X}^T=\mathbf{X}$ is closed and T_R is Hausdorff Chapter7/ReducedCriterion · EpGroupoid.Examples.nonEffectivePart_trivialZ3 Chapter7/ReducedCriterion · EpGroupoid.Examples.t2Space_reducedMor_trivialZ3, the case the first reading had left open (Findings, Chapter 7, items 60, 128, 129) (pp. 321–323).

VII.36 Appendices 7.6.3 and 7.6.4: Zhou's example, and the reduced groupoid of X^1

Appendix 7.6.3 (V7.82) Chapter7/ReducedCriterion · EpGroupoid.Examples.remark_7_2_19_2021. *Zhou's example: $\mathbb{Z}/2$ acting on the retract of Part I's Subsection 1.4.4 by $(s,f)\mapsto(s,-f)$; $\mathrm{cl}(\mathbf{X}^T)\setminus\mathbf{X}^T$ is one point, so T_R is not Hausdorff and LUC fails.* The formalization uses its own example of §VII.16, $\mathbb{Z}/2$ acting by $(t,z)\mapsto(t,\pm z)$ on the retract of Part I's Remark 4.2.2 (2017: 4.3), which has the same features: the action is trivial over $t<0$ and moves points arbitrarily close to the origin over $t>0$. The point $(-1,o_0)$ is a limit of $(-1,(t,0))$, $t<0$, which lie in $\mathbf{X}^T$, and is not in $\mathbf{X}^T$ Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.mem_closure_nonEffectivePart Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isClosed_nonEffectivePart, so by Theorem 7.2.22 LUC fails Chapter7/ReducedCriterion · EpGroupoid.Examples.JumpingRetract.not_isLUC and T_R is not Hausdorff,

as §VII.16 had shown directly Chapter7/ReducedNotHausdorff · EpGroupoid.Examples.JumpingRetract.not_t2Space_reducedMor. $\mathbf{X}^T$ is not computed exactly (gap G.83). The counterexample found in the first reading is thus consistent with the 2021 text: it is of the kind Remark 7.2.8 announces and Appendix 7.6.3 gives, and it fails LUC as Theorem 7.2.22 requires (Findings, Chapter 7, items 107, 129, 130) (p. 348).

Appendix 7.6.4, Theorem 7.6.7 and Corollary 7.6.8 (V7.83) Chapter7/ReducedCriterion · EpGroupoid.theorem_7_6_7_2021. *Under (SLUC), $(\mathbf{X}^T)^1 = (\mathbf{X}^1)^T$, and the identification of $(X^1)_R$ with $(X_R)^1$ is an sc-diffeomorphism; likewise for every i. (7.24).* A morphism of level 1 in $\mathbf{X}^T$ lies in $(\mathbf{X}^1)^T$, since $s = t$ near it in $\mathbf{X}$ gives $s = t$ near it in $\mathbf{X}^1$ Chapter7/ReducedCriterion · EpGroupoid.mem_nonEffectivePart_liftOne_of_levelVal. *The converse* follows the text: for g acting trivially near x in X^1, $\hat{g} = \mathrm{id}$ on $U(x) \cap X_1$ by LUC for X^1, through the level-one form of Γ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.levelOneΓ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.continuousOn_levelOneΓ Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.eventually_of_levelOneΓ_mem Chapter7/ReducedCriterion · EpGroupoid.NaturalRep.levelOneΓ_mem_of_eventually, and on $U(x)$ by density Chapter7/ReducedCriterion · EpGroupoid.levelVal_mem_nonEffectivePart_of_liftOne. The text uses that $U(x) \cap X_1$ is connected in X^1 and dense in $U(x)$; both are a hypothesis Chapter7/ReducedCriterion · EpGroupoid.HasConnectedDenseLevelOneReps, the density being that of X_1 in X (gap G.73). Then $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ Chapter7/ReducedCriterion · EpGroupoid.nonEffectivePart_liftOne_eq, the two relations agree on level 1 Chapter7/ReducedCriterion · EpGroupoid.liftOne_reducedπ_eq_iff, and $[\varphi]_1 \mapsto [\varphi]$ Chapter7/ReducedCriterion · EpGroupoid.reducedLevelOneMap is well defined, injective and continuous with range $\pi(\mathbf{X}_1)$. Of (SLUC) only LUC for X^1 is used. Corollary 7.6.8 takes "(ZHOU) implies LUC for X^1" as a hypothesis Chapter7/ReducedCriterion · EpGroupoid.corollary_7_6_8_2021, which §VII.38 discharges Chapter7/Zhou · EpGroupoid.corollary_7_6_8_2021_of_isZhou. All hypotheses hold for the trivial ep-groupoid of a point Chapter7/ReducedCriterion · EpGroupoid.Examples.pointEpGroupoid Chapter7/ReducedCriterion · EpGroupoid.Examples.isLUC_liftOne_point Chapter7/ReducedCriterion · EpGroupoid.Examples.hasConnectedDenseLevelOneReps_point Chapter7/ReducedCriterion · EpGroupoid.Examples.theorem_7_6_7_2021_point Chapter7/Zhou · EpGroupoid.Examples.corollary_7_6_8_2021_of_isZhou_point. Not formalized: the sc-diffeomorphism, which needs the M-polyfold $(\mathbf{X}^R)^1$, and $i \geq 2$ (gap G.82) (Findings, Chapter 7, item 131) (pp. 348–351).

VII.37 §7.4 (new in the 2021 edition): index lifts, Lemma 7.4.4, the Part I facts, and Tg

§7.4 has no 2017 counterpart. It introduces the Zhou condition and the unique continuation and regularity properties that the reduced groupoid and the index lifts need.

Index lifts, Proposition 7.4.2 and Remark 7.4.3 (V7.84). The lifts X^i are bundled ep-groupoids Chapter7/Zhou · EP.liftOne Chapter7/Zhou · EP.lift, with $X^0 = X$, $X^{i+1} = (X^i)^1 = (X^1)^i$ Chapter7/Zhou · EP.lift_zero Chapter7/Zhou · EP.lift_succ Chapter7/Zhou · EP.lift_succ'. *Under LUC, the property passes from a connected invariant $U(x)$ to a larger $W(x)$, so $U(x)$ may be taken small or large.* By §VII.35 every natural representation has the property under LUC, so the hypotheses on $U(x)$ are not used Chapter7/Zhou · EpGroupoid.proposition_7_4_2_2021 Chapter7/Zhou · EpGroupoid.remark_7_4_3_2021 (Findings, Chapter 7, item 132) (pp. 328–329).

Lemma 7.4.4 (V7.85) Chapter7/Zhou · EpGroupoid.lemma_7_4_4_2021. *If X^i has LUC, so has every X^k, $k \leq i$.* For $i = 1$ the text shows that the points of $U(x) \cap X_1$ near which $\hat{g} = \mathrm{id}$ in X^1 form an open, closed and nonempty subset of the connected $U(x) \cap X_1$, and concludes by density Chapter7/Zhou · EpGroupoid.NaturalRep.eqOn_of_isClosed_liftOne Chapter7/Zhou · EpGroupoid.isLUC_of_isLUC_liftOne. The connectedness of $U(x) \cap X_1$ in X^1 is used without proof, and the density is that of X_1 in X (gap G.73); both are a hypothesis on each X^k, $k < i$ Chapter7/Zhou · EpGroupoid.HasLevelOneConnectedDenseReps (gap G.85), satisfied on subsingletons and by every lift of the point Chapter7/Zhou · EpGroupoid.hasLevelOneConnectedDenseReps_of_subsingleton Chapter7/Zhou · EpGroupoid.Examples.hasLevelOneConnectedDenseReps_lift_pointEP Chapter7/Zhou · EpGroupoid.Examples.lemma_7_4_4_2021_pointEP. In the proof, "$s : W(g) \to W(y)$" for $W(h)$ (possible misprint) (Findings, Chapter 7, item 133) (p. 329).

The Part I facts (V7.86). The proofs of §7.4 use, without stating them, facts about an M-polyfold

that Part I does not prove: X_∞ dense in every X_m (gap G.73) Chapter7/Zhou · MPolyfold.DenseSmooth, $U \cap X_\infty$ connected for connected open U Chapter7/Zhou · MPolyfold.SmoothConnected, maps agreeing near a smooth point in X_∞ have the same tangent there Chapter7/Zhou · MPolyfold.TangentDeterminedSmooth Chapter7/Zhou · MPolyfold.TangentDeterminedSmooth.eq_self, and tangent vectors are approximated by tangent vectors at nearby smooth points, which follows from the openness of $p : TX \to X^1$ (gap G.8) Chapter7/Zhou · MPolyfold.eqOn_tangentFiber_of_mem_closure. They are bundled as one hypothesis Chapter7/Zhou · MPolyfold.SmoothRegular, with the density of $(T_xX)_\infty$ in T_xX separate Chapter7/Zhou · MPolyfold.DenseSmoothTangent, and consequences Chapter7/Zhou · MPolyfold.DenseSmooth.dense Chapter7/Zhou · MPolyfold.DenseSmooth.subset_closure_inter Chapter7/Zhou · MPolyfold.DenseSmooth.exists_smooth. They hold on subsingletons with all points smooth and on the discrete M-polyfold of a point Chapter7/Zhou · MPolyfold.smoothRegular_of_subsingleton Chapter7/Zhou · MPolyfold.smoothRegular_discrete_unit Chapter7/Zhou · MPolyfold.denseSmoothTangent_discrete_unit (gap G.85) (Findings, Chapter 7, item 134) (pp. 331–340).

The tangent Tg (V7.87). For a smooth x and $g \in G_x$, Chapter 7's tangent of a morphism (Definition 7.1.4, $Tt \circ (Ts|T_g\mathbf{X})^{-1}$, without charts, §VII.3) is, on T_xX, the tangent $T\Phi(g)$ of $\hat{g}$, for every natural representation Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_Γ Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_eq_tΦ Chapter7/Zhou · EpGroupoid.NaturalRep.tΦ_eq_tΦ: $t = \Phi(g) \circ s$ on the sheet $\Gamma(g, \cdot)$. At a fixed point z of $\Phi(g)$, $\Gamma(g, z) \in G_z$ acts near z as g Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ_val Chapter7/Zhou · EpGroupoid.NaturalRep.eventually_Φ_eq_Γ Chapter7/Zhou · EpGroupoid.NaturalRep.tangentMor_isoΓ, and lies in T_z iff $\Phi(g) = \mathrm{id}$ near z Chapter7/Zhou · EpGroupoid.NaturalRep.isoΓ_mem_trivialSubgroup_iff. Tg is a set map on the fibre; (ZHOU) asks only that it be the identity, so the linear structure of T_xX (gap G.7) is not needed (Findings, Chapter 7, item 135) (p. 331).

VII.38 §7.4: (ZHOU), (ZHOU'), unique continuation, (SLUC) and (VSLUC)

Definitions 7.4.5 and 7.4.7 (V7.88). *(ZHOU): for every smooth x and $g \in G_x$, $Tg = \mathrm{id}$ on T_xX implies $g \in T_x$* Chapter7/Zhou · EpGroupoid.IsZhou. T_x^∞, the g acting trivially near x in X_∞ Chapter7/Zhou · EpGroupoid.trivialSetInf Chapter7/Zhou · EpGroupoid.trivialSubgroupInf, testable in any natural representation Chapter7/Zhou · EpGroupoid.mem_trivialSetInf_iff Chapter7/Zhou · EpGroupoid.mem_trivialSubgroupInf_iff, contains T_x Chapter7/Zhou · EpGroupoid.trivialSubgroup_le_trivialSubgroupInf. *(ZHOU'): $Tg = \mathrm{id}$ on $(T_xX)_\infty$ implies $g \in T_x^\infty$* Chapter7/Zhou · EpGroupoid.IsZhou'. Both hold when every $T_x = G_x$ Chapter7/Zhou · EpGroupoid.isZhou_of_trivialSubgroup_eq_top Chapter7/Zhou · EpGroupoid.isZhou'_of_trivialSubgroup_eq_top, on subsingletons Chapter7/Zhou · EpGroupoid.isZhou_of_subsingleton Chapter7/Zhou · EpGroupoid.isZhou'_of_subsingleton, for the trivial epgroupoid Chapter7/Zhou · EpGroupoid.isZhou_trivial Chapter7/Zhou · EpGroupoid.isZhou'_trivial and for the trivial $\mathbb{Z}/3$-action, where $T_x \neq 1$ Chapter7/Zhou · EpGroupoid.Examples.isZhou_trivialZ3 Chapter7/Zhou · EpGroupoid.Examples.isZhou'_trivialZ3 (Findings, Chapter 7, items 135, 144) (p. 331).

Lemma 7.4.6 and Proposition 7.4.8 (V7.89) Chapter7/Zhou · EpGroupoid.proposition_7_4_8_2021. *(ZHOU) and (ZHOU') are equivalent. (ZHOU) □ (ZHOU')*: if $Tg = \mathrm{id}$ on the dense $(T_xX)_\infty$, then on T_xX, and (ZHOU) gives $g \in T_x \subseteq T_x^\infty$ Chapter7/Zhou · EpGroupoid.lemma_7_4_6_2021. *(ZHOU') □ (ZHOU)*: the text uses sequences; the formalization shows that the smooth points y of $U(x)$ that $\Phi(g)$ fixes with tangent the identity form an open and closed subset of the connected $U(x)_\infty$ Chapter7/Zhou · EpGroupoid.NaturalRep.mem_tangentU_of_fiber Chapter7/Zhou · EpGroupoid.NaturalRep.tΦ_eq_self_of_eqOn Chapter7/Zhou · EpGroupoid.NaturalRep.fix_and_tangent_of_mem_closure Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_smooth_clopen Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_smooth_tangent; continuity and density give $\Phi(g) = \mathrm{id}$ near x Chapter7/Zhou · eq_of_mem_closure_of_continuousAt Chapter7/Zhou · EpGroupoid.isZhou_of_isZhou'. The Part I facts of §VII.37 are hypotheses, satisfied by the point Chapter7/Zhou · EpGroupoid.Examples.proposition_7_4_8_2021_point. "$z_k \to Y$" for $z_k \to y$ in the proof (possible misprint) (Findings, Chapter 7, item 136) (pp. 331–332).

Proposition 7.4.10 and Corollary 7.4.11 (V7.90) Chapter7/Zhou · EpGroupoid.proposition_7_4_10_2021. *(ZHOU) implies LUC, and then every X^i has LUC.* The text shows that the smooth points near which $\hat{g} = \mathrm{id}$ form an

open and closed subset of $U(x)_\infty$, closedness by approximating tangent vectors at a limit point by tangent vectors at nearby smooth points. The formalization isolates the conclusion as unique continuation through X_∞ `Chapter7/Zhou · EpGroupoid.IsSmoothUC` `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_isZhou`, which gives LUC for X, by the density of X_∞, and for X^1, through Proposition 7.2.21, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ being closed in $\mathbf{X}^1$ `Chapter7/Zhou · EpGroupoid.isLUC_of_isSmoothUC` `Chapter7/Zhou · EpGroupoid.isLUC_liftOne_of_isSmoothUC` `Chapter7/Zhou · EpGroupoid.corollary_7_4_11_2021`. This form for every X^i takes (ZHOU) for every X^i, the conclusion of Proposition 7.4.9, as a hypothesis `Chapter7/Zhou · EpGroupoid.corollary_7_4_11_2021_lift`; with (ZHOU) for X only it is §VII.43 (gap G.84, closed) (Findings, Chapter 7, items 136, 137) (pp. 333–334).

Definition 7.4.12 and Theorem 7.4.13 (V7.91) `Chapter7/Zhou · EpGroupoid.theorem_7_4_13_2021`. *(SLUC), every X^i has LUC* `Chapter7/Zhou · EP.IsSLUC`, *follows from (ZHOU).* Proposition 7.4.9, (ZHOU) for X gives (ZHOU) for every X^i, needs the identification of $(X^i)_\infty$ with X_∞ and of the tangent spaces; this form of the theorem takes (ZHOU) and the Part I facts for every X^i, and Proposition 7.4.9 itself, with the theorem for (ZHOU) on X only, is §VII.43 (gap G.84, closed). They hold for every lift of the point `Chapter7/Zhou · EpGroupoid.Examples.pointEP` `Chapter7/Zhou · EpGroupoid.Examples.lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.smoothRegular_lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.isZhou_lift_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.isSLUC_pointEP` `Chapter7/Zhou · EpGroupoid.Examples.theorem_7_4_13_2021_pointEP` (Findings, Chapter 7, item 137) (p. 334).

Definition 7.4.14, Lemma 7.4.15, Propositions 7.4.16–7.4.18 (V7.92). *(VSLUC): LUC for X_∞, on G_x-invariant open $V(x) \subseteq X_\infty$* `Chapter7/Zhou · EpGroupoid.IsVSLUC`, read with $V(x)$ open in X_∞, inside a natural representation and invariant. It is equivalent to unique continuation through X_∞, given the Part I facts `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_isVSLUC` `Chapter7/Zhou · EpGroupoid.isVSLUC_of_isSmoothUC`, so it gives LUC (Lemma 7.4.15) `Chapter7/Zhou · EpGroupoid.lemma_7_4_15_2021` for X and X^1 `Chapter7/Zhou · EpGroupoid.proposition_7_4_16_2021` by the argument of Proposition 7.4.10, with connectedness of $U(x)_\infty$ in place of the text's maximality argument; (ZHOU) implies it `Chapter7/Zhou · EpGroupoid.isVSLUC_of_isZhou`. This form of Proposition 7.4.17 takes (VSLUC) for every X^i `Chapter7/Zhou · EpGroupoid.proposition_7_4_17_2021`; Proposition 7.4.16 for every X^i, Proposition 7.4.17 for (VSLUC) on X only and Proposition 7.4.18, which needs natural representations of X^N in terms of those of X, are §VII.43 (gap G.84, closed). Witnesses: `Chapter7/Zhou · EpGroupoid.isSmoothUC_of_subsingleton` `Chapter7/Zhou · EpGroupoid.isVSLUC_of_subsingleton` `Chapter7/Zhou · EpGroupoid.isLUC_of_subsingleton_mor` `Chapter7/Zhou · EpGroupoid.isVSLUC_trivial` `Chapter7/Zhou · EpGroupoid.isSmoothUC_trivial` (Findings, Chapter 7, items 136, 137) (pp. 335–336).

VII.39 §7.4: (LR), (SLR), and the example of Appendix 7.6.3

Definitions 7.4.19, 7.4.21 and Lemma 7.4.20 (V7.93). *(LR): inside every neighbourhood of x a natural representation such that every sc-diffeomorphism β of $U(x)$ with $|\beta(y)| = |y|$ equals some $\hat{g}$* `Chapter7/Zhou · EpGroupoid.IsLR`; *(LR'): the same for β onto an open image with $\beta(x) = x$* `Chapter7/Zhou · EpGroupoid.IsLR'`. "Arbitrarily small" is "inside every open $O \ni x$", and $|\beta(y)| = |y|$ is $\pi(\beta(y)) = \pi(y)$. *Lemma 7.4.20: such a β maps $U(x)$ onto $U(x)$* `Chapter7/Zhou · EpGroupoid.lemma_7_4_20_2021`. Two points of $U(x)$ in one orbit are related by the natural action, so where $\beta(y) \in U(x)$, $\beta(y) = \hat{g}(y)$ for some g `Chapter7/Zhou · EpGroupoid.NaturalRep.exists_Φ_eq_of_π_eq`; so $U(x) \cap \beta^{-1}(U(x))$ is covered by the finitely many closed sets $\{\beta = \hat{g}\}$, and connectedness of $U(x)$, instead of the text's paths, gives $\beta(U(x)) \subseteq U(x)$ and then equality; the only point of $U(x)$ in the orbit of x is x `Chapter7/Zhou · EpGroupoid.NaturalRep.eq_self_of_π_eq`. Hence (LR) and (LR') are equivalent `Chapter7/Zhou · EpGroupoid.isLR_iff_isLR'` (Findings, Chapter 7, item 138) (pp. 337–338).

Proposition 7.4.22, Definition 7.4.23 and Theorem 7.4.24 (V7.94) `Chapter7/Zhou · EpGroupoid.proposition_7_4_22_2021`. *(ZHOU) implies (LR), and (SLR), every X^i has (LR)* `Chapter7/Zhou · EP.IsSLR`. The text passes to the reduced ep-groupoid and uses the density of the regular points. The formalization does neither: $U(x)_\infty$ is covered by the closed sets $\{\beta = \hat{g}\}$; the union of their X_∞-interiors is open and dense, and two pieces whose closures meet have $T\hat{g} = T\hat{g}'$ at a common smooth point, so $\hat{g} = \hat{g}'$ by (ZHOU) and LUC `Chapter7/Zhou · EpGroupoid.NaturalRep.eqOn_Φ_of_tangent`; connectedness of $U(x)_\infty$ and density finish `Chapter7/Zhou · EpGroupoid.NaturalRep.exists_eq_Φ_of_isZhou`. This form of Theorem 7.4.24 takes (ZHOU) for every X^i,

and the form for (ZHOU) on X only is §VII.43 (gap G.84, closed) Chapter7/Zhou · EpGroupoid.theorem_7_4_24_2021 Chapter7/Zhou · EpGroupoid.Examples.isSLR_pointEP (Findings, Chapter 7, item 138) (pp. 338–340).

The example of Appendix 7.6.3 and (ZHOU) (V7.95) Chapter7/Zhou · EpGroupoid.Examples.JumpingRetract.not_isZhou. *The example fails the Zhou condition (p. 331).* For the analogue of §VII.36: it is not LUC, and (ZHOU) implies LUC by Proposition 7.4.10, so it fails (ZHOU), given the Part I facts for its object M-polyfold, which are a hypothesis (gap G.83) (Findings, Chapter 7, item 139) (p. 331).

VII.40 Appendix 7.6.5: the boundary of a tame ep-groupoid, Theorem 7.6.9 (topological part)

Transport along index-preserving local homeomorphisms (V7.96). The maps (7.25)–(7.28) are defined through sc-diffeomorphisms of sheets; what they need of them is topological. A map that is a homeomorphism of an open W onto an open set and preserves the corner index Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn Chapter7/TameBoundaryEp · TameBoundary.IsDLocalHomeo Chapter7/TameBoundaryEp · TameBoundary.isDHomeoOn_id carries faces, face-structured neighbourhoods, the morphisms of Chapter 2's L_X and F_X and their extensions to the corresponding objects Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.isFaceIn_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.mem_faceStructuredNbhdsAt_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.isMor_image Chapter7/TameBoundaryEp · TameBoundary.IsDHomeoOn.image_faceExt, so it maps triples (y, V, θ) to triples Chapter7/TameBoundaryEp · TameBoundary.BTriple.map with a class independent of W Chapter7/TameBoundaryEp · TameBoundary.map_cls_eq. For an index-preserving local homeomorphism f, $bf[y, V, \theta] = [f(y), f(V), f(\theta)]$ Chapter7/TameBoundaryEp · TameBoundary.bLift Chapter7/TameBoundaryEp · TameBoundary.bLift_cls is the unique lift through B of this kind Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf.unique Chapter7/TameBoundaryEp · TameBoundary.IsLiftOf.comp Chapter7/TameBoundaryEp · TameBoundary.isLiftOf_id, continuous, open and a local homeomorphism Chapter7/TameBoundaryEp · TameBoundary.continuous_bLift Chapter7/TameBoundaryEp · TameBoundary.isOpenMap_bLift Chapter7/TameBoundaryEp · TameBoundary.isLocalHomeomorph_bLift; B has compact preimages of compact sets Chapter7/TameBoundaryEp · TameBoundary.isCompact_preimage_bMap, and representatives can be centred Chapter7/TameBoundaryEp · TameBoundary.exists_rep_centered. Sc-diffeomorphisms preserve the degeneracy index Chapter7/TameBoundaryEp · MPolyfold.IsScDiffeoOn.isDHomeoOn Chapter7/TameBoundaryEp · MPolyfold.IsLocalScDiffeo.isDLocalHomeo (Findings, Chapter 7, item 140) (pp. 353–355).

(7.25)–(7.29), the boundary groupoid (V7.97) Chapter7/TameBoundaryEp · EpGroupoid.boundaryGroupoid. For a tame ep-groupoid, $\mathbf{X}$ is tame Chapter7/TameBoundaryEp · EpGroupoid.isTame_PMor, and bX, $b\mathbf{X}$ are its tame boundaries Chapter7/TameBoundaryEp · EpGroupoid.cornerObj Chapter7/TameBoundaryEp · EpGroupoid.cornerMor Chapter7/TameBoundaryEp · EpGroupoid.bObj Chapter7/TameBoundaryEp · EpGroupoid.bMor. s, t, ι, u are index-preserving local homeomorphisms Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_src Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_tgt Chapter7/TameBoundaryEp · EpGroupoid.isScDiffeoOn_inv Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_inv Chapter7/TameBoundaryEp · EpGroupoid.isDHomeoOn_unit Chapter7/TameBoundaryEp · EpGroupoid.isDLocalHomeo_unit — u as a continuous section of the local homeomorphism s, onto the open set $u(X)$ — so (7.25), (7.27), (7.28) are their lifts Chapter7/TameBoundaryEp · EpGroupoid.bSrc Chapter7/TameBoundaryEp · EpGroupoid.bTgt Chapter7/TameBoundaryEp · EpGroupoid.bInv Chapter7/TameBoundaryEp · EpGroupoid.bUnit, with the printed formulas Chapter7/TameBoundaryEp · EpGroupoid.bSrc_cls Chapter7/TameBoundaryEp · EpGroupoid.bTgt_cls Chapter7/TameBoundaryEp · EpGroupoid.bInv_cls Chapter7/TameBoundaryEp · EpGroupoid.bUnit_cls. *(7.29).* A class of $b\mathbf{X}$ is determined by its image under B and its source Chapter7/TameBoundaryEp · EpGroupoid.eq_of_bMap_eq_of_bSrc_eq, so $m(a, b)$ is the unique class over $\psi \circ \varphi$ with source $s(b)$ Chapter7/TameBoundaryEp · EpGroupoid.exists_bComp Chapter7/TameBoundaryEp · EpGroupoid.bComp Chapter7/TameBoundaryEp · EpGroupoid.bComp_spec; near φ, $\gamma \mapsto (s|\mathbf{W})^{-1}(t\gamma) \circ \gamma$ agrees with a sheet inverse of s composed with s Chapter7/TameBoundaryEp · EpGroupoid.eventuallyEq_comp_sheet, so the printed class $[\psi \circ \varphi, \mathbf{W}\ {}_s\circ_t\ \mathbf{V}, \sigma\ {}_s\circ_t\ \theta]$ is this one Chapter7/TameBoundaryEp · EpGroupoid.bComp_eq. Off the composable pairs m takes a junk value, as in §VII.1. The groupoid laws hold and B is a functor Chapter7/

TameBoundaryEp · EpGroupoid.boundaryFunctor (Findings, Chapter 7, item 140) (pp. 354–355).

Theorem 7.6.9, topological part (V7.98) Chapter7/TameBoundaryEp · EpGroupoid.theorem_7_6_9_2021. *For a tame ep-groupoid, (1) bX and $b\mathbf{X}$ are metrizable, (3) both maps B are continuous (and sc-smooth), (4) a point of index d has d preimages, (5) $(bX, b\mathbf{X})$ is a tame ep-groupoid; the topology is characterized by the openness of the sets $U(x, V, \theta)$ and the maps Φ.* Formalized: metrizability Chapter7/TameBoundaryEp · EpGroupoid.metrizableSpace_bObj Chapter7/TameBoundaryEp · EpGroupoid.metrizableSpace_bMor from Chapter 2's tame boundary; the characterization; continuity of B and the fibre count; s, t local homeomorphisms Chapter7/TameBoundaryEp · EpGroupoid.isLocalHomeomorph_bSrc Chapter7/TameBoundaryEp · EpGroupoid.isLocalHomeomorph_bTgt, u, ι continuous Chapter7/TameBoundaryEp · EpGroupoid.continuous_bSrc Chapter7/TameBoundaryEp · EpGroupoid.continuous_bTgt Chapter7/TameBoundaryEp · EpGroupoid.continuous_bInv Chapter7/TameBoundaryEp · EpGroupoid.continuous_bUnit, m continuous on composable pairs Chapter7/TameBoundaryEp · EpGroupoid.continuousOn_bComp, and the properness property of Definition 7.1.3 (3), by the compact preimages under B Chapter7/TameBoundaryEp · EpGroupoid.exists_isProperMap_bTgt. "Φ is a homeomorphism onto θ" holds, as we read it, only for V centred at x, as for (2.44) in Chapter 2; in general Φ is an open embedding, the form used. Witness: the trivial ep-groupoid of $[0,\infty)^n \oplus W$, $n \geq 1$, with n classes over the corner Chapter7/TameBoundaryEp · EpGroupoid.bSrc_trivial Chapter7/TameBoundaryEp · EpGroupoid.trivial_quadrant_witness. The rest — (2), the sc-smoothness in (3) and (5), d_{bX} in (4) — depends on 2021 Theorem 2.7.13 (2) and is §VII.42 (Findings, Chapter 7, items 140, 141, 145–147) (p. 355).

VII.41 The 2021 additions: what is not formalized, and what they taught us

- **Theorem 7.6.7 beyond $i = 1$ as sets** (gap G.82): the sc-diffeomorphism $(X^1)_R \cong (X_R)^1$ needs the M-polyfold $(\mathbf{X}^R)^1$, and $i \geq 2$ the lifts of X_R (§VII.36; Findings, Chapter 7, item 131).
- **Zhou's retract** (gap G.83): the analogue has a point of $\mathrm{cl}(\mathbf{X}^T) \setminus \mathbf{X}^T$; $\mathbf{X}^T$ is not computed exactly, and its failure of (ZHOU) is conditional on the Part I facts (§§VII.36, VII.39; Findings, Chapter 7, items 130, 139).
- **X^i, $i \geq 2$, in §7.4** (gap G.84, closed since the second batch of 2021 additions): Propositions 7.4.9 and 7.4.16–7.4.18 and the results for every X^i are §VII.43; of "(ZHOU') is the same for X and X^i" only the direction from X to X^i, the one the proof uses, is formalized (Findings, Chapter 7, items 137, 148–150).
- **The Part I facts** (gap G.85): explicit, satisfiable hypotheses, with the connectedness and density of $U(x) \cap X_1$, taken on every X^i for the results on every level, and the level-connected natural representations of Proposition 7.4.18 (§§VII.37, VII.43; Findings, Chapter 7, items 133, 134, 148–150).

The counterexample and the 2021 text agree. The first reading found that $\mathbf{X}^R$ need not be Hausdorff, on a dimension-jumping retract, and took Hausdorffness and paracompactness as hypotheses of Proposition 7.2.10 (2017: 7.9) and Proposition 7.10 (2017). The 2021 text rewrites §7.2 to the same effect: Remark 7.2.8 says the structure fails in general, Theorem 7.2.14 assumes T_R Hausdorff, Theorem 7.2.22 characterizes it by LUC and the closedness of $\mathbf{X}^T$, and Appendix 7.6.3 gives Zhou's example of the same kind. Paracompactness, which the formalization had left open, holds always (§§VII.34–VII.36; gap G.71, closed).

Hausdorffness sits inside Lemma 7.2.11. Its conclusion at every morphism is equivalent to T_R Hausdorff, so it cannot serve, as we read it, as a step towards Hausdorffness; the paracompactness of Proposition 7.2.12 is proved instead from the openness of π and the properness of (s, t) (§VII.34).

Connectedness replaces sequences. Propositions 7.2.21, 7.4.8, 7.4.10, 7.4.22, Lemmas 7.4.4, 7.4.15 and 7.4.20 are proved by showing a set open and closed in a connected $U(x)$, $U(x)_\infty$ or $U(x) \cap X_1$; what this needs of Part I — density and connectedness of the smooth points, locality of tangents, the openness of $p : TX \to X^1$ — becomes visible as explicit hypotheses (§§VII.35–VII.39).

Tg **needs no linear structure.** (ZHOU) asks that a map be the identity; Chapter 7's chart-free tangent of a morphism suffices (§VII.37).

VII.42 Appendix 7.6.5: Theorem 7.6.9 in full (the boundary ep-groupoid)

Theorem 7.6.9 (2)–(5) rest on Part I's Theorem 2.7.13 (2)–(4): for a tame M-polyfold A, the tame boundary bA carries a tame M-polyfold structure `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold` `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_isTame` whose charts are $p \mapsto \pi(\varphi(Bp))$, π a coordinate projection of a tame chart φ and the face read off the triple; $B : bA \to A$ is sc-smooth `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryMap_isScSmooth` and $d_A(Bp) = d_{bA}(p) + 1$ `Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap` `Chapter2/TameBoundaryStructure · MPolyfold.degIndex_tameBoundaryMap_sub_one`; the structure is characterized by the charts Φ of centred triples, and unique with that property `Chapter2/TameBoundaryStructure · MPolyfold.tameBoundaryPolyfold_unique` `Chapter2/TameBoundaryStructure · MPolyfold.theorem_2_7_13_structure_2021`. The text leaves the verification of Theorem 7.6.9 to the reader, with a sketch: for $[\varphi, \mathbf{V}, \theta]$ choose $\mathbf{V}$ with $s : \mathbf{V} \to V$ an sc-diffeomorphism; in the charts $\Phi_{(\varphi,\mathbf{V},\theta)}$ and $\Phi_{(x,V,s(\theta))}$ the map s becomes $s : \theta \to s(\theta)$, an sc-diffeomorphism of faces. The formalization replaces this sc-diffeomorphism by a computation in the charts of bA, which needs only that the map below is sc-smooth (Findings, Chapter 7, items 145–147).

Maps of tame boundaries (V7.99) `Chapter7/TameBoundaryEpStructure · MPolyfold.isScSmoothOn_of_bMap`. *Let $f : A \to B$ be sc-smooth on an open U and $g : bA \to bB$ continuous on an open $O \subseteq B^{-1}(U)$ with $B \circ g = f \circ B$ on O. Then g is sc-smooth on O. Proof.* Sc-smoothness is local. Near $p \in O$ take the chart $\chi = \pi \circ \varphi \circ B$ of bA at p and $\chi' = \pi' \circ \varphi' \circ B$ of bB at $g(p)$, with φ, φ' tame charts at Bp and $f(Bp)$. On a neighbourhood of $\chi(p)$ in the face, $\chi' \circ g \circ \chi^{-1} = \pi' \circ (\varphi' \circ f \circ \varphi^{-1}) \circ \iota$, with ι the inclusion of the face into the model of φ: indeed $B \circ \chi^{-1} = \varphi^{-1} \circ \iota$ near $\chi(p)$ and $B \circ g = f \circ B$. The right side is a composition of sc-smooth maps, π' and ι being linear. *The lifts.* The lift bf of an sc-smooth index-preserving local homeomorphism (§VII.40) covers f and is continuous, so it is sc-smooth `Chapter7/TameBoundaryEpStructure · MPolyfold.isScSmoothMap_bLift`. For a local sc-diffeomorphism f and a sheet $f : W \to V$ with inverse σ, a lift of σ takes $bf(c)$ back to c for $Bc \in W$, by the uniqueness of lifts `Chapter7/TameBoundaryEpStructure · MPolyfold.isLiftOf_bLift_sheet`; so bf is injective on $B^{-1}(W)$, maps it onto $B^{-1}(V)$, and its inverse, covering σ and continuous, is sc-smooth by the same statement: bf is a local sc-diffeomorphism `Chapter7/TameBoundaryEpStructure · MPolyfold.isLocalScDiffeo_bLift` (Findings, Chapter 7, item 146).

The boundary ep-groupoid (V7.100) `Chapter7/TameBoundaryEpStructure · EpGroupoid.boundaryEpGroupoid`. For a tame ep-groupoid, bX and $b\mathbf{X}$ are paracompact `Chapter7/TameBoundaryEpStructure · EpGroupoid.paracompactSpace_bObj` `Chapter7/TameBoundaryEpStructure · EpGroupoid.paracompactSpace_bMor` and carry the structures of Theorem 2.7.13 (2) for the object and the morphism M-polyfold `Chapter7/TameBoundaryEpStructure · EpGroupoid.bPObj` `Chapter7/TameBoundaryEpStructure · EpGroupoid.bPMor` ($\mathbf{X}$ is tame, §VII.40). Since s, t are local sc-diffeomorphisms of $(X, \mathbf{X})$, their lifts (7.25) are local sc-diffeomorphisms `Chapter7/TameBoundaryEpStructure · EpGroupoid.isLocalScDiffeo_bSrc` `Chapter7/TameBoundaryEpStructure · EpGroupoid.isLocalScDiffeo_bTgt`; u and ι are sc-smooth index-preserving local homeomorphisms, so their lifts (7.28), (7.27) are sc-smooth `Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothMap_bUnit` `Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothMap_bInv`. *The multiplication (7.29).* Condition (4) of Definition 7.1.3 (2017: 7.3) is proved in its sheet form (§VII.1, V7.3): for every sheet σ of $s : b\mathbf{X} \to bX$ over V, $b \mapsto m(\sigma(tb), b)$ is sc-smooth on $t^{-1}(V)$ `Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothCompSheet_bComp`. Near b take a sheet σ_1 of $s : \mathbf{X} \to X$ through $B(\sigma(tb))$; then $b' \mapsto m(\sigma(tb'), b')$ is continuous and covers $\gamma \mapsto m(\sigma_1(t\gamma), \gamma)$, which is sc-smooth by (4) for $(X, \mathbf{X})$ in sheet form `Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet`, so the statement on maps of tame boundaries applies; the sheet form gives (4) `Chapter7/EpGroupoid · isScSmoothComp_iff_sheet` `Chapter7/TameBoundaryEpStructure · EpGroupoid.isScSmoothComp_bComp`. With the groupoid and the properness of §VII.40 this is an ep-groupoid, tame by Theorem 2.7.13 (2) `Chapter7/TameBoundaryEpStructure · EpGroupoid.isTame_boundaryEpGroupoid`. *B is an ep-functor*

Chapter7/TameBoundaryEpStructure · EpGroupoid.boundaryEpFunctor: a functor (§VII.40), sc-smooth on objects and morphisms, so a morphism of $\mathcal{EP}$ (Definition 7.1.7 (2017: 7.7)) Chapter7/EpGroupoid · EpFunctor. *The diagram of the sketch* Chapter7/TameBoundaryEpStructure · EpGroupoid.phiTot_bSrc: for $(\varphi, \mathbf{V}, \theta)$ with $\mathbf{V}$ in a sheet of s, s maps $U(\varphi, \mathbf{V}, \theta)$ into $U(x, s(\mathbf{V}), s(\theta))$ and $\Phi_{(x,s(\mathbf{V}),s(\theta))} \circ s = s \circ \Phi_{(\varphi,\mathbf{V},\theta)}$ there (Findings, Chapter 7, items 145, 146) (p. 355).

Theorem 7.6.9 (V7.101) Chapter7/TameBoundaryEpStructure · EpGroupoid.theorem_7_6_9_full_2021. *For a tame ep-groupoid: (1) bX, $b\mathbf{X}$ are metrizable; (2) they carry the tame M-polyfold structures above, characterized as in Theorem 2.7.13 — for a triple with V centred at x, $U(x, V, \theta)$ is open and $\Phi_{(x,V,\theta)}$ is an sc-diffeomorphism onto the face θ, inverse to the homeomorphism of §VII.40 — and every structure with this property is equivalent to it; (3) both maps B are sc-smooth; (4) $d_X(B(p)) - 1 = d_{bX}(p)$ and $d_{\mathbf{X}}(B(a)) - 1 = d_{b\mathbf{X}}(a)$; (5) the boundary ep-groupoid is tame, with the structure maps (7.25)–(7.29), s, t local sc-diffeomorphisms, u, ι sc-smooth, m sc-smooth on the fibred product; and B is an ep-functor. Proof.* (1) is §VII.40; (2)–(4) are Theorem 2.7.13 (2)–(4) for the object and the morphism M-polyfold; (5) and the ep-functor are the previous paragraph. *Readings.* "Natural" in (2) is read through the characterization and its uniqueness, for representatives with V centred at x (in Part I every representative is covered since, with image $\theta \cap V^x$, §II.37; gap G.78, closed); the hypotheses of the two uniqueness statements are satisfied by the structures of (2) themselves. (4) is stated as printed, with the subtraction of natural numbers (B takes values in ∂X, where $d_X \geq 1$); the additive form is Part I's. *Witness* Chapter7/TameBoundaryEpStructure · EpGroupoid.trivial_quadrant_boundaryEp_witness: the trivial ep-groupoid of the tame quadrant $[0, \infty)^n \oplus W$, $n \geq 1$, is tame, its boundary ep-groupoid is tame with nonempty bX, $b\mathbf{X}$, and the classes over the corner have $d_{bX} = n - 1$ (Findings, Chapter 7, items 145, 147) (p. 355).

VII.43 §7.4 on every lift X^i: Propositions 7.4.9, 7.4.16–7.4.18

The smooth points of X^1 (V7.102) Chapter7/ZhouLevels · MPolyfold.continuous_levelDown. The inclusion $X_{m+1} \to X_m$ is continuous; so the identity $(X^1)_\infty \to X_\infty$ and its inverse are continuous Chapter7/ZhouLevels · MPolyfold.shiftSmoothMap Chapter7/ZhouLevels · MPolyfold.shiftSmoothInv Chapter7/ZhouLevels · MPolyfold.continuous_shiftSmoothMap Chapter7/ZhouLevels · MPolyfold.continuous_shiftSmoothInv, a homeomorphism Chapter7/ZhouLevels · MPolyfold.shiftSmoothHomeomorph. This is the text's "$X_\infty = X^i_\infty$" in Propositions 7.4.9 and 7.4.16, as topological spaces.

Proposition 7.4.9 (V7.102) Chapter7/ZhouLevels · EP.proposition_7_4_9_2021. *If X satisfies (ZHOU), so does every X^i.* The text says (ZHOU') is the same for X and X^i, by $X_\infty = X^i_\infty$ and Proposition 7.4.8. *(ZHOU') passes from X to X^1.* An isotropy element of $x \in X_1$ in X^1 is one in X Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal; a natural representation of X^1 at x with $U'(x) \subseteq U(x) \cap X_1$ acts, on $U'(x)$, by the natural action of X Chapter7/ZhouLevels · EpGroupoid.levelVal_Φ_liftOne, and the tangent of a smooth morphism of $\mathbf{X}^1$ is, under $T(X^1) \to TX$ (Part I's Proposition 2.3.17 for $i = 1$), that of $\mathbf{X}$ Chapter7/ZhouLevels · EpGroupoid.tangentLower_tangentMor_liftOne. So $Tg = \mathrm{id}$ on $(T_x X^1)_\infty$ gives $Tg = \mathrm{id}$ on $(T_x X)_\infty$, (ZHOU') for X gives $\hat{g} = \mathrm{id}$ near x in X_∞, hence near x in $(X^1)_\infty$ Chapter7/ZhouLevels · EpGroupoid.isZhou'_liftOne, and by induction on i for every X^i Chapter7/ZhouLevels · EP.isZhou'_lift. The converse, from X^i to X, is not used and not formalized. *The proposition.* (ZHOU) for X gives (ZHOU') for X by Lemma 7.4.6, which uses $(T_x X)_\infty$ dense in $T_x X$; then (ZHOU') for X^i, and (ZHOU) for X^i by Proposition 7.4.8 on X^i, which uses the Part I facts on X^i (§VII.37). Witness Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_9_2021_pointEP (Findings, Chapter 7, item 148) (p. 333).

Corollary 7.4.11, Theorems 7.4.13 and 7.4.24 for every X^i (V7.102). With Proposition 7.4.9, the forms of §§VII.38–VII.39 that took (ZHOU) for every X^i hold for (ZHOU) on X only: every X^i has LUC Chapter7/ZhouLevels · EP.corollary_7_4_11_all_2021, X has (SLUC) Chapter7/ZhouLevels · EP.theorem_7_4_13_all_2021 and (SLR) Chapter7/ZhouLevels · EP.theorem_7_4_24_all_2021. Witnesses Chapter7/ZhouLevels · EpGroupoid.Examples.theorem_7_4_13_all_2021_pointEP Chapter7/ZhouLevels · EpGroupoid.Examples.theorem_7_4_24_all_2021_pointEP (pp. 334, 340).

Propositions 7.4.16 and 7.4.17 (V7.103) Chapter7/ZhouLevels · EP.proposition_7_4_16_all_2021. *If X has (VS-*

LUC), so does every X^i, and in particular every X^i has LUC; so (VSLUC) implies (SLUC). For $x \in X_1$ smooth, the text's $V(x)$ is taken inside a natural representation $U'(x)$ of X^1 at x, with $U'(x) \subseteq U(x)$: $V'(x) = V(x) \cap U'(x)$, an open invariant subset of $(X^1)_\infty = X_\infty$, and $\hat{g}$ is the same map on it Chapter7/ZhouLevels · EpGroupoid.isVSLUC_liftOne Chapter7/ZhouLevels · EP.isVSLUC_lift. The density of X_∞ in every level passes to X^1 Chapter7/ZhouLevels · MPolyfold.DenseSmooth.shift and so to every X^i Chapter7/ZhouLevels · EP.denseSmooth_lift. Lemma 7.4.15 on every X^i (§VII.38), with `Smooth–Connected` on every X^i, gives LUC; and (SLUC) follows Chapter7/ZhouLevels · EP.proposition_7_4_17_all_2021. Witnesses Chapter7/ZhouLevels · EpGroupoid.Examples.isVSLUC_pointEP Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_17_all_2021_pointEP (Findings, Chapter 7, item 149) (p. 336).

Proposition 7.4.18 (V7.104) Chapter7/ZhouLevels · EP.proposition_7_4_18_2021. *(VSLUC) and (SLUC) are equivalent.* One direction is Proposition 7.4.17. *(SLUC)* □ *(VSLUC)*, the text's proof: take $V(x) = U(x) \cap X_\infty$; an open subset of it on which $\hat{g} = \mathrm{id}$ contains $W' \cap X_\infty$ for a nonempty open W' of some X_N; by density and continuity $\hat{g} = \mathrm{id}$ on W', and LUC for X^N gives $\hat{g} = \mathrm{id}$ on $U(x)$, $U(x) \cap X_N$ being connected "by construction". *The neighbourhood basis.* The inclusion $X_N \to X_0$ through the levels Chapter7/ZhouLevels · MPolyfold.levelDownTo Chapter7/ZhouLevels · MPolyfold.continuous_levelDownTo is continuous, and the sets $O \cap X_\infty$, O open in some X_{N+1}, form a neighbourhood basis of every smooth point Chapter7/ZhouLevels · MPolyfold.exists_level_nhds_smooth. *Natural representations one level up.* An isotropy element of $x \in X_1$ in X is one in X^1 Chapter7/ZhouLevels · EpGroupoid.isotropyLevelMk Chapter7/ZhouLevels · EpGroupoid.isotropyLevelMk_levelVal Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal_mul Chapter7/ZhouLevels · EpGroupoid.isotropyLevelVal_inv. If $U(x) \cap X_1$ is connected in X^1, the natural action of X restricts to $U(x) \cap X_1$ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.liftOneΦ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.levelVal_liftOneΦ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.mapsTo_liftOneΦ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.isScDiffeoOn_liftOneΦ and is a natural representation of X^1 at x Chapter7/ZhouLevels · EpGroupoid.NaturalRep.liftOneRep, with the same $\hat{g}$ Chapter7/ZhouLevels · EpGroupoid.NaturalRep.eqOn_of_liftOneRep. The text's "by construction" is a hypothesis: every $U(x) \cap X_m$ connected in X_m Chapter7/ZhouLevels · EpGroupoid.NaturalRep.LevelConnected, which passes to the restriction Chapter7/ZhouLevels · EpGroupoid.NaturalRep.levelConnected_liftOneRep, so LUC for X^N applies to the restriction of $U(x)$ to X^N, built one level at a time Chapter7/ZhouLevels · EP.eqOn_of_level_smooth. Hence, with the density of X_∞ and level-connected natural representations at every smooth point, (SLUC) gives (VSLUC) Chapter7/ZhouLevels · EP.isVSLUC_of_isSLUC. Witnesses Chapter7/ZhouLevels · EpGroupoid.Examples.levelConnected_pointEP Chapter7/ZhouLevels · EpGroupoid.Examples.proposition_7_4_18_2021_pointEP (gaps G.73, G.85) (Findings, Chapter 7, item 150) (p. 336).

Part VIII. Bundles and covering functors (HWZ Chapter 8, §§8.1–8.3, 8.5, 8.6 (2017 §§8.1–8.5))

Chapter 8 continues Part II; as for Chapter 7, its numbers and pages are those of the book (Springer 2021, pp. 357–400), with the 2017 text's in parentheses (2017 pp. 257–289). It treats bundles over ep-groupoids and covering functors. §8.1 equips the tangent $T(X,\mathbf{X}) = (TX, T\mathbf{X})$ of an ep-groupoid with the structure maps (8.1)–(8.9) and proves that it is an ep-groupoid, tame if X is, with an sc-smooth projection functor to X^1 (Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8)); it describes the morphisms of $T\mathbf{X}$ by pairs $(T\varphi, a)$ (Remark 8.1.10 (2017: 8.1)), proves that the tangent of an sc-smooth functor is an sc-smooth functor and that $T : \mathcal{EP} \to \mathcal{EP}$ is a functor (Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3)), defines natural transformations and natural equivalence (Definition 8.1.13 (2017: 8.1)), proves that the tangent of a natural transformation is one (Proposition 8.1.14 (2017: 8.1)), and that the paracompactness of $|X|$ passes to $|TX|$ (Theorem 8.1.15 (2017: 8.4)). §8.2 defines sc-differential forms on ep-groupoids and the sc-de Rham complex (Definitions 8.2.1, 8.2.2 (2017: 8.2, 8.3)). §8.3 defines strong bundles over ep-groupoids (Definition 8.3.1 (2017: 8.4)) with their functorial picture (pp. 375–377 (2017 pp. 271–272)), strong bundle maps and strong bundle functors (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) and section functors (Definition 8.3.4 (2017: 8.7)). §8.5 (2017 §8.4) defines proper covering functors (Definition 8.5.1 (2017: 8.8)), proves that they compose (Proposition 8.5.2 (2017: 8.2)), describes them near a fibre (Theorem 8.5.3 (2017: 8.5), Remark 8.5.4 (2017: 8.2)), defines covering functors of finite groupoids and geometric lifts (Definition 8.5.5 (2017: 8.9)), constructs the proper covering functor of a geometric lift (Proposition 8.5.7 (2017: 8.3)), shows that every proper covering functor is locally of this form (Theorem 8.5.8 (2017: 8.6)), and defines proper strong bundle covering functors (Definition 8.5.9 (2017: 8.10)). The appendix counts the fibres of a proper covering functor (§8.6.1 (2017 §8.5.1): Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11), Proposition 8.6.4 (2017: 8.4)) and describes strong bundle coverings fibrewise (§8.6.2 (2017 §8.5.2)). This Part retells §8.1 in §§VIII.1–VIII.9 — the tangent data (8.1)–(8.9) and the tangents of maps defined on open sets (§VIII.1), Theorem 8.1.1 (2017: 8.1) and Lemma 8.1.2 (2017: 8.1) (§VIII.2), Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5) (§VIII.3), Lemmas 8.1.7 and 8.1.8 (2017: 8.6 and 8.7) (§VIII.4), Lemma 8.1.9 (2017: 8.8) (§VIII.5), Remark 8.1.10 (2017: 8.1) (§VIII.6), Theorems 8.1.11 and 8.1.12 (2017: 8.2 and 8.3) (§VIII.7), Definition 8.1.13 (2017: 8.1), the equivalence relation $\simeq$ and Proposition 8.1.14 (2017: 8.1) (§VIII.8), Theorem 8.1.15 (2017: 8.4) (§VIII.9); §8.3 in §§VIII.10–VIII.14 — strong bundle maps and the pull-back along s (§VIII.10), Definition 8.3.1 (2017: 8.4) and the fibre maps (§VIII.11), the functorial picture (§VIII.12), Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6) and their equivalence (§VIII.13), Definition 8.3.4 (2017: 8.7) (§VIII.14); §8.5 (2017 §8.4) in §§VIII.15–VIII.22 — Definition 8.5.1 (2017: 8.8) and the degeneracy remark (§VIII.15), Proposition 8.5.2 (2017: 8.2) (§VIII.16), Theorem 8.5.3 (2017: 8.5) and Remark 8.5.4 (2017: 8.2) (§VIII.17), Diff_{sc}, Definition 8.5.5 (2017: 8.9) and the discussion after it (§VIII.18), disjoint unions of M-polyfolds (§VIII.19), Proposition 8.5.7 (2017: 8.3) (§VIII.20), Theorem 8.5.8 (2017: 8.6) (§VIII.21), Definition 8.5.9 (2017: 8.10) (§VIII.22); the appendix in §§VIII.23–VIII.25 — §8.6.1 (2017 §8.5.1) and Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11) (§VIII.23), Proposition 8.6.4 (2017: 8.4) (§VIII.24), §8.6.2 (2017 §8.5.2) (§VIII.25); the wit-

nesses and the counterexample (§VIII.26); what is not formalized, among it §8.2 (§VIII.27); and what the formalization taught us (§VIII.28). Each item carries the number of its comparison item (V8.1–V8.43) in `Exposition/verification/ch08.md`. The additions of the 2021 edition (V8.44–V8.52) are retold in §§VIII.29–VIII.33: §8.1 (§VIII.29), §§8.3–8.4 (§VIII.30), §8.5 (§VIII.31), what is not formalized and what they taught us (§VIII.32), and Proposition 8.1.17 (2) for every $(TX)^i$ (§VIII.33).

What is proved, and what is not. Theorems 8.1.1–8.5.8 (2017: 8.1–8.6) are proved, Theorem 8.5.3 (2017: 8.5) with (5) read with s on both vertical arrows; so are Propositions 8.1.14, 8.5.2 and 8.6.4 (2017: 8.1, 8.2 and 8.4), Proposition 8.5.7 (2017: 8.3) for nonempty D, Lemmas 8.1.2–8.6.3 (2017: 8.1–8.11) (Lemma 8.1.2 (2017: 8.1) in the sheet form its proof uses), Remark 8.1.10 (2017: 8.1) without its linearity claims, and Remark 8.5.4 (2017: 8.2). Every definition of the chapter is formalized except Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3). Three places needed more than a reading. Theorem 8.5.3 (5) (2017: 8.5), as printed with t on the right vertical arrow, does not hold as we read it: the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ violates it in every chart at 0, and the square holds with s (§VIII.17). Definition 8.5.5 (2017: 8.9) allows $D = \emptyset$, and then the functor of Proposition 8.5.7 (2017: 8.3) is surjective on objects only if $U(x) = \emptyset$; Proposition 8.5.7 (2017: 8.3) is proved for nonempty D, which holds where Theorem 8.5.8 (2017: 8.6) uses it (§VIII.20). The degeneracy remark after Definition 8.5.1 (2017: 8.8) uses the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, which Part I has for tame M-polyfolds; it is proved for tame X, Y, and as an inequality in general (§VIII.15). Several omitted steps are supplied: the tangents of maps defined on open sets, which the proofs of §8.1 differentiate without a definition (§VIII.1); the tameness of Tr in Lemma 8.1.9 (2017: 8.8), asserted without argument (§VIII.5); that $\simeq$ is an equivalence relation (§VIII.8); the open embedding $G_x \backslash TU(x) \to |TX|$ in the proof of Theorem 8.1.15 (2017: 8.4) (§VIII.9); the ep-groupoid of the functorial picture (§VIII.12); the equivalence of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6) (§VIII.13); and the "one verifies" of Proposition 8.5.7 (2017: 8.3) and of Theorem 8.5.8 (2017: 8.6) (§§VIII.20, VIII.21). Several proofs take another route: Lemmas 8.1.4–8.1.6 (2017: 8.3–8.5), Theorem 8.1.11 (2017: 8.2) and Proposition 8.1.14 (2017: 8.1) by fibre injectivity instead of differentiating identities (§§VIII.3, VIII.7, VIII.8); condition (4) of Definition 7.1.3 (2017: 7.3) for $T\mathbf{X}$ in Chapter 7's sheet form (§VIII.4); Theorem 8.1.15 (2017: 8.4) by metrizability instead of a closed paracompact cover (§VIII.9); Theorem 8.5.3 (2017: 8.5) through Definition 8.5.1 (3) (2017: 8.8) instead of sheets and natural representations in Y (§VIII.17). Not formalized: Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3) (gap G.74), Lemma 8.1.2 (2017: 8.1) as printed, with $T(\mathbf{X}\,{}_s\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\times_{Tt} T\mathbf{X}$ (gap G.75), the linearity of T_xX and of $T\varphi$ (gaps G.7, G.70), and a few remarks that nothing uses (§VIII.27).

Seven structural choices. (1) *The tangent is Part I's.* TX and $T\mathbf{X}$ are Part I's tangent M-polyfolds and Ts, Tt, Tu, $T\iota$ Part I's tangent maps (§§II.8, II.9); for X in `Type u` they live in `Type (max u 1)`, and so does the tangent functor of Theorem 8.1.12 (2017: 8.3). The fibres T_xX carry no linear structure (gap G.7), so no statement about the linearity of a tangent map is formalized. (2) *The composition of $T\mathbf{X}$ is defined by choice*: total, with a junk value off composable pairs, as Chapter 7's composition, and unique on composable pairs. (3) *Fibre injectivity.* s being a local sc-diffeomorphism, two vectors of $T\mathbf{X}$ with the same base point and the same image under Ts are equal; most identities of §8.1 are proved this way instead of by differentiating identities. (4) *Tangents of maps defined on open sets* are built from the charts, as Part I builds Tf, with a junk value off $p^{-1}(U)$. (5) *Condition (4) in sheet form*, as in Chapter 7 (§VII.2): for $T\mathbf{X}$, for the functorial picture of a strong bundle and for the ep-groupoid of a geometric lift; Lemma 8.1.2 (2017: 8.1) is stated in this form. (6) *Strong bundle maps between strong bundles over M-polyfolds are a reading*: Part I defines strong bundle maps only between local models, and the global notion is a map covering a map of the bases, linear on the fibres, sc-smooth on $[0]$ and on $[1]$ (§VIII.10). (7) *Data, not hypotheses.* "A connected and sufficiently small $U(x)$ admitting the natural G_x-action" in Theorems 8.5.3 and 8.5.8 (2017: 8.5 and 8.6) is a datum, a chart of Theorem 8.5.3 (2017: 8.5), as natural representations are in Chapter 7; Definition 8.5.5 (2017: 8.9) is stated with Mathlib's groupoids, C being `SingleObj G`, and its finiteness enters as instance arguments of the results that use it.

Map.

- `Chapter8/Basic`: the tangent data (8.1)–(8.9), Definitions 8.1.13, 8.5.1, 8.5.5 (2017: 8.1, 8.8, 8.9) and the category Diff_{sc}.

- `Chapter8/TangentLocal`: tangents of maps sc-smooth on open sets, of sc-diffeomorphisms between open sets and of local sc-diffeomorphisms.
- `Chapter8/TangentEp`: Theorem 8.1.1 (2017: 8.1), Lemmas 8.1.2–8.1.8 (2017: 8.1–8.7), Remark 8.1.10 (2017: 8.1), Theorems 8.1.11, 8.1.12 (2017: 8.2, 8.3), Proposition 8.1.14 (2017: 8.1).
- `Chapter8/TangentTame`: Lemma 8.1.9 (2017: 8.8), the tangent of a tame retraction, the tameness clauses of Theorems 8.1.1 and 8.1.12 (2017: 8.1 and 8.3).
- `Chapter8/NatTrans`: natural equivalence is an equivalence relation; whiskering.
- `Chapter8/TangentOrbit`: the action of G_x on $TU(x)$, the orbit space of TX, Theorem 8.1.15 (2017: 8.4).
- `Chapter8/StrongBundleMap`: strong bundle maps between strong bundles over M-polyfolds, the pull-back along s.
- `Chapter8/EpStrongBundle`: Definitions 8.3.1–8.3.4 (2017: 8.4–8.7), the functorial picture, Definition 8.5.9 (2017: 8.10), §8.6.2 (2017 §8.5.2).
- `Chapter8/ProperCovering`: the degeneracy remark after Definition 8.5.1 (2017: 8.8), Proposition 8.5.2 (2017: 8.2), §8.6.1 (2017 §8.5.1) with Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11) and Proposition 8.6.4 (2017: 8.4).
- `Chapter8/ProperCoveringLocal`: Theorem 8.5.3 (2017: 8.5) and Remark 8.5.4 (2017: 8.2).
- `Chapter8/SigmaPolyfold`: disjoint unions of M-polyfolds.
- `Chapter8/GeometricLift`: the discussion after Definition 8.5.5 (2017: 8.9), Proposition 8.5.7 (2017: 8.3), Theorem 8.5.8 (2017: 8.6).
- `Chapter8/NonVacuity`: covering functors and a geometric lift; the counterexample to Theorem 8.5.3 (5) (2017: 8.5) as printed.
- `Chapter8/NonVacuityBundle`: witnesses for Definitions 8.3.1–8.5.1 and 8.5.9 (2017: 8.4–8.8 and 8.10).
- `Chapter8/TangentZhou`: Propositions 8.1.16, 8.1.17 and Definition 8.4.2 (§§VIII.29–VIII.30).
- `Chapter8/Book2021`: Definition 8.3.5, Proposition 8.4.1, Example 8.5.6, Definition 8.5.9 in the 2021 form, Lemma 8.5.10 (§§VIII.30–VIII.31).
- `Chapter8/TangentZhouLevels`: Proposition 8.1.17 (2) for every $(TX)^i$, through $T(X^i) \cong (TX)^i$ as ep-groupoids (§VIII.33).

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid as in Part VII, with source and target s, t, composition $m(\psi, \varphi) = \psi \circ \varphi$, unit $u(x) = 1_x$ and inverse $\iota(\varphi) = \varphi^{-1}$. TX and $T\mathbf{X}$ are Part I's tangent M-polyfolds (§II.8), p the projection of a tangent vector to its base point on level 1, and Tf Part I's tangent map of an sc-smooth f (§II.9). For a vector $v \in T\mathbf{X}$ we write $p(v) \in \mathbf{X}_1$ for its base point, so $p(Ts(v)) = s(p(v))$. A *sheet* of s is an open set $\mathbf{W} \subseteq \mathbf{X}$ on which s is an sc-diffeomorphism onto an open V, its inverse σ a *local inverse* of s; for open $U \subseteq X$, $TU = p^{-1}(U)$. G_x is the isotropy group of x, and a natural representation $(U(x), \Phi, \Gamma)$ is Chapter 7's (§VII.7), with $U(x)$ open and connected. Part I results are cited in the same way, by their 2021 numbers with the 2017 numbers in parentheses; the Lean names carry the preprint numbers (Definition 2.5.2, tame retractions, is 2017 Definition 2.17 and preprint Definition 2.34, §II.14; Proposition 2.5.20, fibred products, is 2017 Proposition 2.15 and preprint Proposition 2.46, §II.17; Proposition 2.6.8, the pull-back strong bundle, is 2017 Proposition 2.16 and preprint Proposition 2.54, §II.20; Proposition 2.4.4, 2017 Proposition 2.7, is `proposition_2_24`, §II.11; Lemma 1.3.4, 2017 Lemma 1.1, is preprint Lemma 1.24, §I.18). Findings are those of `Exposition/Findings.md`, "Chapter 8".

VIII.1 The tangent data (8.1)–(8.9), and tangents of maps on open sets

The tangent data (V8.1). The objects of $T(X, \mathbf{X})$ are the points of TX `Chapter2/TangentBundle · MPolyfold.TangentSpace` `Chapter2/TangentBundle · MPolyfold.tangentPolyfold`, the morphisms those of $T\mathbf{X}$; a vector $(\varphi, h) \in T_\varphi\mathbf{X}$ is viewed as a morphism $Ts(\varphi)(h) \to Tt(\varphi)(h)$ (8.2), so that $s_T = Ts$ and $t_T = Tt$ (8.3) `Chapter8/Basic · EpGroupoid.tSrc` `Chapter8/Basic · EpGroupoid.tTgt`, Part I's tangent maps `Chapter2/TangentMap · MPolyfold.tangentMap`. The unit is $u_T = Tu$ and the inverse $\iota_T = T\iota$ (8.9) `Chapter8/Basic · EpGroupoid.tUnit` `Chapter8/Basic · EpGroupoid.tInv`, and the base point $p : T\mathbf{X} \to \mathbf{X}^1$ is Part I's projection `Chapter8/Basic · EpGroupoid.`

tBase Chapter2/TangentBundle · MPolyfold.tangentProj. *The composition* (8.5), $(\psi,k)\circ(\varphi,h)=(\psi\circ\varphi, Ts(\psi\circ\varphi)^{-1}(Ts(\varphi)(h)))$ for $Ts(\psi)(k)=Tt(\varphi)(h)$ (8.4), needs the inverse of Ts on one fibre. *Step 1, fibre bijectivity.* Since s is a local sc-diffeomorphism, Ts maps every fibre $T_\varphi\mathbf{X}$ bijectively onto $T_{s(\varphi)}X$ (Step 4 of the next paragraph). So two vectors of $T\mathbf{X}$ with the same base point and the same image under Ts are equal Chapter8/TangentEp · EpGroupoid.tangent_ext, and every vector over $s(\varphi)$ is $Ts(v)$ for a v over φ Chapter8/TangentEp · EpGroupoid.exists_tSrc_eq. *Step 2, the choice.* For a composable pair, $s(p\,k)=t(p\,h)$ Chapter8/TangentEp · EpGroupoid.src_tBase_eq, and $k\circ h$ is chosen as a vector over $p(k)\circ p(h)$ with $Ts(k\circ h)=Ts(h)$ Chapter8/Basic · EpGroupoid.tComp Chapter8/TangentEp · EpGroupoid.tComp_spec Chapter8/TangentEp · EpGroupoid.tBase_tComp Chapter8/TangentEp · EpGroupoid.tSrc_tComp; off composable pairs it is the junk value h, which no statement uses. By Step 1 the choice is the only possible one, which is (8.5) in this form Chapter8/TangentEp · EpGroupoid.tComp_eq. *Step 3, the units.* $u_T=Tu$ is (8.8), $1_a=(1_x, Ts(1_x)^{-1}a)$: $Tu(a)$ lies over 1_x, and $Ts(Tu(a))=a$ by the chain rule for $s\circ u=\mathrm{id}$, so Step 1 identifies the two Chapter8/TangentEp · EpGroupoid.tUnit_eq. (8.5) writes $T_{\varphi\circ\psi}\mathbf{X}$ for $T_{\psi\circ\varphi}\mathbf{X}$ (possible misprint) (Findings, Chapter 8, items 1, 2) (pp. 357–359 (2017 pp. 257–258)).

Tangents of maps on open sets (V8.2). The proofs of §8.1 differentiate maps that are defined only on open sets, such as a local inverse $(s|\mathbf{U}(\psi))^{-1}$ and the map α of Lemma 8.1.3 (2017: 8.2), and write TU for the part of TX over an open U; Part I defines Tf only for maps sc-smooth on the whole M-polyfold. Here TU is the open set $p^{-1}(U)$ Chapter8/TangentLocal · MPolyfold.tangentPreimage Chapter8/TangentLocal · MPolyfold.isOpen_tangentPreimage, and it is the tangent of the open sub-M-polyfold U when $U\neq\emptyset$ Chapter8/TangentLocal · MPolyfold.isScDiffeoOn_tangentMap_subtypeVal Chapter8/TangentLocal · MPolyfold.isOpenEmbedding_tangentMap_subtypeVal. *Step 1, the construction.* For f sc-smooth on U, Tf on $p^{-1}(U)$ is built from the charts as Part I builds Tf (§II.9), with $U\cap V\cap f^{-1}(V')$ in place of $V\cap f^{-1}(V')$ Chapter8/TangentLocal · MPolyfold.TangentPre.fimageOn Chapter8/TangentLocal · MPolyfold.tangentMapOn; off $p^{-1}(U)$ it takes a junk value, which exists because a nonempty M-polyfold has points on level 1 (E_1 is dense in E_0 and a nonempty retract meets it) Chapter8/TangentLocal · MPolyfold.exists_mem_levelSet_one Chapter8/TangentLocal · MPolyfold.nonempty_tangentSpace. *Step 2, the calculus.* Tf lies over f Chapter8/TangentLocal · MPolyfold.levelVal_tangentProj_tangentMapOn, agrees with Part I's tangent when f is sc-smooth everywhere Chapter8/TangentLocal · MPolyfold.tangentMapOn_eq_tangentMap, satisfies the chain rule Chapter8/TangentLocal · MPolyfold.tangentMapOn_comp and is sc-smooth on $p^{-1}(U)$ Chapter8/TangentLocal · MPolyfold.tangentMapOn_isScSmoothOn. *Step 3, sc-diffeomorphisms between open sets.* The tangent of an sc-diffeomorphism $U\to V$ is an sc-diffeomorphism $p^{-1}(U)\to p^{-1}(V)$ with the tangent of the inverse as its inverse, by the chain rule Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.tangent Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.tangentMap, and it is bijective on every fibre Chapter8/TangentLocal · MPolyfold.IsScDiffeoOn.bijOn_tangentFiber; this is the step "$Ts : T\mathbf{U}(\varphi)\to TU(x)$ is an sc-diffeomorphism" of Lemma 8.1.8 (2017: 8.7). *Step 4, local sc-diffeomorphisms.* A sheet of Tf around v is $p^{-1}(U)$ for a sheet U of f around $p(v)$, so Tf is a local sc-diffeomorphism Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.tangentMap, bijective on the fibres Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.bijOn_tangentFiber and surjective when f is Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.surjective_tangentMap. Step 4 is Lemma 8.1.7's (2017: 8.6) "by construction" (Findings, Chapter 8, item 3) (used on pp. 358–364 (2017 pp. 258–263)).

VIII.2 Theorem 8.1.1 (2017: 8.1), and Lemma 8.1.2 (2017: 8.1)

Theorem 8.1.1 (2017: 8.1) (V8.3) Chapter8/TangentEp · EpGroupoid.theorem_8_1. *$T(X,\mathbf{X})=(TX,T\mathbf{X})$ with the structure maps (8.3)–(8.9) is an ep-groupoid; it is tame if X is; and the projections form an sc-smooth functor $(p,\mathbf{p}) : T(X,\mathbf{X})\to(X,\mathbf{X})^1$.* The text proves it on pp. 363–365 (2017 pp. 263–264) by collecting Lemmas 8.1.2–8.1.9 (2017: 8.1–8.8), and the formalization does the same. The groupoid laws are Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5) (§VIII.3) Chapter8/TangentEp · EpGroupoid.tangentGroupoidStr; conditions (1), (2) and (4) of Definition 7.1.3 (2017: 7.3) are Lemma 8.1.7 (2017: 8.6) and condition (3) is Lemma 8.1.8 (2017: 8.7) (§VIII.4); together they make the ep-groupoid Chapter8/TangentEp · EpGroupoid.tangentEpGroupoid, and `theorem_8_1` records that it has the M-polyfolds TX, $T\mathbf{X}$, the structure maps of the text, and the base point and source of a composite. The tameness sentence is Lemma 8.1.9 (2017: 8.8) (§VIII.5) Chap-

ter8/TangentTame · EpGroupoid.theorem_8_1_tame. *The projection functor.* $(X,\mathbf{X})^1$ is Chapter 7's lift to level 1 (§VII.22) Chapter7/Lift · EpGroupoid.liftOne, and $(p,\mathbf{p})$ is a functor into it Chapter8/TangentEp · EpGroupoid.tangentProjFunctor: it commutes with sources and targets because $p\circ Ts = s\circ p$ and $p\circ Tt = t\circ p$, with the composition because $p(k\circ h) = p(k)\circ p(h)$ (Step 2 of §VIII.1, the text's "This is obvious"), and with the units because $Tu(a)$ lies over 1_x; the two projections are sc-smooth by Part I (Findings, Chapter 8, items 2, 9) (2017 pp. 258, 263–264).

Lemma 8.1.2 (2017: 8.1) (V8.4) Chapter8/TangentEp · EpGroupoid.lemma_8_1. *As printed*: with $T(\mathbf{X}\,{}_s\!\times_t\mathbf{X})$ identified with $T\mathbf{X}\,{}_{Ts}\!\times_{Tt}T\mathbf{X}$, the composition (8.5) is $(\psi,k)\circ(\varphi,h) = Tm(\psi,\varphi)(k,h)$. **The text's proof** varies φ' near φ, writes the partner as $\psi' = F(\varphi') = (s|\mathbf{U}(\psi))^{-1}(t(\varphi'))$, applies s to $m(F(\varphi'),\varphi')$, which gives $s(\varphi')$, and differentiates this identity at φ; since s is a local sc-diffeomorphism, $Tm(k,h) = Ts(\psi\circ\varphi)^{-1}Ts(\varphi)(h)$. **The formalized statement** is the form the proof uses; the identification of the tangent of the fibred product with the fibred product of the tangents, and Tm on it, are not formalized (gap G.75, §VIII.27). Let $s:\mathbf{W}\to V$ be a sheet with inverse σ (the text's $(s|\mathbf{U}(\psi))^{-1}$), and k,h composable with $p(k)\in\mathbf{W}$. *Step 1.* $g(\varphi') = m(\sigma(t\varphi'),\varphi')$, the text's $m(F(\varphi'),\varphi')$, is sc-smooth on $t^{-1}(V)$: this is Chapter 7's sheet form of condition (4) (§VII.2) Chapter7/EpGroupoid · EpGroupoid.isScSmoothCompSheet. *Step 2,* $k = T\sigma(Tt(h))$. $Tt(h) = Ts(k)$, and $T\sigma$ inverts Ts on $p^{-1}(\mathbf{W})$ (Step 3 of §VIII.1) Chapter8/TangentEp · EpGroupoid.tangentMapOn_srcInv_tSrc; in particular h lies over $t^{-1}(V)$ Chapter8/TangentEp · EpGroupoid.mem_tangentPreimage_tgt. *Step 3,* $k\circ h = Tg(h)$. $Tg(h)$ lies over $g(p\,h) = m(\sigma(s(p\,k)),p\,h) = p(k)\circ p(h)$, and $Ts(Tg(h)) = Ts(h)$ by the chain rule for $s\circ g = s$ on $t^{-1}(V)$, which is the text's identity; by the uniqueness of (8.5) the two vectors agree Chapter8/TangentEp · EpGroupoid.tComp_eq_tangentMapOn. The lemma also restates that $k\circ h$ lies over $p(k)\circ p(h)$ with $Ts(k\circ h) = Ts(h)$. The proof writes $s|\mathbf{U}(\varphi\circ\psi)$ for $s|\mathbf{U}(\psi\circ\varphi)$ (possible misprint) (Findings, Chapter 8, items 5, 111) (2017 p. 259).

VIII.3 Lemmas 8.1.3–8.1.6 (2017: 8.2–8.5): the groupoid laws of $T(X,\mathbf{X})$

Lemma 8.1.3 (2017: 8.2) (V8.5) Chapter8/TangentEp · EpGroupoid.lemma_8_2. $s_T(\Psi\circ\Phi) = s_T(\Phi)$ *and* $t_T(\Psi\circ\Phi) = t_T(\Psi)$. The first is (8.5) itself (Step 2 of §VIII.1). **The text's proof** of the second takes the map $\alpha = t\circ(s|\mathbf{U}(\psi\circ\varphi))^{-1}$, notes $\alpha\circ s = t\circ(s|\mathbf{U}(\psi))^{-1}\circ t$ near φ, and differentiates both sides at φ in the direction h. **The formalized proof** does the same with the tangents of §VIII.1, on a sheet $s:\mathbf{W}\to V$ with inverse σ around $p(k)$. *Step 1.* By Lemma 8.1.2 (2017: 8.1), $k\circ h = Tg(h)$ with $g(\varphi') = m(\sigma(t\varphi'),\varphi')$. *Step 2.* On $t^{-1}(V)$, $t\circ g = t\circ\sigma\circ t$ (the text's identity), so $Tt(Tg(h)) = T(t\circ\sigma\circ t)(h) = Tt(T\sigma(Tt(h))) = Tt(k)$, the last step by Step 2 of Lemma 8.1.2 (2017: 8.1) Chapter8/TangentEp · EpGroupoid.tTgt_tangentMapOn. This is the only identity of §8.1 that is genuinely differentiated; the others follow from fibre injectivity. The proof has "==", ": $\mathbf{U}(\psi\circ\varphi)\to U(x)$" without the map s, and "$Tt)\varphi)(h)$" (possible misprints) (Findings, Chapter 8, items 4, 6) (pp. 359–361 (2017 pp. 259–260)).

Lemma 8.1.4 (2017: 8.3) (V8.6) Chapter8/TangentEp · EpGroupoid.lemma_8_3. *The composition of* $T\mathbf{X}$ *is associative.* **The text's proof** computes both sides of $(\sigma,l)\circ((\psi,k)\circ(\varphi,h))$ with (8.5) and cancels $Ts(\psi\circ\varphi)(Ts(\psi\circ\varphi))^{-1}$. **The formalized proof** compares the two sides by fibre injectivity (§VIII.1): both lie over $p(l)\circ p(k)\circ p(h)$, by the associativity of $\mathbf{X}$, and both have Ts equal to $Ts(h)$, by Lemma 8.1.3 (2017: 8.2) applied twice; composability of the outer pairs is Lemma 8.1.3's (2017: 8.2) target formula (Findings, Chapter 8, item 4) (pp. 360–362 (2017 pp. 260–261)).

Lemma 8.1.5 (2017: 8.4) (V8.7) Chapter8/TangentEp · EpGroupoid.lemma_8_4. $1_q = (1_x, Ts(1_x)^{-1}q)$ *is a morphism* $q\to q$ *and a two-sided unit.* $1_q = Tu(q)$ (Step 3 of §VIII.1). *Step 1,* $1_q : q\to q$. $Ts(Tu(q)) = q$ and $Tt(Tu(q)) = q$ are the chain rule for $s\circ u = \mathrm{id}$ and $t\circ u = \mathrm{id}$ Chapter8/TangentEp · EpGroupoid.tSrc_tUnit Chapter8/TangentEp · EpGroupoid.tTgt_tUnit. The text argues the second through $t\circ(s|\mathbf{U}(1_x))^{-1} = \mathrm{id}$ near x; the identity $t\circ u = \mathrm{id}$ holds everywhere and needs no local inverse. *Step 2, the unit laws.* $k\circ 1_{Ts(k)}$ and k lie over $p(k)\circ 1 = p(k)$ and have the same Ts, and likewise on the left; the uniqueness of (8.5) gives both Chapter8/TangentEp · EpGroupoid.tComp_tUnit Chapter8/TangentEp · EpGroupoid.tUnit_tComp (Findings, Chapter 8, items 4, 6) (2017 p. 261).

Lemma 8.1.6 (2017: 8.5) (V8.8) Chapter8/TangentEp · EpGroupoid.lemma_8_5. $\iota_T(\varphi,k) = (\varphi^{-1}, T\iota(\varphi)k)$ *is a*

two-sided inverse. As in the text, the chain rule for $s \circ \iota = t$ and $t \circ \iota = s$ makes $\iota_T(v)$ a morphism $t_T(v) \to s_T(v)$ `Chapter8/TangentEp · EpGroupoid.tSrc_tInv` `Chapter8/TangentEp · EpGroupoid.tTgt_tInv`. The text then computes $\iota_T(\varphi, k) \circ (\varphi, k)$ with (8.5); here the uniqueness of (8.5) does it: $\iota_T(v) \circ v$ lies over $\varphi^{-1} \circ \varphi = 1_x$ and has Ts equal to $Ts(v)$, which characterizes $1_{Ts(v)}$, and the other side is the same `Chapter8/TangentEp · EpGroupoid.tInv_tComp` `Chapter8/TangentEp · EpGroupoid.tComp_tInv` (Findings, Chapter 8, item 4) (pp. 361–363 (2017 pp. 261–262)).

VIII.4 Lemmas 8.1.7 and 8.1.8 (2017: 8.6 and 8.7): the differential conditions

Lemma 8.1.7 (2017: 8.6) (V8.9) `Chapter8/TangentEp · EpGroupoid.lemma_8_6`. *s_T, t_T are surjective local sc-diffeomorphisms and all structure maps of $T(X, \mathbf{X})$ are sc-smooth. Step 1, s_T and t_T.* "By construction" is Step 4 of the tangents on open sets (§VIII.1): the tangent of a surjective local sc-diffeomorphism is one `Chapter8/TangentEp · EpGroupoid.isLocalScDiffeo_tSrc` `Chapter8/TangentEp · EpGroupoid.isLocalScDiffeo_tTgt` `Chapter8/TangentEp · EpGroupoid.surjective_tSrc` `Chapter8/TangentEp · EpGroupoid.surjective_tTgt`. *Step 2, u_T and ι_T* are Part I's tangents of the sc-smooth u and ι `Chapter8/TangentEp · EpGroupoid.isScSmoothMap_tUnit` `Chapter8/TangentEp · EpGroupoid.isScSmoothMap_tInv`. *Step 3, the multiplication.* **The text** says it is sc-smooth "by inspection", or because $T\mathbf{X}_{s_T}\times_{t_T} T\mathbf{X}$ is the tangent of $\mathbf{X}_s\times_t \mathbf{X}$ and $m_{T\mathbf{X}}$ "is precisely Tm". **The formalized proof** uses Chapter 7's sheet form of condition (4) (§VII.2) `Chapter7/EpGroupoid · IsScSmoothCompSheet`: for a sheet of s_T with inverse σ_T, the map $\Psi \mapsto \sigma_T(t_T\Psi) \circ \Psi$ is sc-smooth on $t_T^{-1}(V_T)$. Near Ψ_0, take a sheet $s : \mathbf{W} \to V$ around the base point of $\sigma_T(t_T\Psi_0)$; on the open set of Ψ with $p(\sigma_T(t_T\Psi)) \in \mathbf{W}$, the map is Tg for the g of Lemma 8.1.2 (2017: 8.1), by Step 3 of Lemma 8.1.2 (2017: 8.1), and Tg is sc-smooth `Chapter8/TangentEp · EpGroupoid.isScSmoothCompSheet_tangent`; sc-smoothness is local `Chapter7/Discrete · MPolyfold.IsScSmoothOn.of_locally`. Chapter 7's equivalence of the two forms gives (4) on the fibred product itself `Chapter7/EpGroupoid · isScSmoothComp_iff_sheet` `Chapter8/TangentEp · EpGroupoid.isScSmoothComp_tangent`, without $T(\mathbf{X}_s\times_t \mathbf{X}) = T\mathbf{X}_{Ts}\times_{Tt} T\mathbf{X}$ (gap G.75) (Findings, Chapter 8, item 7) (2017 p. 262).

Lemma 8.1.8 (2017: 8.7) (V8.10) `Chapter8/TangentEp · EpGroupoid.lemma_8_7`. *Every $a \in TX$ has an open $V' \ni a$ with $t_T : s_T^{-1}(\operatorname{cl} V') \to TX$ proper*, namely $V' = p^{-1}(V)$ for an open V around $p(a)$ as in Definition 7.1.3 (3) (2017: 7.3) for X. **The text's proof** takes (φ_k, h_k) with sources in $\operatorname{cl} TV$ and convergent targets, gets $\varphi_k \to \varphi$ in $\mathbf{X}$ from the properness of X, then in $\mathbf{X}^1$, and concludes that (φ_k, h_k) converges since $Ts : T\mathbf{U}(\varphi) \to TU(x)$ is an sc-diffeomorphism. **The formalized proof** checks Chapter 7's sequential criterion for properness on metrizable spaces (§VII.2) `Chapter7/ProperMaps · MPolyfold.isProperMap_domRestrict_iff_of_isClosed`. *Step 1.* Let Φ_k have $s_T(\Phi_k) \in \operatorname{cl} V'$ and $t_T(\Phi_k) \to b$. The base points have $s(p\,\Phi_k) \in \operatorname{cl} V$, p being continuous, and $t(p\,\Phi_k) \to p(b)$; properness of X gives a subsequence with $p(\Phi_{k_j}) \to \varphi_0$ in $\mathbf{X}$ `Chapter7/ProperMaps · MPolyfold.exists_tendsto_subseq_of_isProperMap_domRestrict`. *Step 2.* Take a sheet $t : \mathbf{W} \to V''$ around φ_0. Its tangent is an sc-diffeomorphism $p^{-1}(\mathbf{W}) \to p^{-1}(V'')$ with continuous inverse $(Tt)^{-1}$ (Step 3 of §VIII.1), and $b \in p^{-1}(V'')$; eventually $p(\Phi_{k_j}) \in \mathbf{W}$, so $\Phi_{k_j} = (Tt)^{-1}(t_T\Phi_{k_j}) \to (Tt)^{-1}(b)$. The text's step "$\varphi_k \to \varphi$ in $\mathbf{X}^1$" is not needed: convergence in $\mathbf{X}$ already puts the φ_k in the sheet, and TX, $T\mathbf{X}$ are metrizable, so the sequential criterion applies. The proof writes $t(\varphi_k, h_k) = Ts(\varphi_k)(h_k)$ for $Tt(\varphi_k)(h_k)$ (possible misprint) (Findings, Chapter 8, item 7) (pp. 362–364 (2017 pp. 262–263)).

VIII.5 Lemma 8.1.9 (2017: 8.8): the tangent of a tame ep-groupoid

Lemma 8.1.9 (2017: 8.8) (V8.11) `Chapter8/TangentTame · EpGroupoid.lemma_8_8`. *If X is tame, so is TX*; tame means that the object M-polyfold has an equivalent tame atlas (Definition 7.1.6 (2017: 7.6), §VII.3) `Chapter7/EpGroupoid · EpGroupoid.IsTame`. **The text's proof** is one sentence: for a tame chart onto a tame retract $O = r(U)$, "$Tr : TU \to TU$ is also a tame retraction". Part I does not state this,

and the text gives no argument; it is proved here from Part I's Definition 2.5.2 (2017: 2.17) (§II.14) `Chapter2/Tame · ScRetraction.IsTame`. The tangent of the retract (O, C, E) is (TO, TC, TE), $TC = C^1 \oplus E$, with $Tr(a, b) = (r(a), Dr(a)b)$ `Chapter2/TangentBundle · ScRetraction.tangentRetraction Chapter2/TangentBundle · LocalModel.tangent`. *Step 1, a presentation of TC.* An sc-isomorphism of TE with $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$, built from a presentation T of C, presents TC as a partial quadrant whose $\mathbb{R}^n$-coordinates at (a, b) are those of a `Chapter8/TangentTame · ScIso.tangentPres Chapter8/TangentTame · tangentSet_eq_tangentPres`; so $d_{TC}(a, b) = d_C(a)$ `Chapter8/TangentTame · degIndex_tangentSet`, and the minimal subspace at (a, b) is $(E_a \cap E_1) \oplus E$ `Chapter8/TangentTame · mem_minimalSubspaceWith_tangentPres`. *Step 2, condition (1)* of Definition 2.5.2 (2017: 2.17) for Tr at (a, b) is condition (1) for r at a, by Step 1 and $Tr(a, b) = (r(a), Dr(a)b)$. *Step 3, condition (2)* at a smooth point $(a, b) \in TO$. With A the sc-complement of $T_aO = Dr(a)E$ in E_a that the tameness of r gives, the complement for Tr is $A' = (A \cap E_1) \oplus A$. With $L = D(Tr)(a, b)$, three facts suffice: L is a projection, since $Tr \circ Tr = Tr$ `Chapter2/Retracts · ScRetraction.scDeriv_comp_self`; its first row is $(Dr(a), 0)$ `Chapter8/TangentTame · ScRetraction.scDeriv_tangentRetraction_fst`; and $L(0, l) = (0, Dr(a)l)$ `Chapter8/TangentTame · ScRetraction.scDeriv_tangentRetraction_inr`. Then $LTE \cap A' = 0$ (if $(k, l) = L(k, l) \in A'$, then $k \in T_aO \cap A = 0$ and $l = Dr(a)l \in T_aO \cap A = 0$) and $LTE + A' = TE$ (split k, then the remainder of l), on every level. The second derivative of r, which enters L, is never needed `Chapter8/TangentTame · ScRetraction.IsTameWith.tangent`. *Step 4, all presentations.* Tameness quantifies over all presentations of TC; Part I's Lemma 1.3.4 (2017: 1.1), the presentation independence of d_C (§I.18), passes from one to all `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame Chapter8/TangentTame · ScRetraction.IsTame.tangentRetraction Chapter8/TangentTame · IsTameRetract.tangent`. *Step 5, from charts to TX.* Around $\alpha \in TX$ the tangent chart of a chart onto a tame retract around $p(\alpha)$ is modelled on the tame retract $T(O, C, E)$ `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`, so these charts form an equivalent tame atlas `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart Chapter8/TangentTame · MPolyfold.IsTame.tangent`. The tameness clauses of Theorem 8.1.1 (2017: 8.1) and Theorem 8.1.12 (2017: 8.3) follow `Chapter8/TangentTame · EpGroupoid.theorem_8_1_tame Chapter8/TangentTame · theorem_8_3_tame`, and the hypothesis is satisfiable: the tangent of the trivial ep-groupoid of $\mathbb{R}^n$ is tame `Chapter8/TangentTame · EpGroupoid.isTame_tangentEpGroupoid_trivial_rn` (Findings, Chapter 8, items 8, 104) (2017 p. 263).

VIII.6 Remark 8.1.10 (2017: 8.1): the morphisms as pairs $(T\varphi, a)$

Remark 8.1.10 (2017: 8.1) (V8.12) `Chapter8/TangentEp · EpGroupoid.remark_8_1`. The text describes $T\mathbf{X}$ a second way: a morphism $\varphi : x \to y$ on level 1 has a linear isomorphism $T\varphi : T_xX \to T_yX$, the tangent of $t \circ (s|\mathbf{U}(\varphi))^{-1}$ at x; the pairs $(T\varphi, a)$, $a \in T_{s(\varphi)}X$, composed by $(T\psi, h) \circ (T\varphi, l) = (T(\psi \circ \varphi), l)$, form a groupoid, and $G(\varphi, h) = (T\varphi, Ts(h))$ is a bijection compatible with the compositions; moreover $t_T \circ (s_T|T_\varphi\mathbf{X})^{-1} = T\varphi$. $T\varphi$ is Chapter 7's $Tt \circ (Ts|T_\varphi\mathbf{X})^{-1}$ (§VII.3) `Chapter7/EpGroupoid · EpGroupoid.tangentMor`. *Step 1, well defined.* $T\varphi(Ts(h)) = Tt(h)$ for every $h \in T_\varphi\mathbf{X}$, which is the last sentence of the remark `Chapter8/TangentEp · EpGroupoid.tangentMor_tSrc`. *Step 2, functorial.* $T(\psi \circ \varphi) = T\psi \circ T\varphi$ on $T_{s(\varphi)}X$: given a, take h over φ with $Ts(h) = a$ (fibre surjectivity) and k over ψ with $Ts(k) = Tt(h)$; then $k \circ h$ lies over $\psi \circ \varphi$ with $Ts = a$, and $Tt(k \circ h) = Tt(k)$ by Lemma 8.1.3 (2017: 8.2) `Chapter8/TangentEp · EpGroupoid.tangentMor_comp`. *Step 3, the map G.* As a bare map $(\varphi, h) \mapsto (T\varphi, Ts(h))$ need not be injective: distinct morphisms can have the same $T\varphi$, as the three morphisms at a point of the trivial $\mathbb{Z}/3$-action on $\mathbb{R}$ (§VII.30), all with $T\varphi = \mathrm{id}$. So the first entry is read as $T\varphi$ labelled by φ `Chapter8/TangentEp · EpGroupoid.remarkG`; then G is injective by fibre injectivity, its range is the set of pairs (φ, a) with $\varphi \in \mathbf{X}_1$ and a over $s(\varphi)$ by fibre surjectivity, composability of (k, h) is $Ts(k) = T\varphi(Ts(h))$ with $\varphi = p(h)$, and $G(k \circ h) = (p(k) \circ p(h), Ts(h))$, the text's computation. Not formalized: that $T\varphi$ and $t_T \circ (s_T|T_xX)^{-1}$ are linear isomorphisms and that $T\varphi$ is the tangent of $\hat{\varphi}$ at x (gaps G.7, G.70); the M-polyfold structure transported along G, and the triples $(a, T\varphi, b)$. The remark writes "$T\varphi : T_xX \to T_yY$" for T_yX, and calls $T\varphi$ the linearization "at φ" on 2017 p. 265 where 2017 p. 264 has "at x" (possible misprints) (Findings, Chapter 8, items 10, 112, 113) (pp. 364–366 (2017 pp. 264–265)).

VIII.7 Theorems 8.1.11 and 8.1.12 (2017: 8.2 and 8.3): the tangent functor

Theorem 8.1.11 (2017: 8.2) (V8.13) `Chapter8/TangentEp · EpFunctor.theorem_8_2`. *For an sc-smooth functor* $(\Phi,\Phi) : X \to Y$, $(T\Phi, T\Phi)$ *is an sc-smooth functor* $TX \to TY$ `Chapter8/TangentEp · EpFunctor.tangent`. **The text's proof** differentiates $\Phi(m(\psi,\varphi)) = m(\Phi\psi,\Phi\varphi)$ on the fibred product, using Lemma 8.1.2 (2017: 8.1) as printed, and differentiates $s\circ\Phi = \Phi\circ s$ and $t\circ\Phi = \Phi\circ t$; it leaves the units to the reader. **The formalized proof.** *Step 1.* The tangent maps are sc-smooth by Part I, and compatibility with sources and targets is the chain rule for $s\circ\Phi = \Phi\circ s$, $t\circ\Phi = \Phi\circ t$, as in the text. *Step 2, the composition, by fibre injectivity in* TY. $T\Phi(k\circ h)$ lies over $\Phi(p(k)\circ p(h)) = \Phi(p\,k)\circ\Phi(p\,h)$, and its Ts is $T\Phi(Ts(k\circ h)) = T\Phi(Ts(h)) = Ts(T\Phi(h))$; the composite $T\Phi(k)\circ T\Phi(h)$ has the same base point and the same Ts. No tangent of the fibred product is needed. *Step 3, the units*, not checked in the text, follow from the chain rule for $\Phi\circ u = u\circ\Phi$ (Findings, Chapter 8, items 4, 11) (pp. 365–367 (2017 pp. 265–266)).

Theorem 8.1.12 (2017: 8.3) (V8.14) `Chapter8/TangentEp · theorem_8_3`. T *is a functor* $\mathcal{EP} \to \mathcal{EP}$, $X \mapsto TX$, $\Phi \mapsto T\Phi$, *and* TX *is tame if* X *is.* On objects it is Theorem 8.1.1 (2017: 8.1) `Chapter8/TangentEp · EP.tangent`, on morphisms Theorem 8.1.11 (2017: 8.2), and the functor laws are Part I's chain rule and the tangent of the identity `Chapter2/TangentMap · MPolyfold.tangentMap_comp` `Chapter2/TangentMap · MPolyfold.tangentMap_id` `Chapter8/TangentEp · tangentFunctor`; $\mathcal{EP}$ is Chapter 7's category (§VII.4) `Chapter7/EpGroupoid · EP`. Since TX lives in `Type (max u 1)`, the functor goes from $\mathcal{EP}$ in that universe to itself, an artifact of Part I's `TangentSpace`. The tameness sentence is Lemma 8.1.9 (2017: 8.8) (§VIII.5) `Chapter8/TangentTame · theorem_8_3_tame` (Findings, Chapter 8, items 2, 11) (2017 p. 266).

VIII.8 Definition 8.1.13 (2017: 8.1), the equivalence relation $\simeq$, and Proposition 8.1.14 (2017: 8.1)

Definition 8.1.13 (2017: 8.1) (V8.15). A natural transformation $\tau : F \to G$ between sc-smooth functors $X \to Y$ is an sc-smooth map $\tau : X \to \mathbf{Y}$ with $\tau(x) : F(x) \to G(x)$ and $\tau(x')\circ F(h) = G(h)\circ\tau(x)$ for every $h : x \to x'$ `Chapter8/Basic · EpNatTrans`; $F \simeq G$ if one exists `Chapter8/Basic · EpFunctor.NaturallyEquivalent` (Findings, Chapter 8, item 12) (2017 p. 266).

The equivalence relation (V8.16) `Chapter8/NatTrans · EpFunctor.naturallyEquivalent_equivalence`. The text states without proof that $\simeq$ is an equivalence relation. Reflexivity is $x \mapsto 1_{F(x)}$ `Chapter8/NatTrans · EpNatTrans.refl`, symmetry $x \mapsto \tau(x)^{-1}$, sc-smooth since ι is, natural by inverting the square of h^{-1} `Chapter8/NatTrans · EpNatTrans.symm`, and transitivity $x \mapsto \tau'(x)\circ\tau(x)$ `Chapter8/NatTrans · EpNatTrans.trans` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.refl` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.symm` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.trans`. *The one analytic point* is the sc-smoothness of $x \mapsto \tau'(x)\circ\tau(x)$ `Chapter8/NatTrans · EpNatTrans.isScSmoothMap_comp_app`. Near x_0, take a sheet $\mathbf{W} \ni \tau'(x_0)$ of s with inverse σ; for $x \in \tau'^{-1}(\mathbf{W})$, $\tau'(x) = \sigma(s\,\tau'(x)) = \sigma(G(x)) = \sigma(t\,\tau(x))$, so $\tau'(x)\circ\tau(x) = m(\sigma(t\psi),\psi)$ at $\psi = \tau(x)$, which is sc-smooth by Chapter 7's sheet form of condition (4), composed with the sc-smooth τ. Whiskering, not in the text, is added `Chapter8/NatTrans · EpNatTrans.whiskerLeft` `Chapter8/NatTrans · EpNatTrans.whiskerRight` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_left` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right` (Findings, Chapter 8, item 12) (2017 p. 267).

Proposition 8.1.14 (2017: 8.1) (V8.17) `Chapter8/TangentEp · proposition_8_1`. *If* $\tau : \Phi \to \Psi$ *is a natural transformation,* $T\tau : TX \to T\mathbf{Y}$ *is a natural transformation* $T\Phi \to T\Psi$; *hence* $\Phi \simeq \Psi$ *implies* $T\Phi \simeq T\Psi$ `Chapter8/TangentEp · EpNatTrans.tangent`. *Step 1, as in the text.* $T\tau$ is sc-smooth by Part I, and the chain rule for $s\circ\tau = \Phi$ and $t\circ\tau = \Psi$ makes $T\tau(a)$ a morphism $T\Phi(a) \to T\Psi(a)$ of TY. *Step 2, naturality.* **The text** differentiates the naturality square $\tau(t\alpha)\circ\Phi(\alpha) = \Psi(\alpha)\circ\tau(s\alpha)$ at $\alpha_0 \in \mathbf{X}^1$, again through Lemma 8.1.2 (2017: 8.1) as printed. **The formalized proof** compares $T\tau(Tt(v))\circ T\Phi(v)$ and $T\Psi(v)\circ T\tau(Ts(v))$ by fibre injectivity in TY: they lie over the two sides of the naturality square at $p(v)$, which are equal, and both have Ts equal to $T\Phi(Ts(v))$, by Lemma 8.1.3 (2017: 8.2) and Step 1. The proof has "$T\tau(x_0(h))$", and

labels the lower arrow of the last diagram $T\tau(x_0)h$ for $T\tau(x_0')h'$ (possible misprints, the second perhaps an artefact of the text extraction) (Findings, Chapter 8, items 4, 13) (pp. 367–372 (2017 pp. 267–268)).

VIII.9 Theorem 8.1.15 (2017: 8.4): the orbit space of TX

Theorem 8.1.15 (2017: 8.4) (V8.18) `Chapter8/TangentOrbit · EpGroupoid.theorem_8_4`. *If $|X|$ is paracompact, so is $|TX|$*; paracompact orbit space is Chapter 7's Definition 7.5.1 (2017: 7.15) (§VII.23) `Chapter7/EpGroupoid · EpGroupoid.HasParacompactOrbitSpace`. **The text's proof**: $|X^1|$ is paracompact (Proposition 7.5.3 (2017: 7.12)); for $z \in |X^1|$ take $x_z \in X_1$ and $U(x_z)$ with the natural G_{x_z}-action such that $TU(x_z)$ is, as a bundle over $U(x_z)^1$, sc-isomorphic to a local bundle model $TO \to O^1$ with an sc-smooth bundle action, so that $|TU(x_z)|$ and $|U(x_z)^1|$ are metrizable; a locally finite closed refinement (A_z) of the cover $(|U(x_z)^1|)$ gives closed paracompact sets $\tilde{A}_z \subseteq |TX|$ covering it locally finitely, and $|TX|$ is regular Hausdorff, so Part I's Proposition 2.7.4 (2017: 2.17) applies. **The formalized proof** proves more, that $|TX|$ is metrizable `Chapter8/TangentOrbit · EpGroupoid.metrizableSpace_tangent_orbitSpace`, along the route of Proposition 7.5.3 (2017: 7.12) (§VII.23): Smirnov's theorem pulled back along the continuous map $|p| : |TX| \to |X^1|$ (§II.7) `Chapter2/Metrizable · metrizableSpace_of_pullback`. *Step 1, the map $|p|$.* The projection functor of Theorem 8.1.1 (2017: 8.1) induces $|p|$ `Chapter8/TangentOrbit · EpGroupoid.tangentOrbitProj Chapter8/TangentOrbit · EpGroupoid.continuous_tangentOrbitProj`, and $|p|^{-1}(\pi_1(U \cap X_1)) = \pi_T(p^{-1}(U))$ for every $U \subseteq X$, by fibre surjectivity of Ts: a vector over a point isomorphic in X^1 to a point of $U \cap X_1$ is isomorphic in TX to a vector over it `Chapter8/TangentOrbit · EpGroupoid.preimage_tangentOrbitProj_image`. *Step 2, the action of G_x on $TU(x)$.* For a natural representation at $x \in X_1$, $TU(x) = p^{-1}(U(x))$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangentU` carries the action $g \mapsto T\Phi(g)$ by the tangents of the natural action, sc-diffeomorphisms of $TU(x)$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tΦ Chapter8/TangentOrbit · EpGroupoid.NaturalRep.isScDiffeoOn_tΦ Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangentMulAction`, with the tangent $T\Gamma(g)$ of the sheet `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tΓ`. Its linearity on the fibres is not formalized (gap G.7), and the local bundle model $TO \to O^1$ is not used. *Step 3, the open embedding.* Two vectors of $TU(x)$ have the same image in $|TX|$ iff they are in one G_x-orbit `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.tangent_π_eq_iff`: a morphism $k : v \to w$ of TX lies over a morphism between points of $U(x)$, so $p(k) = \Gamma(g, s(p\,k))$ by Theorem 7.1.13 (3) (2017: 7.1), and $k = T\Gamma(g)(v)$ by fibre injectivity. So $G_x\backslash TU(x) \to |TX|$ is an open embedding onto $|TU(x)|$, the argument of Proposition 7.1.19 (2017: 7.6) for TX, which the text leaves implicit `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.isOpenEmbedding_tangentOrbitQuotientMap Chapter8/TangentOrbit · EpGroupoid.NaturalRep.range_tangentOrbitQuotientMap`. *Step 4, metrizability.* $TU(x)$ is metrizable as a subspace of TX (Part I's Proposition 2.3.15 (3) (2017: 2.5)) `Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace`, and $G_x\backslash TU(x)$ by the metric $\hat{\rho}$ of Lemma 7.3.2 (2017: 7.4), repeated for a finite group acting by homeomorphisms on a metrizable space; hence $|TU(x)|$ is metrizable `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.metrizableSpace_image_tangentU`, and by Step 1 it is $|p|^{-1}(|U(x)_1|)$ `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.image_tangentU_eq_preimage`, $|U(x)_1|$ being open in $|X^1|$. These are the text's two metrizability claims `Chapter8/TangentOrbit · EpGroupoid.NaturalRep.metrizableSpace_image_liftU`. *Step 5.* $|X^1|$ is metrizable by Proposition 7.5.3 (2017: 7.12) `Chapter7/Lift · EpGroupoid.metrizableSpace_liftOne_orbitSpace`, hence paracompact and Hausdorff; every point of it lies in some $|U(x)_1|$, for a natural representation at a representative (Theorem 7.1.13 (2017: 7.1)) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1`, whose preimage is metrizable by Step 4. The pulled-back Smirnov theorem makes $|TX|$ metrizable, hence paracompact (Stone). The closed sets $\tilde{A}_z$, the regularity of $|TX|$ and Proposition 2.7.4 (2017: 2.17) are not used; the step shared with the text is the metrizability of $|TU(x_z)|$. The theorem is not vacuous `Chapter8/TangentOrbit · EpGroupoid.hasParacompactOrbitSpace_tangent_trivial`. "parcompact" (possible misprint) (Findings, Chapter 8, items 20, 21, 104) (2017 p. 268).

VIII.10 Strong bundle maps, and the pull-back along s

Strong bundle maps and the pull-back (V8.19). §8.3 works with strong bundle maps between strong bundles over M-polyfolds, the structure map $\mu : \mathbf{X}_s\times_P W \to W$ first of all, and with the pull-back of $P : W \to X$ along s; it defines neither notion beyond Part I. *The pull-back* is Part I's Proposition 2.6.8 (2017: 2.16) (§II.20) `Chapter2/StrongBundle · pullbackSpace` `Chapter3/ParamPerturbation · StrongBundle.pullback`, and its paracompactness, which Part I's definition of a strong bundle requires, is proved `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` `Chapter8/StrongBundleMap · StrongBundle.paracompactSpace_pullback`. Added here: its atlas, built from pairs of charts `Chapter8/StrongBundleMap · StrongBundle.pullback_atlas` `Chapter8/StrongBundleMap · PullbackChartData.exists_mem_atlas`; the two projections are sc-smooth `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackProj` `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackProj'`; its universal property on $[0]$, a map into the pull-back being sc-smooth iff its two components are `Chapter8/StrongBundleMap · StrongBundle.isScSmoothOn_to_pullback_iff` `Chapter8/StrongBundleMap · pullbackPair` `Chapter8/StrongBundleMap · StrongBundle.isScSmoothMap_pullbackPair`, sc-smoothness being checked chart by chart `Chapter8/StrongBundleMap · MPolyfold.isScSmoothMap_of_forall_charts`; and over a sheet $s : \mathbf{W} \to V$ with inverse σ the projection $(\varphi, w) \mapsto w$ is an sc-diffeomorphism onto $P^{-1}(V)$, with inverse $w \mapsto (\sigma(Pw), w)$ `Chapter8/StrongBundleMap · pullbackSheetInv` `Chapter8/StrongBundleMap · StrongBundle.isScDiffeoOn_pullbackProj'`, so a local sc-diffeomorphism `Chapter8/StrongBundleMap · StrongBundle.isLocalScDiffeo_pullbackProj'`. *Strong bundle maps.* Part I defines them only between local models (Definition 2.6.1 (2017: 2.22), Definition 2.47 of the preprint, §II.18), and asks the transition maps of a strong bundle atlas to be sc-smooth on $[0]$ and $[1]$. The global notion is a reading `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap`: $\Phi : Y \to Y'$ with $P' \circ \Phi = f \circ P$, linear on the fibres `Chapter8/StrongBundleMap · fiberRestrict`, sc-smooth $Y[0] \to Y'[0]$, mapping $Y_{0,1}$ into $Y'_{0,1}$ and sc-smooth $Y[1] \to Y'[1]$. In the charts the last two conditions say that the local representatives $K[i] \to K'[i]$ are sc-smooth, as Part I asks of transition maps. The preservation of the double filtration, part of Definition 2.6.1 (2017: 2.22), is not assumed but follows `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.mapsTo_doubleLevel`, and so do the continuity of Φ, the boundedness of its fibre maps and the sc-smoothness of the base map $f = P' \circ \Phi \circ 0$ `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.continuous` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.fiberMap` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.isScSmoothMap_base`. Strong bundle maps compose and identities are strong bundle maps `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.comp` `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleMap.id`; an isomorphism is a pair of inverse strong bundle maps `Chapter8/StrongBundleMap · StrongBundle.IsStrongBundleIso`; and the projection of the pull-back to W is a strong bundle map covering s, the identity on each fibre `Chapter8/StrongBundleMap · StrongBundle.isStrongBundleMap_pullbackProj'` `Chapter8/StrongBundleMap · StrongBundle.fiberMap_pullbackProj'`. Strong bundle maps over open subsets of the base are not formalized (§VIII.27) (Findings, Chapter 8, items 30, 31) (used on pp. 373–396 (2017 pp. 270–285)).

VIII.11 Definition 8.3.1 (2017: 8.4): strong bundles over ep-groupoids, and the fibre maps

Definition 8.3.1 (2017: 8.4) (V8.20) `Chapter8/EpStrongBundle · EpStrongBundle`. *A strong bundle over X is a pair (P, μ): a strong bundle $P : W \to X$ over the object M-polyfold and a strong bundle map $\mu : \mathbf{X}_s\times_P W \to W$ covering t, with (1) μ a surjective local sc-diffeomorphism, linear on the fibres; (2) $\mu(1_x, w) = w$; (3) $\mu(\psi \circ \gamma, w) = \mu(\psi, \mu(\gamma, w))$.* The formalization takes the definition as printed, on Part I's pull-back and with the strong bundle maps of §VIII.10; the conditions $s(\psi) = P(w)$ are carried by the points of the pull-back, the pairs $(1_{Pw}, w)$ are points of it `Chapter8/EpStrongBundle · EpGroupoid.unitPair`, and the composite $P \circ \mu$ is $t \circ \pi_1$ `Chapter8/EpStrongBundle · EpStrongBundle.P_μ`. The surjectivity in (1) follows from (2) and is kept, as in the text; μ is continuous `Chapter8/EpStrongBundle · EpStrongBundle.continuous_μ`. The definition is met by every strong bundle over an M-polyfold, as a strong bundle over the trivial ep-groupoid with $\mu(z, w) = w$ `Chapter8/NonVacuityBundle · EpStrongBundle.trivial` (Findings, Chapter 8, items 32, 102) (2017

p. 271).

The fibre maps $\mu(\psi,\cdot)$ (V8.21) `Chapter8/EpStrongBundle · EpStrongBundle.fiberMorEquiv`. *For* $\psi : x \to y$, $\mu(\psi,\cdot) : W_x \to W_y$ *is a linear isomorphism*, stated after the definition without proof. It is proved in the stronger form of a topological linear isomorphism `Chapter8/EpStrongBundle · EpStrongBundle.fiberMor`. *Step 1, linear and bounded.* μ is linear on the fibres of $\mathbf{X}\,{}_s\!\times_P W \to \mathbf{X}$, so $\mu(\psi,\cdot)$ is additive and homogeneous `Chapter8/EpStrongBundle · EpStrongBundle.μ_add Chapter8/EpStrongBundle · EpStrongBundle.μ_smul`, and bounded, the fibre maps of a strong bundle map being bounded (§VIII.10). *Step 2, the inverse* is $\mu(\psi^{-1},\cdot)$: by (3) and (2), $\mu(\psi^{-1},\mu(\psi,w)) = \mu(1_x,w) = w$, and symmetrically `Chapter8/EpStrongBundle · EpStrongBundle.μ_inv_μ Chapter8/EpStrongBundle · EpStrongBundle.μ_μ_inv` (Findings, Chapter 8, item 32) (2017 p. 271).

VIII.12 The functorial picture: the ep-groupoid $(W, \mathbf{W})$

The functorial picture (V8.22) `Chapter8/EpStrongBundle · EpStrongBundle.toEpGroupoid`. *With* $\mathbf{W} = \mathbf{X}\,{}_s\!\times_P W$, $s(\psi,w) = w$, $t(\psi,w) = \mu(\psi,w)$, *units* $(1_x,w)$, *inverses* $(\psi^{-1},\mu(\psi,w))$ *and composition* $(\psi,w')\circ(\gamma,w) = (\psi\circ\gamma,w)$, $(W,\mathbf{W})$ *is an ep-groupoid, and* $(P,\pi_1) : (W,\mathbf{W}) \to (X,\mathbf{X})$ *is an sc-smooth functor.* **The text** says "we can define the ep-groupoid" and gives no proof. **The formalized proof** checks Definition 7.1.3 (2017: 7.3). *Step 1, the groupoid.* The composition is total, with the junk value (γ,w) off composable pairs, and the composite of a composable pair has target $\mu(\psi\circ\gamma,w) = \mu(\psi,w')$ by Definition 8.3.1 (3) (2017: 8.4) `Chapter8/EpStrongBundle · EpStrongBundle.morComp Chapter8/EpStrongBundle · EpStrongBundle.morComp_val Chapter8/EpStrongBundle · EpStrongBundle.μ_morComp`; the inverse is a morphism $\mu(\psi,w) \to w$ by Definition 8.3.1 (2), (3) (2017: 8.4) `Chapter8/EpStrongBundle · EpStrongBundle.morInv`, and the groupoid laws are those of $\mathbf{X}$ in the first entry `Chapter8/EpStrongBundle · EpStrongBundle.toGroupoidStr`. *Step 2, (1) and (2).* $s_{\mathbf{W}}$ is the projection of the pull-back along the local sc-diffeomorphism s, a surjective local sc-diffeomorphism by §VIII.10, and $t_{\mathbf{W}} = \mu$ is one by Definition 8.3.1 (1) (2017: 8.4); the units and inverses are sc-smooth by the universal property of the pull-back. *Step 3, (3), by sequences.* For $w \in W$ take $V(Pw)$ as in Definition 7.1.3 (3) (2017: 7.3) for X and $V = P^{-1}(V(Pw))$. If (ψ_k,w_k) has $w_k \in \operatorname{cl} V$ and $\mu(\psi_k,w_k) \to w'$, then $s(\psi_k) = P(w_k) \in \operatorname{cl} V(Pw)$ and $t(\psi_k) = P(\mu(\psi_k,w_k)) \to P(w')$, so a subsequence has $\psi_k \to \psi$ by the properness of X, and $w_k = \mu(\psi_k^{-1},\mu(\psi_k,w_k)) \to \mu(\psi^{-1},w')$ by the continuity of μ and ι `Chapter8/EpStrongBundle · EpStrongBundle.exists_isProperMap_tgt`. *Step 4, (4)* is proved in Chapter 7's sheet form, through a sheet of s in $\mathbf{X}$ and condition (4) for X `Chapter8/EpStrongBundle · EpStrongBundle.isScSmoothCompSheet_toGroupoidStr`. *Step 5, the functor.* $s_{\mathbf{W}}$, $t_{\mathbf{W}}$ are strong bundle maps covering s, t `Chapter8/EpStrongBundle · EpStrongBundle.isStrongBundleMap_src Chapter8/EpStrongBundle · EpStrongBundle.isStrongBundleMap_tgt`, (P,π_1) is an sc-smooth functor `Chapter8/EpStrongBundle · EpStrongBundle.projFunctor`, and $|P| : |W| \to |X|$ is continuous and surjective `Chapter8/EpStrongBundle · EpStrongBundle.continuous_orbitMap_projFunctor Chapter8/EpStrongBundle · EpStrongBundle.surjective_orbitMap_projFunctor`, since a point isomorphic to $P(w)$ is the image of a point isomorphic to w, namely $\mu(\psi,w)$ `Chapter8/EpStrongBundle · EpStrongBundle.exists_isomorphic_of_isomorphic`. An `EpGroupoid` has its objects and morphisms in one universe, so from here on W lives in the universe of the objects and morphisms of X (Findings, Chapter 8, items 2, 33) (pp. 375–377 (2017 pp. 271–272)).

VIII.13 Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6), and "these are identical notions"

Definition 8.3.2 (2017: 8.5) (V8.23) `Chapter8/EpStrongBundle · EpStrongBundle.IsMap`. *A strong bundle map* $(P,\mu) \to (P',\mu')$ *is a strong bundle map* $\Phi : W \to W'$ *covering the object map of an sc-smooth functor* $\phi : X \to X'$, *with* $\Phi(\mu(\gamma,w)) = \mu'(\phi(\gamma),\Phi(w))$. As printed, with the strong bundle maps of §VIII.10; the identity is one `Chapter8/NonVacuityBundle · EpStrongBundle.isMap_id` (Findings, Chapter 8, item 34) (2017 p. 273).

Definition 8.3.3 (2017: 8.6) (V8.24) `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor`. *A strong bundle functor is a functor* $\Phi : (W,\mathbf{W}) \to (W',\mathbf{W}')$, *linear on the fibres, covering an sc-smooth func-*

tor ϕ and inducing strong bundle maps $W \to W'$ and $\mathbf{W} \to \mathbf{W}'$. "Linear on the fibres", "covers ϕ" and "induces strong bundle maps" are read together as two strong bundle maps in the sense of §VIII.10: on objects, for P and P', covering ϕ on objects, which contains $P' \circ \Phi = \phi \circ P$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.projFunctor_comp`; and on morphisms, for the pull-backs $\mathbf{W} \to \mathbf{X}$ and $\mathbf{W}' \to \mathbf{X}'$, covering ϕ on morphisms. The second condition forces the morphism map into the form $(\gamma, w) \mapsto (\phi(\gamma), \Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.mor_val`. The identity is a strong bundle functor `Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleFunctor_id` (Findings, Chapter 8, item 34) (2017 p. 274).

The equivalence (V8.25) `Chapter8/EpStrongBundle · EpStrongBundle.isMap_iff_strongBundleFunctor`. *$\Phi : W \to W'$ is a strong bundle map of Definition 8.3.2 (2017: 8.5) iff it is the object map of a strong bundle functor of Definition 8.3.3 (2017: 8.6), and the functor, $\Phi(\gamma, w) = (\phi(\gamma), \Phi(w))$, is unique.* The text states that "these are identical notions" and that the functor is recovered from the map. *Step 1, from the functor to the map.* The object map of a strong bundle functor is a strong bundle map by definition, and its compatibility with μ is the functor's compatibility with targets, read through the form of the morphism map: $\Phi(\mu(\gamma, w)) = \Phi(t(\gamma, w)) = t'(\phi(\gamma), \Phi(w)) = \mu'(\phi(\gamma), \Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.isMap`. *Step 2, from the map to the functor.* $(\gamma, w) \mapsto (\phi(\gamma), \Phi(w))$ lands in the pull-back, since $s(\phi(\gamma)) = \phi(Pw) = P'(\Phi(w))$ `Chapter8/EpStrongBundle · EpStrongBundle.liftMor`; it is a functor by the functor laws of ϕ and the compatibility with μ `Chapter8/EpStrongBundle · EpStrongBundle.IsMap.liftFunctor`. It is a strong bundle map of the pull-backs: linear on the fibres `Chapter8/EpStrongBundle · isLinearMap_fiberRestrict_pullbackMap`, sc-smooth on $[0]$ by the universal property of §VIII.10, and on $[1]$ by the universal property of the pull-back on level $[1]$, which is proved here `Chapter8/EpStrongBundle · StrongBundle.pullbackProj'One` `Chapter8/EpStrongBundle · StrongBundle.isScSmoothMap_to_pullback_one` `Chapter8/EpStrongBundle · EpStrongBundle.IsMap.isStrongBundleFunctor_liftFunctor`. *Step 3, uniqueness.* Every strong bundle functor with object map Φ has this morphism map, by the form of Definition 8.3.3's (2017: 8.6) morphism maps `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.eq_liftFunctor` (Findings, Chapter 8, item 34) (pp. 377–379 (2017 pp. 273–274)).

VIII.14 Definition 8.3.4 (2017: 8.7): section functors

Definition 8.3.4 (2017: 8.7) (V8.26). *(1) An sc-smooth section functor of (P, μ) is an sc-smooth section f of P with $f(y) = \mu(\varphi, f(x))$ for every $\varphi : x \to y$; (2) an sc⁺-section functor takes values in $W_{0,1}$ and is sc-smooth into $W[1]$; (3) an sc-Fredholm section functor is an sc-smooth section functor that is an sc-Fredholm section of P. The sc-smooth section functors form a vector space $\Gamma(P, \mu)$, the sc⁺-section functors a subspace $\Gamma^+(P, \mu)$.* The three notions `Chapter8/EpStrongBundle · EpStrongBundle.IsFunctorialSection` `Chapter8/EpStrongBundle · EpStrongBundle.IsScSmoothSectionFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.IsScPlusSectionFunctor` `Chapter8/EpStrongBundle · EpStrongBundle.IsScFredholmSectionFunctor` and their sets `Chapter8/EpStrongBundle · EpStrongBundle.sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.fredholmSectionFunctors` are as printed; (3) uses Part I's Definition 3.1.16 (2017: 3.8) in its germ reading (§III.48) `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection`, and Part I's sections `Chapter2/StrongBundle · StrongBundle.IsScSmoothSection`. *"Vector space"* is read as closure under Part I's fibrewise zero section, sums and constant multiples `Chapter3/StrongBundleSections · fiberZero` `Chapter3/StrongBundleSections · sectionAdd`; no `Module` structure is built. *Step 1, the analytic point.* Part I does not state that the sum of two sc-smooth sections, or a constant multiple of one, is sc-smooth. It is proved in the charts: a section is sc-smooth as soon as its principal parts are, as maps into the local strong bundle models `Chapter8/EpStrongBundle · StrongBundle.isScSmoothSection_of_principal`, and the principal part of a sum is the sum of the principal parts `Chapter8/EpStrongBundle · StrongBundle.IsScSmoothSection.sectionAdd_of_isScSmoothSection` `Chapter8/EpStrongBundle · StrongBundle.IsScSmoothSection.sectionSmul_const` `Chapter8/EpStrongBundle · StrongBundle.isScSmoothSection_fiberZero`. *Step 2, functoriality* is preserved because each $\mu(\varphi, \cdot)$ is linear (§VIII.11). So Γ contains the zero section and is closed under sums and constant multiples `Chapter8/EpStrongBundle · EpStrongBundle.fiberZero_mem_sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.sectionAdd_mem_sectionFunctors` `Chapter8/EpStrongBundle · EpStrongBundle.`

sectionSmul_mem_sectionFunctors, and so is $\Gamma^+ \subseteq \Gamma$, the values in $W_{0,1}$ and the sc-smoothness into $W[1]$ being preserved in the same way Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors_subset Chapter8/EpStrongBundle · EpStrongBundle.fiberZero_mem_plusSectionFunctors Chapter8/EpStrongBundle · EpStrongBundle.sectionAdd_mem_plusSectionFunctors Chapter8/EpStrongBundle · EpStrongBundle.sectionSmul_mem_plusSectionFunctors; the sc-Fredholm section functors lie in Γ Chapter8/EpStrongBundle · EpStrongBundle.fredholmSectionFunctors_subset. All three sets are inhabited by non-zero sections: over a trivial ep-groupoid every section is functorial, and the graph section of a smooth map $\mathbb{R}^n \to \mathbb{R}^k$ lies in Γ and is sc-Fredholm (§VIII.26) (Findings, Chapter 8, items 35, 102, 103) (pp. 378–384 (2017 pp. 274–275)).

VIII.15 Definition 8.5.1 (2017: 8.8), and the degeneracy remark after it

Definition 8.5.1 (2017: 8.8) (V8.27) Chapter8/Basic · EpFunctor.IsProperCovering. *An sc-smooth functor $F : Y \to X$ is a proper covering functor if (1) F is a surjective local sc-diffeomorphism on objects; (2) every $F^{-1}(x)$ is finite, and x has a neighbourhood $U(x)$ with $F^{-1}(U(x))$ the disjoint union of neighbourhoods $U(y)$, $y \in F^{-1}(x)$, each mapped sc-diffeomorphically onto $U(x)$; (3) $L_F : \varphi \mapsto (F(\varphi), s(\varphi))$ is an sc-diffeomorphism $\mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$.* In (2) $U(x)$ is open and the local inverses $\sigma(y)$ of F on the $U(y)$ are part of the data; (3) is stated on Part I's fibred-product sub-M-polyfold of $\mathbf{X} \times Y$ (Proposition 2.5.20 (2017: 2.15), Proposition 2.46 of the preprint, §II.17) Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct Chapter8/Basic · EpFunctor.fibredMap, as Chapter 7 states Definition 7.1.3 (4) (2017: 7.3). The identity functor of every ep-groupoid is a proper covering functor Chapter8/NonVacuityBundle · EpFunctor.isProperCovering_id (Findings, Chapter 8, items 50, 102) (2017 p. 275).

The degeneracy remark (V8.28). *For $\varphi \in \mathbf{Y}$, $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(F(\varphi), s(\varphi)) = \frac{1}{2}\, d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi))$.* **The text** argues that L_F is an sc-diffeomorphism and that $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$, with $d_{\mathbf{X}}(F(\varphi)) = d_Y(s(\varphi)) = d_{\mathbf{Y}}(\varphi)$, F and s being local sc-diffeomorphisms. **The formalized proof** splits the claims. *Step 1, in general.* $d_{\mathbf{Y}}(\varphi) = d_{\mathbf{X}\,{}_s\!\times_F Y}(L_F(\varphi))$ and $d_{\mathbf{X}}(F(\varphi)) = d_{\mathbf{Y}}(\varphi)$, by Part I's Proposition 2.4.4 (2017: 2.7), the invariance of d under sc-diffeomorphisms (§II.11) Chapter2/DegeneracyIndex · proposition_2_24 Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_fibredMap Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_mor. *Step 2, the product formula.* Part I's formalization has $d_{X\times Y} = d_X + d_Y$ for tame factors Chapter2/FibredProduct · MPolyfold.isTame_prod and in general only $d_{X\times Y} \le d_X + d_Y$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le, and Part I's Proposition 2.5.20 (2017: 2.15) states its formula $d = \frac{1}{2} d_{X\times Y}$ for tame X, Y. So $d_{\mathbf{X}\times Y}(F(\varphi), s(\varphi)) = 2\, d_{\mathbf{Y}}(\varphi)$, and with it the remark's second equality, is proved for tame X and Y ($\mathbf{X}$ is then tame, s being a local sc-diffeomorphism) Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_prod_fibredMap, and in general as the inequality $\le$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.degIndex_prod_fibredMap_le. As we read it, the remark needs X and Y tame, which the text does not state; whether the equality holds without tameness is not settled here (Findings, Chapter 8, item 51) (pp. 379–385 (2017 pp. 275–276)).

VIII.16 Proposition 8.5.2 (2017: 8.2): composites of proper covering functors

Proposition 8.5.2 (2017: 8.2) (V8.29) Chapter8/ProperCovering · EpFunctor.IsProperCovering.proposition_8_2. *If $F : Y \to X$ and $H : Z \to Y$ are proper covering functors, so is $F \circ H$.* **The text's proof** says that (1) and (2) "follow easily", and for (3) writes $L_{F\circ H} = \tilde{K} \circ L_H$ with $\tilde{K}(\psi, z) = (F(\psi), z)$, whose inverse is $\tilde{L}(\varphi, z) = (L_F^{-1}(\varphi, H(z)), z)$. **The formalized proof.** *Step 1, (1)* holds since composites of surjective local sc-diffeomorphisms are such, and the fibre of $F \circ H$ over x is the finite union of the fibres of H over the points of $F^{-1}(x)$. *Step 2, (2).* The even covering of F at x has to be shrunk: take $U(x)$ for F and, for each $y \in F^{-1}(x)$, the evenly covered neighbourhood $U'(y)$ of H; the set $U_0 = \bigcap_y F(U(y) \cap U'(y))$, a finite intersection, is an open neighbourhood of x with $(F|U(y))^{-1}(U_0) \subseteq U'(y)$, and over U_0 the sheets of $F \circ H$ are those of H over the $U'(y)$, cut down to the preimages of the sets $(F|U(y))^{-1}(U_0)$

Chapter8/ProperCovering · EpFunctor.IsProperCovering.evenlyCovered_comp. *Step 3, (3)*, through the text's $\tilde{K}$ and $\tilde{L}$, which are inverse to each other Chapter8/ProperCovering · EpFunctor.IsProperCovering.isScDiffeo_fibredMap_comp. Their sc-smoothness, as maps between sub-M-polyfolds of products, uses Part I's Proposition 2.3.19 (2) (2017: 2.6) (a map into a sub-M-polyfold is sc-smooth as soon as it is sc-smooth into the ambient M-polyfold), proved again privately, as in Chapter 7. The proof has "fo" for "for" (possible misprint) (Findings, Chapter 8, item 52) (pp. 384–386 (2017 pp. 276–277)).

VIII.17 Theorem 8.5.3 (2017: 8.5) (Proper Covering I), and Remark 8.5.4 (2017: 8.2)

Theorem 8.5.3 (2017: 8.5) (V8.30) Chapter8/ProperCoveringLocal · EpFunctor.theorem_8_5. *Let $F : Y \to X$ be a proper covering functor and $x \in X$. For a connected, sufficiently small open $U(x)$ admitting the natural G_x-action: (0) $F^{-1}(U(x))$ is the disjoint union of connected open $U(y) \ni y$, $y \in F^{-1}(x)$, with $F : U(y) \to U(x)$ sc-diffeomorphisms; (1) $\mathbf{U}(x) = \{\varphi \mid s(\varphi), t(\varphi) \in U(x)\}$ is the disjoint union of the components $\mathbf{U}(g)$, $g \in G_x$, on which s, t are sc-diffeomorphisms onto $U(x)$; (2) the morphisms from $U(y)$ to $U(y')$ form the disjoint union of connected open $\mathbf{U}(\psi_0) \ni \psi_0$, $\psi_0 : y \to y'$, with (3) $s : \mathbf{U}(\psi_0) \to U(y)$ and $t : \mathbf{U}(\psi_0) \to U(y')$ sc-diffeomorphisms, (4) $F : \mathbf{U}(\psi_0) \to \mathbf{U}(F(\psi_0))$ an sc-diffeomorphism, and (5) a commutative square of sc-diffeomorphisms with top arrow F, left arrow s, right arrow t and bottom arrow $F : U(s(\psi_0)) \to U(t(F(\psi_0)))$.* The data form a chart Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart: a natural representation R of G_x on $U(x)$ (Chapter 7's datum, §VII.7, with $U(x)$ open and connected), the sets $U(y)$ and $\mathbf{U}(\psi_0)$, and (0)–(5) as fields; $\mathbf{U}(g)$ is read as the connected component of $\mathbf{U}(x)$ at g, which contains $F(\psi_0)$ by Remark 8.5.4 (2017: 8.2) (a). "Sufficiently small" is proved in two forms: charts with $U(x)$ inside every open $V \ni x$ (`theorem_8_5`), and one open $W \ni x$ such that every natural representation on a connected $U(x) \subseteq W$ is the R of a chart Chapter8/ProperCoveringLocal · EpFunctor.theorem_8_5_of_naturalRep.

The text's proof starts from $U'(x)$ and the $U'(y)$ of Definition 8.5.1 (2) (2017: 8.8), takes connected $V(y) \subseteq U'(y)$ with natural representations of the G_y and, for each $\psi : y \to y'$, sheets $V(\psi)$ with $\hat{\psi} = t \circ (s|V(\psi))^{-1} : V(y) \to V(y')$ (8.15), so that the morphisms between points of $\bigcup V(y)$ form the disjoint union of the $V(\psi)$. It shrinks to a G_x-invariant connected $U(x)$, puts $U(y) = (F|V(y))^{-1}(U(x))$, shows that $\hat{\psi}$ maps $U(y)$ onto $U(y')$, and sets $\mathbf{U}(\psi) = (s|V(\psi))^{-1}(U(y))$; (4) comes from $F = (s|\mathbf{U}(g))^{-1} \circ F \circ s$ on $\mathbf{U}(\psi_0)$, and (5) is called obvious. **The formalized proof** uses Definition 8.5.1 (3) (2017: 8.8) instead of natural representations in Y: the sheets $\mathbf{U}(\psi_0)$ are the images of continuous sections of s given by L_F^{-1}. *Step 1, the input* Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data: $U'(x)$ with the $U'(y)$ and local inverses $\sigma(y)$ of F (Definition 8.5.1 (2) (2017: 8.8)), the continuous inverse L_F^{-1} of the sc-diffeomorphism L_F (Definition 8.5.1 (3) (2017: 8.8)), and a natural representation R with $U(x) \subseteq U'(x)$; one $W = U'(x)$ serves for every such R Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.exists_data. *Step 2, (0).* $U(y) = U'(y) \cap F^{-1}(U(x)) = \sigma(y)(U(x))$ Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.U is open, connected as the image of the connected $U(x)$, and mapped sc-diffeomorphically onto $U(x)$. *Step 3, (1)* involves only R and is Chapter 7's description of $\mathbf{U}(x)$ by the sheets $\Gamma(g, U(x))$ (§VII.8) Chapter7/NaturalAction · EpGroupoid.NaturalRep.iUnion_image_Γ Chapter7/NaturalAction · EpGroupoid.NaturalRep.connectedComponentIn_morOn. *Step 4, the sheets.* For $\psi_0 : y \to y'$ with $g = F(\psi_0) \in G_x$, put $\tau(z) = L_F^{-1}(\Gamma(g, F(z)), z)$ for $z \in U(y)$ Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.τ and $\mathbf{U}(\psi_0) = \tau(U(y))$ Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.sheet. τ is a continuous section of s with $\tau(y) = \psi_0$, and a continuous section of a local sc-diffeomorphism over an open set is sc-smooth, being locally a local inverse Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.isScSmoothOn_of_section Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScSmoothOn_τ; this gives s in (3). *Step 5, t in (3).* $F(t(\tau z)) = t(\Gamma(g, Fz)) = \Phi(g)(Fz) \in U(x)$, so $t \circ \tau$ maps $U(y)$ into $F^{-1}(U(x))$, the disjoint union of the open $U(y'')$; since $U(y)$ is connected and $t(\tau(y)) = y'$, it maps into $U(y')$ Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.mapsTo_of_isPreconnected Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.tgt_τ_mem. This is the counterpart of the text's

claim that $\widehat{\psi}$ maps $U(y)$ onto $U(y')$; there $t \circ \tau = \sigma(y') \circ \Phi(g) \circ F$, an sc-diffeomorphism with inverse $\sigma(y) \circ \Phi(g)^{-1} \circ F$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.isScDiffeoOn_congr_left` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScDiffeoOn_hat`. *Step 6, (2).* A morphism ψ from $U(y)$ to $U(y')$ has $F(\psi) \in \mathbf{U}(x)$, so $F(\psi) = \Gamma(g, F(s\psi))$ for one $g \in G_x$ by (1), and $\psi = L_F^{-1}(\Gamma(g, F(s\psi)), s\psi)$ lies in the sheet of the morphism ψ_0 out of y with $F(\psi_0) = g$; distinct ψ_0 give disjoint sheets by the injectivity of L_F, and the sheets are the connected components `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.connectedComponentIn_eq_of_iUnion`. *Step 7, (4)* is the text's formula: $F = \Gamma(g, \cdot) \circ F \circ s$ on $\mathbf{U}(\psi_0)$, an sc-diffeomorphism onto $\Gamma(g, U(x)) = \mathbf{U}(g)$ with inverse $\tau \circ \sigma(y) \circ s$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.isScDiffeoOn_mor_sheet`. *Step 8, (5)* is the functoriality $F(s\psi) = s(F\psi)$ (the field `comm_src`), together with the companion square $F(t\psi) = t(F\psi)$ (the field `comm_tgt`); the arrows are sc-diffeomorphisms by (0), (1), (3), (4). The chart is assembled `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringLocal.Data.toChart`, and charts exist inside every V because natural representations do (Theorem 7.1.13 (2017: 7.1)) `Chapter7/NaturalRep · EpGroupoid.theorem_7_1`. The finiteness of $F^{-1}(x)$ and Definition 8.5.1 (1) (2017: 8.8) are not used.

(5) as printed. With t on the right arrow and the corner $U(t(F(\psi_0)))$, the square says $F(s(\psi)) = t(F(\psi))$ on $\mathbf{U}(\psi_0)$. A functor gives $F(s(\psi)) = s(F(\psi))$, and $F(\psi) = \Gamma(g, F(s\psi))$ has $t(F(\psi)) = \Phi(g)(F(s\psi))$, which differs from $F(s\psi)$ wherever $\Phi(g)$ does not fix $F(s\psi)$. So (5) as printed does not hold as we read it: for the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$, a proper covering functor, every chart at 0 contains, in the sheet of $\psi_0 = (-1, 0)$ `Chapter8/NonVacuity · CoveringExamples.reflectionψ₀`, a $\psi = (-1, z)$ with $F(s\psi) = z \neq -z = t(F\psi)$ `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_five_as_printed`, and such charts exist `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_five_as_printed_nonvacuous`. The square holds with s on both vertical arrows (the corner $U(s(F(\psi_0))) = U(x)$ is the same set), which suggests t as a possible misprint for s. Possible misprints in the proof: "$g \in G_y$" and "G_y-invariant" for G_x, "$F(z') = s(F(\varphi))$" for $t(F(\varphi))$, "admitting admitting", and "morphismsm" in (2) (Findings, Chapter 8, items 53, 54, 55) (pp. 385–388 (2017 pp. 277–279)).

Remark 8.5.4 (2017: 8.2) (V8.31). *(a) A morphism between points of $F^{-1}(x)$ is mapped into G_x; (b) everything in Theorem 8.5.3 (2017: 8.5) is determined by the choice of $U(x)$.* (a) is functoriality `Chapter8/ProperCoveringLocal · EpFunctor.remark_8_2_a`, and defines the element of G_x `Chapter8/ProperCoveringLocal · EpFunctor.isotropyMor`. (b) is read as: two charts with the same $U(x)$ have the same Φ and Γ on $U(x)$, the same $U(y)$ and the same $\mathbf{U}(\psi_0)$ `Chapter8/ProperCoveringLocal · EpFunctor.remark_8_2_b`. *Step 1.* On a connected $U(x)$, $\Gamma(g, \cdot)$ is the continuous section of s through g, and two such sections agree (§VII.9) `Chapter7/NaturalAction · EpGroupoid.NaturalRep.Γ_eqOn`; $\Phi(g) = t \circ \Gamma(g, \cdot)$ follows. *Step 2.* The $U(y)$ and the $\mathbf{U}(\psi_0)$ are the connected components of $F^{-1}(U(x))$ at y and of the morphisms from $U(y)$ to $U(y')$ at ψ_0, so they are determined by $U(x)$ `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart.connectedComponentIn_U` `Chapter8/ProperCoveringLocal · EpFunctor.ProperCoveringChart.connectedComponentIn_sheet`. The remarks on the naturality of Theorem 7.1.13's (2017: 7.1) construction have no further formal content (Findings, Chapter 8, item 56) (pp. 385–387 (2017 pp. 277–278)).

VIII.18 Diff_{sc}, Definition 8.5.5 (2017: 8.9), and the discussion after it

Diff_{sc} and Definition 8.5.5 (2017: 8.9) (V8.32). *Diff_{sc} has the M-polyfolds as objects and the sc-diffeomorphisms as morphisms. For finite groupoids C, with one object, and D, a functor $f : D \to C$ is a covering functor if $\psi \mapsto (f(\psi), s(\psi))$ is a bijection $\mathbf{D} \to C \times_f D$; a geometric lift of f is a pair of functors $L_D : D \to \mathrm{Diff}_{sc}$, $L_C : C \to \mathrm{Diff}_{sc}$ with a natural transformation $\tau : L_D \to L_C \circ f$.* Diff_{sc} is the category of bundled M-polyfolds of one universe `Chapter8/Basic · MPolyfoldCat` with Part I's sc-diffeomorphisms, bundled with their inverses `Chapter8/Basic · ScDiffeo` `Chapter8/Basic · ScDiffeo.id` `Chapter8/Basic · ScDiffeo.comp` `Chapter8/Basic · ScDiffeo.ext`. C is Mathlib's `SingleObj G` for a group G, its automorphism group, and D a Mathlib groupoid, both in `Type`; then $C \times_f D = G \times D$ `Chapter8/Basic · coveringMap` `Chapter8/Basic · IsCoveringFunctor` `Chapter8/Basic · GeometricLift`. The covering property says that a $g \in G$ lifts uniquely at every y `Chapter8/GeometricLift · IsCoveringFunctor.existsUnique_lift` `Chapter8/GeometricLift · IsCoveringFunctor.lift`. The finiteness ("finitely many objects and morphisms") is not part of the two structures; it is carried as instance

arguments by the results that use it, Proposition 8.5.7 (2017: 8.3) and Theorem 8.5.8 (2017: 8.6). $D = \emptyset$ is allowed, as in the text `Chapter8/GeometricLift · isCoveringFunctor_of_isEmpty` (§VIII.20). The definition is not vacuous in either direction: the identity of `SingleObj G` is a covering functor, and for non-trivial G the functor sending everything to 1 is not `Chapter8/NonVacuity · CoveringExamples.isCoveringFunctor_id` `Chapter8/NonVacuity · CoveringExamples.not_isCoveringFunctor_one` (Findings, Chapter 8, items 57, 100) (pp. 387–389 (2017 pp. 279–280)).

The discussion after Definition 8.5.5 (2017: 8.9) (V8.33). *With $U(x) = L_C(\star)$, $U(y) = L_D(y)$, the maps $\tau(y) : U(y) \to U(x)$ and $\varphi_{yy'}(\psi) = L_D(\psi)$ are sc-diffeomorphisms, $\varphi_{yy''}(\psi \circ \gamma) = \varphi_{y'y''}(\psi) \circ \varphi_{yy'}(\gamma)$, G_y acts on $U(y)$ and G_x on $U(x)$, and $\tau(y)(h * z) = f(h) * \tau(y)(z)$.* The text states these as consequences of the definition; they are proved. $U(x)$ `Chapter8/GeometricLift · GeometricLift.Ux` and $\tau(y)$ `Chapter8/GeometricLift · GeometricLift.τy` `Chapter8/GeometricLift · GeometricLift.isScDiffeoOn_τy`; $\varphi_{yy'}$ `Chapter8/GeometricLift · GeometricLift.φ` `Chapter8/GeometricLift · GeometricLift.isScDiffeoOn_φ` is functorial because L_D is `Chapter8/GeometricLift · GeometricLift.φ_comp` `Chapter8/GeometricLift · GeometricLift.φ_id` `Chapter8/GeometricLift · GeometricLift.φ_inv_φ`. The actions are $h * z = \varphi_{yy}(h)(z)$ and $g * z = L_C(g)(z)$ `Chapter8/GeometricLift · GeometricLift.smul_def` `Chapter8/GeometricLift · GeometricLift.smul_def'`, sc-smooth `Chapter8/GeometricLift · GeometricLift.isScSmoothMap_smul`; the action of G_x is a left action through the composition of `SingleObj G`, in which $g \gg h = hg$. The naturality of τ at a morphism, $\tau(y')(\varphi_{yy'}(\psi)(z)) = f(\psi) * \tau(y)(z)$ `Chapter8/GeometricLift · GeometricLift.τy_φ`, specializes to the equivariance `Chapter8/GeometricLift · GeometricLift.τ_equivariant`. Functoriality is stated with Mathlib's $\gamma \gg \psi$ for the text's $\psi \circ \gamma$, and G_y is Mathlib's monoid `End y` (Findings, Chapter 8, item 57) (pp. 388–390 (2017 pp. 280–281)).

VIII.19 Disjoint unions of M-polyfolds

Disjoint unions (V8.34). *In the proof of Proposition 8.5.7 (2017: 8.3), "$Y = \bigsqcup U(y)$ possesses a natural M-polyfold structure".* The text does not describe it. It is read as the structure whose charts are those of the summands, transported along the inclusions `Chapter8/SigmaPolyfold · MChart.pushforward` `Chapter8/SigmaPolyfold · MPolyfold.sigmaChart` `Chapter8/SigmaPolyfold · MPolyfold.sigma`, for any index type. *Step 1, compatibility.* Charts of different summands have disjoint domains, and Part I's compatibility then asks for the sc-smoothness of maps between empty retracts, which holds, so nothing is added `Chapter8/SigmaPolyfold · MChart.compatible_of_disjoint`; the charts of the union are exactly the transported charts `Chapter8/SigmaPolyfold · MPolyfold.isChart_sigmaChart_iff`. A transported chart is extended by 0 off its image, so no summand needs to be nonempty. *Step 2, paracompactness* is proved summand by summand, without metrizability `Chapter8/SigmaPolyfold · paracompactSpace_sigma`. *Step 3, summand-wise properties.* The summands are open and sc-diffeomorphic to the X_i `Chapter8/SigmaPolyfold · MPolyfold.isScDiffeoOn_sigmaMk`; sc-smoothness, local sc-diffeomorphisms, the degeneracy index and tameness are checked summand by summand `Chapter8/SigmaPolyfold · MPolyfold.isScSmoothMap_sigma_iff` `Chapter8/SigmaPolyfold · MPolyfold.isLocalScDiffeo_sigma_iff` `Chapter8/SigmaPolyfold · MPolyfold.degIndex_sigma` `Chapter8/SigmaPolyfold · MPolyfold.isTame_sigma_iff`. *Step 4, products.* Products distribute over the union `Chapter8/SigmaPolyfold · MPolyfold.isScDiffeoOn_sigmaProdDistrib`, and $G \times X \cong \bigsqcup_G X$ for discrete G `Chapter8/SigmaPolyfold · isScDiffeoOn_prod_discrete_sigma`, which identifies the text's $\mathrm{mor}(y, y') \times U(y)$ with the model of §VIII.20 (Findings, Chapter 8, item 58) (2017 p. 281).

VIII.20 Proposition 8.5.7 (2017: 8.3): the proper covering functor of a geometric lift

Proposition 8.5.7 (2017: 8.3) (V8.35) `Chapter8/GeometricLift · proposition_8_3`. *A covering functor $f : D \to C$ with a geometric lift gives a proper covering functor $F : Y \to X$, with $X = G_x \ltimes U(x)$, $Y = \bigsqcup_y U(y)$, $\mathbf{Y} = \bigsqcup_{y,y'} \mathrm{mor}(y, y') \times U(y)$, $F|U(y) = \tau(y)$ and $F(\psi, z) = (f(\psi), \tau(y)(z))$.*

The text's proof defines X, Y and F, says that "one verifies" that Y is an ep-groupoid and F a functor, and shows that (8.16), L_F, is injective, surjective and a local sc-diffeomorphism. **The formalized proof** is for nonempty D. *Step 1, X* is Chapter 7's translation groupoid (§VII.30) `Chapter7/NonVacuity · EpGroupoid.Examples.actionEpGroupoid` `Chapter8/GeometricLift · geometricLiftTarget`, with G discrete and lifted by `ULift` to the universe of $U(x)$; a finite G makes it an ep-groupoid. *Step 2, Y.* Objects and morphisms are disjoint unions (§VIII.19) `Chapter8/GeometricLift · GeometricLift.YObj` `Chapter8/GeometricLift · GeometricLift.YMor` `Chapter8/GeometricLift · GeometricLift.yObjPolyfold` `Chapter8/GeometricLift · GeometricLift.yMorPolyfold`, $\mathrm{mor}(y,y') \times U(y)$ being modelled as $\bigsqcup_{\psi:y\to y'} U(y)$. The composition $(\psi,z')\circ(\gamma,z) = (\psi\circ\gamma,z)$ is total, with a junk value off composable pairs `Chapter8/GeometricLift · homComp` `Chapter8/GeometricLift · GeometricLift.sourceStr`. On the summand of ψ, s is the identity of $U(y)$ and t is $\varphi_{yy'}(\psi)$, so both are local sc-diffeomorphisms `Chapter8/GeometricLift · GeometricLift.isLocalScDiffeo_ySrc` `Chapter8/GeometricLift · GeometricLift.isLocalScDiffeo_yTgt`. "One verifies" is carried out: t is proper, finitely many summands each being mapped homeomorphically onto a summand, so Definition 7.1.3 (3) (2017: 7.3) holds with $V = Y$ `Chapter8/GeometricLift · GeometricLift.isProperMap_yTgt`; and (4) holds in the sheet form, since near (ψ_0,z_0) the map $\psi\mapsto\sigma(t\psi)\circ\psi$ is $(\psi_0,z)\mapsto(\gamma_0\circ\psi_0,z)$, the summand index γ_0 of $\sigma(t\psi)$ being locally constant `Chapter8/GeometricLift · GeometricLift.isScSmoothCompSheet_y` `Chapter8/GeometricLift · geometricLiftSource`. *Step 3, F* is a functor by the naturality of τ (§VIII.18) `Chapter8/GeometricLift · geometricLiftFunctor`. *Step 4, Definition 8.5.1 (1), (2) (2017: 8.8).* F is a local sc-diffeomorphism, each $\tau(y)$ being one on an open summand `Chapter8/GeometricLift · geometricLiftFunctor_isLocalScDiffeo`, and surjective since $D \neq \emptyset$ `Chapter8/GeometricLift · geometricLiftFunctor_surjective`; its fibres have one point in each summand and are finite `Chapter8/GeometricLift · geometricLiftFunctor_finite_fiber`, and (2) holds with $U(x)$ the whole object space of X, the sheets being the summands `Chapter8/GeometricLift · geometricLiftFunctor_evenlyCovered`. *Step 5, (8.16).* $L_F(\psi,z) = ((f(\psi),\tau(y)z),(y,z))$ is bijective because f is a covering functor. The text deduces the sc-diffeomorphism from "bijection and local sc-diffeomorphism"; here the inverse is shown sc-smooth directly: near $((g_0,q_0),(y_0,z_0))$ it is $((g_0,q),(y_0,z))\mapsto(\psi_0,z)$, ψ_0 the lift of g_0 at y_0 `Chapter8/GeometricLift · GeometricLift.fibredInv` `Chapter8/GeometricLift · GeometricLift.isScSmoothMap_fibredInv` `Chapter8/GeometricLift · geometricLiftFunctor_isScDiffeo_fibredMap`. *D nonempty.* Definition 8.5.5 (2017: 8.9) allows $D=\emptyset$, and then $Y=\emptyset$ while $U(x)$ is arbitrary: F is surjective on objects iff $D\neq\emptyset$ or $U(x)=\emptyset$ `Chapter8/GeometricLift · geometricLiftFunctor_surjective_iff`. The text does not say so; the hypothesis is used for the surjectivity only, and holds in Theorem 8.5.8 (2017: 8.6). The proposition is not vacuous: $\mathbb{Z}/2$ acting on $\mathbb{R}$ by reflection is a geometric lift of the identity of `SingleObj` $\mathbb{Z}/2$ (§VIII.26). The proof's "$F : X \to Y$" is read as $F : Y \to X$ (possible misprint) (Findings, Chapter 8, items 59, 101) (pp. 389–391 (2017 pp. 281–282)).

VIII.21 Theorem 8.5.8 (2017: 8.6) (Proper Covering II)

Theorem 8.5.8 (2017: 8.6) (V8.36) `Chapter8/GeometricLift · EpFunctor.theorem_8_6`. *Let $F : Y \to X$ be a proper covering functor, $x \in X$, C the full subcategory on x and D the full subcategory on $F^{-1}(x)$. Then $f = F|D : D \to C$ is a covering functor, and for $U(x)$ as in Theorem 8.5.3 (2017: 8.5) there are a geometric lift $A : K \to L$ of f and fully faithful open embeddings $\Phi : L \to X$, $\Psi : K \to Y$ whose images are the full subcategories on $U(x)$ and $F^{-1}(U(x))$, with $F\circ\Psi = \Phi\circ A$.* **The text's proof** puts $L_C(g) = \Phi(g)$, $L_D(y) = U(y)$, $L_D(\psi) = t\circ(s|\mathbf{U}(\psi_0))^{-1}$, $\tau(y) = F|U(y)$, says "one verifies" that this is a geometric lift, takes A from Proposition 8.5.7 (2017: 8.3) and puts $\Phi(g,z) = \Gamma(g,z)$, $\Psi(\psi,z) = (s|\mathbf{U}(\psi_0))^{-1}(z)$. **The formalized proof** follows it, and carries out the "one verifies" through F being injective on each $U(y)$. *Step 1, f.* C is `SingleObj` G_x and D the full subcategory on $F^{-1}(x)$ `Chapter8/GeometricLift · GroupoidStr.FullOb` `Chapter8/GeometricLift · EpFunctor.FiberCat`; F sends its morphisms into G_x `Chapter8/GeometricLift · EpFunctor.fiberIsotropy` `Chapter8/GeometricLift · EpFunctor.restrictFiber`. f is a covering functor because L_F is a bijection and a morphism ψ with $s(\psi)\in F^{-1}(x)$ and $F(\psi)\in G_x$ has $t(\psi)\in F^{-1}(x)$ `Chapter8/GeometricLift · EpFunctor.covering_restrict`. D is finite by Definition 8.5.1 (2) (2017: 8.8) and Proposition 7.1.12 (2017: 7.4) `Chapter8/GeometricLift · EpFunctor.fintypeFiberCat`, and nonempty by the surjectivity of F `Chapter8/GeometricLift · EpFunctor.nonempty_fiberCat`. *Step 2, the lift.* For a chart of Theorem 8.5.3 (2017: 8.5), let σ_ψ be the inverse of s on $\mathbf{U}(\psi)$ `Chapter8/GeometricLift · EpFunctor.`

ProperCoveringChart.σSrc. By (1) and (4) of Theorem 8.5.3 (2017: 8.5), $F(\sigma_\psi(z)) = \Gamma(f(\psi), F(z))$ Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.mor_σSrc, so $\tau(y') \circ L_D(\psi) = \Phi(f(\psi)) \circ \tau(y)$, the naturality of τ. The functoriality of L_D follows: $t\sigma_{\gamma \gg \psi}(z)$ and $t\sigma_\psi(t\sigma_\gamma(z))$ lie in the same $U(y'')$ and have the same image $\Phi(f(\psi))\Phi(f(\gamma))F(z)$ under F, which is injective on $U(y'')$ Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.hat_comp. The sets $U(x)$, $U(y)$ become objects of Diff_{sc} as open sub-M-polyfolds Chapter8/GeometricLift · MPolyfoldCat.ofOpen Chapter8/GeometricLift · ScDiffeo.ofOpen, and the geometric lift is assembled Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.geometricLift; A is its functor of Proposition 8.5.7 (2017: 8.3), a proper covering functor since $D \neq \emptyset$. *Step 3, Φ and Ψ.* Φ is the inclusion on objects and Γ on morphisms Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.Φ; Ψ is the inclusion on objects and $\Psi(\psi, z) = \sigma_\psi(z)$ Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.Ψ, which preserves compositions and units by the unique lifting of Definition 8.5.1 (3) (2017: 8.8): $\Psi(\psi, z') \circ \Psi(\gamma, z)$ and $\Psi(\gamma \gg \psi, z)$ have the same source and the same image under F. "Fully faithful open embedding" is read as full and faithful on the underlying groupoids, and an injective local sc-diffeomorphism on objects and on morphisms Chapter8/GeometricLift · EpFunctor.IsFullyFaithfulOpenEmbedding Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Φ Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.isFullyFaithfulOpenEmbedding_Ψ; the images are the full subcategories, on objects and on morphisms Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Φ_obj Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Φ_mor Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Ψ_obj Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.range_Ψ_mor, the last two by Theorem 8.5.3 (0) and (2) (2017: 8.5). *Step 4, the square.* $F(\Psi(\psi, z)) = F(\sigma_\psi(z)) = \Gamma(f(\psi), F(z)) = \Phi(A(\psi, z))$ Chapter8/GeometricLift · EpFunctor.ProperCoveringChart.comp_Ψ. *Readings.* The theorem is stated for ep-groupoids whose objects and morphisms are in Type, since Definition 8.5.5 (2017: 8.9) is; "a sufficiently small $U(x)$ as in Theorem 8.5" is any chart of Theorem 8.5.3 (2017: 8.5), and such charts exist inside every open $V \ni x$, which `theorem_8_6` restates; "a geometric lift $A : K \to L$" is a geometric lift with its functor A of Proposition 8.5.7 (2017: 8.3); and since the morphisms of D are the ψ_0 themselves, "$\psi \in \mathbf{U}(\psi_0)$" in the definition of L_D is read as $\psi = \psi_0$. Possible misprints: "there exists exists", "$\tau(t)$" for $\tau(y)$, and "$A(\psi, z) = (f(z), F(z))$" for $(f(\psi), F(z))$, as in Proposition 8.5.7 (2017: 8.3) (Findings, Chapter 8, items 2, 60) (2017 pp. 283–284).

VIII.22 Definition 8.5.9 (2017: 8.10): proper strong bundle covering functors

Definition 8.5.9 (2017: 8.10) (V8.37) Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering. *Let (P, μ) be a strong bundle over Y, (Q, ν) one over X, and $A : W \to V$ a strong bundle map with $Q \circ A = F \circ P$. A is a proper strong bundle covering functor if (1) A is surjective and a local strong bundle isomorphism covering a proper covering functor F; (2) over the even coverings of Definition 8.5.1 (2) (2017: 8.8), $A : W|U(y) \to V|U(x)$ are strong bundle isomorphisms; (3) $(\psi, w) \mapsto ((F(\psi), A(w)), w)$ is a strong bundle isomorphism $\mathbf{W} \to \mathbf{V}\,{}_s\!\times_A W$ covering $L_F : \mathbf{Y} \to \mathbf{X}\,{}_s\!\times_F Y$. Readings.* A is taken in the functorial picture, as a strong bundle functor covering F (Definition 8.3.3 (2017: 8.6); equivalently its object map is a strong bundle map of Definition 8.3.2 (2017: 8.5), §VIII.13). "$A : W|U \to V|U'$ is a strong bundle isomorphism" means that $F : U \to U'$ is an sc-diffeomorphism and $A : P^{-1}(U) \to Q^{-1}(U')$ an sc-diffeomorphism on $[0]$ and on $[1]$ Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn; its inverse is then linear on the fibres, so the fibre maps are topological linear isomorphisms Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn.fiberEquiv. The finiteness of $F^{-1}(x)$ in (2) is that of Definition 8.5.1 (2) (2017: 8.8) and is not repeated. *The strong bundle of (3).* The text does not say which strong bundle structure $\mathbf{V}\,{}_s\!\times_A W \to \mathbf{X}\,{}_s\!\times_F Y$ carries. It is read as the pull-back of $P : W \to Y$ along the sc-smooth projection $\mathrm{pr}_2 : \mathbf{X}\,{}_s\!\times_F Y \to Y$ Chapter8/EpStrongBundle · EpFunctor.fibredSnd Chapter8/EpStrongBundle · EpFunctor.isScSmoothMap_fibredSnd, on Part I's fibred-product sub-M-polyfold. Its points $((\varphi, y), w)$ with $P(w) = y$ correspond to the text's $((\varphi, v), w)$ with $v = A(w)$ Chapter8/EpStrongBundle · EpStrongBundle.fibredBundleEquiv; under the correspondence the projection is the text's $((\varphi, v), w) \mapsto (\varphi, P(w))$ Chapter8/EpStrongBundle · EpStrongBundle.pullbackProj_fibredBundleEquiv, and the

map of (3) is $(\psi, w) \mapsto ((F(\psi), s(\psi)), w)$ Chapter8/EpStrongBundle · EpStrongBundle.fibredLift, which the correspondence turns into the text's $((F(\psi), A(w)), w)$ Chapter8/EpStrongBundle · EpStrongBundle.fibredBundleEquiv_symm_fibredLift and which covers L_F Chapter8/EpStrongBundle · EpStrongBundle.pullbackProj_fibredLift. The definition is satisfiable: the identity of every strong bundle over an ep-groupoid is a proper strong bundle covering functor, with (3) on this reading (§VIII.26). The closing sentence, "all structural statements about proper covering functors lift to the strong bundle situation", is not formalized beyond §8.6.2 (2017 §8.5.2) (Findings, Chapter 8, items 70, 102, 103) (2017 p. 285).

VIII.23 Appendix §8.6.1 (2017 §8.5.1): the morphisms over a fibre, and Lemmas 8.6.1–8.6.3 (2017: 8.9–8.11)

The morphisms over a fibre (V8.38). *For a proper covering functor $F : Y \to X$ and $x \in X$, $\psi \mapsto (F(\psi), s(\psi))$ is a bijection $\mathrm{mor}(F^{-1}(x)) \to G_x \times F^{-1}(x)$, so $\#\mathrm{mor}(F^{-1}(x)) = \#G_x \cdot \#F^{-1}(x)$; $F^{-1}(x)$ splits into isomorphism classes $\Theta_1, \dots, \Theta_{q'}$ and points of one class have isomorphic isotropy groups.* $\mathrm{mor}(F^{-1}(x))$ is the set of morphisms between points of $F^{-1}(x)$. *Step 1, the bijection.* A morphism between points of $F^{-1}(x)$ is mapped into G_x (Remark 8.5.4 (2017: 8.2) (a)) Chapter8/ProperCovering · EpFunctor.isotropyOfMorOn, and by Definition 8.5.1 (3) (2017: 8.8) every (g, y) has exactly one lift Chapter8/ProperCovering · EpFunctor.IsProperCovering.bijective_fibredMap Chapter8/ProperCovering · EpFunctor.IsProperCovering.existsUnique_lift, whose target lies again in $F^{-1}(x)$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.morOnFiberEquiv. *Step 2, the count.* $\#$ is the cardinality of finite sets: the fibre is finite by Definition 8.5.1 (2) (2017: 8.8), the isotropy groups by Proposition 7.1.12 (2017: 7.4) (§VII.6) Chapter7/Basic · EpGroupoid.proposition_7_4, hence so is $\mathrm{mor}(F^{-1}(x))$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.finite_morOn_fiber Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_morOn_fiber. *Step 3, the classes* are indexed by their images in $|Y|$: $\Theta_c = F^{-1}(x) \cap \pi^{-1}(c)$ Chapter8/ProperCovering · EpFunctor.fiberClass Chapter8/ProperCovering · EpFunctor.fiberIndex Chapter8/ProperCovering · EpFunctor.fiberClasses Chapter8/ProperCovering · EpFunctor.mem_fiberClass_iff; they cover the fibre and are pairwise disjoint Chapter8/ProperCovering · EpFunctor.iUnion_fiberClass Chapter8/ProperCovering · EpFunctor.pairwiseDisjoint_fiberClass, with $\Theta_c = \emptyset$ for $c \notin \pi(F^{-1}(x))$, and isomorphic points have conjugate, hence equinumerous, isotropy groups Chapter8/ProperCovering · EpFunctor.nat_card_isotropy_eq_of_mem_fiberClass. With this indexing the correspondence $\Theta_j \leftrightarrow \Theta'_j$ of Lemma 8.6.2 (2017: 8.10) is "the same index" (Findings, Chapter 8, item 71) (2017 p. 286).

Lemma 8.6.1 (2017: 8.9) (V8.39) Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_9. $\#G_x \cdot \#\Theta_j = \#\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} = \#G_{y_0} \cdot (\#\Theta_j)^2$, *and* $\#G_x = \#G_{y_0} \cdot \#\Theta_j$ *for* $y_0 \in \Theta_j$. **The text's proof** restricts the bijection to (8.17), $\{\psi \mid s(\psi), t(\psi) \in \Theta_j\} \to G_x \times \Theta_j$, and, choosing $\psi_y : y_0 \to y$, uses the bijections $\mathrm{mor}(y_0, y_0) \to \mathrm{mor}(y, y')$, $\psi \mapsto \psi_{y'} \circ \psi \circ \psi_y^{-1}$. **The formalized proof** does the same. *Step 1.* The set of the lemma is $\mathrm{mor}(\Theta_j)$ Chapter8/ProperCovering · EpFunctor.morOn_fiberClass_eq, and (8.17) is a bijection because $t(\psi) \in \Theta_j$ is automatic for $s(\psi) \in \Theta_j$ and $t(\psi) \in F^{-1}(x)$ Chapter8/ProperCovering · EpFunctor.IsProperCovering.morOnClassEquiv. *Step 2.* The text's bijections assemble into one, $\mathrm{mor}(\Theta_j) \cong (\Theta_j \times \Theta_j) \times G_{y_0}$ Chapter8/ProperCovering · EpFunctor.nonempty_morOnClass_equiv_isotropy. *Step 3.* The third claim divides by $\#\Theta_j \geq 1$. (8.17) writes "$\psi \to (F\psi), s(\psi))$" for $\psi \mapsto (F(\psi), s(\psi))$ (possible misprint) (Findings, Chapter 8, items 71, 72) (pp. 396–398 (2017 pp. 286–287)).

Lemma 8.6.2 (2017: 8.10) (V8.40) Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_10. *For isomorphic x, x' the isomorphism classes of $F^{-1}(x)$ and $F^{-1}(x')$ correspond one to one, and isomorphic $y \in F^{-1}(x)$, $y' \in F^{-1}(x')$ lie in corresponding classes Θ_j, Θ'_j for exactly one j.* **The text** argues before the lemma by lifting a morphism $x \to x'$ through Definition 8.5.1 (3) (2017: 8.8). **The formalized proof** does the same: every $y \in F^{-1}(x)$ is the source of a lift of $\varphi : x \to x'$, whose target lies in $F^{-1}(x')$ and is isomorphic to y Chapter8/ProperCovering · EpFunctor.IsProperCovering.exists_isomorphic_of_isomorphic. So the index sets in $|Y|$ agree Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberIndex_eq, and with the classes indexed by their images in $|Y|$ the unique common index of y and y' is their common image. As a bijection between

the sets of classes the correspondence is `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_10_equiv` (Findings, Chapter 8, items 71, 73) (2017 p. 287).

Lemma 8.6.3 (2017: 8.11) (V8.41) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.lemma_8_11`. *For isomorphic x, x', $\#\Theta_j = \#\Theta'_j$ and $\#F^{-1}(x) = \#F^{-1}(x')$.* As in the text: $\#\Theta_j = \#G_x/\#G_{y_0}$ by Lemma 8.6.1 (2017: 8.9), $\#G_x = \#G_{x'}$, and $\#G_{y_0} = \#G_{y'_0}$ for $y_0 \in \Theta_j$, $y'_0 \in \Theta'_j$, isomorphic points having conjugate isotropy groups; the division is by $\#G_{y_0} \geq 1$. For an index outside both index sets both classes are empty. The fibre is the disjoint union of its classes `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_fiber_eq_sum` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.ncard_fiber_eq_of_isomorphic`. Possible misprints: after Lemma 8.6.2 (2017: 8.10), "the third claim of Lemma 8.10" for that of Lemma 8.6.1 (2017: 8.9), which has the claim $\#G_x = \#G_{y_0} \cdot \#\Theta_j$; and in the lemma, $\sum_{i=1}^{q} \#\Theta_j$ sums over i with summands indexed by j (Findings, Chapter 8, item 73) (2017 p. 287).

VIII.24 Proposition 8.6.4 (2017: 8.4): the number of sheets

Proposition 8.6.4 (2017: 8.4) (V8.42) `Chapter8/ProperCovering · EpFunctor.IsProperCovering.proposition_8_4`. *$\#F^{-1}(x)$ is constant on the preimage of every connected component of $|X|$, and $x \mapsto \#F^{-1}(x)$ is an sc-smooth functor $X \to \mathbb{N}$.* **The text** argues that under small perturbations of x "the structure of isomorphism classes stays the same". **The formalized proof** uses only the local constancy of $\#F^{-1}(x)$. *Step 1.* For $x' \in U(x)$, Definition 8.5.1 (2) (2017: 8.8) gives the bijection $y \mapsto (F|U(y))^{-1}(x')$ from $F^{-1}(x)$ onto $F^{-1}(x')$, the $U(y)$ being disjoint and covering $F^{-1}(U(x))$; so $\#F^{-1}$ is locally constant on X `Chapter8/ProperCovering · EpFunctor.IsProperCovering.eventually_ncard_fiber_eq` `Chapter8/ProperCovering · EpFunctor.IsProperCovering.isLocallyConstant_ncard_fiber`. *Step 2.* By Lemma 8.6.3 (2017: 8.11) it descends to $|X|$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberCardOrbit`, where it is locally constant in the quotient topology, a level set upstairs being open `Chapter8/ProperCovering · EpFunctor.IsProperCovering.isLocallyConstant_fiberCardOrbit`, hence constant on connected components. *Step 3, the functor.* $\mathbb{N}$ is Lean's, with 0, as the trivial ep-groupoid of the discrete M-polyfold `Chapter7/EpGroupoid · EpGroupoid.trivial` `Chapter7/Discrete · discretePolyfold`; a morphism φ goes to the unit of $\#F^{-1}(s\varphi) = \#F^{-1}(t\varphi)$, and the object map is sc-smooth because it is locally constant `Chapter8/ProperCovering · EpFunctor.IsProperCovering.fiberCardFunctor`. The values are ≥ 1, F being surjective, which matches the text's $\mathbb{N} = \{1, 2, \dots\}$ `Chapter8/ProperCovering · EpFunctor.IsProperCovering.one_le_ncard_fiber` (Findings, Chapter 8, item 74) (2017 p. 288).

VIII.25 Appendix §8.6.2 (2017 §8.5.2): the structure of strong bundle coverings

§8.6.2 (2017 §8.5.2) (V8.43) `Chapter8/EpStrongBundle · EpStrongBundle.section_8_5_2`. *For a proper strong bundle covering functor A over F, $A(\varphi, w) = (F(\varphi), A_{s(\varphi)}(w))$ with topological linear isomorphisms $A_y : W_y \to V_{F(y)}$, and $\mu_X(F(\varphi)) \circ A_{s(\varphi)} = A_{t(\varphi)} \circ \mu_Y(\varphi)$: $y \mapsto A_y$ is a natural transformation $\mu_Y \to \mu_X \circ F$ of functors into* Ban. **The text** writes the morphism map in this form and computes $t(A(\varphi, w)) = A(t(\varphi, w))$. **The formalized proof** does the same. *Step 1.* The morphism map has the form $(\varphi, w) \mapsto (F(\varphi), A(w))$, as for every strong bundle functor (§VIII.13) `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering.mor_val`, and the fibre maps A_y are topological linear isomorphisms by Definition 8.5.9 (1) (2017: 8.10), the local strong bundle isomorphisms having linear inverses `Chapter8/EpStrongBundle · EpStrongBundle.IsProperStrongBundleCovering.fiberEquiv`. *Step 2.* A strong bundle functor commutes with the targets, $\nu(F(\varphi), A(w)) = A(\mu(\varphi, w))$ `Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor.μ_comm`, which is the text's computation. The natural transformation is stated as its naturality identity; the fibres $Q^{-1}(F(s\varphi))$ and $Q^{-1}(s(F\varphi))$ are identified through their values in V, and the category Ban is not formalized. Only the strong bundle functor part of Definition 8.5.9 (2017: 8.10) enters the identity (Findings, Chapter 8, item 75) (2017 pp. 288–289).

VIII.26 Witnesses and counterexamples

Every hypothesis of the chapter is shown satisfiable, and the one statement that fails as printed is refuted (Findings, Chapter 8, items 100–105).

Covering functors (Definition 8.5.5 (2017: 8.9)). The identity of `SingleObj G` is a covering functor for every group G `Chapter8/NonVacuity · CoveringExamples.isCoveringFunctor_id`; for non-trivial G the functor sending every morphism to 1 is not `Chapter8/NonVacuity · CoveringExamples.not_isCoveringFunctor_one`; the empty groupoid gives a covering functor (§VIII.20).

A geometric lift, and Proposition 8.5.7 (2017: 8.3) applied. $\mathbb{Z}/2 = \mathbb{Z}^\times$ acting on $\mathbb{R}$ `Chapter8/NonVacuity · CoveringExamples.rCat` by $y \mapsto \pm y$ `Chapter8/NonVacuity · CoveringExamples.reflectionDiffeo Chapter8/NonVacuity · CoveringExamples.reflectionHom` is a geometric lift of the identity of `SingleObj` $\mathbb{Z}^\times$, with $L_D = L_C$ and $\tau = 1$ `Chapter8/NonVacuity · CoveringExamples.reflectionLift`. Proposition 8.5.7 (2017: 8.3) makes it a proper covering functor onto the translation groupoid $\mathbb{Z}/2 \ltimes \mathbb{R}$, with non-trivial isotropy at 0 `Chapter8/NonVacuity · CoveringExamples.isProperCovering_reflectionLift`.

Identities and trivial strong bundles. The identity functor of every ep-groupoid is a proper covering functor `Chapter8/NonVacuityBundle · EpFunctor.isScDiffeoOn_fibredMap_id Chapter8/NonVacuityBundle · EpFunctor.isProperCovering_id`. Every strong bundle over an M-polyfold is a strong bundle over its trivial ep-groupoid with $\mu(z, w) = w$ (Definition 8.3.1 (2017: 8.4)) `Chapter8/NonVacuityBundle · EpStrongBundle.trivial`; the identity is a strong bundle map and a strong bundle functor (Definitions 8.3.2, 8.3.3 (2017: 8.5, 8.6)) `Chapter8/NonVacuityBundle · EpStrongBundle.isMap_id Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleFunctor_id`, and the identity of every strong bundle over an ep-groupoid is a proper strong bundle covering functor `Chapter8/NonVacuityBundle · StrongBundle.isStrongBundleIsoOn_id Chapter8/NonVacuityBundle · EpStrongBundle.isStrongBundleIso_fibredLift_id Chapter8/NonVacuityBundle · EpStrongBundle.isProperStrongBundleCovering_id`. Over a trivial ep-groupoid every section is functorial `Chapter8/NonVacuityBundle · EpStrongBundle.isFunctorialSection_trivial`, so Γ, Γ^+ and the sc-Fredholm section functors are Part I's sc-smooth, sc⁺- and sc-Fredholm sections `Chapter8/NonVacuityBundle · EpStrongBundle.mem_sectionFunctors_trivial Chapter8/NonVacuityBundle · EpStrongBundle.mem_plusSectionFunctors_trivial Chapter8/NonVacuityBundle · EpStrongBundle.mem_fredholmSectionFunctors_trivial`. The witnesses use that maps of pull-backs $(z, y) \mapsto (g(z), \Phi(y))$ are strong bundle maps `Chapter8/NonVacuityBundle · pullbackMap Chapter8/NonVacuityBundle · StrongBundle.isStrongBundleMap_pullbackMap`.

A strong bundle over an ep-groupoid in `Type`. Part I's trivial strong bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ `Chapter3/NonVacuity · TrivialBundle.bundle` over the trivial ep-groupoid of $\mathbb{R}^n$ `Chapter8/NonVacuityBundle · TrivialBundle.epBundle` has an ep-groupoid as functorial picture `Chapter8/NonVacuityBundle · TrivialBundle.epBundleEpGroupoid`, its identity is a proper strong bundle covering functor `Chapter8/NonVacuityBundle · TrivialBundle.epBundle_isMap_id Chapter8/NonVacuityBundle · TrivialBundle.epBundle_isProperStrongBundleCovering`, $0 \in \Gamma^+$ `Chapter8/NonVacuityBundle · TrivialBundle.fiberZero_mem_plusSectionFunctors_epBundle`, and for a smooth $\phi : \mathbb{R}^n \to \mathbb{R}^k$ the graph section lies in Γ and is sc-Fredholm `Chapter8/NonVacuityBundle · TrivialBundle.graphSection_mem_sectionFunctors_epBundle Chapter8/NonVacuityBundle · TrivialBundle.graphSection_mem_fredholmSectionFunctors_epBundle`, from Part I's results on graph sections `Chapter3/NonVacuity · TrivialBundle.isScSmoothSection_graphSection`.

Tangent witnesses. The tangent of the trivial ep-groupoid of $\mathbb{R}^n$ is tame `Chapter8/TangentTame · EpGroupoid.isTame_tangentEpGroupoid_trivial_rn` (Lemma 8.1.9 (2017: 8.8); the trivial ep-groupoid itself is tame by Chapter 7 `Chapter7/EpGroupoid · EpGroupoid.isTame_trivial_rn`), and the tangent of the trivial ep-groupoid of any M-polyfold has a paracompact orbit space `Chapter8/TangentOrbit · EpGroupoid.hasParacompactOrbitSpace_tangent_trivial` (Theorem 8.1.15 (2017: 8.4); its hypothesis is Chapter 7's `Chapter7/EpGroupoid · EpGroupoid.hasParacompactOrbitSpace_trivial`).

Theorem 8.5.3 (5) (2017: 8.5) as printed. For the identity functor of $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§VII.30) `Chapter7/NonVacuity · EpGroupoid.Examples.reflectionEpGroupoid`, every chart of Theorem 8.5.3 (2017: 8.5) at 0 has, in the sheet of $\psi_0 = (-1, 0)$, a morphism ψ with $F(s(\psi)) \neq t(F(\psi))$ `Chapter8/NonVacuity · CoveringExamples.`

`theorem_8_5_five_as_printed`, and such charts exist `Chapter8/NonVacuity · CoveringExamples.theorem_8_5_five_as_printed_nonvacuous` (§VIII.17).

What the witnesses do not exercise. "A sufficiently small $U(x)$" is a chart datum, and charts exist inside every open $V \ni x$, so part (c) of Theorem 8.5.8 (2017: 8.6), which quantifies over all charts, is not vacuous; no hypothesis of the form "for all compatible charts" is made in the chapter. But every proper covering functor constructed has one-point fibres $F^{-1}(x)$ (identities, and the functor of the reflection lift, whose D has one object), so the even coverings of Definitions 8.5.1 (2) and 8.5.9 (2) (2017: 8.8 and 8.10) are met with one sheet and the counting of §8.6.1 (2017 §8.5.1) only with $q = 1$, $\#\Theta_1 = 1$; no proper strong bundle covering functor other than identities is constructed, and Γ^+ is witnessed by the zero section only (Findings, Chapter 8, item 105).

VIII.27 What is not formalized

- **Definitions 8.2.1 and 8.2.2 (2017: 8.2 and 8.3)** (§8.2, pp. 368–375 (2017 pp. 268–270); gap G.74): sc-differential forms on ep-groupoids, the direct limit $\Omega^*_{ep,\infty}(X)$ and the sc-de Rham complex. They need the Whitney sums $\oplus^k TX$, the linear structure of T_xX (gap G.7), linear maps $T\varphi$ (gap G.70), the levels X^i for $i \geq 2$ (gap G.6, closed since: §II.32, §II.33), and the sc-differential forms, Lie bracket and exterior derivative of Part I's §4.4, which are not formalized either (gaps G.37–G.39; the Lie bracket on an M-polyfold is formalized since, §IV.29, gap G.38 closed). Like Part I's §4.4, the two definitions have no V-item. Definition 8.2.1 (2017: 8.2) writes "$T\varphi : T_xX \to T_yY$" for T_yX, and the argument that $d\omega$ is compatible with morphisms has $t(\varphi_0)$ for $t(\varphi)$ (possible misprints) (Findings, Chapter 8, item 110).
- **Lemma 8.1.2 (2017: 8.1) as printed** (2017 p. 259; gap G.75): the identification $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ and $m_{T\mathbf{X}} = Tm$ on it, also used in the proof of Lemma 8.1.7 (2017: 8.6). The sheet form is what the proofs use, and condition (4) for $T\mathbf{X}$ is proved without the identification, so nothing depends on the gap (§§VIII.2, VIII.4; Findings, Chapter 8, item 111).
- **The linearity claims of Remark 8.1.10 (2017: 8.1)** (pp. 364–366 (2017 pp. 264–265); gaps G.7, G.70): that $T\varphi$ and $t_T \circ (s_T|T_xX)^{-1}$ are linear isomorphisms, and that $T\varphi$ is the tangent of $\hat{\varphi}$ at x; the M-polyfold structure transported along G and the triples $(a, T\varphi, b)$, not given a gap since nothing depends on them (§VIII.6; Findings, Chapter 8, items 112, 113). Gap G.70 is taken up in part: $T\varphi$ is now well defined and functorial; G.7 remains open (Findings, Chapter 8, item 114).
- **The local bundle model in the proof of Theorem 8.1.15 (2017: 8.4)** (2017 p. 268): the sc-isomorphism of $TU(x_z)$ with $TO \to O^1$ and the linearity of the bundle action of G_{x_z} (gap G.7); not needed by the proof here (§VIII.9; Findings, Chapter 8, items 21, 113).
- **Theorem 8.5.3 (5) (2017: 8.5) as printed** (2017 p. 277): it does not hold as we read it; the square with s on both vertical arrows and the companion square with t on both are proved (§VIII.17; Findings, Chapter 8, item 54).
- **Proposition 8.5.7 (2017: 8.3) for $D = \emptyset$** (2017 p. 281): the functor is then not surjective on objects unless $U(x) = \emptyset$; the proposition is proved for nonempty D (§VIII.20; Findings, Chapter 8, item 59).
- **The degeneracy remark for non-tame X, Y** (pp. 379–385 (2017 pp. 275–276)): only $d_{\mathbf{X}\times Y} \leq 2\, d_{\mathbf{Y}}$ along the image of L_F (§VIII.15; Findings, Chapter 8, item 51).
- **Remarks without formal content**: the naturality remarks of Remark 8.5.4 (2017: 8.2) (b), strong bundle maps over open subsets of the base, the closing sentence of Definition 8.5.9 (2017: 8.10), "the structure of isomorphism classes stays the same" before Proposition 8.6.4 (2017: 8.4), and the category Ban of §8.6.2 (2017 §8.5.2); none is used elsewhere in the chapter, so none is given a gap (Findings, Chapter 8, item 113).

VIII.28 What the formalization taught us

Fibre injectivity replaces calculus on fibred products. The text proves the groupoid laws of $T(X, \mathbf{X})$, Theorem 8.1.11 (2017: 8.2) and Proposition 8.1.14 (2017: 8.1) by differentiating identities on $\mathbf{X}\,{}_s\!\times_t \mathbf{X}$, which needs the tangent of a fibred product. Since s is a local sc-diffeomorphism, a vector of $T\mathbf{X}$ is determined by its base point and its image under Ts; base points are computed in $\mathbf{X}$ and images under Ts by Lemma 8.1.3 (2017: 8.2). Only one identity of §8.1, the target formula of Lemma 8.1.3 (2017: 8.2), is genuinely differentiated, and the identification $T(\mathbf{X}\,{}_s\!\times_t \mathbf{X}) = T\mathbf{X}\,{}_{Ts}\!\times_{Tt} T\mathbf{X}$ is never needed (§§VIII.3, VIII.7, VIII.8).

The tangent calculus of §8.1 lives on open sets. Every proof of §8.1 differentiates a map defined only near a point: a local inverse of s, the map α of Lemma 8.1.3 (2017: 8.2), $\widehat{\psi}$. Part I differentiates only globally sc-smooth maps. Tangents of maps on open sets, with the chain rule and the tangents of local sc-diffeomorphisms, are the infrastructure the chapter rests on, and Lemma 8.1.7's (2017: 8.6) "by construction" is exactly their last property (§VIII.1).

Definition 8.5.1 (3) (2017: 8.8) carries Theorem 8.5.3 (2017: 8.5). The text builds the sheets $\mathbf{U}(\psi_0)$ from natural representations of the groups G_y in Y and never invokes condition (3). With (3), the sheets are images of the continuous sections $z \mapsto L_F^{-1}(\Gamma(g, Fz), z)$ of s, and connectedness of $U(y)$ does the rest; neither the finiteness of the fibre nor condition (1) is used. The same condition counts the morphisms over a fibre in §8.6.1 (2017 §8.5.1) and gives the unique lifting behind Theorem 8.5.8 (2017: 8.6) (§§VIII.17, VIII.21, VIII.23).

A functor commutes with s, not with "s on one side and t on the other". Theorem 8.5.3 (5) (2017: 8.5) as printed equates $F(s(\psi))$ with $t(F(\psi))$; a non-trivial element of G_x moves points, so the square fails for the reflection of $\mathbb{R}$ in every chart at 0. The two squares that functoriality gives, with s on both sides and with t on both sides, are what the proofs downstream use (§VIII.17).

Empty groupoids and non-tame products. Two statements needed a hypothesis that the text does not state: Proposition 8.5.7 (2017: 8.3) needs $D \neq \emptyset$, because Definition 8.5.5 (2017: 8.9) admits the empty groupoid, over which the functor of Proposition 8.5.7 (2017: 8.3) is not surjective; and the degeneracy remark after Definition 8.5.1 (2017: 8.8) needs tameness, the only form of the product formula $d_{\mathbf{X}\times Y} = d_{\mathbf{X}} + d_Y$ that Part I proves. In both cases the hypothesis holds where the text uses the statement (§§VIII.15, VIII.20).

Metrizability is the shorter road to paracompactness, again. As for Proposition 7.5.3 (2017: 7.12), Theorem 8.1.15 (2017: 8.4) follows from Smirnov's theorem pulled back along $|TX| \to |X^1|$: $|TX|$ is metrizable, which is stronger than paracompact, and the closed paracompact sets $\tilde{A}_z$, the regularity of $|TX|$ and Proposition 2.7.4 (2017: 2.17) drop out. What remains of the text's argument is the metrizability of $|TU(x_z)|$, and for it the open embedding $G_x \backslash TU(x) \to |TX|$, which the text leaves implicit (§VIII.9).

Strong bundles over ep-groupoids need more of Part I than Part I states. A global notion of strong bundle map, the atlas and universal property of the pull-back on both levels $[0]$ and $[1]$, the sum of sc-smooth sections, and the strong bundle structure on $\mathbf{V}\,{}_s\!\times_A W$ are all used in §8.3 and Definition 8.5.9 (2017: 8.10) without being defined or proved in Part I. With them in place, the claims of §8.3 that the text states without proof, the functorial picture and the identity of Definitions 8.3.2 and 8.3.3 (2017: 8.5 and 8.6), are routine (§§VIII.10–VIII.14, VIII.22).

VIII.29 The 2021 additions to §8.1: Propositions 8.1.16 and 8.1.17, the Zhou condition and LUC for TX

The Springer edition (2021) adds Propositions 8.1.16, 8.1.17, Definition 8.3.5, §8.4, Example 8.5.6, condition (0) of Definition 8.5.9 and Lemma 8.5.10; Sections VIII.29–VIII.33 retell them, with the 2021 numbers and pages (V8.44–V8.52). The conditions of Chapter 7's §7.4 are those of §§VII.35–VII.39.

Proposition 8.1.16 (V8.44) Chapter8/TangentZhou · EpGroupoid.proposition_8_1_16_2021. *If X satisfies (ZHOU), so does TX.* **The text's proof**: an isotropy element of a smooth $b \in TX$ over x is a pair (g,h), $g \in G_x$; differentiating (8.16) and using that $T\tau$ is onto, $T(g,h) = \mathrm{id}$ gives $Tg = \mathrm{id}$, so $g \in T_x$ by (ZHOU), and then $(g,h) \in T_b$. **The formalized proof.** *Step 1.* A smooth point of TX lies over a smooth point of X Chapter8/TangentZhou · MPolyfold.levelVal_tangentProj_mem_smoothSet, and the base $p(k)$ of an isotropy element k of b is an isotropy element of $p(b)$. *Step 2.* The step "$T(g,h) = \mathrm{id}$ implies $Tg = \mathrm{id}$" needs the second tangent bundle $T(TX)$ and $T\tau$, which are not formalized; it is the hypothesis Chapter8/TangentZhou · EpGroupoid.TangentZhouReflect (gap G.88). *Step 3.* (ZHOU) for X puts $p(k)$ in $\mathbf{X}^T$, and if $s = t$ near $p(v)$ then $Ts = Tt$ near v, so $k \in (T\mathbf{X})^T$ Chapter8/TangentZhou · EpGroupoid.mem_nonEffectivePart_tangent. As in Chapter 7, Tg is a set map on the fibre, no linear structure is used (§VII.37). For every trivial ep-groupoid the tangent of a morphism is the identity and the hypothesis holds Chapter8/TangentZhou · EpGroupoid.tangentMor_trivial Chapter8/TangentZhou · EpGroupoid.tangentZhouReflect_trivial Chapter8/TangentZhou · EpGroupoid.Examples.proposition_8_1_16_2021_point (Findings, Chapter 8, item 115) (p. 369).

Proposition 8.1.17 (V8.45) Chapter8/TangentZhou · EpGroupoid.proposition_8_1_17_2021_one. *(1) If X^1 has LUC, so has TX, with $(T\mathbf{X})^T = p^{-1}((\mathbf{X}^1)^T)$. (2) (SLUC) for X gives (SLUC) for TX. (1).* If $v \in (T\mathbf{X})^T$, then $Ts = Tt$ on a neighbourhood O of v, and $s = t$ on the open set $p(O)$, p being open: $p(v) \in (\mathbf{X}^1)^T$. Conversely, $(\mathbf{X}^1)^T = (\mathbf{X}^T)^1$ by Appendix 7.6.4 (§VII.36), and Step 3 of Proposition 8.1.16 applies Chapter8/TangentZhou · EpGroupoid.nonEffectivePart_tangent_eq. So $(T\mathbf{X})^T$ is closed as a preimage of the closed $(\mathbf{X}^1)^T$ (Proposition 7.2.21 for X^1), and TX has LUC by Proposition 7.2.21 for TX. The hypotheses are those of Appendix 7.6.4 — $U(x) \cap X_1$ connected in X^1 and dense in $U(x)$ (the density is gap G.73) — and the openness of $p : T\mathbf{X} \to \mathbf{X}^1$ (gap G.8), satisfied by the point Chapter8/TangentZhou · EpGroupoid.isOpenMap_tangentProj_of_subsingleton Chapter8/TangentZhou · EpGroupoid.Examples.proposition_8_1_17_2021_one_point. *(2)* for $(TX)^0 = TX$ is (1) Chapter8/TangentZhou · EpGroupoid.proposition_8_1_17_2021_two; for every $(TX)^i$ it needs $(TX)^i = T(X^i)$, which is §VIII.33 (gap G.88, closed for (2)) (Findings, Chapter 8, items 116, 124) (p. 371).

VIII.30 Definition 8.3.5, Proposition 8.4.1 and Definition 8.4.2: auxiliary norms, paracompactness, local regularity

Definition 8.3.5 (V8.46) Chapter8/Book2021 · EpStrongBundle.IsAuxNorm. *An auxiliary norm for a strong bundle over an ep-groupoid is a continuous $N : W_{0,1} \to [0,\infty)$, a complete norm on every fibre, with $h_k \to 0_x$ whenever $P(h_k) \to x$ and $N(h_k) \to 0$, and invariant: $N(h) = N(k)$ if $h = \mu(\varphi, k)$.* The first three conditions are Part I's auxiliary norm (Definition 5.1.1) on the underlying strong bundle, so N lives on its level-one space; values in $[0,\infty)$ follow from (1). In (3), $h \in W_{0,1}$ is given and $h = \mu(\varphi,k)$ is an equation in W; it is not vacuous, since $\mu(\varphi,k) \in W_{0,1}$ for $k \in W_{0,1}$ Chapter8/Book2021 · EpStrongBundle.exists_oneVal_eq_μ. Over the trivial ep-groupoid every auxiliary norm of Part I is one Chapter8/Book2021 · EpStrongBundle.exists_isAuxNorm_trivial, and $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ has one Chapter8/Book2021 · TrivialBundle.exists_isAuxNorm_epBundle (Findings, Chapter 8, item 117) (p. 379).

Proposition 8.4.1 (V8.47) Chapter8/Book2021 · EpStrongBundle.proposition_8_4_1_2021_two_imp_one. *$|X|$, $|W[0]|$ and $|W[1]|$ are paracompact together. (2) □ (1).* $\mu(\varphi, 0_{s(\varphi)}) = 0_{t(\varphi)}$ Chapter8/Book2021 · EpStrongBundle.μ_fiberZero, so the zero section induces $|0| : |X| \to |W[0]|$ Chapter8/Book2021 · EpStrongBundle.zeroOrbitMap, continuous Chapter8/Book2021 · EpStrongBundle.continuous_zeroOrbitMap, with left inverse $|P|$ Chapter8/Book2021 · EpStrongBundle.leftInverse_zeroOrbitMap, hence a topological embedding Chapter8/Book2021 · EpStrongBundle.isEmbedding_zeroOrbitMap. The text concludes as a closed subset of a paracompact space; the formalized proof goes through metrizability: $|W[0]|$ paracompact is metrizable by Theorem 7.3.1 and Smirnov's theorem (§VII.23), so is its subspace $|X|$, hence paracompact. $W[1]$ is not built as an ep-groupoid, and (1) □ (2), (3) needs G_x-equivariant bundle charts with invariant metrics; not formalized (gap G.86) (Findings, Chapter 8, item 118) (p. 380).

Definition 8.4.2 (V8.48) Chapter8/TangentZhou · EpStrongBundle.IsLR. *The strong bundle has the local regularity property if every x has arbitrarily small $U(x)$ such that every strong bundle isomorphism B of $W|U(x)$ with*

$|B(w)| = |w|$ *is* $w \mapsto g * w$ *for some* $g \in G_x$, the natural lift $g * w = \mu(\Gamma(g, P(w)), w)$ `Chapter8/TangentZhou · EpStrongBundle.natLift`. As for Definition 8.5.9 (§VIII.22), the strong bundle isomorphism is read as a strong bundle map $W \to W$ that is Part I's strong bundle isomorphism on $W|U(x)$, "arbitrarily small" as "inside every open $O \ni x$", and $|B(w)| = |w|$ as equality of orbits in $(W, \mathbf{W})$, as for (LR) (§VII.39). Every strong bundle over a trivial ep-groupoid has it `Chapter8/TangentZhou · EpStrongBundle.isLR_trivial`. Proposition 8.4.3, effective X with (LR) gives the property, needs the openness of P and the lift of (LR) to the bundle; not formalized (gap G.86) (Findings, Chapter 8, item 119) (p. 382).

VIII.31 Example 8.5.6, Definition 8.5.9 in the 2021 form, and Lemma 8.5.10

The remark after Definition 8.5.5 and Example 8.5.6 (V8.49) `Chapter8/Book2021 · example_8_5_6_2021`. *For a covering functor* $f : D \to C$, $f : G_d \to G_{f(d)}$ *is injective* `Chapter8/Book2021 · IsCoveringFunctor.injective_map_end` — by the bijectivity required in Definition 8.5.5 — *but need not be surjective*: D with two objects and one morphism between any two `Chapter8/Book2021 · Example856D`, mapped to $\mathbb{Z}^\times$ as a one-object category with $f(\varphi) = f(\varphi^{-1}) = -1$ `Chapter8/Book2021 · example856Functor`, is a covering functor whose G_a, G_b are trivial while $G_x \cong \mathbb{Z}/2$ (Findings, Chapter 8, item 120) (p. 388).

Definition 8.5.9, 2021 form (V8.50) `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021`. *A strong bundle map* $A : W \to V$ *over an ep-functor* F *with* $Q \circ A = F \circ P$ *is a proper strong bundle covering functor if (0)* $\tau(F(\psi), A(w)) = A(\mu(\psi, w))$ *and (1)–(3) of the 2017 definition hold.* It is stated for the object map, (0) being the strong bundle map property of Definition 8.3.2 (2017: 8.5); (1)–(3) are those of §VIII.22. The reading of §VIII.22 had taken 2017's "strong bundle map A" as a strong bundle functor, which contains (0); the two are equivalent `Chapter8/Book2021 · EpStrongBundle.isProperStrongBundleCovering2021_iff Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering.isProperStrongBundleCovering2021`: the functor is determined by its object map, on morphisms $(\psi, w) \mapsto (F(\psi), A(w))$. Identities are witnesses `Chapter8/Book2021 · EpStrongBundle.isProperStrongBundleCovering2021_id Chapter8/Book2021 · TrivialBundle.epBundle_isProperStrongBundleCovering2021` (Findings, Chapter 8, items 70, 121) (p. 394).

Lemma 8.5.10 (V8.51) `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.lemma_8_5_10_2021`. $B : W \to F^*V$, $w \mapsto (P(w), A(w))$, *is a strong bundle isomorphism over* Y *for* $\mu_{F^*V}(\varphi, (y, v)) = (t(\varphi), \mu_X(F(\varphi), v))$, *with (8.23)* $B \circ \mu_Y = \mu_{F^*V} \circ B$, *and* $A = \bar{A} \circ B$. F^*V is Part I's pull-back, whose paracompactness always holds, and μ_{F^*V} is its formula `Chapter8/Book2021 · EpStrongBundle.pullbackμ`. *Step 1.* B lands in F^*V `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_val`, and (8.23) is (0) `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleB_μ`. *Step 2, bijective.* On the fibre over y, A is the linear isomorphism $A_y : W_y \to V_{F(y)}$ of a local strong bundle isomorphism of Definition 8.5.9 (1) at y, so B is bijective `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bijective_bundleB`. *Step 3, the inverse.* Near every point, $B^{-1}(y, v) = \Psi(v)$ for the inverse Ψ of a local strong bundle isomorphism of Definition 8.5.9 (1), whose base map is injective `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.eq_of_isStrongBundleIsoOn Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.bundleBInv_eq_of`, so B and B^{-1} are strong bundle maps `Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleB Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.mapsTo_oneSet_bundleBInv Chapter8/Book2021 · EpStrongBundle.IsProperStrongBundleCovering2021.isStrongBundleMap_bundleBInv`. "Strong bundle isomorphism over the ep-groupoid Y" is read as a strong bundle isomorphism $W \to F^*V$ covering the identity, with (8.23); that μ_{F^*V} makes F^*V a strong bundle over the ep-groupoid Y is not proved, only its formula is used (gap G.89) (Findings, Chapter 8, item 122) (p. 395).

VIII.32 The 2021 additions: what is not formalized, and what they taught us

- **Proposition 8.1.16 without `TangentZhouReflect`** (gap G.88): it needs $T(TX)$ (§VIII.29; Findings, Chapter 8, item 115). Proposition 8.1.17 (2) for every $(TX)^i$, the other part of G.88, is formalized (§VIII.33); of $(TX)^i = T(X^i)$ the compatibility with unit, inverse and composition is not, nor the linearity on the fibres (gap G.7) (Findings, Chapter 8, item 124).
- **Proposition 8.2.3** (p. 373; gap G.87): the Poincaré lemma for sc-differential forms on ep-groupoids, with the forms of §8.2 (gap G.74) (Findings, Chapter 8, item 123).
- **Proposition 8.4.1 (1) □ (2), (3), and Proposition 8.4.3** (gap G.86): equivariant bundle charts, $W[1]$ as an ep-groupoid, the openness of P (§VIII.30; Findings, Chapter 8, items 118, 119).
- F^*V **as a strong bundle over the ep-groupoid** Y in Lemma 8.5.10 (gap G.89) (§VIII.31; Findings, Chapter 8, item 122).

The tangent inherits the Chapter 7 conditions through the base. Both propositions of §8.1 reduce to Chapter 7: a vector of $T\mathbf{X}$ is in $(T\mathbf{X})^T$ as soon as its base is in $(\mathbf{X}^1)^T$, and the openness of p gives the converse. What remains is the passage from $T(g,h)$ to Tg, which needs the second tangent bundle (§VIII.29).

Condition (0) was already in the reading. The 2021 Definition 8.5.9 adds (0); the first reading had read 2017's "strong bundle map" as a strong bundle functor, which contains it, and the two forms are equivalent (§VIII.31).

VIII.33 Proposition 8.1.17 (2) for every $(TX)^i$

Equivalences of ep-groupoids (V8.52) `Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv`. The text's "naturally $(TX)^i = T(X^i)$" is used only to carry LUC from $T(X^i)$ to $(TX)^i$. For that it suffices to have sc-diffeomorphisms of the object and of the morphism M-polyfolds commuting with source and target; an sc-diffeomorphism is a homeomorphism `Chapter8/TangentZhouLevels · MPolyfold.ScEquiv.toHomeomorph`. Such equivalences compose and contain the identities `Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.refl` `Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.trans`, pass to the index raises `Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.shift`, map $\mathbf{X}^T$ onto $\mathbf{X}'^T$, since $s = t$ near a morphism is preserved `Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.nonEffectivePart_eq`, and so transport LUC, by the characterization of Proposition 7.2.21 ($\mathbf{X}^T$ closed) `Chapter8/TangentZhouLevels · EpGroupoid.EpEquiv.isLUC`. The compatibility with unit, inverse and composition is not formalized, since LUC does not use it.

$T(X^i) \cong (TX)^i$ (V8.52) `Chapter8/TangentZhouLevels · EP.tangentLiftEpEquiv`. Part I's Proposition 2.3.17 for $i = 1$, on the object and the morphism M-polyfold, gives an equivalence $T(X^1) \cong (TX)^1$, commuting with s and t by the naturality of the inclusion $T(X^1) \to TX$ `Chapter8/TangentZhouLevels · EpGroupoid.tangentShiftEpEquiv`. By recursion, composing with the index raise of the previous equivalence, $T(X^i) \cong (TX)^i$ for every i, for the bundled tangent ep-groupoid `Chapter8/TangentZhouLevels · EP.tangentEP`, as Part I's Φ_i for M-polyfolds `Chapter2/TangentLiftAll · BundledMPolyfold.tangentLiftEquiv` `Chapter2/TangentLiftAll · proposition_2_3_17_all_2021`.

Proposition 8.1.17 (2) (V8.52) `Chapter8/TangentZhouLevels · EP.proposition_8_1_17_two_all_2021`. *If X has (SLUC), so has TX.* **The text's proof**: (SLUC) gives LUC for every $X^{i+1} = (X^i)^1$; by (1) every $T(X^i)$ has LUC; and $(TX)^i = T(X^i)$. **The formalized proof** is the same: (1) applied to X^i `Chapter8/TangentZhouLevels · EP.isLUC_tangent_lift`, then the equivalence. The hypotheses of (1) — natural representations with $U(x) \cap (X^i)_1$ connected in $(X^i)^1$ and dense in $U(x)$ (gap G.73), and the openness of $p : T(\mathbf{X}^i) \to (\mathbf{X}^i)^1$ (gap G.8) — are taken on every X^i. They hold for the point ep-groupoid and all its lifts `Chapter8/TangentZhouLevels · EpGroupoid.hasConnectedDenseLevelOneReps_of_subsingleton` `Chapter8/TangentZhouLevels · EpGroupoid.Examples.proposition_8_1_17_two_all_2021_pointEP` (gap G.88, closed for (2)) (Findings, Chapter 8, item 124) (p. 371).

Part IX. Branched ep⁺-subgroupoids (HWZ Chapter 9, §§9.1–9.5 and §9.6.1 (2017 §§9.1–9.5))

Chapter 9 continues Part II; its numbers and pages are those of the book (Springer 2021, pp. 401–458), with the 2017 text's in parentheses (2017 pp. 291–332). It introduces the objects that will carry the solution sets of perturbed sc-Fredholm sections with symmetries: functors $\Theta : X \to \mathbb{Q}^+$ on an ep-groupoid that are, near every point of their support, finite weighted sums of indicator functions of properly embedded smooth finite-dimensional submanifolds. §9.1 gives the definition (Definition 9.1.2 (2017: 9.1)), the local closedness of the orbit support (Lemma 9.1.5 (2017: 9.1)), symmetric local branching structures (Lemma 9.1.7 (2017: 9.2)) and the dimensional decomposition (Propositions 9.1.11 (2017: 9.2) and 9.1.12, the latter new in 2021 with its proof in Appendix §9.6.1). §9.2 attaches to Θ a section T_Θ of formal sums of tangent spaces (Theorem 9.2.1 (2017: 9.1)), proved through good and bad points and a graph representation of the sheets near a bad point (Proposition 9.2.8, Lemmas 9.2.9–9.2.11 (2017: 9.3, 9.6–9.8)); it defines the tangent $T\Theta$ (Definition 9.2.12 (2017: 9.6)) and, through the position of a submanifold relative to ∂X (Lemma 9.2.13 (2017: 9.9)), the boundary $\partial\Theta$ (Proposition 9.2.15, Definition 9.2.16 (2017: 9.4, 9.7)). §9.3 orients Θ (Definition 9.3.2 (2017: 9.8)), compares orientations at good points (Lemma 9.3.4 (2017: 9.10)), gives an example (Example 9.3.5 (2017: 9.1)) and the boundary orientation (Definitions 9.3.6–9.3.10). §9.4 studies essential points and sets of measure zero of the sheets and decomposes an oriented structure (Propositions 9.4.9, 9.4.10 (2017: 9.5, 9.6)). §9.5 integrates sc-differential forms and proves Stokes' theorem; of it, this part covers the sets of measure zero in $|\mathrm{supp}(\Theta)|$ and the σ-algebras $L(S)$, $L(\partial S)$ (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16)) and the isotropy factors of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (2017: Lemmas 9.18, 9.19), not the integrals.

What is proved, and what is not. All definitions of §§9.1–9.4 and §9.6.1, and those of §9.5 that do not involve forms, are formalized. Theorem 9.2.1 (2017: 9.1) is proved for every branched ep⁺-subgroupoid of an ep-groupoid whose object M-polyfold is $\mathbb{R}^m$, and on a general M-polyfold under the graph representation of Lemmas 9.2.9–9.2.11 and the local constancy of the dimension (Remark 9.1.1), taken as hypotheses. Proposition 9.1.12 is proved in the same two forms, as are Lemma 9.6.4, Proposition 9.6.5 and Lemma 9.3.4 (the last for n-dimensional Θ, with the level-1 continuity of frames in general). Lemmas 9.1.5, 9.1.7, 9.2.3, 9.2.5, 9.2.7 (1), 9.3.7, 9.4.2, 9.4.6, 9.4.8, 9.6.1 and 9.6.2, Propositions 9.2.15, 9.3.9, 9.4.9, 9.4.10 and the remarks with formal content are proved, several in a stronger form. Five places needed more than a reading. The uniqueness of Proposition 9.1.11 (2017: 9.2) does not hold as we read it (§IX.4); Lemma 9.2.13 (2017: 9.9) is read at points of ∂X, since its printed converse does not hold as we read it at interior points (§IX.12); the characterization in Definition 9.2.16 (2017: 9.7) is not unique as we read it (§IX.13); Lemma 9.3.4 (2017: 9.10) does not hold as we read it without equidimensionality, and two claims of Example 9.3.5 (2017: 9.1) do not either (§IX.15); Lemma 9.4.4 (2017: 9.13) needs sheets of one dimension (§IX.17), and so does Lemma 9.5.2 (2017: 9.16), whose proof rests on two sheets inducing the same null sets on their overlap (§IX.19). Not formalized: the global part of Proposition 9.1.11, Remark 9.1.4, Proposition 9.1.9, Lemma 9.4.3, the identification of $T_y\partial M$ with the outward direction, and in §9.5

the forms, integrals and measures, with Theorems 9.5.4, 9.5.5 and 9.5.15–9.5.18 (§IX.24).

Six structural choices. (1) *Tangent sets, linear algebra in charts.* Part I's T_xX carries no linear structure (gap G.7). The tangent space T_yM of a submanifold is the set of tangent vectors fixed by the tangent of the retraction that presents M, and $\mathrm{Gr}(x)$ is generated by such sets; in a chart the image of T_yM is a linear subspace of the right dimension, and every argument that needs linear algebra is carried out there (§IX.6). (2) *Functors as functions.* Θ is a function on objects with $\Theta(s(\varphi)) = \Theta(t(\varphi))$, valued in $\mathbb{Q}$, nonnegativity a field. (3) *Dimension labellings.* Without the local constancy of the dimension a submanifold could a priori have several dimensions; the splitting by dimension takes labellings $\dim : I \to \mathbb{N}$ as hypotheses, which are unique when they exist (§IX.1). (4) *Slices instead of sequences.* Proposition 9.2.8 is used in the text to count the weights of a class of tangent sheets by the values of Θ at good points; the formalization counts them on one finite slice transverse to the class (§IX.9). The same device proves Proposition 9.6.5 (§IX.22). (5) *Hypotheses with witnesses on $\mathbb{R}^m$.* The facts of the text's §9.2 that need level-0 sc-calculus inside a partial quadrant are hypotheses on a general M-polyfold (`HasGraphChartAt`, `HasLocallyConstantDimAt`, `HasDomainInvarianceAt`, `HasContinuousLevelOneOn`), each proved for every ep-groupoid over $\mathbb{R}^m$, where the main results then hold without hypothesis (§IX.23). (6) *Two kinds of sheet geometry.* §9.4 is formalized twice: for Mathlib manifolds with corners, where Definitions 9.4.1, 9.4.5 and Lemmas 9.4.2, 9.4.6 are proved, and for sheets in an M-polyfold, where Lemmas 9.4.4 and 9.4.6 are the fields of a structure `SheetGeometry`, satisfiable by graphs in $\mathbb{R}^2$ (§IX.17). The null sets of §9.5 are data in the same way, a structure `SheetNullSets` (§IX.19).

Map.

- `Chapter9/Basic`: functors, supports, Definition 9.1.2 (2017: 9.1), Lemma 9.1.5 (2017: 9.1), Definitions 9.1.6, 9.1.8 (2017: 9.2, 9.3), the indicator of a point.
- `Chapter9/Symmetric`: the restriction remark and Lemma 9.1.7 (2017: 9.2).
- `Chapter9/Propositions`: Remark 9.1.1, n-dimensional Θ, Remark 9.1.3 (b), the statements and the local construction of Propositions 9.1.11 and 9.1.12, the counterexample to the uniqueness of 9.1.11, submanifolds of $\mathbb{R}^n$.
- `Chapter9/GoodPoints`: Definitions 9.2.2, 9.2.6, Lemmas 9.2.3, 9.2.5, 9.2.7 (1) (openness), Remark 9.2.4.
- `Chapter9/Tangent`: T_yM, $\mathrm{Gr}(X)$, Theorem 9.2.1 under the independence, Definition 9.2.12.
- `Chapter9/TangentSlicesBasic`, `TangentSlicesRn`, `TangentSlices`: the chart form of T_yM, the graph representation and tangent slices, Lemma 9.2.7 (1) (density) and (2), the case of $\mathbb{R}^m$.
- `Chapter9/AppendixSlices`: one graph chart for all dimensions, Proposition 9.6.5, Proposition 9.1.12 and the independence of Theorem 9.2.1 on $\mathbb{R}^m$ without hypotheses.
- `Chapter9/Appendix`: Lemmas 9.6.1, 9.6.2, 9.6.4, Definition 9.6.3, Proposition 9.1.12 under `TopMultiplicity`.
- `Chapter9/BoundaryPosition`, `BoundaryPositionConverse`: Lemma 9.2.13, Remark 9.2.14, Definition 9.3.6, Lemma 9.3.7.
- `Chapter9/Boundary`: Proposition 9.2.15, Definition 9.2.16, Remark 9.2.17.
- `Chapter9/OrientationTangent`, `OrientationBasic`, `OrientationExample`, `OrientationUniqueness`, `OrientationBoundary`, `OrientationRn` (index `Chapter9/Orientation`): §9.3.
- `Chapter9/BranchingGeometryMeasure`, `BranchingGeometryManifold`, `BranchingGeometry`: §9.4.
- `Chapter9/Measure`: Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2, the remark after Definition 9.5.3, witnesses.
- `Chapter9/MeasureEffective`: the isotropy and set parts of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13.

Throughout, $X = (X, \mathbf{X})$ is an ep-groupoid as in Part VII, s, t its source and target, $\pi : X \to |X|$ the orbit map, X^∞ the full subcategory of smooth objects, G_x the isotropy group of x with a natural representation (Φ, Γ) on a neighbourhood $U(x)$ (§VII.4). A *submanifold* is a smooth finite-dimensional submanifold in the sense of Part I (§IV.2): the image, near each of its points, of an sc$^+$-retraction. For $\Theta : X \to \mathbb{Q}$, $\mathrm{supp}(\Theta) = \{\Theta > 0\}$ and $|\mathrm{supp}(\Theta)| = \pi(\mathrm{supp}(\Theta))$. A local branching structure at x is

$(U(x),(M_i)_{i\in I},(\sigma_i)_{i\in I})$ as in Definition 9.1.2; we write $I_y=\{i\mid y\in M_i\}$ and $M_U=\bigcup_i M_i$.

IX.1 Definition 9.1.2 (2017: 9.1), and the dimension of a submanifold (Remark 9.1.1 (2017: 9.1))

Definition 9.1.2 (2017: 9.1) (V9.1) `Chapter9/Basic · EpGroupoid.IsBranchedEpSubgroupoid` `Chapter9/Basic · EpGroupoid.LocalBranchingStructure`. A branched ep⁺-subgroupoid is a functor $\Theta\geq 0$ `Chapter9/Basic · EpGroupoid.IsFunctorial` with $\mathrm{supp}(\Theta)\subseteq X_\infty$ `Chapter9/Basic · EpGroupoid.supp`, $|\mathrm{supp}(\Theta)|$ paracompact `Chapter9/Basic · EpGroupoid.orbitSupp`, and a local branching structure at every point of the support. The formalization reads the paracompactness of the induced topology as a `ParacompactSpace` instance on the subtype $|\mathrm{supp}(\Theta)|$, and "the inclusion $M_i\to U(x)$ is proper" as `IsProperMap` of the inclusion of subtypes. For a closed M_i in $U(x)$ the latter is automatic `Chapter9/Symmetric · isProperMap_inclusion_iff`. Witnesses: the zero functor `Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_zero` and the indicator of a smooth point of $\mathbb{R}^n$, a 0-dimensional submanifold `Chapter9/Basic · rnPolyfold_isSmoothSubmanifold_singleton` `Chapter9/Basic · EpGroupoid.localBranchingStructure_indicator` `Chapter9/Basic · EpGroupoid.isBranchedEpSubgroupoid_indicator`. The 2017 text refers in (4) to "(ii) and (iii)", the 2021 text to (2) and (3).

The dimension (V9.2). Remark 9.1.1 says that $m\mapsto \dim T_mM$ is locally constant. With T_mM a set of tangent vectors, the dimension is measured in a chart: M has dimension k at a if the tangent space at the chart image of a of the sub-retract presenting M is k-dimensional `Chapter9/Propositions · MPolyfold.HasDimAt`. *Step 1, chart independence.* A germ of sc-diffeomorphisms between retracts restricts to a linear isomorphism of the tangent spaces of sub-retracts `Chapter9/Propositions · finrank_span_tangentSpace_eq`, so the value is the same for every chart and every presentation `Chapter9/Propositions · MPolyfold.finrank_tangentSpace_chart_change` `Chapter9/Propositions · MPolyfold.HasDimAt.finrank_eq`; it is unique `Chapter9/Propositions · MPolyfold.HasDimAt.unique` and exists at every point `Chapter9/Propositions · MPolyfold.IsSmoothSubmanifold.exists_hasDimAt`. *Step 2, invariance.* It is unchanged by intersecting with an open set and by sc-diffeomorphisms `Chapter9/Propositions · MPolyfold.hasDimAt_inter_isOpen_iff` `Chapter9/Propositions · MPolyfold.hasDimAt_image_iff`. *Step 3, local constancy* is not proved on a general M-polyfold; it is the hypothesis `HasLocallyConstantDimAt` `Chapter9/AppendixSlices · MPolyfold.HasLocallyConstantDimAt`. On $\mathbb{R}^m$, T_yM is the range of the projection $Ds(y)$ of the retraction, whose rank is its trace, continuous in y and integer valued `Chapter9/AppendixSlices · MPolyfold.hasLocallyConstantDimAt_rn`. Where local constancy is not available the dimensions of the sheets enter as labellings; all sheets of one structure pass through x, so a labelling is unique `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dim_unique`.

n-dimensional Θ. The 2021 text says the sheets "can be taken" n-dimensional (2017: are n-dimensional), and this is the formal reading: some local branching structure at each point of the support has only n-dimensional sheets `Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.IsOfDim` `Chapter9/Propositions · EpGroupoid.IsNDimensional`. The zero functor is n-dimensional for every n `Chapter9/Propositions · EpGroupoid.isNDimensional_zero`; the footnote, that an n-dimensional $\Theta\neq 0$ is not m-dimensional for $m\neq n$, is not formalized (gap G.93).

Submanifolds of $\mathbb{R}^n$ (V9.3). The witnesses of the chapter are built from closed submanifolds of $\mathbb{R}^n$. The image of a smooth idempotent s is one, of dimension rank $Ds(a)$ `Chapter9/Propositions · rnPolyfold_isSmoothSubmanifold_range` `Chapter9/Propositions · rnPolyfold_hasDimAt_range`. The graph of a smooth $\beta:\mathbb{R}\to\mathbb{R}$ is the image of $(s,t)\mapsto(s,\beta(s))$ `Chapter9/Propositions · graphRetraction` `Chapter9/Propositions · rnPolyfold_isSubmanifoldOfDim_graph2`. Finite weighted sums of closed submanifolds, and indicators of locally closed ones, are branched ep⁺-subgroupoids of the trivial ep-groupoid `Chapter9/Propositions · EpGroupoid.isBranchedEpSubgroupoid_sum_rn` `Chapter9/Propositions · EpGroupoid.isNDimensional_sum_rn` `Chapter9/Propositions · EpGroupoid.isBranchedEpSubgroupoid_indicator_rn`, whose orbit space is $\mathbb{R}^n$, so the paracompactness is free `Chapter9/Propositions · EpGroupoid.paracompactSpace_orbitSupp_trivial_rn`.

IX.2 Remark 9.1.3 (b) (2017: 9.2 (b)), Lemma 9.1.5 (2017: 9.1), Definitions 9.1.6 and 9.1.8 (2017: 9.2 and 9.3)

Remark 9.1.3 (b) (2017: 9.2 (b)) (V9.4) `Chapter9/Propositions · EpGroupoid.remark_9_1_3_b`. The text takes β_1, β_2 with prescribed signs, the graphs G_1, G_2 with weight 1, and $\Theta(p) = \#\{i \mid p \in G_i\}$ `Chapter9/Propositions · EpGroupoid.thetaGraphs` `Chapter9/Propositions · EpGroupoid.thetaGraphs_eq`, and says that the graphs of $\gamma_1 = 0$ and of some γ_2 positive off 0 describe Θ too. *Step 1.* The sign conditions are the structure `Remark913Hyp` `Chapter9/Propositions · EpGroupoid.Remark913Hyp`, satisfied by $\beta_1(s) = e(s)$, $\beta_2(s) = e(-s)$ with Mathlib's `expNegInvGlue` e `Chapter9/Propositions · EpGroupoid.remark913Hyp_expNegInvGlue`. *Step 2.* For the second structure to describe the same Θ, γ_2 must be $\beta_1 + \beta_2$ `Chapter9/Propositions · EpGroupoid.Remark913Hyp.eq_add_of_thetaGraphs_eq`, which is smooth and positive off 0 `Chapter9/Propositions · EpGroupoid.Remark913Hyp.γ₂_pos`. The two structures are `Chapter9/Propositions · EpGroupoid.Remark913Hyp.structureβ` and `Chapter9/Propositions · EpGroupoid.Remark913Hyp.structureγ`. *Step 3.* "Not unique, even close to x" is read as: no sheet of the first structure agrees with a sheet of the second on any neighbourhood of the origin, since each of β_1, β_2 differs from 0 and from $\beta_1 + \beta_2$ at points arbitrarily close to 0.

Lemma 9.1.5 (2017: 9.1) (V9.5) `Chapter9/Basic · EpGroupoid.lemma_9_1_5`. $|\mathrm{supp}(\Theta)|$ *is locally closed in* $|X|$. **The text's proof** takes a neighbourhood with the natural G_x-action, the properness property and a local branching structure, and shows that the image of $\bigcup_i M_i$ is closed in the image of $U(x)$. **The formalized proof.** *Step 1.* For a functor the support is saturated, $\pi^{-1}(|\mathrm{supp}(\Theta)|) = \mathrm{supp}(\Theta)$ `Chapter9/Basic · EpGroupoid.preimage_π_orbitSupp`. *Step 2.* On $U(x)$ the support is $\bigcup_i M_i$ `Chapter9/Basic · EpGroupoid.LocalBranchingStructure.mem_supp_iff`, a finite union of sets closed in $U(x)$ `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.isClosed_M` `Chapter9/Basic · EpGroupoid.LocalBranchingStructure.isClosed_preimage_val_supp`, so $\mathrm{supp}(\Theta)$ is locally closed in X `Chapter9/Basic · EpGroupoid.isLocallyClosed_supp`. *Step 3.* π is open (Chapter 7), and the image of the locally closed saturated set is locally closed. The properness property of the text is not used.

Definitions 9.1.6 and 9.1.8 (2017: 9.2 and 9.3) (V9.6, V9.8) are read literally `Chapter9/Basic · EpGroupoid.IsClosedBranched` `Chapter9/Basic · EpGroupoid.IsCompactBranched` `Chapter9/Basic · EpGroupoid.IsOrbifoldType` `Chapter9/Basic · EpGroupoid.IsManifoldType`; the indicator of a point is compact and of manifold type `Chapter9/Basic · EpGroupoid.isCompactBranched_indicator` `Chapter9/Basic · EpGroupoid.isManifoldType_indicator`. Proposition 9.1.9 and Remark 9.1.10 (natural manifold and orbifold structures on $|\mathrm{supp}(\Theta)|$) and Remark 9.1.4 (metrizability, the compactness criterion) are not formalized (gaps G.95, G.94).

IX.3 The restriction remark, and Lemma 9.1.7 (2017: 9.2)

The restriction (V9.7). Before Lemma 9.1.7 the text replaces M_i by $N_i = M_i \cap V(x_0)$ for $V(x_0) \subseteq U(x)$ and notes that some N_i may be empty. Definition 9.1.2 asks $x \in M_i$, which the N_i need not satisfy at x_0; the remark is formalized for an open $V \subseteq U(x)$ containing x itself, which is the case the symmetrization uses `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.restrict`. The N_i are submanifolds `Chapter9/Symmetric · MPolyfold.IsSmoothSubmanifold.inter_isOpen`, and they are properly embedded in V because they are closed in $U(x)$.

Lemma 9.1.7 (2017: 9.2) (V9.7) `Chapter9/Symmetric · EpGroupoid.lemma_9_1_7` `Chapter9/Symmetric · EpGroupoid.lemma_9_1_7'`. *At* $x \in \mathrm{supp}(\Theta)$ *there is a local branching structure on the domain of a natural representation of* G_x*, with a* G_x*-action on the index set,* $g * M_i = M_{g(i)}$ *and* $\sigma_{g(i)} = \sigma_i$*; it can be found inside every open* $V \ni x$. **The text's proof** sets $J = I \times G_x$, $M_{(i,g)} = g * M_i$, $\sigma_{(i,g)} = \sigma_i/|G_x|$ and computes $\sum_{y \in g*M_i} \sigma_{(i,g)} = \frac{1}{|G_x|} \sum_g \Theta(g^{-1} * y) = \Theta(y)$ by functoriality. **The formalized proof** follows it `Chapter9/Symmetric · EpGroupoid.LocalBranchingStructure.symmetrize` `Chapter9/Symmetric · EpGroupoid.SymmetricBranchingStructure`. *Step 1.* Restrict the given structure to the domain $U \subseteq V \cap U(x)$ of a natural representation (Theorem 7.1.13, §VII.4). *Step 2.* $\Phi(g)(M_i \cap U)$ is a submanifold properly embedded in U through x, since $\Phi(g)$ is an sc-diffeomorphism of U fixing

x Chapter9/Symmetric · MPolyfold.IsSmoothSubmanifold.image_of_isScDiffeoOn; the text uses this without saying so. *Step 3.* The weights: $\Theta(\Phi(g)(y)) = \Theta(y)$ Chapter9/Symmetric · EpGroupoid.IsFunctorial.apply_Φ and $y \in \Phi(g)(A) \iff \Phi(g)^{-1}(y) \in A$ Chapter9/Symmetric · EpGroupoid.NaturalRep.mem_image_Φ_iff turn the text's computation into a sum over G_x with $|G_x| =$ `Nat.card` of the finite group. *Step 4.* G_x acts on J by left multiplication on the second factor Chapter9/Symmetric · EpGroupoid.mulActionProdLeft.

IX.4 Proposition 9.1.11 (2017: 9.2): the local construction, and the uniqueness

Proposition 9.1.11 (2017: 9.2) (V9.9) Chapter9/Propositions · EpGroupoid.Proposition_9_1_11. *A branched ep⁺-subgroupoid determines a unique sequence $\Theta_0, \Theta_1, \ldots$ of branched ep⁺-subgroupoids with Θ_k k-dimensional and $\Theta = \sum_k \Theta_k$, locally finite.* The formal statement: for every branched Θ there is exactly one family (Θ_k) with `IsDimDecomposition` Chapter9/Propositions · EpGroupoid.IsDimDecomposition, every Θ_k branched and k-dimensional, at each object finitely many nonzero and Θ their sum, and every such family is locally finite Chapter9/Propositions · EpGroupoid.IsLocallyFiniteFamily.

The local construction. The text splits the indices of a symmetric structure by the dimension of M_i and puts $\Theta_k(y) = \sum_{i \in I_k, y \in M_i} \sigma_i$. Formally, for a labelling dim, Θ_k Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart has the k-dimensional local branching structure $(M_i)_{\dim i = k}$ Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartStructure Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartStructure_isOfDim and $\Theta = \sum_k \Theta_k$ on $U(x)$ Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.eq_sum_dimPart; $\Theta_k = 0$ if no sheet has dimension k Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart_eq_zero, and $\Theta_n = \Theta$ for an n-dimensional structure Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPart_eq_of_isOfDim. For a symmetric structure G_x preserves dimension, since $\Phi(g)$ is an sc-diffeomorphism Chapter9/Propositions · EpGroupoid.SymmetricBranchingStructure.dim_smul, so Θ_k is G_x-invariant Chapter9/Propositions · EpGroupoid.SymmetricBranchingStructure.dimPart_Φ. That the local Θ_k does not depend on the structure is Proposition 9.1.12 (§IX.5).

The uniqueness, as we read it, does not hold Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn. On the trivial ep-groupoid of $\mathbb{R}^2$, let L be the x-axis Chapter9/Propositions · EpGroupoid.axis2 and $\Theta = 1_L$. *Family A:* $\Theta_0 = 1_{\{0\}}$, $\Theta_1 = 1_{L \setminus \{0\}}$ Chapter9/Propositions · EpGroupoid.decompA Chapter9/Propositions · EpGroupoid.decompA_isDimDecomposition. *Family B:* $\Theta_0 = 0$, $\Theta_1 = 1_L$ Chapter9/Propositions · EpGroupoid.decompB Chapter9/Propositions · EpGroupoid.decompB_isDimDecomposition. In family A, $L \setminus \{0\}$ is a 1-dimensional submanifold properly embedded in the open set $\mathbb{R}^2 \setminus \{0\}$, which is all Definition 9.1.2 asks at its points, and $1_{\{0\}}$ is 0-dimensional; both families satisfy (1) and (2). The proof of the text constructs family B. A condition that singles it out, for instance that each Θ_k be represented near every point by a subfamily of a local branching structure of Θ, is a possible repair; it and the global gluing of the local Θ_k (across morphisms, with the paracompactness of $|\mathrm{supp}(\Theta_k)|$ and local finiteness) are not formalized (gap G.96).

IX.5 Proposition 9.1.12: the statement and its cases

Proposition 9.1.12 (V9.10) Chapter9/Propositions · EpGroupoid.Proposition_9_1_12, new in the 2021 edition. *For two local branching structures of Θ on $U(x)$ there is an open G_x-invariant $V(x) \subseteq U(x)$ on which the dimensional decompositions agree.* The formal conclusion is `DimPartsAgreeNear`: an open $V \ni x$ in both domains on which $\Theta_k = \Theta'_k$ for every k Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear. "G_x-invariant, with the usual properties" is obtained by shrinking V to the domain of a natural representation Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.DimPartsAgreeNear.exists_naturalRep. *The equidimensional case* is immediate, since then $\Theta_n = \Theta$ for both Chapter9/Propositions · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_isOfDim. The general case is proved in

Appendix §9.6.1 (§§IX.21–IX.22): under the hypothesis `TopMultiplicity` Chapter9/Appendix · EpGroupoid.proposition_9_1_12_of_topMultiplicity, under the graph representation Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_of_hasGraphChartAt, and without hypothesis on every ep-groupoid over $\mathbb{R}^m$ Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_rn. Mixed structures exist: $\Theta = 1 + \delta_0$ on $\mathbb{R}^1$ has sheets of dimensions 0 and 1 Chapter9/Appendix · exists_mixed_rn_one, and on $\mathbb{R}^2$ two labelled structures of one Θ with sheets of dimensions 1 and 0 have different sheets and agreeing dimensional parts Chapter9/AppendixSlices · EpGroupoid.Remark913Hyp.exists_mixed_rn_two.

IX.6 The category $\mathrm{Gr}(X)$, and the tangent spaces T_yM

T_yM (V9.12). Part I's Remark 4.2.2 (preprint: 4.14) defines $T_yM = Tr(y)(T_yX)$ for an sc$^+$-retraction r with image M near y. Here T_yM is the set of tangent vectors at y fixed by the tangent of r Chapter9/Tangent · MPolyfold.submanifoldTangent. *Step 1.* It does not depend on the retraction Chapter9/Tangent · MPolyfold.mem_submanifoldTangent_iff Chapter9/Tangent · MPolyfold.submanifoldTangent_congr, lies in the fibre over y Chapter9/Tangent · MPolyfold.submanifoldTangent_subset_tangentFiber, is local Chapter9/Tangent · MPolyfold.submanifoldTangent_inter_isOpen, and is mapped onto $T_{f(y)}f(M)$ by an sc-diffeomorphism f Chapter9/Tangent · MPolyfold.submanifoldTangent_image. *Step 2, the chart form.* In a chart the second components of the images of T_yM form a set Chapter9/TangentSlicesBasic · MPolyfold.chartTangent equal to the tangent space of the chart model of M Chapter9/TangentSlicesBasic · MPolyfold.chartTangent_eq, a linear subspace Chapter9/TangentSlicesBasic · MPolyfold.exists_submodule_chartTangent of dimension $\dim_y M$ Chapter9/TangentSlicesBasic · MPolyfold.finrank_chartTangent; a chart change acts on it by a continuous linear map Chapter9/TangentSlicesBasic · MPolyfold.exists_clm_tangentChartMap_snd Chapter9/TangentSlicesBasic · MPolyfold.chartTangent_chart_change, a linear isomorphism Chapter9/TangentSlicesBasic · MPolyfold.exists_linearEquiv_chartTangent. So the generators T_xM_i of $T_\Theta(x)$ are linear subspaces in every chart Chapter9/TangentSlicesBasic · EpGroupoid.LocalBranchingStructure.exists_submodule_of_mem_support, which is what the text means by "finite-dimensional linear subspaces of T_xX".

$\mathrm{Gr}(X)$ (V9.11). $\mathrm{Gr}(x)$ is the additive group of finitely supported $\mathbb{Q}$-valued functions on subsets of T_xX Chapter9/Tangent · EpGroupoid.Gr, a morphism $\varphi : x \to y$ acts by the image under its tangent $T\varphi$ (Chapter 7) Chapter9/Tangent · EpGroupoid.grPush, with $T1_x = \mathrm{id}$ Chapter9/Tangent · EpGroupoid.tangentMor_unit_of_eq and $T\varphi$ injective on the fibre Chapter9/Tangent · EpGroupoid.tangentMor_injOn; the push-forward is functorial Chapter9/Tangent · EpGroupoid.grPush_comp Chapter9/Tangent · EpGroupoid.grPush_unit. The objects (x, a) Chapter9/Tangent · EpGroupoid.GrObj form a category over the full subcategory X^∞ Chapter9/Tangent · EpGroupoid.SmoothCat with the projection π Chapter9/Tangent · EpGroupoid.grProj.

IX.7 Good points: Definitions 9.2.2 and 9.2.6 (2017: 9.4 and 9.5), Lemmas 9.2.3 and 9.2.5 (2017: 9.3 and 9.4), Remark 9.2.4 (2017: 9.4)

Definition 9.2.2 (2017: 9.4) (V9.15) Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.IsGoodPoint. $y \in M_U$ Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.MU is good if there is an open $V(y) \subseteq U(x)$ on which all sheets through y coincide. The text states this for equidimensional Θ; no proof below uses equidimensionality, so the definition and Lemmas 9.2.3 and 9.2.5 are stated for every Θ with a local branching structure.

The intrinsic form Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.isGoodPoint_iff. *For $y \in U(x)$: y is good iff $y \in \mathrm{supp}(\Theta)$ and Θ is constant on $\mathrm{supp}(\Theta)$ near y. Step 1.* Near y take V in the complement of the sheets not through y Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.nbhdAt Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.isOpen_nbhdAt; there $\Theta(z) \le \Theta(y)$ Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.le_of_mem_nbhdAt, with equality iff z lies on all sheets through y Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.eq_iff_of_mem_nbhdAt. *Step 2.* If all sheets through y coincide on V, Θ is $\Theta(y)$ on $\mathrm{supp}(\Theta) \cap V$; conversely, if Θ is constant on the support near y, every

point of a sheet through y near y lies on all of them, by Step 1.

Lemma 9.2.3 (2017: 9.3) (V9.16) `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.lemma_9_2_3` follows at once, for any two structures, at possibly different points, whose domains contain y: the intrinsic form does not mention the structure. This is the text's argument, in which $Q(z) = V(y) \cap W(y)$ is presumably $Q(y)$.

Remark 9.2.4 (2017: 9.4) (V9.17). (a) `Chapter9/GoodPoints · EpGroupoid.remark_9_2_4_a` is Lemma 9.2.3 with the good set of Definition 9.2.6 `Chapter9/GoodPoints · EpGroupoid.mem_goodSet_iff`. (b) `Chapter9/GoodPoints · EpGroupoid.remark_9_2_4_b`: near a good y, $\Theta = \Theta(y) \cdot 1_M$ with $M = M_i \cap V(y)$ for any $i \in I_y$, properly embedded in $V(y)$ `Chapter9/GoodPoints · EpGroupoid.LocalBranchingStructure.restrictAt`.

Lemma 9.2.5 (2017: 9.4) (V9.18) `Chapter9/GoodPoints · EpGroupoid.lemma_9_2_5`. *$s(\varphi)$ is good iff $t(\varphi)$ is good.* The text transports local branching structures along morphisms. The formal proof uses the intrinsic form and the sheet of φ: on a sheet $\mathbf{W}$ of s around φ, $\hat{\varphi} = t \circ (s|\mathbf{W})^{-1}$ is an sc-diffeomorphism between open neighbourhoods of $s(\varphi)$ and $t(\varphi)$ with $\Theta \circ \hat{\varphi} = \Theta$ `Chapter9/GoodPoints · EpGroupoid.Sheet.apply_hat`, so "Θ constant on the support near the point" passes from one to the other `Chapter9/GoodPoints · EpGroupoid.src_mem_goodSet_of_tgt`.

Definition 9.2.6 (2017: 9.5) (V9.19) `Chapter9/GoodPoints · EpGroupoid.goodSet` `Chapter9/GoodPoints · EpGroupoid.badSet`. Both sets are saturated `Chapter9/GoodPoints · EpGroupoid.preimage_π_image_goodSet` `Chapter9/GoodPoints · EpGroupoid.preimage_π_image_badSet`, by Lemma 9.2.5. The text cites "Remark 9.2.4 (c)" here; the remark has (a) and (b), and Lemma 9.2.5 is what the descent to $|X|$ uses.

IX.8 Lemma 9.2.7 (2017: 9.5): good points are dense and accumulate on every sheet

Lemma 9.2.7 (1) (2017: 9.5 (1)) (V9.20). The openness of the good set in $\mathrm{supp}(\Theta)$ is the definition `Chapter9/GoodPoints · EpGroupoid.lemma_9_2_7_1`. **The text** derives the density from (2). **The formalized proof** needs neither (2) nor equidimensionality `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_1_dense`. *Step 1.* Given an open $W \subseteq U(x)$ meeting the support, shrink W one sheet at a time so that $M_U \cap W$ either lies in M_j or misses it `Chapter9/TangentSlices · exists_isOpen_subset_inter_eq_empty_or_subset`: if $M_U \cap W \not\subseteq M_j$, a point of $M_U \cap W$ off the closed set M_j has a neighbourhood in W missing M_j, and the sheets already decided stay decided. *Step 2.* On the final W every sheet through a point of $M_U \cap W$ contains $M_U \cap W$, so all sheets through it agree on W and it is good. So the good set is dense `Chapter9/TangentSlices · EpGroupoid.dense_goodSet` and the bad set nowhere dense `Chapter9/TangentSlices · EpGroupoid.isNowhereDense_badSet`.

Lemma 9.2.7 (2) (2017: 9.5 (2)) (V9.21) `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_2`. *For an n-dimensional structure at x and every i, good points of M_i converge to x.* **The text's proof** fixes i_0, sets $\Sigma_i = M_i \cap M_{i_0} \cap D_k(x)$ and either finds a point of $M_{i_0} \cap D_k(x)$ off all Σ_i or moves a point off the boundaries of those Σ_i with interior; such a point is "clearly" good. **The formalized proof** does this with Step 1 above applied inside M_{i_0}. The point found is good only if $M_{i_0} \subseteq M_i$ near it forces $M_{i_0} = M_i$ near it, which is invariance of domain for submanifolds of equal dimension, a fact of Part I that is not formalized; it is the hypothesis `HasDomainInvarianceAt` `Chapter9/TangentSlices · MPolyfold.HasDomainInvarianceAt`. On $\mathbb{R}^m$ it holds: an n-dimensional submanifold contained in another near a point is open in it, by the inverse function theorem `Chapter9/TangentSlices · MPolyfold.hasDomainInvarianceAt_rn` `Chapter9/TangentSlices · EpGroupoid.lemma_9_2_7_2_rn`. The statement holds at every x, not only at bad points. In Remark 9.1.3 (b) the origin is bad `Chapter9/TangentSlices · EpGroupoid.Remark913Hyp.zero_mem_badSet` and is approached by good points on each graph `Chapter9/TangentSlices · EpGroupoid.Remark913Hyp.exists_seq_goodSet_structureβ`.

IX.9 Proposition 9.2.8 and Lemmas 9.2.9–9.2.11 (2017: 9.3 and 9.6–9.8): graph charts and tangent slices

The text. At a bad point x the indices with the same $T_xM_i = L$ form a class Ξ. Locally, $x = 0$ in a partial quadrant C; $L \cap C$ has interior in L (Lemma 9.2.9); the M_i, $i \in \Xi$, are graphs $a + B_i(a)$ over a cone around a unit h_0 in that interior, with an sc-complement Z of L (Lemma 9.2.10), through the inversion of a map Φ on the cone (Lemma 9.2.11). Proposition 9.2.8 then splits $\Xi = I^1 \sqcup \cdots \sqcup I^p$ with sequences of good points $x_k^e \to x$ lying on M_i exactly for $i \in I^e$, and the proof of Theorem 9.2.1 counts $\sum_{i\in\Xi} \sigma_i = \sum_e \Theta(x_k^e)$.

What the count needs. One finite set F near x that meets each M_i, $i \in \Xi$, exactly once and misses the other sheets, so that $\sum_{i\in\Xi} \sigma_i = \sum_{y\in F} \Theta(y)$: a *tangent slice* `Chapter9/Tangent · MPolyfold.HasTangentSlicesAt`. The sequences of Proposition 9.2.8 are one way to produce such sets; a slice needs only one good direction $h_0 \in L$, off the finitely many other tangent spaces. So Proposition 9.2.8 is formalized in this slice form (V9.22).

The graph representation as a hypothesis. Lemmas 9.2.9–9.2.11 rest on sc-calculus at level 0 inside a partial quadrant (sc-complements, the mean value formula for sc⁺-retractions, an inverse function theorem on a cone), which Part I does not formalize. Their content in a chart is the hypothesis `IsGraphChartAt`: every n-dimensional submanifold through x is, along each admissible ray th of its tangent space, met by the affine fibre through th, with secants uniformly approximated by T_xM `Chapter9/TangentSlicesBasic · MPolyfold.IsGraphChartAt` `Chapter9/TangentSlicesBasic · MPolyfold.HasGraphChartsAt`. Rays replace the cones of Lemma 9.2.10, and the admissible directions play the role of C. *From graphs to slices* `Chapter9/TangentSlicesBasic · MPolyfold.hasTangentSlicesAt_of_hasGraphChartsAt`: with Q a projection onto L, the slice is $\{y \mid Q(\varphi y - \varphi x) = th_0,\ \|\varphi y - \varphi x - th_0\| < \varepsilon t\}$ for small t; it meets each sheet of the class once, by the graph property, and misses the other sheets, since their tangent spaces do not contain h_0 `Chapter9/TangentSlicesBasic · exists_mem_ball_forall_not_mem` `Chapter9/TangentSlicesBasic · eq_zero_of_proj_eq_zero`. *On* $\mathbb{R}^m$ the graph representation holds at every point and in every dimension `Chapter9/TangentSlicesRn · MPolyfold.isGraphChartAt_rn` `Chapter9/TangentSlicesRn · MPolyfold.hasGraphChartsAt_rn` `Chapter9/TangentSlicesRn · MPolyfold.hasTangentSlicesAt_rn`: a submanifold is near x the image of a smooth retraction s, and the inverse function theorem for $Q \circ s$ restricted to L gives the graph `Chapter9/TangentSlicesRn · exists_mem_graph_of_hasStrictFDerivAt` `Chapter9/TangentSlicesRn · MPolyfold.rn_exists_retraction`.

One chart for all dimensions. For Proposition 9.6.5 and the independence of Theorem 9.2.1 without equidimensionality, the sheets of all dimensions through x must be compared in one chart: `HasGraphChartAt` asks one chart and one set of admissible directions for all of them `Chapter9/AppendixSlices · MPolyfold.HasGraphChartAt` `Chapter9/AppendixSlices · MPolyfold.HasGraphChartAt.hasGraphChartsAt`, which holds on $\mathbb{R}^m$ `Chapter9/AppendixSlices · MPolyfold.hasGraphChartAt_rn`. In it, every class has slices inside every neighbourhood `Chapter9/AppendixSlices · MPolyfold.exists_slice_of_isGraphChartAt`, which can be chosen to miss finitely many given submanifolds of smaller dimension `Chapter9/AppendixSlices · MPolyfold.exists_generic_slice`. For Lemma 9.3.4 good points must be found near the slice, which a ray does not provide; the ball form `HasGraphBallAt` asks the graph property over a ball of foot points `Chapter9/OrientationRn · MPolyfold.HasGraphBallAt` `Chapter9/OrientationRn · MPolyfold.HasGraphBallAt.hasGraphChartsAt`, again true on $\mathbb{R}^m$ `Chapter9/OrientationRn · hasGraphBallAt_rn`.

IX.10 Theorem 9.2.1 (2017: 9.1): the section T_Θ

Theorem 9.2.1 (2017: 9.1) (V9.13) `Chapter9/Tangent · EpGroupoid.theorem_9_2_1`. *There is a section functor* $T_\Theta : X^\infty \to \mathrm{Gr}(X)$, $\pi \circ T_\Theta = \mathrm{Id}$, *with* $T_\Theta(x) = \sum_i \sigma_i T_xM_i$ *for every local branching structure at* $x \in \mathrm{supp}(\Theta)$, *the zero sum off the support, and* $T_\Theta(\varphi) = (T_\Theta(s(\varphi)), \varphi)$. The theorem is proved in two parts. *Part 1, the section, given the independence.* Write $\sum_i \sigma_i T_xM_i$ as an element of $\mathrm{Gr}(x)$

Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum. If any two structures at x give the same sum Chapter9/Tangent · EpGroupoid.HasIndependentTangentSums, $T_\Theta(x)$ is that sum for a chosen structure, and 0 off the support Chapter9/Tangent · EpGroupoid.tangentValue Chapter9/Tangent · EpGroupoid.tangentValue_eq. For a morphism φ, a structure at $s(\varphi)$ is transported to $t(\varphi)$ by $\widehat{\varphi}$ Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.transport, which maps the sheets' tangent sets by $T\varphi$ Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_transport Chapter9/Tangent · EpGroupoid.tangentMor_eq_tangentMapOn_hat; by the independence $T\varphi_*(T_\Theta(s(\varphi))) = T_\Theta(t(\varphi))$ Chapter9/Tangent · EpGroupoid.grPush_tangentValue, which makes T_Θ a functor Chapter9/Tangent · EpGroupoid.tangentSection.

Part 2, the independence (V9.14). *At a good point* there is no hypothesis: near it all sheets through it coincide, so the sum is $\Theta(y) \cdot T_yM$ for every structure Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_isGoodPoint Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_subset_goodSet. *Equidimensional structures* Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_isOfDim: group the sheets of both structures through x by tangent set; for each class L, a tangent slice F (§IX.9) meeting each sheet of either structure with tangent space L once gives $\sum_{T_xM_i=L}\sigma_i = \sum_{y\in F}\Theta(y) = \sum_{T_xM'_j=L}\sigma'_j$. With sheets of one dimension a slice for one structure is one for the other. *In general* the text reduces to the equidimensional case through the dimensional decomposition, and so does the formalization: if the dimensional parts of two structures agree near x (Proposition 9.1.12), their tangent sums agree dimension by dimension Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_sum_dimPartStructure Chapter9/Tangent · EpGroupoid.LocalBranchingStructure.tangentSum_eq_of_dimPartsAgreeNear Chapter9/Tangent · EpGroupoid.hasIndependentTangentSums_of_dimPartsAgreeNear. With one graph chart for all dimensions and the local constancy of the dimension, which supplies the labellings, the independence follows Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_of_hasGraphChartAt_of_hasLocallyConstantDimAt; restricting a structure to a neighbourhood where the dimensions are constant is Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.exists_restrict_labelled. On every ep-groupoid over $\mathbb{R}^m$ both hypotheses hold, so T_Θ exists for every branched Θ Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn. In Remark 9.1.3 (b) the two structures have the same tangent sum at the bad origin Chapter9/TangentSlicesRn · EpGroupoid.Remark913Hyp.tangentSum_structureβ_eq_structureγ. The text writes "it suffices to prove the proposition" (p. 410) for the theorem and closes with "This completes the proof of Theorem 9.2.8" (p. 418), presumably Theorem 9.2.1; in $\sum_{i\in I|y\in M_j}\sigma_iT_yM_i$ (p. 418) the index is presumably i.

IX.11 Definition 9.2.12 (2017: 9.6): the tangent $T\Theta$

Definition 9.2.12 (2017: 9.6) (V9.23) Chapter9/Tangent · EpGroupoid.tangentTheta. $T\Theta(h) = 0$ over the complement of the support and $T\Theta(h) = \sum_{h\in TM_i}\sigma_i$ over $U(x)$, with $h \in TM_i$ read as $h \in T_{p(h)}M_i$ Chapter9/Tangent · EpGroupoid.tangentSubmanifold. The formal definition evaluates $T_\Theta(p(h))$ at the tangent sets containing h Chapter9/Tangent · EpGroupoid.grEval, so it does not depend on a structure by construction; that it agrees with the text's local formula for every structure Chapter9/Tangent · EpGroupoid.tangentTheta_eq is the text's "the local definitions fit together", and comes from the independence through the identity $\sum_{h\in T_yM_i}\sigma_i = \sum_{h\in T_yM'_j}\sigma'_j$ Chapter9/Tangent · EpGroupoid.sum_tangent_eq_sum_tangent. $T\Theta \geq 0$ Chapter9/Tangent · EpGroupoid.tangentTheta_nonneg and it is a functor on the tangent ep-groupoid of Chapter 8 Chapter9/Tangent · EpGroupoid.isFunctorial_tangentTheta.

Is $T\Theta$ a branched ep⁺-subgroupoid? The text says so without discussion. It needs that the TM_i are smooth submanifolds of TX of smooth points, closed in $TU(x)$, and that $|\mathrm{supp}(T\Theta)|$ is paracompact; these are the fields of `TangentBranchingHyp` Chapter9/Tangent · EpGroupoid.TangentBranchingHyp, under which the theorem holds Chapter9/Tangent · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta Chapter9/Tangent · EpGroupoid.tangentBranchingStructure. *Closedness* is proved Chapter9/TangentSlices · EpGroupoid.isClosed_tangentSubmanifold. *Paracompactness* follows from that of $|X|$ Chapter9/TangentSlices · EpGroupoid.paracompactSpace_orbitSupp_tangentTheta. So only the smoothness of the TM_i remains Chapter9/TangentSlices · EpGroupoid.tangentBranchingHyp_of_isSmoothSubmanifold Chapter9/TangentSlices · EpGroupoid.

isBranchedEpSubgroupoid_tangentTheta_of_isSmoothSubmanifold. TM_i is a sub-M-polyfold of TX, the image of the retraction Ts Chapter9/AppendixSlices · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold_tangent Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.isSubMPolyfold_tangentSubmanifold; that Ts is an sc⁺-retraction of TX is not formalized (gap G.99). On $\mathbb{R}^m$ the independence is free and only that field remains Chapter9/AppendixSlices · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_rn; the tangent of a line in $\mathbb{R}^2$ is a sub-M-polyfold of $T\mathbb{R}^2$ Chapter9/AppendixSlices · rnPolyfold_isSubMPolyfold_tangent_graph2. The full hypothesis has the zero functor as witness Chapter9/Tangent · EpGroupoid.tangentBranchingHyp_zero Chapter9/Tangent · EpGroupoid.isBranchedEpSubgroupoid_tangentTheta_zero. The text's introduction (p. 408) calls $T\Theta$ a "branched ep-subgroupoid", Definition 9.2.12 an ep⁺-subgroupoid.

IX.12 Lemma 9.2.13 (2017: 9.9) and Remark 9.2.14 (2017: 9.5): the position of M relative to ∂X

Reduced tangent position in a chart (V9.24). $T^R_x X$ is not a subspace of a vector space in the project; in a chart φ with $\varphi(x) = z$ in a partial quadrant, "$T_xM \subseteq T^R_x X$" is the inclusion of the chart image of T_xM in the reduced tangent space E_z Chapter9/BoundaryPosition · IsReducedTangentAt Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_reduced, equivalently in the partial quadrant or the partial cone at z Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_partialQuadrant Chapter9/BoundaryPosition · isReducedTangentAt_iff_subset_partialCone. It does not depend on the chart Chapter9/BoundaryPosition · MPolyfold.isReducedTangentAt_chart_change Chapter9/BoundaryPosition · MPolyfold.isTangentReducedAt_iff_forall, and defines `IsTangentReducedAt` Chapter9/BoundaryPosition · MPolyfold.IsTangentReducedAt. $d_M(x)$ is the degeneracy index of the induced M-polyfold structure Chapter9/BoundaryPosition · MPolyfold.subDegIndex, at most $d_X(x)$ Chapter9/BoundaryPosition · MPolyfold.subDegIndex_le.

Lemma 9.2.13 (2017: 9.9). *For $x \in M$: (1) $x \in \partial X$ and $d_M(x) = 0$ iff (2) $T_xM \subseteq T^R_x X$.* **As printed (2) □ (1) does not hold at interior points**: for a point p of $\mathbb{R}^n$, $\{p\}$ is in reduced tangent position at p and $p \notin \partial\mathbb{R}^n$ Chapter9/BoundaryPosition · MPolyfold.not_lemma_9_2_13_two_one. The lemma is about $x \in M \cap \partial X$, where the text uses it, and is formalized there. *(1) □ (2)* Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13 Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13_chart: if $d_M(x) = 0$, Part I's Theorem 2.5.? (preprint: 2.31) gives a chart of M without boundary near x, whose tangent space lies in the reduced tangent space of X; the boundary hypothesis is not used, and the implication holds off ∂X too Chapter9/BoundaryPosition · MPolyfold.lemma_9_2_13_of_not_mem_boundary. *(2) □ (1)* Chapter9/BoundaryPositionConverse · MPolyfold.lemma_9_2_13_two_one. **The text's proof** builds a complement Y and a map F and asserts that the image is M near 0. **The formalized proof.** *Step 1.* In a chart, M is the image of an sc⁺-retraction s near z, with $T_zM \subseteq E_z$. *Step 2,* "the image is M near 0", which the text does not spell out: the projection p onto T_zM along a complement is injective on M near z, by a level-1 contraction with the mean value inequality Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_injOn_proj. *Step 3.* By Mathlib's inverse function theorem for $p \circ s$ on T_zM, $\chi = p(\cdot - z)$ is a chart of $M \cap G$ onto an open subset of $T_zM \subseteq E_z$, with inverse $s \circ (z + \cdot) \circ \Psi^{-1}$, a model without boundary Chapter9/BoundaryPositionConverse · ScRetraction.IsScPlus.exists_flat. *Step 4.* So $d_M(x) = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.subDegIndex_eq_zero_of_isTangentReducedAt Chapter9/BoundaryPositionConverse · MPolyfold.degIndex_eq_zero_of_isReducedTangentAt. The complement Y and the map F are not needed. Together: the equivalence at $x \in \partial X$ Chapter9/BoundaryPositionConverse · MPolyfold.lemma_9_2_13_iff, and at every $x \in M$: $d_M(x) = 0 \iff T_xM \subseteq T^R_x X$ Chapter9/BoundaryPositionConverse · MPolyfold.subDegIndex_eq_zero_iff. The witness is a point of $\mathbb{R}^n$; a witness at a boundary point needs a submanifold API for quadrants, not yet available.

Remark 9.2.14 (2017: 9.5) (V9.25). ∂M is the boundary of the induced M-polyfold Chapter9/BoundaryPosition · MPolyfold.mem_boundary_subPolyfold_iff. *$x \in \partial M$ iff $x \in \partial X$ and $T_xM \not\subseteq T^R_x X$* Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_iff: □ with $d_M(x) \geq 1$ from Lemma 9.2.13 Chapter9/BoundaryPosition · MPolyfold.remark_9_2_14; □ since $\partial M \subseteq \partial X$ Chapter9/BoundaryPosition · MPolyfold.remark_9_2_14_mem_boundary and a point of M in reduced position has

$d_M = 0$ Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_not_isTangentReducedAt Chapter9/BoundaryPositionConverse · MPolyfold.remark_9_2_14_not_mem_boundary_iff.

IX.13 Proposition 9.2.15 (2017: 9.4), Definition 9.2.16 (2017: 9.7) and Remark 9.2.17 (2017: 9.6): the boundary $\partial\Theta$

The boundary of a sheet. $y \in \partial M$ is characterized through T_yM (Remark 9.2.14), so the set of y with $y \in \partial M_i$ Chapter9/Boundary · MPolyfold.submanifoldBoundary Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_iff is read off the tangent sets: $y \in \partial M$ iff $y \in \partial X$ and T_yM is not in reduced position Chapter9/Boundary · MPolyfold.IsReducedTangentSet Chapter9/Boundary · MPolyfold.isReducedTangentSet_iff. It is local and invariant under sc-diffeomorphisms Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_inter_isOpen_iff Chapter9/Boundary · MPolyfold.mem_submanifoldBoundary_image_iff.

Proposition 9.2.15 (2017: 9.4) (V9.26). *(1)* The boundary weight $\sum_{y\in\partial M_i}\sigma_i$ is the part of $T_\Theta(y)$ on tangent sets not in reduced position Chapter9/Boundary · EpGroupoid.boundaryPart Chapter9/Boundary · EpGroupoid.filter_tangentValue_eq; so, under the independence of Theorem 9.2.1, it is the same for any two structures whose domains contain y Chapter9/Boundary · EpGroupoid.proposition_9_2_15_1. *(2)* If $d_X(y) = 0$, no tangent set is out of reduced position, so the weight and the formal sum vanish for every structure, with no independence Chapter9/Boundary · EpGroupoid.proposition_9_2_15_2. *(3)* $\partial\Theta$, 0 off ∂X and the boundary part of T_Θ on it Chapter9/Boundary · EpGroupoid.boundaryTheta, is a functor Chapter9/Boundary · EpGroupoid.isFunctorial_boundaryTheta, nonnegative, supported in ∂X and given by the local formula on every structure Chapter9/Boundary · EpGroupoid.boundaryTheta_eq Chapter9/Boundary · EpGroupoid.proposition_9_2_15_3. (1) and (2) need neither $x \in \operatorname{supp}(\Theta) \cap \partial X$ nor that Θ be branched.

Definition 9.2.16 (2017: 9.7) (V9.27). *$\partial\Theta$ is uniquely characterized by (1)* $\operatorname{supp}(\partial\Theta) \subseteq \partial X$ *and (2) the local formula at points of* $\operatorname{supp}(\partial\Theta)$ Chapter9/Boundary · EpGroupoid.IsBoundaryCharacterized. $\partial\Theta$ has (1) and (2) Chapter9/Boundary · EpGroupoid.isBoundaryCharacterized_boundaryTheta. **The uniqueness, as we read it, does not hold**: (2) is asked only on $\operatorname{supp}(F)$, so $F = 0$ has (1) and (2) for every Θ Chapter9/Boundary · EpGroupoid.isBoundaryCharacterized_zero, and for $\Theta = 1$ on $[0,\infty)$, where $\partial\Theta(0) = 1$ Chapter9/Boundary · boundaryTheta_halfLine_one, both 0 and $\partial\Theta$ do Chapter9/Boundary · not_unique_isBoundaryCharacterized_halfLine. With (2) asked at the points of $\operatorname{supp}(\Theta)$ and $F = 0$ off $\operatorname{supp}(\Theta)$, $F = \partial\Theta$ Chapter9/Boundary · EpGroupoid.eq_boundaryTheta. Before the definition the text writes "In view of Proposition 9.2.13" (2017: "Proposition 9.9", the number of Lemma 9.2.13 there), presumably Proposition 9.2.15, and in it "branched ep⁺-groupoid".

Remark 9.2.17 (2017: 9.6) (V9.28). $\partial\Theta \le \Theta$ Chapter9/Boundary · EpGroupoid.boundaryTheta_le, and $\partial\Theta$ is not Θ restricted to ∂X: for the indicator of the corner 0 of $[0,\infty)$, $\Theta(0) = 1$ and $\partial\Theta(0) = 0$ Chapter9/Boundary · boundaryTheta_halfLineCorner, while for $\Theta = 1$ on $[0,\infty)$, whose sheet is all of $[0,\infty)$ Chapter9/Boundary · MPolyfold.HasCornerStructure.isSmoothSubmanifold_univ Chapter9/Boundary · EpGroupoid.isBranchedEpSubgroupoid_one, $\partial\Theta(0) = 1$ Chapter9/Boundary · EpGroupoid.boundaryTheta_one. For the indicator of an interior point $\partial\Theta = 0$ Chapter9/Boundary · EpGroupoid.boundaryTheta_indicator_of_mem. The witnesses satisfy the independence Chapter9/Boundary · EpGroupoid.hasIndependentTangentSums_indicator_of_mem. The remark's "In Section 9.5, Definition 9.3.6" refers to §9.3.

IX.14 Orientations: $\widehat{\mathrm{Gr}}(X)$, Definition 9.3.2 (2017: 9.8) and Remark 9.3.3 (2017: 9.8)

Oriented tangent sets (V9.29). An orientation of a finite-dimensional vector space is a class of bases; for a tangent set it is a set of frames, lists of vectors of the set, whose chart images Chapter9/OrientationTangent · MPolyfold.chartVec Chapter9/OrientationTangent · MPolyfold.chartVec_injOn are the positive bases of an orientation of the chart image Chapter9/OrientationTangent · MPolyfold.OrTangent Chapter9/OrientationTangent · MPolyfold.chartFrames Chapter9/OrientationTangent · MPolyfold.

IsChartOrientation Chapter9/OrientationTangent · MPolyfold.IsOrientedAt. A chart change acts linearly Chapter9/OrientationTangent · MPolyfold.exists_clm_chartVec, so this does not depend on the chart Chapter9/OrientationTangent · MPolyfold.IsOrientedAt.isChartOrientation Chapter9/OrientationTangent · MPolyfold.isOrientedAt_iff_forall; every tangent set with linear chart image has exactly two orientations Chapter9/OrientationTangent · MPolyfold.exists_two_orientations, and sc-diffeomorphisms and morphisms push them forward Chapter9/OrientationTangent · MPolyfold.OrTangent.push Chapter9/OrientationTangent · MPolyfold.IsOrientedAt.push_tangentMapOn Chapter9/OrientationTangent · EpGroupoid.isOrientedAt_push_tangentMor. $\widehat{\mathrm{Gr}}(x)$ is generated by oriented tangent sets Chapter9/OrientationTangent · EpGroupoid.GrHat Chapter9/OrientationTangent · EpGroupoid.grHatPush Chapter9/OrientationTangent · EpGroupoid.GrHatObj Chapter9/OrientationTangent · EpGroupoid.grHatProj, and forgetting the orientation is a functor over X^∞ Chapter9/OrientationTangent · EpGroupoid.grForget Chapter9/OrientationTangent · EpGroupoid.grForgetFunctor Chapter9/OrientationTangent · EpGroupoid.grForgetFunctor_comp_grProj, adding the two orientations of a set to the set Chapter9/OrientationTangent · EpGroupoid.grForget_single_add_single. Remark 9.3.1 (2017: 9.7), on an earlier notion of orientation, has no formal content.

Definition 9.3.2 (2017: 9.8) (V9.30) Chapter9/OrientationBasic · EpGroupoid.IsOrientation. An orientation of a submanifold is a continuous choice of orientations of its tangent sets Chapter9/OrientationBasic · MPolyfold.SubmanifoldOrientation, continuity being local constancy of the chart frames; an oriented local branching structure carries one per sheet Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure, with oriented sum $\sum_{y\in M_i}\sigma_i T_y(M_i,o_i)$ Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure.orientedSum forgetting to the tangent sum Chapter9/OrientationBasic · EpGroupoid.OrientedBranchingStructure.grForget_orientedSum. An orientation of Θ assigns to each smooth object an element of $\widehat{\mathrm{Gr}}$, naturally in morphisms, forgetting to T_Θ, and locally an oriented sum; the "lifts T_Θ" clause uses T_Θ, hence the independence of Theorem 9.2.1 Chapter9/OrientationBasic · EpGroupoid.IsOrientation.mk', which on $\mathbb{R}^m$ is free Chapter9/OrientationRn · EpGroupoid.isOrientation_sum_trivial_rn. It defines a section functor Chapter9/OrientationBasic · EpGroupoid.orientedSection Chapter9/OrientationBasic · EpGroupoid.orientedSection_comp_grHatProj Chapter9/OrientationBasic · EpGroupoid.orientedSection_comp_grForgetFunctor_obj. Witnesses: $\Theta = 0$ Chapter9/OrientationBasic · EpGroupoid.isOrientation_zero, and finite weighted sums of oriented closed submanifolds of $\mathbb{R}^n$ Chapter9/OrientationBasic · EpGroupoid.isOrientation_sum_rn.

Remark 9.3.3 (2017: 9.8) (V9.31) Chapter9/OrientationBasic · EpGroupoid.remark_9_3_3. For the indicator of a point p, $T_q = q\cdot(\{p\},+) + (1-q)\cdot(\{p\},-)$, $q\in(0,1)\cap\mathbb{Q}$, are pairwise distinct orientations: a 0-dimensional tangent set has two orientations, and any split of the weight 1 between them is an orientation.

IX.15 Lemma 9.3.4 (2017: 9.10) and Example 9.3.5 (2017: 9.1)

Lemma 9.3.4 (2017: 9.10) (V9.32). *Two orientations of Θ that agree at good points are the same.* **Without equidimensionality, as we read it, the lemma does not hold** Chapter9/OrientationRn · lemma_9_3_4_mixed_dim_rn Chapter9/OrientationUniqueness · lemma_9_3_4_mixed_dim: on $\mathbb{R}^2$ let $\Theta = 1_{\mathbb{R}\times 0} + 1_{(0,0)}$ Chapter9/OrientationUniqueness · mixedTheta Chapter9/OrientationUniqueness · mixedTheta_isBranched. The origin is bad Chapter9/OrientationUniqueness · zero_not_mem_goodSet_mixed, the good points are the other points of the line, and two orientations that orient the line the same way and the point $\{0\}$ oppositely agree at all good points and differ at the origin. The lemma is formalized for n-dimensional Θ Chapter9/OrientationUniqueness · EpGroupoid.IsOrientationOfDim.

The text's proof takes, at a bad x, the sequences of good points of Proposition 9.2.8 for the first structure and compares the orientations of the two structures along them. As we read it, this needs that the same sequences serve the second structure, which the text does not show, and that frames of different sheets can be compared along a sequence. **The formalized proof** Chapter9/OrientationUniqueness · EpGroupoid.lemma_9_3_4. *Step 1.* Its hypothesis `HasOrientedGoodPointsAt` Chapter9/OrientationUniqueness · MPolyfold.HasOrientedGoodPointsAt asks, for each class of sheets of both structures with one tangent set at x, good points near x on every sheet of the class, on which the orientations of the sheets can be compared; this is

Proposition 9.2.8 applied to both families together. Under it, the oriented sums at x agree class by class, which is the lemma. *Step 2,* from the ball graph representation (§IX.9) Chapter9/OrientationRn · MPolyfold.hasOrientedGoodPointsAt_of_hasGraphBallAt: over a ball of foot points in the class tangent space each sheet of the class is a graph, the good points are dense in it (Lemma 9.2.7 (1)), and the relative orientations of the sheets are locally constant. *Step 3,* local constancy: the relative orientation of two frame fields is the sign of a determinant Chapter9/OrientationRn · frameDet Chapter9/OrientationRn · orientation_eq_iff_frameDet Chapter9/OrientationRn · MPolyfold.frames_eq_iff_frameDet, continuous if the chart frames are continuous on level 1 Chapter9/OrientationRn · MPolyfold.continuousOn_chartVec_of_continuousOn Chapter9/OrientationRn · MPolyfold.SubmanifoldOrientation.exists_isOpen_tangent_eq_iff. This continuity is the hypothesis `HasContinuousLevelOneOn` Chapter9/OrientationRn · MPolyfold.HasContinuousLevelOneOn, true on $\mathbb{R}^m$ Chapter9/OrientationRn · hasContinuousLevelOneOn_rn. So Chapter9/OrientationRn · EpGroupoid.lemma_9_3_4_of_hasGraphBallAt holds in general under the two hypotheses, and on every ep-groupoid over $\mathbb{R}^m$ without them Chapter9/OrientationRn · EpGroupoid.lemma_9_3_4_rn Chapter9/OrientationRn · hasOrientedGoodPointsAt_rn. Witness in dimension 0: Chapter9/OrientationUniqueness · hasOrientedGoodPointsAt_rn_zero.

Example 9.3.5 (2017: 9.1) (V9.33) Chapter9/OrientationExample · example_9_3_5 Chapter9/OrientationRn · example_9_3_5_rn. Take $\beta^{++}(s) = e(s) + e(-s)$ with $e =$ `expNegInvGlue`, $\beta^{--} = -\beta^{++}$, β^{+-} equal to β^{++} for $s \le 0$ and to β^{--} for $s > 0$, and β^{-+} the other way Chapter9/OrientationExample · betaPP Chapter9/OrientationExample · betaMM Chapter9/OrientationExample · betaPM Chapter9/OrientationExample · betaMP. The two pairs of graphs cover each point equally often Chapter9/OrientationExample · example_9_3_5_count, which defines Θ Chapter9/OrientationExample · example935Theta Chapter9/OrientationExample · example935Theta_isBranched. Orienting the graphs by $\pm(1, \beta')$ Chapter9/OrientationExample · graphOrientation Chapter9/OrientationExample · chartVec_image_graph2 gives $\hat{T}^1$ from G^{++}, G^{--} and $\hat{T}^2$ from G^{+-}, G^{-+} Chapter9/OrientationExample · exampleT1 Chapter9/OrientationExample · exampleT2. Verified as stated: they coincide at the origin Chapter9/OrientationExample · example_9_3_5_eq_origin, differ at $(s, \beta^{++}(s))$ for $s > 0$ Chapter9/OrientationExample · example_9_3_5_ne_of_pos, and no signs for the second structure reproduce $\hat{T}^1$ near the origin Chapter9/OrientationExample · example_9_3_5_cannot_coincide. **Two claims do not hold as we read them.** "Otherwise different": they also coincide wherever $x < 0$, where $G^{+-} = G^{++}$ and $G^{-+} = G^{--}$ with the same orientations Chapter9/OrientationExample · example_9_3_5_eq_of_neg. "Four possibilities to orient": the four sign choices for the first structure give four different orientations Chapter9/OrientationExample · orientedLift1_injective, and $\hat{T}^2$ is none of them Chapter9/OrientationExample · exampleT2_ne_orientedLift1, so there are at least five with weights 1, and infinitely many with fractional weights, as in Remark 9.3.3. The good points distinguish $\hat{T}^1$ from $\hat{T}^2$, in line with Lemma 9.3.4 Chapter9/OrientationRn · exists_goodSet_exampleT1_ne_exampleT2. A classification of all orientations of Θ is not formalized (gap G.101).

IX.16 Definition 9.3.6 (2017: 9.10), Lemma 9.3.7, and the boundary orientation ((9.11), Definition 9.3.8, Proposition 9.3.9, Definition 9.3.10)

Definition 9.3.6 (2017: 9.10) (V9.34) Chapter9/BoundaryPosition · EpGroupoid.TameBranchingStructure Chapter9/BoundaryPosition · EpGroupoid.IsTameBranched. A tame structure has a natural representation on its domain, the properness property, and sheets with tame induced structures in the sense of Part I's Definition 2.5.18. Witnesses: $\Theta = 0$ Chapter9/BoundaryPosition · EpGroupoid.isTameBranched_zero and a point of $\mathbb{R}^n$ Chapter9/BoundaryPosition · EpGroupoid.isTameBranched_indicator Chapter9/BoundaryPosition · MPolyfold.hasTameInducedStructure_of_degIndex_eq_zero. "In particular every M_i has an equivalent structure of a manifold with boundary with corners" is not formalized (gap G.102).

Lemma 9.3.7 (V9.35) Chapter9/BoundaryPosition · MPolyfold.lemma_9_3_7, new in the 2021 edition. *For tame Y and submanifolds M_1, M_2 of one dimension with tame induced structures, $T_yM_1 = T_yM_2$ implies $d_{M_1}(y) = d_{M_2}(y)$.* **The text's proof** describes T_yM in a tame chart of M by the cone C_yM and shows that $T_0O \cap C$ is the reduced tangent space. **The formalized proof.** For a submanifold with tame induced structure,

$d_M(y)$ is the codimension of the reduced part of the chart image of T_yM, which depends only on that linear space and on the partial quadrant of the chart of Y `Chapter9/BoundaryPosition · MPolyfold.subDegIndex_eq_quotDim_of_hasTameInducedStructure` (Remark 2.5.9 in every chart). So equal tangent spaces give equal indices `Chapter9/BoundaryPosition · MPolyfold.subDegIndex_eq_of_tangentSpace_eq` `Chapter9/OrientationBoundary · MPolyfold.subDegIndex_eq_of_submanifoldTangent_eq`, with neither "Y tame" nor "of the same dimension". The text's "$T_0O \cap C = T_0^R O$" (a cone against a subspace) is read through the linearity of the chart image of T_0M.

Definition 9.3.8 (2017: 9.9) (V9.36) `Chapter9/OrientationBoundary · MPolyfold.boundaryFrames`. *For an oriented (M, o), $y \in \partial M$ with $d_M(y) = 1$: a basis $(a_1, \dots, a_{n-1})$ of $T_y\partial M$ is positive iff $(e, a_1, \dots, a_{n-1})$ is positive, e outward.* Linear algebra first: the convention orients a hyperplane K `Chapter9/OrientationBoundary · exists_boundary_orientation`, by the determinant criterion `Chapter9/OrientationBoundary · exists_basis_orientation_iff_det_pos` `Chapter9/OrientationBoundary · det_cons_smul_add`. In a tangent set: for $K \subseteq T_yM$ with $(m-1)$-dimensional chart image and $e \notin K$, the frames orient K `Chapter9/OrientationBoundary · MPolyfold.isOrientedAt_boundaryFrames`, and $e \mapsto ce + k$, $c > 0$, $k \in K$, does not change them `Chapter9/OrientationBoundary · MPolyfold.boundaryFrames_eq`; both orientations occur, on a graph `Chapter9/OrientationBoundary · exists_boundaryFrames_graph`. $T_y\partial M$ and e are data here: that $T_y\partial M = T_y^R M$ and the outward half-space are determined by M through the cones C_yM is not formalized (gap G.103).

(9.11) and Proposition 9.3.9 (V9.37), new in the 2021 edition. Split the oriented sum at y by $\ell = d_{M_i}(y)$ `Chapter9/OrientationBoundary · EpGroupoid.degreePart` `Chapter9/OrientationBoundary · EpGroupoid.degreePart_orientedSum`; by Lemma 9.3.7 tangent sets of different degree never coincide `Chapter9/OrientationBoundary · MPolyfold.HasSubDegree` `Chapter9/OrientationBoundary · MPolyfold.SubmanifoldOrientation.hasSubDegree_tangent_iff`, so two oriented structures with the same oriented sum at y have the same degree-ℓ parts for every ℓ `Chapter9/OrientationBoundary · EpGroupoid.eq_9_11`. Applying a boundary operation $T_y(M, o) \mapsto T_y(\partial M, o_{\partial M})$, a function `bd` of the oriented tangent set, to the degree-1 part gives (9.12), which therefore does not depend on the structure `Chapter9/OrientationBoundary · EpGroupoid.proposition_9_3_9`. **Definition 9.3.10 (2017: 9.11)** (V9.38) is (9.12) with 0 off the support `Chapter9/OrientationBoundary · EpGroupoid.boundaryLift`; the condition $y \in \partial M_i$ is implied by $d_{M_i}(y) = 1$. The continuity of $o_{\partial M}$ along ∂M and its extension to the faces (Figure 9.1) are not formalized (gap G.103); Remark 9.3.11, on the non-tame case, has no formal content.

IX.17 §9.4: essential points and sets of measure zero (Definitions 9.4.1, 9.4.5, 9.4.7, Lemmas 9.4.2–9.4.8 (2017: 9.12–9.14, 9.11–9.15))

For Mathlib manifolds. Definitions 9.4.1 and 9.4.5 are about a smooth finite-dimensional manifold M with boundary and corners, formalized for Mathlib manifolds with corners. *Definition 9.4.1 (1)* (V9.39): $T_z^K M$ is the span of the unit limits $\varphi(z_k)/|\varphi(z_k)|$, $z_k \to z$ in $K \setminus \{z\}$, in a chart φ centred at z. In a chart it is the span of Mathlib's tangent cone of the chart image of K `Chapter9/BranchingGeometryManifold · chartImage` `Chapter9/BranchingGeometryManifold · span_tangentConeAt_chartImage_eq_map` `Chapter9/BranchingGeometryMeasure · span_unitDirections` `Chapter9/BranchingGeometryMeasure · unitDirections` `Chapter9/BranchingGeometryManifold · relTangent`, and a chart change maps it by the derivative `Chapter9/BranchingGeometryManifold · relTangent_eq_map`, which is the text's "independent of the choice of φ". *Definition 9.4.1 (2)*: z is essential if it is a limit of points of K where $T^K M$ is everything `Chapter9/BranchingGeometryManifold · relTangent_eq_top_iff` `Chapter9/BranchingGeometryManifold · fullSet` `Chapter9/BranchingGeometryManifold · essentialSet` `Chapter9/BranchingGeometryManifold · essentialSet_eq`. For two crossing lines in $\mathbb{R}^2$ the essential set is $\{0\}$ `Chapter9/BranchingGeometryManifold · crossLines` `Chapter9/BranchingGeometryManifold · essentialSet_crossLines` `Chapter9/BranchingGeometryManifold · nonEssential_crossLines`. *Lemma 9.4.2 (2017: 9.11)* (V9.40), "obvious from the definition": K^e is a closure taken in K `Chapter9/BranchingGeometryManifold · lemma_9_4_2` `Chapter9/BranchingGeometryManifold · lemma_9_4_2_open`. *Definition 9.4.5 (2017: 9.13)* (V9.43) `Chapter9/BranchingGeometryManifold · IsMeasureZero`: null in every chart; countable unions `Chapter9/BranchingGeometryManifold · isMeasureZero_iUnion`, a chart cover suffices `Chapter9/BranchingGeometryManifold · isMeasureZero_of_forall_mem`, and $\mathbb{R}^2$ is not null `Chap-`

ter9/BranchingGeometryManifold · not_isMeasureZero_univ. *Lemma 9.4.6 (2017: 9.14)* (V9.44) Chapter9/BranchingGeometryManifold · lemma_9_4_6, "proved similarly as Lemma 2.8 in [57]": *Step 1.* At a point where $T_z^K M \neq T_z M$ there is a unit direction v off the span, and points of K near z lie outside a cone around v Chapter9/BranchingGeometryMeasure · eventually_abs_le_of_tangentConeAt. *Step 2.* A cone has positive density at its vertex Chapter9/BranchingGeometryMeasure · exists_pos_measure_cone_lt, so z is not a Lebesgue density point of K; by Lebesgue's density theorem the set of such z is null Chapter9/BranchingGeometryMeasure · addHaar_setOf_span_tangentConeAt_ne_top Chapter9/BranchingGeometryManifold · isMeasureZero_setOf_relTangent_ne_top. *Step 3.* A non-essential point has a neighbourhood in K of such points, by Lemma 9.4.2. The statement holds for every K, not only for $K(i,i')$.

For sheets in an M-polyfold. The sheets M_i are submanifolds of an M-polyfold, which carry no Mathlib manifold structure in the project. So the essential points and null sets of the sheets are the data of a `SheetGeometry` Chapter9/BranchingGeometry · MPolyfold.SheetGeometry, with Lemma 9.4.4 (`ess_comm`), Lemma 9.4.6 (`isNull_diff_ess`) and the note of Definition 9.4.7 (`submanifoldTangent_eq`) as fields; it is satisfiable by pieces of graphs in $\mathbb{R}^2$, with the essential sets and null sets of the Mathlib part read in the chart $z \mapsto z_0$ Chapter9/BranchingGeometry · graphSheets Chapter9/BranchingGeometry · graphSheetGeometry (gap G.105).

Lemma 9.4.4 (2017: 9.13) (V9.42). *Essential for $K(i,i') \subseteq M_i$ iff for $K(i,i') \subseteq M_{i'}$*, "left to the reader". **As we read it, it needs sheets of one dimension** Chapter9/BranchingGeometry · lemma_9_4_4_mixed_dim: $\mathbb{R} \times 0$ as a subset of $\mathbb{R}$ is essential at every point, as a subset of $\mathbb{R}^2$ nowhere, since its relative tangent space is a line. For an n-dimensional Θ the text's reduction to $T_0^D N$ in a chart applies.

Definition 9.4.7 (2017: 9.14) (V9.45). $i \sim_y i'$ through a chain of links $y \in K(i_{p-1}, i_p)^e$ Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.IsLink Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.YEquiv is an equivalence relation Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.equivalence_yEquiv Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.ySetoid; equivalent sheets have the same tangent set at y Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.submanifoldTangent_eq_of_yEquiv; $\#P_y$ is Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.numClasses.

Lemma 9.4.8 (2017: 9.15) (V9.46) Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.lemma_9_4_8. $\{y \in M_U \mid \#P_y \geq 2\}$ *is null.* "Null in M_U" is null in every sheet through the point Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.IsNullIn. The text decomposes M_U by (I_y, P_y) and concludes that the bad part lies in $\bigcup_{j,j'} K(j,j')^{ne}$. As we read it, $K(j,j')^{ne}$ is null only in M_j and $M_{j'}$; inside a sheet M_i through y the proof uses $K(i,j)^{ne}$ for j not y-equivalent to i: if y had two classes, the class of i omits some $j \in I_y$, so $y \in K(i,j)$ is not essential. Hence $M_U \setminus M_U^*$ is null Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.MStar Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.isNullIn_diff_MStar.

IX.18 Property (T), Propositions 9.4.9 and 9.4.10 (2017: 9.5 and 9.6)

Property (T) (V9.41) Chapter9/BranchingGeometry · EpGroupoid.OrientedBranchingStructure.PropertyT: the orientation relation of two sheets with the same tangent space is the same at every common point with the same tangent space; the first structure of Example 9.3.5 has it Chapter9/BranchingGeometry · propertyT_example935B1. Under (T) the sign map $K(i,i')^e \to \{\pm 1\}$ is constant Chapter9/BranchingGeometry · EpGroupoid.tangent_eq_iff_of_mem_ess. Lemma 9.4.3, that (T) holds after shrinking, is not formalized (gap G.104): it needs the convergence of oriented frames of two sheets in a chart, the continuity of the change-of-frame determinant along $M_i \cap M_{i'}$.

Proposition 9.4.9 (2017: 9.5) (V9.47) Chapter9/BranchingGeometry · EpGroupoid.proposition_9_4_9. *Off a null set $M_U = M_U^* = \bigsqcup_J M_U^J$, and on M_U^J: $\Theta = \sigma_J$, $T_\Theta = \sigma_J T^J$, $\hat{T}_\Theta = a_J(T^J, o_J^a) + b_J(T^J, o_J^b)$, $a_J + b_J = \sigma_J$, $o_J^a = -o_J^b$. Step 1.* The null set is Lemma 9.4.8; the pieces are the level sets of $y \mapsto I_y$ Chapter9/BranchingGeometry · MPolyfold.SheetGeometry.MStarPart. *Step 2.* On M_U^J all sheets of J are y-equivalent, so they have one tangent set T^J and $T_\Theta(y) = \sigma_J T^J$ Chapter9/BranchingGeometry · EpGroupoid.LocalBranchingStructure.

tangentValue_eq_sum, which uses the independence of Theorem 9.2.1. *Step 3.* By (T) the orientations of the sheets of J at y fall into two opposite classes; $o^a_J = -o^b_J$ is "$o^a_J \neq o^b_J$ on the same tangent set", since it has exactly two orientations Chapter9/BranchingGeometry · MPolyfold.OrTangent.eq_or_eq_or_eq_of_isOrientedAt Chapter9/BranchingGeometry · MPolyfold.exists_ne_isOrientedAt; if one class is empty its orientation is the other one. The footnote's "measure zero" in M_U refers forward to Definition 9.5.1 and Lemma 9.5.2 (p. 437), consistently as we read it.

Proposition 9.4.10 (2017: 9.6) (V9.48) Chapter9/BranchingGeometry · EpGroupoid.proposition_9_4_10. *Two oriented structures on $U(x)$ with (a) the same weight sums and (b) the same oriented sums agree off a null set, piece by piece, as in (1)–(5).* **The text** takes the union B'' of the two structures, which describes 2Θ, applies Proposition 9.4.9 to B, B', B'' and intersects. **The formalized proof** takes $M^\dagger_U = M''^*_U$ for the union only, not intersected with M^*_U and M'^*_U, which (1)–(5) do not need. The pieces A_p are cut out by the finitely many values of $y \mapsto (I_y, I'_y, J_+(y), J'_+(y))$, with $J_\pm(y)$ the two orientation classes at y, so Property (T) is not used. The two structures of Example 9.3.5, all signs $+$, satisfy (a) and (b) Chapter9/BranchingGeometry · example935B1 Chapter9/BranchingGeometry · example935B2 Chapter9/BranchingGeometry · example935_hypotheses_9_4_10 Chapter9/BranchingGeometry · example935_MU_eq; at the origin G^{++} and G^{--} are tangent but not 0-equivalent, so the set of Lemma 9.4.8 is not empty Chapter9/BranchingGeometry · example935_two_le_numClasses.

IX.19 §9.5: sets of measure zero and the σ-algebras $L(S)$, $L(\partial S)$ (Definitions 9.5.1, 9.5.3, 9.5.14, Lemma 9.5.2 (2017: Definitions 9.15–9.17, Lemma 9.16))

The setting of §9.5. Θ is oriented, tame, compact and n-dimensional, and $S = |\mathrm{supp}(\Theta)|$. Measure zero in a sheet M_i is that of Definition 9.4.5, which the project has for Mathlib manifolds (§IX.17) but not for submanifolds of an M-polyfold. So, as in §9.4, the null sets of the sheets are data: a system `SheetNullSets` on a class $\mathcal{S}$ of sheets Chapter9/Measure · SheetNullSets, with null sets closed under subsets, countable unions Chapter9/Measure · SheetNullSets.isNull_iUnion_countable and finite unions Chapter9/Measure · SheetNullSets.isNull_union, local, induced alike by two sheets of $\mathcal{S}$ on their overlap (`isNull_transfer`), and a nonempty sheet not null in itself. A §9.4 sheet geometry with these properties is a `SheetMeasureGeometry` Chapter9/Measure · MPolyfold.SheetMeasureGeometry Chapter9/Measure · MPolyfold.SheetMeasureGeometry.toSheetNullSets, and then null preimages are §9.4's "null in M_U" Chapter9/Measure · EpGroupoid.LocalBranchingStructure.hasNullPreimages_iff_isNullIn. *The covering criterion* Chapter9/Measure · SheetNullSets.isNull_of_cover: a subset C of a sheet $N \in \mathcal{S}$ is null in N if near each of its points it is covered by finitely many sheets of $\mathcal{S}$ in which it is null; by locality and the transfer property.

Definition 9.5.1 (2017: 9.15) (V9.49) Chapter9/Measure · EpGroupoid.IsMeasureZeroSupp. $A \subseteq |X|$ has measure zero if every $x \in \mathrm{supp}(\Theta)$ has a tame local branching structure (Definition 9.3.6, the text's "suitable $U(x)$") with sheets in $\mathcal{S}$ whose preimages $M_i \cap \pi^{-1}(A)$ are null in M_i Chapter9/Measure · EpGroupoid.LocalBranchingStructure.HasNullPreimages. The variants ask this of some local branching structure Chapter9/Measure · EpGroupoid.IsMeasureZeroSuppAny, or of every structure with sheets in $\mathcal{S}$ Chapter9/Measure · EpGroupoid.IsMeasureZeroSuppAll; the standing assumption is that tame structures with sheets in $\mathcal{S}$ exist Chapter9/Measure · EpGroupoid.HasTameSheetsIn. Compactness and orientation are not used.

Lemma 9.5.2 (2017: 9.16) (V9.50) Chapter9/Measure · EpGroupoid.lemma_9_5_2. *The notion does not depend on the choice of the local branching structures.* **The text** cites the literature, without proof. **The formalized proof.** *Step 1, the core* Chapter9/Measure · EpGroupoid.isMeasureZeroSuppAll_of_any. Suppose some structure at every point has null preimages of A, and let B' be any structure with sheets in $\mathcal{S}$ and M'_j one of its sheets. Each $z \in M'_j \cap \pi^{-1}(A)$ lies in the support, so a structure B at z with null preimages exists. Near z, $M'_j \subseteq \mathrm{supp}(\Theta) \cap U_B = \bigcup_i M_i$, so $M'_j \cap \pi^{-1}(A)$ is covered near z by the finitely many M_i, in which it is null. The covering criterion gives that it is null in M'_j. *Step 2.* A tame structure

is a structure Chapter9/Measure · EpGroupoid.isMeasureZeroSuppAny_of_isMeasureZeroSupp, and under the standing assumption every point has a tame one with sheets in $\mathcal{S}$; the three notions agree. *Step 3.* The sets of measure zero form a σ-ideal Chapter9/Measure · EpGroupoid.isSigmaIdeal_isMeasureZeroSuppAll Chapter9/Measure · EpGroupoid.isSigmaIdeal_isMeasureZeroSupp, and $S \neq \emptyset$ is not null, since a sheet through a point of the support is not null in itself Chapter9/Measure · EpGroupoid.not_isMeasureZeroSuppAny_orbitSupp Chapter9/Measure · EpGroupoid.not_isMeasureZeroSupp_orbitSupp.

The transfer property, as we read it, needs one dimension Chapter9/Measure · isNull_transfer_mixed_dim. Step 1 uses that a set null in the sheets M_i is null in M_j' on their overlap. For sheets of different dimensions this does not hold: the axis $\mathbb{R} \times 0$ has Lebesgue measure zero in the sheet $\mathbb{R}^2$ and is not null in itself, a 1-dimensional sheet; so no single system contains the planar null sets and those of graphs. The lemma is therefore proved for structures whose sheets lie in one class, as do the n-dimensional structures of an n-dimensional Θ. That *every* structure of an n-dimensional Θ has n-dimensional sheets follows from Proposition 9.1.12, which is proved on $\mathbb{R}^m$ and under the graph representation (§§IX.5, IX.22), not on a general M-polyfold (gaps G.97, G.106).

Definition 9.5.3 (2017: 9.16) and the remark after it (V9.51). *Step 1, a general fact.* For a family $\mathcal{N}$ of subsets, the σ-algebra generated by the Borel sets and $\mathcal{N}$ Chapter9/Measure · borelNullSigma Chapter9/Measure · borel_le_borelNullSigma Chapter9/Measure · measurableSet_borelNullSigma_of_mem is, if $\mathcal{N}$ is a σ-ideal Chapter9/Measure · IsSigmaIdeal, the family $\{A \mid A \,\Delta\, B \in \mathcal{N} \text{ for a Borel } B\}$ Chapter9/Measure · borelNullCompletion Chapter9/Measure · measurableSet_borelNullSigma_iff: that family is a σ-algebra, by $\bigcup_k A_k \,\Delta\, \bigcup_k B_k \subseteq \bigcup_k (A_k \,\Delta\, B_k)$ Chapter9/Measure · symmDiff_iUnion_subset, contains the Borel sets and $\mathcal{N}$, and lies in every σ-algebra containing both. *Step 2.* $L(S)$ is this σ-algebra on the subspace S for the sets of measure zero Chapter9/Measure · EpGroupoid.measureZeroSets Chapter9/Measure · EpGroupoid.lebesgueSigma, a σ-ideal under the standing assumption Chapter9/Measure · EpGroupoid.isSigmaIdeal_measureZeroSets. *Step 3.* The text's remark that $L(S)$ is obtained by adding the sets of measure zero is read as the completion: $K \in L(S)$ iff $K \,\Delta\, B$ has measure zero for a Borel B Chapter9/Measure · EpGroupoid.remark_9_5_3; sets of measure zero belong to $L(S)$ Chapter9/Measure · EpGroupoid.measurableSet_lebesgueSigma_of_isMeasureZeroSupp. The form $K = B \cup N$ would need Borel null hulls of null sets, which an abstract system does not provide.

Definition 9.5.14 (2017: 9.17) and the paragraph before it (V9.52). $\partial S = |\mathrm{supp}(\partial\Theta)| \subseteq S$ Chapter9/Measure · EpGroupoid.orbitSupp_boundaryTheta_subset. $(n-1)$-dimensional measure zero in the boundaries ∂M_i Chapter9/Boundary · MPolyfold.submanifoldBoundary is a second system on a class of boundaries, and K has measure zero in ∂S if at every point of $\mathrm{supp}(\partial\Theta)$ a tame structure has boundary preimages null in the ∂M_i Chapter9/Measure · EpGroupoid.IsMeasureZeroBoundary Chapter9/Measure · EpGroupoid.IsMeasureZeroBoundaryAll. *"A well-defined notion"* Chapter9/Measure · EpGroupoid.isMeasureZeroBoundaryAll_of: the proof of Lemma 9.5.2 applies to the boundaries. By the local formula for $\partial\Theta$ (Proposition 9.2.15), which holds for every structure given the independence of Theorem 9.2.1, a point of $\partial M_j'$ lies in $\mathrm{supp}(\partial\Theta)$, and a point of $\mathrm{supp}(\partial\Theta)$ lies in some ∂M_i of any structure at it, a positive sum having a nonzero term Chapter9/Measure · EpGroupoid.exists_mem_of_sum_pos; so the boundary preimage in $\partial M_j'$ is covered near each point by the ∂M_i of a structure with null boundary preimages. $L(\partial S)$ is the σ-algebra generated by the Borel sets of ∂S and these sets Chapter9/Measure · EpGroupoid.boundaryMeasureZeroSets Chapter9/Measure · EpGroupoid.lebesgueSigmaBoundary. The description of $(n-1)$-dimensional measure zero face by face, and a witness with $\partial S \neq \emptyset$, are not formalized (gap G.108).

Witnesses (V9.53). *Tools.* Measure zero of Definition 9.4.5 is local Chapter9/Measure · isMeasureZero_image_of_local, holds for subsingletons Chapter9/Measure · isMeasureZero_of_subsingleton, and on $\mathbb{R}^2$ is Lebesgue measure zero Chapter9/Measure · isMeasureZero_rn2_iff. On the trivial ep-groupoid of $\mathbb{R}^n$ every local branching structure restricts to a tame one, on the domain of a natural representation Chapter9/Measure · exists_tameBranchingStructure_trivial_rn Chapter9/Measure · hasTameSheetsIn_trivial_rn, and π is injective Chapter9/Measure · injective_π_trivial_rn. *Example 9.3.5.* The graph sheets of §9.4 carry a `SheetMeasureGeometry` Chapter9/Measure · graphSheetMeasureGeometry Chapter9/Measure · graphNull; for the Θ of the two graphs G^{++}, G^{--}, which touch at 0 Chapter9/Measure · theta935 Chapter9/Measure · isBranchedEpSubgroupoid_theta935 Chapter9/Measure · hasTameSheetsIn_theta935 Chapter9/Measure · zero_mem_supp_theta935, the point $|0|$ has measure

zero Chapter9/Measure · isMeasureZeroSupp_theta935_zero, S does not Chapter9/Measure · not_isMeasureZeroSupp_theta935, and both are in $L(S)$ Chapter9/Measure · lebesgueSigma_theta935. *The plane.* For $\Theta = 1$ on $\mathbb{R}^2$, with the open subsets of $\mathbb{R}^2$ as sheets and Lebesgue null sets Chapter9/Measure · planeSheets Chapter9/Measure · planeNull Chapter9/Measure · planeSheet Chapter9/Measure · isSmoothSubmanifold_univ_rn2 Chapter9/Measure · thetaPlane Chapter9/Measure · isBranchedEpSubgroupoid_thetaPlane Chapter9/Measure · mem_supp_thetaPlane Chapter9/Measure · hasTameSheetsIn_thetaPlane, some member of $L(S)$ is not Borel Chapter9/Measure · exists_mem_lebesgueSigma_not_borel_thetaPlane: every subset of the null axis is in $L(S)$, there are $2^{\mathfrak{c}}$ of them, and only $\mathfrak{c}$ Borel sets.

IX.20 §9.5: the isotropy factors (Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 (2017: Lemmas 9.18, 9.19))

The measures μ_ω of Theorems 9.5.4 and 9.5.5 are local weighted integrals of a form over the sheets, divided by $\#G_x$ or $\#G_x^{\text{eff}}$. Forms and integrals are not formalized (gaps G.74, G.80, G.107), so the measure identities of Lemmas 9.5.9–9.5.11 and Proposition 9.5.13 are not stated; what their proofs rest on, the orders of the isotropy groups and the decompositions of the preimages, is.

Lemma 9.5.9 (2017: 9.18) (V9.54). **The text's proof** identifies $H = \{g \mid g * y = y\}$ with G_y, writes $G = \bigcup_{r\in R} rH$, chooses $V(y)$ invariant under H and disjoint from its other translates, notes that $\tilde{M}_j = V \cap M_j$ is a branching structure on V, decomposes the preimage of K as $\bigcup_r \varphi_r(\tilde{K})$, and uses $\#R = \#G/\#H$. **The formalized part.** *Step 1,* $\#R \cdot \#G_y = \#G_x$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_isotropy: the stabilizer H of y for the action of G_x on $U(x)$ is isomorphic to G_y by the natural representation (Chapter 7) Chapter9/MeasureEffective · EpGroupoid.NaturalRep.mem_stabilizer_iff_Φ, and R is `Quotient.out` on G_x/H. *Step 2, the slice* Chapter9/MeasureEffective · EpGroupoid.NaturalRep.exists_slice: inside any neighbourhood of y there is an open V with $\varphi_g(V) = V$ for $g \in H$ and $\varphi_g(V) \cap V = \emptyset$ for $g \notin H$: separate y from $\varphi_g(y) \neq y$ for the finitely many $g \notin H$, and make the neighbourhood H-invariant. *Step 3, the preimage.* $|\varphi_g(z)| = |z|$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.π_Φ, so φ_g maps $M_U = \operatorname{supp}(\Theta) \cap U$ to itself, Θ being a functor Chapter9/MeasureEffective · EpGroupoid.NaturalRep.mapsTo_Φ_MU; for $K \subseteq |M_U \cap V|$, $M_U \cap \pi^{-1}(K) = \bigcup_g \varphi_g(M_U \cap V \cap \pi^{-1}(K))$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion, the same with g running over the representatives r Chapter9/MeasureEffective · EpGroupoid.NaturalRep.MU_inter_preimage_eq_iUnion_quotient, and the union over R is disjoint Chapter9/MeasureEffective · EpGroupoid.NaturalRep.pairwise_disjoint_image_Φ_out. The restriction $\tilde{M}_j$ is §IX.3's `restrictAt`: the project's structures pass through their base point, so at y the sheets through y are kept.

Lemma 9.5.10 (V9.55), new in the 2021 edition. *Step 1.* $\#T_x \cdot \#G_x^{\text{eff}} = \#G_x$ for the subgroup T_x acting trivially near x Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_mul_card_trivialSubgroup. *Step 2.* Under LUC for the given $U(x)$ (Definition 7.2.18, `NaturalRep.HasLUC`), T_x is the kernel of the representation Chapter9/MeasureEffective · EpGroupoid.NaturalRep.kerΦ_eq_trivialSubgroup, hence lies in every stabilizer Chapter9/MeasureEffective · EpGroupoid.NaturalRep.kerΦ_le_stabilizer, and corresponds to T_y under $H \cong G_y$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.stabilizerEquivIsotropy_mem_trivialSubgroup_iff, so $\#T_y = \#T_x$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_trivialSubgroup_eq. *Step 3, (9.23).* Dividing Step 1 of Lemma 9.5.9 by $\#T_x = \#T_y$ gives $\#R \cdot \#G_y^{\text{eff}} = \#G_x^{\text{eff}}$ Chapter9/MeasureEffective · EpGroupoid.NaturalRep.card_quotient_stabilizer_mul_card_effIsotropy.

Lemma 9.5.11 (2017: 9.19) (V9.56). The orders $\#G_x$ and $\#G_x^{\text{eff}}$ are invariant along morphisms, by conjugation with φ (Lemma 7.2.6) Chapter9/MeasureEffective · EpGroupoid.card_isotropy_eq_of_mor Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_of_mor. A functor is constant on orbits Chapter9/MeasureEffective · EpGroupoid.IsFunctorial.eq_of_π_eq, so for structures on U, U' and any map $\hat{\varphi}$ of U onto U' with $|\hat{\varphi}(z)| = |z|$, $|M_U| = |M_{U'}|$ Chapter9/MeasureEffective · EpGroupoid.image_π_MU_eq; the maps φ_g of a natural representation are such maps Chapter9/MeasureEffective · EpGroupoid.NaturalRep.Φ_hyps. The "change

of variables" in the integrals is not stated.

Proposition 9.5.13 (V9.57), new in the 2021 edition. *(1)* For effective X, $T_x = \{1\}$, so $\#G_x^{\mathrm{eff}} = \#G_x$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_of_isEffective and the normalizing factors of the two integrals agree. *(2)* The reduced ep-groupoid X_R (Chapter 7) has the object M-polyfold of X, with Hausdorff and paracompact morphism space as instances (Hausdorff under LUC); functors Chapter9/MeasureEffective · EpGroupoid.isFunctorial_reduced_iff and local branching structures Chapter9/MeasureEffective · EpGroupoid.LocalBranchingStructure.toReduced of X and X_R coincide, and $\#G_x^{\mathrm{eff}} = \#G_x^R = \#(G_x^R)^{\mathrm{eff}}$ Chapter9/MeasureEffective · EpGroupoid.card_effIsotropy_eq_reduced.

IX.21 Appendix §9.6.1: Lemmas 9.6.1 and 9.6.2, Definition 9.6.3 and Lemma 9.6.4

In place of "measure zero." The text's proofs of Lemmas 9.6.1, 9.6.2 and 9.6.4 remove sets of smaller dimension or not containing a sheet near the point, "of measure zero", and move a point slightly. The formalization uses a topological covering lemma instead, with no Lebesgue measure and no Baire category. *Step 1.* If $A \cap W \subseteq N$ for an open $W \ni a$, then $\dim_a A \leq \dim_a N$, by the inclusion of the chart tangent spaces Chapter9/Appendix · MPolyfold.HasDimAt.le_of_inter_subset. *Step 2*, the covering lemma Chapter9/Appendix · MPolyfold.exists_mem_forall_notMem: a nonempty open piece of a k-dimensional submanifold is not covered by finitely many submanifolds of smaller dimension closed in an open set, by induction on the family: removing one closed set of smaller dimension leaves a nonempty open piece, by Step 1. *Step 3*, "moving z slightly": a point of the piece off all the given sets Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsGeneric Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.exists_isGeneric.

Lemma 9.6.1 (V9.58) Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_1. *Two structures at x have the same top dimension* Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.topDim. If the first has top dimension k and the second $\ell < k$, a generic point of a k-dimensional sheet of the first has $\Theta > 0$ by the first structure and $\Theta = 0$ by the second. So $\dim_\Theta(x)$ is well defined Chapter9/Appendix · EpGroupoid.localTopDim Chapter9/Appendix · EpGroupoid.localTopDim_eq, $-\infty$ off the support Chapter9/Appendix · EpGroupoid.localTopDim_eq_bot.

Lemma 9.6.2 (V9.59) Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_2. *The unions M_U^k of the top-dimensional sheets agree on $U \cap U'$* Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.topUnion Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.mem_topUnion_iff_of_mem: a generic point of $M_U^k \setminus M_{U'}^k$ would have $\Theta > 0$ by the first structure and, off the lower-dimensional sheets of the second, $\Theta = 0$.

Definition 9.6.3 (V9.60) Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsTopGoodPoint Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.IsTopBadPoint, as in the text. At points top-good for both structures the top parts agree, with no hypothesis Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_isTopGoodPoint: near such a point the top part is $\sigma \cdot 1_M$ for both, and σ is read off Θ at a generic point of M.

Lemma 9.6.4 (V9.61). *Top-goodness does not depend on the structure.* **The text's proof**: near a top-good z, $\Theta = \sigma \cdot \delta_M + \Theta_0 + \cdots + \Theta_{k-1}$; a point of M near z off the lower-dimensional sets has $\Theta = \sigma$, and "similarly as good points", z is top-good for the second structure. **As we read it, this needs that the top weight of the second structure at y is σ**, an argument the text does not give: the second structure could a priori have, through y, several top-dimensional sheets with tangent spaces that differ. **The formalized proof** makes this step the hypothesis `TopMultiplicity` Chapter9/Appendix · MPolyfold.TopMultiplicity: at a point where Θ equals σ on generic points of a k-dimensional sheet, the top weight of any structure is σ. Under it the lemma holds Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.lemma_9_6_4 Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.isTopBadPoint_iff, the top parts agree on $U \cap U'$ Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_topMultiplicity, and Proposition 9.1.12 follows by the descending induction of p. 454: subtract the top parts Chapter9/Appendix · EpGroupoid.LocalBranchingStructure.subOn and repeat Chapter9/Appendix · EpGroupoid.

LocalBranchingStructure.dimPartsAgreeNear_of_topMultiplicity Chapter9/Appendix · EpGroupoid.proposition_9_1_12_of_topMultiplicity. Witnesses: $k = 0$ on every M-polyfold Chapter9/Appendix · MPolyfold.topMultiplicity_zero, and $k \geq n$ on $\mathbb{R}^n$, where an n-dimensional submanifold is a neighbourhood of each of its points Chapter9/Appendix · rnPolyfold_mem_nhds_of_hasDimAt Chapter9/Appendix · topMultiplicity_rn_of_le; so Proposition 9.1.12 holds on $\mathbb{R}^0$ and $\mathbb{R}^1$ unconditionally Chapter9/Appendix · proposition_9_1_12_rn_zero Chapter9/Appendix · proposition_9_1_12_rn_one. The step itself is proved from the graph representation in §IX.22 Chapter9/AppendixSlices · MPolyfold.topMultiplicityAt_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.lemma_9_6_4_of_hasGraphChartAt. In the proof the top-good point is called z and then y.

IX.22 Proposition 9.6.5, the formula on p. 457, and the proof of Proposition 9.1.12

Proposition 9.6.5 (V9.62) Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.proposition_9_6_5. *At a top-bad x, a class Ξ of top-dimensional indices with one tangent space splits as $I^1 \sqcup \cdots \sqcup I^p$ with sequences of top-good points $x_j^e \to x$ lying on M_i exactly for $i \in I^e$.* In (1) the text writes "$I_k = I^1 \sqcup \cdots \sqcup I^p$"; the I^e being subsets of Ξ, it reads as $\Xi = I^1 \sqcup \cdots \sqcup I^p$. **The slice form.** Under `HasGraphChartAt` at x, for every class and every neighbourhood W of x there is a finite top slice in W Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.IsTopSlice: it meets each sheet of the class once, misses the other top-dimensional sheets and all lower-dimensional ones, using a generic slice (§IX.9). Top slices in every W replace the top-good sequences.

The formula on p. 457 Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_sum_slices. $\Theta_k(x) = \sum_\Xi \sum_{y \in \text{slice}} \Theta(y)$; for one class the slice gives $\sum_{i \in \Xi} \sigma_i$ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.sum_class_eq_sum_of_isTopSlice. The text's $\Theta_k(x_e^k)$ uses points x_e^k not introduced, presumably points x_j^e of the sequences. **The independence of the right-hand side**, which the text asserts while Ξ, I^e and x^e come from the first structure, is an argument the text does not give. It is proved by choosing one slice for the sheets of both structures, off all lower-dimensional sets of both, so that both class weights are the same sum of values of Θ Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.sum_class_eq_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPart_topDim_eq_of_hasGraphChartAt. The same grouping proves `TopMultiplicityAt` Chapter9/AppendixSlices · MPolyfold.TopMultiplicityAt Chapter9/AppendixSlices · MPolyfold.topMultiplicity_of_hasGraphChartAt, on $\mathbb{R}^m$ in all dimensions Chapter9/AppendixSlices · MPolyfold.topMultiplicity_rn. A per-dimension graph representation is not enough here: the lower-dimensional sheets through y must be avoided in the chart in which the top sheets are graphs, which is why one chart for all dimensions is asked.

The proof of Proposition 9.1.12. With the top parts agreeing near x, subtract them from both structures and repeat in lower dimensions Chapter9/AppendixSlices · EpGroupoid.LocalBranchingStructure.dimPartsAgreeNear_of_hasGraphChartAt Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_of_hasGraphChartAt; on $\mathbb{R}^m$ without hypothesis Chapter9/AppendixSlices · EpGroupoid.proposition_9_1_12_rn. "This completes the proof of Proposition 9.6.5" (p. 457) refers to Proposition 9.1.12, Proposition 9.6.5 being stated without proof.

IX.23 Witnesses, counterexamples, and the hypotheses on $\mathbb{R}^m$

Hypotheses taken on a general M-polyfold, with their witnesses. The independence of Theorem 9.2.1, `HasIndependentTangentSums`: every Θ over $\mathbb{R}^m$ Chapter9/AppendixSlices · EpGroupoid.hasIndependentTangentSums_rn, the indicator of a point Chapter9/Tangent · hasIndependentTangentSums_indicator. Tangent slices `HasTangentSlicesAt`: $\mathbb{R}^m$ Chapter9/TangentSlicesRn · MPolyfold.hasTangentSlicesAt_rn and $\mathbb{R}^0$ Chapter9/Tangent · hasTangentSlicesAt_rn_zero. Graph charts `IsGraphChartAt`, `HasGraphChart–`

sAt, HasGraphChartAt, HasGraphBallAt: every point of $\mathbb{R}^m$ (§IX.9). The local constancy of the dimension: $\mathbb{R}^m$ Chapter9/AppendixSlices · MPolyfold.hasLocallyConstantDimAt_rn. Invariance of domain: $\mathbb{R}^m$ Chapter9/TangentSlices · MPolyfold.hasDomainInvarianceAt_rn. TopMultiplicity: $k = 0$, $k \geq n$ on $\mathbb{R}^n$, and all k on $\mathbb{R}^m$ in the pointwise form (§§IX.21–IX.22). HasOrientedGoodPointsAt: $\mathbb{R}^m$ Chapter9/OrientationRn · hasOrientedGoodPointsAt_rn. HasContinuousLevelOneOn: $\mathbb{R}^m$ Chapter9/OrientationRn · hasContinuousLevelOneOn_rn. TangentBranchingHyp: the zero functor (§IX.11). SheetGeometry: graphs in $\mathbb{R}^2$ Chapter9/BranchingGeometry · graphSheetGeometry. Property (T): Example 9.3.5 Chapter9/BranchingGeometry · propertyT_example935B1. Remark913Hyp: expNegInvGlue Chapter9/Propositions · EpGroupoid.remark913Hyp_expNegInvGlue. SheetNullSets and SheetMeasureGeometry: graphs and open subsets of $\mathbb{R}^2$ Chapter9/Measure · graphSheetMeasureGeometry Chapter9/Measure · planeNull; HasTameSheetsIn: every structure on $\mathbb{R}^n$ Chapter9/Measure · hasTameSheetsIn_trivial_rn (§IX.19).

Unconditional on $\mathbb{R}^m$. For every ep-groupoid whose object M-polyfold is $\mathbb{R}^m$ (any morphisms): Proposition 9.1.12, the independence and hence Theorem 9.2.1, Lemma 9.2.7 (2), Lemma 9.3.4 for n-dimensional Θ, Lemma 9.6.4 and Proposition 9.6.5. What remains on $\mathbb{R}^m$ is the smoothness of the TM_i in Definition 9.2.12 and the paracompactness of $|X|$.

Counterexamples (each to a statement as we read it): the uniqueness in Proposition 9.1.11 Chapter9/Propositions · EpGroupoid.not_proposition_9_1_11_rn; Lemma 9.2.13 (2) $\Rightarrow$ (1) off ∂X Chapter9/BoundaryPosition · MPolyfold.not_lemma_9_2_13_two_one; the uniqueness in Definition 9.2.16 Chapter9/Boundary · not_unique_isBoundaryCharacterized_halfLine; Lemma 9.3.4 without equidimensionality Chapter9/OrientationRn · lemma_9_3_4_mixed_dim_rn; "otherwise different" and "four possibilities" in Example 9.3.5 Chapter9/OrientationExample · example_9_3_5_eq_of_neg Chapter9/OrientationExample · exampleT2_ne_orientedLift1; Lemma 9.4.4 for sheets of different dimensions Chapter9/BranchingGeometry · lemma_9_4_4_mixed_dim; the transfer of null sets in Lemma 9.5.2 across dimensions Chapter9/Measure · isNull_transfer_mixed_dim.

IX.24 What is not formalized

- Remark 9.1.1 on a general M-polyfold, the footnote to Definition 9.1.2 and "locally equidimensional" (G.93); Remark 9.1.4 (G.94); Proposition 9.1.9 and Remark 9.1.10 (G.95); the global part of Proposition 9.1.11 and a repaired uniqueness (G.96).
- The graph representation of Lemmas 9.2.9–9.2.11 on a general M-polyfold, and Proposition 9.2.8 as printed (G.97); invariance of domain for Lemma 9.2.7 (2) (G.98); the TM_i as smooth submanifolds of TX (G.99).
- The level-1 continuity of frames for Lemma 9.3.4 (G.100); a classification of the orientations in Example 9.3.5 (G.101); the corner structure of a tame M_i (G.102); $T_y\partial M = T_y^R M$, the outward direction, the continuity of $o_{\partial M}$ and Figure 9.1 (G.103).
- Lemma 9.4.3 (G.104); Lemmas 9.4.4 and 9.4.6 for M-polyfold sheets (G.105).
- §9.5: Lebesgue null sets in the sheets of a general M-polyfold, and Lemma 9.5.2 for structures with sheets of several dimensions (G.106); branched integration, Theorems 9.5.4 and 9.5.5, Definition 9.5.6, (9.21)–(9.22), and the measure identities of Lemmas 9.5.7, 9.5.9–9.5.11, Remark 9.5.8 and Proposition 9.5.13 (G.107); $(n-1)$-dimensional measure zero face by face and a witness with $\partial S \neq \emptyset$ (G.108); the canonical boundary measures, Theorems 9.5.15 and 9.5.16 (G.109); Stokes' theorem, Theorems 9.5.17 and 9.5.18, and Remark 9.5.20 (G.110). They need the sc-differential forms of §8.2 and Part I (G.74, G.80). Remarks 9.5.12 and 9.5.19 discuss open questions. §§9.6.2–9.6.3 are open questions.

IX.25 What the formalization taught us

The definition of a branched ep$^+$-subgroupoid is local and says nothing about how the local pictures fit together, and three statements that read as uniqueness claims, Proposition 9.1.11, Definition 9.2.16 and Lemma 9.3.4, turned out to depend on exactly that: an indicator can be split into a point and a punctured

line, a characterization asked only on its own support is met by zero, and a lower-dimensional sheet through a bad point escapes the good points. In each case the repair is a condition that ties the object to the local branching structures of Θ (decompositions by subfamilies, (2) asked on $\operatorname{supp}(\Theta)$, equidimensionality). The analytic heart of §9.2, the graph representation of tangent sheets near a bad point, is a statement about one chart in which sheets of all dimensions are graphs; separated from the counting it serves, it becomes a single hypothesis, true on $\mathbb{R}^m$ by the inverse function theorem, from which Proposition 9.1.12, Theorem 9.2.1, Lemma 9.6.4 and Proposition 9.6.5 follow. Two steps the text leaves to "similarly" (Lemma 9.6.4) or asserts (the independence on p. 457) needed that both structures be compared on one common slice. Finally, "measure zero" in §9.6.1 is not needed: the arguments only use that a piece of a k-dimensional submanifold is not covered by finitely many thinner ones, which is topology. In §9.5 the opposite happens: the measure-theoretic part needs only that the null sets of the sheets form a local σ-ideal compatible on overlaps, and that compatibility is exactly where dimensions must agree; and of the isotropy computations that normalize the integrals, everything except the integrals themselves is finite group theory.

Part X. Equivalences and localization (HWZ Chapter 10, §§10.1–10.5 and Appendix §§10.6.1–10.6.2 (2017 §§10.1–10.5))

Chapter 10 continues Part II; its numbers and pages are those of the book (Springer 2021, pp. 461–520), with the 2017 text's in parentheses (2017 pp. 333–374). It studies the sc-smooth functors between ep-groupoids that should become invertible once ep-groupoids are taken as models of polyfolds. §10.1 defines equivalences (Definition 10.1.1 (2017: 10.1)), characterizes them as the equivalences of categories that are local sc-diffeomorphisms on objects (Lemma 10.1.2 (2017: 10.1)), builds the refinement along an open covering (Proposition 10.1.3 (2017: 10.1)), shows that natural equivalence, the two-out-of-three property and the tangent functor respect equivalences (Definition 10.1.4, Propositions 10.1.5, 10.1.6, Theorem 10.1.7 (2017: 10.2, 10.2, 10.3, 10.1)) and, mostly new in 2021, that the properties of Chapter 7 are invariant (Remarks 10.1.8, 10.1.11, 10.1.12, Propositions 10.1.9, 10.1.10, 10.1.13, 10.1.14, 10.1.16–10.1.19, Example 10.1.15). §10.2 constructs the weak fibred product (Definition 10.2.1 (2017: 10.3), Theorem 10.2.2 (2017: 10.2), Remark 10.2.3 (2017: 10.2)). §10.4 carries all this to strong bundles: strong bundle equivalences (Definition 10.4.1 (2017: 10.10)), the pull-back and push-forward of section functors (Propositions 10.4.2–10.4.4, Theorem 10.4.5 (2017: 10.7–10.9, 10.8)), natural transformations of strong bundle functors (Definitions 10.4.8, 10.4.10, Remark 10.4.9 (2017: 10.11, 10.12, 10.4)) and the category $\mathcal{SEP}$ (Definition 10.4.11 (2017: 10.13)). §10.3, treated here after §10.4 in the order of the verified edition, inverts the equivalences: diagrams $X \overset{F}{\leftarrow} A \overset{\Phi}{\rightarrow} Y$ with F an equivalence, their refinement and equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4, 10.3.6, Definition 10.3.7 (2017: 10.4–10.7, 10.3–10.5)), the composition of classes (Theorem 10.3.8 (2017: 10.3), Appendix §10.6.1), generalized isomorphisms (Definition 10.3.9, Theorem 10.3.10 (2017: 10.8, 10.4), Appendix §10.6.2), the category $\mathcal{EP}(E^{-1})$ with the functors i and $|\cdot|$ (Proposition 10.3.11, Definition 10.3.12, Theorems 10.3.14, 10.3.15 (2017: 10.6, 10.9, 10.5, 10.6)) and the invariance of the properties of Chapter 7 under generalized isomorphisms (Propositions 10.3.16, 10.3.17, Remark 10.3.18, Theorem 10.3.19, Definition 10.3.20, Theorem 10.3.21). §10.5, treated after §10.3, repeats the localization for strong bundles: diagrams $W \overset{\Phi}{\leftarrow} W'' \overset{\Psi}{\rightarrow} W'$ with Φ a strong bundle equivalence, their classes, the maps $[d]$, $|[D]|$ and the homomorphisms γ (pp. 504–506), the category $\mathcal{SEP}(F^{-1})$ and generalized strong bundle isomorphisms (Definitions 10.5.1, 10.5.2 (2017: 10.14, 10.15), Remark 10.5.3), the push-forward of section functors (Theorem 10.5.4 (2017: 10.9), Remark 10.5.5) and the local regularity property (Proposition 10.5.6, Theorem 10.5.7 (2017: 10.10)).

What is proved, and what is not. All definitions of §§10.1, 10.2 and 10.4 are formalized. Lemma 10.1.2 (with its converse), Propositions 10.1.3, 10.1.5, 10.1.6, 10.1.13, 10.1.14, 10.1.16, 10.1.17, 10.1.18, Theorem 10.1.7, Remarks 10.1.11 and 10.1.12, Example 10.1.15, all claims of Theorem 10.2.2 and Remark 10.2.3, Propositions 10.4.2–10.4.4, Theorem 10.4.5 without its Fred part, Remark 10.4.9 and the equivalence of Definitions 10.4.8 and 10.4.10 are proved, several in a stronger form; Proposition 10.1.19 is reduced to

Proposition 10.1.9 for the lifts. Two results the text quotes or leaves to the reader are proved: Proposition 10.1.5 and the fullness and faithfulness of TF in Theorem 10.1.7; the sc-smoothness of the push-forward, "easily verified" in the text, is proved too. Not formalized: the subcategory $\mathcal{EP}_{lreg}$ (G.111), Remark 10.1.8 with Appendix §10.6.3 (G.112), Propositions 10.1.9 and 10.1.10 (G.113, G.114), Proposition 10.1.19 unconditionally (G.115), the comparison of the M-polyfold structures of Theorem 10.2.2 with structures induced from triple products (G.116), the Fred part of Theorem 10.4.5 (G.121) and Propositions 10.4.6, 10.4.7 (G.122). In §10.3 all definitions are formalized, and (E1)–(E3), Lemmas 10.3.2, 10.3.4, 10.3.6, Theorems 10.3.8, 10.3.10, 10.3.14, 10.3.15, Propositions 10.3.11, 10.3.17, Remark 10.3.18 (its non-uniqueness claim through the example of Remark 10.5.5) and Theorem 10.3.21 for (LUC), (SLUC), (ZHOU) are proved, with the proofs of the appendix; not formalized are the universal property of the localization (G.117), Proposition 10.3.16 with the (LR), (SLR) parts of Theorem 10.3.21 (G.118) and Theorem 10.3.19 (G.120); G.119, the non-uniqueness in Remark 10.3.18, is closed by §X.23. In §10.5 the diagrams, their refinements, $[d]$, $|[D]|$, the square $|\mathcal{D}|$, γ, Definition 10.5.2 and Remark 10.5.5 (with an example) are formalized, Definition 10.5.1 as a quiver, Remark 10.5.3 for the base and total spaces, Theorem 10.5.4 without its Fred part, and Proposition 10.5.6 as a reduction; not formalized are the transitivity of the equivalence of diagrams with the composition in $\mathcal{SEP}(F^{-1})$ and the strong bundle Theorem 10.3.10 (G.123), (10.24) made precise with q, Q (G.124), the Fred part of Theorem 10.5.4 (G.125), Proposition 10.5.6 itself (G.126) and Theorem 10.5.7 (G.127).

Six structural choices. (1) *Equivalences as a structure.* `EpFunctor.IsEquivalence` has the four conditions of Definition 10.1.1 as fields, with full and faithful stated on the sets of morphisms; Mathlib's `Functor.IsEquivalence` of the underlying functor is derived (§X.1). (2) *Indexed coverings.* The refinement $X_{\mathcal{U}}$ takes a family $(U_i)_{i\in\iota}$ of open sets; objects and morphisms are open subsets of products with the discrete ι (§X.2). (3) *Iterated fibred products.* The weak fibred product is built from Part I's fibred products, twice, so that its M-polyfold structures come with their universal properties; the admissibility, F or G a local sc-diffeomorphism on objects, is all that the construction uses (§X.9). (4) *Total pull-backs and push-forwards.* Φ^* and Φ_* are defined on all maps, by the zero of the fibre where needed, and the push-forward with one choice of x and σ made for all; all statements are about section functors, where the choice does not matter (§X.12). (5) *Diagrams one universe up.* A diagram contains its middle ep-groupoid as an object of $\mathcal{EP}$, so the classes of diagrams form a type in the universe of the objects of $\mathcal{EP}$ and $\mathcal{EP}(E^{-1})$ is a large category like $\mathcal{EP}$; compositions are written in Mathlib's order, d.comp e for the text's $[e]\circ[d]$ (§X.16). (6) *Strong bundle classes for the generated relation.* Without the weak fibred product of strong bundles, having a common refinement is not known to be transitive; the generalized strong bundle maps are the classes of the equivalence relation it generates, every invariant is proved invariant under it, and $\mathcal{SEP}(F^{-1})$ is a quiver (§X.21).

Map.

- `Chapter10/Equivalence`: local sc-diffeomorphisms at a point, Definition 10.1.1, Lemma 10.1.2, Definition 10.1.4, Propositions 10.1.5, 10.1.6, Theorem 10.1.7, the pair groupoid of $\mathbb{Z}$ over the point.
- `Chapter10/EquivalenceRefinement`: open coverings and Proposition 10.1.3, the covering of $\mathbb{Z}/2\ltimes\mathbb{R}$.
- `Chapter10/EquivalenceReduced`: Remark 10.1.12, Propositions 10.1.13, 10.1.14, 10.1.16, 10.1.18.
- `Chapter10/EquivalenceExamples`: Remark 10.1.11, Example 10.1.15.
- `Chapter10/EquivalenceZhou`: Proposition 10.1.17 and the reduction of Proposition 10.1.19.
- `Chapter10/WeakFibredProduct`: Definition 10.2.1 and the first claim of Theorem 10.2.2.
- `Chapter10/WeakFibredProductTheorem`: the projections, Theorem 10.2.2, Remark 10.2.3, witnesses.
- `Chapter10/StrongBundleEquivalence`: Definition 10.4.1, the pull-back and the push-forward, Propositions 10.4.2–10.4.4, Theorem 10.4.5.
- `Chapter10/StrongBundleEquivalenceNat`: Definitions 10.4.8, 10.4.10, 10.4.11, Remark 10.4.9.
- `Chapter10/StrongBundleEquivalenceExamples`: the strong bundle equivalence over $-\mathrm{id}$ and its uses.
- `Chapter10/Localization`: diagrams, Definitions 10.3.1, 10.3.3, 10.3.5, 10.3.7, Lemmas 10.3.2, 10.3.4,

10.3.6, Theorem 10.3.8 with Appendix §10.6.1, the diagonal of Appendix §10.6.2.
- `Chapter10/LocalizationCategory`: (E1), Definitions 10.3.9, 10.3.12, Theorem 10.3.10 with Appendix §10.6.2, Proposition 10.3.11, Theorems 10.3.14, 10.3.15, witnesses.
- `Chapter10/LocalizationProperties`: Proposition 10.3.17, Remark 10.3.18, Definition 10.3.20, Theorem 10.3.21.
- `Chapter10/StrongBundleLocalization`: diagrams of strong bundles, their refinements and classes, $[d]$, $|[D]|$, the square $|\mathcal{D}|$, γ, Definitions 10.5.1, 10.5.2, Remark 10.5.3.
- `Chapter10/StrongBundleLocalizationSections`: the push-forward and pull-back, Theorem 10.5.4, the reduction of Proposition 10.5.6.
- `Chapter10/StrongBundleLocalizationExamples`: the relation of the homomorphisms γ, the witnesses over $-\mathrm{id}$, Remark 10.5.5 over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$.

Throughout, $X = (X, \mathbf{X})$, Y, Z are ep-groupoids as in Part VII, s, t the source and target, $|X|$ the orbit space and $|F| : |X| \to |Y|$ the induced map, G_x the isotropy group, T_x its subgroup acting trivially near x and $\mathbf{X}^T$ the non-effective part (§VII.3). Functors are sc-smooth functors (`EpFunctor`), and natural transformations those of Chapter 8 (`EpNatTrans`).

X.1 Definition 10.1.1 (2017: 10.1) and Lemma 10.1.2 (2017: 10.1)

Definition 10.1.1 (2017: 10.1) (V10.1) `Chapter10/Equivalence · EpFunctor.IsEquivalence` `Chapter10/Equivalence · EP.equivalences`. The four fields are fullness and faithfulness on the sets of morphisms, $|F|$ a homeomorphism, and F a local sc-diffeomorphism on objects. Fullness and faithfulness are those of Mathlib for the underlying functor `Chapter10/Equivalence · GroupoidFunctor.full_iff` `Chapter10/Equivalence · GroupoidFunctor.faithful_iff`, and so is essential surjectivity `Chapter10/Equivalence · GroupoidFunctor.essSurj_iff`. *Step 1, essential surjectivity.* For $y \in Y$, $|F|$ is onto, so $[y] = [F(x)]$ for some x, that is a morphism $F(x) \to y$ exists `Chapter10/Equivalence · EpFunctor.IsEquivalence.essSurj` `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso`. *Step 2, the morphism sets.* Every $\psi : F(x) \to F(x')$ is $F(\varphi)$ `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_mor_eq` and F is injective on morphisms with the same ends `Chapter10/Equivalence · EpFunctor.IsEquivalence.mor_injective`. *Step 3, the class E.* The identity is an equivalence `Chapter10/Equivalence · EpFunctor.IsEquivalence.id`, composites are `Chapter10/Equivalence · EpFunctor.IsEquivalence.comp` (the composite of homeomorphisms and of local sc-diffeomorphisms, with $|G \circ F| = |G| \circ |F|$ `Chapter10/Equivalence · GroupoidFunctor.toFunctor_comp`), so `EP.equivalences` is a multiplicative class. Witness: the functor from the pair groupoid of $\mathbb{Z}$ (one morphism between any two integers) to the point `Chapter10/Equivalence · EpGroupoid.Examples.pairToPoint` `Chapter10/Equivalence · EpGroupoid.Examples.isEquivalence_pairToPoint`, a local sc-diffeomorphism because $\mathbb{Z}$ is discrete `Chapter10/Equivalence · EpGroupoid.Examples.isLocalScDiffeo_discrete`.

Local sc-diffeomorphisms at a point. The local property of Definition 10.1.1 (3) is localized `Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt` `Chapter10/Equivalence · MPolyfold.isLocalScDiffeo_iff_forall` `Chapter10/Equivalence · MPolyfold.IsLocalScDiffeo.at`, with the rules used below: composition `Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.comp`, agreement near the point `Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.congr`, and the two cancellation rules: if $g \circ f$ and g are local sc-diffeomorphisms at x and $f(x)$, so is f at x `Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.of_comp_left`; if $g \circ f$ and f are, so is g at $f(x)$ `Chapter10/Equivalence · MPolyfold.IsLocalScDiffeoAt.of_comp_right`.

Lemma 10.1.2 (2017: 10.1) (V10.2) `Chapter10/Equivalence · EpFunctor.lemma_10_1_2` `Chapter10/Equivalence · EpFunctor.isEquivalence_iff`. *Step 1, $|F|$ is bijective.* For a functor of groupoids, $|F|$ is injective when F is full ($F(x) \cong F(x')$ lifts to $x \cong x'$) `Chapter10/Equivalence · GroupoidFunctor.injective_orbitMap_of_full`, surjective exactly when F is essentially surjective `Chapter10/Equivalence · GroupoidFunctor.surjective_orbitMap_iff`, hence bijective for an equivalence of categories `Chapter10/Equivalence · GroupoidFunctor.bijective_orbitMap_of_isEquivalence`. *Step 2, $|F|$ is a homeomorphism.* $|F|$ is continuous by the universal property of the quotient `Chapter10/Equivalence · EpFunctor.continuous_orbitMap`; it is open because F is a local sc-diffeomorphism, hence open, on objects and the orbit maps are open (Proposition 7.1.8 (2017:

7.1)) Chapter7/Basic · EpGroupoid.proposition_7_1 Chapter10/Equivalence · EpFunctor.isOpenMap_orbitMap. *Step 3, the converse.* An equivalence is full, faithful and essentially surjective (§X.1), so an equivalence of categories Chapter10/Equivalence · EpFunctor.IsEquivalence.isEquivalence_toFunctor. *Not invertible.* The functor $\mathbb{Z} \to \mathrm{pt}$ above has no left inverse on objects Chapter10/Equivalence · EpGroupoid.Examples.not_exists_leftInverse_pairToPoint.

X.2 Proposition 10.1.3 (2017: 10.1): the refinement $X_{\mathcal{U}}$

The construction Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.objSet Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.morSet Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.groupoidStr. A covering is a family of open sets U_i with every object in some U_i; the trivial covering by X is one Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.univ. The text indexes the covering by itself and writes (U, x); the index type is the formal reading. The objects (i, x), $x \in U_i$, form an open subset of $\iota \times X$ Chapter10/EquivalenceRefinement · isOpen_of_discrete_left, the morphisms (i, φ, j) one of $(\iota \times \iota) \times \mathbf{X}$, and both get the open-subset M-polyfold structures Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.objPolyfold Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.morPolyfold, which need a nonempty set: the object set of X is assumed nonempty (for an empty X there is nothing to prove). Products with a discrete space keep the paracompactness and metrizability Chapter10/EquivalenceRefinement · paracompactSpace_discrete_prod.

$X_{\mathcal{U}}$ is an ep-groupoid Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.epGroupoid. *Step 1, local sc-diffeomorphisms.* On a product with a discrete factor a map that is a local sc-diffeomorphism on each sheet is one Chapter10/EquivalenceRefinement · isLocalScDiffeo_sheet Chapter10/EquivalenceRefinement · isLocalScDiffeo_snd_discrete; $s(i, \varphi, j) = (i, s(\varphi))$ and $t(i, \varphi, j) = (j, t(\varphi))$ are local sc-diffeomorphisms Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isLocalScDiffeo_src Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isLocalScDiffeo_tgt, through the restriction rules for open subsets Chapter10/EquivalenceRefinement · MPolyfold.isScSmoothOn_restrictOpen_iff Chapter10/EquivalenceRefinement · MPolyfold.isScSmoothMap_restrict Chapter10/EquivalenceRefinement · MPolyfold.isLocalScDiffeoAt_restrict. *Step 2, unit, inverse, multiplication.* They are those of X on each sheet, sc-smooth by the continuity on the discrete factor Chapter10/EquivalenceRefinement · isScSmoothOn_discrete_of_continuousOn Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothMap_unit Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothMap_inv Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.isScSmoothCompSheet. *Step 3, properness.* At (i, x) take a neighbourhood V_0 of x with the properness property of X and an open $V(x)$ with $\mathrm{cl}\, V(x) \subseteq U_i \cap V_0$; then t is proper on $s^{-1}(\mathrm{cl}(\{i\} \times V(x)))$ Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.exists_isProperMap. The proof uses ultrafilters in place of the text's sequences; the index j of a limit is determined by the sheet, which is clopen.

Proposition 10.1.3 (2017: 10.1) (V10.3) Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.proposition_10_1_3. $F(i, x) = x$, $F(i, \varphi, j) = \varphi$ Chapter10/EquivalenceRefinement · EpGroupoid.OpenCover.functor is a local sc-diffeomorphism on objects, faithful because a morphism of $X_{\mathcal{U}}$ is determined by its ends and its φ, full because $\psi : x \to x'$ lifts to (i, ψ, j), and essentially surjective because every x lies in some U_i; Lemma 10.1.2 applies. The text writes the morphism as (U, φ, v) in the formula for F; we read (U, φ, V) (possible misprint). *Witness.* The covering $\{x > -1\}$, $\{x < 1\}$ of $\mathbb{Z}/2 \ltimes \mathbb{R}$ Chapter10/EquivalenceRefinement · EpGroupoid.Examples.reflectionCover gives an equivalence Chapter10/EquivalenceRefinement · EpGroupoid.Examples.isEquivalence_reflectionCover that is not injective on objects Chapter10/EquivalenceRefinement · EpGroupoid.Examples.not_injective_reflectionCover_obj.

X.3 Definition 10.1.4 (2017: 10.2), Propositions 10.1.5 and 10.1.6 (2017: 10.2 and 10.3)

Definition 10.1.4 (2017: 10.2) (V10.4) Chapter10/Equivalence · definition_10_1_4. The definition is Chapter 8's Definition 8.1.13 Chapter8/Basic · EpNatTrans Chapter8/Basic · EpFunctor.NaturallyEquivalent; the claims

after it, that $\simeq$ is an equivalence relation and compatible with composition, are Chapter 8's `Chapter8/NatTrans · EpFunctor.naturallyEquivalent_equivalence` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_left` `Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right`. A natural transformation of ep-groupoids is a natural isomorphism of the underlying functors, every morphism being invertible `Chapter10/Equivalence · EpNatTrans.toNatIso`.

Proposition 10.1.5 (2017: 10.2) (V10.5) `Chapter10/Equivalence · EpFunctor.proposition_10_1_5` `Chapter10/Equivalence · EpFunctor.IsEquivalence.of_natTrans`. The text quotes it from the literature. *Step 1.* $\tau : X \to \mathbf{X}'$ is continuous `Chapter10/Equivalence · EpNatTrans.continuous_app` and, since $s \circ \tau = F$ with F and s local sc-diffeomorphisms, a local sc-diffeomorphism `Chapter10/Equivalence · EpNatTrans.isLocalScDiffeo_app`. *Step 2.* $G = t \circ \tau$ is a composite of local sc-diffeomorphisms `Chapter10/Equivalence · EpNatTrans.isLocalScDiffeo_obj`. *Step 3.* The underlying functors are naturally isomorphic, so G is an equivalence of categories with F, and Lemma 10.1.2 applies. The symmetric statement follows from the symmetry of $\simeq$.

Proposition 10.1.6 (2017: 10.3) (V10.6) `Chapter10/Equivalence · EpFunctor.proposition_10_1_6`. *Step 1, categories.* Φ is an equivalence of categories since F and $\Phi \circ F$ are (Mathlib's two-out-of-three). *Step 2, objects in the image.* At $F(x)$, $\Phi \circ F$ and F are local sc-diffeomorphisms at x, so Φ is one at $F(x)$ (the right cancellation rule). *Step 3, transport.* The text's argument stops here; an arbitrary y receives a morphism $\sigma : F(x) \to y$ `Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso`, and the local sc-diffeomorphism property of a functor passes from $s(\sigma)$ to $t(\sigma)$ `Chapter10/Equivalence · EpFunctor.isLocalScDiffeoAt_obj_tgt`: near σ, $t \circ \tau = \Phi \circ t \circ \dots$ is written with a local inverse τ of s, and on morphisms Φ is a local sc-diffeomorphism where it is one on objects `Chapter10/Equivalence · EpFunctor.isLocalScDiffeoAt_mor` `Chapter10/Equivalence · EpFunctor.isLocalScDiffeo_mor`. *Step 4.* Lemma 10.1.2. The text's $|\Phi| : |X| \to |Y|$ in the proof is read $|\Phi| : |Y| \to |Z|$ (possible misprint).

X.4 Theorem 10.1.7 (2017: 10.1): the tangent of an equivalence

Theorem 10.1.7 (2017: 10.1) (V10.7) `Chapter10/Equivalence · EpFunctor.theorem_10_1_7`. TF is Chapter 8's tangent functor `Chapter8/TangentEp · EpFunctor.tangent`; the text proves the homeomorphism and the local sc-diffeomorphism property and leaves full and faithful to the reader. Here all goes through Lemma 10.1.2. *Step 1, local sc-diffeomorphism.* The tangent of a local sc-diffeomorphism is one (Part I). *Step 2, faithful.* A tangent morphism (φ, v) is determined by φ and its source vector `Chapter8/TangentEp · EpGroupoid.tangent_ext`, and F is faithful. *Step 3, full.* For $(\psi, w) : TF(x, a) \to TF(x', a')$, $\psi = F(\varphi)$ by fullness of F; φ lifts to a tangent morphism with source (x, a) `Chapter8/TangentEp · EpGroupoid.exists_tSrc_eq'`, whose image under TF has the same source as (ψ, w), and the targets agree since TF is injective on the tangent fibre at x' `Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.injOn_tangentFiber`. *Step 4, essentially surjective.* For (y, b) take $\psi : F(x) \to y$, lift ψ^{-1} with source (y, b) and use that TF maps T_xX onto $T_{F(x)}Y$ `Chapter8/TangentLocal · MPolyfold.IsLocalScDiffeo.tangentFiber_subset_image`, then invert `Chapter8/Basic · EpGroupoid.tInv`.

X.5 Remarks 10.1.8 and 10.1.11 (2017: 10.1), Propositions 10.1.9 and 10.1.10 (2017: 10.4)

Not formalized. The full subcategory $\mathcal{EP}_{lreg}$ of p. 461 (G.111); Remark 10.1.8 (new in the 2021 edition) and the example of Appendix §10.6.3, which needs a weighted Sobolev sc-retract (G.112); Proposition 10.1.9 (new in the 2021 edition), the invariance of (LR) (Definition 7.4.19, Chapter 7's `IsLR` `Chapter7/Zhou · EpGroupoid.IsLR`), whose proof transports a natural representation along F and a morphism $F(x) \to y$, for which the Chapter 7 API has no transport (G.113); Proposition 10.1.10 (2017: 10.4), which assumes (LR) and uses 10.1.9 (G.114).

Remark 10.1.11 (2017: 10.1) (V10.8) `Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11`. The ep-groupoid is $\mathbb{Z}/2$ acting on $\mathbb{R}^2$ by $z \mapsto -z$ `Chapter10/EquivalenceExamples · EpGroupoid.Examples.pointReflectionEpGroupoid`, sc-smooth by `Chapter10/EquivalenceExamples · EpGroupoid.Examples.isScSmoothMap_`

units_smul_rn; $G = -\mathrm{id}$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.negFunctor is an equivalence Chapter10/EquivalenceExamples · EpGroupoid.Examples.isEquivalence_negFunctor with $|\mathrm{id}| = |G|$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.orbitMap_id_eq_negFunctor. *Step 1.* $\alpha_z = (-1, z)$ is an sc-smooth family $F(z) \to G(z)$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11_family: the conclusion of Proposition 10.1.10 (1) holds in the example. *Step 2.* A family with $\alpha_0 = 1_0$ and $\alpha_z : z \to -z$ near 0 must take the value $(-1, z)$ at every $z \neq 0$, because the only morphism $z \to -z$ is the group element -1 Chapter10/EquivalenceExamples · EpGroupoid.Examples.rn_eq_zero_of_eq_neg Chapter10/EquivalenceExamples · EpGroupoid.Examples.not_eventually_eq_neg; continuity at 0 fails. Hence no sc-smooth family exists either Chapter10/EquivalenceExamples · EpGroupoid.Examples.remark_10_1_11_scSmooth.

X.6 Remark 10.1.12, Propositions 10.1.13 and 10.1.14 (2017: 10.5), Example 10.1.15 (2017: 10.1)

Remark 10.1.12 (V10.9) Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.remark_10_1_12. *Step 1.* $g \mapsto F(g)$ is a homomorphism $G_x \to G_{F(x)}$ Chapter10/EquivalenceReduced · EpFunctor.isotropyHom Chapter10/EquivalenceReduced · EpFunctor.isotropyHom_val, bijective for an equivalence (fullness and faithfulness at (x, x)) Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.bijective_isotropyHom. *Step 2.* $g \in T_x$ iff $F(g) \in T_{F(x)}$ Chapter10/EquivalenceReduced · EpFunctor.isotropyHom_mem_trivialSet_iff, since F is a local sc-diffeomorphism on objects and on morphisms, and the local actions correspond. *Step 3.* Conjugation by $\psi : F(x) \to y$ Chapter7/Groupoid · GroupoidStr.isotropyConj maps $T_{F(x)}$ onto T_y Chapter7/Effective · EpGroupoid.isotropyConj_mem_trivialSet_iff. The text's $G_x \to Y_y$ is read $G_x \to G_y$ (possible misprint).

Proposition 10.1.13 (V10.10) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_13. *Step 1,* $\mathbf{X}^T$ *along* F. For a local sc-diffeomorphism on objects, $\varphi \in \mathbf{X}^T$ iff $F(\varphi) \in \mathbf{Y}^T$ Chapter10/EquivalenceReduced · EpFunctor.mor_mem_nonEffectivePart Chapter10/EquivalenceReduced · EpFunctor.mem_nonEffectivePart_of_mor, that is $\mathbf{X}^T = F^{-1}(\mathbf{Y}^T) \cap \{s = t\}$ Chapter10/EquivalenceReduced · EpFunctor.nonEffectivePart_eq. *Step 2, (1).* So F_R is well defined Chapter10/EquivalenceReduced · EpFunctor.reducedMap Chapter10/EquivalenceReduced · EpFunctor.reducedMap_π, continuous Chapter10/EquivalenceReduced · EpFunctor.continuous_reducedMap and open, F being open on morphisms and the quotient maps open Chapter10/EquivalenceReduced · EpFunctor.isOpenMap_reducedMap; only the local sc-diffeomorphism property of F is used. *Step 3, closedness.* For an equivalence, $\mathbf{X}^T$ is closed iff $\mathbf{Y}^T$ is Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.isClosed_nonEffectivePart_iff. From Y to X: $\mathbf{X}^T$ is the preimage of a closed set intersected with a closed set. From X to Y: a $\psi \in \mathrm{cl}(\mathbf{Y}^T)$ at y is conjugated, by a local inverse of t at a morphism $\sigma : F(x) \to y$, into a morphism at $F(x)$ in $\mathrm{cl}(F(\mathbf{X}^T))$, with the continuity of the composition at a point Chapter10/EquivalenceReduced · EpGroupoid.continuousAt_comp; closures replace the text's sequences. *Step 4, (2).* Theorem 7.2.22 says $\mathbf{X}^R$ is Hausdorff iff $\mathbf{X}^T$ is closed Chapter7/ReducedCriterion · EpGroupoid.theorem_7_2_22_2021; with Step 3 this gives (2), the route of the text.

Proposition 10.1.14 (2017: 10.5) (V10.11) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_14. With both reduced morphism spaces Hausdorff the reduced ep-groupoids of Chapter 7 exist Chapter7/Reduced · EpGroupoid.reducedEpGroupoid. F_R is F on objects and F_R of Step 2 on morphisms Chapter10/EquivalenceReduced · EpFunctor.reducedGroupoidFunctor Chapter10/EquivalenceReduced · EpFunctor.reduced; it is a functor since $[\cdot]$ is Chapter10/EquivalenceReduced · EpFunctor.reducedπ_eq_of_mor, sc-smooth since the M-polyfold structure of $\mathbf{X}^R$ is the one for which the quotient map is a local sc-diffeomorphism, full and faithful by Remark 10.1.12, and $|F_R| = |F|$. By Proposition 10.1.13 (2) either Hausdorff hypothesis implies the other; both are kept, as in the text.

Example 10.1.15 (2017: 10.1) (V10.12) Chapter10/EquivalenceExamples · EpGroupoid.Examples.example_10_1_15. $X = \mathbb{Z}/4 \ltimes \mathbb{R}$ with $1, 3$ acting by -1 Chapter10/EquivalenceExamples · EpGroupoid.Examples.z4LineAction Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X, $Y = \mathbb{Z}/4 \ltimes \mathbb{R}^2$ by quarter turns Chapter10/EquivalenceExamples · EpGroupoid.Examples.z4PlaneAction Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y, Φ to the origin Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Functor. *Step 1.* 2 acts trivially on $\mathbb{R}$, not on $\mathbb{R}^2$ Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X_

two_mem_nonEffectivePart Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y_two_not_mem_nonEffectivePart. *Step 2.* Hence $(1,x)$ and $(3,x) = (2,x)\circ(1,x)$ have the same class Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015X_reducedπ_one_eq_three, their images not Chapter10/EquivalenceExamples · EpGroupoid.Examples.example1015Y_reducedπ_one_ne_three, so no map of sets f with $f([\varphi]) = [\Phi(\varphi)]$ exists. The text speaks of an induced map of the reduced ep-groupoids; the statement here needs no topology.

X.7 Propositions 10.1.16–10.1.19: (LUC), (ZHOU), (SLUC), (SLR)

Proposition 10.1.16 (V10.13) Chapter10/EquivalenceReduced · EpFunctor.proposition_10_1_16. (LUC) Chapter7/ReducedCriterion · EpGroupoid.IsLUC is the closedness of $\mathbf{X}^T$ (Proposition 7.2.21, part of Theorem 7.2.22) Chapter7/ReducedCriterion · EpGroupoid.proposition_7_2_21_2021, and the closedness passes along equivalences (§X.6, Step 3). The text cites Theorem 7.2.22 here and in Proposition 10.1.13; the citation is the theorem on p. 323.

Proposition 10.1.17 (V10.14) Chapter10/EquivalenceZhou · EpFunctor.proposition_10_1_17. *Step 1, smooth points.* x is smooth iff $F(x)$ is, and iff a $y \cong F(x)$ is Chapter10/EquivalenceZhou · EpGroupoid.mem_smoothSet_src_iff Chapter10/EquivalenceZhou · EpGroupoid.mem_smoothSet_tgt_iff. *Step 2, Y to X.* For $g \in G_x$ with $Tg = \mathrm{id}$ on T_xX: $T(F(g))\circ TF = TF\circ Tg$ Chapter10/EquivalenceZhou · EpFunctor.tangentMor_mor and TF is onto $T_{F(x)}Y$, so $T(F(g)) = \mathrm{id}$, $F(g) \in T_{F(x)}$ by (ZHOU) for Y, and $g \in T_x$ by Remark 10.1.12 Chapter10/EquivalenceZhou · EpFunctor.IsEquivalence.isZhou_of_isZhou_right. *Step 3, X to Y.* For $k \in G'_y$ with $Tk = \mathrm{id}$ and $\psi : F(x) \to y$, $k' = \psi^{-1}k\psi \in G'_{F(x)}$ has $Tk' = \mathrm{id}$ by the composition rule for T Chapter8/TangentEp · EpGroupoid.tangentMor_comp; $k' = F(g)$ with $g \in G_x$ (Remark 10.1.12), $Tg = \mathrm{id}$ by the injectivity of TF on T_xX, so $g \in T_x$, $k' \in T_{F(x)}$ and $k \in T_y$ Chapter10/EquivalenceZhou · EpFunctor.IsEquivalence.isZhou_of_isZhou. The text argues with natural representations instead; it writes $U(F(y))$ and "$x \in U(x)$" in (b), read $U(F(x))$ and $a \in U(x)$ (possible misprints).

Proposition 10.1.18 (V10.15) Chapter10/EquivalenceReduced · EP.proposition_10_1_18. *Step 1.* The restriction of F to level 1 is an sc-smooth functor $X^1 \to Y^1$ Chapter10/EquivalenceReduced · EpFunctor.liftOne, an equivalence when F is Chapter10/EquivalenceReduced · EpFunctor.IsEquivalence.liftOne (orbit spaces of X^1 carry the level-1 topology, and $|F^1|$ is a homeomorphism by Lemma 10.1.2 again). *Step 2.* By induction, every $X^i \to Y^i$ Chapter10/EquivalenceReduced · EP.exists_isEquivalence_lift over Chapter 7's lifts Chapter7/Zhou · EP.lift. *Step 3.* Proposition 10.1.16 at each level. The text cites Proposition 10.1.9 at Step 3; we read Proposition 10.1.16 (possible misprint).

Proposition 10.1.19 (V10.16) Chapter10/EquivalenceZhou · EP.isSLR_iff_of_isLR_iff. Steps 1–2 as above, then Proposition 10.1.9 at each level, taken as a hypothesis (G.113). The hypothesis holds when the two ep-groupoids agree Chapter10/EquivalenceZhou · EP.isLR_iff_lift_self. The unconditional statement is G.115.

X.8 Definition 10.2.1 (2017: 10.3): the weak fibred product

Definition 10.2.1 (2017: 10.3) (V10.17) Chapter10/WeakFibredProduct · WeakFibredProduct.groupoid. *Objects.* $\{(\phi,z) \mid t(\phi) = G(z)\} \subseteq \mathbf{Y}\times Z$ Chapter10/WeakFibredProduct · WeakFibredProduct.objRight, then $\{(x,(\phi,z)) \mid F(x) = s(\phi)\}$ Chapter10/WeakFibredProduct · WeakFibredProduct.objSet Chapter10/WeakFibredProduct · WeakFibredProduct.Obj, with projections and extensionality Chapter10/WeakFibredProduct · WeakFibredProduct.Obj.ext. *Morphisms.* The same with $\mathbf{X}$, $\mathbf{Z}$ and $F\circ s$, $G\circ s$ Chapter10/WeakFibredProduct · WeakFibredProduct.morRight Chapter10/WeakFibredProduct · WeakFibredProduct.morSet Chapter10/WeakFibredProduct · WeakFibredProduct.Mor Chapter10/WeakFibredProduct · WeakFibredProduct.Mor.ext. *Structure maps.* The conjugate $G(k)\circ\phi\circ F(h)^{-1}$ Chapter10/WeakFibredProduct · WeakFibredProduct.conj has the ends $F(t(h))$, $G(t(k))$ Chapter10/WeakFibredProduct · WeakFibredProduct.src_conj Chapter10/WeakFibredProduct · WeakFibredProduct.tgt_conj; it is the target's middle entry Chapter10/WeakFibredProduct · WeakFibredProduct.tgt, s keeps ϕ Chapter10/WeakFibredProduct · WeakFibredProduct.src, the unit

is $(1_x, \phi, 1_z)$ Chapter10/WeakFibredProduct · WeakFibredProduct.unit and the inverse $(h^{-1}, \mathrm{conj}, k^{-1})$ Chapter10/WeakFibredProduct · WeakFibredProduct.inv. The text prints $\iota(h,\psi,k) = (h^{-1}, G(k)\circ\phi\circ F(h)^{-1}, k^{-1})$; we read $\psi = \phi$ (possible misprint). The product of a composable pair is $(h\circ h', \phi', k\circ k')$ Chapter10/WeakFibredProduct · WeakFibredProduct.compOf, extended to a total map that returns its first argument off composable pairs Chapter10/WeakFibredProduct · WeakFibredProduct.comp. *Groupoid laws.* By the rules $\mathrm{conj}(h\circ h', \phi', k\circ k') = \mathrm{conj}(h, \mathrm{conj}(h', \phi', k'), k)$ Chapter10/WeakFibredProduct · WeakFibredProduct.conj_comp, conj of units Chapter10/WeakFibredProduct · WeakFibredProduct.conj_unit and of inverses Chapter10/WeakFibredProduct · WeakFibredProduct.conj_inv, with ϕ recovered from the target Chapter10/WeakFibredProduct · WeakFibredProduct.conj_recover Chapter10/WeakFibredProduct · WeakFibredProduct.conj_comp_mor; a morphism is a loop iff its ends agree Chapter10/WeakFibredProduct · WeakFibredProduct.src_eq_tgt_iff.

X.9 Theorem 10.2.2 (2017: 10.2): $X \times_Y Z$ is an ep-groupoid

The admissibility Chapter10/WeakFibredProduct · WeakFibredProduct.Admissible. The construction uses of F (or of G) only that it is a local sc-diffeomorphism on objects, which every equivalence is Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Admissible.of_isEquivalence_left Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Admissible.of_isEquivalence_right. *Step 1, sub-M-polyfolds.* Part I's fibred product of an sc-smooth map with a local sc-diffeomorphism is a sub-M-polyfold (Proposition 2.5.20); $t : \mathbf{Y} \to Y$ is one, so $\mathbf{Y}_t \times_G Z$ is a sub-M-polyfold of $\mathbf{Y} \times Z$ Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_objRight Chapter10/WeakFibredProduct · WeakFibredProduct.PA, and its projection to Z is a local sc-diffeomorphism Chapter10/WeakFibredProduct · MPolyfold.isLocalScDiffeo_snd_fibredProduct; then the second fibred product with F or with that projection Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_objSet Chapter10/WeakFibredProduct · WeakFibredProduct.PObj, and the same for morphisms Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_morRight Chapter10/WeakFibredProduct · WeakFibredProduct.isSubMPolyfold_morSet Chapter10/WeakFibredProduct · WeakFibredProduct.PB Chapter10/WeakFibredProduct · WeakFibredProduct.PMor. The text says the sets are sub-M-polyfolds of the triple products; the agreement of the two structures is not formalized and not used (G.116). *Step 2, the projections.* x, ϕ, z, h, ϕ, k are sc-smooth Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_x Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_z Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_k, and a map into $X \times_Y Z$ is sc-smooth when its entries are Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothOn_obj_of Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothOn_mor_of. For F admissible, $(x,\phi,z) \mapsto z$ and $(h,\phi,k) \mapsto k$ are local sc-diffeomorphisms ((10.9), (10.10)) Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_z Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_k; for G admissible the projections to X and $\mathbf{X}$ are Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_x Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_h Chapter10/WeakFibredProduct · MPolyfold.isLocalScDiffeo_fst_fibredProduct. *Step 3, s and t.* $\pi_Z \circ s = s_Z \circ \pi_{\mathbf{Z}}$ and $\pi_Z \circ t = t_Z \circ \pi_{\mathbf{Z}}$ with three local sc-diffeomorphisms make s, t ones Chapter10/WeakFibredProduct · MPolyfold.IsLocalScDiffeo.of_comm Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_src Chapter10/WeakFibredProduct · WeakFibredProduct.isLocalScDiffeo_tgt; their sc-smoothness uses that of the conjugate Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_conj Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_src Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_tgt. *Step 4, unit, inverse, multiplication* Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_unit Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothMap_inv Chapter10/WeakFibredProduct · WeakFibredProduct.isScSmoothComp: entries are multiplications of X and Z and projections, sc-smooth on the composable pairs Chapter10/WeakFibredProduct · EpGroupoid.isScSmoothOn_comp_of. *Step 5, properness.* For open $U \ni x$, $V \ni z$ with the properness property in X and Z, $W = \{x' \in U, z' \in V\}$ has t proper on $s^{-1}(\mathrm{cl}\, W)$ Chapter10/WeakFibredProduct · WeakFibredProduct.exists_isProperMap: subsequences of h_n, then of k_n, converge, and $\phi_n = G(k_n)^{-1} \circ (G(k_n)\circ\phi_n\circ F(h_n)^{-1}) \circ F(h_n)$ converges by continuity of the multiplication Chapter10/WeakFibredProduct · EpGroupoid.tendsto_comp Chapter10/WeakFibredProduct · WeakFibredProduct.continuous_conj. The text writes $t : s^{-1}(U) \to X$; Definition 7.1.3 (3) asks for the closure, which is the reading. No admissibility is needed here. *Conclusion* (V10.18) Chapter10/

WeakFibredProduct · WeakFibredProduct.epGroupoid.

X.10 Theorem 10.2.2 (2017: 10.2): the projections, and Remark 10.2.3 (2017: 10.2)

The functors Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projLeft Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projRight Chapter10/WeakFibredProductTheorem · WeakFibredProduct.projLeft_obj. $\sigma(x,\phi,z) = \phi$ is a natural transformation $F \circ \pi_X \to G \circ \pi_Z$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.natTrans Chapter10/WeakFibredProductTheorem · WeakFibredProduct.natTrans_app: its naturality is the definition of the target, and it is sc-smooth as a projection. Hence $F \circ \pi_X \simeq G \circ \pi_Z$ whenever $X \times_Y Z$ is defined Chapter10/WeakFibredProductTheorem · WeakFibredProduct.naturallyEquivalent.

F an equivalence (V10.19) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projRight. *Step 1, faithful.* A morphism over k is (h,ϕ,k) with $F(h)$ determined by k and the ends, and F is faithful. *Step 2, full.* For $k : z \to z'$ between (x,ϕ,z) and (x',ϕ',z'), $\phi'^{-1} \circ G(k) \circ \phi : F(x) \to F(x')$ is $F(h_0)$ by fullness of F, and (h_0,ϕ,k) lies over k. The text writes the composite as $\phi' \circ G(k) \circ \phi^{-1}$, which does not compose; we read $\phi'^{-1} \circ G(k) \circ \phi$ (possible misprint). *Step 3, essentially surjective.* $G(z)$ receives $\psi : F(x) \to G(z)$ Chapter10/Equivalence · EpFunctor.IsEquivalence.exists_iso, and (x,ψ,z) lies over z. *Step 4.* Lemma 10.1.2 with Step 2 of §X.9. *The rest of the theorem* Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2: an equivalence preserves the degeneracy index Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.degIndex_obj, so $d(x,\phi,z) = d_Z(z)$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.degIndex_eq_z, and pulls back tameness Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.isTame_of_isTame and the paracompactness of the orbit space, through $|\pi_Z|$ Chapter10/WeakFibredProductTheorem · EpFunctor.IsEquivalence.hasParacompactOrbitSpace. *Witness.* $\mathbb{Z} \to$ pt and $\mathrm{id}_{\mathrm{pt}}$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Examples.theorem_10_2_2_pairToPoint.

G an equivalence, and both (V10.20) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.isEquivalence_projLeft Chapter10/WeakFibredProductTheorem · WeakFibredProduct.degIndex_eq_x Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2_right Chapter10/WeakFibredProductTheorem · WeakFibredProduct.theorem_10_2_2_both. The symmetric argument with ϕ^{-1}. The admissibility is a proposition, so the ep-groupoids built from F and from G coincide and the two cases combine.

Remark 10.2.3 (2017: 10.2) (V10.21) Chapter10/WeakFibredProductTheorem · WeakFibredProduct.remark_10_2_3. Both projections are equivalences and $\tau = \sigma$ is natural. *Witness.* id and $-$id on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ Chapter10/WeakFibredProductTheorem · WeakFibredProduct.Examples.remark_10_2_3_negFunctor. The sentences on "smaller models" have no formal content.

X.11 Definition 10.4.1 (2017: 10.10): strong bundle equivalences

The orbit space at level $[1]$ Chapter10/StrongBundleEquivalence · EpStrongBundle.oneOrbitSetoid Chapter10/StrongBundleEquivalence · EpStrongBundle.OneOrbitSpace Chapter10/StrongBundleEquivalence · EpStrongBundle.oneOrbitMap. $W[1]$ is not built as an ep-groupoid; $|W[1]|$ is the quotient of $W_{0,1}$ with its level-1 topology by the orbit relation of W, and a strong bundle functor induces a continuous map of these quotients Chapter10/StrongBundleEquivalence · EpStrongBundle.continuous_oneOrbitMap.

Definition 10.4.1 (2017: 10.10) (V10.22) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence. A strong bundle functor Chapter8/EpStrongBundle · EpStrongBundle.IsStrongBundleFunctor with ϕ an equivalence, Φ an equivalence at level $[0]$, $|\Phi|$ a homeomorphism of $|W[1]|$, and local strong bundle isomorphisms on objects and morphisms Chapter8/EpStrongBundle · StrongBundle.IsStrongBundleIsoOn. Clause (1) at level $[1]$ is read in this form: fullness

and faithfulness at $[1]$ are those at $[0]$ restricted, and the local sc-diffeomorphism property at $[1]$ follows from (2) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_one Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isLocalScDiffeo_obj. *Consequences.* $P' \circ \Phi = \phi \circ P$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.covers; $\Phi_x : W_x \to W'_{\phi(x)}$ is a linear isomorphism, the local strong bundle isomorphism at x restricted Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.P_inv Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.fiberEquiv, so Φ is injective on fibres Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.eq_of_obj_eq. *Class.* The identity Chapter10/StrongBundleEquivalence · EpStrongBundle.isStrongBundleEquivalence_id; composites Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.comp, with strong bundle functors Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleFunctor.comp and local strong bundle isomorphisms composing on open sets Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.restrict Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.comp Chapter10/StrongBundleEquivalence · StrongBundle.IsStrongBundleIsoOn.exists_comp; isomorphisms with an inverse strong bundle functor Chapter10/StrongBundleEquivalenceNat · StrongBundle.isStrongBundleIsoOn_univ_of_inverse Chapter10/StrongBundleEquivalenceNat · EpFunctor.isEquivalence_of_inverse Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.isStrongBundleEquivalence_of_inverse. *Witness.* Over $-\mathrm{id}$ of $\mathbb{R}^n$ Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negMap, the projection of the pull-back of the trivial bundle Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negBundle Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negEpBundle, with inverse Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negInv Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.isStrongBundleMap_negInv, is a strong bundle equivalence Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.isStrongBundleEquivalence_pullbackProj' not over the identity Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExa trivialNeg_obj_ne_id.

X.12 The pull-back and the push-forward, (10.15)–(10.17) and (10.22)

Pull-back (V10.23) Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback. $\Phi^* f'(x)$ is the unique $w \in W_x$ with $\Phi(w) = f'(\phi(x))$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.obj_pullback Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_eq, a section Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.P_pullback Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isSection_pullback, functorial when f' is Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pullback ((10.17), by the injectivity of Φ on fibres and of $\mu'(\sigma, \cdot)$ Chapter10/StrongBundleEquivalence · EpStrongBundle.μ_injective_snd).

Push-forward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.preObj Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.preMor Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_def. For x' choose x and $\sigma : \phi(x) \to x'$ (essential surjectivity of ϕ) and set $\Phi_* f(x') = \mu'(\sigma, \Phi(f(x)))$, a section Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.P_pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isSection_pushforward. *(10.22).* For a section functor f and any y, $\sigma' : \phi(y) \to x'$, the value is $\mu'(\sigma', \Phi(f(y)))$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq: $\sigma'^{-1} \circ \sigma : \phi(x) \to \phi(y)$ is $\phi(\tau)$ by fullness, $f(y) = \mu(\tau, f(x))$ by functoriality, and Φ intertwines μ and μ' Chapter10/StrongBundleEquivalence · EpStrongBundle.μ_mk_congr. Hence $\Phi_* f$ is a section functor Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isFunctorialSection_pushforward, and $\Phi^* \Phi_* f = f$, $\Phi_* \Phi^* f' = f'$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_pushforward Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_

pullback.

(10.15) Chapter10/StrongBundleEquivalence · EpStrongBundle.exists_μ_eq_of_orbitMap_eq Chapter10/StrongBundleEquivalence · EpStrongBundle.orbitMap_eq_of_forall_exists. $|\Phi| = |\Psi|$ iff every $\Psi(w)$ is $\mu'(\sigma, \Phi(w))$ for some σ.

X.13 Propositions 10.4.2–10.4.4 (2017: 10.7–10.9)

Proposition 10.4.2 (2017: 10.7) (V10.24) Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_2 Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pullback_eq_of_orbitMap_eq. With $g = \Psi^* f'$, (10.15) gives $\sigma : \phi(x) \to \psi(x)$ with $\mu'(\sigma, \Phi(g(x))) = \Psi(g(x)) = f'(\psi(x))$; functoriality of f' gives $f'(\psi(x)) = \mu'(\sigma, f'(\phi(x))) = \mu'(\sigma, \Phi(\Phi^* f'(x)))$; the injectivity of $\mu'(\sigma, \cdot)$ and of Φ_x gives $g(x) = \Phi^* f'(x)$. sc-smoothness is not used. The text's "$f' : W' \to X'$" is read $f' : X' \to W'$ (possible misprint).

Proposition 10.4.3 (2017: 10.8) (V10.25) Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_3 Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.pushforward_eq_of_orbitMap_eq. With $\sigma : \phi(x) \to \psi(x)$ from (10.15) at $f(x)$, and (10.22) for Ψ applied to the composite of σ^{-1} with the choice for Φ.

Proposition 10.4.4 (2017: 10.9) (V10.26) Chapter10/StrongBundleEquivalence · EpStrongBundle.proposition_10_4_4. The pull-back identity pointwise from the uniqueness in the definition; the push-forward identity from (10.22) for $\Psi \circ \Phi$.

X.14 Theorem 10.4.5 (2017: 10.8)

Theorem 10.4.5 (2017: 10.8) (V10.27) Chapter10/StrongBundleEquivalence · EpStrongBundle.theorem_10_4_5. *Step 1, pull-back.* Near x_0, $\Phi^* f'$ agrees with $\Phi_U^{-1} \circ f' \circ \phi$ for the local strong bundle isomorphism Φ_U Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.eqOn_pullback, so it is sc-smooth Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pullback and sc^+ when f' is Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pullback, smoothness being a local property in charts Chapter10/StrongBundleEquivalence · MPolyfold.isScSmoothOn_of_forall_charts. *Step 2, push-forward.* The text says the sc-smoothness is easily verified. Near x_0', with a local inverse τ of t' at the chosen $\sigma_0 : \phi(y_0) \to x_0'$ and a local inverse g_1 of ϕ at y_0, (10.22) gives $\Phi_* f = \mu' \circ (\tau, \Phi \circ f \circ g_1 \circ s' \circ \tau)$ Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.exists_pushforward_local, sc-smooth Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScSmoothSection_pushforward. At level $[1]$ the pair is sc-smooth into the pull-back's level $[1]$ by its universal property on an open set Chapter10/StrongBundleEquivalence · StrongBundle.isScSmoothOn_to_pullback_one, so sc^+ is preserved Chapter10/StrongBundleEquivalence · EpStrongBundle.IsStrongBundleEquivalence.isScPlusSection_pushforward. *Step 3.* With §X.12 and §X.13, bijections between Chapter 8's sets of section functors Chapter8/EpStrongBundle · EpStrongBundle.sectionFunctors Chapter8/EpStrongBundle · EpStrongBundle.plusSectionFunctors, independent of Φ within $|\Phi|$. The text's second display, $\Phi^* : \Gamma(P', \mu') \to \Gamma(P, \mu)$ in the Γ^+ line, is read with Γ^+ (possible misprint). *Not formalized:* the sc-Fredholm part (G.121). *Witness* Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.bijOn_pushforward_example.

X.15 Definitions 10.4.8, 10.4.10 and 10.4.11, Remark 10.4.9 (2017: 10.11–10.13, 10.4)

Definition 10.4.8 (2017: 10.11) (V10.28) Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatEquiv. T

is data with $T(w) : \Phi(w) \to \Psi(w)$ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.mem and naturality, over an EpNatTrans τ. Reflexivity Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.refl; a witness with $\Phi \neq \Psi$ Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.strongBundleNatEquiv_example.

Remark 10.4.9 (2017: 10.4) (V10.29). *Step 1.* A morphism of $\mathbf{W}'$ is (σ, w') with $\sigma \in \mathbf{X}'$; $T(w)$ lies over $\tau(P(w))$ and starts at $\Phi(w)$, so $T(w) = (\tau(P(w)), \Phi(w))$ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.T_val Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.T_eq and $\Psi(w) = \mu'(\tau(P(w)), \Phi(w))$ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.obj_eq. The text prints the second line of (10.23) with μ; we read μ'. *Step 2.* T and Ψ on objects are determined by Φ and τ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.unique, and Ψ on morphisms too, if Ψ is a strong bundle functor Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.unique_functor. *Step 3.* T is a strong bundle map covering τ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.isStrongBundleMap_T.

Definition 10.4.10 (2017: 10.12) (V10.30) Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.IsStrongBundleNatTransMap. Definition 10.4.8 gives it by Remark 10.4.9 Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.isStrongBundleNatTransMap; conversely the naturality of T follows from (10.23) and the functoriality of Φ, Ψ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.IsStrongBundleNatTransMap.toNatTrans. $\Phi \simeq \Psi$ implies $|\Phi| = |\Psi|$ Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatTrans.orbitMap_eq Chapter10/StrongBundleEquivalenceNat · EpStrongBundle.StrongBundleNatEquiv.orbitMap_eq.

Definition 10.4.11 (2017: 10.13) (V10.31) Chapter10/StrongBundleEquivalenceNat · SEP Chapter10/StrongBundleEquivalenceNat · SEP.Hom Chapter10/StrongBundleEquivalenceNat · SEP.Hom.ext'. A morphism is a pair (ϕ, Φ) with Φ a strong bundle functor covering ϕ; Φ alone determines ϕ only on the image of P. The strong bundle equivalences form a multiplicative class Chapter10/StrongBundleEquivalenceNat · SEP.equivalences, mapped into E by the forgetful functor Chapter10/StrongBundleEquivalenceNat · SEP.toEP Chapter10/StrongBundleEquivalenceNat · SEP.toEP_map_mem_equivalences. *Witness* Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEP Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.epSEP Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEPHom Chapter10/StrongBundleEquivalenceExamples · StrongBundleEquivalenceExamples.negSEPHom_mem_equivalences.

X.16 §10.3: (E1)–(E3), diagrams and their equivalence (Definitions 10.3.1, 10.3.3, 10.3.5, Lemmas 10.3.2, 10.3.4 (2017: Definitions 10.4–10.6, Lemmas 10.3, 10.4))

(E1)–(E3) (V10.32). (E1): an isomorphism of $\mathcal{EP}$ has an inverse sc-smooth functor, so it is full, faithful, a homeomorphism on orbit spaces and an sc-diffeomorphism on objects Chapter10/LocalizationCategory · EPLocalization.isEquivalence_of_isIso. (E2) and (E3) are §X.1 and Proposition 10.1.5 Chapter10/Equivalence · EpFunctor.IsEquivalence.comp Chapter10/Equivalence · EpFunctor.proposition_10_1_5.

Diagrams (V10.33) Chapter10/Localization · EPLocalization.Diagram. The middle ep-groupoid is an object of $\mathcal{EP}$, so that a diagram, and later its class, lives one universe above the object and morphism types; no class-sized collection occurs. The weak fibred product of §X.9 is made an object of $\mathcal{EP}$ when admissible Chapter10/Localization · EPLocalization.wfp.

Definition 10.3.1 (2017: 10.4) Chapter10/Localization · EPLocalization.Diagram.IsRefinement Chapter10/Localization · EPLocalization.Diagram.IsRefinement.refl. Both examples after the definition: precomposition with an equivalence H Chapter10/Localization · EPLocalization.Diagram.precomp Chapter10/Localization · EPLocalization.Diagram.isRefinement_precomp, with H itself refining and the natural equiva-

lences equalities Chapter10/Localization · EpFunctor.NaturallyEquivalent.of_eq; and naturally equivalent legs, with $H = 1_A$ Chapter10/Localization · EPLocalization.Diagram.isRefinement_of_naturallyEquivalent.

Lemma 10.3.2 (2017: 10.3) (V10.34) Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_2 Chapter10/Localization · EPLocalization.Diagram.IsRefinement.trans. For $H : d' \to d$ and $H' : d'' \to d'$, $H \circ H'$ is an equivalence (§X.1) and $F \circ H \circ H' \simeq F' \circ H' \simeq F''$, by the compatibility of $\simeq$ with composition Chapter8/NatTrans · EpFunctor.NaturallyEquivalent.comp_right and its transitivity; the same for Φ.

Definition 10.3.3 (2017: 10.5) and Lemma 10.3.4 (2017: 10.4) (V10.35, V10.36) Chapter10/Localization · EPLocalization.Diagram.IsCommonRefinement Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_4. *Step 1.* For W refining d by H and d' by H', and W' refining d' by K and d'' by K', H' and K are equivalences into A', so the weak fibred product $W \times_{A'} W'$ is defined and its projections are equivalences (Theorem 10.2.2). *Step 2.* The diagram (10.11) over $W \times_{A'} W'$ refines d by $H \circ \pi_W$; it refines d'' by $K' \circ \pi_{W'}$, because $F' \circ H' \circ \pi_W \simeq F' \circ K \circ \pi_{W'}$ by the natural transformation σ of the weak fibred product (§X.10) and Lemma 10.3.2. A refinement is equivalent to the refined diagram Chapter10/Localization · EPLocalization.Diagram.IsRefinement.equiv.

Definition 10.3.5 (2017: 10.6) (V10.37) Chapter10/Localization · EPLocalization.Diagram.Equiv Chapter10/Localization · EPLocalization.Diagram.setoid Chapter10/Localization · EPLocalization.Diagram.equiv_iff. Reflexivity and symmetry are immediate Chapter10/Localization · EPLocalization.Diagram.Equiv.refl Chapter10/Localization · EPLocalization.Diagram.Equiv.symm, transitivity is Lemma 10.3.4 Chapter10/Localization · EPLocalization.Diagram.Equiv.trans. *Witness.* pt $\leftarrow$ $\mathbb{Z}$-pair $\to$ pt, both legs the equivalence of §X.1, is equivalent to 1_{pt} through the refinement by that equivalence, which is not an isomorphism Chapter10/LocalizationCategory · EPLocalization.Examples.equiv_pairToPoint_id.

X.17 Lemma 10.3.6 (2017: 10.5), Definition 10.3.7 (2017: 10.7) and Theorem 10.3.8 (2017: 10.3) with Appendix §10.6.1

Lemma 10.3.6 (2017: 10.5) (V10.38) Chapter10/Localization · EPLocalization.Diagram.lemma_10_3_6. $|d| = |\Phi| \circ |F|^{-1}$, with $|F|^{-1}$ the inverse of the homeomorphism of Definition 10.1.1 (2) Chapter10/Localization · EPLocalization.Diagram.orbitMap Chapter10/Localization · EPLocalization.Diagram.continuous_orbitMap Chapter10/Localization · EPLocalization.Diagram.orbitMap_orbitMap_F Chapter10/Localization · EPLocalization.Diagram.orbitMap_eq_of. For $H : d' \to d$, $|F'| = |F \circ H| = |F| \circ |H|$ Chapter10/Localization · EpFunctor.orbitMap_comp Chapter10/Localization · EpFunctor.orbitMap_id, since naturally equivalent functors have the same orbit map Chapter10/Localization · EpFunctor.NaturallyEquivalent.orbitMap_eq, and likewise $|\Phi'| = |\Phi| \circ |H|$; the factors $|H|$ cancel. Equivalent diagrams have a common refinement, hence the same map Chapter10/Localization · EPLocalization.Diagram.Equiv.orbitMap_eq.

Definition 10.3.7 (2017: 10.7) (V10.39) Chapter10/Localization · EPLocalization.Diagram.orbitMapQ Chapter10/Localization · EPLocalization.Diagram.orbitMapQ_mk. When Φ is an equivalence, $|\Phi| \circ |F|^{-1}$ is a composite of homeomorphisms Chapter10/Localization · EPLocalization.Diagram.isHomeomorph_orbitMap.

The composite Chapter10/Localization · EPLocalization.Diagram.comp Chapter10/Localization · EPLocalization.Diagram.id. For d and $e : Y \xleftarrow{G} B \xrightarrow{\Psi} Z$, the weak fibred product of Φ and G is admissible since G is an equivalence Chapter10/Localization · EPLocalization.Diagram.admissible, and π_A is an equivalence (Theorem 10.2.2), so $F \circ \pi_A$ is one. `d.comp e` is the text's $[e] \circ [d]$.

Theorem 10.3.8 (2017: 10.3) (V10.40) Chapter10/Localization · EPLocalization.Diagram.theorem_10_3_8, the proof of Appendix §10.6.1. *Step 1, the functor L.* For refinements $H : d' \to d$, $K : e' \to e$ with natural transformations $\tau_{A'} : \Phi' \to \Phi \circ H$ and $\tau_{B'} : G' \to G \circ K$, $L(a', \phi, b') = (H(a'), \tau_{B'}(b') \circ \phi \circ \tau_{A'}(a')^{-1}, K(b'))$ ((10.28), (10.30)) Chapter10/Localization · EPLocalization.mapMid Chapter10/Localization · EPLocalization.mapObj Chapter10/Localization · EPLocalization.mapMor Chapter10/Localization · EPLocalization.wfpMap is an sc-smooth functor, entry by entry, with the bookkeeping identity Chapter10/Localization · GroupoidStr.comp_comp_inv_comp_inv. In the text's naturality of $\tau_{B'}$ the letter τ_A appears; we read $\tau_{B'}$ (possible misprint). *Step 2, L*

is an equivalence Chapter10/Localization · EPLocalization.isEquivalence_wfpMap. $\pi_A \circ L = H \circ \pi_{A'}$ holds on the nose Chapter10/Localization · EPLocalization.projLeft_comp_wfpMap Chapter10/Localization · EPLocalization.projRight_comp_wfpMap; π_A, $\pi_{A'}$ and H are equivalences, and the two-out-of-three rule on the other side from Proposition 10.1.6 Chapter10/Localization · EpFunctor.IsEquivalence.of_comp_left applies. The text checks directly that L is full, faithful and a local sc-diffeomorphism. *Step 3, the refinement.* $(F \circ \pi_A) \circ L = F \circ H \circ \pi_{A'} \simeq F' \circ \pi_{A'}$, and $(\Psi \circ \pi_B) \circ L = \Psi \circ K \circ \pi_{B'} \simeq \Psi' \circ \pi_{B'}$ Chapter10/Localization · EPLocalization.Diagram.IsRefinement.comp; the text writes the second as "$(\Phi \circ \pi_B) \circ L \simeq \Psi' \circ \pi_{B'}$", read with Ψ (possible misprint). With common refinements on both sides, the class of the composite depends only on the classes Chapter10/Localization · EPLocalization.Diagram.Equiv.comp. *Step 4, associativity.* The text identifies $(A \times_Y A') \times_Z A''$ and $A \times_Y (A' \times_Z A'')$; here the reassociation $((a, \phi, a'), \psi, a'') \mapsto (a, \phi, (a', \psi, a''))$ is a functor Chapter10/Localization · EPLocalization.assocObj Chapter10/Localization · EPLocalization.assocMor Chapter10/Localization · EPLocalization.assocFunctor, an equivalence by two out of three Chapter10/Localization · EPLocalization.projLeft_comp_assocFunctor Chapter10/Localization · EPLocalization.isEquivalence_assocFunctor, and a refinement with identity natural transformations Chapter10/Localization · EPLocalization.Diagram.isRefinement_assoc Chapter10/Localization · EPLocalization.Diagram.equiv_assoc. *Step 5, units.* $X \xleftarrow{\pi_X} X \times_X A \xrightarrow{\Phi \circ \pi_A} Y$ refines d by π_A, with $F \circ \pi_A \simeq \pi_X$ by $\tau(x, \phi, a) = \phi$ Chapter10/Localization · EPLocalization.Diagram.isRefinement_id_comp, and symmetrically on the other side Chapter10/Localization · EPLocalization.Diagram.isRefinement_comp_id. The text's proof announces $[1_X] \circ [d] = [d]$ for $d : X \to Y$, which in its order of composition is $[d] \circ [1_X] = [d]$, writes $\tau(a, \phi, a)$ for $\tau(x, \phi, a)$ and "$\psi = \varphi$" for $\psi = \phi$ (possible misprints).

X.18 Definitions 10.3.9 and 10.3.12, Theorem 10.3.10 (2017: Definitions 10.8, 10.9, Theorem 10.4) with Appendix §10.6.2

Definition 10.3.12 (2017: 10.9) (V10.44) Chapter10/LocalizationCategory · EPLocalization.EPLoc. The objects are a copy of those of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.toEP_of Chapter10/LocalizationCategory · EPLocalization.EPLoc.of_toEP, the morphisms the quotients of diagrams Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk, with $[d] \gg [e] = [d.\mathrm{comp}\, e]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_comp and $\mathrm{id} = [1_X]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.id_eq; the category laws are Theorem 10.3.8. $[d] = [d']$ iff d and d' are equivalent Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_eq_homMk_iff, and every morphism is a $[d]$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_surjective.

Definition 10.3.9 (2017: 10.8) (V10.41) Chapter10/LocalizationCategory · EPLocalization.EPLoc.IsGeneralizedIso, the isomorphisms of the category Chapter10/LocalizationCategory · EPLocalization.EPLoc.isGeneralizedIso_iff_isIso.

Theorem 10.3.10 (2017: 10.4) (V10.42) Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_10, the proof of Appendix §10.6.2. *Step 1, Φ an equivalence.* The reversed diagram $Y \xleftarrow{\Phi} A \xrightarrow{F} X$ Chapter10/LocalizationCategory · EPLocalization.Diagram.symm. The composite $X \xleftarrow{F \circ \pi_1} A \times_Y A \xrightarrow{F \circ \pi_2} X$ is refined by $X \xleftarrow{F} A \xrightarrow{F} X$ through the diagonal $a \mapsto (a, 1_{\Phi(a)}, a)$ Chapter10/Localization · EPLocalization.diagObj Chapter10/Localization · EPLocalization.diagMor Chapter10/Localization · EPLocalization.diagFunctor, an equivalence since $\pi_1 \circ \Delta = 1_A$ Chapter10/Localization · EPLocalization.isEquivalence_diagFunctor Chapter10/Localization · EPLocalization.Diagram.isRefinement_comp_symm, and that diagram refines 1_X by F Chapter10/Localization · EPLocalization.Diagram.isRefinement_id; so both composites are units Chapter10/LocalizationCategory · EPLocalization.Diagram.equiv_comp_symm Chapter10/LocalizationCategory · EPLocalization.Diagram.equiv_symm_comp. *Step 2, the converse.* Let $[b] = [Y \xleftarrow{G} B \xrightarrow{\Psi} X]$ be an inverse. A diagram $X \leftarrow C \xrightarrow{\Theta} X$ equivalent to 1_X has Θ an equivalence (Propositions 10.1.5 and 10.1.6 applied to a common refinement) Chapter10/LocalizationCategory · EPLocalization.Diagram.isEquivalence_Φ_of_equiv_id;

so $\Psi \circ \pi_B$ and $\Phi \circ \pi_{A'}$ are equivalences. *Step 3, (C1).* Φ is an equivalence of categories Chapter10/LocalizationCategory · EPLocalization.isEquivalence_toFunctor_of_comp. Faithfulness is checked on morphisms (h, ϕ, k) with $k = 1_b$; in the text's injectivity argument $A \times_X B$ is read $A \times_Y B$ and $\mathrm{mor}_Y(\Psi(a), \Psi(a'))$ as $\mathrm{mor}_X(\Psi(b), \Psi(b'))$, and in the essential surjectivity "between $\Phi(a)$ and x" as "and y" (possible misprints). *Step 4, (C2).* The text defines $\Gamma(b') = (b', (s|U)^{-1}(b'), \ldots)$, read with $(s|U)^{-1}(\Psi(b'))$ (possible misprint), and then uses that $\Gamma' : U' \to A$ is a local sc-diffeomorphism; Γ' is a composite with Ψ, so as we read it this needs Ψ to be a local sc-diffeomorphism, the other half of (C2), and the step does not hold as we read it. The formal argument: a local inverse κ of the equivalence $\Phi \circ \pi_{A'}$ at (b, ϕ, a) gives the local sc-smooth right inverse $\pi_{A'} \circ \kappa$ of Φ with value a at $\Phi(a)$; Φ is injective near a by a local section $a' \mapsto (a', \phi(a'), b(a'))$ of the local sc-diffeomorphism $\pi_A : A \times_Y B \to A$ and the local injectivity of $\Psi \circ \pi_B$, s and G Chapter10/LocalizationCategory · MPolyfold.IsLocalScDiffeoAt.exists_injOn; an injective map with such a right inverse is a local sc-diffeomorphism Chapter10/LocalizationCategory · MPolyfold.isLocalScDiffeoAt_of_injOn Chapter10/LocalizationCategory · EPLocalization.isLocalScDiffeo_obj_of_comp. *Step 5.* Lemma 10.1.2 Chapter10/LocalizationCategory · EPLocalization.Diagram.isEquivalence_Φ_of_equiv. *(2).* An inverse is unique, so every inverse is the class of the reversed diagram Chapter10/LocalizationCategory · EPLocalization.EPLoc.isIso_homMk_iff Chapter10/LocalizationCategory · EPLocalization.EPLoc.isIso_homMk Chapter10/LocalizationCategory · EPLocalization.EPLoc.inv_homMk.

X.19 Proposition 10.3.11, Theorems 10.3.14 and 10.3.15 (2017: Proposition 10.6, Theorems 10.5, 10.6)

Proposition 10.3.11 (2017: 10.6) (V10.43) Chapter10/LocalizationCategory · EPLocalization.EPLoc.proposition_10_3_11. $||[d]||$ on the quotient Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap_homMk Chapter10/LocalizationCategory · EPLocalization.EPLoc.continuous_orbitMap Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitMap_id. The computation before the proposition: an object (a, ϕ, b) of $A \times_Y B$ has $\phi : \Phi(a) \to G(b)$, so $|\Phi(a)| = |G(b)|$, and $|d.\mathrm{comp}\, e| = |e| \circ |d|$ Chapter10/Localization · EPLocalization.Diagram.orbitMap_comp.

Theorem 10.3.14 (2017: 10.5) (V10.45) Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_14. *Step 1.* $d_\Phi = X \xleftarrow{1} X \xrightarrow{\Phi} Y$ Chapter10/LocalizationCategory · EPLocalization.Diagram.ofFunctor and i Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor_obj Chapter10/LocalizationCategory · EPLocalization.EPLoc.localizationFunctor_map. *Step 2, functoriality.* The composite (10.13) $X \xleftarrow{\pi_X} X \times_Y Y \xrightarrow{\Psi \circ \pi_Y} Z$ refines $d_{\Psi \circ \Phi}$ by π_X, since $\Phi \circ \pi_X \simeq \pi_Y$ (§X.10) Chapter10/LocalizationCategory · EPLocalization.Diagram.isRefinement_ofFunctor_comp. The text states the refinement in the other direction; with Definition 10.3.1 (the refining equivalence goes out of the refinement) it is this one, and the equality of classes is the same. *Step 3, inverses.* For $F \in E$, Theorem 10.3.10 with the legs 1 and F Chapter10/LocalizationCategory · EPLocalization.EPLoc.isInvertedBy_localizationFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.inv_localizationFunctor_map, and every generalized map is $i(\Phi) \circ i(F)^{-1}$ Chapter10/LocalizationCategory · EPLocalization.EPLoc.homMk_eq. *Witness.* The pair groupoid of $\mathbb{Z}$ and the point as objects of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.Examples.pairEP Chapter10/LocalizationCategory · EPLocalization.Examples.pointEP Chapter10/LocalizationCategory · EPLocalization.Examples.pairToPointEP Chapter10/LocalizationCategory · EPLocalization.Examples.isEquivalence_pairToPointEP: the functor is not an isomorphism of $\mathcal{EP}$ Chapter10/LocalizationCategory · EPLocalization.Examples.not_isIso_pairToPointEP, its image is invertible Chapter10/LocalizationCategory · EPLocalization.Examples.isGeneralizedIso_pairToPoint, with the inverse of Theorem 10.3.14 Chapter10/LocalizationCategory · EPLocalization.Examples.inv_localizationFunctor_map_pairToPoint. *Not formalized.* The universal property mentioned on p. 483 (G.117).

Theorem 10.3.15 (2017: 10.6) (V10.46) Chapter10/LocalizationCategory · EPLocalization.EPLoc.orbitFunctor Chapter10/LocalizationCategory · EPLocalization.EPLoc.theorem_10_3_15. A functor into Mathlib's `TopCat` by Proposition 10.3.11.

X.20 Propositions 10.3.16, 10.3.17, Remark 10.3.18 (2017: 10.3), Theorem 10.3.19 (2017: 10.7), Definition 10.3.20, Theorem 10.3.21

The reduction `Chapter10/LocalizationProperties · EPLocalization.EPLoc.exists_diagram_of_isGeneralizedIso` `Chapter10/LocalizationProperties · EPLocalization.EPLoc.iff_of_isGeneralizedIso`. A generalized isomorphism is some $[d]$, and by Theorem 10.3.10 its Φ is an equivalence; a property invariant under equivalences passes from X to A to Y, as on p. 492.

Proposition 10.3.17 (V10.47) `Chapter10/LocalizationProperties · EPLocalization.EPLoc.proposition_10_3_17`, by Proposition 10.1.16. **Theorem 10.3.21** (V10.50) `Chapter10/LocalizationProperties · EPLocalization.EPLoc.theorem_10_3_21`, for (LUC), (SLUC), (ZHOU) by Propositions 10.1.16, 10.1.18, 10.1.17; the text leaves the proof to the reader. Proposition 10.3.16 and the (LR), (SLR) parts rest on Propositions 10.1.9 and 10.1.19 (G.118), Theorem 10.3.19 on Propositions 10.1.10 and 10.3.16 (G.120); as we read it, the proof's step on p. 494 gives a family α_z, not a natural transformation; the $\mathbb{Z}/3$ example of Remark 10.5.5 (§X.23) suggests that the statement does not hold as we read it; a Lean check is pending.

Remark 10.3.18 (2017: 10.3) (V10.48) `Chapter10/LocalizationProperties · EPLocalization.OrbitChart` `Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel`. Reflexive by 1_X `Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.refl`, symmetric by the inverse `Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.symm` and transitive by the composite `Chapter10/LocalizationProperties · EPLocalization.OrbitChart.Rel.trans`, each with Proposition 10.3.11 `Chapter10/LocalizationProperties · EPLocalization.OrbitChart.setoid`. *Witness* `Chapter10/LocalizationProperties · EPLocalization.OrbitChart.rel_pairEP_pointEP`. The metrizability of Z is not used. The non-uniqueness of f is shown by the base classes of the example of Remark 10.5.5 (§X.23; G.119, closed).

Definition 10.3.20 (V10.49) `Chapter10/LocalizationProperties · EPLocalization.EPreg` `Chapter10/LocalizationProperties · EPLocalization.EPLocReg`, Mathlib full subcategories, with the point as an object of both `Chapter10/LocalizationProperties · EPLocalization.Examples.pointReg` `Chapter10/LocalizationProperties · EPLocalization.Examples.pointLocReg`. Remark 10.3.13 has no formal content.

X.21 §10.5: diagrams of strong bundles, $[d]$, $|[D]|$ and γ

Diagrams and refinements (V10.51) `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram` `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement`. A diagram is a middle object of $\mathcal{SEP}$ with a strong bundle equivalence Φ and a morphism Ψ of $\mathcal{SEP}$. A refinement is a strong bundle equivalence H with $\Phi \circ H \simeq \Phi'$ and $\Psi \circ H \simeq \Psi'$, natural equivalence of morphisms of $\mathcal{SEP}$ being Definition 10.4.8 `Chapter10/StrongBundleLocalization · SEP.NatEquiv`, an equivalence relation compatible with composition on the left `Chapter10/StrongBundleLocalization · SEP.NatEquiv.comp_left`. The two examples `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.isRefinement_precomp` `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.isRefinement_of_natEquiv` and the strong bundle Lemma 10.3.2 `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.trans` follow: $H \circ H'$ is a strong bundle equivalence and $\Phi \circ H \circ H' \simeq \Phi' \circ H' \simeq \Phi''$. *The classes.* Having a common refinement `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv` is reflexive and symmetric `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.refl` `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.symm`; its transitivity would be the strong bundle Lemma 10.3.4, through the weak fibred product of strong bundles, which is not constructed (G.123). $[D]$ is the class for the generated equivalence relation `Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap` `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.cls`; every invariant below is proved invariant under `Equiv`, hence defined on the classes, and a refinement has the class of the refined diagram `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.equiv` `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.cls_eq_cls_of_equiv`.

$[d]$ **and** $|[D]|$ (V10.52) `Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.base` `Chapter10/`

StrongBundleLocalization · SBLocalization.SDiagram.total Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.base Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.orbitMap. Forgetting the bundles gives the diagram d of the base ep-groupoids, and the total spaces as ep-groupoids $(W, \mathbf{W})$ Chapter10/StrongBundleLocalization · SEP.totalEP give a diagram of total spaces; refinements and equivalences pass to both Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.IsRefinement.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.base Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.total, since a natural equivalence of strong bundle functors covers one of the base functors and is one of the total-space functors. $|[D]|$ is §10.3's orbit map $|\Psi| \circ |\Phi|^{-1}$ of the diagram of total spaces Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap, continuous Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.continuous_orbitMap and the same for equivalent diagrams by Lemma 10.3.6 Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.Equiv.orbitMap_eq.

The square $|\mathcal{D}|$ (V10.53) Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap_square Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.orbitMap_square. Every orbit of $|W|$ is $|\Phi(w)|$ for some $w \in W''$, $|\Phi|$ being onto; then $|[D]|(|\Phi(w)|) = |\Psi(w)|$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.orbitMap_orbitMap_Φ, $|P'(\Psi(w))| = |\psi(P''(w))|$ since Ψ covers ψ, and $|[d]|(|\phi(P''(w))|) = |\psi(P''(w))|$ by Lemma 10.3.6. The printed $|X| \to |X|$ is read with X'.

γ (V10.54) Chapter10/StrongBundleLocalization · EPLocalization.Diagram.exists_isotropyData Chapter10/StrongBundleLocalization · EPLocalization.Diagram.isotropyHom. The data: x'' and a from the essential surjectivity of ϕ, b from $|[d]|(|x|) = |x'|$. γ is the composite of the conjugation by a^{-1}, the inverse of the isomorphism $G_{x''} \to G_{\phi(x'')}$ of the equivalence ϕ (Remark 10.1.12), ψ on isotropy groups and the conjugation by b, with its values Chapter10/StrongBundleLocalization · EPLocalization.Diagram.isotropyHom_spec; for ψ an equivalence each factor is bijective Chapter10/StrongBundleLocalization · EPLocalization.Diagram.bijective_isotropyHom. The germ q and its lift Q are not formalized (G.124).

X.22 Definitions 10.5.1 and 10.5.2 (2017: 10.14 and 10.15), Remark 10.5.3

Definition 10.5.1 (2017: 10.14) (V10.55) Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.idHom Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.localizationPrefunctor. The objects are a one-field structure around $\mathcal{SEP}$, the morphisms the classes; without a composition (G.123) this is a quiver, with the identities $[1_W]$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.id and the prefunctor $\Psi \mapsto [1, \Psi]$ Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.ofHom, which maps 1_W to $[1_W]$ Chapter10/StrongBundleLocalization · SBLocalization.SEPLoc.localizationPrefunctor_map_id.

Definition 10.5.2 (2017: 10.15) (V10.56) Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.IsGeneralizedIso: some representative has Ψ in F (Φ is in F for every diagram). The reversed diagram Chapter10/StrongBundleLocalization · SBLocalization.SDiagram.symm of such a representative, and $[1, \Psi]$ for Ψ in F, give examples Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.isGeneralizedIso_symm Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.isGeneralizedIso_ofHom. *Witness* Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.negDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isGeneralizedIso_negDiagram: $[1, P']$ for the projection $P' : (-\mathrm{id})^* W' \to W'$ of §X.11, between two different strong bundles.

Remark 10.5.3 (V10.57) Chapter10/StrongBundleLocalization · SBLocalization.GenSBMap.remark_10_5_3. For a representative with Φ, Ψ in F, ϕ and ψ are equivalences and Φ, Ψ are equivalences of the total spaces (Definition 10.4.1); Theorem 10.3.10 makes $[d]$ and the class of the diagram of total spaces generalized isomorphisms, and Lemma 10.3.6 makes $|[D]|$ a homeomorphism. The strong bundle version of Theorem 10.3.10 itself needs the composition (G.123).

X.23 Theorem 10.5.4 (2017: 10.9), Remark 10.5.5, Proposition 10.5.6

The push-forward along a diagram (V10.58) Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pullback. For Φ, Ψ in F, $D_*f = \Psi_*\Phi^*f$ with §10.4's pull-back and push-forward, and D^*g is the push-forward along the reversed diagram. That they map section functors to section functors, are mutually inverse and do the same on sc$^+$-section functors Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.mapsTo_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.invOn_pullback_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.bijOn_pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.bijOn_pushforward_plus comes from Theorem 10.4.5. *The text's step* Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward_eq_of_isRefinement: for a refinement by H, $\Psi_*\Phi^*f = \Psi_*H_*H^*\Phi^*f = (\Psi\circ H)_*(\Phi\circ H)^*f = \Psi'_*\Phi'^*f$, by Theorem 10.4.5 for H, Proposition 10.4.4, and Propositions 10.4.2, 10.4.3 for the naturally equivalent pairs, which have the same orbit maps. *Through the orbit spaces.* Step 1: $|\mu(\sigma,w)| = |w|$ Chapter10/StrongBundleLocalizationSections · EpStrongBundle.π_μ, so a section functor takes isomorphic objects to one orbit Chapter10/StrongBundleLocalizationSections · EpStrongBundle.π_eq_π_of_isFunctorialSection, and two section functors with $|f(x)| = |g(x)|$ for all x agree: a morphism $(\sigma, f(x)) : f(x) \to g(x)$ has $\sigma \in G_x$, and $f(x) = \mu(\sigma, f(x))$ since f is functorial Chapter10/StrongBundleLocalizationSections · EpStrongBundle.eq_of_forall_π_eq. Step 2: $|(D_*f)(x')| = |[D]|(|f(x)|)$ whenever $|[d]|(|x|) = |x'|$ Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.π_pushforward. Step 3: $|[d]|$ is determined by $|[D]|$ through the square, $|P|$ being onto Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.base_orbitMap_eq_of_orbitMap_eq. Hence diagrams with the same $|D|$ have the same push-forward Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pushforward_eq_of_orbitMap_eq and, reversed, the same pull-back Chapter10/StrongBundleLocalizationSections · SBLocalization.SDiagram.pullback_eq_of_orbitMap_eq.

Theorem 10.5.4 (2017: 10.9) (V10.59) Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.theorem_10_5_4. $[D]_*$ and $[D]^*$ use a representative with Ψ in F chosen once Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.rep Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pushforward Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pullback; every other such representative has the same $|D|$, an invariant of the class, hence the same push-forward and pull-back Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pushforward_eq Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.pullback_eq. The rest is the previous paragraph for the chosen representative, the last assertion with Step 3. The Fred part is G.125. *Witnesses* Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.bijOn_pushforward_negDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.pushforward_foldDiagram: $W' \xleftarrow{P'} (-\mathrm{id})^*W' \xrightarrow{P'} W'$ refines $1_{W'}$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isRefinement_foldDiagram Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.cls_foldDiagram, so its push-forward is that of $[1_{W'}]$, the identity Chapter10/StrongBundleLocalizationExamples · SBLocalization.SDiagram.pushforward_id.

Remark 10.5.5 (V10.60) Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.remark_10_5_5. The positive half is the last assertion of Theorem 10.5.4 Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.pushforward_inv_eq_pushforward_id; for the negative half the text gives no example. *The invariant* Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.IsotropyRel: $g \in G_x$ and $g' \in G_{x'}$ are related by d if $g = a\circ\phi(k)\circ a^{-1}$ and $g' = b\circ\psi(k)\circ b^{-1}$ for some z, $k \in G_z$, a, b; the relation contains the graph of every γ Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.isotropyRel_isotropyHom, and it does not change under refinement, through the conjugations by the natural transformations Chapter10/StrongBundleLocalizationExamples · EpNatTrans.isotropyHom_eq Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.IsRefinement.isotropyRel_iff, nor under equivalence Chapter10/StrongBundleLocalizationExamples · EPLocalization.Diagram.Equiv.isotropyRel_

iff. *The example.* $W = \mathbb{R} \times \mathbb{R}^k \to \mathbb{R}$ over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ with $\mu((g,x),w) = w$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3EpBundle, and Inv over $(g,x) \mapsto (g^{-1},x)$, a functor since $\mathbb{Z}/3$ is abelian and a strong bundle equivalence as its own inverse Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3InvHom Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.z3InvHom_mem_equivalences. Inv is the identity on objects, so $|[\mathrm{Inv}]| = |[1_W]|$ Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.orbitMap_ofHom_z3InvHom and the base orbit maps agree Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.base_orbitMap_ofHom_z3InvHom. For $[1_W]$ the generator of $G_0 = \mathbb{Z}/3$ is related to itself Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.isotropyRel_id; for $[\mathrm{Inv}]$ a related pair is (k,k^{-1}) up to conjugation, which is trivial in $\mathbb{Z}/3$, and the generator is not its own inverse Chapter10/StrongBundleLocalizationExamples · StrongBundleLocalizationExamples.not_isotropyRel_inv. So the base classes differ, and with them the classes. By Remark 10.5.3 the base classes are two distinct generalized isomorphisms with the same orbit map: the non-uniqueness claimed in Remark 10.3.18 (G.119, closed).

Proposition 10.5.6 (V10.61) Chapter10/StrongBundleLocalizationSections · SBLocalization.GenSBMap.iff_of_isGeneralizedIso. A representative with Φ, Ψ in F carries a property invariant under strong bundle equivalences from W to W'' to W'. For the local regularity property this needs Proposition 10.4.6 (G.122), so Proposition 10.5.6 itself is G.126; the text's "see Proposition 10.4.7" is read as Proposition 10.4.6. Theorem 10.5.7 rests on Proposition 10.4.7 and on the weak fibred product of strong bundles (G.127); as for Theorem 10.3.19, the example of Remark 10.5.5 suggests that it does not hold as we read it; a Lean check is pending. The 2017 Theorems 10.7 and 10.10, without local regularity, do not hold as we read them, by the example.

X.24 Witnesses and the hypotheses

Hypotheses taken, with their witnesses. The nonempty object set in Proposition 10.1.3: the covering of $\mathbb{Z}/2 \ltimes \mathbb{R}$ (§X.2). `Admissible`: every equivalence (§X.9), and the pair groupoid of $\mathbb{Z}$ over the point, and $\pm$id on $\mathbb{Z}/2 \ltimes \mathbb{R}^2$ (§X.10). Proposition 10.1.9 for the lifts, in Proposition 10.1.19: the identity (§X.7). The two Hausdorff hypotheses of Proposition 10.1.14 are the text's. §§10.3 and 10.5 take no hypothesis.

Every definition has a non-identity witness. Equivalences: $\mathbb{Z} \to$ pt, the refinement of $\mathbb{Z}/2 \ltimes \mathbb{R}$, $-$id of $\mathbb{Z}/2 \ltimes \mathbb{R}^2$; the weak fibred product: §X.10; strong bundle equivalences, push-forwards, natural equivalences of strong bundle functors, morphisms of $\mathcal{SEP}$: the bundle over $-$id (§§X.11–X.15); equivalent diagrams, generalized isomorphisms that do not come from isomorphisms, equivalent pairs (X,β) and objects of $\mathcal{EP}_{reg}$: the pair groupoid of $\mathbb{Z}$ and the point (§§X.16–X.20); refinements of diagrams of strong bundles, generalized strong bundle isomorphisms and their push-forwards: the bundle over $-$id, and distinct classes with the same orbit map: the bundle over $\mathbb{Z}/3 \times \mathbb{R} \rightrightarrows \mathbb{R}$ (§§X.22, X.23).

X.25 What is not formalized

- §10.1: $\mathcal{EP}_{lreg}$ (G.111); Remark 10.1.8 and Appendix §10.6.3 (G.112); Propositions 10.1.9 (G.113) and 10.1.10 (G.114); Proposition 10.1.19 without the hypothesis (G.115).
- §10.2: the agreement of the iterated sub-M-polyfold structures with those induced from the triple products (G.116).
- §10.3: the universal property of the localization (G.117); Proposition 10.3.16 and the (LR), (SLR) parts of Theorem 10.3.21 (G.118); Theorem 10.3.19 (G.120). The non-uniqueness in Remark 10.3.18 (G.119) is closed by §X.23. Remark 10.3.13 has no formal content.
- §10.4: the Fred part of Theorem 10.4.5 (G.121); Propositions 10.4.6 and 10.4.7 (G.122).
- §10.5: the transitivity of the equivalence of diagrams of strong bundles, the composition in $\mathcal{SEP}(F^{-1})$ and the strong bundle Theorem 10.3.10 (G.123); (10.24) with generalized vertical maps

and the local expressions q, Q (G.124); the Fred part of Theorem 10.5.4 (G.125); Proposition 10.5.6 beyond the reduction (G.126); Theorem 10.5.7 (G.127).

X.26 What the formalization taught us

The equivalences of §10.1 are the sc-smooth versions of equivalences of categories, and most proofs of the section are categorical arguments plus one local fact: a functor which is a local sc-diffeomorphism on objects at $s(\varphi)$ is one at $t(\varphi)$. The text uses this fact silently in Proposition 10.1.6, where the argument covers only the image of F; with it the proof is two lines. Proposition 10.1.5, quoted without proof, and the fullness and faithfulness of TF, left to the reader, are short once tangent morphisms are known to be determined by their base and source vector. The invariance results new in 2021 split into two kinds: those about $\mathbf{X}^T$ ((LUC), the reduced groupoids, (SLUC)) go through Theorem 7.2.22 and one closure argument; those about natural representations ((LR), (SLR), Proposition 10.1.10) need a transport of natural representations that Chapter 7 does not have, and remain open here, and with them Proposition 10.3.16 and Theorem 10.3.19. In §10.2 the weak fibred product needs much less than an equivalence: one local sc-diffeomorphism on objects for the structure, nothing for properness. In §10.3 the two places where the text argues with spaces, "the same space" for the associativity and the direct check that L is an equivalence, become functors and two-out-of-three arguments, and the local half of the hard direction of Theorem 10.3.10 needs a different route, through a local right inverse and local injectivity. In §10.4 the push-forward is well defined on section functors because of fullness (10.22), and once (10.22) is in hand its sc-smoothness is a local formula with two local inverses; Propositions 10.4.2–10.4.4 do not need sc-smoothness at all. In §10.5 the push-forward along a generalized strong bundle isomorphism is best seen through the orbit spaces: a section functor is determined by the orbits of its values, so the push-forward depends only on $||[D]||$, which gives the independence of the representative and the last assertion of Theorem 10.5.4 at once; and the example the text does not give for Remark 10.5.5 needs an invariant finer than the orbit maps, the relation of the isotropy homomorphisms γ, which also settles the non-uniqueness left informal in Remark 10.3.18.

Concordance

Every declaration cited in this book, by file, with its line in the source. 6702 declarations in 348 files. The web edition also shows each exact formal statement.

Polyfold/Chapter1/Book2021.lean. `remark_1_1_2_2021` (63); `remark_1_1_2_2021_openMapping` (82); `proposition_1_1_11_2021` (107).

Polyfold/Chapter1/ChainRule.lean. `IsSc1With.norm_sub_le_of_segment` (68); `IsSc1With.exists_delta` (95); `IsSc1With.hasFDerivWithinAt_comp` (125); `IsSc1With.tangentMapWith_comp` (245); `IsSc1With.comp` (256); `theorem_1_21'` (269); `theorem_1_21` (281); `IsSc1.tangentMap_congr` (289); `IsSck.congr` (298); `IsSck.comp` (306); `IsScSmooth.comp` (317); `IsSck.comp_of_isRelOpenInQuadrant` (323).

Polyfold/Chapter1/Degeneracy.lean. `quadrantDeg` (60); `degIndexWith` (65); `coordMap` (77); `coordMap_surjective` (82); `finrank_quotient_ker_coordMap` (91); `mem_ker_coordMap` (99); `eq_zero_of_eventually_nonneg_mul` (110); `eventually_add_smul_mem_quadrant` (122); `map_ker_coordMap_le` (141); `quadrantDeg_eq_of_equiv` (158); `quadrantDeg_zero` (177); `lemma_1_24` (184); `lemma_1_24_dim` (204); `degIndex` (226); `degIndex_eq` (229); `degIndex_quadrant` (237).

Polyfold/Chapter1/ExpositionLemmas.lean. `ScScale.finiteDimensional_of_surjective_incl` (73); `ScBanach.smooth_eq_top_of_finiteDimensional` (94); `ScBanach.toBase_bijective_of_finiteDimensional` (100); `ScBanach.isoConstOfFiniteDimensional` (109); `ScBanach.isoConstOfFiniteDimensional_apply` (115); `ScBanach.IsScSplitting.isTopCompl` (126); `ScBanach.IsScSplitting.scProj` (134); `ScBanach.IsScSplitting.scProj_apply` (150); `ScBanach.IsScSplitting.scProj_idem` (154); `ScBanach.IsScSplitting.range_scProj` (159); `ScBanach.IsScSplitting.ker_scProj` (167); `ScOperator.idem_lvl` (175); `ScOperator.levelSub_range_eq` (180); `ScBanach.isScSplitting_of_idem` (196); `ScBanach.isScSplitting_iff_exists_scProj` (234); `ScBanach.inclBound` (254); `ScBanach.one_le_inclBound` (256); `ScBanach.norm_incl_le` (262); `ScBanach.norm_toBase_le` (265); `ScBanach.norm_inclK_le` (275); `ScBanach.exists_smooth_mem_coset` (293); `ScBanach.mk_mem_quotient_smooth_iff` (363); `continuousOn_apply_iff_uniformOnCompact` (390); `IsSc1With.deriv_uniformOnCompact` (440); `ScBanach.lvlSet_const` (456); `isSc1_iff_contDiffOn_const` (467); `isSc1_iff_fderivWithin_const` (481); `isScSmooth_iff_contDiffOn_const` (492); `isSc1_iff_contDiffOn_of_isRelOpenInQuadrant` (506); `isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant` (512).

Polyfold/Chapter1/FredholmIndex.lean. `linIndex` (93); `HasFiniteIndex` (97); `finite_of_finite_quotient` (103); `finite_of_finite_ker_range` (111); `finiteDimensional_ker_of_comp` (137); `finiteDimensional_coker_of_comp` (143); `HasFiniteIndex.comp_and_linIndex` (153); `HasFiniteIndex.comp` (215); `linIndex_comp` (218); `HasFiniteIndex.equiv` (231); `linIndex_equiv` (238); `HasFiniteIndex.of_finiteRange_sub_id_and_linIndex` (250); `linIndex_eq_zero_of_finiteRange_sub_id` (306); `HasFiniteIndex.of_bijective_add_and_linIndex` (313); `HasFiniteIndex.of_comp_of_comp` (333); `fredholmIndex` (351); `IsFredholm.hasFiniteIndex` (355); `finiteDimensional_range_sub_id_of_equiv` (362); `bijective_one_add_of_norm_lt` (369); `hasFiniteIndex_comp_add_of_norm_lt` (384); `fredholmIndex_add_of_norm_lt` (401); `fredhol-`

mIndex_add_isCompactOperator (439); ScBanach.finrank_levelSub_of_le_smooth (468); ScFredholmSplitting.C_le_smooth (480); ScFredholmSplitting.fredholmIndex_lvl (488); IsScFredholm.fredholmIndex_lvl (498); scIndex_eq_fredholmIndex (505); proposition_1_10_index (509); proposition_1_10_index_lvl (517).

Polyfold/Chapter1/FredholmPerturbation.lean. isStrictMap_of_isClosed_range (62); isFredholm_of_banach (73); isCompactOperator_of_finiteDimensional_range (84); finiteDimensional_ker_one_add (103); exists_bound_below_on_compl (130); isClosed_range_one_add (181); exists_unit_far (215); pick (243); pick_notMem (246); chain (255); chain_mono (260); chain_eq_sup (265); chain_ne_top (276); chain_isClosed (285); coFG_range_one_add (292); isFredholm_one_add (333); isCompactOperator_sub_one_of_equiv (344); isFredholm_add_isCompactOperator (354).

Polyfold/Chapter1/Levels.lean. exists_delta_of_continuousOn (117); continuousOn_clm_comp_of_isCompactOperator (144); contDiffOn_clm_comp_of_isCompactOperator (170); ScBanach.shift_lvlSet (310); ScBanach.shift_tangentSet (320); ScBanach.shift_inclK_mem_lvlSet (331); ScBanach.incl_shift_inclK (337); ScBanach.tangent_inclK (343); IsSc0.levelMap_shift (354); IsSc0.shift (364); IsScDomain.shift (377); levelDeriv (389); toBase_levelDeriv (394); levelDeriv_zero (400); IsSc1With.exists_levelDeriv (409); IsSc1With.toBase_levelDeriv (432); IsSc1With.levelMap_tangentMapWith (439); IsSc1With.continuousOn_levelDeriv (455); IsSc1With.continuousOn_levelDeriv_clm (468); IsSc1With.hasFDerivWithinAt_level (483); IsSc1With.contDiffOn_level (592); IsSc1With.levelDeriv_eq (604); IsSc1Levels (644); IsSc1Levels.continuousOn_clm (656); IsSc1Levels.contDiffOn (662); IsSc1Levels.toBase_apply (674); IsSc1Levels.isSc1With (704); IsSc1With.isSc1Levels (734); proposition_1_16 (744); IsSc1With.shift (752); IsSck.shift (769); IsScSmooth.shift (789); IsSck.contDiffOn_levels' (802); IsSck.contDiffOn_levels (856); IsSck.contDiffOn_levels_of_le (862); IsSck.contDiffOn_base (868); IsScSmooth.contDiffOn_levels (874); proposition_1_19 (884); corollary_1_20 (919); corollary_1_20_base (932); IsScSmooth.of_contDiffOn_levels (943); IsScSmooth.of_contDiffOn_finiteDimensional (954).

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